GE Medical Systems

gehealthcare.com

Technical Publication

Direction 2152922-100
Revision 22

Book 1 of 5

Pages 1 - 216

GE Medical Systems
CT/i System Service Manual - Advanced

Legal Notes, TOC, Chapters 1, 2, 3, 4 & 5

Safety, Image Quality, Alignments, Checks & Theory

LEGAL NOTES

2152922-100, REVISION 22

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IMPORTANT PRECAUTIONS

LANGUAGE

WARNING

- **THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.**
- **IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.**
- **DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.**
- **FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.**

AVERTISSEMENT

- **CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.**
- **SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.**
- **NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.**
- **LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.**

WARNUNG

- **DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.**
- **FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.**
- **VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.**
- **WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.**

AVISO

- **ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.**
- **SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.**
- **NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.**
- **LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.**

ATENÇÃO

- **ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.**
- **SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.**
- **NÃO TENTE REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.**
- **O NÃO CUMPRIMENTO DESTE AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A‘ CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.**

AVVERTENZA

- **IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.**
- **SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.**
- **SI PROCEDA ALLA MANUTENZIONE DELL’APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.**
- **NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL’ADDETTO ALLA MANUTENZIONE, ALL’UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.**

警告

このサービスマニュアルには英語版しかありません。
ＧＥＭＳ以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。
この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意:

本维修手册仅存有英文本。
非 GEMS 公司的维修员要求非英文本的维修手册时，
客户需自行负责翻译。
未详细阅读和完全了解本手册之前，不得进行维修。
忽略本注意事项会对维修员，操作员或病人造成触
电，机械伤害或其他伤害。

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "damage in shipment" written on all copies of the freight or express bill before delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14-day period.

To file a report:

- Call 1-800-548-3366 and use option 8.
- Fill out a report on http://us44hdd21/sctq/InstallFulfill/InstalFulfillment.htm
- Contact your local service coordinator for more information on this process.

Rev. Jan. 5, 2005

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

LITHIUM BATTERY CAUTIONARY STATEMENTS

CAUTION Risk of Explosion

Danger of explosion if battery is incorrectly replaced. Replace only with the same or equivalent type recommended by the manufacturer. Discard used batteries according to the manufacturer’s instructions.

ATTENTION Danger d’Explosion

Il y a danger d’explosion s’il y a replacement incorrect de la batterie. Remplacer uniquement avec une batterie du même type ou d’un type recommandé par le constructeur. Mettre au rébut les batteries usagées conformément aux instructions du fabricant.

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives.

GE personnel, please use the GEMS PQR Process to report all omissions, errors, and defects in this publication.

End of Section

Revision History

Revision	Date	Reason for change
0	February 1996	Initial draft.
1	May 1996	Content Update
2	June 1997	General content update, Changes for 3.6 software, moved LFC into separate direction, added to theory section, modified alignments section, added index to replacements section, added replacement test & verification chapter.
3	August 1997	Updated content for changes for 4.0 software (Addition of Performix tube, HEMRC Assembly & associated boards, G5 Collimator, Anode transformer tank, HEMRC replacement parts, updated Functional Maps, & added HEMRC assembly to theory section). Also, updated kV board, CTVRC Control board, mA board, and added G2 Collimator boards, G2 Axial board, plus other general content corrections. Grouped subsystem hardware into their own chapters.
4	October 1997	Updated and added host related information. Added HV trouble shooting. Up-dated Service menus.
5	May 1998	CT/i (Octane) Information added. CQA974837 resolved.
6	September 1998	Fixed CQA: 984450, 981084, 983370, 965042, 984520, 983610, 981550, 984050, 983259, 975830
7	November 1998	Fixed CQA: 983271, 985456, 983968, 985454, 985453, 983055, 985572 Updated Screens (TS, PM, etc.) in Chapter 7
8	May 1999	Fixed CQA 986892, 990088, 992955, 993301, 993464, 991094, 992543, 993740, 992185(New Tube Change Flow Chart), Publication re-formatted into 5 Books.
9	August 1999	Fixed CQA 994314, 994677, 995131 and 995132
10	November 1999	Added troubleshooting to section 4 of table chapter. Fixed CQA 997878. Updated POR procedure to reflect the need for multiple X_F and X_R measurements.
11	February 2000	CQA 998095 Removed general steps from proprietary signal to noise check in HV and X-Ray chapter. CQA 998167 Image Calibration chapter Calibration process A and B corrected with when scaled cal should be selected. CQA 998166 Section 11.7 Table 2-10 corrected technique. CQA 998173 Section 12.6 Table 2-19 added specifications.
12	September 2000	PCN 199808, Additions to Gantry, Chapter 11 to reflect S/A style slipring updates. CQA 1001234 Metric Hardware Cross Reference Added CTi 6.x/4/x Changes
13	October 2000	A/B audit. Fixed CQA 1002415.

Revision	Date	Reason for change
14	November 2000	Fixed CQA 1006816.
15	September 2001	CQA 1008025 - Added jumper settings for 9.2 GB drives CQA 1013915 - Corrected line tap connection table (Table 13-2) Fixed SPR CTCge55549 Added new BIT3 board information
16	October 2001	CQA 10110189 - Added part number for Push Force Gauge to Section 4.3.1 of Chapter 12.
17	August 2002	Added “Appendix A - Torque” Chapter 2: Added Section 17.4 - Artifacts Caused by Collimator Grease - G5 Collimator
18	November 2002	Chapter 9: Updated Section 28.0 - Anode or Cathode Inverter Chapter 12: Enhanced Longitudinal Encoder Asm replacement.
19	January 2004	Ctcge 79574
20	September 2004	CQA 13007222
21	June 2005	PSR 13040894: In Section 22.0 - 46-309500G1 X-Ray Tube Replacement of Chapter 9, added step to 22.4 - Install New Tube.
22	December 2005	PQR 13052511: In Section 9.2.3 of Chapter 11, corrected in-lb torque value for collimator lock bolts.

Table of Contents

Preface
Publication Conventions

Please become familiar with the conventions used within this publication before proceeding.

Section 1.0
Safety & Hazard Information

1.1 Text and Character Representation

Within this publication, different paragraph and character styles have been used to indicated potential hazards. Paragraph prefixes, such as hazard, caution, danger and warning, are used to identify important safety information. Text (Hazard) styles are applied to the paragraph contents that is applicable to each specific safety statement. Words describe the type of potential hazard that may be encountered and are placed immediately before the paragraph it modifies. Safety information will normally include:

- Type of potential Hazard
- Nature of potential injury
- Causative condition
- How to avoid or correct the causative condition

EXAMPLES OF HAZARD STATEMENTS:

DANGER EXCESSIVE VOLTAGE CRUSH POINT

DANGER IS USED WHEN A HAZARD EXISTS WHICH WILL CAUSE SEVERE PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- **ELECTROCUTION**
- **CRUSHING**
- **RADIATION**

WARNING ROTATING EQUIPMENT BARE WIRES

WARNING IS USED WHEN A HAZARD EXISTS WHICH COULD OR CAN CAUSE SERIOUS PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- Potential for shock
- Exposed wires
- Failure to Tag and lockout system power could allow for un-command motion.

CAUTION Pinch Points Loss of Data Sharp Objects

Caution is used when a hazard exists which can or could cause minor injury to self or others if instructions are ignored. They include for example:

- **Loss of critical patient data**
- **Crush or pinch points**
- **Sharp objects**

NOTICE Equipment Damage Possible

Notice is used when a hazard is present that can cause property damage but has absolutely <u>no personal injury risk</u>.They can include:

- Disk drive will crash
- Internal mechanical damage, such as to the x-ray tube
- Coasting the rotor through resonance.

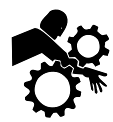
It is important that the reader not ignore hazard statements in this document.

1.2 Graphical Representation

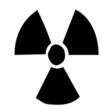
Important information will always be preceded by the exclamation point ⚠ contained within a triangle, as seen throughout this chapter. In addition to text, several different icons (symbols) may be used to make you aware of specific types of hazards that could possibly cause harm.

ELECTRICAL

MECHANICAL

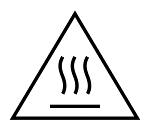


RADIATION

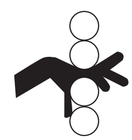


LASER

HEAT

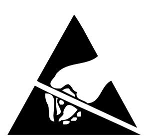


PINCH

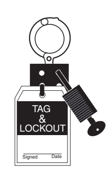


Some others make you aware of specific procedures that should be followed.

AVOID STATIC ELECTRICITY



TAG AND LOCK OUT



WEAR EYE PROTECTION

Section 2.0
Publication Conventions

2.1 Standard Paragraphs and Character Styles

Prefixes are used to highlight important non-safety related information. Paragraph prefixes (such as Purpose, Example, Comment and Note) are used to identify important but non-safety related information. Text styles are also applied to text within each paragraph modified by the specific prefix.

EXAMPLES OF PREFIXES USED FOR GENERAL INFORMATION

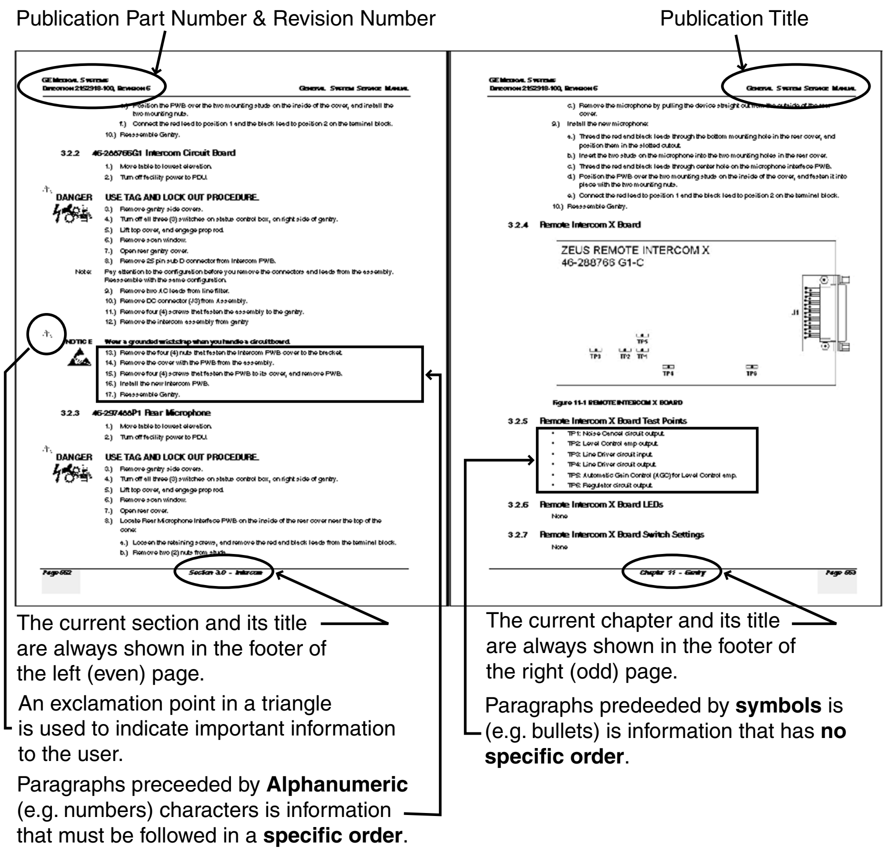
Purpose: Introduces and provides meaning as to the information contained within the chapter, section or subsection, Such as used at the beginning this chapter for example.

Note: Conveys information that should be considered important to the reader.

Example: Used to make the reader aware that the paragraph(s) that follow are examples of information possibly stated previously.

Comment: Represents "additional" information that may or may not be relevant.

2.2 Page Layout



Headers and footers in this publication are designed to allow you to quickly identify your location. The document's part number and revision number appears in every header on every page. Odd numbered page footers indicate the current chapter, its title and current page number. Even page footers show the current section and its title, as well current page number.

2.3 Computer Screen Output and Input

Within this publication different character styles are used to indicate computer input and output text. Character (input, output, and variable) styles are used and applied to the text within a paragraph so as to indicated direction. Computer screen output and input is also formatted using mono (fixed width) spaced fonts.

Example: Fixed Output

`This paragraph denotes computer screen fixed output. It's output is fixed from the sense that it does not vary from application to application.`

Example: Variable Output

This paragraph denotes computer screen output that is variable. Its output varies from application to application. Variable output is sometimes found placed between greater than and lesser than operators. For example: <variable_ouput>

Example: Fixed Input

This paragraph denotes fixed input. It's typed input that will not vary from application to application.

Example: Variable Input

This paragraph denotes computer input that can vary from application to application. Variable input will normaly be found placed between greater than and lesser than operators. For example: <variable_input>

End of Preface

Chapter 1
Information Sources, Quality & Safety

Section 1.0
Operational and Service Materials

This section lists the “Customer Operating and Service Material” that ship with each various HSA CT/i system. Additional copies are available through your local GE sales and service representative.

1.1 CT/i General Publications

GE’s CT/i Operational and Service publications include:

GE Part Number	Publication Title
2142878-100	Operator's Reference Manual (HSA CT/i)
2152916-100	Preinstallation Manual (preinstallation kit, only)
2152915-100	Safety Guidelines Manual (HSA CT/i)
2152918-100	System Manual (HSA CT/i)
2168070-100	HSA CT/i Load From Cold Procedures
46-018308	Safety Video Tape (HSA CT/i)
2152914-100	HSA CT/i Applications Precautions (HSA CT/i)
46-018316	FRU Manual (HSA CT/i)
46-018303	HiSpeed Advantage Drawings and Diagrams Vol.1 & 2
2152921-100	IRIX Command Guide (HSA CT/i)
2181737-100	HSA CT/i Option Installation

TABLE 1-1 GENERAL CT/I PUBLICATIONS

1.2 Customer Software

Operating and basic service software required for the operation, calibration, service and maintenance of the HSA CT/i system:

HSA CT/I 3.X, 4.X AND 5.X RELEASES

- IRIX Operating System Software CDROM
- Load From Cold CDROM: Contains application and service softwares.

1.3 Advanced CT/i Service Publications

In addition to the Service publications shipped with the system, additional manuals are available to GE Service personnel and licensed in-house service customers. Licensed in-house customers should contact your GE sales or service representative for the publications listed in table Table 1-2. Please make periodic inquires about these publications because they can change frequently as they are updated.

Licensed in-house customers should obtain CD-ROM 2152912 and publication 2152912-5. Please contact your GE sales or service representative from whom you purchased your service license from if you have not received the fore mentioned documents.

Advanced Class documentation parts list:

GE Part Number	Publication Title
2152922-100	Advanced System Service Manual in paper format.
46-018318	CT HiSpeed Functional Interconnect in paper format.
2136597-100	HiSpeed CT/i Advanced System Installation in paper format.
2152912	Advanced Service Information CD-ROM. Requires a personal computer attached to a CD-ROM reader. The CT/i system can be used to view this CD-ROM.
2152912-5	Installation Instructions for 2152912 in paper. Necessary to install and use CD-ROM 2152912.

TABLE 1-2 ADVANCED CT/I PUBLICATIONS

1.4 HSA Service Publications

The following table, Table 1-3, lists HSA Service publications necessary in maintaining HSA Service quality. Publications are numerically ordered and their GEMS classification provided under the column type.

Direction Number	Document Title	Type
2106475	Mounting Template	A
2108300	Fast Tube Change Procedure	C
2112035	Uninterrupted Power Supply Systems	A
2126799	Installation of Gantry Control Panel	A
2131408-100	CT HSA Patient Table (CT1)	A
2131409-100	CT HSA Gantry (CT2)	A
2131411-100	CT HSA Power Distribution Unit (PDU)	A
2135524-100	Gantry Controls Metal-Free Cradle	A
2135898-100	Metal-Free Cradle & Accessories Upgrade Kit	A
2136221-100	CT HP-DAS Technical Manual	C
2136597-100	HiSpeed CT/i Advanced System Install	C
2138987-100	Listing of Current CT Publications - General	A
2138988-100	Listing of Current CT Publications - Proprietary	C
2142877-100	Operator CBT (CDROM)	A
2142878-100	CT/i Operator Reference Manual (English)	A
2149733	CT Planned Maintenance	A
2149870-100	Supplemental User Manual for PDF CD-ROMS	A

Table 1-3 Applicable HSA Service Publications

Direction Number	Document Title	Type
2152911	CT Service Information	A
2152912	CT Service Information for newest scanners	C
2152914-100	CT/i Applications Tips & Precautions	A
2152915-100	CT/i Safety Guidelines	A
2152916-100	CT/i Pre-install	A
2152918-100	CT/i General Service Manual	A
2152921-100	IRIX Command Guide	A
2152922-100	CT/i Advanced System Manual	C
2152926-100	CT/i General Systems Install	A
2154688-100	CT/i Upgrade Manual	A
2159827-100	Z.x HSA to CT/i Upgrade Manual	A
2160288-100	CT/i Service Tips	A
2160574-100	Service Intro. Planner CT/i 3.4	A
2168070-100	CT/i Load From Cold Procedures	A
2181737-100	CT/i Option Installation (SGI Memory)	A
46-000030 F4874X	Installation Feedback Form F4874 (replaces 46-018079)	A
46-000030 F4879X	Form 4879X HHS Data Record for CT HiSpeed Advantage	A
46-018303	CT HiSpeed Systems,Vol. 1 & 2; (dwgs, diags, bds & schematics)	A
46-018310	PDU Door Diagram	A
46-018311	Table Door Diagram	A
46-018312	CT HiSpeed Advantage Gantry Door Diagram (ct2)	A
46-018316	CT HiSpeed Systems field replaceable units (FRU)	A
46-018318	CT HiSpeed Systems Functional Interconnect Drawings	C
46-018328	CT HiSpeed Advantage (PDU-AB Servo Amplifier)	A

Table 1-3 Applicable HSA Service Publications (Continued)

Section 2.0 Safety and System Quality Considerations

2.1 System Dangers and Precautions

MECHANICAL AND ELECTRICAL HAZARDS

1.) Become thoroughly familiar with all potential hazards to avoid injury. read direction 46-018302, CT Hispeed Advantage Safety Guidelines Manual or view the 46-018308 CT Hispeed Advantage Safety Video prior to servicing the gantry and HV subsystems.

2.) Only qualified service personnel should remove any covers or panels.

3.) turn off both the loop contactor and gantry HVDC (550) enable switch before you access the gantry.

RE-CHECK CALIBRATIONS

Always use and verify that the calibration of the HV Bleeder tool you use is current. HV miscalibration can lead to customer dissatisfaction, premature loss of an X-ray tubes or other system damage.

KEEP IT CLEAN

DAS CLEANLINESS Any dirt on the surface increases leakage current on the DAS filter or converter cards and causes the DAS to fail the drift spec. Always wear white cotton gloves, or use the board extractor, to insert or remove DAS boards. Fingerprints on the board cause trouble in high humidity environments.

AVOID STATIC ELECTRICITY

Wear a wrist strap to remove or replace electronic components and/or transfer them to or from an anti-static container. Never place boards on non-conductive surfaces, such as white bench tops, plastic covered books or non-conductive plastic bags.

2.2 General Safety Requirements

- Wear Safety glasses at all times.
- Use approved lifting methods.
- Observe Ryerson Road Standard Safety Protocol.

2.3 Torque Wrenches and Specifications

Many service operations on this CT scanner require a calibrated torque wrench. The use of a torque wrench may appear complicated because there are several standards and metrics. Conversion factors and charts are provided in Appendix A, to help simplify the task. Appendix A also provides additional information on torque and the proper use of torque wrenches.

2.4 Metric Hardware Cross Reference

Socket Head Cap and Thread Pitch	Hex Key Size Nominal	Hex Head Cap and Thread Pitch	Socket Wrench Size Nominal
M1.6 x 0.35	1.5mm	N/A	N/A
M2 x 0.4	1.5mm	N/A	N/A
M2.5 x 0.45	2.0mm	N/A	N/A
M3 x 0.5	2.5mm	N/A	N/A
M4 x 0.7	3.0mm	N/A	N/A
M5 x 0.8	4.0mm	M5 x 0.8	8.0mm
M6 x 1.0	5.0mm	M6 x 1.0	10.0mm
M8 x 1.25	6.0mm	M8 x 1.25	13.0mm
M10 x 1.5	8.0mm	M10 x 1.5	16.0mm
M12 x 1.75	10.0mm	M12 x 1.75	18.0mm

Table 1-4 American Standard Metric Hex/Socket Head Cap Screws to Tool Cross Reference

Socket Head Cap and Thread Pitch	Hex Key Size Nominal	Hex Head Cap and Thread Pitch	Socket Wrench Size Nominal
M14 x 2.0	12.0mm	M14 x 2.0	21.0mm
M16 x 2.0	14.0mm	M16 x 2.0	24.0mm
M20 x 2.5	17.0mm	M20 x 2.5	30.0mm
M24 x 3.0	19.0mm	M24 x 3.0	36.0mm
M30 x 3.5	22.0mm	M30 x 3.5	46.0mm
M36 x 4.0	27.0mm	M36 x 4.0	55.0mm
M42 x 4.5	32.0mm	M42 x 4.5	65.0mm
M48 x 5.0	36.0mm	M48 x 5.0	75.0mm

Table 1-4 American Standard Metric Hex/Socket Head Cap Screws to Tool Cross Reference

2.5 Slip Ring Considerations

Avoid contact, inhalation and ingestion of slip ring debris whenever you work with slip ring components. Take the following precautions when you handle slip ring material:

1.) Wear Neoprene or nitrile gloves to limit irritation and ingestion of metallic dust.
   - Do NOT remove gloves near an exposed slip ring. The powder inside the gloves can contaminate the ring.
   - Gloves: Large (Qty 100) 46-194427P347
   - Gloves: XL (Qty 100) 46-194427P348

2.) Use a HEPA (High Efficiency Particulate Air) vacuum cleaner to remove residual brush debris.
   - HEPA vacuum Cleaner: 46-297933P1
   - HEPA filter: 46-297948P1

3.) Use the HEPA vacuum cleaner to remove all existing brush debris from the brush blocks, brackets and slip ring covers before you service the slip ring brush assemblies.

4.) Use the HEPA vacuum cleaner to remove all existing brush debris from the gantry base and floor after you reassemble the slip ring covers.

5.) Wash your hands thoroughly after you service any slip ring components.

2.6 ESD and Device Handling

2.6.1 Electrostatic Discharge (ESD)

The circuit boards and disk drives for this system contain densely populated electronic components which are expensive and electrically sensitive. An electrostatic discharge (ESD) between 100 to 1000 V may damage a component. This is substantially less than the 3000 V discharge needed to feel any static. The ESD may cause an immediate failure, or it may weaken components to produce future, intermittent problems.

2.6.2 Proper ESD Handling

Always use the ESD strap pro-actively, see Table 1-5. Put the board or drive inside an anti-static bag or approved container before it is handled by a non-grounded person, moved from the grounded (ESD safe) area, or stored. Always place the board or drive top side up on a flat surface when it is unmounted. Never handle the part outside its anti-static container unless the surrounding surfaces and you are grounded. Discharge the outside of the container before transferring the part.

Pro-Active Action	Procedure
Turn power OFF	Turn power OFF before you touch, insert or remove parts containing electronic components.
Use wrist strap	Unless you are working near a live 30 V or more circuit, ground your wrist to the specially designed ground plug on the unit before you touch any parts. This includes connecting cables to a drive, board, device, or bulkhead. **Test your strap while wearing it with a specially designed meter.** If it fails, it may be due to dry skin; apply lotion to your wrist and test again. Throw away any strap that is more than three months old.
Don't let anything but your grounded hand touch the electronic FRU	Do not let your sleeve, tie, pen, Styrofoam cup, plastic manual binder or clothing touch the circuit board or disk drive. Wearing cotton clothes and shoes with rubber like soles may lessen how much ESD you generate walking across the room. Working in a room where relative humidity is under 20% can generate electrostatic voltages of 7000 to 35,000 Volts. However it only takes 100 V to destroy an EEPROM.
Use proper handling	Handle circuit boards, disk drives, or any electronic part as little as possible. Place them on an anti-static workbench pad or in a static dissipative bag that you have grounded. Do not stack them. **Store Circuit boards should be stored in an anti-static container**. Pink, blue, or clear poly bags do NOT give protection from external sources of ESD. If you have an anti-static box, you can use the box as a static free work surface once you ground it.
Treat failed parts the same as good	You don't want to add to the expense, complication and future unreliability of a part by allowing it to be repeatedly zapped. Treat failed parts with proper ESD handling.
Use a special vacuum	When you use a vacuum, be sure it is the type that prevents electrostatic buildup.

Table 1-5 Actions that Reduce the Chances of ESD damage

2.7 For Electromagnetic Compatibility (EMC) Systems

For CE Compliance on systems with EMC parts, it is critical that all covers, screws, shielding, gaskets, mesh, clamps, are in good condition, installed tightly without skew or stress. Proper installation following all comments noted in this service manual is required in order to achieve full EMC performance.

2.7.1 EMC

ElectroMagnetic Compatibility (EMC) describes an electronic system that curbs the electromagnetic influence between electronic systems. This means it minimizes how much electromagnetic energy it emits or conducts into the surroundings so that this energy is not dangerous nor distorts its own or another system's operation. It means it minimizes the electromagnetic interference from itself or other electronic systems.

2.7.2 EMI

ElectroMagnetic Interference (EMI) describes the energy that is emitted or conducted from an operating electronic system. This energy can be in many forms. It can be radio frequency (RF) waves, magnetic fields, electrical potential variations, electrical current leakage.

2.7.3 Maintaining EMC Compliant

- Only use power and signal wiring provided or specified by GE Medical Systems. Never use an adaptor to connect a power source plug. Do not change cable length or material. Use of cables not properly shielded and grounded may result in the equipment causing or responding to radio frequency interference in violation of the European Union Medical Device Directive (CE mark) and FCC regulations.
- Use the peripherals specified by GE Medical Systems.
- Install the scanner, peripherals, and replacement parts only as detailed in the manuals. Use CE certified parts.
- Reinstall all hardware before returning the scanner to clinical use.

2.8 CE Compliance

To maintain CE Compliance, please observe the following:

- Use only CE marked components for hubs, transceivers, peripherals, modems.
- Make sure transceiver is LOCKed into place on bulkhead (Ethernet) AUI connector.
- It is recommended that coax wire be used to connect system ethernet to site hub.
- FIBER OPTIC IS BEST for problem sites but expensive and requires an optical HUB.
- If un-shielded twisted pair (UTP) is used, wrap a ferrite ring or clamp to cable.

weight	<= 18 grams
resistance	>= 110 Ohms @ 25 MHz,
	>= 225 Ohms @ 100 MHz

2.9 Replacement Test and Verification

2.9.1 Introduction

The Component Replacement and Verification Requirement matrix found at the beginning of hardware chapters defines the minimum system retest required when you replace or adjust a component. The definition of replacement or adjustment includes reseating or swapping parts.

2.9.2 Verification Test Philosophy

This Component Replacement and Verification Requirement matrix provides testing instructions, in the event you replace or readjust a component during system installation or operation. This matrix instruction is “targeted” to test components, but is not intended as a test for failing components. The Component Replacement and Verification Requirement matrix ensures the replaced/adjusted components function properly with the rest of the System. All tests do stress the “target” components most probable failure mode.

Occasionally, additional tests were added to ensure quality/reliability levels after the replacement/adjustment of a component. The nature or frequency of a particular failure often determines whether you have additional test to run. (Example: Did a correction change the symptoms without completely fixing the problem, or did you really fix an intermittent failure?).

2.9.3 Verification Test Instructions

1.) The matrix lists the minimum number of tests that should be run after component adjustments or replacement.

Additional tests may be performed if deemed necessary.

2.) The component tests listed in the matrix may be run in parallel with the normal test sequence.

In this situation, don't consider the listed tests as additional tests (this applies to all System Tests).

3.) The “Perform System Functional Test” listed in the Verification Test column applies to the System Scanning Test per PROTOCOL LIST 20.8 and only doing the first 6 series of scans!

End of Chapter

Chapter 2
Image Calibration

Section 1.0
Introduction

This chapter explains what you must do to assure you get the best image quality possible. It contains descriptions of the Calibration Procedure, Initial Tests, System Functional tests and the Image Series which you run before you turn the system over to the customer. The Image Series sections instruct you to scan system phantoms, record the means and standard deviations, and evaluate the image quality of the resulting images.

CALIBRATION AND THE SYSTEM PERFORMANCE CHECK

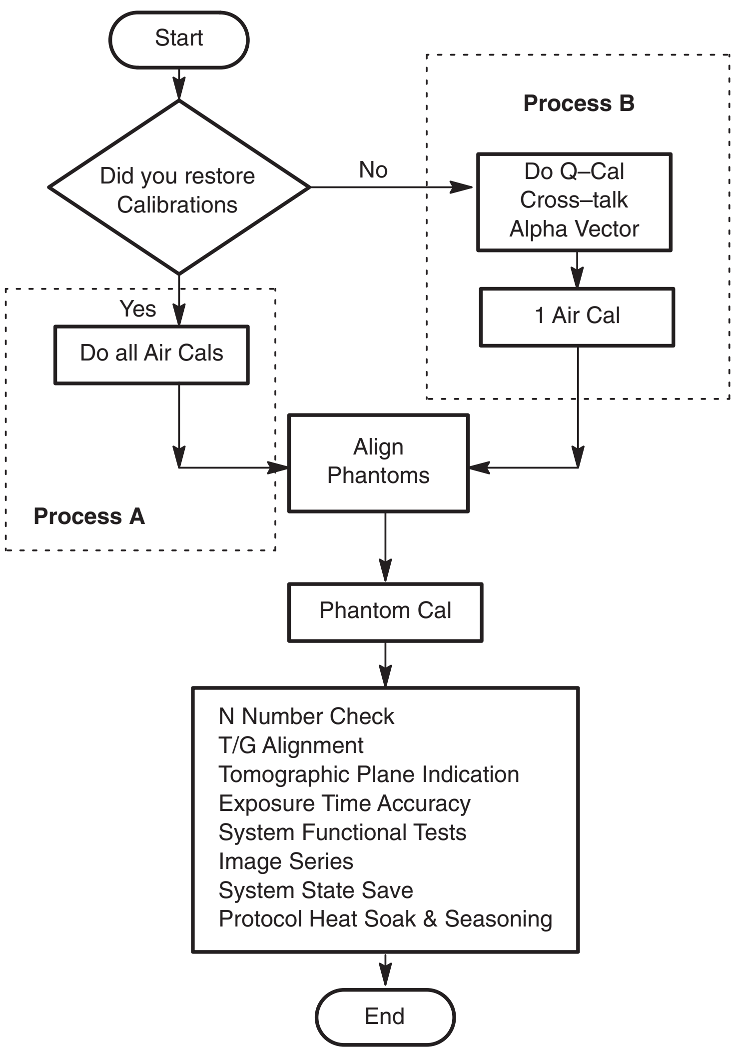


Figure 2-1 System Test Overview

Section 2.0
The Calibration Process

New scanners are delivered with a System State MOD that holds the unique values for the hardware characterizations and imaging calibration. Restore these files from the MOD. If you no longer have these files or the major system hardware has changed, you may need to create new calibration files.

2.1 Prepare the QA Phantom

When the Quality Assurance phantom is new, it requires someone to attach the correct label to it.

1.) Fill the Quality Assurance (QA) phantom with distilled water.
2.) Locate the multi-language sticker packet in the QA phantom shipping box.
3.) Attach the sticker with the customer's language to the face of the phantom hanger bracket.

2.2 Check for the Presence of Cal Files

If your system has a factory supplied state MOD, you should have loaded the system calibration files during the Installation. **To Restore System State,** refer to Section 13.0 on page 80 of this manual.

To determine whether or not the system contains calibration files:

1.) Select SERVICE
Select UTILITIES

```
Editor
Calculator
Calendar
Shell

Tube Display
```
Cal Analysis
```
Scan Analysis
DD File Analysis
Image Analysis
Install Options
Verify Options
Verify Security

Application Shutdown
```

2.) Select CAL ANALYSIS
3.) Select START... to open the Cal List/Select window.

IF a Cal list appears (as shown below), the factory supplied Calibration files are present. After you complete this section, proceed to 2.3 on page 61.

-or-

IF the system does NOT display a list of calibrations, proceed to 2.4 on page 62.

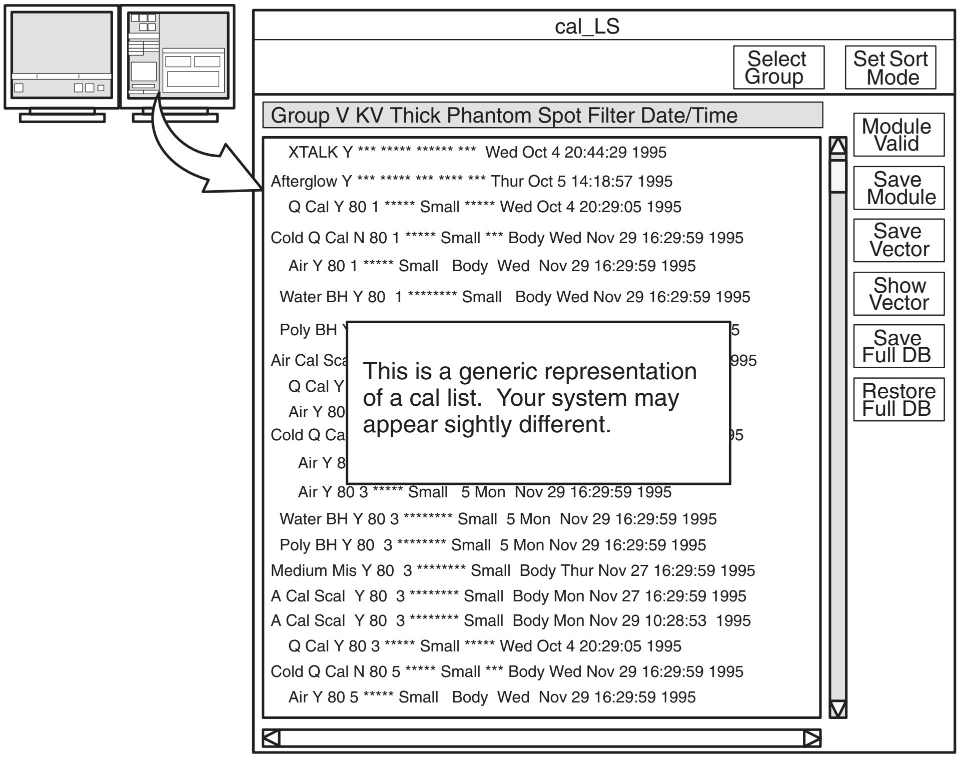


Figure 2-2 Example of a Calibration List

4.) Select DISMISS
5.) Select SHUTDOWN
6.) Display the **Toolchest** menu, and select SYSTEM DOWN.

Note: Do not log out — **Do NOT select LOG OUT at this time. Log out may inadvertently delete software in the `/usr/g` directory.**

7.) After the system completes the shutdown, restart the system.

2.3 Calibration Process "A" — "When NO Cal Files Exist"

1.) Select DAILY PREP  to warm up the tube.

2.) Select SCANNER UTILITIES. 

3.) Select DETAILED CALIBRATION 

.

4.) Refer to Figure 2-3. Select the following screen parameters:

a.) `Calibration Type:` AIR

b.) `KV:` SELECT ALL KV SETTINGS.

c.) `Aperture:` 1MM, 3MM, 5MM, 7MM, 10MM.

Note: **Do NOT select aperture SCALED**

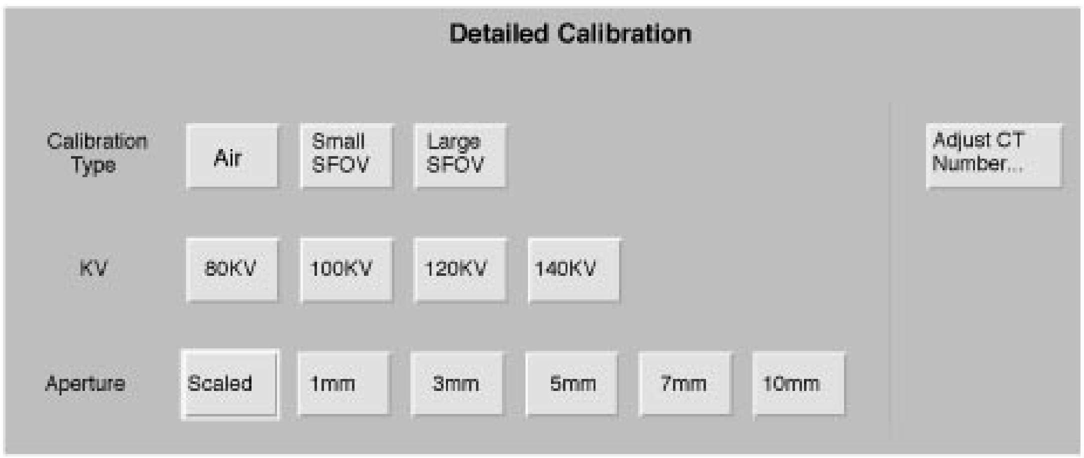


Figure 2-3 Detailed Calibration screen

5.) Run the selected Air Calibrations.

6.) When the calibration process completes, continue to the Phantom Self Calibration procedure, on page 64.

2.4 Calibration Process "B" — "When Cal Files Exist"

Follow this calibration process when the system does not have the factory a set of calibration files from the System State MOD.

1.) Do Q-Cals, use Section Section 19.0 for reference, if required.

2.) Do Cross-talk calibration, use Section Section 20.0 for reference, if required.

3.) Do Alpha Vector calibration, use Section Section 21.0 for reference, if required.

4.) Do Hot ISO* calibration, if available.

5.) Use the user select option, (NOT SmartCal), to acquire 1 Air Cal at 120kV/1mm/large focal spot.

6.) Use a bubble level and the Z-axis knob on the phantom holder to level the phantom.

7.) Refer to Figure 2-4. Acquire Air and Phantom Calibrations, with the following parameters:

a.) `Calibration Type:` AIR, SMALL FOV, LARGE FOV

b.) `KV:` SELECT ALL THE KV SETTINGS FOR SITE USE.

c.) `Aperture:` SCALED (equivalent to the RP SmartCal)

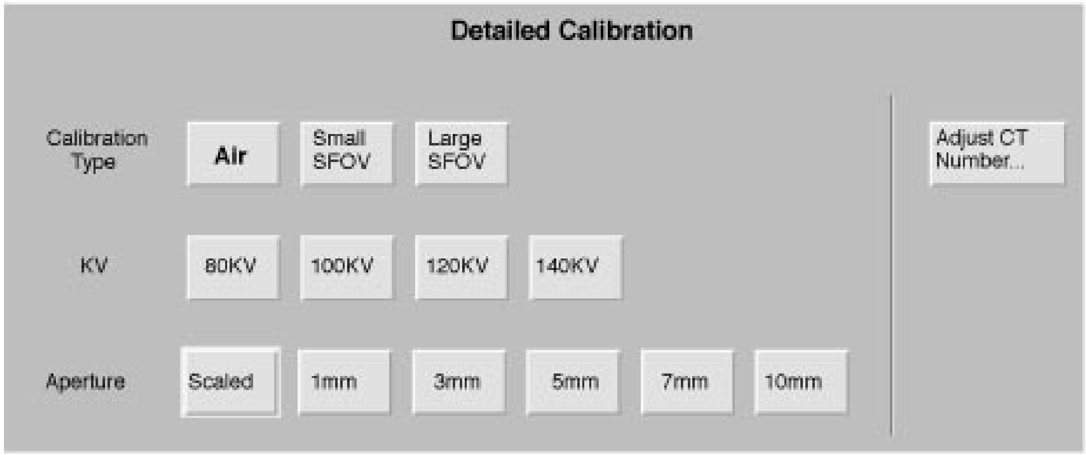


Figure 2-4 Detailed Calibration screen

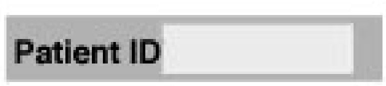
8.) When the calibration process completes, continue to the Phantom Self Calibration procedure, on page 64.

Section 3.0
How to Scan with Protocols

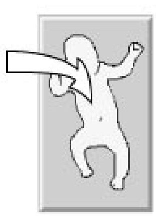
Manufacturing, Installation and Service protocols exist under Infant area 20. Use these to help you prescribe System Tests. If you know the Protocol Number, you may enter it on the Exam Rx Screen in the Protocol field. This describes how to select it graphically.

1.) Select NEW PATIENT.

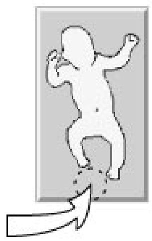
2.) Enter `Patient ID`.



3.) Click left on the “Infant” anatomical figure icon



4.) Click left on the area below the infant's feet (inside circle).



5.) Select a protocol from the list, to display the corresponding view edit screen. Optional Method: Enter the Protocol Number into the Protocol Number Field on the Exam Rx Screen.

Figure 2-5 Protocol List (Example)

Note: Check screen technique

Please read on-screen instructions carefully before preforming any scanning test. Verify the stated technique is selected before scanning and filling out a data sheet. If you do not, a incorrect protocol may be prescribe that is not correct for your system type. This may lead to a perceived image problem that only exists because an inappropriate technique was specified.

Section 4.0 'N' Number Check

Run the small and large phantom self calibrations before you acquire the Image Series, or **any time you doubt the N# values**. The phantom self calibration procedure modifies 48cm poly phantom and 20cm water phantom CT numbers until water has a CT number equal to zero. After a successful self calibration, any water phantom scanned against a small or large CAL has a **CT number** near zero.

1.) Select EXAM RX

2.) Place the QA phantom on the phantom holder and align the water section with the laser lights.

3.) Select SCANNER UTILITIES
4.) Select CENTER PHANTOM from the Scanner Utilities Menu and center the Phantom.
5.) Select the Service Protocol 20.9. **-OR-** Manually select the scan parameters detailed in Table 2-1 and 2-2.

PHANTOM	SCAN FOV	KV	MA	THICKNESS	SCAN TIME	# OF SCANS	DISPLAY FOV
20	S	80	170	10	4	2	25cm
20	S	100	240	10	4	2	25cm
20	S	120	200	10	4	2	25cm
20	S	140	140	10	4	2	25cm

Table 2-1 SMALL CAL N #

PHANTOM	SCAN FOV	KV	MA	THICKNESS	SCAN TIME	# OF SCANS	DISPLAY FOV
20	L	120	200	10	4	2	25cm
20	L	140	140	10	4	2	25cm

Table 2-2 LARGE CAL N #

6.) Complete the scans.
7.) Proceed to section Section 5.0, and use the worksheets in Table 2-3 and Table 2-6 to record the values, and adjust the N numbers.

Section 5.0
How to Analyze Calibration Data

1.) Select SERVICE
2.) Select UTILITIES
3.) Select IMAGE ANALYSIS
4.) Select START
5.) When the Browser opens, select an Exam/Series to analyze.
6.) Select MANUAL ROI
7.) A viewer opens with a 14cm X 14cm box located in the middle of the image.
8.) Click the right mouse button once to measure and display the ROI.

To examine subsequent images: Click the right mouse button 2X to close the image and result window, and select a new image from the browser.

9.) Use the Table 2-3 worksheet record the Mean values for each technique.
N# adjustment instructions begin with Step 10.

Small CAL	80kV	100kV	120kV	140kV
Exam#				
Slice 1				
Slice 2				
2 Slice Average				
N#				

Table 2-3 Phantom Self Calibration Worksheet - Small Cals

Large CAL	80kV	100kV	120kV	140kV
Exam#	n/a	n/a		
Slice 1	n/a	n/a		
Slice 2	n/a	n/a		
2 Slice Average	n/a	n/a		
N#	n/a	n/a		

Table 2-4 Phantom Self Calibration Worksheet - Large Cals

Use a 14 X 14 cm box; CTn = CTc-AvXc, where:

- CTn = New CT number
- CTc = Current CT number
- AvXc = 2 slice avg. mean of the 14cm X 14cm box.

Specification for the 2 Slice Average = 0.0 ±1.5

10.) Select SET/ADJUST N#S

11.) When the N# adjustment tool opens, type/enter the values you recorded in Table 2-3, into the table on the screen.

a.) Start a new exam to accept the values you entered into the table.

b.) Repeat the process until the N numbers meet the specification.

c.) Fill in the following N Number Summary Table.

Scan FOV	80kV	100kV	120kV	140kV
Small				
Large	n/a	n/a		

Table 2-5 "N" Number Summary

Specification for the 2 Slice Average= 0.0 ±1.5

Section 6.0
System Tests

System Tests, described in the following sections, exercise all aspects of the system. You want to use them to assure system integrity, before turnover to the Customer. Although the Means, Standard Deviation and Resolution specifications do not apply during system functional tests, treat any artifact or image anomaly as a failure.

If you encounter a failure during the system tests:

- Record any evidence of artifacts, such as rings, streaks, shading, cupping, noise or center artifacts.
- Correct artifacts, System Test or Image Series failures when they occur. Any delay in repairs could increase the number of retests.

Section 7.0
Table/Gantry Alignment Procedure

1.) Drive the table to its highest elevation.
2.) Check the relationship between the cradle and the cradle drive end:
   a.) Set the gap between the angled rollers, to center the cradle over the drive end.
   b.) Make sure the cradle does NOT touch either angled roller.
3.) To adjust the cradle, loosen the six cradle mounting bolts.
4.) Turn on the alignment lights.
5.) Advance the end of the cradle to the internal light.
   - The cradle end should be parallel to the light, assuming proper alignment light function.
   - If necessary, adjust the table to bring the cradle end into parallel alignment with the light.
6.) Align the front edge of the cradle with the laser light.
7.) Select the Service Protocol 20.4. **-or-** Manually select the scan parameters in Table 2-6.

Perpendicular Alignment

SCAN TYPE	DIRECTION	KV	MA	SFOV	THICKNESS	SCAN TIME	START LOC.	ALGORITHM
Axial	Head First	120	40	Large	5mm	2sec	S0	Bone
Axial	Head First	120	40	Large	5mm	2sec	I1000	Bone

Table 2-6 Perpendicular Alignment Scan Parameters

8.) Use over/under multi-image display to view both images at the same time.
9.) Orient the vertical cursor to 90 degrees, and use it to verify that the left and right edges of the cradle fall on a straight line. If the edged do NOT line up:
   a.) Loosen the table anchors at locations #5, #6 and #7.
   b.) Move the table half the distance.
   c.) Return to step 7. Repeat the procedure until the table edges line up.
   d.) Tighten the anchors, and torque to 75 ±6 N-m (55 ±5 ft.-lbs).
   e.) Reinstall all the table parts you removed to access the anchors, including covers, ground straps and the center support bar.

Section 8.0
Tomographic Plane Indication

1.) Place the QA phantom on the phantom holder.
2.) Turn ON the internal alignment lights, and drive the phantom into the gantry opening, until the black line on the phantom lines up with the internal laser lights.
3.) Verify that BOTH internal axial lasers line up along the black line on the QA phantom. If not, check table/gantry, cradle, and/or laser alignment.
4.) Center the phantom in the scan plane with the Calibration program.
5.) Select the Service Protocol 20.5 **-or-** Manually select the scan parameters in Table 2-7.

Tomo Plane Indication

SCAN TYPE	KV	MA	SFOV	THICKNESS	SCAN TIME	START LOC.	END LOC.	ALGORITHM
Axial	120	100	Small	1mm	2.0sec	I5.0	S5.0	Bone detail

Table 2-7 Tomographic Plane Indication Scan Parameters

6.) Display the image series, and locate the scan plane indicator, the longest bar in the bar pattern on the right side of the phantom.

The right side of the phantom corresponds to the side of the image labeled **L** on the display screen.

7.) On the HHS Data Sheet, record the scan location (shown on the image annotation) of the image with the darkest scan plane indicator (darkest long bar).
8.) If your system meets all the installation and alignment specifications, the image at scan location zero (S0.0) should contain the scan plane indicator. If scan location S1.0 or scan location I1.0 has the darkest bar, the system still meets the specification.
   - The scan plane deviation should equal S0.0 ±1.0mm.
   - If necessary, adjust the internal alignment light position to meet the S0.0 ±1.0mm requirement.
9.) Repeat the Tomographic Plane Indication test with the external alignment lights.
   a.) Use the external alignment light, and press the external landmark.
   b.) Verify the external light lines up along the black line on BOTH the left and right sides of the QA phantom.
   c.) The scan plane indication must fall within the S0.0 ±1.0mm specification.
10.) Check the box on **Form 4879**.

Section 9.0
Exposure Time Accuracy

1.) Turn on Monitor Enable to read the scan times.
   a.) Select END EXAM on the Exam Rx desktop to enable Diagnostic scanning.
   b.) Display the **Service** Desktop Manager.
   c.) Select SYSTEM INTEGRATION.

d.) Select DIAGNOSTIC DATA COLLECTION.

e.) Select the DDC protocol called `bleedersetup`.

f.) Select SYSTEM INTEGRATION.

g.) Select GENERATOR CALIBRATION.

h.) Select START to open the Generator Installation and Verification menu.

i.) Select BLEEDER SETUP (DDC) to open the Diagnostic Data Collection menu.

j.) Select MONITOR ENABLE (DDC) to display the scan times in the message log. Do NOT Dismiss, or close, the DDC window, because it turns OFF the monitor enable function.

2.) Toggle to the Exam Rx desktop.

3.) Select the Service Protocol 20.6 **-or-** Manually select the scan parameters in Table 2-8.

Exposure Time Accuracy

SCAN TYPE	KV	MA	SFOV	THICKNESS	SCAN TIME	START LOC.	END LOC.	PITCH
Scout	120	40	–	–	–	I10	S10	–
Scout	120	40	–	–	–	I75	S75	–
Scout	120	40	–	–	–	I300	S300	–
Axial	120	40	Large	1 mm	0.6 sec	S0		–
Axial	120	40	Large	1 mm	1.0 sec	S0		–
Axial	120	40	Large	1 mm	4.0 sec	S0		–
Helical	120	40	Large	10 mm	–	I145	S145	1:1

Table 2-8 Exposure Time Accuracy Scan Parameters

4.) Complete the scans, and record the scan times displayed in the message log, on **Form 4879**.

5.) When complete, toggle to the Service Desktop, and DISMISS the DDC window to turn OFF the Monitor Enable function.
Failure to turn OFF Monitor Enable fills the message log with kV, mA and scan times.

Section 10.0 System Scanning Test

Use the System Scanning Test to verify hardware functionality. Review images for visible artifacts, and review the message log for unacceptable errors.

1.) Place the QA phantom on the cradle.

- Drive the table to an elevation of 100.
- Align the black line on the phantom with the internal laser lights.

⚠ **NOTICE Avoid Detector Damage**

Never scan above 50 mA without first placing a phantom in the field of view. Levels in excess of 50 mA can cause temporary radiation damage to the detector that lasts several hours. If you acquire image series cals with a radiation damaged detector, the cals may cause artifacts in subsequent image series scans.

2.) Select Service Protocol 20.8. **-or-** Manually select the scan parameters in Table 2-9.

System Scan/Cust Qual Reli

Scan Type	kV	mA	SFOV	Thickness	Scan Time	Start Loc.	End Loc.	Tilt/Pitch
Scout	120	40	-	-	-	S200	I800	0°
Scout	120	40	-	-	-	S200	I800	90°
Cine	120	50	Large	10mm	30sec	S0	S0	1.0
Axial	120	50	Large	10mm	1.0sec	S0	S0	I30
Axial	120	50	Large	10mm	1.0sec	S0	S0	S30
Helical Full	120	50	Large	5mm	30sec	S70	I75	1:1
Helical Plus*	120	50	Large	5mm	30sec	S70	I75	1:1

Table 2-9 System Scanning Test scan parameters

3.) Complete the scans.

Section 11.0 Image Series Scan Protocol

The person who acquires an image series has certain responsibilities. They are to review the images and verify that they meet the specifications listed on data sheets, see the data sheets beginning on page 74. Responsibilities also include measuring means and standard deviation, and keeping a record of failures that occur during image series.

11.1 Recommended Scan Parameters

Unless otherwise stated, use the following scan parameters during the Image Series acquisition:

- Scan FOV equal to display FOV (Field of View)
- 512x512 matrix size
- Peristaltic ON

11.2 Failure Criteria

Consider *any* image series scan that does not meet specifications as failing. For means and standard deviations, 90% of the slices must pass.

Any failure on a particular technique requires at least 10 additional slices to evaluate effectively.

11.3 Leveling the Phantom

The phantom must be level. Systems with metal-free cradles have a phantom holder with a perpendicular adjustment (Z-axis) knob on it. **Each time you change phantoms**, make sure you use a bubble level, and the Z-axis knob on the phantom holder, to level the phantom.

11.4 Image Troubleshooting

If you loaded the factory supplied phantom calibrations on your system, and you followed the Z-Align and Air Calibration procedures in this manual, the image series should pass.

If you encounter image failures with the factory supplied cal files, consult the Image troubleshooting flowchart in Figure 2-11, on page 91.

11.5 Data Recording: Means and Standard Deviation

Any failure on a particular technique requires at least a 10 additional slices to evaluate effectively. For means and standard deviations, 90% of the slices must pass.

- Record **means** to two decimal places, and round to the nearest one-tenth, (one decimal place) when you compare the resulting values to the spec.
- Record **standard deviations** to two decimal places, then round off to one decimal place, to compare it to the spec.
- **Average standard deviations**: Use two decimal places to average the values, then round off to one place.

Before you record the means and standard deviations, check the image data sheets to determine whether to average the means and standard deviations, or record them slice by slice. Make sure you record all the required Image data on the HHS data sheets.

11.6 Image Series Definitions

Xc - Mean CT number for the specified center coordinates of the phantom image.

AvXc - Average Mean CT number for the center of the phantom image: Average the mean CT value for all specified center coordinates of all slices in an exam.

Xo - Mean CT number for the outside of the phantom image: Average the mean CT value for all specified outside coordinates of one slice.

AvXo - Average outside mean CT number for the number of slices in an exam.

AvSDc - Average image noise about the center image coordinate (measured as the standard deviation) of all slices in an exam.

AvSDo - Average image noise (standard deviation) for the outside of a phantom: Average of all outside coordinates of all the slices in an exam.

11.7 Image Series for CT/i System

1.) Place the 48cm phantom on the phantom holder.
   - Align phantom with the internal laser lights.
   - Center the phantom with the Calibration program.
2.) Select the Second Series of the Service Protocol 20.10 **-or-** Manually select the scan parameters in Table 2-10.

   Image Series 48cm

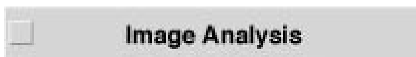
SCAN TYPE	PHANTOM	KV	MA	SFOV	THICKNESS	SCAN TIME	START LOC.	# OF SCANS
Axial	48cm	120	400	Large	10mm	4.0sec	S0	4

Table 2-10 CT/i Image Series Scan Parameters

3.) Record the data on the data sheet, and verify the images meet specifications.
4.) Proceed to section 11.8, to analyze the image series.

11.8 Analyze Image Series

1.) Display the Service **Utilities** menu.
2.) Select IMAGE ANALYSIS.



3.) When the Browser opens, select the 48cm Exam.
4.) Select SERIES MEANS to open a viewer and results window.
5.) The results window reports the series means data.
Record the results data on the data sheet, and verify it meets all specifications.

11.9 QA Image Series

1.) Place the QA phantom, 46-241852G1, on the phantom holder.
- Align with the black line with the laser lights.
- Center with the Calibration program.

2.) Select the Service Protocol 20.12 **-or-** Manually select the scan parameters in Table 2-11.

SCAN TYPE	PHANTOM TYPE	KV	MA	SFOV	THICKNESS	SCAN TIME	# OF SCANS	COMMENTS
Axial	QA#1	120	170	Small	10mm	2.0sec	4	S0.0
Axial	QA#1	120	300	Small	10mm	1.0sec	4	Bone Retro w/ DFOV = 15cm
Axial	QA#1	120	300	Small	10mm	1.0sec	4	S0.0
Axial	QA#2	120	170	Small	10mm	2.0sec	4	S35.0 Peristaltic OFF
Axial	QA#3	120	170	Small	10mm	2.0sec	4	S50.0

Table 2-11 Image Series QA scan Parameters

3.) Retrospectively reconstruct the first QA#1 scan with the Bone algorithm, and 15cm Display FOV, for later analysis.
Use the parameters in Table 2-11 to reconstruct the image.
4.) Record the data on the data sheet, and verify the images meet specifications.
5.) Proceed to section Section 12.0, to analyze the QA image series.

Section 12.0
Analyze QA Image Series

Return to the **Image Analysis** tool.

1.) Select the QA Exam from the Image Browser.
2.) Select QA#1 to open a viewer and results window.
3.) Refer to Figure 2-6, to adjust the image window:
   a.) Press and hold the middle mouse button.
   b.) Move the mouse in the horizontal and vertical directions to adjust the window width and Window level, until the line pair becomes visible.

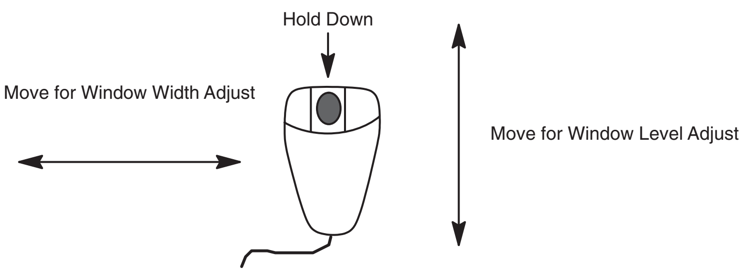


Figure 2-6 Window Width/Level Adjust with Mouse

4.) Refer to Figure 2-7. To position the ROI box, and analyze:
   a.) Move: Position the mouse cursor over the box. Press and hold the left mouse button, and move the mouse to reposition the box.
   b.) Accept: Click the right mouse button once, to accept the current position.
   c.) Accept: Toggle the right mouse button to accept the remaining ROIs.

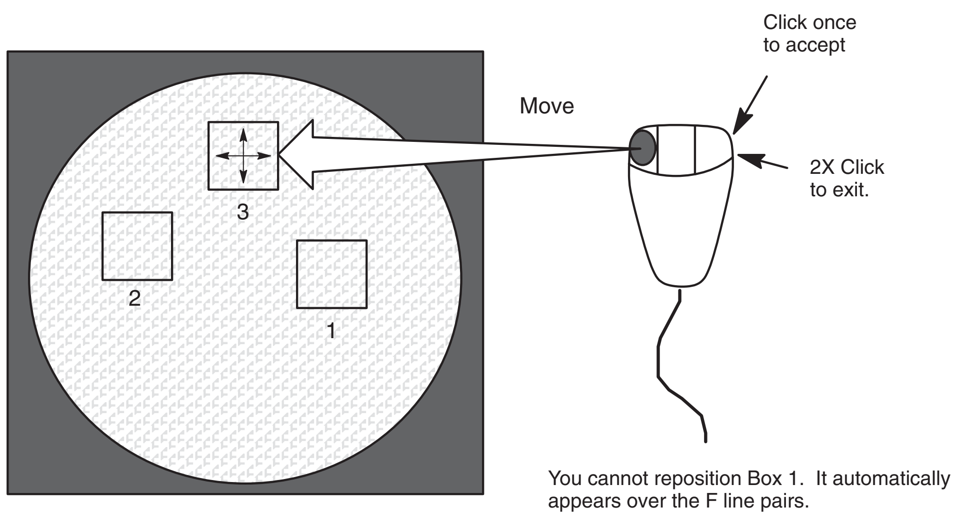


Figure 2-7 Reposition ROI with Mouse

5.) System calculates and displays the MTF and Contrast Scale in the Image Analysis Results window.
6.) Double click the right mouse button to exit.
7.) The software automatically evaluates the QA#2 & 3.
Select the Exam/Series and the QA#2 or QA#3 Softkeys to execute.
8.) Record the data on the data sheet, and verify the scans meet all specifications.

12.1 Image Data Sheet — Scan Parameters

Scan parameters: 48cm/L/10mm/120KV/400mA/4sec

EXAM # # OF SLICE	AVXC		AVXO	AVXO-AVXC		AVSDO	COMMENTS	SCAN TECH APROV
SPECS	—	—	—	±8.5	—	< 14.0	—	—

Table 2-12 Image Data Sheet - Scan Parameters (48cm/L/10mm/120KV/400mA/4sec)

Box Size: 45 x 45 pixels
Center Coordinates:256, 256
Outside Coordinates:256, 60
452, 256
256, 452
60, 256
Image Acceptance/Date: _______________
Certified Image Reviewer:_______________
Artifact Limits:

Artifact	Limits
Rings:	48/L; 30 to 36 counts 42/L; 15 to 18 counts 1mm; NA
Band:	8.0 counts
Band Radius:	0 to 23.5cm
Clump:	48/L; 3.0 SIGMA 42/L; 2.2 SIGMA 1mm; N/A
Center Spot:	N/A
Center Artifact:	N/A
Smudge:	48/L; 6.8 counts 1mm; 14.0 counts 42/L; 6.1 counts 1mm; 12 counts
Streaks:	4.0 counts

FIGURE 2-8 ARTIFACT LIMITS

12.2 Image Data Sheet — QA#1 Scan

QA#1 scan parameters: S/10mm/120KV/170mA/2 sec Scan at 0mm- Use this scan data for following Bone Retro. Record data on **Form 4879**.

EXAM # # OF SLICE	MTF	MTF 4 SLICE AVERAGE	CONTRAST SCALE	COMMENTS/ ARTIFACTS	SCAN TECH APROV
		—			
		—			
		—			
SPECS	—	0.68 to 1.0	110.0 to 130.0	—	—

Table 2-13 Image Data Sheet - QA#1 (S/10mm/120KV/170mA/2 sec/Bone Retro)

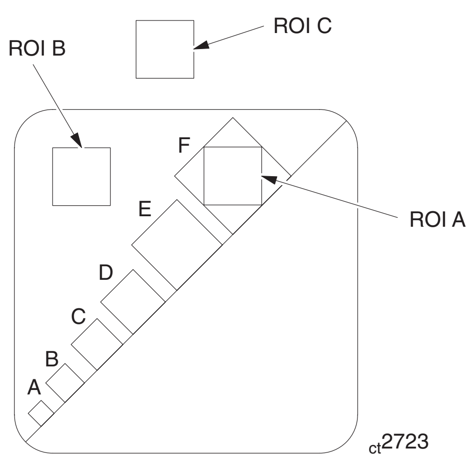
Box Size: 17 x 17 pixels

Image Acceptance/Date: _______________

Certified Image Reviewer:_______________

FOR REFERENCE ONLY:

Use the manual method for calculating MTF and Contrast Scale only when you cannot access Tool QA#1.



Contrast Scale = mean of B–mean of C.

SD_A = SD of ROI Box over F slits.

$SD_{ave} = (SD_B + SC_C)/2$

Modulation = $\sqrt{SD_A^{\ 2} - SD_{ave}^{\ 2}}$

MTF = 2.2 (Modulation/Contrast Scale)

Figure 2-9 Calculating MTF Manually

12.3 Image Data Sheet — QA#1 Scan

QA#1 scan parameters: S/10mm/120KV/170mA/2 sec/Bone Retro/DISPLAY FOV=15cm Scan at 0mm

EXAM # # OF SLICE	LINE PATTERNS VISIBLE			COMMENTS/ ARTIFACTS	SCAN TECH APROV
SPECS	B,C,D,E,F			—	—

Table 2-14 Image Data Sheet - QA#1 (S/10mm/120KV/170mA/2 sec/Bone Retro/FOV15cm)

Image Acceptance/Date: _______________

Certified Image Reviewer:_______________

QA#1 scan parameters: S/10mm/120KV/340mA/1 sec Scan at 0mm Record results on **Form 4879**.

EXAM # # OF SLICE	MTF	MTF 4 SLICE AVERAGE	CONTRAST SCALE	COMMENTS/ ARTIFACTS	SCAN TECH APROV
		—			
		—			
		—			
SPECS	—	0.65 to 1.00	110.0 to 130.0	—	—

Table 2-15 Image Data Sheet - QA#1 (S/10mm/120KV/340mA/1 sec)

Box Size: 17 x 17 pixels

Image Acceptance/Date: _______________

Certified Image Reviewer:_______________

12.4 Image Data Sheet — QA#2 Scan

QA#2 scan parameters: S/10mm/120KV/170mA/2 sec/Peristaltic OFFScan at 35mm (S35.0)

EXAM# AND SLICE#	VISIBLE HOLES (VIEW AT WINDOW OF 20)	CONTRAST FACTOR		COMMENTS/ ARTIFACTS	SCAN TECH APROV
SPECS	See Below	2.0 to 12.0		—	—

Table 2-16 Image Data Sheet - QA#2 (S/10mm/120KV/170mA/2 sec/Peristaltic OFF)

CONTRAST FACTOR	VISIBLE NUMBER OF HOLES: LOWER LIMIT*	VISIBLE NUMBER OF HOLES: UPPER LIMIT*	SMALLEST VISIBLE HOLE: SIZE
2.00 to 3.99	2	5	7.5mm
4.00 to 7.99	3	5	5.0mm
8.00 to 12.00	4	5	3.0mm

Table 2-17 Image Data Sheet - QA#2 (S/10mm/120KV/170mA/2 sec/Peristaltic OFF) Limits

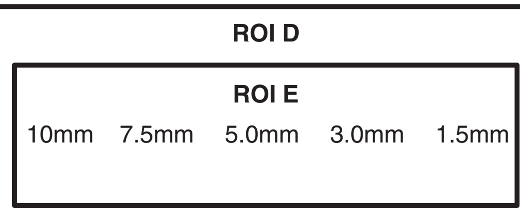
Required number of visible holes depends on the contrast factor and 2 out of the 4 scans taken must meet this specification.

Image Acceptance/Date: _______________

Certified Image Reviewer:_______________

FOR REFERENCE ONLY:

Use the manual method for calculating Contrast Factor only when you cannot access Tool QA#2.



ROI D 49 X 9 pixels x=256, y=200
ROI E 49 X 9 pixels x= 256, y=235
Contrast Factor = Mean E – Mean D

Figure 2-10 Calculating Contrast Factor Manually

12.5 Image Data Sheet — QA#3 Scan

QA#3 scan parameters: S/10mm/120KV/170mA/2 sec Scan at 50mm Record data on **Form 4879.**

Exam # # of Slice	Mean	Std Dev	Average Std Dev (average of 4 slices)	Comments/ Artifacts	Scan Tech Aprov
			—		
			—		
			—		
SPECS	0.0 ± 1.5	—	See Below*	—	—

Table 2-18 IMAGE DATA SHEET -- QA#3 S/10mm/120KV/170mA/2 sec

Box Size: 51 x 51 pixels at Coordinate 256,256

If tube has < 5000 scans, spec = 3.00 to 3.60.

If tube has > 5000 scans, spec = 3.00 to 3.80.

Image Acceptance/Date: _______________

Certified Image Reviewer:_______________

12.6 Image Data Sheet — QA#3 Evaluation

QA#3 S/10mm/120KV/170mA/2 sec (Step 5.5.1) Do not scan. Evaluate images from previous step.

Exam # # of Slice	AvXc	AvXo	AvXo-AvXc			Comments/Artifacts	Scan Tech Aprov
SPECS	0.0 ± 1.5	—	± 1.5			—	—

Table 2-19 IMAGE DATA SHEET -- QA#3 S/10mm/120KV/170mA/2 sec Evaluation

Box Size: 31 x 31 pixels
Center Coordinates:256, 256
Outside Coordinates:256, 95
417, 257
256, 417
95, 256

ARTIFACT LIMITS:

Artifact	Limit
Rings:	4.0/4.8 cts
Band:	2.8 cts
Band Radius:	0 – 8.5 cm
Clump:	N/A
Center Spot:	N/A
Center Artifact:	3.5 Std. Dev.
Smudge:	2.2 counts
	1mm; 4.0 counts
Streaks:	4.0 cts

Table 2-20 QA#3 Artifact Limits

Image Acceptance/Date: _______________
Certified Image Reviewer:_______________

Section 13.0 System State MOD

The procedure that follows tells how to create, update or restore the System State. The MOD used must have a UNIX file system. Any MOD formatted for images cannot be used.

Note: DO NOT SAVE State after you reload software UNTIL you restore the REAL State; new software puts system defaults on the disk. If you save state before you restore, you will be saving defaults.

Prevent this MOD from being labeled as an IMAGE ARCHIVE MOD because this step will format the MOD differently without mentioning there are system files on it. To investigate an IMAGE MOD, use a Shell and the DOS MODE commands shown on page 274.

1.) Load a new or the system MOD into the drive on the front of the console. An MOD that has been labeled for IMAGE ARCHIVE cannot be used because it has a different format than a system MOD.
2.) Select SERVICE
3.) Select PM
4.) Select SYSTEM STATE to open the System State Save/Restore menu.
5.) Select ALL or the items that are appropriate to archive.
   - Protocols — Scan, DDC, Archive
   - Characterization — Table, Gantry, InSite
   - Calibration — Tube, HV, Gantry, DAS, Detector, SW
   - Configuration — Network, Filming, Timezones, InSite
   - Auto Voice
   - Display Preferences
6.) Select SAVE or RESTORE
7.) When the archive operation completes, the LED on the drive will stop flashing and it will be safe to select FILE and QUIT from the pull down menu.
8.) Remove the MOD. Mark it so that no one will use it for IMAGE ARCHIVE.

RESTORE DISPLAY PREFERENCES

To restore the Display Preferences, bring applications down. Use the Service Utility called **Application Shutdown**. In a shell, enter: **`sysStateABB -restoreProtoFilm`**

`-saveProtoScan`	`-saveInfo`	`-restoreProtoArchive`
`-saveProtoDDC`	`-saveVoice`	`-restoreCharact`
`-saveProtoFilm`	`-saveAll`	`-restoreCal`
`-saveProtoArchive`	`-restoreProtoScan`	`-restoreVoice`
`-saveCharact`	`-restoreProtoDDC`	`-restoreAll`
`-saveCal`	`-restoreProtoFilm`	

Section 14.0 Tube Heat Soak and Seasoning

1.) If you ran the Tube Heat Soak and Seasoning during Generator Calibration, skip this section.
2.) Place a 48cm phantom in the beam, to eliminate X-Ray damage to the detector.

Ignore the image quality of the reconstructed images.

3.) Use the Service Heat Soak and Seasoning **scan protocol,** located under CT/i Install.

-or-

Manually select the scan parameters in Table 10-11.

4.) Select the 256 stat recon parameter, to decrease the reconstruction time.

5.) Turn OFF the auto film and auto archive functions.

6.) Acquire the series.

kV	mA	Time	ISD	# of Slices	Aperture
120	200	4	15 sec	40	1 mm
5.5 second inter-group delay					
120	40	1	1 sec	10	1 mm
2.2 second inter-group delay					
140	40	1	1 sec	20	1 mm

Table 2-21 Heat Soak and Seasoning Scan Parameters

Section 15.0 Thermal Test

Start the thermal test with a cold X-Ray tube. Allow the tube to cool at least 60 minutes before starting this test.

1.) Place the QA phantom on the phantom holder.
- Align the water section of the phantom with the internal laser lights.
- Use the Calibration program to center the phantom.

2.) Select the Service Generic Protocol 20.1 **or** Manually select the scan parameters in Table 2-22.

Service Generic Scan

SCAN TYPE	PHANTOM	KV	MA	SFOV	THICKNESS (MM)	SCAN TIME (SEC)	START LOC.	# OF SCANS
Axial	QA	120	200	Small	1	4.0	S0	1
Axial	QA	120	200	Small	10	4.0	S0	20
Axial	QA	120	200	Small	1	4.0	S0	1

Table 2-22 Thermal Test Scan Parameters

3.) Record the Std Dev data on the following data sheet, and verify the images meet specifications.

THERMAL TEST DATA SHEET

Exam #______________________	
1mm______________	
10mm_____________	
10mm_____________	
10mm_____________	
10mm_____________	
10mm_____________	1mm slice difference_________________
10mm_____________	
10mm_____________	10mm slice difference________________
10mm_____________	
10mm_____________	
10mm_____________	
10mm_____________	
10mm_____________	
10mm_____________	< 4.0 cts YES______________
10mm_____________	
10mm_____________	technique_______________________
10mm_____________	
10mm_____________	
10mm_____________	technician_______________________
10mm_____________	
10mm_____________	Date_______________________
1mm______________	

Section 16.0
Cradle Incrementation Test

Use cradle incrementation test to verify cradle position agrees with the commands sent from the OC and Gantry display is within SPEC over the 300 mm range.

1.) Cover the cradle with a blanket, or protective material.
2.) Load the cradle with a least 34kg (75 lbs) of weight.
3.) Extend a metric tape measure at least 300mm, and tape it on the cradle top, next to one of the edges.
4.) Turn on the patient alignment lights.
5.) Drive the cradle into the gantry, and align a convenient point on the tape measure, such as 50mm, with the internal axial lights.
6.) Use the scan RX menu to acquire a scan at location I100 mm.
Do not worry about technique and image quality during the cradle incrementation test.
7.) Turn on the patient alignment lights.
8.) Record the distance of cradle movement from your starting point.
9.) Repeat this procedure at the following scan locations:
    - I200 mm
    - I300 mm
    - 0 mm
10.) Verify your recorded measurements fall within the minimum and maximum specifications listed in Table 2-23.

SCAN LOCATION	MINIMUM DISTANCE (MM)	RECORDED DISTANCE	MAXIMUM DISTANCE (MM)
I100	99.75		100.25
I200	199.75		200.20
I300	299.75		300.25
S0	-0.25		0.25

Table 2-23 Cradle Incrementation Specs

Tech__________________________ Date __________

Section 17.0 Artifacts

17.1 Artifacts Defined

17.1.1 Center Smudge

17.1.1.1 Definition

A dark or light area, with no defined edges, located near the center of the reconstruction.

17.1.1.2 Method of Measurement (includes 12.5 cm / 5 inch)

- 48cm/L and 42cm/L images: position a 13x13 pixel ellipse over the smudge, and measure the mean of the smudge.
- Other techniques: position a minimum 13x13 pixel ellipse (169 pixels), over the smudge, and measure the mean of the smudge, If necessary increase the size of the ellipse to approximate the size of the smudge.

17.1.2 Reference Mean

- 48cm/L and 42cm/L techniques: position a 41x41 pixel ellipse about the center of the reconstruction circle, and use it to measure the reference, background mean.
- Other techniques: Use an ellipse at least 4 times larger than the ellipse used to measure the mean of the smudge, to measure the reference, background mean.

17.1.2.1 Failure

The technique fails when the difference between the mean of the smudge and the reference exceeds the corresponding limit value for that technique.

17.1.2.2 Application

10MM	7MM	5MM	3MM	1MM
48/L	—	—	—	48/L
42/L	—	—	—	42/L
—	35/M	35/M	35/M	—
35/L	—	—	—	—
20/S	—	—	—	20/S
12.5/S	—	12.5/S	—	—

Table 2-24 Center Smudge Techniques

If either the 48/L or 42/L pass the test for a given kV and aperture, consider both the 48/L and 42/L having passed that technique.

17.1.2.3 Failure Rate

80% of all scans at a given technique must meet the spec.

17.1.3 Rings

17.1.3.1 Definition

A dark or light circle, or partially closed circle, approximately 1 to 3 pixels in width.

17.1.3.2 Method of Measurement

Ring mean value: Deposit two ellipses bordering the ring or partial ring.

- Take care not to include pixels outside the ring within the ellipses. (You may magnify an image up to 3X to help determine if a ring or partial ring exists.)
- Most rings consist of a dark and light ring pair; measure one color ring at a time.

48cm/L and 42cm/L techniques: the ring must equal 0.5cm radius (1.0cm diameter), or greater.

- Measure rings smaller than 0.5cm radius for clump.
- Rings must have an arc of 30° or greater to be considered a ring.
- Bone detail images must have a ring with at least a180° arc to fail.

17.1.3.3 Background mean value

Position a 2cm x 2cm (or larger) box over an unmagnified image, so it includes the ring or partial ring, and measure the mean.

17.1.3.4 Failure

The technique fails when the difference between the ring mean and the background mean exceeds the corresponding limit value for that technique. The IA Program spec limit equals 1.2 times the spec limit of the manual measurement.

17.1.3.5 Application

10MM	7MM	5MM	3MM	1MM
48/L	—	—	—	48/L
42/L	—	—	—	42/L
—	35/M	35/M	35/M	—
35/L	—	—	—	—
20/S	—	—	—	20/S
12.5/S	—	12.5/S	—	—

Table 2-25 Ring Techniques

If either the 48/L or 42/L pass the test for a given kV and aperture, consider both the 48/L and 42/L as having passed that technique.

17.1.3.6 Failure Rate

48/L and 42/L techniques: 80% of all slices must meet the spec at a given technique

All other technique: No more than one ring every 250 slices

17.1.4 Streaks (General)

17.1.4.1 Definition

Straight dark or light lines, of any length, across an image, usually 1 to 3 pixels in width.

17.1.4.2 Method of Measurement

Mean of the Streak:

- Use the trace to outline the streak in a magnified or unmagnified image, then measure distance or deposit cursors to measure.
- Most streaks consist of a dark and light streak pair, measure only one color streak at a time.
- Measure the mean of the streak.

Background mean: Measure the mean on both sides of the streaks.

17.1.4.3 Failure

The technique fails when the difference between the mean of the streak and the background mean exceeds the corresponding limit value for that technique.

17.1.4.4 Application

All images fall under the streak spec requirements.

17.1.4.5 Failure Rate

It takes 100 streak free scans after the initial streak, to clear the streak failure. A repeat streak at the same location with 250 scans or less between streaks requires corrective action.

17.1.5 Streaks Caused by High Voltage Disturbances

17.1.5.1 Definition

Same as for **Streaks** above, except that High Voltage disturbances cause these streaks, which result in missing DAS views. Often streaks of this origin occur in multiples, and appear to emanate from a common point. We refer to multiple streaks of this type as `fan beam' streaks.

17.1.5.2 Method of Measurement

Use CD plot to look for missing views. A failure occurs when each group of 3 or fewer contiguous missing views creates streaks in the corresponding image.

17.1.5.3 Application

All images fall under the “Streaks Caused by High Voltage Disturbances” spec requirements.

17.1.5.4 Failure Rate

Run the Heat Soak/Seasoning on systems that fail either of the following specs:

- 48cm 120kv/400ma/1mm/1sec scans: Ten images may contain a total of one group of 1 to 3 contiguous missing views that cause streaks.
- All other techniques: One group of 1 to 3 contiguous missing views that cause streaks, allowed in 250 scans.

17.1.6 Center Artifact

17.1.6.1 Definition

A dark or light, well defined spot, 4 pixels in size, located in the center 16 pixels of the reconstruction circle.

17.1.6.2 Method of measurement

- Reference mean and standard deviation: Measure the mean and standard deviation of a 41 x 41 pixel box centered about X=256 and Y=256.
- Center Artifact Average (Ave.) The average of the 4 center pixel located at X (256,257) and Y (256,257) or the average of any four pixel square box which includes any of the four center pixels.

17.1.6.3 Failure

The technique fails when the difference between the Center Artifact Average and the Reference Mean exceeds the corresponding limit value for that technique.

17.1.6.4 Application

10MM	7MM	5MM	3MM	1MM
—	35/M	35/M	35/M	—
35/L	—	—	—	—
20/S	—	—	—	20/S

Table 2-26 Center Artifact Techniques

17.1.6.5 Failure Rate

80% of all scans at a given technique must meet the spec.

17.1.7 Center Spot

17.1.7.1 Definition

A dark or light area, consisting of 3 to 25 pixels near the center of reconstruction, that has no defined edges.

Two types of failures can cause this artifact:

- Center Spot: Compare the mean of a 5x5 pixel box near the center to a 21x21 pixel box about the center.
- Max pixel requirement: A positive center spot places a maximum limit on the maximum value(s) allowed in the 5x5 pixel box.

17.1.7.2 Method of Measurement

- Background mean: Measure the mean of a 21x21 pixel box centered at X=256, Y=256.
- Center spot mean: Measure the means of all 5x5 pixel boxes which include pixel X=256, Y=256 (25 5x5 boxes exist).
- Max Pixel value: Measure the most positive pixel in each of the 5x5 pixel boxes.
- Subtract the background mean from the center spot mean.

17.1.7.3 Failure

The technique fails when the difference between the Center Artifact Average and the Reference Mean exceeds the corresponding limit value for that technique.

100kV and 140kV Failure

- The center spot fails when the difference equals or exceeds 3.5
- The center spot fails when the difference equals or falls below -3.5.
- The Max pixel fails when the difference equals or exceeds 1.8 and the Max Pixel value equals or exceeds 4.0.

120kV Failure: (5mm and 10mm only)

- The center spot fails when the difference equals or exceeds 3.2
- The center spot fails when the difference equals or falls below -3.2.
- The Max pixel fails when the difference equals or exceeds 1.5 and the Max Pixel value equals or exceeds 4.0.

120kV failure: (1mm only)

- The Max pixel fails when the difference equals or exceeds 2.4 and the Max Pixel value equals or exceeds 6.4.
- No Center Spot for 1mm scans

80kV Failure

- The center spot fails when the difference equals or exceeds 3.5
- The center spot fails when the difference equals or falls below -3.5.
- No Max Pix for 80kV scans

17.1.7.4 Application

10MM	7MM	5MM	3MM	1 MM (MAX PIXEL ONLY)
12.5/S	—	12.5/S	—	12.5/S

Table 2-27 Center Spot Techniques

17.1.7.5 Failure Rate

90% of all scans at a given technique must meet the spec.

17.1.8 Clump

17.1.8.1 Definition

Light or dark areas of varying intensity, consisting of 3 or more contiguous pixels at the center of reconstructions

17.1.8.2 Method of Measurement

Reference mean and Standard Deviation (Std. Dev.): Measure the mean and standard deviation of a 41x41 pixel box centered about X=256 and Y=256.

Find all pixels in a 9x9 pixel box, centered about X=256, Y=256, that fall outside the limits of the reference mean plus/minus the value of the reference std. dev. times the limit number given for a technique.

17.1.8.3 Failure

A failing clump consists of three or more pixels that touch, and exceed the limit on the same side of the reference mean (more positive or more negative)

17.1.8.4 Application

10MM	5MM	3MM	1MM
48/L	48/L	48/L	—
42/L	42/L	42/L	—

Table 2-28 Clump Techniques

If at any technique, one of the above passes, consider that technique as having passed.

17.1.8.5 Failure Rate

80% of all scans at a given technique must meet the spec.

17.1.9 Banding

17.1.9.1 Definition

A dark or light circular area, greater than 3 pixels in width. The band may have less than 360 degrees of arc.

17.1.9.2 Method of Measurement

MEAN OF BAND

- Position two non-overlapping boxes, approximately the width of the band, over the worst portion of the band. (Use one box when the band occurs near the center of the image, and/or space is limited.)
- Use a square or rectangular box with an area of at least 50 pixels.
- The mean of the band equals the average of the box means.

BACKGROUND MEAN

- Measure the mean of a box inside the radius of the band, and the mean of a box outside the radius of the mean, on a radial through center of reconstruction.
- Use the average of the inside and outside means as the background mean.
- Use the same box size you used to measure the mean of the band. (Position both boxes on one side of the band, when the band occurs near the center of the image, and/or space is limited.)

The image fails for band if the difference between the background mean and the mean of the band exceeds the corresponding spec limit for that technique.

17.1.9.3 Application

10MM	7MM	5MM	3MM	1MM	RADIUS*
48/L	—	—	—	48/L	0-24.0
35/L	—	—	—	—	0-15.8
—	35/M	35/M	35/M	—	0-17.5
20/S	—	—	—	20/S	0-8.5
12.5/S	—	12.5/S	—	—	0-5.1

Table 2-29 Banding Techniques

* All location radii in cm.

17.1.9.4 Failure Rate

80% of all scans at a given technique must pass the band requirements.

17.1.10 Visual Acceptability of images

In addition to the artifacts described in this section, review several techniques for visual acceptability. Manufacturing Staging reviews the 12.5/S scans and the Tommy Phantom for visual artifacts.

- 80% of all slices at a given technique must pass the corresponding visual artifact inspection.
- “Repeating artifacts” do not include artifacts that move or change color.

17.2 Image Artifact Troubleshooting

Use this flowchart to troubleshoot image problems on a system with factory supplied calibration. This flowchart assumes you followed the Z-Align and Calibration procedures in this manual.

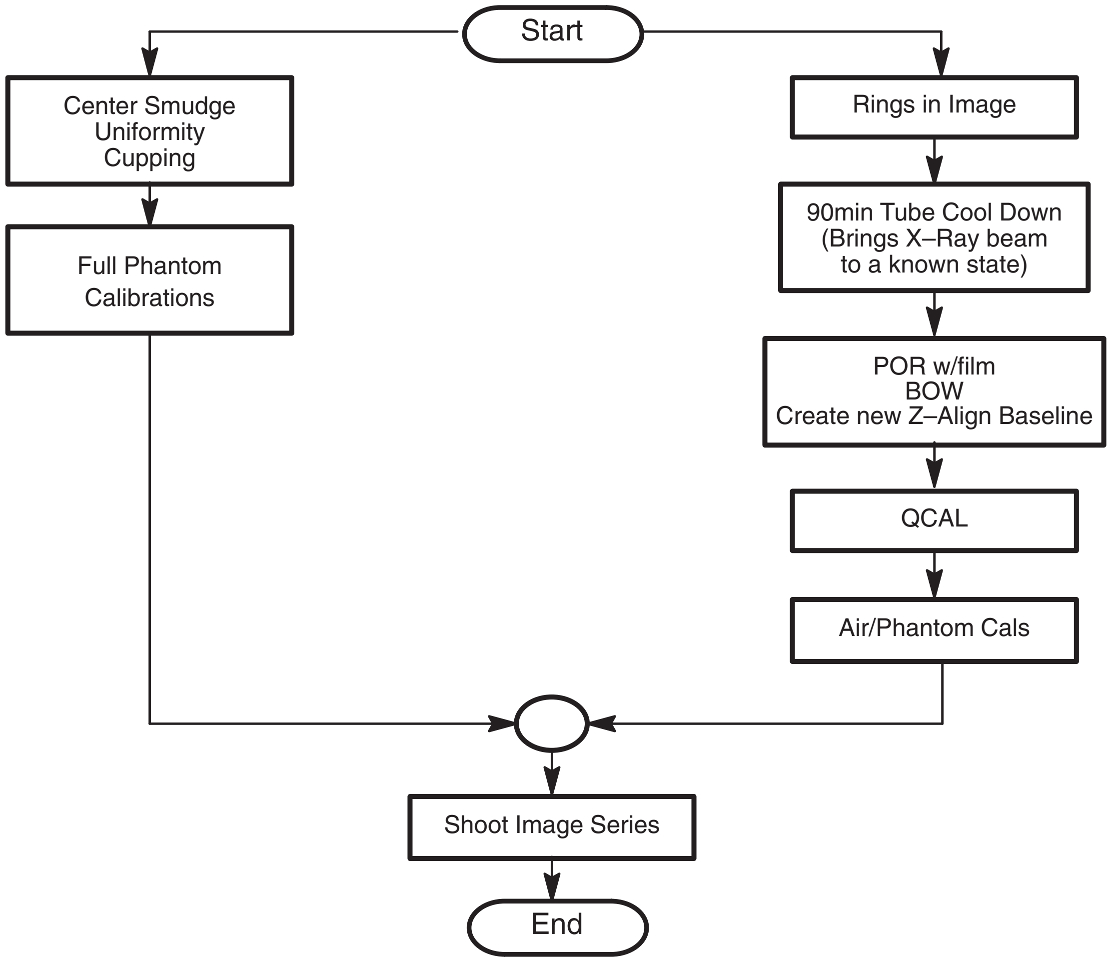


Figure 2-11 Image Troubleshooting Flowchart

17.3 Quantifying and Testing for Artifacts

17.3.1 Streak Test

1.) Place the 48cm phantom on the phantom holder.
- Align the phantom to the internal laser lights.
- Use the Calibration program to center the phantom.

2.) Select the Service Protocol 20.2 **-or-** Manually select the scan parameters in Table 2-30.

SCAN TYPE	KV	MA	SFOV	THICKNESS	SCAN TIME	# OF SCANS
Axial	120	200	Large	5mm	1.0sec	50

Table 2-30 Streak Test Scan Parameters

3.) Evaluate all images for streaks or rings.

4.) Fix all streak or ring failures.

`All 50 slices ring and streak free ? YES _______`

17.3.2 Image Analysis Program

The Image Analysis tool, located in **Service, System Integration, Image Analysis**, calculates the means and series means of a designated group of images.

- The tool positions the first ROI in the center of the phantom, and the remaining ROIs around the outside of the phantom.
- The means tool calculates the means and standard deviation in 5 designated locations on every image in the series.
- The series means tool provides averages for individual images, as well as the total average of all the image ROIs in an exam.

To access the means tool or series means tool:

1.) Display the Service Desktop Manager.

2.) Select SYSTEM INTEGRATION

3.) Select IMAGE ANALYSIS

4.) Select the exam/series from the browser.

5.) The system automatically calculates and displays the means.

17.3.3 Artifact Measurement

Use the Image Analysis tool in this procedure to measure all artifacts, except bands. Follow the procedure in Section 17.1 to measure Bands.

Note: Security Key Required

You must insert a security key to access these diagnostic tools.

Use the following sequence of softkey selections to execute all artifact analysis programs. Each artifact analysis program provides a set of on-screen prompts, to guide you through the corresponding tool application. Refer to Section 17.3.1 and 17.1 for additional artifact tests, descriptions, and information.

1.) Display the Top Level Screen.

2.) Select UTILITIES

3.) Select IMAGE ANALYSIS

4.) Select the exam/series from the browser.

5.) Select SMUDGE, CA, BAND, RING... etc.

6.) Follow on screen prompts, to display the results of each calculation.

Section 17.1 contains Artifact definitions.

17.4 Artifacts Caused by Collimator Grease - G5 Collimator

Image Artifacts have been generated and reported on some systems due to the contamination of the bowtie and the primary copper filter. This contamination is from the lubricating grease used on the filter positioning drive screw assembly.

The following information may apply in general to all 46-296300G5 Collimators.

17.4.1 Inspection Process

The inspection takes less than 5 minutes. It consists of simply examining the Copper primary filter.

17.4.1.1 Required Tool

Bright Flashlight

17.4.1.2 General Inspection Procedure

1.) Remove Mylar Scan Window
2.) Launch Diagnostic Data Collection, DDC, from the Service Desktop.
    a.) Select DIAGNOSTICS.
    b.) Select DIAGNOSTIC DATA COLLECTION.
    c.) Select POSITION TUBE.
    d.) Enter **`180`** and execute.
    e.) Select STATIC X-RAY OFF.
        i.) `Filter` AIR.
        ii.) `Slice Collimation` ***`Largest Aperture`***, 4 x 5.00 for example.
    f.) Select ACCEPT RX.
3.) Using a flashlight, inspect the primary copper filter by looking down into the collimator output port. See Figure 2-12.

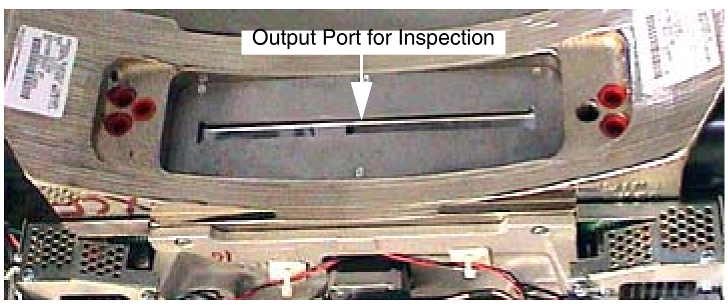


Figure 2-12 Collimator Output Port

Comment: *The copper filter should be clean, dent and scratch free. Discoloration is acceptable.*

Figure 2-13 Extremely Contaminated Copper Filter

If contamination is visible, proceed to Cleaning Process, below.

17.4.2 Cleaning Process

This procedure details the steps necessary to remove the contamination without removing the X-Ray Tube. The entire process will take approximately 3 hours. Tube/Collimator Alignments do not need to be performed. If you wish, you can check the alignments after completing the cleaning process. Any adjustments will require a complete Detailed Phantom Calibration.

Note: If you are at this step during a tube change you must perform a complete Tube Alignment and Detailed Phantom Calibration.

NOTICE Do not check tube alignments if contamination is present. You will get false results. Perform Tube Alignment checks only after the contamination has been removed.

17.4.2.1 Required Tools

- 3 mm Hex key for 3/8” drive
- Phillips #0 screwdriver
- Phillips #2 screwdriver
- Vacuum Cleaner or Tape
- Field Torque Wrench Kit 46-268445G1
- ESD Kit 2220482
- Aero Duster Spray 2226685
- Collimator Cleaning Kit 2339299
  Collimator Cleaning Kit contains the following items:
  - Aero Duster Spray System2335064
  - Alcohol Pads 46-183039P1
  - 91% pure Alcohol 46-183000P164
  - Cleaning Swabs 2339300
  - Loctite 242 10CC 46-170686P2
  - Service Note T-1449 2339305-100

17.4.2.2 Cleaning Procedure Flowchart

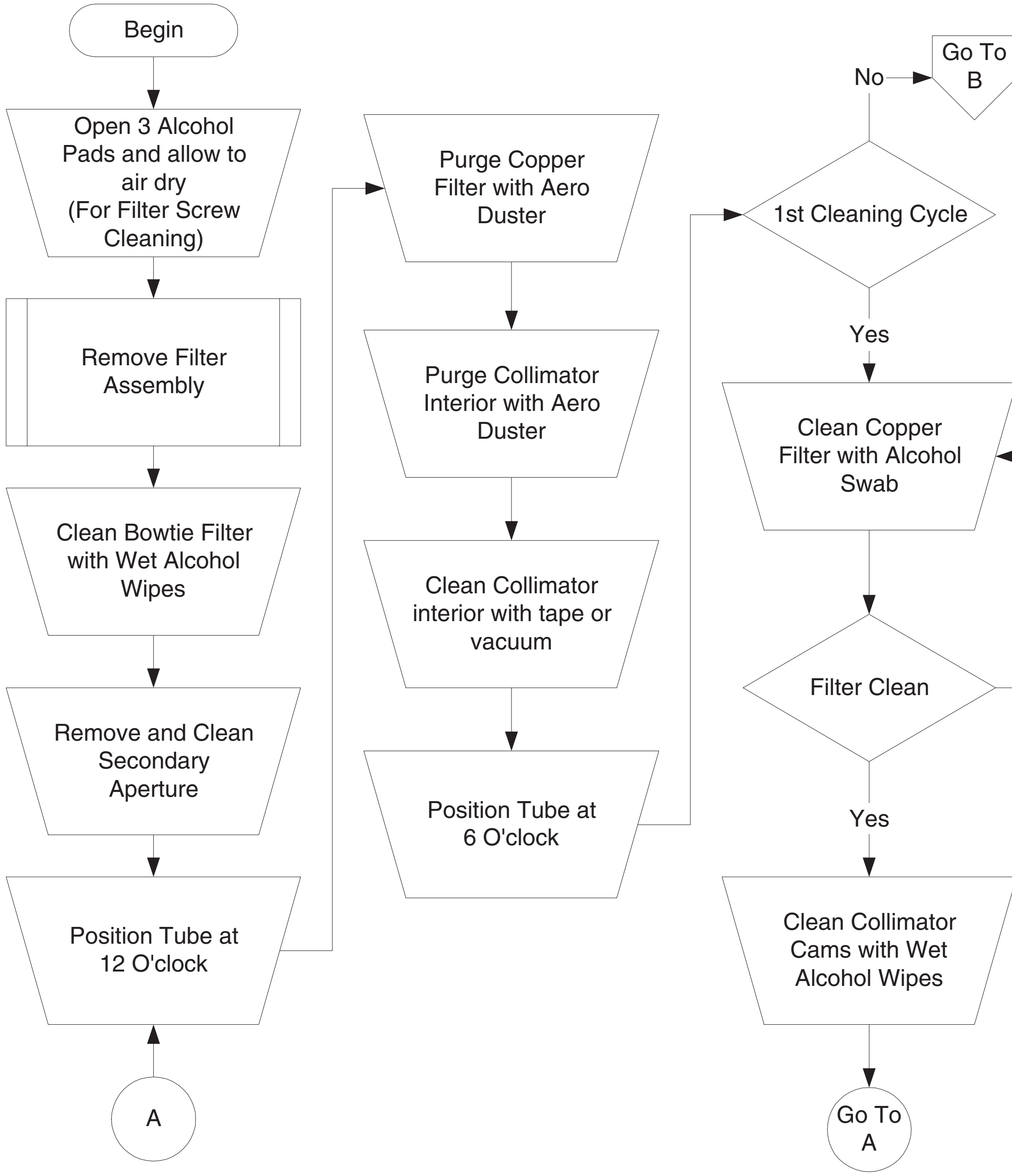


Figure 2-14 Clean Process Flowchart

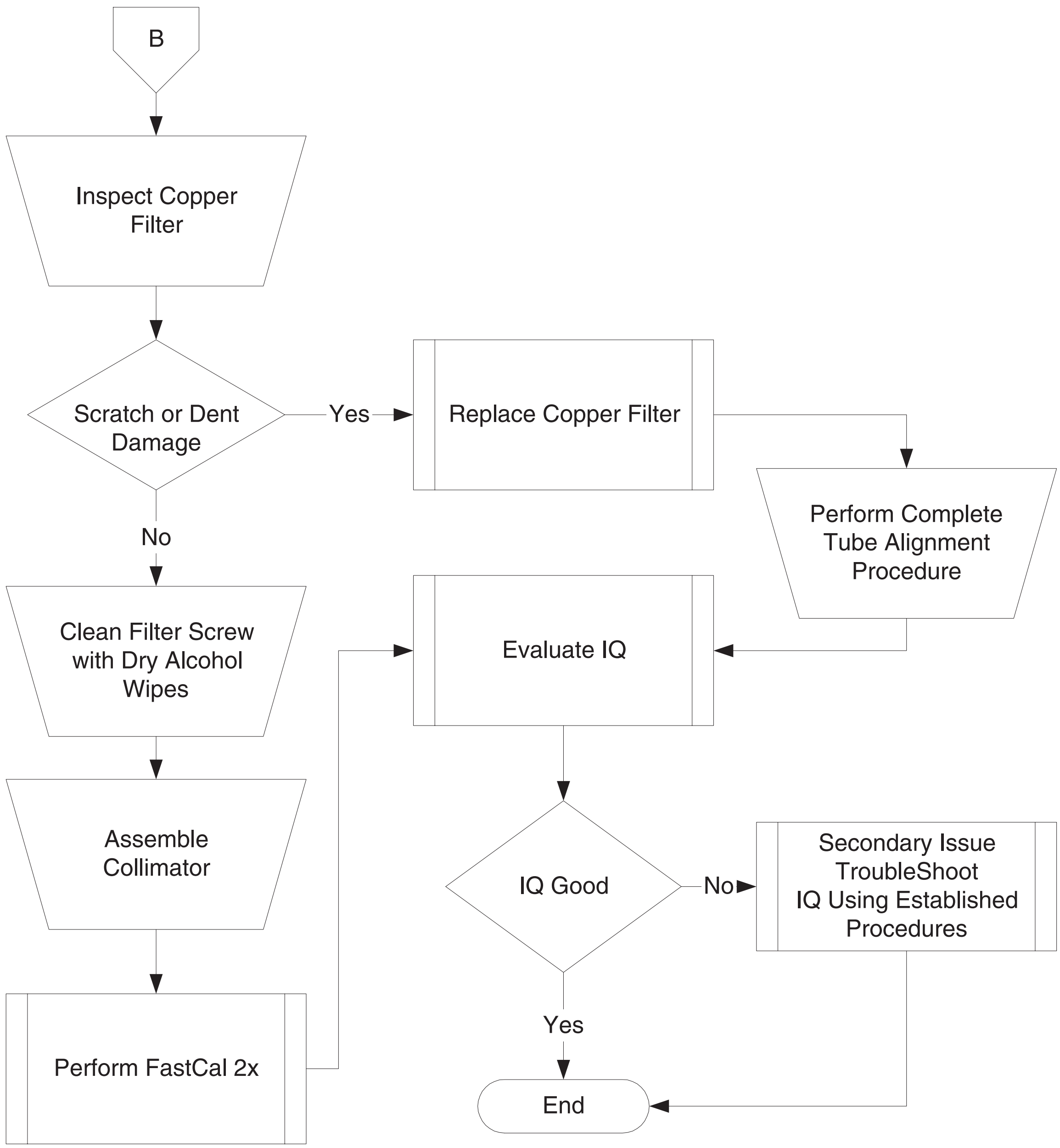


Figure 2-15 Collimator Cleaning Flowchart (continued)

17.4.2.3 Cleaning Procedure Details

1.) Remove the Gantry Covers as needed.

WARNING **RISK OF ELECTRICAL SHOCK. FOLLOW PROPER LOCKOUT/TAGOUT PROCEDURES.**

2.) Perform Gantry Power Lockout/Tagout procedures.

3.) Open three (3) Lint Free Alcohol Pads, unfold and allow to air dry.

4.) Position the gantry with the Collimator at six-o'clock.

5.) Remove Collimator Filter Assembly.

6.) Clean Bowtie Filter with fresh, wet alcohol pads.

Clean until pads are no longer soiled.

7.) Remove Secondary Aperture and Output Window. See Figure 2-16.
- Take care not to lose the six (6) screws.
- Take care not to damage or nick the lead aperture.

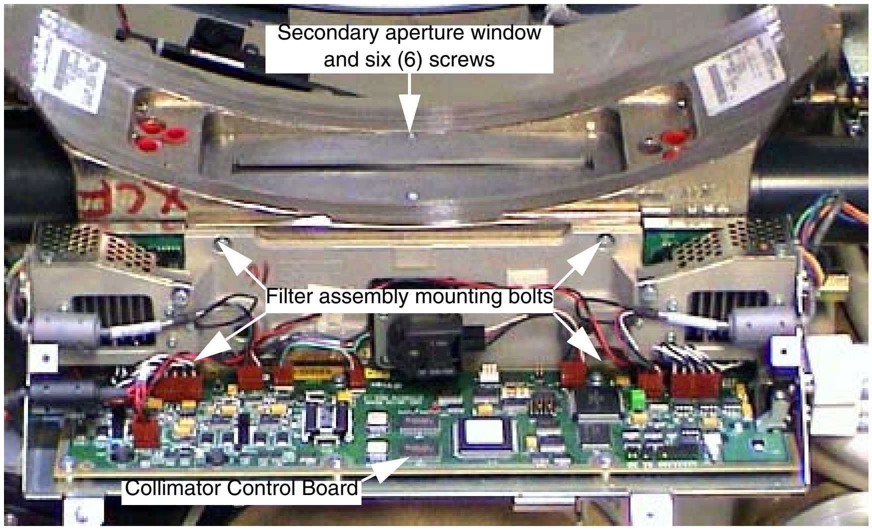


Figure 2-16 Collimator Assembly

a.) Use fresh, wet alcohol pads to clean the window and output port.

b.) Inspect output port and carefully remove any metal or lead that protrudes into the x-ray beam path.

8.) Rotate gantry so collimator is at 6 o'clock. See Figure 2-17.

a.) Using the Aero Duster and nozzle, blow out debris from the Copper Filter chamber.

b.) Using the Aero Duster and nozzle, blow out debris from the Collimator Interior.

c.) Clean collimator interior with vacuum cleaner or tape to remove any attached grease to metal particles.

⚠ **NOTICE Potential equipment damage**

Do not use the metal end of the vacuum hose. This can scratch the collimator cams. Use non-metallic accessories supplied with the vacuum cleaner.

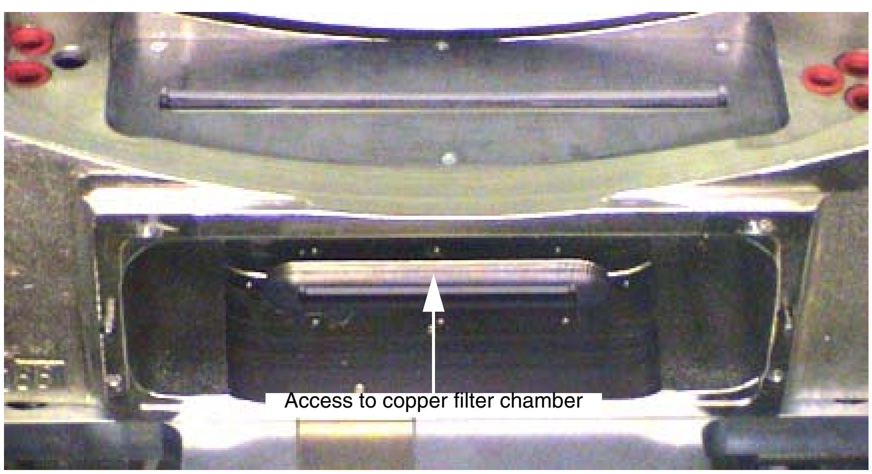


Figure 2-17 Cleaning Collimator Interior

9.) Rotate gantry so collimator is at 12 o'clock and repeat step 8 cleaning.

10.) Rotate gantry so collimator is at 6 o'clock and repeat step 8 cleaning.
This is to ensure all loose particles are removed from Copper Filter Chamber.

11.) Use clean swab, wet with alcohol, to clean the Copper Filter. See Figure 2-18.

a.) Cut swab to 7.5 cm length (3 inches).

b.) Wet lint-free foam head with alcohol. Squeeze excess alcohol from head.

NOTICE Potential equipment damage

Too much alcohol can dissolve glue that secures lead lining in place. This type of damage will result in intermittent artifacts and require collimator replacement.

c.) Carefully insert swab into copper filter chamber, and wipe filter clean.

NOTICE Potential equipment damage

Use extreme to care not dent or scratch the copper filter. Such damage will require replacement of the copper filter, resulting in a complete tube change procedure.

d.) Remove swab and inspect copper filter. Repeat with clean swabs as necessary until clean.

Figure 2-18 Swabs, Pure Alcohol and Alcohol Pads

12.) Using fresh, wet alcohol pads, clean the Collimator Cams.
Rotate the Cams using the motor shaft on each side of the collimator.

NOTICE Potential equipment damage

Use care to not scratch or bend the cams. Do not allow cams to contact each other while rotating by hand. Damage can result in tracking errors or excessive patient dose. This would require collimator replacement.

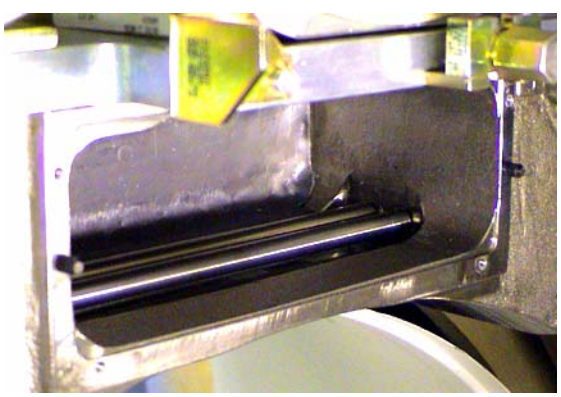


Figure 2-19 Cleaning Collimator Cams

13.) Using the dry lint free alcohol pads from step 3, clean the Bowtie Filter assembly positioning screw. See Figure 2-20.

   a.) Remove only excess grease from the drive nut.

      * Remove only accumulated grease that may dislodge.

      * The grease should lightly coat the screw thread, not fill it.

   b.) Position the filter using a flat blade screwdriver.

NOTICE Potential equipment damage

Do not remove filter off of the positioning drive screw.

Do not crush home switch with filter assembly.

Either action will require replacement of the Filter Assembly.

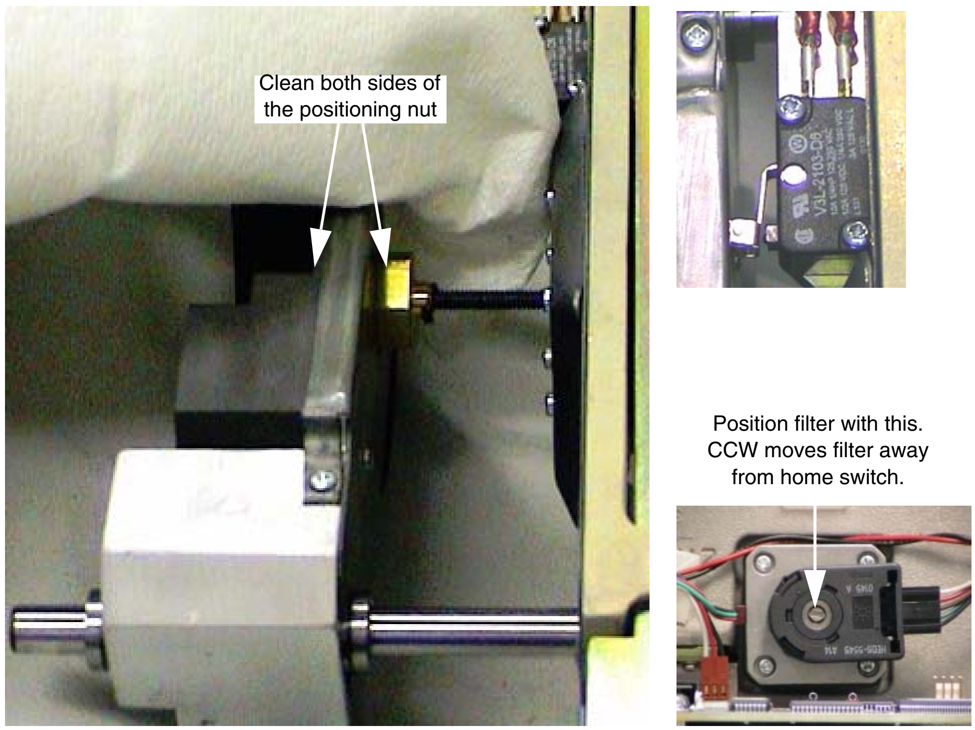


Figure 2-20 Filter Position Screw

14.) Assemble collimator.

   a.) Four (4) Filter Assembly bolts. Torque to 3 ± 0.3 N-m (26.5 lbf-in).

   b.) Six (6) Secondary Aperture screws.

   Use Loctite 242. Take care not to damage the lead window.

15.) Restore gantry power and perform a hardware reset.

17.4.3 IQ Evaluation

NOTICE **Allow DAS and Detector to warm up for 1 hour.**

1.) Perform Fastcal, twice.

2.) Perform IQ or 1x Image Series, per Section 11.0 of this chapter (beginning on page 70).

3.) Evaluate Image Quality and ensure system meets specifications both numerical and visually.

   Perform Ct Number Adjustment if necessary.

17.4.4 Additional Information

Image Quality Testing may fail for one or more of the following reasons:

- Tube Alignments were performed with contamination present.
  - Check and adjust tube alignments as necessary.
  - Perform Detailed Phantom Calibrations and CT Number Adjustment.
- Detailed Calibrations were performed with contamination present.

  Reload Phantom Calibrations from Saved State and perform Fastcal twice or perform Detailed Phantom Calibrations and CT Number Adjustment.
- Beam Obstruction may be present on Tube Output Port or chamber between Tube and Copper Filter.

  Remove Tube and Inspect this area for beam obstructions. Clean or replace parts as needed.
- Component failure within the Image Chain in addition to the collimator contamination.

  Troubleshoot accordingly.

Section 18.0
Phantom Replacement Verification

This table explains what you should do to verify that a new phantom is acceptable.

COMPONENTS	TASK	VERIFICATION TEST
48 cm or 35 cm	Scan two slices at: 120kV/200ma/ 10mm/4sec	Look for voids or hard spots in the images.
QA		Tomographic plane indication (internal lights only). Acquire QA scans for image quality checks.

Table 2-31 Phantom Verification

```
Image Acceptance/Date:_______________
Certified Image Reviewer:_______________
Image Acceptance/Date:_______________
Certified Image Reviewer:________________
```

Section 19.0
Q-Cal

Note: Run the Q-Cal procedure before you calibrate and scan. The Q-Cal function destroys all old phantom calibrations.

19.1 Preparation (Important)

Turn ON DAS power at least two hours prior to starting calibration

Start Q-Cal with a very cold tube. Do NOT take any exposures for at least one and one half hours before you start Q-Cal. (This also means NO tube warmup). Complete tests that do not interfere with tube cooling during this wait.

Make sure the mylar window centers on the patient alignment lights. An improperly centered mylar window could cause Q-Cal to fail, because one of the mylar window's metal bands could shade the detector.

1.) Select SERVICE
2.) Select REPLACEMENT PROCEDURES
3.) Select CALIBRATION
4.) Select Q-CAL
5.) Select these Cal Types in this order number of scans
    a.) Cold Qcal 40 scans
    b.) Heat Qcal 30 scans
    c.) Hot Qcal 40 scans

Note: Recommended: In order to increase the chance that the Image Series passes, repeat Q-Cal if more than 96 hours have elapsed since you completed the last Q-Cal.

19.2 When To Run Q-Cal

Note: In order to increase the likelihood a successful Image Series, repeat Q-Cal if more than 96 hours has elapsed since the last one.

The original Q-Cal remains valid until you make a change. Q-Cal should be run any time after you have adjusted or replaced any of the following components.

- Adjust or replace the Collimator
- Replace the X-Ray tube (Fast Tube Change does not always require Q-Cal)
- Replace the Detector
- Move or replace one or more ADC Boards.
- Adjust Plane of Rotation (POR)
- Adjust Beam on Window (BOW)
- Adjust Isocenter

19.3 Errors - Bad Error Path in Calibration Occasionally

Measures are being taken to fix this problem.

19.3.1 Problem - Error Message During Processing of Cal Vectors

There is a known *infrequent* bad error path in Calibration where the IG Board can get stuck and not process cal vectors. The calMain process has not been programmed to reset the IG board when it encounters this error condition.

19.3.2 Symptom - Error #250007

In the following examples, look for error 250007 which is associated with: “Qcal Processing Error: Stream Error”.

Example: #1

```
Tue Jul 30 02:41:45 1996 Error: 250007
Suite: ST5A Host: SBC0 Process: iglsd
File: iglsd_sm.m 1.15 Method: iglsd_state_machine() Line: 641
IGLSD state machine recieved bad event(0,current state(2).
seqNum=(99)
Tue Jul 30 02:41:51 1996 Error: 126151
```

```
Suite: ST5A Host: SBC0 Process: calMain
File: QcalJob.m 1.16 Method: jobDone Line: 624
Qcal processing error: Stream error.
Major Function: Cal Processing;
Minor Function: Cal Data Processing.
```

Example: #2

```
Tue Jun 18 12:32:36 1996 Error: 250007
Suite: RP11 Host: SBC0 Process: iglsd
iglsd_sm.m 1.15 iglsd_state_machine() 641
IGLSD state machine recieved bad event (0), current state (2).
seqNum=(1)
Tue Jun 18 12:32:37 1996 Error: 200300013
rhapby11 cal_composite
cal_comp_events.c 294
The cal_proc process returned a bad status of 116
Tue Jun 18 12:32:42 1996 Error: 126150
Suite: RP11 Host: SBC0 Process: calMain
XtalkJob.m 1.15 jobDone 794
Crosstalk processing error:Stream error.
Major Function:CalProc;
Minor Function: Xtalk.
```

19.3.3 Solution - Restart System

If you get the 250007 error you will need to do a shutdown/start-up before continuing calibration in order to clear the error. Notice that 250007 is a Support level error message.

Section 20.0 Calibrate Crosstalk

Run **crosstalk** when you have a new detector to calibrate, or you are rebuilding the scanner's calibration files.

1.) Select SERVICE
2.) Select SYSTEM INTEGRATION
3.) Select CALIBRATION
4.) Select START...
5.) Select Cal Type — number of scans

Select Cal Type	number of scans
4 - Xtalk Air	4 scans

6.) Reference the table to report the table height to the Gantry Display.
7.) Position the table as low as possible.
8.) Adjust the phantom holder as high as possible, with respect to the table.
9.) Place the Crosstalk phantom on the phantom holder, and use the alignment lights to center it.
10.) Raise the table 180 mm from the centered position.
11.) Acquire the Crosstalk phantom scans.
12.) Select Cal Type — number of scans

Select Cal Type	number of scans
5 - Xtalk Phantom	8 scans

Section 21.0
Generate Alpha Vector Cal

1.) Select ALPHA VECTOR to open a Shelltool.

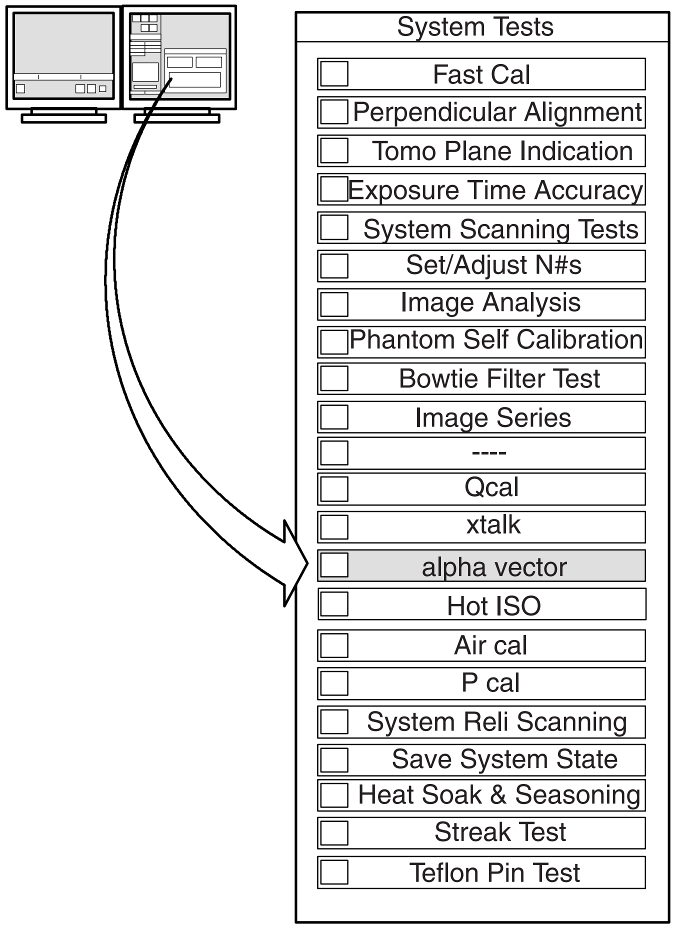


Figure 2-21 Alpha Vector Cal

2.) Select Cal Type — number of scans

Select Cal Type	number of scans
6 - Afterglow	2 scans

Note: Do **NOT** exit this program until the rotor comes down, so the software can calculate and install the afterglow vector.

3.) When the rotor stops, select SHUTDOWN to shut down the system.

4.) When shutdown completes, restart the system.

Section 22.0
Scan GE Performance Phantom

Optional Image Series for use with the GE Performance Phantom 2102582

PHANTOM	KV	MA	SFOV	THICKNESS	SCAN TIME	# OF SCANS	COMMENTS
GE PP#1	120	170	Small	10mm	2sec	4	S0.0 Peristaltic OFF
GE PP#1	120	170	Small	10mm	2sec	4	S0.0 Peristaltic ON Edge Retro (**see below)
GE PP#1	120	340	Small	10mm	1sec	4	S0.0 Peristaltic OFF
GE PP#1	120	340	Small	10mm	1sec	4	S0.0 Peristaltic ON Edge Retro (**see below)
GE PP#2	120	170	Small	10mm	2sec	4	S35.0 Peristaltic OFF

Table 2-32 GE Performance Phantom (2102585) Image Series Scan Parameters

**Edge Retro Display FOV=10cm, FOV Center = L22, A22

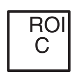
22.1 Image Data Sheet — GE Performance Phantom “Optional”

Location__________ Date(s) Scanned________________

GE Performance Phantom #1/S/10mm/120kV/170mA/2 sec/Peristaltic OFF
Scan at 0mm- Use this scan data for the Edge Retrospective reconstruction

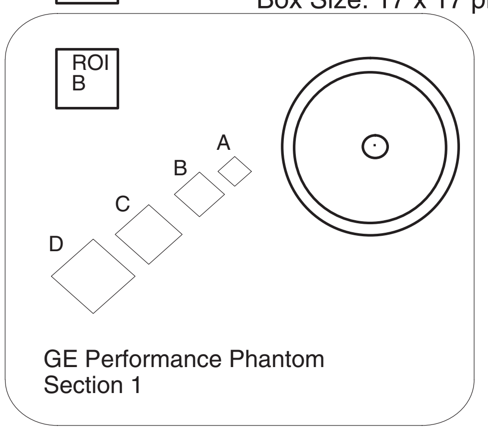
Exam# # of Image	MTF 50%	MTF 10%	MTF**4 Slice Average 50%	MTF**4 Slice Average 10%	Contrast Scale	Comments/ Artifacts	Scan Tech Aprov
			–	–			
			–	–			
			–	–			
SPECS	–		50%≥4.0	10%≥6.0	110.0 to 130.0	–	–

Table 2-33 GE Performance Phantom #1/S/10mm/120kV/170mA/2 sec/Peristaltic OFF



Contrast = Mean B – Mean C
Box Size: 17 x 17 pixels

**Round Average to Nearest Tenth
**Only 4 Slice Average Must Pass



Section A slot size = 0.40mm
Section B slot size = 0.50mm
Section C slot size = 0.65mm
Section D slot size = 0.80mm

GE Performance Phantom #1/S/10mm/120KV/170mA/2 sec
Edge Retro: DFOV=10cm, FOV CENTER = L22, A22
Scan at 0mm (S0.0)

Exam # # of Image	MTF 50%	MTF 10%	MTF** 4 Slice Average 50%	MTF** 4 Slice Average 10%	Comments/ Artifacts	ScanTech Aprov
			–	–		
			–	–		
			–	–		
SPECS			50%≥8.5	10%≥13.0	–	–

Table 2-34 GE Performance Phantom #1/S/10mm/120KV/170mA/2 sec

22.2 Image Data Sheet — GE Performance Phantom “Optional”

Location__________ Date(s) Scanned________________

GE Performance Phantom #1/S/10mm/120KV/340mA/1 sec/Peristaltic OFF

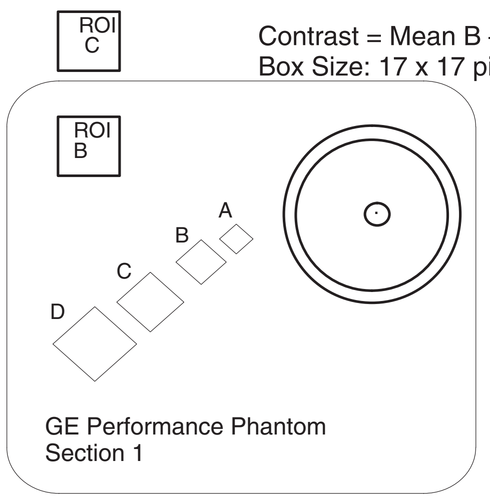
Scan at 0mm- Use this scan data for Edge Retrospective reconstruction.

Exam # # of Image	MTF 50%	MTF 10%	MTF**4 Slice Average 50%	MTF**4 Slice Average 10%	Contrast Scale	Comments/ Artifacts	Scan Tech Aprov
			–	–			
			–	–			
			–	–			
SPECS	–		50%≥4.0	10%≥6.0	110.0 to 130.0	–	–

Table 2-35 GE Performance Phantom #1/S/10mm/120KV/340mA/1 sec/Peristaltic OFF

Contrast = Mean B – Mean C
Box Size: 17 x 17 pixels

**Round Average to Nearest Tenth
**Only 4 Slice Average Must Pass



Section A slot size is .40mm
Section B slot size is .50mm
Section C slot size is .65mm
Section D slot size is .80mm

GE Performance Phantom #1/S/10mm/120KV/170mA/2 sec

Edge Retro: DFOV=10cm, FOV CENTER = L22, A22

Scan at 0mm (S0.0)

Exam # # of Image	MTF 50%	MTF 10%	MTF** 4 Slice Average 50%	MTF** 4 Slice Average 10%	Comments/Artifacts	Scan Tech Aprov
			–	–		
			–	–		
			–	–		
SPECS	–		50%≥8.5	10%≥13.0	–	–

Table 2-36 GE Performance Phantom #1/S/10mm/120KV/170mA/2 sec

22.3 Image Data Sheet — GE Performance Phantom “Optional”

Location__________ Date(s) Scanned_______________

GE Performance Phantom #2/S/10mm/120KV/170mA/2 sec

Scan at 35mm PERISTALTIC OFF

Exam # # of Slice	Contrast Factor	Size of Smallest Visible Hole* (View at window of 20)	Comments / Artifacts	Scan Tech Aprov
SPECS	–	If Contrast Factor is 2-3.99, must see 6mm or smaller If Contrast Factor is 4-7.99, must see 5mm or smaller If Contrast Factor is 8-12.0, must see 3mm or smaller	–	–

Table 2-37 GE Performance Phantom #2/S/10mm/120KV/170mA/2 sec

Required number of visible holes depends on the contrast factor. Two of the 4 scans taken must meet this specification.

TO ANALYZE:

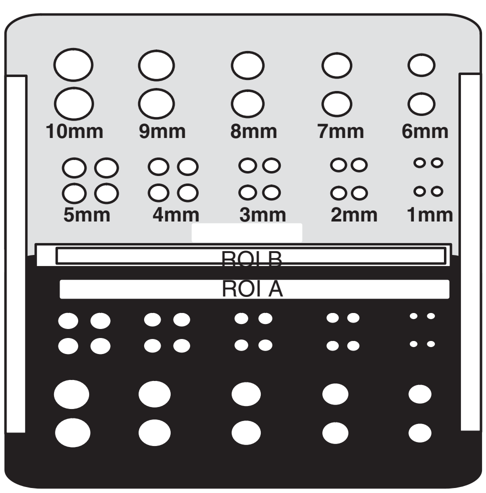
1.) Record Contrast Factor
Contrast Factor - IROI A – ROI BI

	Box Location	Box Size
ROI A	x=256; y=276	205 x 11 pixels
ROI B	x=256; y=256	205 x 11 pixels

2.) Set window width to 20.

3.) Record the size of the smallest hole you can see in the bottom half of the phantom. If necessary, adjust the viewing distance to see the holes.

4.) Verify the smallest visible hole size meets spec.



**GE Performance Phantom
Section 2**

End of Chapter

Chapter 3 System Alignments

Section 1.0 Overview

This section describes the procedures used to align the X-Ray tube unit, collimator and detector to specification. Follow these procedures to align the X-Ray system.

- Plane of Rotation – Adjust tube into or out of the gantry (Z-axis).
- Beam on Window – Adjust detector into or out of the gantry (Z-axis).
- Isocenter Alignment – Adjust tube up/down (X-axis).
- Center Body Filter and System Sag – Adjust collimator and filter up/down (X-axis). Calculate Sag parameters. (no adjustments).
- Radial Alignment – Adjust detector CW or CCW.

Note: Use the Large focal spot for all alignments, except ISO, which uses both Large and Small spots.

IMPORTANT

You must record the data collected in this chapter on **FORM F4879**, located in Chapter 11 of the System Installation Manual (2152926-100) Retain this data with all other HHS data collected at the site.

Section 2.0 Gantry Geometry Definitions

Note: See Figure 3-1 during the following discussion.

THETA

(Also known as ISO) Along the tangent to Gantry rotation, on the rotating structure. Move the X-Ray tube to the 180° (6 o'clock) position to equal 0° Theta.

You can adjust the tube unit and collimator in the Theta direction.

Z AXIS

Parallel to the axis of Gantry rotation, in the direction of cradle motion.

- You can adjust the tube unit and detector in the Z-axis direction.
- Positive Z Axis direction: Move the tube/detector toward the table
- Negative Z Axis direction: Move the tube/detector away from the table

RADIAL

Along the radius, from the center of Gantry rotation, in and out from Gantry isocenter.

You can adjust the detector in the Radial direction.

Note: You cannot adjust the tube or collimator in the Radial direction. The Gantry's design fixes this dimension.

Figure 3-1 Gantry Geometry

Section 3.0 Introduction

Use this procedure to make sure the X-Ray beam focuses directly upon the detector window. The flowchart in Figure 3-2 shows an overview of the process to be followed for a Tube change.

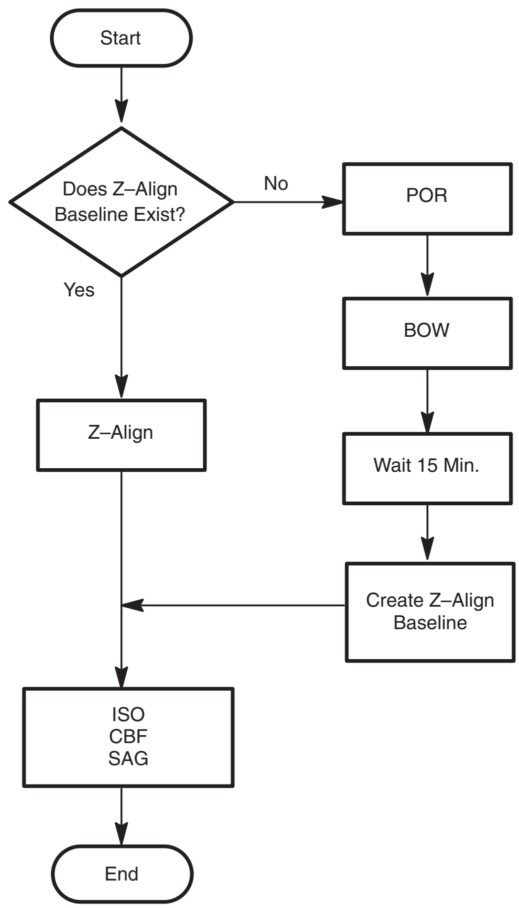


Figure 3-2 Tube Change Flowchart

Section 4.0
Advanced Tube Alignment

Follow this procedure to ensure that the X-Ray beam generated from the tube focuses directly upon the detector window. The flowchart in Figure 3-2 shows an overview of this section.

1.) Display the Service Desktop Manager.
2.) Select REPLACEMENT PROCEDURES.
3.) Select TUBE CHANGE from the Replacement Procedures, Figure 3-3, Figure 3-4, & Figure 3-5.
4.) Follow through the menu items, in order to complete all items required during a tube change.

- Items below ----, Figure 3-4 are not required for a FAST Tube Change, but might be required if you encounter problems while completing any of the tasks above ---- in the menu list.
- The square at the left of the menu item changes color after you complete the task. You can reenter completed tasks.

TUBE CHANGE SCREENS

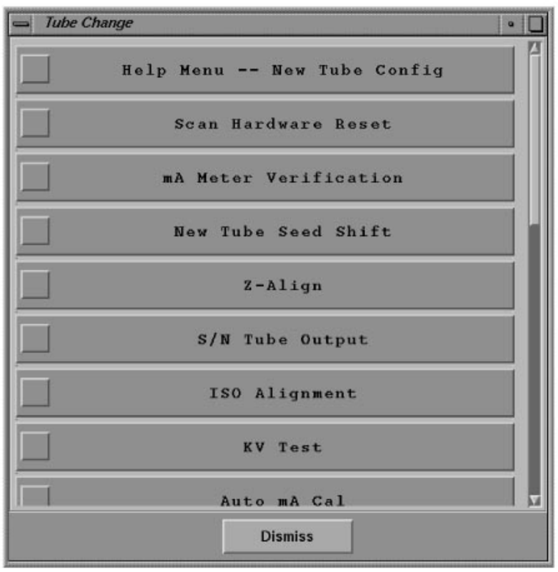


Figure 3-3 Replace Tube Unit Screen 1

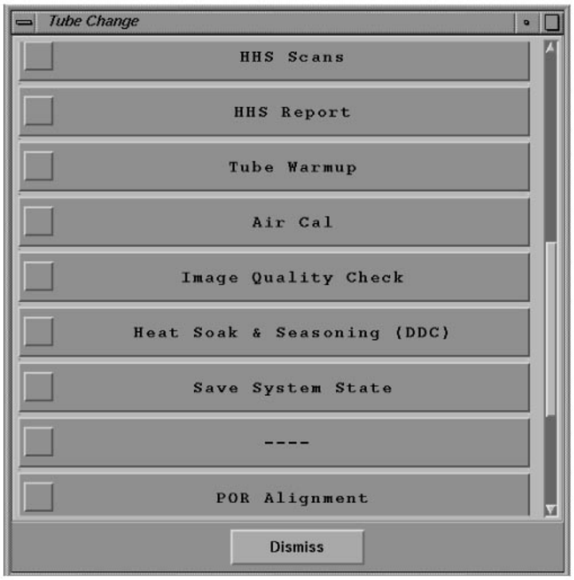


Figure 3-4 Replace Tube Unit, Screen 2

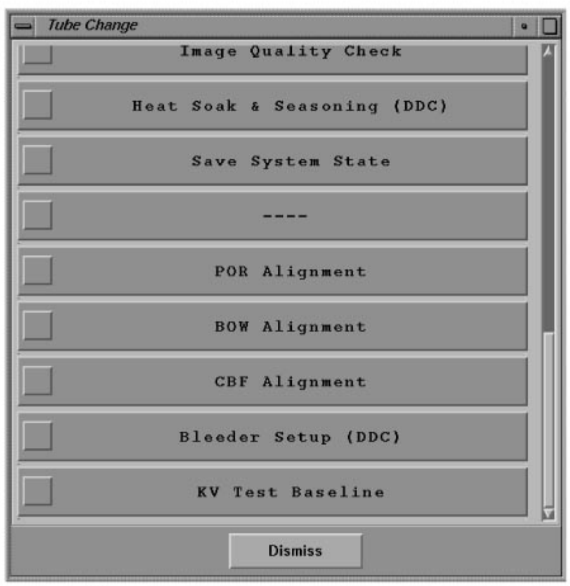


Figure 3-5 Replace Tube Unit, Screen 3

Section 5.0 Required Tools

TOOL	TOOL
Polaroid type 52 film	Film developer
Caliper	Masking tape
Socket set	Torque wrench
BOW alignment tools	48cm Phantom
Radial alignment tools	Dial indicator mount
Radial dial indicators	Dial indicators
System Installation Manual (2136597-100) (Copy of **FORM F4879** from Chapter 1)	

Figure 3-6 Required Tools

Section 6.0 Z Alignment

Use the Z-Align tool to speed up system installation and tube changes. The Z-Align procedure verifies that your POR and BOW still meet specifications, relative to the last alignment of the system. Acquire a Z-Align baseline scan to take advantage of this feature.

Note: Start this procedure with a cold X-Ray tube unit.

- If you acquired any low technique scans prior to beginning the Plane of Rotation, wait at least 5 minutes before you start this alignment procedure.

  -or-
- If you recently acquired image scans, and want to re-check the Plane of Rotation, wait at least 90 minutes, to allow the tube unit to cool. (Wait 90 minutes, if the tube unit had more than 25 Kilojoules exposure, [KV x mA x Sec ³1000] within the last 30 minutes.)

1.) Select SERVICE.
2.) Select SYSTEM INTEGRATION.
3.) Select SYSTEM ALIGNMENTS.
4.) Select START.
5.) Select Z-ALIGN.
6.) Check for the presence of a Z-Align reference baseline.
    a.) A gray Z-Align softkey means no baseline exists.
    Exit this function, and proceed to the Plane of Rotation (POR) adjustment, in section Section 7.0.
    -or-
    b.) A black Z-Align softkey means a baseline exists; proceed with the following steps.
7.) Position the tube to 0 degrees.
8.) Refer to Figure 3-14. Select Z-ALIGN
9.) Press START SCAN when it flashes.
10.) The screen displays either:

a.) `The Tube is Aligned in the Z-Direction. No movement is needed.`
Exit and proceed to ISO alignment, in section Section 11.0, on page 122.
-or-
b.) Fail – Move the tube the distance and direction indicated on the screen.
- Move the tube, and repeat the Z-Align Air Scan.
- Continue this process until the Z-Align Tool reports “The Tube is aligned in the Z direction. No movement is needed”, then proceed to the ISO Alignment, on page 122.

11.) Check the corresponding box on **FORM 4879**, to indicate that you used Z-Align.

Section 7.0
Plane of Rotation (POR)

7.1 Tools Required

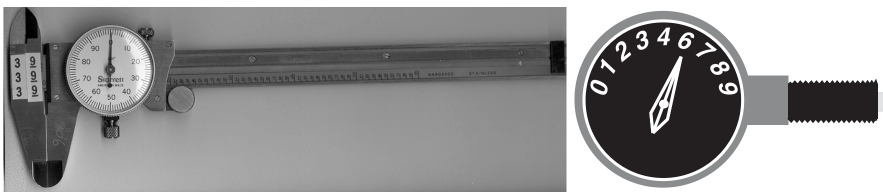


Figure 3-7 Vernier Caliper & Dial Gauge Indicator

- Calibrated Vernier Caliper (millimeters or inches).
- Type 52 Polaroid Film (see Figure 3-8)
- Type 52 Polaroid Film Developer
- Dial gauge Indicator

7.2 Procedure

A 10 mm and 1 mm exposure are taken to measure POR alignment. Remember to always preform POR measurements using a cold X-Ray tube unit!

- If you have acquired any low technique scans prior to beginning the Plane of Rotation, wait at least 5 minutes before you start this procedure.

  -or-
- If you recently acquired image scans, and want to re-check the Plane of Rotation, wait at least 90 minutes, to allow the tube unit to cool. (*Wait 90 minutes, if the tube unit had more than 25 Kilojoules exposure, [KV x mA x Sec ÷ 1000] within the last 30 minutes.*)

7.2.1 10 mm Exposure

1.) Mount the system’s phantom holder, and its 48cm phantom onto its holder.

2.) Refer to Figure 3-11 in the instruction that follow. Attach a “*Polaroid type 52*" film on the outside edge of the 48cm phantom at the 3:00 o’clock position.

Note: Only the film should be projecting into the Gantry bore. The phantom is used only to position and hold the film in the gantry bore.

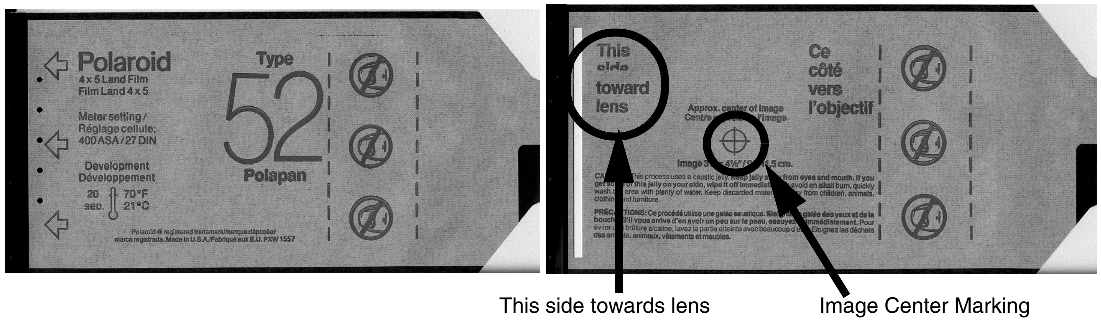


Figure 3-8 Polaroid Film, Type 52

Orient the side of the film side marked **"This side toward lens"** towards isocenter, see Figure 3-8. When exposed and developed, the film shows the alignment of the two beams with respect to the table, as viewed from the X-Ray tube in the 3:00 o'clock position.

3.) Advance the cradle and rotate the phantom if necessary, while using the alignment lights, to position the film's center marking on the alignment light marks.

4.) Select POR ALIGNMENT

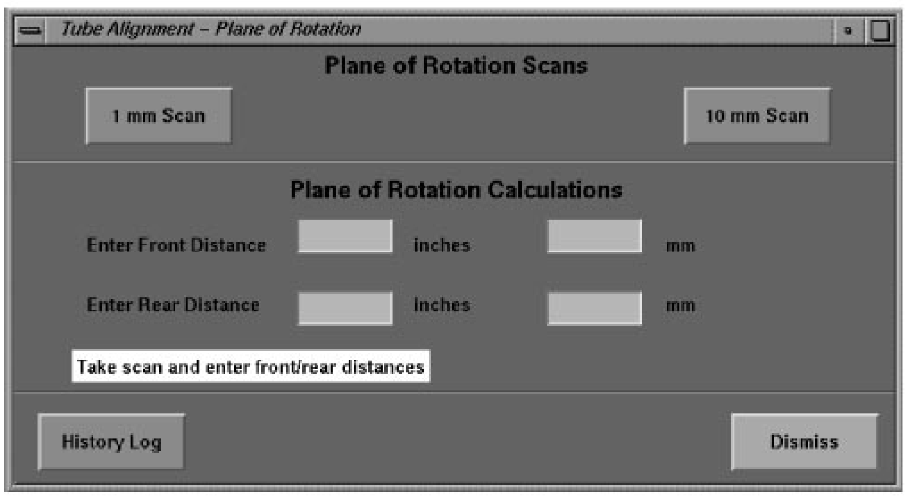


Figure 3-9 Advanced Service POR Screen

5.) Select 10MM SCAN

6.) Select ACCEPT

7.) Press START SCAN when it flashes.

8.) Remove the exposed film and immediately mark the outside of the film nearest the table. In Figure 3-11, table side is marked by the "T" on the film. Now go and develop the exposed film. After the film is developed, transfer the table marking to the film itself while keeping the orientation correct.

9.) Verify that the film's:

- narrow (white beam) slit lies within (between) the wider (gray) X-Ray slit.
- edges in both Z direction are equally well defined by the exit slit of the collimator. The edges of the narrow beam should be much sharper than the wide beam. If a difference in edge definition exists, check for gross Z misalignment. (*Mis-alignment of the slit in the tube's collimator adapter is a common cause of fuzzy film edges.*)

10.) Refer to Figure 3-10 and Figure 3-11 for the following instructions. Use a Vernier Calipers to measure the width of the 2 wider (dark gray) slits. They're the dark gray slits that extend past the edges of the narrower (off-white) slit and to the blackest part of the film.

a.) Take 3 measurements to obtain an average value for X_F. X_F is the side of the film closest to the table. Using the same film measurement, take 1 X_F measurement at the top of the film, another near the middle and another near the bottom. Add these 3 X_F distances together and divide this sum by 3 (n), to obtain an average value for X_F. Use this average as X_F.

$$X_f = (X_{f1} + X_{f2} + X_{f3}) / (n)$$

Note: It's important that you take multiple measurements. The more measurements you take, the more accurate the measurement. There is less likelihood of measurement error.

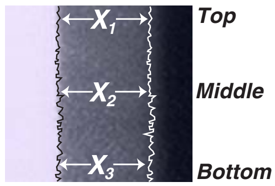


Figure 3-10 XF and XR measurement points

b.) Now take 3 measurements to obtain an average value for X_R. Take 1 X_R measurement at the top of the film, another near the middle and another near the bottom of the film. Add these 3 values together and divide the sum by 3 to obtain an average for X_R.

$$X_r = (X_{r1} + X_{r2} + X_{r3}) / (n)$$

c.) Use these resultant values for X_F (front distance) and X_R (rear distance) for all further calculations. The difference in center lines between the wide and narrow beam equals:

$$\text{Difference} = (X_{f_{avg}} - X_{r_{avg}}) / 2$$

11.) To meet the "*cold*" tube conditions and maintain the perpendicularity specification of the fan beam as the tube heats, the specifications are:

HSA TUBE:

X_F must be slightly less than X_R

(X_F-X_R)÷2, *must fall in the range of 0 to -0.010I (0 to -.25mm)*

PERFORMIX TUBE:

X_F must be slightly greater than X_R

(X_F-X_R)÷2, *must fall in the range of 0 to +0.010I (0 to +.25mm)*

12.) To calculate the amount of required movement:

a.) Type/enter the XF and XR values on the POR screen to automatically calculate and display the distance.

b.) To calculate the distance *without* the automatic program: Subtract X_F form X_R to find the difference; add 0.02 to this value and then multiply by 0.26 to determine the amount of tube shift required. $\text{TubeShift} = (X_f - X_r + 0.02) \times 0.26$

Note: A positive tube shift value means "*move the tube toward the table*", and negative tube shift value means "*move the tube away from the table.*"

13.) To adjust the tube:

a.) Barely loosen the large circular knurled nut.

b.) Loosen the four (4) mounting bolts located on the Gantry/Tube mounting bracket.

c.) Reposition tube, and tighten the bolts and nut.

Recommended: Install a dial indicator on the bracket attached to the collimator mounting bracket and bearing, on the adjacent Gantry surface. Zero the indicator before you loosen the mounting bolts, then observe the amount of indicator shift as you loosen the mounting bolts. Take this shift into consideration when you adjust the tube, so you obtain the correct adjustment distance after you tighten the mounting bolts.

14.) If the 10 mm beam width exposure did not meet specification after adjusting the tube, wait 5 minutes and repeat steps 2 thru 10.

HSA TUBE:

- Measure the exposure. Verify that $(X_F - X_R) \div 2$ = 0 to -0.01” (0 to -.25mm).
- Wait 5 minutes, then proceed to the 1.0mm aperture exposure.

PERFORMIX TUBE:

- Measure the exposure. Verify that $(X_F - X_R) \div 2$ = 0 to +0.01” (0 to +.25mm).
- Wait 5 minutes, then proceed to the 1.0mm aperture exposure.

7.2.2 1 mm Exposure

15.) Select POR ALIGNMENT

16.) Select 1MM SCAN

17.) Select ACCEPT

18.) Press START SCAN when it flashes.

19.) Remove and develop the film pack.

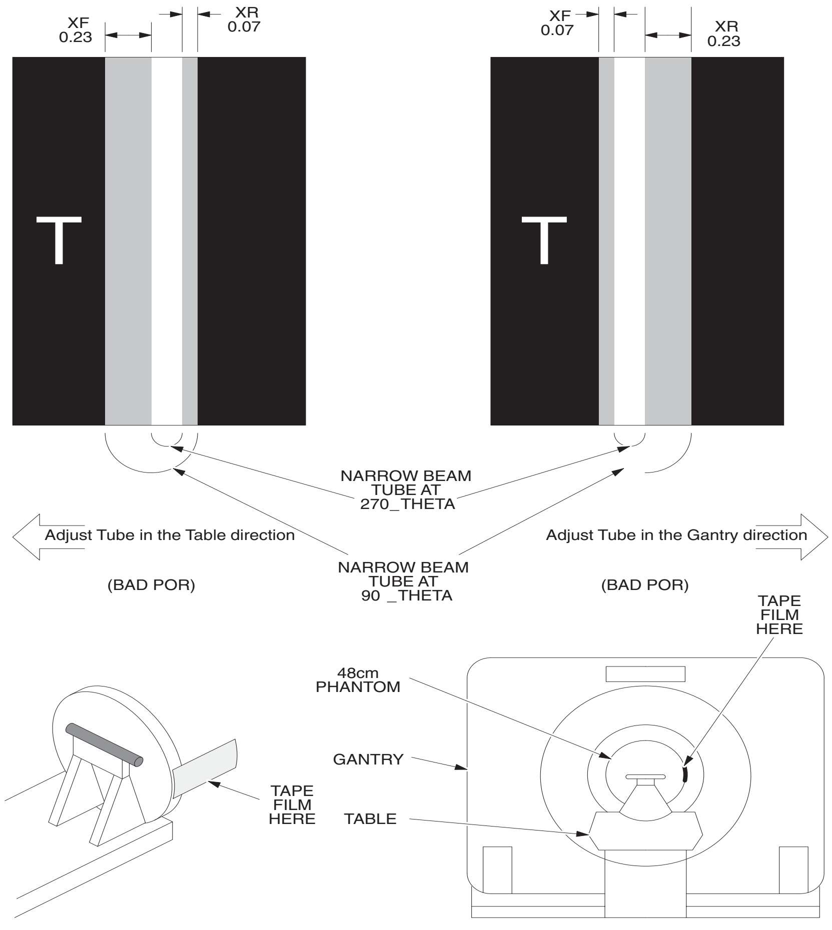
20.) Inspect the resulting film: (You do not have to measure the beam widths.)

Make sure you can see the edges of the wide beam on either side of the narrow beam

21.) When the Plane of Rotation meets specification, tighten the four tube mounting bolts to a torque of 25 $\pm$2 ft. lbs.

Note: Hand-tighten the circular knurled nut to prevent the Z adjust lever from rattling loose.

22.) Record the completion of POR on **FORM F4879**.



Techniques

FILTER	APERATURE	KVP	mA	SCAN SPEED	X—RayON	X—RayOFF
Body	10mm	80	40	4 sec.	0	360

Figure 3-11 PLANE OF ROTATION FILM INTERPRETATION

Section 8.0
X-Ray Beam on Detector Window (BOW)

This procedure checks and adjusts the position of the detector window so that it intercepts the entire width of the x-ray fan beam under worst case conditions. This procedure offsets the beam to account for the fan beam shift toward the table that occurs with heating of the x-ray tube.

⚠ WARNING **The gantry contains electrical and mechanical hazards. Make sure you turn off both the loop contactor and gantry HVDC (550) enable switch BEFORE you access the gantry. Also make sure you read direction 46-018302, CT HiSpeed Advantage Safety Guidelines manual or view 46-018308 CT HiSpeed Advantage Safety video prior to servicing GANTRY subsystems.**

1.) Start this procedure with a cold X-Ray tube unit.
   - If you acquired any low technique scans prior to beginning the Beam on Window test, wait at least 5 minutes before you start this alignment procedure.

   -or-

- If you recently acquired image scans, and want to re-check the BOW, wait at least 90 minutes, to allow the tube unit to cool, if the tube unit had more than 25 Kilojoules exposure, [KV x mA x Sec ÷1000] within the last 30 minutes.

2.) Remove any objects from the Detector Field of view.
3.) Turn ON the Axial loop contactor switch.
4.) Select BOW ALIGNMENT. Refer to Figure 3-12.
5.) Select POSITION THE TUBE; 0 degrees
6.) Select TAKE AIR SCAN

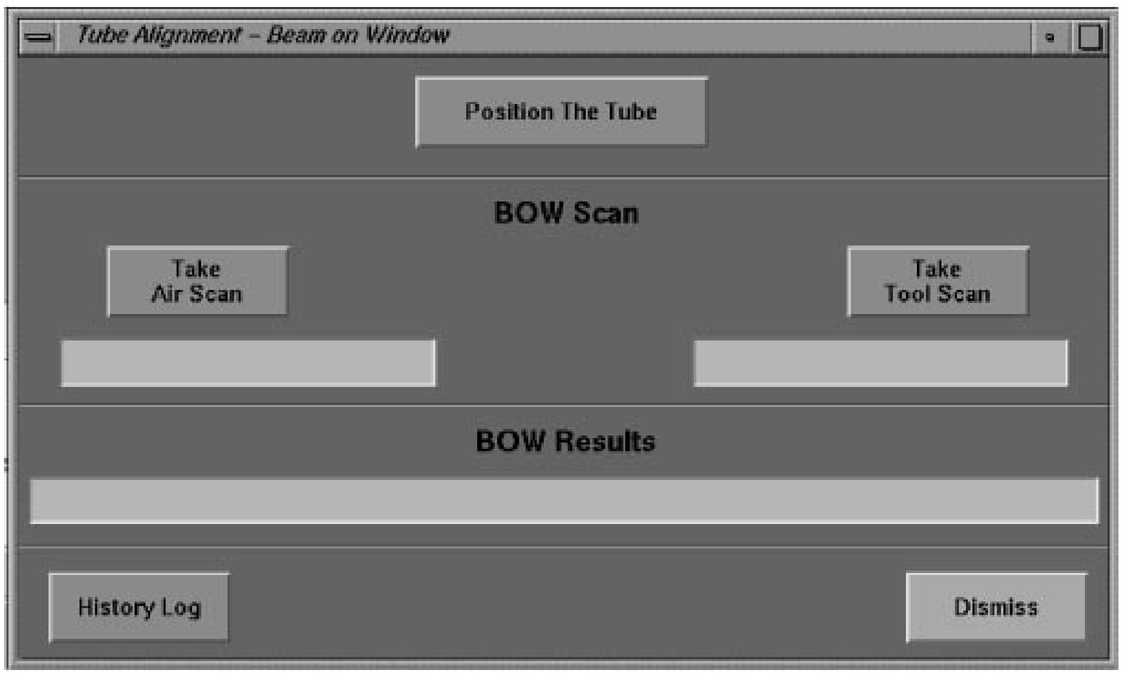


Figure 3-12 ADVANCED BOW SCREEN

Section 9.0
Install Alignment Hardware

After the Air Scan completes, install the BOW alignment tools (masking plates).

1.) Turn OFF the Axial loop contactor switch.
2.) Turn OFF the Gantry HVDC (550) enable switch.
3.) Make sure the X-Ray tube remains in the 12 o'clock position, (0° Gantry position).

Note: Take care when handling the precision tooling unit.

4.) Refer to Figure 3-13. (Critical for Detector proper alignment)
   a.) Position the masking plates in the 3 locations specified in the illustration.
   b.) Make sure the masking plates lay flat, and come in direct contact with the Detector Window.
   c.) Orient the shorter length sides of the trapezoidal openings of the tools toward the Gantry mounting plate.
5.) Turn ON the Axial loop contactor switch.

6.) Turn ON the Gantry HVDC (550) enable switch.

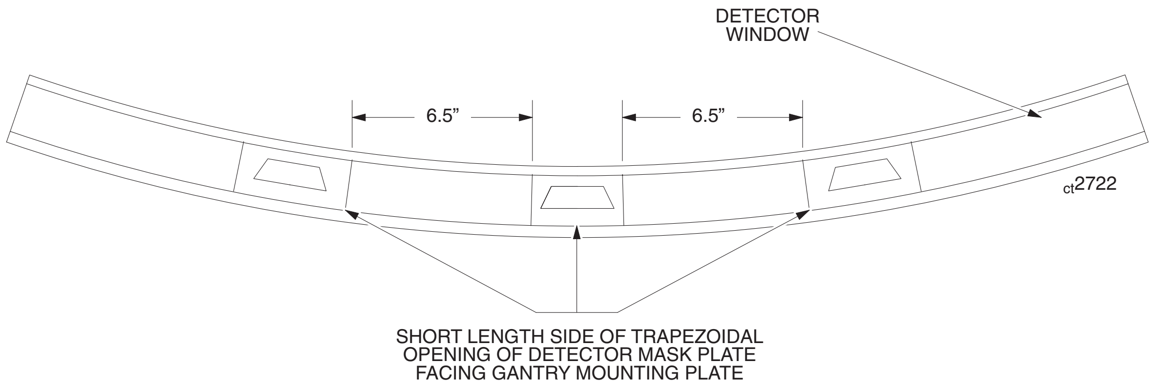


Figure 3-13 Detector Masking Plate Orientation (HILIGHT SYSTEM)

SCAN TOOL AND ADJUST DETECTOR

1.) Select BOW TOOL SCAN
2.) Press START SCAN when it flashes.
3.) Upon completion of the BOW tool scan, the system automatically calculates and displays the required amount of adjustment.
   - If the system displays the message, `The detector is within alignment`, proceed to step 6.
   - To adjust the detector, proceed to step a.
4.) Refer to Figure 3-20, on page 127.
   a.) Loosen all three detector mounting nuts, even if you plan to adjust only one or two.
   b.) Loosen the adjustment locking nuts *only* on the mounts which require adjustment.
   c.) Use the locking nut flats as a reference, and turn the mounting studs by the specified number of flats, to bring the detector into alignment.

Note: Turn the mounting studs CW to move the detector toward the Gantry

   d.) Hold the adjustment stud with a wrench, and re-tighten the adjustment locking nut.
   e.) Re-tighten the detector mounting nuts. Torque to 25 ft.-lbs.
5.) Check adjustment with BOW program, and repeat until the system displays the message `Detector is within alignment`.
6.) Record the BOW result on **FORM 4879**.
7.) Select DISMISS to close the BOW Alignment window.
8.) Turn OFF the loop contactor switch
9.) Turn OFF the Gantry HVDC (550) enable switch.
10.) Remove the window alignment hardware from the detector.
11.) Turn ON the loop contactor switch in Gantry.
12.) Turn ON the Gantry HVDC (550) enable switch.

Section 10.0
Create Z Alignment Baseline

GE designed the Z-Align tool to decrease system installation and tube change times. Acquire a Z-Align baseline scan, to take advantage of this feature. Make sure the system POR and BOW fall within specifications *before* you create the Z-Alignment Baseline.

1.) Start this procedure with a cold X-Ray tube unit.
    Wait at least 15 minutes after the last BOW exposure.
2.) Select REPLACEMENT PROCEDURES
3.) Select TUBE CHANGE
4.) Refer to Figure 3-14. Select Z-ALIGN.
5.) Position the tube to 0 degrees.
6.) Select CREATE BASELINE

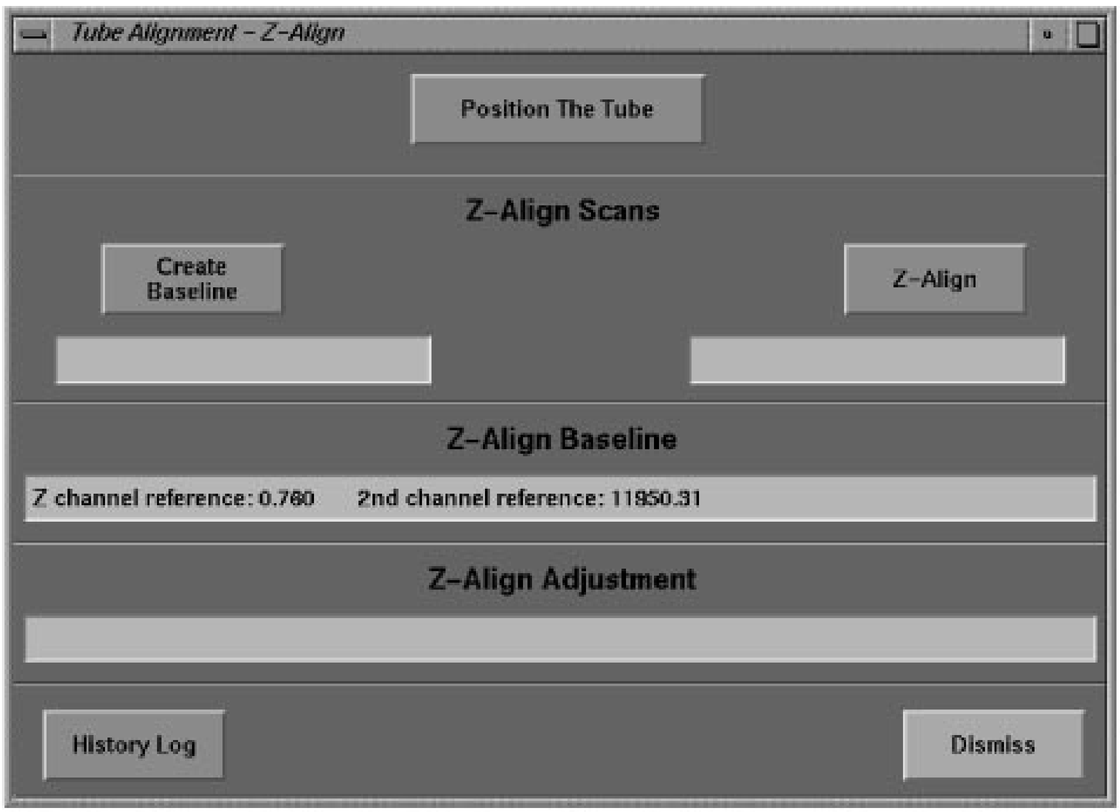


Figure 3-14 ADVANCED Z-ALIGN SCREEN

7.) Select ACCEPT
8.) Press START SCAN when it flashes.
9.) The system acquires the Z-Alignment Baseline scan and displays the Z-Align data on the screen.
10.) Exit the Z-Align screen and proceed to the ISO Alignment.

Section 11.0
Isocenter (ISO)

This procedure aligns the tube focal spot with the detector center.

1.) Start this procedure with a cold X-Ray tube unit.
   - If you acquired any low technique scans prior to beginning the Isocenter test, wait at least 5 minutes before you start this alignment procedure.
   -or-
   - If you recently acquired image scans, and want to re-check the ISO, wait at least 90 minutes to allow the tube unit to cool. (Wait 90 minutes, if the tube unit had more than 25 Kilojoules exposure, [kV x mA x Sec ÷1000] within the last 30 minutes.)
2.) Select ISO ALIGNMENT

Figure 3-15 ISOCENTER PROGRAM

3.) Remove any objects from the Detector’s Field of view.
4.) Drive the Gantry to the 0 degree (upright) Tilt position
5.) Enable the Gantry Loop Contractor switch.
6.) Select TAKE AIR SCANS
7.) Select ACCEPT
8.) Press START SCAN when it flashes.
9.) After the system creates the AIR CAL files:
   a.) Follow the proper safety protocol to access the Gantry
   b.) Attach a round, 1/8” diameter metal shaft (0.125 inch shaft diameter) to the end of the table nearest the Gantry, or on phantom holder.
   c.) Turn ON the laser alignment lights.
   d.) Advance and adjust the table to position the metal shaft 1.4" up and 1.4" right of Isocenter.
   e.) Make sure the shaft remains level (perpendicular to the scan plane.)

Note: IMPORTANT: Always use a round 1/8” diameter metal shaft. Never position the metal shaft directly on ISO center, always offset it.

The Software program that calculates tube movement for ISO Center looks for a Sinusoidal waveform generated by the off center metal shaft. A metal shaft, positioned at Isocenter, generates a waveform consisting of a straight line.

10.) Select TAKE PIN SCAN

The system displays a message that describes the distance and direction required to bring the tube into alignment. Continue to adjust the tube position, and acquire ISO air and pin scans, until the software displays a tube-in-alignment message. The tube adjustment process begins with step 12.

Note: Replace the tube if the distance between the large and small focal spots exceeds 0.16 channels.

The system ISO equals the average of the large and small ISO values. This value should equal 373.75 ±0.02 channels.

Whenever the system ISO falls outside the 373.75 ±0.1 channel window, create a new small and large AIR CAL.

11.) Turn “off” the Axial loop contactor and manually rotate the tube to the 3 o’clock position.

12.) Select the Starrett #25-141 gauge (0.001 inch to 0.250 inch), attached to the nonmagnetic holding fixture, and bolt the gauge and its holding fixture to the special tube mounting bracket.

13.) Zero out gauge before you loosen the mounting bolts.

14.) Loosen the four mounting bolts (9/16”) that fasten the tube unit to the Gantry mounting plate.

15.) Move the tube in the recommended distance and direction, as measured by the ISOCENTER gauge.

- Turn the vertical adjustment screw clockwise turn to move the tube up.
- Turn the vertical adjustment screw counterclockwise to move the tube down.

16.) Tighten the tube mounting bolts, and torque to 25 ±2 ft. lbs.

17.) Repeat ISO Scans until the ISOCENTER meets the specification.

Note: If ISO falls outside 373.75 ±0.1 channels, you do not have to wait between tube adjustment and the test scan.

If ISO projects between ±0.02 and ±0.1 Channels, wait 5 minutes after you adjust the tube before you scan.

If ISO projects within 373.75 ±0.02 channels, wait 5 minutes then repeat the scan to verify the results remain within spec.

18.) Wait 5 minutes, then recheck POR with the 10mm aperture, to verify the plane of rotation remains unchanged.

- If the plane of rotation appears to change, DO NOT adjust the tube. Wait 30 minutes and repeat the POR check.
- If the second check also shows a change in POR, adjust the tube to bring it into specification, and check the X-Ray beam to Detector Window and ISO Alignments to make sure they still meet spec.
- ISO does not effect the detector Radial Alignment.
- You must always finish the tube alignments with ISO.

19.) Record the final ISO value on **FORM 4879**.

Section 12.0
CBF (Center Body Filter) and SAG

This procedure aligns the center of the body filter, coincident with isocenter alignment, and measures the change in alignment during a rotating scan. It reports the peak amplitude difference in channel numbers, as well as the hysteresis, or difference in channel numbers between first and last views of a scan.

1.) Select CBF AND SAG ALIGN
2.) Move the table out of the detector field of view.
Make sure nothing obstructs the attenuation of the X-Ray beam to detector window.
3.) Refer to Figure 3-16. Select TAKE CBF/SAG SCANS.
4.) Press START SCAN when it flashes.
5.) Upon completion of the scan, the system calculates and displays the average centroid value for CBF, collimator movement recommendations, if any, and the SAG value.

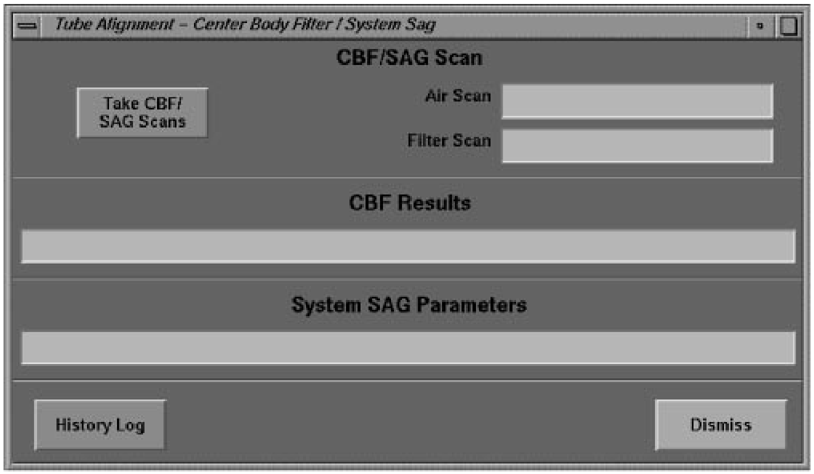


Figure 3-16 ADVANCED CBF/SAG SCREEN

6.) If the calculated CBF value fails to meet the specification of 373.75 $\pm$2, use the following procedure to adjust the collimator by the displayed amount:
   a.) Mount the gauge and its holding fixture to the special mounting bracket of the collimator.
   b.) Zero out the gauge.
   c.) Loosen the four collimator mounting bolts, located in the four slotted holes on the collimator.
   d.) Adjust the 3/4 inch bolt, located at the top of the collimator, to move the collimator up or down by the specified distance.
      - Turn the bolt clockwise to move the collimator down.
      - Turn the bolt counterclockwise to move the collimator up.
   e.) Tighten the collimator bolts, and torque the four lock bolts to 10 $\pm$1.0 ft.lbs. (120 $\pm$12 in.lbs.).
7.) Wait 5 minutes between exposures.
8.) Repeat the CBF scans, and collimator adjustment, until CBF falls within the 373.75 $\pm$0.2CH specification.
9.) Verify SAG meets the CBF peak-to-peak channel difference of $\leq$1.1 channels
10.) Remove the gauge and holding fixture from the Gantry.

Section 13.0
Radial Alignment (Cold Tube)

CAUTION **Start with a cold tube unit. If you executed any low technique scans prior to beginning radial alignment, wait at least 5 minutes before you start this alignment procedure. If you executed image scanning, and want to re-check radial alignment, wait at least 90 minutes to allow the tube unit to cool, if the tube unit has greater than 25 Kilojoules exposure KV x mA x Sec / 1000 within the last 30 min. Failure to heed this caution may result in inaccurate radial alignment.**

Note: You do not have to check Radial alignment when you replace the tube.

1.) Roughly adjust the ISO Alignment before you check radial alignment.
2.) Use the Isocenter procedure, on page 122, to move the tube within 0.005" of the correct position (the average ISO value of the large and small focal spot).
    - Don't try to attain an ISO Alignment value of 373.75 ±0.02 channels, at this time. Proceed to the Radial Alignment as soon as ISO falls within 0.005" of the specified range.
3.) Select RADIAL ALIGNMENT. Refer to Figure 3-17.

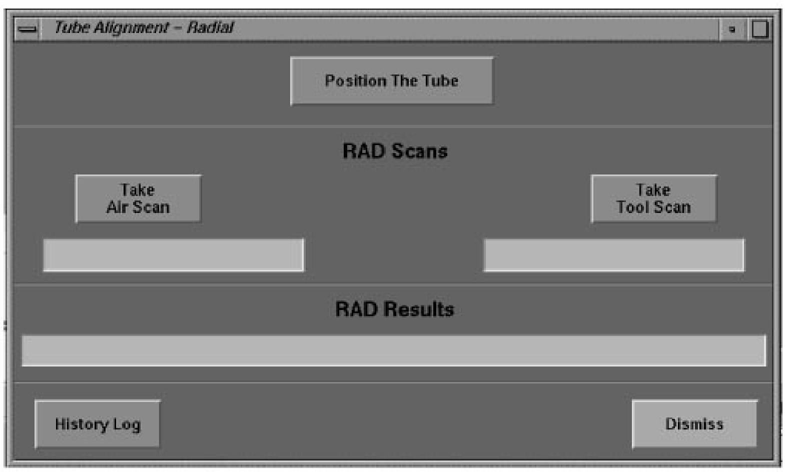


Figure 3-17 ADVANCED RADIAL ALIGNMENT SCREEN

4.) Remove any objects from the Detector Field of view.
5.) Select TAKE AIR SCAN
6.) Press START SCAN when it flashes.

13.1 Tool Placement and Scan

1.) Select POSITION THE TUBE, and type/enter the value of `180°x`.
2.) After the tube reaches the 180° position:
    a.) Turn OFF the axial enable switch.
    b.) Turn OFF the Gantry HVDC (550) enable switch.
3.) Open the front gantry cover; leave it open for the rest of the radial alignment test.
4.) Refer to Figure 3-18. Install the rotational alignment tools on the detector.
    a.) The pin must project into the X-Ray beam.

b.) Place the tool which contains the pin on the detector, but do not tighten its holding screw.

c.) Position the other tool over the pin, and insert the opposite end into the locating holes on the detector, and finger tighten the holding screws.

d.) Make sure the tube remains in the 180° position.

e.) Turn ON the gantry HVDC (550) enable switch. (You can leave the axial enable switch in the OFF position for this test.)

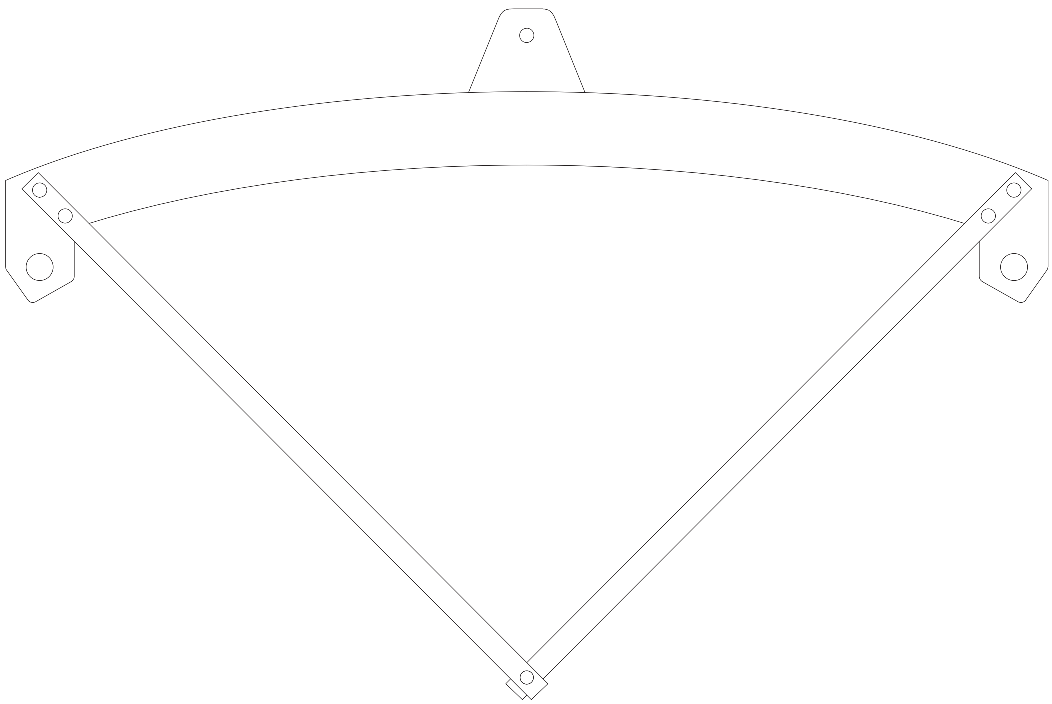


Figure 3-18 RADIAL ALIGNMENT TOOL

5.) Select TAKE TOOL SCAN

6.) When the scan completes, the system automatically calculates and displays the average centroid value, and recommended distance to rotate the detector.

- If the displayed value falls within 376.4 $\pm$0.2 channels, proceed to the final ISO Alignments.
- If the value falls outside the specification, proceed to the adjustments.

13.2 Adjustment

Note: The laser light may need to be removed to do the adjustment with the dial indicator.

Requirement: Start with the cam in the center of its range of motion.

Rotate the detector if the displayed value falls outside the 376.4 $\pm$0.2 channel specification.

1.) Refer to Figure 3-19. Use the rotational adjustment hardware to install a dial indicator on the side of the detector.

2.) Set the dial indicator to 0.000".

3.) Refer to Figure 3-20, on page 127.

a.) Loosen all three sets of the large detector mounting nuts slightly, until they feel finger tight.

b.) Loosen the allen head rotational adjustment locking screws.

4.) Refer to the Average Centroid value display, and use the adjustment cam to rotate the detector in the specified distance and direction.

5.) After you complete the adjustment:

a.) Tighten the allen head rotational adjustment locking screws.

b.) Torque the inside detector mounting nuts to 25 ft.-lbs.

c.) Hold the inside nut to keep it from turning, while you torque the outside nuts to 25 ft.-lbs.

Note: If you don't have a torque wrench, tighten the nuts until finger tight, then tightened an additional 1/2 flat with a wrench.

6.) Repeat the radial alignment procedure, to verify the adjustment.

7.) Record the actual Average Centroid value measurement on **FORM 4879**.

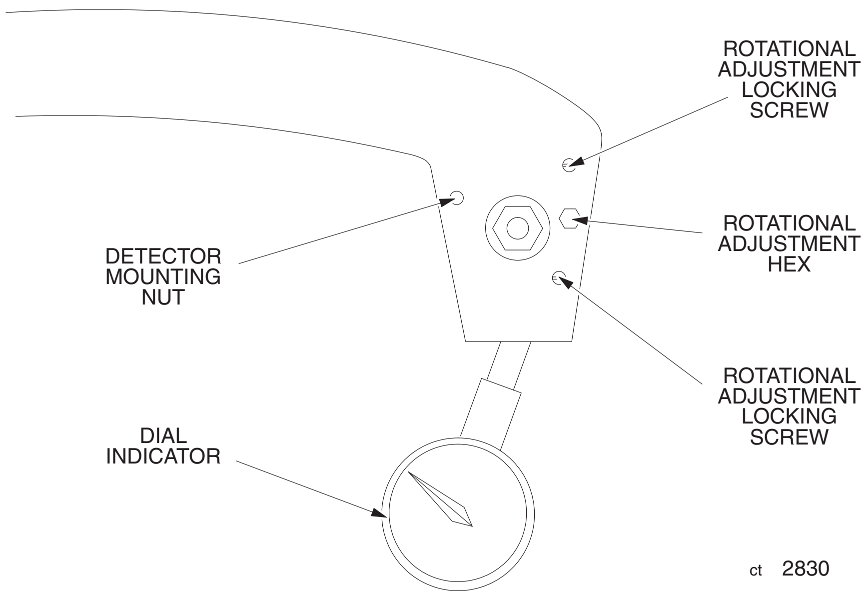


Figure 3-19 ROTATIONAL HARDWARE

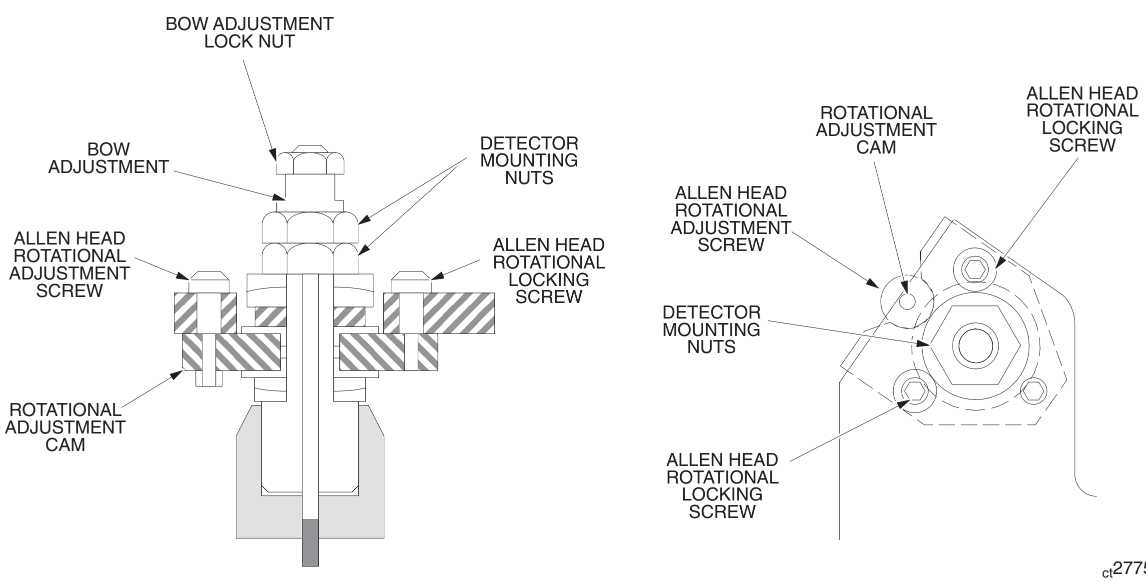


Figure 3-20 MOUNTING/ADJUSTMENT HARDWARE (DETAIL)

Section 14.0 Signal to Noise Output Check

1.) Select S/N TUBE OUTPUT
2.) Select TAKE SCAN
3.) Press START SCAN when it flashes.
4.) Verify that `test passed` appears on the screen.

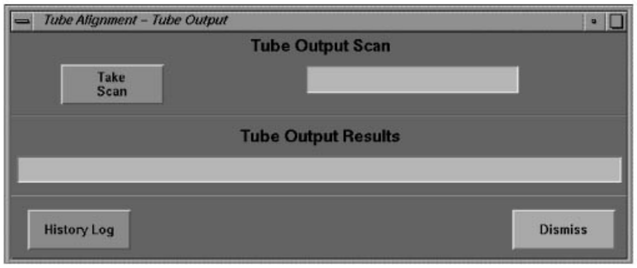


Figure 3-21 TUBE OUTPUT CHECK

Section 15.0 Detector Output Check

1.) Select DETECTOR OUTPUT
2.) Select TAKE SCAN
3.) Press START SCAN when it flashes.
4.) Verify the displayed results pass.

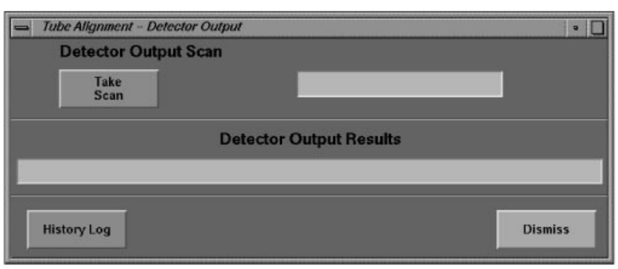


Figure 3-22 Detector Output Screen

End of Chapter

Chapter 4
Functional Checks

DANGER **DANGEROUS ELECTRICAL & MECHANICAL HAZARDS THAT CAN CAUSE SEROUS INJURY WHENEVER YOU SERVICE THE TABLE AND GANTRY.**

- **BEFORE YOU CHECK THE POWER SUPPLIES INSIDE THE GANTRY OR REMOVE THE ROTOR CONTROLLER COVER: TURN "OFF" THE GANTRY'S 120 VAC ENABLE SWITCH.**
- **HIGH VOLTAGE DC IS PRESENT IN THE HEMRC ASSEMBLY, WHENEVER GANTRY 120 VAC IS ENABLED.BEFORE YOU PERFORM ANY SERVICE OPERATIONS, VERIFY WHICH TYPE OF HVDC SUPPLY THE SYSTEM USES. THE DESCRIPTION FOR THE OUTPUT OF THE HIGH VOLTAGE DC POWER SUPPLY IN THE HSA CT/I SYSTEM IS HVDC. THIS COULD BE EITHER A REGULATED 550 VDC OR AN UN-REGULATED OUTPUT VOLTAGE THAT VARIES BETWEEN 450 TO 750 VDC DEPENDING UPON INPUT LINE VOLTAGE AND THE OUTPUT LOAD ON THE UNREGULATED HVDC SUPPLY.**
- **READ DIRECTION 2152915-100, HSA CT/I SAFETY GUIDELINES MANUAL, OR VIEW 46-08308 CT HISPEED ADVANTAGE SAFETY VIDEO, PRIOR TO SERVICING THESE SUBSYSTEMS.**

WARNING **TUBE HOIST CAN CAUSE INJURY IF IMPROPERLY SECURED.**

PREVENT THE HOIST FROM ROLLING OFF THE BACK END OF THE HOIST BOOM WHEN A DAS, TUBE OR DETECTOR IS INSTALLED OR REPLACED. PLACE THE HAND KNOB, USED TO HOLD THE FRONT GANTRY SHROUD, INTO THE HOLE ON TOP OF THE GANTRY HOIST BOOM, WHERE IT ATTACHES TO THE GANTRY.THIS WILL PREVENT THE HOIST FROM SLIDING OFF THE END.

Section 1.0
Input (PDU) Power

WARNING **VERIFY THAT ALL PERSONNEL ARE CLEAR AND REMAIN CLEAR OF THE SYSTEM WHEN TURNING ON THE WALL POWER.**

1.) Turn OFF all three switches on the gantry status display box.
2.) Turn OFF all three power switches in the table.
3.) Set the PDU HVDC enable key to DISABLE.
4.) Turn ON main wall power.

Section 2.0
Gantry Display Test

1.) Turn ON the gantry 120 VAC Enable Switch on the gantry status display box.
2.) Turn ON the table power or recycle power to the table for the gantry display to start its power up self-test.
3.) Make sure the gantry display goes through the power-up self-test. The display continues to cycle through its self-test until it completes the hardware reset and download.
4.) TURN ON "X-RAY DRIVES" power by pressing the RESET button on the gantry mounted table control.

NON-ADJUSTABLE SUPPLIES	ADJUSTABLE SUPPLIES
STC, OBC +24	ETC, STC, OBC, +5, ±15
Table (Display, Emergency Stop) +24	Table (Cradle Drive, Relays) +24
Detector Heater +26	Data Communication +12
Filament Supply +30	Das +5, ±5
Tilt/Elevation +170	Collimator +38.5

Table 4-1 OBC, STC AND TABLE POWER SUPPLIES

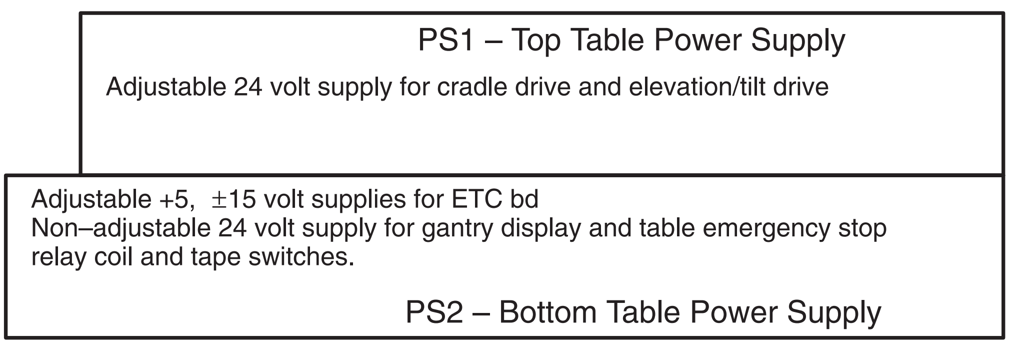


Figure 4-1 TABLE POWER SUPPLIES (LEFT VIEW)

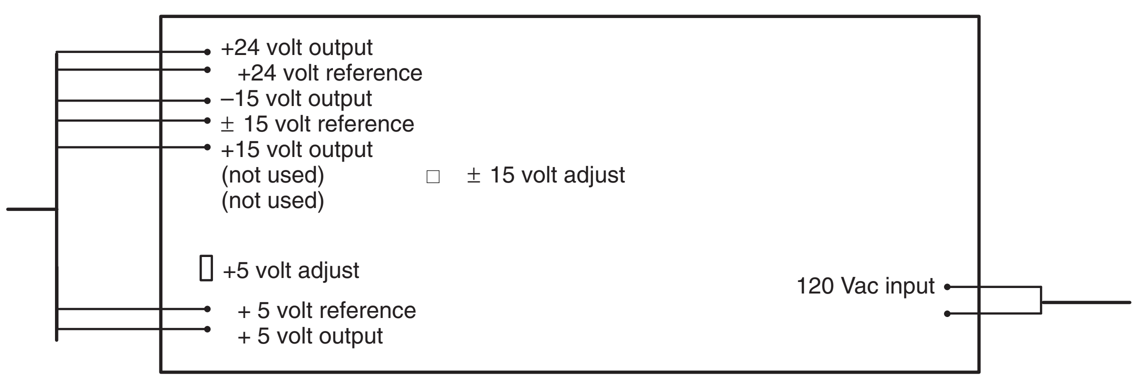


Figure 4-2 OBC, STC AND BOTTOM TABLE POWER SUPPLIES (TOP VIEW)

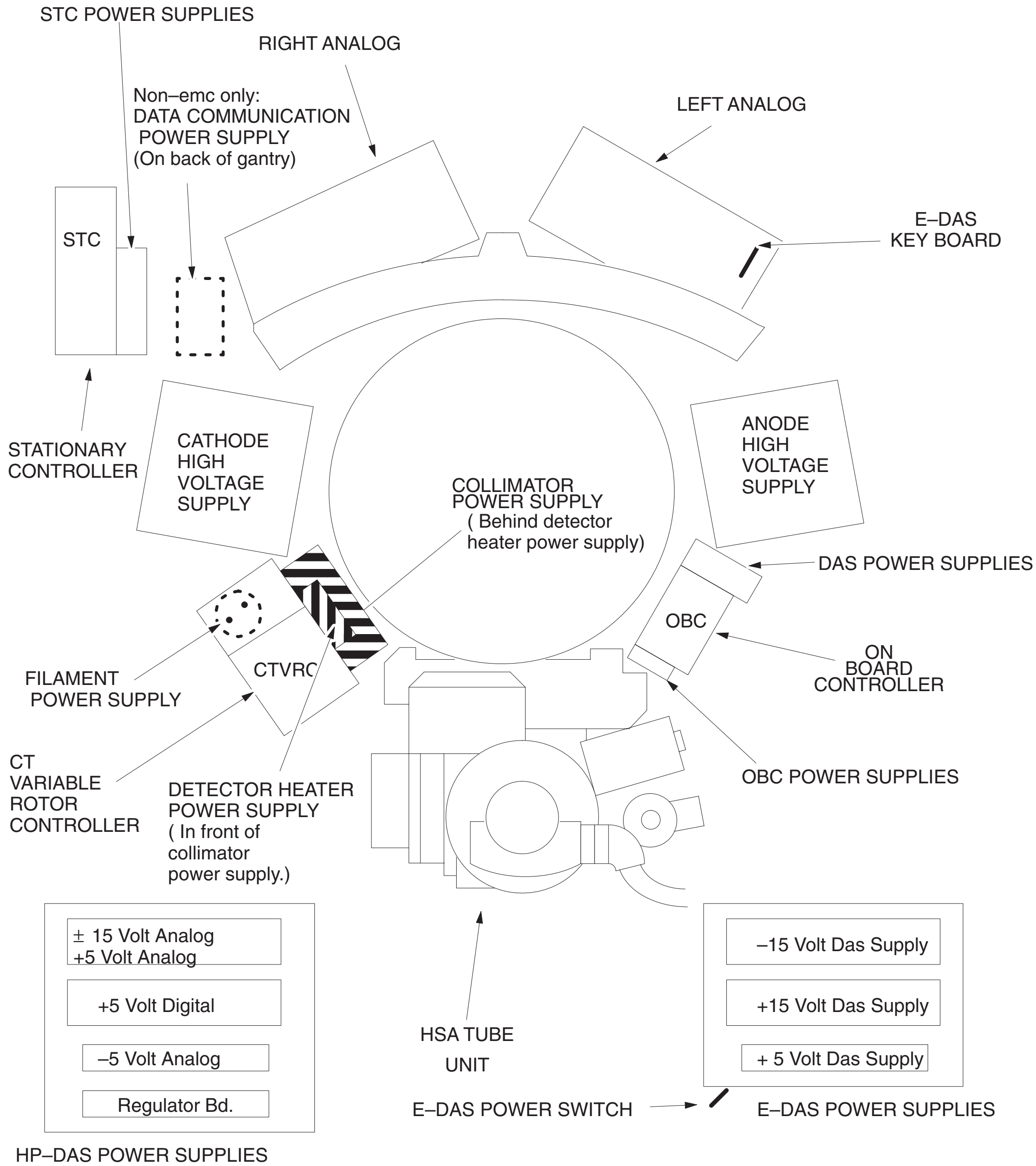


Figure 4-3 GANTRY POWER SUPPLIES WITH HSA TUBE

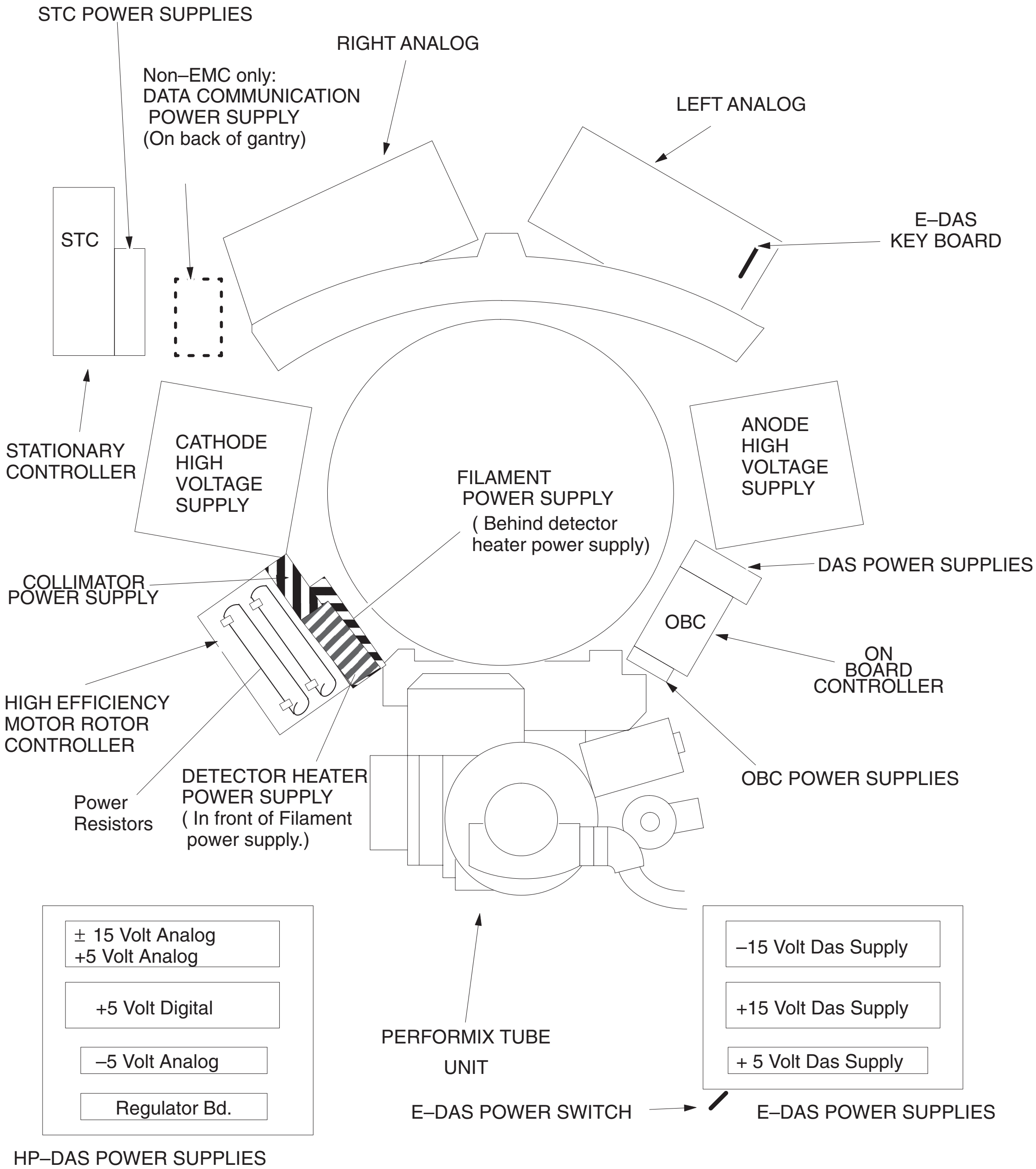


Figure 4-4 GANTRY POWER SUPPLIES WITH PERFORMIX TUBE

Section 3.0 Power Supplies

3.1 ETC Power Supplies

Use a DVM to verify the table power supply voltages at the following test points on the ETC bd:

ETC Board Test Points	Specifications
21 WRT 18	+15vdc (±0.25V)
11 WRT 18	-15vdc (±0.25V)
25 WRT 24	+24vdc (±0.2V)
1 WRT 2	+5vdc (±0.25V)
WRT means "With Respect To"	

Table 4-2 ETC Power Supplies

3.2 STC Power Supplies

1.) Turn OFF the axial drive enable and HVDC enable switches on the gantry status display.
2.) Turn ON gantry 120VAC enable switch on the status display box.
3.) Turn ON STC power switch on the STC.
4.) Use a DVM to verify the STC power supply voltages at the following test points on the Axial bd:

Axial Board Test Points	Specifications
36 WRT 40	+15vdc (±0.25V)
38 WRT 40	-15vdc (±0.25V)
37 WRT 42	+24vdc (±0.2V)
39 WRT 41	+5vdc (±0.25V)
WRT means "With Respect To"	

Table 4-3 STC Power Supplies

3.3 OBC Power Supplies

1.) Turn ON OBC power switch on the OBC.
2.) Use a DVM to verify the OBC power supply voltages at the following test points on the Collimator I/O bd:

I/O Board Test Points	Specifications
TP14 WRT TP12	+5vdc (±0.25V)
TP15 WRT TP17	+15vdc (±0.25V)
TP16 WRT TP17	-15vdc (±0.25V)
TP20 WRT TP21	+24vdc (±0.2V)
WRT means "With Respect To"	

Table 4-4 OBC Power Supplies

3.4 HP-DAS Power Supplies

DAS power supplies can be checked using DAS Tools or DDC. If all power supply readings fall within indicated ranges, there is no need for adjustment or voltage measurement with a multimeter. Normal DAS Tool readings are:

+5V digital	4.80 to 5.20V	
+15V analog	14.3 to 15.7V	
-15V analog	-14.3 to -15.7V	
+5V analog	4.85 to 5.10V	5.8 to 6.1 at the supply
-5V analog	-4.85 to -5.10V	-5.8 to -6.1 at the supply

Test Points	Specifications
+5 DIGITAL	5 +/- 0.2 V digital
±15 WRT GND	±15vdc (±0.25V) analog
+5 WRT GND	+5vdc (±0.25V) analog
+5 WRT GND	+5vdc (±0.25V) digital
-5 WRT GND	-5vdc (±0.25V) analog
WRT means "With Respect To"	

Table 4-5 HP-DAS Test Points

3.5 E-DAS Power Supplies

1.) Turn on the DAS power switch on the DAS power supply assembly.
2.) Use a DVM to verify the DAS power supply voltages at the following test points on the DAS CAL/AUX bd:

Test Points	Specifications
+15 WRT GND	+15vdc (±0.25V)
-15 WRT GND	-15vdc (±0.25V)
+5 WRT GND	+5vdc (±0.25V)
WRT means "With Respect To"	

Table 4-6 E-DAS Power Supplies

3.6 Data Communication Power Supply

1.) Push back the rear gantry shroud.
2.) Use a DVM to verify the data communications power supply voltage at the following location:

Test Points	Specifications
Communications power supply CN2 connector Red wire WRT to black wire	+11.75vdc to +12.25vdc
WRT means "With Respect To"	

Table 4-7 Data Communications Power Supply

3.7 Detector Heater Power Supply

Use a DVM to verify the detector heater power supply voltage at the following location:

Test Points	Specifications
Across cap on detector heater	+25 vdc to +30 vdc power supply

Table 4-8 Detector Heater Power Supply

3.8 Filament Power Supply

Use a DVM to verify the filament power supply voltage at the following location:

Test Points	Specifications
Across cap on filament power supply	+28 vdc to +37 vdc

Table 4-9 Filament Power Supply

3.9 Tilt/Elevation Power Supplies +170vdc

No adjustment required

3.10 Collimator Power Supply

Use a DVM to verify the collimator power supply voltage at the following location:

Test Points	Specifications
Collimator power supply +out WRT -out	+38.0vdc to 39.0vdc

Table 4-10 Collimator Power Supply

Note: Turn off the gantry 120 Vac enable switch. Install the rotor controller cover. Turn on the gantry 120 Vac enable switch.

Section 4.0
Axial Motion

4.1 Axial Encoder Check

Turn the Gantry by hand until it passes through the home flag. Verify:

- DS270(CHA) and DS271(CHB) AX bd LEDs toggle.
- DS320(CHC) toggles on and off once while DS321(home flag) is high.

4.2 Axial Brake Check

Toggle the axial drive enable switch on the status display box. Listen to the brake; it should energize and de-energize.

Note: The brake may not toggle if the system underwent a hardware reset since the last time you turned on gantry AC power. If the brake doesn't toggle, use the 120 Vac enable switch on the gantry status display box to turn gantry 120 Vac power off, then on. Then, toggle the axial drive enable switch on the status display box. You should now hear the brake as it energizes and de-energizes

Make sure:

- When you turn off the axial drive enable, the switch pilot light turns off and the brake releases. (You can easily rotate the Gantry by hand.)
- When you turn on the axial drive enable switch, the switch pilot light turns on and the brake energizes.

Turn on all three switches on the gantry status display box.

Section 5.0 Intercom

Note: The Intercom board is set at the factory and should not require re-adjustment.

At install time, make sure the Gantry Cable has enough slack to pull the cable through to the front of cabinet. Recommended: 1 Meter (3 feet).

Make sure the intercom's talk and listen functions work. If you hear feedback, or if the intercom does not seem loud enough with the console intercom controls set to maximum:

1.) Close gantry covers.
2.) Remove the Intercom Board from the Console:
    a.) Shutdown system software, and turn off power to the computer/console.
    b.) Remove the console/computer front cover.
    c.) IF you correctly cabled the gantry during installation, remove the two screws that fasten the intercom carrier assembly to the console, and remove the assembly from the console/computer. Go to step 4.
3.) If insufficient cable to the back of the intercom assembly exists:
    a.) Disconnect keyboard cable and gantry cable from interconnect board.
    b.) Disconnect interface board power, and console i/f cable.
    c.) Remove intercom/interface board assembly.
    d.) Pull gantry interface cable through to the front of the unit.
    e.) Route the keyboard cable to the front of the console.
4.) Place the assembly in front of console, and re-connect the cables.

HOW TO ADJUST INTERCOM BOARD

1.) Verify Jumpers at:
    a.) JP4, over pin 1 and 2 (pin 1 is the leftmost pin) Connectors located on the left side of the board.
    b.) JP5 jumper set.
    c.) JP1, JP2, JP3, JP6, JP7, JP8 do NOT have jumpers.
2.) Verify pot settings at:

a.) R3 – 0 turns CW (console speaker) more turns CW equals more volume.
b.) R5 – 4 turns CW (table speaker) more turns CW equals more volume.
c.) R10 – 3 turns CCW (min gantry volume - user preference) fewer turns CCW equals less minimum volume.
d.) R100 – 10.25 turns CW (autovoice threshold).
e.) R82 and R16 are non CT/i functions - no adjustment required.

HOW TO RE-INSTALL INTERCOM BOARD

1.) Install intercom/interface board assembly.
2.) Re-attach all cables and re-install front panel.
3.) Power up and test intercom and autovoice functions.
4.) Verify sound levels of intercom at console and gantry.
5.) Verify autovoice level.

Section 6.0 Alignment Lights (Accuracy)

DANGER **VERIFY ALL PERSONNEL IS CLEAR OF THE SYSTEM AND THE GANTRY ROTATES FREELY TO 180 DEGREES.**

1.) Press the alignment light button, on the gantry-mounted table controls, to position the gantry.
2.) Press the alignment light button on the gantry-mounted table controls again, to turn the lights off.
3.) Turn OFF the axial drive enable and HVDC enable switches, on the gantry status display box.
4.) Use the switch on the gantry control assembly to manually turn on the alignment lights.

Locate the gantry control assembly, near the collimator. You can reach the switch through the opening in the center of the gantry, between the front and back shrouds.

WARNING **WHEN YOU OPERATE THE ALIGNMENT LIGHTS, NEVER STARE AT THE LASER BEAMS BECAUSE THIS CAN CAUSE PERMANENT EYE DAMAGE.**

LASER LIGHT

6.1 Internal Axial Lights

Place a sheet of plain white paper over the output port of each light and verify that the two lines of laser light coincide with each other.

Do not try to adjust the internal axial lasers on the CT/i system to shine “down” on the collimator.

6.2 External Axial to Internal Axial Distance

Raise the table to its highest elevation. Extend the cradle until both the internal and external laser lights shine on the cradle. Place a metric rule on the right edge of the cradle, and measure the distance from the internal axial laser line to the external axial line, generated by each laser. Verify this distance equals 310.0 mm ±±1.0mm. Place the rule on the left edge of the cradle and measure again.

Extend the cradle until both internal and external laser lines shine on it. Lower the table to the lowest elevation. Verify the 310.0 mm ±1.0mm distance between the internal and external lights, on both edges of the cradle, as above.

6.3 Coronal Lights

Place a sheet of plain white paper at the left side of the patient opening, in front of the coronal laser light. Verify that the two coronal lines coincide with each other. Check the right side in the same way.

Place the paper in the center of the Gantry opening, and use a level to verify that the coronal lines are horizontal.

6.4 Alignment Light Visualization

H.H.S. requirements state that lights used to define the tomographic plane must be visible under ambient light conditions, up to 500 lux. To verify:

1.) Turn the scan room lights on to their brightest normal level. Do not add localized spot lights to increase the brightness level.
2.) Raise the table to its highest position, advance the cradle into the gantry, and turn on the alignment lights.
3.) Center the back of your hand over the cradle, and hold it in the alignment light. Make sure you can see the external axial alignment lights on your hand. You only have to see the axial lights, not the sagittal or coronal lights.
4.) Repeat the procedure with the internal axial lights.

If you cannot see the external or internal axial lights under the conditions described above, obtain a DIGAPHOT model #3300 or 3303 light meter; measure the ambient light intensity at the cradle surface at the external and internal alignment light locations.

Note: Footcandles x 10.76 = lux.

If the light reading(s) exceed 500 lux, reduce the room lighting to the 500 lux level, and repeat steps 3 and 4. If the light meter readings equal 500 lux or less, replace the laser light(s) and/or their power supplies.

Section 7.0 Emergency Stop

Use the gantry-mounted (or table-mounted) control pushbuttons to advance the cradle about two feet from the home position.

1.) Press one of the emergency stop switches on the Gantry (or table if available). Verify the following:
    a.) The Gantry/Table doesn't move.
    b.) The Reset Light on Gantry mounted controls flashes slowly.
2.) Depress one of the tilt buttons, to verify the emergency stop disables tilt.
3.) Depress one of the table elevation buttons, to verify the emergency stop disables table elevation.
4.) Depress one of the cradle drive buttons, to verify the emergency stop disables the cradle drive.
5.) Move the cradle to the home position, to verify the emergency stop released the cradle clutch.
6.) Make sure the cradle latches securely in the home position.
7.) Press the reset switch on the gantry-mounted table controls (or on the REM box, if available), to turn on X-RAY DRIVES POWER. Make sure the tilt, elevation, and cradle drive now work.
8.) Repeat steps 1 through 7 with the other Gantry (or table, if available) emergency stop switch.

9.) Repeat steps 1 through 7 with the four table tape switches.
10.) Repeat steps 1 through 7 with the console emergency stop switch.

Section 8.0 Collimator

1.) Select TROUBLESHOOTING from the Service Desktop Manager.
2.) Select KV & MA softkey.
3.) Select COLLIMATOR & FILTER softkey. See Figure 4-5.
4.) Depending upon the current downloaded firmware, the scan subsystem may or may not have to download new firmware. If asked to download now, select yes.
5.) Select 1MM softkey.
6.) Select AIR softkey.
7.) Select ACCEPT softkey.
8.) Select 180 DEGREE to set Tube Position.
9.) Select RUN. Verify monitor messages: `Current mode = at position`
`All power supplies are OK` Look in the collimator and make sure you see the1mm aperture centered in the collimator.
10.) Select 1MM to deselect the 1mm aperture setting.
11.) Select the 3MM softkey.
12.) Select the ACCEPT softkey.
13.) Select RUN. Verify monitor messages: `Current mode = at position`
`All power supplies are OK` Look in the collimator and make sure you see the 3mm aperture centered in the collimator.
14.) Repeat the procedure for the 5mm, 7mm, 10mm, and closed positions.

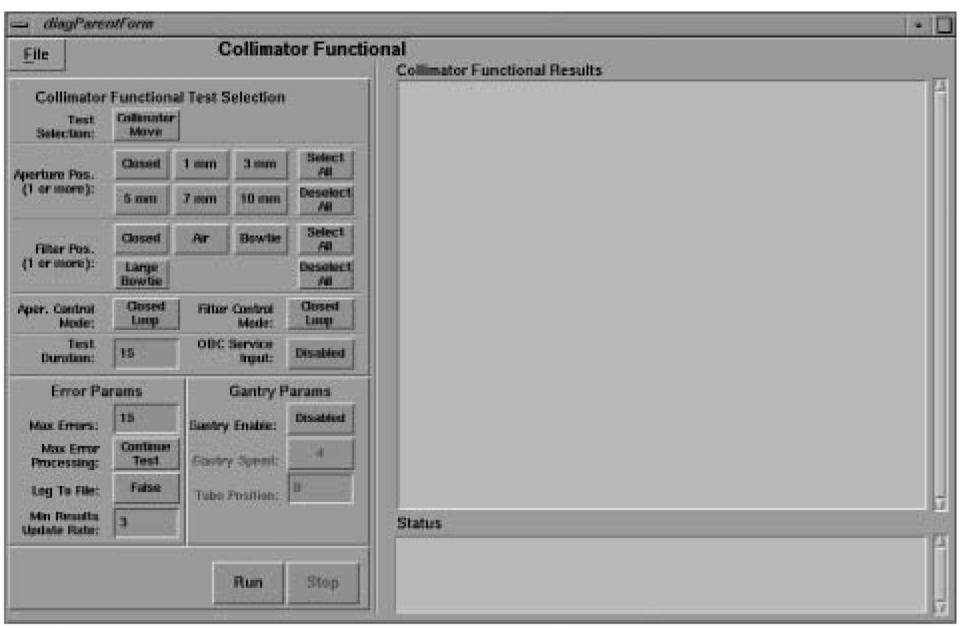


Figure 4-5 COLLIMATOR FUNCTIONAL TEST SCREEN

Section 9.0 Mechanical Characterization Procedure

Figure 4-6 shows the Service Desktop Selection Menu. Use this menu to access the Mechanical Characterization Program. Use this tool to set up the characterization tables for the system firmware. The system firmware reads feedback devices to track various mechanical characteristics of the machine. The Characterization tool tells the system firmware how the values of these feedback devices relate to the actual machine characteristics. Use the Characterization tool to define a reference value for each of the feedback devices. The system stores the characterization data in a series of characterization and configuration files.

The characterization tool provides the user with a step-by-step process to build the characterization files. The tool prompts the user for a response, reads the appropriate feedback device, and stores the value in the corresponding characterization file. The tool also provides a mechanism to download the files to the firmware, and permits the user to view the files.

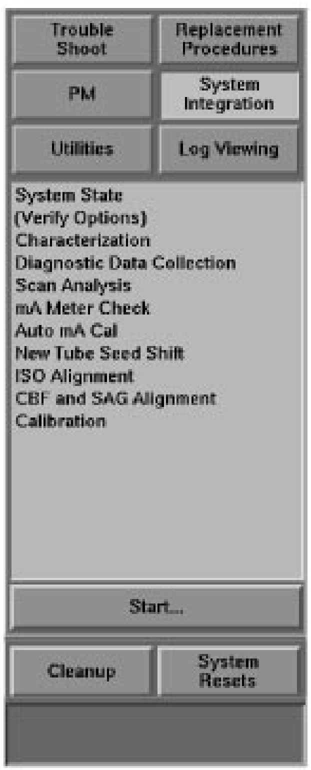


Figure 4-6 SELECT CHARACTERIZATION

9.1 Characterization Inputs (Soft/Manual Entry)

In order to update *typed in* values for mechanical characterization, you must use **TAB**+**ENTER** to enter and update EACH entry.

9.2 Mechanical Characterization Screen

Select the CHARACTERIZATION menu item (Figure 4-6) to display the screen shown in Figure 4-7. Use this tool to characterize:

- Gantry Tilt Position
- Table Elevation
- Table Cradle
- Collimator

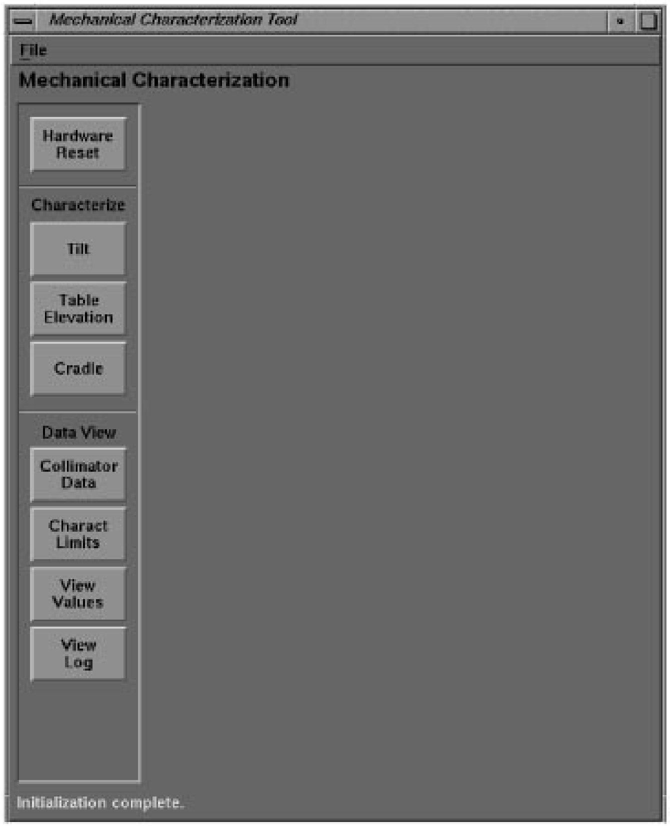


Figure 4-7 MECHANICAL CHARACTERIZATION SCREEN

9.3 Gantry Tilt Position

Select TILT and CHARACTERIZE in sequence, then follow the Tilt Characterization instructions displayed on the monitor screen. (See Figure 4-8.)

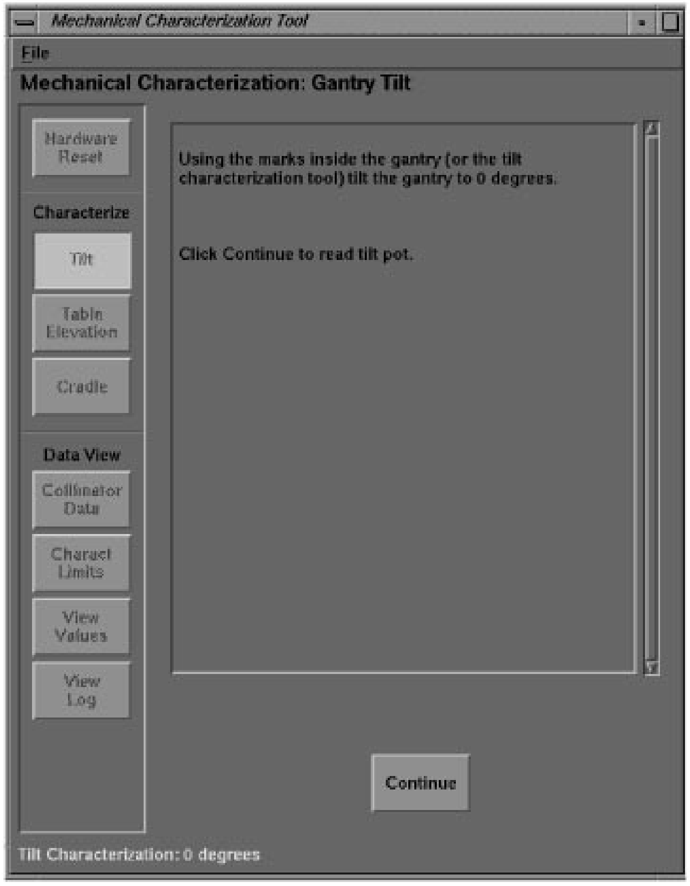


Figure 4-8 GANTRY TILT CHARACTERIZATION

9.4 Table Elevation Characterization

Measure the distance from the top of the weldment plate to the rear leg upper pivot pin on the table, to accurately position the elevation.

Note: Do not use elevation heights or measurement positions depicted on the monitor display. Follow the procedure, described below, to obtain the elevation heights and measurement positions.

1.) Remove the upper right and lower right table covers.
2.) Move the measurement plate, at the right rear of the table, to its “out” position, and tighten the mounting screws.
3.) Move the measurement plate on the right side of table to its “out” position, and tighten the mounting screws.
4.) Select TABLE ELEVATION and CHARACTERIZE in sequence, then follow the Characterization instructions displayed on the monitor screen. (See Figure 4-9)

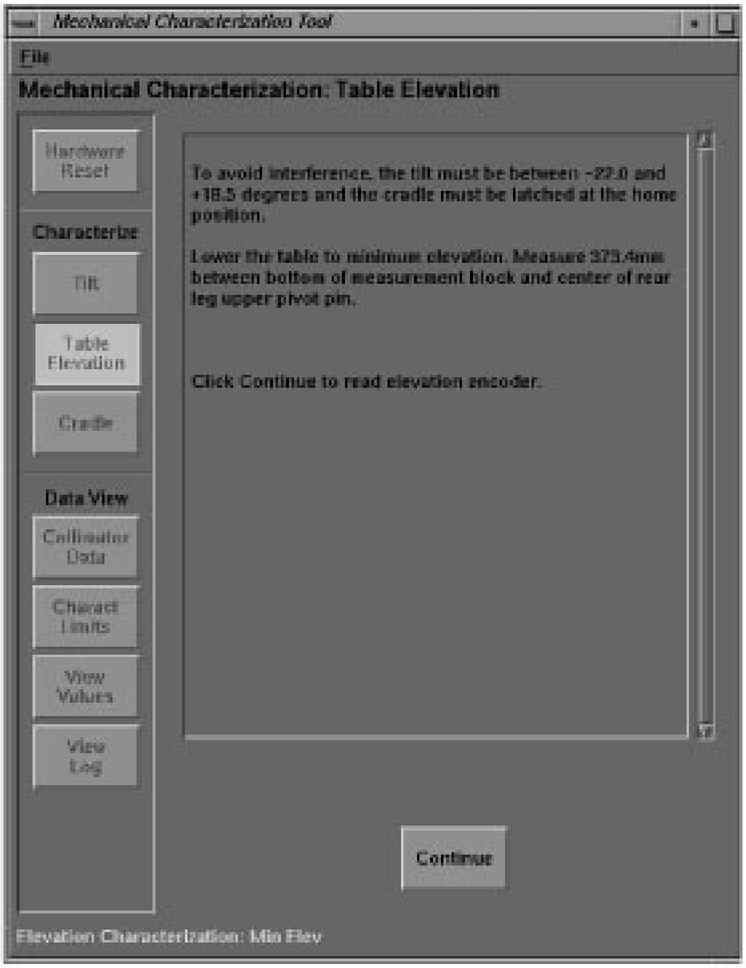


Figure 4-9 TABLE ELEVATION CHARACTERIZATION

Note: When you measure elevation distance, measure as vertically as possible. Measure between the bottom surface of the appropriate measurement block (same plane as the top of the weldment plate) and the center of the rear leg upper pivot pin, per Figure 4-10. This measurement is critical. Incorrect measurements cause elevation characterization failure.

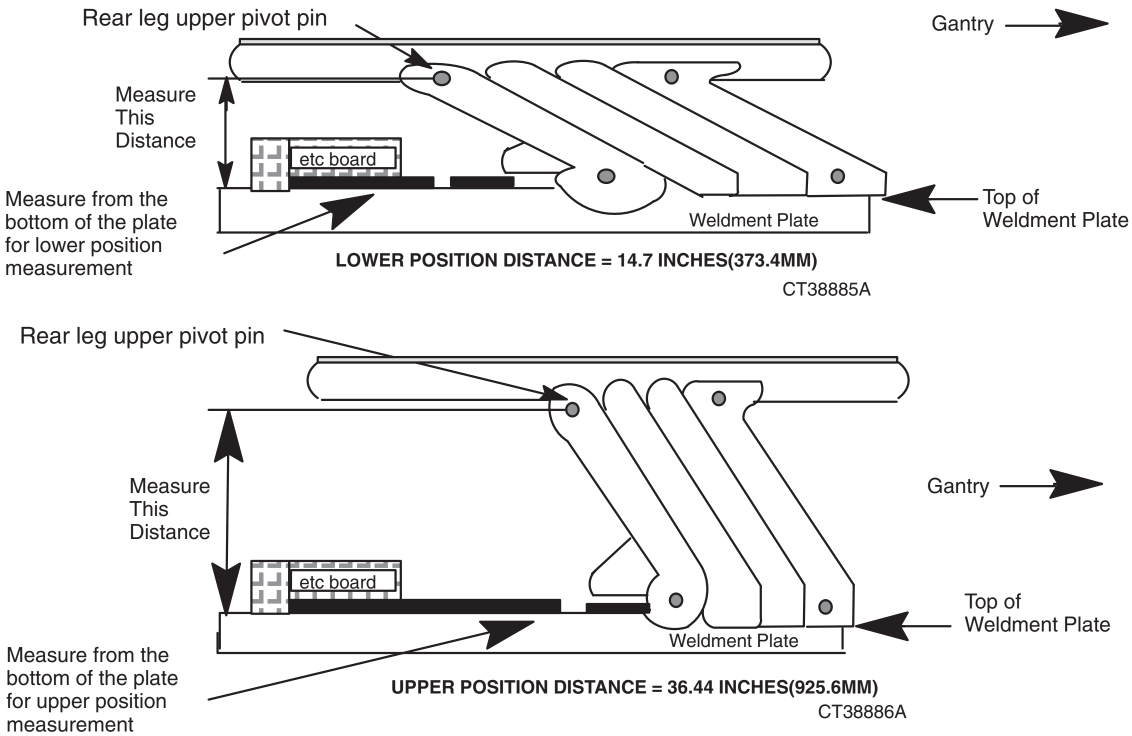


Figure 4-10 ELEVATION DISTANCE MEASUREMENT

9.5 Cradle Characterization

Select CRADLE and CHARACTERIZE in sequence, then follow the Characterization instructions displayed on the monitor screen. (See Figure 4-11.)

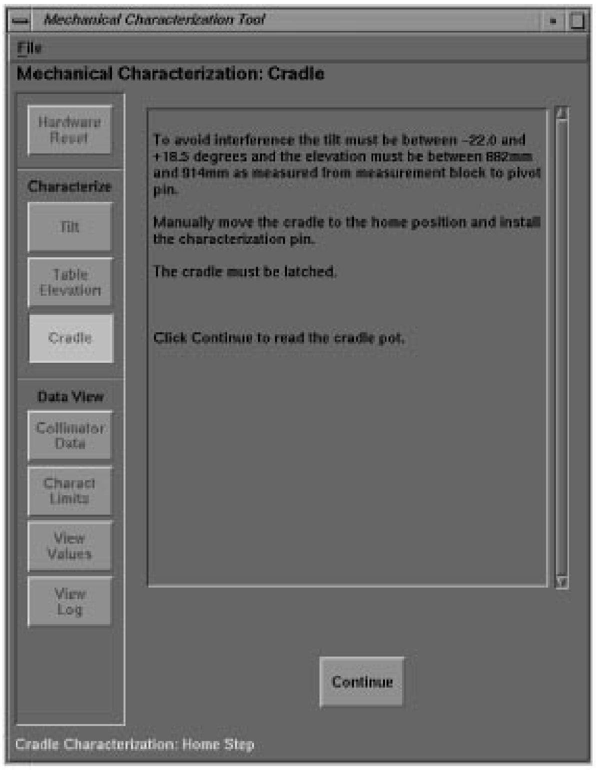


Figure 4-11 CRADLE CHARACTERIZATION

9.6 Collimator Characterization

The characterization tool permits the user to view and update the current collimator characterization data.

1.) Select the COLLIMATOR DATA softkey shown in Figure 4-12.
2.) Enter the characterization values listed on the collimator characterization label.
3.) Select UPDATE TABLE and verify a “`successful characterization`” message is displayed.
4.) If the collimator characterization value(s) update is not successful:
    a.) Verify that the correct values have been updated.
    b.) Make sure the scanning hardware is up.

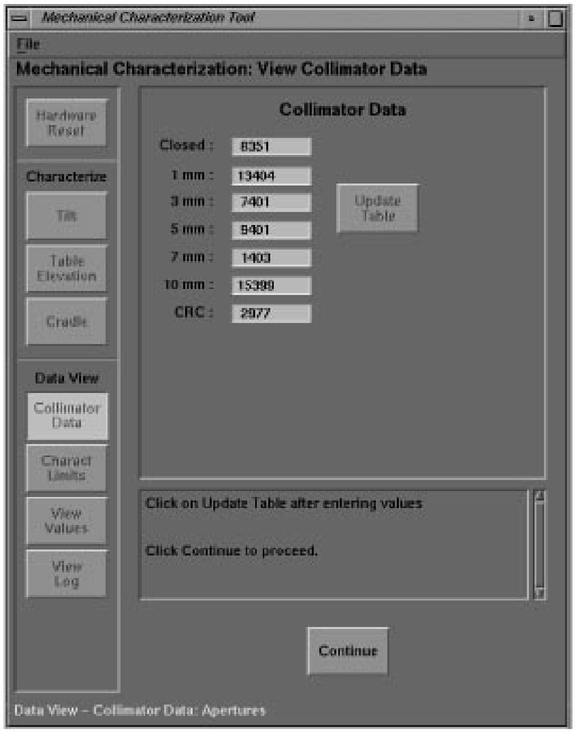


Figure 4-12 COLLIMATOR DATA

9.7 Hardware Reset

Select the HARDWARE RESETS softkey on the Characterization Screen to download the controller firmware and the new characterization values. (see Figure 4-7)

9.8 Characterization Limits

Select the CHARACT LIMITS softkey (Figure 4-13) to display the limit values for each of the characterized functions in storage in a disk resident configuration file. The limit values:

- Gantry Tilt: Minimum = -30°/Maximum= 30°
- Table Cradle: Minimum = 0 inches/Maximum= 67.3 inches (171 cm.)
- Table Elevation: Minimum = 1.0 inch (2.54 cm.)/Maximum = 22.7 inches (57.66 cm.)

Note: Measure table elevation limit distances from isocenter.

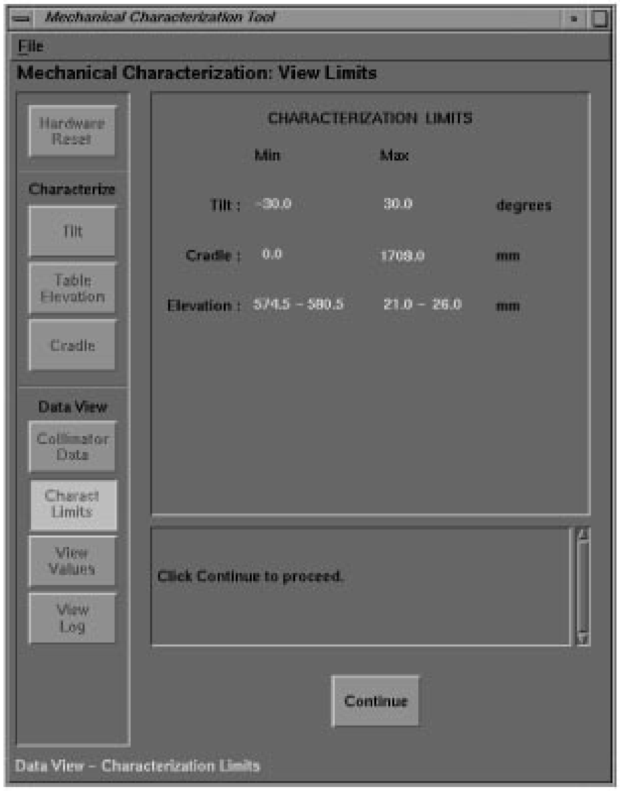


Figure 4-13 CHARACTERIZATION LIMITS

9.9 View Values

The system stores all characterization values in disk files. Select the VIEW VALUES softkey, shown in Figure 4-14, to display the current disk resident table containing the characterization values for each of the functions (table elevation, gantry tilt, etc.). In addition to the characterization values for gantry tilt, table elevation and table cradle position, the table displays the max/min limits for encoder counts (elevation and cradle) and the limits for the cradle and tilt pot outputs (in counts). The data shows the most recent change for each.

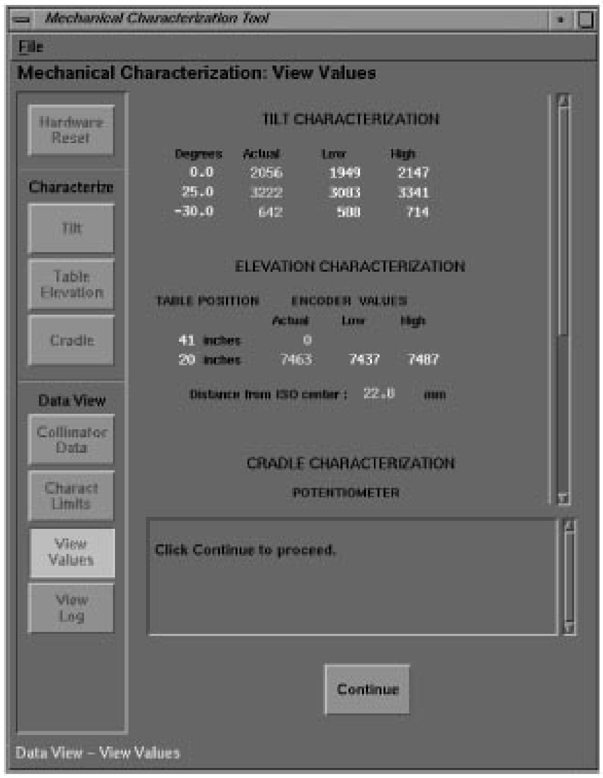


Figure 4-14 VIEW VALUES SOFTKEY

9.10 View Log

The View Log feature displays event messages and time of occurrence. Select the VIEWLOG softkey to display the characterization log. The characterization tool maintains a log to provide a characterization program history. The tool logs the following events:

- Characterization program entry and exit
- Characterization file modified
- Characterization file downloaded

End of Chapter

Chapter 5 System Theory

Section 1.0 Introduction

The HiSpeed CT/i service systems were designed to provide maximum assistance to both experienced and unexperienced service personnel.

The concept of Functional Error Messages and a Functional Breakdown of the scanner started with the HiSpeed Advantage family of scanners and extends to the HiSpeed CT/i in a more defined form. Here you will find the system's major functions described with their minor functions under them.

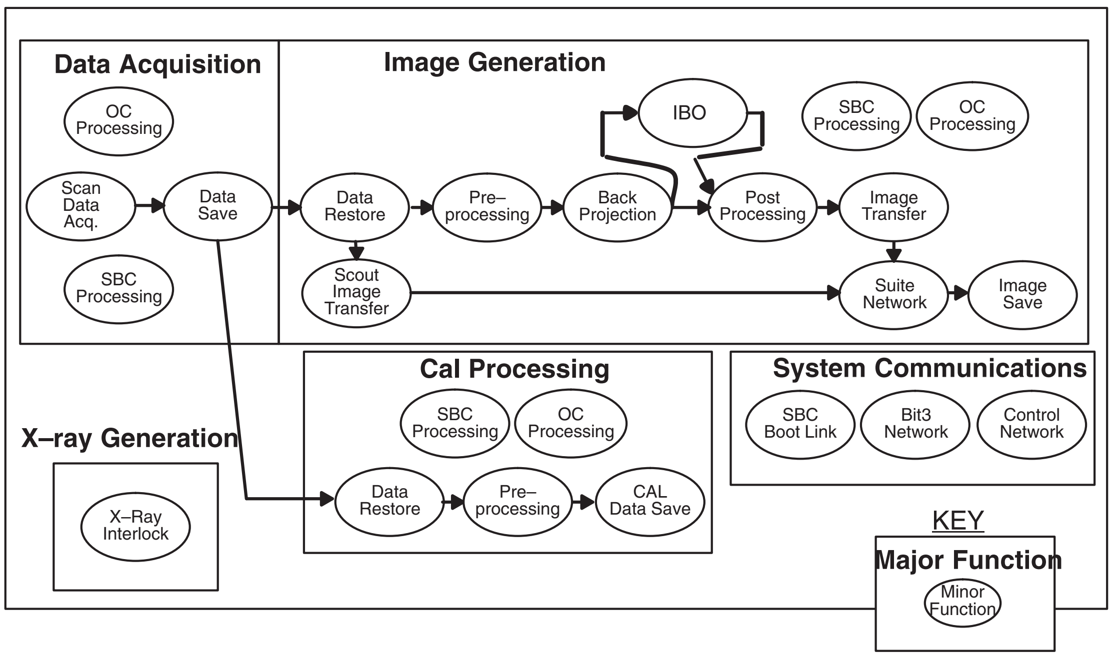


Figure 5-1 Example of System Functions Map

Section 2.0 Data Acquisition *(Major Function)*

2.1 Scan Data Acquisition (Minor Function)

The software “captures”, filters and converts Analog data samples from the detector, into digital signals in the DAS. Then, the system transmits the data from the DAS, through the OBC, to the RCOM taxi interface. The data crosses the slip-ring, and enters the RPSCOM taxi receive interface. From there, the system transmits the data over a high speed taxi interface to the FEP (Front-End Processing) board. The FEP converts the data from 16 bit DAS floating point to 32 bit integer, and sends it into an input FIFO. Two DSP chips read the data and perform view compression, offset correction and any view dependent scan data corrections. The DSPs also perform scout construction processing during scout scans. Once processed, the FEP transfers the data into the VME FIFO, and software on the FEP interrupts the SBC, which indicates the presence of data for saving.

2.2 Scan Data Save (Minor Function)

When the SBC receives notification that data is ready to be saved, it initiates a DMA transfer of the data from the FEP VME FIFO to the SBC's SCSI Disk Controller. The disk controller writes the data to the correct location on the High Speed SCSI Disk. The disk controller manages the writing of front-end processed axial scan data, diagnostic axial scan data (from DDC) and processed scout data.

2.3 Scan Data Trigger Generation (Minor Function)

2.3.1 DAS Trigger Generation Function

The axial control board is responsible for interfacing the axial drive control firmware of the gantry STationary Controller (STC) microprocessor to the DAS trigger generation functions. These interfaces include the gantry encoder, the Axial Control board and the ETC.

The axial control board generates the 984 Hz DAS trigger signal. This frequency is constant regardless of which scan speed is selected. It is also programmable to accommodate future CT applications and geometries. The DAS trigger signal uses inputs from gantry encoder, table, and x-ray command circuitry to coordinate the DAS trigger signal with gantry and table motion. The DAS trigger function produces both offset triggers for DAS offset characterization and view triggers for actual scan data acquisition. The trigger circuitry supports 4 scan modes: static, scout, axial, and helical. The modes and offset or view triggers output are selected by firmware.

The actions required to generate triggers and coordinate starting them at a desired position are too fast for firmware to carry out. Thus, firmware sets up the hardware by pre-programming the modes and parameters before the triggers are actually generated.

Scout and helical modes require a synch pulse from the table to coordinate the start of triggers. Static and axial modes do not need any information from the table. Scout scans use a fixed clock input reference to generate the triggers. Static mode also uses a fixed clock reference. Helical and axial use the gantry encoder signal as a reference to generate triggers. This way triggers and gantry position are correlated.

2.3.2 DAS Trigger Circuitry

Before the gantry can take a scan, several trigger circuit parameters must be programmed into the axial board. These parameters include: reference input, x-ray to trigger delay, DAS output rate, scan mode, and scan duration.

The trigger circuitry has four modes: offsets, scout scan, static scan, and axial scan. The trigger circuit uses either an encoder or a 3 MHZ oscillator pulse train as its time base reference. The DAS output rate can be arbitrarily programmed by downloading counter divide and multiply values into a phase-locked-loop circuit. The phase-locked-loop can be inserted or bypassed in the trigger circuit chain. For scout, static, and axial scan modes, the duration of the scan must be programmed by downloading a trigger count value (how many DAS triggers during the scan) into a trigger count register.

2.3.3 DAS Trigger Mode Register

The DAS trigger mode register configures the axial drive hardware to operate in the static, scout, axial, helical, or offset scan modes.

2.3.4 PLL 82C54 Counters

The DAS trigger circuit contains three 82C54 counters that multiply and divide axial encoder frequency to generate the 984 Hz DAS trigger rate. The 82C54 counters must be programmed with the following divide values:

2.3.5 X-ray to First DAS Trigger Delay

The x-ray command signal must precede the occurrence of the first DAS trigger by 10 msec to allow the high-voltage subsystem some x-ray ramp up time. This delay assures that the tube is at full kv and ma before the DAS begins taking data. The delay between x-ray command and first DAS trigger is programmed into an 82C54 counter.

2.3.6 DAS Trigger Counter (68230)

The DAS trigger circuit contains a 68230 programmable counter that is used to control the length of a scan by counting the number of triggers that the DAS trigger circuit generates. The counter is preprogrammed with (984*n-1) count (n=1,2,3,4).

2.3.7 DAS Trigger Status Timer

An 82C54 counter is gated by the output trigger signal. By programming this counter, the output of the counter can be used as a status to determine whether the DAS trigger frequency is present. The counter can also measure frequency to determine the actual output frequency of the DAS triggers.

2.4 Detector Heater (Minor Function)

The I/O Board monitors and controls the temperature of the Detector. The Detector Heater temperature is monitored by a thermistor mounted in the center of the detector. The output of this thermistor goes to the I/O Board and the output of the I/O Board is a FET switch which turns ON/OFF, a solid state relay that turns ON/OFF a Power Supply which supplies approximately +24 VDC to the Detector Heater that is attached around the circumference of the Detector. The set point for turning on the Detector Heater is 34 degrees C.

In addition to the detector temperature being available to the MUX on the I/O Board, is also is sent to the DAS via an BNC connector. Software converts this reading to a temperature whenever a Read Header is requested from software. Since the counts for the detector temperature are part of the Offsets collected prior to a scan, the detector temperature is available at any time and can be seen by looking at the header of any scan file.

2.5 SBC Processing (Minor Function)

During data acquisition, SBC software coordinates the following activities needed to acquire DAS data:

- Processes the prescription received from the OC
- Distributes the OC prescription to the STC and FEP
- Records the completion information for the prescription

The SBC software also provides the following support functions:

- Scan database operations, such as scan file allocation and database updates etc.
- Adds an entry to one of the queues which control subsequent processing.

2.6 OC Processing (Minor Function)

Prescription software runs on the OC and performs the following **data acquisition** functions:

- Collects and validates the operator's prescription information through the Scan Rx user interface
- Reserves image space for the prescription
- Verifies the presence of a sufficient number of available scan data files for the prescription
- Sends the prescription to the SBC

Section 3.0 Image Generation *(Major Function)*

3.1 Data Restore (Minor Function)

The Data Restore function coordinates the transfer of axial scan data from the High Speed SCSI Disk to the IG. IG software initiates the Data Restore function after it receives a request to process scan data (either generate an image, generate a calibration vector or generate a DD file). The IG software interrupts the SBC when it needs to transfer data from the High-Speed SCSI Disk to the IG. The SBC initiates a DMA transfer of the data (including a calibration module, offset vector, and “Front-End Processed” scan data) from the High-Speed SCSI Disk, through the SBC SCSI Disk Controller, to the shared memory in the IG.

3.2 Preprocessing (Minor Function)

Preprocessing begins with a transfer of scan data from the shared memory on the IG. The IG processes the data to generate filtered projections for normal axial images, PPSCANs, and views vs. channels (VVC). Preprocessing also includes any communication between the SBC and the IG, needed to prepare for, and clean-up after, these processing steps.

3.3 Back Projection (Minor Function)

Once the filtered projections are generated (for normal axial images) they are Back Projected using the eight IG DSP's, the Pixel Requestor Unit, and the Attenuation Pipeline. Completed image pixels are stored one or the other of the IG Image Memories.

3.4 IBO (Minor Function) (Minor Function)

After the IG processes the data to generate filtered projections, the Iterative Bone Option (IBO) processing steps occur. The IG sends two simultaneous backprojections (one SFOV image and one DFOV image) into image memory. The IG reduces the SFOV image, and applies a threshold, before it pulls the SFOV image from the IG image memory into DSP Local memory. The IG software running on the DSPs re-projects, IBO processes and filters the SFOV image. Upon completion of the second filtering, the IG back projects the SFOV projections on top of the previously back projected DFOV image, which readies the projections for post processing.

3.5 Post Processing (Minor Function)

After back projection generates an image memory full of pixels, the IG performs additional processing, which includes the use of the DSPs for image compression. The additional processing occurs “in place”. In other words, the post processing output returns to the same image memory locations its input occupied.

During PPSCAN and views vs. channels, the filtered projections/images transfer directly from IG memory to the OC, in the same way as normal axial images.

3.6 Image Transfer (Minor Function)

Upon completion of post processing, the SBC software in control of reconstruction transfers the compressed axial image from the image memory on the IG into SBC memory. The SBC prepares the image for broadcast to the OC computer and any Independent Consoles (ICs) within the suite.

3.7 Suite Network (Minor Function)

The SBC broadcasts scout and axial images from SBC memory, across the ethernet (Bit3 Interface), to the OC computer. The image goes to shared memory on the OC computer. The system also uses the suite (ethernet/Bit3) network as the communication path between OC and SBC processing functions.

3.8 Image Save (Minor Function)

The image leaves shared memory, and travels to the OC magnetic disk. The disk controller saves each image to the OC database and raw partition. When the user selects AutoView, the software also takes the image out of shared memory, writes to the IP bulk memory, and displays it on the screen.

3.9 Scout Image Transfer (Minor Function)

The Scout Image Transfer function takes processed scout data out of its “temporary” buffer on the High Speed SCSI Disk and transfers it into the SBC memory. The SBC broadcasts scout images across the ethernet (Bit3 Link), to the OC computer. This normally occurs concurrently with scout acquisition. If the transfer to the OC fails, the operator may choose to retry the transfer.

3.10 SBC Processing (Minor Function)

During image generation, software on the SBC coordinates the activities necessary to generate images from scan data which includes: managing the reconstruction data flow (including initiating the image transfer and broadcast on the suite network), and controlling multiple reconstructions. The SBC software also provides support functions including: scan database operations (e.g. reading scan file information, database updates etc.), and management of the queues (e.g. reading, and deleting queue entries, etc.) which control image generation.

3.11 OC Processing (Minor Function)

Scan Prescription software runs on the OC (SGI host) and controls image generation functions (SRU), such as:

- Collects and validates the prescription information from the operator, through the Scan Rx or Recon Rx user interface
- Reserves image space (SCSI drives and cables) for the prescription
- Sends the prescription to the SBC (BIT3-BIT3)

Section 4.0 System Communications *(Major Function)*

4.1 SBC Boot Link (Minor Function)

The serial line connection is using the “Specialix” (INDIGO) or the “Serial Port Expander” (OCTANE) between the OC and the SBC:

- Boots the SBC when power is on, but UNIX is down
- Is used by the “`cu`” utility during installation
- Can be used to monitor the SBC boot output

During UNIX boot up, the computer probes for a disk before UNIX comes on-line. The high speed SCSI disk power-up takes approximately 20 to 30 seconds.

1.) After full power is applied to the Computer subsystems:

   a.) Both the SBC and OC CPUs complete their power up diagnostics.

   b.) The OC and SBC begin auto-booting UNIX.

   c.) The OC boot-up script contains a script to boot the SBC, if it fails the first boot attempt.

2.) If the OC already has UNIX booted, but Applications is down (the SBC not booted):

   A program within the Applications start-up script polls to see if the SBC is booted. If not, the script uses the “`cu`” serial connection to manually boot the SBC.

4.2 Control Network (Minor Function)

The control network has a bi-directional communication path, which means components send *and* receive information. The SBC sends data packets across the ethernet cable, to the CPUs on the control LAN (STC, ETC and OBC). These packets may contain prescription information, statuses and other commands necessary to control the scanning hardware.

When downloading the controllers, the ETC, STC and OBC read the “download” files from the SBC's local SCSI disk, across the ethernet cable, to the designated CPU.

4.3 Suite Network (Minor Function)

After the scout or axial image reaches the SBC memory, the SBC broadcasts it across the ethernet, to the OC computer and any IC's within the suite. The image goes to the shared memory on the OC computer. The system also uses the suite (ethernet) network as the communication path between the OC and SBC processing functions.

4.4 Slipring Communications (Minor Function)

The slipring provides for a means to transfer bi-directional data across the slipring’s rotating interface.

Control and status information travels in both directions across the rotating interface. Scan Data travels from the rotating side to the stationary side and ultimately to the reconstruction subsystem.

4.4.1 DAS Data Transfer

The first task of the Gantry Communication subsystem is to connect the Data Acquisition Subsystem with the Image Generation subsystem through a high speed serial channel. It is the function of this subsystem to coordinate the flow of DAS data and ensure that it arrives at the Image Generation subsystem error-free.

4.4.2 CPU Communications

The second task of the subsystem is to connect the CPU in the STC with the CPU in the OBC through the slipring serial channel. Inter-CPU communication normally takes place over a serial Ethernet connection, so the only differences with communicating over the slip ring channel are the DAS Triggers and the Exposure Command.

4.4.3 Scan Control Commands

The third and final task of the subsystem is to pass two scan control commands from the stationary to the rotating side of the gantry. A command is a time sensitive signal also known as a hard-line or virtual wire. The two commands which are transmitted over the slip ring channel are the DAS TRiggers and the Exposure Command.

4.5 DAS Serial Control (Minor Function)

The DAS requires an RS422 interface for its serial link. The output of the OBC CPU Board is RS232. Thus, an RS232 to RS422 conversion is necessary. This conversion takes place on the Gentry I/O Board.

The Gentry I/O Board also has some diagnostics provisions built in that allows loop-back on the serial link input as well as the serial link output.

4.6 Final Scan / Autovoice Control (Minor Function)

Final Scan Control and Autovoice control are implemented using: Hard Switches on the keyboard assembly, an Hardkey Processor in the Keyboard assembly, hard wiring between the Keyboard Assembly and the STC, and the STC CPU Board.

The Hardkey processor monitors the status of the “Start Scan” and “Stop Scan” Push-button on the keyboard assembly. Upon closure of the Start Scan Switch Contacts, the Hardkey Processor sends a start scan message to the STC indicating that the scan should start.

A watchdog timer running on the hard-wire link between the Hardkey Processor and the STC ensures that the links integrity has not been compromised. A watchdog time out due to a broken connection or a processor malfunction at either end will cause an error condition that will terminate any application scan in progress and log appropriate error messages.

This communications link is also used to signal the start of Autovoice messages, i.e. scanner ready to scan.

4.7 Autovoice / Intercom (Minor Function)

The Intercom function selects and amplifies the sound of a patient's voice while the patient is laying on the CT table. The Intercom also amplifies the voice of the system operator while he or she is at the operator's console.

4.7.1 Components

The Intercom / Autovoice function consists of the following components and the wiring that connects them: Gantry microphones, OC microphone, OC Intercom board, Computer Audio Board (piggybacked on the Motherboard), Gantry Intercom board, OC and remote volume controls, OC Push-To-Talk Button, OC speaker and Table speaker.

4.7.2 Gantry MIcrophones and Speakers

Patient speech is received by the microphones mounted in the Gantry and processed in the Gantry Intercom board before being passed along to the console for further amplification. The amplified signal drives a speaker within the console for communication with the operator.

In the Table - Gantry room, Pressure Zone Microphones (PZM) are used to improve the Signal (speech) to Noise (background) ratio. These microphones are located within the back and front of the gantry cone to provide optimum coupling of speech information.

4.7.3 Console Microphone and Speaker

The console microphone receives the operator's voice and sends it to a pre-amp located in the Intercom board. A switch on the intercom board selects either the OC microphone or the Audio Output from the computer. This signal is amplified and then sent to the speaker in the patient vicinity. The operator can adjust the volume for both the console and the patient speakers.

4.7.4 Where autovoice is stored

Autovoice information is stored on the system disk and is selectable by the operator for playback within scan prescriptions. Autovoice messages are recorded using the OC microphone. A switch on the intercom board is used to connect the OC microphone to the record input of the computers audio section. Software on the OC computer stores the digital output of the audio section as a file on disk.

4.7.5 Speech frequencies

The important part of this speech is found in the frequency spectrum from 600 Hz to 10KHz. A patient's speech power, into the microphones, changes over a wide range as the Table and patient move through the gantry during a scan. Also, the frequency content of background noises caused by cooling fans cycling on and off, gantry rotation starting and stopping, air conditioning cycling on and off etc. can vary between 60 Hz and 20 KHz.

4.7.6 Intercom Board

A two stage pre-amp circuit provides AC coupling and a 10K Ohm load for the microphones. Two inverting inputs and two non-inverting inputs provide noise cancelling.

A Voice Operated Switch (VOX) enables the intercom connection between patient and console whenever the patient speaks. This VOX has a limited bandwidth, rapid attack and slow decay and responds only to speech, not the more constant background noise.

A band-pass filter which excludes frequencies below 500 Hz and above 5 KHz feeds two rectifier circuits. One rectifier circuit has a rapid response to incoming signals (speech). The second rectifier circuit has a slower response to incoming signals (background).

4.7.7 ALC

Automatic Level Control (ALC) adjusts the intercom amplifier gain for variations in speech power due to patient condition, age, sex and position. The ALC stage converts a wide range of signal level to a comparatively uniform signal level (50 to 100 mV). R/C networks in this stage shape the system response to favor voice frequencies. When a loud noise is sensed, a higher than normal DC voltage controls the variable gain element within the ALC chip. This keeps amplifier gain within the specified limits.

4.8 Security Link

The system provides for a security mechanism to control access to certain programs on the scanner. This mechanism makes use of a commercial hardware security key commonly referred to as a dongle.

The system communicates with the security hardkey via a standard serial RS-232 communications port on the OC Computer. Software running on the OC Computer will query the hardware key if present and determine what the access privileges that are allowed with that key installed.

If the key query finds no key, an expired key, or an invalid key, access privileges remain at the General Class Level.

If the key query finds a key that is valid and not expired, access appropriate for that key type will be allowed.

4.9 Scan Control Network

The Scan Control Network provides a network communications path between the SBC Scan Recon Control (SRC) in the console and the hardware controllers in the gantry.

The network is used to send scan control commands, status and data. This network uses standard ethernet protocols and operates over a 10Base2, Thin-net topology.

During scanning subsystems resets, firmware of the correct type, diagnostic or application is “downloaded” from the SBC to the Scan Control CPUs (Heurikon Boards) using the SCU Network.

Section 5.0
X-Ray Generation *(Major Function)*

5.1 X-Ray Exposure Interlocks (Minor Function)

Primary Abort mechanism: If the SBC or the FEP detects an error during data acquisition, software on the SBC sends a command to software on the STC to stop X-Rays.

- If the primary mechanism breaks down, both the SBC and FEP software have access to a register/memory location on the FEP board that connects to an abort relay.
- The software that detects the failure of the “primary” abort mechanism, writes to this FEP register, which in turn opens the abort relay connected to the PDU. The abort relay turns off X-Rays and the 550VDC.
- A number of other subsystems, including the PDU and gantry, have a part in the “X-Ray exposure interlock chain.”

5.2 Final Exposure Command

5.3 DC HV Supply Backup Contactor Interlock

The Back-Up Contactor is a three-phase 480 VAC contactor. Contact closure supplies power to the HVDC power supply whether it be a DCRGS or Un-regulated type. The back-up contactor provides a means to remotely inhibit x-ray generation by turning off the voltage supply to the HVDC Power Supply.

5.4 Table – Gantry Sync

AXIAL SYNCHRONIZATION

The table and gantry subsystems require a hardware sync line which is driven by the ETC and received by the Axial Control Board in the gantry. This function is used for Scout Scans to indicate to the trigger control logic on the Axial Board that the cradle has reached start of scan position. The trigger control logic only looks for this signal when Scout Scan Mode is enabled on the Axial Board. The signal also interrupts the processor (CPU) which controls the Axial Board in the STC. A noisy line of a missed-wired line could cause spurious interrupts to the STC.

The line is driven deferentially and received on the Axial Board through an opto-isolator. An output port on the DUART is used to generate the sync signal. The DUART must be initialized to configure the output port (OP2). Once the device is programmed, the output can be toggled high and low by writing 04H to address 0FFAE5CH to set the output high, and then writing 04H at address 0FFAE5EH to reset the output low again.

5.5 Tube Rotor Control –HSA Tube

5.5.1 Tube Rotor Control

The Rotor Control Board is located in the OBC and communicates with the OBC CPU via the VMEBus. The Rotor Control Board provides the PWM generation necessary for the gate drives to the CTVRC power module. Current feedback from the CTVRC module is utilized for closed loop stator current regulation to minimize X-Ray Tube thermal effects on anode rotating speed. Three modes of operation exist: Acceleration, Run and Brake. Braking of the X-Ray Tube anode is accomplished by electrically reversing the phase polarity of one of the CTVRC Inverters. The power amplifier drive frequency is controlled from the Rotor Control Board and is continuously variable between 90 and 180 hz (anode speeds of 5200 to 10,400 rpm). Actual operating speed is programmed by firmware from the OBC CPU.

Protection circuitry for Stator Open Circuit, Stator Short Circuit and DC Rail shoot-through are provided in addition to normal stator current monitoring by firmware during patient scanning.

5.5.2 CTVRC Power Module

The CTVRC Power Module utilizes two variable frequency IGBT half bridge inverters. The inverters are digitally synchronized in the Rotor Control Board to provide the required 2-phase quadrature drive. DC Power for the two inverters comes from the DC Rail HVDC buss. The outputs of the inverters are transformer coupled to maximize tube spit immunity. A resonant tuned output scheme is utilized and normal operation is both above and below resonance. The DC Rail and capacitor voltages are sensed and divided down to low voltages and are provided to the rotor control board DC rail monitor.

5.5.3 Tube Motor

The X-Ray Tube contains the quadrature stator windings that forms half of the X-Ray Tube motor which spins the anode of the tube.

The second half of the X-Ray Tube motor is made up of the Anode Rotor assembly inside the X-Ray Tube Insert.

5.6 Tube Rotor Control – PERFORMIX Tube

5.6.1 Tube Rotor Control

The Rotor Control Board is located in the OBC and communicates with the OBC CPU via the VME Bus. The Rotor Control Board performs three main functions. It provides an interface between the OBC and the HEMRC, it provides for HVDC Bus voltage monitoring, and it provides a CAN interface between the OBC and future subsystems. The rotor control firmware resides on the OBC CPU board and communicates with the HEMRC Control board which controls the acceleration, run, and deceleration cycles of the rotating anode.

The rotor motor (HEM) is an induction motor, but also is a generator. The frequency of the voltage supplied by the HEMRC is variable. When the HEMRC frequency is higher than the motor mechanical frequency, the HEM is a motor that accelerates or holds constant the Anode rotor speed. When the HEMRC frequency is lower than the HEM rotor mechanical frequency, the HEM becomes a generator which converts energy stored in the anode rotation to 3 phase current which is forced back through the HEMIT into the HEMRC where it is converted to charge stored in the HEMRC's DC buss.

Protection circuitry for Stator Open Circuit, Stator Short Circuit and DC Rail shoot-through are provided in addition to normal stator current monitoring by firmware during patient scanning.

5.6.2 HEMRC Interface Board

The HEMRC Interface board provides the input means for the system to monitor the HVDC Bus. The resistors R1 through R5 form the input network of a differential amplifier circuit located on the HEMRC Control Board. The output of this network drives a set of fault detectors and can be read by the OBC CPU for monitoring of bus status.

HVDC is applied to the board at TB1 and TB2, passed through fuses F1 and F2 and output to the AC Drive at J1. Fuses F1 and F2 provide isolation between the HVDC bus and the AC Drive in the event of a component failure. An LED, (DS1) is provided to indicate to service personnel when voltage is present.

The Chopper circuit is used to help dissipate excess energy in the AC Drive's internal DC bus. During braking of the X-ray tube rotor the system HVDC bus is turned off. This allows U1 to enable the chopper circuitry. As the tube decelerates, its motor acts as a generator and some of the kinetic energy is converted to current. This current is channelled by the AC Drive into its internal DC bus. As a result, the voltage on the bus begins to rise. If the bus voltage exceeds 810 volts the drive will disable itself and abort the braking process. The tube is then allowed to coast. The chopper is provided to limit the bus voltage to approximately 750 volts and, thereby, prevent the tube from coasting.

5.6.3 HEMRC AC Drive

The HEMRC AC Drive is at the heart of the Rotor Control Function. It contains its own microprocessor, power supplies and a three phase full bridge inverter. It communicates with the CT/i system through the OBC CPU via a CAN (Controller Area Network) serial bus. A derivative of the customized vendors AC Drive protocol is utilized for maximum speed and efficiency of communication.

The OBC controls all sequence operations of the drive via firmware. The drive's internal cpu controls lower level detail functions and fault protection. In addition to the CAN communication, a number of hardware lines are used to control the drive. These include enable, start, stop, at speed, and fault signals. The drive's isolated 12V supply is also returned to the OBC and used to power opto-isolation of these lines.

During operation the output of the drive is a 3 phase voltage produced by variable pulse width switching of the drive's IGBT inverter. The peak voltage of this output equals either the HVDC (High Voltage DC) bus voltage or the rectified 380 volts from isolation transformer T1, whichever is greater. However, independent of bus voltage, the drive uses PWM switching to maintain the commanded RMS 3 phase output voltage and frequency. The OBC firmware modifies the commands to the drive as required to supply the current needed for acceleration, run and deceleration of the X-ray tube.

5.6.4 HEMIT

Conventional X-ray tubes use a stator assembly which is electrically near ground potential. Isolation from anode potential is achieved within the tube insert via the gap between anode stem and stator. The High Efficiency Motor used in the Performix tube does not have this relatively large gap. Therefore, the stator is electrically tied to the anode, riding on anode potential.

The HEMIT (High Efficiency Motor Isolation Transformer) is used to provide the necessary high voltage isolation between the AC Drive and the tube stator. The HEMIT is actually three single phase transformers configured as a delta-wye three phase bank and packaged inside the Anode HV Supply. A stator filter board is used in the primary circuit to minimize spit energy coupled back to the AC Drive on the stator cable. Stator power is delivered to the tube via the Anode HV cable.

5.6.5 Tube Motor

The X-Ray Tube contains the 3 phase stator windings that forms half of the X-Ray Tube motor which spins the anode of the tube.

The second half of the X-Ray Tube motor is made up of the Anode Rotor assembly inside the X-Ray Tube Insert.

5.7 kV Loop

The KV Loop function regulates the KV output of the system to the anode and cathode of the X-ray tube.

5.7.1 KV Board (46-321064G1)

The KV Loop function regulates the KV output of the system to the anode and cathode of the X-ray tube.

The components involved in this function are the KV control board, two high frequency inverters, and two high voltage supplies, each consisting of a transformer, rectifier and filter, which are connected to the anode and cathode of the X-ray tube. The KV output is regulated by the KV control board, which monitors the KV using voltage dividers in the high voltage supplies and maintains the KV by adjusting the operating frequency of the two inverters.

The KV control board is located in the OBC card rack and communicates with the CPU via the VME bus.

The KV command DAC generates the analog KV set point that is summed with the cathode KV feedback signal that originates in the cathode HV supply. The product is modified, forming Vcnt, which determines both the operating frequency of the inverters through a voltage controlled oscillator (VCO), and the phase shift for the cathode inverter. The output of the VCO is processed by the phase shifting circuit, and then drives a set of fiber optic transmitters which function as commands to the cathode IGBT gate drivers. Anode KV is also fed back to the KV control board and is combined with the KV command to form an anode KV error signal. This controls the anode phase shifter which independently regulates the anode KV. These signals drive a set of fiber optic drivers which are the commands to the anode gate drivers.

5.7.2 KV Board (46-321198G1 or 2143147)

The KV Loop function regulates the KV applied to the anode and cathode of the X-ray tube.

The components involved in this function are the KV control board, two high frequency “H” bridge inverters, and two high voltage supplies. The KV control board is located in the OBC card rack and communicates with the CPU via the VME bus. The inverters are located on their respective HV supplies and each supply consists of a HV transformer, rectifier and filter assembly, and built-in HV divider. The cathode supply also contains filament transformers. The supplies are connected to the anode and cathode of the X-ray tube via HV cables.

The KV is regulated by the KV control board which monitors the KV sensed by the voltage dividers in the high voltage supplies. KV is maintained by adjusting the operating frequency and duty cycle of the two inverters as follows:

The KV command DAC develops an analog KV command that is inverted and summed with the anode and cathode KV feedback signals. The resultant error is integrated and summed with a Preset DAC command which is KV and mA dependent. This signal, VCNT, is used to determine both the operating frequency, through a voltage controlled oscillator (VCO), and the average of both inverters duty cycle. Each inverters duty cycle is determined by phase shifting one half of its “H” bridge. Two balance circuits further modify the duty cycle of each inverter as required to keep anode KV approximately equal to cathode KV.

The output of the VCO and phase shifting circuits are processed by a pair of PALs and used to drive a set of fiber optic transmitters located on the OBC backplane. The fiber optics carry the commands to the IGBT gate drivers located on the inverters.

5.7.3 HV Supplies

Each high voltage supply consists of a high frequency transformer with voltage doublers connected to the secondaries. These are stacked to provide the high voltage required by the X-ray tube. The KV feedback divider resistors are connected to the high voltage outputs and the ground ends of the doubler stacks are grounded through small resistors for the mA feedbacks.

5.7.4 Inverters

The function of the inverters is to convert HVDC into a +/- DC Voltage square wave of a frequency and duty cycle determined by the KV Control Board. This square wave is then applied to a series resonant circuit consisting of an inductor and a capacitor. This circuit can be thought of as a tuned circuit which reduces the output of the inverter as it is de-tuned above the resonant frequency of 19KHz.

At higher mA, the circuit Q is high and the tuning is very sharp, characterized by rapidly falling KV as the frequency increases. At lower mA, the Q is lower and the tuning becomes less selective, requiring a larger increase in frequency to lower KV. At very low mA, frequency control is lost and the KV is controlled by phase shifting the inverters, which lowers their duty cycle and has the effect of lowering the average voltage of the square wave output.

5.8 mA Loop

The filament/mA control function is a closed loop filament and closed loop mA control. During the period of time during pre-exposure, and for the first 4 ms of the X-ray exposure, the filament/mA control is open loop on filament current. This filament current is a function of anticipated mA and KV, and is characterized during Generator Characterization to provide the correct filament heating that results in the requested mA for the requested KV. During the remainder of the exposure, the filament/mA control function is closed loop, regulating mA by adjusting the filament current.

The filament/mA control function provides 10 - 440 mA capability with 5% accuracy over the entire range, depending upon system configuration. Protection circuitry for; filament open circuit, mA imbalance, over-current, and inverter faults is provided in addition to monitoring of mA by firmware during exposures.

The mA Control Board contains both the digital/analog control circuitry and the filament inverter. The closed loop filament control uses filament feedback internally from the filament inverter. The closed loop mA control uses mA feedback from both anode and cathode HV supplies, but is closed loop on only anode mA to regulate the filament inverter. The filament inverter is a parallel resonant AC current supply with a 16KHz switching frequency.

The Cathode HV Supply contains the large and small focal spot filament transformers which provide HV isolation between the HV Filaments and the Low Voltage Filament Inverter. In addition, both the Anode and Cathode HV Supplies have an externally mounted Interface Measurement circuit board. This board contains the mA Sensor Resistor (10 Ohms), through which the actual tube mA flows. mA feedback to the mA Control Loop is sensed at this resistor.

Dual filaments/dual focal spots are provided to improve imaging below the 24 KW Level. Large focal spot selection is required for techniques above 24KW.

FILAMENT POWER SUPPLY

The power supply for the filament inverter is an unregulated 30VDC supply.

5.9 Tube Cooling

X-RAY TUBE COOLING CONTROL

The X-Ray Tube Pump and Fans are controlled by the Gentry I/O Board in the OBC. The output of the Gentry I/O Board drives a normally closed relay on the Laser Control Assembly. The X-Ray Tube Pump and fans are turned ON under the following conditions.

- Software Command
- A Watchdog Error
- A Tube Temperature/Pressure Error
- A Tube Casing Temperature error (Note that there is no longer a Tube Casing Thermistor installed.)

Software sets the tube Cooling ON bit whenever the system is initialized, and leaves it on for 60 seconds after a scan. When diagnostic firmware is downloaded, this bit is also set.

5.10 Tube Identifiers

The purpose of the X-Ray Tube Identification function is to identify the X-Ray tube type that is mounted on the CT gantry. This identification is performed by accessing the four identification jumpers located on the X-Ray tube.

The components involved in this function are the Gentry I/O board located in the OBC, the OBC backplane, the X-Ray tube, and the wiring connecting these components.

The Gentry I/O board provides +24 VDC to the X-Ray tube for use by the tube ID jumpers and the tube pressure switch. The four tube ID jumpers can be inserted (short circuit) or left out (open circuit). +24 VDC returns through these four jumpers to the Gentry I/O board for tube identification. Up to 16 different tube types can be identified with this implementation.

The tube IDs can be accessed through the Gentry I/O board bits called TUBEID1, TUBEID2, TUBEID3, and TUBEID4.

5.11 Tube Pressure Sense

The X-Ray Tube Unit contains a Normally Closed pressure sense switch wired in series with the Tube Unit Thermal Switch. Either of these switches opening will terminate exposures and prevent further exposures until they re-close.

5.12 X-Ray Collimation and Filtration

X-Ray Collimation is used to shape the X-Ray Beam to the desired shape for scanning. The beam attributes affected are; Beam Width (Scan FOV), Beam Thickness (Slice Thickness), Beam Profile (Beam Intensity Shaping).

5.12.1 Collimator Control Board

The Collimator board performs seven main functions in the Gantry. These functions are;

- VME Interface to the On-Board-Controller (OBC) CPU board.
- Collimator aperture control registers and logic.
- Collimator filter control registers and logic
- Voltage Regulation of the 5 Volt power used by the aperture and filter feedback optical encoder.
- Electrical interface to the aperture and filter stepper amplifiers.
- Electrical interface to the aperture and filter optical encoders.
- Voltage monitoring of the encoder 5 volt power and also the 38 volt supply used by the aperture and filter drives.

The Collimator Board is located in the A1 slot of the OBC chassis. The OBC chassis is located on the rotating portion of the CT Gantry.

The Collimator board provides the control electronics for the CT Collimator Aperture Drive and Filter Drive, and also provides the electronic interface to the remainder of the CT system.

Position commands and position read back are made by the OBC CPU via the VME bus. The aperture and filter position are controlled via two stepper motors. The stepper motors are driven by the stepper drive modules mounted on the collimator assembly. Each rising edge of the respective aperture or filter “Step Pulse” signals will cause the corresponding stepper motor to increment or decrement one step as determined by the logic state of the “Step Dir” signals.

Position feedback is provided by two phase optical encoders mechanically linked to the stepper motors. In the case of the aperture, the encoder is mounted on the opposite end of the same shaft as the motor. The aperture size is determined by slits machined, at various angles, into the mandrel linking the encoder and the motor. The aperture encoder has 5,000 lines or 20,000 quadrature pulses per revolution. The aperture stepper motor uses 50,000 micro-steps per revolution.

The filter encoder is mechanically linked to the stepper motor by the chain and sprocket mechanical assembly which screws the filter block in and out of the X-Ray beam. The filter motor and encoder have over 15 turns of travel. A home switch is provided in addition to the encoder position feedback for the filter control. Home position is defined to be the logical end of the home switch and the presence of the encoder “Z” pulse.

5.12.2 X-Ray Filtration

The X-Ray Tube Unit and Primary Collimator provide a fixed amount of “Beam Hardening FIltration” in the X-Ray Beams path.

In addition to the fixed filtration a shaped filter is used in the X-Ray Beam to optimize photon flux across the patient cross-section. The filter is thinner in the middle than at the edges allowing more X-Ray energy to pass through the center of the scan field of view where the thickest portion of anatomy is likely to be.

Optional filter options optimize the beam profile for head and body scans by using different shape filters.

5.13 Filament Select

Selection of either the large or small focal spots is controlled from the mA Control Board by switching the Filament Select Relay, located on the OBC Backplane. The filament drive current is routed by the Filament Select Relay to either the large or small focal spot transformers in the Cathode HV Supply.

5.14 Rotating HVON Control

5.14.1 HVON Sense (KV Board 46-321064G1)

The purpose of the HVON sense function is to sense and report when Hi-Voltage (KV) is on. Hi-Voltage on is defined to be when either cathode or anode KV is above 75% of commanded value. When this condition occurs, HVON is asserted and the X-Ray lights are turned on. The circuitry that senses and detects this condition is located on the KV Control Board in the OBC. HVON is an interrupt driven transition that signals the OBC CPU whenever a transition from low to high, or, from high to low occurs.

HVON is also transmitted to both the Gentry I/O Board and the mA Control Board. In the Gentry I/O Board, the HVON signal is used for an input to the back-up timer. When either HVON or Expose Command is asserted, the Back-Up Timer starts. When both HVON and Exposure Command are low, the back-up timer stops.

In the mA Control Board, the HVON signal is used to open and close the mA Control Loop during tube spits. When a tube spit is sensed by the KV Control Board, HVON goes low. The mA board senses this and open the Filament/mA Control Loop into open loop mode and regulates the filament current. When the spit is completed and the HVON is asserted by the KV Board, the mA Control Board senses this and closes the Filament/mA Loop on mA current again as in normal operations.

5.14.2 HVON Sense (KV Board 46-321198G1)

The purpose of the HVON sense function is to detect when Hi-Voltage (KV) is present and alert the system accordingly. The information is used to:

- Control the X-Ray ON lights and console indicators.
- Control the back-up exposure timer on the I/O board.
- Enable the mA open loop control.

For purposes of controlling the X-Ray ON lights, etc., HV is defined as present whenever the cathode or anode KV is above 10 kV. When this condition occurs, a signal on the KV Control board, HVND, is asserted and the X-Ray lights are turned on. HVND is an interrupt driven transition that signals the OBC CPU whenever a transition from low to high, or, from high to low occurs.

For purposes of controlling the back-up timer and mA loop, HV is defined as present whenever the sum of cathode and anode kV is >75% of the commanded kV. When this condition occurs, a different signal on the KV Control board, HVON, is asserted and transmitted to both the Gentry I/O Board and the mA Control Board. On the Gentry I/O Board, when either HVON or Expose Command is asserted, the Back-Up Timer runs. When both HVON and Exposure Command are low, the back-up timer stops.

On the mA Control Board, the HVON signal is used to open and close the mA Control Loop. This prevents filament surges during tube spits. When a tube spit is sensed by the KV Control board, HVON goes low. The mA board senses this and opens the mA closed loop and regulates the filament current to the preset value. When spit recovery is completed, HVON is again asserted by the KV board and the mA Control board senses this and closes the mA loop again.

5.15 Rotating Backup Timer

The Backup Timer function sends an interrupt signal to the CPU in the event that X-Rays have not been turned OFF within a certain period of time. This time period is a function of exposure time, but it is not a fixed percentage. For example, for scans <= 500 msec., the exposure time is set for 108% of selected scan time plus 100 msec., and for scans >500 msec., but <=4000 msec., the exposure time is set for 108% of selected scan time, and for scans >4000 msec. the exposure time is set for 105% of selected scan time.

There are two timer modes selectable by software. The longest one is selectable when scans of 30 sec. or longer are selectable by the operator.

Section 6.0 Calibration Processing *(Major Function)*

6.1 Data Restore, Cal Data (Minor Function)

The Data Restore function (SBC under host control) coordinates the transfer of axial scan data from the High Speed SCSI Disk to the IG Board.

- IG software initiates the Data Restore function after it receives a request to process scan data (in this case to generate a calibration vector).
- The IG software interrupts the SBC, which prompts the SBC to transfer data from the High-Speed SCSI Disk to the IG.
- The SBC initiates a DMA transfer of the data (which includes an offset vector, and “Front-End Processed” scan data) from the High-Speed SCSI Disk, through the SBC SCSI Disk Controller, to the shared memory in the IG.

6.2 Cal Data Processing (Minor Function)

- Scan data transfers from shared memory into DSP Local Memory.
- The IG processes the data to generate the requested calibration vector(s). Cal Data processing also includes any communication between the SBC and the IG, needed to prepare for, and clean-up after, these processing steps.

6.3 Cal Data Save (Minor Function)

The SBC receives notification that the requested calibration vector(s) have been produced. The SBC initiates a DMA transfer of the data from the IG shared memory, and saves the vector(s) to the High Speed SCSI Disk.

6.4 SBC Processing (Minor Function)

During cal processing, software on the SBC coordinates calibration generation activities, including:

- Generation of calibration vectors from scan data
- Management of the calibration data flow (e.g. returns calibration status to the Cal Rx software on the OC)
- Provides support functions which control calibration vector generation, such as scan database operations (e.g. reads scan file information, cal database updates etc.), and manages the

queues (e.g. reads, and deletes queue entries, etc.)

6.5 OC Processing (Minor Function)

Prescription software runs on the OC, and performs cal processing functions, including:

- Collects and validates the operator's prescription information, through the Cal Rx user interface
- Validates the calibration scan lists, and adds prerequisite scans, as necessary
- Sends the prescription to the SBC

This Major Function has been combined with Cal Processing.

6.6 Data Restore, DD File (Minor Function)

The Data Restore function coordinates the transfer of axial scan data from the High Speed SCSI Disk to the SBC.

- SCSI software initiates the Data Restore function after it receives a request to process scan data (in this case to generate a DD file).
- The SBC transfers the data (which includes a calibration module, if one exists, an offset vector, and scan data) from the High-Speed SCSI Disk, through the SBC SCSI Disk Controller, to SBC memory.

6.7 Diagnostic Data Save (Minor Function)

The SBC produces the requested DD file(s), transfers the data to the High Speed SCSI Disk controller, and saves them on the SCSI disk.

6.8 SBC Processing, DD File (Minor Function)

During scan data analysis, SBC software coordinates the activities that generate DD files from scan data, including:

- Management of the tools that process data flow.
- The SBC software also provides support functions which control DD file generation, such as scan database operations (e.g. reads scan file information, DD file database updates etc.), and queue management (e.g. reads, and deletes queue entries, etc.).

6.9 OC Processing, DD File (Minor Function)

Prescription software runs on the OC, and performs such scan data analysis functions as the collection and validation of the operator's prescription information, through the Scan Analysis and Tool Rx user interface.

Section 7.0
Patient Positioning *(Major Function)*

7.1 Patient Loading

Patient loading is provided by latching the cradle at the home position and allows for quick patient access by the Cradle Release function. A mechanical latch is located at the cradle home position which automatically locks the cradle when it is at the home position. Home Position is when the cradle is at the extreme end of travel away from the gantry, and in its locked position. The locked condition is maintained during power off or when the cradle is released. Releasing the cradle disengages the motor from the drive roller and allows the cradle to travel freely. The cradle can be released through the table/gantry side push-button. Minimum height of the table is 20.26 Inches (51.46 cm) to assist in patient loading.

7.2 Patient Scanning

Patient Scanning is the ability of the table to move the patient throughout the scan plane. The cradle itself is a structure made of a high strength skin surrounding a foam core. These materials have low X-Ray absorption properties which are necessary to minimize image artifacts. Scan modes include; Scout, Axial, Cine, and Helical. For Scout Scans, the cradle moves a prescribed distance at a constant speed in or out of the Gantry, while X-Rays are on and Gantry is stationary. During this scan a hardware sync is generated to the axial control board, located on the STC, which indicates that the table has reached the start of the scan position. This tells the axial control to start scan triggers. During Axial Scans and Cine Scans, the table remains stationary while the gantry rotates and the X-Rays are on. In a scan series, the table indexes to the next prescribed position during the inter-scan delay period. For a Helical Scan, the table moves at a constant speed while the Gantry rotates with the X-Rays on.

7.3 Patient Alignment Lights (Minor Function)

The purpose of the Patient Alignment Lights function is to control the axial, sagital and coronal laser alignment lights.

The components providing this functionality are all on the rotating side of the CT gantry, and include the Gentry I/O board in the OBC, the OBC Backplane, the Laser Control Assembly on the back of the OBC, the three Laser Display Assemblies located around the rotating section of the gantry, and the wiring connecting these components.

The System detects a command from software or from the Table / Gantry Operator Controls to turn on the Laser Alignment Lights and several actions take place.

A signal is sent to the STC Axial Control Board to move the gantry rotating frame to a position of 180 degrees. This allows the Laser Lights to project through the gantry front cover openings.

The Patient Alignment Lights are enabled by the command bit AXLTSON, located on the Gentry I/O board. The status bit ALR-STAT and the ALIGNLTS ON LED are each also on the Gentry I/O board, and both are available to verify command circuitry status.

The Gentry I/O board supplies continuous +24VDC and PGND to a DC/DC converter located on the Laser Control board in the Laser Control Assembly. When the Gentry I/O board receives the AXLTSON command, it provides an approximately +8VDC voltage drop across the DC/DC converter's control input. This enables the DC/DC converter. The converter switches on and sends +12VDC and +12 RTN to the three Laser Display Assemblies to turn them on. Each Laser Display Assembly has a power converter to supply the correct energy for the lasers. Disabling AXLTSON turns off the DC/DC converter control input and the three Patient Alignment Lights. An alignment light service switch is provided on the Laser Control Assembly for use in adjusting the alignment lights. This switch disables the AXLTSON command and provides a continuous 8 VDC to the DC/DC converter.

7.4 Patient Scan Plane Angle (Tilt) (Minor Function)

Tilt motion control provides the control and drive to the Gantry Tilt motor. Tilt allows varied scan angles.

The Tilt Angle range is from +30 Degrees (toward the table) to -30 Degrees (away from the table), moving at an average speed of 1 Degree per Second.

7.5 Patient Vertical Position (Minor Function)

Patient positioning provides both horizontal and vertical positioning of the patient. Longitudinal motion of the cradle provides horizontal positioning through the scan plane.

Vertical motion provides for centering patient within the scan field of view. Elevating causes both vertical and horizontal motion due to the arc traversed during table motion. This results in increased scannable range as the table is moves up.

Patient positioning is done manually through the table/gantry side operator controls.

Alignment lights, located in the gantry, assist in precise positioning of the patient. These are operated through the table/gantry operator controls.

7.6 Patient Longitudinal Position (Minor Function)

Patient positioning provides both horizontal and vertical positioning of the patient. Longitudinal motion of the cradle provides horizontal positioning through the scan plane.

Patient positioning is done manually through the table/gantry side operator controls.

Longitudinal motion can also be controlled with console push-button used to advance the patient to the next scan position.

Alignment lights, located in the gantry, assist in precise positioning of the patient. These are operated through the table/gantry operator controls.

Section 8.0
Axial Control *(Major Function)*

8.1 Axial Loop Contactor Interlock (Minor Function)

The Axial Loop Contactor Minor function controls the output of the Axial Servo Amplifier connections to the Axial Drive Motor. The Axial Loop Contactor is a DC Rated relay that connects or disconnects the servo amplifier drive to the axial servo motor. This function also provides status monitoring and reporting of the loop contactor. There are interlocks in the loop contactor control for; gantry front cover open and an Axial Enable/Disable switch in the Gantry Service Display/Service Control box on the right side of the gantry.

⚠ **DANGER** **Disable the Axial Drive Mechanism before servicing the gantry interior. The Axial Loop Contactor Status LED Indicator on the Gantry Service Control Box indicates status of the Axial Loop control circuit only, not the state of the Axial Loop**

Contactor. A failure in the Loop Contactor could leave the Servo Amplifier connected to the axial drive motor at all times.

8.1.1 Axial Loop Contactor Power Circuit

When the PDU power is turned on, the PDU ON relay contacts close, allowing the +24V to flow out to the Gantry bulkhead. The current goes through relay K457 on the Axial Control board in the STC, through the Axial Loop Contactor Enable Switch on the Gantry Status Display Box, and through the Gantry Front Cover Interlock Switch, and returns to PDU servo assembly. The +24V then energizes the coil of the servo amplifier power pilot relay. This closes a set of N.O. contacts on the pilot relay, allowing 120VAC from the PDU Power Transformer to go through the servo amplifiers overload relay N.C. contacts and the Axial Loop Contactor Relay Coil. Both the Overload Relay and the Axial Loop Contactor Relay are located on the Servo Amplifier Assembly. The 120 VAC energizes the Axial Loop Contactor Relay.

8.1.2 Axial Loop Contactor Read back Circuit

When the Axial Loop Contactor Relay turns on, it closes a set of N.O. contacts that permits the flow of +24V through the Loop Contactor Relay contacts, back to the Gantry bulkhead. The current enters the Axial Board at and flows into an opto-isolator. The opto- isolator energizes the LPCNTRRB read back signal.

8.1.3 Loop Contactor Circuit

Relay K457 on the Axial board is energized when it receives the LOOPCONT command from data bit FFAE06. The N.O. contacts close, allowing +24V from the PDU ON relay on the PDU's Relay Control Board to flow through the K457 relay on the Axial board. The 24V continues into the Gantry Status Display Box, where it enters the Axial Loop Contactor Enable Switch at S3-1. If the switch is closed, the current continues out of the Gantry Display Box to the Front Cover Interlock Switch. If the front cover is also closed, the current can go through the interlock switch and back through the Gantry Status Display Box and the Axial Control Board to the PDU at J18. From J18 the current goes through the coil of relay K57 on the PDU's Relay Control Board and to ground. Relay K57's N.O. contacts close, allowing the 120VAC power to the Axial Loop Contactor coil. Also, if the Axial Loop Contactor Enable Switch is closed, +24V flows from the Axial Board into S3-2 of the Gantry Status Display Box and lights the Axial Loop Contactor Enable LED.

8.1.4 Axial Brake Circuit

Driver ULN2003A on the Axial Board receives the AXBRAKE command from data bit FFAE06 and enables 24V to the Axial Loop Contactor Enable Switch on the Gantry Display box. If the Enable Switch is in the ON position, the current can continue out of the Gantry Display Box to J1 of the Axial Brake and energize the Axial Brake Relay Coil. The current returns through the Gantry Display Box to the Axial Board and to ground.

8.1.5 Remote Axial C-Pulse Indicator Circuit

The ULN2003A driver connected at J3-B9 and J3-B10 on the Axial Board receives the CH-C signal coming ultimately from the Axial Encoder. When the driver senses CH-C, it enables 24V and sends this current through J3 and out to the Gantry Display Box, where it comes in through J1 and lights the Remote Axial C-Pulse LED. The current returns via J1 and J3 to the Axial Board and back to the driver.

8.1.6 Gantry Status Display Box

Axial Brake Circuit - The Axial Loop Contactor Enable Switch receives +24V from the Axial Board at S3-3. If the Enable Switch is in the ON position, the current continues out of the Gantry Display

Box to J1 of the Axial Brake and energizes the Axial Brake Relay Coil. The current returns through the Gantry Display Box to the Axial Board and to ground.

Front Cover Interlock Circuit - +24V arrives from the Axial board on J1 pin 3 and goes through the Axial Loop Contactor Enable Switch if that switch is closed. The current then goes out of the Gantry Display Box via pin 7 of J1, and from there, to the Front Cover Interlock Switch. If the front cover is closed, the 24V goes through the switch and back into the Gantry Display Box at J1-6. the 24V goes out of the Gantry Display Box through J1 pin 4, and back into the Axial Board.

8.2 Axial Servo Control Loop (Minor Function)

The hardware involved in the axial servo drive consists of the Axial Control Board, the Axial Servo Amp, the Axial drive motor, Axial motor brake, Axial gantry encoder, Axial drive belt and home flag.

The axial drive motor is a permanent magnet, DC servo motor. The axial brake engages the motor shaft and is meant as a static brake to hold the gantry still once it has been positioned by the axial drive. The brake's friction is not sufficient to hold the gantry still against the full accelerating force of the motor and amplifier. Should the brake fail while the gantry is in motion, the gantry will continue to rotate until halted by firmware. The brake cannot hold the gantry still while the tube or inverters are being changed. When servicing the rotating base, the gantry should be locked using the locking pin mechanism.

The axial brake is released when the Axial Drive Enable Switch is in the “disable” position. This allows the gantry to be rotated by hand without fighting the friction of the brake.

The encoder is directly coupled to the back of the Axial motor. It is an incremental encoder and provides 2048 counts per rotation. The quadrature information from the encoder is decoded on the Axial board and used for gantry position and speed control as well as DAS trigger generation.

The motor is coupled to the rotating bearing with a steel reinforced, Kevlar belt, with a gear ratio of 13 motor rotations to 1 gantry bearing rotation.

Section 9.0
Operator I/O *(Major Function)*

9.1 Table/Gantry Side Operator Interface w/Foot Pedals (Minor Function)

MANUAL OPERATOR CONTROLS

The Table/Gantry side push-button and foot switches located on each side of the table/gantry give the operator manual control of the of drive operations for Patient Positioning.

9.2 Gantry Display (Minor Function)

Centered on top of the Gantry, directly above the table opening, is the Gantry Display Board. Easily observable by the operator, this board gives patient position information along with certain status indicators.

The Gantry Display Board is controlled via an RS-232 interface located on the ETC (Enhanced Table Controller) circuit board. A DUART on the ETC provides the serial interface and the data is transmitted to and from the display and the ETC. Power for the Gantry Display Board is provided by the +24V from the quad (24 Hour) power supply located at the base of the table. Firmware communicates position and other status information through this interface.

9.3 Site X-Ray On Light (Minor Function)

The axial board provides a set of 24V/40mA contacts to control a 24V relay in the PDU. The 24V PDU relay closes a set of 110 VAC contacts that drive the site X-ray ON Light. The 24V relay contacts are closed (x-ray light turned on) whenever the axial control board commands x-rays to the OBC. The x-ray on light's relay is commanded by a register written to by firmware. The bits of the register will be cleared in the event of a power failure, CPU fault, or watchdog/board fault condition.

9.4 Gantry X-Ray On Light (Minor Function)

The axial board must drive a 24V 100mA LED mounted in the rear of the Gantry. This light, separate from the Gantry display, will indicate the presence of the axial board's x-ray command and will coincide with the PDU x-ray ON light signal.

The light is turned on and off by firmware writing to address 0FFAE08h bit 9. The light is controlled by firmware rather than hardware so that it can be coordinated with actual x-ray state and not just the x-ray exposure command. STC firmware receives LAN notification from the OBC when x-rays turn on and off. Firmware must coordinate the state of the x-ray exposure command and the actual state of hardware on the OBC. The x-ray light command register will be reset and the light turned off by an axial board reset.

The Gantry x-ray light is driven from the axial control board by three ULN2003 darlingtion transistors tied in parallel. The Gantry x-ray light should be used in conjunction with the site x-ray on light pilot relay. This relay, located in the PDU, is also controlled by the axial board. It is used by site personnel to wire a 120VAC x-ray light indicator in the CT suite.

9.5 Touch Screen (Optional) (Minor Function)

9.6 Image Video (Minor Function)

9.7 Input Devices (Minor Function)

9.7.1 Bar Code Scanner, Option

The Bar Code Scanner option allows bar coded patient information to be read into the scanner. Typical information type for the bar codes include: Patient ID #, and Requisition #.

The bar code scanner used is an off-the-shelf product that provides keyboard input emulation to the scanners computer. Rather than character output from the bar code scanner, the scanner outputs the equivalent of normal keyboard matrix key positions. In other words the output of the scanner is equivalent to, third key, second row of the keyboard. This allows any language keyboard to be used in conjunction with the bar code scanner.

Normally the computer keyboards output is a switch position on the keyboard. A character map in the computer translates the keyboard switch position into a printable (or non-printable) character that the computer then uses. By using a different character map in the computer software, we can change the interpretation of the keyboard switch positions which is what we do with other language keyboards. The keyboard may have different legends on the key caps, but the output for the location is the same, i.e. third key, second row, no matter the language. It’s the character map in the computer software that “changes” the language.

The bar code reader has the same output as a keyboard (it’s on a “Y” cable with the keyboard) in that it's output is the equivalent to third key, second row. This allows the bar code reader to work with any language keyboard configuration.

One implication of this implementation is that the bar code will be interpreted using whatever character map is in use on the scanners computer. If the bar codes are generated on a computer that is using a different language character map the output to the scanners screen maybe incorrectly mapped.

9.7.2 ConnectPro HIS/RIS Interface Option

ConnectPro works in conjunction with the Patient Information Management Feature on the scanner. The option allows queries to be made of the Hospitals Information Management system to return patient schedule information based upon user entered criteria such as: dates, times, modality, etc...

The information returned from the query is inputted to the Patient INformation Management application were it may be edited, or otherwise manipulated. This information may then be directly entered into the Exam Prescription.

9.7.2.1 Bar Code Reader Only

Use the hand-held bar code scanner to directly input information to the Exam Rx patient information screen. The Bar Code Reader hardware option reads information from the paper schedule or patient wrist strap bar codes.

9.7.2.2 HIS/RIS Interface Software *with* Bar Code Reader

Use the HIS/RIS software option to network query patient information directly to the CT/i Exam Rx desktop from the Hospital Information System (HIS) and/or Radiology Information Systems (RIS) database. This option kit also contains the hand-held bar code reader.

Section 10.0 System Monitoring *(Major Function)*

10.1 Mains Under voltage (Minor Function)

Monitors the AC Mains input to the system and reports an error if the level drops below a predetermined level.

10.2 DC Rail Monitor (Minor Function)

The HVDC Rail voltage is monitored for correct status and range during system operation. Voltage detected at an inappropriate time or at an incorrect level will be reported to the system control CPU and will result in an message being recorded in the system message log. Inappropriate levels detected during scanning will cause the current exposure cycle to be terminated with messages reported in the system log.

10.3 Gantry Temperature Sensors (Minor Function)

10.3.1 Thermistor Interface

The I/O Board monitors thermistors from the detector, previously described, the Gantry Ambient, the OBC Ambient and both the Anode and Cathode H.V. Power Supplies. The acceptable limits for various temperatures are:

Gantry Ambient: <40 Degrees C.,

The Gantry Ambient Thermistor is located on the Gantry casting between the Filament Power Assembly and the Tube and has the following characteristics:

TEMPERATURE	RESISTANCE	VOLTAGE READ BY MUX
25 Deg. C.	5000 ohms	3.18 volts
40 Deg. C	2663 ohms	5.39 volts

Table 5-1 Gantry Ambient Thermistor - Gantry Ambient: <40 Degrees C.

H.V. P/S's: <55 Degrees C.

In systems shipped before April 1993, the H.V. P/S thermistor is located inside of each tank and has the following characteristics:

TEMPERATURE	RESISTANCE	VOLTAGE READ BY MUX
25 Deg. C.	5000 ohms	3.18 volts
55 Deg. C	1493 ohms	6.77 volts

Table 5-2 Gantry Ambient Thermistor - H.V. P/S's: <55 Degrees C.

In April 1993, the thermistors were deleted from the HV Power Supplies. The gentry I/O board (46-288512G1-J) was modified to report gantry ambient temperature when the HV power supply is measured.

OBC Ambient: <55 Degrees C.

The OBC Ambient Thermistor is located inside of the OBC Card Rack on a metal plate near the exit port of the rack and near the CPU board. This thermistor has the following characteristics:

TEMPERATURE	RESISTANCE	VOLTAGE READ BY MUX
25 Deg. C.	10000 ohms	3.58 volts
55 Deg. C	2894 ohms	8.83 volts

Table 5-3 Gantry Ambient Thermistor - OBC Ambient: <55 Degrees C.

10.3.2 Detector Temperature

The Detector Heater Thermistor is located near the center of the Detector and has the following characteristics:

TEMPERATURE	RESISTANCE	VOLTAGE READ BY MUX
25 Deg. C.	10000 ohms	3.58 volts
34 Deg. C	6742 ohms	5.50 volts

Table 5-4 Detector Heater Thermistor (Typical Values)

The following table summarizes acceptable limits for the detector Temperature for various system functions and shows what is to happen in the event that the temperatures are outside of the limitations.

DETECTOR TEMPERATURE LIMITS	WHEN SAMPLED	ACTION TAKEN IF OUTSIDE LIMITATION
34 +/-1 C	Before CALS.	If this limit is exceeded, the error log will contain the actual temperature and a message will be posted to the operator requiring confirmation to continue CALS.
>30.5 C and <38 C	Before a Scan	If this limit is exceeded, the error log will contain the actual temperature. No messages will be posted to the operator.
>38 C, but the Gantry Temp is <= 40 C.	During a scan, every 5 sec.	The error log will contain the actual temperature of the Detector along with some messages to that effect and also a message will be posted to the Operator that “Detector Heater Control may have failed.” every 5 sec. Scanning will be allowed.
>38 C. and the Gantry Temp is >40 C.	During a scan, every 5 sec.	The error log will contain the actual temperature of the Detector along with some messages to that effect and also a message will be posted to the Operator that “Gantry Cooling may have failed - wait 20 minutes and retry”. Scanning will not be allowed.
>30.5 C and <38 C	During Diagnostic, every 5 sec.	If this limit is exceeded, the error log will contain the actual temperature.

Table 5-5 Detector Heater Thermistor (Acceptable Limits)

10.4 Rotating Power Supply Monitor (Minor Function)

A/D INTERFACE (46-321198G1)

The I/O Board has an 8 channel Mux that feeds a 12 bit A/D. The KV board, mA board and the CTVRC board each have dual 8 Channel Mux. The outputs from each of these feed into Mux inputs on the I/O Board. Separate 8 Channel Mux for the thermistors and Power Supplies feed into the main Mux on the I/O Board.

KV MUX

The signals that are on the **KV Mux** are;

MUX 1	MUX 2
Total KV	5V ref
Cathode KV	5V ref
Anode KV	PCNT (Average percent of duty cycle)
KV Cmd.	APH (Anode Duty Cycle)

Table 5-6 KV Mux Signals

MUX 1	MUX 2
10V. VREF	CPH (Cathode Duty Cycle)
VCNT (Inv. Freq.)	INVFRE (Inverter Frequency)
Cathode Inv. Current	+5V ref
Anode Inv. Current	Signal Ground

Table 5-6 KV Mux Signals (Continued)

MA MUX

The signals that are on the **mA Mux** are;

MUX #1	MUX #2
Anode mA	FILCT
Cathode mA	15VFB
SGND.	FILCMD
FIL DMD	mA BAL
FIL FB	CATH
MA DMD	VCC
+5V.REF1	+5V.REF2
30V.FB	ANOD+

Table 5-7 Ma Mux Signals

CTVRC MUX

The signals that are on the **CTVRV Mux** are;

MUX 1	MUX 2
DVC (Total Rail Volt.)	+5V ref (not used)
DCHI (Cap. Volt)	WCUR
DCLO (Cap. Volt)	STMP
CURRREF	CLREF
PWREF	+5V ref (not used)
LCUR	+5V ref (not used)
RCUR	+5V ref (not used)
+10V. REF.	Signal Ground

Table 5-8 CTVRC Mux Signals

GENTRY I/O TEMPERATURE MUX(S)

The signals that are on the **Gentry I/O Temperature Mux** are;

- Tube Casing (Not used)
- OBC Ambient
- Anode Inverter. (Not used)
- Cathode Inverter. (Not used)
- Anode HV P/S (approx. Gantry ambient)
- Cathode HV P/S (approx Gantry ambient)

- Gantry Ambient, Detector Heater

The signals that are on the **Gentry I/O Power Supply Mux** are;
+5V., +10V., +12V., -12V., +15V., -15V., +24V., SGND.

Two other signals that are on the Gentry I/O Mux are;

- Service
- Gnd.

10.5 Rotating DC Reference Monitor (Minor Function)

Section 11.0 System Power Control *(Major Function)*

11.1 HV Test Mode (Minor Function)

The HV test mode function is a diagnostic/test functional available in the HVDC power supply. The HV test mode switches the output voltage of the HVDC power supply to approximately 50 VDC to provide a lower voltage test capability for troubleshooting.

DANGER **THIS HV TEST MODE CREATES A LETHAL OUTPUT VOLTAGE. SINCE THE OUTPUT VOLTAGE OF 50 VDC IS FLOATING ON THE 480 VAC LINE, PEAK VOLTAGES WITH RESPECT TO GROUND OF OVER 600 VOLTS CAN BE PRESENT.**

Switching between the HV test mode and the normal HVDC output of the HVDC power supply can be accomplished in two ways:

1.) Normally, diagnostics controls and switches the output of the HVDC power supply through a control bit on the Axial Control Board in the STC. This control bit is hard-wired to the control electronics in the PDU, and electrically switches the HVDC power supply between the two possible output voltages.

2.) Manual control of the HV Test Mode may be accomplished two different ways depending upon system configuration:
   - For systems that utilize a DCRGS as the HVDC Power Supply, there is a dip switch provided on the DCRGS regulator control board that manually switches between the two possible output voltages. This switch overrides the command coming from the Axial Control Board.
   - For system that utilize an un-regulated HVDC power supply, there is a control switch on the PDU control/interface circuit board.

11.2 Emergency Stop (Minor Function)

The PDU provides control to stop motion and X-Rays in an emergency. The console and table contain switches which allow the operator to stop table and gantry motion and X-Ray output. These switches will stop X-Rays even during normal scan, when interruption will result in an incomplete scan that does not produce a useful image. Operation of “Emergency Stop” or “Drives Off” does not cause loss of computer power or loss of previously saved images.

This operation does not remove all power from the system and should not be used to remove all power to the system. Use the System Emergency Off button to turn off the system mains disconnect.

11.3 24 Hour Gantry 120VAC (Minor Function)

11.4 24 Hour Control Power 120VAC (Minor Function)

11.5 DC Rail Control (Minor Function)

11.6 System On-Off Control (Minor Function)

Section 12.0 System Control *(Major Function)*

12.1 Scan Control (Minor Function)

The SGI host commands the SBC according to the user's scan requests. The host stays in communication with the SBC as it controls the Table and Gantry hardware during a scan.

12.2 Tube Cooling Monitoring (Minor Function)

12.3 Scan Database Manager (Minor Function)

12.4 Peripherals Control (Minor Function)

12.5 Network Interface (Minor Function)

12.6 System Security (Minor Function)

12.7 OC Processing (Minor Function)

The SGI host translates user requests to commands for the rest of the system.

Section 13.0
Data (Image) Management *(Major Function)*

13.1 Image Save (Minor Function)

13.2 Image Restore (Minor Function)

13.3 Filming (Minor Function)

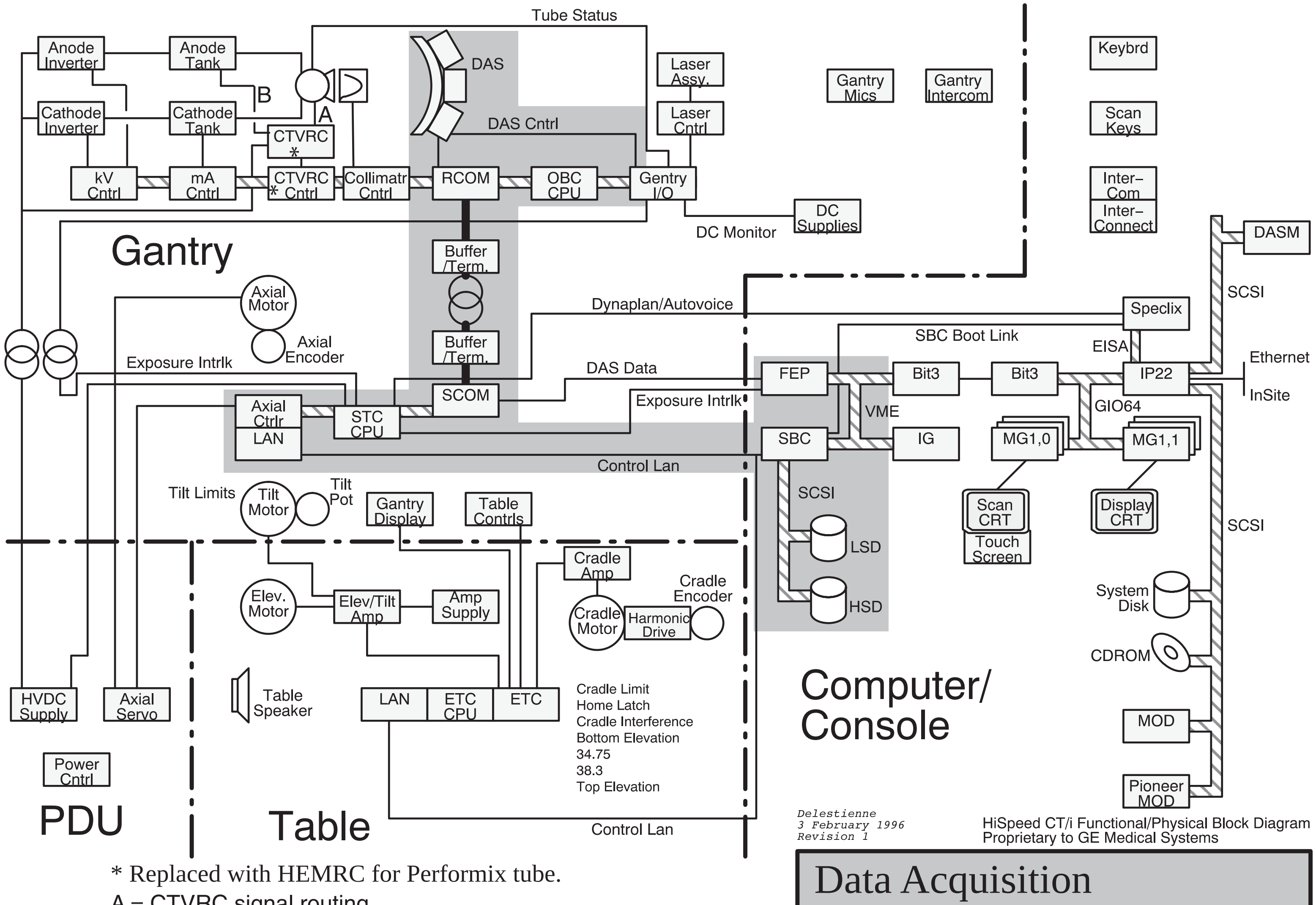
13.4 Archive Save / Archive Restore (Minor Function)

Section 14.0
Display Image Processing *(Major Function)*

IMAGE PROCESSING (MINOR FUNCTION)

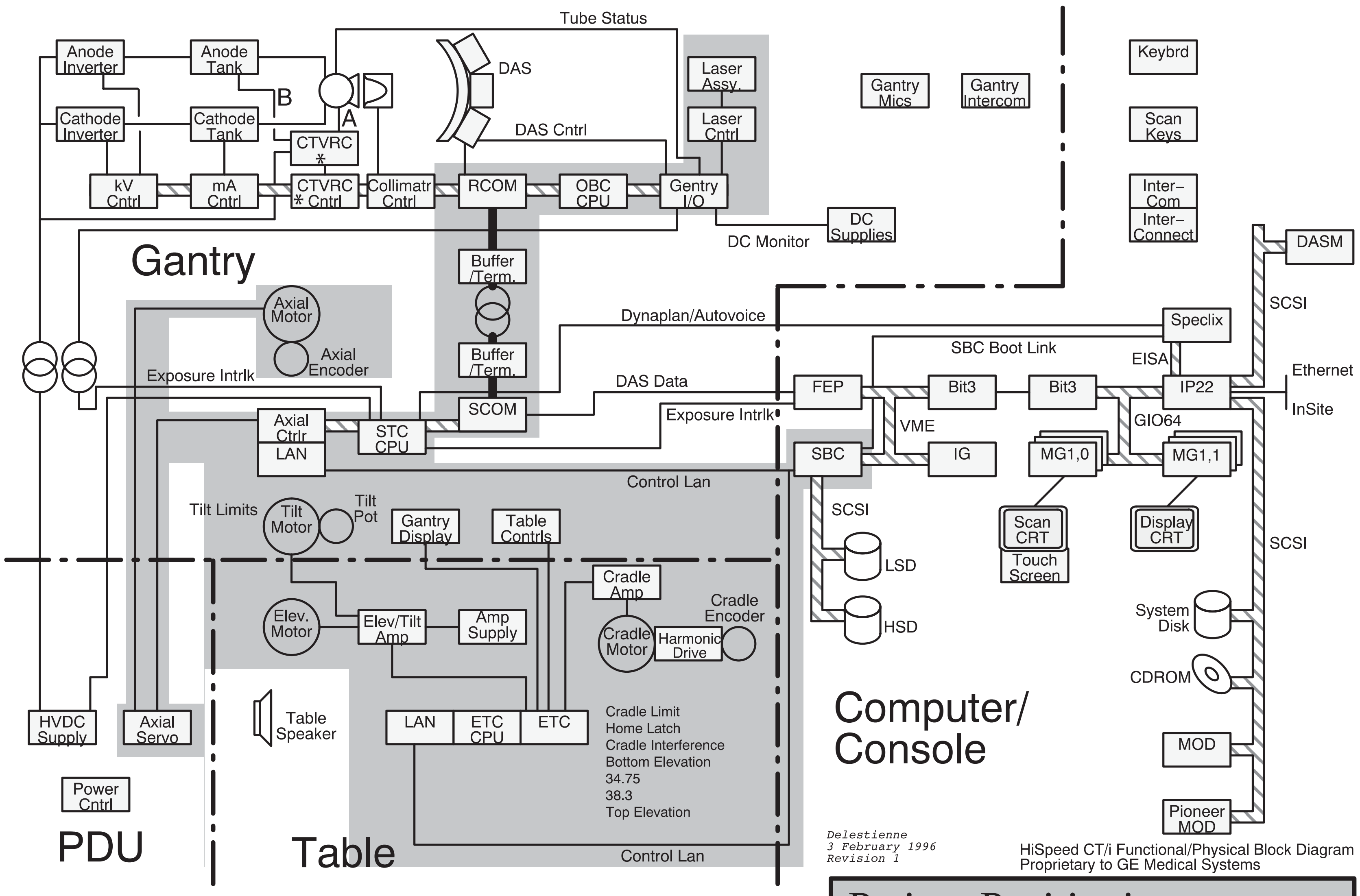
Section 15.0
System Function Maps

15.1 CT/i (Original)



* Replaced with HEMRC for Performix tube.
A = CTVRC signal routing
B = HEMRC signal routing

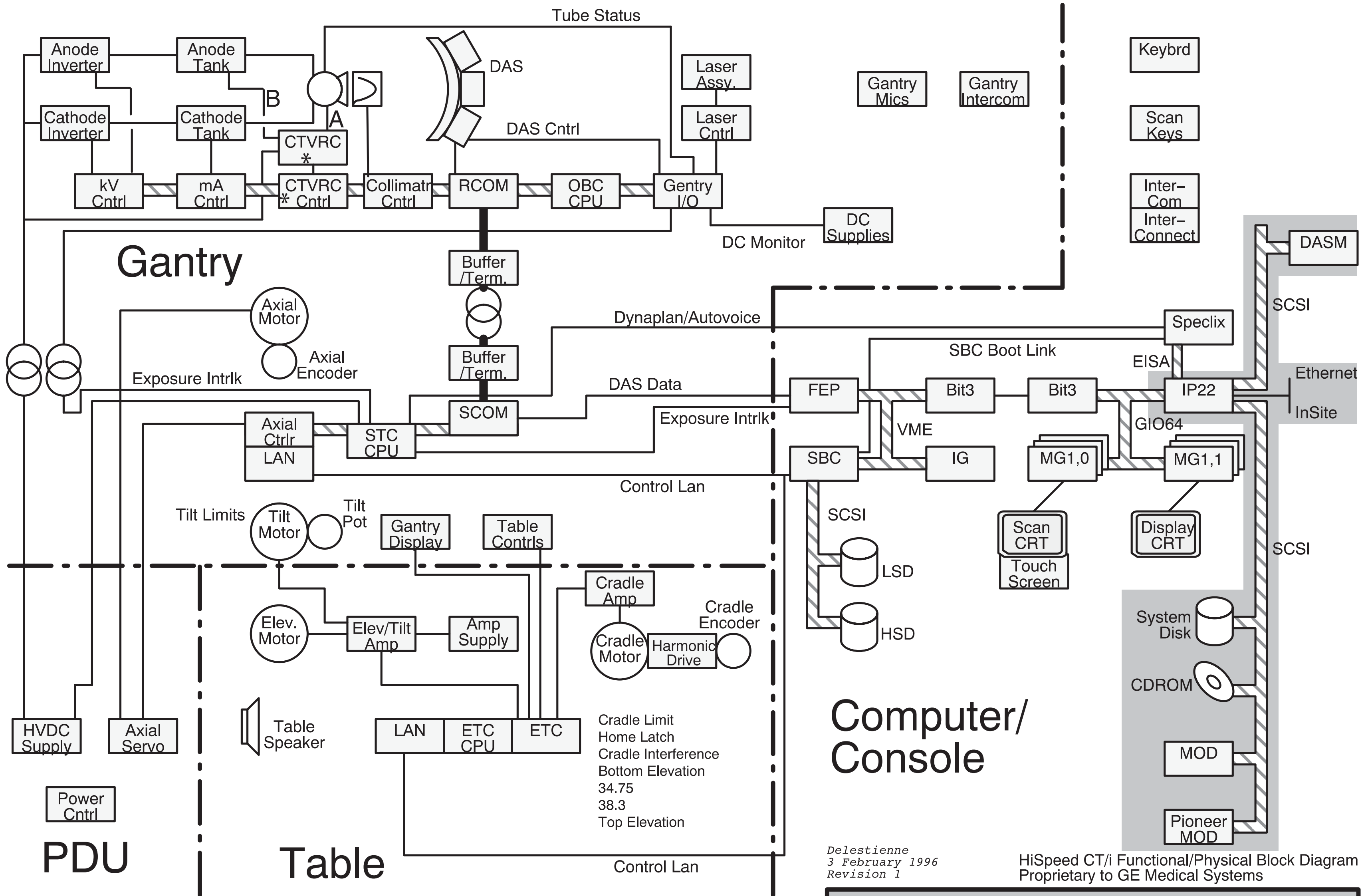
Figure 5-2 Functional Map, Data Acquisition



Patient Positioning

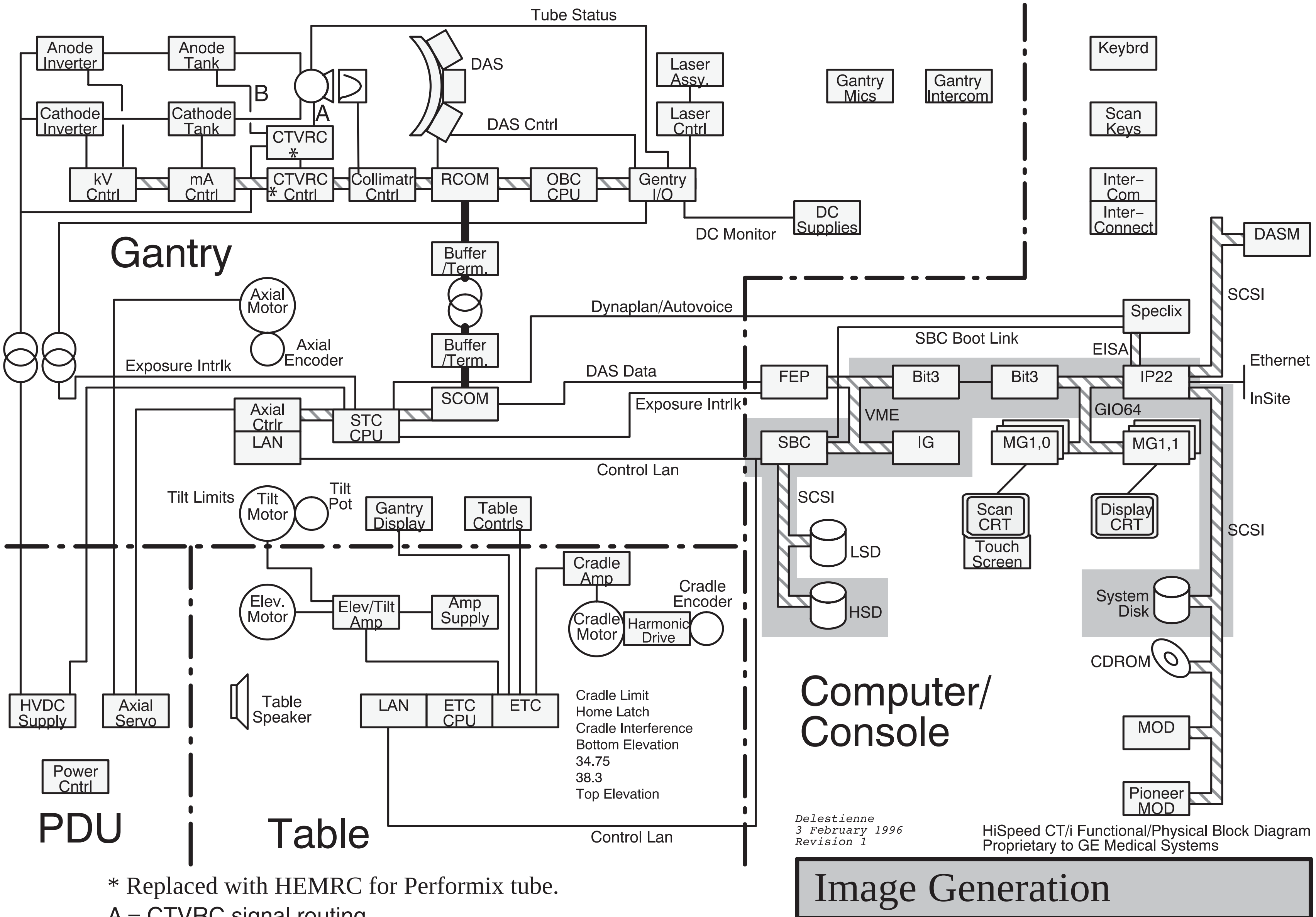
* Replaced with HEMRC for Performix tube.
A = CTVRC signal routing
B = HEMRC signal routing

Figure 5-3 Functional Map, Patient Positioning



* Replaced with HEMRC for Performix tube.
A = CTVRC signal routing
B = HEMRC signal routing

Figure 5-4 Functional Map, Data Management

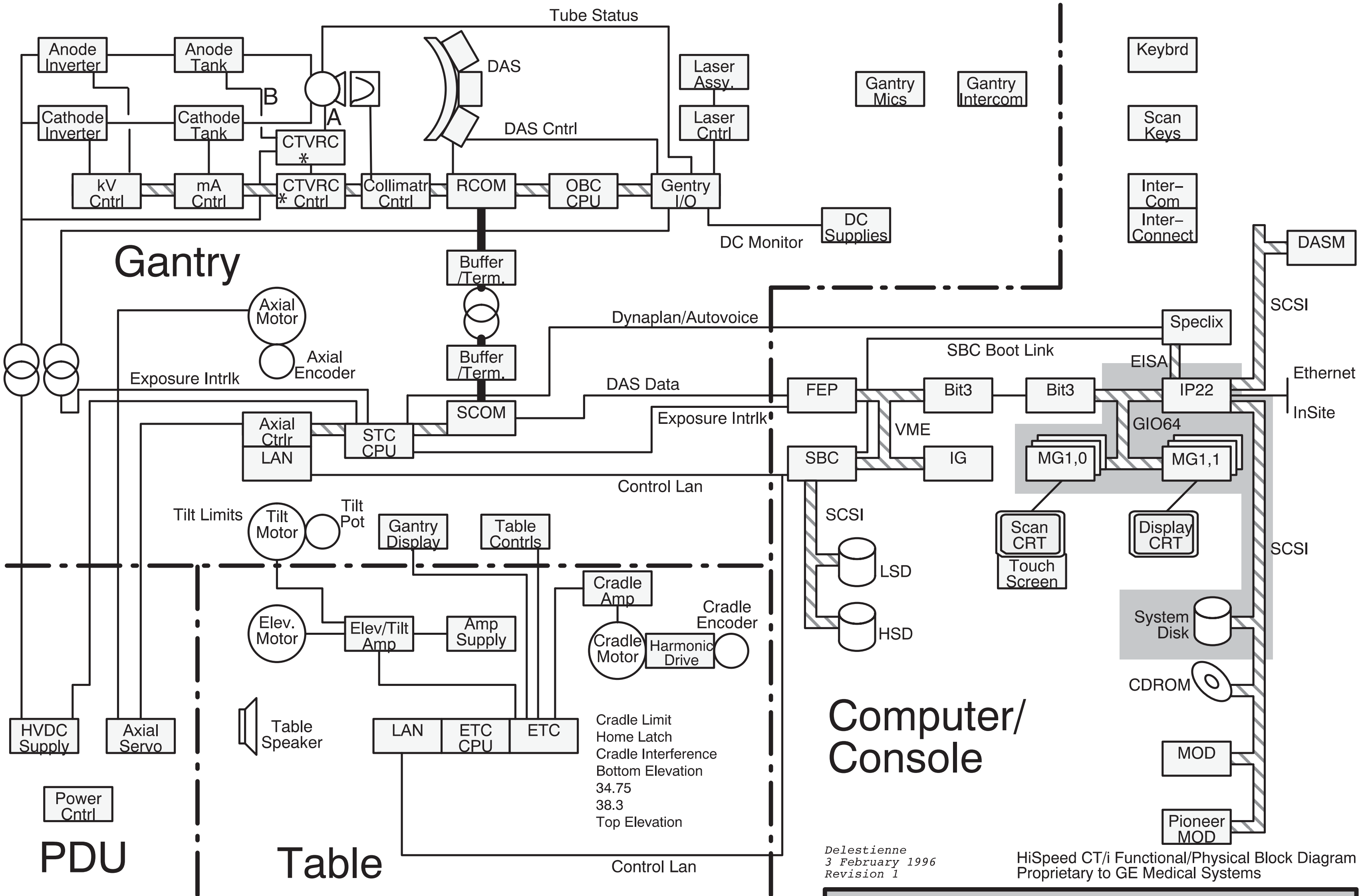


* Replaced with HEMRC for Performix tube.

A = CTVRC signal routing

B = HEMRC signal routing

Figure 5-5 Functional Map, Image Generation

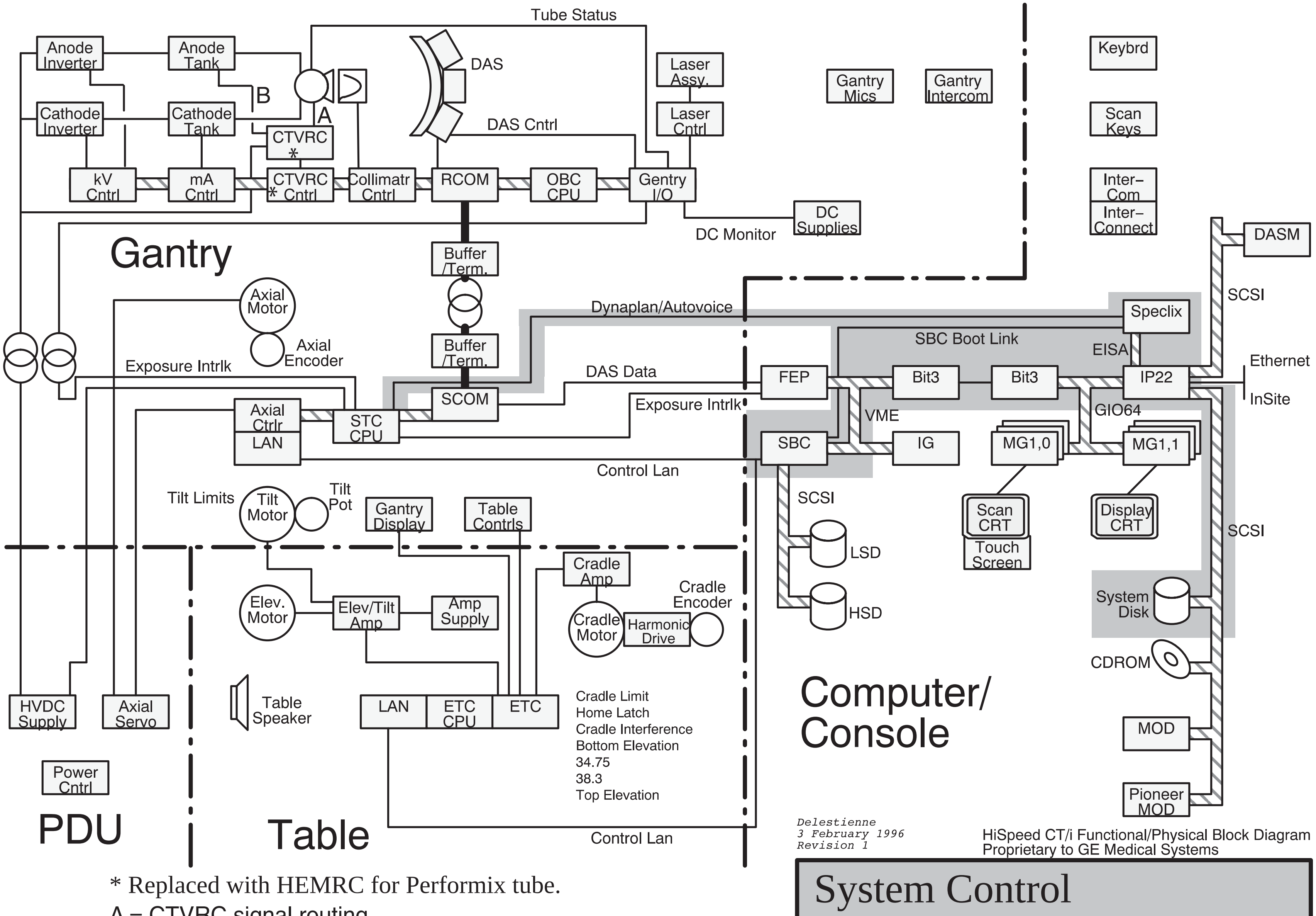


* Replaced with HEMRC for Performix tube.

A = CTVRC signal routing

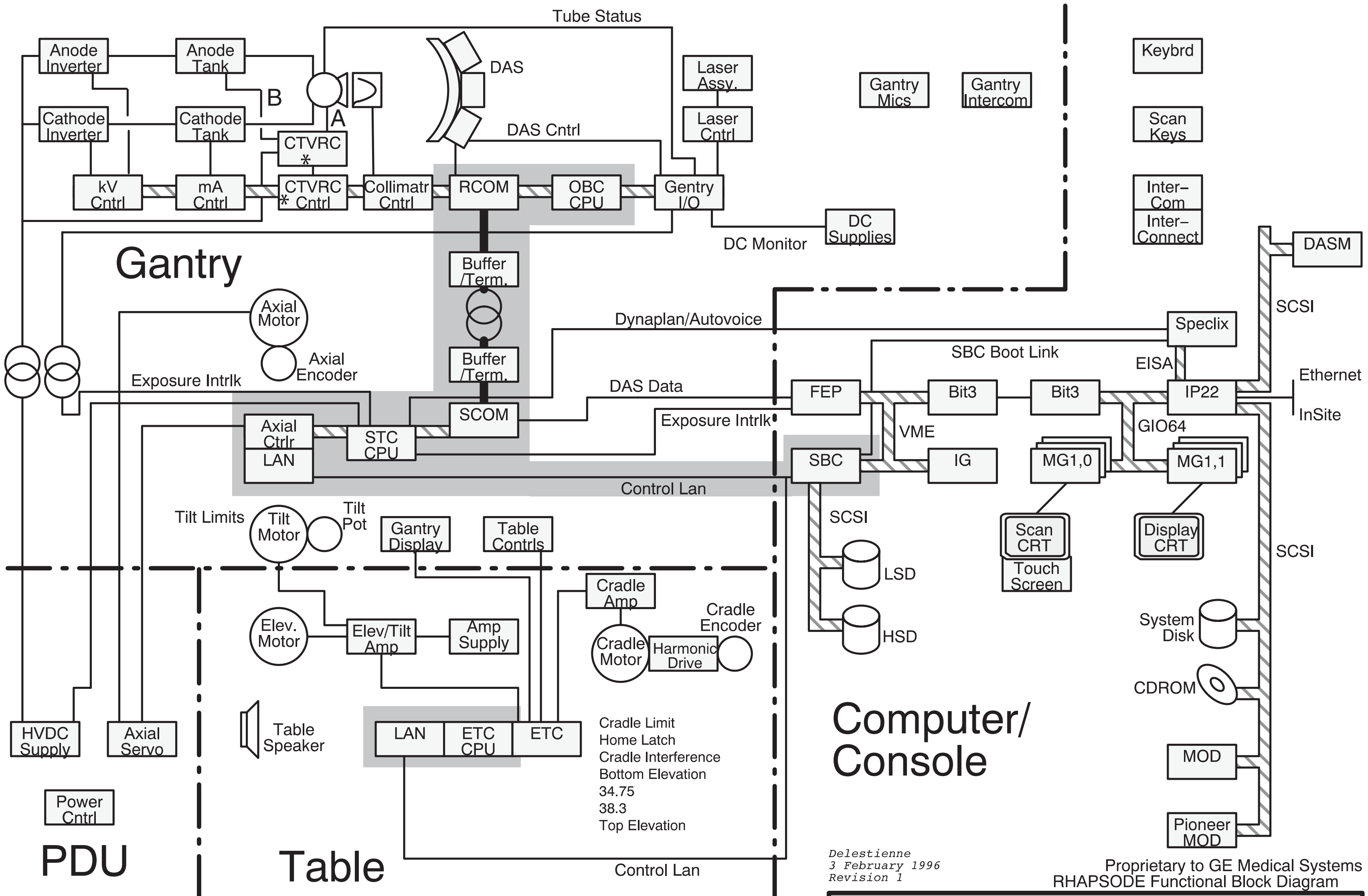
B = HEMRC signal routing

Figure 5-6 Functional Map, Display Processing



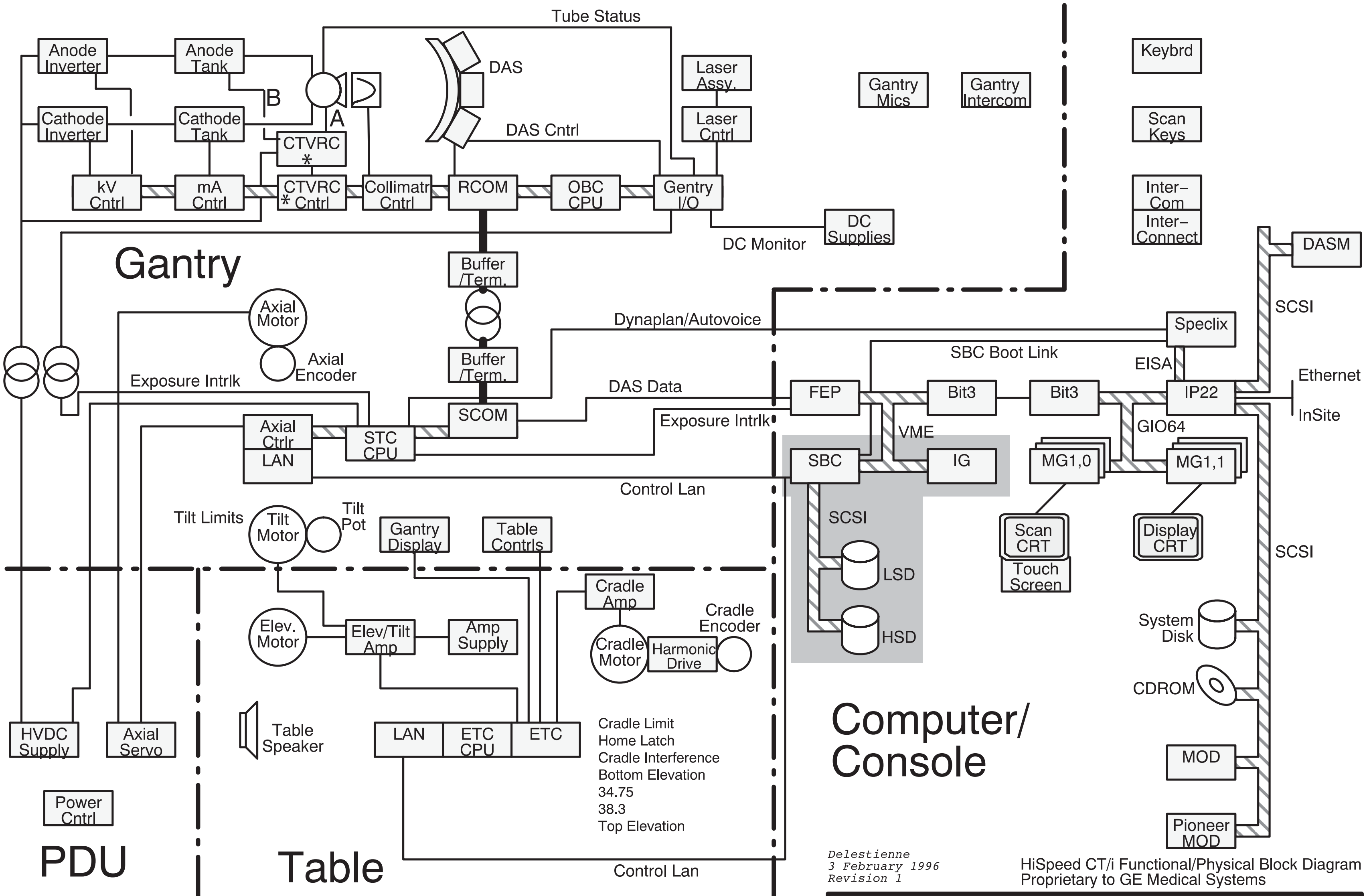
* Replaced with HEMRC for Performix tube.
A = CTVRC signal routing
B = HEMRC signal routing

Figure 5-7 Functional Map, System Control



* Replaced with HEMRC for Performix tube.
A = CTVRC signal routing
B = HEMRC signal routing

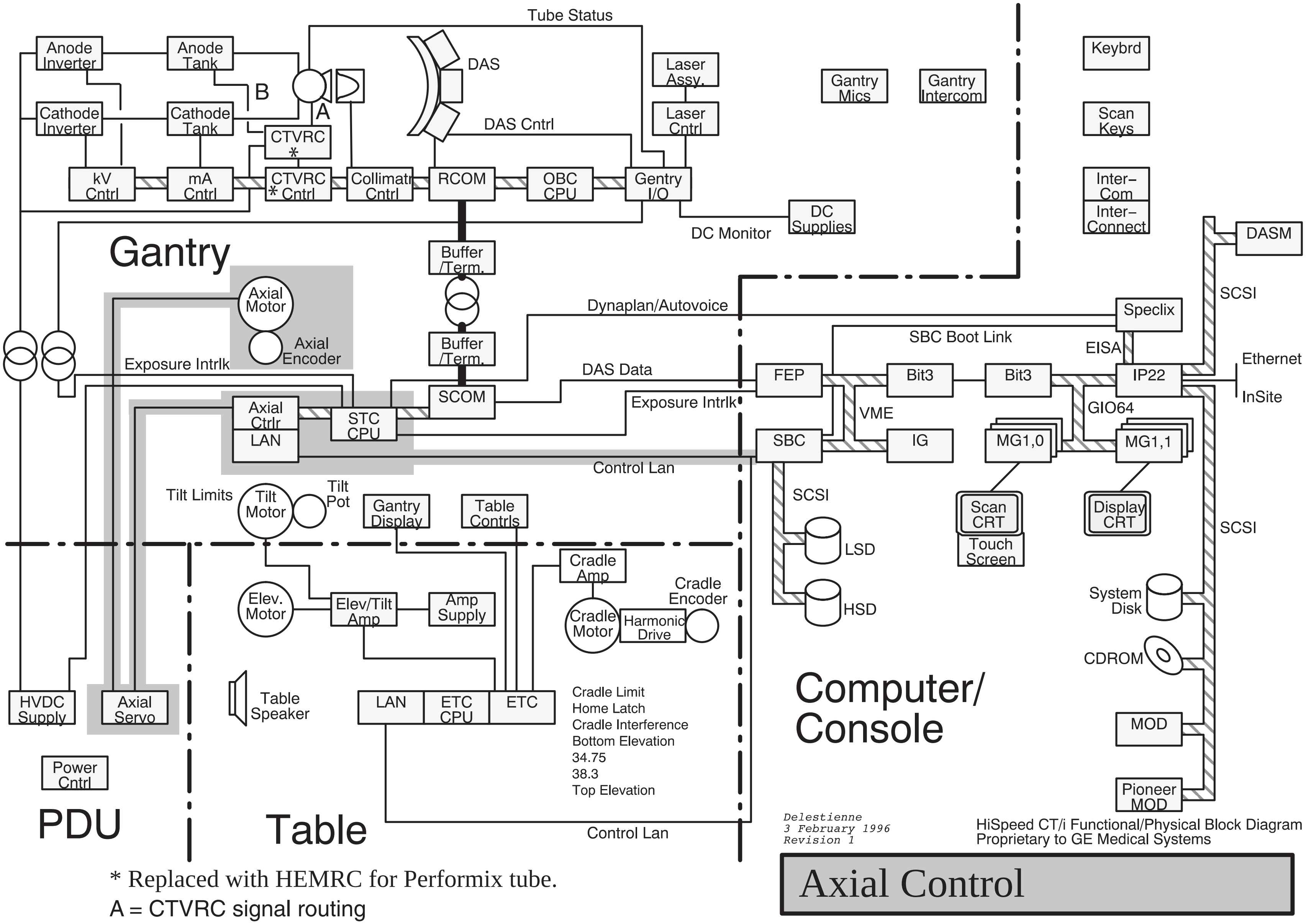
Figure 5-8 Functional Map, System Communications



CAL Processing

* Replaced with HEMRC for Performix tube.
A = CTVRC signal routing
B = HEMRC signal routing

Figure 5-9 Function Map, CAL Processing

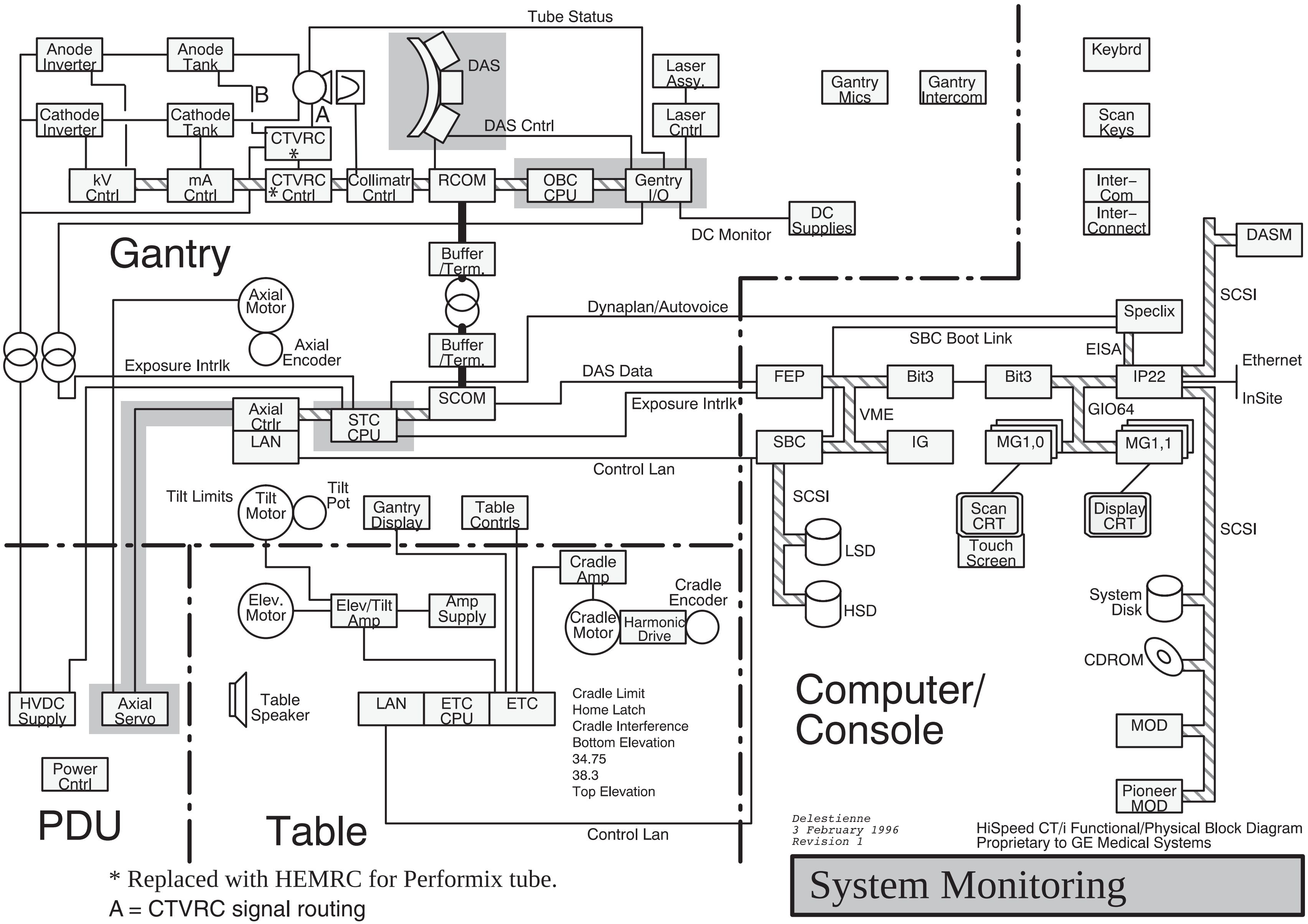


* Replaced with HEMRC for Performix tube.

A = CTVRC signal routing

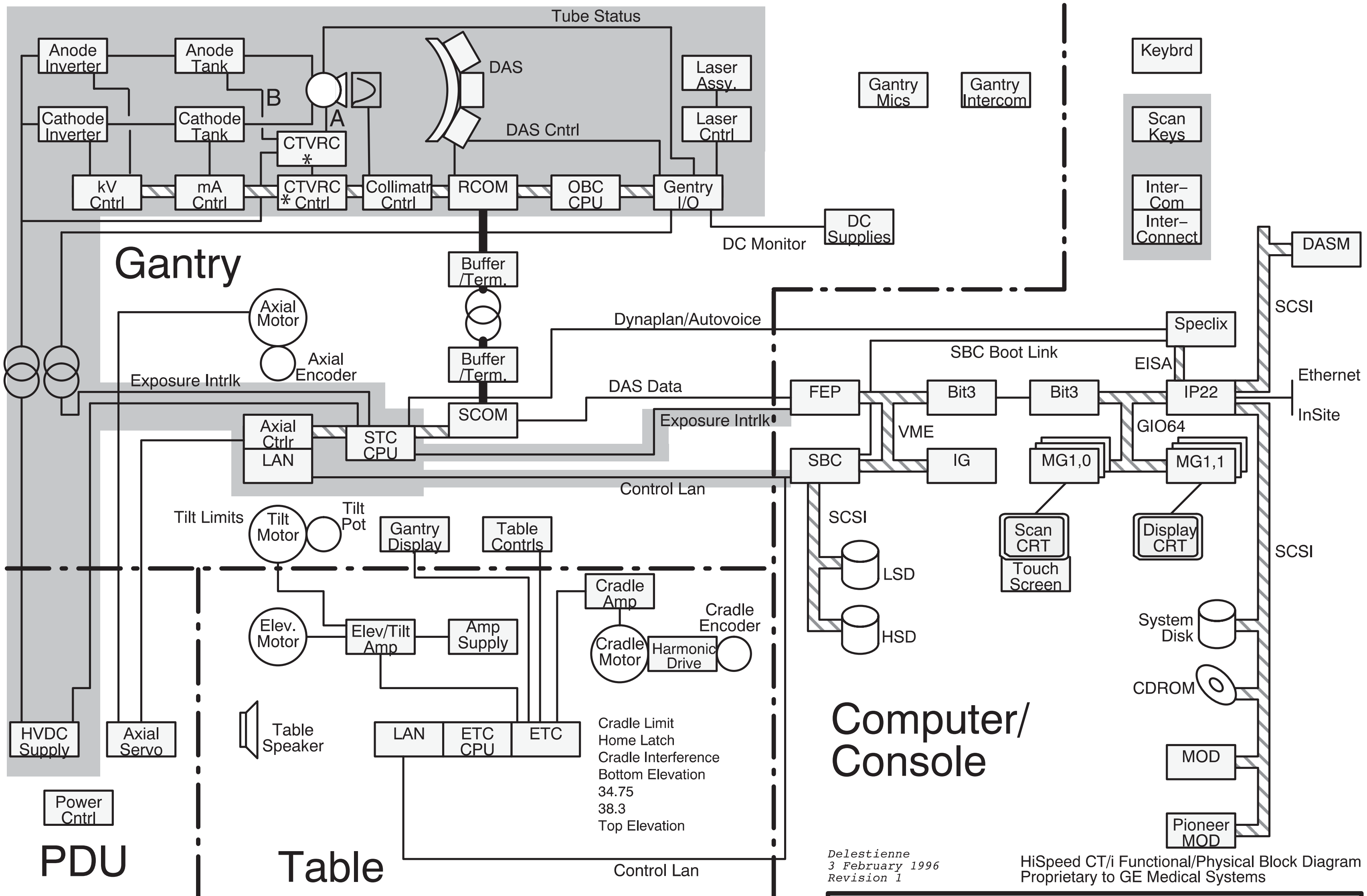
B = HEMRC signal routing

Figure 5-10 Function Map, Axial Control



* Replaced with HEMRC for Performix tube.
A = CTVRC signal routing
B = HEMRC signal routing

Figure 5-11 Function Map, System Monitoring



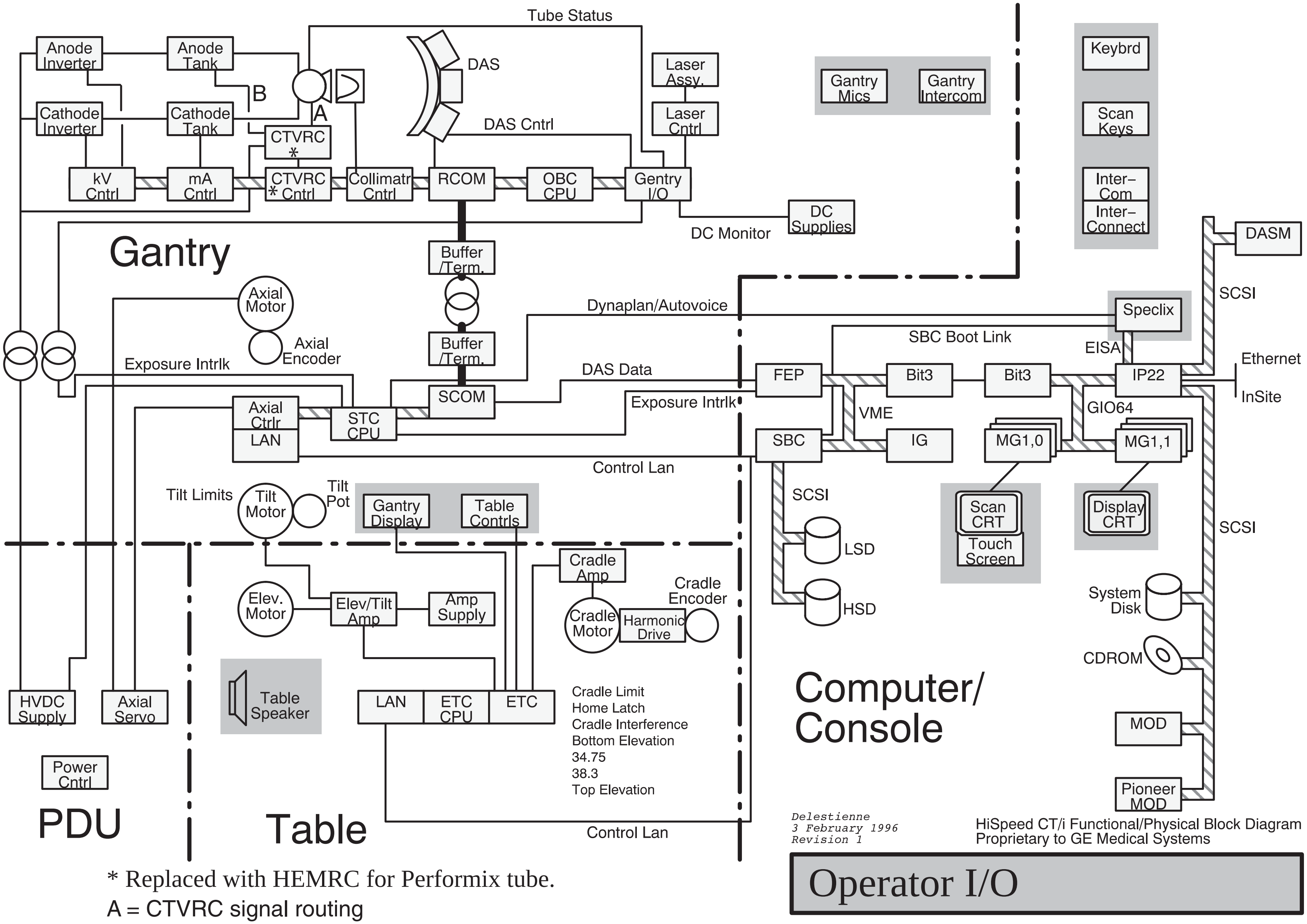
X–RAY Generation

* Replaced with HEMRC for Performix tube.

A = CTVRC signal routing

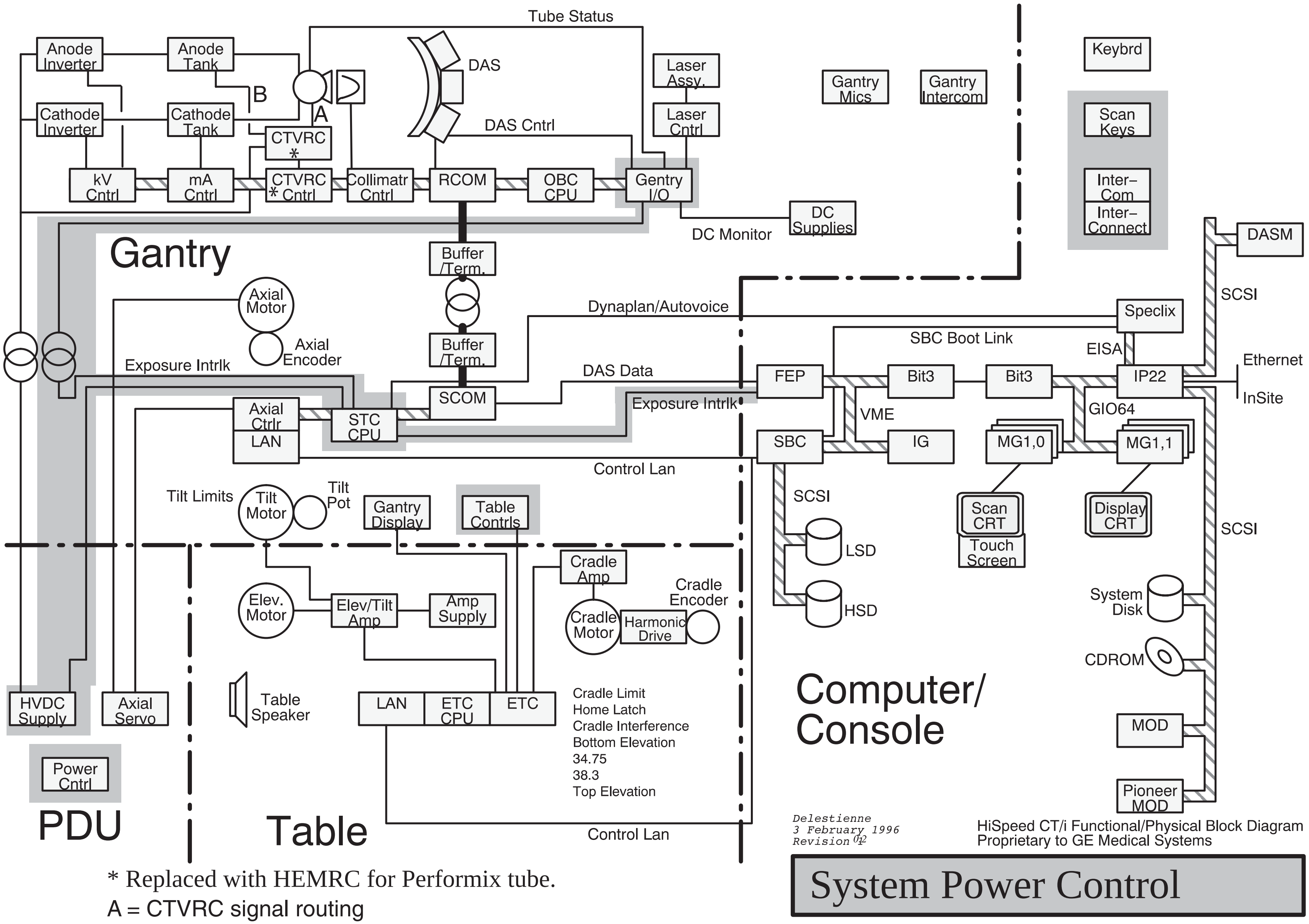
B = HEMRC signal routing

Figure 5-12 Function Map, X-Ray Generation



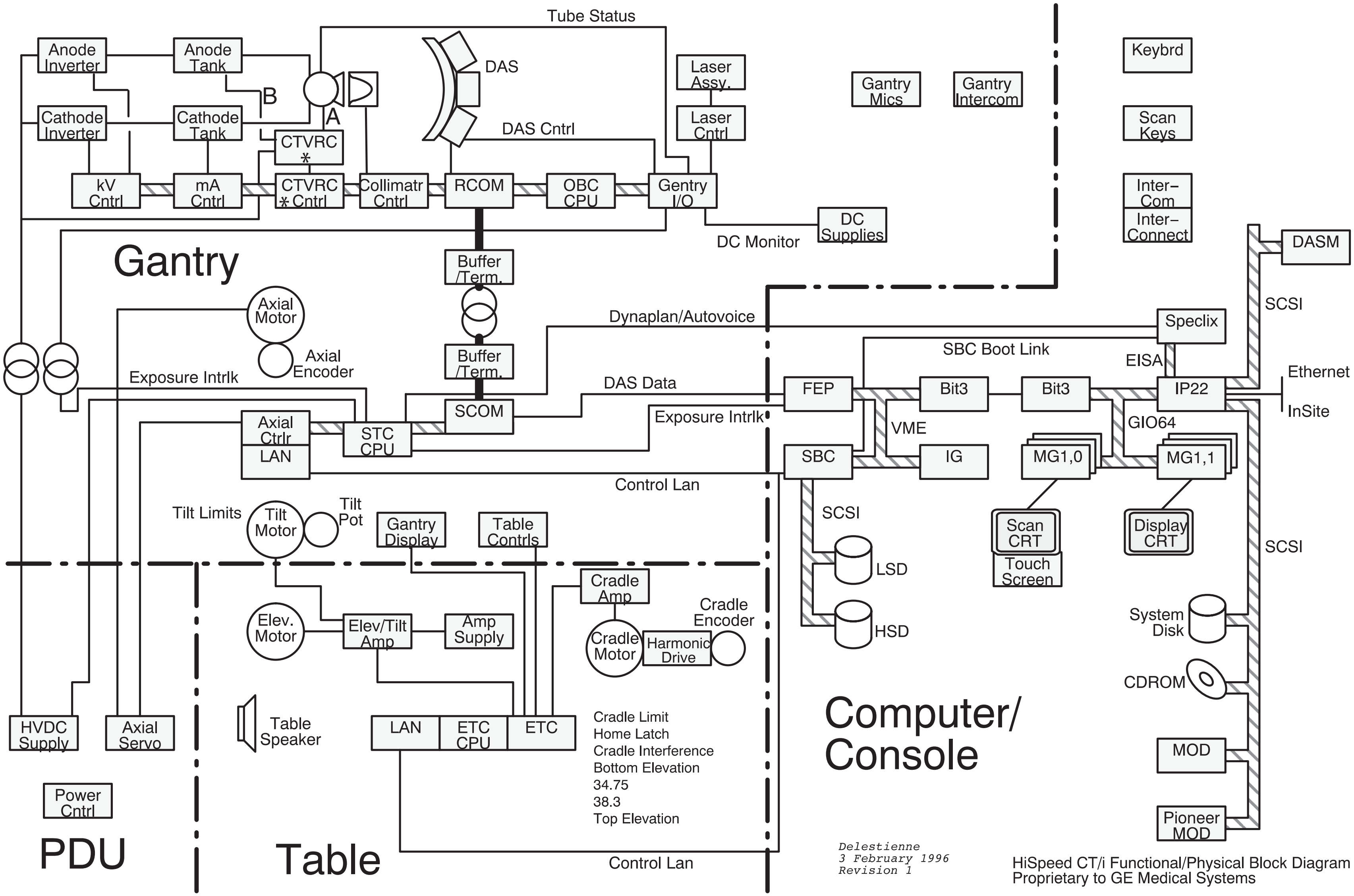
* Replaced with HEMRC for Performix tube.
A = CTVRC signal routing
B = HEMRC signal routing

Figure 5-13 Function Map, Operator I/O



* Replaced with HEMRC for Performix tube.
A = CTVRC signal routing
B = HEMRC signal routing

Figure 5-14 Function Map, System Power Control



* Replaced with HEMRC for Performix tube.

A = CTVRC signal routing

B = HEMRC signal routing

Figure 5-15 HiSpeed CT/i Simplified System Block Diagram

15.2 CT/i Version 5.x

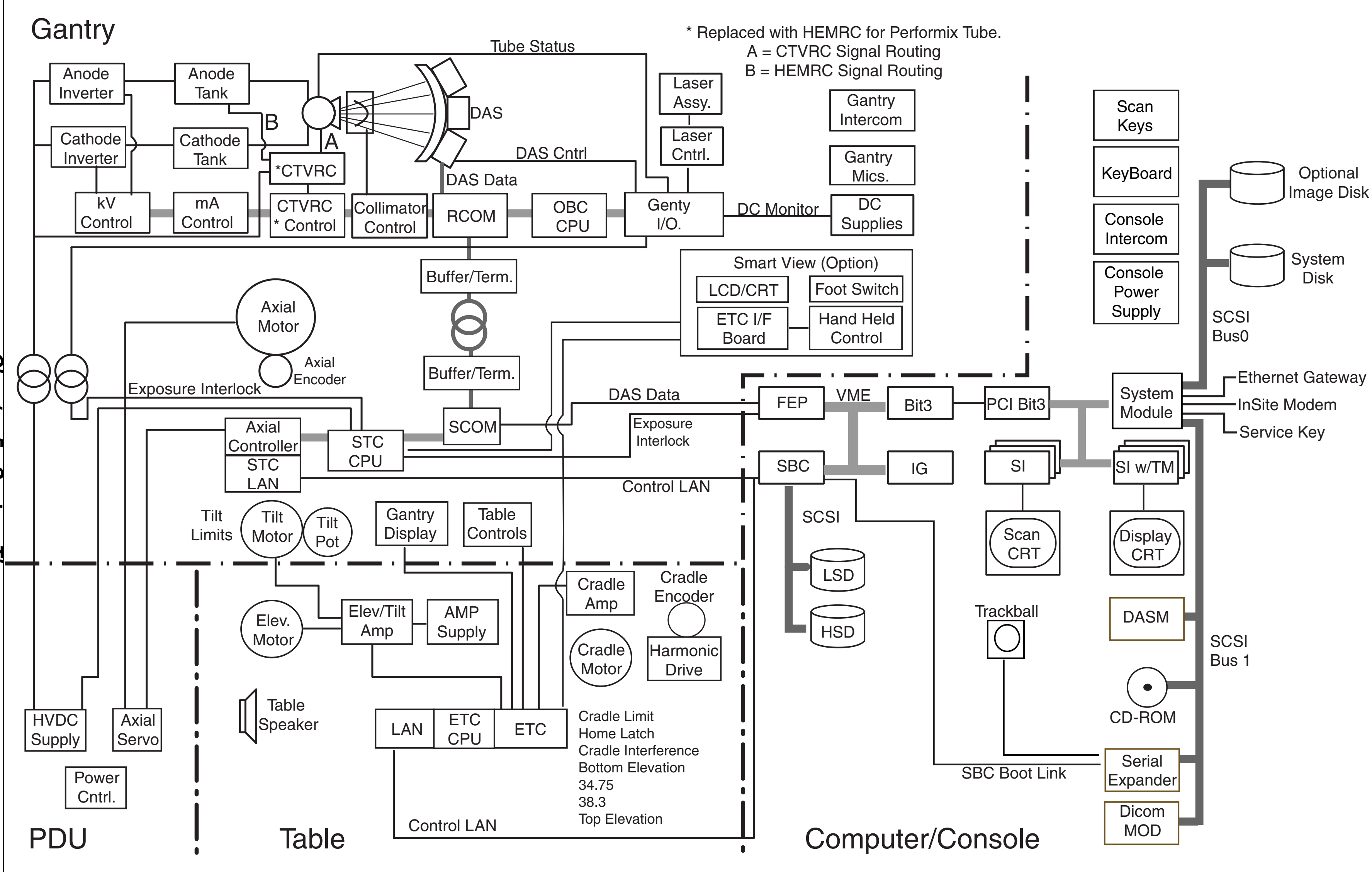


Figure 5-16 CT/i Version 5.x CT/i Simplified System Block Diagram

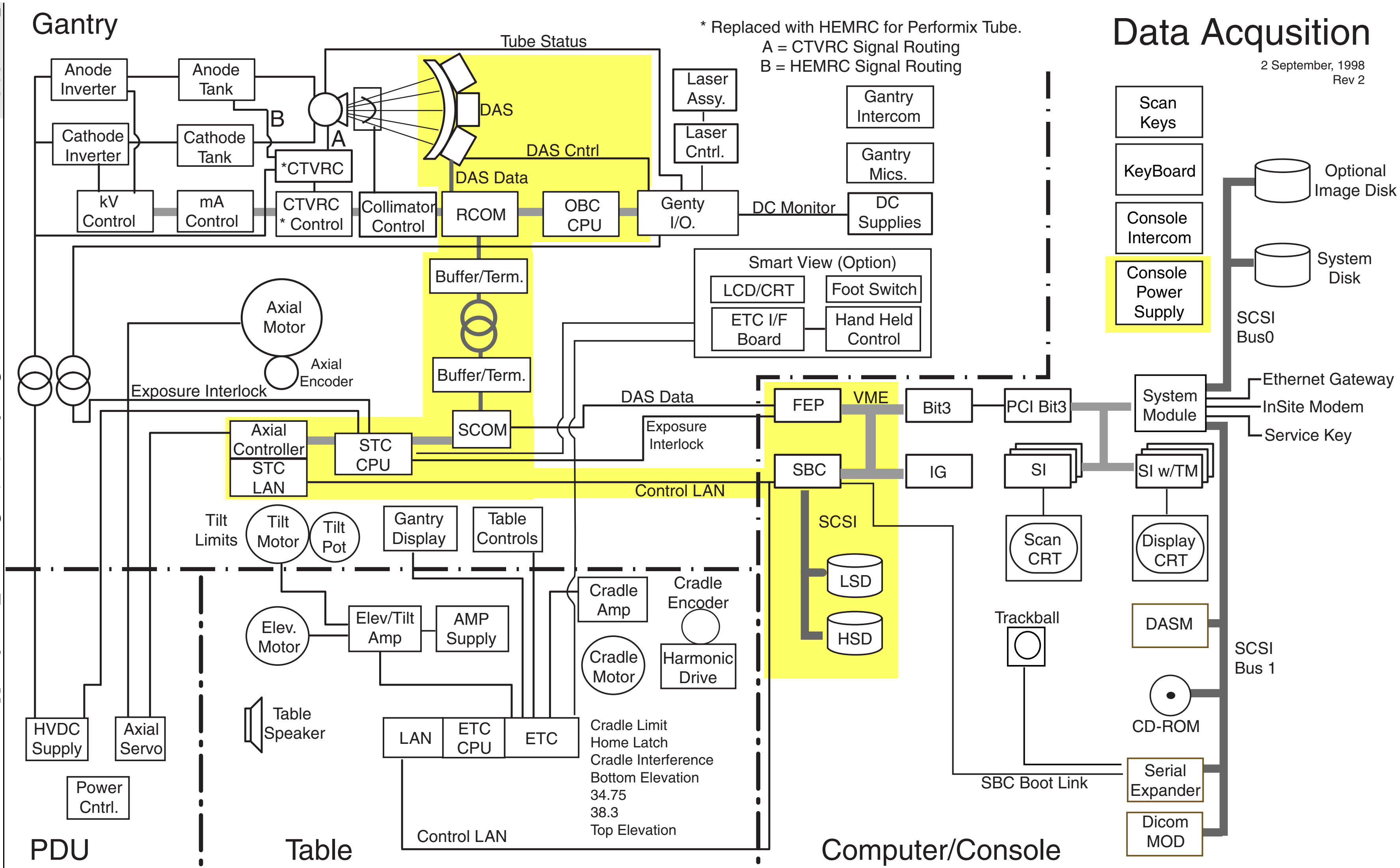


Figure 5-17 Function Map, Data Acquisition

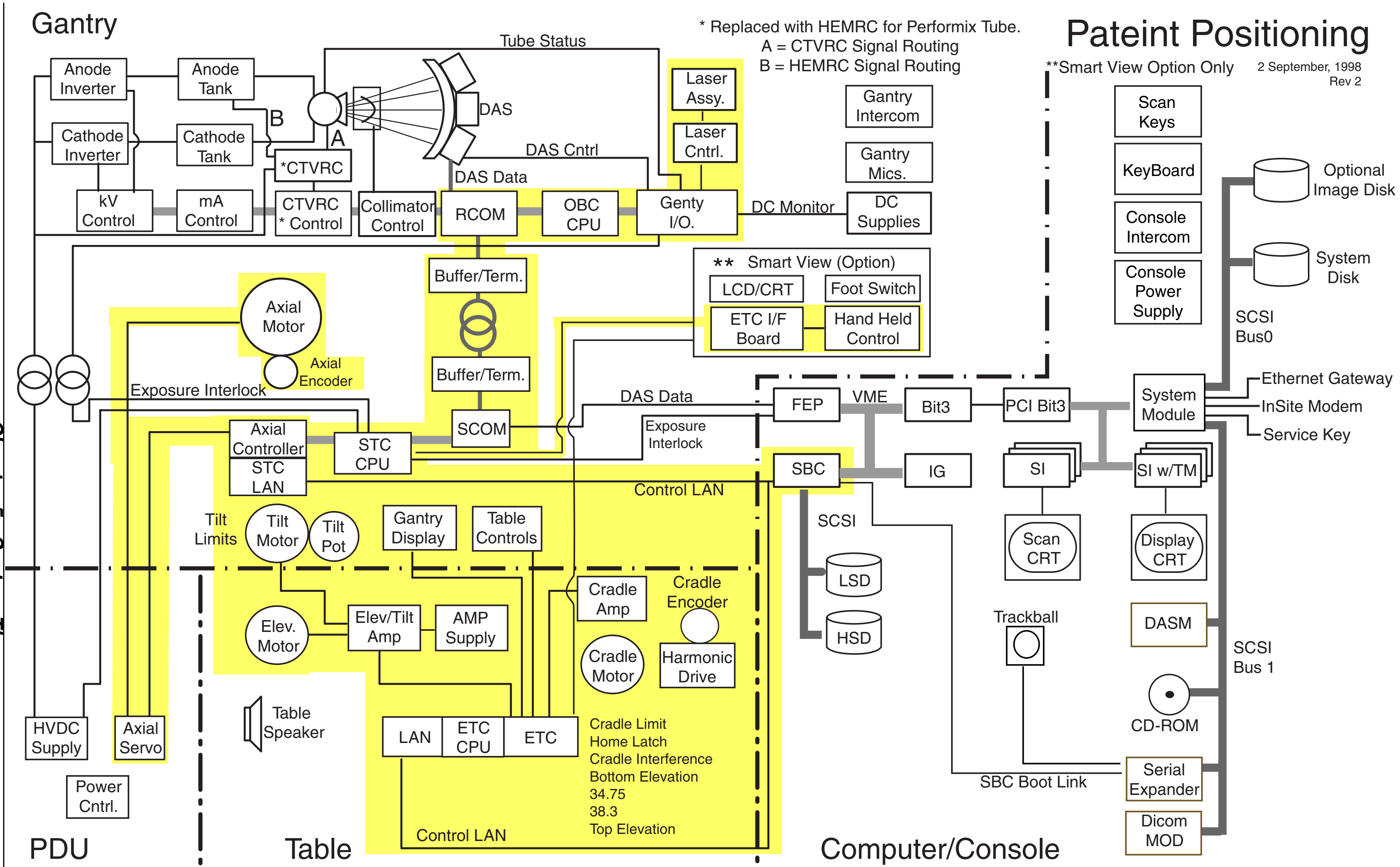


Figure 5-18 Function Map, Patient Positioning

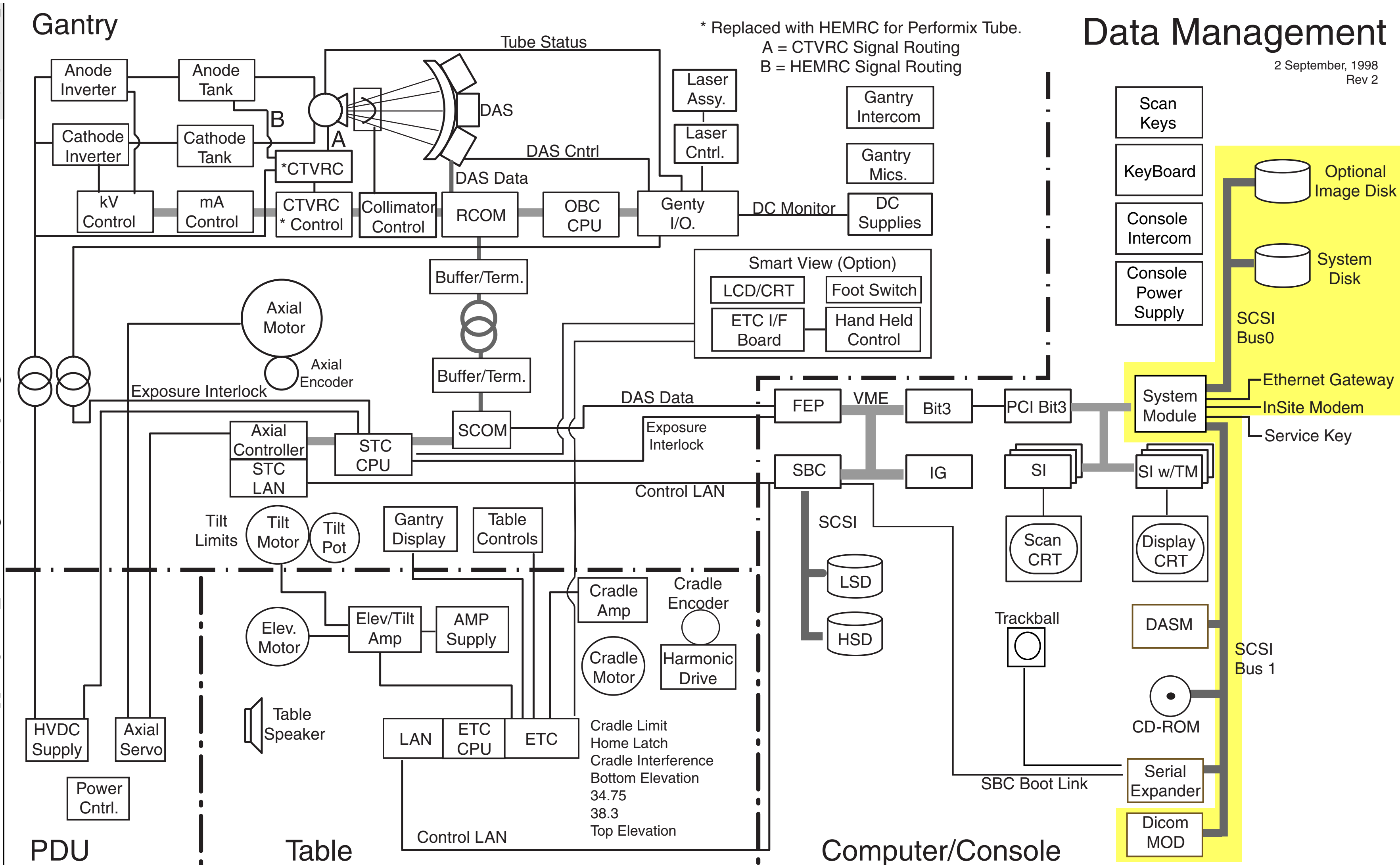


Figure 5-19 Function Map, Data Management

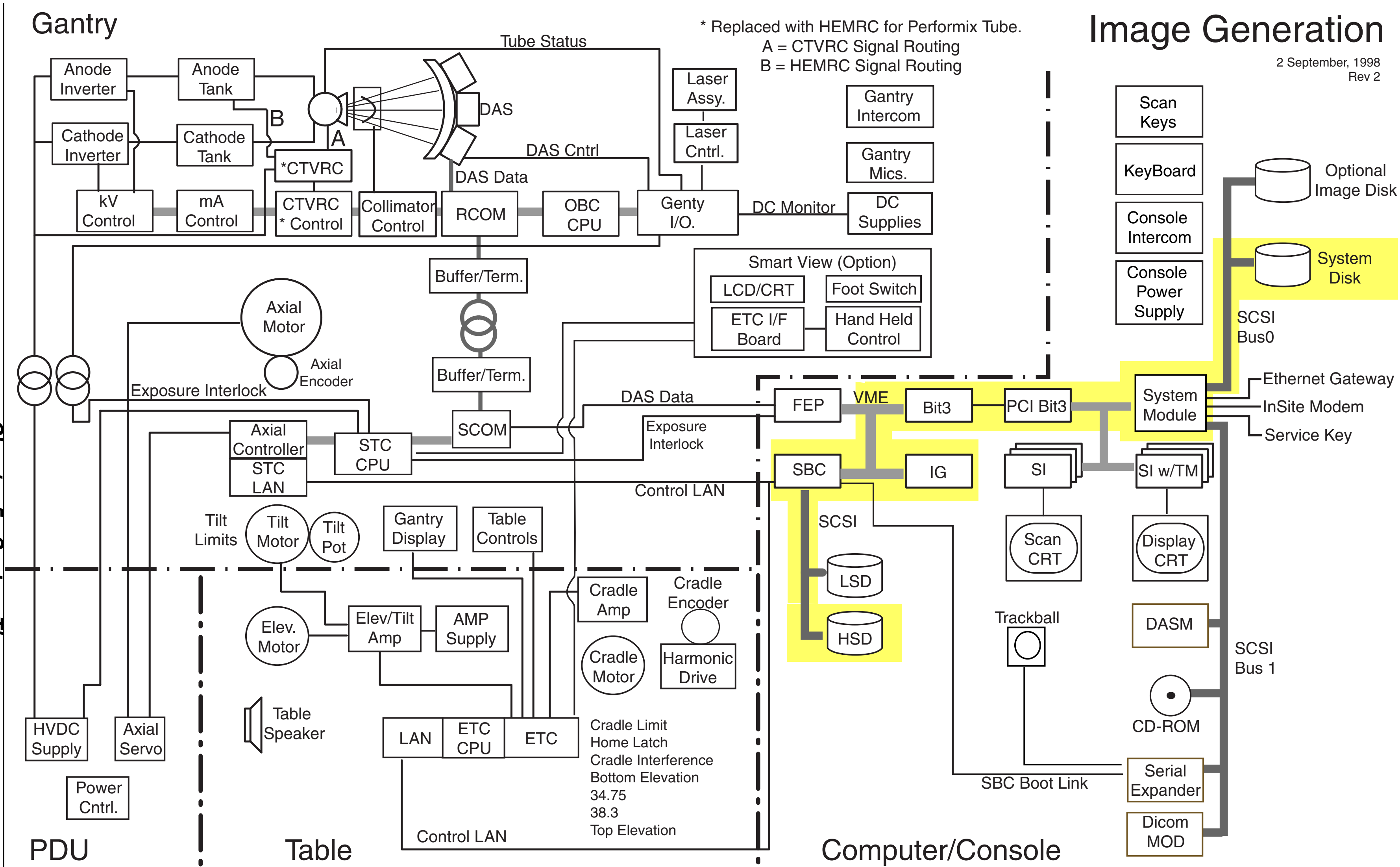


Figure 5-20 Function Map, Image Generation

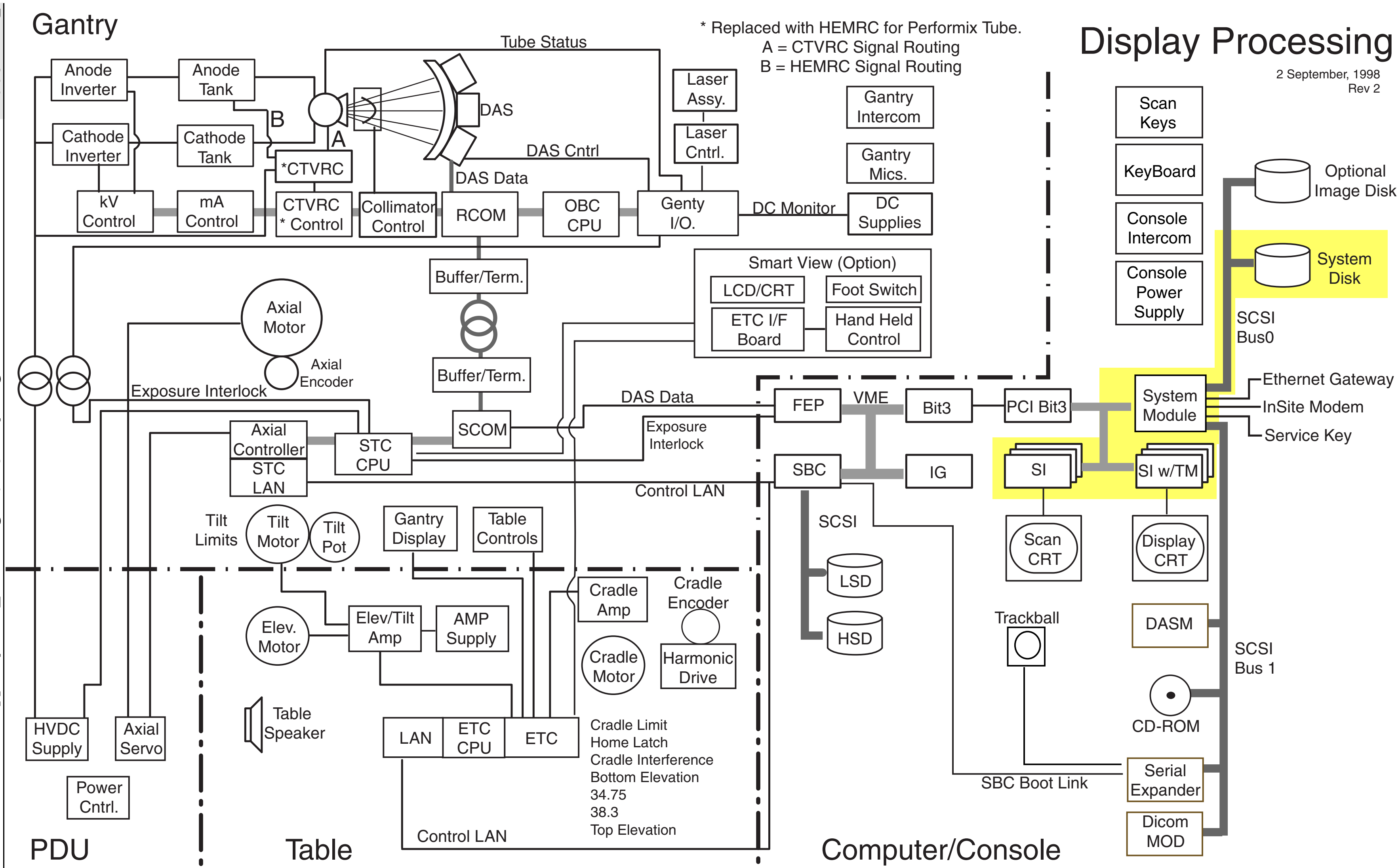


Figure 5-21 Function Map, Display Processing

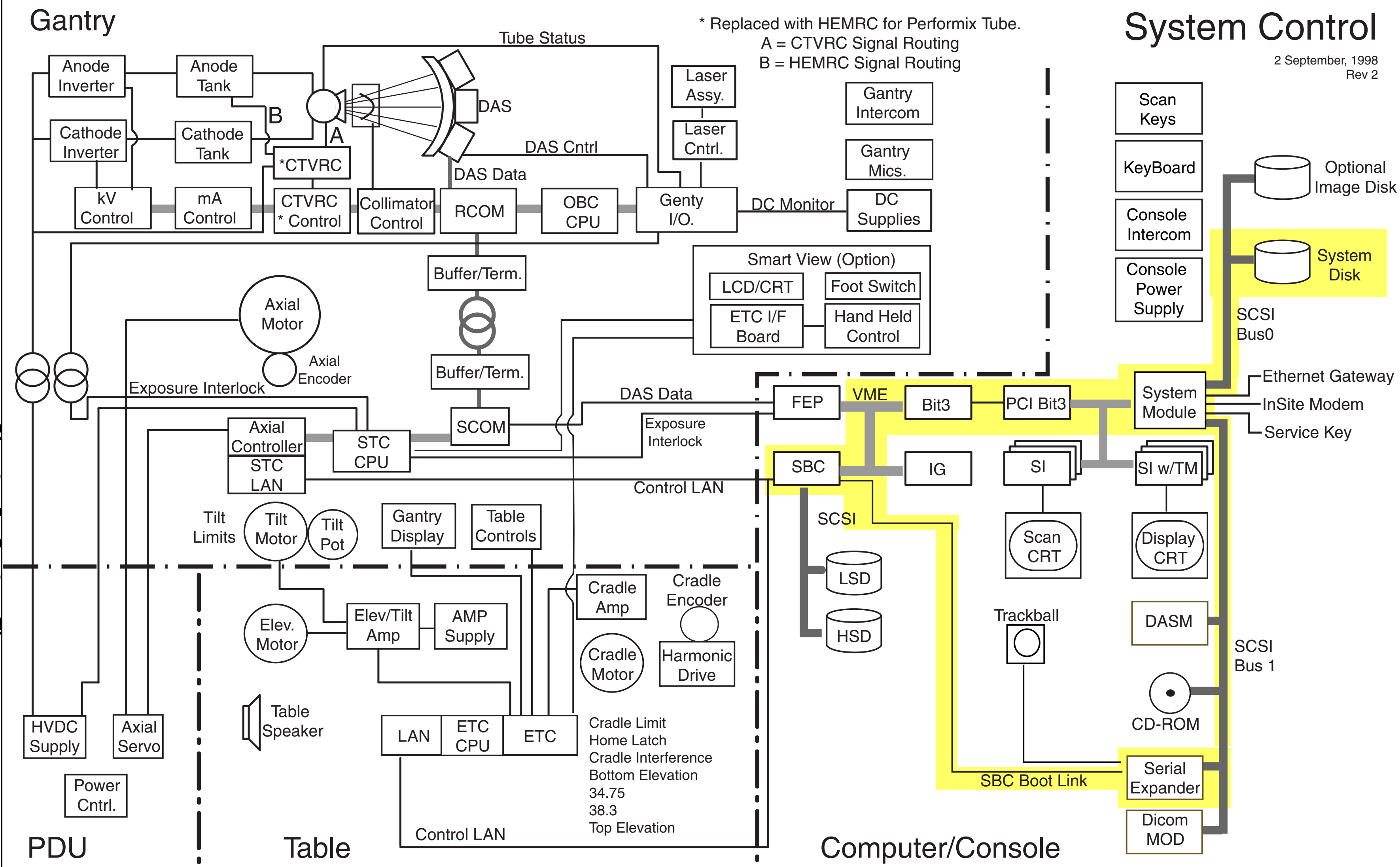


Figure 5-22 Function Map, System Control

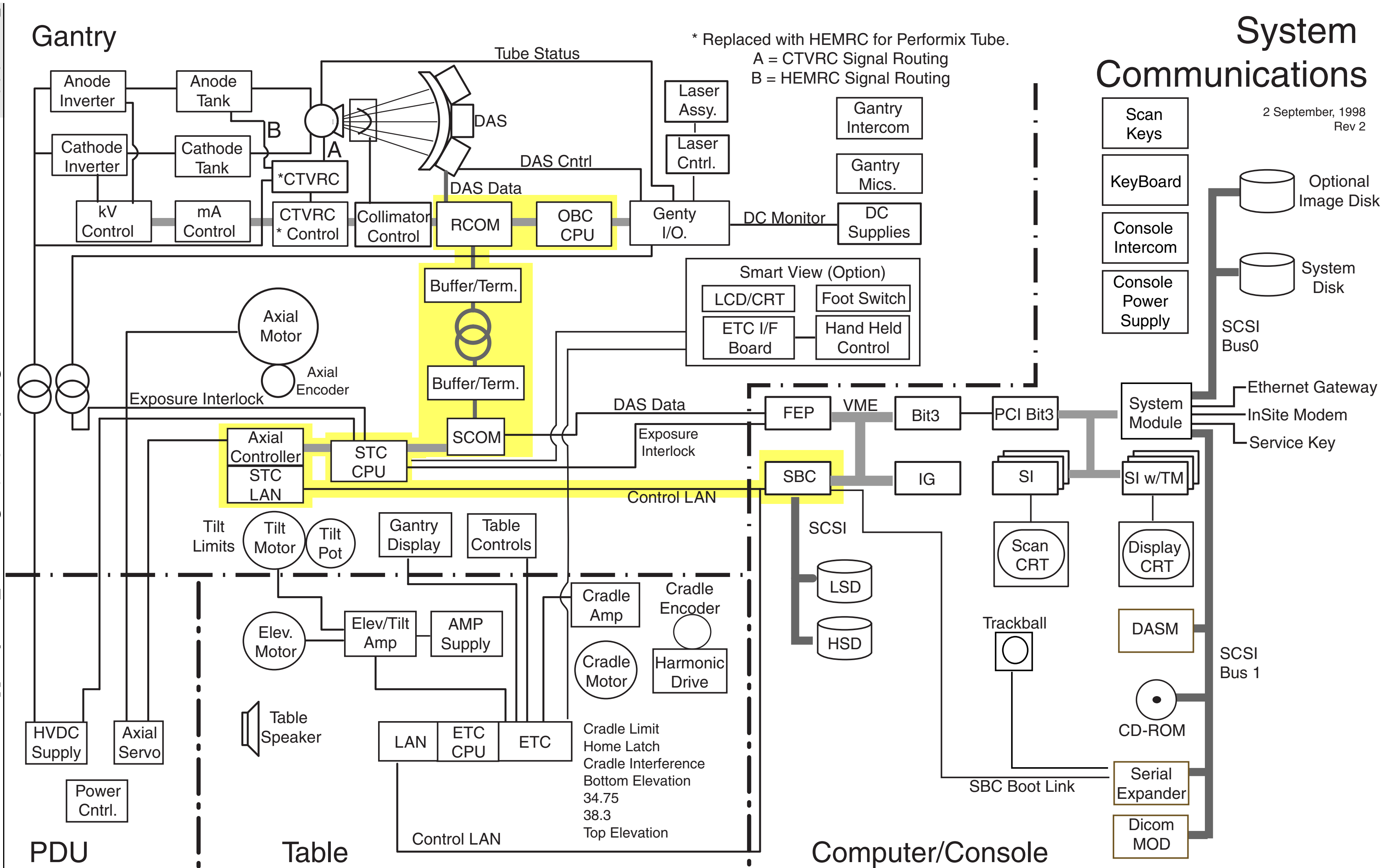


Figure 5-23 Function Map, System Communications

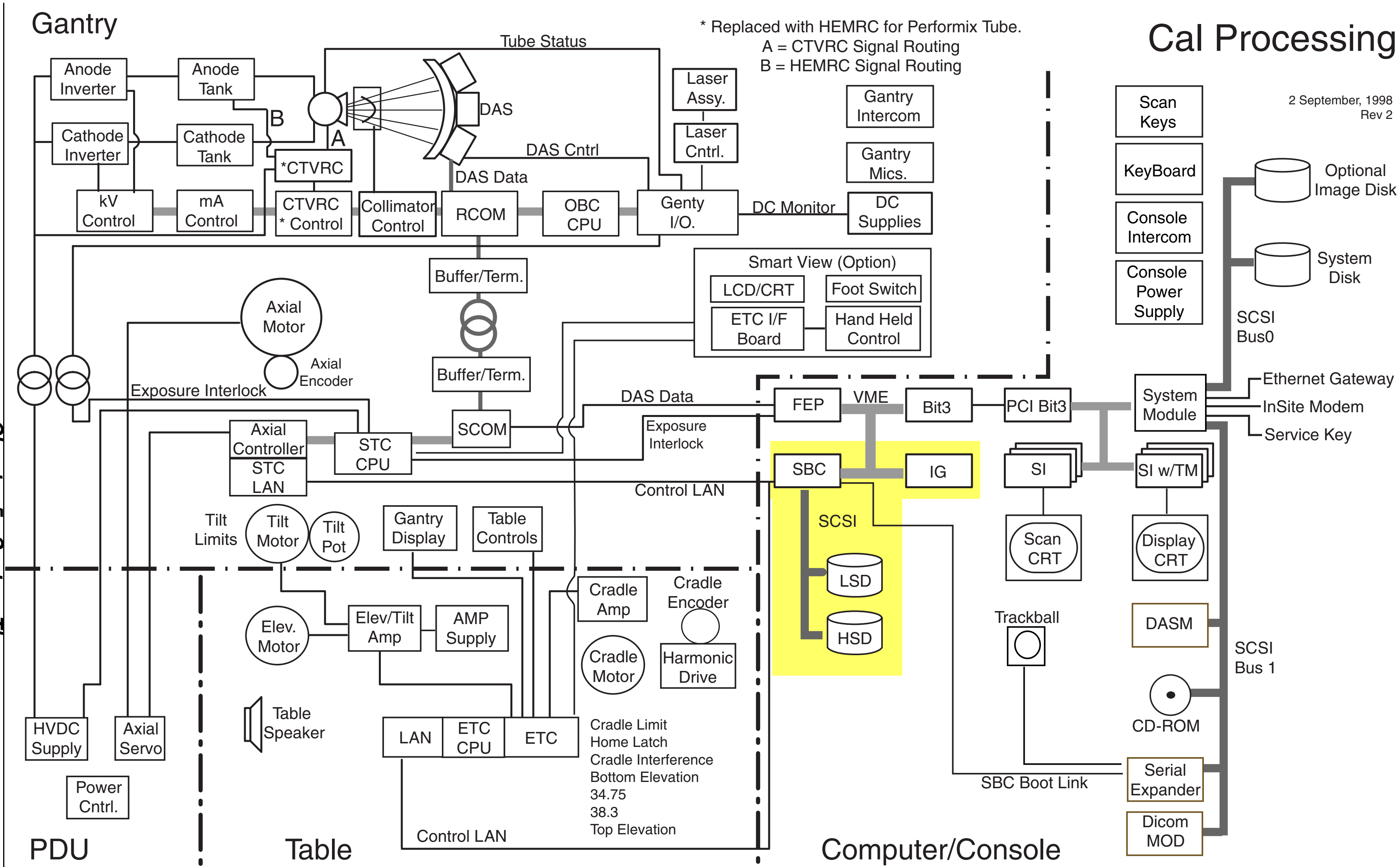


Figure 5-24 Function Map, Cal Processing

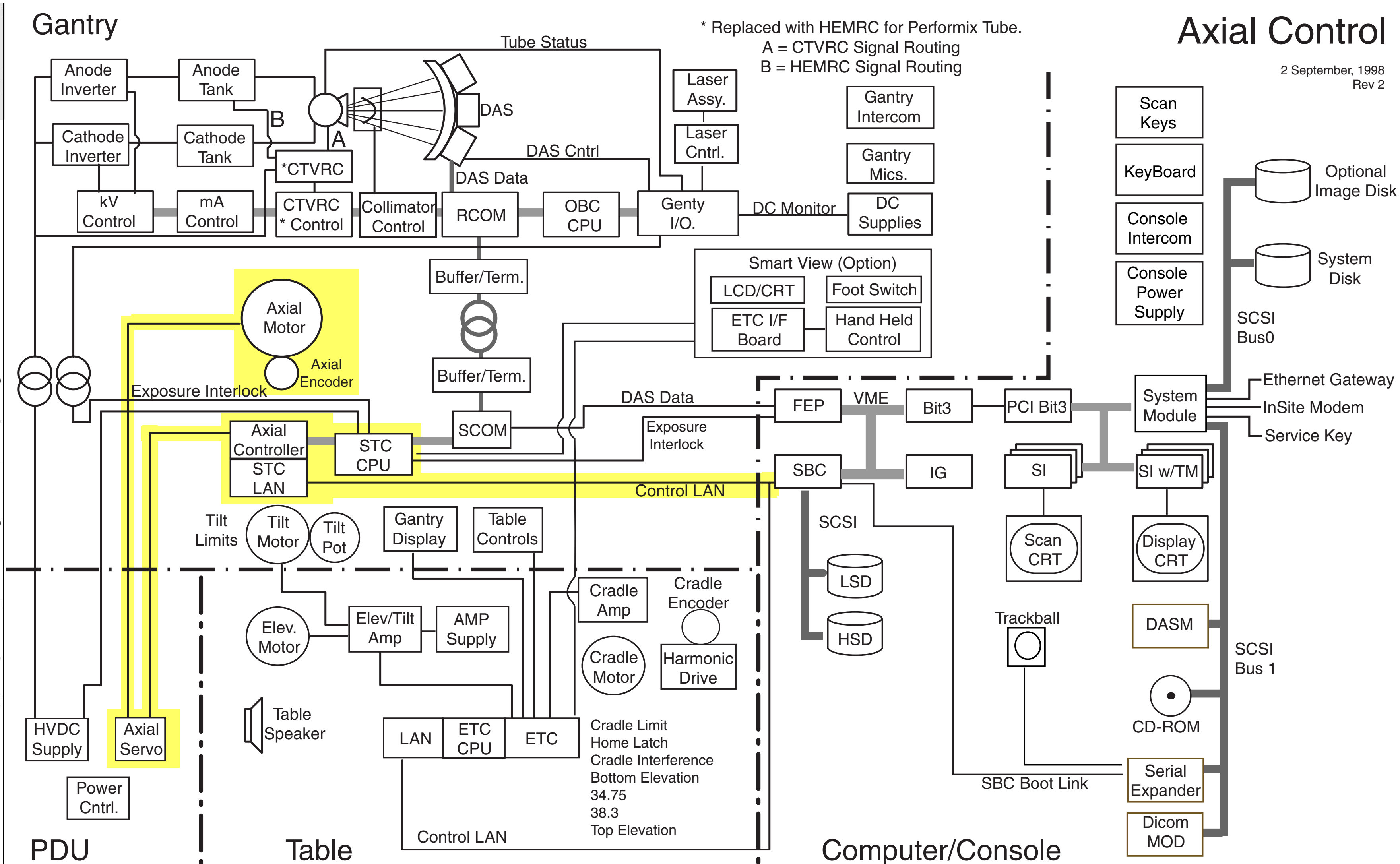


Figure 5-25 Function Map, Axial Control

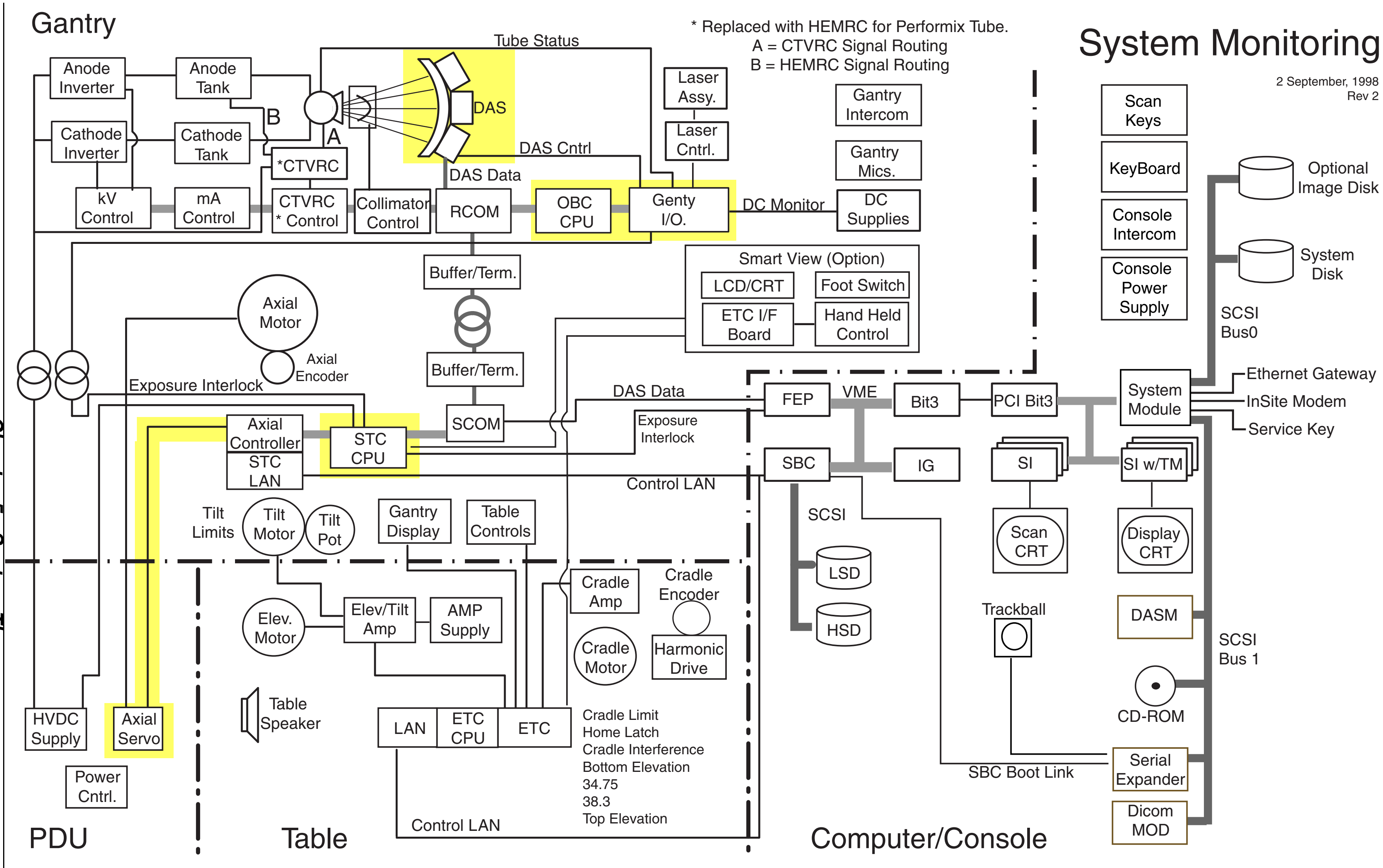


Figure 5-26 Function Map, System Monitoring

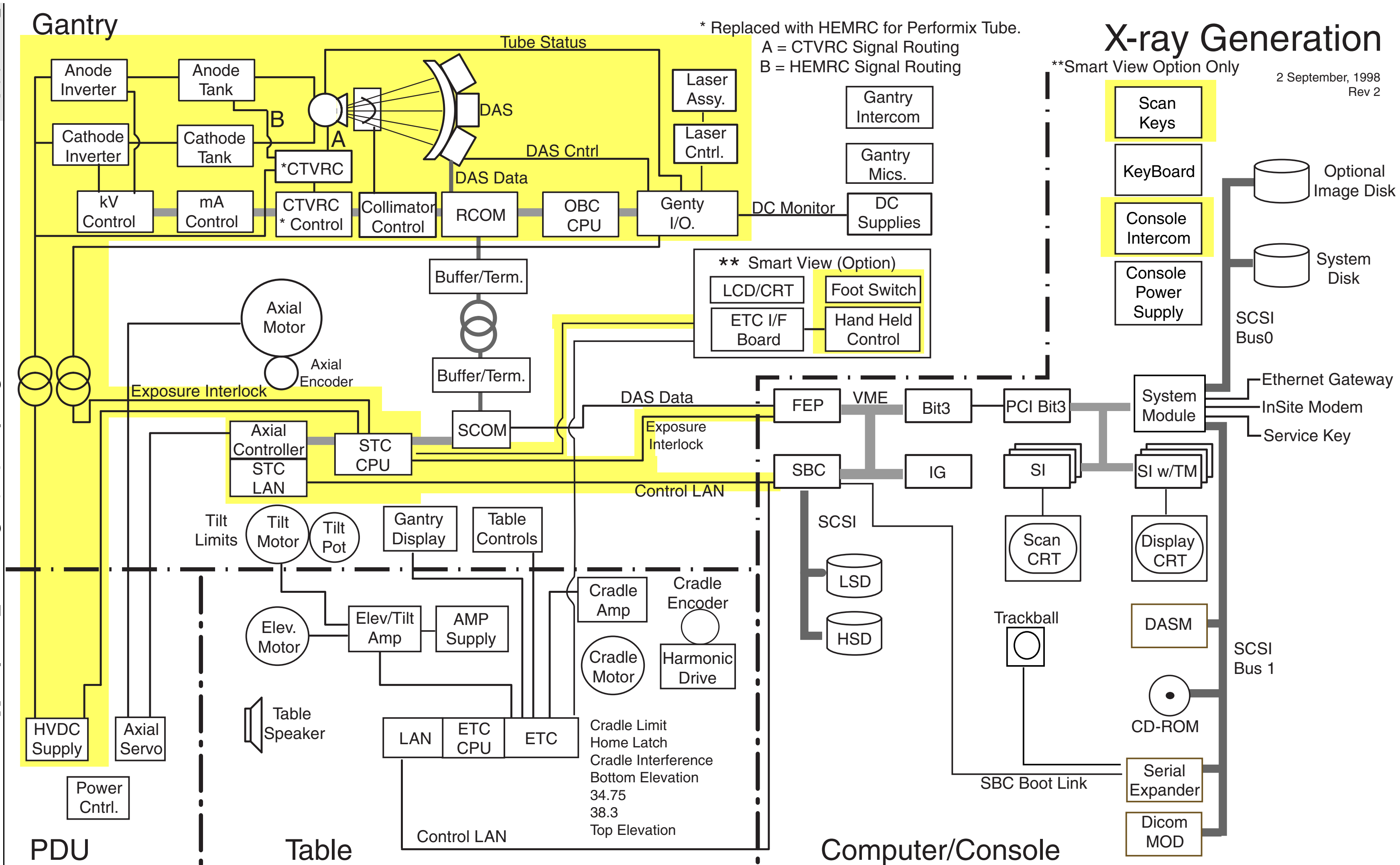


Figure 5-27 Function Map, X-Ray Generation

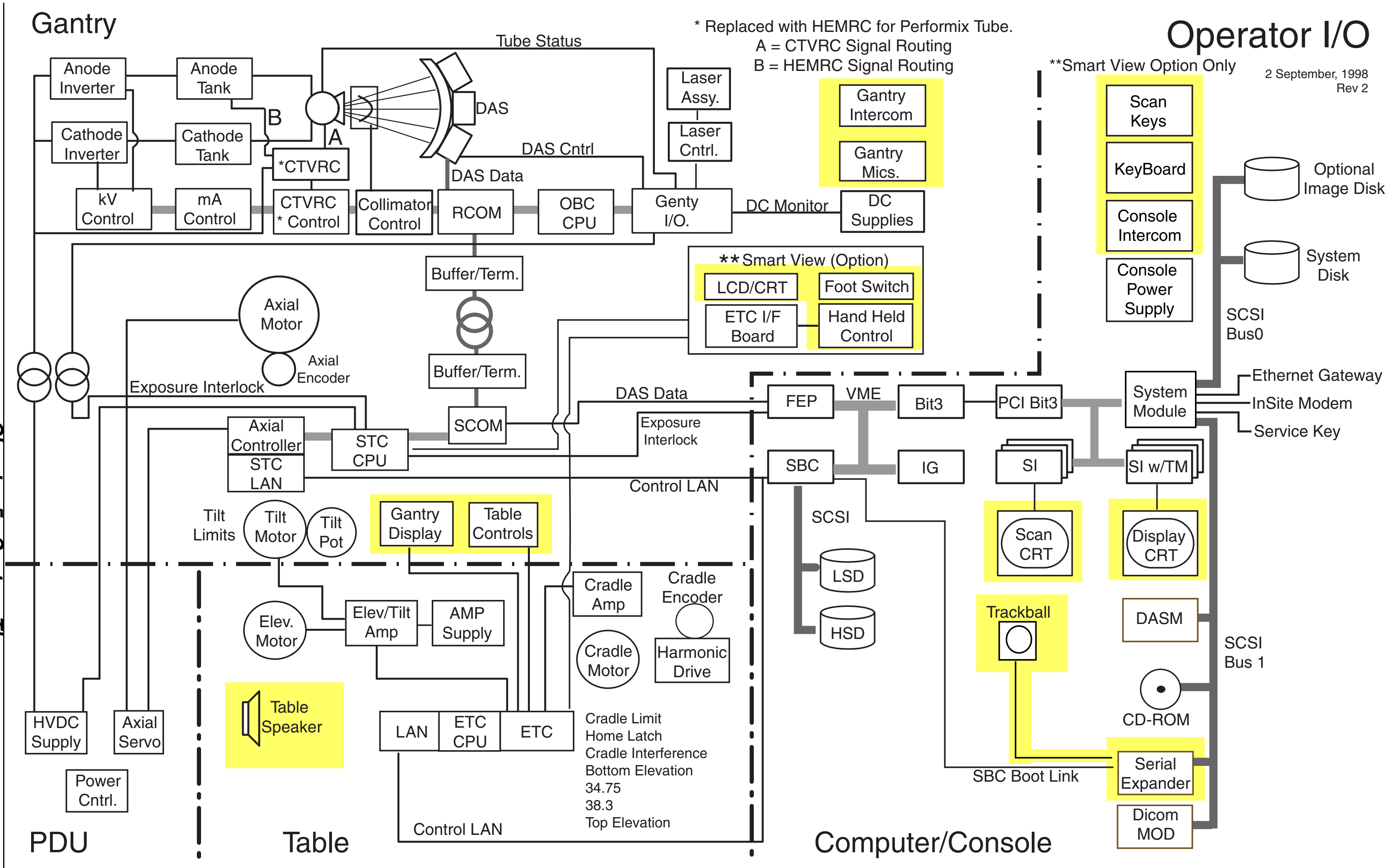


Figure 5-28 Function Map, Operator I/O

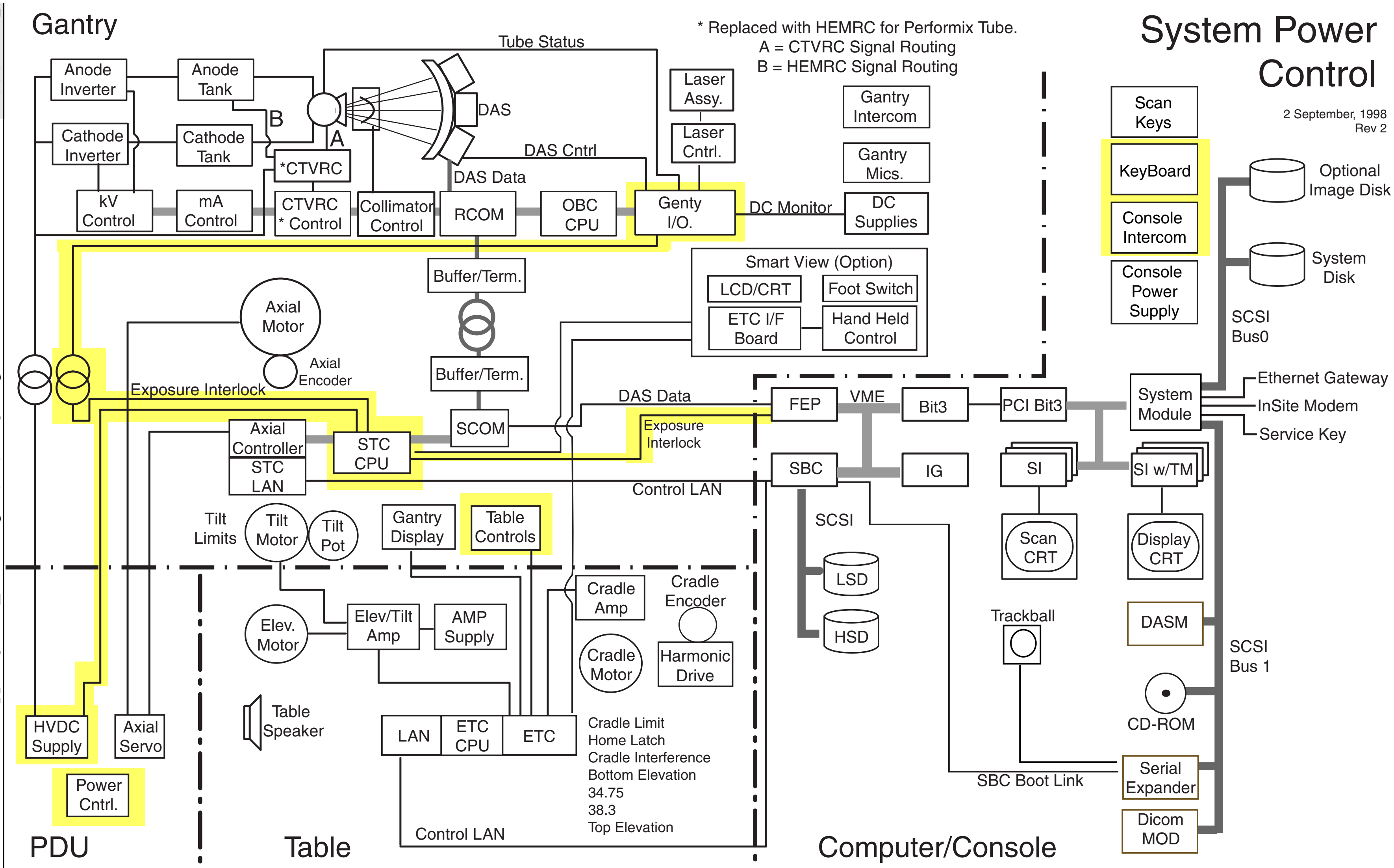


Figure 5-29 Function Map, System Power Control

Section 16.0
System Architectures

16.1 CT/i NexGen (Octane™) System Architecture

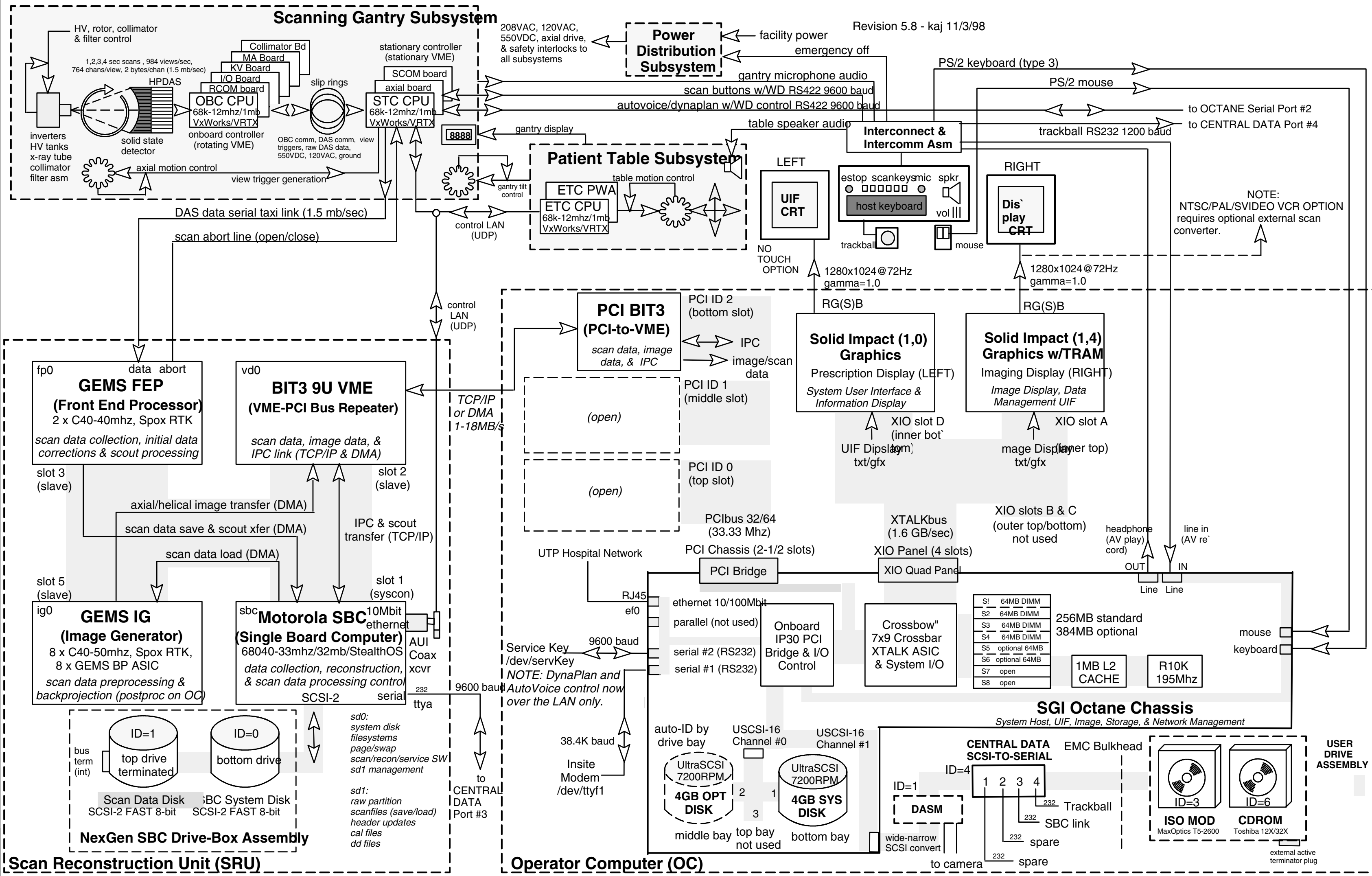


Figure 5-30 CT/i NexGen (Octane) System Architecture Diagram

16.2 CT/i (Indigo™) System Architecture

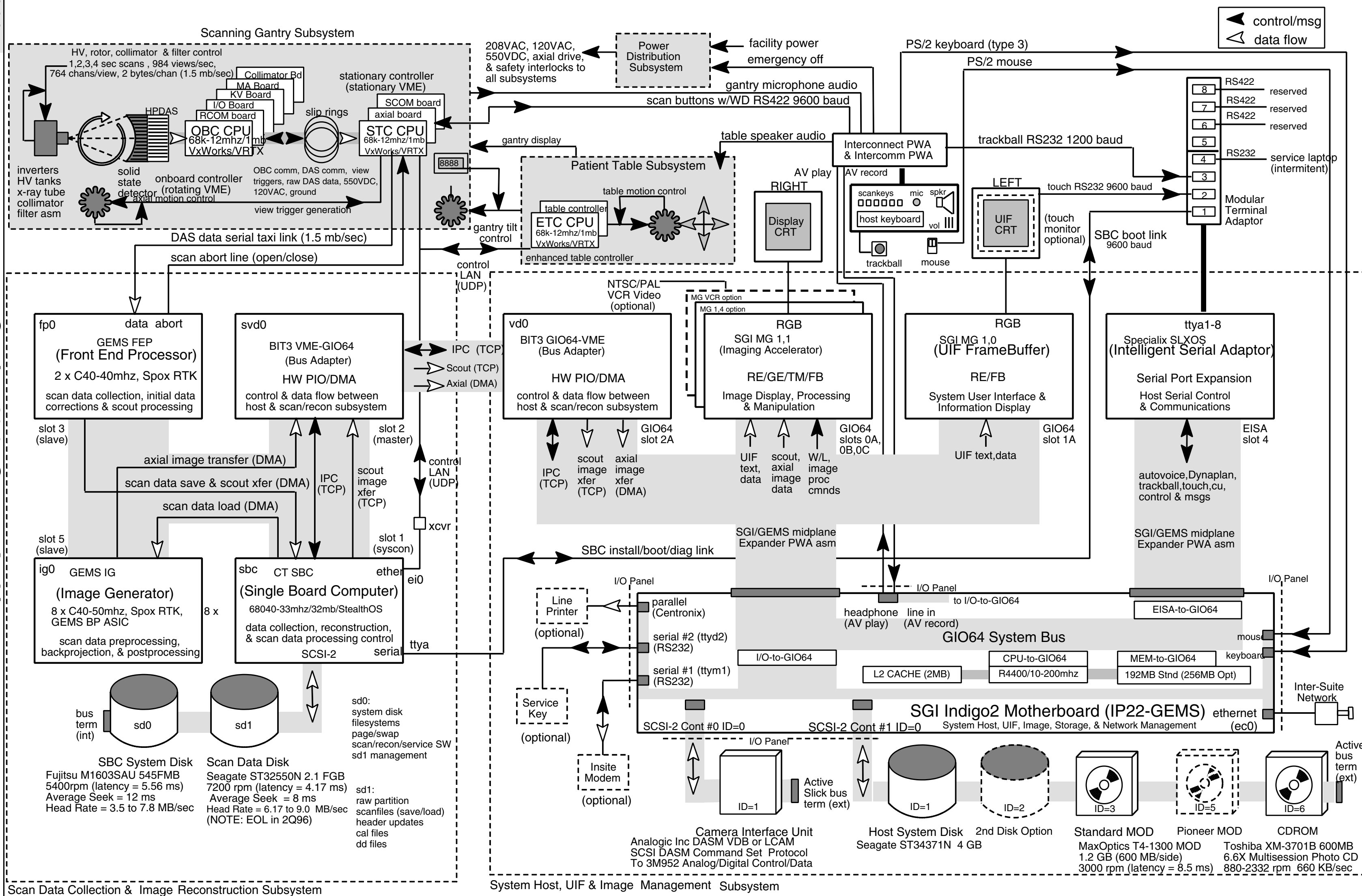


Figure 5-31 CT/i (Indigo2) System Architecture Diagram

16.3 CT/i Indigo2 SCSI Peripheral Bus

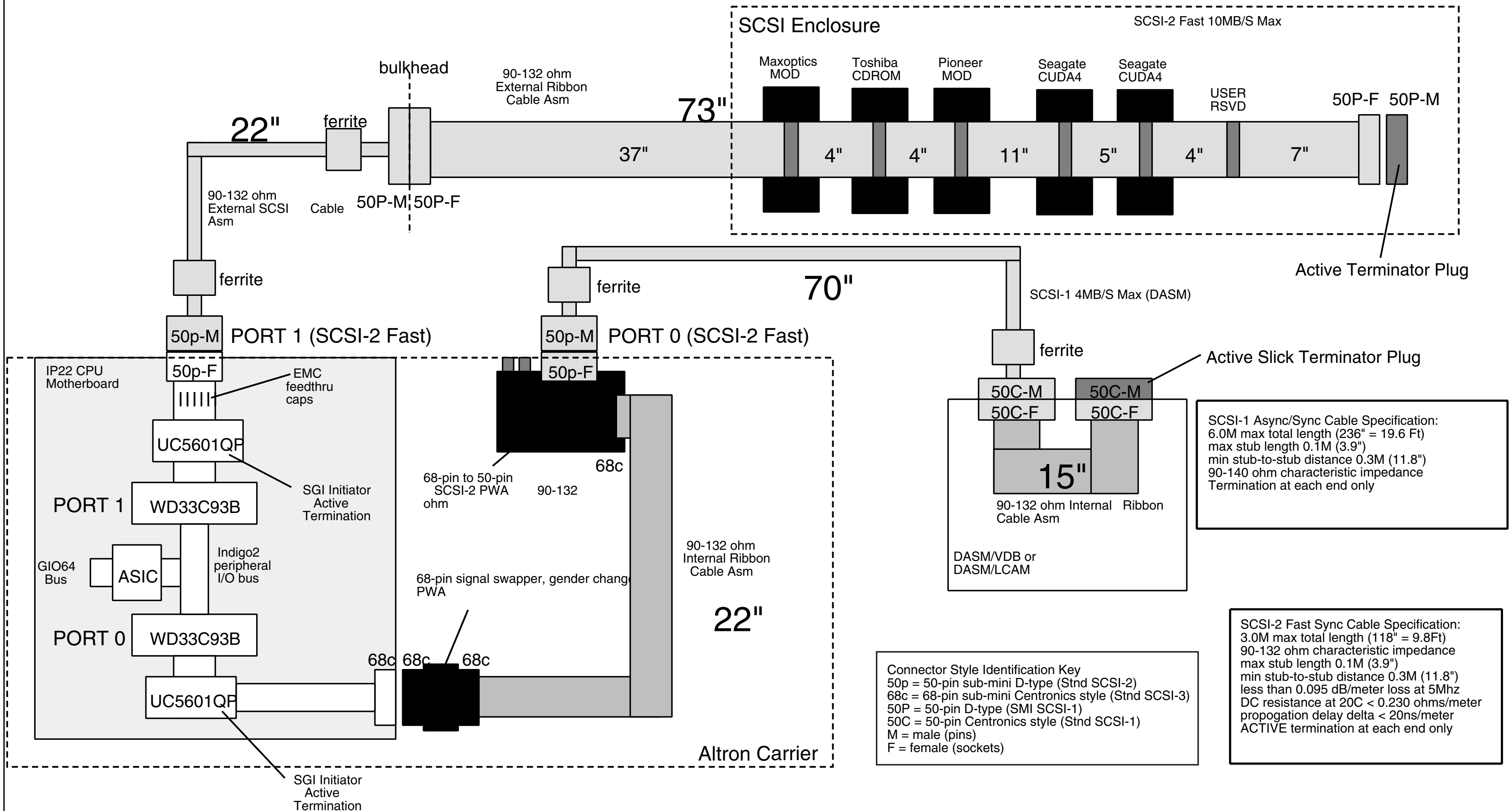


Figure 5-32 CT/iIndigo2 SCSI Peripheral Bus Diagram

16.4 CT/i Filming Architecture

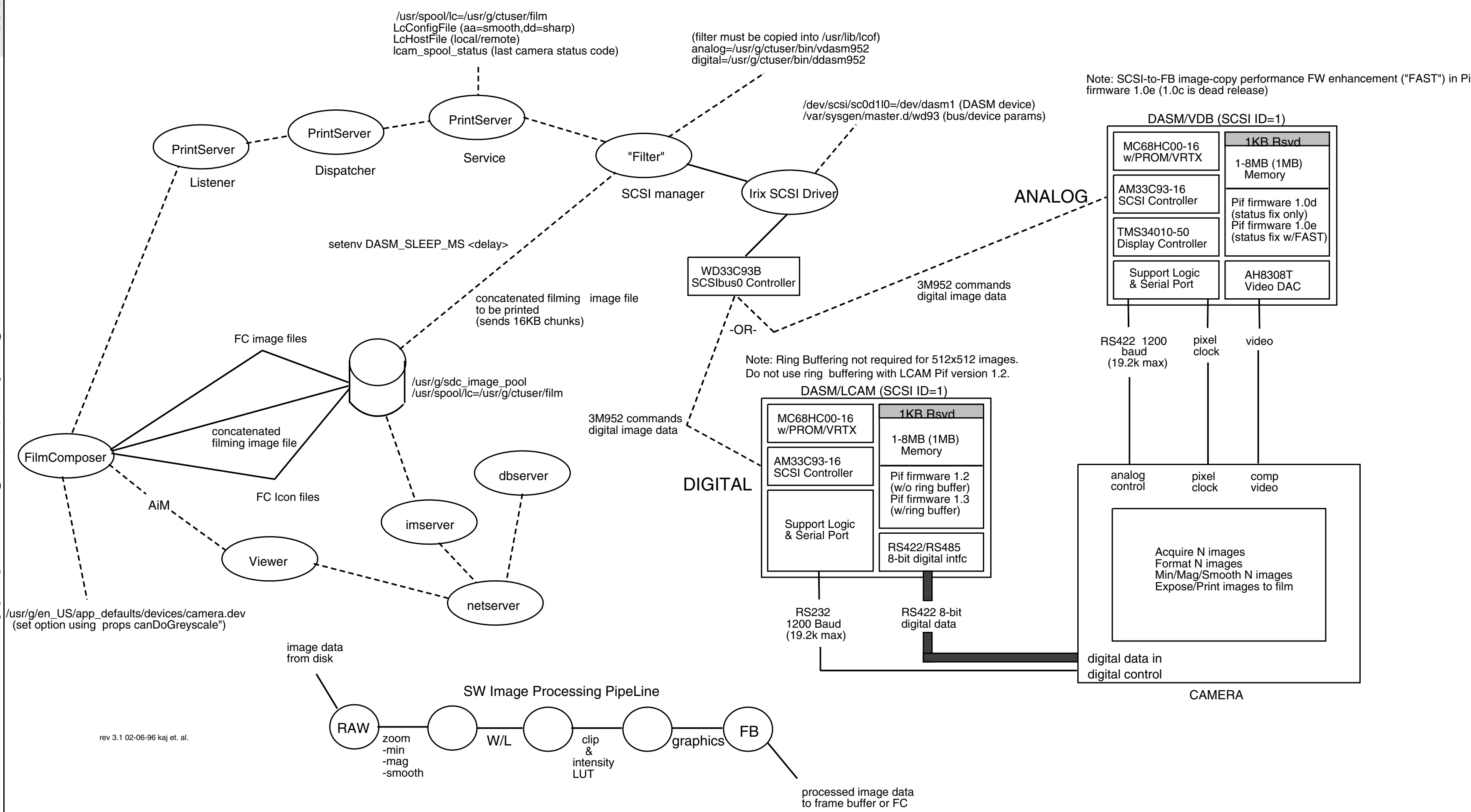


Figure 5-33 CT/i Filming Architecture Diagram

Section 17.0
CT/i Processes

17.1 CT/i Service Desktop Processes

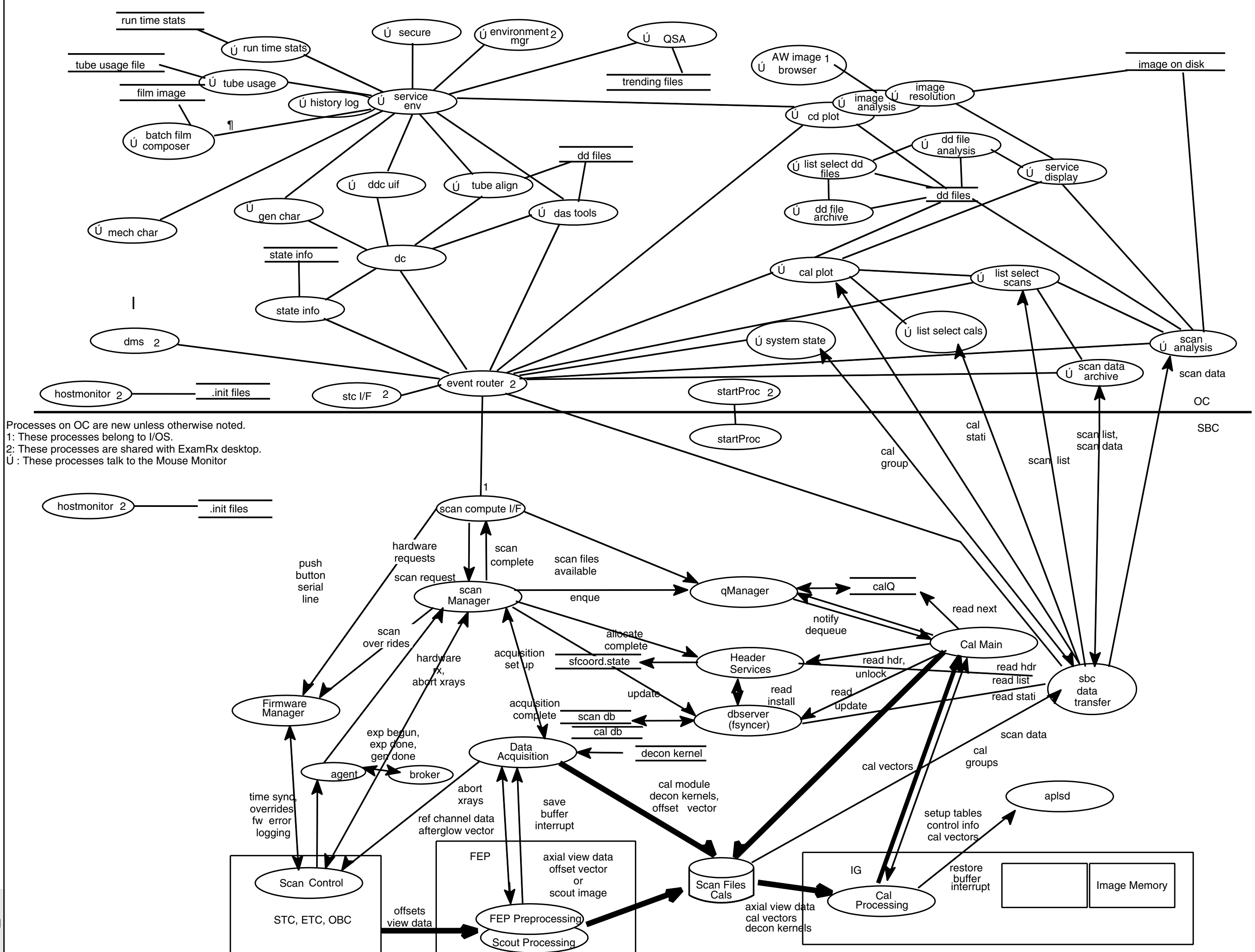


Figure 5-34 CT/i Service Desktop Processes Diagram

17.2 CT/i Exam Rx Processes

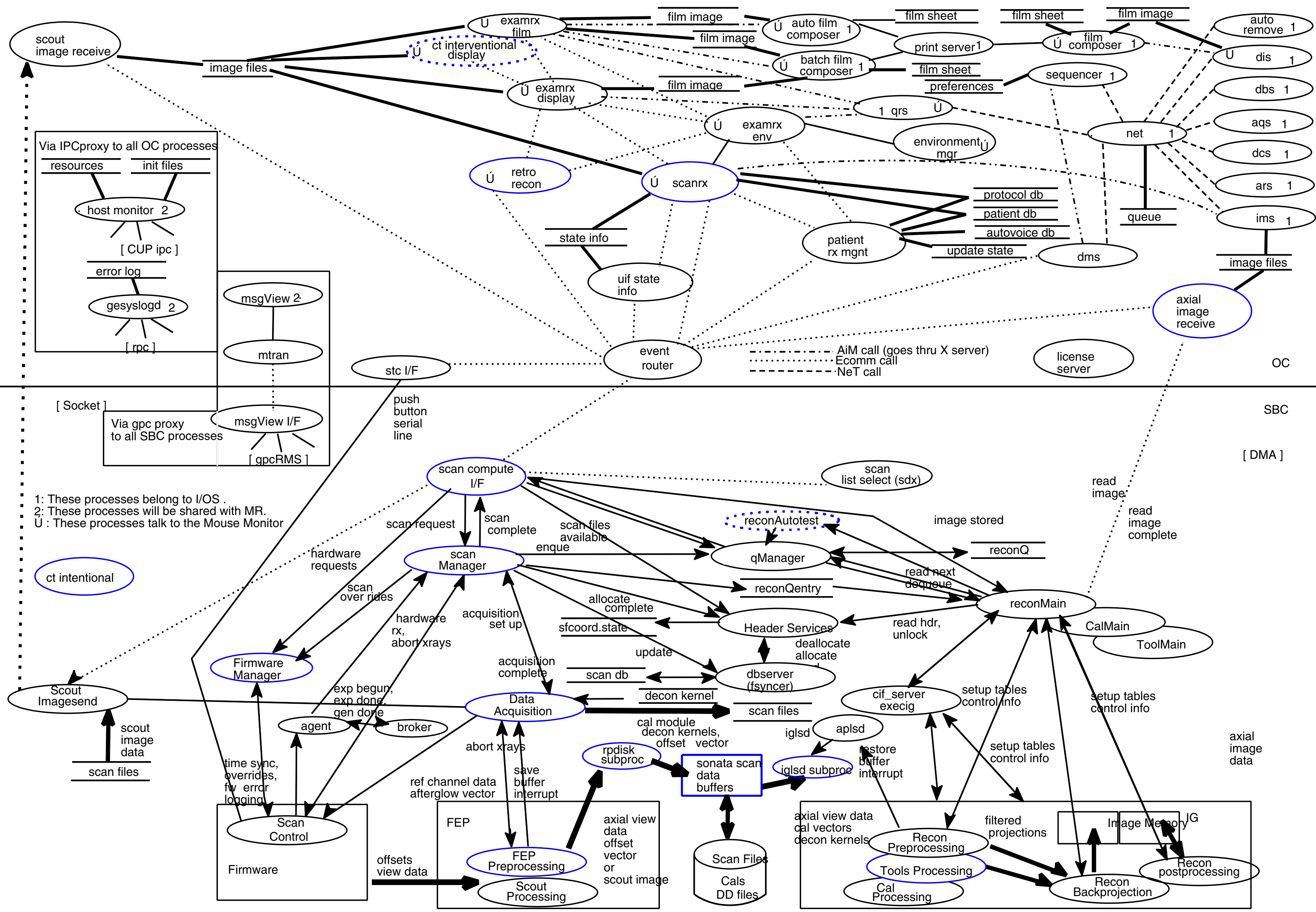


Figure 5-35 CT/i Exam Rx Processes Diagram

Section 18.0
Effects on Image Quality (Hardware & Settings)

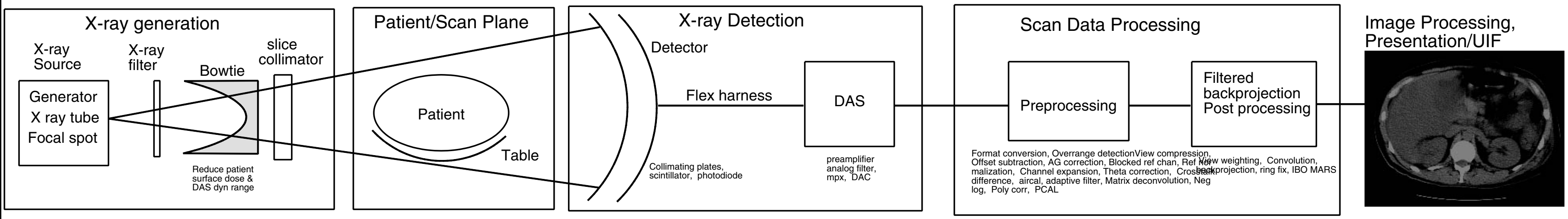


Figure 5-36 Image Quality Chain Diagram

X-RAY SOURCE	FILTERS, COLLIMATOR & GANTRY	PATIENT	DETECTOR	DAS	PRE PROCESSING	FILTERED BACK PROJECTION	IMAGE
FS size, FS iso position error	gantry alignments	position in SFOV, anatomy, patient motion	aperture (1 & 2mm), crosstalk,	Scan time		Algorithm, DFOV, image filters	Spatial resolution-radial
FS size, FS iso position error	gantry alignments	position in SFOV, anatomy, patient motion	aperture, crosstalk, temporal response	Scan time, Analog filter	view compression, primary speed correction	Algorithm, DFOV, helical algo, image filters	Spatial resolution-azimuthal
FS length, sag, POR error	Slice aperture, gantry tilt	motion, helical				Helical algo, 1/2 scan algo	Spatial resolution-Z-axis
KV, mA, X-ray intensity, scan time, off focus radiation	Slice aperture, bowtie, X-ray filtration	size, position, shape, dose, scatter, atten, partial vol	QDE, primary speed, crosstalk		primary speed correction, view compression	Algorithm	LCD (Quantum Noise)
low mA and KV	small slice	large size, dense anatomy	gain, noise @ low signal,	Noise @ low signal	primary speed correction		LCD (Non Quantum noise)
kv, off focus radiation	X-ray filtration	size, anatomy, contrast media	Scatter collimation				Tissue contrast

Table 5-9 Image Quality - Effects of Hardware and System Settings

X-RAY SOURCE	FILTERS, COLLIMATOR & GANTRY	PATIENT	DETECTOR	DAS	PRE PROCESSING	FILTERED BACK PROJECTION	IMAGE
FS position error (Z-thermal, X, & sag)	nicks, cracks, scratches, dirt, oil	position in SFOV, anatomy	Beam drift, Z-axis gain uniformity, spectral error, collimators, BOW, offset drift, rad damage, temporal response	linearity, offset drift, Analog filter response	Offset sub, AG corr, Q-cal, Theta corr, Crosstalk corr, Air cal, Poly corr, PCAL	Ring fix	Rings, Bands, arcs
FS position error (Z-thermal, X, & sag)	nicks, cracks, scratches, dirt, oil	position in SFOV, anatomy	Beam drift, Z-axis gain uniformity, spectral error, collimators, BOW, offset drift, rad damage, temporal response, xtalk	linearity, offset drift, Analog filter response, over ranged data, gain drift	Offset sub, AG corr, Q-cal, Theta corr, Crosstalk corr, Air cal, Poly corr, PCAL		Center artifacts
Spits, generator ripple, FS position change	encoder error, bearing run-out, smart start, X-ray filter	movement, beam hardening, partial volume, bone, helical pitch	Flex harness connections	Flex harness connections	data transfer	Peristaltic weighting, IBO, MARS, half scan algo	Streaks
KV error	CBF alignment, X-ray filter	position in SFOV, partial volume, bone, metal	blocked reference channels,	Over-ranged data	Air cal, Poly corr, PCAL, DOC errors	IBO for bone beam hardening	Image CT No uniformity
off focus radiation		position, size, shape	temporal response		AG corrr, matrix decon		CT No uniformity @ edge
FS isocenter position error, FS size	gantry geometry, isocenter alignment	anatomy, cradle	aperture, pitch	# of views		algorithm, no of views	Multiple fine streak patterns
FS anode wobble (Z)		patient Z anatomy	EMI	EMI			Multiple broad streak patterns (tire track)

Table 5-9 Image Quality - Effects of Hardware and System Settings (Continued)

GE MEDICAL SYSTEMS

GE MEDICAL SYSTEMS-AMERICAS: FAX 414.544.3384
P.O. BOX 414; MILWAUKEE, WISCONSIN 53201-0414, U.S.A.

GE MEDICAL SYSTEMS-EUROPE: FAX 33.1.40.93.33.33
PARIS, FRANCE

GE MEDICAL SYSTEMS-ASIA: FAX 65.291.7006

GE Medical Systems

gehealthcare.com

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GE Medical Systems CT/i System Service Manual - Advanced

Chapters 6 & 7

System Information & Software Tools

Chapter 6
System Information

Section 1.0
Operating System (OS)

1.1 Time and Date

If the timezone is wrong or if the OC timezone doesn't match the SBC one, you must first run reconfig on the OC to select the correct one then enter the reconfig command at the SBC level to make sure they match.

1.) On the Service Desktop, select UTILITIES –> SHUTDOWN APPLICATIONS.
2.) Open a Unix Shell and become <root> in the OC window.
3.) Enter: **su**

    Enter the super user (root) password, default password is **#bigguy**)
4.) Enter: **setdate**

    You will now be presented with a series of date questions. Enter time specific values.

    The month is ? ***<MM>***

    The day is ? ***<DD>***

    The hour is ? ***<HH>***

    The minute is ? ***<mm>***

    The year is ? ***<YYYY>***

    MM is month (01-12), ***DD*** is day (01-31), ***HH*** is hour (00-23), ***mm*** is minutes (0-59), ***YYYY*** is the year.
5.) Close the shell by typing: **exit**
6.) Type:**st** to restart application software.

Note: Synchronize OC & SBC times

The plot function in Smart Prep may not work if the SBC and OC do not have the same time. Setting the timezone correctly is imperative to being able to synchronize OC and SBC time. Use reconfig to set the timezone.

1.2 User Choices

1.2.1 Screen Saver

You can turn the screen saver on or off, and select the screen saver that appears for the current session. To do so, open a Unix shell and type: **`ssaver`**

The SGI GUI for doing this will open.

1.2.2 Mouse

You can adjust the acceleration and click speed of the mouse and switch operation of the buttons. To do so, open a Unix shell and type: **`mouse`**

The SGI GUI for doing this will open.

1.2.3 Keyboard Language

If a new keyboard is not set to the site's language, one need only press the **ESC** key while the SGI host is booting to get its System Maintenance Menu. Then select the last item on this menu to get the *Keyboard Layout* choices. Select the desired language, like US for USA English.

KEYBOARD LAYOUT CHOICE	LANGUAGE	SUPPORTED BY CT
BE	Belgian	
DE	German	X
de_CH	Swiss German	
DK	Danish	
ES	Spanish	
FI	Finnish	
FR	French	X
fr_CH	Swiss French	
GB	Great Britain	
IT	Italian	X
NO	Norwegian	
PT	Portuguese	
SE	Swedish	X
US	United States	X

Table 6-1 Keyboard Choices (Language)

1.3 Computer/Console Power-Up & Initialization

1.3.1 Indigo Host bootup

Each day the host creates a new `SYSLOG` to reports its bootup and hardware problems. The current one is `/usr/var/SYSLOG`. Yesterday's log is `SYSLOG.0`. The previous day has a higher number. The host keeps a week's worth of its messages. The SGI program called `sysmon` can be used to sort or filter SGI informational messages or critical errors.

Example:
Typical output

```
xx: sys_y unix: IRIX Release 6.2 IP22 Version 03131015 System V
xx: sys_y unix: Copyright 1987-1996 Silicon Graphics, Inc.
xx: sys_y unix: All Rights Reserved.
xx: sys_y unix:
```

```
xx: sys_y unix: Specialix SLXOS, Version 1.01, ID 1.146
xx: sys_y unix: (C)1991-1994 Specialix Research Ltd.
xx: sys_y unix: NOTICE: BIT3 Model 608 board found at GIO slot 3
xx: sys_y unix: SLXOS: Detected EISA SI/XIO host card in slot 4.
xx: sys_y unix: SLXOS: One host card, One module, 8 ports.
xx: sys_y unix: NOTICE: Start mounting filesystem: /
xx: sys_y unix: NOTICE: Starting XFS recovery on filesystem:
      / (dev: 128/272)
xx: sys_y unix: NOTICE: Ending XFS recovery for filesystem:
       / (dev: 128/272)
xx: sys_y unix: NOTICE: Start mounting filesystem: /usr
xx: sys_y unix: NOTICE: Ending clean XFS mount for filesystem: /usr
xx: sys_y unix: NOTICE: Start mounting filesystem: /usr/g
xx: sys_y unix: NOTICE: Ending clean XFS mount for filesystem: /usr/g
xx: sys_y unix: NOTICE: Start mounting filesystem:
      /usr/g/sdc_image_pool
xx: sys_y unix: NOTICE: Ending clean XFS mount for filesystem:
      /usr/g/sdc_image_pool
xx: sys_y unix: NOTICE: Start mounting filesystem: /data
xx: sys_y unix: NOTICE: Ending clean XFS mount for filesystem: /data
xx: sys_y routed[169]: IP_ADD_MEMBERSHIP ALLHOSTS:
      No buffer space available
xx: sys_y routed[169]: setsockopt(IP_ADD_MEMBERSHIP RIP):
      No buffer space available
xx: sys_y sendmail: starting
xx: sys_y MAKEDEV_tape: Syntax error
xx: sys_y last message repeated 3 times
xx: sys_y MAKEDEV_tape: install: malformed device number .
xx: sys_y MAKEDEV_tape: *** Error code 255 (bu21)
xx: sys_y MAKEDEV_tape: sh[15]: test: argument expected
xx: sys_y last message repeated 3 times
xx: sys_y MAKEDEV_tape: UX:make: ERROR:
      `tape' not remade because of errors (bu14)
xx: sys_y sendmail[743]:
      starting daemon (950413.SGI.8.6.12): SMTP+queueing@00:15:00
xx: sys_y timeslave[254]: sbc will not tell us the date
xx: sys_y timeslave[254]: send(ICMP): Connection refused
xx: sys_y mediad: Cannot contact objectserver.  Query timed out.
xx: sys_y Xsession: ctuser: login
xx: sys_y access control disabled, clients can connect from any host
xx: sys_y timeslave[254]: Date measurements from sbc are working again.
xx: sys_y timeslave[254]: Time measurements from sbc are working again.
xx: sys_y timeslave[254]: date changed from 10/21/97 13:19:15
```

1.3.2 Octane Host bootup

The Octane host creates a `SYSLOG` text file/report during bootup to log the status of major processes. The `SYSLOG` can be found in directory `/var/admin/`. The contents of a typical ASCII text `SYSLOG` file gives you a good indication of which processes started successfully or unsuccessfully.

For problem identification and resolution, it is normal practice to execute the **hinv** command to obtain additional information specific to your systems' hardware configuration, see page 376. Together this information will give you a good idea of your Host computer's health.

What follows is a typical example of the key communications that allow a Octane host to function properly. Actual system hardware (i.e. disk drives) identified will depend upon your specific hardware configuration.

Example: SYSLOG File

```
...
Jan xx:bay18 unix: STS: Config device ST-1400A
```

Comment: *Here above, the disk drive model ST-1400A (model depends on hardware installed) has been detected and is attached/detached through the Central Data Serial Box. PCI devices are searched for next.*

```
Jan xx:bay18 unix: Attaching BIT3 MV617 PCI Card, rev 54
```

Comment: *As you can see above, a BIT3 PCI board has been detected and attached. Now the mounting of the filesystem takes place.*

```
Janxx:bayx unix: NOTICE:Start mounting filesystem: /
Janxx:bayx unix: NOTICE:Starting XFS recovery on filesystem:/(dev:0/349)
Janxx:bayx unix: NOTICE:Ending XFS recovery for filesystem:
/(/hw/node/xtalk/15/pci/0/scsi_ctlr/0/target/1/lun/0/disk/partition/0/
block)
Janxx:bayx unix: NOTICE:Start mounting filesystem:/usr
Janxx:bayx unix: NOTICE:Ending clean XFS mount for filesystem: /usr
Janxx:bayx unix: NOTICE:Start mounting filesystem:
      /usr/g/sdc_image_pool3
Janxx:bayx unix: NOTICE:Ending clean XFS mount for filesystem:
      /usr/g/sdc_image_pool3
Jan xx:bayx unix: NOTICE:Start mounting filesystem:
      /usr/g/sdc_image_pool2
Jan xx:bayx unix: NOTICE: Ending clean XFS mount for filesystem:
      /usr/g/sdc_image_pool2
Jan xx:bayx unix: NOTICE: Start mounting filesystem: /data
Jan xx:bayx unix: NOTICE: Ending clean XFS mount for filesystem: /data
```

Comment: *Filesystem mounted. Now its time to attach the IP.*

```
Jan xx:bayx routed[214]: IP_ADD_MEMBERSHIP ALLHOSTS:
      No buffer space available
Jan xx:bayx routed[214]: setsockopt(IP_ADD_MEMBERSHIP RIP):
      No buffer space available
```

Comment: *In this example, the IP was allocated no buffer space.*

```
Jan xx:bayx timeslave[366]: recvfrom(date read)=-1:
      Connection refused
```

Comment: *As seen in the lines above, the SBC in the reconstruction system is not up yet and a process in the Octane computer has reported it as the connection being refused. The SBC may not have completed booting up completely.*

1.4 IRIX File System

The host uses the XFS file system rather than EFS. The major features of XFS include:

- full 64-bit file capabilities (files larger than 2 GB)
- rapid and reliable recovery after system crashes because of the use of journaling technology
- efficient support of large, sparse files (files with "holes")
- integrated, full-function volume manager, the XLV Volume Manager

- extremely high I/O performance that scales well on multiprocessing systems
- guaranteed-rate I/O for multimedia and data acquisition uses
- compatibility with existing applications and with NFS
- user-specified filesystem block sizes ranging from 512 bytes up to 64 KB
- small directories and symbolic links of 156 characters or less take no space

XFS uses database journaling technology to provide high reliability and rapid recovery. Recovery after a system crash is completed within a few seconds, without the use of a filesystem checker such as the fsck command. Recovery time is independent of filesystem size. XFS is designed to be a very high performance filesystem. Under certain conditions, throughput exceeds 100 MB per second.

1.4.1 Same commands as efs system

Most filesystem commands, such as **du**, **dvhtool**, **ls**, **mount**, **prtvtoc**, and **umount**, work with XFS filesystems as well as EFS filesystems. A few commands, such as **df**, **fx**, and **mkfs** have additional features for XFS.

1.4.2 New xfs commands

The filesystem commands clri, fsck, findblk, and ncheck are not used with XFS filesystems. For backup and restore, the standard IRIX commands **Backup**, **bru**, **cpio**, **Restore**, and **tar** can be used **for files less than 2 GB in size**. To dump XFS filesystems, the new command **xfsdump** must be used instead of dump. Restoring from these dumps is done using **xfsrestore**.

1.5 ETC, STC & OBC “Heurikon” CPU - Power-up Self Tests

Located in the Table, gantry stationary and rotating subsystems is the ETC, STC and OBC CPU’s. Following the commands, these boards are responsible for controlling their respective subsystem hardware. They preform integrity of themselves using built in self-tests.

The Heurikon CPU boards perform self tests each time they are powered up. The self tests are slightly different depending on the CPU (ETC, STC or OBC) assignment.

To perform a power up self test, the CPU board must be powered down and remain powered down for approximately 90 seconds. To do this manually to the STC and OBC, you should turn off power to the entire STC or OBC assembly. The ETC is powered down by turning off the 24 Hr. Table switch.

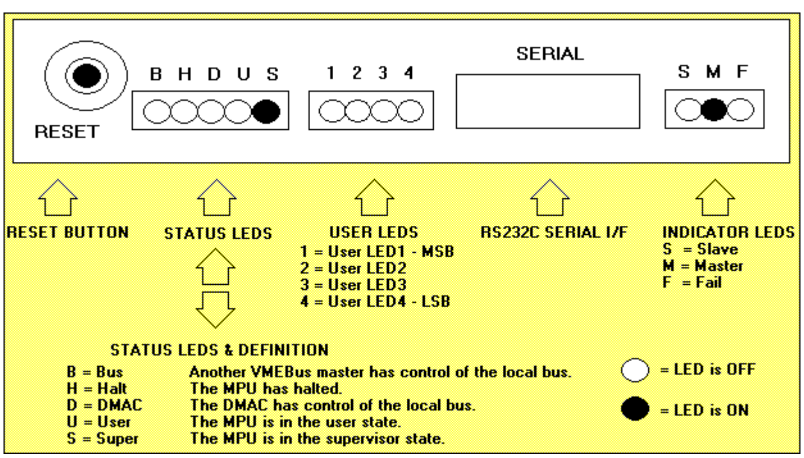


Figure 6-1 Heurikon CPU leds

When the CPU boards are powered up, they will automatically (unless there is a power problem) execute the self tests. If self test error conditions are found, the CPU's User LEDs (1thru 4) will report the condition(s). If multiple conditions are present, the LEDs will repeatedly cycle through all the conditions.

1.5.1 Power Up Self Test Results

1.5.1.1 LED Descriptions

On Power-up, the STC, OBC and ETC controllers display the results of their self tests. Power must remain off to a controller for at least 60SEC or Self-test may not be run or results may be inaccurate. This is because the dynamic ram retains the CPON information. In this event, power-up tests are bypassed and the results of the last. power-up test displayed on LEDs

1 2 3 4	HEX	LED ASSIGNMENT
◆◆◆◆	F	Not Available - - Do Not Use for a test
◆◆◆◇	E	HK68/VE Basic (Motorola Instruction Set and Prom Checksum Test)
◆◆◇◆	D	HK68/VE RAM Verification Test
◆◆◇◇	C	HK68/VE CIO Unit Test
◆◇◆◆	B	SCA_LAN Basic (Rd & Wr Registers and Internal Loopbacks)
◆◇◆◇	A	SCA_LAN External Loopback (on the wire)
◆◇◇◆	9	SCA_LAN TDR (on the wire)
◆◇◇◇	8	SCOM Basic (RD & Wr Registers and Interrupts)
◇◆◆◆	7	SCOM VME Fifo Loopback (on board test)
◇◆◆◇	6	SCOM AP Fifo Loopback (on board test)
◇◆◇◆	5	SCOM EXTENDED DIAGNOSTICS (off board test with RCOM)
◇◆◇◇	4	spare for GE future use
◇◇◆◆	3	spare for GE future use
◇◇◆◇	2	spare for GE future use
◇◇◇◆	1	spare for GE future use
◇◇◇◇	0	Not Available - - Do Not Use for a test

Legend: ◆ = LED ON, ◇ = LED OFF, 1 = Led MSB, 4 = Led LSB

Figure 6-2 STC Self-Test LED Outputs

1 2 3 4	HEX	LED ASSIGNMENT
◆◆◆◆	F	Not Available - - Do Not Use for a test
◆◆◆◇	E	HK68/VE Basic (Motorola Instruction Set and Prom Checksum Test)
◆◆◇◆	D	HK68/VE RAM Verification Test
◆◆◇◇	C	HK68/VE CIO Unit Test
◆◇◆◆	B	SCA_LAN Basic (Rd & Wr Registers and Internal Loopbacks)
◆◇◆◇	A	SCA_LAN External Loopback (on the wire)
◆◇◇◆	9	SCA_LAN TDR (on the wire)
◆◇◇◇	8	SCOM Basic (RD & Wr Registers and Interrupts)
◇◆◆◆	7	SCOM VME Fifo Loopback (on board test)
◇◆◆◇	6	SCOM AP Fifo Loopback (on board test)
◇◆◇◆	5	SCOM EXTENDED DIAGNOSTICS (off board test with RCOM)
◇◆◇◇	4	spare for GE future use
◇◇◆◆	3	spare for GE future use
◇◇◆◇	2	spare for GE future use
◇◇◇◆	1	spare for GE future use
◇◇◇◇	0	Not Available - - Do Not Use for a test

Legend: ◆ = LED ON, ◇ = LED OFF, 1 = Led MSB, 4 = Led LSB

Figure 6-3 OBC Self-Test LED Outputs

1 2 3 4	HEX	LED ASSIGNMENT
◆◆◆◆	F	Not Available - - Do Not Use for a test
◆◆◆◇	E	HK68/VE Basic (Motorola Instruction Set and Prom Checksum Test)
◆◆◇◆	D	HK68/VE RAM Verification Test
◆◆◇◇	C	HK68/VE CIO Unit Test
◆◇◆◆	B	SCA_LAN Basic (Rd & Wr Registers and Internal Loopbacks)
◆◇◆◇	A	SCA_LAN External Loopback (on the wire)
◆◇◇◆	9	SCA_LAN TDR (on the wire)
◆◇◇◇	8	SCOM Basic (RD & Wr Registers and Interrupts)
◇◆◆◆	7	SCOM VME Fifo Loopback (on board test)
◇◆◆◇	6	SCOM AP Fifo Loopback (on board test)
◇◆◇◆	5	SCOM EXTENDED DIAGNOSTICS (off board test with RCOM)
◇◆◇◇	4	spare for GE future use
◇◇◆◆	3	spare for GE future use
◇◇◆◇	2	spare for GE future use
◇◇◇◆	1	spare for GE future use
◇◇◇◇	0	Not Available - - Do Not Use for a test
Legend:		◆ = LED ON, ◇ = LED OFF, 1 = Led MSB, 4 = Led LSB

Figure 6-4 ETC Self-Test LED Outputs

ETC, STC & OBC (Heurikon) Tests

FUNCTION	LEDS	DESCRIPTION
Initialization	(F: • • • •)	Setup interrupt vectors & CIO
Failure	(E: • • • o)	CPU HALTS
Processor/PROM Checksum	(E: • • • o)	68000 Instruction set check (ram used) ROM Verified using CRC16 based polynomial
Failure	(E: • • • o)	CPU HALTS
Ram Verification -	(D: • • o •)	Each word of memory R/W 16 times
Failure	(E: • • • o)	CPU HALTS
CIO Verification	(C: • • o o)	Checks interrupts, timers, counters (no VME)
Failure	(E: • • • o)	CPU HALTS
• = "on" o = "off"		

Table 6-2 HEURIKON BOARD RELATED LED READOUTS

At this point the type of node (ETC, STC or OBCR) determines the tests that are run.

ETC - VME/LAN TESTS

FUNCTION	LEDS	DESCRIPTION
LAN controller tests	(B: • o • •)	Checks module present, controller & internal loops
Failure	(B: • o • •)	Flashes, possibly with other failures
LAN External loop-back	(A: • o • o)	Checks wire/ termination
Failure	(A: • o • o)	Flashes, possibly with other failures
TDR test	(9: • o o •)	Checks wire/ termination
Failure	(9: • o o •)	Flashes, possibly with other failures
• = "on" o = "off"		

Table 6-3 ETC VME & LAN RELATED LED READOUTS

STC - VME/LAN TESTS

FUNCTION	LEDS	DESCRIPTION
LAN controller tests	(B: • o • •)	Checks module present, controller & internal loops
Failure	(B: • o • •)	Flashes, possibly with other failures
LAN External loop-back	(A: • o • o)	Checks wire/ termination
Failure	(A: • o • o)	Flashes, possibly with other failures
TDR test	(9: • o o •)	Checks wire/ termination
Failure	(9: • o o •)	Flashes, possibly with other failures
• = “on” o = “off”		

Table 6-4 STC VME/LAN RELATED LED READOUTS

SCOM/Communications Test

FUNCTION	LEDS	DESCRIPTION
Module Present test	(8: • o o o)	Checks for Presence of TAXI
Failure	(8: • o o o)	Flashes, possibly with other failures
VME FIFO test	(7: o • • •)	Checks VME path using loop-back
Failure	(7: • o o o)	Flashes, possibly with other failures
AP FIFO test	(6: o • • o)	Checks DAS path using loop-back
Failure	(6: • o o o)	Flashes, possibly with other failures
• = “on” o = “off”		

Table 6-5 SCOM/COMMUNICATIONS RELATED LED READOUTS

OBC (OBCR) - RCOM/Communications Test

FUNCTION	LEDS	DESCRIPTION
Module Present test	(8: • o o o)	Checks for Presence of TAXI
Failure	(8: • o o o)	Flashes, possibly with other failures
VME FIFO test	(7: o • • •)	Checks VME path using loop-back
Failure	(7: • o o o)	Flashes, possibly with other failures
DAS FIFO test	(6: o • • o)	Checks DAS path using loop-back
Failure	(6: • o o o)	Flashes, possibly with other failures
• = “on” o = “off”		

Table 6-6 OBCR RCOM/COMMINCATIONS REALETD LED READOUTS

1.5.1.2 Obtaining & Interpreting Power-up Self Test Results

Results of the power-up self tests can be obtained using `nbsclient`. See Section 29.0 on page 327 for additional information on `nbsclient`.

Typical Power Up Self Tests Results - No Errors

```
[NBS, OBCR]: d 2400
002400:  4350 4f4e 0000 71e0  0000 0000 0000 0000   *CPON..q.........*
002410:  0000 0000 0000 0000  0000 0000 0000 0000   *................*
002420:  0000 0000 0000 0000  0000 0000 0000 0000   *................*
002430:  0000 0000 0000 0000  0000 0000 0000 0000   *................*
002440:  0000 0000 0000 0000  0000 0000 0000 0000   *................*
002450:  0000 0000 0000 0000  0000 0000 0000 0000   *................*
002460:  0000 0000 0000 0000  0000 0000 0000 0000   *................*
002470:  0000 0000 0000 0000  0000 0000 0000 0000   *................*
[NBS, OBCR]:
```

SCA (LAN) & INT (Internal Board) Tests

ERROR OUTPUT

```
[NBS, OBCR]: d 2400
002400:  4350 4f4e 0000 71e0  0000 0000 0000 0000   *CPON..q.........*
002410:  0000 0000 0000 0000  0000 0000 0000 0000   *................*
002420:  0000 0000 0000 0000  0000 0000 0000 0000   *................*
002430:  0000 0000 0000 0000  0000 0000 0000 0000   *................*
002440:  0000 0000 0000 0000  0000 0000 0000 0000   *................*
002450:  0000 0000 0000 0000  0B10 0000 0000 0000   *................*
002460:  0000 0000 0000 0000  /      /0 0000 0000   *................*
002470:  0000 0000 0000 0000/      / 00 0000 0000   *................*
                          /         \
                        /             \
          First Error: 0Bxx  0Bxx : Second Error
```

ERROR TRANSLATION

0B01 : SCA Module not present

0B02 : SCA ISBX address register

0B03 : SCA ID PAL

0B04 : SCA 82560 registers

0B05 : SCA local memory test

0B06 : SCA 82592 Diagnostic command execution

0B10 : INT loop couldn't initalize interface

0B11 : INT loop couldn't enable DMA channels

0B12 : INT loop failed to TX test packet

0B13 : INT loop failed to RX test packet

0B14 : INT loop data integritiy error

0B15 : INT loop bus error

EXT (External CPU) Tests

ERROR OUTPUT

```
[NBS, OBCR]: d 2400
002400:  4350 4f4e 0000 71e0  0000 0000 0000 0000   *CPON..q.........*
002410:  0000 0000 0000 0000  0000 0000 0000 0000   *................*
002420:  0000 0000 0000 0000  0000 0000 0000 0000   *................*
002430:  0000 0000 0000 0000  0000 0000 0000 0000   *................*
002440:  0000 0000 0000 0000  0000 0000 0000 0000   *................*
002450:  0000 0000 0A01 0000  0000 0000 0000 0000   *................*
002460:  0000 0000 /      /0  0000 0000 0000 0000   *................*
002470:  0000 000/      / 00  0000 0000 0000 0000   *................*
                /       \
              /           \
First Error: 0Axx  0Axx: Second Error
```

ERROR TRANSLATION

0A10 : EXT loop couldn't initalize interface

0A11 : EXT loop couldn't enable DMA channels

0A12 : EXT loop failed to TX test packet

0A13 : EXT loop failed to RX test packet

0A14 : EXT loop data integritiy error

0A15 : EXT loop bus error

TDR Tests

ERROR OUTPUT

```
[NBS, OBCR]: d 2400
002400:  4350 4f4e 0000 71e0  0000 0000 0000 0000   *CPON..q.........*
002410:  0000 0000 0000 0000  0000 0000 0000 0000   *................*
002420:  0000 0000 0000 0000  0000 0000 0000 0000   *................*
002430:  0000 0000 0000 0000  0000 0000 0000 0000   *................*
002440:  0000 0000 0000 0000  0000 0000 0000 0000   *................*
002450:  0000 0000 0000 0000  0000 0000 0000 0000   *................*
002460:  /       / 0000 0000  0000 0000 0000 0000   *................*
002470: /       /  0000 0000  0000 0000 0000 0000   *................*
       /        \
      /          \
09xx    09xx : Second Error
```

ERROR TRANSLATION

```
0901 : TDR couldn't initalize interface
0902 : TDR transceiver problem
0903 : TDR open circuit
0904 : TDR short circuit
0905 : TDR command excution failed
```

Tests 6 & 7

ERROR OUTPUT

```
[NBS, OBCR]: d 2400
002400:  4350 4f4e 0000 71e0  0000 0000 0000 0000   *CPON..q.........*
002410:  0000 0000 0000 0000  0000 0000 0000 0000   *................*
002420:  0000 0000 0000 0000  0000 0000 0000 0000   *................*
002430:  0000 0000 0000 0000  0000 0000 0000 0000   *................*
002440:  0000 0000 |Test 6 |  |Test 7 | 0000 0000   *................*
002450:  0000 0000 0000 0000  0000 0000 0000 0000   *................*
002460:  0000 0000 0000 0000  0000 0000 0000 0000   *................*
002470:  0000 0000 0000 0000  0000 0000 0000 0000   *................*
```

ERROR TRANSLATION

0X01 : Module not present
0X02 : Extented test wait
0X03 : FIFO empty status error
0X04 : VME FIFO status indicator absent
0X05 : interupt mask register not zero
0X06 : interupt mask register not all ones
0X07 : transmit FIFO not empty
0X08 : FIFO half status error
0X09 : Receive FIFO empty/not empty and should be
0X11 : RX FIFO full status not indicated
0X12 : RX FIFO full status indicated
0X13 : RX FIFO full status and word count disagree
0X14 : RX FIFO half status not indicated
0X15 : RX FIFO half status indicated
0X16 : RX FIFO half status and word count disagree
0X17 : RX FIFO empty status not indicated
0X18 : RX FIFO empty status indicated
0X19 : RX FIFO empty status and word count disagree
0X1A : RX FIFO data integrity error
0X21 : TX FIFO full status not indicated
0X22 : TX FIFO full status indicated
0X23 : TX FIFO full status and word count disagree
0X24 : TX FIFO half status not indicated
0X25 : TX FIFO half status indicated
0X26 : TX FIFO half status and word count disagree
0X27 : TX FIFO empty status not indicated
0X28 : TX FIFO empty status indicated
0X29 : TX FIFO empty status and word count disagree
0X2A : TX FIFO data integrity error
0X31 : AP FIFO full status not indicated

`0X32` : AP FIFO full status indicated
`0X33` : AP FIFO full status and word count disagree
`0X34` : AP FIFO empty status not indicated
`0X35` : AP FIFO empty status indicated
`0X36` : AP FIFO empty status and word count disagree
`0X41` : DAS FIFO full status not indicated
`0X42` : DAS FIFO full status indicated
`0X43` : DAS FIFO full status and word count disagree
`0X51` : SCOM/RCOM illegal board status
`0X52` : Missed interrupt
`0X53` : Missed FIFO half full interrupt
`0X54` : Missed FIFO full interrupt
`0X55` : IRQ6 interrupt status error

`Note:` `X` indicates the test that failed

1.6 Indigo Boot Environment & Control

1.6.1 Indigo Command Monitor

The Command (PROM) Monitor program controls the boot environment for all Silicon Graphics workstations. With the Command Monitor, you can boot and operate the CPU under controlled conditions, run the CPU in Command Monitor mode, and load programs like the operating system kernel or special debugging and execution versions of the kernel.

PROM stands for Programmable Read-Only Memory. Most PROM chips are installed at the factory with software programmed into them that allows the CPU to boot and allows you to perform system administration and software installations. The PROMs are not part of your disk or your operating system; they are the lowest level of access available for your system. You cannot erase them or bypass them.

The SGI computers for CT/i systems use the ARCS PROM. ARCS stands for Advanced RISC Computing Standard. This PROM provides a graphical interface and allows mouse control of booting and execution. ARCS systems also support the use of the keyboard.

1.6.2 Entering the Indigo Command (PROM) Monitor

Shutdown then restart the system, or if the system is already off, turn it on. Press **ESC** or click the STOP FOR MAINTENANCE button. Select the fifth item on the following menu:

```
System Maintenance Menu
1   Start System
2   Install System Software
3   Run Diagnostics
4   Recover System
5   Enter Command Monitor
6   Select Keyboard Layout
```

1.6.3 Indigo Command Monitor (Command Summary)

COMMAND SUMMARY

COMMAND	WHAT IT DOES	EXAMPLE
auto	Boots default operating system (no arguments). This has the same effect as choosing Start System from the PROM Monitor initial menu.	**auto**
boot	Boots the named file with the given arguments.	**boot *[-f][-n] pathname***
date	Displays or sets the date and time.	**date *[mmddhhmm[ccyy|yy][.ss]]***
eaddr	Prints the Ethernet address of the built-in Ethernet controller.	**eaddr**
exit	Leaves Command Monitor and returns to the PROM menu.	**exit**
help	Prints a Command Monitor command summary.	**help *[command] ? [command]***
hinv	Prints an inventory of known hardware on the system. Some optional boards may not be known to the PROM monitor.	**hinv**
init	Partially restarts the Command Monitor, noting changed environment variables.	**init**
ls	List files on a specified device.	**ls*devicename***
off	Turns off power to the system.	**off**
pathname	Given a valid file pathname, the system attempts to find and execute any program found in that path.	***pathname***
printenv	Displays the current environment variables.	**printenv *[env_var_list]***
resetenv	Resets all environment variables to default.	**resetenv**
resetpw	Resets the PROM password to null (no password required).	**resetpw**
setenv	Sets environment variables. Using the -p flag makes the variable setting persistent, that is, the setting remains through reboot cycles.	**setenv *[-p] variablevalue***
single	Boots the system into single-user mode.	**single**
unsetenv	Un-sets an environment variable.	**unsetenvvariable**
version	Displays Command Monitor version.	**version**

Table 6-7 Command Monitor (Command Summary)

1.7 Host Computer Devices

1.7.1 Host Devices

HOST DEVICES

INDIGO2	HOST DEVICE	OCTANE	COMMENTS
`/dev/ttym1`	serial port 1	`/dev/ttym1`	used for InSite modem (with PPP)
`/dev/console`	serial port 1	`/dev/console`	if EPROM configured for serial console(d)
`/ec0`	ethernet	`/ef0`	kernel ethernet device
`/dev/plp`	printer	`/dev/plp`	Centronics parallel printer port
`/dev/audio`	audio in/out		system audio and AutoVoice record/play
`/dev/keybd`	keyboard	`/dev/keybd`	the PS/2 keyboard (type 3)
`/dev/mouse`	mouse	`/dev/mouse`	the PS/2 mouse

Table 6-8 Host Devices - Filesystem Names

1.7.2 Devices on High Speed Bus

These are the boards used for graphics and/or communications.

Note: Head Assignment

If the board controlling the primary monitor is removed, the secondary board and monitor become the primary head by default.

Because the boards are interchangeable, this feature is useful in determining whether one board is good or possibly defective. If one of the monitors is blank or faulty, you can use the **`/usr/gfx/gfxinfo`** command to see which boards the host recognizes; swap their locations.

COMMAND	GIO64 (INDIGO)	XIO (OCTANE)	COMMENTS
`/dev/gfx`	MG 1,1	SI with TRAM	Primary Head is controlled by first recognized gfx
`/dev/gfx`	MG 1,0	Solid Impact	1.) Secondary Head is controlled by next recognized gfx. 2.) On Octane, the SI board with a TRAM Module must be installed in the top graphic board location, for proper display performance.
`/vd0`	GIO64 BIT3	PCI Bit3	host side kernel device to SBC VMEbus

Table 6-9 High Speed Devices - Filesystem Names

1.7.3 SCSI Devices

SCSI DEVICES

INDIGO2	SCSI DEVICE	OCTANE	COMMENTS
/dev/dsk/dks1d1sZ	Primary system disk	/dev/dsk/dks0d1s*Z*	where *Z* is the partition number *Z* = 0,1,3,5,6,7
/dev/dsk/dks1d2sZ	Additional system disk	/dev/dsk/dks0d2sZ	
/dev/scsi/sc0d1l0	DASM (VDB or LCAM)	/dev/scsi/sc1d1l0	the /dev/dasm1 device gets linked to this
/dev/scsi/sc1d3l0	MaxOptics Image MOD	/dev/scsi/sc1d3l0	current image archive device
/dev/scsi/sc1d5l0	Pioneer MOD		an obsolete archive device
not applicable	Serial Expander	/dev/scsi/sc1d4l0 /dev/scsi/sc1d4l1	
/dev/scsi/sc1d6l0	CDROM	/dev/scsi/sc1d6l0	the LFC and CBT device

Table 6-10 SCSI Devices - Filesystem Names

You can use the **scsistat** command to obtain information about SCSI devices.

In a shell, enter the command and hit enter: **scsistat** ENTER

Example: Typical scsistat command for **Indigo only** output.

```
{ctuser@rhap13}[1] scsistat
scsistat: Must be super user to run.
{ctuser@rhap13}[2] su
Password:
{ctuser@rhap13}[1] scsistat
 Device 1 1  Disk      SEAGATE  ST15150N         FW Rev: 0023
 Device 1 3  Optical   Maxoptix T4-1300          FW Rev: 0.10
 Device 1 5  Optical   PIONEER  DE-C7101         FW Rev: 0500
 Device 1 6  CD-ROM    TOSHIBA  CD-ROM XM-3601TA FW Rev: 0725
```

The general form of the SGI SCSI devices output listing is:

disk partition as a filesystem = /dev/dsk/dks*X*d*YZ*

or

generalized SCSI device = /dev/scsi/sc*X*d*YZ*

where:

X is the **SCSI controller channel** (0 = SCSI bus0, 1= SCSI bus 1l)

Y is the **unit number** (OC disk is unit 1, MAX is unit 3 and CDROM is unit 6)

Z is the **partition ID** (filesystem s0, s1, s2,...), volume (vol), or other (l0)

1.8 Using a Parallel Printer On Indigo Only - For ASCII Text Files Only

This procedure can be used to print a hardcopy of the HHS data located in the file /usr/g/service/log/gencal.hhs_scan.report

As ROOT, from a shell window, enter the following commands in order exactly as shown below.

ADDING A TEXT ONLY PRINTER TO SOFTWARE

COMMANDS (ADD TEXT-ONLY PRINTER)	COMMENTS/EXPLANATION
`/usr/lib/lpshut`	make sure print spooler is down
`/usr/lib/lpadmin -ptemp1 -v/dev/plp -mdumb`	create dumb text printer "temp1"
`/usr/lib/lpadmin -dtemp1`	make printer "temp1" the default
`enable temp1`	enable the printer "temp1"
`/usr/lib/accept temp1`	enable print jobs to "temp1"
`/usr/lib/lpsched`	start the print job scheduler

Table 6-11 Commands for Adding a Text Only Printer to the Software

Then, to print any text file to the BubbleJet, simply enter (where ***<filename>*** is a local text file or a complete path to a text file)

PRINTING A TEXT FILE

COMMAND (PRINT TEXT FILE)	COMMENTS/EXPLANATION
lp *<filename>*	print ***filename*** to printer

Table 6-12 Command for Printing a Text File

A header page with the username and the filename will be printed at the front of every print job that is sent to the line printer. The printer named "temp1" will persist as the default printer until you do a LFC or remove it with the following command (as ROOT):

COMMAND (REMOVE PRINTER)	COMMENTS/EXPLANATION
`/usr/lib/lpadmin -xtemp1`	removes the printer "temp1"

Table 6-13 Command for Removing the Printer from Software

1.9 IRIX < Man > Pages - Help for Commands

The **man** pages are a group of computer system software and hardware information arranged or accessed by topic.

On the SGI computer, man page information may be obtained in one of two ways.

- Open a Unix Shell and enter:

  su - ENTER

  <root password> ENTER

  man *<command>* ENTER

Comment: *<command> is the software command or topic of interest. When the man command is used, help information will be displayed in the Unix shell about that command.*

  man -k *<keyword>*

Comment: *<keyword> will be used to search for appropriate man page subjects. The Unix shell displays a list of appropriate topics.*

- Open a Unix shell

  su root ENTER

  <root password> ENTER

  xman ENTER

Comment: *"xman" will start a windowed, interactive interface for the man pages. Select the topic of interest from the window and the information will appear.*

Note: Use the "MAN" Pages — Use the information in the man pages with some caution! Not all commands or information contained in the man pages are applicable to the HiSpeed CT/i configuration.

Section 2.0 Applications and Features

2.1 Fast Recon using the CT/i (Octane) Computer

Fast Recon is an option if the CT/i scanner has a NexGen (Octane) computer. Fast Recon cannot be done by the Indigo2 computer. Fast Recon will also require an options MOD (MaxOptics) to be loaded to enable the feature. Once installed Fast Recon does not require any special keystrokes or button pushes; it will simply be in place and enabled at all times while reconstruction is active.

CT Fast Recon introduces a new reconstruction algorithm for Standard and Soft recons.

CT/i Fast Recon is made possible by:

- Post processing done by the host process Axial Receive rather than the ReconMain process on the SBC with its IG (Image Generator) board
- Post processing runs in parallel with other simultaneous IG operations
- ViewLoop optimizations for Standard and Soft Recon modes
- Modified recon kernel

CT Fast Recon can perform differently in different reconstruction modes. Different host CPU loading scenarios are possible as post processing now competes for CPU cycles with other traditional CT/i host based software processes like ExamRxDisplay and IOS. Process priorities have been designed to maintain the current times needed to display. Auto View may skip images to catch up to the current scan when the many ways the system uses to copy, send or modify images slows down display.

2.2 Camera (Systems with 4.1 and 5.3 Software, or Higher)

2.2.1 Camera Installation and Configuration Files

The system supports either DASM Laser or network DICOM Print type cameras. Configuring the System for camera and its parameters is done from the SERVICE DESKTOP, UTILITIES menu, and selecting CAMERA INSTALLATION.

Note: Camera Installation is no longer a part of RECONFIG.

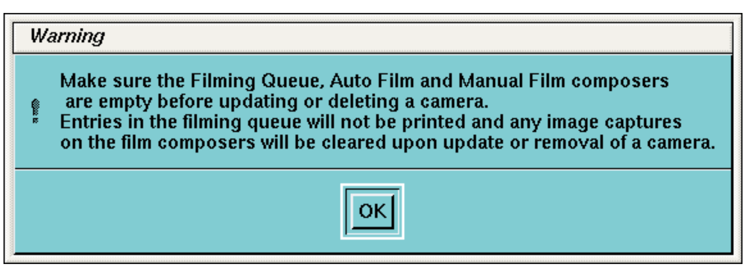


Figure 6-5 Camera Configuration Warning

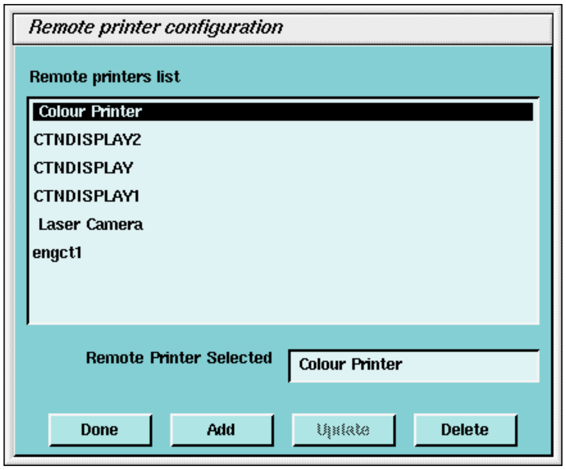


Figure 6-6 Remote Printer Configuration

2.2.2 DASM Laser Camera

A DASM Laser Camera is a camera connected to the CT system through a DASM (either Analog or Digital). The CT System connects to the DASM via the Host Computer SCSI Bus, and provides either Analog Video (Analog DASM) or Digital Video (Digital DASM) and control & command signals to the Laser Camera.

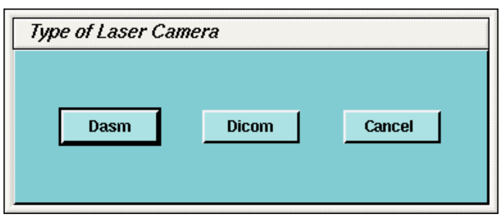


Figure 6-7 Type of Laser Camera (DASM)

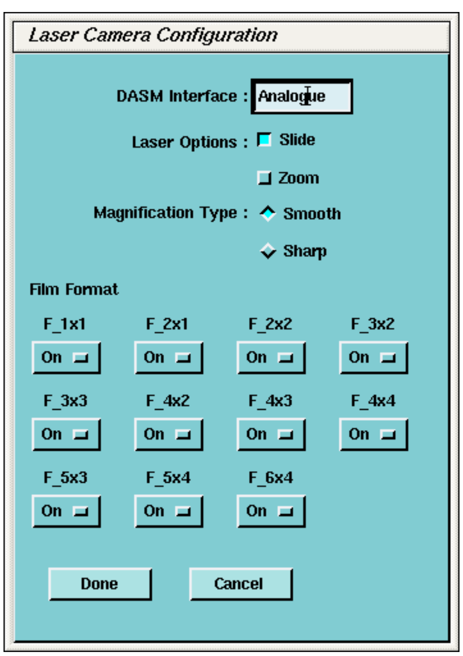


Figure 6-8 Laser Camera Configuration

Figure 6-8 is an example of the required configuration parameters for a DASM Laser Camera.

1.) The **DASM Interface** is selected automatically. It is a good idea to verify the pre-set information as camera models do change over time.

Note: The DASM Interface type is automatically sensed according to the type of DASM installed.

2.) Two **Options** are available with a Laser Camera, **Slides** and **Zoom**. Setting this option allows the option to be enabled or disabled at the application level. However, before selecting **Slides** or **Zoom**, be sure that the customer's camera supports these options.

3.) Camera manufacturers provide two **Film** resolution options for cameras. The **Smooth** resolution blurs the image, while the **Sharp** resolution makes the image "pixelly". To film good images and have them look like images filmed by other GE HiSpeed and HiLight systems, use the following camera settings:

Kodak:	Smooth
Dupont/Sterling:	Smooth
3M/Imation (Laser Camera:	Sharp
3M/Imation (Dry View):	Smooth
Agfa:	Smooth

If you hear from the site that the images on film are "too pixelly", chances ar that the film has been set to sharp; you need to set it to smooth. And vice-versa.

2.2.2.1 Filming Quality

All cameras on the market provide various adjustment "algorithms" so that the image quality on films match the quality as seen on display monitors. Your technologists and radiologists will not compromise the film quality. Since your camera service vendor will be at your site to setup the new CT/i port on your camera, it will be best for you to ensure that the new films from CT/i matches in quality with the films that your radiologists are used to reviewing.

Please make that a top priority.

2.2.2.2 Filming Error and Status logs

During Laser Camera Print filming, the system writes to two camera log files, `lclog` and `prslog`. When a print job starts, the Laser Camera status information is logged to `~ctuser/logfiles/lclog`. The print job information is logged to `~ctuser/logfiles/prslog`.

LCLOG:

Location on OC: `/usr/g/ctuser/logfiles/lclog`

Description - This logfile contains Laser Camera print filming sequence and Printer status information for the most recent print session job. Each time a new print job is performed, the status information for that latest job will overwrite the previous one.

Example: `lclog`

An example of a `lclog` output follows:

```
Successful Camera Initialization
User_Msg... CODE----> 301
#301fname = /usr/g/ctuser/film/img41a000QY
arg_copies = 1arg_format = 4x3_fidddasm952 interface was
loaded...Set_Vendor_Bits...LcSyscall: cmd 30
scsisleep duration=100000000nsLcgetResponse: ready 1
Set_12_Line_Border pass...
LcSyscall: cmd a4
LcgetResponse: ready 1
Clear_Alarm...LcSyscall: cmd 85
```

```
LcgetResponse: ready 1
Request_Status Called...LcSyscall: cmd 96
LcgetResponse: ready 1
LcSysrep: resp ``STA,1,RDY''
LcSysrep(RQS): status160->STA,1,RDY
LcSysrep: RDYLcSysrep():ALM 1,log_msg: code = 1 (logged_error = 0)
LcSyscall: cmd 82
LcgetResponse: ready 1
LcSysrep: resp ``PAS''
LcSysrep(ALI): status160->PAS
Allocate_Device OK
opening data file /usr/g/ctuser/film/img41a000QYSet_Greyscale...
LcSyscall: cmd a5
LcgetResponse: ready 1
      Start of Print Job
STATISTICS************START PRINTING FILM****************/usr/g/ctuser/
film/img41a000QYLcSyscall: cmd 90
LcgetResponse: ready 1
LcSysrep: resp ``PAS''
LcSysrep(MAT): status160->PAS
LcSyscall: cmd a3
LcgetResponse: ready 1
LcSysrep: resp ``PAS''
LcSysrep(LUT): status160->PAS
LcSyscall: cmd 9f
LcgetResponse: ready 1
LcSysrep: resp ``PAS''
LcSysrep(WIM): status160->PAS
Lc_clear_all:CMI...LcSyscall: cmd 86
LcgetResponse: ready 1
LcSysrep: resp ``PAS''
LcSysrep(CLR): status160->PAS
      Start of Image Acquisition Process
lc_load_and_acquire: img->image_sx=512lc_load_and_acquire: img-
>image_sy=512lc_load_and_acquire: img->image_psize=0.000000main : zoomd =
0, zoomh = 0
set_zoomd : ...set_zoomf...loading file /usr/g/ctuser/film/
img41a000QYxxL952_vdbSetFormat pass...format = 12 zoom = 0.000000
set_zoomd : ...set_zoomf...L952_vdbSetFormat leavingOLD SYTLE IMAGE
ACQUISITION, NO RING BUFFERINGentering rbL952_lcamStore: fname = /usr/g/
ctuser/film/img41a000QY, num_imgs=12, hdrlen =632entering
rbL952_lcamStore: image_sx = 512,image_sy =512, image_deep=
8rbL952_lcamStore: nbchuncks = loop_var = 16 nblocks= 512 image_size
=262144ACQUIRE IMAGE... Image_ID = 1
LcSyscall: cmd 84
LcgetResponse: ready 0
LcgetResponse: ready 1
LcSysrep: resp ``PAS''
LcSysrep(AQU): status160->PAS
```

```
ACQUIRE IMAGE... Image_ID = 2
```

Comment: *The above 6 steps are repeated for each succeeding Image Acquisition. Image acquisition completed, begin Printing*

```
LcSysrep(AQU): status160->PAS
L952_vdbPrint: format = 12 print_copies = 1Define_Zone: format =
12set_zoomd : ...set_zoomf...xxDefine_Zone: format = 12 nb_zone = 4
nb_image_line 3xxDefine_Zone: set_zoomd = 0.000000 , set_zoomf =
0.000000scan_ssparam = dd
xxDefine_Zone scanned ssparam = ddxxDefine_Zone: images_id[0] =
1xxDefine_Zone: images_id[1] = 2xxDefine_Zone: images_id[2] =
3Request_Status Called...LcSyscall: cmd 96
LcgetResponse: ready 1
LcSysrep: resp ``STA,1,RDY''
LcSysrep(RQS): status160->STA,1,RDY
LcSysrep: RDYLcSysrep():ALM 1,log_msg: code = 1 (logged_error = 0)
LcSyscall: cmd 8b
LcgetResponse: ready 1
LcSysrep: resp ``PAS''
LcSysrep(DZO): status160->PAS
xxDefine_Zone scanned ssparam = ddxxDefine_Zone: images_id[0] =
5xxDefine_Zone: images_id[1] = 6xxDefine_Zone: images_id[2] =
7Request_Status Called...LcSyscall: cmd 96
LcgetResponse: ready 1
LcSysrep: resp ``STA,1,RDY''
LcSysrep(RQS): status160->STA,1,RDY
LcSysrep: RDYLcSysrep():ALM 1,log_msg: code = 1 (logged_error = 0)
LcSyscall: cmd 8b
LcgetResponse: ready 1
LcSysrep: resp ``PAS''
LcSysrep(DZO): status160->PAS
xxDefine_Zone scanned ssparam = ddxxDefine_Zone: images_id[0] =
9xxDefine_Zone: images_id[1] = 10xxDefine_Zone: images_id[2] =
11Request_Status Called...LcSyscall: cmd 96
LcgetResponse: ready 1
LcSysrep: resp ``STA,1,RDY''
LcSysrep(RQS): status160->STA,1,RDY
LcSysrep: RDYLcSysrep():ALM 1,log_msg: code = 1 (logged_error = 0)
LcSyscall: cmd 8b
LcgetResponse: ready 1
LcSysrep: resp ``PAS''
LcSysrep(DZO): status160->PAS
xxDefine_Zone scanned ssparam = ddxxDefine_Zone: images_id[0] =
13xxDefine_Zone: images_id[1] = 14xxDefine_Zone: images_id[2] =
15Request_Status Called...LcSyscall: cmd 96
LcgetResponse: ready 1
LcSysrep: resp ``STA,1,RDY''
LcSysrep(RQS): status160->STA,1,RDY
LcSysrep: RDYLcSysrep():ALM 1,log_msg: code = 1 (logged_error = 0)
LcSyscall: cmd 8b
```

```
LcgetResponse: ready 1
LcSysrep: resp ``PAS''
LcSysrep(DZO): status160->PAS
Lc_set_up_to_print call...nbcopies= 1LcSyscall: cmd 9a
LcgetResponse: ready 1
LcSysrep: resp ``STC''
LcSysrep(STP): status160->STC
LcSysrep(STP): STC OK...
EXPOSE...
LcSyscall: cmd 8d
LcgetResponse: ready 1
LcSysrep: resp ``PTC''
LcSysrep(EXP): status160->PTC
camscan: (17:75 to 8b)  0x72 EXP 0xd  0x6 EOE 0xd & 0xa  0xb  0xc PTC 0xd
& 0xa  0xb  0xc  0xa
camscan: dcr's 1
LcSysrep(EXP): DCR...print_copies = 0User_Msg... CODE----> 353
#353
```

Comment: *End of Print Job*

```
STATISTICS:******END OF THIS JOB /usr/g/ctuser/film/img41a000QY Ready To
Print new*******Release_Device...LcSyscall: cmd 95
LcgetResponse: ready 1
User_Msg... CODE----> 350
#350
```

PRSLOG:

Location - OC: `/usr/g/ctuser/logfiles/prslog`

Description - This is a running history log of print server initialization and shutdowns, and print jobs that are started and completed.

Example: `prslog` **Example prslog output:**

Comment: *Successful print server initialization:*

```
MESSAGE from Process 1639>> Tue Aug 18 13:10:38 1998  [Server]>
initialization in progress for port PRSserver
MESSAGE from Process 1639>> Tue Aug 18 13:10:38 1998  [Server]>
...initialization completed for port PRSserver
MESSAGE from Process 1674>> Tue Aug 18 13:11:02 1998  [PRSserver]> Hello,
I'm the print server, still alive on host engbay13
```

Comment: *Successful print jobs running:*

```
MESSAGE from Process 1799>> Tue Aug 18 13:15:56 1998  [PRSserver]> Print
job started
MESSAGE from Process 1799>> Tue Aug 18 13:16:21 1998  Printed Ex: 1472 Se:
103 Im: 1
MESSAGE from Process 1799>> Tue Aug 18 13:16:21 1998  [PRSserver]>
Completed print job: Ex: 1472 Se: 103 Im: 1
MESSAGE from Process 1817>> Tue Aug 18 13:16:50 1998  [PRSserver]> Print
job started
```

```
MESSAGE from Process 1817>> Tue Aug 18 13:17:16 1998  Printed Ex: 1472 Se: 103 Im: 25
MESSAGE from Process 1817>> Tue Aug 18 13:17:16 1998  [PRSserver]> Completed print job: Ex: 1472 Se: 103 Im: 25
MESSAGE from Process 1825>> Tue Aug 18 13:17:41 1998 [PRSserver]> Print job started
MESSAGE from Process 1825>> Tue Aug 18 13:18:06 1998 Printed Ex: 1472 Se: 103 Im: 49
MESSAGE from Process 1825>> Tue Aug 18 13:18:06 1998 [PRSserver]> Completed print job: Ex: 1472 Se: 103 Im: 49
MESSAGE from Process 1831>> Tue Aug 18 13:18:33 1998 [PRSserver]> Print job started
MESSAGE from Process 1831>> Tue Aug 18 13:18:59 1998 Printed Ex: 1472 Se: 103 Im: 73
MESSAGE from Process 1831>> Tue Aug 18 13:18:59 1998 [PRSserver]> Completed print job: Ex: 1472 Se: 103 Im: 73
```

Comment: *Print server shutdown from Applications being brought down:*

```
MESSAGE from Process 1639>> Tue Aug 18 14:48:34 1998 [Server]> Caught signal : 2
MESSAGE from Process 1674>> Tue Aug 18 14:48:35 1998 [Server]> Caught signal : 2.
MESSAGE from Process 1639>> Tue Aug 18 14:48:41 1998 [Server]> terminated
```

Comment: *Successful print server initialization:*

```
MESSAGE from Process 1598>> Tue Aug 18 14:53:43 1998 [Server]> initialization in progress for port PRSserver
MESSAGE from Process 1598>> Tue Aug 18 14:53:43 1998 [Server]> ...initialization completed for port PRSserver
MESSAGE from Process 1636>> Tue Aug 18 14:54:09 1998 [PRSserver]> Hello, I'm the print server, still alive on host engbay13
MESSAGE from Process 1902>> Tue Aug 18 15:15:40 1998 [PRSserver]> Print job started
MESSAGE from Process 1902>> Tue Aug 18 15:16:06 1998 Printed Ex: 1476 Se: 2 Im: 1
MESSAGE from Process 1902>> Tue Aug 18 15:16:06 1998 [PRSserver]> Completed print job: Ex: 1476 Se: 2 Im: 1
MESSAGE from Process 1926>> Tue Aug 18 15:17:10 1998 [PRSserver]> Print job started
MESSAGE from Process 1926>> Tue Aug 18 15:17:35 1998 Printed Ex: 1476 Se: 2 Im: 16
MESSAGE from Process 1926>> Tue Aug 18 15:17:35 1998 [PRSserver]> Completed print job: Ex: 1476 Se: 2 Im: 16
```

2.2.3 DICOM Print Camera

A DICOM Print Camera is a network camera that has a hostname and IP Address connected on the Hospital Network (Ethernet Connection) from the CT System. The CT System uses TCP/IP network protocol to communicate and send DICOM Images in packets to the Camera for filming. Refer to DICOM Terms on page 260 for a glossary of DICOM terms and definitions associated with DICOM Print. Figure 6-9 below is an example of the required configuration parameters for a DICOM Print Camera:

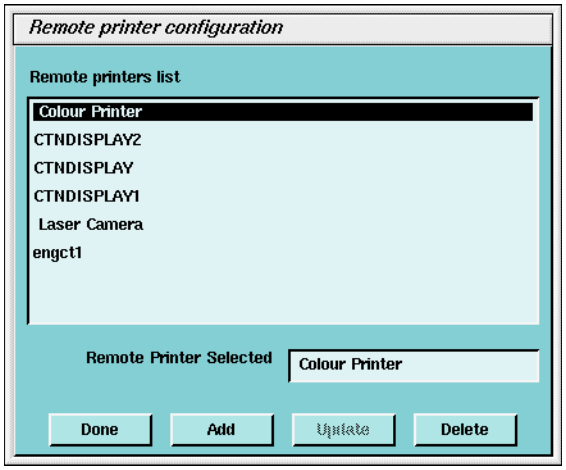


Figure 6-9 Remote Printer Configuration

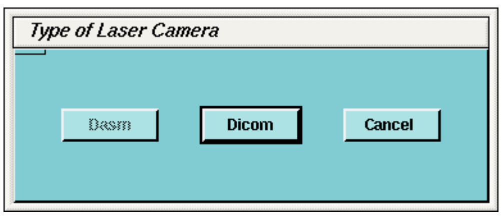


Figure 6-10 Type of Laser Camera (No DASM)

1.) Set up the Network Parameters

Note: To determine the correct DICOM Camera Network parameters (IP Address, Hostname, AE Title, Port Number, and Comments) contact the Hospital's Network Administrator.

- **Device Name** - Name of printer
- **Host Name** - DICOM Print Server host name as defined by the network.
- **IP Address** - DICOM Print Server IP Address as defined by the network.
- **Application Title** - DICOM Print Server Application Entity Title as defined by the server.
- **TCP/IP Listen Port** - DICOM Print Server TCP/IP Listen Port as defined by the server.
- **Comments** - (Optional) Comments to be used by the DICOM Print Server.

2.) Medium Type selects the type of film to be used, either Blue Film or Clear Film.

3.) The Destination parameter selects the final location for the film output, either Magazine or Processor.

4.) Orientation selects the film orientation; currently only the Portrait option is supported.

5.) The Magnification Type parameter selects the algorithm used to interpolate pixels to provide the necessary film resolution. This parameter should be set in conjunction with the camera manufacturer to make the best possible image. The settings are:

- None - No interpolation. This option is not supported by all camera vendors.
- Replicate - Adjacent pixels are interpolated, which results in images describes as "pixelly". This algorithm is not usually preferred.
- Bilinear - A first order interpolation of pixels is used which results in images described as blurred. This algorithm is not usually preferred.
- Cubic - A third order interpolation is used with a large number of possible formulations. Camera manufacturers define parameters, called smoothing type, to set coefficients used in the algorithm. The implementation of these coefficients is camera manufacturer

dependent.

6.) The valid **Film Formats** are determined by the camera manufacturer (for example, IMATION does not support 4x6, 2x4, or 1x2; AGFA does not support 2x4). Also note that the DICOM Print convention is to designate film formats by column x row (e.g. 12-on-1 film is 3x4).

The Network Parameters entered in the Camera Installation GUI (including Camera Hostname, IP Address, AE Title, Port Number, and Comment) are written to **`/usr/g/ctuser/Prefs/SdCPHosts`** file on the OC.

The settings information entered in the Camera Installation GUI is written to **`/usr/g/ctuser/app-defaults/devices/camera.dev`** file on the OC.

A second screen, Figure 6-11, with image quality and time-out information parameters for filming sessions, comes up after selecting ADVANCED. The Figure 6-11 below is an example of the required image quality and time-out parameters for a DICOM Print Camera:

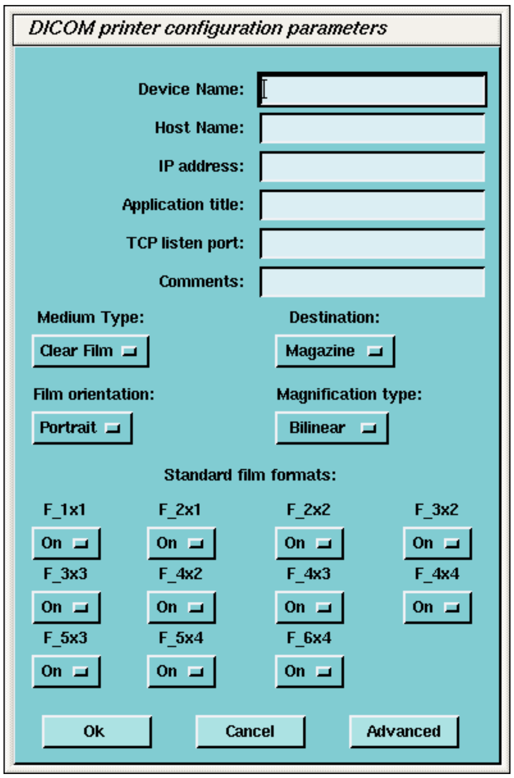


Figure 6-11 DICOM Print Camera Image Quality & Time-out Settings

The image quality parameters are saved on the OC in
`/usr/g/ctuser/app-defaults/devices.camera.dev` file.
The time-out parameters are saved on the OC in
`/usr/g/ctuser/app-defaults/print/dprint.cfg` file.

Note: To determine the correct camera settings, contact the Camera Service representative and review the Camera Manufacturer's DICOM Conformance Statement. The detailed DICOM Conformance Statement for HiSpeed CT/i is available as Direction 2162114-100. You may need to refer to a copy of this document as you are working with the camera manufacturer's representative, to correctly set up the DICOM Print Camera settings.

2.2.3.1 Sample camera.dev File Contents

EXAMPLE FOR AN AGFA DICOM PRINT CAMERA

Enter the following:

1.) {ctuser@bayXX}[5] **`cd /usr/g/ctuser/app-defaults/devices`**
2.) {ctuser@bayXX}[6] **`cat camera.dev`**

`camera.dev` **Contents listing**	**Description**
`set dName {Dicom Camera}`	Sets the name that appears in manual composer and scanRx autofilm setup
`set dType digital`	Refers to 1 of {postscript, analogue, or digital}. For DICOM the dType is set to digital.
`set dQueueType DICM`	Refers to the Job type; can be {LP=postscript, LC=laser camera, or DICM} for DICOM print cameras.
`set ctype {Imation Print Server}`	DICOM camera type selected during Camera Installation
`set dQueueName dicom`	Sets the name that appears in the Filming Queue
`set defaultFormat  4x3_fid`	Default format for manual composer selected during Camera Installation
`set dHostName agfacamera`	Hostname of the print server entered during Camera Installation
`set dAppTitle DRYSTAR`	AE title of the print server entered during Camera Installation
`set medType BLUE`	Medium type selected during Camera Installation - can be one of BLUE FILM, CLEAR FILM, or PAPER - this element is sent during NCREATE of the Film Session.
`set destination MAGAZINE`	The destination for printed film selected in during Camera Installation. Can be one of MAGAZINE, PROCESSOR - this element is sent during NCREATE of the Film Session.
`set filmOrientation PORTRAIT`	The Orientation of image boxes on film selected during Camera Installation can be one of LANDSCAPE or PORTRAIT - this element is sent during NCREATE of the Film Box. Note: Only Portrait is supported in 1st release CT/i.
`set magType CUBIC`	The magnification of film selected during Camera Installation. Can be one of REPLICATE, BILINEAR, CUBIC, or NONE - this element is sent during NCREATE of the Film Box.

Table 6-14 camera.dev contents listing

`camera.dev` **Contents listing**	**Description**

`set minDensity 5`
`set maxDensity 300`
`set borderDensity BLACK`
`set emptyDensity BLACK`
`set smoothType 140`
`set configuration`
`PERCEPTION_LUT=200`
	These elements are sent during the NCREATE of the Film Box and will set the following values:
	• sets minimum Optical Density - film and camera type dependent.
	• sets Maximum Optical Density - film and camera type dependent.
	• sets border density
	• sets empty image density
	• sets the smoothing type when magType is set to CUBIC
	• sets configuration info. This value differs for all camera vendors who typically define the LUT for contrast.

`pformat 1x1_fid`
`pformat 2x1_fid`
`pformat 2x2_fid`
`pformat 3x2_fid`
`pformat 3x3_fid`
`pformat 4x3_fid`
`pformat 5x3_fid`
`pformat 4x4_fid`
`pformat 5x4_fid`
	Manual film composer and auto film formats
	Note: DICOM defines film format as column vs. row, as opposed to GE's Laser film format definition of row vs. column.

Table 6-14 camera.dev contents listing (Continued)

2.2.3.2 Sample SdCPHosts File Contents (DICOM Print only):

Enter the following:

1.) {ctuser@bayXX}[2] **`cd /usr/g/ctuser/Prefs`** ENTER
2.) {ctuser@bayXX}[3] **`cat SdCPHosts`** ENTER

`SdCPHosts` **Contents Listing Example**	**Description**
`3.7.52.164`	The IP address of the camera, entered during Camera Installation
`camera`	The hostname of the camera, entered during Camera Installation
`PRINTSCP`	The AE (Applications Entity) Title, entered during Camera Installation
`2106`	The TCP Listen Port number, entered during Camera Installation
`ctn display`	A comment entered in the network comment field of Camera Installation.

Table 6-15 SdCPHosts contents listing example

2.2.3.3 Save System State

Once the camera is installed, the settings now stored in the configuration files: camera.dev, sdCPHosts, and dprint.cfg need to be saved. Save the parameters to your System State MOD. Run |System State| and select |Camera Preferences|, |Save|.

2.2.3.4 Filming Image Quality Setup

Note: It is very important that the camera limits are clearly understood from the camera manufactures Conformance Statement, and work closely with the Camera Field Engineer when setting up min. and max. density and configuration.

The parameters that directly effect Filming Image Quality in the camera.dev file are:

- set minDensity
- set maxDensity
- set smoothType - Used only when Mag type is set to Cubic.
- set configuration - This value sets the min. & max density curve range. Camera manufacturer dependent.

Density Setup Tips with Blue Film Type

The starting min. and max density settings vary with camera and film type, and configuration settings.

Note: If the configuration is set to 200, and maxDensity 300, films will be quite dark. Bottom line, the higher the density and config LUT, the darker the film.

See Table 6-16, "Density Values," on page 246, for some suggested settings for the AGFA camera. **For other camera models, refer to the camera manufactures conformance statement and consult with the camera FE.**

Camera Type	Media Type	Film Type	Suggested Starting	
			Minimum Density	Maximum Density
AGFA Drystar 2000	Blue Film	TS Blue Base Low Speed High Density	17	185
AGFA Drystar 2000	Blue Film	TS Blue Base Fast Speed Normal Density	18	229
AGFA Drystar 2000	Blue Film	DT Blue Base Normal Speed High Density	24	300
AGFA Drystar 2000	Clear Film	TS Clear Base Low Speed High Density	5	173
AGFA Drystar 2000	Clear Film	TS Clear Base Fast Speed Normal Density	6	217
AGFA Drystar 3000	Blue Film	DT Blue Base Normal Density	23	300
AGFA Drystar 3000	Clear Film	DT Clear Base Normal Density	6	300

Table 6-16 Density Values

RECOMMENDATIONS

1.) If the Hospital already has the camera in-use in laser mode, make sure you use these values as the start-point. You may want to take a number of films before you change out the hardware and use them for comparison afterwards.
2.) Setup the DICOM Print Camera, and use the initial starting point. Setup to look as good as the camera FE and GE CT FE can make it.
3.) Assume that before the DICOM Print install is complete the films have been approved by the appropriate Hospital Staff. This means some time (up to 4 hours) must be allocated for the Camera FE, CT FE and site to work together. If it is possible, the camera manufacturer can create a film with multiple contrasts for the Doctors to pick from.

2.2.3.5 Troubleshooting DICOM Print Camera Problems

LOG OF ERROR AND FILMING STATUS

During DICOM Print filming, the system writes to two camera logfiles, `dcplog` and `prslog`. When a print job starts, the dicom information is logged to `~ctuser/logfiles/dcplog`. The print job information is logged to `~ctuser/logfiles/prslog`. The called AE `title/host/IP/port number` is taken from `~ctuser/Prefs/SdCPHosts` file.

---

dcplog

This logfile contains dicom print filming sequence and Printer status information for the most recent print session job. Each time a new print job is performed, the status information for that latest job will overwrite the previous one.

1.) Printer Status Area in the dcplog report

The Printer Status area in the log report will either be NORMAL, WARNING, or FAILURE. In the event of a WARNING or FAILURE, the Status Info field attempted to identify the root cause.

NORMAL - print job was successful, no problems.

WARNING - one of three conditions can happen:

a.) The job aborts and the status info field indicates SUPPLY FULL, RECEIVER FULL, or FILM JAM. (See part 3 below for FILM JAM example.
b.) The job continues and Warning is posted to the operator if Status Info field reports SUPPLY LOW.
c.) The job continues and a Warning is *not posted to the operator, but the message is put in dcplog file.*

Note: What gets reported is dependent upon the camera type and the camera server's ability to report it!

FAILURE - the print job has aborted, see Status Info field for more information.

2.) dcplog Sample Output of a Successful 1on1 Film Job.

Note: The output is broken up into sections with key film session actions in bold, and an explanation indented and in italics:

Example: dcplog

The dcplog output includes the use of the following acronyms

- **SCP** = Service Class Provider: The camera, a receiver of images.
- **SCU** = Service Class User: The OC scanner, has ability to send images

calling AE title

`dcm_bind: AETitle = engbay26_DCP`

called AE title - hostname - IP address port number of printer `map_app_title`: title IMN host engctn1 ip-addr 3.7.52.164 port 2104

Print SCU (on the OC) requests an association with print SCP (print server at camera) using the IP address, port number and AE title. The SCU proposes abstract syntaxes (in this case print service class) along with transfer syntaxes used for each syntax and PDU transfer rate.

EstablishAssoc: DCM_OPEN_REQ Action success

The Print SCP responds with an association acceptance. If the association has been accepted, the Dicom parameters (host, IP, AE, port correctly configured):

EstablishAssoc: OPEN_CONF received

The SCU sends an NGET request to the SCP for printer status:

Starting the print session

The SCP returns an NGET response status and printer status to the SCU:

Note:

- IF the SCP returns a `NORMAL` status to the SCU, the job continues.
- If the SCP returns an `ERROR` status to the SCU, the print job will fail.
- If the SCP returns a `WARNING` status, the job may fail or continue depending on status info. See Table 6-17 on page 248 for list of supported status.

```
NgetService: Event Received: DCM_NGET_END
NgetService: Event Received: DCM_DATA
```

PRINTER STATUS	
SOP uid	
Instance uid	
Printer status	NORMAL
status info	
printer_name	advt
manufacturer	AGFA
model	ADVT
device serial number	123456
software version	Version 2.0
AETitle	IMN

Table 6-17 Printer Events

Comment: *The SCU sends an NCREATE request to the SCP to create the film session. The Film session presentation consists of copies, priority of job, medium type, and film destination:*

```
NcreateService: NCREATE BEG Action Success
NcreateService: DCM DATA Action Success
NcreateService: Waiting for Event
NcreateService: Event received: DCM_RETURN_BUFF
NcreateService: Waiting for Event
```

Comment: *The SCP returns Ncreate RSP status to SCU along with instance uid for film session:*

```
NcreateService: Event received: DCM_NCREATE_END
NcreateService: Status is : 0
film session instance uid 1.3.51.1
film session instance uid 1.3.51.1
```

```
filmbox ref sop uid 1.3.51.1
```

Comment: *The SCU sends NCreate RQ to the SCP to create the film box. The presentation includes film format, orientation, magnification, film size:*

```
NcreateService: NCREATE BEG Action Success
NcreateService: DCM DATA Action Success
NcreateService: Waiting for Event
```

Comment: *The SCP returns NCREATE RSP status to the SCU along with referenced sop instance uid for film box and referenced SOP instance uids for each image box:*

```
NcreateService: Event received: DCM_RETURN_BUFF
NcreateService: Waiting for Event
NcreateService: Event received: DCM_NCREATE_END
NcreateService: Waiting for Event
NcreateService: Event received: DCM_DATA
filmbox instance uid 1.3.51.1.1
```

Comment: *SCU sends NSET RQ to the SCP to set the image box. The presentation includes instance uid, image position on the film, number of Rows, Columns, Bits, and image pixel data:*

```
NsetService: Event Received: DCM_RETURN_BUFF
```

Comment: *SCP returns NSET RSP status to the SCU along with affected sop instance uid for image box:*

```
NsetService: Event Received: DCM_NSET_END
Image Attributes set
```

Comment: *The SCU and SCP repeat the NSET RQ and NSET RSP for the image boxes until all images have been sent to the SCP. When all images have been sent, the SCU, it sends NACTION RQ to the SCP to print the film box with instance uid generated during the NCREATE. The SCP returns NACTION RSP to print the film:*

```
NactionService: Event received: DCM_NACTION_END
NactionService: Event received: DCM_DATA
parse_data_set returned status 0x0
Film Box sent to printer N - Action
```

Comment: *The SCU sends NDELETE RQ to the SCP to delete the film box with instance uid generated during the NCREATE. SCP returns NDELETE RSP to delete the film box and returns the sop Instance of the film job:*

```
NdeleteService: Event Received : DCM_NDELETE_END
Film box instance deleted
ref SOP C uid 1.2.840.10008.5.1.1.14
ref SOP I uid 1.3.51.1.1.1.1
Print Session successfully completed
```

Comment: *The SCU sends RELEASE RQ to the SCP to release the association:*

```
req close assoc
CloseAssoc: DCM_CLOSE_REQ Action Success
```

Comment: *The SCP returns RELEASE RP to release association:*

```
close accepted
```

Dcplog example of a print job leading up to a Film Jam:

```
{ctuser@engbayXX}[17] cd /usr/g/ctusr/logfiles
{ctuser@engbayXX}[18] more dcplog
_[40;1H_[K# DICOM print_scu pid: 5463
```

```
print_scu -aIMN -hcamera -c1 -f1x1_fid -p/usr/g/ctuser/film/
img21a0017f -d/usr/g/ctuser/app-defaults/devices/camera.dev
dcm_bind: AETitle = engbay26_DCP
map_app_title: title IMN host camera ip-addr 3.7.52.164 port 2104
EstablishAssoc: DCM_OPEN_REQ Action success
EstablishAssoc: OPEN_CONF received
Starting the print session
NgetService: Event Received : DCM_NGET_END
NgetService: Event Received : DCM_DATA
```

PRINTER STATUS	
SOP uid	
Instance uid	
Printer status	WARNING ¨
status info	FILM JAM ¨
printer_name	advt
manufacturer	AGFA
model	ADVT
device serial number	123456
software version	Version 2.0
Warning	Media jam. Failed during the print session, status -1. Job stopped here.
CloseAssoc	DCM_CLOSE_REQ Action Success

Table 6-18 Printer FILM JAM

prslog

LOCATION	DESCRIPTION
OC: `/usr/g/ctuser/logfiles/prslog`	This is a running history log of print server initialization and shutdowns, and print jobs are started and completed.
Example prslog output	
Successful print server initialization:	
MESSAGE from Process 1639>> Tue Aug 18 13:10:38 1998	[Server]> initialization in progress for port PRSserver
MESSAGE from Process 1639>> Tue Aug 18 13:10:38 1998	[Server]> ...initialization completed for port PRSserver
MESSAGE from Process 1674>> Tue Aug 18 13:11:02 1998	[PRSserver]> Hello, I'm the print server, still alive on host engbay13
Successful print jobs running:	
MESSAGE from Process 1799>> Tue Aug 18 13:15:56 1998	[PRSserver]> Print job started
MESSAGE from Process 1799>> Tue Aug 18 13:16:21 1998	Printed Ex: 1472 Se: 103 Im: 1
MESSAGE from Process 1799>> Tue Aug 18 13:16:21 1998	[PRSserver]> Completed print job: Ex: 1472 Se: 103 Im: 1
MESSAGE from Process 1817>> Tue Aug 18 13:16:50 1998	[PRSserver]> Print job started
MESSAGE from Process 1817>> Tue Aug 18 13:17:16 1998	Printed Ex: 1472 Se: 103 Im: 25
MESSAGE from Process 1817>> Tue Aug 18 13:17:16 1998	[PRSserver]> Completed print job: Ex: 1472 Se: 103 Im: 25
MESSAGE from Process 1825>> Tue Aug 18 13:17:41 1998	[PRSserver]> Print job started
MESSAGE from Process 1825>> Tue Aug 18 13:18:06 1998	Printed Ex: 1472 Se: 103 Im: 49
MESSAGE from Process 1825>> Tue Aug 18 13:18:06 1998	[PRSserver]> Completed print job: Ex: 1472 Se: 103 Im: 49
MESSAGE from Process 1831>> Tue Aug 18 13:18:33 1998	[PRSserver]> Print job started
MESSAGE from Process 1831>> Tue Aug 18 13:18:59 1998	Printed Ex: 1472 Se: 103 Im: 73
MESSAGE from Process 1831>> Tue Aug 18 13:18:59 1998	[PRSserver]> Completed print job: Ex: 1472 Se: 103 Im: 73
Print server shutdown from Applications being brought down:	
MESSAGE from Process 1639>> Tue Aug 18 14:48:34 1998	[Server]> Caught signal : 2

Table 6-19 prslog Output

LOCATION	DESCRIPTION
MESSAGE from Process 1674>> Tue Aug 18 14:48:35 1998	[Server]> Caught signal : 2.
MESSAGE from Process 1639>> Tue Aug 18 14:48:41 1998	[Server]> terminated
Successful print server initialization:	
MESSAGE from Process 1598>> Tue Aug 18 14:53:43 1998	[Server]> initialization in progress for port PRSserver
MESSAGE from Process 1598>> Tue Aug 18 14:53:43 1998	[Server]> ...initialization completed for port PRSserver
MESSAGE from Process 1636>> Tue Aug 18 14:54:09 1998	[PRSserver]> Hello, I'm the print server, still alive on host engbay13
MESSAGE from Process 1902>> Tue Aug 18 15:15:40 1998	[PRSserver]> Print job started
MESSAGE from Process 1902>> Tue Aug 18 15:16:06 1998	Printed Ex: 1476 Se: 2 Im: 1
MESSAGE from Process 1902>> Tue Aug 18 15:16:06 1998	[PRSserver]> Completed print job: Ex: 1476 Se: 2 Im: 1
MESSAGE from Process 1926>> Tue Aug 18 15:17:10 1998	[PRSserver]> Print job started
MESSAGE from Process 1926>> Tue Aug 18 15:17:35 1998	Printed Ex: 1476 Se: 2 Im: 16
MESSAGE from Process 1926>> Tue Aug 18 15:17:35 1998	[PRSserver]> Completed print job: Ex: 1476 Se: 2 Im: 16

Table 6-19 prslog Output (Continued)

3.) A known error that's reported from Nget, with Imation Cameras, that should not be troubleshot follow:

The prslog reports:

`MESSAGE from Process 2059 >> Wed Aug 19 10:03:48 1998 [PRSserver]> Print job started`

`Message from Process 2060 >> NgetService: N-GET response received with failure/warning Status`

`Message from Process 2060 >> AETitle: IMN_PrintServer`

`Message from Process 2060 >> Print Session successfully completed`

The dcplog reports:

`NgetService: N-GET response received with failure/warning Status ¨the known error.`

PRINTER STATUS	
SOP uid	:
Instance uid	
Printer status	NORMAL
status info	

Table 6-20 Imation Print Report

PRINTER STATUS	
printer_name	IMN_LaserImager
manufacturer	Imation
model	M8700
device serial number	
software version	1.5b4
AETitle	IMN_PrintServer

Table 6-20 Imation Print Report

What Imation supports:

Imation supports the following six elements/attributes:

```
>  (0x21100010, CS, ”NORMAL”)  # Printer Status  OK
>  (0x21100020, CS, ””)  # Printer Status Info  OK
>  (0x21100030, LO, ”IMN_LaserImager”)  # Printer Name  OK
>  (0x00080070, LO, ”Imation”)  # Manufacturer  OK
>  (0x00081090, LO, ”M8700”)  # Manufacturer’s Model Name  OK
>  (0x00181020, LO, ”1.5b4”)  # Software Versions  OK
```

The Bug:

Nget is requesting status from these three additional elements that are not supported:

```
>  (0x00181000, LO, ””)  # Device Serial Number
>  (0x00181200, DA, ””)  # Date of Last Calibration
>  (0x00181201, TM, ””)  # Time of Last Calibration
```

The Fix:

Instruct Camera FE to disable the above three elements that are not supported.

4.) Communication and Network Error Troubleshooting

The most common types of network errors that can occur with DICOM Print are a

- DCM Network error
- DCM Protocol error.

DCM Network errors:

Successful network communications to the camera is dependent upon having the correct IP Address and Port Number configured. Any errors associated with the network will be logged as a “DCM Network Error” in the “type” field in the dcplog report as shown in the example below. Use ping and snoop to discover the root cause, covered in the troubleshooting steps below.

Example: dcplog w/DCM Network Error

Example of dcplog with a DCM Network Error:

```
{ctuser@engbayXX}[3] cd /usr/g/ctusr/logfiles
{ctuser@engbayXX}[4] more dcplog
_[40;1H_[K# DICOM print_scu pid: 5498
print_scu -aIMN -hcamera -c1 -f1x1_fid -p/usr/g/ctuser/film/
img22a0017f -d/usr/g/ctuser/app-defaults/devices/camera.dev
dcm_bind: AETitle = engbay26_DCP
map_app_title: title IMN host camera ip-addr 3.7.52.164 port 2104
EstablishAssoc: DCM_OPEN_REQ Action success
```

Errors logged beyond this point of failure may be a result of this Error:

DCM kernel lower level error:

```
type = 508 -- DCM network error    ¨ ERROR
code = 114 -- lost transport connection
ul_code = 52, reason = 0, source = 0, reject = 0
filename = kernel/D_assoc.c  line = 3051
Failed to contact printer, status 114
```

Troubleshooting a DCM Network Error:

1.) Verify correct IP Address and Port Number are correct in the Install Camera GUI.

Note: If the IP Address and Port Number are correct, the remote application (camera server) may not be running.

2.) Verify Applications restarted after running Install Camera from Service Desktop Utilities.

3.) Verify on the OC in `/usr/g/ctuser/SdCPHosts` the IP Address and Port Number are correct.

Enter the following:

```
{ctuser@bayXX}[2] cd  /usr/g/ctuser/Prefs
{ctuser@bayXX}[3] cat SdCPHosts
3.7.52.164 camera IMN 106 ctn display
```

4.) Ping to the camera's IP address, and check for packet loss. A successful ping indicates a good physical connection and IP Address. Port number can still be bad; proceed to next step.

Example of successful ping:

```
{ctuser@engbayXX}[5] ping 3.7.52.164
PING 3.7.52.164 (3.7.52.164): 56 data bytes
64 bytes from 3.7.52.164: icmp_seq=0 ttl=255 time=0.927 ms
64 bytes from 3.7.52.164: icmp_seq=1 ttl=255 time=1.079 ms
64 bytes from 3.7.52.164: icmp_seq=2 ttl=255 time=1.090 ms
64 bytes from 3.7.52.164: icmp_seq=3 ttl=255 time=1.070 ms
64 bytes from 3.7.52.164: icmp_seq=4 ttl=255 time=1.048 ms
64 bytes from 3.7.52.164: icmp_seq=5 ttl=255 time=1.073 ms
64 bytes from 3.7.52.164: icmp_seq=6 ttl=255 time=1.199 ms
----3.7.52.164 PING Statistics----
7 packets transmitted, 7 packets received, 0% packet loss
round-trip min/avg/max = 0.927/1.069/1.199 ms
```

5.) If you are unable to successfully ping the camera, use the snoop tool to monitor what is going on with communication packets during a print job. Snoop will read the number of responses from the server while attempting to do a print job. In Example A below, there is only one summary line being reported, (one outbound), and NO inbound response indicating the remote camera host (engctnl) cannot be reached. Refer to Snoop on page 258 for snoop and its usage.

Example: A **snoop -SVta** *3.7.52.164* ←where *3.7.52.164* , in this case, is the camera's <IP address>

```
Using device ef0 (promiscuous mode)
14:46:19.250400engbay26 -> engctn1length:58 ETHER Type=0800 (IP), size = 58 bytes
14:46:19.250400engbay26 -> engctn1length:58  IP D=3.7.52.164 S=3.7.52.151 LEN=44, ID=57050
```

```
14:46:19.250400engbay26 -> engctn1length:58  TCP D=2106 S=1192 Syn
Seq=1001039841 Len=0 Win=16384
```

6.) If there are only two summary lines, (Example B) one outbound and one inbound, this indicates that we can successfully ping the remote camera host, (IP Address is good) but the remote application is either not running (i.e. the machine is up, the application that acts as the print server is not running), or the wrong port number is being used. Refer to Snoop on page 258 for snoop and its usage.

Example: B

snoop -SVta *3.7.52.164* ¨ where *3.7.52.164,* in this case is the camera <IP address>

```
Using device ef0 (promiscuous mode)
14:46:19.250400engbay26 -> engctn1length:58  ETHER Type=0800 (IP),
size = 58 bytes
14:46:19.250400engbay26 -> engctn1 length:58  IP  D=3.7.52.164
S=3.7.52.151 LEN=44, ID=57050
14:46:19.250400engbay26 -> engctn1length:58  TCP D=2106 S=1192 Syn
Seq=1001039841 Len=0 Win=16384
________________________________
14:46:19.251971engctn1 -> engbay26length:60  ETHER Type=0800 (IP),
size = 60 bytes
14:46:19.251971engctn1 -> engbay26length:60  IP  D=3.7.52.151
S=3.7.52.164 LEN=40, ID=10027
14:46:19.251971engctn1 -> engbay26length:60  TCP D=1192 S=2106 Rst
Ack=1001039842 Win=0
```

Example C below shows what would be logged in the dcplog with incorrect port number problem. This is really a tcp initialization error, attempting to open an association, the remote host is up and running but the port number is wrong. Note: this same error can also be caused by the remote application (camera server) not running.

Example: C

```
{ctuser@engbayXX}[17] cd /usr/g/ctusr/logfiles
{ctuser@engbayXX}[18] more dcplog
# DICOM print_scu pid: 2523
print_scu -aIMN -hengctn1 -c1 -f1x1_fid -p./1on1 -d./camera.dev
dcm_bind: AETitle = engbay26_DCP
map_app_title: title IMN host engctn1 ip-addr 3.7.52.164 port 2106
EstablishAssoc: DCM_OPEN_REQ Action success
Errors logged beyond this point of failure may be a result of this
Error:
DCM kernel lower level error:
type = 508 -- DCM network error ¨ ERROR
code = 114 -- lost transport connection
ul_code = 52, reason = 0, source = 0, reject = 0
filename = kernel/D_assoc.c  line = 3051
Failed to contact printer, status 114
```

DCM Protocol Error

A DCM Protocol Error indicates a problem with calling parameters when trying to open an association. They can be caused by having an incorrect AE Title configuration. The errors reported by a print server are only as good as the dicom implementation of that server. The Imation server will accept any called AE title. The Kodak mlp190 will accept any called AE title. The AGFA however, requires the AE to match. The following is an example of what will be reported in the dcplog with an incorrect AE Title on an AGFA system in (Example D), and what snoop is reporting, in (Example E).

Example: D

```
> cd /usr/g/ctuser/logfiles
> more dcplog
print_scu -aIMN1 -hengctn1 -c1 -f1x1_fid -p./1on1 -d./camera.dev
# DICOM print_scu pid: 2492
print_scu -aIMN1 -hengctn1 -c1 -f1x1_fid -p./1on1 -d./camera.dev
dcm_bind: AETitle = engbay26_DCP
map_app_title: title IMN1 host engctn1 ip-addr 3.7.52.164 port 2106
EstablishAssoc: DCM_OPEN_REQ Action success
Errors logged beyond this point of failure may be a result of this
Error:
DCM kernel lower level error:
type = 507 -- DCM Protocol error  ERROR
code = 166 -- invalid pdu parameter value
ul_code = 37, reason = 0, source = 0, reject = 0
filename = kernel/D_assoc.c  line = 3051
DCM kernel lower level error:
type = 503 -- DCM Kernel integrity errors
code = 136 -- error with the dicom upper layer
ul_code = 22, reason = 0, source = 0, reject = 0
filename = kernel/D_assoc.c  line = 500
Fatal DCM error: 136
dcm_deinit: Kernel Deinit Failed
Failed to contact printer, status 166
```

Example: E

The number of packets, outbound and inbound with length of ~60 and ~500 indicates that the remote application is running, but it is not allowing the scu (Service Class User, i.e. the OC) to open an association. This also indicates the IP Address and Port Number is correct.

```
engbay26 2# snoop -SVta <camera IP address>
```

Using device ef0 (promiscuous mode)

```
15:10:36.357083    engbay26 -> engctn1     length:   58  ETHER Type=0800
(IP), size = 58 bytes
15:10:36.357083    engbay26 -> engctn1     length:   58  IP  D=3.7.52.164
S=3.7.52.151 LEN=44, ID=59135
15:10:36.357083    engbay26 -> engctn1     length:   58  TCP D=2106 S=1209
Syn Seq=1188358241 Len=0 Win=16384
________________________________
15:10:36.358280     engctn1 -> engbay26    length:   60  ETHER Type=0800
(IP), size = 60 bytes
15:10:36.358280     engctn1 -> engbay26    length:   60  IP  D=3.7.52.151
S=3.7.52.164 LEN=44, ID=37125
15:10:36.358280     engctn1 -> engbay26    length:   60  TCP D=1209 S=2106
Syn Ack=1188358242 Seq=1847802416 Len=0 Win=8760
________________________________
15:10:36.358390    engbay26 -> engctn1     length:   54  ETHER Type=0800
(IP), size = 54 bytes
15:10:36.358390    engbay26 -> engctn1     length:   54  IP  D=3.7.52.164
S=3.7.52.151 LEN=40, ID=59137
15:10:36.358390    engbay26 -> engctn1    length:  54  TCP D=2106 S=1209
Ack=1847802417 Seq=1188358242 Len=0 Win=16060
________________________________
```

```
15:10:36.361533    engbay26 -> engctn1     length:  456  ETHER Type=0800
(IP), size = 456 bytes
15:10:36.361533    engbay26 -> engctn1     length:  456  IP  D=3.7.52.164
S=3.7.52.151 LEN=442, ID=59138
15:10:36.361533   engbay26 -> engctn1    length:  456  TCP D=2106 S=1209
Ack=1847802417 Seq=1188358242 Len=402 Win=16060
________________________________
15:10:36.412509     engctn1 -> engbay26    length:   60  ETHER Type=0800
(IP), size = 60 bytes
15:10:36.412509     engctn1 -> engbay26    length:   60  IP  D=3.7.52.151
S=3.7.52.164 LEN=40, ID=37126
15:10:36.412509    engctn1 -> engbay26   length:  60  TCP D=1209 S=2106
Ack=1188358644 Seq=1847802417 Len=0 Win=8760
________________________________
15:10:36.424127     engctn1 -> engbay26    length:   64  ETHER Type=0800
(IP), size = 64 bytes
15:10:36.424127     engctn1 -> engbay26    length:   64  IP  D=3.7.52.151
S=3.7.52.164 LEN=50, ID=37127
15:10:36.424127    engctn1 -> engbay26   length:  64  TCP D=1209 S=2106
Ack=1188358644 Seq=1847802417 Len=10 Win=8760
________________________________
15:10:36.424376    engbay26 -> engctn1     length:   64  ETHER Type=0800
(IP), size = 64 bytes
15:10:36.424376    engbay26 -> engctn1     length:   64  IP  D=3.7.52.164
S=3.7.52.151 LEN=50, ID=59141
15:10:36.424376   engbay26 -> engctn1    length:  64  TCP D=2106 S=1209
Ack=1847802427 Seq=1188358644 Len=10 Win=16060
________________________________
15:10:36.428902     engctn1 -> engbay26    length:   60  ETHER Type=0800
(IP), size = 60 bytes
15:10:36.428902     engctn1 -> engbay26    length:   60  IP  D=3.7.52.151
S=3.7.52.164 LEN=40, ID=37128
15:10:36.428902     engctn1 -> engbay26    length:   60  TCP D=1209 S=2106
Fin Ack=1188358654 Seq=1847802427 Len=0 Win=8760
________________________________
15:10:36.428975    engbay26 -> engctn1     length:   54  ETHER Type=0800
(IP), size = 54 bytes
15:10:36.428975    engbay26 -> engctn1     length:   54  IP  D=3.7.52.164
S=3.7.52.151 LEN=40, ID=59143
15:10:36.428975   engbay26 -> engctn1    length:  54  TCP D=2106 S=1209
Ack=1847802428 Seq=1188358654 Len=0 Win=16060
________________________________
```

Note: If the AE title is correct, the server may have a security feature that requires that the local host be registered on the remote host.

Image Packet Transfer, Output From *snoop*

This is an excerpt from a snoop output representing actual image packets, (length ~1514), being transferred to the camera:

```
12:19:58.436211    engbay26 -> engctn1     length: 1514  ETHER Type=0800
(IP), size = 1514 bytes
12:19:58.436211    engbay26 -> engctn1     length: 1514  IP  D=3.7.52.164
S=3.7.52.151 LEN=1500, ID=38793
12:19:58.436211   engbay26 -> engctn1    length: 1514  TCP D=2106 S=1511
Ack=3095191028 Seq=1815234494 Len=1460 Win=16060
________________________________
```

```
12:19:58.436256    engbay26 -> engctn1     length: 1514  ETHER Type=0800
(IP), size = 1514 bytes
12:19:58.436256    engbay26 -> engctn1     length: 1514  IP  D=3.7.52.164
S=3.7.52.151 LEN=1500, ID=38794
12:19:58.436256   engbay26 -> engctn1    length: 1514  TCP D=2106 S=1511
Ack=3095191028 Seq=1815235954 Len=1460 Win=1606
```

Snoop

Snoop (`snoop`) is the troubleshooting tool that monitors all the communication and image packets inbound and outbound to the camera during a print job (depending on which switch settings you use). The packet size length is important in understanding what is being transferred. A length size of < 500 indicates requests and responses between the scanner and the print server. These are from the NGET (printer status), and NCREATE (film session and film box). A series of packet lengths of about 1500 indicates an image transfer in progress. This applies to both dicom print and dicom send.

Steps for Starting a Snoop Session	
Step	**Comment**
1. Open up a Unix shell	From Desktop, select Unix Shell
2. Become root.	**`su -`**
3. Start the snoop session in the shell and set it up to display outgoing and incoming packets.	**`snoop -SVta <camera ip address>`**
4. Send a DICOM Print job to the camera	In ImageWorks desktop, display an image and drag/drop the image into the film composer and Print it.
5. Observe the output packets of data being sent and received.	Length sizes < 500 = communication request between the scanner and the print server. Length sizes ~1500 = the image packet size being sent.

Table 6-21 Snoop Session Steps

The following examples show common uses of snoop. See Number 3 below for a description of snoop usage and switch descriptions. Typical use examples:

1.) How to display outgoing and in-going packets:

```
{ctuser@bayXX}[3] su  -
password
bayxx 1# snoop -SVta  <camera's ip address>
```

Comment: *Using device ef0 (promiscuous mode)*

```
15:00:18.606959     engbay26 -> engctn1      length:   58  ETHER
Type=0800 (IP), size = 58 bytes
15:00:18.606959     engbay26 -> engctn1      length:   58  IP
D=3.7.52.164 S=3.7.52.151 LEN=44, ID=59593
15:00:18.606959     engbay26 -> engctn1      length:   58  TCP D=2104
S=3565 Syn Seq=1295817451 Len=0 Win=16384
________________________________
15:00:18.608481      engctn1 -> engbay26     length:   60  ETHER
Type=0800 (IP), size = 60 bytes
```

```
15:00:18.608481      engctn1 -> engbay26     length:   60  IP
D=3.7.52.151 S=3.7.52.164 LEN=40, ID=33153
15:00:18.608481     engctn1 -> engbay26     length:   60  TCP D=3565
S=2104 Rst Ack=1295817452 Win=0
```

2.) How to display incoming packets only:

```
{ctuser@bayXX}[3] su -
password
bayxx 1#  snoop -SPVta  <camera ip address>
Using device ef0 (promiscuous mode)
14:58:54.506391      engctn1 -> engbay26     length:   60  ETHER
Type=0800 (IP), size = 60 bytes
14:58:54.506391      engctn1 -> engbay26     length:   60  IP
D=3.7.52.151 S=3.7.52.164 LEN=40, ID=14589
14:58:54.506391     engctn1 -> engbay26     length:   60  TCP D=3563
S=2104 Rst Ack=1285065404 Win=0
```

3.) Usage for snoop:

```
[ -a ]   # Listen to packets on audio
[ -d device ]# settable to le?, ie?, bf?, tr?
[ -s snaplen ]# Truncate packets
[ -c count ]# Quit after count packets
[ -P ]   # Turn OFF promiscuous mode
[ -D ]   # Report dropped packets
[ -S ]   # Report packet size
[ -i file ]# Read previously captured packets
[ -o file ]# Capture packets in file
[ -n file ]# Load addr-to-name table from file
[ -N ]   # Create addr-to-name table
[ -t  r|a|d ]# Time: Relative, Absolute or Delta
[ -v ]   # Verbose packet display
[ -V ]   # Show all summary lines
[ -p first[,last] ]# Select packet(s) to display
[ -x offset[,length] ]# Hex dump from offset for length
[ -C ]   # Print packet filter code
```

For additional information, refer to the manual page for snoop by opening a Unix shell, and entering the following commands at the prompt:

```
> su  -
> password
> man snoop
```

DICOM Terms

Configuration - The DICOM Print Configuration Information field is controlled by the Camera Manufacturer. It is typically used to set information on the Look-up Table to be used to convert the inputted digital image data to the hardcopy film output (since the range of valid data for the input may not match the range for the output data); however, it is not limited to this purpose. The string field is defined by the Camera Manufacturer and is currently up to 1024 bytes. The value is equivalent to working the contrast on a image monitor.

Density - A film term which represents the pixel value at a particular point on the film. Empty Density is the pixel representation of a blank image frame on a film. Border Density is the pixel representation of the area outside of the image frames on the film. Minimum Density is the minimum pixel representation to be used within an image, while Maximum Density is the maximum pixel representation to be used within an image. The last two values are equivalent to working the brightness on a image monitor. The range and effect of the last two density parameters are Camera Manufacturer dependent.

DICOM - Acronym for Digital Imaging and Communication in Medicine. This standard is a detailed specification for transferring medical images and related information between computers.

Magnification Type

Images from the CT scanner are digitized at a low resolution and are then printed at a higher resolution. To accomplish this, images are interpolated prior to being printed. A number of techniques may be used to perform the image interpolation. The most common techniques are:

- <u>Replication:</u> This is the simplest method of interpolation (zero order interpolation). In this case adjacent data is used to calculate the fill data. The resultant images are typically extremely blocky and contain jagged edges.
- <u>Bilinear:</u> Also known as first order (linear) interpolation, this technique consists of fitting straight lines through adjacent data points to determine intermediate points. The resultant images are somewhat blurred.
- <u>Cubic:</u> Third order (cubic) interpolation is usually the favored technique. There are a large number of possible formulations for cubic interpolation. Each differs by the coefficients used in the process. The Camera Manufacturers use a second parameter called a Smoothing Type to set the coefficients. The implementation of the coefficient is Camera Manufacturer dependent. The cubic interpolation presents the smoothest version of interpolation when compared to replication or bilinear interpolation.

Service Class - Represents a specific application feature by defining a set of related SOP classes (DICOM Print).

Smoothing Type - A value used in conjunction with the Magnification Type. It is only relevant when the magnification type is set to Cubic. Smoothing is used to set the coefficients for the formulation of the interpolation. The valid values and meaning of the Smoothing Type parameter are controlled by the DICOM Print Manufacturer. For example, Imation expects a smoothing factor of 0 to 15, while Agfa expects a smoothing factor of VR type 0, or falling within the range of 100 to 299.

SCP - Acronym for Service Class Provider. This is the Service Class server. (In the case of DICOM Print, this is the DICOM Print Camera).

SCU - Acronym for Service Class User. This is the Service Class client. (In the case of DICOM Print, this is the CT Scanner).

SOP - Acronym for Service Object Pair. This term is used in DICOM to specify the capabilities of a DICOM entity. The entity is defined by the union of the Information Object Definition (IOD) (e.g. CT image) and the DICOM Message Service Element (DIMSE) Services (e.g. store).

2.3 User Informational Tools

2.3.1 The CBT and Hard Disk Space it Requires

Computer Based Training (CBT) software is provided with the system to assist the operator in learning. By using the CBT and operator reference manual, users can quickly obtain the necessary skills to operate this CT scanner in an efficient and effective manner.

During installation, the CBT makes safe but minor changes to the system disk. The CBT creates/ modifies user information files (e.g. for bookmarks). To do this, the CBT software requires predefined hard disk area and an IRIX path name to access it.The following directory is added:
`/usr/g/cbt`

This directory contains the necessary cbt startup files and executables. This directory is also used and should be used to store the CBT bookmarks as necessary.

2.3.2 Adobe(TM) Acrobat(TM) Electronic Documentation Viewer

Adobe's Acrobat Reader has to be pre-installed on the system and should be used when necessary to view documents in portable document format (PDF).

To access the Acrobat reader on a CT/i system, open an UNIX shell (on either monitor) and at the prompt, type: **`geacroread`** . If asked, accept the Acrobat Reader License Agreement and continue. Normally, a window is posted displaying the top level CDROM contents (listed as selectable filenames or directories) which can be selected and opened by the user. Acrobat Reader will recognize a Table of Contents, for some CDROMs which will automatically open and display.

An online Guide for using Acrobat Reader can be found using the File menu and selecting: Help-Reader.pdf. Additional information can be found on Adobe Systems WEB page. Adobe's WEB address is **`http://www.adobe.com/`**

Some CT Product Documentation CDROMs will contain a file named user_man.pdf which will also provide help in using Acrobat Reader.

To access proprietary documentation, you will be asked to supply the proper password before the document will be displayed.

2.4 CT/i' "SMART" Features

2.4.1 SmartBeam

2.4.1.1 SmartBeam Theory

Special hardware and software filters for SmartBeam can enable lower techniques on body scans. The SmartBeam filter reduces body technique factors by 20% to 30% while it maintains previous levels of image quality.

- The SFOV prescription determines filter selection. (No additional operator intervention required.)
- The Calibration procedures remain the same, but the presence of the SmartBeam filter increases the required number of scans per kV. The system automatically adjusts the screens to show the correct number of scans.

The SmartBeam bowtie filter has two profiles:

- The head portion of the filter provides the same x-ray attenuation of heads and small bodies as the previous filter configuration.
- The body portion of the filter decreases x-ray attenuation of the large body SFOV, which results in acceptable image quality at lower doses.

The SmartBeam collimator, 46-296300G4 is the forward production collimator for HiSpeed CT/i Systems. The G4 collimator will eventually be the only collimator FRU part for the installed base of G1, G2 and G3 collimators.

The system requires an option key (MOD) to enable operation of the SmartBeam filter on the HiSpeed CT/i System.

Note: Collimator Version

For HiSpeed CT/i Systems with G1, G2 or G3 collimators, you must install G4 collimator if system software is R 3.5, or either G4 or G5 if R3.6 or later, in order to be compatible with SmartBeam.

If SmartBeam is added to the HiSpeed CT/i, you must re-calibrate the system after the collimator change. Failure to update calibration could result in Image Quality and dose issues. If software is configured for SmartBeam but the hardware cannot support it, your scanner will not work. Either un-install SmartBeam or get a newer collimator.

Model Number	Comments
46-296300G1	Original HiSpeed Advantage Collimator
46-296300G2	Aperture modified to reduce x-ray scatter; backwards compatible with G1
46-296300G3	EMC compatible and excess parts removed from collimator assembly; backwards compatible with G1 and G2
46-296300G4	More durable Aluminum Carbon (AlC) SmartBeam filter replaces Teflon filter in collimator; backwards compatible with G1, G2 and G3 collimators
46-296300G5	New filter drive mechanism, and higher torque aperture stepping motor. **NOT BACKWARD COMPATIBLE**!

Table 6-22 HSA Collimator History

A G2 Collimator Board was introduced with R3.6 to resolve Aperture Motion without Command errors at a 0.8 second scan period.The G5 collimator is also being introduced. The filterType value in the `scanhardware.cfg` file is “1” for G1, G2, G3 collimators, “2” for a G4 collimator and “3” for a G5 collimator. Also the firmware characterization file `collimator.char` has four more fields.

2.4.1.2 Built-in Protocol Files/SmartBeam

Your new CT/i system comes with built-in protocols that will be used by the technologists. There are Adult protocols and Pediatric protocols. Within the Adult protocols, for large body scanning only, we have sent protocols for both Smartbeam and non-Smartbeam option systems. The differences is in the dose provided to the patient.

Before turning the system over for patient scanning, do the following, so that the appropriate protocols are used by your system. Determine if you have the SmartBeam option installed. On the Service Desktop, on the Utilities menu, run “Verify Options”. If SmartBeam shows up in the list, SmartBeam has been installed and is functional.

Systems Without SmartBeam Option

Steps to remove the Smartbeam protocols:

1.) Go to Protocol Management and Select the region 3 on the Adult body by clicking around the arrow for the region. These are the C-spine protocols
2.) Select the protocol name with (S/B) on it. This is a smartbeam protocol.
3.) Press “DELETE” at the bottom of the menu and confirm.
4.) Press DONE.
5.) Repeat steps 1–4 for Regions 3–9 from the Adult body.

Systems With SmartBeam Option

Steps to remove the non-Smartbeam protocols:

1.) Go to Protocol Management and Select the region 3 on the Adult body. These are the chest? protocols 3. Select the 1st protocol, i.e. the one that does not have (S/B) on it. This is for non-smartbeam option systems.
2.) Press "DELETE" from the bottom of the menu, and confirm.
3.) Also, edit the name of the smartbeam protocol. Place the cursor at the end of the smartbeam protocol name, and delete the end characters "(S/B)".
4.) Press DONE.
5.) Repeat steps 1–4 for each of the Regions 3–9 from the Adult body.

2.4.2 SmartScan

SmartScan is included with CT/i; the SmartScan software automatically lessens mA when it shoots x-ray through the short axis of the patient. You'll find more about SmartScan and DD files on page 324.

2.4.3 SmartPrep

SmartPrep enables the user to monitor contrast enhancement in the patient so imaging can occur in the optimal enhancement time window. The operator takes a single, low dose, scan before contrast is injected. This *baseline* image is used to place up to three ***R**egions **O**f **I**nterest* to be monitored during the contrast monitoring phase. The doctor can see when contrast reaches the optimal enhancement level then command the exam to begin. This can minimize patient exposure to contrast media and radiation.

2.4.4 SmartTrend

SmarTrend is run automatically in CT/i to proactively collect air and bowtie scans during Daily Prep. The data is used to establish short and long term trends which would be used to evaluate system performance and detect center spot and other subtle image artifacts.

2.4.4.1 Introduction

- The feature proactively collect air/bowtie scans during Daily Prep (Fast Cal).
- The data will establish short and long term trends used to evaluate system performance and detect center spot and other image artifacts which.
- If the analyzed data is out of specification, a message will be displayed to inform the operator. The results will also be checked as part of the ProDiags platform and an automatic dial out to the Automatic Support Center (ASC) will be created (InSite systems only).

2.4.4.2 Data Collection Method:

1.) Air scans (120 kV/4 sec/10 mm/60 mA) are run with and without the bow-tie filter.
2.) Data is offset corrected and normalized.
3.) Current data is compared with previous day's data (short term trend).
4.) The data is also compared with a baseline (long term trend).
5.) Compare ratio with previous day's result and with a baseline.

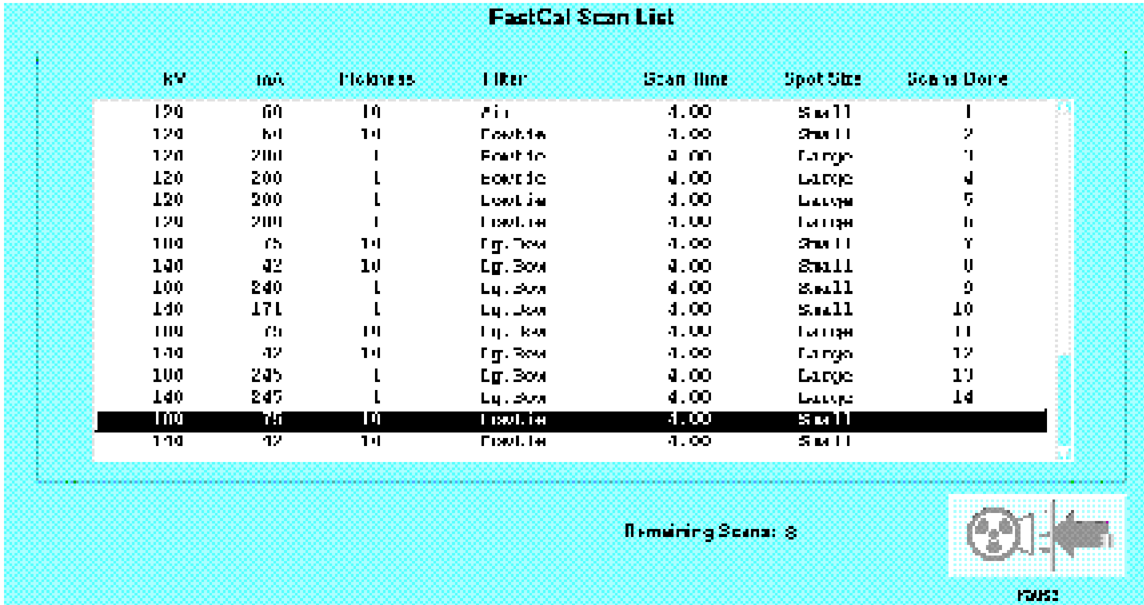


Figure 6-12 FastCal Scan List 1

2.4.4.3 Establishing A Smart Trend Baseline

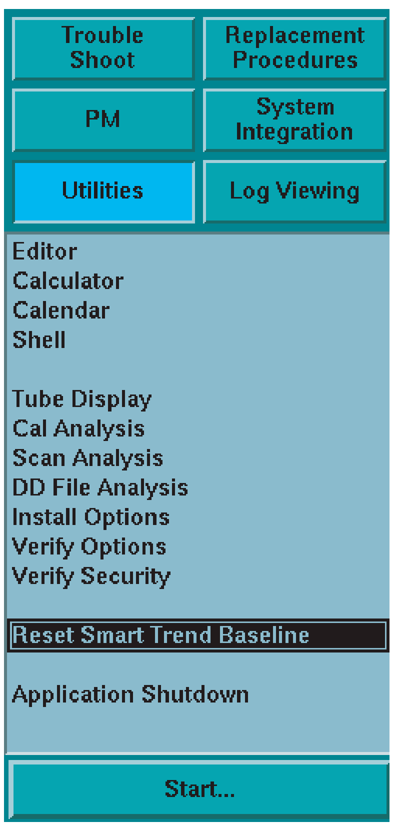


Figure 6-13 Resetting SMART TREND Baseline from Utility Menu

A new baseline must be established when:

- the x-ray tube is replaced,

- the collimator is replaced,
- the detector is replaced,
- or a DAS converter card is replaced.

Initiate a new Smart Trend baseline by pressing the RESET SMART TREND BASELINE button on the Utility Menu of the Service Desktop. Then run a successful Fast Cal using the Daily Prep software.

2.4.4.4 DAS Gain Data Storage

BASE LINE STORAGE

- The first storage must be invoked by the field engineer during tube change, collimator change, detector change, realignment, or DAS board changed.
- Store offset corrected normalized msd files of both scans every 30 days.

DAILY STORAGE

Store offset corrected normalized msd files of both scans every day.

Process Name	Path and/or File(s)	Comments
dasgain	`/usr/g/insite/ProDiags/qsa/results`	All data stores
	`dd.air_baseline.msd`	Dasgain creates this baseline file for the air scan, when run for the very first time. This baseline file gets reset every 30 days by default.
	`dd.bow_yesterday.rat`	A copy of today's gain corrected bowtie scan's dd file.
	`dd.air_yesterday.msd`	On the second day, dasgain saves a copy of today's air scan msd file and calls it as yesterdays air scan file.
	`dd.air_st.rat, dd.bow_st.rat`	short term ratio operations are done for both air and bowtie scans.
After the ratio operation, today's bowtie scan msd file will be saved as yesterdays file as before. If any of the limit checks fails, the corresponding error file is also saved. From the third day onwards, dasgain performs the short term and long term ratios, and limit checks on the air scan, and short term ratio and limit checks on the bowtie scans.		
	`dd.air_lt.rat`	A long term ratio file for air scan is also saved in addition to the files saved on the second day.
	`dasGain.dat`	The dasgain also creates a data file in the same directory which gives information regarding when the processing and baseline creation are done last. This information is used to block off any dasgain operations which might happen in case the user run Fast Cal more than once in a day, and to reset the baseline after the specified number of days.

Table 6-23 DAS Gain Files

Process Name	Path and/or File(s)	Comments
	Dasgain_*.lmt	The limit files are different for each limit check operation. They are present in the config directory with the file names.
The prodiags application qsa dials out when it finds that one of the over limit files has been created in /usr/g/insite/ProDiags/qsa/results. Qsa also saves the air msd files for the purpose of trending. The file dd.air_yesterday.msd is saved as dd.air_yesterday1.msd.gz, dd.air_yesterday2.msd.gz,... and so on up until dd.air_yesterday7.msd.gz, after which it rolls over.		

Table 6-23 DAS Gain Files (Continued)

Note: All “DD” (Diagnostic Data) files can be reviewed using the DD File Analysis tool. However, since this tool does not read the qsa directory (/usr/g/insite/ProDiags/qsa/results), the “DD” files must be copied into the /data directory so they can be analyzed using the DD Tool.

2.4.4.5 Smart Trend Configuration

The various operations performed by dasgain can be configured from a config file, dasGain.cfg, located in the config directory /usr/g/config. The following are the configurable parameters in dasgain.

Parameter	Description
Delete_files	This option, if set, will make the program to delete the input scan files and all the intermediate files created during the processing. By default, this option is enabled.
Reset_baseline	This will force the program to reset the baseline every time you run Fast Cal. This is disabled by default.
First_chnl	This gives the first channel for which dasgain processing is to be done. Default value is 3.
Last_chnl	This gives the last channel for which dasgain processing is to be done. Default value is 746.
Dasgain_interval	This is the minimum interval (in hours) at which dasgain processing is to be done. Default value is 20.
Baseline_interval	This is the maximum duration (in days) for which a baseline is kept valid. Default value is 30. This means that the baseline will be reset every 30 days.
Disable_dasgain	This option, if set, will disable all dasgain processing. By default, dasgain processing is enabled.
Disable_air_lmt_chk	This option, if set will disable the limit checks for air scan file. By default, air scan limit check is enabled.
Disable_bow_lmt_chk	This option, if set will disable the limit checks for bow scan file. By default, bow scan limit check is enabled.

Table 6-24 DASGAIN Parameter

2.4.4.6 Message Logging

Startup and ending messages are posted into the Session Log and ProDiags history log.

The following messages are also posted into the ProDiags history log.

- When a new baseline is created.

- When the baseline is expired. Currently, the baseline is set to expire every 30 days. A new baseline will be created every 30 days. However, this also can be configured by changing the entry for baseline_interval in the config file `/usr/g/config/dasGain.cfg.`
- When smart trend processing is not done. If a processing is already done during the day, the next time the user try to run it again, the processing is skipped. The user can enable dasgain processing every time he runs it, by editing the config file dasGain.cfg file and changing the entry for `dasgain_interval`.

The SessionLog is updated in the file `/usr/g/service/state/ssw.srv_activity.ref` and the ProDiags history is stored in the file `/usr/g/service/state/ssw.ProDiags.hist`.

2.4.5 DC Cal

The Data Acquisition Subsystem (DAS) converts analog output signals from the X-Ray Detector array to digital signals for transmission to the Scan Recon Unit (SRU). DC Cal tests will exercise functions of DAS to verify its proper operation and help diagnose malfunctioning DAS converter cards. The tests include checking absolute and differential linearities and are performed as a part of Daily Prep procedure.

2.4.5.1 Introduction

- The feature collects DC cal data during Daily Prep (Fast Cal) to evaluate the DAS linearity.
- If the analyzed data is out of specification, a message will be displayed to inform the operator. The results will also be checked as part of the ProDiags platform and an automatic dial out to the Automatic Support Center (ASC) will be created (InSite systems only).

2.4.5.2 User Interface

The operator will see an additional screen with DC Cal scans after the completion of normal FastCal scans.

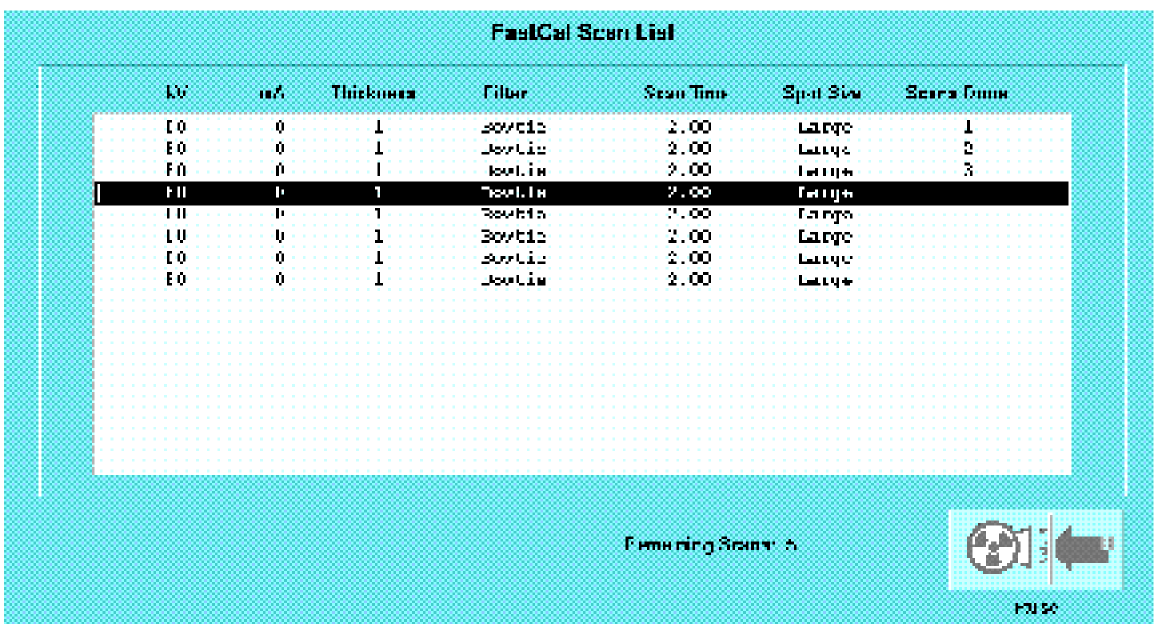


Figure 6-14 FastCal Scan List 2

2.4.5.3 Data Acquisition

- Applies to HP DAS only.
- Data acquisition consists of two scans using DAS mode.
- Scan parameters are 80 kV/ 0 mA/ 2 sec/ 1 mm/ bow tie/ large spot/ 1:1 compression / offset correct.

2.4.5.4 Data Processing

The absolute linearity checks for the absolute counts in the reference CAL (CAL 7), and the ratios of each cal with the reference CAL. The differential linearity checks for the channel to channel difference for each of the absolute linearity ratios.

2.4.5.5 DC Cal Data Storage

Process Name	Path and/or File(s)	Comments
dccal	`/usr/g/insite/ProDiags/qsa/results`	The process will create appropriate error files with “means” data for all channels / levels in case of test failure(s).
	`dasdc_absolute.cal<level>.err` `dasdc_diff.cal<level>.err`	The naming convention for error files will be saved with”.err” extension. <level> ranges from 0 to 7.
	`/usr/g/config/dccal.cfg`	The dc cal scans can be enabled and disabled by editing the config file.
	`/data`	Where input scan data is stored.
	`/usr/g/config/HPDasTools.cfg`	DC Cal test limits also used by DAS Tools in the Service Desktop.

Table 6-25 DC Cal Data Storage Files

2.4.5.6 Message Logging

A message will be put into the Session Log and the ProDiags history log, stating when the DC Cal processing began and ended.

2.4.6 Troubleshooting Smart Trend & DC Cals

Note: Applicable to Systems with 4.1 and 5.x SW only

If SmartTrend reports an error, the following steps should be followed to investigate the problem: Remember SmartTrend is just a program that is monitoring the systems DAS/Detector subsystem. It looks at daily values of dasgains and other factors and compares them with previously measured values. If the program notices changes in day to day values then it flags a message that the DAS/ Detector subsystem could have a problem. Normal troubleshooting procedures for the DAS/ Detector should be used. Do not troubleshoot the program or use the program as a troubleshooting tool. The SmartTrend baseline can be reset once confidence has been established that there is nothing wrong with the DAS/Detector subsystem, using the DAS tools that are available.

1.) Check the GE Message Log to determine the type of test(s) that exceeded the specified limits. The Log will indicate the path and name of the relevant error file(s). Does the Smart Trend baseline need to be reset?
2.) Look for more failure details in the `*.err` files (using the unix *more* command is one option for viewing the file). The error files will indicate the relevant Exam/Series/Image number. It will also list the actual and expected results which fell outside the specified limits.

To review the files for future use they need to be renamed or moved or they will be overwritten.

3.) Use Scan Analysis or DD Analysis to examine the scan file(s) listed in the Message Log for abnormalities.

All "DD" (Diagnostic Data) files can be reviewed using the DD File Analysis tool. However, since this tool does not read the qsa directory `/usr/g/insite/ProDiags/qsa/results`, the "DD" files must be copied into the `/data` directory so they can be analyzed using the DD Tool.

4.) Refer to the DAS/Detector troubleshooting procedures for further testing/verifying methods. See Chapter 10 on page 701

2.4.6.1 User Message Box

The following dialog is posted to the user, if any of the limit checks fails.

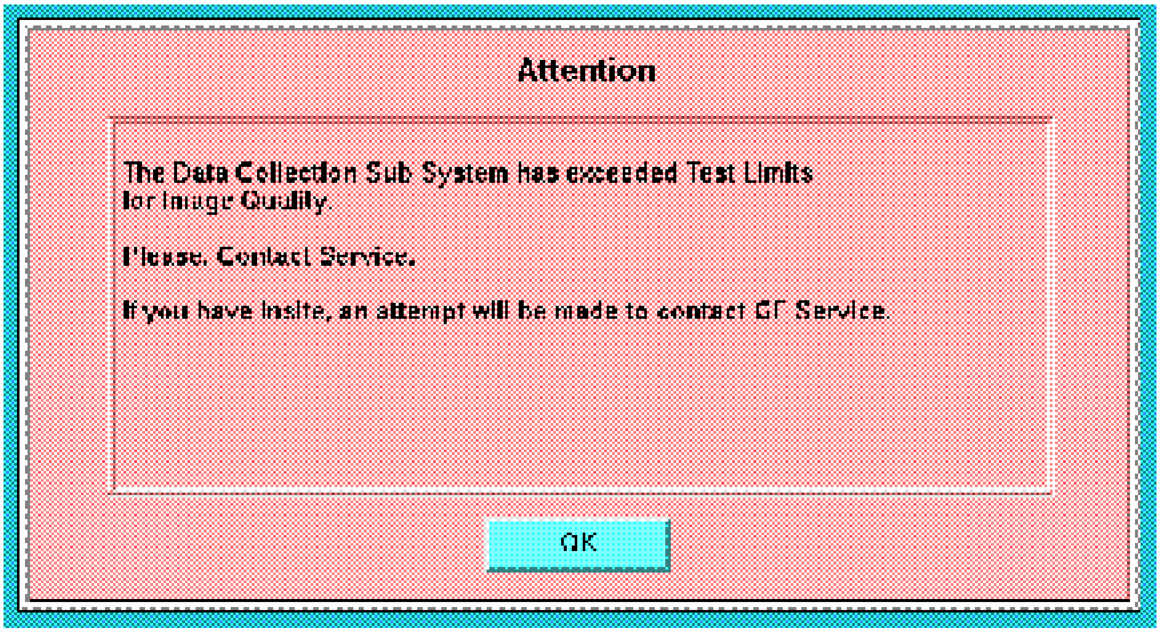


Figure 6-15 User Message Box

2.4.6.2 SMART TREND Messages

SESSION LOG

Only Tool startup and ending messages go into the session log.

```
Wed Apr  8 17:05:53 1998  PRDIMSG Smart Trend Processing, BEGIN
Wed Apr  8 17:05:57 1998  PRDIMSG Smart Trend Processing: END
```

PRODIAGS HISTORY LOG

All the messages related to Smart Trend activity are posted into the prodiags history log. When you run Smart Trend on a system for the first time, it creates the baseline. Here are some samples:

```
Fri Mar 27 10:18:14 1998    Pro Diags: Smart Trend Processing Started.
Fri Mar 27 10:18:14 1998    Smart Trend: 85 days elapsed since last
baseline
Fri Mar 27 10:18:14 1998
```

```
        Smart Trend: last baseline was done on Thu Jan  1 00:00:00 1998
        85 days have elapsed since the last baseline
        new baseline will be created this time
Fri Mar 27 10:18:18 1998    Smart Trend: New baseline will be created -
                                         -for air scan files.
Fri Mar 27 10:18:18 1998    Smart Trend: air scan limit checks disabled
Fri Mar 27 10:18:18 1998    Smart Trend: New baseline created for -
                                         -air scan files.
Fri Mar 27 10:18:18 1998    Pro Diags: Smart Trend Processing completed
```

The previous message will come when you run Smart Trend for the first time on a system. By default, it assume that the last time baseline was created on 1 Jan. 1998.

It is designed to reset the baseline every 30 days. A similar message will appear if the baseline exceeds 30 days. A new baseline will be created every 30 days.

Smart Trend is configured to run only once in a day (Once every 20 hours. Can be changed by editing the config file.)If you try to run it again, within the same day, you will get a message like:

```
Fri Mar 27 10:42:50 1998    Pro Diags: Smart Trend Processing Started.
Fri Mar 27 10:42:50 1998
        Smart Trend: last processing was done at Fri Mar 27 10:18:18 1998
        min duration before next dasgain is 20 hours
        hence Smart Trend processing is skipped this time
Fri Mar 27 10:42:50 1998    Pro Diags: Smart Trend Processing completed
```

During a normal run, the following messages are posted.

```
Fri Apr  3 17:51:49 1998    Pro Diags: Smart Trend Processing Started.
Fri Apr  3 17:51:49 1998   Smart Trend: 7 days elapsed since last baseline
Fri Apr  3 17:51:53 1998    Pro Diags: Smart Trend Processing completed
```

In case of some limit check errors, the log will look something like:

```
Wed Apr  8 17:05:53 1998    Pro Diags: Smart Trend Processing Started.
Wed Apr  8 17:05:53 1998   Smart Trend: 12 days elapsed since last baseline
Wed Apr  8 17:05:57 1998   Smart Trend: Limit Check failed for Air Scan, -
                                         -Long Term Ratio, Channel to -
                                         -Channel Difference Data
Wed Apr  8 17:05:57 1998   Smart Trend: Limit Check failed for Air Scan, -
                                    -Long Term Ratio, Trend Removed Data
Wed Apr  8 17:05:57 1998   Smart Trend: Limit Check failed for Air Scan, -
                                         -Long Term Ratio, Trend Data
Wed Apr  8 17:05:57 1998    Pro Diags: Smart Trend Processing completed
```

Sample Error Log Messages

Messages get posted into the error log ONLY when there is a failure in limit check. Each line will come as a separate message in the error log and will include the name and path of the error (*.err) file which was created as a result. The messages look like:

```
Smart Trend: Limit Check failed for Air Scan, Short Term Ratio, Trend
Removed Data
Look for details in error file : /usr/g/insite/ProDiags/qsa/results/
dasgain_air_st_hp.err
```

The error file can be displayed using the "more" unix command. The file will show the description of the DAS Gain operation, the image reference, and the actual and expected results for each detector channel related to the failure.

```
Description : Air Scan, Short Term Ratio, Trend Removed Data
        Exam Number   : 65001
        Series Number : 11
        Image Number  : 91
===================================================
Channel no  Actual Data    Min Limit   Max Limit
---------------------------------------------------
       469     1.003167     0.997000     1.003000
       470     1.004288     0.997000     1.003000
       471     1.006006     0.997000     1.003000
       472     1.007453     0.997000     1.003000
       473     1.008718     0.997000     1.003000
       474     1.010555     0.997000     1.003000
       475     1.012821     0.997000     1.003000
       476     1.014022     0.997000     1.003000
       477     1.016002     0.997000     1.003000
       478     1.017027     0.997000     1.003000
       479     1.018715     0.997000     1.003000
       480     1.019103     0.997000     1.003000
       481     0.984435     0.997000     1.003000
       482     0.872350     0.997000     1.003000
       483     0.858800     0.997000     1.003000
       484     0.995074     0.997000     1.003000
       485     1.018806     0.997000     1.003000
       486     1.018834     0.997000     1.003000
       487     1.017343     0.997000     1.003000
       488     1.015438     0.997000     1.003000
       489     1.014128     0.997000     1.003000
       490     1.012628     0.997000     1.003000
       491     1.011010     0.997000     1.003000
       492     1.009319     0.997000     1.003000
       493     1.007040     0.997000     1.003000
       494     1.005386     0.997000     1.003000
       495     1.004121     0.997000     1.003000
       496     1.003649     0.997000     1.003000
===================================================
```

Additional- Sample Error Messages

Smart Trend: Limit Check failed for Air Scan, Short Term Ratio, Trend Data

Look for details in error file: /usr/g/insite/ProDiags/qsa/results/dasgain_air_st_lp.err

```
Smart Trend: Limit Check failed for Air Scan, Short Term Ratio, Channel
to Channel Difference Data
Look for details in error file : /usr/g/insite/ProDiags/qsa/results/
dasgain_air_st_c2c.err
```

```
Smart Trend: Limit Check failed for Bowtie Scan, Short Term Ratio, Trend
Data
Look for details in error file : /usr/g/insite/ProDiags/qsa/results/
dasgain_bow_st_lp.err
```

2.4.6.3 DC CAL Messages

SESSION LOG

Only Tool startup and ending messages go into the session log.

```
Wed Apr  8 19:02:00 1998  PRDIMSG DC Cal Processing: BEGIN
Wed Apr  8 19:02:15 1998  PRDIMSG DC Cal Processing: END
```

PRODIAGS HISTORY LOG

This is the place where majority of the messages end up. During a normal operation, here is what the history log look like:

```
Wed Apr  8 18:35:03 1998    Pro Diags: DC Cal Processing Started.
Wed Apr  8 18:35:21 1998    Pro Diags: DC Cal Processing completed.
```

When limit errors occur, a message will be posted into the history log indicating which CAL LEVEL has the limit error. This will look something like:

```
Wed Apr  8 19:02:00 1998   Pro Diags: DC Cal Processing Started.
Wed Apr  8 19:02:15 1998   DC Cal: Differential Linearity Error detected -
      -in CAL LEVEL 0
Wed Apr  8 19:02:15 1998   DC Cal: Analog Linearity Error detected -
      -in CAL LEVEL 1
Wed Apr  8 19:02:15 1998   Pro Diags: DC Cal Processing completed
```

ERROR LOG

Messages get posted into error log ONLY when there is a failure in limit check. Each line will come as a separate message in the error log and will include the name and path of the error (*.err) file which was created as a result. The messages look like:

```
DC Cal: Differential Linearity Error detected in CAL LEVEL 0
DC Cal: Analog Linearity Error detected in CAL LEVEL 1
DC Cal: Differential Linearity Error detected in CAL LEVEL 2
DC Cal: Analog Linearity Error detected in CAL LEVEL 3
DC Cal: Differential Linearity Error detected in CAL LEVEL 4
DC Cal: Analog Linearity Error detected in CAL LEVEL 5
DC Cal: Differential Linearity Error detected in CAL LEVEL 6
```

2.4.7 SmartView

SmartView is also known as **CT INTERVENTIONAL**. With the SmartView software and hardware option, the Octane host, the Hand and Foot Controls and the cradle side monitor enable a clinician to single handily and quickly biopsy a patient.

2.5 Magneto Optical Disk (MOD) Archival & Retrieval

2.5.1 How to Label a Maxoptics MOD for system files

If you have an MOD upon which you want to put system files, this is different from image files; you prepare the MOD by making a file system on it. System State and DD File Analysis will detect this condition and prepare the MOD in the drive for you. To prep a system MOD under other circumstances, open a unix shell. Type: **`mkfsMOD`**

(formatting takes about 3-5 min). OR, use **`FSST; 8; 3`**

2.5.2 Save Scan Files to MOD

1.) Go to Service Desktop, click UTILITIES, click SCAN ANALYSIS.
2.) Select EXAM.
3.) Press SAVE SCAN DATA.
    This step moves Scans from the scan database and makes unix files on the system disk in directory `/data`.
4.) Go to Tool Chest menu and click on SAVE SCAN FILES. This shows list of scan files saved.
5.) Hilight on or more of the scan files from the list.
6.) Click SAVE TO MOD.
    This step now moves the scan (unix) files to your MaxOptics MOD. You can append scan files to a used MOD.

2.5.3 Restore Scan Files From MOD

1.) Place MOD in drive, Click RESTORE SCAN FILES from the Toolchest menu.
2.) Select the scan files.
3.) Click RESTORE.
4.) This copies the scan files from the MOD straight into the Scan Database.

2.5.4 Reserve/Release Scan Data

This function allows you to lock scan data in the database so that a particular scan is saved for future reconstruction. Otherwise, eventually all of the 450 scan data buffer will be overwritten with new scan data.

This feature is very crude at this state, and tediously (1 scan at a time), but allows you to accomplish what needs to be done.

1.) Click the RESERVE/RELEASE button from the Toolchest menu.
2.) Select RESERVE OR RELEASE
3.) Select RESERVE option, or the RELEASE option.

Note that:

- Column 1 is an index number, but also the number for identifying the scan file
- Column 2 is the station
- Column 3 is Exam #
- Column 4 is Series #
- Column 5 is Scan #
- Column 6 is 0 (ignore, does not change)

4.) Enter an index # to reserve or release the scan file.

There will be no status update or feedback. Once reserved, make sure you remember to release it as soon as you don't need it anymore. Make sure you record the exam, series, scan number you reserve so you know which to release.

2.5.5 Archive Media Content

MODs labeled (formatted) for storing images have a DOS like structure. MODs formatted for software have a UNIX structure. There are some DOS MODE commands in `/usr/g/bin` to help you view and copy files between the Image Archive media and the system. The size of DICOMDIR indicates how much space images are taking on the MOD. You must use Image Works to DETACH then do another dmls in a shell to see an updated size.

`dmls`	list files of current directory
`dmcd` ***<path>***	change to the directory identified by path
`dmcat props`	show content of the file props which tells you the properties of that media
`dmcat stat`	show content of the file stat which shows last time media was used
`dmcpin -b` ***<dosname> <unixname>***	copy file on media to the system

ABOUT THE PIONEER MOD

NexGen software will have READ-ONLY support of Pioneer MOD's.

2.5.6 System Reset after Restore System State

When the characterization files are restored, the scanning hardware needs to be reset. System State will prompt the user for a reset and automatically do it if the user decides to proceed.

If the system is not reset the restored characterization values will not be used until the system hardware is reset.

2.6 Tele-radiology (Framegrabber Type) Systems

Some Genesis based systems have teleradiology (TR) systems that framegrab the Genesis GFB video (512 x 512 50/60Hz). CT/i *DOES NOT* directly support this type of TR. The CT/i RGB color display video is a much larger format at a much higher pixel frequency, not to mention that its RGB! GE Sales Reps have been told to *NOT* promise any direct compatibility with framegrabbing TR systems (DICOM 3.0 TR systems may work depending on the DICOM implementation but GEMS does not and cannot validate all the various TRs.)

In the framegrabber case, a high quality (300Mhz bandwidth) video splitter/amplifier (as listed above) is needed to intercept and re-drive the display CRT RGB video. Composite grayscale would then be available on Green #2 (1280x1024 pixels at 72Hz). Any framegrabber hardware attempting to capture this signal must be capable of a 140Mhz pixel rate. This also involves TR system configuration parameters. The TR capture software may also need upgrading to deal with 1280x1024 and/or “crop” the signal. The TR remote display software may need upgrading to view the larger format. The image transmission times to the remote TR may be up to 4 times as long. GEMS will supply all technical information necessary to assist TR suppliers in making their systems work with CT/i but *GEMS cannot be responsible for this third party TR equipment, software, or compatibility with CT/i.*

2.7 Touch Config Defaults (Indigo2 Systems Only)

The following are the proper touch config default values for the original CT/i console. If for some reason, the touch screen does not work, please verify these values are correct in the touch “Setup” menu.The Octane host based console does not support the touch monitor feature.

If during touch “SETUP” the “RESTORE DEFAULTS” is pushed, touch will not work until the proper values are restored as shown below. The “RESET DEFAULT” settings are wrong.

Touch Configuration Defaults	
Jitter Control	8
Press Threshold	5
Release Threshold	3
Mouse Button 1 Emulation	Normal
Mouse Button 2 Emulation	Shift
Mouse Button 3 Emulation	Ctrl
Serial Device Name	/dev/ttya2
Display Name	:0.0
Display Width Range	Low, 0, High, 1279
Display Height Range	Low, 0, High, 1023

Table 6-26 Touch Config Defaults

2.8 Networking

2.8.1 Host/SBC Network

VALIDATING OC/SBC NETWORK CONNECTION

There are two command line executables that can be used to check OC and SBC network configuration and status. They are **ifconfig** and **netstat**. They can be run from the host or the SBC.

2.8.1.1 ifconfig

The command **ifconfig** can be used to verify that the network interface is running and is correctly configured on *your system only*. The interface is defined as running when it has been probed, attached and started by the OS (host or SBC). There are several devices that are important to host and SBC network operation. On the host side they are the gateway (**ef0**) and the BIT3 (**vd0**) devices. On the SBC side they would be the control LAN (**ei0**) and BIT3 (**vd0**) devices. Use the **ifconfig** as follows to get configuration data about your network. At a command line on the OC, type **ifconfig** followed by the device you want to inspect. If your connected to host use **ef0** or **vd0**. If your connected to the SBC use **ei0** or **vd0**. An example of the **ifconfig** usage on the host follows:

Example: Using the **ifconfig** command to check the host network

```
>>ifconfig ef0
ef0:flags=1c63<UP,BROADCAST,NOTRAILERS,RUNNING,FILTMULTI,MULTICAST,CKSUM
inet 3.7.52.150 netmask 0xfffffc00 braodcast 3.7.52.0
```

IP addresses (e.g. *3.7.52.150*) will vary and depend on your own network configuration

```
>>ifconfig vd0
vd0:flags=8e3<UP,BROADCAST,NOTRAILERS,RUNNING,NOARP,MULTICAST>
inet 192.2.100.1 netmask 0xfffffc00 braodcast 192.2.100.255
```

2.8.1.2 netstat

The command **netstat** can be used to obtain network status about your network configuration *on your system*. At a command line on the OC, type **nestst** followed by the appropriate argument. Using the **-i** argument, you can obtain status on your system's network. Using the **-r** argument, you can obtain status on the devices routed by your network (e.g. an external suite). An example of the **netsat** usage initiated from the host using both arguments follows:

Example: Using the **netstat** command to check the network status

```
>>netstat -i
Name  Mtu  Network   Address    Ipkts   Ierrs  Opkts   Oerrs  Coll
ef0   1500 3.7.52    rhap25     655083  0      258478  1      141141
vd0   4336 192.2.100ct01_oc0    19178   30     20406   53     0
lo0   8304 loopback  localhost  965831  0      965831  0      0
>>netstat -r
192.2.100  ct01_oc0   0xffffff00      U      83     195    vd0
```

2.8.2 DICOM

2.8.2.1 Configuring the DICOM Network

Use the Gateway Host name for the Application Entity (AE) Title, the Gateway IP number for the DICOM Address and Port 104 for the CT/i scanner.

The CT/i DICOM configuration is set in `/usr/g/config/WLdcm.cfg`

`WLdcm` means Work List Server (software) for DICOM. Unsuccessful transfers are logged to the GE Error Log from WLServer. The most recent WLrsp.binx file with the biggest number in `/usr/g/config` is usually the one that failed to transfer.

2.8.2.2 How to add stations to network:

1.) Select Network from Image Works, go to Select Remote Host from the pull down menu.
2.) Select Add. Enter the IP address, station name, network protocol you want to use.
3.) Save.

2.8.2.3 DICOM Port Number

- Genesis stations (HiLight, HiSpeed): 104
- Non-Genesis stations: 4006

This lo0 entry also must be present in file `/etc/hosts` or the network will not work.

```
127.0.0.1        localhost
```

2.8.3 CT/i Image Networking Compatibilities with 3.6 and Later Software

The purpose of this section is to identify the networking compatibility between:

- CT/i to CT/i systems
- CT/i to AW systems
- CT/i to Genesis systems
- CT/i to 3rd party systems

The primary networking functions are:

- Query & Retrieve
- Send
- Receive

For the following discussion, see Figure 6-16.

Query - CT/i can query another CT/i, another Genesis system or any 3rd party image server that is a query provider.

Retrieve - CT/i can retrieve from another CT/i, another genesis system or any 3rd party image server that is a retrieve provider

Send - CT/i can receive images sent from another CT/i system, another Genesis system or an AW system.

Receive - CT/i can send images to anther CT/i system, another Genesis system or an AW system, and 3rd party image server that is a receive provider.

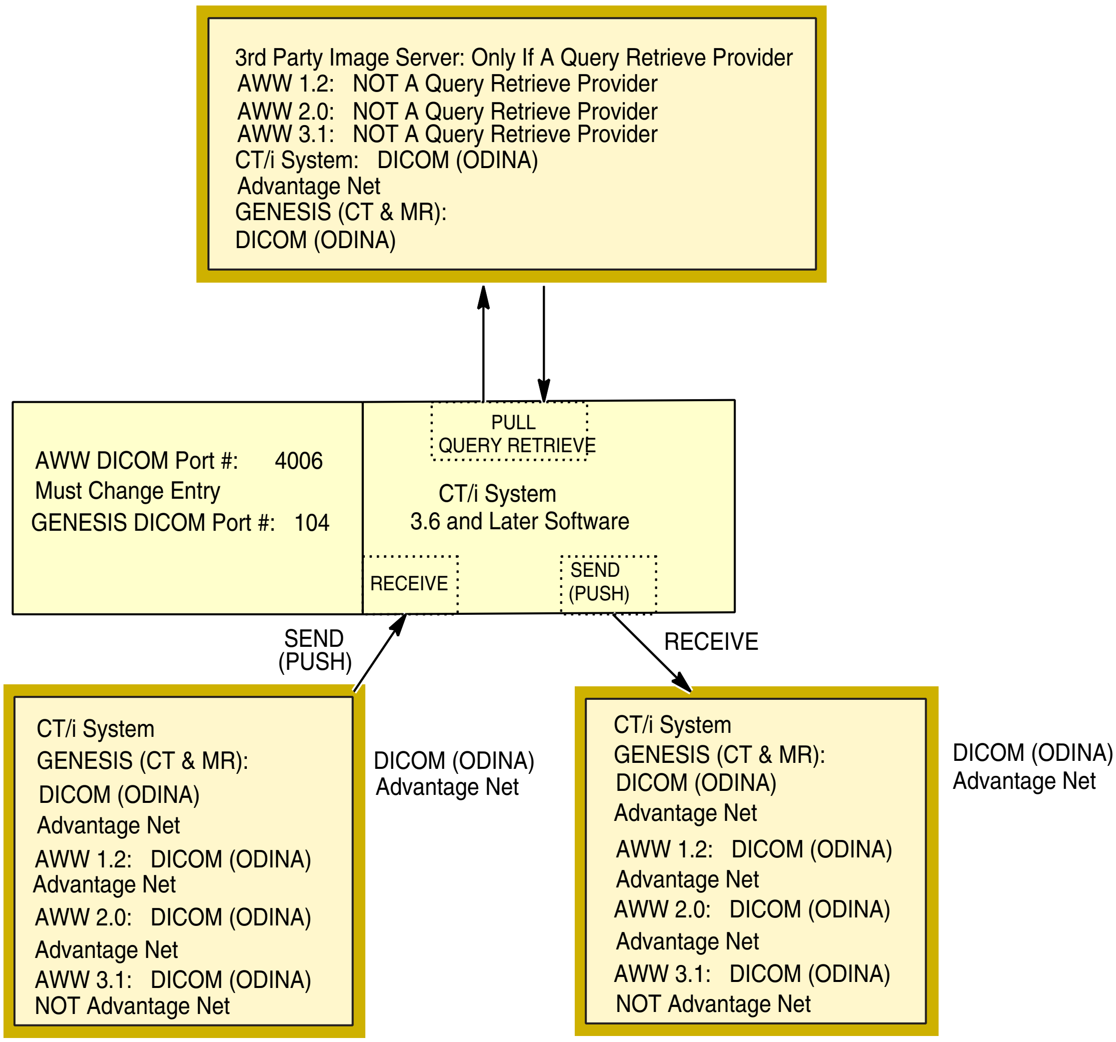


Figure 6-16 Networking Compatibility Diagram

2.8.4 2.8.4 Procedure to Create or Add a Static Route to the CT/i System

OVERVIEW

This procedure is used to turn off the routing daemon (if it is not already off), and add a default network route (static route) on a CT/i System that is part of a Hospital Network.

This applies to all HiSpeed CT/i software version 3.6 and above. The typical application is to connect a CT/i System to a network which uses a router or static routing instead of RIP.

Note that CT/i software version 5.3 and above (Octane computer) relies on static routing for the OC-SBC Bit-3 communication link (see the note on page 278 below).

PROCEDURE

It is recommended that you discuss your site's specific needs with the Network Administrator before performing this procedure. If you need assistance performing these steps, please contact the Network Support Group at the OnLine Center.

Note: Please be aware that if this procedure is performed on a system, it will need to be performed again following a software reload. Prior to performing a software reload, ensure that changes to the files addressed in this procedure are documented.

1.) Open a UNIX shell and switch user to root:

    `su -` (and enter the root password)

2.) Change directory as follows:

    `cd /etc/config`

3.) Create a backup copy of the static-route.options file:

    `cp static-route.options static-route.options.lfc`

4.) Determine the desired static route IP address(es) from the site's Network Administration. Add these desired static routes to the static-route.options file. It is preferred to use the “jot” text editor to modify the file, as “jot” is an X-Windows screen editor with an intuitive user interface.

    `jot static-route.options`

5.) Add the desired route address(es) at the end of the file, using the following syntax:

    `$ROUTE $QUIET add default` ***`www.xxx.yyy.zzz`*** (where this is the IP Address of the default router, provided by the site)

    or

    `$ROUTE $QUIET add -net` ***`www.xxx.yyy.zzz`*** (where this is the IP Address of the network/subnetwork, provided by the site)

    or

    `$ROUTE $QUIET add` ***`www.xxx.yyy.zzz`*** (where this is the IP Address of a specific host, provided by the site)

Note: Version 5.3 and above on Octane

If your system is operating with software version 5.3 and above (Octane computer), there will be an entry in this file for the OC-SBC route as follows:

`$ROUTE $QUIET add 192.2.100.1 192.2.100.2`

6.) Save the changes to the static-route.options file using the FILE pulldown menu, then exit “jot”

7.) Verify the entries made to the static-route.options file by typing:

    `more static-route.options`

8.) Reboot the system for the changes to take effect.

2.9 Error Messages (Firmware)

2.9.1 Using the Message Fields for Troubleshooting

The fields in the run time error serve as road signs for fault isolation and are used as shown here.

2.9.2 Message Log Layout

```
Date  Time
Suite:Suite name Host:Host name Proc:Process name Error:Error #
(File: File name Method: Method name Line#: Location ID ) does not appear
for service view level
Function: Major : Minor Function
Scan / Image Type: Scan/image Type Scan: Scan Specifics Image: Image
Specifics
```

```
Exception Level: level/flag Ticks: ticks (This line, firmware only)
Log Series: #####
service lines:
```

Comment: About Service Lines — *Specific and detailed information concerning the error including (but not limited to): - Additional problem description and cause information. Information indicating what was supposed to happen, what did happen, how was it detected (example is: expected value vs. actual value, name of file attempted to open, etc.) including any information concerning abnormal conditions related to the error. Network input/output information. Compute que information.*

2.9.3 Using Error Messages

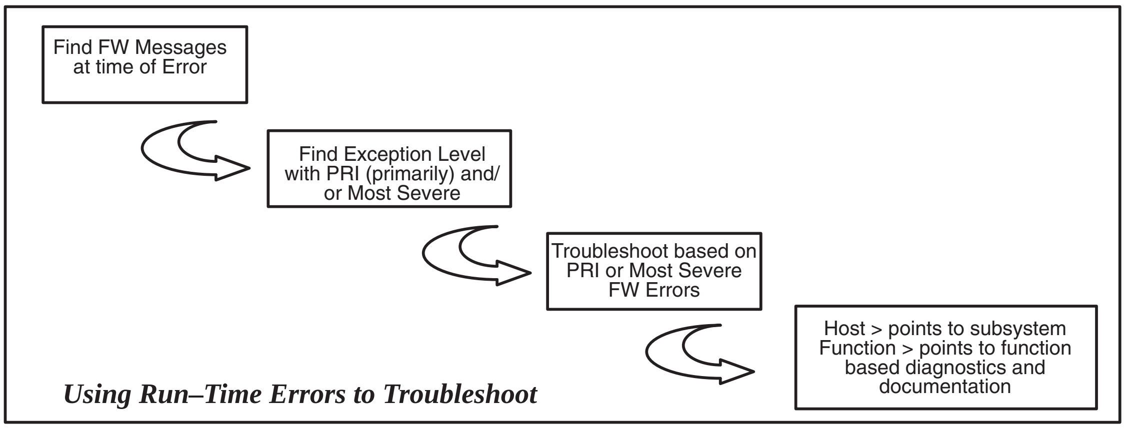


Figure 6-17 Utilizing Error Messages for Troubleshooting

Error Log Field descriptions are summarized below:

- `Date` and `Time` – date and time that the error message enter the error message log, NOT the time the error occurred.
- `Suite name` – name of suite.
- `Host name` – This is the name of the host processor (OC, STC, ETC, OBC) of the firmware which detected the error.
- `Process name` – Name of process which detected the error.
- `Error Number` – Unique error identification. Can be used for sorting, searching, and counting specific errors in error log. Can also be used as a pointer in documentation.
- `File name`, `Method name`, `Line #` – Does not appear when using the service view level
- `Function` – hierarchical breakdown of system functions uses major and minor functions.
- `Scan` – exam, series, scan # for raw file.
- `Image` – exam, series, image # for image file.
- `Type` – Scan type, that is, axial, scout, helical, cine, diagnostic.
- Exception Levels:
  - `Primary` = first error reported in a group of error messages.
  - `Secondary` = rest of the errors in a group.
  - `Most Severe` = message that has the most severe scanner exception handling consequences in a group of errors.
  - `Soft` = errors that have no effect on exception handling, but indicate system parameters that are beginning to approach the boundaries of satisfactory performance.
- `Ticks` – Ticks will indicate real error occurrence order (clocks are synchronized).
- `Log Series` – relates all errors recorded as part of a verification series.

2.10 Keyboard Shortcuts

FUNCTION	KEY
print image	**F1**
print page	**F2**
print mid	**F3**
print series	**F4**
screen save	**F8**
level up	↑ cursor key up (1)
level down	↓ cursor key down
width up	→ cursor key right
width down	← cursor key left
The arrow keys are just right of the **ALT** Graph key. With **NUM LOCK** off, numeric keypad arrows work same way.	
scroll image (nudge image)	shift + cursor keys
page up (in viewer)	**PAGE UP** (2)
page down (in viewer)	**PAGE DOWN**
With **NUM LOCK** off, **PG UP** and **PG DN** do same thing, with auto-repeat	
toggle graphic editing (in viewer)	**SHIFT** + **LEFT MOUSE**
magnify (in viewer)	**SHIFT** + **MIDDLE MOUSE**
cut graphics objects (in viewer) copy graphics objects (in viewer) paste graphics objects (in viewer)	**L6** **L8** **L10**
bring window to front	front (**L5**)
kill application/window	stop (**L1**)
iconify	props (**L3**)
delete last point of free-hand drawing (in viewer)	**BACKSPACE**

Table 6-27 Keyboard Shortcuts

Chapter 7
Software Tools

Section 1.0
Desktop Control Introduction

Use the mouse to access and operate these diagnostics and tools from the right hand display monitor, or open a shell, and type/enter a Unix command line. The system displays the Service Desktop Manager along the left hand side of the right side display monitor, as shown in Figure 7-2.

Use the mouse to make screen selections on the service desktop.

Typical mousebutton functions:

- Press Mousebutton one to select
- Press Mousebutton two to extend a selection.
- Press Mousebutton three to access pop-up menus.

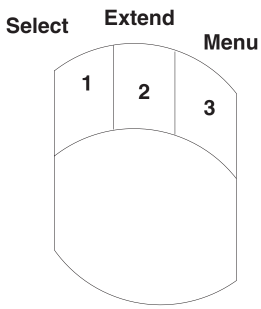


Figure 7-1 Mousebutton Definitions

Section 2.0
Service Desktop Main Menu

The service desktop (Figure 7-2) is the entry point for all service tools and diags. The desktop is designed with six major functional menu areas each with its own purpose. They are:

GENERAL SERVICE

- *Troubleshooting* Provides a list of key tools/diags required for system troubleshooting and performance measurement (See Section 15.0 on page 290).
- *Replacement Procedures* Displays a list of tools/diags required for checkout of replaced items (Section 17.0 on page 294).
- *PM* (Planned Maintenance) Set of files to view and tools to run during PM visit (Section 22.3 on page 304).
- *System Integration* Use to set up and align the system (Section 22.4 on page 305).
- *Utilities* Displays a set of product utilities (Cal/Scan/Image/DD analysis) plus some useful system utilities (Section 20.0 on page 296).

- *Log Viewing* Displays system log information (Section 22.6 on page 307).

ADVANCED SERVICE

- *Advanced Replacement Procedures* Set of key replacement step-by-step menus. Provides a custom set of Tools and Diagnostics for the user. Page 294.
- *Advanced Troubleshooting* Organized by Major Function, to provides access to system troubleshooting and performance measurement software. Page 292.
- *Advanced PM* (Planned Maintenance) Set of files to view and tools to run during PM visit. Page 295.
- *Advanced System Integration* Use to set up and align the system. Page 296.
- *Advanced Log viewing* Displays system log information. Page 299.

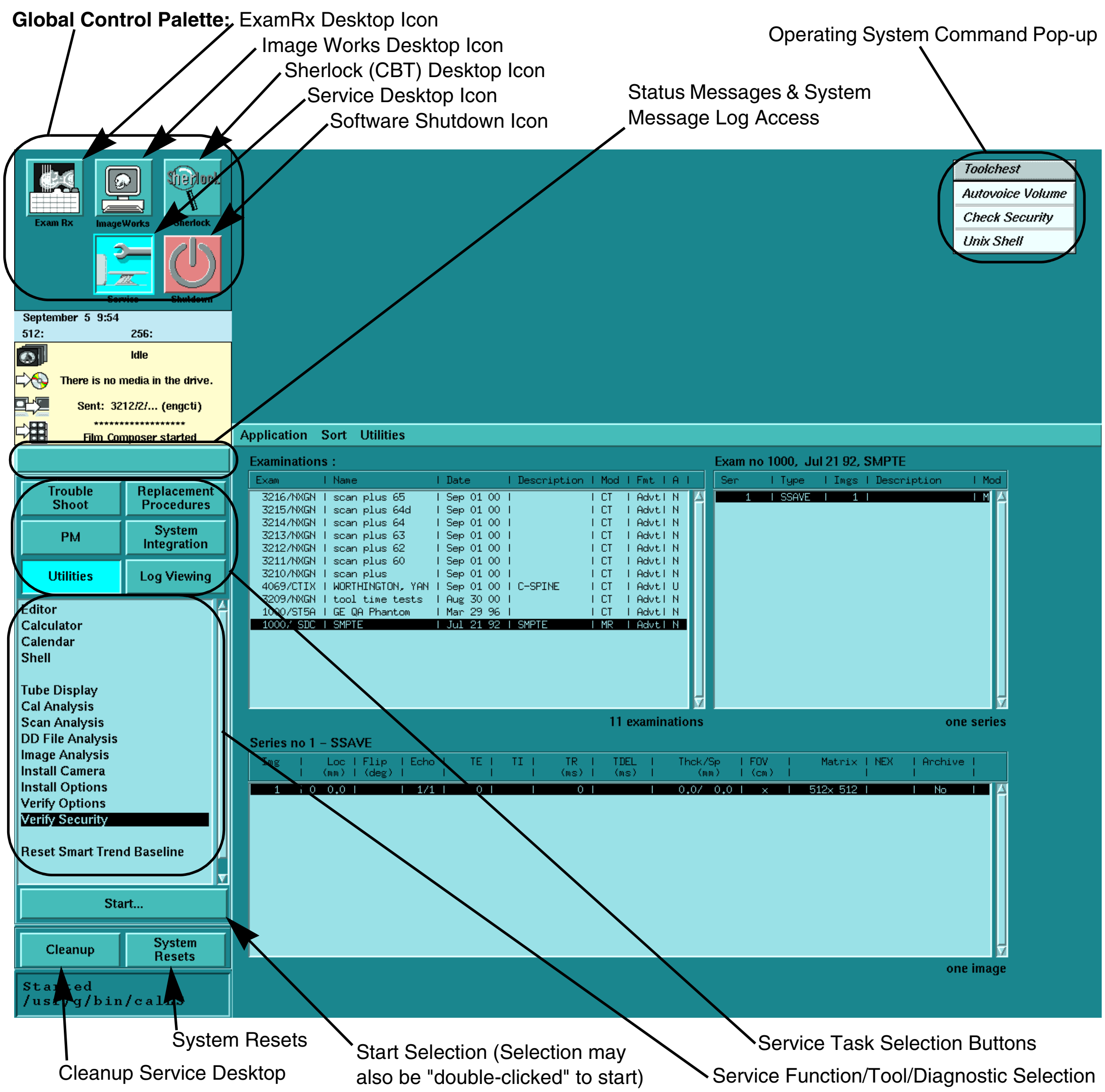


Figure 7-2 Service Desktop, Display Screen Overview

Note: The figure above is provided as an example only, and may not accurately reflect your screen.

Section 3.0 Menu Function Descriptions

The first half of this chapter briefly describes the basic service diagnostics and tools menus. The second half describes their procedures.

The HSA CT/i product has five distinct desktops, one of which is the Service Desktop. The user may move between desktops with the touch of a button on the Global Control Palette, which is always visible on all desktops. When changing desktops, the palette below the Global Control Palette is replaced with the appropriate desktop specific Control Palette. (See Figure 7-2) Switching desktops does not modify the current view of a desktop. Even though it may no longer be visible, it is still in the same state as when the switch occurred.

The users of the Service Desktop have different needs than the technologists, radiologists, doctors and other users of the system. Therefore, the functionality for the Service Desktop is different than those of the other desktops. Windows can be resized, iconified, overlapped and scrolled. This allows for greater flexibility for the user, especially in the area of trouble shooting where access to many different functions may be needed at the same time.

Section 4.0 Access - Advanced Service

Proprietary Class C LightSpeed software is distributed on a Class C CDROM, and is not part of the standard Class A/B software that is factory loaded or field installed during a software Load From Cold process. This Class C software must be loaded on the system before either the local service-person or an InSite engineer can use these proprietary tools.

Once Class C LightSpeed software is loaded on the system, the system grants access, if the software detects the presence of a service key (in-house, by license or GE Service), or if InSite is connected. For InSite sessions, the software enables an encrypted environmental variable and detects if the InSite user is valid. When InSite is running applications that cause table or gantry motion or enable high voltage, on-site confirmation is required before the tests are performed (see Table 7-1).

		ADVANCED CLASS C SOFTWARE LOADED ON THE SYSTEM?	
		YES	NO
INSITE RUNNING?	YES	• Remote Class C Access • Local Class C Access w/ Security Key • Local Class A/B Access w/o Security Key • Local Confirmation Required for Remote Motion or HV Tests	• Remote Class A/B Access Only • Local Class A/B Access Only • Local Confirmation Required for Remote Run Motion or HV Tests
	NO	• Local Class C Access w/ Security Key • Local Class A/B Access w/o Security Key	Local Class A/B Access with or without Security Key.

Table 7-1 Advanced Materials Access Matrix

The Service Desktop controls access to the Class C software. The task selection buttons remain consistent regardless of whether Class C software of the service key are installed. However, the tool lists displayed in the left navigational area of the Service Desktop may be different, as well as the contents of tool capabilities from the displayed list. InSite users will always see the Class C version of the Service Desktop, as long as Class C software is loaded on the system.

Section 5.0
Security Key Verification

1.) On a system that hasn't had a security key installed since re-boot, enter the service desktop. The non-proprietary menus should be available.

2.) On the **Utilities** menu select “**Verify Security**.” A shell should come up with the text “Security Hard Key is not present.”

3.) Insert the security key. Select “**Check Security**” from the root menu. Wait 15 sec and switch desktops. The proprietary menus should be available.

4.) Repeat step (2). The text should say “Security Level: Proprietary” or “Security Level: class M” depending upon the type of security key installed. The service key's expiration date should also be posted.

Section 6.0
Procedural User Interface

The Service Desktop contains a mixture of Tools and Diagnostics to be used by a Service Engineer. The main philosophy behind the user interface for the Service Desktop is to provide a **procedural** approach to servicing the scanner. There is no longer the concept of separate Tools and Diagnostics screens. All the necessary tools and diagnostics are available at the same time for the procedure at hand, whether it be trouble shooting, replacing a part or performing routine maintenance.

See Figure 7-3 and Figure 7-4 for the Service Desktop Control Palette. Selecting one of the buttons at the top of the Palette will cause a new list to be displayed in the middle of the Palette. In Figure 7-3, REPLACEMENT PROCEDURES has been selected and the list contains software elements that are needed to preform Replacement Related Procedures.

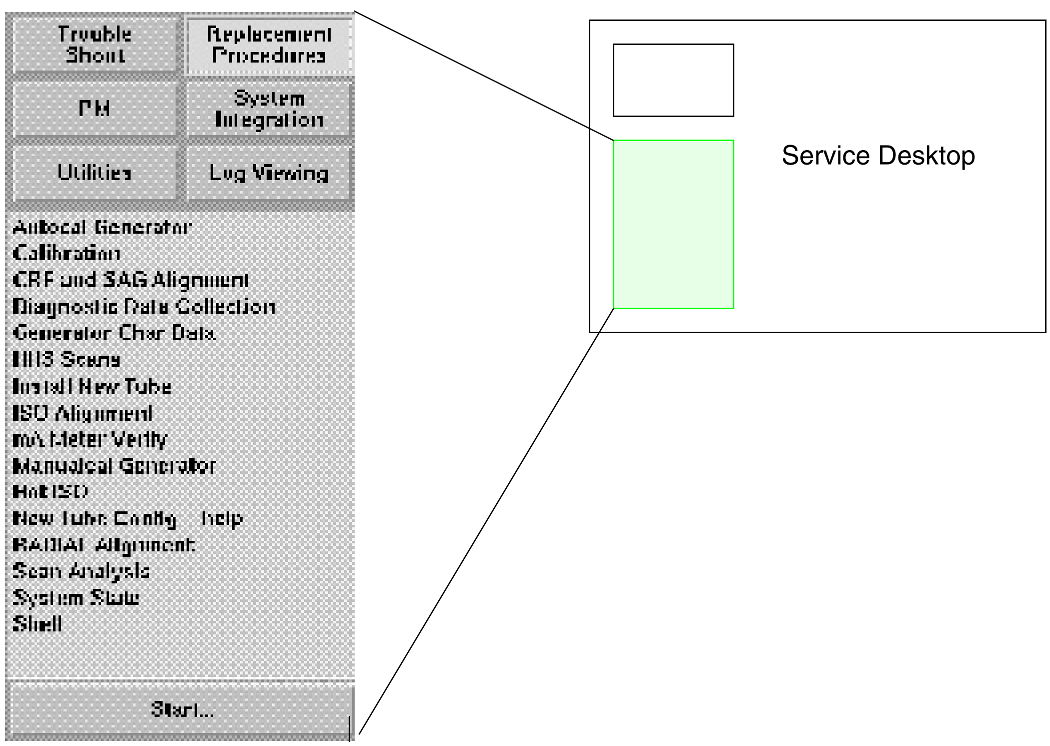


Figure 7-3 Service Desk-Top Control Palette (General Service)

In Figure 7-4, REPLACEMENT PROCEDURES has been selected and the list contains software elements that are needed to preform Replacement Related Procedures. Major items that can be installed or replaced. An item can be selected from this list which then causes another screen to open. In the example, HV SUBSYSTEM CHANGE has been selected and Figure 7-5 shows an example of the HV Subsystem Change Menu.

The Menu Screen in Figure 7-5 shows a list of procedures that may need to be performed for replacements in the HV Subsystem. The order that the items are in the menu, will usually indicate the order that the procedures would normally be completed in.

The small ‘square’ to the left of the menu item description will change colors after that item has been selected and exited from. This can be used as a reminder as to which procedures have been used.

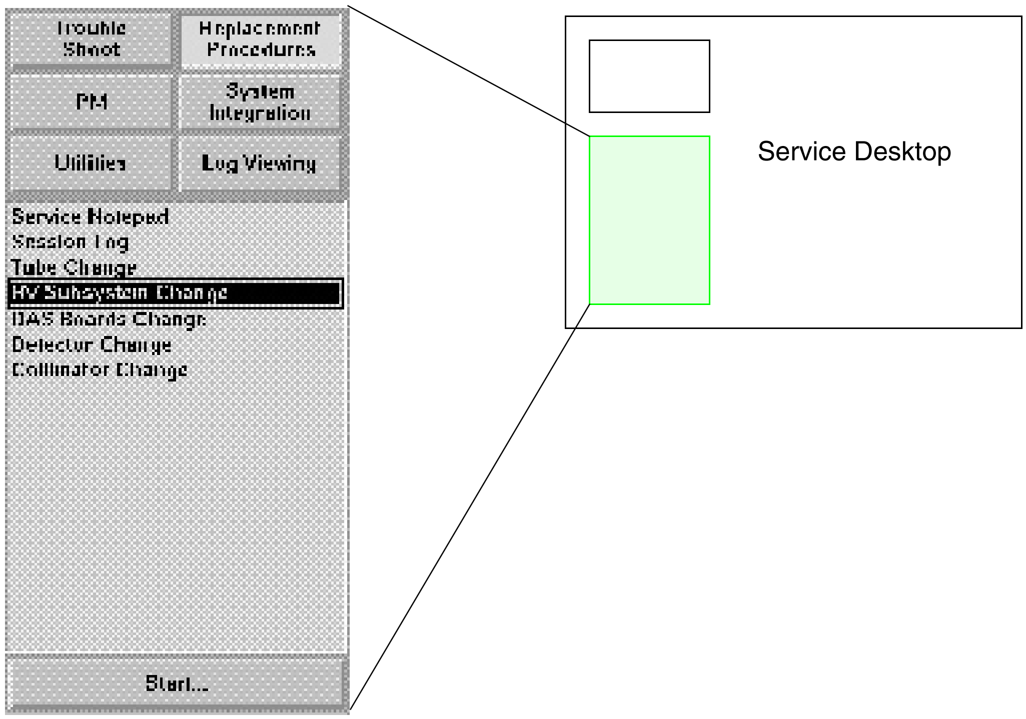


Figure 7-4 Advanced Service Desktop Control Palette Example

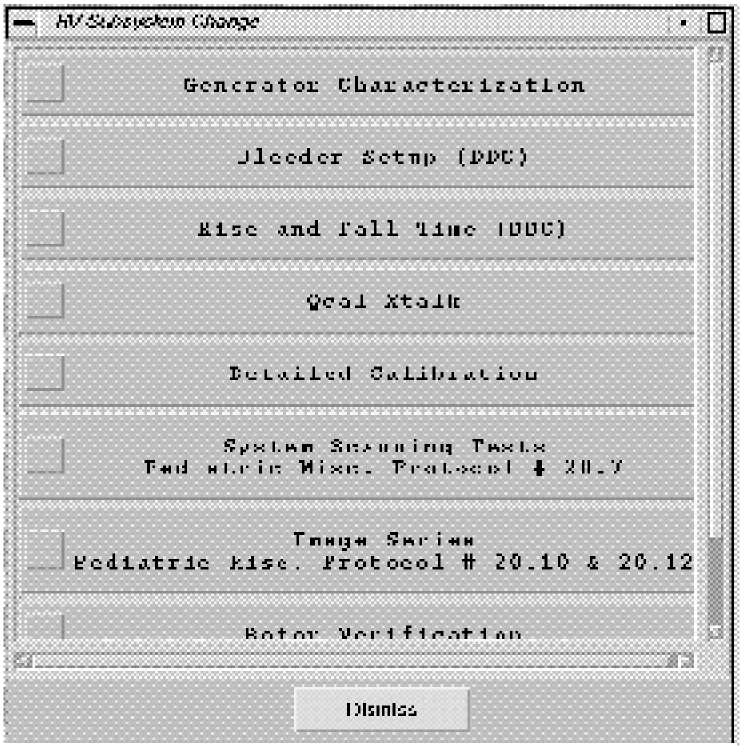


Figure 7-5 Next Menu Appears from the first choice (Advanced Service)

Section 7.0 Smart Downloads

The HSA CT/i is the next step in the evolution of the HSA Family of Scanners. The SBC and scan chassis remains roughly the same as the RP2.x. As in RP2.x, many of the CT/i diagnostics will need diagnostics firmware to be loaded into the STC, OBC and ETC. In RP2.x, entering the Diagnostics layer of screens automatically loaded this firmware – a 90 second delay. Leaving the Diagnostics screens caused the same delay to reload the application firmware back into the controllers. With the new CT/i Service Desktop strategy of allowing access to both tools and diagnostics on the same screen, we run the risk of causing this time delay a lot more frequently.

To address this concern, we have introduced the concept of “smart downloads”. This means that the appropriate firmware will only be loaded into controllers when absolutely necessary. Not all diagnostics need the diagnostics firmware and not all tools need the application firmware downloaded.

Refer to Section 16.0 for information as to how the screens are organized to assist in determining when downloading of the controllers will occur.

Section 8.0 Desktop Inter-Operability

To reduce the need for the Service Engineer to switch to other desktops, some applications from other desktops are also included on the Service Desktop (such as basic display and manipulation of patient image data).

Section 9.0 Service Desktop Management

Change desktops by selecting the corresponding desktop icon from the top left area of the right display monitor. See Figure 7-2.

Launch, or start each service tool or diagnostic one of two ways.

- Highlight the tool, then select the START... button, near the bottom of the desktop.
- Double click the mouse on the tool to invoke it.

The CLEANUP button on the bottom of the desktop “cleans up” any previously opened windows, and restores the desktop to its original state.

Note: CLEANUP reloads firmware

If you ran diagnostics that required diagnostic firmware, the CLEANUP button will also reload the application firmware.

The DISMISS button cleans up, then returns to the Service Desktop trouble shoot menu.

The SYSTEM RESETS button displays the reset menu for various product or application firmware.

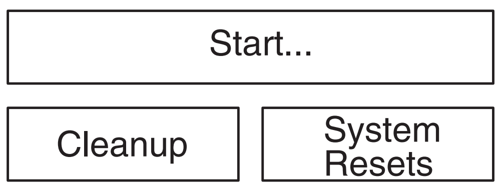


Figure 7-6 Desk Top Management Button

Section 10.0 Service Desktop Clean Up

When the Service Engineer is done with the system, rather than having to close or cancel each application still visible on the Service Desktop, a CLEANUP button is available which returns the desktop to an initial state.

The CLEAN UP Button should be selected whenever the User is done with the Service Desktop or whenever it is desired to get the Desktops back to a known state.

Section 11.0 Exit the Service Desktop

Select CLEAN UP**,** then the Desktop Icon, to exit the Service Desktop.

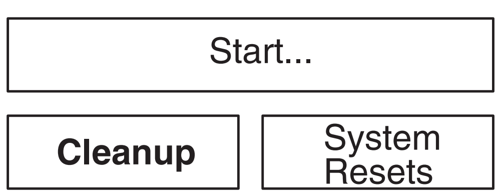


Figure 7-7 Desk Top Management Button - Clean Up

Also see Figure 7-2.

Section 12.0
System Resets

Use the SYSTEM RESETS choice to reach the menu that enables you to reset the software processes that control the hardware. If the Applications ScanMgr is stuck, use the Scan Reset. If the Applications Reconstruction is stuck, use the Recon (Image Generator) Reset. Use the choices under Diagnostics if the system needs a reset during Service tests.

1.) Select SYSTEM RESETS from the Service Desktop Manager (SDM).

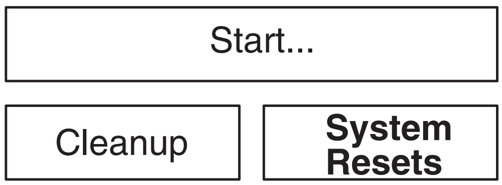


Figure 7-8 Desk Top Management Button - System Resets

2.) Select the reset from the displayed screen, then select, RUN.
The Status box displays the status of the selected reset.

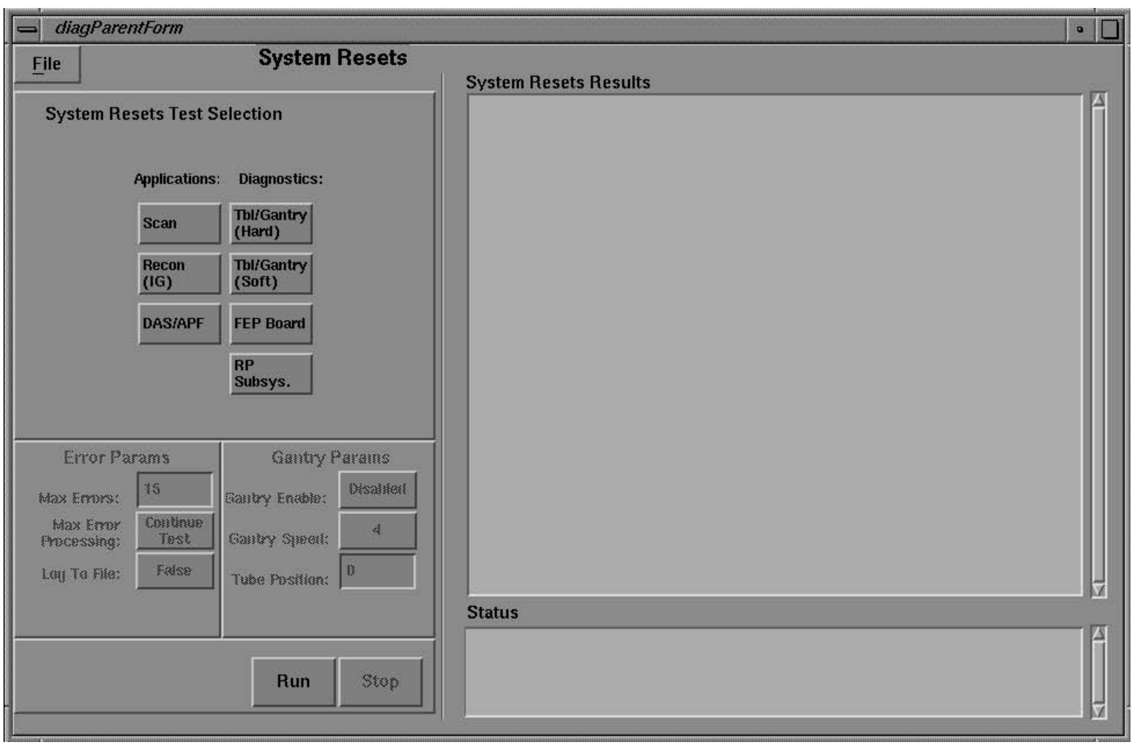


Figure 7-9 System Resets Screen

Note: RUN checks for the existence of firmware

When you select RUN, the system checks for the firmware software in the scan subsystem. If required, the system downloads the firmware, after it posts a message asking you to confirm the firmware download.

The CLEAN UP Button should be selected whenever the User is done with the Service Desktop or whenever it is desired to get the Desktops back to a known state.

Section 13.0 Session Log

The intent of Session Log is for Service Engineers to communicate their activities such as: “I replaced the KV board” or “I performed quarterly PM.” It also records whenever the system is re-booted or re-configured.

Session Log has a simple user interface that allows the user to record some text with their name and a date. There is a simple search capability. The file that holds this text is saved and restored with System State.

Section 14.0 Trouble Shoot Major Functions Menu

Each major function contains three separate sections. This division allows the user to see which tools/diags/utilities require application firmware, *or* diagnostic firmware, *or* operate independent of either firmware. This helps you determine the most efficient course of action when you troubleshoot a problem.

The Service Software divides into three areas:

- Diagnostic Firmware Needed
- Application Firmware Needed
- No Firmware Needed

The replacement procedures in this area walk the user through the procedural steps normally found in the service manual. These procedures are organized to help the user step through each test and improve the speed and efficiency of each replacement.

- Collimator Change:

  This procedure directs the user through the steps necessary to test the replacement of the collimator. The steps consist of different tools, linked together in step-by-step fashion, to aid in the verification of correct system operation after the FRU change. As each test completes, the software shades the corresponding box, to help you track the testing progress.
- DAS Board Change:

  These tests walk the user through verification of a DAS Board change, in the same manner as above.
- Detector Change:

  The detector change step-by-step procedure involves several steps of the alignment procedure, as well as calibration.
- High Voltage Subsystem Change:

  The High Voltage Subsystem Change menu contains a collection of the most frequently used tools when a change is made in the HV Subsystem.
- Tube Change:

  Specific steps for these change procedures guide the user through the normal checkout phases after he or she changes the X-Ray tube.

The Log Viewing menu provides access to various system log files.

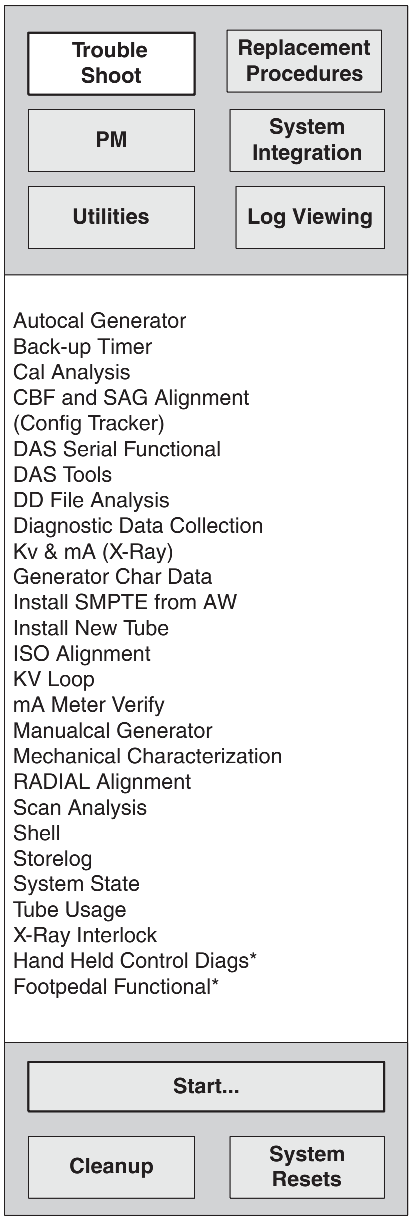
See Figure 7-16.

Section 15.0
Trouble Shoot Menu

1.) Display the **Service** Desktop.

2.) Select TROUBLE SHOOT to display the HSA CT/i system top level Troubleshoot menu. Refer to Figure 7-10 and Figure 7-11.

Note: Please refer to Screens on page 301.



CT/i Pro Systems with SmartView Option

Figure 7-10 General Service Desktop Troubleshooting Menu

TROUBLE SHOOT CHOICES

Note: Parentheses in table indicate future features

Parentheses surrounding a name on the menu indicates it is a planned feature that is not yet implemented. If you select it, a Unix Shell Tool will probably open.

Use the Trouble Shoot menu to access the following tools and diagnostics:

TOOL	DESCRIPTION
Autocal Generator	Automatically updates the x-ray generator characterization files.
Back-up Timer	Activates the exposure backup timer
Cal Analysis	Use to examine calibration information.
CBF and SAG Alignment	Use to check the Center Body Filter (CBF) and System Angular Geometry (SAG) Alignments for the focal spot, relative to the collimator and detector.
(Config Tracker)	Not yet available. Gathers information about the system configuration.
DAS Serial Functional	Use to test the serial communications link between the OBC and DAS
DAS Tools	Use to exercise and verify all scan data acquisition functions, like **microphonics.**
DD File Analysis	Use to view and analyze the diagnostic data files, cal, image or scan files.
Diagnostic Data Collection	Performs many service scanning functions and tests. Gathers system data with or without X-Ray, rotating or not.
KV & mA	Use to perform *X-ray Functional Tests*.
Generator Char Data	Use to examine the X-Ray generator Characterization files.
Install SMPTE	Use to install the SMPTE pattern, and display it as a patient image.
Install New Tube	Updates the system resident tube information file.
ISO Alignment	Tube ISO Alignment.
KV Loop	Tests the kV board.
mA Meter Verify	Verifies the mA metering circuit adjustments.
Manualcal Generator	Use to manually adjust the X-Ray generation characterization files.
Mechanical Characterization	Use to set-up the mechanical characterization files.
RADIAL Alignment	Use to measure the focal spot/detector radial alignment.
Scan Analysis	Use to List/Select and examine scan data.
Shell	Opens a Unix Shell Window where you can enter IRIX or UNIX commands.
Storelog	If Apps are shutdown first, it can store log files to MOD, then it removes those files from the system disks making more disk space available. If the host finds it needs more disk space when it boots, it will run storelog to make room.

Table 7-2 Tools & Diags Accessible through the Troubleshooting Menu

TOOL	DESCRIPTION
System State	Save and restore system configuration and calibration information to MOD if Applications are Shutdown (under Utilities). *Prevent this MOD from being labeled as an IMAGE ARCHIVE MOD because this step will format the MOD differently without mentioning there are system files on it. To investigate an IMAGE MOD, use a Shell and the DOS MODE commands shown on page 274* *DO NOT SAVE State after you reload software UNTIL you restore the REAL State; new software puts system defaults on the disk. If you save state before you restore, you will be saving defaults.*
Tube Usage	Displays X-Ray tube related information for current and previous X-Ray tubes.
X-Ray Interlock	Tests the exposure interlocks.
Hand Held Control Diags	Use to run Hand Held Control diagnostics (used in CT/i Pro systems with the SmartView scanning option).
Foot pedal Functional	Use to run Foot pedal functional testing (this FRU is used in CT/i Pro systems with the SmartView scanning option).

Table 7-2 Tools & Diags Accessible through the Troubleshooting Menu (Continued)

Section 16.0
Advanced Troubleshooting Menu

Note: Please refer to Screens on page 301.

The Advanced Troubleshooting menu provides access to a list of Major Function elements of the HiSpeed CT/i system. Select one of these Major Functions, and double click or select START to display a window that contains the tools and diagnostics used when troubleshooting this functional area of the system. Refer to Chapter 5 for Block diagrams that map the system functions to the physical hardware involved.

Figure 7-11 shows the Image Quality, System Communications, and System Power Control Troubleshooting screens. Each window has four areas:

1.) The first column lists tools and diagnostics that require the download of diagnostic firmware to the scan control sub-system.
2.) The second column lists tools and diagnostics that require the download of application firmware to the scan control sub-system.
3.) The third column lists tools and diagnostics that do not require the download of any firmware to the scan control sub-system.
4.) The bottom of the screen provides three functions related to the operation of the desktop window.
    - CLEANUP removes any windows left open from the service session and stops any processes left running.
    - DISMISS removes the corresponding window from the display, but does not stop any software you started from these menus.
    - START SELECTED TASK starts the highlighted software process in the window (provides the same action as double-clicking a menu selection).

Grouping the Tools and Diagnostics associated with troubleshooting a functional area of the system provides a convenient reminder of the available system software related to this function.

Note: Parentheses surrounding a name on the menu indicates it is a planned feature, one not yet implemented. If you select it, a Unix Shell Tool will probably open.

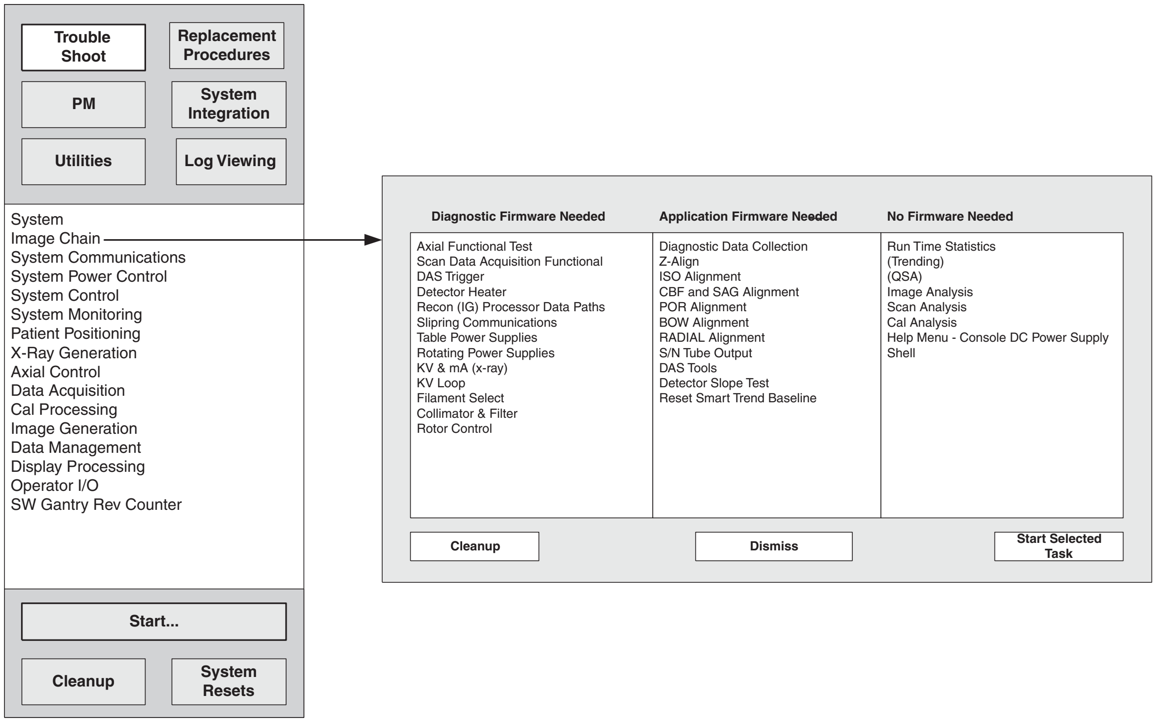


Figure 7-11 Advanced Service Troubleshooting Menus

Section 17.0
Replacement Procedures

Note: Please refer to Screens on page 301.

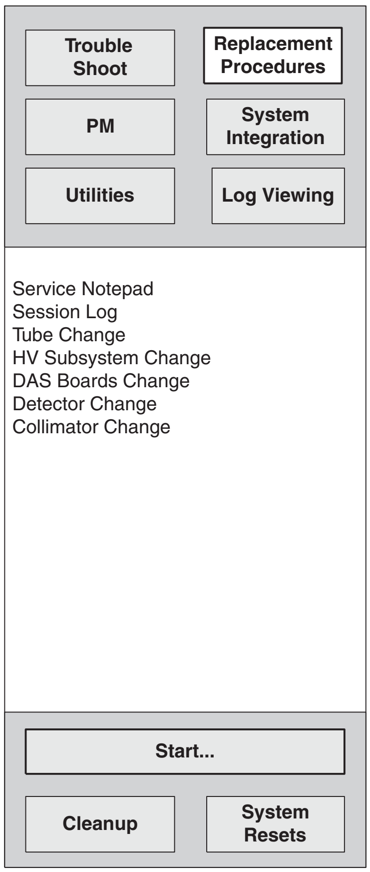


Figure 7-12 Advanced Replacement Procedures Menu

SERVICE NOTEPAD

Use to create a message for GE InSite and other Field Engineers. It will appear on the Health Page.

SESSION LOG

Displays FE comments, InSite comments, Tool and Diagnostic Start Messages and what images InSite looked at. Can be configured not to do any of these.

TUBE CHANGE

Use this option to display a list of tools and tests required when replacing the X-Ray Tube.

HV SUBSYSTEM CHANGE

Use to display a list of tools and tests required when replacing a High Voltage sub-system FRU.

DAS BOARDS CHANGE

Use this option to display a list of tools and tests required when replacing a DAS board.

Section 18.0
PM Information Menu

Note: Please refer to Screens on page 301.

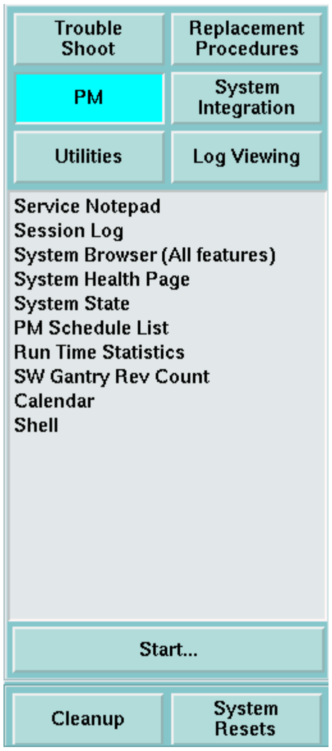


Figure 7-13 Advanced PM Menu

SERVICE NOTEPAD

Use to create a message for GE InSite and other Field Engineers. It will appear on the Health page.

SESSION LOG

Displays FE comments, InSite comments, Tool and Diagnostic Start Messages and what images InSite looked at. Can be configured not to do any of these.

SYSTEM HEALTH PAGE

Summarizes several key system performance parameters. Presents Service Notes from the user. Can be configured not to show any of these items.

SYSTEM STATE

Use this utility to save or restore configuration, characterization, calibration and protocol files. Use it with a system formatted (mkfsMOD) MOD during system reconfiguration or software load.

PM SCHEDULE LIST

Presents a visual checkout of completed PM item and access to PM activities. They are listed under categories Console/Computer, Gantry, HV, Tube, Hardkey/Keyboard, System, Images. System State and Storelog are two activities that are offered here too.

RUN TIME STATISTICS

Shows key system statistics such as tube spits, rotor up-time and various system temperature readings. This tool allows the Field Service or OLC Engineer to save this information into a text file for retrieval via InSite.

Section 19.0 System Integration Menu

Note: Please refer to Screens on page 301.

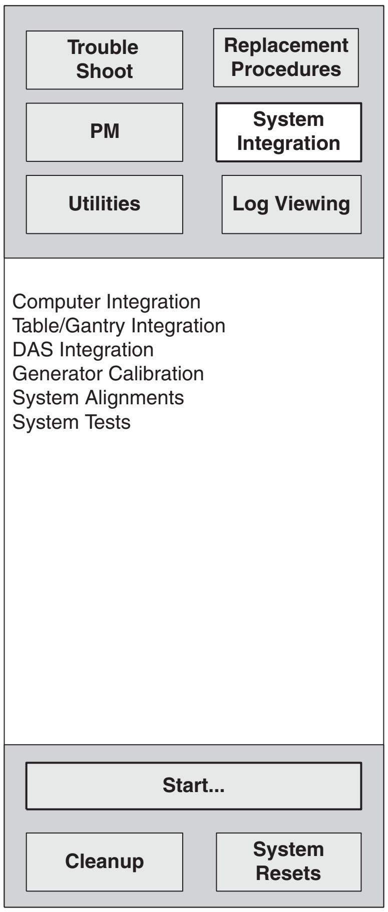


Figure 7-14 Advanced System Integration Menu

Section 20.0 Utilities Menu

Note: Please refer to Screens on page 301.

EDITOR

Opens a text editor that enables you to view file structure and file content. The default location is `/usr/g/bin`. Default operation is View Only which is the safest way to use this tool. If you change a system process file you'll have to reload software.

SHELL

Presents a window that enables you to enter IRIX (OC) and UNIX (SBC) commands. Example: Enter: `hinv` to get the same information that the OC Hardware Info menu item offers. Another good Shell tool is FSST.

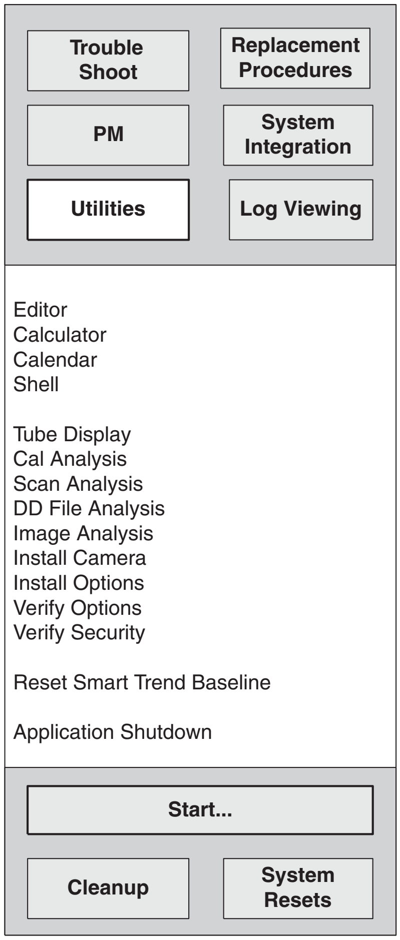


Figure 7-15 Advanced Utilities Menu

TUBE DISPLAY

Shows you the X-ray Tube's serial and model numbers, its meter reading, and install date.

CAL ANALYSIS

Puts calibration data into a UNIX file you can review, store and restore.

SCAN ANALYSIS

Enables you to view and analyze scan data.

DD FILE ANALYSIS

Use to view and analyze the diagnostic data files, cal, image or scan files.

IMAGE ANALYSIS

Use it to evaluate images for artifacts.

INSTALL CAMERA

Use this option to set up a laser or Dicom camera.

VERIFY OPTIONS

See what options the host thinks it has.

VERIFY SECURITY

Use to force the system to check the security level when you want to change that level. You will have to toggle some buttons to get the Service Menus to change.

RESET SMART TREND BASELINE

Creates a new baseline file for the Smart Trend tool (CT/i software version 5.x only). This is required in the event that a x-ray tube, detector or DAS converter card is replaced.

APPLICATION SHUTDOWN

Stops the scanning level of software but keeps the oc and sbc responsive to Irix or Unix commands and GE scripts. Apps should be down to run reconfig, storelog, spltst, newTu, and to check the sbc disk drives.

Section 21.0
Log Viewing Menu

Note: Please refer to Screens on page 301.

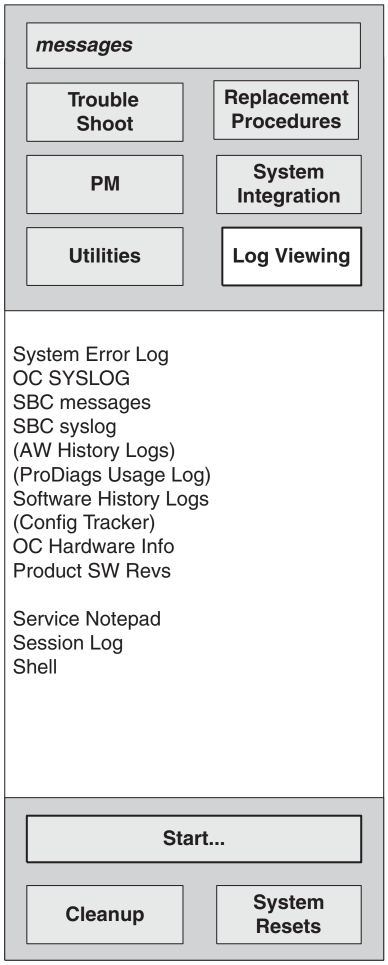


Figure 7-16 Advanced Log Viewing Menu

MESSAGES

Presents the current messages. Offers a button that lets you show all the messages as a LOG with the option of filtering that LOG to just see SERVICE messages.

SYSTEM ERROR LOG

Presents errors generated by failing application processes. It takes about a minute to appear.

OC SYSLOG

This system log viewer displays error messages stored in `/var/adm/SYSLOG`. These messages consist of the SGI IRIX host error messages that indicate bootup problems, SGI hardware failures or warnings.

SBC MESSAGES

Presents messages from the SBC UNIX operating system and its device drivers. The disk drives for the SBC still use the efs filesystem while the OC uses the xfs filesystem.

SBC SYSLOG

Presents messages generated by the scan control hardware and reconstruction processes that can occur while the host is working. **Parentheses mean that application is not yet available.**

OC HARDWARE INFO

Lists important information about the SGI host.

PRODUCT SW REVS

Lists the various software versions used throughout the system.

SERVICE NOTEPAD

Use this to create a message for GE InSite and other Field Engineers. It will appear on the Health page.

SESSION LOG

Displays FE comments, InSite comments, Tool and Diagnostic Start Messages and what images InSite looked at. Can be configured not to do any of these.

SHELL

Presents a window that enables you to enter IRIX and UNIX commands, start scrips that perform a series of commands, or programs. Press **ALT+F12** to exit the shell when it is no longer needed.

Section 22.0 Screens

GENERAL SERVICE

ADVANCED SERVICE

22.1 Troubleshooting Screens - General Service

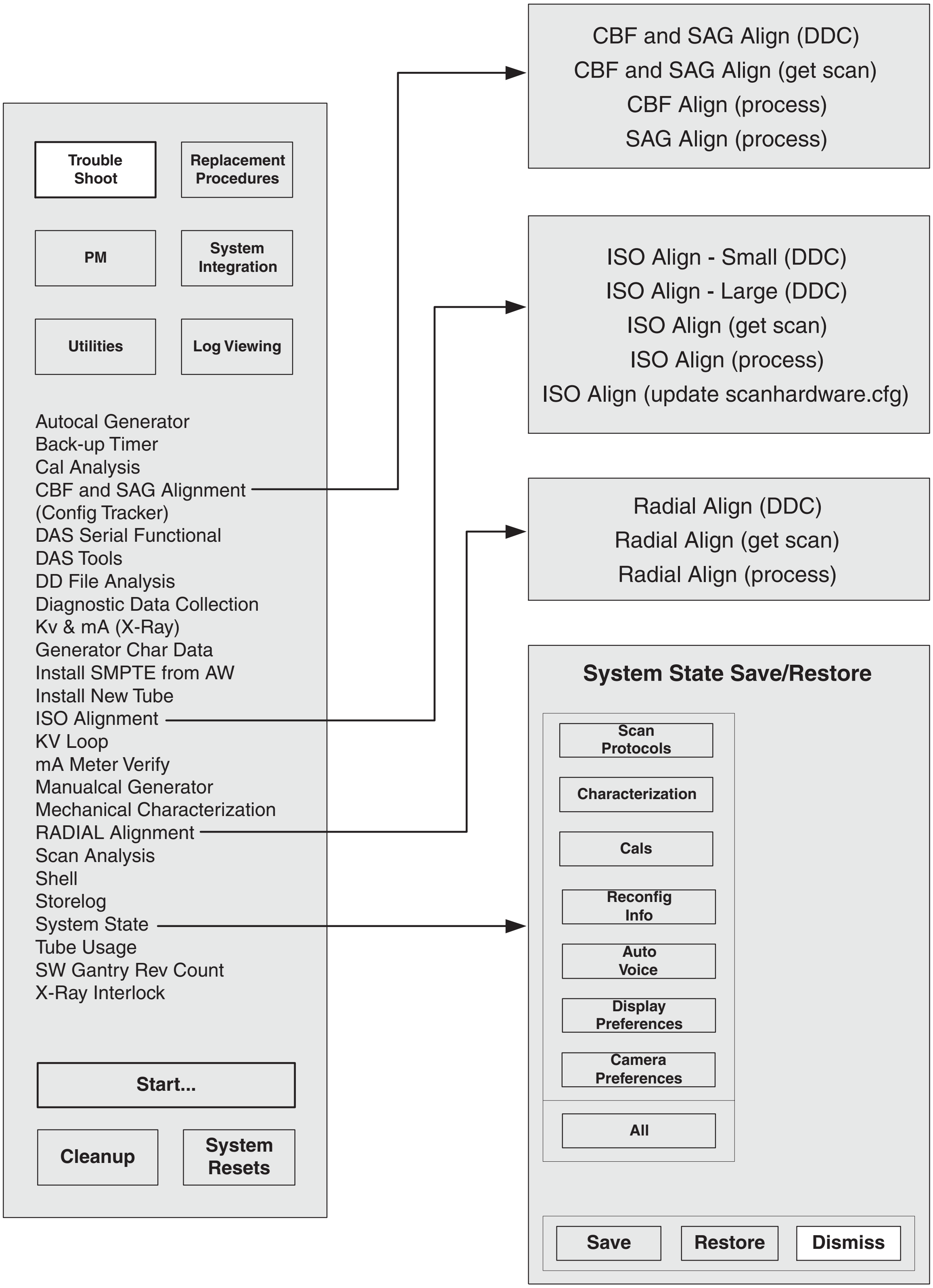


Figure 7-17 Troubleshooting Menus - General Service

22.2 Replacement Procedures - General Service

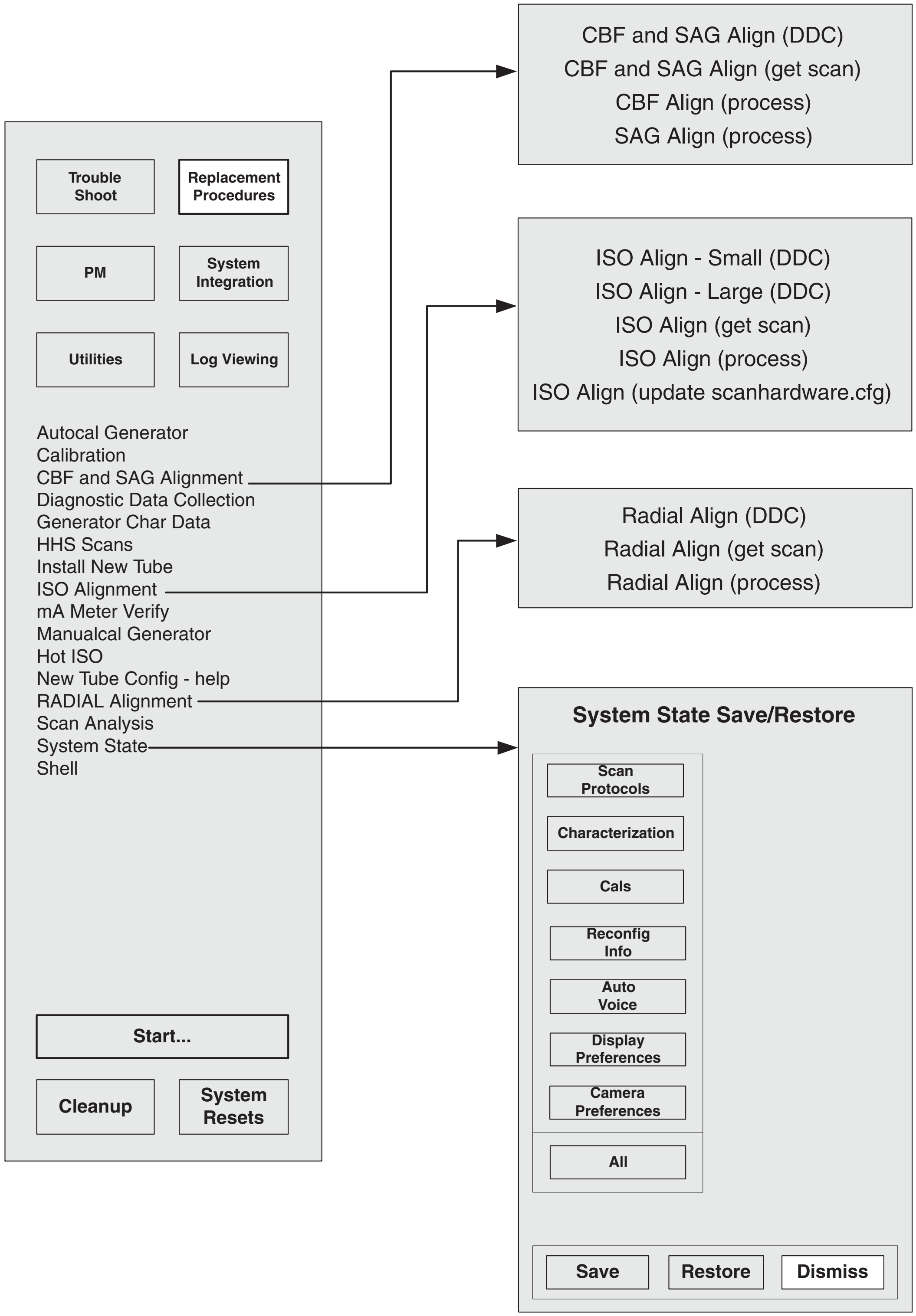


Figure 7-18 Replacement Procedures Menus - General Service

22.3 Planned Maintenance - General Service

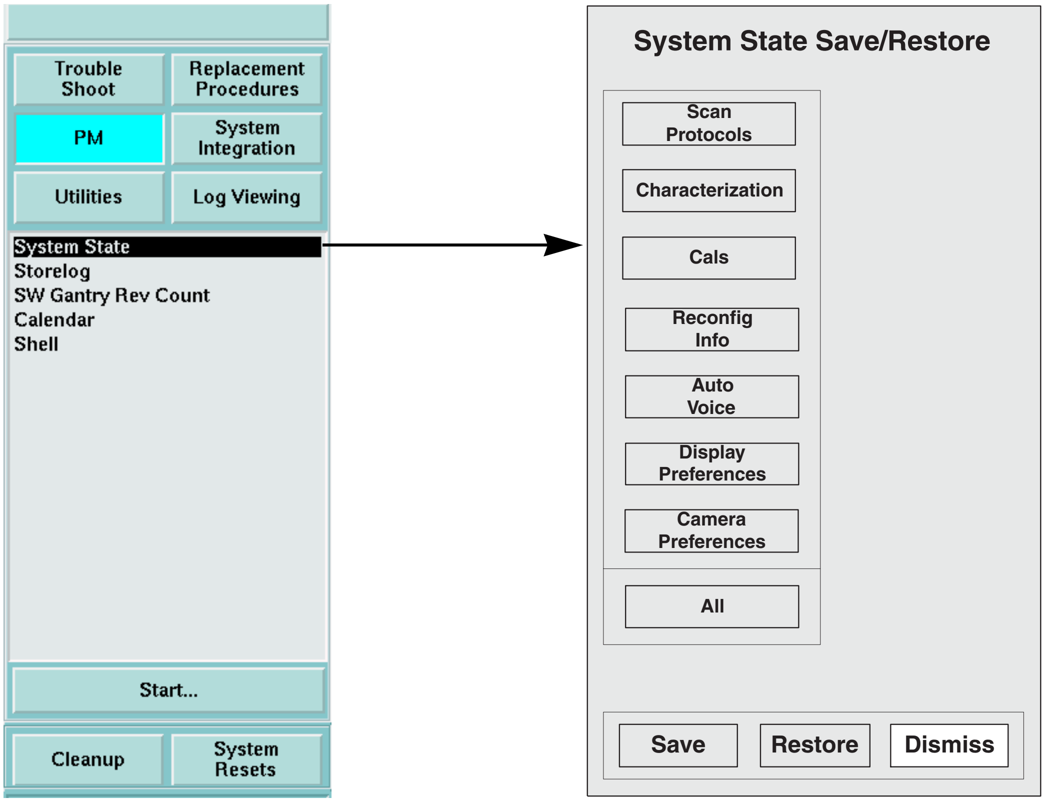


Figure 7-19 Planned Maintenance Menus - General Service

22.4 System Integration - General Service

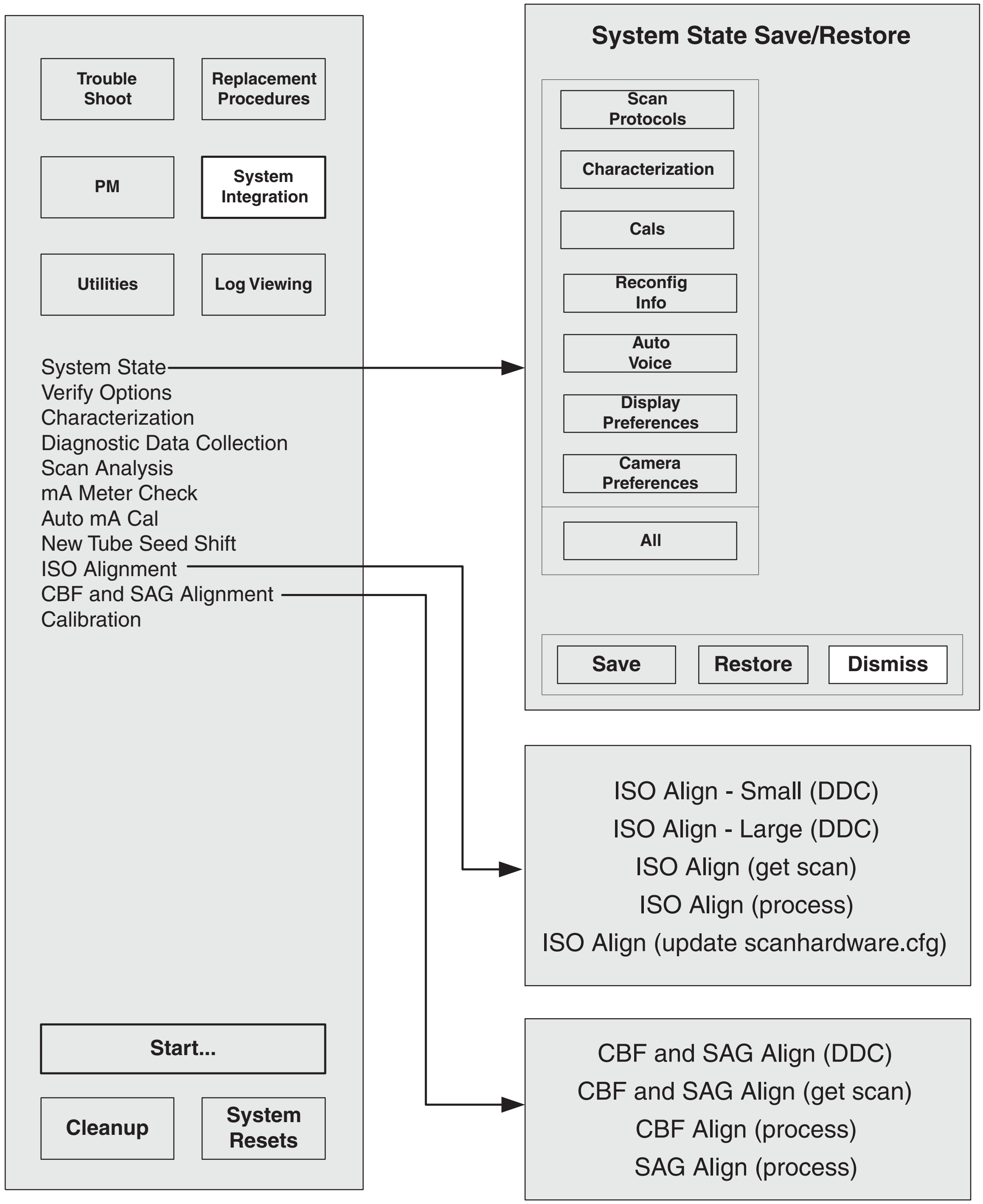


Figure 7-20 System Integration Menus - General Service

22.5 Utilities - General Service

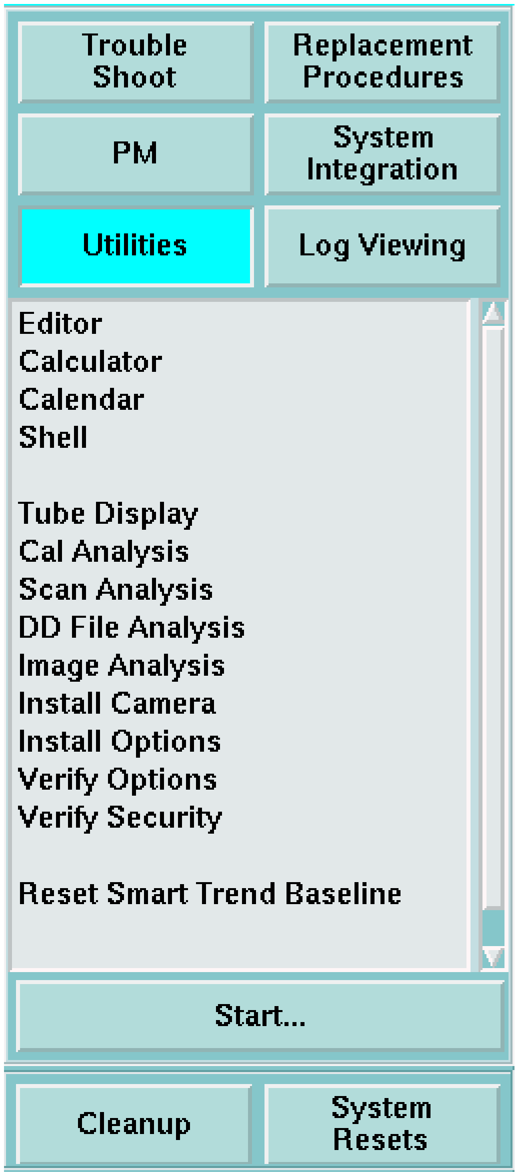


Figure 7-21 Utilities Menu - General Service

22.6 Log Viewing - General Service

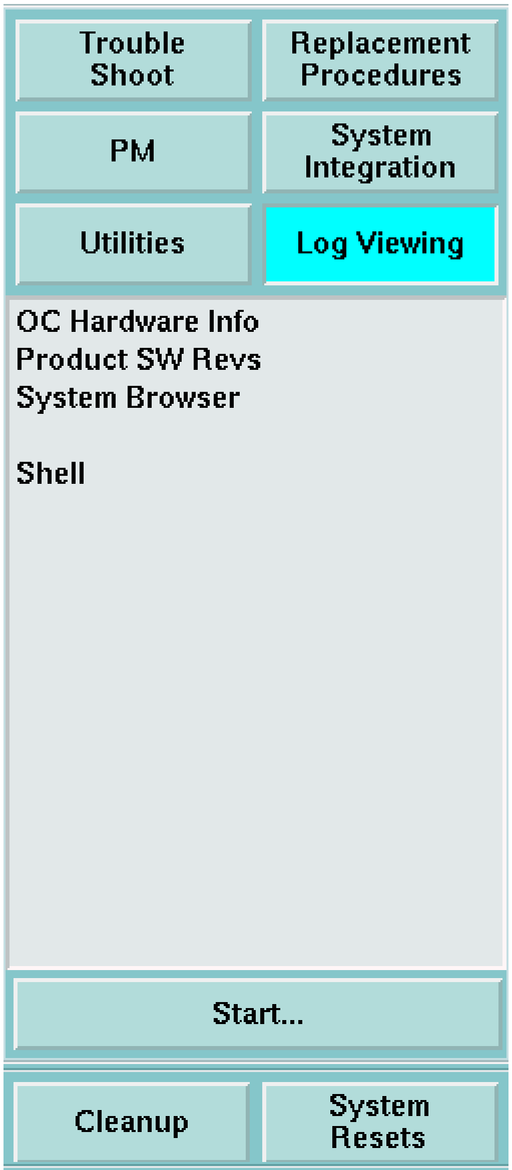


Figure 7-22 Log Viewing Menu - General Service

22.7 Troubleshooting Screens - Advanced Service

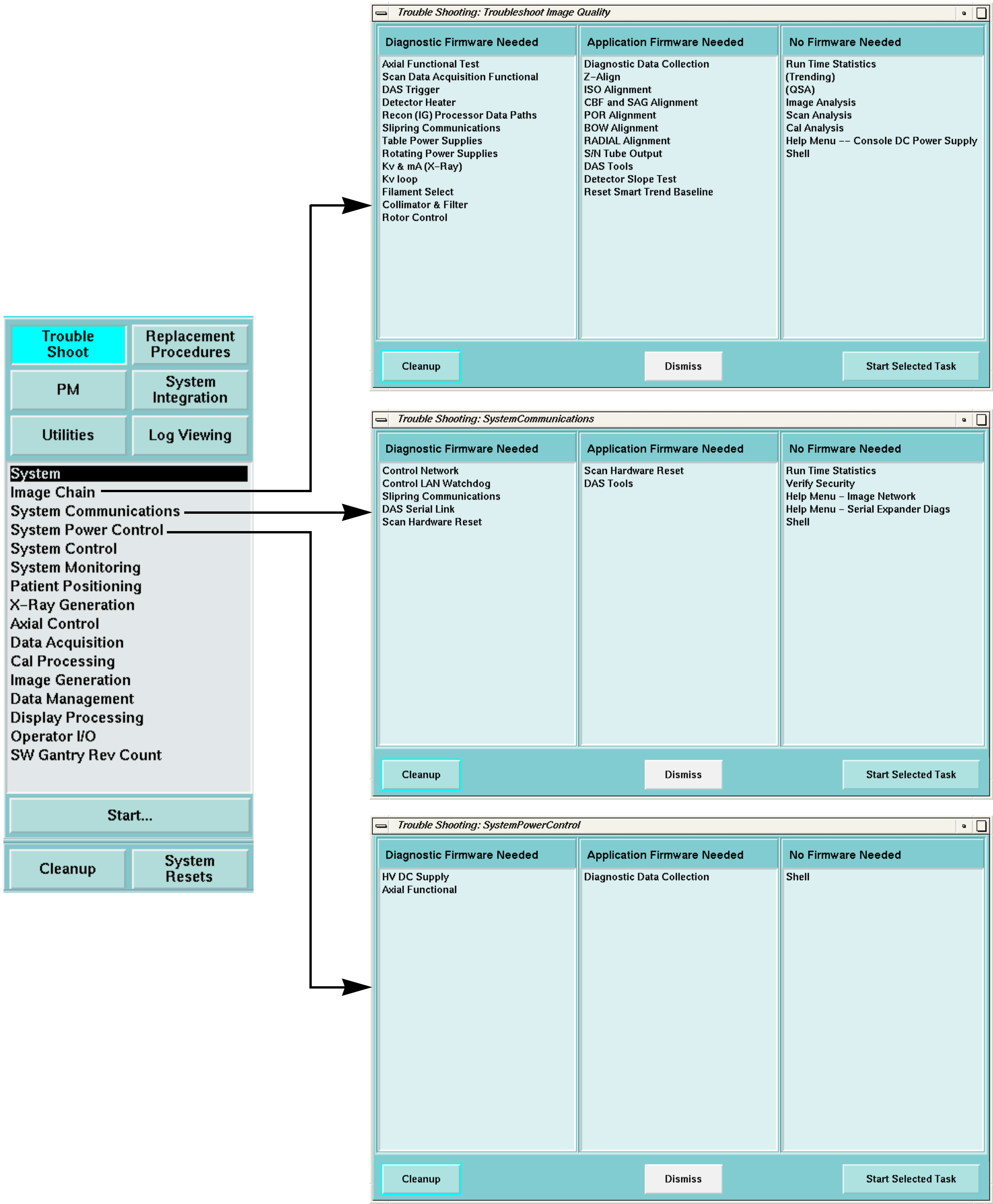


Figure 7-23 Image Chain, System Communications, and System Power Control Troubleshoot Menus - Advanced Service

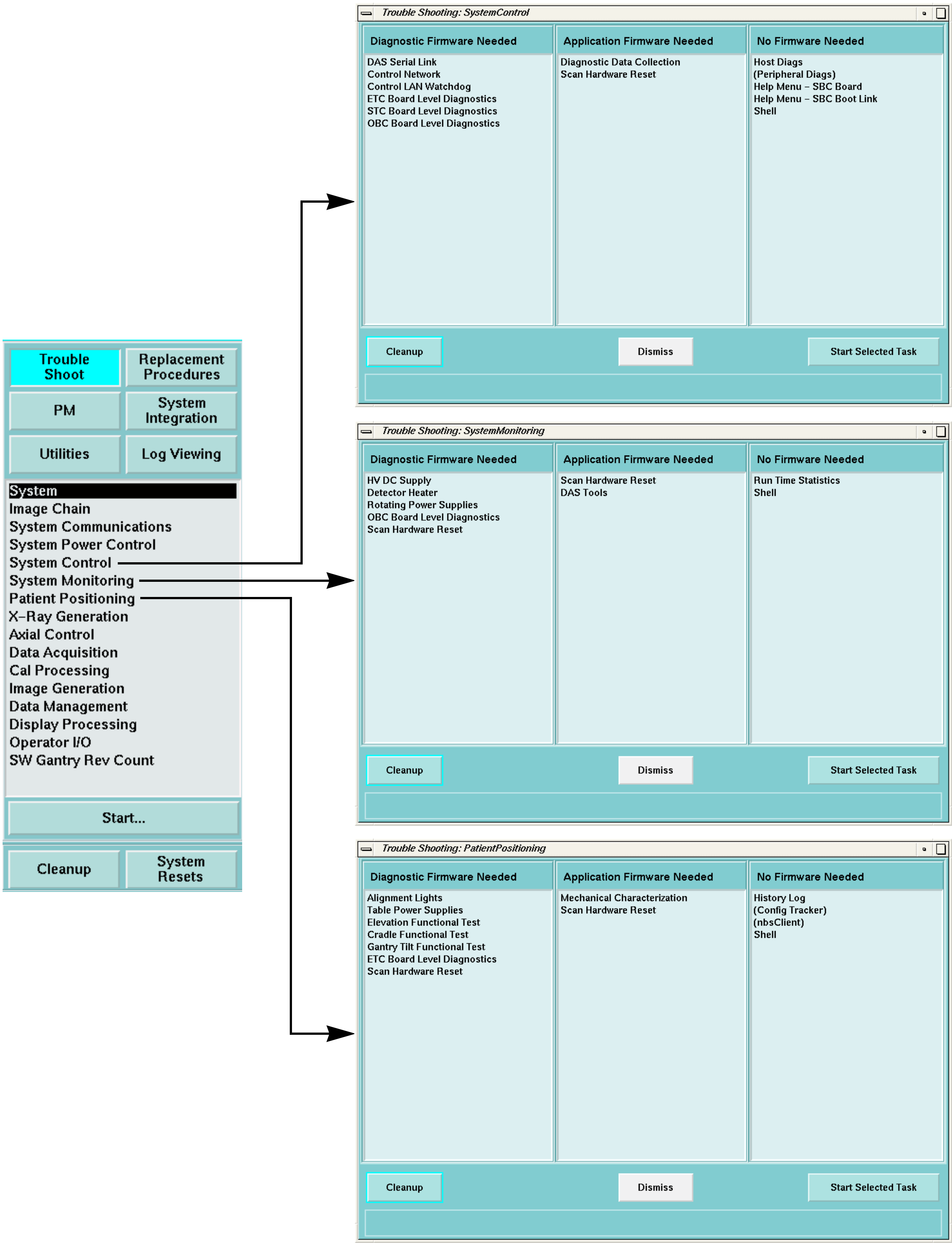


Figure 7-24 System Control, System Monitoring, and Patient Positioning Troubleshoot Menus - Advanced Service

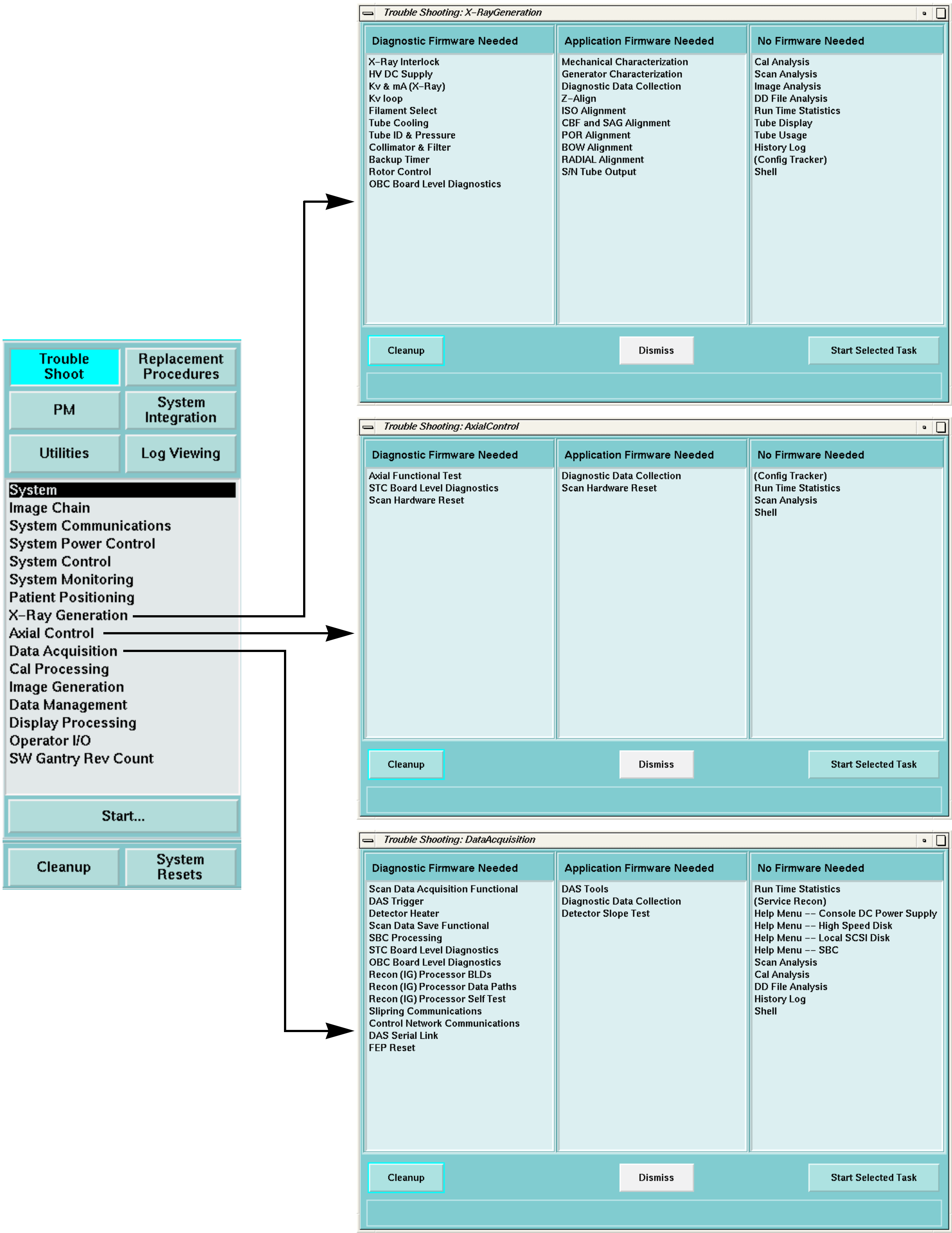


Figure 7-25 X-Ray Generation, Axial Control, and Data Acquisition Troubleshoot Menus - Advanced Service

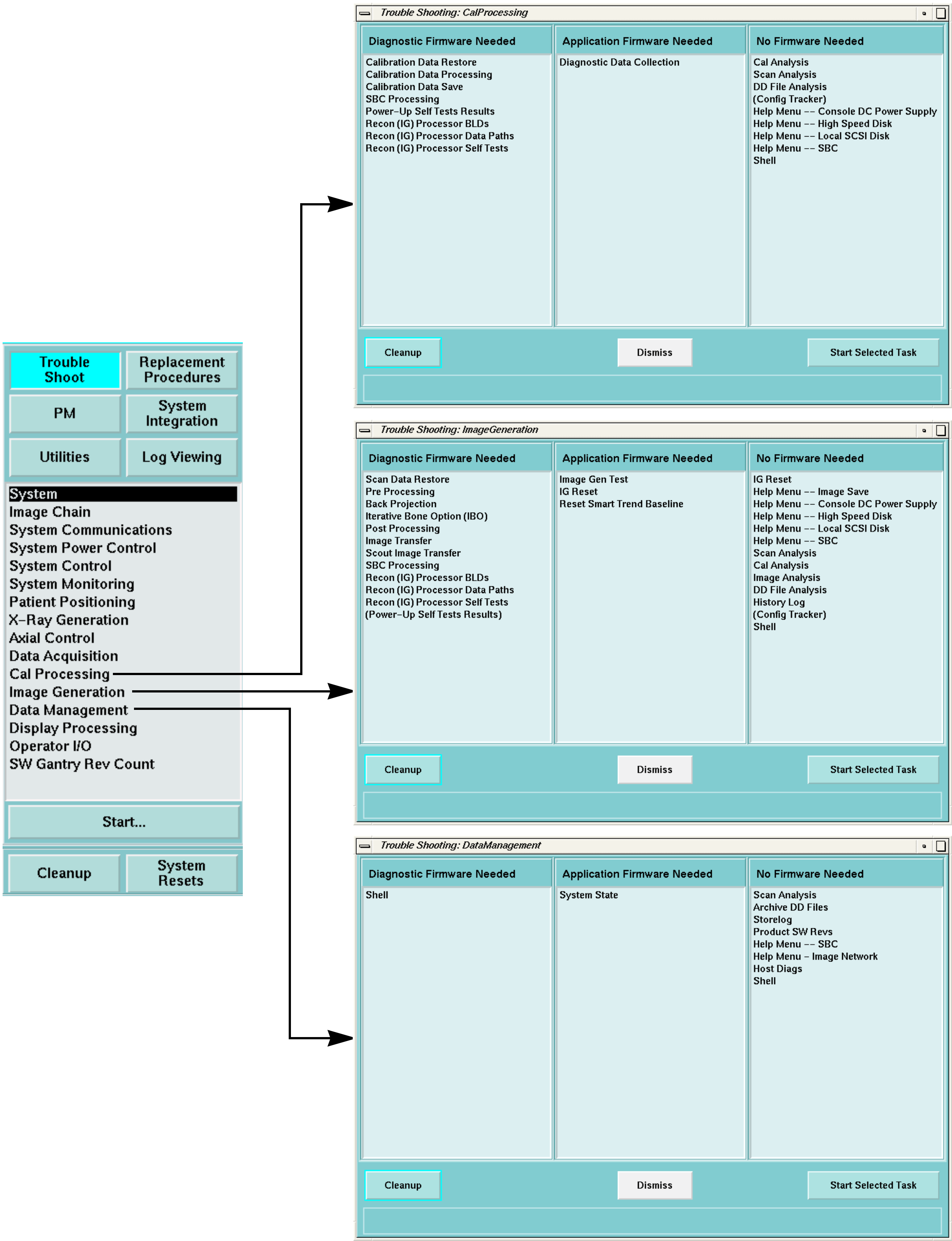


Figure 7-26 Cal Processing, Image Generation, and Data Management Troubleshoot Menus - Advanced Service

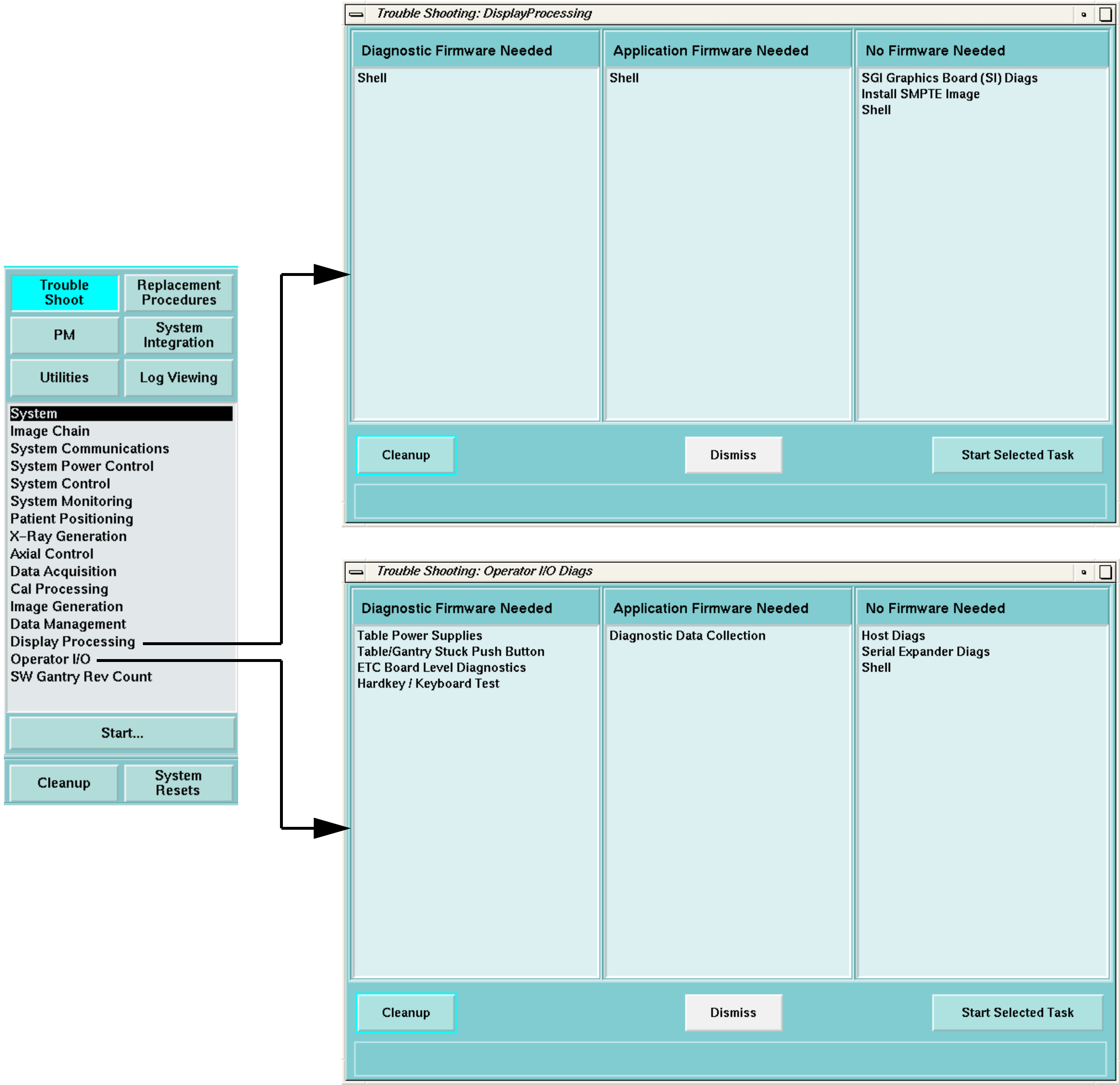


Figure 7-27 Display Processing and Operator I/O Troubleshoot Menus - Advanced Service

22.8 Replacement Procedures - Advanced

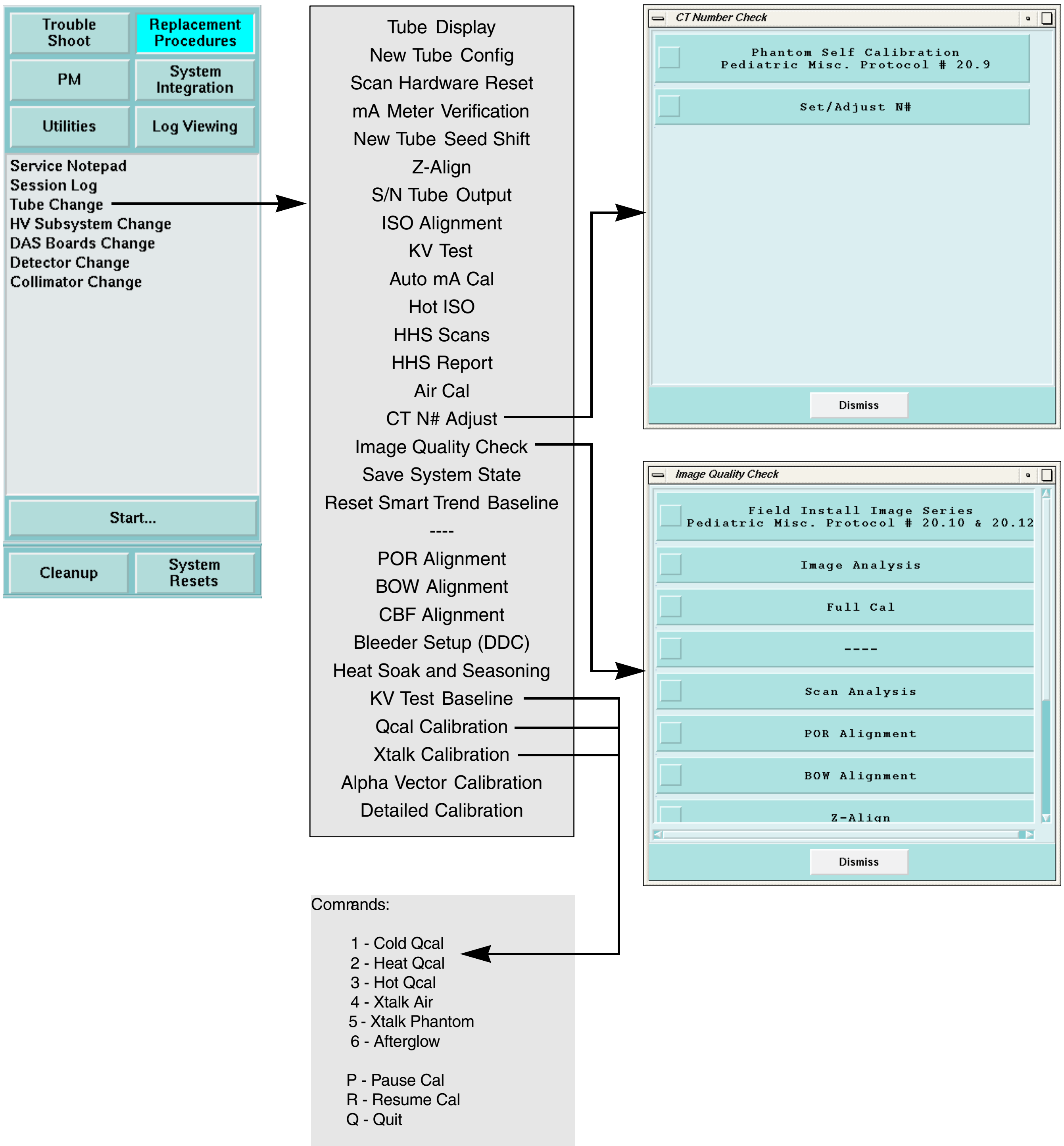


Figure 7-28 Tube Change Replacement Procedures Menus - Advanced Service

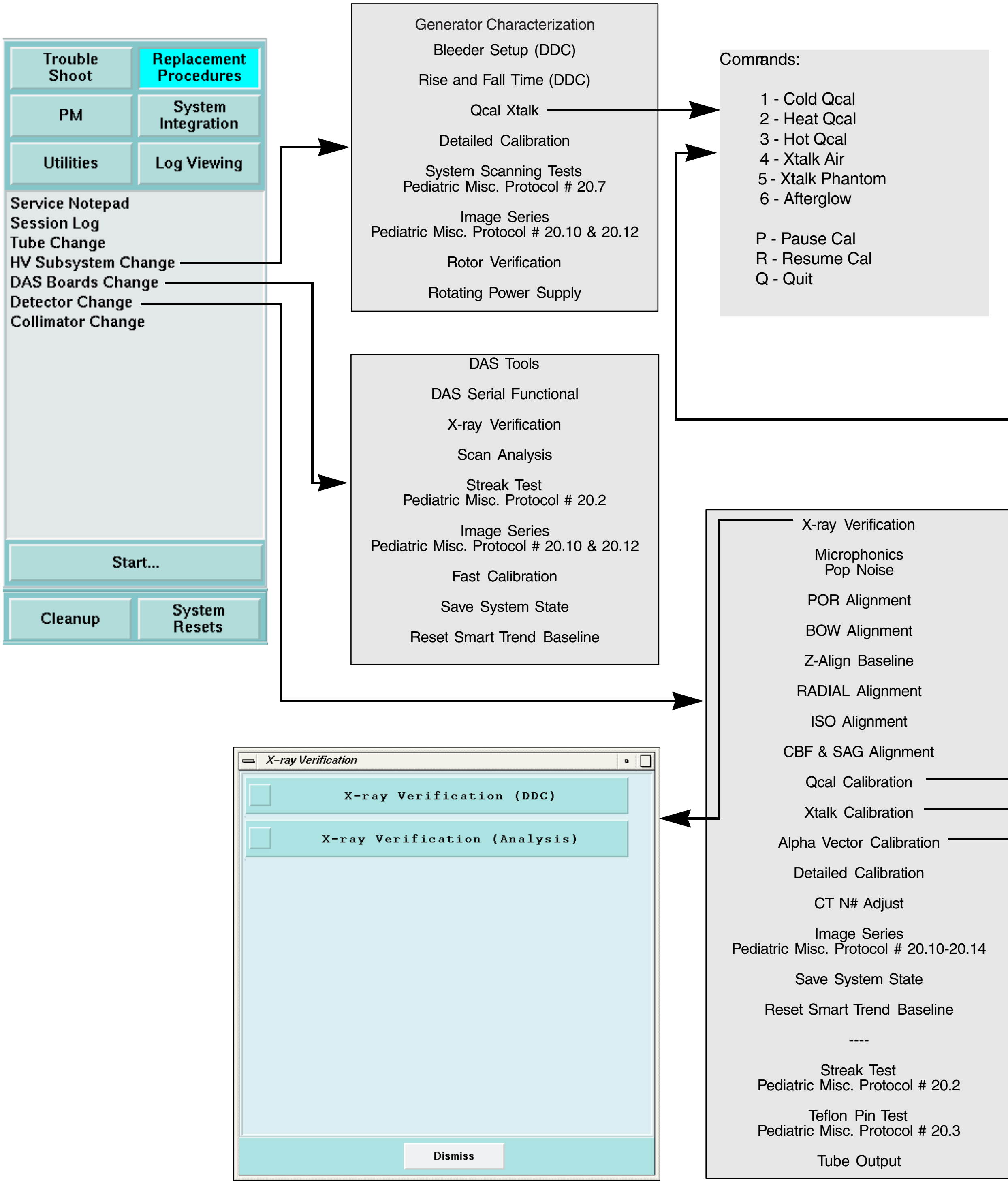


Figure 7-29 HV Subsystem Change, DAS Boards Change, and Detector Change Replacement Procedures Menus - Advanced Service

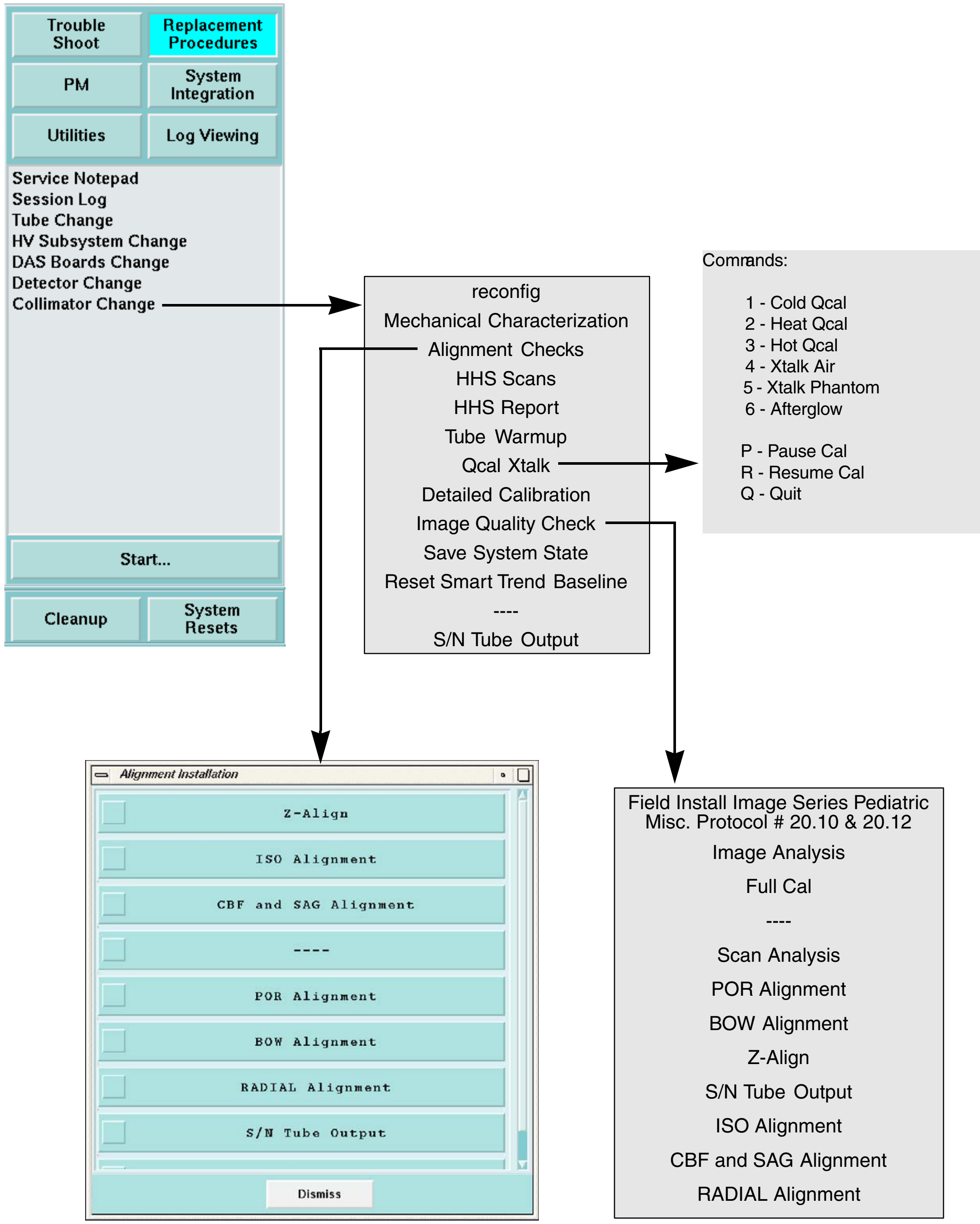


Figure 7-30 Collimator Change Replacement Procedures Menus - Advanced Service

22.9 Planned Maintenance - Advanced

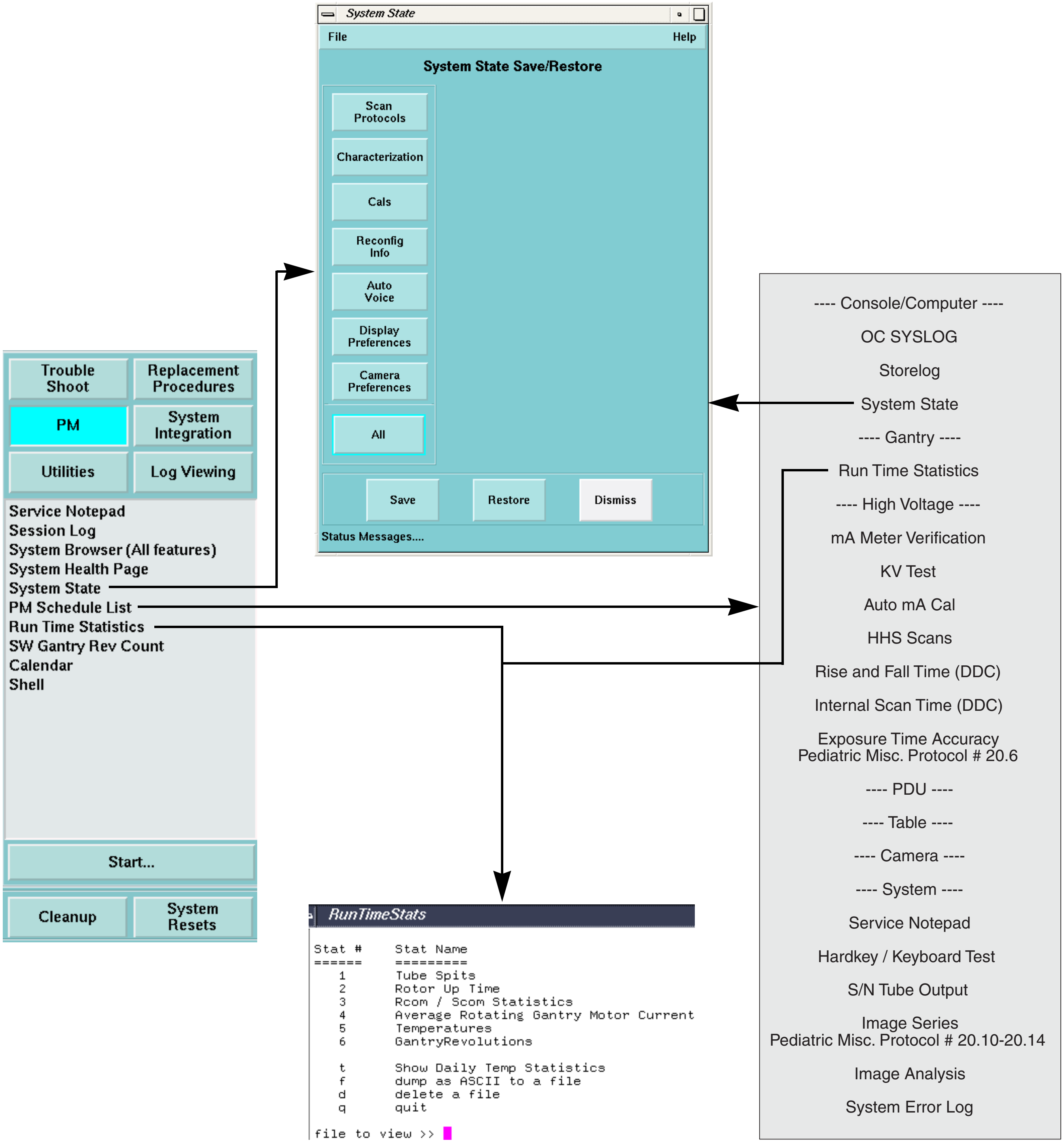


Figure 7-31 System State, PM Schedule List, and Run Time Statistics PM Menus - Advanced Service

22.10 System Integration - Advanced

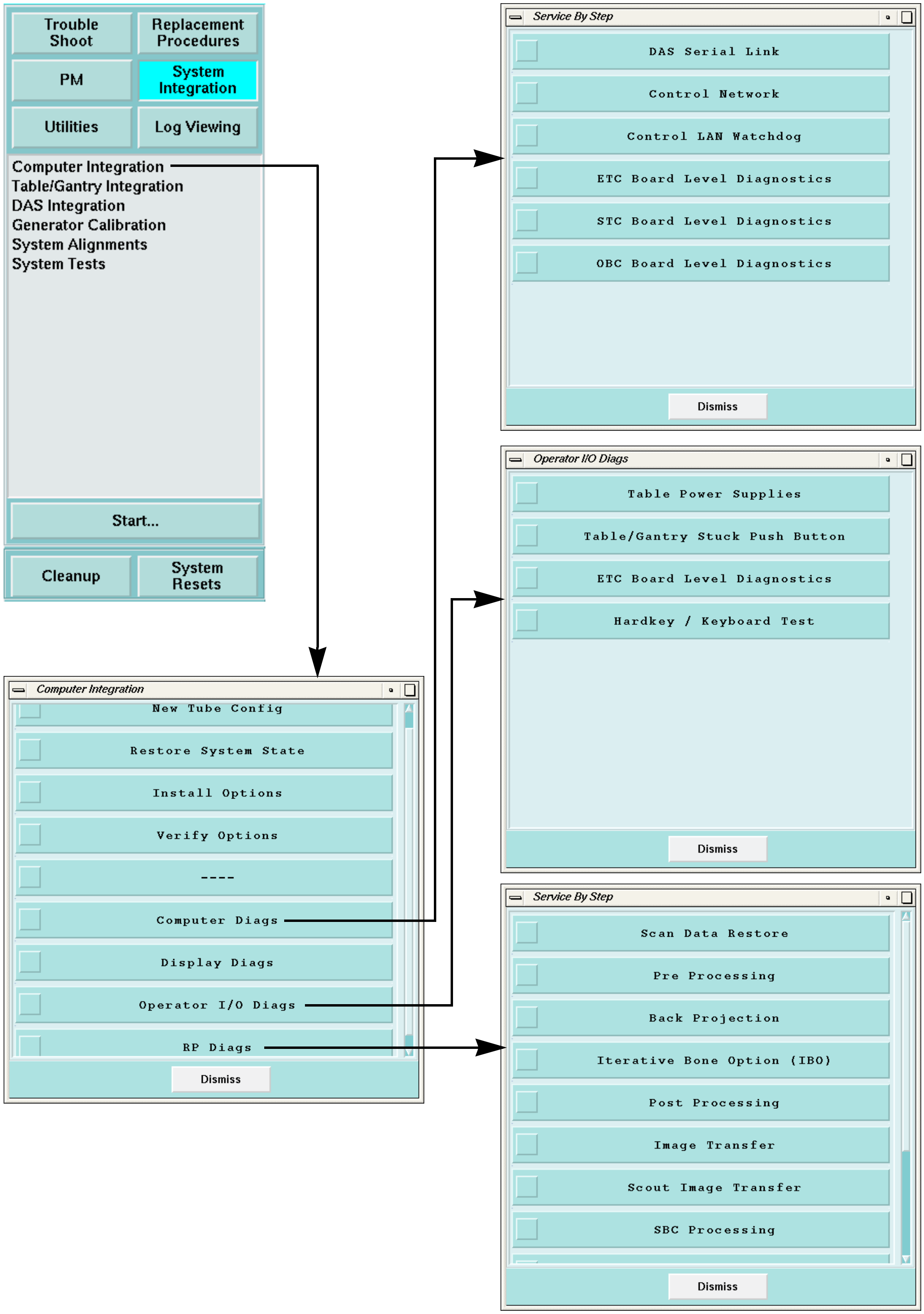


Figure 7-32 Computer Integration Menus - Advanced Service

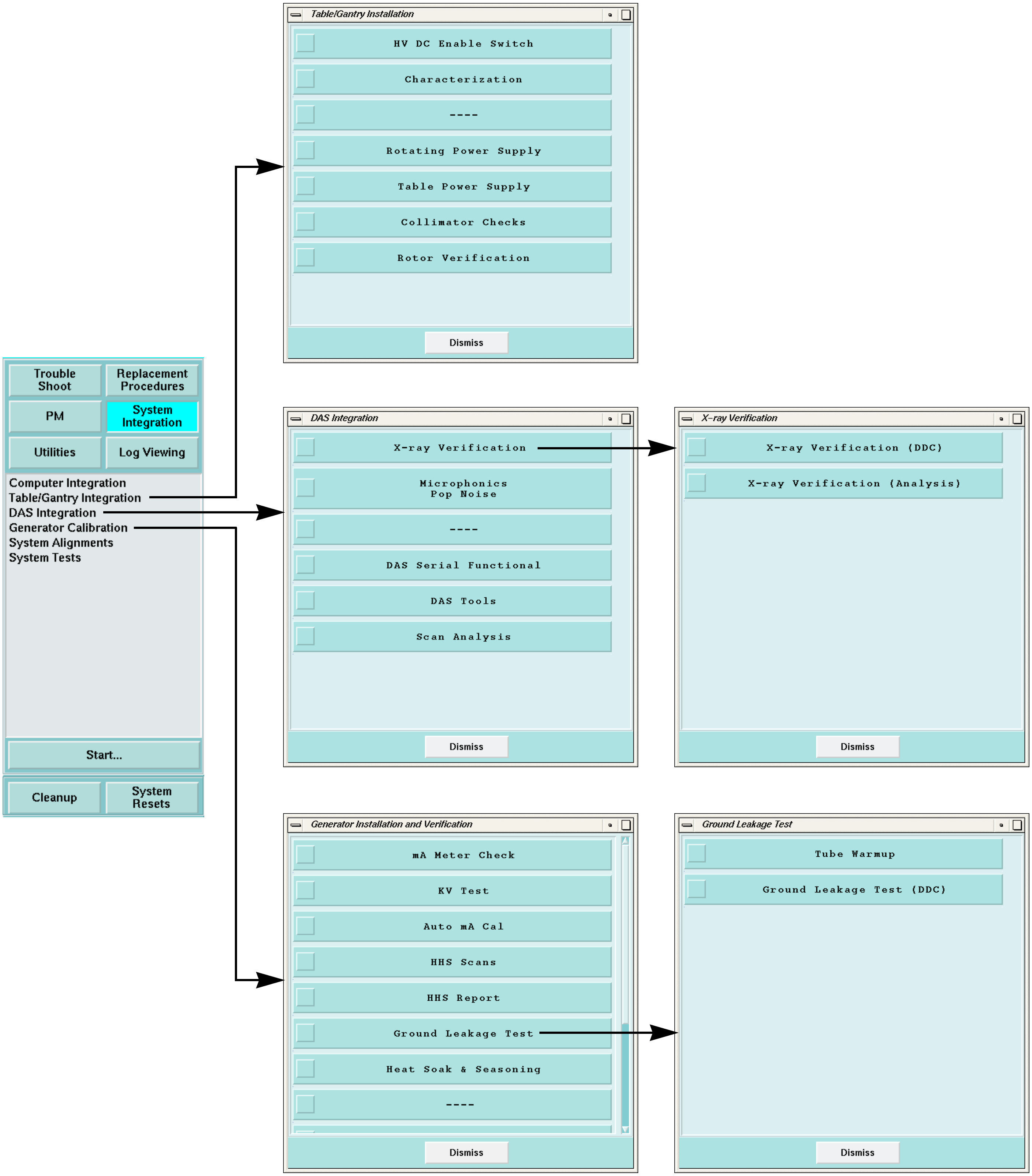


Figure 7-33 Table/Gantry Integration, DAS Integration, and Generator Calibration Menus - Advanced Service

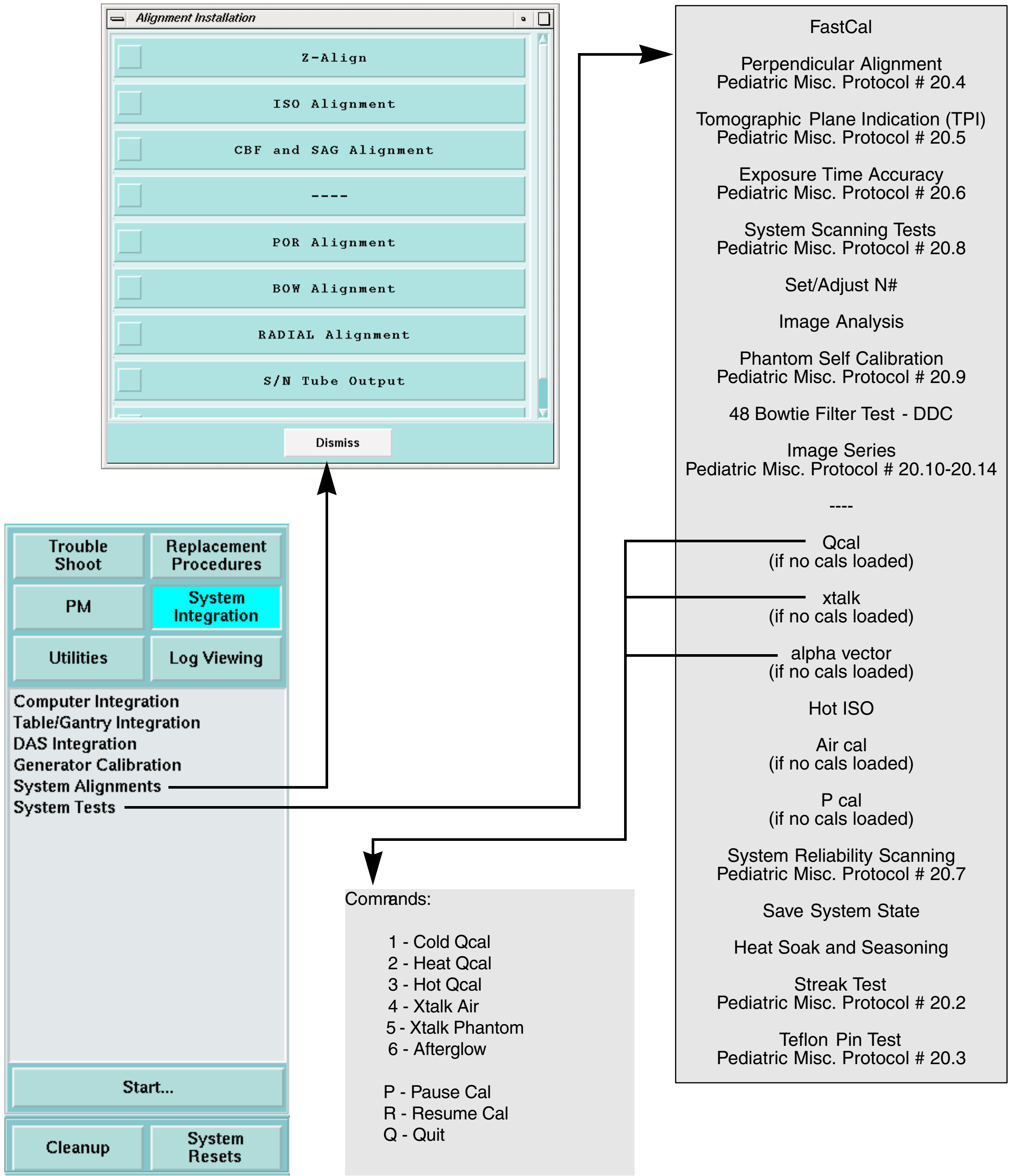


Figure 7-34 System Alignments and System Tests Menus - Advanced Service

22.11 Utilities - Advanced

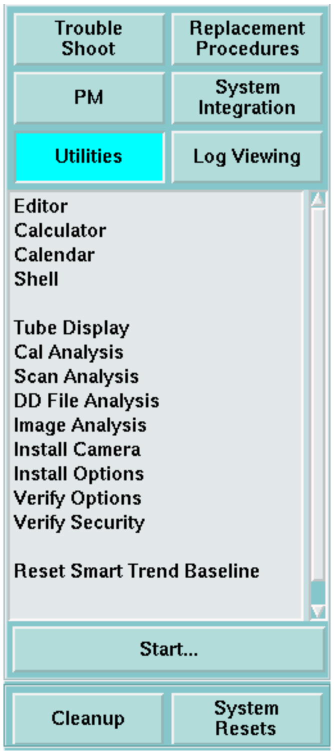


Figure 7-35 Utilities Menu - Advanced Service

22.12 Log Viewing - Advanced

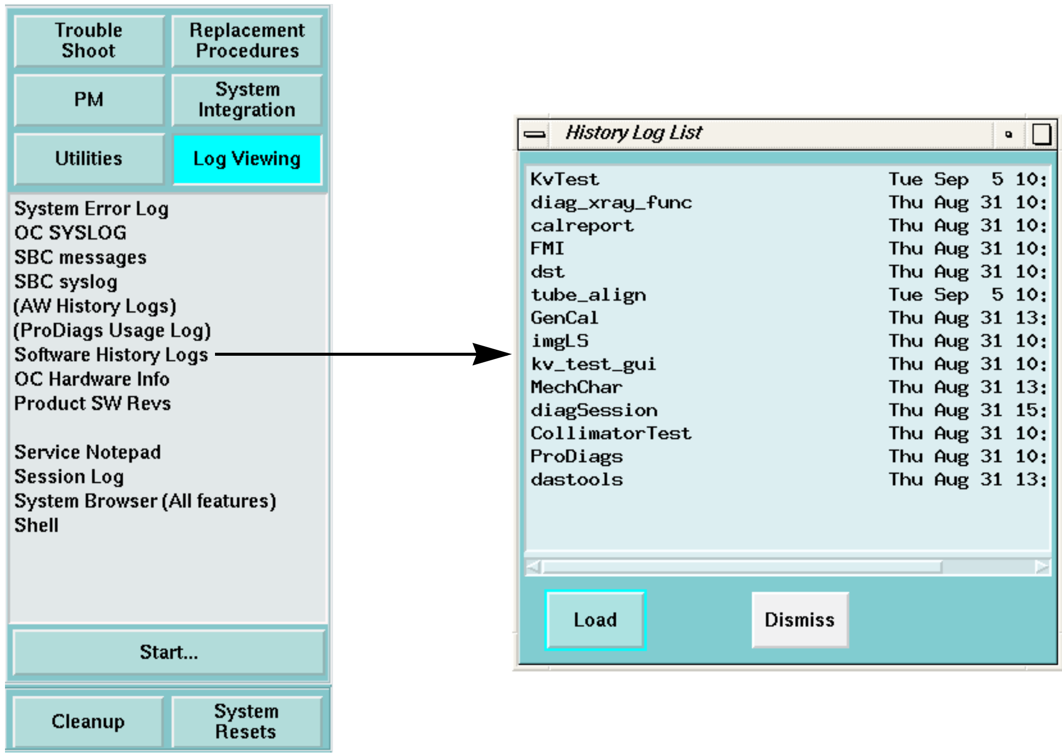


Figure 7-36 Software History Logs - Advanced Service

Section 23.0 Firmware

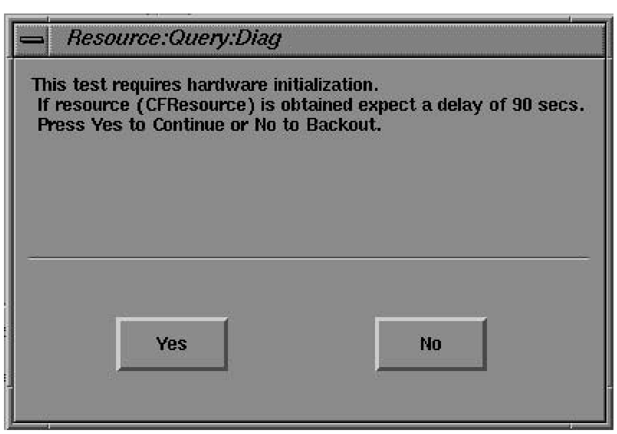


Figure 7-37 Firmware Download Query Pop-Up

Some diagnostic tests require no firmware, some require application firmware, others require diagnostic firmware. If a selected test finds that application firmware is loaded and it needs diagnostic, -OR- if application is loaded and diagnostic is needed, you will have to wait for a FIRMWARE DOWNLOAD to take place.

Section 24.0 Applications Shutdown

Some commands that you will enter into a shell require that Applications Software is shutdown. Some scripts include this step and some do not. Things that require application GE software to be down:

- Reconfig
- Restart applications software for whatever reason, i.e. recover from some bug
- Some diagnostic tests, like Touch Diags
- storelog
- Depending upon software revision maybe changetube or newTu
- Specialx diagnostics
- Some Disk checks on SBC

Section 25.0 System Shutdown

⚠ **NOTICE Potential for loss of data** — **Because of the way in which the operating system software makes use of disk caching, follow the recommended shutdown procedure to give the system a chance to write any information in the cache buffers to the disk before you turn OFF power.**

Always shutdown the system software, to prevent file corruption problems. Use the recommended procedure to minimize the chance that the system leaves any files on the system disks in a bad state.

1.) Select SHUTDOWN to stop scanner applications software.

   A script starts that synchronizes the operating system file structure, and halts the operating system on both the OC host computer and the Scan subsystems SBC computer.

2.) When the screen states that it is through, power off the console power switch.

3.) Turn off the System Mains Disconnect to remove all system power.

Figure 7-38 Service Desktop Manager, Shutdown

Section 26.0 Access the Unix Shell

The shell offers you a way to use Unix or Irix commands and run certain Irix or GE programs. The shell accepts your commands, checks the syntax, calls the appropriate software tasks and takes control when other commands finish. The shell is interactive because it responds to your instruction and reports errors, status and results.

Some helpful shell commands

- `ezlog`
- `FSST`
- `hinv`
- `setdate` if R4.0 or later
- `su root; #bigguy; xman`

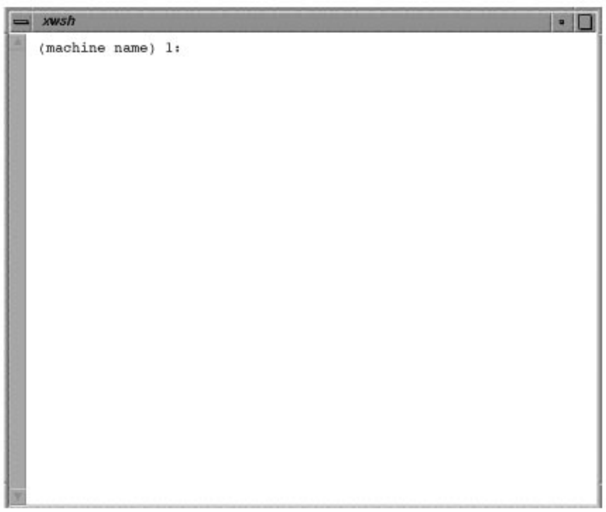


Figure 7-39 Unix Shell

Section 27.0 Tool Selections

You access installation, characterization and calibration tools from the Service Desktop Replacement Procedures Menu:

- *Mechanical Characterization* – Chapter 4, Table/Gantry Maintenance, describes this procedure.
- *Generator Characterization* – Chapter 9, HV and X-Ray, describes.
- *Tube Alignment* – Chapter 3, System Alignment, describes this procedure.
- *Scan Analysis* – Chapter 3, System Alignment, describes this procedure.
- *Diagnostic Data Collection (DDC)* – Chapter 3, System Alignment, describes this procedure. For information on SmartScan use during DDC, see below.

Section 28.0
How to Access DDC

1.) Select REPLACEMENTS PROCEDURES.
2.) Select DIAGNOSTIC DATA COLLECTION
   - Use DDC to collect DAS data with and without X-Ray and/or Rotation.
   - Use the Scan Analysis tool to examine collected data.

28.1 Run SmartScan DDC

The SmartScan feature varies the mA during tube rotation, depending on the patient attenuation profiles. For detailed information, see the operator manuals.

The DDC Smart Scan selection accepts variable mA-specific field entries during the prescription of a view-compressed Rotating X-Ray On or Stationary X-Ray On scan. You can also reconstruct Rotating X-Ray On scans.

Scan and image headers include fields for SmartScan parameters. These fields appear on the display, when you use the List/Select header option. The new scan image header information introduced by SmartScan contains:

- SmartScan: on or off
- Smart mA: computed effective mA (equivalent to technic mA)
- mA phase: min @ 0 degrees or max @ 0 degrees
- mA modulation:% of max reduction (See Definitions, Section 28.2, that follows)
- mA clip: the minimum mA value for the image

Tools Scanning supports stationary and rotating SmartScan modes.

- Stationary X-Ray On scans: Manually select the modulation percentage, phase, simulated start angle, and simulated gantry rotation speed.
- Rotating X-Ray On scans: Select the modulation percentage and phase.

Use **Recon Rx** to list/select and reconstruct certain rotating view-compressed X-Ray On DDC scans. Scan data contains additional parameters that define the modifications for SmartScan. You can view these fields in the Scan Data header, from the List/Select header option.

As part of SmartScan, the enhanced DDC accepts variable mA-specific fields, when you prescribe view-compressed Rotating X-Ray On and Stationary X-Ray On scans. You can also reconstruct Rotating X-Ray On scans with the appropriate number of views.

28.2 Definitions

% MODULATION – The percentage of reduction in maximum mA.

PHASE OF WAVEFORM – The phase or waveform orientation represents the mA waveform's relationship to the gantry position. (See Figure 7-40.) This value indicates whether the mA equals the minimum (1) or maximum (0) value when the gantry reaches 0 degrees during a scan. For example, if the phase equals 1, the prescribed mA equals 300, and the modulation equals 50%, then the mA equals 150 when the gantry reaches the 0 degree position.

CLIPPING LEVEL – The maximum modulation allowed for the CT/i hardware equals 50%, and the minimum allowable patient scanning current equals 40 mA, even though 100% mA modulation is possible otherwise. An mA modulation of more than 50% is acceptable, but will be clipped at the clipping level imposed by the hardware.

28.3 Rotating X-Ray On Scan

At the DDC screen, Select ROTATING X-RAY ON to display the Rotating X-Ray On Screen.

Select SMARTSCAN to enable the function.

When you select SmartScan, the software displays an additional phase softkey and a modulation data field.

Choose a percent (%) modulation. If higher than 50%, the circuitry clips mA modulation at 50%. Choose the phase between the maximum mA or minimum mA at 0 degrees on the gantry.

Type/enter the run description and compression factor. Type/enter the scan time, scan speed, the number of scans, the ISD, the X-Ray On position (see Figure 7-41, plots of channel data from scan analysis), the mA, and the kV.

Note: Do not use 40ma — Do NOT use the default value, 40 mA, for SmartScan, because this current has no available modulation. (40 mA is the lowest threshold value for SmartScan). Enter the “%” of modulation and the phase that you want.

Choose the cal vector, FOV, filter, spot size, and aperture. Type/enter the protocol name. Then accept the Rx.

When the scan finishes, examine the database headers and verify that the correct SmartScan parameters exist.

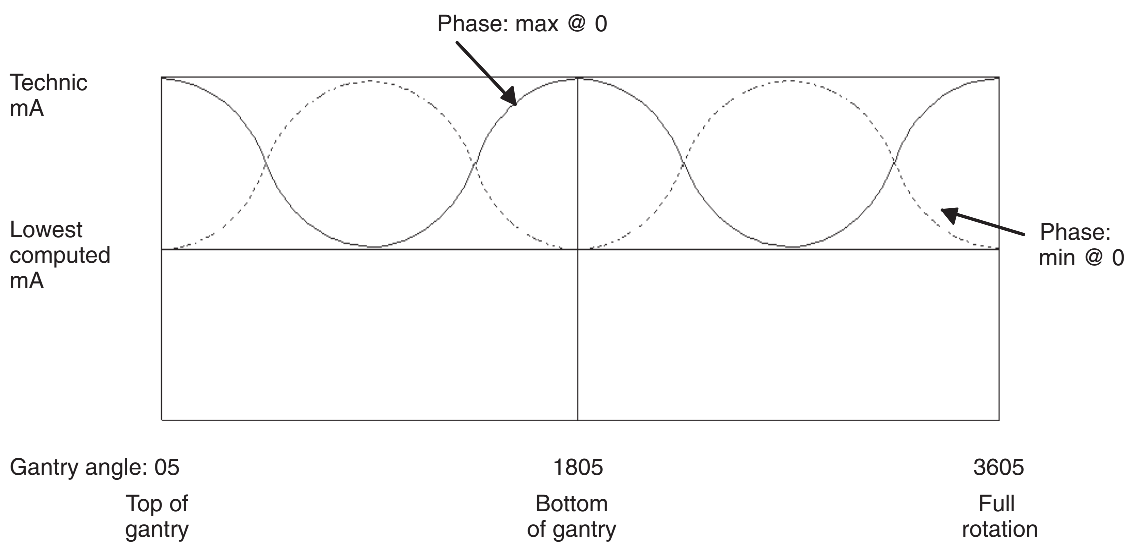


Figure 7-40 ma Variation vs. Gantry Position for Rotating X-ray On (X-ray On Position 0°)

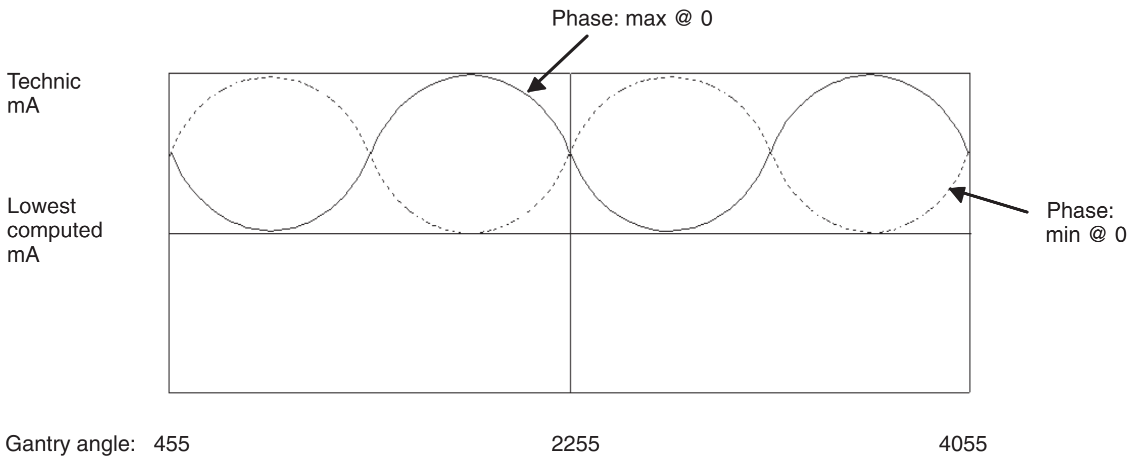


Figure 7-41 mA Variation vs: Gantry Angle for Rotating X-ray On (X-ray On Position 45°)

28.4 Stationary X-Ray On Scan

When you select SmartScan, additional fields for the modulation, simulated gantry speed, and phase appear. For a Stationary X-Ray On scan, the phase refers to the starting angle in the mA waveform, independent of the tube position in the gantry.

Type/enter or select the description, compression factor, scan time, number of scans, ISD, and mA.

Do NOT use the default value, 40 mA, for SmartScan, because no modulation is available at that current. (40 mA is the lowest threshold value for SmartScan).

Enter the kV, modulation, simulated gantry speed, and phase. Choose a phase between 0 and 180 degrees. See Figure 7-42 to select the waveform display frequency.

Type/enter or select the cal filter, FOV, filter, spot size, aperture, and protocol name. Then accept the Rx.

When the scan finishes. examine the database headers and verify that the correct SmartScan parameters exist.

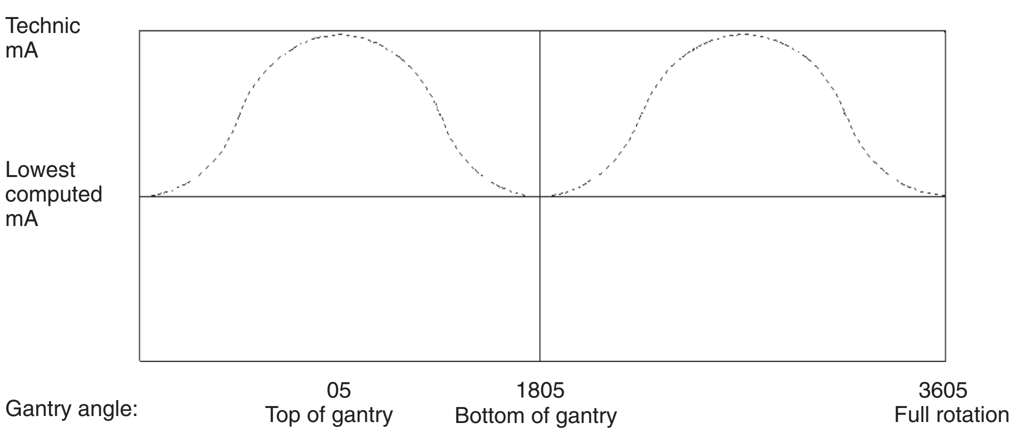


Figure 7-42 mA Variations vs. Gantry Angle For Stationary X-ray On (1 Sec. Scan Time, 1 Sec. Simulated Gantry Speed)

28.5 Reconstruct DDC images

Use ReconRx screens to list/select the DDC-acquired SmartScan data. Use ReconRx to reconstruct the DDC-acquired data into images.

DDC scans appear in Recon Rx List/Select when they consist of:

- Rotating X-Ray On scans
- Full rotation scans consisting of 984 views
- Scans that have a corresponding cal file

28.6 Other SmartScan Notes

- You may force SmartScan on round phantoms, but noise will increase because the software applies insufficient mA to the tube in one of the orthogonal directions (vertical or horizontal).
- CTDI measurements remain the same for SmartScan on or off, since the test uses circular phantoms, and SmartScan leaves the mA unmodulated.
- For non-round objects (e.g. flat phantoms placed on the table), you can verify mA modulation with SmartScan on. Scan with applications. After you take complete orthogonal Scouts, with the default SmartScan on, the system chooses the appropriate modulation, and displays it in the header information for resulting scan image.
- You may take noise measurements on images acquired in DDC (Rotating X-Ray On) or with reconstructed SmartScan patient scans.

Section 29.0 nbsClient

The nbsClient network boot server enables you to review the Scan Control Network CPU Boards statuses and activity.

Follow the list of steps below to connect to the STC, OBC and/or ETC CPU Board controllers.

At the Operators Console console:

1.) Open an UNIX shell on the right-hand display.
2.) Type: **`rsh sbc`** <u>**ENTER**</u> to start a remote shell on the SBC
3.) type **`nbsClient`** ***`<controller>`*** <u>**ENTER**</u>
`<controller>` = **stc** or **etc** or **obcr**
<u>**CNTRL+C**</u> Logs you out of the nbsClient session.

Note: The following applies to the controllers:

- You can only access the controllers for a short time before they log you out. So get the info, then press CNTRL+C to exit the session.
- Staying logged into the controllers for too long a period can cause errors, so keep the sessions as short as possible

```
============================================================
List of nbsClient commands for controllers
============================================================

?                        - print this list
@                        - boot (load and go)
```

Table 7-3 nbs Client Commands

```
p                       - print boot params
c                       - change boot params
l                       - load boot file
g adrs                  - go to adrs
d adrs[,n               - display memory
m adrs                  - modify memory
f adrs, nbytes, value   - fill memory
e                       - print fatal exception
a                       - print value of PC
r type                  - reboot, type = 'soft' or 'hard'
s device [c]            - Print[clear] SCA or R/SCOM driver
                              statistics
t cmd                   - Run diag, cmd = led value(s) of HK
                              tests
u TID                   - Print TCB info for specified TID
v TID                   - Summarize TCB info, TID = 0 => all
w TID                   - Summarize stack usage, TID = 0 => all
x TID                   - Print a stack trace of TID
y                       - dump the error log
z                       - pipe the error log to the console
```

Table 7-3 nbs Client Commands (Continued)

```
$dev(0,procnum)host:/file h=# e=# b=# g=# u=usr [pw=passwd] f=#
=============== =============
```

29.1 Typical ETC session

```
genesis @ BAYA_SBC0 2: nbsClient etc
Remote console connected to host etc
     Tricord System Network Boot Server
       Copyright 1989
CPU: Heurikon HK68/VE
Version: 4.0.1
Creation date: Thu May 30 14:06:51 CDT 1991
[NBS, ETC]: ? <==== Gives list of commands

?                         - print this list
@                         - boot (load and go)
p                         - print boot params
c                         - change boot params
l                         - load boot file
g adrs                    - go to adrs
d adrs[,n]                - display memory
m adrs                    - modify memory
f adrs, nbytes, value     - fill memory
e                         - print fatal exception
a                         - print value of PC
r type                    - reboot, type = 'soft' or 'hard'
s device [c]              - Print[clear] SCA or R/SCOM driver
                            statistics
t cmd                     - Run diag, cmd = led value(s) of HK tests
u TID                     - Print TCB info for specified TID
```

Table 7-4 NBS Client ETC Commands

```
v TID                        - Summarize TCB info, TID = 0 => all
w TID                        - Summarize stack usage, TID = 0 => all
x TID                        - Print a stack trace of TID
y                            - dump the error log
z                            - pipe the error log to the console
```

Table 7-4 NBS Client ETC Commands (Continued)

```
$dev(0,procnum)host:/file h=# e=# b=# g=# u=usr [pw=passwd] f=#
available boot devices: sca ex <==== NOTE: sca is the device we use
[NBS, ETC]:
[NBS, ETC]: a
Current PC = 0x3ae88 <==== NOTE: Running in ram because 0x3????
   Running in ROM PC = 0xf???? (? is don't care)
[NBS, ETC]: y <==== NOTE: displays and empty's log left in ram
8: *** Creation of Posting Agent: OK ****

>  495<t8:     -->   Calling Communication Create Functions
>  495<t8:           etc_comm_init: OK
>  495<t8:           intf_comm_init: OK
>  496<t8:           cmc_comm_init: OK
>  508<t8:           cr_comm_init: OK
>  508<t8:           emc_comm_init: OK
>  513<t8:           el_comm_init: OK
>  519<t8:           tc_comm_init: OK
>  525<t8:           gd_comm_init: OK
>  525<t8:           tmc_comm_init: OK
>  530<t8:           tilt_comm_init: OK
>  530<t8:     ***   Communication Init: OK ****
>  534<t8:     -->   Calling Exception Handler Init Functions
>  540<t8:           exl_comm_init: OK
>  550<t8:           exl_task_init: OK
>  550<t8:     ***   Exception Handler Init: OK ****
>  550<t8:     -->   Calling Quick Hardware Test Functions
>  3513<t8:          syncitest: OK
>  3514<t8:    ===   | ETC board revision: 46-264368G01 L
>  3525<t8:          etc board check: OK
>  3525<t8:          Intf Matrix test: OK
>  3525<t8:    ***   Quick Hardware Tests: OK ****
>  3525<t8:    -->   Calling Hardware Init Functions
>  3526<t8:          etcHwInit: OK
>  3526<t8:          etcDuartInit: OK
>  3527<t8:          etcPowerSupply: OK
>  3527<t8:    ***   Hardware Initialization: OK ****
>  3720<t8:    ***   Create Tasks Command From Host: OK ****
>  3720<t8:    -->   Task Create Functions
>  3724<t8:          cmc_task_init: OK
>  3726<t8:          emc_task_init: OK
>  3727<t8:          tmc_task_init: OK
>  3728<t8:          etc_task_init: OK
>  3729<t8:          intf_task_init: OK
>  3734<t8:          cr_task_init: OK
>  3737<t8:          tc_task_init: OK
```

Table 7-5 NBS Client ETC Error Log Dump

```
> 3741<t8:           el_task_init: OK
> 3745<t8:           tilt_task_init: OK
> 3799<t8:           gd_task_init: OK
> 3799<t8:     ***   ALM Task Creation: OK ****
> 3799<t8:     -->   Calling Watchdogging/Monitoring Init
                     Functions
> 3801<t8:           create_cpu_watchdog: OK
> 3804<t8:           create_memory_watchdog: OK
> 3806<t8:           start heartbeat: OK
> 3806<t8:     ***   Watchdogging/Monitoring Init: OK ****
> 3809<t8:     **********************************************
> 3809<t8:     *     ETCScan 10.22: SUCCESSFUL INITIALIZATION *
> 3809<t8:     **********************************************
> 4506<t1e:          gantry display fw rev is 46-296341G01 Rev 0
> 380811<t17:        etc in mail: PAUSE rcvd
> 421452<t17:        etc in mail: PAUSE rcvd
> 782681<t17:        etc in mail: PAUSE rcvd
> 789761<t17:        etc in mail: PAUSE rcvd
> 862566<t17:        etc in mail: PAUSE rcvd
```

Table 7-5 NBS Client ETC Error Log Dump (Continued)

```
[NBS, ETC]:
[NBS, ETC]: v 0 <==== NOTE: displays tasks and status

NAME          ENTRY  TID    PRI  Status PC   SP       ERRN                 DELAY
------------  -----  -----  ---  ------ ---  -------  -----   -----        -----
excTask       3dffa         2    0           PEND     ac22    fc596        3006a  0
cmcLm628IsrT  17228         11          10   QPEND    308c8   e2d94        0      0
emc_cmd_task  21e04         12          20   QPEND+T  308c8   e239e        0      5
cmc_cmd_task  12d4a         10          25   QPEND+T  308c8   e3782        0      10
tmc_cmd_task  1bb6c         13          30   QPEND+T  308c8   e199a        0      0
cpudogTask    3c0ea         21          48   DELAY    ac22    d2fe8        0      2
netTask              433fc       4      50   PEND     ac22    f9024        0      0
agent                32c48       9      52   PEND             ac22  ff7e4  0      0
exl_in_mail          27eec       a      53   QPEND    ac22    eccc6        0      0
applCtrlTask         2480c       8      53   QPEND    ac22    ee930        10001  0
exl_lan_wdog  299de         f           54   READY    30814   e73e6        0      0
logTask              423f2       3      56   PEND             ac22  fb166  0      0
netBootServe  3a7a0         7           60   READY    49198   f40a2        d0003  0
serialBootSe         3a98c       6      60   PEND             ac22  f6210  0      0
portmapd             56638       5      100  PEND             ac22  f7af0  16     0
exl_verify_w         29876       e      101  QPEND    308c8   e85e2        0      0
exl_notify_t         285ea       b      102  QPEND    308c8   ebb32        0      0
exl_verify_t         29696       c      103  QPEND    308c8   ea97c        0      0
interference         276a8       14     150  QPEND    308c8   e0faa        0      0
etc_in_mail          11a9c       17     159  QPEND    ac22    de186        0      0
cradle_exec          10710       16     160  QPEND    308c8   df3ac        0      0
tc_exec              18c60       19     170  QPEND    308c8   dbe4e        0      0
tc_in_mail_t         191b4       18     175  QPEND    ac22    dcfe4        0      0
elev_exec            1faa4       1b     180  QPEND    308c8   d9a98        0      0
tilt_exec            1aace       1d     190  QPEND    308c8   d7704        0      0
cradle_in_ma  dd6c          15          195  QPEND    ac22    e054c        0      0
elev_in_mail         20344       1a     197  QPEND    ac22    dac40        0      0
tilt_in_mail         1b2f8       1c     198  QPEND    ac22    d88a4        0      0
gd_indicator         25b0c       1e     200  QPEND    308c8   d653c        0      0
gd_in_mail_t         26998       1f     201  QPEND    ac22    d5318        0      0
gd_drive_mai  26690         20          205  READY    3089c   d4188        0      0
exl_logging_         290ec       d      245  QPEND    308c8   e979e        0      0
hk_heartbeat  24d48         23          252  READY    3089c   d1b1c        0      0
memdogTask    3c544         22          253  READY    3a6c6   d2500        0      0
idle          2e72e         1           254  READY    2e72e   fcc58        0      0
```

Table 7-6 NBS Client ETC Tasks and Status

```
[NBS, ETC]:
[NBS, ETC]: s sca0 <==== NOTE: LAN statistics

Receive Statistics:
       input packets    7104    input errors        0
       alignment errs   0       CRC ERRORS          0
       done interrupt   0       ia match            0
       eof interrupts   0       length errors       0
       long frame       0       missed packets      0
       no ia match      0       no end of pack      0
       no SFD detect    0       overrun error       0
       collisions       0       ok frames           0
       short frames     0
```

Table 7-7 NBS Client ETC LAN Statistics

```
Transmit Statistics:
       output packets    7061    output errors         0
       collisions        0
       tx commands       0       packs deferred        0
       done interrupt    0       long frames           0
       heart beat        0       tx interrupts         0
       lost packets      0       lost crs              0
       lost cts          0       late collision        0
       no local mem      0       re-transmits          0
       dma underruns     0
[NBS, ETC]:
```

Table 7-7 NBS Client ETC LAN Statistics (Continued)

```
============================
```

29.2 Typical STC session

```
genesis @ BAYA_SBC0 3: nbsClient stc
Remote console connected to host stc
     Tricord System Network Boot Server
       Copyright 1989
CPU: Heurikon HK68/VE
Version: 4.0.1
Creation date: Thu May 30 14:06:51 CDT 1991
[NBS, STC]: a
Current PC = 0x32308 <==== NOTE: Running in ram because0x3????
                  Running in ROM PC = 0xf???? (? is don't care)
[NBS, STC]: y <==== NOTE: displays and empty's log left in ram
46-264806G01 P

> 1123<t8:      =====|      SCA Hardware Revision: 46-264832G01 A
> 1138<t4:      sr0:        protocol initialization complete
> 1544<t8:                  axial board check: OK
> 1544<t8:      ***         Quick Hardware Tests: OK ****
> 1544<t8:      -->         Calling Hardware Init Functions
> 1545<t8:                  stc_interrupt_init: OK
> 1545<t8:                  AX_duartInit: OK
> 1545<t8:                  AX_pitInit: OK
> 1546<t8:                  sc_hdw_init: OK
> 1546<t8:      ***         Hardware Initialization: OK ****
> 1723<t8:      ***         Create Tasks Command From Host: OK ****
> 1723<t8:      -->         Task Create Functions
> 1736<t8:                  cp_task_init: OK
> 1785<t8:                  sc_task_init: OK
> 1805<t8:                  ax_task_init: OK
> 1809<t8:                  sarq_task_init: OK
> 1810<t8:      ***         ALM Task Creation: OK ****
> 1810<t8:      -->         Calling Watchdogging/Monitoring Init
                            Functions
> 1813<t8:                  create_memory_watchdog: OK
> 1816<t8:                  create_cpu_watchdog: OK
> 1818<t8:                  start heartbeat: OK
```

Table 7-8 NBS Client STC Error Log Dump

```
> 1818<t8:      ***        Watchdogging/Monitoring Init: OK ****
> 1821<t8:      **********************************************
> 1821<t8:      *          STCScan 10.20: SUCCESSFUL INITIALIZATION
> 1821<t8:      **********************************************
> 2034<t38:                SARQtask: rcvd OK
> 61770<t2d:    Scan 0:    Scan Cycle=308850 mS
> 266822<t2a:   scReRefMgr  --     End-of-Series command received--
                                   Sending commands.
> 266832<t2a:   scReRefMgr  --     Waiting for replies.
> 268590<t2a:   scReRefMgr  --     Re-reference complete.
> 326124<t2a:   scReRefMgr  --     End-of-Series command received--
                                   Sending commands.
> 326133<t2a:   scReRefMgr  --     Waiting for replies.
> 326501<t2a:   scReRefMgr  --     Re-reference complete.
> 361223<t2a:   scReRefMgr  --     End-of-Series command received--
                                   Sending commands.
> 361232<t2a:   scReRefMgr  --     Waiting for replies.
> 361600<t2a:   scReRefMgr  --     Re-reference complete.
> 380829<t1e:   scInMail[With Xray] --  PAUSING current scan.
> 389273<t2a:   scReRefMgr  --     End-of-Series command received--
                                   Sending commands.
> 389281<t2a:   scReRefMgr  --     Waiting for replies.
> 390733<t2a:   scReRefMgr  --     Re-reference complete.
> 421471<t1e:   scInMail[With Xray] --  PAUSING current scan.
> 782693<t1e:   scInMail[With Xray] --  PAUSING current scan.
> 789767<t1e:   scInMail[With Xray] --  PAUSING current scan.
> 862578<t1e:   scInMail[With Xray] --  PAUSING current scan.
> 947623<t2a:   scReRefMgr  --     End-of-Series command received--
                                   Sending commands.
> 947633<t2a:   scReRefMgr  --     Waiting for replies.
> 948315<t2a:   scReRefMgr  --     Re-reference complete.
```

Table 7-8 NBS Client STC Error Log Dump (Continued)

```
[NBS, STC]:
[NBS, STC]: v 0 <==== NOTE: displays tasks and status
```

NAME	ENTRY	TID	PRI	STATUS	PC	SP	ERRN	DELAY
excTask	3547a	2	0	PEND	ac2	fc596	3006a	0
ax_cntl_law_	19b9a	32	3	QPEND+T	27d48	b7bb2	0	4
scScnTmrMgr	1255c	2b	5	PEND	27b70	c0128	0	0
scTrigSeqenc	117c4	2d	6	READY	27c6c	bdcae	0	0
cpudogTask	3356a	3a	48	DELAY	ac22	afbd4	0	1
netTask	3a87c	4	50	PEND	ac22	f9024	0	0
agent	2a0c8	9	52	PEND	ac22	ff7e4	0	0
exm_in_mail_	22932	10	53	QPEND	ac22	e468e	0	0
exl_in_mail_	1fbdc	a	53	QPEND	ac22	eb1ee	0	0
applCtrlTask	1e6d4	8	53	QPEND	ac22	ee6ac	10001	0
exm_lan_wdog	23ba4	14	54	QPEND	27d48	dffa0	0	0
exl_lan_wdog	216ce	f	54	READY	27c6c	e590e	0	0
vap_rx	222a0	16	55	PEND	ac22	ddad4	0	0
logTask	39872	3	56	PEND	ac22	fb166	0	0

Table 7-9 NBS Client STC Tasks and Status

NAME	ENTRY	TID	PRI	STATUS	PC	SP	ERRN	DELAY
-------------	------	----	---	-------	------	-----	-----	-----
netBootServe	31c20	7	60	READY	40618	f4c2c	d0003	0
serialBootSe	31e0c	6	60	PEND	ac22	f5f8c	0	0
portmapd	4dab8	5	100	PEND	ac22	f7af0	16	0
exm_verify_w	241721	15	101	QPEND	27d48	dedda	0	0
exl_verify_w	21566	e	101	QPEND	27d48	e6b0a	0	0
exm_notify_t	232ba	11	102	QPEND	27d48	e3526	0	0
exl_notify_t	202da	b	102	QPEND	27d48	ea05a	0	0
exm_verify_t	24476	12	103	QPEND	27d48	e2352	0	0
exl_verify_t	21386	c	103	QPEND	27d48	e8ea4	0	0
cp_rcv	16ea4	1b	110	QPEND	27d48	d232a	0	0
cp_xmit	1712c	18	111	QPEND	27d48	d58a0	0	0
scTransition	13148	2c	122	READY	27c6c	bef00	0	0
scOffsetStrt	12338	26	122	READY	27c6c	c5c28	0	0
scCPBEnableT	10044	22	122	READY	27c6c	ca4f8	0	0
scAVoiceTmr	12958	21	122	READY	27c6c	cb72c	0	0
scReRefMgr	13ff8	2a	124	READY	27c6c	c1350	0	0
scStartPbLog	106f2	29	124	READY	27c6c	c2578	0	0
scAutoVLogic	f9ce	20	125	DELAY	27d1c	cc988	0	12
scMinOffsetT	1026c	25	126	READY	27c6c	c6e54	0	0
scRxMgr	f1b4	1f	127	READY	27c6c	cdb6c	0	0
scPrepTmr	12f94	27	128	READY	27c6c	c4a00	0	0
scDrvResumeT	14dbe	24	129	READY	27c6c	c8080	0	0
scLongDlySer	12c18	23	129	READY	27c6c	c9298	0	0
ax_error_tas	1b646	2f	140	READY	27c6c	bb118	0	0
ax_hard_init	181f2	34	141	READY	27c6c	b57f8	0	0
ax_commander	17eb6	30	142	QPEND	27d48	b9ef8	0	0
ax_in_mail_t	18a8a	31	143	QPEND	ac22	b8d28	0	0
ax_outmail_t	1b410	33	144	PEND	27b70	b69f8	0	0
sarq	52c10	38	150	QPEND	ac22	b1020	0	0
scScanDataQu	fb8a	28	170	READY	27c6c	c37a4	0	0
scInMail	dc50	1e	171	QPEND	ac22	ced5a	0	0
scMainsMonit	14962	2e	172	READY	27c6c	bc3d8	0	0
ax_pause_tas	1c266	36	191	READY	27c6c	b3468	0	0
ax_pos_req_t	1cad6	35	192	PEND	27b70	b4640	0	0
gantry_disab	1db42	37	195	READY	27d1c	b22a8	0	0
cp_in_mail	15ff8	17	210	QPEND	ac22	d698e	0	0
cp_key	1565c	1a	213	QPEND	27d48	d34d6	0	0
cp_led	15c14	1c	214	QPEND	27d48	d114a	0	0
cp_autovoice	15158	19	214	PEND	27b70	d46bc	0	0
cp_watchdog	16bac	1d	230	READY	27d1c	cff8c	0	0
exm_logging_	2362a	13	240	QPEND	27d48	e1170	0	0
exl_logging_	20ddc	d	245	QPEND	27d48	e7cc6	0	0
hk_heartbeat	1ec10	3b	252	READY	27d1c	af1e4	0	0
memdogTask	339c4	39	253	READY	31b48	b05c8	0	0
idle	25b6a	1	254	READY	25b6a	fcc58	0	0

Table 7-9 NBS Client STC Tasks and Status (Continued)

```
[NBS, STC]: s sr0  <==== NOTE: Slip Ring statistics

sr0:Receive Statistics:
    INIT's received           =  3     INITACK's received        = 2
    ACK's received            = 53     NAK's received            = 0
    DATA's received           = 6312
sr0:Transmit Statistics:
    INIT's transmitted        = 3      INITACK's transmitted     = 3
    ACK's transmitted         = 6221   NAK's transmitted         = 0
    DATA's transmitted        = 6322
sr0:Overrun/Underrun Statistics:
    Receive overruns          = 0      Receive underruns         = 5
    Transmit overruns         = 0
sr0:Checksum/Sequence Error Statistics:
    Header checksum errors    = 0      Data checksum errors      = 0
    Sequence errors           = 0
sr0:Timeout Error Statistics:
    INIT timeouts             = 0      DATA timeouts             = 0
sr0:Interrupt Statistics:
    Level 6 interrupts        = 6376   DAS/AP FIFO full          = 0
    TAXI link failures        = 0
sr0:Miscellaneous Statistics:
    Black hole occurrences    = 0      Total words thrown away   = 0
    Most words thrown away    = 0      Errored DATA frames       = 0
[NBS, STC]:
```

Table 7-10 NBS Client STC Slip Ring Statistics

```
[NBS, STC]: s sca0 <==== NOTE: LAN statistics

Receive Statistics:
     input packets      6534       input errors       0
     alignment errs                  0              crc errors      0
     done interrupt                  0              ia match        0
     eof interrupts                  0              length errors      0
     long frame         0      missed packets        0
     no ia match        0      no end of pack        0
     no SFD detect      0      overrun error         0
     collisions         0      ok frames             0
     short frames       0
Transmit Statistics:
     output packets            6530              output errors      0
     collisions         0
     tx commands        0      packs deferred        0
     done interrupt            0                 long frames     0
     heart beat         0      tx interrupts         0
     lost packets       0      lost crs    0
     lost cts           0      late collision        0
     no local mem       0      re-transmits          0
     dma underruns      0
[NBS, STC]:
```

Table 7-11 NBS Client STC LAN Statistics

============================

29.3 Typical OBC session

```
genesis @ BAYA_SBC0 4: nbsClient obcr  <==== NOTE: OBC hostname is obcr
Remote console connected to host obcr
     Tricord System Network Boot Server
       Copyright 1989
CPU: Heurikon HK68/VE
Version: 4.0.1
Creation date: Thu May 30 14:06:51 CDT 1991
[NBS, OBCR]: a
Current PC = 0x34dac <==== NOTE: Running in ram because 0x3????
             Running in ROM PC = 0xf???? (? is don't care)
[NBS, OBCR]: y <==== NOTE: displays and empty's log left in ram

>  3098<t8:             IObrd: VME memory WRITE tests.
>  3099<t8:             IObrd: fault latch tests.
>  3099<t8:            IOitest: OK
>  3099<t8:   =====|    KVbrd Rev: 46-288904G01 A |=====
>  3099<t8:             KVbrd: VME memory READ tests.
>  3099<t8:             KVbrd: VME memory WRITE tests.
>  3100<t8:             KVbrd: fault latch tests.
>  3100<t8:             KVbrd: bit pattern tests.
>  3100<t8:            KVitest: OK
>  3100<t8:   =====|    MAbrd Rev: 46-288886G01 A |=====
>  3101<t8:             MAbrd: VME memory READ tests.
>  3101<t8:             MAbrd: VME memory WRITE tests.
>  3101<t8:             MAbrd: fault latch tests.
>  3101<t8:             MAbrd: bit pattern tests.
>  3101<t8:            MAitest: OK
>  3102<t8:   =====|    CTVRCbrd Rev: 46-288858G01 B |=====
>  3102<t8:             CTVRCbrd: VME memory READ tests.
>  3102<t8:             CTVRCbrd: VME memory WRITE tests.
>  3103<t8:             CTVRCbrd: fault latch tests.
>  3103<t8:            CTVRCitest: OK
>  3103<t8:   =====|    Collimator Board Revision: 46-288856G01 B
>  3104<t8:            COLitest: OK
>  3104<t8:   =====|    RARQbrd Rev: 46-288306G01 D |=====
>  3123<t4:   sr0:      protocol initialization complete
>  3127<t8:   ping OK
>  3128<t8:             RARQitest: OK
>  3128<t8:   ***       Quick Hardware Tests: OK ****
>  3128<t8:   -->       Calling Hardware Init Functions
>  3128<t8:            COLPowerSup: OK
>  3128<t8:            gen_hw_init: OK
>  3129<t8:            obc_isr_init: OK
>  3129<t8:   ***      Hardware Initialization: OK ****
>  3302<t8:   ***      Create Tasks Command From Host: OK ****
>  3302<t8:   -->      Task Create Functions
>  3311<t8:             col_task_init: OK
>  3344<t8:             gen_task_init: OK
>  3346<t8:             al_task_init: OK
>  3349<t8:             das_task_init: OK
```

Table 7-12 NBS Client OBC Error Log Dump

```
>  3352<t8:          rarq_task_init: OK
>  3352<t8:   ***    ALM Task Creation: OK ****
>  3352<t8:   -->    Calling Watchdogging/Monitoring Init Functions
>  3354<t8:          create_cpu_watchdog: OK
>  3358<t8:          create_memory_watchdog: OK
>  3360<t8:          start heartbeat: OK
>  3360<t8:   *** Watchdogging/Monitoring Init: OK ****
>  3365<t8:   ******************************************
>  3366<t8:   * OBCbuild 10.12: SUCCESSFUL INITIALIZATION *
>  3366<t8:   ******************************************
>  3502<t26:  On scan 0 DAS RESET executed
>  3712<t26:  On scan 0 Establish scanning default state executed.
>  3728<t26:  On scan 0 DAS START executed.
>  4016<t27:  RARQ task received OK from SARQ
```

Table 7-12 NBS Client OBC Error Log Dump (Continued)

[NBS, OBCR]:
[NBS, OBCR]: **v 0** <==== NOTE: displays tasks and status
[NBS, OBCR]:

NAME	ENTRY	TID	PR	STATUS	PC	SP	ERRNO	DELAY
excTask	37f1e	2	0	PEND	ac22	fc596	3006a	0
cpudogTask	3600e	28	48	DELAY	ac22	c6050	0	16
netTask	3d320	4	50	PEND	ac22	f9024	0	0
agent	2c9ac	9	52	PEND	ac22	ff7e4	0	0
exl_in_mail_	20864	a	53	QPEND	ac22	ebf72	1000a	0
applCtrlTask	1f35c	8	53	QPEND	ac22	ee220	10001	0
exl_lan_wdog	22356	f	54	READY	2a550	e6692	1000a	0
logTask	3c316	3	56	PEND	ac22	fb166	0	0
netBootServe	346c4	7	60	READY	430bc	f47a0	d0003	0
serialBootSe	348b0	6	60	PEND	ac22	f5b00	0	0
portmapd	5055c	5	100	PEND	ac22	f79ec	16	0
exl_verify_w	221ee	e	101	QPEND	2a62c	e788e	0	0
exl_notify_t	20f62	b	102	QPEND	2a62c	eadde	1000a	0
exl_verify_t	2200e	c	103	QPEND	2a62c	e9c28	1000a	0
gen_cal	1bf04	1c	104	PEND	2a470	d4120	0	0
gen_xray_mon	13ebc	1b	105	QPEND	2a62c	d5368	0	0
col_in_mail	e93a	10	120	QPEND	ac22	e1c32	0	0
das_in_mail	1d6dc	26	150	QPEND	ac22	c891a	0	0
al_in_mail	1cfa4	25	150	QPEND	ac22	c9d0c	0	0
gen_abort	15e90	1d	150	QPEND	2a62c	d2f0c	0	0
rarq	26f84	27	155	QPEND	ac22	c6a14	0	0
gen_filament	1bbf0	1a	158	PEND	2a470	d659c	0	0
gen_rotor	171aa	19	159	PEND	2a470	d7790	0	0
gen_control	12ccc	17	160	QPEND	2a62c	d9c06	0	0
gen_config	11878	16	160	PEND	2a470	dade2	0	0
gen_in_mail	f8d8	15	165	QPEND	ac22	dc2c6	0	0
col_home	dbc0	14	181	READY	2a550	dd5d4	0	0
col_control	e5e8	11	182	PEND	2a470	e0b3a	0	0
gen_query	14484	18	200	PEND	2a470	d88fc	0	0
col_aperture	d1d0	13	200	PEND	2a470	de7a4	0	0
col_filter	d460	12	200	PEND	2a470	df980	0	0

Table 7-13 NBS Client OBC Tasks and Status

```
gen_pump        1baa4  23   205  DELAY     2a550  cc1f4     0      336779
gen_rotor_ti    1b874  22   205  READY     2a550  cd40a     0      0
gen_technic_    1a6b8  20   205  READY     2a550  cf760     0      0
gen_meter       1ca1c  24   206  PEND      2a470  caf8e     0      0
gen_thermal_    195a0  21   210  DELAY     2a600  ce652     0      68
gen_tube_tra    18de4  1f   210  READY     2a600  d0abc     0      0
gen_power_mo    19040  1e   210  READY     2a600  d1d04     0      0
exl_logging_    21a64  d    245  QPEND     2a62c  e8a4a     1000a  0
hk_heartbeat    1f898  2a   253  READY     2a600  c4b84     0      0
memdogTask      36468  29   253  READY     345f2  c5568     0      0
idle            2846a  1    254  READY     2846a  fcc58     0      0
[NBS, OBCR]:
```

Table 7-13 NBS Client OBC Tasks and Status (Continued)

```
[NBS, OBCR]:s sr0   <==== NOTE: Slip Ring statistics

sr0: Receive Statistics:
     INIT's received           = 3     INITACK's received         = 3
     ACK's received            = 6276  NAK's received             = 0
     DATA's received           = 6498
sr0: Transmit Statistics:
     INIT's transmitted        = 4     INITACK's transmitted      = 3
     ACK's transmitted         = 89    NAK's transmitted          = 0
     DATA's transmitted        = 6503
sr0: Overrun/Underrun Statistics:
     Receive overruns          = 0     Receive underruns          = 250
     Transmit overruns         = 0
sr0: Checksum/Sequence Error Statistics:
     Header checksum errors    = 0      Data checksum errors      = 0
     Sequence errors           = 0
sr0: Timeout Error Statistics:
     INIT timeouts             = 0     DATA timeouts              = 17
sr0: Interrupt Statistics:
     Level 6 interrupts        = 13031 DAS/AP FIFO full           = 0
     TAXI link failures        = 0
sr0: Miscellaneous Statistics:
     Black hole occurrences    = 0     Total words thrown away    = 0
     Most words thrown away    = 0     Errored DATA frames        = 0
[NBS, OBCR]:
```

Table 7-14 NBS Client OBC Slip Ring Statistics

Section 30.0
Tube Warm Up and FastCal Operation

The Tube Warmup and FastCal were modified for HSA CT/i 3.6 software release to support a preferred fast cal, the QSA (Smart Trend and DC Cal) feature and the Detector Slope Test feature (DST).

It is to mandatory to perform the warmup2 scans each time FastCal is executed. Previously, warmup2 was part of GenCal which could be cancelled to enter into Fastcal. In such cases, the tube is not warm enough to run Fastcal. This means warmup2 scans will be included as part of FastCal and will be removed from GenCal section.

Tube Warmup has been changed to only include the scans required to bring the tube to a safe operating point. This minimizes the amount of time and heat required for tube warmup. The additional heating scans and GenCal scans have been moved to FastCal.

FastCal has been modified to optimize the GenCal and FastCal performance by heating the tube such that it is at a proper operating point during cals. FastCal will consist of the following:

- Warmup 1 scans required to put the tube at approximately 20 - 40% heat storage for GenCal
- GenCal scans (AutomA Cal and Ductility Scans) and the associated processing.
- Warmup2 scans required to put the tube at approximately 40% to 70% of heat storage for FastCal.
- The original FastCal scans and processing, QSA scans and processing, DST scans and processing. See Figure 7-43**.**

30.1 Scan Sequence

Tube Warm Up for the HSA Tube will first perform the following scans:

- 1-80kV/50mA/2sec
- 1-100kV/50mA/2sec ISD 2.2
- 1-100kV/100mA/21sec ISD 1
- 1-120kV/200mA/16sec ISD 4

Approximately 46 seconds

Tube warm up for the Performix 1x tube will perform the following scans

:

Scans	Technique	ISD
1	80kv/50ma/10 second	
1	100kv/80ma/10 second	2.2 second
1	120kv/120ma/10 second	2.2 second
1	120kv/200ma/10 second	1.0 second

Table 7-15 Performix Tube Warm Up Scans

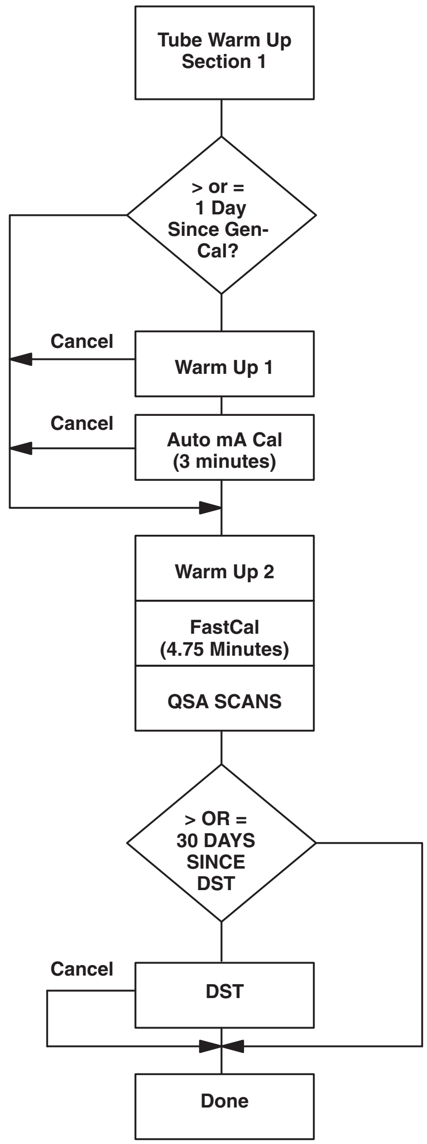


Figure 7-43 Tube Warmup and FastCal Flow Chart

30.2 Warm Up and FastCal Operation

Note: The heating scans in Tube Warmup and FastCal are specific to the tube type HSA (MX_165CT_I).

30.2.1 Warmup 1 and Auto mA Cal

A new process called ‘`gencalgui -fastca`” was implemented which executes the Warmup I scans and Auto mA Cal scans and processing. This process is invoked from CalRx in the FastCalMgr process before the original FastCal scans and processing.

This process checks the tube temperature before the Warmup I scans. The tube target temperature must be 150 degrees Celsius for the HSA tube. If the tube temperature is not warm enough the feature will exit and the operator will be directed to do Tube Warmup.

It is sufficient to calibrate the generator once a day, 18 hours will pass before the Auto mA Cal scans will be executed again. When Auto mA Cal is executed from FastCal, the time is logged. If more than 18 hours have elapsed since the timestamp, the warmup and GenCal scans will be executed on the next run of FastCal.

If the operator cancels Warmup I or Auto mA Cal, the FastCal process is then started. (Note: The timestamp will have been updated and the warmup GenCal scans will not run during FastCal for another day.)

30.2.2 Warmup 2, FastCal and QSA

First the Warmup II scans will be executed. These will be added to the FastCal protocols, so the operator will have additional scans executed with FastCal scanning.

Then the original FastCal scanning and processing will occur during this process.

Upon implementation of preferred fast cal, only the scans at kVs specified during reconfig will executed with each aperture. The operator preferences will be saved in a config file.

The QSA scans are automatically executed as part of the FastCal scans. The QSA processing is kicked off upon completion of the QSA scanning.

In CT/i 5.x, QSA scans are enhanced to include short and long term trending of air and bowties scan data. In addition, DC Cals are preformed to evaluate DAS performance. See Chapter 6,Section 2.0 for more details.

See Chapter 6, 2.4.4 on page 263 and 2.4.5 on page 267, for more details on Smart Trend and DC Cals.

30.3 Detector Slope Test

After FastCal, CalRx checks to see if the Detector Slope Test needs to run. If it does, the dst process starts. If the detector Slope Test does not need to be run then CalRx will terminate normally.

FastCal determines if it is time to run the Detector Slope Test by comparing the current date to a date stored in a reference file. If the current date is greater than the date stored in this file, the Detector Slope Test shall be run. If the current date is less than that stored, the Detector Slope Test will not be run automatically.

30.4 Preferred FastCal

Preferred FastCal allows the selection of a default set of FastCal scans that are a subset of the entire calibration set. This allows the site to tailor the FastCal scans that will be run daily based upon their use of scan techniques. The Preferred FastCal set is configured using the reconfig utility script.

A new configuration file for preferred fastcal will be created by reconfig in the /usr/g/config directory with file name PreferFastCal.cfg.

If the PreferFastCal.cfg file does not exist, FastCal will use all KVs by default.

Section 31.0
Scan Analysis Overview

The scan analysis feature allows users to have interactive access to scan files collected on the SBC. 2-Dimensional Displays will be viewed through a viewer while 1-Dimensional vectors will be viewed through the plotter. Raw scan or offset-corrected data can be processed and presented to users via VVC, CDPlot or MSDPlot functions.

Note: ROI definition — The concept of an ROI is extended to include point-oriented and plot-oriented ROIs, in addition to the conventional area-oriented ROIs (i.e. box, band, ring, etc).

31.1 Starting Scan Analysis

Upon starting the Scan List Select window, users can highlight an exam > series > scan, and perform the desired analysis feature by pressing any of the following buttons:

MEANS AND STDEV: Provides Means and Standard Deviation Plot analysis of a scan file. The plotter is started to display the means vector and the standard deviation vector, computed across the entire scan, in two separate windows. Cursor reporting is provided.

VVC: Provides Views-vs-Channel analysis of a scan file. The viewer is started to display the 2-D scan with cursor reporting (see page 343). The following ROIs are available: cross-hair, horizontal line, vertical line, cdplot box. Processing of ROIs is described on page 343.

OFFSET CORRECTED: Similar to VVC but first applies offset correction to a raw scan file.

CD PLOT: Provides Convolved Data Plot analysis of a scan file. The tool will first apply the prep and filter processes to the selected scan file. (On a non-propriety system, the filter step is skipped). The viewer is then started to display the resulting filtered projection file with cursor reporting (see page 343). The following ROIs are available: cdplot box, horizontal line, vertical line, cross-hair. Processing of ROIs is described on page 343.

SHOW SCAN HEADER: Displays header info of a scan file.

SHOW CAL VECTORS: Plots all calibration vectors in a scan file via the plotter.

SAVE SCAN DATA: Saves scan file from SBC database to the “/data” disk partition, converted to iq file format. This is useful for later archiving.

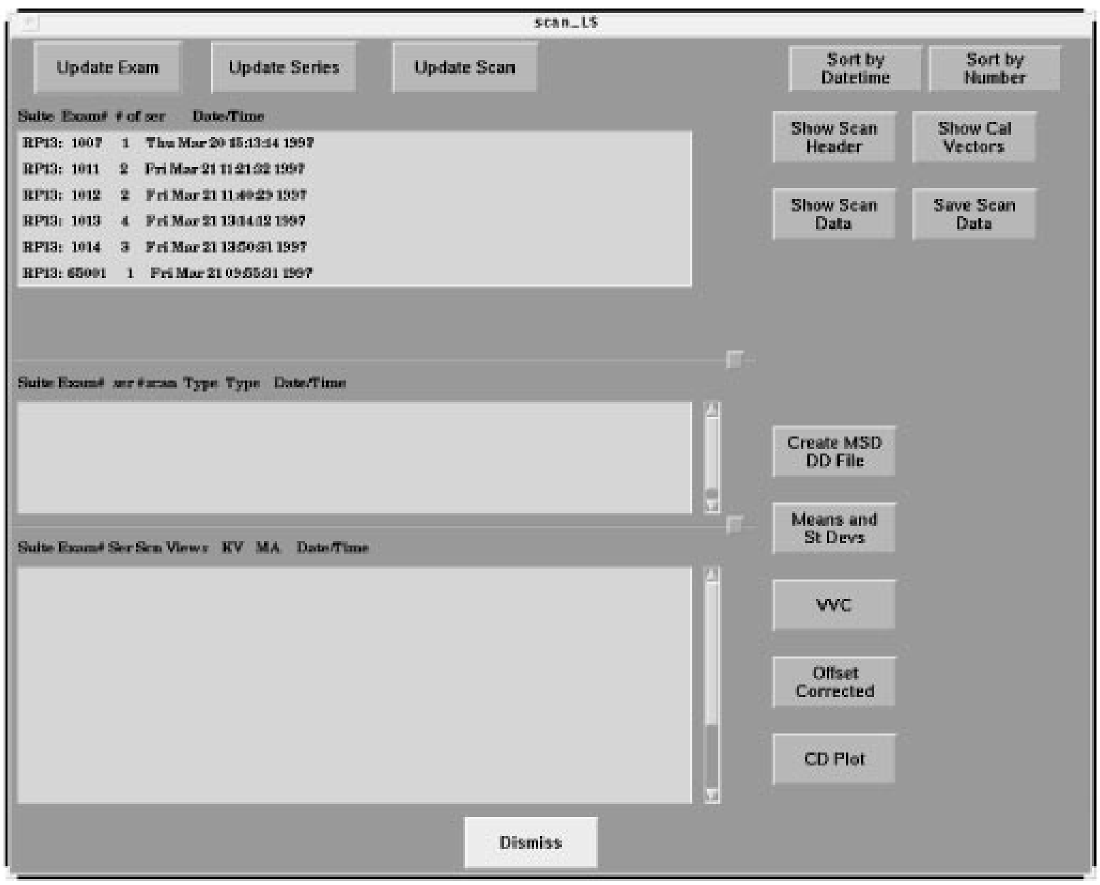


Figure 7-44 Scan Analysis User Interface

31.2 Scan Analysis Viewer Interactions

31.2.1 Mouse Behavior

The viewer displays scan data in a window. Users can interact with it using the mouse. Moving the mouse around in the display will provide interactive cursor reporting (see 31.2.2). When the cursor is moved near the edge of or into an ROI, the cursor appearance will change to reflect what will happen if the adjust-ROI button is dragged. These special cursors are referred to as adjustment cursors. The following mouse interaction is provided:

- Adjust-ROI (MB1]:

  Dragging the left mouse button, while the adjustment cursors are appearing, will adjust either the location or size of an ROI. When button is pressed or clicked, adjustment cursor will jump to corresponding edge or center of ROI.
- Adjust-W/L [MB2]:

  Dragging the middle mouse button changes the window leveling. Behavior is similar to AW, with ramping such that amount of change is exponentially proportional to how far mouse is dragged from start of press. However, unlike AW, W/L updates are much slower so that <u>a slower drag is recommended</u>.
- Cycle-ROI [MB1 Á MB2]:

  Clicking the middle button while holding the left will cycle through the available ROI types.
- Generate Postscript [MB1 Á MB3]:

  Clicking the right button while holding the left will output the scan in a postscript file in /data directory. (This postscript feature is also available in the plotter).
- Accept [MB3]:

  Clicking the right button once will indicate acceptance of the ROI location and size.
- Exit-Viewer [MB3 x2]:

  Double-clicking the right button will terminate the viewer.

31.2.2 Cursor Reporting

- Channel Reporting:

  Depending on scan file type, the viewer will report the channel/view numbers along with X, Y coordinates, as the mouse cursor is moved.
- W/L Reporting:

  Whenever the Adjust-W/L mouse button is pressed (or clicked), the current window-level values are displayed.
- ROI-Location/Size Reporting:

  Whenever the Adjust-ROI mouse button is pressed (or clicked), the current ROI location and dimension are displayed.

Cursor reporting within the Plotter is described below.

31.2.3 Plotter Interactions

The viewer will start up the plotter when any of the plot-oriented ROI is accepted.

After the viewer starts up a plot, the view window is de-activated. Cursor reporting of X/Y is available by moving the mouse inside the plot window. A click of MB3 (right button) will exit the plotter and return control to the viewer. Clicking the right button while holding the left will output the plot in a postscript file in /data.

31.2.4 ROI Processing

The processing carried out by the viewer whenever an ROI is accepted depends on the type of ROI:

- Point-Oriented ROI:

  The pixel value at X,Y is reported. *(ROIs: cross-hair)*

- Plot-Oriented ROI:

  This type of ROIs appears in **dotted linestyle** and always generates a plot via the plotter. For cdplot ROI, the means vector of the region is first plotted, and upon exit of this plot, the standard deviation vector is then plotted. *(ROIs: horizontal/vertical dotted lines, cdplot)*

- Area-Oriented ROI:

  This type of ROIs appears in **solid linestyle** and is always associated with the statistics (means, stdev, max, min, area) of the area enclosed. *(ROIs: box, ellipses, band, ring, donut, point-to-point line,)*

31.2.5 Known Scan Analysis Limitations

When interacting with the viewer or plotter, avoid closing the window via the window frame pulldown. Instead, use the right (MB3) mouse button to exit cleanly. (Otherwise, some processes will be left behind, until the time the GUI is “dismissed” or the desktop is “cleaned up.”

CDPlot analysis of helical scans is not supported. However, VVC or Offset-Corrected functions are available to troubleshoot helicals.

When the plotter/viewer does not start up or the screen dims after startup, the problem could be because of colormap issues. Check the log files and exit other concurrent plotters, viewers, or service applications and try again.

Section 32.0
DD FIle List Select and DD Math

This section covers the use of Scan Analysis (scanLS) and DD Analysis (ddLS) to make use of the ddmath functions.

32.1 DD File List Select Overview

DD math is a means for the user to apply mathematical operations: add, subtract, multiply, and divide to dd files, and calculate the channel to channel difference or ratio of means vs. standard deviation vectors of a dd file. It allows the user to specify the scaling factor for the output vector, and provides three output modes: dd file, plot, and view numbers.

DD math is part of the dd analysis user interface. Scan Analysis is used to generate dd files that may then be manipulated and or examined using dd File Analysis (ddListSelect).

32.2 DD Files Generation

There are eighteen different DD file types of six orientations. The orientations are View, Channel, RTS, CAL, Elements, and Header.

Channel oriented Means and Standard Deviation type dd files can be created from iq scan files in the Scan Analysis application.

32.3 DD Math Functions

DD math consists of the following functions:

- Add
- Subtract
- Multiply
- Divide
- Channel to Channel difference
- Ratio of means vs. standard deviation

32.3.1 Add, Subtract, Multiply, Divide

Applies add, subtract, multiply, and divide between vectors in two dd files.

The output file is a dd file with one of the following suffixes:

- `.add`
- `.dif`
- `.mul`
- `.rat`

Operations can be performed on dd files in View orientation, Channel orientation, RTS orientation, and Cal orientation.

Currently, no dd type restrictions are applied to operations between dd files, as long as the dd vectors have the same number of elements. If one file has a single vector and the other file has multiple vectors the mathematical operation will be applied multiple times using the single vector. Otherwise the mathematical operation will be applied component wise for the number of vectors in each file.

32.3.2 Channel to Channel Difference

Applies the following calculation to the data from the data set(s) in the dd files for View, RTS or Cal orientation.

(X2-X1), (X3-X2), (X4-X3),...,(Xn-Xn-1)

The output is channel to channel dd file with extension: `.c2c`

32.3.3 Ratio of means vs. standard deviation

Takes a MSD (means and standard deviation) or RTS (real time statistics) type of dd file, calculates the ratio of data in means vector (1st set) to data in standard deviation vector (2nd set). The output file is a Ratio type of dd file with the extension: `.rat`

32.4 DD Math Output Mode

Three output modes are supported in dd math:

- Plot

  Plot will plot the output dd vector using an on screen vector display.
- View Numbers

  View Numbers will display the dd vector numerical values on the screen and the user can perform numerical searches in the window.
- DD File

  Allows the user to specify the output dd file name with a full path or the file basename.

If only base name is provided the program will use the default prefix and suffix for the output file. The created dd file will be shown in the dd file list.

32.5 Creating dd Files With Scan Analysis

Refer to Figure 7-45.

The user may select file(s) from the List Select Viewer on the left side of the User Interface and then select the CREATE MSD DD FILE button to generate msd dd file(s) from the selected scan file(s). An message window will popup to inform the user of the name of each dd file created. The created dd files will use the following naming convention:

`dd.suit#.exam#.series#.scan#.msd` the output dd file will be created in directory `/data`

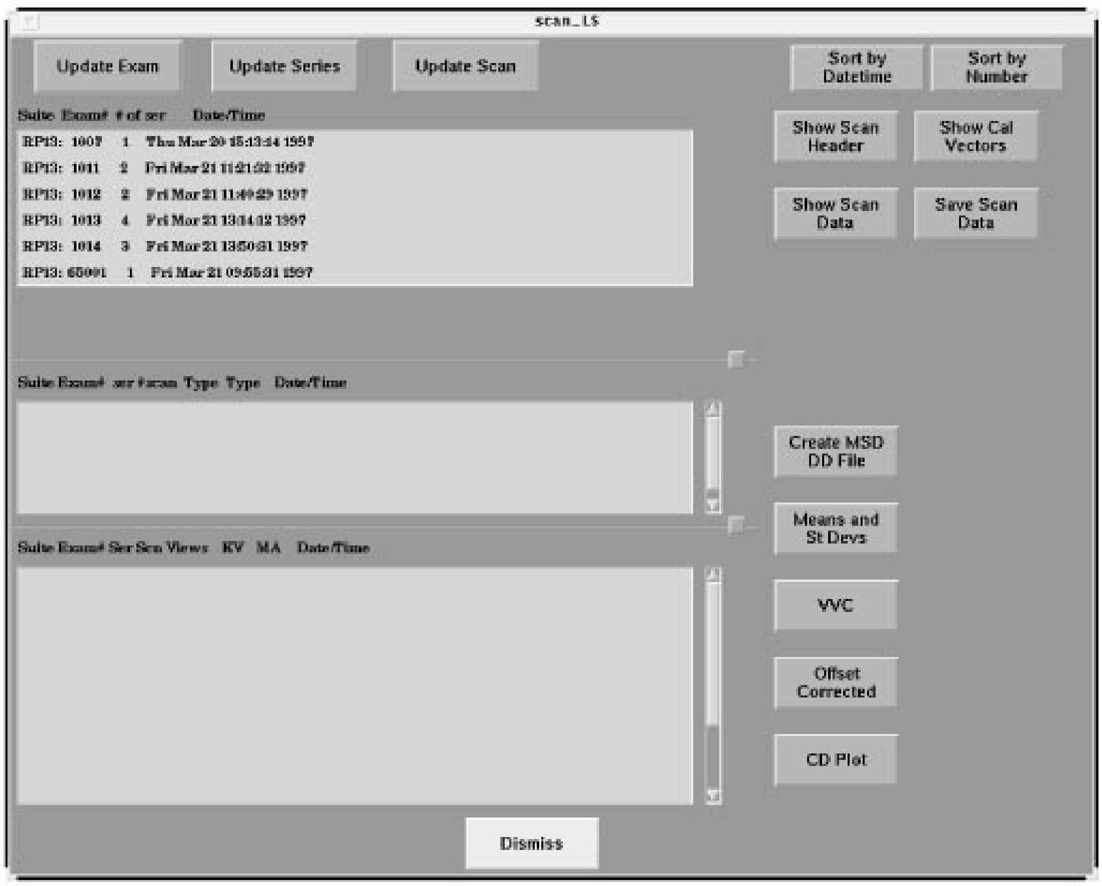


Figure 7-45 Scan Analysis User Interface

32.6 DD Analysis User Interfaces

The DD math operation panel and a set of the dd math operation buttons are part of the ddLS screen.

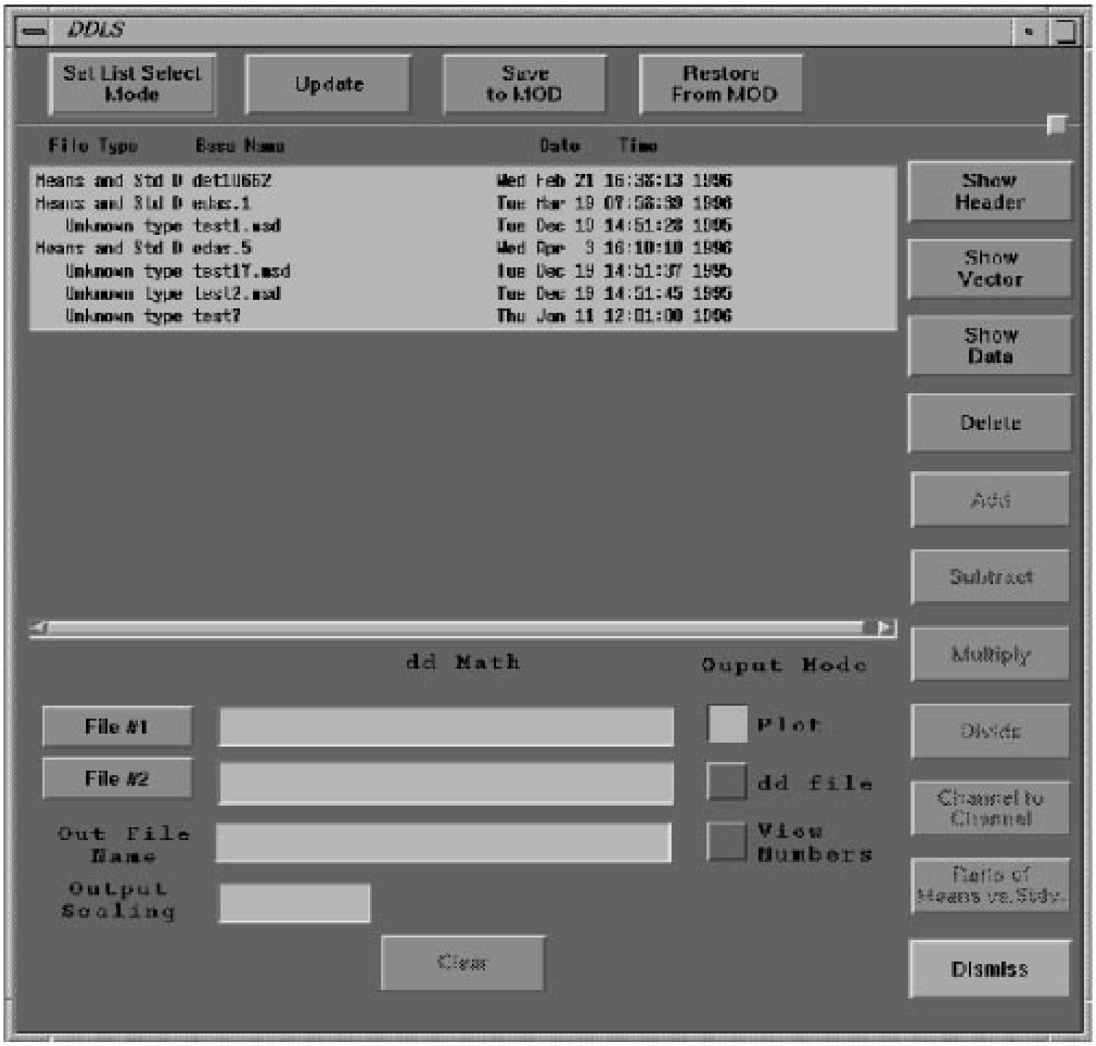


Figure 7-46 DD List Select User Interface

32.6.1 Functions in ddLS User Interface

The ddLS supports the following functions for various file types.

- Save to MOD
- Restore From MOD
- Show Header
- Show Vector
- Show Data
- DD math operations
- Set list select mode
- Update

The user can perform these functions, except dd math operations, by simply selecting one or more files in the list select window, and clicking the function button.

The following file types are supported in the ddLS user interface.

- DD File
- Cal File
- Data File
- Scan File
- Image File

32.6.2 File Operations

- DD Math Operations

  Perform: add, subtract, multiply, divide, and channel to channel difference operations on dd files. These operations are only available for dd file types.
- Show Header

  Displays the header information of the selected file(s) in the show header window for the following file types: DD Files, Image Files, and Scan Files
- Show Vector

  Plots the vector(s) of the selected files in the display window for the following file types: DD Files and Cal Files
- Show Data

  Prints the numerical data of the dd vector(s) to the display window(s). For Image file types and scan file types, it will display the VVC plots of the selected files.
- Save/Restore to/from MOD

  Saves the selected files to the MOD and Restores all the dd files from `/MOD/ddfiles` to `/data` directory.
- Set list select mode:

  Pops up the DDLS dialog window and allows the user to set the list select file type.
- Update:

  Refreshes the display in the ddLS user interface.

32.7 DD Math Operations in ddLS

The default ddLS screen is shown in the Figure 7-47. The dd math operation buttons will be insensitive if no files are selected into the dd math operation panel.

The user may start dd math operation(s) by selecting the file(s) and putting them into the selection field by clicking the button FILE #1 or FILE #2. If the selected file is not a dd file, the application will not put it into the dd math operation field. A message window will pop up and ask user to select a dd file.

If only one file is selected and it is of the file type RTS dd file or MSD dd file, both CHANNEL TO CHANNEL and RATIO OF MEANS VS. STDV BUTTONS will become sensitive. If the selected file is not of the type MSD or RTS, only CHANNEL TO CHANNEL button will become sensitive.

When two dd files are selected, the ADD, SUBTRACT, MULTIPLY and DIVIDE buttons become sensitive and CHANNEL TO CHANNEL and RATIO OF MEANS VS. STDV button will be insensitive.

The user can specify the output file name when the dd file output mode is set. Otherwise a default dd file name will be provided.

The default output scaling factor is 1.0. The user can set the scaling factor to any real number.

When the dd math operation buttons are sensitive, the user can select the desired button to start the dd math operation.

The CLEAR button will clear the dd math operation fields, reset dd math operation buttons and reset the output mode to the default mode — Plot mode.

32.8 Limitations for ddLS

Button sensitivity on the ddLS user interface will change according to the selected file type(s). All unsupported function buttons will be insensitive.

When list select mode is set to Cal Files, only cal vector files will be listed in the DD Analysis list scroll window.

SHOW HEADER and SHOW DATA will be insensitive since these functions are not supported for cal vector files.

When list select mode is set to Data file, files in the directory `/usr/g/service/state` will be listed in the list select window. SHOW HEADER, SHOW DATA and SHOW VECTOR will be insensitive.

For Image Files and Scan Files, SHOW VECTOR is not available.

32.9 DD File List Select Mode

Figure 7-47 DD List Select File Type User Interface

The buttons in both of the Major File types panel and the dd minor types panel are radio buttons. The DD minor type toggle buttons will be insensitive unless the DD FILES button is set. The PROCEED button will close the interface and list the files of the selected file type in the dd analysis scroll window. The CANCEL button will pop down the dialog window, and will not update the dd analysis scroll window.

32.10 DD List Select Functional Block Diagram

Figure 7-48 is a software function diagram for dd analysis. It is only intended to give an overview of the steps in the process.

In the dd math case, the user starts by selecting the file of interest and using the FILE #1 and FILE #2 push buttons to put them into the dd math panel. During this process file type checking is performed to check if the selected file(s) are dd files. The user can then select the output mode, specify the output dd file name (if dd file output mode is set) or the output scaling factor. The user can then start the dd math operation by selecting the dd math operation buttons, which collect the user input and start the ddmath process which performs the mathematical calculations. When ddmath is done, the process manager will decide if it needs to start the plot vector or show data processes based on the users request(s).

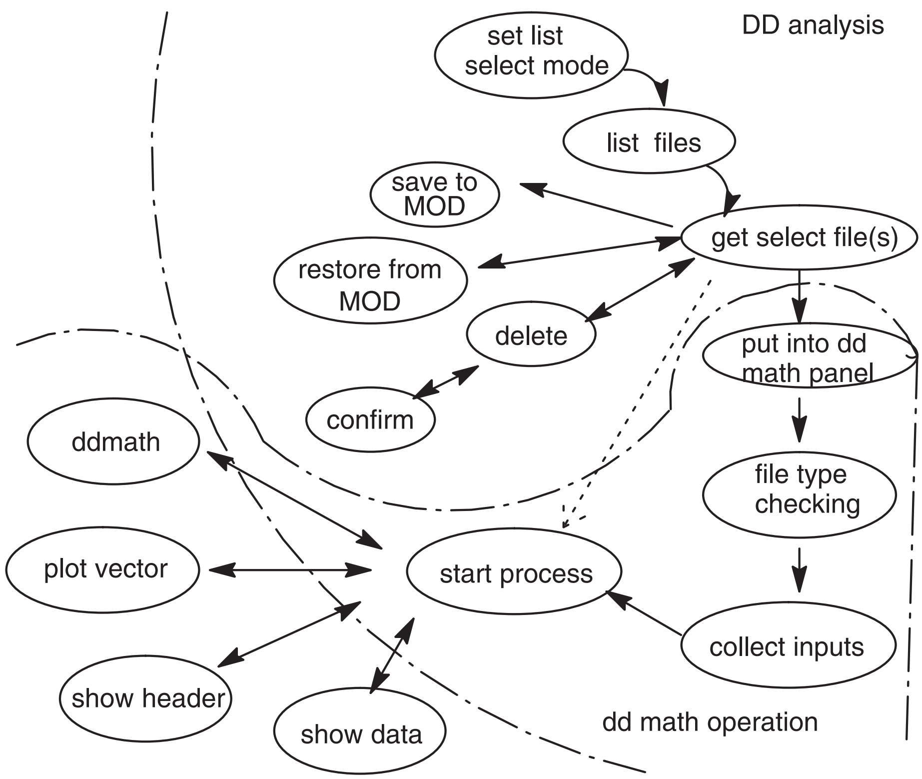


Figure 7-48 DD File Functional Block Diagram

32.10.1 DD List Select Data Flow Diagrams

Figure 7-49 shows the data flow in the dd math operation process. When the dd analysis interface is brought up, based on the file selection mode, the default file path of this file type is searched, and the files of the selected type are listed in the dd analysis user interface. When the user starts any ddmath operations, the application will collect the input from the dd analysis user interface, pack it into a command buffer and pass it to the process manager. The process manager starts the process which creates the dd file in directory `/data` and plots the vector or the number out in the display window(s).

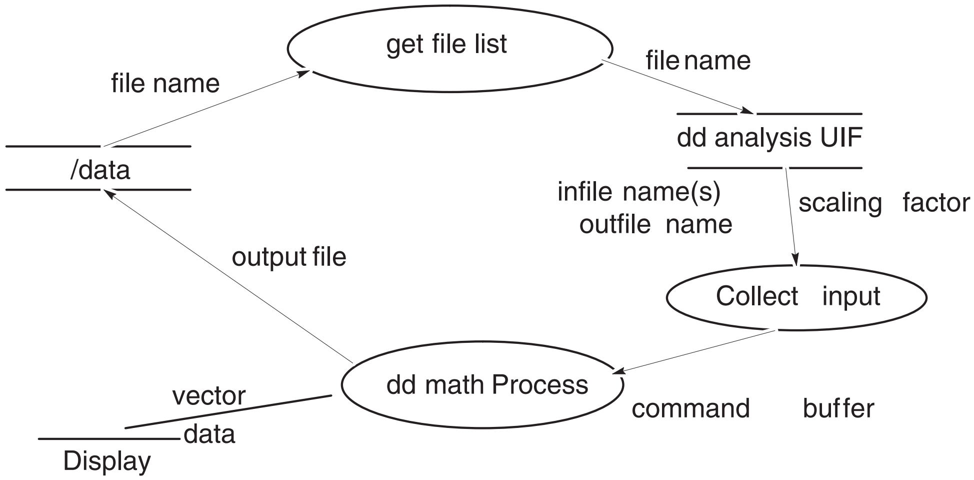


Figure 7-49 DD Math Data Flow

32.10.2 DD List Select Processing

Two different file type checks are implemented in the ddLS. First, when the user selects file(s) from the list select scroll window for the dd math operation, the software verifies that the selected file is a dd file type. If not, a pop-up message window will direct the user to select a dd file.

Second, when the user selects the dd math operation button, the software will verify that the file type of the two selected files are valid for the dd math operation. If the two selected files have the same number of elements and a valid file orientation, View, Channel or Cal Orientation, the operation will proceed.

When the user selects file(s) for the dd math operation and the output dd file mode is set, a default output file basename will appear in the output file name text field. The user can edit the output file name and may also specify the full path of the output file name or specify the base name of the output file and software will add the prefix and suffix to the output name.

The default output base name will be `ddmath****`, the default full name of the output file will be `dd.ddmath****`.(action suffix)

Where the action suffix is dependent on the dd math operation. For example `.add` will be the suffix for Add operation, `.dif` for subtract, `.rat` for Divide or Ratio of Means vs. StDev.

If ddmath fails, it will display an error message to inform user of the failure and removes the output dd file if dd File output mode is not set and no other processes need the output dd file.

If plot vector completes successfully, it will remove the output dd file if necessary.

Section 33.0
X-Ray Tube Heat Soak And Seasoning

33.1 Heat Soak and Seasoning Overview

The Heat Soak and Seasoning program drives the scanner to perform a set of scans, which when taken in the proper sequence and with the correct time intervals between scans can extend the life of CT tubes.

Heat Soak and Seasoning is a program that is run by the service person whenever deemed necessary by observing the number of tubespits during actual scan operations. The entire protocol consists of three phases which will be executed in sequence.

This program seasons a new tube by first performing a tube heat soak. This process is done to remove any undissolved gases in order to minimize the occurrence of mA overloads.

After the heat soak, a high voltage stability test (Seasoning) is performed to verify that the tube is stable. Real time feedback of high voltage stability is provided to the user in order to determine if the current technique scans must be repeated. Because some of the scans used in this procedure are not used in normal patient scanning, special calibration scans are needed to determine the parameters needed to make these scans.

The Heat Soak and Seasoning procedure can be thought of as a series of alternating calibrations and scans. The particular sequence and parameters differ with the tube's type. In some cases, tube cooling delays need to be determined so that scans can be completed without need for extra cooling delays.

33.1.1 Tube Warmup

The Tube warmup phase of the Tube Heat Soak and Seasoning procedure raises the temperature of the tube slowly in order to prepare it for the high power scans that will follow. This phase is very important to minimize target damage when it is suddenly subjected to high energy input.

33.1.2 Heat Soak

New tubes may have undissolved gases that could render the tube unusable due to excessive arcing. The purpose of the heat soak phase is to redissolve gases in the tube at high temperatures in order to minimize the occurrence of current overloads. (i.e. tube spits).

The Heat Soak procedure consists of three sub-phases which are Heat input, Anode Soak and casing Soak. The Heat input and Anode Soak scans are performed in a dynamic series so that additional tube cooling is not necessary during the two sub-phases.

- Heat Input

  This phase heats the target up to maximum heat storage. This heats the target and other parts of the tube to maximum temperature for proper degassing. At the same time, the tube "getter", a chemical which absorbs gases, is heated up to its activation temperature to absorb the gases in the tube.

- Anode Soak

  This phase maintains the target at maximum heat storage and maximum temperature to continue the degassing and absorption of gases by the getter.

- Casing Soak

  In this phase, the tube unit is heated up to the casing heat storage limit. This heats the oil to a point where gases can be reabsorbed by the oil.

33.1.3 High Voltage Seasoning

High Voltage seasoning eradicates any small micron sized particles that may be in the tube insert. Left in the tube, these particles can become charged and thus, cause arcing. In addition, this segment can be used as troubleshooting tool by the service person to verify that tube operation is stable prior to customer use.

Interruptions during the automated scans are allowed only during the Seasoning phase. if interrupted, scanning can be resumed from the previous scanning station till completion. The state entered on an interruption is called the manual mode.

33.1.4 Hot ISO

Prior to R3.6 software, the ISO alignment was done cold and the focal spot position data used by the system is this cold position data. However, clinically the tube is typically at 50-90% heat storage for most of the scans done on a normal day. Therefore, better system performance (IQ) can be achieved if the focal spot position values are set at a point equivalent to when the tube is warm.

Hot ISO requires that the tube be heated to near maximum capacity so that the total drift of the focal spot can be measured. Heat Soak and Seasoning (HSS) heats up the new tube to near maximum storage for tube Seasoning. ISO scans are added between the heating scans and the season scans of the HSS feature with minimal impact on both tube change time and HSS.

The ISO values that result from Hot ISO requires four new fields in ScanHardware.cfg (two to store the drift values of each of the spots (small & large) and two to store the ISO values of the cold spots (small & large). These values will also have to be stored in the INFO file for Save/Restore of system state.

33.2 Tube Heat Soak and Seasoning Protocols

The Tube Heat Soak and Seasoning protocol for the HSA MX_165_CT_I tube is given below. Refer to page 574 for more detail about the Performix tube and other x-ray tube replacement procedures. All scans are done with the tube's large focal spot. The scan time and interscan delays specified in the protocols listed must be rigidly enforced. The protocol for the Performix tube is similar.

The execution of the scans listed must follow certain rules. The tube warm up stage is optional depending on the temperature of the tube. The next five scan groups (Heat Input to Casing Soak 2) must be executed without interruptions (if possible). If the protocol is stopped during this period, it is recommended that this entire group of protocols be executed again. During Seasoning, execution may be stopped at the operator's discretion and can be either resumed or started once again from the previous scan group.

MX_165_CT_I Scan Group	kV	mA	# of Scans	ISD (sec.)	Scan Time (sec.)	Pre Group Delay (sec.)	Static / Rotate (4.0sec)
Warm Up	80	100	15	2	2	2	S
Heat Input	80	300	24	2	3	2	R
Anode Soak 1	80	300	25	5	3	3	R
Anode Soak 2	80	300	9	1	1	2	R
Casing Soak 1	80	300	90	12	2	60	R
Casing Soak 2	80	300	10	7	1	7	R
Seasoning 1	90	50	5	5	0.1	5	R
Seasoning 2	100	50	5	5	0.1	5	R
Seasoning 3	110	50	5	5	0.1	5	R
Seasoning 4	120	50	5	5	0.1	5	R
Seasoning 5	130	50	5	5	0.1	5	R
Seasoning 6	135	50	5	5	0.1	5	R
Seasoning 7	140	50	10	5	0.1	5	R
Seasoning 8	145	50	10	5	0.1	5	R
Seasoning 9	150	50	15	5	0.1	5	R

Table 7-16 MX 165 CT HEAT SOAK AND SEASONING PROTOCOL

33.3 Running Heat Soak And Seasoning

When the feature is started, the interface window pops up. Refer to Figure 7-50. It lists all the scan sets to be executed for the entire procedure.

The default protocol is loaded at the time of start-up. This is also the protocol that is loaded on pressing the Warm Up and HSS (Heat Soak And Seasoning) button on the Heat Soak and Seasoning Window. Refer to Figure 7-50.

The Heat Soak and Season button skips the first scan set (Heat Input) and lists the rest of the protocol. This is desired in case the tube is already warm and a ductility warmup is not needed.

Season button loads Seasoning 1 to Seasoning 9 scan sets.

After the required protocol is loaded, the Accept button is pressed which makes the three protocol buttons insensitive. The scan set being executed is highlighted as shown in Figure 7-50.

The Back One button, Tube Spits this series, Tube Spits last scan and the boxes that shows the number of tube spits are insensitive until the Seasoning stage. During Seasoning, the number of tube spits during each scan is shown in the Tube Spits last scan box and the Tube Spits this series box keeps track of the spits in that series.

Dismiss can be pressed at any time to exit this feature. Dismiss is not active during execution until the cancel button is pressed from the activation screen (see below).

An Activation Screen pops up during execution of the Heat Soak And Seasoning Scans when the scans are being done. It lists the Kv and mA applied during each scan, the current scan of a series, the number of scans left and provides buttons to pause and quit the application.

The Back One button in the Heat Soak seasoning interface becomes sensitive to user input in the Seasoning phase upon pressing the Cancel button of the activation screen. At this point, scanning may be resumed by pressing the Back One button of the Heat Soak and Seasoning main interface. The scan sets from the previous scan group are loaded and displayed on the Heat Soak and Seasoning main interface and scanning starts afresh.

Pressing the Pause button of the activation screen during scanning causes the scanning to pause and the options are to Resume and to Cancel. Cancel gives the option of going back one if in the Seasoning mode (as explained above), or the application can be quit.

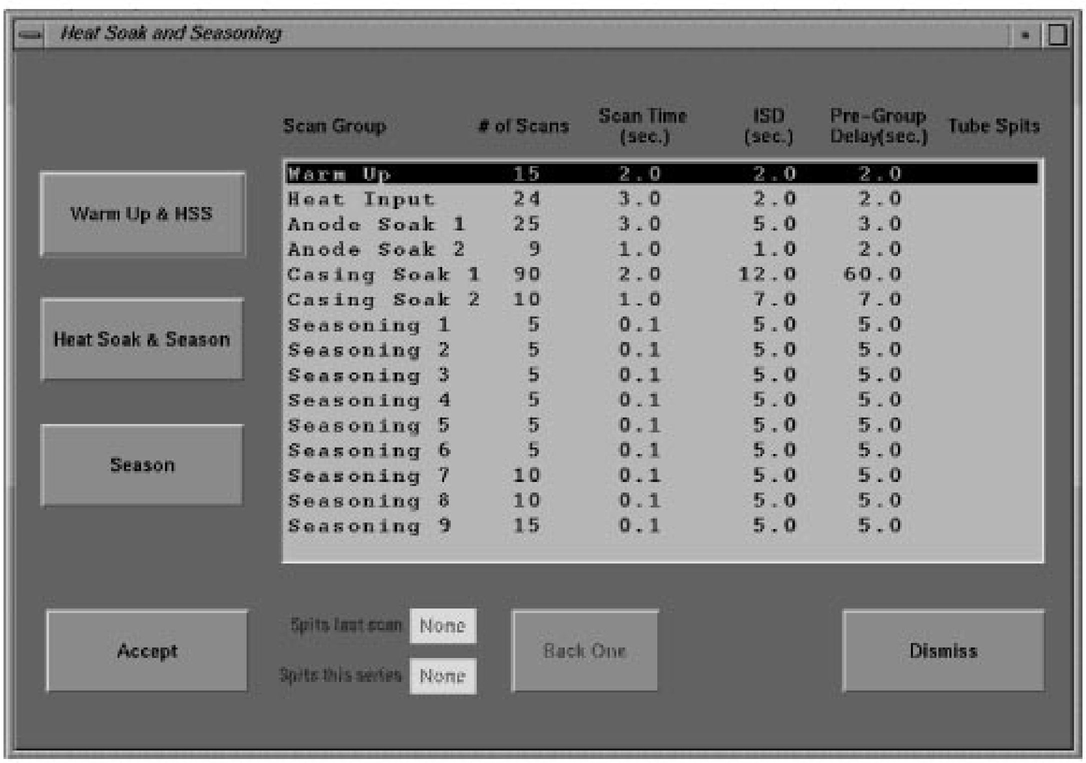


Figure 7-50 Heat Soak And Seasoning User Screen

Section 34.0
Exposure Backup Timer Functional Test

Use this test to verify the backup timer operation (i.e., timer activates, timer counts down to zero and backup contactor de-energizes).

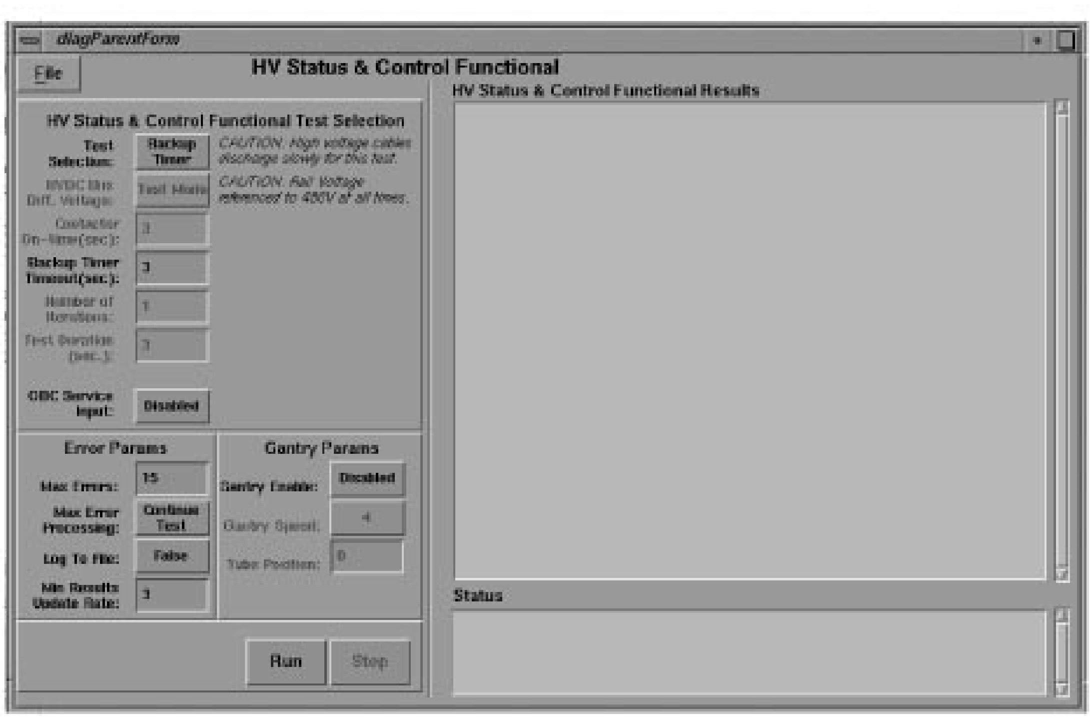


Figure 7-51 Backup Timer Diagnostic Screen

The Gentry I/O Board contains the backup timer. The software loads the scan time +5% into the backup counter before the start of exposure. The extra 5% gives the backup contactor time to energize. The backup timer begins counting down when the system detects the HV ON or Exposure Command. If either of these conditions persist after the timer counts down to zero, it sends a level 1 interrupt to the CPU and disables the backup contactor.

The read and write verification requires the operation of the clock and clock select circuits. This diagnostic tests both the 488.28 Hz and 1953 Hz clocks. The diagnostic simulates an exposure, and verifies that the circuit generates a backup timer interrupt.

The system posts a test status message to the screen while it runs the corresponding test.

The Backup Timer Timeout defaults to three seconds which should provide enough time to verify operation of the backup timer.

1.) Select RUN

- The results window indicates the progress of the test, and not the state of the hardware.
- The screen information updates one line at a time, as each step completes.
- If a failure occurs, the system posts an inverted video error message indicating a test abort after the failing step.

Note: Failure to read the correct rail voltage does not cause test suspension.

Section 35.0
X-Ray Interlock Functional Test

This function tests the ability of the X-Ray interlock to disable an exposure. The test opens and closes the STC and FEP Board interlock relays and verifies the state of the Gentry I/O interlock sensor. In the event of a fault, the test allows the user to loop on this condition indefinitely, for troubleshooting purposes.

Select the Troubleshoot Menu X-RAY GENERATION X-RAY INTERLOCK softkey (Figure 7-52) to access the X-Ray Interlock Functional Test.

Note: When making selections

- You may select other tests from this screen by clicking mousebutton one on the test selection softkey or by clicking mouse button three over the test selection softkey, to display the following pop-up selection menu:
- When you select Run, the system checks the scan subsystem for resident firmware. If the system does not detect the firmware, it posts a message to inform you that it needs to download firmware. It prompts you to select |Yes| to download the firmware.

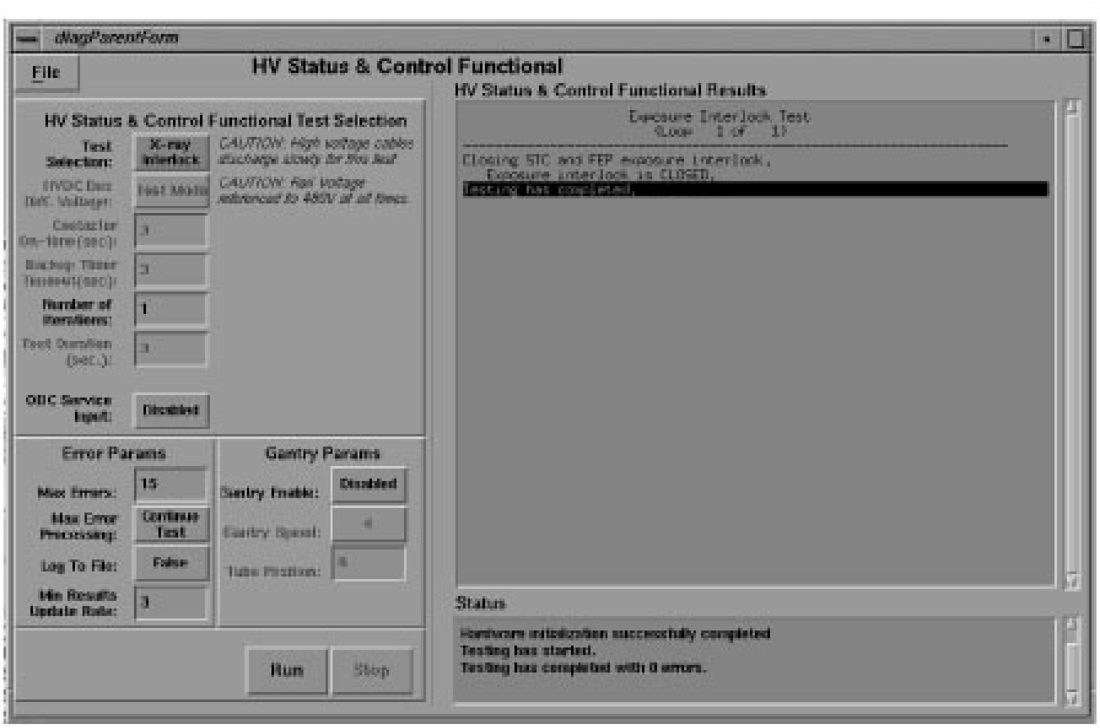


Figure 7-52 X-Ray Interlock Functional Test results example

Section 36.0
X-Ray Exposure Manual Test

Use the SERVICE > TROUBLESHOOTING > KV & MA > X-RAY FUNCTIONAL TEST to test the scanner's ability to generate accurate kV and mA. It assumes your baseline is accurate. Test this baseline with a bleeder at least once a year.

Figure 7-53 shows the **X-Ray Functional Test screen**. Input ranges are:

- **KV:** 60 to 140KV in 1KV steps
- **mA:** 40 to 400mA in 1mA steps (10 to 440 mA with Performix tube and CRPDU)
- **Duration:** 1.0 to 10 Seconds in 0.1 second steps
- **Iterations:** 1 to 100
- **ISD:** 1 to 60 Seconds in 1 second steps

Select RUN and wait for the Scan Start button on the console keypad to illuminate. Press the Scan Start button, when lit, to initiate the scan.

The X-Ray Functional Test Results screen output consists of HV statistics. The data displayed was taken 1007ms into the exposure and was posted to the screen. (“_” indicates an unknown value)

- *Average:* the average value taken over the duration of the exposure.
- *Selected:* the value prescribed by the user.
- *Last Sample:* the last value read before the screen updated. The Last Sample exposure duration displays the data collection time, **in milliseconds, from the start of exposure**.

Data displayed in the Last Sample column represents the last sample of HV statistics taken on or before 1007 milliseconds after the start of the exposure.

Figure 7-53 represents the screen at the end of the exposure. You can tell the exposure has ended because the Last Sample exposure duration equals or exceeds the Selected exposure duration value.

Note: Backup timer determines exposure duration

The backup timer determines the exposure duration. This timer stops counting after the system removes the Exposure Command and HV ON status, which means the last exposure could have occurred later than indicated.

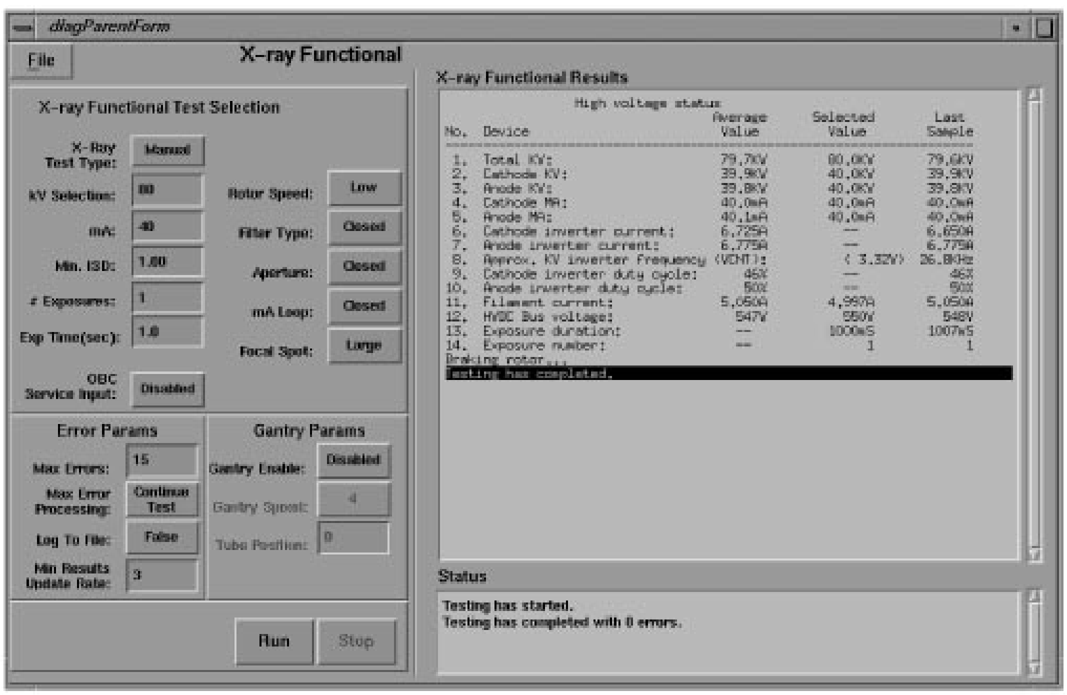


Figure 7-53 X-Ray Functional Test Results Example

Section 37.0
Image Generation Testing

37.1 Test Description

The Image Generation Test validates the image reconstruction hardware and software. Testing consists of creating images from test data stored on the High Speed Disk and validating their checksums. Errors detected by this diagnostic should be the same as those detected during patient scanning since the same image reconstruction software is utilized in both situations. Table 7-17 and 7-18 list the scan files and images used by this test.

Note: Images are NOT saved or displayed!

37.2 Test Initialization

37.2.1 Security Check

This is an Advanced diagnostic. Testing is permitted only if the service key is present or if executed remotely by InSite.

37.2.2 Check/Load Scan Data Files

The Scan Database must contain the Image Generation Test scan files before testing can begin. If these files are not present, the test loads the scan file from the OC's `/usr/g/service/tools/added_tools/Image_Gen_Test` directory into the Scan Database. This takes approximately 4 minutes to complete.

37.2.3 Create Test Error Log

The test error messages can be found in the `imageGen.log` file. This file is renamed to `imageGen.log.last` and a new `imageGen.log` file is created each time the test is initialized. This file/log can be examined via the "Display Test Error Log" selection on the main menu or at a later time by executing **`vf -e imageGenLog`** at the UNIX prompt within the test directory.

37.2.4 Read Test Protocol File

The `ImageGen.RATpro` file is read by the Image Generation Test at initialization. This file contains the protocols and image checksums used by the test.

37.3 Test Execution

The Image Generation Test is just an interface to the application image reconstruction environment. This is the same environment used for patient scanning. Therefore, the errors detected during this test should be the same as those found during patient scanning. Errors found by the application code are logged to the System Error Log. Errors found during test initialization and checksum verification are logged to the Test Error Log.

37.4 Test Termination

The QUIT option on the test main menu halts further testing and removes the shell window. The scan files used by this test remain on the disk until overwritten by another scan file.

37.5 Test Coverage

The hardware and software required to create images is verified by this test. Hardware includes the High Speed Disk, SBC, and IG board.

Scan File	Exam	Series	Scan	Scan FOV	Rotation
1.	64,999	1	1	Large	Normal
2.	64,996	1	1	Ped Head	Normal
3.	64,997	1	1	Large	Normal
4.	64,998	2	1	Large	Helical

Table 7-17 SCAN FILES USED BY IMAGE GEN TEST

Image No.	Scan File	Series	SFOV	DFOV	Recon Algorithm	Special Processing	Target
1.	1	Axial	Large	48.0	Soft	-	-
2.	1	Axial	Large	48.0	Standard	-	-
3.	1	Axial	Large	48.0	Standard	256 matrix	-
4.	1	Axial	Large	40.0	Detail	-	-
5.	1	Axial	Large	30.0	Sharp	-	-
6.	1	Axial	Large	20.0	Bone	-	-
7.	1	Axial	Large	10.0	Edge	-	-
8.	1	Axial	Large	5.0	Standard	256 matrix	-
9.	1	Axial	Large	38.0	Lung	-	-
10.	1	Axial	Large	48.0	Standard	-	Targeted
11.	1	Axial	Large	48.0	Standard	-	Targeted
12.	1	Axial	Large	48.0	Standard	-	Targeted
13.	1	Axial	Large	48.0	Standard	-	Targeted
14.	1	Axial	Large	48.0	Standard	Post Off	-
15.	2	Axial	Pediatric	25.0	Standard	IBO	-
16.	2	Axial	Pediatric	25.0	Standard	IBO Segment	-
17.	2	Axial	Pediatric	15.0	Standard	IBO 256 matrix	Targeted
18.	2	Axial	Pediatric	15.0	Bone	IBO	Targeted
19.	3	Axial	Large	48.0	Standard	Peristaltic	-
20.	3	Axial	Large	48.0	Standard	Segment	-
21.	3	Axial	Large	48.0	Standard	Segment	-
22.	3	Axial	Large	48.0	Standard	Segment	-
23.	4	Helical	Large	48.0	Standard	Helical	-
24.	4	Helical	Large	48.0	Sharp	Delta Start, Helical	-
25.	4	Helical	Large	48.0	Lung	Delta Start, Segment	-

Table 7-18 IMAGES CREATED BY IMAGE GEN TEST

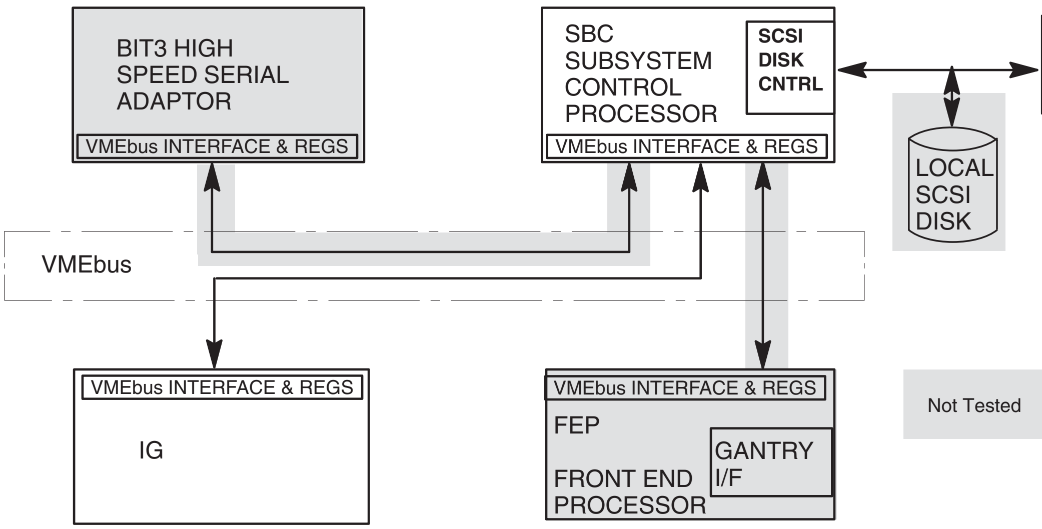


Figure 7-54 Test Hardware Functional Block Diagram

37.6 IG Test Usage

The Image Generation Test is started from the **Troubleshooting Menu** which contains a "Image Generation" choice. The absence of this choice indicates the Advance Service is not loaded. Insert the Service Key and select Check Security.

Before selecting **Image Generation**, bring up the System Message Log since this can **NOT** be done while the test is running. If the System Message Log is brought up during the test, the window displaying the test information will be lost in the background. To prevent this from happening, follow these instructions:

1.) If test is running and the System Message Log is not displayed, exit test.
2.) Press the Message Status Box at the top of the screen to display messages.
3.) Press the VIEW LOG RIGHT button to view the System Message Log on the right side of the screen.
4.) Press the Message Status Box again to put messages in the background.
5.) Press the IMAGE GEN TEST button to execute this test.
6.) Press the Message Status Box again to display the System Message Log in the foreground.

Touch the IMAGE GEN TEST button to execute the start-up script, which brings up the test window on the left side of the screen. The following message appears if the test scan files are not loaded onto the scan database:

```
Loading scan data into the database.
This will take up to 4 minutes. Please Wait.
```

Once the test scan data is loaded, the main test menu is displayed. The main test menu and the results screens are described below:

The following is the menu displayed if the scan data is successfully loaded to the High Speed Disk and the security check passes.

```
Image Generation Main Test Menu

   1. Execute Image Generation Test
   2. Display Test Error Log
   3. Restart Recon
   4. Help Screen
   5. Quit

   Enter Selection [1-5]->
```

Figure 7-55 Image Gen Test Main Menu

A description for each menu option follows:

- Execute Image Generation Test – This option executes the Image Generation Test after the number of passes are entered. A valid entry for the number of passes is from 1 to 9999. The default value is 1. Each pass takes approximately 1 minute to complete.

Note: Before executing the test, the Recon Status Box located at the top of the screen should display an “Idle” state. This state indicates the Image Reconstruction Process is ready to create images. Other possible states are “Active” and “Shutdown”. An “Active” state indicates the Reconstruction Process is busy creating images. You should wait for these images to complete before continuing. If a “Shutdown” state is indicated, the Image Reconstruction Process has been halted usually due to an error condition. Restart the process by selecting option 3, “Restart Recon”, before beginning the test.

- Display Test Error Log – This option displays the Test Error Log. A help screen is available by pressing **H** on the key board. Press **H** again to remove this screen or **Q** to return to the main menu.

Note: If executing this test remotely via a modem and the screen is garbled, try increasing the size of the window.

- Restart Recon – This option restarts the Image Reconstruction Process. This should be performed any time a “Shutdown” status is indicated in the Recon Status Box.
- Help Screen – View the help screen via the UNIX “more” command. (Refer to Figure 7-56.)
- Quit – This option terminates the test.

37.7 IG Results Screen

Once the test executes, a results screen appears:

```
Image Generation Test Results

Queued Images Completed: 20 of 20
Passes Completed: 1 of 2
Failed Images: 0
Retried Completed Images: 0

Testing complete
```

Screen Output	Description
`Queued Images Completed`	This field indicates the number of images that the Image Reconstruction Process was requested to create and the number of images created thus far.
`Passes Completed`	This field contains the pass count and the number of passes requested.
`Failed Images`	These are the number of images which failed the checksum test. More information concerning what images failed and the checksum found can be found in the Test Error Log.
`Retried Completed Images`	This field indicates the number of times the Image Reconstruction Process detected an error while creating an image. This process automatically resets the hardware and tries again during this test and normal application use.

Table 7-19 IG Results Screen Interpretation

Note: More information concerning why the reconstruction failed can be found in the System Message Log.

37.8 Error Messages

Two types of error logs are used for this diagnostic: System Message Log and the Test Error Log. Both logs should be examined after the test completes. Even if no errors are detected with the image, errors may be present in the log.

37.8.1 System Message Log

The System Message Log is the primary log used by the application code for logging error messages. This log should have information concerning Retried Completed Images or a reconstruction shutdown.

37.8.2 Test Error Log

The Test Error Log is used by the test itself for logging checksum and implementation errors. A checksum error indicates the checksum of the image just created does not match the expected value. The cause of this error is probably due to a defective IG board. Implementation errors are mostly due to software errors.

37.9 Image Gen Help Screen

```
TEST DESCRIPTION
The Image Generator Test validates the reconstruction hardware and
software. Testing consists of creating images from test data stored
on the High Speed Disk and validating their checksums. Errors detected
by this diagnostic should be the same as those detected during patient
scanning since the same application reconstruction software is
utilized in both situations. Two error logs are used by this
diagnostic: the System Error Log and the Test Error Log. The System
Error log is the primary log used by the application code for logging
error messages. The Test Error Log is used by the test itself for
logging checksum and implementation errors. Both logs should be
examined after running the test.
Part of test initialization is to check the Scan Database for the
“Image Gen Test” scan files. If not present, the scan data is copied
from the local SCSI disk to the Scan Database. This takes
approximately 4 minutes. Once complete, the Image Generation Test Main
Menu is brought up. Select from the following options:
Execute Test - This option executes the Image Generation Test after
the number of passes are entered. Each pass takes approximately 1
minute to complete.
View Test Log - This option displays the Test Error Log via a utility
similar to view. A help screen is available by hitting the h key which
takes several seconds to be display. The help screen or the log can
be exited by hitting the q key.
Help Screen - View this screen.
Quit - This will terminate the test. The scan files are left on the
system. These files should be deleted if NO further testing is
planned.
TEST OPERATING NOTES
The Recon Status Box located at the top of the screen indicates the
state of the test. At no time should “Shutdown” be displayed here. If
so, a “Restart & Resume” must be performed. This is accomplished by
pressing the Recon Status box, then the Queue Mangmnt button, and
finally, the Restart & Resume button. If the problem persists, run the
IG BLD.
The Scan Database must have room for the test scan files. The test
will NOT run if no room is available. This can be corrected by
releasing unnecessary scan files.
```

Figure 7-56 Image Gen Test Help Screen

Section 38.0 rhapSnap

WHAT RHAPSNAP SAVES

Core files from the OC and SBC

`/usr/g/service/log/core*`

`/usr/g/bin/core*`

`/usr/tmp/core*`

UNIX kernel core files from the OC and SBC

`/var/adm/crash/*`

`/usr/g/service/log` from the OC and SBC

`/var/adm/*SYSLOG*` files from the OC and SBC

Install log files from the OC

`/var/adm/install*`)

SDC log files from the OC

`/usr/g/ctuser/logfiles/sdclog`

ScanRx info files from the OC

`/usr/g/service/log/exam*.protocol`

`/usr/g/service/log/exam*.scan.request`

``/usr/g/service/log/gesys_`uname -n`.log``

Miscellaneous information such as disk space, process status and showprods.

Queue directory

`/usr/g/queue` from the SBC

Genesis scan database from the SBC

`/usr/g/data_management`

SDC log from the OC

`/usr/g/ctuser/logfiles/sdclog`

Install files from the OC

`/var/adm/install*`

Miscellaneous files

`/usr/g/bin/*.timers*` from the SBC and

`/usr/g/en_US/app_defaults/archive/SCSI.fol`

`/usr/g/en_US/app_defaults/devices/camera.dev`

`/usr/g/config/INFO`

`/usr/g/bin/*.timers*` from the OC

GE MEDICAL SYSTEMS

GE MEDICAL SYSTEMS-AMERICAS: FAX 414.544.3384
P.O. BOX 414; MILWAUKEE, WISCONSIN 53201-0414, U.S.A.

GE MEDICAL SYSTEMS-EUROPE: FAX 33.1.40.93.33.33
PARIS, FRANCE

GE MEDICAL SYSTEMS-ASIA: FAX 65.291.7006

GE Medical Systems

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Chapter 8

Console (Host Computer & Scan Recon)

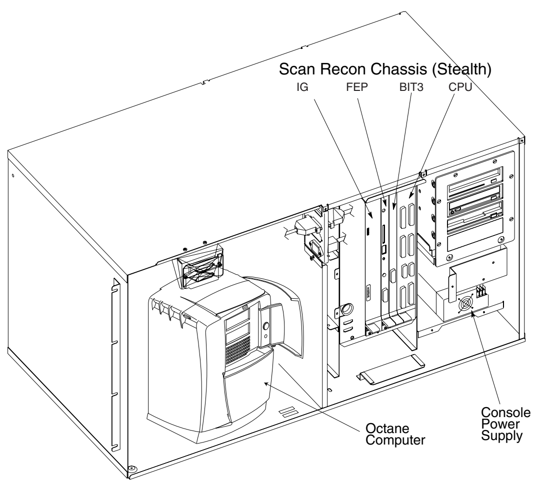
Chapter 8 Console

Section 1.0 Introduction

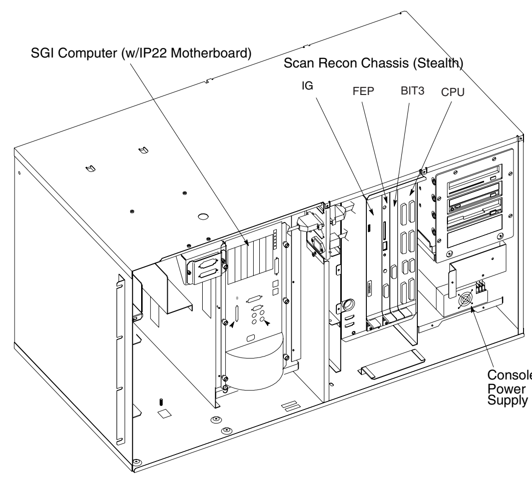
This chapter contains three very important sections:

- Section 2.1 describes the Host computer for the CTi console, the Octane™ from Silicon Graphics, Inc. (SGI®). It is one of the most important parts of the CT/i console.
- Section 2.2 describes the original host computer for the CT/i console, the Indigo2 model from SGI.
- Section 3.0 describes the SBC whose hardware is the same for both consoles but with software differences.

OCTANE HOST, SEE PAGE 370



INDIGO HOST, SEE PAGE 419



SCAN RECON SYSTEM (COMMON TO BOTH OCTANE & INDIGO), SEE PAGE 500

For architecture diagrams, see CT/i NexGen (Octane™) System Architecture on page 207 and CT/i (Indigo™) System Architecture on page 208.

Section 2.0
Host

2.1 CT/i (Octane™) Host Computer

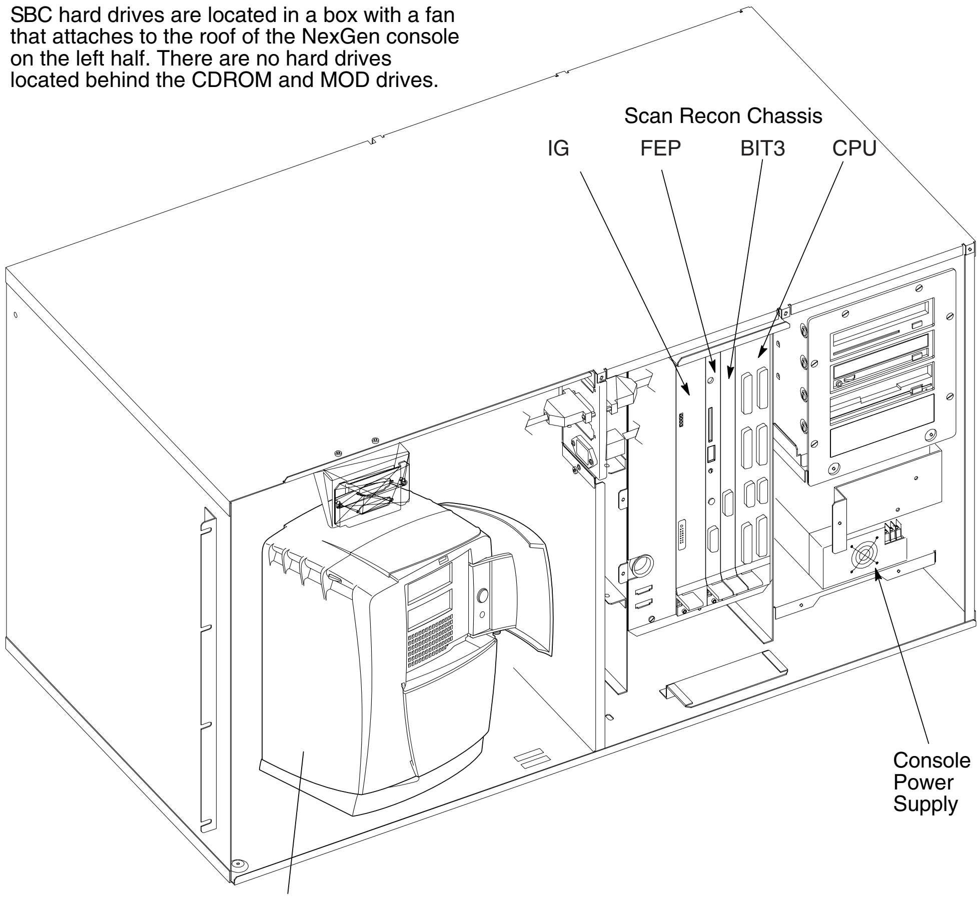


Figure 8-1 CT/i NexGen Console

2.1.1 CT/i Host (Octane) Overview

CAUTION **The OCTANE computer runs very hot. Wait at least five minutes after removing power before you touch components inside it.**

The newest, alternate CT/i Host Computer is the **SGI Octane** workstation; it features:

- A slid-out **System Module** with a singleR10000 processor brick running at 195 MHz
- DIMM Memory: Two 64 MB **DIMM modules** in the first bank and two 32 MB DIMMs in the second. Giving the system a total of **192MB**
- A Unique **System ID module** containing system Ethernet number. Which gets imprinted on option MODs

The Octane also features XIO architectures. **XIO boards** are high speed boards produced by Silicon Graphics and used for graphics, networking, disk interface, and video boards. They reside in the XIO module and are directly connected to the **frontplane**.

- A BIT3 PCI module to provide high speed transfer of status and commands between SGI host and VMEbus in the Scan Recon Computer
- A graphics module that allows the Solid Impact (SI) to become the primary or secondary head
- **TRAM Options** that allow memory to be added to the **Solid Impact (SI) Graphics module**
- A slide-out OC **Power Supply**
- Slide-out **hard drives** with slot dependent IDs. The bottom slot is SCSI ID 1, the one above it is SCSIID 2, and the top slot assigns SCSI ID 3

2.1.2 About the CT/i (Octane) host

The host configures the devices it controls, as it boots. IRIX 6.4 assigns/configures at bootup what devices it sees. Unsupported devices may not work with the OCTANE workstation and may even cause problems with supported devices.

Octane architecture has a seven port crossbar switch which acts as a packet switching router. This means the computer subsystems can communicate at very high speeds, in a very predictable manner.

Octane integrates powerful compute resources via a high-bandwidth, low-latency memory system. Five dedicated processing blocks access this main memory. These processing blocks include the CPU, the imaging engine, the graphics engine, the compression engine, the video system, and I/O. All of these processing elements access data from a single ultra-high-speed unified memory bank.

2.1.2.1 Main (DIMM) Memory

The Octane host uses DIMM memory rather than SIMM. DIMM stands for Dual In-line Memory Module. DIMM has signal and power pins on each side, to support the two memory chips used inside the module. The pins on SIMMs, although on both sides of the chip too, are connected to the same memory chip. A DIMM allows for a 128-bit data path by interleaving memory on alternating memory access cycles. SIMMs only have a 64-bit data path.

Octane reallocates memory as needed. And rather than copy data from one subsystem to another, the processing blocks can simply exchange pointers, reducing the performance penalty imposed by copying data.

Octane's system memory is made up of DIMMs (Dual Inline Memory Modules) which use synchronous DRAM (SDRAM) technology - the fastest memory currently available. Each DIMM fills one of two slots in a bank. Memory must always be added in increments of two DIMMs of the same type and density. Four banks of memory are supported.

2.1.2.2 Small Computer System Interface (SCSI)

The host contains two wide SCSI buses, one for internal devices and one for external devices. The wide (Ultra SCSI, 16-bit) buses run at 40 MB per second (peak theoretical rate). The narrow (Fast SCSI, 8-bit) narrow devices run at 10 MB per second (peak theoretical rate). Additionally:

- The length of combined SCSI cables determines how many devices can be daisy- chained.
- Only external SCSI devices must have their addresses.

2.1.2.3 Peripheral Component Interconnect (PCI)

CT/i supports the Peripheral Component Interconnect (PCI) expansion bus at 266 MB per second. This bus is used to communicate with the scan recon engine (SBC), over the BIT3 link (using a PCI BIT3 Bus Adapter Card).

2.1.2.4 Octane Connectors and Controls

FRONT (SEE FIGURE 8-2)

- Status LED
- Reset switch
- Power on

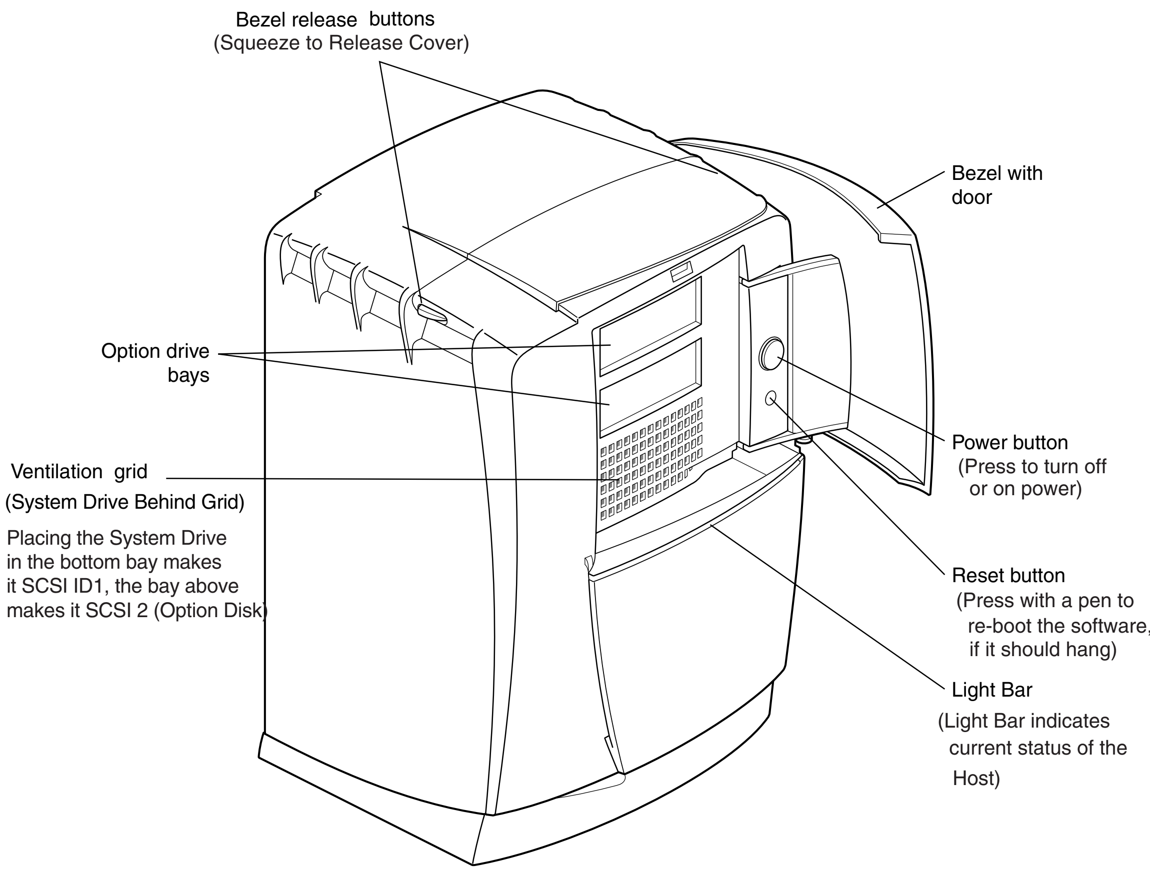


Figure 8-2 Front View of the Octane Computer

REAR (SEE FIGURE 8-3)

- 2 serial 460kbps (DB-9)
- Monitor (HD15) PC style
- Ultra Fast/Wide SCSI (H-den 68)
- 10BaseT/100BaseTX Ethernet (RJ45)
- 1284 parallel (C miniature)
- Mouse/keyboard (2 x MDIN6)

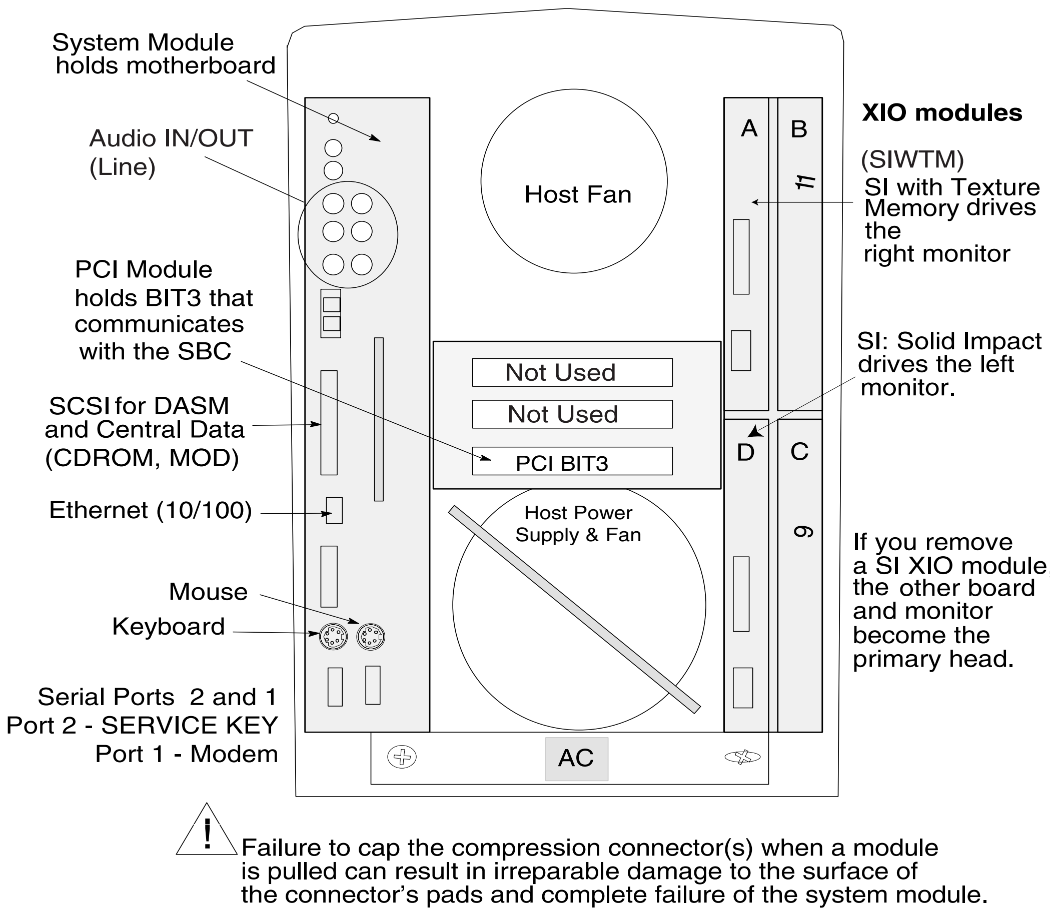


Figure 8-3 Rear View of Octane Computer

2.1.2.5 Octane Serial Expansion

Serial Devices

OCTANE ONBOARD PORTS

Filesystem DeviceName	Port	Link (Apps.) Name	Type
/dev/ttyd1	Port 1 (modem/InSite or Laptop)		RS232
/dev/ttyd2	Port 2 SERVICE KEY	/dev/servkey	RS232

Table 8-1 On-board Serial Port Filesystem Assignments (Octane)

Octane Expansion Ports (Effective CT/i 5.3)

The Octane uses a Central Data Serial Expander attached to the host's SCSI output to expand the number of available serial ports. The Touch monitor is no longer supported and so only four additional ports are needed on the CT/i “Octane” console.

Filesystem DeviceName	Port	Link (Apps.) Name	Type
/dev/ttyd040	1	Not Used	RS232/RJ45
/dev/ttyd041	2	Not Used	RS232/RJ45
/dev/ttyd042	3	/dev/ttyal "SBC"	RS232/RJ45
/dev/ttyd043	4	/dev/input/trakb	RS232/RJ45

Table 8-2 Expansion Serial Port, Through Central Data Box, Assignments (Octane)

2.1.2.6 Octane Hardware Precautions

General Precautions

Please observe the following precautions:

- Place a cap on all Octane modules or XIO board compression connectors, before moving.
- Place a cap on the optical digital ports when the cables are not connected.
- Remove power before you open the chassis or connect cables other than keyboard, mouse, and audio cables.
- Re-boot the system after you reattach the keyboard or mouse to get it to be recognized.
- Plug in all cables completely.
- Practice good ESD prevention when performing hardware tasks on any electronic component.
- Never block the cooling vents nor fail to return all covers that enable good air flow.
- Don't move the host while it is running nor within one minute of powering it off. This might damage the disk drive. Don't move the console while an MOD is in its drive because this may damage the MOD drive.
- Don't place liquids or food near the keyboard.
- Don't dangle the mouse by its cable.
- Drives can be easily damaged. Handle them carefully; do not drop or handle roughly.

Compression Connectors

NOTICE Potential for Damage **Failure to follow these instructions can result in irreparable damage to the surface of the connector's pads, which may result in intermittent or complete failure of the system module, PCI module or XIO modules.**

The OCTANE workstation uses compression connectors to connect the system module, the *PCI module* and the *XIO modules* to the frontplane.

Each compression connector has 96 pads and two halves. One half is on the frontplane of the chassis; the other is on the system module, PCI module, or XIO board. Each pad on a frontplane connector is a flat gold-plated surface. *Each pad on the system module, PCI module or XIO board is composed of hundreds of tiny bristles. When a bristled pad is pressed into a gold-plated pad, a connection is created for one signal.*

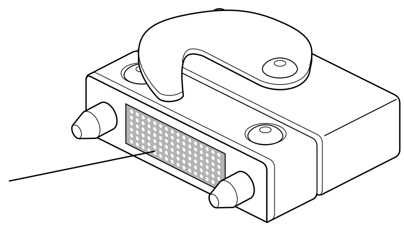


Figure 8-4 Control Connector

The bristled pads attract and hold dust, lint, grease, powder, and dirt. The presence of these substances clogs or damages the bristles and prevents them from making proper contact with the system's frontplane.

NOTICE Avoid Damage **To avoid damaging the system module, PCI module or XIO modules, follow these guidelines whenever it is outside the computer.**

- **Do not touch the pads of the compression connector with anything. The bristles might be damaged.**
- **Whenever the module or board is not in the chassis, put the protective cap over the compression connector and put the module or board in an antistatic bag. Close (fold over) the open end of the bag to minimize exposure.**
- **Do not put anything (not even water) onto the pads, except as specified in the cleaning instructions below.**
- **Before laying the board on a surface, make sure that the surface is free of dust, lint, powder, metal filings, oil, water, and so on.**
- **Do not blow dust, dirt, or powder anywhere near the board when it is not inside its protective bag.**
- **Do not use a cleaning product that contains any of the following ingredients: halogenated hydrocarbons, aromatic hydrocarbons, ethers, sulphur, ketones, or solvents of any kind. These substances cause irreparable damage to the connector's surface.**

Note: Avoid Pollutants Some pollutants can irreversibly damage (corrode or chemically alter) the pad surfaces. Although cleaning may remove the pollutant, it does not repair damage incurred by this contact.

A compression connector should never need to be cleaned if you keep the protective cover on whenever the module or board is not in the chassis. However, if the connector becomes dirty, follow the instructions below for removing pollutants.

REMOVING POLLUTANTS

Hold a can of dry compressed inert gas so that the tip of the applicator is one to two inches away from the first row of pads at the topmost edge of the connector and at a slight angle so that the spray hits each pad and flows downward. You will be spraying at the pads on one row, but in the direction of the next row of pads. Do not allow the applicator to touch the pads. Start spraying. As you spray, move the spray along the side of the connector until the entire first row has been sprayed. Move down to the next row. Repeat until all the rows have been sprayed.

2.1.3 Diagnosing (Octane) Host Computer Hardware Problems

`diagnostic(s): (OC) hinv, ide, ide fe`

`error log(s): (OC) /var/adm/SYSLOG*`

There are number of ways that Octane hardware problems can be identified and diagnosed. They range from doing a hardware inventory, diagnostic testing, to executing simple command line executables.

HARDWARE INVENTORY

Using the **`hinv`** software command, a listing of the hardware devices that the host computer can communicate with or not communicate with is displayed.

POWER-ON TESTS

Power on tests run automatically whenever the host computer is powered on or reset. They test the motherboard, the memory modules, and graphics boards. Fault notification is done through light bar LED codes and Error Messages in the OC `/var/adm/SYSLOG`, or on the console monitor (CRT).

INTERACTIVE DIAGNOSTIC ENVIRONMENT (IDE)

IDE offers more in depth tests of the SGI hardware. Fault reporting is done through error messages. Quickly interrupt the scanner's bootup, select **Stop for Maintenance.** More details follow on the next page.

CONFIDENCE TESTS

Use the SGI Confidence Tests to test:

- **keyboard** (alpha-numeric keys only),
- **CD-ROM** (place a CD inside first),
- **monitor** (use to adjust convergence) or
- **mouse**

These tests are run from the operating system level. In a shell, enter: **`confidence`**

OR

On the Service Menu, with a Service key in the console port, select:

TROUBLE SHOOT > SYSTEM > DISPLAY CRT > CONFIDENCE TESTS

2.1.3.1 Hardware Inventory <hinv>

From a shell tool, enter the hardware inventory command: > **`hinv`**

A typical output listing of the hinv command follows. Remember that the following report may differ from the one you receive slightly. The **`hinv`** command reports your current hardware configuration. A typical output report from the **`hinv`** command follows:

Example: **`hinv`** command

1.) `1 195 MHZ IP30 Processor`

Line 1 - Identifies the system as having a standard single 195MhzProcessor Module “CPU brick” which is plugged into the Octane IP30 System Module.

OPTIONAL- Dual 195Mhz PM can be used to replace the single processor PM. Line 1 will change and indicate a dual processor is installed.

2.) `CPU: MIPS R10000 Processor Chip Revision: 2.7`

3.) `FPU: MIPS R10010 Floating Point Chip Revision: 0.0`

Line 2 - MIPS R10000 Microprocessor IC (inside the PM). Note that revision levels may change over time. You should not become alarmed if yours is different.

Line 3 - MIPS R10010 Floating Point Co-processor IC (inside the PM). Note that revision levels may change over time. You should not become alarmed if yours is different.

4.) `Main memory size: 192 Mbytes`

Line 4 - Standard Main Memory configuration. The Octane computer utilizes Dual Inline Memory Modules (DIMM) using Error Correction Code (ECC). The Octane IP30 has 8 DIMM connectors arranged in (4) banks of (2). They are labeled S1 thru S8. DIMM’s must be installed in like-pairs into connectors S1-S2, S3-S4, S5-S6, and S7-S8. (2 x 64MB) DIMM’s are normally in connectors S1 and S2. (2 x 32MB) DIMM’s are normally in connectors S3 and S4. Please see page 407, Host memory (DIMM) replacement procedures for more details.

OPTIONAL - 256MB memory adds (2 x 32MB) DIMM’s into memory connectors S5 and S6.

5.) `Instruction cache size: 32 Kbytes`

6.) `Data cache size: 32 Kbytes`

Lines 5 & 6 - The MIPS R10K primary instruction cache & data cache memory is inside the R10K microprocessor in the PM (CPU brick).

7.) `Secondary unified instruction/data cache size: 1 Mbyte`

The CPU secondary cache memory is located inside the PM (CPU brick).

8.) `Integral SCSI controller 0: Version QL1040B (rev. 2)`

Line 8 - The Octane IP30 has (2) Ultra-SCSI Wide controller chips (QLOGIC). Controller #0 runs the (3) internal Octane drive bays.

9.) `Disk drive: unit 1 on SCSI controller 0`

Line 9 - This is the 4GB Ultra-SCSI Wide system disk drive (ID 1 on SCSIbus 0) in Octane bay #1 (lowest drive bay).

OPTIONAL - 4GB Ultra-SCSI Wide second image disk drive (ID 2 on SCSIbus 0) in Octane bay #2 (middle bay).

10.) `Integral SCSI controller 1: Version QL1040B (rev. 2)`

Line 10 - The Octane IP30 has 2 Ultra-SCSI Wide controller chips (QLOGIC). This is controller#1 which runs the external SCSIbus. All devices on the CT/i Octane external SCSIbus are 8-bit (narrow) SCSIbus devices. A Wide-to-Narrow SCSI cable is used.

11.) `Disk drive: unit 1 on SCSI controller 1`

Line 11 - This is the Analogic DASM (device ID 1 on SCSIbus 1) on the Octane external SCSIbus.

12.) `Optical disk: unit 3 on SCSI controller 1`

Line 12 - This is the Maxoptix T5-2600 MOD drive (device ID 3 on SCSIbus 1) on the Octane external SCSIbus.

13.) `Comm device: unit 4 on SCSI controller 1`

Line 13 - This is the Central Data SCSI-to-Serial port expander module (device ID 4 on SCSIbus 1) on the Octane external SCSIbus.

14.) `Comm device: unit 4, lun 1 on SCSI controller 1`

Line 14 - The Central Data SCSI-to-Serial port expander module utilizes 2 logical units to manage the expanded serial ports.

15.) `CDROM: unit 6 on SCSI controller 1`

Line 15 - This is currently the Toshiba XM3701B 12X CDROM drive (device ID 6 on SCSIbus 1) on the Octane external SCSIbus.

16.) `IOC3 serial port: tty1`

Line 16 - This is the 9-pin D-type native serial port #1 on the IP30. RS232 device /dev/ttym1 for remote boot or PPP connections. This is also the Octane primary console port via device /dev/ttyd1 (can be used for laptop control via firmware command).

17.) `IOC3 serial port: tty2`

Line 17 - This is the Octane's 9-pin D-type native serial port #2 on the IP30. Used as RS232 for the SERVICE KEY.

18.) `IOC3 parallel port: plp1`

Line 18 - (The Octane's native Centronics parallel port on the IP30 is not used in CT/i)

19.) `Graphics board: SI with texture option`

Line 19 - This is an SGI Solid Impact (SI) graphics card in the top left location of the XIO quad module. This SI card has a 4MB Texture Ram (TRAM) module and must be connected to the right (image display) CRT of CT/i.

20.) `Graphics board: SI`

Line 20 - This is an SGI Solid Impact (SI) graphics card in the lower left location of the XIO quad module. This SI card does not have Texture Ram (RAM) and is connected to the left (operator interface) CRT of CT/i.

21.) `Integral Fast Ethernet: ef0, version 1`

Line 21 - This is the FAST ethernet (100 megabit/sec) device mode of the Octane IP30 native 10/100 megabit (auto-sensing) ethernet chip.

22.) `Integral Ethernet: et0, IO0`

Line 22 - This is the 10 megabit/sec device mode of the Octane IP30 native 10/100 megabit (auto-sensing) ethernet chip.

23.) `Iris Audio Processor: version RAD revision` *`12.0`*`, number 1`

Line 23 - This is the Octane IP30 native audio processor chip. CT/i uses this for recording and playing Autovoice digital audio files.

24.) `PCI card, bus 0, slot 232, Vendor 0x0, Device 0x0`

Line 24 - This is the PCI BIT3 card in the Octane PCI expansion chassis slot #0 (top slot). It is BIT3 model 617-1 (PCI side).

Note: See SYSLOG — Together, hinv and SYSLOG text, page 221, can be useful when troubleshooting or attempting to find solutions to a hardware problem.

2.1.3.2 Power On Tests

Power-Up Sequence - Overview

The Octane computer follows a sequential power up procedure. After power to the Octane computer applied, a LED on the front of the SGI chassis turns on. While the motherboard is running the power-up self-test the LED is RED.

If it finds a fault, it will enter an entry in the SYSLOG if it can. Knowing where it halted in the boot up can also prove to be a clue to the problem.

When the LED first lights on power on, the monitor will display:

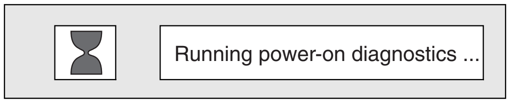


Figure 8-5 Power-on Diagnostics Notifier

After all of the power-on tests pass, the Octane LED will turn white and the “Starting Up System” pop-up window will appear on the monitor.

This is when you can access SGI diagnostics and its host command line. Press the **ESC** key or click on the "Stop for Maintenance" box if you want to access the SGI firmware.

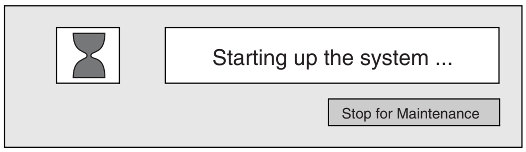


Figure 8-6 Starting up the system notifier

If you don't interrupt, after a few seconds the System Is Coming Up pop-up will appear.

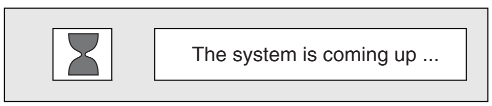


Figure 8-7 The System is coming up notifier

Using the Light Bar LEDs to Troubleshoot

The light bar LEDs on the front of the Octane computer provide useful diagnostic information. If a problem occurs during computer initialization, the Octane computer will report it using the LEDs on the light bar. In this section, failure symptoms are described, as well as their possible causes and remedies.

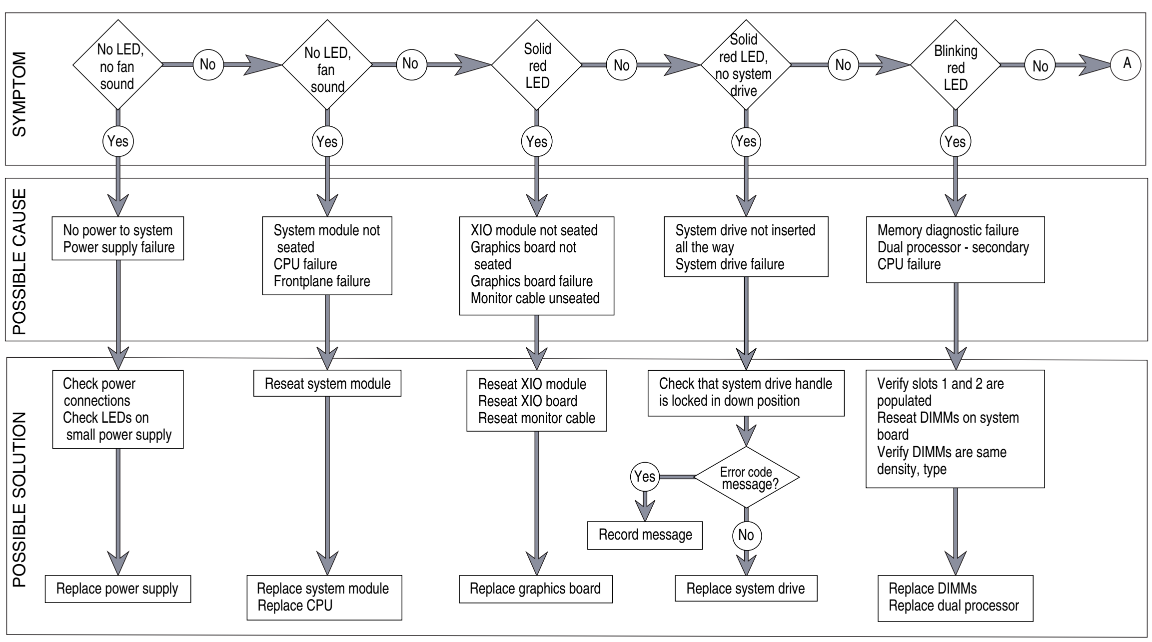


Figure 8-8 Interpreting the Light Bar LEDs

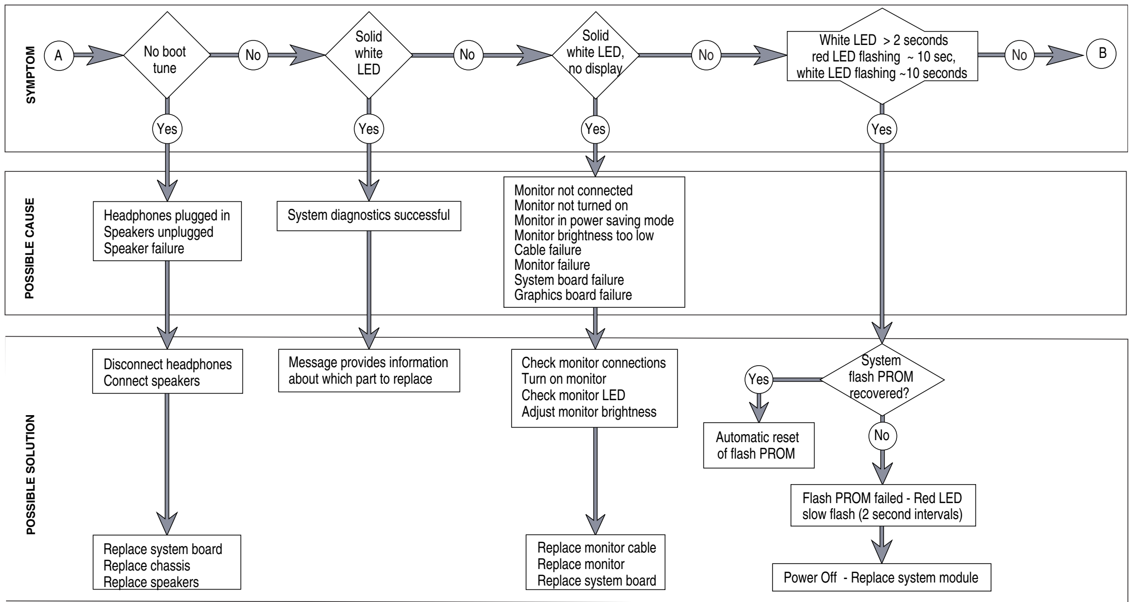


Figure 8-9 Interpreting the Light Bar LEDs (Part A)

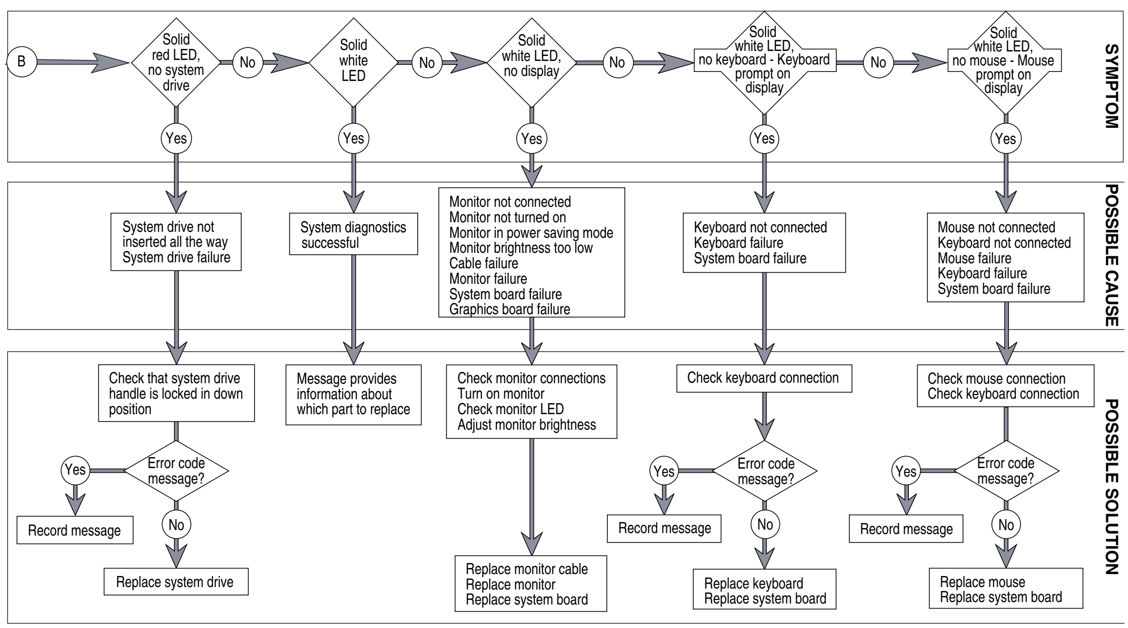


Figure 8-10 Interpreting the Light Bar LEDs (Part B)

CPOP Connector LEDs

Located immediately behind the lower right front cover are 7 green LEDs. These LEDs are visible with the front cover remove and by looking through the holes located in the lower front right of the chassis, see Figure 8-11. The LEDs you will see are attached to the back of the front plane circuit board, as viewed through the holes in the lower right area of the chassis next to the DB15 connector. There is 2 columns; 1 column consisting of 4 LEDs and another with 3 LEDs. They are depicted in Table 8-3 and Table 8-4 that way.

These LED's indicate whether the XIO modules are properly seated and have been detected by hardware. Below, is a brief description of these 7 green LEDs follows.

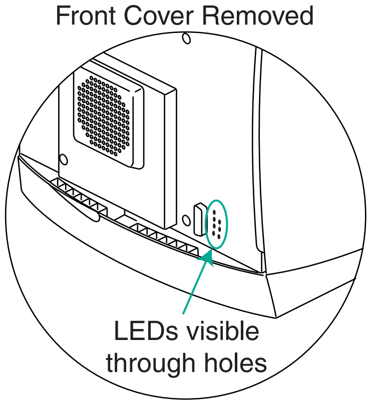


Figure 8-11 CPOP Connector LEDs

Description	Column 1	Column 2	Description
Base IO	OFF		
Quad A	OFF	OFF	PCI
Quad C	OFF	OFF	Quad B
Quad D	OFF	OFF	Heart

Table 8-3 CPOP Connector LEDS - Generic Application

Description	Column 1	Column 2	Description
System Module	ON		
Quad A (SI w/ TM)	ON	ON	PCI Chassis
Quad C (None in CT/i)	OFF	OFF	Quad B (None in CT/i)
Quad D (SI)	ON	ON	Heart ASCI

Table 8-4 CPOP Connector LEDS - CT/i Specific Application

Base IO	Main System Module is seated/detected OK
Quad A	Top left XIO quad module is seated/detected OK
Quad C	Lower right XIO quad module is seated/detected OK
Quad D	Lower left XIO quad module is seated/detected OK
PCI	PCI chassis/ASIC seated/detected OK
Quad B	Top right XIO quad module seated/detected OK
Heart.	Heart memory control ASIC on System Module status OK

The purpose of these LEDS is *NOT* “diagnostic” in nature but to tell you that you've got the modules seated correctly and things are basically “alive”. In the case of the Heart ASIC, it is a “status OK” indicator.

2.1.3.3 Command Monitor

The procedure for entering Command Monitor follows:

1.) Start or shutdown the system as needed. If the host computer is off, turn it on and proceed to (Step 2.) If the host computer is up and running, bring it down appropriately. After a few seconds, the screen will clear and you'll see a notifier like the one shown in Figure 8-14, Select the RESTART button.

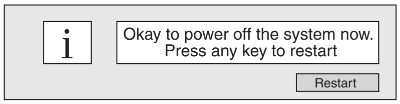


Figure 8-12 Okay to Power Off System - Notification Screen

Note: If the system is malfunctioning and you cannot communicate with it using the mouse or keyboard, then press the Reset Switch on the front Chassis.

2.) Click on STOP FOR MAINTENACE, using the mouse immediately or you can press the **ESC** key. You only have three to five seconds to preform this action. See figure Figure 8-15 .

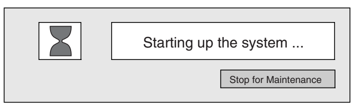


Figure 8-13 Maintenance Option Menu

3.) The following Host Maintenance menu appears. Click on the menu selection ENTER COMMAND MONITOR.

```
Start System
Install System Software
Run Diagnostics
Recover System
Enter Command Monitor
Select Keyboard Layout
```

4.) At the command monitor >> prompt, enter:

`ide fe`

Comment: normally 20 minutes

This will run a verbose version of the automated Octane diagnostics. IDE is loaded from the system disk directory **`/stand/ide`** or from the IRIX operating CD-ROM if installed.

To exit ide, press: **ESC**

To interrupt: **CTL+C**]

To test just the motherboard SCSI interface, enter: **`scsi`**

To test the memory modules, enter: **`memtest`**

To test the motherboard audio, enter: **`audio`**

To test the motherboard FPU, enter: **`fpu`**

For help while in ide, press:**h**

5.) Watch for messages.

If the diagnostics find a problem, you will see a message similar to:

```
ERROR: Failure detected on the CPU module
```

or

a message indicating a failure with other Octane parts. Generally if these test run without error for 15 minutes, the IP30 motherboard and its basic functionality are good.

6.) To stop the tests, press the **ESC** key.

This will halt execution of the test and return you to the command monitor prompt. A list of available commands may be viewed by typing **`?`** or **`help`**.

2.1.3.4 Interactive Diagnostic Environment (IDE))

IDE diagnostics are powerful tools that can be used to check Octane computer functionality more thoroughly. When you power on the system, basic Octane power-on tests check hardware. The IDE tests give a greater depth of host testing. The default set of IDE tests takes about 30 minutes to run. In which the program stops the tests to report any failures to the screen/head.

IDE Tests

AVAILABLE TESTS

There are two primary forms of testing available: comprehensive and FRU level. To determine which diagnostic tests are available to you, type "**`help_ide`**" at the prompt followed by a carriage return.

```
ide>> help_ide
ip30
regular_tests
extended_tests
frontplane
pm
gfx
tmezz
memory
```

Using the **`help_ide`** command, a list of the available diagnostic tests that can be executed can be displayed. Regular and extended test are comprehensive whereas stand-alone are more selective. Comprehensive test generally provide the broadest range of coverage. These are the **`regular_test`** and **`extended_test`** as listed above. All of the others are targeted to the FRU level or isolate specific functionality.

EXECUTION TIMES OF AVAILABLE TESTS

The following are some of the approximate execution times that have been observed. They are approximate because execution depends on memory and processor installed.

TEST NAME	DURATION (APPROXIMATE)
`ip30`	8 seconds
`regular_tests`	22 minutes
`extended_tests`	37 minutes
`frontplane`	0 minutes
`pm`	10 seconds
`gfx`	19 minutes
`tmezz`	70 seconds
`memory`	19 minutes

Table 8-5 Octane IDE Execution Times (approximately)

IDE Execution

IDE Diagnostics are initiated during the Octane Computer boot-up process. Diagnostics begin execution immediately upon entering the diagnostic environment. If you wish to interrupt that testing, use the control C key stroke sequence to halt and return you to the ide diagnostic prompt. Entering exit at the prompt will return you the boot sequence.

The procedure for entering IDE diagnostics follows:

1.) Start or shutdown the system as needed. If the host computer is off, turn it on and proceed to (Step 2.). If the host computer is up and running, bring it down appropriately. After a few seconds, the screen will clear and you'll see a notifier like the one shown in Figure 8-14, Select the RESTART button.

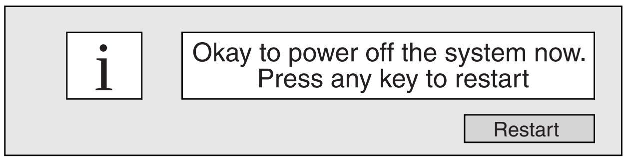
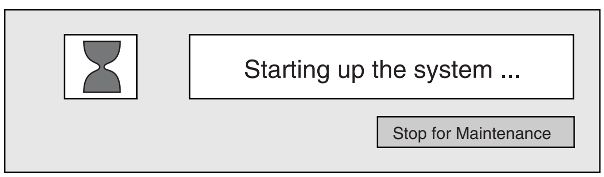


Figure 8-14 Okay to Power Off System - Notification Screen

Note: If the system is malfunctioning and you cannot communicate with it using the mouse or keyboard, then press the Reset Switch on the front Chassis.

2.) Click on STOP FOR MAINTENANCE, using the mouse immediately. You only have three to five seconds to preform this action. See figure Figure 8-15



.

Figure 8-15 Maintenance Option Menu

3.) The following Host Maintenance menu appears. Click on the menu selection "RUN DIAGNSTICS".

```
Start System
Install System Software
Run Diagnostics
Recover System
Enter Command Monitor
Select Keyboard Layout
```

Notice that the screen blanks and the following text is displayed.

```
        Starting diagnostic program ...
     Press ESC to return to the menu
```

4.) If you have not pressed **ESC** to exit out of program, a default set of tests will begin automatically. This default test program is the same as the test named **`regular_tests`** when using the **`help_ide`** command. The following screen output is displayed for approximately 3 minutes or until you press and hold the control and C keys simultaneously.

```
             Starting diagnostic program...
           Press <Esc> to return to the menu.
SGI Version 6.4 IP30 IDE field  Feb  5, 1997
                       System: IP30
```

```
                Processor: 195 Mhz R10000, with FPU
       Primary I-cache size: 32 Kbytes
       Primary D-cache size: 32 Kbytes
       Secondary cache size: 1024 Kbytes
                Memory size: 192 Mbytes
                    Network: ef0 ethernet (100/10 base-T)
                  SCSI Disk: scsi(0)disk(1)
                  SCSI Disk: scsi(1)disk(1)
                SCSI Device: Controller 1 ID 3
                SCSI Device: Controller 1 ID 4
                 SCSI CDROM: scsi(1)cdrom(6)
                      Audio: RAD Audio Processor
                    PCI Bus: XTALKPCI (13)
                 PCI Device: slot 1 vendor 0x108a part 0x1 rev 84
                   Graphics: SI with texture option
                   Graphics: SI
 Field included scripts (help_ide to get this again):
  ALL:    regular_tests
          extended_tests
  FRUs:   ip30   frontplane   memory
          pm (processor module)
          gfx   tmezz
  subsystems: serial (loopback)   ethernet (loopback)
  useful scripts:        help_ide   help_mem
    This test takes approximately 12 minutes on a good
    dual processor machine with 128MB memory and MXI graphics.
```

Comment: Loopback connector not provided with CT/i system

After approximately 3 minutes, the display shows the following text:

```
Testing Graphics in X10 Slot 10
```

`Testing Graphics in X10 Slot 10` is reported because of the graphics card in place at that location.

Both heads (monitors) should then go dark. Within a few minutes, graphics tests begin running on the left display head. This continues for approximately 8 minutes. You will notice that the cursor in lower left corner of the monitor will turn colors and flash, giving you an indication of processor activity.

Next, the graphics test will start on the right head,. During this time the display start off blank, then flash colors and eventually display graphic patterns. This will continue for approximately 10 minutes on the right display head. Total Test time is approximately 20 minutes. When Diagnostic Tests are complete, the following text will be displayed on the monitor.

```
Testing TMEZZ option card.
TREZZ board test PASSED
All graphics test have passed
TEST RESULTS:
Processor Module test completed.
Memory tests completed.
CPU tets completed.
Audio tests completed.
```

```
SCSI test completed.
Graphics board 0 tests completed.
Graphics board 1 tests completed.
ide>>
```

5.) To exit IDE Diagnostics and return to computer boot-up, enter **exit** on the IDE command line followed by a carriage return.

```
ide>> exit
```

Running Tests Stand-alone from the IDE prompt

To execute IDE diagnostic test, you must have initiated the IDE diagnostics utility and be at the `ide>>` diagnostic prompt.

"REGULAR_TESTS"

Enter "**regular_tests**" followed by a carriage return. The following is displayed

```
ide>> regular_tests
This test takes approxiamately 12 minutes on a good
dual processor machine with 128MB memory and MXI graphics.
```

After approximately 3 minutes of testing, the monitor will blank. This occurs because of graphics test being executed. The left head (MG 1,0) begins first and is run for 8 minutes, then the test switches to the right head (MG 1, 4) for 10 minutes.

After approximately 21 minutes, **regular_test** will complete and the ide prompt displayed.

```
Testing TMEZZ option card.
TREZZ board test PASSED
All graphics test have passed
TEST RESULTS:
Processor Module test completed.
Memory tests completed.
CPU tets completed.
Audio tests completed.
SCSI test completed.
Graphics board 0 tests completed.
Graphics board 1 tests completed.
ide >>
```

"IP30"

At the `ide>>` prompt, enter "**ip30**" followed by a carriage return

```
ide>> ip30
```

This test takes approximately 8 seconds to execute and finish.

"TMEZZ"

At the `ide >>` prompt, enter **tmezz** followed by a carriage return.

```
ide>> tmezz
```

This test runs for approximately 70 seconds. Notice that the monitor will blank and upon completion of the test and the `ide>>` prompt will again be re-displayed.

"GFX"

At the `ide >>` prompt, enter **`gfx`** followed by a carriage return.

```
ide>> gfx
Testing TMEZZ option card.
TREZZ board test PASSED
All graphics test have passed
ide >>
```

If the test executes successfully, the text output above will be displayed. This testing takes approximately 19 minutes.

"EXTENDED_TESTS"

At the `ide >>` prompt, enter **`extended_tests`** followed by a carriage return.

```
ide>> extended_tests
```

Immediately the following this command the following is displayed.

```
The extended tests takes approxiamately 25 minutes on a good
dual processor machine with 128MB memory and MXI graphics.
```

The first 19 minutes test the graphics system. Graphics (MGI 1,0) testing begins on the left head (monitor) first and lasts for 9 minutes. Then graphics (MG 1, 4) testing begins on the right head and lasts 10 minutes.

The last 20 minutes test the rest of the system. After approximately 39 minutes of execution the following screen text will be displayed indicating that testing has completed successfully.

```
All graphics test have passed
TEST RESULTS:
Processor Module test completed.
Memory tests completed.
CPU tets completed.
Audio tests completed.
SCSI test completed.
Graphics board 0 tests completed.
Graphics board 1 tests completed.
ide >>
```

"MEMORY"

At the `ide >>` prompt, enter **`memory`** followed by a carriage return.

```
ide>> memory
```

If no errors are found, the following screen output will be displayed followed by the ide command line prompt.

```
EEC test
ide>>
```

IDE Commands

IDE diagnostics provide a number of commands in addition to diagnostics test programs. These commands can be used to manipulate the diagnostic environment and move data into and out of memory, registers and files. At the ide prompt `ide>>`, enter the question mark character "**?**" followed by a carriage return.

```
ide>> ?
```

```
           IDE Generic Commands
errlog     - errlog [ "on"|"off" ]
runcached  - runcached [ "on"|"off" ]
qmode      - qmode [ "on"|"off" ]
c_on_error - c_on_error [ "on"|"off" ]
xenv       - xenv
clear_log  - clear all pass/fail info in IDE’s diagnostic result log
dump_log   - display IDE’s diagnostic result log
printregs  - printregs
boot       - boot [-f FILE] [-n] [ARGS]
dump       - dump [-(b|h|w)] [-(o|d|u|x|c|B)] ADDR1:ADDR2|ADDR#COUNT
echo       - echo ["STRING"|VAL ...]
exit       - exit [VAL]
fill       - fill [-(b|h|w)] [-v VAL] ADDR1:ADDR2|ADDR#COUNT
g          - g [-(b|h|w)] ADDRESS
help       - type help for help usage
hinv       - inventory
ls         - filelist
p          - put
printenv   - printenv [ENV_VAR_LIST]
printf     - printf "FORMAT" [ARG1 ARG2...]
quit       -
read       -
setenv     - setenv ENV_VAR STRING
source     - source SOURCE_PATH
spin       - spin [[-c COUNT] [-v VAL] [-(r|w)[+](b|h|w) ADDR]]*
unsetenv   - unsetenv ENV_VAR
version    - version
wait       - Wait for <CR> to continue (usage: wait ["message"])
symbols    - symbols [ -l | -k KEY | -t
{cmds|udefs|diags|vars|globals|ints|strs|sets|setops|debug|all} ]
ide_delay  - ide_delay { -u USECS|-m MSECS|-s SECS }
exec       - exec -f FN1[,FN2..] { [-v VID1[,VID2..]]|[-s
SET1[,SET2..]] } [-a ARG0[,ARG1..]]

           IDE Architecture-Specific Commands:

           IDE CPU-set Commands:
create_set - create_set -s SET[,SET,...] [-(a|e|c COPYSET)]
copy_set   - copy_set -s SRCSET,DESTSET1[,DESTSET2...]
display_set - display_set -s SET[,SET,...]
add_cpu    - add_cpu -s SET[,SET,...] -v VID1[,VID2,...]
del_cpu    - del_cpu -s SET[,SET,...] -v VID1[,VID2,...]
set_union  - set_union -s SET,SET,RESULTSET
set_differ - set_differ -s SET,SET,RESULTSET
set_inter  - set_inter -s SET,SET,RESULTSET
set_equal  - set_equal -s SET,SET
set_inequal - set_inequal -s SET,SET
```

```
set_inclusion - set_inclusion -s SUBSET,SUPERSET
cpu_in      - cpu_in -s SET -v VID
set_empty   - set_empty -s SET
set_exists  - set_exists -s SET

            IP30 Diagnostics:
help_mem    - Print memory diagram
help_impact - Print gfx HW diagram (2 args are 10/20 and 0/1)
emfail      - Produces failure message
ismp        - Returns true on MP
slavecpu    - Returns slave CPU valid if ismp is true
led         - Set led color
wait        - Wait for <CR> to continue (usage: wait ["message"])
chg_volt    - utility to do voltage margining
buffon      - Turn on message buffering
buffoff     - Turn off message buffering
ttyprint    - Turn on/off message echoing on serial console.
scache1     - Secondary cache misc test
ct        - Cache Thrasher Test, usage: ct [debug_level [seed_in_hex]]
fpu         - Floating Point Unit test
lpackd      - LinPack tests
tlb         - TLB
utlb        - UTLB miss exception
ldram_bkend - Low DRAM test
lkh_bkend   - Knaizuk Hartmann Low DRAM test
memtest     - CPU memory test (arg is number between 0 and 9)
ecctest     - ECC corner case memory testing
hr_regs     - Heart register read-write test
x_regs      - Xbow register read-write test
x_acc       - Xbow register access test
br_regs     - Bridge register read-write test
ioc3_regs   - IOC3 register read-write test
enet        - Enet registers tests
duart_regs  - IOC3 register read-write test
rtc_regs    - rtc_regs (destructive test!)
```

2.1.3.5 DIMM Memory - Checking for Faults

DIMM errors appear in the OC error log file called SYSLOG located in the directory `/var/adm/`. Hard (unrecoverable) memory errors will cause an SGI operating system (Irix) PANIC. Usually, a PANIC message will be posted to a screen window and logged in /var/adm/SYSLOG. The offending module will be identified by its socket number.

A bootup failure messages indicating "`PANIC: CPU parity error interrupt`"may mean a bad module in the first bank. If the system will not re-boot after a hard memory error `PANIC`, it is probably because the **Octane host needs the first memory bank to be in good working in order for boot up**. To eliminate this possibility, swap all modules in the first bank with those in the second. For the Octane host, this means swap S3 and S4 with the modules in S1 and S2 (see Figure 8-20 or Figure 8-16). Prior to doing this, check that all DIMMs are correctly seated in their slots.

To view just the critical host errors, open a shell and type: **`sysmon`**

To view all entries by IRIX type: **`/var/adm/SYSLOG`** for today's entries

or

`/var/adm/SYSLOG.0` for yesterday's.

Example: **`/var/adm/SYSLOG.1, .../SYSLOG.2, .../SYSLOG.x`**

SYSLOG Enter: **`ezlog`** to access several logs for the entire scanner.

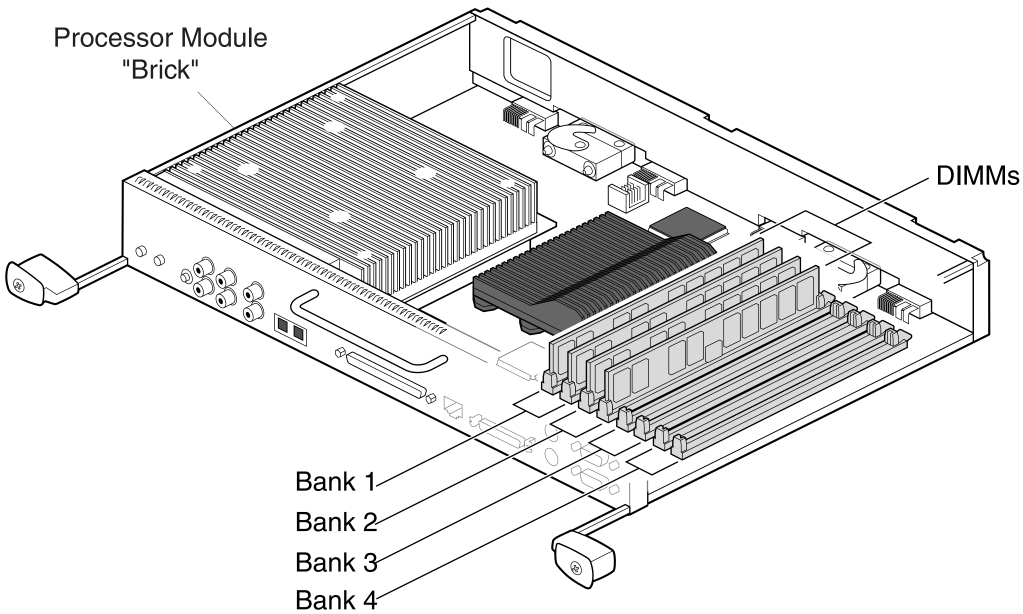


Figure 8-16 Octane System Module

To do a more complete test, interrupt **ESC** bootup, Enter Command Monitor and type:

`ide memtest`

2.1.3.6 Octane Graphics Boards

`diagnostic(s): (OC) hinv, ide, ide fe`

`error log(s): (OC) /var/adm/SYSLOG*`

SI (Solid Impact)

The Octane graphics board that controls the primary "head", the monitor normally on the right is in slot A. It's called SI with Texture Memory or IMPACTSR with 4 TRAM.

You can troubleshoot these boards by trying the secondary board in the primary slot, but at least one of the two must be installed in the primary slot and working to boot the system. The SI with Texture Memory must be in Slot A to run the scanner applications.

Refer to page 409 for replacement procedure.

Octane Graphics System Hardware

To view what components of the graphics system the SGI host currently sees, enter this command in a shell: **`/usr/gfx/gfxinfo`** You should see something that looks similar to the following example

Example: "gfxinfo" command

```
Graphics board 0 is "IMPACTSR" graphics.
    Managed (":0.1") 1280x1024
    Product ID 0x2, 1 GE, 1 RE, 4 TRAMs
    MGRAS revision 1, RA revision 0
    HQ rev B, GE11 rev B, RE4 rev C, PP1 rev A,
    VC3 rev A, CMAP rev E, Heart rev D
    unknown, assuming 19" monitor (id 0xf)
    Channel 0:
    Origin = (0,0)
    Video Output: 1280 pixels, 1024 lines, 72.24Hz (1280x1024_72)
Graphics board 1 is "IMPACTSR" graphics.
    Managed (":0.1") 1280x1024
    Product ID 0x2, 1 GE, 1 RE, 0 TRAM
    MGRAS revision 1, RA revision 0
    HQ rev B, GE11 rev B, RE4 rev C, PP1 rev A,
    VC3 rev A, CMAP rev E, Heart rev D
    unknown, assuming 19" monitor (id 0xf)
    Channel 0:
    Origin = (0,0)
    Video Output: 1280 pixels, 1024 lines, 72.24Hz (1280x1024_72)
```

2.1.3.7 Diagnosing BIT3 Subsystem on CT/I 5.x (Octane)

`diagnostic(s): (OC) hinv, ifconfig, netstat, ping, spray, mvdtest`

`error log(s): (OC) /var/adm/SYSLOG`

If you suspect BIT3 subsystem problems (PCI BIT3 card, BIT3 OC cable, BIT3 SBC cable, or VME BIT3 card), the following tests/checks can help confirm functionality and/or help isolate the FRU. Although not foolproof in every failure mode case, it's very dependable for most typical problems.

All of these checks can be done from the OC only, with a non-working or unknown BIT3 subsystem, and with application SW down.

Diagnostic Steps

Follow the steps below in the order suggested by the results of each test.

1.) Confirm BIT3 HARDWARE communications (WITHOUT relying on any applications software or network/reconfig parameters):

Note: THIS TEST SHOULD ONLY BE RUN ON AN IDLE SYSTEM (NO SCANNING/RECON).

This test performs data transfers between the OC and SBC using the entire BIT3 subsystem (both boards and cables). The above runs 100 passes of data across the BIT3 and checks the results. This test does not rely on any network parameters (IP#'s, hostnames) existing or being correct. The “Transfer rate” shown above is only typical for an idle system but this may vary (you're only looking for write/read errors which may indicate a BIT3 hardware problem). This should only be run on an idle system or you may get read/write errors due to contention on the VMEbus by scan/recon (normal).

```
{ctuser@engbay18}[1] cd /usr/etc
{ctuser@engbay18}[2] mvdsrate 30000200 0f 1000 1000
......pass 1000
```

```
Transfer rate = 8419.076172 Kbytes/sec  elapsed time = 950223 usec
***** End of Test *****
{ctuser@engbay18}[3]
```

Comment: *IF THIS STEP PASSES, THEN PROCEED TO STEP 2 BELOW (BIT3 HARDWARE GOOD). IF THIS STEP FAILS, THEN PROCEED TO STEP 3 BELOW (SUSPECT BIT3 HARDWARE).*

2.) Check the BIT3 NETWORK communications (this relies on the correct IP numbers and hostnames being properly configured and re-configured):

Note: USE 'CONTROL C' KEYS TO STOP THE PING AT ANY TIME.

The '**ping**' command does simple ICMP echo packets between network hosts. The above results are typical with Octane BIT3 on an idle system. If the '**ping**' times out (no response) but **mvdsrate** runs (as in step 1), then you most likely have a network setup (reconfig) problem. This would be typical during/after load-from-cold when network parameters are entered incorrectly (or “accidently” changed to incorrect values).

```
{ctuser@engbay18}[3] ping sbc
PING ct18_sbc0 (192.18.18.2): 56 data bytes
64 bytes from 192.18.18.2: icmp_seq=0 ttl=255 time=1.534 ms
64 bytes from 192.18.18.2: icmp_seq=1 ttl=255 time=1.877 ms
64 bytes from 192.18.18.2: icmp_seq=2 ttl=255 time=1.804 ms
----ct18_sbc0 PING Statistics----
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 1.534/1.706/1.877 ms
{ctuser@engbay18}[4]
```

Comment: *IF THIS STEP PASSES, THE BIT3 HARDWARE/NETWORK IS NOT THE PROBLEM. IF THIS STEP FAILS, CHECK RECONFIG OC-SBC NETWORK SETTINGS/PARAMETERS.*

3.) On the OC, check that the PCI BIT3 board was probed/attached by the Irix device driver successfully at the last bootup as follows:

The messages below indicate that the PCI BIT3 board was functional enough to allow the device driver to detect it, read/write it's registers over the PCIbus, and that the board reported a “good” state.

```
{ctuser@engbay18}[5] more SYSLOG* |grep BIT3
Jan 11 11:42:29 2A:bay1 unix: Attaching BIT3 MV617 PCI Card, rev 56
Jan 11 14:53:15 2A:bay1 unix: Attaching BIT3 MV617 PCI Card, rev 56
Jan 12 07:23:02 2A:bay1 unix: Attaching BIT3 MV617 PCI Card, rev 56
Jan  9 15:09:12 2A:bay1 unix: Attaching BIT3 MV617 PCI Card, rev 56
Jan 10 07:16:38 2A:bay1 unix: Attaching BIT3 MV617 PCI Card, rev 56
```

Comment: *Repeats for all occurrences of Irix bootup/probe/attach by date/time*

Also, when the PCI BIT3 card powers up and configures itself successfully, a green LED lights up on the rear of the PCI BIT3 card. This LED can be seen by carefully sliding out the Octane service tray and looking into the rear cooling vents of the Octane PCI chassis at the back of the Octane. If this PCI BIT3 card green LED does not light when console/Octane power is applied, you very likely have a BIT3 PCI card or PCI card/chassis seating problem.

Comment: *IF THIS STEP PASSES, PROCEED TO STEP 4 BELOW. IF THIS STEP FAILS, THERE IS A PCI BIT3 BOARD, SGI PCI CHASSIS, OR A PCI BOARD/CHASSIS SEATING PROBLEM.*

4.) Open a shell and '**cu sbc**' into the SBC, check that the VME BIT3 board was probed/attached by the **VMUNIX** device driver successfully at the last bootup as follows:

```
{ctuser@engbay}[1] cu sbc
Connected
CT_sbc0 login: root
Password:
Jan 12 08:30:45 CT_sbc0 login: ROOT LOGIN console
Last login: Wed Jan 12 08:20:28 on console
SunOS Release 4.1.1_U1 (GOS_IG) #4: Thu Jun 18 15:22:19 CDT 1998
root @ CT_sbc0 1:
root @ CT_sbc0 1: cd /var/adm
root @ CT_sbc0 4: more messages* |grep "vmunix: svd0"
 Jan 10 14:32:35  vmunix: svd0 at vme16d16 0x2000 vec 0xf0 vec 0xff
   Jan 11 11:43:49 CT_sbc0 vmunix: svd0 at vme16d16 0x2000 vec 0xf0
vec 0xff
   Jan 12 07:22:47 CT_sbc0 vmunix: svd0 at vme16d16 0x2000 vec 0xf0
vec 0xff
   root @ CT_sbc0 5:
```

Comment: *Repeats for all occurrences of Irix bootup/probe/attach by date/time.*

Also, when the VME BIT3 board powers up and configures itself correctly, the green READY LED will light up on the front faceplate of the VME BIT3 board. If this green READY LED does not light up when console/VME power is applied, you very likely have VME BIT3 board or VME board/chassis seating problem.

Comment: *IF THIS STEP PASSES, YOU MAY HAVE A BIT3 CABLE/SEATING PROBLEM. IF THIS STEP FAILS, THERE IS A VME BIT3 BOARD, JUMPERS, OR SEATING PROBLEM.*

OTHER USEFUL INFORMATION

A.) You can check that the BIT3 OC-SBC network device is configured correctly using the following commands on the OC and/or SBC as shown:

a.) CHECK BIT3 NETWORK DEVICE RUNNING WITH CORRECT NETWORK PARAMETERS ON OC

Make sure the device is “RUNNING” and that the “inet”, netmask, and broadcast parameters are set correctly (use 'reconfig' on the OC and SBC if necessary to correct these). The LFC defaults are shown.

```
{ctuser@engbay18}[7] ifconfig vd0
vd0: flags=8e3<UP,BROADCAST,NOTRAILERS,RUNNING,NOARP,MULTICAST>
inet 192.18.18.1 netmask 0xffffff00 broadcast 192.18.18.255
{ctuser@engbay18}[8]
```

b.) CHECK BIT3 NETWORK DEVICE RUNNING WITH CORRECT NETWORK PARAMETERS ON SBC

Make sure the device is “RUNNING” and that the “inet”, netmask, and broadcast parameters are set correctly (use 'reconfig' on the OC and SBC if necessary to correct these). The LFC defaults are shown

```
{ctuser@engbay18}[8] cu sbc
Connected
CT18_sbc0 login: root
```

```
Password:
Jan 12 08:16:50 CT18_sbc0 login: ROOT LOGIN console
Last login: Mon Jan 10 15:51:31 from CT18_oc0
SunOS Release 4.1.1_U1 (GOS_IG) #4: Thu Jun 18 15:22:19 CDT 1998
root @ CT18_sbc0 1: ifconfig vd0
vd0: flags=e1<UP,NOTRAILERS,RUNNING,NOARP>
 inet 192.18.18.2 netmask ffffff00
root @ CT18_sbc0 2:
```

B.) You can check the current status of network communications on the OC or SBC using the following commands:

a.) CHECK NETWORK DEVICE STATUS ON THE OC

```
{ctuser@engbay18}[1] netstat -i
Name Mtu   Network     Address      Ipkts Ierrs    Opkts Oerrs  Coll
ef0  1500  3.7.52      engbay18     24540     1    15004     0  3375
vd0  4336  192.18.18    ct18_oc0     3197     0     2849    16     0
lo0  8304  loopback    localhost    79004     0    79004     0     0
{ctuser@engbay18}[2]
```

"`ef0`" is the hospital/gateway ethernet network.
"`vd0`" is the OC to SBC BIT3 dedicated subnetwork (output "errors" can be
"`normal`" on the BIT3 due to VMEbus busy retries during scan/recon).
"`lo0`" is the host loopback pseudo-device.
"`ppp0`" is the InSite PPP serial port network device.
"`Network`" is the IP base number of the network/subnet.
"`Address`" is the hostname.
"`Ipkts`" is the number of network packets received since the last bootup.
"`Ierrs`" is the number of network receive errors since the last bootup.
"`Opkts`" is the number of network packets transmitted since the last bootup.
"`Oerrs`" is the number of network transmit errors since the last bootup.
"`Coll`" is the number of network collisions (there are normal since this is how ethernet works when multiple nodes "negotiate" for the cable.

b.) CHECK NETWORK DEVICE STATUS ON THE SBC

```
{ctuser@engbay18}[2] cu sbc
Connected
CT18_sbc0 login: root
Password:
Jan 12 08:20:28 CT18_sbc0 login: ROOT LOGIN console
Last login: Wed Jan 12 08:16:50 on console
SunOS Release 4.1.1_U1 (GOS_IG) #4: Thu Jun 18 15:22:19 CDT 1998
root @ CT18_sbc0 1: netstat -i
Name  Mtu  Net/Dest     Address   Ipkts  Ierrs Opkts  Oerrs Collis
Queue
```

```
ei0   1500 192.9.220.0   SBCdLAN     5822    0    5875    0    0      0
vd0   4336 192.18.18.0   CT18_sbc0   8306    0    8756    0    0      0
lo0   1536 127.0.0.0     localhost   2197    0    2197    0    0      0
root @ CT18_sbc0 2:
```

"`ei0`" is the control LAN between SBC and STC/ETC and OBC (via STC).

"`vd0`" is the OC to SBC BIT3 dedicated subnetwork.

"`lo0`" is the host loopback pseudo-device.

"`Net/Dest`" is the IP base number of the network/subnet.

"`Address`" is the hostname.

"`Ipkts`" is the number of network packets received since the last bootup.

"`Ierrs`" is the number of network receive errors since the last bootup.

"`Opkts`" is the number of network packets transmitted since the last bootup.

"`Oerrs`" is the number of network transmit errors since the last bootup.

"`Collis`" is the number of network collisions (there are normal since this is how ethernet works when multiple nodes "negotiate" for the cable.

"`Queue`" is the number of packets waiting in the queue.

2.1.3.8 Host SCSI Bus 0

```
diagnostic(s): (OC) scsistat, hinv, fx
error log(s): (OC) /var/adm/SYSLOG*
```

GENERAL OC SCSI BUS 0 INFO

- SCSIbus 0 is the INTERNAL drive bays of the Octane host computer.
- There are NO JUMPERS on the System Disk or Optional Disk sled assemblies.
- The Optional Disk is a customer purchased option for more image space.
- The System Disk drive in the BOTTOM DRIVE BAY is auto configured as SCSI ID 1.
- The Optional Disk in the MIDDLE DRIVE BAY is auto configured as SCSI ID 2.
- The TOP DRIVE BAY is auto configured as SCSI ID 3 and is currently not used.
- ALWAYS check the `/var/adm/SYSLOG*` if you suspect SCSI device errors.

USING THE IRIX 'hinv' COMMAND

The Irix 'hinv' command shows all devices that were detected the last time the Octane system was BOOTED ONLY. The System Disk and Optional Disk (if present) show up in the 'hinv' output list as follows:

(other output not shown)

```
Integral SCSI controller 0: Version QL1040B (rev. 2) <--- BUS 0 (internal)
Disk drive: unit 1 on SCSI controller 0             <--- SYSTEM DISK
Disk drive: unit 2 on SCSI controller 0             <--- OPTIONAL DISK
(other output not shown)
```

USING THE GEMS 'scsistat' COMMAND

The GEMS 'scsistat' command issues a *LIVE* SCSI inquiry command to both SCSIbus controllers (0 and 1) and all probed/functioning devices will respond. You should be 'root' to run this command. The System Disk and Optional Disk (if present) show up in the 'scsistat' output list as follows:

(other output not shown)

```
 Device 0 1  Disk  SGI  QUANTUM XP34550W FW Rev: LXY4  <---SYSTEM DISK
 Device 0 2  Disk  SGI  QUANTUM XP34550W FW Rev: LXY4  <---OPTIONAL DISK
```

(other output not shown)

Note: Manufacturer, model number, and firmware revision may differ from the example above. Several drives are approved by SGI/GEMS.

USING THE IRIX 'fx' SCSI UTILITY

The Irix 'fx' SCSI utility can be used to test or exercise almost any SCSI device. To non-destructively test the System Disk or the Optional Disk, follow the example below EXACTLY until you are comfortable with 'fx'. This utility is safe when "used as directed". THIS UTILITY IS CAPABLE OF DESTROYING ALL SOFTWARE/DATA IMMEDIATELY ON ANY SCSI DEVICE IF IT IS USED IN SPECIAL EXPERT MODES NOT DOCUMENTED HERE. **DO NOT** EXPERIMENT WITH THIS UTILITY! ALWAYS RESPOND "NO" IF 'FX' ASKS YOU TO "UPDATE THE VOLUME LABEL"!

IRIX 'fx' READ-ONLY TEST EXAMPLE

This example will READ every data block on the system disk. If there are any errors after several retries, the block in question will be remapped to a good spare sector (block) and the data will be recovered (if possible).

This example can be used to test most SCSI devices (not DASM or Central Data Serial Expander) by using the correct ctlr# and drive#. Note that the MOD or CDROM require media installed to run this test.

EXAMPLE SESSION OUTPUT	COMMENTS/ NOTES
{ctuser@engbay24}[1] su	MUST BE ROOT
Password:	
{ctuser@engbay24}[1] fx	ENTER FX UTILITY
fx version 6.4, Sep 17, 1997	
fx: "device-name" = (dksc)	USE DEFAULT
fx: ctlr# = (0)	CONTROLLER#
fx: drive# = (1)	DEVICE SCSI ID#
fx: lun# = (0)	USE DEFAULT LUN#
...opening dksc(0,1,0)	
IMPORTANT: 'fx' may ask you to "update the label" when exiting but you should ALWAYS respond/enter 'NO' to this question or you could change/corrupt the label on your system disk, option disk, or MOD media.	

Table 8-6 Example Output Session

EXAMPLE SESSION OUTPUT	COMMENTS/ NOTES
fx: partitions in use detected on device	DISK IS MOUNTED
fx: devname seq owner state	
fx: /dev/rdsk/dks0d1s7 5 xfs already in use	
fx: /dev/rdsk/dks0d1s6 4 xfs already in use	
fx: /dev/rdsk/dks0d1s5 3 xfs already in use	
fx: /dev/rdsk/dks0d1s3 2 xfs already in use	
fx: /dev/rdsk/dks0d1s0 1 xfs already in use	
fx: Warning: this disk appears to have mounted filesystems.	
Don't do anything destructive, unless you are sure	
nothing is really mounted on this disk.	
...drive selftest...OK	
Scsi drive type == SGI QUANTUM XP34550WLXY4	DEVICE MODEL#
----- please choose one (? for help, .. to quit this menu)-----	
[exi]t [d]ebug/ [l]abel/	
[b]adblock/ [exe]rcise/ [r]epartition/	
fx> exe	EXERCISE DEVICE
----- please choose one (? for help, .. to quit this menu)-----	
[b]utterfly [r]andom [st]op_on_error	
[e]rrlog [se]quential [m]iscompares	
fx/exercise> se	USE SEQUENTIAL
fx/exercise/random: modifier = (rd-only)	READ ONLY MODE
fx/exercise/random: starting block# = (0)	STARTING BLOCK#
fx/exercise/random: nblocks = (8888543)	#BLOCKS TO TEST
fx/exercise/random: nscans = (1)	#PASSES TO RUN
random pass 1: scanning [0, 8888543] (8888543 blocks)	
0%	% TEST COMPLETE
IMPORTANT: 'fx' may ask you to “update the label” when exiting but you should ALWAYS respond/enter 'NO' to this question or you could change/corrupt the label on your system disk, option disk, or MOD media.	

Table 8-6 Example Output Session (Continued)

EXAMPLE SESSION OUTPUT			COMMENTS/ NOTES
(use 'CONTROL C' to stop the testing at any time)			'CTRL C' ABORTS
----- please choose one (? for help, .. to quit this menu)-----			
[b]utterfly	[r]andom	[st]op_on_error	
[e]rrlog	[se]quential	[m]iscompares	
fx/exercise> ..			GO UP 1 MENU LEVEL
----- please choose one (? for help, .. to quit this menu)-----			
[exi]t	[d]ebug/	[l]abel/	
[b]adblock/	[exe]rcise/	[r]epartition/	
fx> exi			EXIT FX UTILITY
IMPORTANT: 'fx' may ask you to "update the label" when exiting but you should ALWAYS respond/enter 'NO' to this question or you could change/corrupt the label on your system disk, option disk, or MOD media.			

Table 8-6 Example Output Session (Continued)

THE SGI IRIX /var/adm/SYSLOG

The Irix SCSI device driver will ALWAYS detect and log an error message in /var/adm/SYSLOG* if there are any SCSIbus or SCSI device hardware errors. If there are no SCSI errors logged here, the SCSIbus and devices are GOOD. Note that attempting to mount/attach/label the MOD without media inserted can result in NORMAL "error" messages in the SYSLOG. SCSI error messages may contain one of the following device ID forms:

```
/dev/dsk/dks(X)d(Y)s(Z)(example: /dev/dsk/dks0d1s0 = system disk root)
/dev/rdsk/dks(X)d(Y)s(Z)(example: /dev/rdsk/dks0d1s1 = system disk swap)
/dev/scsi/sc(X)d(Y)l(Z)(example: /dev/scsi/sc1d6l0 = CDROM drive)
```

where,

(X) is the host CPU SCSIbus controller/channel# (0=drive bays,1=external bus) and (Y) is the SCSI device ID#:

```
1=System Disk, 2=Option Disk, 3=Maxoptix MOD,4=Central Data Serial,
5=Pioneer MOD, 6=CDROM)
```

(Z) is the device partition# being accessed when the error occurred (0-8)

SCSI device error messages always contain the SCSIbus# and SCSI ID#. Messages will also be posted for "normal" events like trying to access a removable media device without media or with bad media (MOD or CDROM). Messages will also be posted if any SCSI device retries occur. If more than 1 SCSI device has errors, you may have a general SCSIbus problem

(check cables, connectors, terminator, device jumpers, DC voltage levels, cooling fans, etc.).

2.1.3.9 Host SCSI Bus 1

```
diagnostic(s): (OC) scsistat, hinv, fx
error log(s): (OC) /var/adm/SYSLOG*
```

GENERAL OC SCSI BUS 1 INFO

- SCSIbus 1 is the EXTERNAL SCSIbus from the Octane host computer.
- SCSIbus 1 contains all other CT/i SCSI and removable media devices.
- All devices on SCSIbus 1 HAVE JUMPERS and are SCSI-1 or SCSI-2 8-bit devices.

USING THE IRIX 'hinv' COMMAND

The Irix 'hinv' command shows all devices that were detected AT BOOTUP ONLY.

The devices on SCSIbus 1 show up in the list as follows:

```
Integral SCSI controller 1: Version QL1040B (rev. 2)
Disk drive: unit 1 on SCSI controller 1        <--- DASM VDB OR LCAM
Optical disk: unit 3 on SCSI controller 1            <--- MAXOPTIX MOD
Comm device: unit 4 on SCSI controller 1             <--- CENTRAL DATA
Comm device: unit 4, lun 1 on SCSI controller 1      <--- CENTRAL DATA
CDROM: unit 6 on SCSI controller 1                   <--- CDROM DRIVE
```

Note: The Central Data serial device normally shows two lines as above.

USING THE GEMS 'scsistat' COMMAND

The 'scsistat' command issues a live SCSI inquiry command to both SCSIbus controllers and all probed/functioning devices will respond. The SCSIbus 1 devices show up in the list as follows:

```
Device 1 1  Disk      ANALOGIC DASM-LCAM-3M     FW Rev: 1.3
Device 1 3  Optical   Maxoptix T5-2600          FW Rev: A6.5
Device 1 4  Comm      CenData  ST-1400B         FW Rev: V6.4
Device 1 6  CD-ROM    TOSHIBA  CD-ROM XM-6201TA FW Rev: 0167
```

Note: Device manufacturers, model numbers, and firmware versions (FW Rev) may differ from the above example due to obsolescence and cost reductions.

IMPORTANT: If the DICOM (currently Maxoptix) MOD drive has attached a DICOM archive media, then Device 1 3 information above will indicate “Unknown...EXCLUSIVELY_OPEN” instead of the device type, model number, and firmware version. This is normal whenever archive has media “attached”. Detach media to show device info.

USING THE IRIX 'fx' SCSI UTILITY

The Irix 'fx' SCSI utility can be used to test or exercise almost any SCSI device. To use 'fx' to non-destructively test the disk drives, CDROM, or MOD, follow the example shown in the “OC SCSI BUS 0" help text but substitute the appropriate controller (ctlr#) and device ID (drive#) when prompted for those parameters. To be sure that you don't conflict with any application software, such as Archive, shut down CT applications software only (using the service desktop utility) and run 'fx' tests from any Irix shell script as 'root'.

For the Maxoptix MOD or CDROM drives, insert compatible media and perform the read-only test as described in the “OC SCSI BUS 0" help text.

THE SGI IRIX /var/adm/SYSLOG

The Irix SCSI device driver will ALWAYS detect and log an error message in /var/adm/SYSLOG* if there are any SCSIbus or SCSI device hardware errors. The Central Data SCSI-serial expander device may retry occasionally due to SCSIbus loading. This results in occasional SCSIbus reset messages due to device 4 time outs. As long as Trackball and '**`cu sbc`**' work, this is OK.

If there are no SCSI errors logged here, the SCSIbus and devices are GOOD. SCSI error messages may contain one of the following device ID forms:

```
/dev/dsk/dks(X)d(Y)s(Z) (example: /dev/dsk/dks0d1s0 = system disk root)
/dev/rdsk/dks(X)d(Y)s(Z)(example: /dev/rdsk/dks0d1s1 = system disk swap)
/dev/scsi/sc(X)d(Y)l(Z)(example: /dev/scsi/sc1d6l0 = CDROM drive)
```

where,

(X) is the host CPU SCSIbus controller/channel# (0=drive bays, 1=external bus) and (Y) is the SCSI device ID#.

where,

1=System Disk, 2=Option Disk, 3=Maxoptix MOD, 4=Central Data Serial, and 5=Pioneer MOD, 6=CDROM

(Z) is the device partition# being accessed when the error occurred (0-8)

SCSI device error messages always contain the SCSIbus# and SCSI ID#.

Messages will also be posted for “normal” events like trying to access a removable media device without media or with bad/corrupted media (MOD or CDROM are removable media). Also, occasional device 4 (Central Data) time outs/resets are Ok as long as the retries work (i.e. - Trackball and SBC serial line work OK).

Messages will also be posted if any SCSI device retries occur. If more than 1 SCSI device has errors, you may have a general SCSIbus problem (cables, connectors, terminator, device jumpers, DC voltage levels, cooling fans, etc.).

2.1.3.10 OC System Disk

For details: See “Host SCSI Bus 0” on page 395.

2.1.3.11 OPTION IMAGE DISK

For details: See “Host SCSI Bus 0” on page 395.

2.1.3.12 DICOM MOD

For details: See “Host SCSI Bus 1” on page 399.

2.1.3.13 CD-ROM Drive

For details: See “Host SCSI Bus 1” on page 399.

2.1.3.14 Ethernet Gateway

```
diagnostic(s): (OC) ifconfig, netstat, ping, spray
error log(s): (OC) /var/adm/SYSLOG*
```

GENERAL ETHERNET GATEWAY INFO

The “ethernet gateway” is not technically a “gateway” but is named as such. The Octane built-in ethernet is auto sensing 10/100 megabit/sec UTP ethernet. To achieve a 100 megabit/sec connection, ALL network equipment as well as the destination machine (and software) MUST be capable of supporting 100 megabit (FAST) ethernet mode. The “100 megabit/second” is the theoretical maximum and is seldom attained in the real world. Generally, due to system overhead and several other network factors, 100mbit can be anywhere from 2-5 times faster than 10mbit (it is NOT “10 times faster”).

The Octane ethernet connector is a standard UTP network RJ45 8-pin connector. The RJ45 must be connected to an appropriate network hub, repeater, or switch. A powered RJ45-to-coaxial transceiver option can be used for legacy coaxial networks but you will be limited to 10 megabit/sec only and some of these transceivers/converters may require a “swapped/reversed” cable or have a switch to reverse XMT/RCV polarity (most all network “hubs”, “switches”, and “repeaters” do this automatically but NOT all “transceivers/converters”).

USING THE 'ifconfig' COMMAND

The **'ifconfig'** command shows you the current state of a network interface device. Some examples are shown below.

Example: "ifconfig" on hospital connection or gateway

Example of 'ifconfig' on the “hospital connection” or “gateway” network interface (Octane built-in ethernet device):

```
engbay12 >>> ifconfig ef0
ef0:
flags=1c63<UP,BROADCAST,NOTRAILERS,RUNNING,FILTMULTI,MULTICAST,CKSUM
inet 3.7.52.142 netmask 0xfffffc00 broadcast 3.7.55.255
```

Example: "ifconfig" on OC-to-SBC BIT3

Example of 'ifconfig' on the OC-to-SBC BIT3 “dedicated subnetwork”:

```
  engbay12 >>> ifconfig vd0
    vd0: flags=8e3<UP,BROADCAST,NOTRAILERS,RUNNING,NOARP,MULTICAST>
            inet 192.12.12.1 netmask 0xffffff00 broadcast 192.12.12.255
```

Note: The various parameters shown above are set ONLY by using 'reconfig' and entering/changing the “network” settings page when loading or when system is reconfigured.

USING THE 'netstat' COMMAND

The **'netstat'** command provides status of the network devices on your system (using the -i switch) or can list the network routing tables that have been assembled by your system based on broadcasted network information (using the -r switch).

Example: "netstat -i" on hospital connection or gateway

Example of 'netstat -i' on the OC “hospital connection” or “gateway” network interface (Octane built-in ethernet device):

```
engbay18 1# netstat -i
Name Mtu   Network     Address      Ipkts Ierrs      Opkts Oerrs    Coll
ef0  1500  3.7.52      engbay18    797475     0     594940     1  315407
vd0  4336  192.18.18   ct18_oc0    180420     0     169749  1298       0
lo0  8304  loopback    localhost  3431581     0    3431581     0       0
```

“Name” is the network device name (ef0 = built-in ethernet, vd0 = BIT3 subnet, lo0 is the local loopback device). “Network” is the IP network or subnetwork that the device is currently configured for. “Address” is the hostname that is associated with the device. “Ipkts” and “Opkts” are the number of input and output packets received or sent on the device since the device was started (or since the system was booted). “Ierrs” and “Oerrs” are the number of packet errors on the device. Errors can be caused by network cables/equipment or the network devices themselves (the BIT3 may normally have many errors on the output side because of the way it accesses the VMEbus depending on how heavily a given system is used).

Example: "netstat -i" command

Example of using the 'netstat -r' command to look at network routing tables:

```
engbay1 3# netstat -r
Routing tables
Internet:
Destination     Gateway          Netmask    Flags   Refs    Use  Interface
default          medctc1us                   UG          1         5  ef0
3.1.116          medctc2us        0xfffffc00 UG          0         0  ef0
3.1.148          medctc2us        0xfffffc00 UG          0         0  ef0
3.1.228          medctc2us        0xfffffc00 UG           0
(etc, etc, etc)
```

Routing tables may go on and on depending how extensive the network is. You can always pipe the output through '**grep**' to see specific routing information that your interested in.

```
engbay1 4# netstat -r |grep engbay1
3.7.52       engbay1       0xfffffc00 U     17        11  ef0
heliosmfg1  localhost                UGHS   3         3  lo0
224          heliosmfg1    0xf0000000 US     1        10  ef0
heliosmfg1 5#
```

USING THE 'ping' COMMAND

The 'ping' command can be useful for establishing that a basic network connection exists between 2 hosts. It's a low level echo response that can be dependent on host loading (packets can be dropped). Do not use ping statistics for network reliability assessment. The Irix **'ping'** command automatically repeats every second and shows turnaround time. Use 'CONTROL C' to stop the 'ping' at any time.

Example: ’ping’ command

Example of 'ping' COMMAND between a CT/i OC and the CT/i SBC:

```
rhap1 2# ping sbc
PING rh01_sbc0 (192.2.100.2): 56 data bytes
64 bytes from 192.2.100.2: icmp_seq=0 ttl=255 time=1 ms
64 bytes from 192.2.100.2: icmp_seq=1 ttl=255 time=1 ms
64 bytes from 192.2.100.2: icmp_seq=2 ttl=255 time=1 ms
----rh01_sbc0 PING Statistics----
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 1/1/1 ms
rhap1 3#
```

This system was “idle” (not scanning, archiving, networking, etc.) and this ping confirms that the OC-to-SBC BIT3 link is “working”. A 1 second turnaround on between “idle” systems on an “quiet” network is fairly typical but this can vary widely. Again, “dropped packets” between busy systems on a busy network is typical and varies widely.

USING THE 'spray' COMMAND

The 'spray' command can be useful in certain situations but it can also be deceiving. The default spray command sends 1162 packets that are 86 bytes long with NO delay between packets. The network and CPU performance, as well as loading and OS type, causes varying “dropped packets” between different systems at different times (dropped packets are usually NOT hardware errors).

The examples below show a 'spray' between two different remote machines with drastically different results. The network and all machines are operating just fine in both cases.

Example:

```
engbay1 7# spray 3.7.52.150
sending 1162 packets of lnth 86 to engbay25 ...
        no packets dropped by engbay25
        3135 packets/sec, 269655 bytes/sec
engbay1 8#
engbay1 16# spray ninja
sending 1162 packets of lnth 86 to ninja ...
        1046 packets (90.017%) dropped by ninja
        11 packets/sec, 992 bytes/sec
engbay1 17#
```

THE SGI IRIX /var/adm/SYSLOG

The Irix network device driver will ALWAYS report errors in /var/adm/SYSLOG* if there are hardware related problems in the local machine network interface, network cable, hub, or transceiver if present. Some of these typical hardware errors may look something like one of the following examples:

Example:
possible causes

```
Apr 22 15:36:26 2A:heliosmfg1 unix: ef0: link fail - check ethernet cable
CHECK ETHERNET CABLE AND CONNECTIONS FROM OC TO NETWORK EQUIPMENT

Apr 22 15:36:26 2A:heliosmfg1 unix: ef0: remote hub fault
CHECK THE NETWORK EQUIPMENT THAT SERVICES YOUR OC NETWORK CONNECTION

Apr 22 15:36:26 2A:heliosmfg1 unix: ef0: rx SSRAM hardware parity error
IF THIS ERROR CONTINUES AND CUSTOMER HAS NETWORK SYMPTOMS, THE OCTANE
IP30 SYSTEM MODULE SHOULD BE REPLACED (ONBOARD ETHERNET).

Apr 22 15:36:26 2A:heliosmfg1 unix: ef0: excessive collisions
THIS COULD BE AN EXTREMELY BUSY NETWORK OR COULD INDICATE PHYSICAL
PROBLEMS WITH THE NETWORK ITSELF (TRANSCEIVER, CABLE, HUB, ROUTER, ETC.)
```

All network device drivers (ec0, ef0, vd0, du0, etc.) and network services (routed, sockd, ppp, portmapper, rlogin, ftp, telnet, etc) will post message entries to the console window and in /var/adm/SYSLOG. These are often difficult to sort through and make sense of depending on the circumstances. Some “error” messages may be “normal” or “useless” and others may be quite meaningful. So, it's difficult to deal with them here. Contact OLC/engineering as needed.

2.1.4 Replacement Procedures

After removing or installing any component, it's a good idea to verify the CPOP LEDs provide the desired values. See "CPOP Connector LEDs" on page 380.

Note: When replacing components in the Octane/OC, it may be necessary to press the Reset Button on the front panel of the Octane to start the OC up again. See Figure 8-2 on page 372

2.1.4.1 Octane Hard Drive

The Octane hard drive is supplied by SGI. *Place the primary system drive in the bottom bay* on the front of the computer. If a second system disk is added, place it in the middle bay. The host assigns their SCSI IDs. The primary disk is SCSI ID 1 (`dks0d1sN`). The one inserted in the middle becomes SCSI ID 2 (`dks0d2sN`).

Note: Top slot not used — Currently, top slot in expansion bay not used (filled).

HOW TO REPLACE OR INSTALL AN OCTANE HARD DRIVE

1.) Shutdown system and remove power.
2.) Press both bezel release buttons on front upper sides.
3.) Tilt the cover forward and lift to remove it.

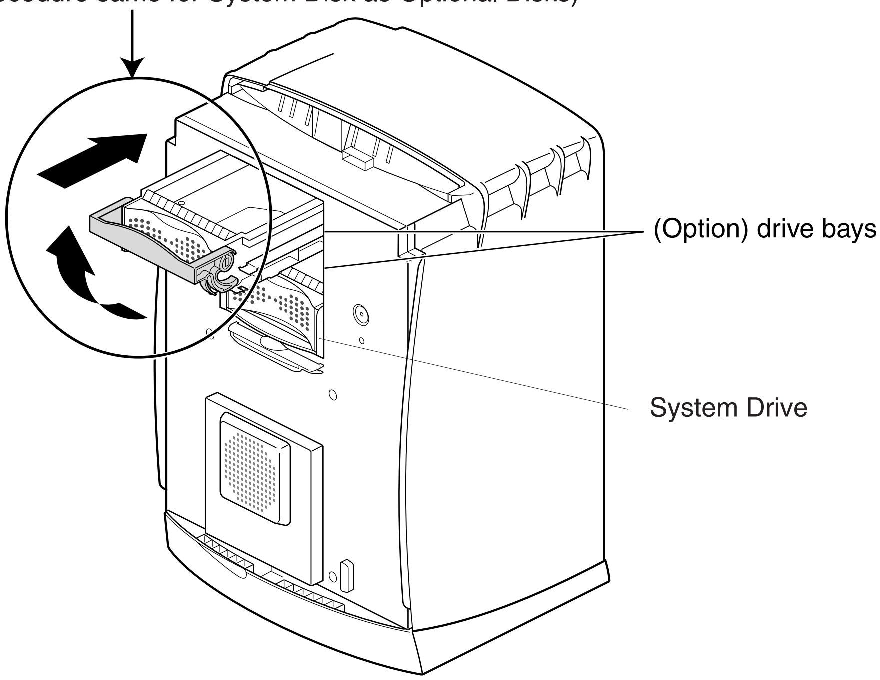
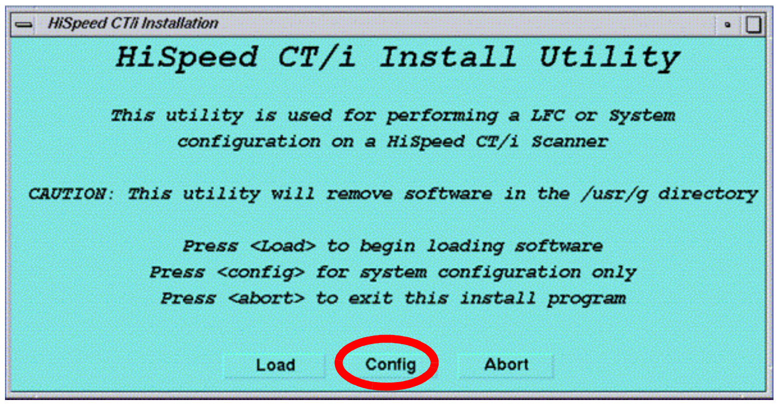


Figure 8-17 CT/i (Octane) System Drive

4.) Lift the drive's handle to the horizontal position and *gently* slide it into the bay. *Pushing a drive in with too much force can damage it. Seat the drive carefully but firmly.*
5.) When it is flush with the chassis, rotate the handle down to lock it.
6.) If needed, remove the plastic panel for a new bay if adding a disk. Keep it in case it is needed later. Snap a saved panel to the cover if you permanently remove a hard drive from a bay. *This insures proper air flow. Do not remove a drive unless you have a replacement or a cover for the bay.*
7.) Re-power the system and use hinv or the Service Utility called System Hardware to verify that the host recognizes the hard drive(s).

CT/I OC Disk Replacement	System Disk	Image Disk
	Follow the LFC process in CT/I LFC Procedures for Versions 4.4, 4.5, 6.1, 6.2 and 6.3	Follow the LFC process found in CT LFC Procedures to regen the image database. Section 2.8

8.) In the HiSpeed CT/i Install Utility window, click on CONFIG.



9.) A pop up message will query if you wish to load the INFO file from the System State MOD; if you have a System State MOD, select YES. Another pop-up message will appear. Place the System State MOD in the drive and select OK.

You will now go through each of the Reconfig screens and verify the information. You may make updates as needed. If you did not have a System State MOD, you will need to enter the appropriate information.

2.1.4.2 Octane Light Bar

To replace the Light or LED Module, remove the front cover, then squeeze both top and bottom wings of the light together at both ends, gently and evenly pull straight out.

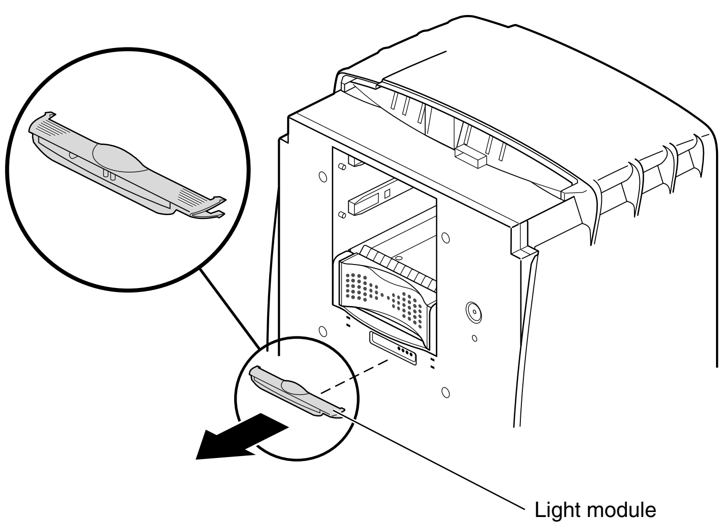


Figure 8-18 CT/i (Octane) Light Bar

2.1.4.3 Octane System Module

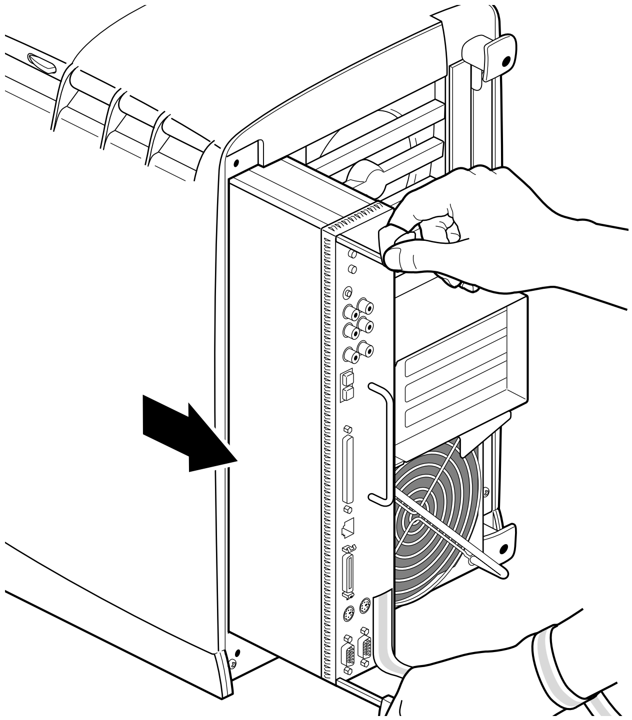


Figure 8-19 Removing the System Module

Removing the System Module

1.) Shutdown system and remove power.

2.) Remove the console's front cover. Pull out platform upon which the computer rests. Release its tie down strap if present.

3.) Remove the locking bar (if applicable)

NOTICE **Wear a grounded ESD wrist strap. Place removed electronic parts on or in an antistatic surface.**

CAUTION **Wait five minutes after power is off before you continue. Let it cool.**

4.) Loosen the captive screw in the sliding handles on the top and bottom.

5.) Pull both handles at the same time until they are fully extended.

6.) Grasp the module handle with your left hand and place your right hand against the top of the computer's back. Pull the module out without allowing the delicate connectors on its back edge to touch anything.

7.) Place the system module on an antistatic surface with the component side up.

8.) Place a cap on each compression connector.

System ID Module

The **System ID Module** can be seen inside computer after the System Module is pulled. It holds the pre-programmed Ethernet address for the Octane computer. It is a small circular disk held by a metal retaining clip. See page 415.

Octane Processor

A Single Processor (brick) is held by four screws; the Dual Processor by six. The Single Processor is placed closest to the panel of audio connectors. Take care to align connectors in the System Module with those on the Processor.

Octane Host (DIMM) Memory

The CT/i (Octane) host uses *192 MB of CPU memory*. DIMM sockets S1 and S2 each contain a 64MB module for a 128 MB bank. Sockets S3 and S4 each contain a 32MB DIMM for a 64MB bank. This yields the 192MB.

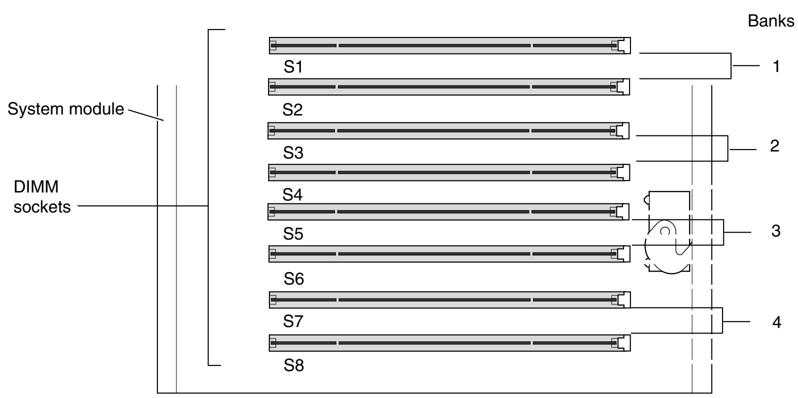


Figure 8-20 CT/i (Octane) Host - DIMM Socket and Bank Identification

There are eight DIMM sockets and four banks. A bank must be completely filled with identical modules or empty. Bank 1 (sockets 1 and 2) must always be filled. Banks are filled sequentially; when Bank 1 is full, fill Bank 2, Bank 3 and then Bank 4. This means you must not skip a bank. See.Figure 8-20 .

NOTICE

- **Memory modules are extremely sensitive to static electricity. Use an ESD wrist strap and handle with care. Also be aware that the heat sinks inside the computer become very HOT.**

Avoid Touching Bristles

- **The DIMMs are located near a delicate compression connector. Be extremely careful not to touch the compression connector gold bristles.**

REPLACING DIMMS

1.) Shut system down and remove power.
2.) Wait 5 minutes after powering off the workstation to allow the heat sinks to cool
3.) Attach a wrist strap, then remove the system module from the chassis. Withdraw the system module and place it on a flat, dry, antistatic surface.
4.) Locate the DIMMs you want to remove.

NOTICE **Do not touch the connector near the DIMM removal levers.**

5.) As shown in Figure 8-21 , press down on the latch at (A), near the end of the DIMM socket. The DIMM partially ejects from the socket. It can then be remove as shown in (B).

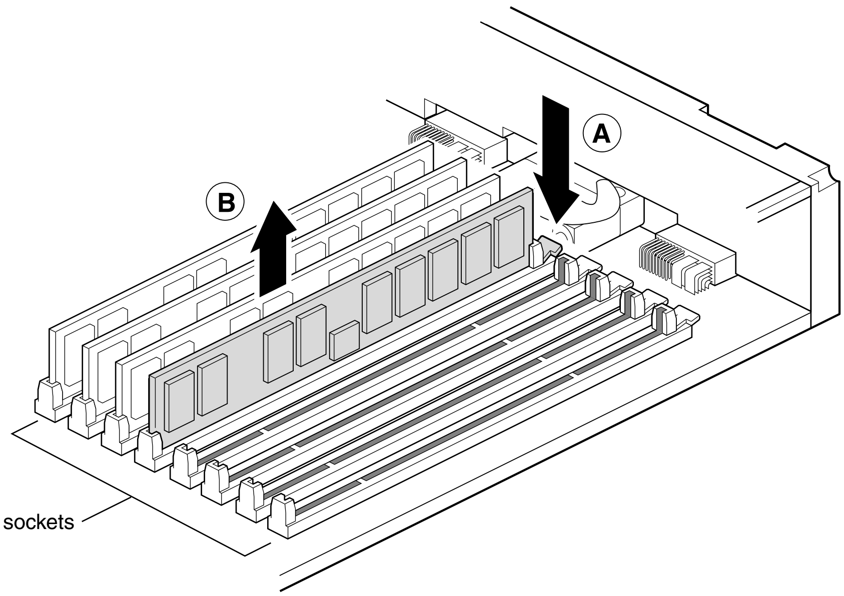


Figure 8-21 CT/i (Octane) DIMM Removal

INSTALLING DIMMS

1.) Insert the DIMM into the socket, gently but firmly. You hear a click as it is seated, and the latch on the end of the socket moves up.DIMMs are notched on the bottom so that they cannot be inserted incorrectly. See Figure 8-22 .

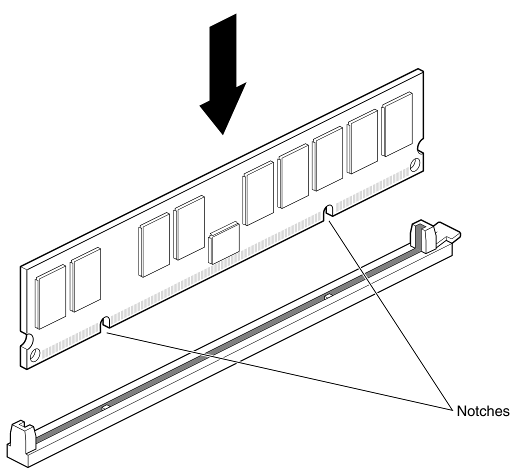


Figure 8-22 CT/i (Octane) Host DIMM Installation

2.) Check to be sure both sockets in the bank are full. DIMMs must be installed in pairs.You have finished installing memory and are ready to replace the system module.

COMMON MISTAKES:

- DIMM Sockets not populated Correctly - Both sockets in a DIMM bank must be either empty or populated. If you are removing one DIMM and not replacing it immediately, also remove the other DIMM in the bank and replace it when you install a new DIMM.

- DIMM not seated properly - Before replacing a memory module, check that all are seated correctly in their slots, the first bank has two DIMMs that are exactly the same. The second bank, if used, has two DIMMs that are exactly the same.
- Incorrect memory combinations - Memory is installed correctly when it is vertical and perpendicular to the motherboard, and the latches on the both sides fit snugly around it. If the memory module appears to be leaning, wear an ESD wrist strap and push it into a vertical position.

2.1.4.4 Octane XIO Module

Before removing a graphics board or TRAM, you must power off the OCTANE workstation, wait 5 minutes for the heat sinks to cool, attach a wrist strap, and not allow the compression connectors to touch anything. Test for heat before touching any of the parts.

XIO Module

1.) Bring down the system.
2.) Power off console
3.) Unplug the power cord.
4.) Power off the monitor by pressing the power button.
5.) Wait 5 minutes before removing the XIO module.
6.) Remove all the cables from the XIO module.

NOTICE **The components inside the OCTANE workstation are extremely sensitive to static electricity; you must wear the wrist strap while replacing parts inside the workstation.**

7.) When you remove the XIO module, the compression connectors on the back of the XIO module (XIO boards) are accessible and easily damaged. All XIO graphics boards have compression connectors, and most XIO option boards do.

NOTICE Avoid Damage **The compression connectors on each XIO board are very delicate and easily damaged. Do not touch or bump the gold bristled pad.**

CAUTION **The heat sinks on the XIO boards become very hot. Wait 5 minutes after powering off the OCTANE workstation before you remove the XIO module. Test before touching any of the XIO boards.**

8.) Loosen the two captive screws in the XIO module handles with the supplied Phillips screwdriver until the screws are disconnected from the chassis.
9.) Grasp the handles and pull until the XIO module protrudes about an inch from the chassis. The handles and XIO module move out about one inch before the I/O panels move.
10.) Continue to pull on the handles until the XIO module releases from the workstation.Grasp the XIO module along its length, and support the base of the module with your hand as you remove it from the chassis.
11.) The handle area protrudes when the XIO module is out of the chassis. When protruding, the identification slots for the XIO boards, D and A, B and C, are visible.

Note: Do not push on the handle area after you have removed the XIO module. The XIO module locks to the workstation only if the handle area is protruding.

(12) Place the XIO module on a flat, antistatic surface.

XIO Components

1.) Before you remove a graphics board or TMRAM, place a cap on the XIO compression connector to prevent accidental damage.

NOTICE Avoid Damage **Never touch the gold (front) surface of the XIO compression connector. Touching it could damage the connector. Place a protective cap on XIO compression connector to prevent damage when components are removed from the OCTANE workstation.**

2.) Using a Phillips screwdriver, remove the screws from the graphics board. The OCTANE/SI graphics board attaches with 4 screws. The TMRAM is attached at the back with one nylon screw.

Note: Only use nylon screws to attach the TMRAM to its base or it will not work properly.

3.) Grasp the graphics board on the I/O panel and on the side of the board with no connectors and lift.

4.) With the same side facing up, place the board on a clean, antistatic surface.

Octane XIO Module

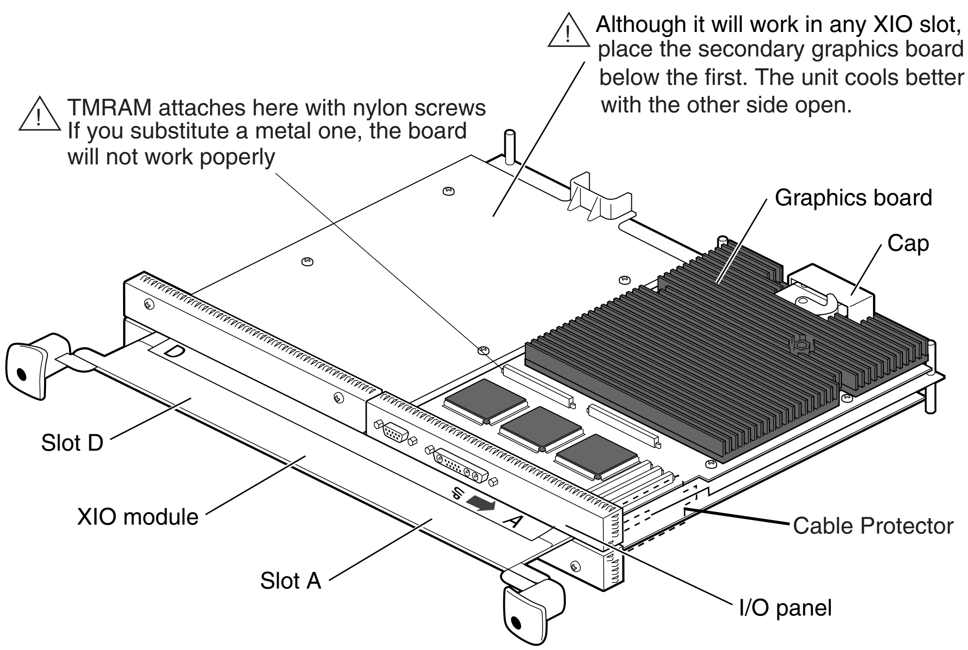


Figure 8-23 Inside the Octane XIO module

Installing a Graphics Board, Option Board, or Blank Panel

1.) Have the XIO module lying on its side with the handles facing you.

2.) The OCTANE/SI graphics board with TMRAM always goes in slot A. The secondary SI goes into slot D beneath slot A.

3.) Orient the board so the component side is up.

4.) Place the graphics board on the standoffs.

5.) Replace the screws, tightening the board to the standoffs.

Note: Be sure all the XIO slots are filled with a graphics board or blank panels. The system will not cool properly if any of the slots are empty.

Note: To have the host report what graphics board information it sees, type the following in a shell:

```
/usr/gfx/gfxinfo
```

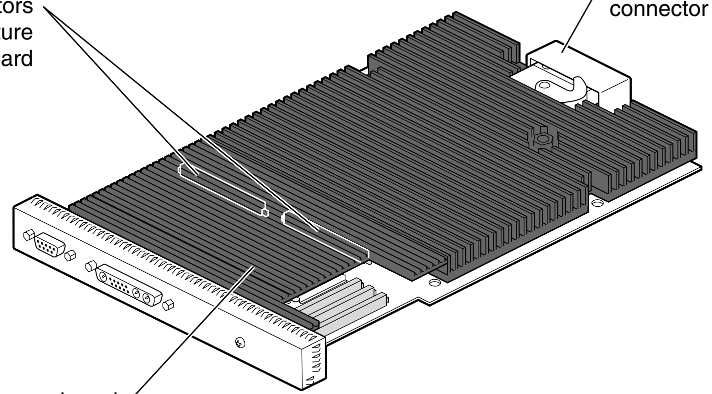


Figure 8-24 Octane Solid Impact board with Texture Memory

Installing the XIO Module

NOTICE **Please observe the following:**

- **Place the XIO module with the graphics boards facing toward the inside of the workstation. The boards may be damaged if placed the other way.**
- **Wear an antistatic wrist strap.**

6.) Slide the XIO module into guides on the top and bottom of the workstation.

7.) Before you insert the XIO module, make sure the handle portion protrudes in a locked position from the I/O panels. If the handles are flush with the I/O panels, the XIO module will stop during insertion. Pull out the handles until the sliding portion of the XIO module is rigid, then continue inserting the XIO module into the chassis.

8.) Grasp the handle area while supporting the XIO module, and slide the module into the chassis.

9.) Use the handles to push the XIO module into a locked position. (The I/O panels are nearly flush with the workstation when properly inserted, however, there is a slight variation in the depth of the boards.)

10.) Tighten the captive screws in the handles.

11.) Remove the wrist strap.

12.) Reconnect all XIO cables to the XIO module.

2.1.4.5 Octane PCI Module

Preparation

1.) Shutdown system and remove power.

2.) Remove the console’s front cover. Pull out platform upon which the computer rests.Release its tie down strap if present.

NOTICE **Wear a grounded ESD wrist strap. Place removed electronic parts on an anti-static surface.**

3.) Remove any cables from the PCI module.

4.) Loosen two captive screws.

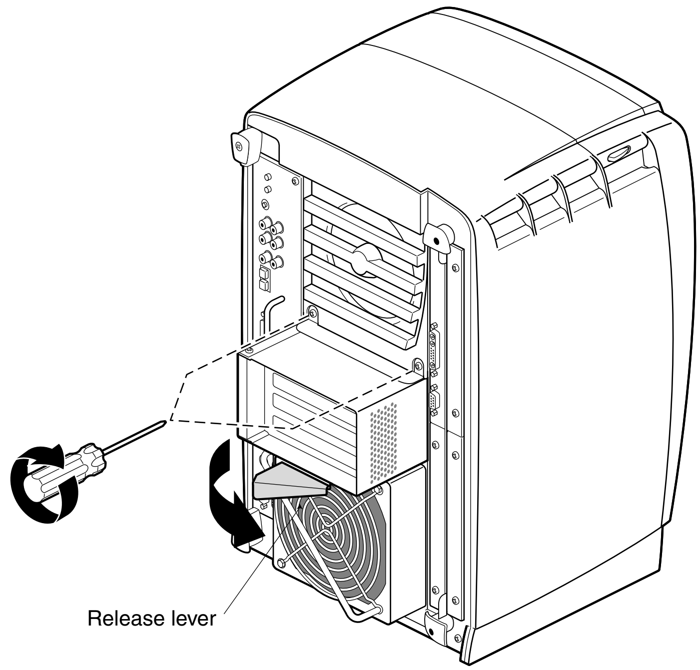


Figure 8-25 CT/i (Octane) Host - Releasing the PCI Module

5.) Pull out release lever along bottom of module.

6.) Slide out the module, taking care not to allow the compression connector to touch anything, then cap that connector once the module is resting on an antistatic surface.

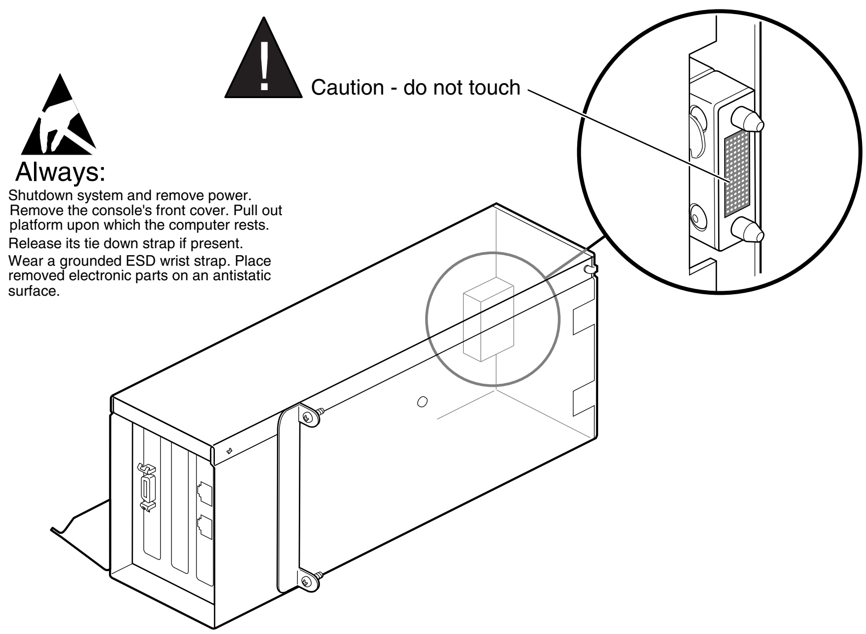


Figure 8-26 CT/i (Octane) Host - Placing the PCI Module for Board. Removal

PCI Card Removal

7.) Lie the module on its right side and loosen the two screws that hold the left side access cover. Then lift and remove this cover.

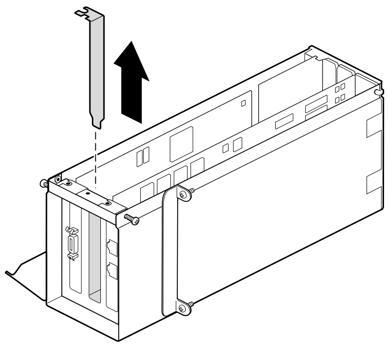


Figure 8-27 CT/i (Octane) Host -Removing screws securing the PCI board

NOTICE The PCI card is extremely sensitive to static electricity.

8.) The PCI card resides in the slot closest to the *bottom;* its PCI slot 1(BIT3). Unscrew the board from the front panel. You can also expand the I/O door. See Figure 8-28 . The I/O door expands open if necessary.

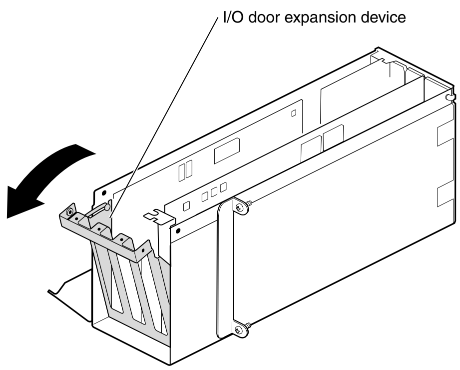


Figure 8-28 PCI Module I/O Expansion Door

Note: Any slots without cards require a panel to ensure good air flow, as shown in Figure 8-27 .

9.) To re-install, reverse the previous steps.

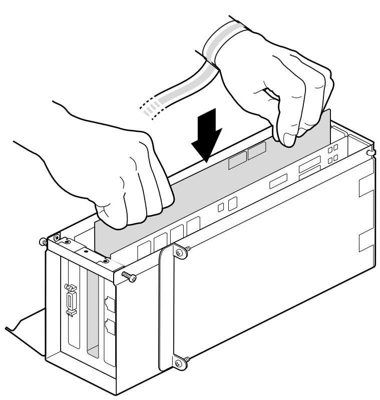


Figure 8-29 Installing the PCI Board into the PCI Module

2.1.4.6 Octane BIT3 PCIBus Board

CAUTION Potential Equipment Damage **Wear a grounded ESD wrist strap. Place removed electronic parts on an antistatic surface or in an antistatic bag.**

1.) Remove screw from BIT3 board.
2.) Pivot I/O door out from the module.
3.) Remove BIT3 board from the PCI module.
    Any slots without cards require a panel, to ensure good air flow.
4.) To install new board, reverse the previous steps.

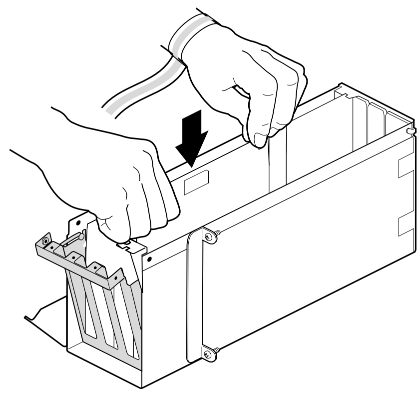


Figure 8-30 Installing the new BIT3 board

2.1.4.7 Octane Power Supply

1.) Shutdown system and remove power.
2.) Remove the console's front cover. Pull out platform upon which the computer rests. Release its tie down strap if present.
3.) **Remove the PCI module**, refer to page 413.

NOTICE **Wear a grounded ESD wrist strap. Place removed module on an antistatic surface.**

Note: The power supply is grounded while its power cord is plugged in. Just have power off.

4.) Use a phillips screwdriver to loosen the two captive screws at the base of the power supply module.
5.) Grasp the handle, pull it out then unplug the power cord.
6.) Reverse these steps to reinstall.

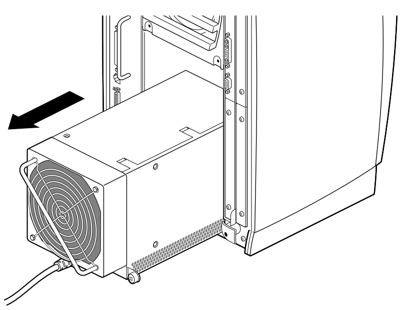


Figure 8-31 Octane Power Supply

2.1.4.8 Octane Frontplane Module

Follow this procedure to replace the **System ID Module**, the **Fan Module**, or the **Frontplane Module**.

Note: Save your ID Module

If the Frontplane Module is replaced, move the System ID Module from the old unit to the new. See Octane Frontplane Module on page 416.The new front plane module does not come with a system ID module.

1.) Shutdown system and remove power.
2.) Remove the console's front cover. Pull out platform upon which the computer rests. Release its tie down strap if present.

NOTICE **Wear a grounded ESD wrist strap. Place removed electronic parts on an antistatic surface.**

3.) Remove the System Module. Refer to page 406
4.) Remove the XIO Module. Refer to page 409
5.) Remove the PCI Module. Refer to page 413
6.) Remove the Octane Power Supply. Refer to page 414
7.) Squeeze both buttons on upper front sides of Octane computer, then tilt forward and lift to remove its front cover.
8.) Remove all Octane Disk Drives. Refer to page 404
9.) Remove the Light Module. Squeeze both top and bottom wings of the light together at both ends, gently and evenly pull straight out.
10.) Loosen the six captive screws that hold the frontplane module to the chassis.

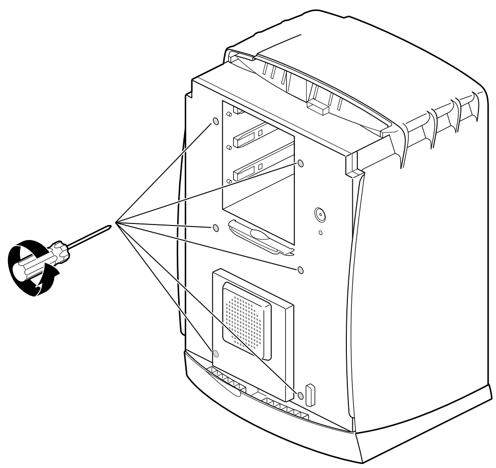


Figure 8-32 Loosen The System Module from the Chassis

11.) Place your hand inside the drive bay and lift the module from the base.
12.) Place it face down on an antistatic surface. Avoid touching any components.

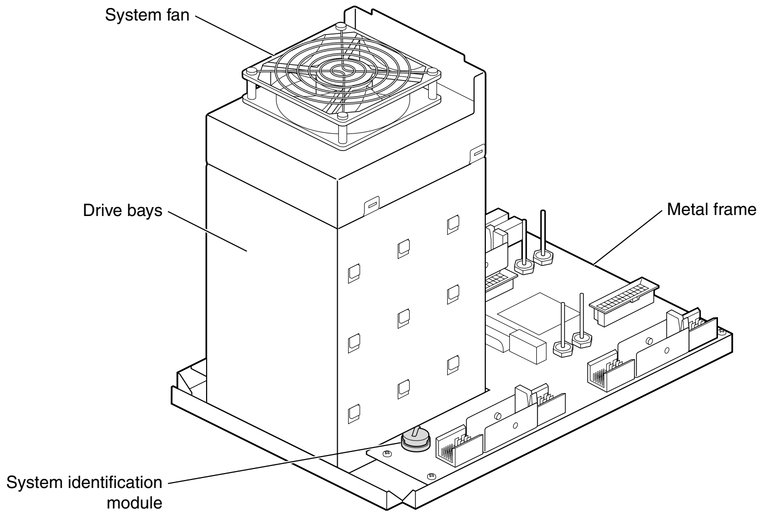


Figure 8-33 Octane Frontplane Module

Now you can replace the System ID module or the Fan. The **System ID Module** holds the pre-programmed Ethernet address for the Octane computer. It is a small circular disk held by a metal retaining clip next to the drive bay.

Power Supply Fan

13.) Use a flat headed screwdriver to separate the four tabs holding the fan module to the back of the drive bay.

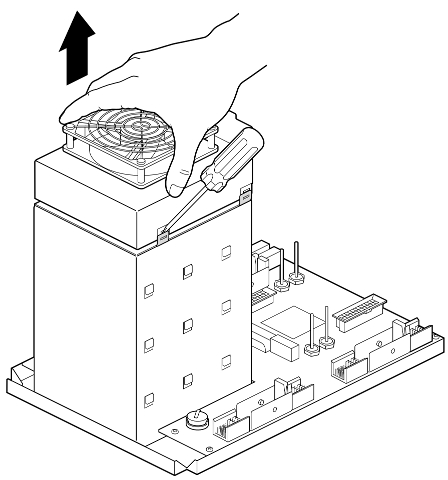


Figure 8-34 Releasing the PS Fan

14.) Disconnect the power connector under the fan

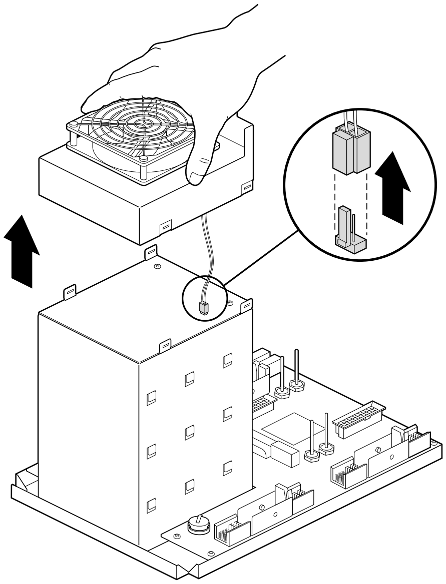


Figure 8-35 Disconnection Electrical Connections from PS Fan

System Identification Module

Remove the system identification module only when replacing the frontplane module.

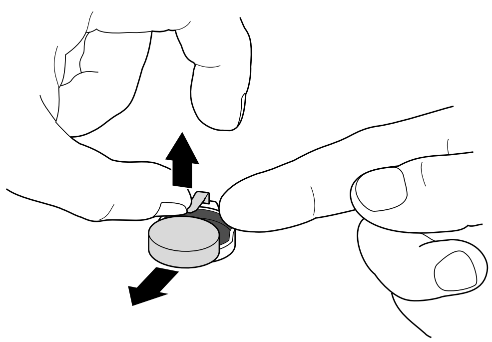


Figure 8-36 Removing the System ID Module

1.) Remove the system identification module.
   a.) Lift up on the metal retaining clip.
   b.) Slide the system identification module to the side and out.
2.) Place the system identification module on the new frontplane.
   a.) Lift up on the retaining clip.
   b.) Slide the system identification module under the clip.

2.1.5 Customer Purchased Options

2.1.5.1 Options MOD

System software options are deployed through a special MOD. If you happen to lose that MOD, please contact your OLC representative who will contact the proper CT Manufacturing personnel to procure a new options MOD. Also, note that if the **Full-House I/O board**, small board with serial, keyboard, mouse, and ethernet connectors for Indigo2, or the Octane **System ID module,** small disk inside System Module slot, is replaced, a new options MOD will be required.

2.1.5.2 2nd Host Disk Option - Installation

Refer to Section 8.30 on page for location of jumpers when the host is an Indigo2. With the system shutdown and the power off, install it behind CDROM and MOD.

If it's an addition to the Octane, simply slide it into the middle or top slot with the power off.

Re-power the system, reload software or open a shell and run this script:
/usr/g/scripts/install2Disk

Use:SERVICE, LOG VIEWING, OC HARDWARE INFO to verify host recognizes the new disk. On the original console, it will look like this:

EXAMPLE:
OC HW INFO

```
Integral Ethernet: ec0, version 1
Integral SCSI controller 1: Version WD33C93B, revision D
CDROM: unit 6 on SCSI controller 1
Optical disk: unit 5 on SCSI controller 1
Optical disk: unit 3 on SCSI controller 1
* Disk drive: unit 2 on SCSI controller 1
Disk drive: unit 1 on SCSI controller 1
Integral SCSI controller 0: Version WD33C93B, revision D
Graphics board: Solid Impact
Graphics board: High Impact
```

2.2 CT/i (Indigo™ 2) Host Computer

NOTICE **CIRCUIT BOARDS ARE VERY STATIC SENSITIVE! Static sensitive components may be damaged if not handled in a static free environment. Take appropriate precautions. (e.g. wear properly grounded wrist strap) when handling all boards.**

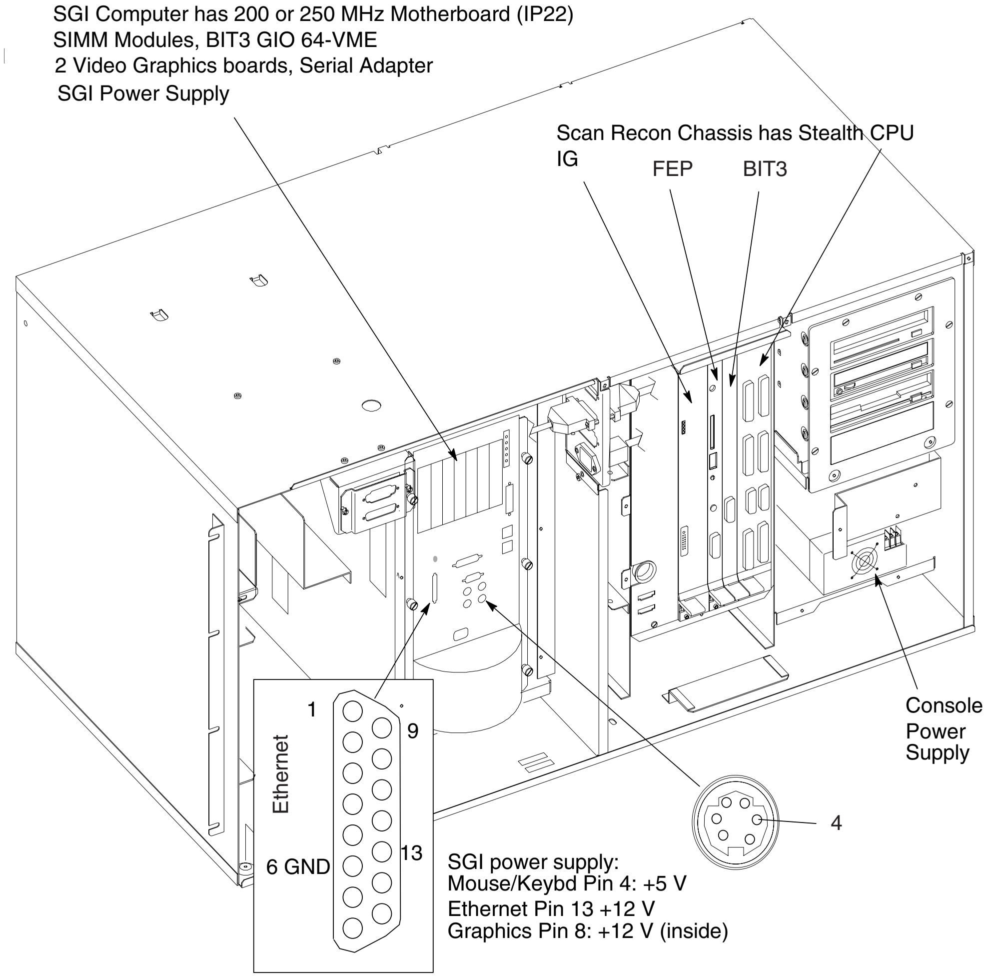


Figure 8-37 CT/i Console Assembly

2.2.1 CT/i Indigo2 Host Overview

The HiSpeed CT/i Host Computer (2115457-34) is located inside the console and has the following features:

- IP22 (motherboard: 250 MHz Processor w/2 M cache has 12 SIMM sockets (3 banks of four modules)
- Memory: 64 or 128MB comprised of four 16 or 32 SIMM modules respectively; each bank must be completely filled with modules of the same size; the first bank must be filled before the next; one bank may hold four 32 MB SIMMs and another four 16 MB modules
- DIP Op Adds four 16MB modules to the final bank
- Ethernet card: Full House I/O determines system ethernet number which gets imprinted on Option MODs
- Midplane: holds the Bit3, two graphics boards, and the Specialix; if host LED not lit and power is good, attempt to re-seat this assembly

- Serial Driver (Specialix) supports direct TTY with software flow control and modem ports with hardware flow control
- Mardi Gras 1,0 graphics supports OC monitor, and Mardi Gras 1,1 supports CT SBC display
- Bit3 provides high speed transfer of status/commands between SGI host and VMEbus of the Scan Recon Computer
- SGI Power Supply powers Midplane and fan

2.2.2 About the CT/i (Indigo2) Host

2.2.2.1 OC System Disk

ALWAYS USE THE ESD STRAP

Put the drive inside an anti-static bag or approved container before it is handled by a non-grounded person, moved from the grounded (ESD safe) area or stored.

Never handle the drive outside its anti-static bag unless the surrounding surfaces and you are grounded.

Discharge the outside of the anti-static bag first before transferring the drive.

NEVER ALLOW A DRIVE TO BE DROPPED

Just a 12 MM. or 1/2 inch drop can damage a drive.

Always place the drive top side up on a grounded, padded surface when it is unmounted.

Never place anything on top of it.

Never move the drive while it has power.

Seagate ST318418N Barracuda Drive

FE NOTE

All Jumper Setting for This Drive Should Be The Same As The Drive You Are Replacing

HANDLING PRECAUTIONS/ELECTROSTATIC DISCHARGE PROTECTION

- Disc drives are fragile. Do not drop or jar the drive and handle the drive only by the edges or frame.
- Drive electronics are extremely sensitive to static electricity. Keep the drive in its antistatic container until you are ready to install it. Wear a wrist strap and cable connected to ground. Discharge static from all items near or that will contact the drive. Never use an ohmmeter on any circuit boards.
- Turn off the power to the host system during installation.
- Always use forced-air ventilation when operating the drive.
- Use caution when troubleshooting a unit that has voltages present.
- Do not disassemble the drive; doing so voids the warranty.
- Return the entire drive for depot service if any part is defective.
- Do not apply pressure or attach labels to circuit board or drive top.

DRIVE CHARACTERISTICS

ST318418N

Formatted capacity	19.924 Gbytes
Total # of data blocks	36,088,282 (0251C800h)
Disc rotation +/-5%	7,200 RPM
Operating voltages	+5V +12V +5V +12V

INSTALLATION INSTRUCTIONS

1.) Set the SCSI ID

Determine which SCSI IDs are already being used in the system and then assign this disc drive a SCSI ID that isn't already being used. Use the J6 connector located on the front of the drive to set the SCSI ID (see Figure 8-38).

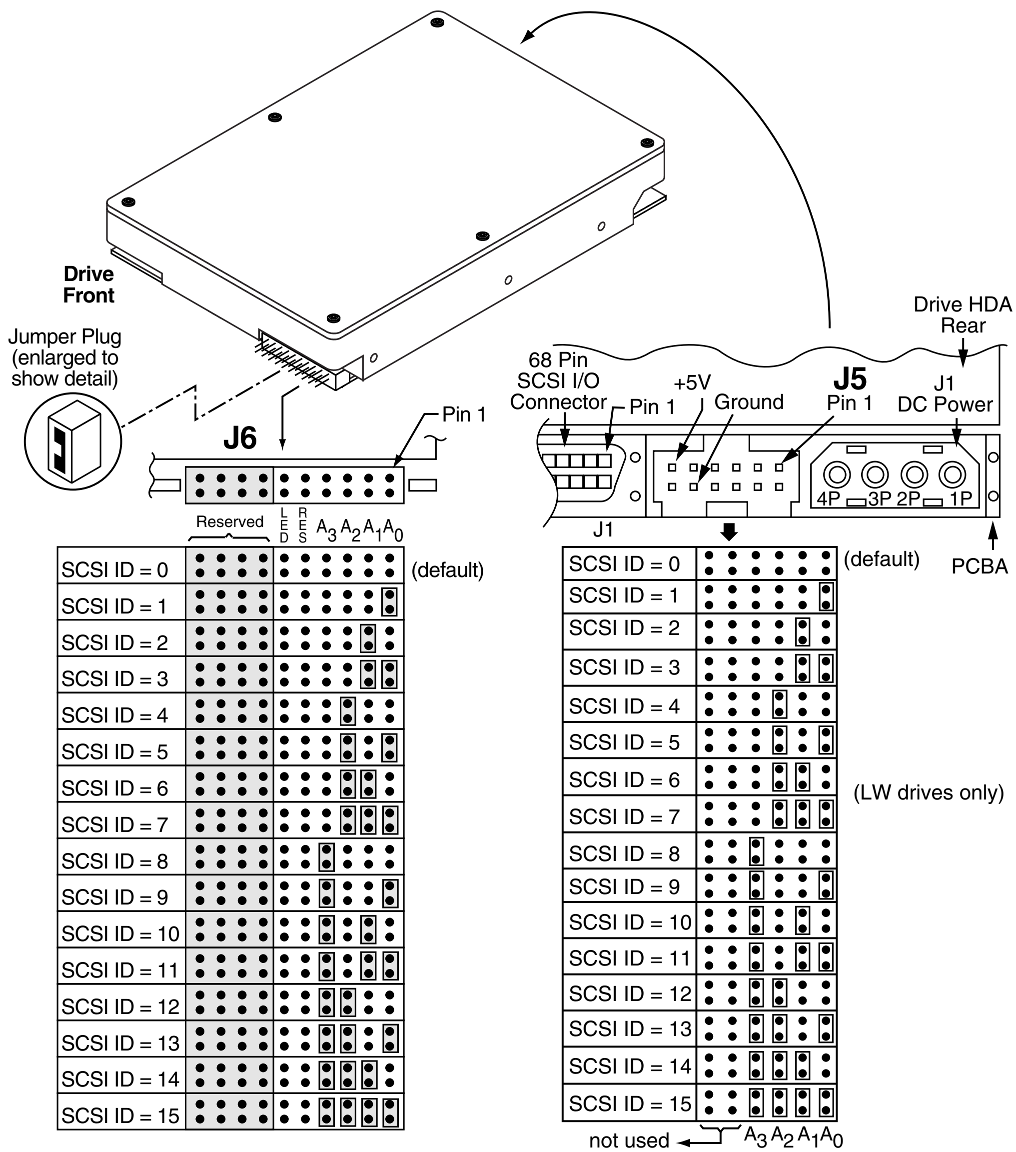


Figure 8-38 Setting the SCSI ID (ST318418N)

- Most Barracuda® 36ES2 drives are factory set with the SCSI ID set at 0. If this is the only SCSI drive in your system and there are no other SCSI devices on the bus (cable), you can leave this drive's SCSI ID set to 0 and proceed to the next step.
- The host system's SCSI controller usually uses SCSI ID 7.
- If you have an LW model drive, the ID may be set using either J6 or J5 (located on the rear of the drive).
- Some systems provide a cable designed to connect to the J5 jumper block on the drive to remotely set the ID. You can connect this cable to J5 and use the host-provided remote switch to set the SCSI ID.

2.) Configure termination

If you are installing the drive in a system that has other SCSI devices installed, terminate only the end devices on the SCSI bus (cable). N models have non-removable internal terminators that you can enable using J2 pins 15 and 16. These terminators are enabled in the default configuration.

To disable these internal terminators, simply remove the jumper from J2 pins 15 and 16. See Figure 8-39.

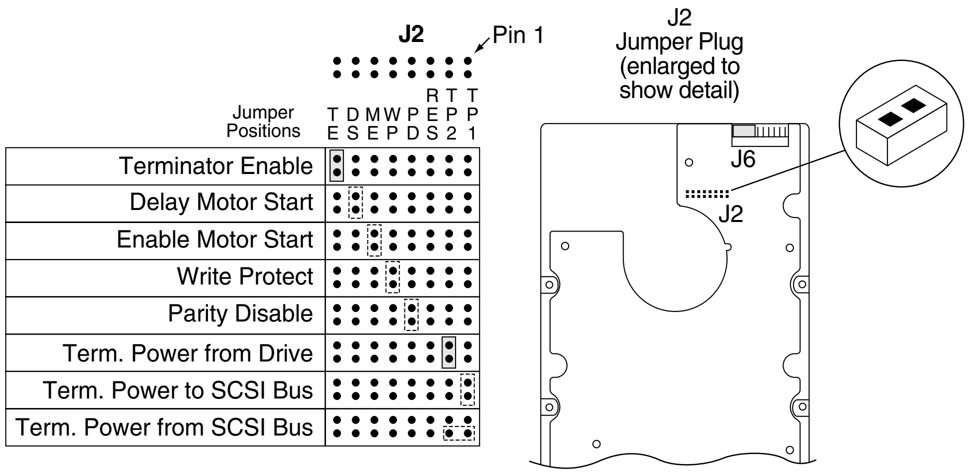


Figure 8-39 N Option Select Jumpers (ST318418N)

LW models do not have internal terminators or any other way of adding internal termination on the drive. You must provide external termination when termination is required. This is normally done by adding an inline terminator on the end of the cable.

- Use active (ANSI SCSI-2 Alternative 2) single-ended terminators when terminating a bus operating in single-ended mode.
- Use SPI-2-compliant active low voltage differential terminators when terminating a SCSI Ultra2 bus operating in LVD mode.
- The host adapter is normally on the other end of the bus and internally terminated. You can configure your bus with another device on the other end if you remove termination from the host adapter.

3.) Configure terminator power

Terminators have to get power from some source. The default configuration results in the drive not supplying termination power to the bus. You should normally leave this drive set at this default unless your host system requires the drive to supply termination power to the bus. To configure this drive to supply termination power to the bus, place a jumper on J2 pins 1 and 2 as shown in Figure 8-40.

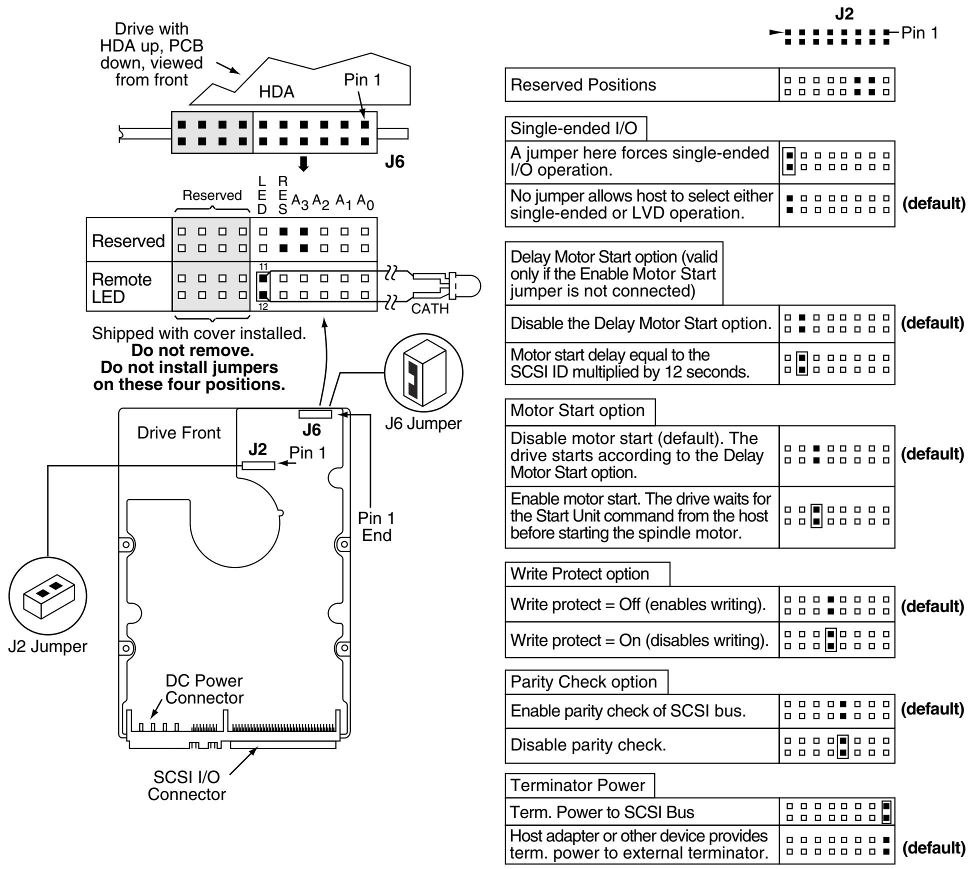


Figure 8-40 LW Option Select Jumpers (ST318418N)

ST39216N (2226715-2) OC Software - 9.2GB

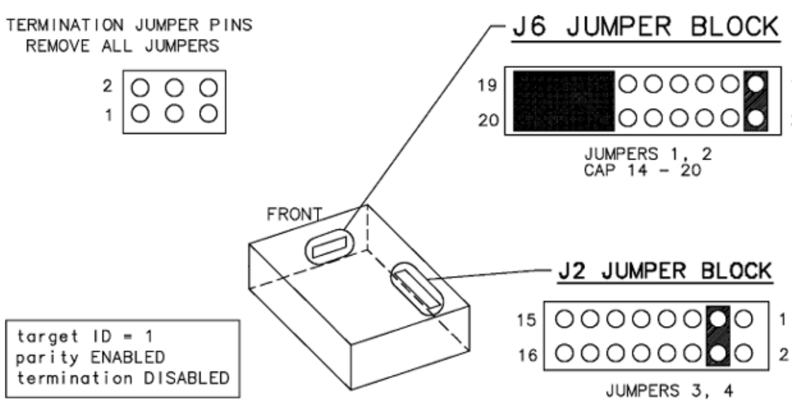


Figure 8-41 ST39216N, 2226715-2 OC (9.2GB) System Disk Jumpers

ST34371N (2180997) OC Software - 4GB

The ST34371N is an OC Computer System Disk for CT/i forward systems, starting with the 3.6 Software release. The ST34371N can be used as a replacement part disk for CT/i Systems running revision 3.6 or later software.

The ST34371N Disk is not compatible with CT/i Software Revisions prior to the 3.6 Release.

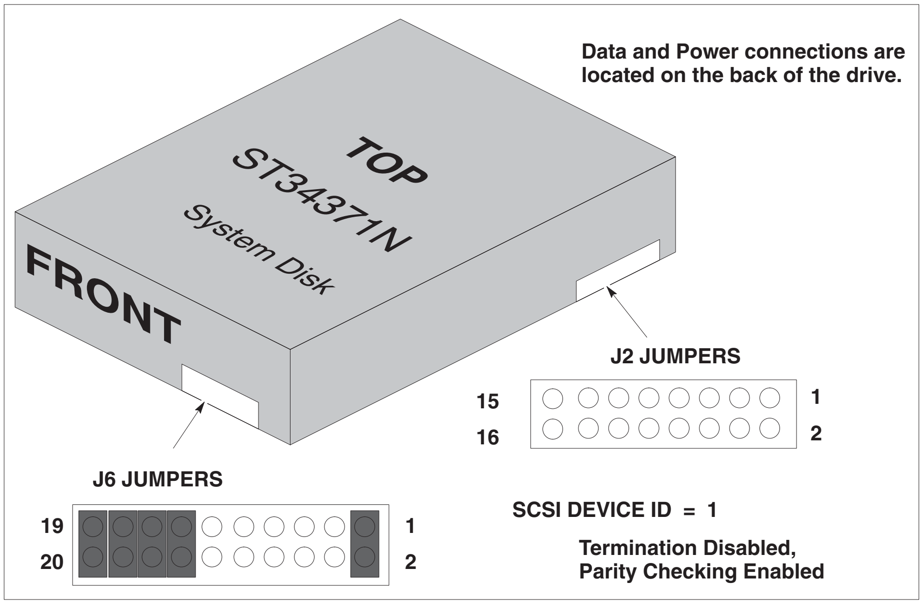


Figure 8-42 ST34371N/2180997 OC (4GB) System Disk Switches

ST15150N (2143386) OC Software - 4GB

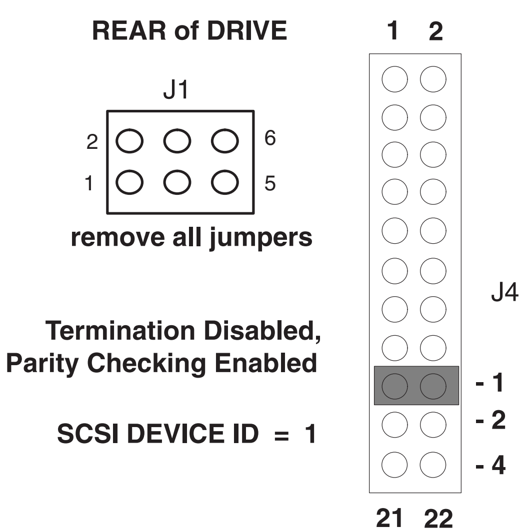


Figure 8-43 ST15150/2143386 System Disk Switches

2.2.2.2 MaxOptics Drive

The MaxOptics drive must be configured with a SCSI ID of 3. The jumpers must be ON for positions 10, 2, 1. Refer to page 233 for more information about the SGI controlled SCSI devices.

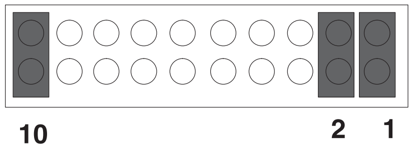


Figure 8-44 MaxOptics Drive Jumpers

MaxOptix MOD Jumper and Switch Settings and Compatibility

COMPATIBILITY NOTE

The MaxOptix T4-1300 drive is not compatible with the T5-2600 drive. If the system has the T4-1300 drive, is should be replaced with a T4-1300 drive, not upgraded to a T5-2600 drive. Do NOT upgrade systems with a T5-2600 drive if the system has a T4-1300 drive. T4-1300 drives are not repairable items. They are available from parts depots.

2140444-2 DESCRIPTION

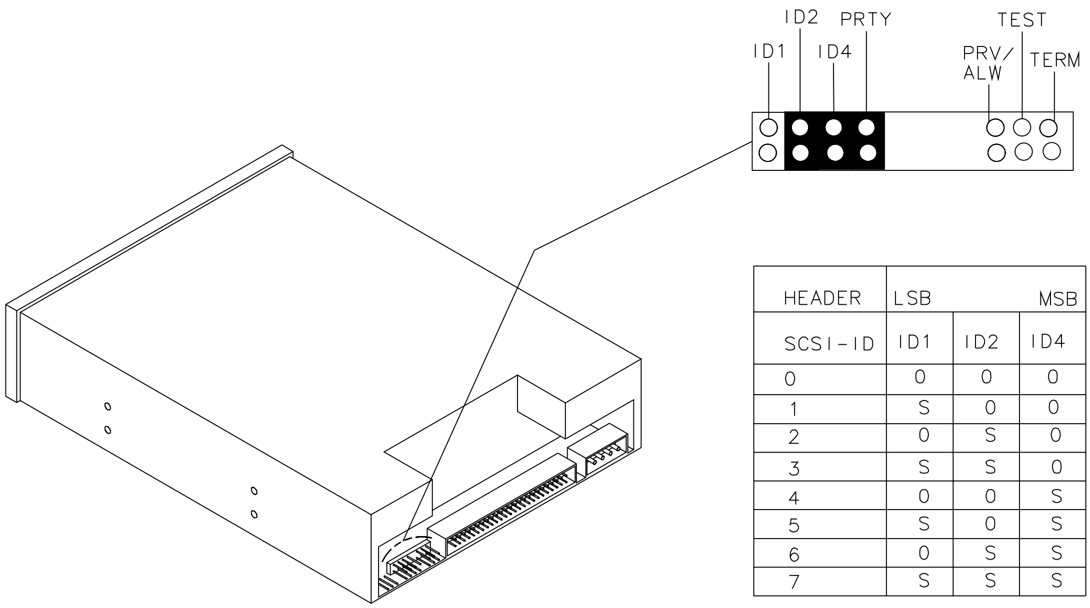
This is the T4-1300 MaxOptix drive. It is available in parts depots. It is a repairable drive. It is used on systems with an Indigo 2 host. Do not replace this drive with a T5-2600 drive (2206624-5). It requires different firmware and software version.

2206624-2 DESCRIPTION

This is an early MaxOptix drive that uses version 6.0 firmware. It is used only on Indigo 2 systems, not systems with the Octane Host. This drive can only be used on systems with 4.0 software and thee 4.0.1 patch. The patch part number is 2208397.

2206624-5 DESCRIPTION

This is the T5-2600 MaxOptix drive with 6.5 firmware. It is compatible with Octane host systems. Do NOT upgrade systems with a T4-1300 drive to the T5-2600 drive. It uses different firmware and software versions. This drive is used on systems with 4.1 software or systems with Octane hosts. Even if the correct software version is available on the system, attempt to replace a T4-1300 drive with another T4-1300 drive.



HEADER	LSB		MSB
SCSI-ID	ID1	ID2	ID4
0	0	0	0
1	S	0	0
2	0	S	0
3	S	S	0
4	0	0	S
5	S	0	S
6	0	S	S
7	S	S	S

0 = OPEN S = SHORT

Figure 8-45 CD-ROM Jumper View

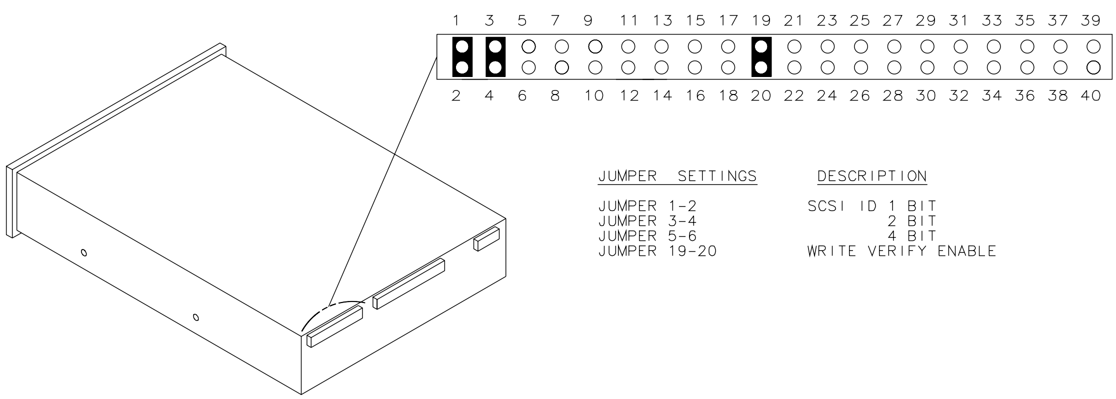


Figure 8-46 MOD Jumper View

2.2.2.3 Bit3 GIO64 Interface Board, 2124215-2

For diagnostic information, See "Diagnosing BIT3 Subsystem on CT/i 3.X/4.X (INDIGO2)" on page 442.

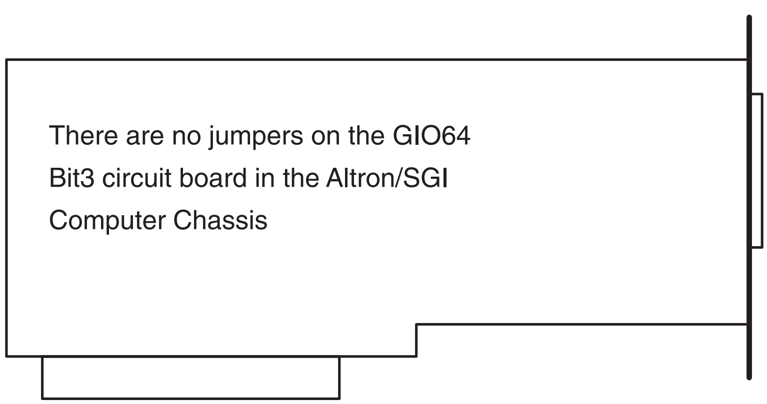


Figure 8-47 Bit3 CT/i GIO64 Board Jumpers

2.2.2.4 SIMM Memory

The Indigo host uses SIMM memory. SIMM stands for Single In-line Memory Module. Signal and power pins on SIMMs lie along both sides of the module and are connected to the same internal memory chip. Because of this, SIMMs have a 64-bit data path.

2.2.2.5 Hard Disk Controller

The SGI host controls the host hard drive which has the IRIX operating system and the GE applications software. It also controls the Magneto Optical Drive, the CDROM and the SCSI interface to the DASM. The SBC controls the two hard drives for the Scan Recon Unit. One drive hold SBC software and the other holds the scan files, cal files and DD files.

2.2.2.6 CT/i (Indigo2) Ethernet Number Location

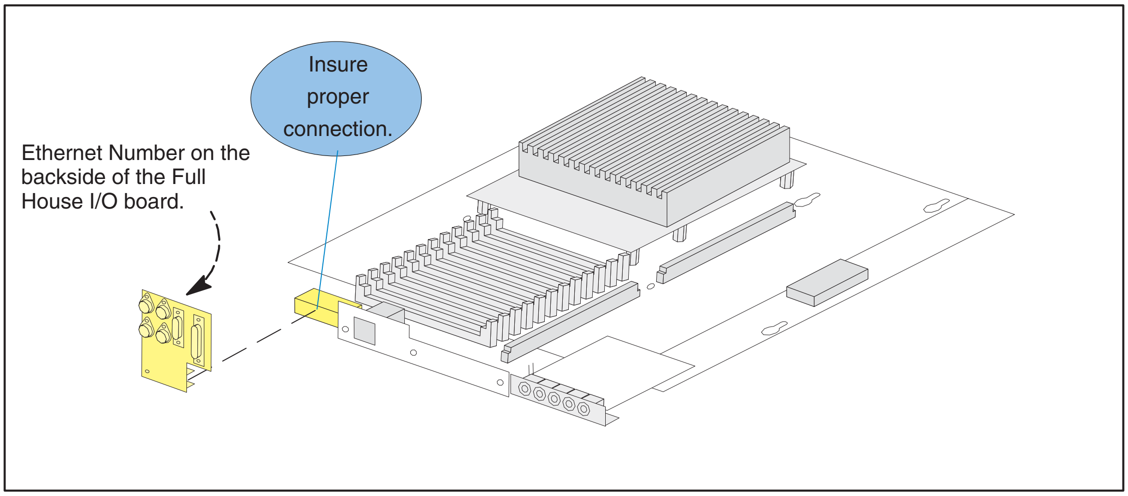


Figure 8-48 SGI Ethernet Number

2.2.3 Diagnosing (Indigo2) Host Computer Hardware Problems

2.2.3.1 Indigo - Minimum Boot Configuration

The minimum Irix bootup configuration for the SGI in the carrier box is:

- the SGI power supply
- the SGI Indigo2 GEMS-IP22 motherboard
- at least 1 full bank of memory SIMMs (minimum 32MB)
- the midplane assembly
- at least 1 MG board and CRT
- the SGI system disk or CDROM with LFC media
- a keyboard - The keyboard is not required for a successful pass of the “power-on” LED self-test.

Note: All SCSI peripherals are powered by the console power supply so it is also needed to use any SGI SCSI peripheral.

For Irix bootup, you can remove the Specialx board, one of MG boards, the BIT3, and all but one full bank of memory SIMMs. The MOD, PMOD, DASM, optional disk, are not needed by Irix. If booting the system disk, the CDROM is not required. If booting from CDROM, the system disk is not needed. If only running the PROM monitor, no peripherals are needed.

2.2.3.2 Checking "Power-On" Self-test Results

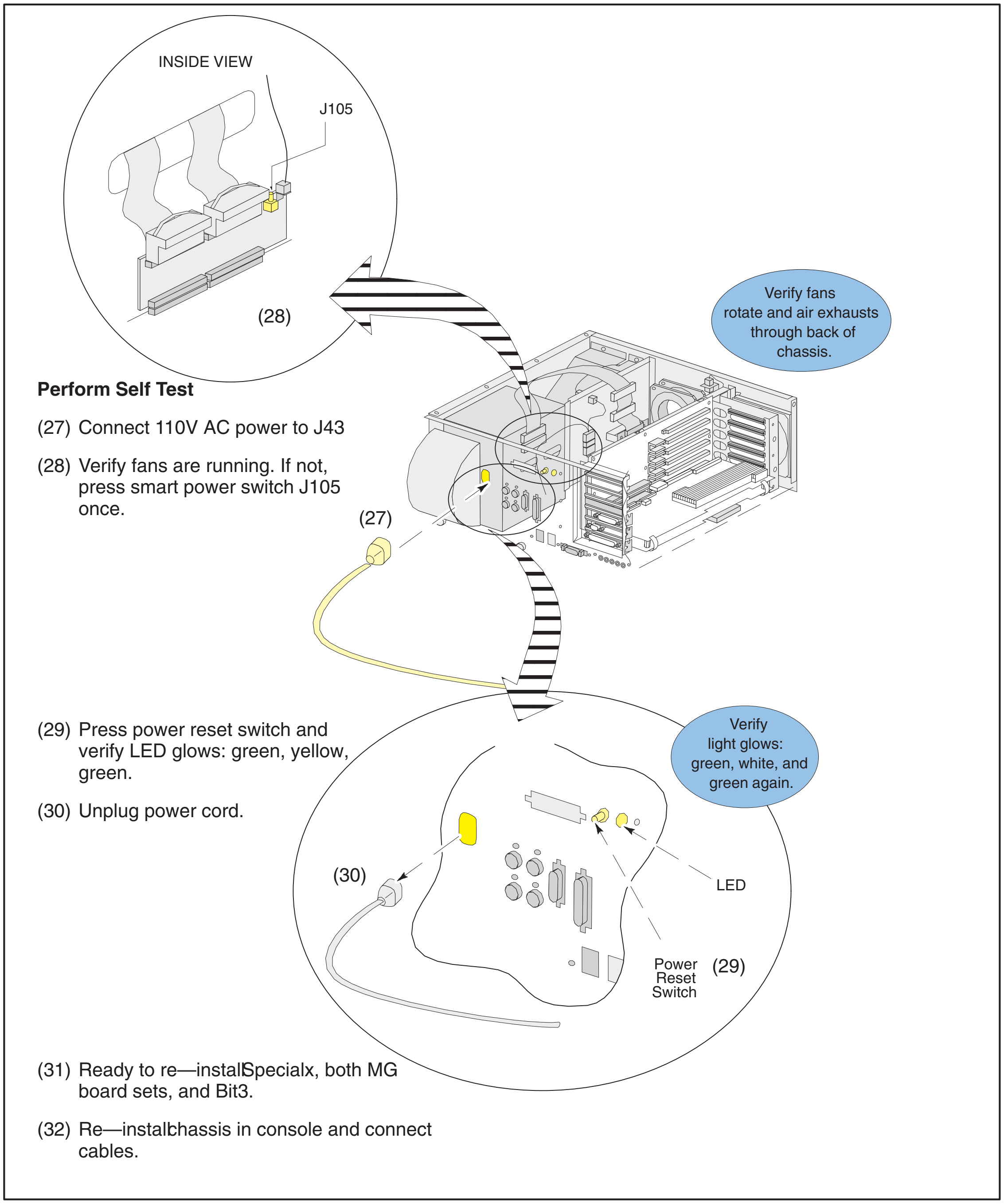


Figure 8-49 SGI IP22 - Minimum Configuration for Power-on Self Test

2.2.3.3 Understanding "Power On" Tests

Power on tests run automatically whenever the host computer is reset. They test the motherboard, the memory modules, and graphics boards. Fault notification is done through LED codes and Error Messages that are placed in the OC `/var/adm/SYSLOG` message file.

When you apply power, the host computer goes through the following steps:

1.) The LED on the front of the SGI computer blinks green, then turns yellow, then the host starts its power-on diagnostics which take about five seconds.

2.) When power-on diagnostics are done, the disk drives spin up. This takes about 15 seconds for each hard disk.

3.) When the host is ready, its LED turns green.

4.) The CPU then boots IRIX. Then application software is started unless you selected **Stop For Maintenance** before it disappeared.

5.) If you press the ESC key after the message about bringing up the system, you should be able to watch most of the bootup messages. They disappear when the monitor systems are being initialized and synchronized.

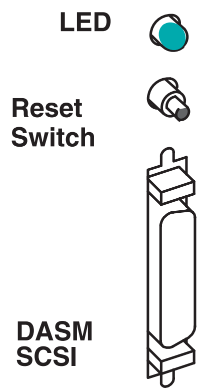


Figure 8-50 SGI CPU LED, Reset Switch and DASM Connector Locations

Where the bootup stops is a clue to where a problem may lie.

If the monitor stays blank, attach a terminal at 9600 baud to top port under the keyboard table. The messages that the host reports as it boots may help you diagnose the problem.

Below is a list of host symptoms, the states they indicate and possible solutions to fix a host problem.

Symptom	System State	Cause
Solid Green LED	NORMAL. System Menus Displayed. Boot successful.	Diagnostics Successful
No LED	No Power to unit	AC Power Cord Out. Power supply failure. Check for fan operation with a tie wrap.
Solid YELLOW LED	HW problem	CPU Board or Graphics board failure
Flashing YELLOW LED	Memory Diagnostic Failure	SIMM Failure. Check Error Messages for faulty SIMM.
Flashing YELLOW LED	Hardware Error Message	Check Error Message and follow suggestions in message(s).
No Boot Tune	No Audio	Cables not plugged into SGI Chassis correctly, faulty speaker, faulty connection to keyboard assembly, cable problem at intercom/ interconnect board. Faulty CPU board. Volume set too low.

Table 8-7 POWER-ON TEST RESULTS, PROBABLE CAUSES

Symptom	System State	Cause
No LED on OC system disk Release bulkhead, pull tray, bottom disk	No system boot.	Disk drive SCSI cable not seated correctly, disk drive power cable not connected, defective VME chassis (console) power supply. Check other drives in User Drive Assembly for LEDs on. See page 420
Diagnostic error messages, no desktops displayed.	Diagnostic failure, no graphics.	CPU failure. Graphics Board Failure. Monitor Failure. See below.

Table 8-7 POWER-ON TEST RESULTS, PROBABLE CAUSES (Continued)

No Visible Light From the LED with No Error Message

During power-on, if the system LED doesn't light up and no error message appears on the screen, follow these steps:

1.) Disconnect the power cable from the back of the system.

2.) Remove the top cover from the system and make sure the midplane is firmly seated by pressing down on top of it. Practice ESD prevention.

    The midplane is the circuit board assembly that the graphics, Bit3 and Specialx boards plug into. It's okay to press quite firmly on the top of the midplane.

3.) Check to make sure the internal power connectors are attached correctly to the system board.

4.) Replace the top cover and reattach the power cable to the front of the chassis.

5.) Check all cables.

    Make sure the cables are securely connected and the power cable is plugged into an outlet that works.

6.) Adjust the brightness control knob on the monitor to test for adequate light on the screen.

7.) Make sure you've pressed the power switch to turn on the system.

If all of the cables are connected and the symptoms remain, you may have a faulty power supply.

Blinking Yellow LED with No Error Message

If the LED keeps blinking and no error message appears on the screen, one or more memory SIMMs may be faulty. Check that all SIMMs are seated all the way into the sockets. Then power on the system again.

If the LED still blinks and the screen is blank, remove all but one set of four SIMMs. Then power on the system again.

If the LED still blinks and the screen is blank, swap the removed SIMMs with the SIMMs still in the chassis. Then power on the system again.

If the LED still blinks and the screen is blank, replace the IP22 motherboard.

Blinking Yellow LED with an Error Message

If the LED stays blinking and a message appears on the screen, one or more pieces of hardware may be faulty. Below is a list of messages and where to go from there:

- If you see this message:

  `No usable memory found. Make sure you have a full bank (4SIMMs)`

1.) Check that all of your SIMMs are seated all the way into the sockets.

2.) Power the system on again. If you get the same message, the SIMMs are faulty. Try swapping the two banks of SIMMs and see if the symptoms change.

- If you see this message:

  `Check or replace: SIMM#`

  1.) Make sure the indicated SIMM is seated all the way into its socket. The # represents the SIMM number that failed the test. A SIMM is installed correctly when it is vertical and perpendicular to the CPU baseboard, and when the latches on the sides of the SIMM fit snugly around it. If the SIMM appears to be leaning, push it into a vertical position. Refer to page 464.

  2.) Power on the system again. If you get the same message, the SIMM is faulty.

- If you see this message:

  `Memory is not usable. Check or replace all SIMMs.`

  1.) Check that all of your SIMMs are seated all the way into the sockets and that they are installed in the correct slots.

  2.) Power on the system again. Try removing all but one bank of SIMMs.

  3.) Power on the system again. Try swapping the removed SIMMs with those still in the motherboard.

  4.) Power on the system again. If you get the error message, then all SIMMs are faulty or the IP22 mother board is faulty.

Solid Yellow LED with No Error Message

If the YELLOW LED stays on without blinking or turning green, the memory is working but some other part is faulty. If you heard a dissonant piano chord sound from the speaker, the graphics board has failed the test.

First, check to make sure the graphics board is installed properly.

1.) Remove the graphics board and then reinstall it.

2.) Power on the system. A message should appear on the monitor.

If no message is displayed on the monitor, the graphics board may actually be defective. To be sure:

1.) Make sure the monitor is plugged in and turned on and the brightness control knob on the monitor is turned on to provide adequate light to the screen.

2.) Check to make sure that the system disk cable is seated at the SGI chassis and at the disk drive.

3.) Try powering off and on again.

If an error message is still not displayed, the graphics board is most likely defective.

Solid YELLOW LED with an Error Message

If the LED stays YELLOW and a message appears on the screen, an SGI part is faulty.

- If you see this message:

  `Check or replace: Graphics board`

  The host graphics board has failed. First, make sure the board is seated properly.

  1.) Remove the graphics board and reinstall it.

  2.) Power on the system. If you get the same error message, the graphics board is faulty.

  3.) Swap the graphics boards.

4.) Power on the system. If you get the same error message, the graphics board is faulty.

5.) Replace the graphics board.

- If you see this message:

  `Check or replace: CPU module`

  The CPU module (brick) has failed. First, make sure the module is seated properly.

  1.) Remove the CPU module and reinstall it.

  2.) Power on the system. If you get the same error message, you have a faulty CPU module.

  3.) Replace the IP22 motherboard. Refer to page 464.

- If you see this message:

  `Check or replace: CPU base board`

  1.) Re-seat all of the IP22 mother board electrical connections.

  2.) Power on the system. If you get the same error message, you have a faulty IP22 motherboard.

  3.) Replace the IP22 motherboard. Refer to page 464.

If the screen looks unusual:

- Make certain that all cables are firmly seated and the brightness setting on the monitors is set to provide adequate light to the screen.
- If the screen has lines through it, dots, dashes, mottled appearance, the graphics board is likely at fault.

Note: Due to the way that the Sony monitors are designed, two faint horizontal lines dividing the screen into thirds are normally visible.

Green LED but the Keyboard Doesn't Work

If the keys on the keyboard don't work, the keyboard is failing.

1.) Shut down your system and make sure that the keyboard cable is firmly connected to the keyboard (not the mouse) connector.

Be certain to check all of the intermediate connections at the keyboard assembly, the interconnect circuit board, RF bulkhead, and at the SGI Chassis.

Note: Some early production systems have had a problem with the RF filter that is in series with the keyboard cable at the RF bulkhead. Symptoms of this problem can look like a keyboard or mouse failure with no keystrokes or mouse activity registering on the system. Elimination of the series filter and replacement with a Ferrite Core has resolved this problem on later production systems.

Note: For troubleshooting purposes the in-line serial filter can be bypassed by unplugging the cables from the filter at the RF bulkhead and connecting one cable to the other directly. Restore the filter after troubleshooting.

2.) Power on the system again.

If the keyboard still doesn't work, the keyboard is faulty.

3.) Replace the keyboard.

Green LED but the Mouse Doesn't Work

If the mouse doesn't work, the mouse may be faulty.

1.) Reference the arrows on the bottom of the mouse, remove the mouse ball retention plate. Remove, wash and dry the ball. Remove accumulated dirt on the internal rollers. A tweezers works well.

2.) Shut down your system and make sure the mouse cable is firmly connected to the mouse (not the keyboard) connector on the SGI chassis.

3.) Power on the system again.

4.) If the mouse still doesn't work, you have a faulty mouse.

5.) Replace the mouse.

Note: The mouse used on the CT/i system is an IBM compatible "Bus" mouse. As an alternate test, the mouse will operate in Windows as a two button mouse if plugged into the field IBM laptop mouse port.

2.2.3.4 Checking "OS Boot" System Parameters

SGI IP22 MOTHERBOARD REPLACEMENT (SEE FIGURE 8-49)

From the SGI Command Monitor (firmware) prompt, use **`printenv`** to check for the following parameters:

- `SystemPartition=scsi(1)disk(1)rdisk(0)partition(8)`
- `OSLoadPartition=scsi(1)disk(1)rdisk(0)partition(0)`
- `OSLoader=sash`
- `OSLoadFilename=/unix`

Note: Only if any of these are NOT set, then set it using the **`setenv`** command from the Command Monitor (firmware) prompt (>) EXACTLY AS SHOWN:

```
>   setenv SystemPartition scsi(1)disk(1)rdisk(0)partition(8)
>   setenv OSLoadPartition scsi(1)disk(1)rdisk(0)partition(0)
>   setenv OSLoader sash
>   setenv OSLoadFilename /unix
```

The parameters will be stored in non-volatile memory "forever" or until the IP22 is swapped out.

2.2.3.5 Peripheral Confidence Tests

Use the SGI Confidence Tests to test the host to host peripheral interface and host peripheral ability to function. Use these to test or adjust the parameters for the:

- **keyboard** (alpha-numeric keys only),
- **CDROM** (place a CD inside first),
- **monitor** (use to adjust convergence) or
- **mouse**

These tests are run from the operating system level. In a shell, enter: **`confidence`**

OR

on the Service Menu, with Service key in console port, select:

TROUBLE SHOOT > SYSTEM > DISPLAY CRT > CONFIDENCE TESTS

2.2.3.6 CT/i (Indigo2) Hardware Inventory (hinv)

```
Iris Audio Processor: version A2 revision 1.1.0
1 250 MHZ IP22 Processor
FPU: MIPS R4000 Floating Point Coprocessor Revision: 0.0
CPU: MIPS R4400 Processor Chip Revision: 6.0
On-board serial ports: 2
On-board bi-directional parallel port
```

```
Data cache size: 16 Kbytes
Instruction cache size: 16 Kbytes
Secondary unified instruction/data cache size: 2 Mbytes on Processor 0
Main memory size: 192 Mbytes
EISA bus: adapter 0
Integral Ethernet: et0, IO0
Integral Ethernet: ec0, version 1
Integral SCSI controller 1: Version WD33C93B, revision D
  CDROM: unit 6 on SCSI controller 1
  Optical disk: unit 5 on SCSI controller 1
  Optical disk: unit 3 on SCSI controller 1
  Disk drive: unit 1 on SCSI controller 1
Integral SCSI controller 0: Version WD33C93B, revision D
Graphics board: Solid Impact
Graphics board: High Impact
```

2.2.3.7 CT/i (Indigo2) Host Power-Up Diagnostics

After power to the SGI computer is turned on, the LED on the front of the SGI chassis will turn on. While the motherboard is running the power-up self-test, that LED will be Yellow.

If it finds a fault, it will enter an entry in the SYSLOG if it can. Knowing where it halted in the boot up can also prove to be a clue to the problem.

When the LED first lights on power on, the monitor will display:

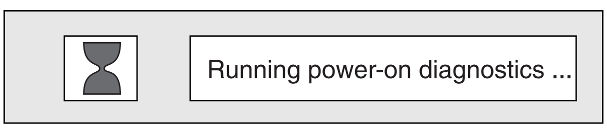


Figure 8-51 Power-on Diagnostics Notification

After all of the power-on tests pass, the SGI LED will turn Green and the “Starting Up System” pop-up window will appear on the monitor.

This is you opportunity to access SGI diagnostics and its host command line. Press the **ESC** key or click on the Stop for Maintenance box if you want to access the Indigo software.

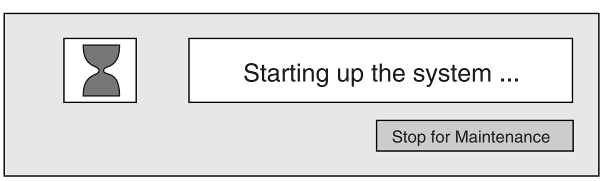


Figure 8-52 Starting up the system notifier

If you don't interrupt, after a few seconds the System Is Coming Up pop-up will appear.

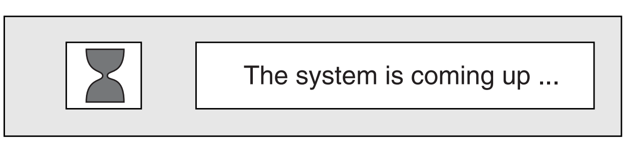


Figure 8-53 The System is coming up notifier

2.2.3.8 Interactive Diagnostic Environment (IDE) Tests

IDE offers more in depth tests of the Indigo hardware. Fault reporting is done through error messages.

When you power on the system, the Indigo computer preforms power-on tests to check the most basic of hardware functionality. IDE tests designed to provide a more comprehensive set of tests that provide greater and more intense hardware testing. IDE tests take approximately 100 minutes to complete. If allowed to run to completion the default set of tests may take several hours due in large part to the amount of memory that is installed on the motherboard. The host stops the tests to report any failures on the screen.

The SGI IP22 CPU also runs a PROM based self-test on power-up or reset. Any errors about the host hardware appear in `/usr/adm/SYSLOG`.

Running IDE Tests:

1.) Shut down the system

After a few seconds, the screen clears and when you see the notifier shown in Figure 8-14, select the RESTART button.

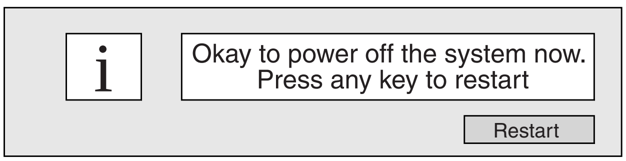


Figure 8-54 Okay to power off notifier

Note: If the system is malfunctioning and you cannot communicate with it using the mouse or keyboard, then press the Reset Switch on the front of the SGI (Altron) Chassis.

2.) When you see the notifier shown below, quickly select the STOP FOR MAINTENANCE button or press **ESC**. You only have three to five seconds.

If you cannot reach this screen, either the IP22 board or graphics card may be faulty.

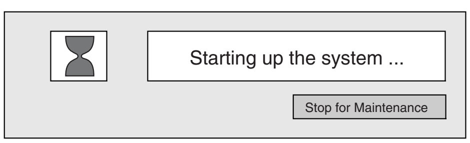


Figure 8-55 Starting up the system notifier

The Host Maintenance menu appears.

```
Start System
Install System Software
Run Diagnostics
Recover System
Enter Command Monitor
Select Keyboard Layout
```

3.) Start diagnostics by either selecting RUN DIAGNOSTICS or ENTER COMMAND MONITOR. The former is automatic; the latter can be more selective and informative.

Once you start SGI IDE, you should see a message similar to:

```
SGI Version 6.2 IP22 IDE field
```

If you do not see a message like this, you cannot run the IDE diagnostics. You may have a faulty OC disk.

4.) If you selected ENTER COMMAND MONITOR, at the >> prompt, enter: **`ide fe`**

Since IDE boots from the system disk or CDROM, the *following items must work well enough to run IDE test:*

- SGI power supply
- IP22 motherboard
- memory(1 full bank)
- midplane
- MG board (1)
- System disk or CDROM.

This will run a verbose version of the automatic SGI diags.

To exit ide, press: **ESC**

To interrupt ide: **CTL+C**]

To test just the motherboard SCSI interface, enter: **`scsi`**

To test the memory modules, enter: **`memtest`**

To test the motherboard audio, enter: **`audio`**

To test the motherboard FPU, enter: **`fpu`**

For help while in ide, press: **h**

If you selected RUN DIAGNOSTICS, the character on the last line (next to the cursor) shows a spinning combination of slashes and dashes while the tests are running. Also, the LED on the front of the SGI Chassis will blink slowly throughout the testing process.

Included in the automatic diagnostics is a series of graphics tests. During these tests, the screen my blank and display various patterns or images.

5.) Watch for messages.

If the diagnostics find a problem, you will see a message similar to:

`ERROR: Failure detected on the CPU module`

or a message indicating a failure with other SGI parts.

CAUTION **Practice good ESD prevention.**

Replace or swap the indicated parts to resolve or verify failures.

Note: If this error occurs in the first few minutes of operation it likely indicates a hardware problem exists.

6.) To stop the IDE tests, press the **ESC** key.

This will halt execution of the test and return you to the IDE command monitor prompt. A list of available commands may be viewed by typing **`?`** or **`help`**.

Known Indigo IDE Bug (Cache Parity Error)

Generally, if IDE test run without error for at least 15 minutes, the IP22 motherboard and basic computer functionality are good. Although **a known bug in the IDE diagnostic can cause a `Cache Parity Error` to be reported.** This usually occurs after an hour or more of execution.

Example: `ECC/PARITY ERROR ON THE SYSAD BUS DURING DATA REFERENCE, SECONDARY CACHE DATA-FILED ERROR`

`NESTED EXCEPTION #1 at ERROR EPC:88521640; FIRST EXCEPTION AT PC:884cd820`

This error appears to be the result of a diagnostic bug and does not appear to indicate any true hardware problem under these conditions.

Because this ERROR can lock up the system, **RESET** may be required to exit and reboot the system.

2.2.3.9 CT/i Won't Boot Correctly After Power Cycle

System will not Boot after a Power Cycle. Pressing the Reset Button on the Indigo computer allows the computer to continue boot cycle with errors. After software start–up system operation appears normal until next power cycle on the Indigo computer.

The following error occurs after IRIX has started to boot up, approximately 2 seconds after the "Stop for maintenance" icon disappears.

```
Panic: Timeout table overflow.
Tune ncallout, callout_himark and
reserve_ncallout to higher values.
```

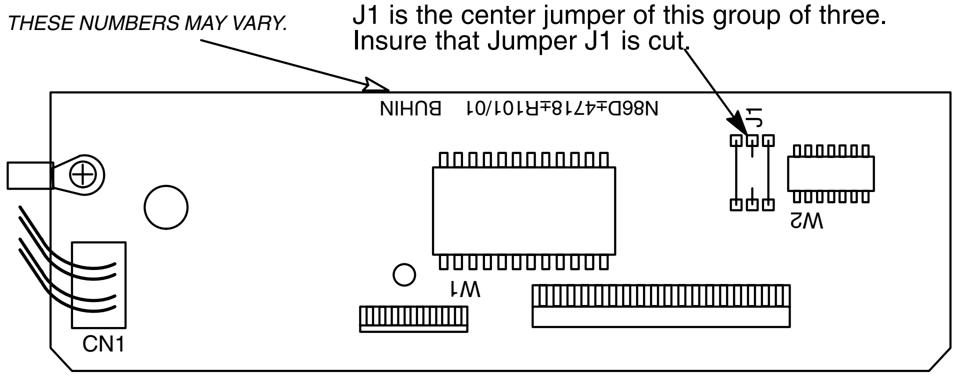
PROBLEM - Wrong PS2 Compatibility Mode

The Fujitsu Keyboard used in the Keyboard Assemblies (English – 2114561–2, French –2114561–27, German – 2114561–30, Scandinavian – 21145621–33) is operating in the wrong 'PS2' compatibility mode. Normally during the power start–up process the SGI computer will identify and configure various hardware components that it is connected to, including the keyboard. Pressing the Reset Button on the SGI computer will force the computer to attempt to boot–up with a default keyboard protocol if possible. This is why the computer can start in most cases even though the keyboard is set to use an incorrect operation mode.

SOLUTION - Cut Jumper

Cut jumper 'J1' on the Fujitsu Keyboard inside the Keyboard Assembly as described here.

1.) Stop system software by selecting the 'Pink' Shutdown button on the Display Monitor.
2.) The system will post a message on the Display Monitor when it is safe to turn off the computer power.
3.) Turn Off the Console Power Switch.
4.) Use a Static Wrist Strap and Observe Static Precautions.
5.) Disconnect the Keyboard Cable at the Keyboard Assembly.
6.) Turn the Keyboard Assembly over and remove 13 (thirteen) screws and save.
7.) Holding the top and bottom halves of the Keyboard Assembly together, turn the Assembly back over.
8.) Lift the top cover of the Keyboard Assembly off of the Bottom half and rotate out of the way. Cables from the Hard keys and Speaker will connect the Top and Bottom Halves.
9.) Remove 4 (four) screws and washers that hold the Fujitsu Keyboard to the bottom half of the Keyboard assembly and save.
10.) Flip the Fujitsu Keyboard over, while insuring that the cables or components are not damaged.



Circuit Board on the bottom of the Fujitsu Keyboard Assembly

Figure 8-56 Fujitsu Keyboard Assembly Circuit Board

11.) Check jumper J1. Refer to Figure 8-56. If J1 is not cut use a pair of wire cutters and cut Jumper labeled 'J1' on the circuit board attached to the bottom of the Fujitsu Keyboard.

NOTICE **Be careful not to leave any loose hardware or pieces of the jumper inside the keyboard assembly.**

12.) Reverse steps 10. through 5.

13.) Turn on Console Power Switch and let the system auto–start.

14.) The 'Panic' error message below, should NOT appear:

```
Panic: Timeout table overflow.
Tune ncallout, callout_himark and
reserve_ncallout to higher values.
```

15.) After the system completes start–up, check keyboard, hard key, intercom and auto–voice functions.

2.2.3.10 Full House I/O Ethernet Number

Note: Re-programming the Ethernet number in the FullHouse I/O board is potentially very dangerous to network sanity. Each and every Ethernet number should be unique as assigned by the manufacturer of the device.

We have experienced duplicate ethernet numbers and this is VERY difficult to trace/troubleshoot the resulting symptoms (unlike the duplicate IP# case, no machines warn/notify of duplicate ethernet numbers).

To burn/re-burn the SGI host ethernet number into the FH-I/O serial EEPROM, use the following command from the "Command Monitor" firmware prompt:

`setenv -f eaddr` ***XXXXXXXX***

(where ***X*** is a UNIQUE and REGISTERED ethernet number as issued by SGI, the manufacturer and holder of the registered ethernet numbers)

The ethernet number, and other parameters, can then be viewed using the **`printenv`** command from the firmware prompt or the command **`nvram`** from the Irix shell prompt.

The only reason anyone would ever have to use this capability would be if GPO sent a spare FH-I/O board that was received unprogrammed from the SGI spares group (this could happen).

2.2.3.11 Indigo2 (SIMM) Memory Errors

SIMM Memory

The IP22 motherboard memory must be configured correctly for it to function properly. There are 12 SIMM sockets across three banks on the IP22 motherboard. A bank must be completely filled with identical modules to work correctly or be empty. The systems is shipped with 192 *MB of CPU memory*.

SIMM sockets 12, 11, 10, and 9 each contain a 32MB SIMM (making up 128MB) and SIMM sockets 8, 7, 6, and 5 each contain a 16MB SIMM (making up 64MB).

The SIMM socket closest to the CPU "brick" is SIMM #12 and the SIMM socket farthest from the CPU "brick" is SIMM #1. SIMM sockets are arranged to accommodate up to 3 banks of 4 SIMMs each.

The DIP (Deterministic Image Performance) option adds another 4 16MB SIMMs into SIMM sockets 4, 3, 2, and 1 resulting in total system CPU main memory of 256MB and fully populating all available SIMM sockets in the SGI IP22 motherboard.

Types of SIMM Memory Faults

NOTICE **Remember that these components are expensive and very susceptible to electrostatic damage. Practice ESD prevention.**

SOFT ERRORS (OPERATING SYSTEM STILL BOOTS)

SIMM errors appear in the OC error log. Soft (recoverable) memory errors are logged to the SGI system error log (`/var/adm/SYSLOG`). An occasional SIMM error is normal. These are tolerated but may result in hard errors eventually (system `PANIC`). The offending module is identified by its socket location which is silk screened on the motherboard. A high soft error frequency should be pro-actively replaced.

To view just the critical host errors, open a shell and type: **`sysmon`**

To view all entries by IRIX type: **`/var/adm/SYSLOG`** for today's entries

or

`/var/adm/SYSLOG.0` for yesterday's.

Enter: **`ezlog`** to access several logs for the entire scanner.

To do a more complete test, interrupt **ESC** bootup, Enter Command Monitor and type:

`ide memtest`

Before replacing a memory module, check that the SIMMs are seated correctly in their slots. Memory is installed correctly when it is vertical and perpendicular to the motherboard, the latches on the sides fit snugly around it. If the memory module appears to be leaning, wear an ESD wrist strap and push it into a vertical position.

Reassemble the host computer and power up the system. If the error message in the OC error log still says: `Check or replace SIMM#`

You have a faulty SIMM.

HARD ERRORS (UNRECOVERABLE)

SIMM memory errors will cause an SGI operating system (Irix) PANIC. Usually, a PANIC message will be posted to a screen window and logged in `/var/adm/SYSLOG`. The offending SIMM will be identified by its SIMM socket number. If the system will not reboot after a hard SIMM memory error PANIC, it is probably because the bad SIMM is one of the first 4 where Irix is trying to boot into. To eliminate this possibility, swap all 4 SIMMs between the first 2banks (swap SIMMs in sockets 12,11,10,9 with SIMMs in sockets 8,7,6,5). (see below for more SIMM socket information)

It's possible that if a SIMM (either bank) fails that the rest of the bank is no longer "visible" or that the entire memory bus is locked. Try removing an entire bank at a time in these instances to determine which bank contains the offending SIMM. We have run systems on 64MB or 128MB but system performance/simultaneity may be seriously hampered and this is not recommended for extended periods. We've seen bootup failure messages indicating "`PANIC: CPU parity error interrupt`" that turned out to be a bad SIMM in the first bank.

There are 12 SIMM sockets on the motherboard. The SIMM socket closest to the CPU "brick" is SIMM #12 and the SIMM socket farthest from the CPU "brick" is SIMM #1. SIMM sockets are arranged to accommodate up to 3 banks of 4 SIMMs each. Each bank MUST be populated with the same type of SIMM (i.e. – 16MB or 32MB) and all sockets in a bank must be populated. In the standard CT/I configuration, SIMM sockets 12, 11, 10, and 9 will each contain a 32MB SIMM (making up 128MB) and SIMM sockets 8, 7, 6, and 5 will each contain a 16MB SIMM (making up 64MB). This results in total system CPU main memory of 192MB. The DIP (Deterministic Image Performance) option adds another 4 16MB SIMMs into SIMM sockets 4, 3, 2, and 1 resulting in total system CPU main memory of 256MB and fully populating all available SIMM sockets in the SGI IP22 motherboard.

Hard (unrecoverable) memory errors will cause an SGI operating system (Irix) PANIC. Usually, a PANIC message will be posted to a screen window and logged in `/var/adm/SYSLOG`. The offending module will be identified by its socket number.

Bootup failure messages indicating "PANIC: CPU parity error interrupt "may mean a bad module in the first bank. If the system will not reboot after a hard memory error PANIC, it is probably because the host needs the first bank to be good in order to boot. To eliminate this possibility, swap all modules in the first bank with those in the second. For Indigo, this means to swapping SIMMs in sockets 12,11,10,9 with SIMMs in sockets 8,7,6,5.

If a module in any bank fails, the rest of the bank may no longer be "visible" or the entire memory bus may lock. Try removing an entire bank at a time in these instances to determine which bank contains the offending module. Systems will operate with 64MB or 128MB but system performance and simultaneity is seriously hampered; this is not recommended for extended periods.

TROUBLESHOOTING MEMORY PROBLEMS

Hard Failures

A SIMM hard failure will prevent the SGI Computer from booting or crash the system if it occurs while the system is running.

Check the right monitor. It may have a message identifying the failing SIMM number. If there is no message, check that the LED on the front of the SGI computer is flashing amber. If it is NOT flashing, you may have another problem. If it is flashing, try booting 1 bank at a time to identify the failing SIMM.

If the system will get to UNIX, halt it at the UNIX level. (Do not run applications software at this time.) Note the time that the error occurred. Open a UNIX shell and type:

```
> su -
> #bigguy
> cd /var/adm
> jot SYSLOG
```

Use the"FIND binoculars to search for the word `PANIC` (use capitals). Look for the SIMM number associated with the "`PANIC`" errors to tell which SIMM to replace.

Obviously, "`PANIC`" errors which occurred at the same time as the system crashed, caused the crash. Other "`PANIC`" errors probably caused other system crashed.

If you want, use the “FIND” binoculars to search for the word SIMM. SIMM parity errors occur occasionally, but if they are not associated with a “PANIC” error, the system corrected the error without a problem.

Soft Errors

Take the system to the UNIX level. (Do not run application software at this time.

Note the time that the error occurred.

Open a UNIX shell and type:

```
> su -
> #bigguy
> cd /var/adm
> jot SYSLOG
```

Use the “FIND” binoculars to search for the word PANIC (use capitals). Look for the SIMM number associated with the “PANIC” errors to tell which SIMM to replace.

Obviously, “PANIC” errors which occurred at the same time as the system crashed, caused the crash. Other “PANIC” errors probably caused other system crashed.

If you want, use the “Find” binoculars to search for the word SIMM. SIMM parity errors occur occasionally, but if they are not associated with a “PANIC” error, the system corrected

2.2.3.12 Indigo2 Graphics Boards Errors

MG (Mardi Gras)

There is an SGI MG1,0 for the UIF (left) CRT monitor (a.k.a. “head”) and an SGI MG1,1 for the DISPLAY (right) head. The MG1,0 is basically a depopulated single card version of the MG1,1, which is a two card sandwich. The MG1,0 does NOT contain the required Geometry Engine (GE) or Texture Memory (TRAM) to run the display head so the boards are NOT interchangeable for CT/i display applications but you can swap them temporarily to just check basic SGI Irix dual head detection and operation problems. Place the single card in the top slot for a GIO pair. Its operating location is the top slot of GIO 1 pair, the fifth slot from the bottom. Place it in slot three for troubleshooting.

If the board attached to the primary monitor is removed, the secondary board and monitor by default become the primary head.

To view what components of the graphics system the SGI host currently sees, enter this command in a shell: **/usr/gfx/gfxinfo**

To test the MG1,1, leave the boards as they are normally and run **ide**, refer to page. It takes about 2.5 hours and does not test the board in the GIO 1 bus. However this board rarely fails. To test it, put it in the top GIO bus 0 slot, third from bottom, and run ide.

The optional TRAM and VCR video upgrades planned for CT/i are for use ONLY on the MG1,1 board and will not plug into the MG1,0.

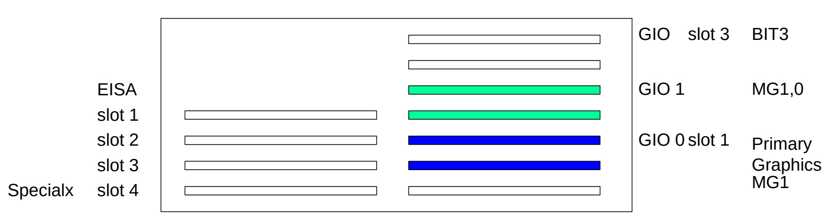


Figure 8-57 The GIO and EISA slots in the Midplane

Troubleshooting The Graphics System

To view what components of the graphics system the SGI host currently sees, enter this command in a shell: **/usr/gfx/gfxinfo**

You should see something that looks similar to the following example:

Example: "gfxinfo" Command

```
Graphics board 0 is "IMPACT" graphics.
        Managed (":0.1") 1280x1024
        Product ID 0x0, 1 GE, 1 RE, 1 TRAM
        MGRAS revision 3, RA revision 5
        HQ rev A, GE11 rev B, RE4 rev A, PP1 rev A,
        VC3 rev A, CMAP rev D, MC rev C
        unknown, assuming 19" monitor (id 0xf)
        Channel 0:
        Origin = (0,0)
        Video Output: 1280 pixels, 1024 lines, 72.24Hz (1280x1024_72)
Graphics board 1 is "IMPACT" graphics.
        Managed (":0.0") 1280x1024
        Product ID 0x1, 1 GE, 1 RE, 0 TRAMs
        MGRAS revision 1, RA revision 0
        HQ rev A, GE11 rev @, RE4 rev A, PP1 rev A,
        VC3 rev A, CMAP rev D, MC rev C
        unknown, assuming 19" monitor (id 0xf)
        Channel 0:
        Origin = (0,0)
        Video Output: 1280 pixels, 1024 lines, 72.24Hz (1280x1024_72)
```

2.2.3.13 Diagnosing BIT3 Subsystem on CT/i 3.X/4.X (INDIGO2)

If you suspect BIT3 subsystem problems (GIO BIT3 card, BIT3 OC cable, BIT3 SBC cable, or VME BIT3 card), the following tests/checks can help confirm functionality and/or help isolate the FRU. Although not foolproof in every failure mode case, it's very dependable for most typical problems.

All of these checks can be done from the OC only, with a non-working or unknown BIT3 subsystem, and with application SW down.

Diagnostic Steps

Follow the steps below in the order suggested by the results of each test.

1.) Confirm that you do not have BIT3 HARDWARE communications (WITHOUT relying on any applications software or network/reconfig parameters):

Note: THIS TEST SHOULD ONLY BE RUN ON AN IDLE SYSTEM (NO SCANNING/RECON).

This test performs data transfers between the OC and SBC using the entireBIT3 subsystem (both boards and cables). The above runs 100 passes of data across the BIT3 and checks the results. This test does not rely on any network parameters (IP#'s, hostnames) existing or being correct. The “Transfer rate” shown above is only typical for an idle system but this may vary (you're only looking for write/read errors which may indicate a BIT3 hardware problem).This should only be run on an idle system or you may get read/write errors due to contention on the VMEbus by scan/recon (normal).

```
{ctuser@rhap1}[1] cd /usr/etc
```

```
{ctuser@rhap1}[2] mvdsrate 30000200 0f 1000 1000
......pass 1000
Transfer rate = 7130.982910 Kbytes/sec  elapsed time = 1121865 usec
***** End of Test *****
{ctuser@rhap1}[3]
```

Comment: *IF THIS STEP PASSES, THEN PROCEED TO STEP 2 BELOW (BIT3 HARDWARE GOOD). IF THIS STEP FAILS, THEN PROCEED TO STEP 3 BELOW (SUSPECT BIT3 HARDWARE).*

2.) Check the BIT3 NETWORK communications (this relies on the correct IP numbers and hostnames being properly reconfigured)

Note: USE **CONTROL C** KEY TO STOP THE PING AT ANY TIME.

The '**ping**' command does simple ICMP echo packets between network hosts. The below results are typical with Indigo2 BIT3 on an idle system. If the '**ping'** times out (no response) but mvdsrate runs (as in step 1), then you most likely have a network setup (reconfig) problem. This would be typical during/after load-from-cold when network parameters are entered incorrectly (or "accidentally" changed to incorrect values)

```
{ctuser@rhap1}[3] ping sbc
PING rh01_sbc0 (192.2.100.2): 56 data bytes
64 bytes from 192.2.100.2: icmp_seq=0 ttl=255 time=1 ms
64 bytes from 192.2.100.2: icmp_seq=1 ttl=255 time=1 ms
64 bytes from 192.2.100.2: icmp_seq=2 ttl=255 time=1 ms
----rh01_sbc0 PING Statistics----
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 1/1/1 ms
{ctuser@rhap1}[4]
```

Comment: *IF THIS STEP PASSES, THE BIT3 HARDWARE/NETWORK IS NOT THE PROBLEM. IF THIS STEP FAILS, CHECK RECONFIG OC-SBC NETWORK SETTINGS/PARAMETERS.*

3.) On the OC, check that the GIO BIT3 board was probed/attached by the Irix device driver successfully at the last bootup as follows:

The messages below indicate that the GIO BIT3 board was functional enough to allow the device driver to detect it, read/write it's registers over the GIObus, and that the board reported a "good" state.

Also, when the GIO BIT3 card powers up and configures itself successfully, a green LED lights up on the top corner of the GIO BIT3 card. Unfortunately, this LED can only be easily seen with Indigo2 chassis removed and the cover off. If this GIO BIT3 card green "RDY" (ready) LED does not light when the console (Altron Indigo2 chassis) power is applied, you very likely have a BIT3 GIO card or GIO card/midplane/motherboard seating problem.

```
{ctuser@rhap1}[4] cd /var/adm
{ctuser@rhap1}[5] more SYSLOG* | grep BIT3
Aug 14 11:03:00 5A:rhap1 unix: NOTICE: BIT3 Model 608 board found at
GIO slot 3
Aug 13 02:10:04 5A:rhap1 unix: NOTICE: BIT3 Model 608 board found at
GIO slot 3
Aug 12 07:16:50 5A:rhap1 unix: NOTICE: BIT3 Model 608 board found at
GIO slot 3
Aug 11 01:19:32 5A:rhap1 unix: NOTICE: BIT3 Model 608 board found at
GIO slot 3
```

```
Aug 11 05:09:29 5A:rhap1 unix: NOTICE: BIT3 Model 608 board found at
GIO slot 3
Aug  8 07:25:28 5A:rhap1 unix: NOTICE: BIT3 Model 608 board found at
GIO slot 3
{ctuser@rhap1}[6]
```

Comment:
- *Repeats for all occurrences of Irix bootup/probe/attach by date/time)*
- *IF THIS STEP PASSES, PROCEED TO STEP 4 BELOW. IF THIS STEP FAILS, THERE IS A GIO BIT3 BOARD OR SEATING PROBLEM.*

4.) Open a shell and `rlogin` into the SBC, check that the VME BIT3 board was probed/attached by the **VMUNIX** device driver successfully at the last bootup as follows:

```
{ctuser@rhap1}[6] cu sbc
Connected
rh01_sbc0 login: root
Password:
Last login: Thu Aug 13 07:01:08 from rh01_oc0
SunOS Release 4.1.1_U1 (GOS_IG) #4: Thu Jun 18 15:22:19 CDT 1998
genesis @ rh01_sbc0 1: cd /var/adm
genesis @ rh01_sbc0 2: more messages* |grep "vmunix: svd0"
Aug 10 12:30:26  vmunix: svd0 at vme16d16 0x2000 vec 0xf0 vec 0xff
Aug 11 05:36:05 rh01_sbc0 vmunix: svd0 at vme16d16 0x2000 vec 0xf0 vec
0xff
Aug 12 07:21:11 rh01_sbc0 vmunix: svd0 at vme16d16 0x2000 vec 0xf0 vec
0xff
Aug 13 02:11:13 rh01_sbc0 vmunix: svd0 at vme16d16 0x2000 vec 0xf0 vec
0xff
Aug 14 07:53:17 rh01_sbc0 vmunix: svd0 at vme16d16 0x2000 vec 0xf0 vec
0xff
Aug 15 01:51:32 rh01_sbc0 vmunix: svd0 at vme16d16 0x2000 vec 0xf0 vec
0xff
genesis @ rh01_sbc0 3:
```

Comment: *Repeats for all occurrences of Irix bootup/probe/attach by date/time.*

Also, when the VME BIT3 board powers up and configures itself correctly, the green READY LED will light up on the front faceplate of the VME BIT3 board. If this green READY LED does not light up when console/VME power is applied, you very likely have VME BIT3 board or VME board/chassis seating problem.

Comment: *IF THIS STEP PASSES, YOU MAY HAVE A BIT3 CABLE/SEATING PROBLEM. IF THIS STEP FAILS, THERE IS A VME BIT3 BOARD, JUMPERS, OR SEATING PROBLEM.*

OTHER USEFUL INFORMATION

A.) You can check that the BIT3 OC-SBC network device is configured correctly using the following commands on the OC and/or SBC as shown:

a.) CHECK BIT3 NETWORK DEVICE RUNNING WITH CORRECT NETWORK PARAMETERS ON OC

Make sure the device is "RUNNING" and that the "inet", netmask, and broadcast parameters are set correctly (use 'reconfig' on the OC and SBC if necessary to correct these). The LFC defaults are shown.

```
{ctuser@rhap1}[7] ifconfig vd0
vd0: flags=8e3<UP,BROADCAST,NOTRAILERS,RUNNING,NOARP,MULTICAST>
       inet 192.2.100.1 netmask 0xffffff00 broadcast 192.2.100.255
{ctuser@rhap1}[8]
```

b.) CHECK BIT3 NETWORK DEVICE RUNNING WITH CORRECT NETWORK PARAMETERS ON SBC

Make sure the device is "RUNNING" and that the "inet", netmask, and broadcast parameters are set correctly (use 'reconfig' on the OC and SBC if necessary to correct these). The LFC defaults are shown.

```
{ctuser@rhap1}[8] cu sbc
Connected
rh01_sbc0 login: root
Password:
Aug 15 02:14:13 rh01_sbc0 login: ROOT LOGIN console
Last login: Sat Aug 15 02:04:00 on console
SunOS Release 4.1.1_U1 (GOS_IG) #4: Thu Jun 18 15:22:19 CDT 1998
root @ rh01_sbc0 1: ifconfig vd0
vd0: flags=e1<UP,NOTRAILERS,RUNNING,NOARP>
        inet 192.2.100.2 netmask ffffff00
root @ rh01_sbc0 2:
```

B.) You can check the current status of network communications on the OC or SBC using the following commands:

a.) CHECK NETWORK DEVICE STATUS ON THE OC

```
{ctuser@rhap1}[1] netstat -i
Name Mtu   Network     Address      Ipkts  Ierrs   Opkts Oerrs   Coll
ec0  1500  3.7.52      rhap1        628953    19   216336     0 149444
vd0  4336  192.2.100   rh01_oc0       3435     0     3904     2      0
lo0  8304  loopback    localhost    480577     0   480577     0      0
ppp0 1500  (pt-to-pt)  olc-pm1           0     0        0     0      0
{ctuser@rhap1}[2]
```

"ec0" is the hospital/gateway ethernet network.

"vd0" is the OC to SBC BIT3 dedicated subnetwork (output "errors" can be "normal" on the BIT3 due to VMEbus busy retries during scan/recon).

"lo0" is the host loopback pseudo-device.

"ppp0" is the InSite PPP serial port network device.

"Network" is the IP base number of the network/subnet.

"Address" is the hostname.

"Ipkts" is the number of network packets received since the last bootup.

"Ierrs" is the number of network receive errors since the last bootup.

"`Opkts`" is the number of network packets transmitted since the last bootup.

"`Oerrs`" is the number of network transmit errors since the last bootup.

"`Coll`" is the number of network collisions (there are normal since this is how Ethernet works when multiple nodes "negotiate" for the cable.

b.) CHECK NETWORK DEVICE STATUS ON THE SBC

```
{ctuser@rhap1}[8] cu sbc
Connected
rh01_sbc0 login: root
Password:
Last login: Sat Aug 15 02:12:23 from rh01_oc0
SunOS Release 4.1.1_U1 (GOS_IG) #4: Thu Jun 18 15:22:19 CDT 1998
genesis @ rh01_sbc0 1: netstat -i
Name  Mtu  Net/Dest     Address     Ipkts  Ierrs  Opkts  Oerrs Collis
Queue
ei0   1500 192.9.220.0  SBCdLAN     4495      0   5658      0      0     0
vd0   4336 192.2.100.0  rh01_sbc0   2498      0   2433      0      0     0
lo0   1536 127.0.0.0    localhost   893       0   893       0      0     0
genesis @ rh01_sbc0 2:
```

"`ei0`" is the control LAN between SBC and STC/ETC and OBC (via STC).

"`vd0`" is the OC to SBC BIT3 dedicated subnetwork.

"`lo0`" is the host loopback pseudo-device.

"`Net/Dest`" is the IP base number of the network/subnet.

"`Address`" is the hostname.

"`Ipkts`" is the number of network packets received since the last bootup.

"`Ierrs`" is the number of network receive errors since the last bootup.

"`Opkts`" is the number of network packets transmitted since the last bootup.

"`Oerrs`" is the number of network transmit errors since the last bootup.

"`Collis`" is the number of network collisions (there are normal since this is how ethernet works when multiple nodes "negotiate" for the cable.

"`Queue`" is the number of packets waiting in the queue.

2.2.3.14 Serial Communications

Serial Devices:

SGI PORTS (EFFECTIVE CT/I 4.1)

FILESYSTEM NAME	PORT	DEVICE
`/dev/ttym1`	RS232 Serial Port **1 out of SGI**	MODEM (InSite)
`/dev/ttyd2`	RS232 Serial Port **2 out of SGI**	SERVICE KEY

Table 8-8 Serial Port Filesystem Names/Assignments (Indigo)

INDIGO PORTS (EFFECTIVE CT/I 4.1)

SERIAL PORT	EISA BUS (INDIGO) SPECIALIX BRD. W/I INDIGO
`/dev/ttya1`	RS232 link to **SBC console port**
`/dev/ttya2`	RS232 **touch panel input**
`/dev/ttya3`	RS232 **trackball input**
`/dev/ttya4`	RS232 port for **laptop**
`/dev/ttya5`	RS422 NO LONGER USED
`/dev/ttya6-ttya8`	RS422 not assigned

Table 8-9 Serial Port Filesystem Names/Assignments (Indigo)

Checking Serial Communication with a Loop-back Test

1.) Shutdown GE CT applications. Select -Utilities- then -Applications Shutdown-
2.) Connect the RXD and TXD signals (#2 and #3) on the port to be tested with a loop-back plug or wire clip.
3.) Open a UNIX shell and log in as root. Enter: **`su -`**
4.) Enter: **`/usr/g/bin`**
5.) Type the command: **`slxtst`**
6.) Choose test no. 3 from the menu.
7.) Specify input ports as **`/dev/ttya`*`N`***, where ***`N`*** represents the port to be tested. Acceptable values are:
    `1` to `5` if Indigo

CT/i Specialx Serial Ports Information

The Indigo2 host uses the Specialix card to increase the number of available serial ports.

The Specialx serial device expansion ports have been assigned as shown below.

```
DevicePort# Type  CT/i system assignment
/dev/ttya1 1 RS232C SBC Install/Boot/Service link
/dev/ttya2 2 RS232C ELO Touch Panel interface
/dev/ttya3 3 RS232C MicroSpeed Trackball interface
/dev/ttya4 4 RS232C GEMS Service Laptop (intermittent)
/dev/ttya5 5 RS422 NO LONGER USED
/dev/ttya6 6 RS422 (reserved)
/dev/ttya7 7 RS422 (reserved)
/dev/ttya8 8 RS422 (reserved)
```

This information is provided for reference only. There should be no need under normal circumstance to manually make changes to any of the Specialx parameters.

The man pages are installed by the INSTALL script on top of GE software loads. The "`slxos`" man page is attached for additional usage details (see also "`slxcfg`", "`slximg`", and "`slxinfo`" man pages).

```
SLXOS - Specialx SI/XIO Intelligent serial I/O device driver.
The SLXOS device driver supports up to four Specialx SI or XIO EISA host
cards, each fitted with up to four TA modules. Each SI TA-4 or TA-8 module
```

can support four or eight serial, either RS232, RS422 or RS423, using DB25 connectors. All XIO MTA modules support eight ports, either RS232 DB25, RS422 DB25, RS232 RJ45 or seven RS232 DB25 and one DB25 Centronics-style parallel port. A fully populated SLXOS system has 128 ports.

The SLXOS driver supports direct and modem ports. Direct (tty) ports are named /dev/ttyaX, where X is the port number in the range 1–128. Direct ports do not honor modem signals, and as such are useful for connecting terminals locally to a system. Modem ports are named /dev/ttyAX, where X is the port number in the range 1–128. Modem ports honor modem signals, and are used for both dial-in and dial-out applications. By default, modem ports all use hardware flow control, and direct ports use software flow control. However this can be overridden using termio and ermios alls, see ioctl(2), termios(3t), termio(7) and stty(1). Note that at the time of writing there is no way to enable/disable hardware flow control using the stty(1) program.

When terminals are used with software flow control at high baud rates, a problem can arise with the use of the IXANY flag. This flag allows any received character to re-start transmission after an XOFF (^S) character has been received by the port to stop transmission. This means that users who type ahead on their terminals can interfere with the flow control mechanism, and suffer data loss. The IXANY flag is set either using stty(1) or via the termios(2) interface, and several applications set it without consulting the user or providing a mechanism to disallow this. This software flow control feature can be disabled using a special feature of the driver. See slxcfg (1) for information on enabling and disable IXANY operation.

The following speeds are supported:

SI			XIO		
stty rate	xmit rate	receive rate	stty rate	xmit rate	receive rate
50	57600	57600	50	57600	57600
75	75	75	75	75	75
110	110	110	110	115200	115200
134	75	1200	134	75	1200
150	150	150	150	150	150
200	1200	75	200	1200	75
300	300	300	300	300	300
600	600	600	600	600	600
1200	1200	1200	1200	1200	1200
1800	1800	1800	1800	1800	1800
2400	2400	2400	2400	2400	2400
4800	4800	4800	4800	4800	4800
9600	9600	9600	9600	9600	9600
19200	19200	19200	19200	19200	19200
38400	38400	38400	38400	38400	38400

Table 8-10 CT/i Specialx Serial Port Rates

Notice that setting 134 baud will give split 1200/75 operation, and 200 baud will give split 75/1200 operation. These speeds can be set independently for the receiver and transmitter if required - see termio(7). Setting the input speed to 134 baud has a different effect to setting the output speed to 134 baud - the first case will select a receive clock of 75 baud, the second case will select a transmit clock of 1200 baud, in accordance with the above table.

One major advantage of the SLXOS driver is the transparent print driver. This allows a printer attached to the AUX. connector of a terminal on a SLXOS port to be used as a regular printer. Such transparent printers are accessed through the device node /dev/ttyaXp where X is the number of the port to which the terminal with the auxiliary printer is attached. To function correctly, the transparent print driver has to be told about the type of terminal being used. See slxcfg (1) and the guide to installation and operation for information about setting up terminal types and transparent print control parameters.

The Unix window size support ioctls TIOCSWINSZ and TIOCGWINSZ are supported. The defaults for these are 80x24 characters, 0x0 pixels, indicating a text only VDU system.

For full termio and termios functionality, SLXOS has to be used with the standard line discipline module. If this line discipline is not used then some features (such as TCFLSH, TCSBRK) may not function as anticipated. For experienced programmers, the functionality of TCFLSH and TCSBRK can be obtained by using M_BREAK and M_FLUSH messages. The line discipline is configured into the autopush configuration file /etc/slxos.ap when SLXOS is installed. For more information on this topic, please refer to autopush(1M) manual page.

Related Files:

/usr/etc/slxos/slxos.ap	– contains autopush module configuration
/usr/etc/slxos/print.slx	– contains information about ports
/usr/etc/slxos/printcap.slx	– contains information about term types
/etc/default/slxos	– contains default portinformation
/dev/ttya*X*	– direct tty line
/dev/ttyA*X*	– modem line
/dev/ttya*X*p	– transparent print port

See Also

slxcfg(1), slximg(1), slxinfo(1), autopush(1M), getty(1M), stty(1), close(2), ioctl(2), open(2), poll(2), read(2), setsid(2), setpgid(2), termios(2), write(2), login(4), fcntl(5), streamio(7), termio(7), tty(7)

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2.2.3.15 Hard Drives

Power-On/Auto Tests

When you turn ON power, the hard drives perform basic hardware and operation self-tests. If the diagnostic detects a hard error, it steadily flashes the green LED on the front of the drive. Do NOT confuse the steady flashing of a hard error with the intermittent blinking that occurs during normal Disk I/O activity.

When New Drives are Installed

When a hard drive is new, software needs to be loaded on it. If a *Second hard drive option* is added, a LFC or the **install2Disk** script at **/usr/g/scripts** will get it to work once its jumper identifies it as SCSI 2, and it's attached to the host's SCSI chain.

Indigo2 Host Disk “fx “Boot Utility

Note: FX CAN BE VERY DANGEROUS TO DISK DATA; IT CAN EASILY DESTROY SYSTEM SOFTWARE! ONLY USE IT IF COMFORTABLE WITH IT AND ONLY AS INDICATED!

Although rare, a CT/i SGI system disk may develop a bad block which might look like one of the below messages in the /var/adm/SYSLOG:

```
May 21 05:52:38 2A:bay6 unix: dks1d1s0 (/): Media error: No addr mark found
in ID field (asc=0x12, asq=0x0), (data byte 160), Block #8351285 (8355803)
May 21 05:52:38 2A:bay6 unix: dks1d1s0 (/):   retrying request
May 21 08:12:04 2A:bay6 unix: dks1d1s0 (/):Media error: ID CRC or ECC error
(asc=0x10, asq=0x0), (data byte 160), Block #8351285 (8355803)
May 21 08:12:04 2A:bay6 unix: dks1d1s0 (/):   retrying request
```

Two different block locations are reported by fsck, the second number being inside parenthesis. The first number reported is the bad block relative to the disk partition in which it was found, the second number is the bad block relative to the entire disk.

To repair and boot the system, it may be necessary to boot the fx disk utility from disk (or from CDROM if the disk boot of fx fails) with **boot /stand/fx** from the SGI firmware command monitor prompt (interrupt bootup with **ESC** then select Command Monitor). It can be used to label bad blocks. It could be used to try reformatting the drive to repair it. Doing so will erase the system software and require a Load From Cold (LFC).

When **fx** asks you for “special privileges” answer **yes**. Enter the correct device information (as shown below). Then go into the **badblock** menu and use the **addblock** function to enter the SECOND number reported by **fsck** (*the one in parenthesis*).

```
> fx version 5.3, Oct  9, 1995
> fx: "device-name" = (dksc)
> fx: ctlr# = (0) 1
> fx: drive# = (1)
> fx: lun# = (0)
> ...opening dksc(1,1,0)
> ...controller test...OK
> Scsi drive type == SEAGATE ST15150N        0017
> ----- please choose one (? for help, .. to quit this menu)-----
> [exi]t              [d]ebug/            [l]abel/            [a]uto
> [b]adblock/         [exe]rcise/         [r]epartition/      [f]ormat
> fx> b
>
> ----- please choose one (? for help, .. to quit this menu)-----
> [a]ddbb          [s]howbb
> fx/badblock> a
>
> please enter a bn from 0 to 8387666
> fx/badblock/addbb: add badblock = 123456
```

```
> trying to save the data ...adding bad block (123456)
> rewrote saved data OK
>
> please enter a bn from 0 to 8387666
> fx/badblock/addbb: add badblock = (USE CONTROL'C' AT THIS POINT TO EXIT)
> ----- please choose one (? for help, .. to quit this menu)-----
> [a]ddbb         [s]howbb
> fx/badblock> ../exi
> ----------------------------------------------------------------
```

It is very likely that the "trying to save the data" portion above will fail and retry many times but fx will eventually just slip the bad block anyway. This means you will have corrupted data on the disk which may or may not be a problem; there are thousands of unused and unnecessary files on Unix/ Irix systems and one of these may be the one that becomes corrupt. In the worst case, a LFC may be necessary after the bad block "repair."

There may also be other scenarios where a similar procedure will be necessary such as a run-time panic with a system disk "medium error" which will also report the bad block number. The details may vary slightly but the concept is the same (*record bad block#, use fx to slip it*).

2.2.3.16 Error Messages

The following table illustrates some of the potential error messages you might encounter.

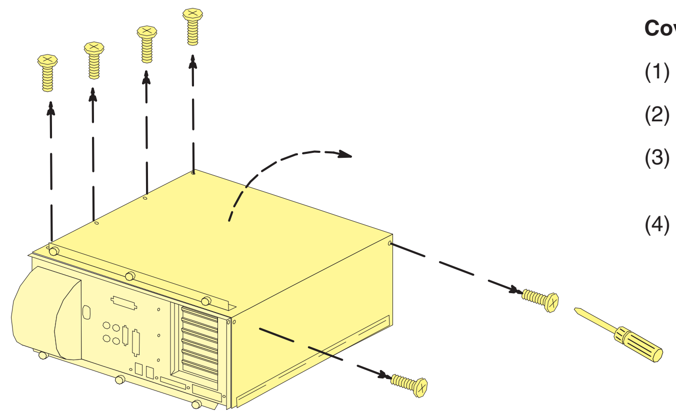
Error Message	Meaning
ALERT: ec0: no carrier: check Ethernet cable	Bad network cable, connection, transceiver (make sure heartbeat is off); rarely the motherboard
BIT3 board found at GIO slot 3	The host's BIT3 board, also known as vd0, used as a network device to transfer commands to SRC and DMA device to transfer scout images passed its Boot Up test
CPU cache parity error or exception error	Replace IP22 motherboard
Detected EISA SI/XIO host card in slot4	Specialix board passed Boot Up Test
dks#d#s# unit attention - retries exhausted	Bad SCSI cable, device or connection; bad terminator, bad DASM (0d1=DASM;1d1=OC; 1d2=OC optional disk; 1d3=MaxOPtics; 1d5=Pioneer; 1d6=CDROM)
ecc/parity error on the sysad bus during data reference, secondary cache data-filed error	An SGI IDE diagnostic bug which does not appear to indicate any hardware problems when it appears after the test has run for an hour. This ERROR may also lock up the system. A RESET is required to exit and reboot the system.
File:Table is full	Shut down the software and perform a System Down so the SBC reboots. Bring applications up and immediately run ImageGenTest. When done, again shut down the software and perform a System Down to clean up the file descriptors on the SBC.

Error Message	Meaning
`Graphics error`	Indicates a graphics MG board failure ONLY if it happens continuously, hourly, maybe daily
`Host LED: No LED and host fan NOT working`	Power problem: check connections, host power supply LEDs, probably a bad host power supply if AC is available
`Host LED: No LED but fan and OC disk (GRN LED flashes) are working`	CPU not working; reseat the SGI midplane or NexGen system module; check internal connections, replace the module or processor
`NG Host LED: solid WHT`	System diagnostics have identified failing part; if you do not see the message on the screen, review OC SYSLOG to find what part
`NG Host LED: solid WHT and keyboard not working`	Check keyboard connection and OC SYSLOG for message, record what it says; problem may be bad keyboard or system board
`NG Host LED: solid WHT and mouse not working`	Check mouse connection and OC SYSLOG for message, record what it says; problem may be bad mouse or system board
`NG Host LED:solid RED LED and no system drive`	Check seating of SCSI cables, if error message in OC SYSLOG, record what it says, system drive bad
`Host SGI LED: blinking yellow`	SIMM failure; check OC SYSLOG to determine which one(s).
`Host SGI LED: solid yellow`	IP22 motherboard or power failure
`Media error: No addr mark found in ID field`	Damaged filesystem on scsi disk; use fx or reload software (1d1=OC; 1d2=OC optional disk; 1d3=MaxOPtics; 1d5=Pioneer; 1d6=CDROM)
`Memory is not usable. Check or replace all SIMMs.`	Check that all of your SIMMs are seated all the way into the sockets and that they are installed in the correct slots. Remove all but one bank of SIMMs. Power on the system again. Swap removed SIMMs with those now in the motherboard. Power on the system again. If you get the same error message with each bank, then all SIMMs are faulty or the IP22 mother board is faulty.
`Memory Parity Error in SIM S#; code:30`	Solitary occurrences are caused by static or slight power glitches, are correctable and no reason to replace SIMM
`mgras: CTXSW timeout`	Indicates a graphics MG board failure ONLY if it happens continuously, hourly, maybe daily
`mgras:CFIFO timeout`	Indicates a graphics MG board failure ONLY if it happens continuously, hourly, maybe daily

Error Message	Meaning
NESTED EXCEPTION #1 at ERROR EPC:88521640; FIRST EXCEPTION AT PC:884cd820	An SGI IDE diagnostic bug which does not appear to indicate any hardware problems when it appears after the test has run for an hour. This ERROR may also lock up the system. A RESET is required to exit and reboot the system.
No Media	Operator Error usually
OBC level1 interrupt with no fault detected.	A severe error was detected on the MA, IO, KV, CTVRC, or RCOM board. If this error occurs at power up intermittently, it can be ignored. If this error occurred during a scan, check that the OBC cover is well connected, check the boards listed above and the OBC CPU and backplane for problems.
PANIC: CPU parity error interrupt	A SIMM in the primary SGI memory bank is probably bad. Swap SIMMs in sockets 12,11,10,9 with SIMMs in sockets 8,7,6,5. If the system boots, then order a replacement for the primary bank.
PANIC: IRIX Killed due to Memory Error in SIMM S#	Replace bad SIMM
savecore: pb # : ALERT:Memory Parity Error in SIM S3	Replace bad SIMM
SCSI bus reset	Bad SCSI cable or connection; bad terminator, bad DASM (0d1=DASM; 1d1=OC; 1d2=OC optional disk;1d3=MaxOptics; 1d5=Pioneer, 1d6=CDROM)
Service uif_scan_rx received unknown message with event code 3243 from service event_router	If something obstructed the x–ray beam during fastcal, the software does not handle it well until the second attempt. This message is benign.
Write of View Data into iq file failed	Rerun the Afterglow cal.

2.2.4 CT/i Host (Indigo) Replacement Procedures

2.2.4.1 Indigo 2 Chassis Cover Removal/Installation



Cover Removal

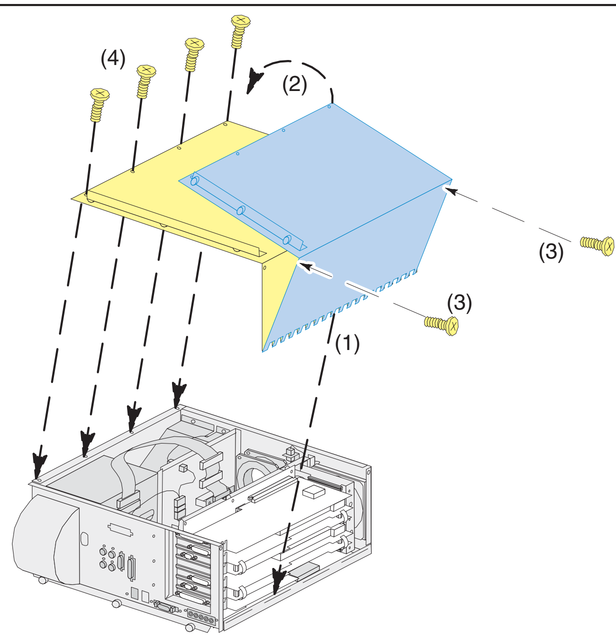
(1) Remove 6 screws from cover.

(2) Lay chassis on right side.

(3) Rotate cover up and lift finger tabs out of chassis.

(4) Set cover aside.

Figure 8-58 SGI Chassis Cover Removal



SGI Chassis Cover Installation

(1) Insert finger tabs into slots in chassis as shown (at an angle).

(2) Rotate the cover closed.

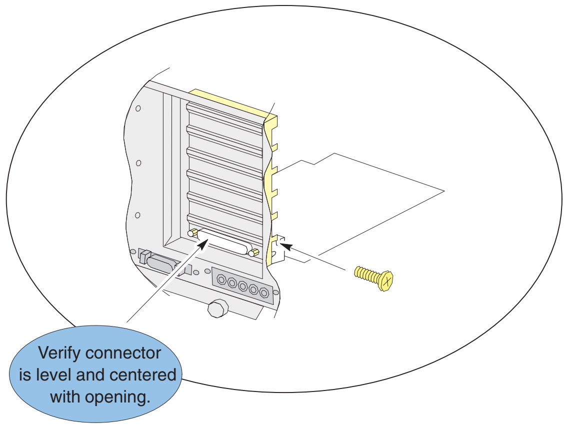
(3) Install 2 screws to draw chassis closed.

(4) Install four screws at.

(5) Tighten all 6 screws.

Figure 8-59 SGI Chassis Cover Installation

2.2.4.2 Specialix Board Removal/Replacement, 2139035-2



Removal

Specialix Board is in the bottom Slot in the SGI Chassis.

(1) Remove screw holding Connector Bracket to chassis

(2) Remove Board from slot.

Installation

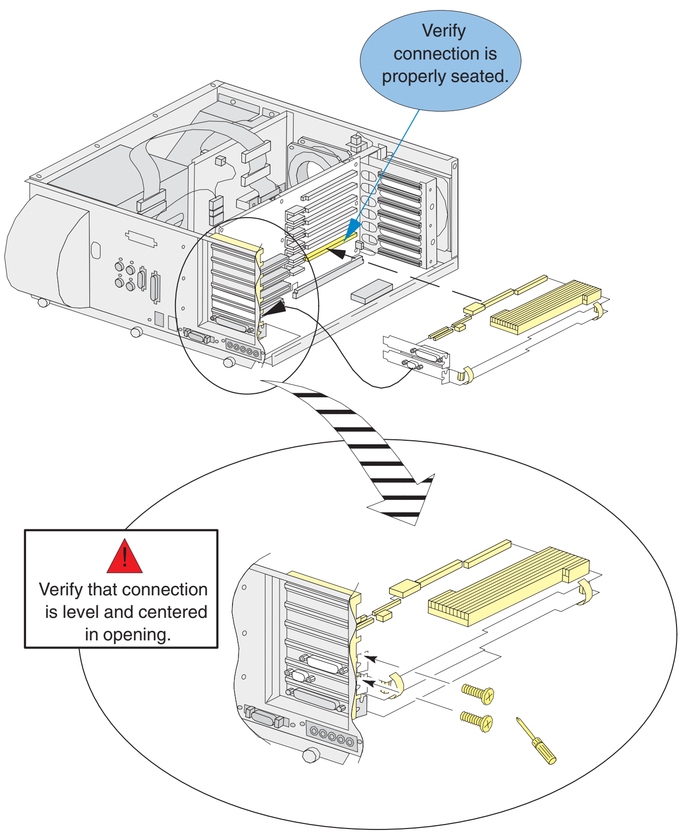
(1) Insert circuit board into bottom slot and check seating in midplane electrical connector.

(2) Verify that the Specialix MTA cable connector is level and centered in the opening.

(3) Install screw holding connector bracket to chassis.

Figure 8-60 Specialix Board

2.2.4.3 Mardi Gras 1,1 Replacement, 2115457-16



Mardi Gras 1,1 Graphics Board Assembly Removal

(1) Remove screw 2 screws holding connector bracket to chassis.

(2) Use nylon finger loops to pull board set from midplane.

Mardi Gras 1,1 Graphics Board Assembly Installation

(1) Insert MG1,1 board assembly into the midplane slots 2nd and 3rd from the bottom (just above the Specialx Serial Bd).

(2) Check that cable connectors are level and centered in the openings.

(3) Install and tighten 2 screws holding the connector bracket to the chassis.

Figure 8-61 Mardi Gras 1,1 Removal and Installation

2.2.4.4 SIMM Removal and Installation

CAUTION **Practice good ESD prevention.**

1.) The Indig2 chassis cover must be removed to access memory SIMMs. Remove the Indigo2 cover. Follow the instructions in Figure 8-62 by replacing the word “install” with “remove”, and reversing the sequence. By starting at step 5 and finishing with step 1.

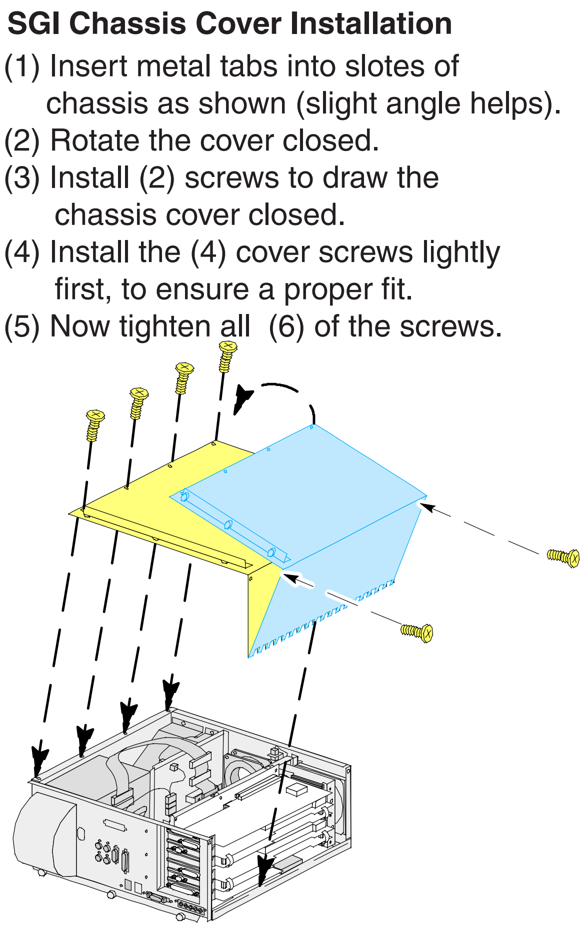


Figure 8-62 Indigo2 Chassis Cover Removal

Following the steps as described for SIMM removal shown in Figure 8-64 by replacing the defective SIMM modules located on the Indigo2 mother board.

Note: Remember that SIMMs must be installed in groups of four (4). See Figure 8-63. These groups must also be of the same memory capacity and type. (e.g., four 16Mb SIMMs in slots 1, 2, 3, and 4.)

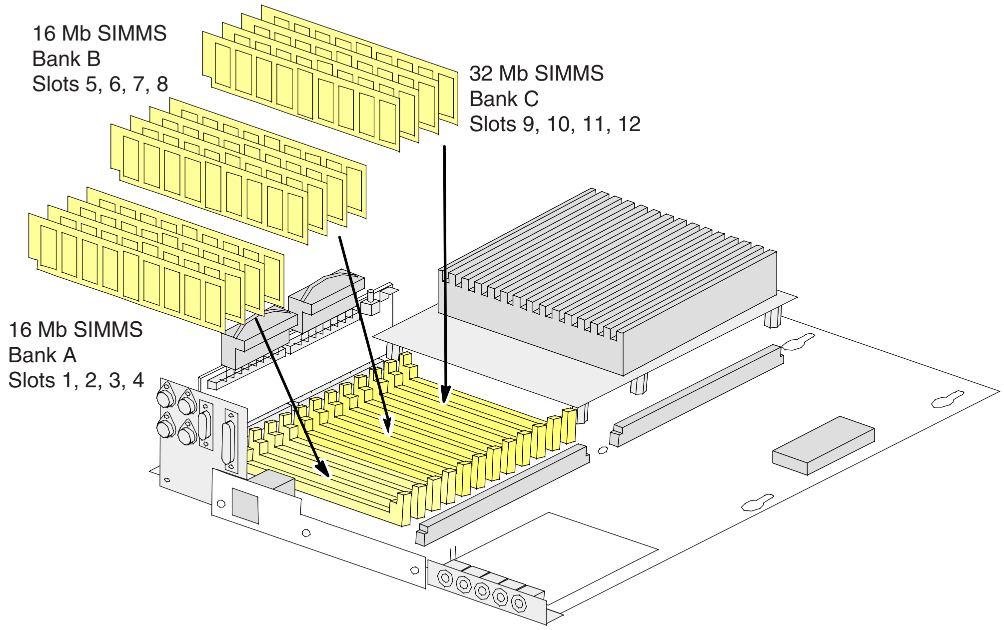


Figure 8-63 Indigo2 SIMM Memory Configuration

Remove the defective SIMM(s)

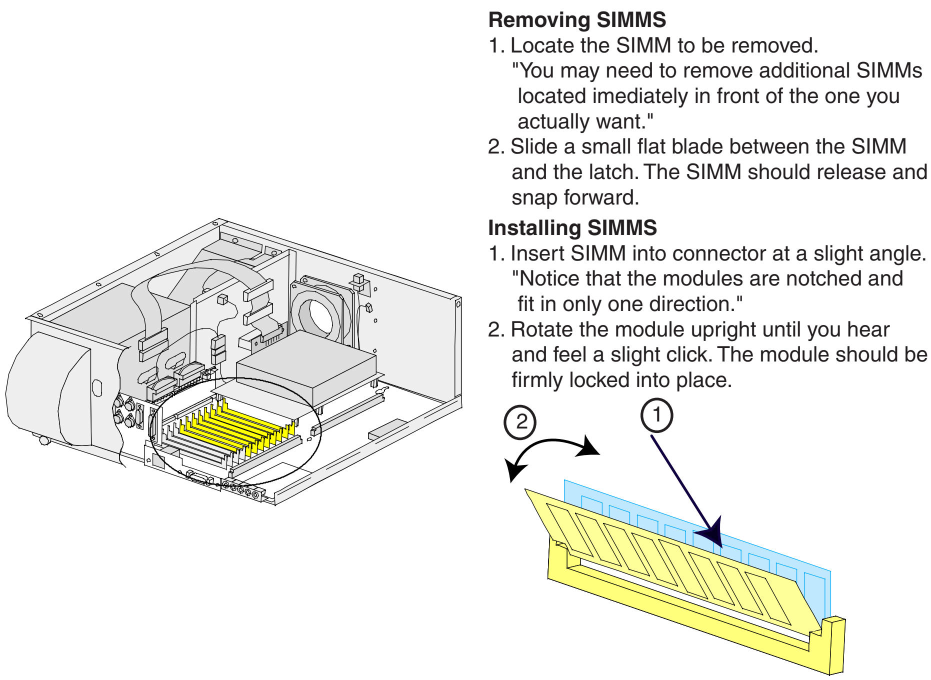


Figure 8-64 Indigo2 SIMM Removal and Installation

2.) Install new SIMM(s)

3.) Re-install the SGI CPU Chassis cover and slide the SGI CPU Chassis into the console base.

Note: If any cables where disconnected while removing the SGI Chassis from the console. Re-connect cables at this time, noting connector labels on cables and chassis.

2.2.4.5 Mardi Gras Texture Memory Option Replacement

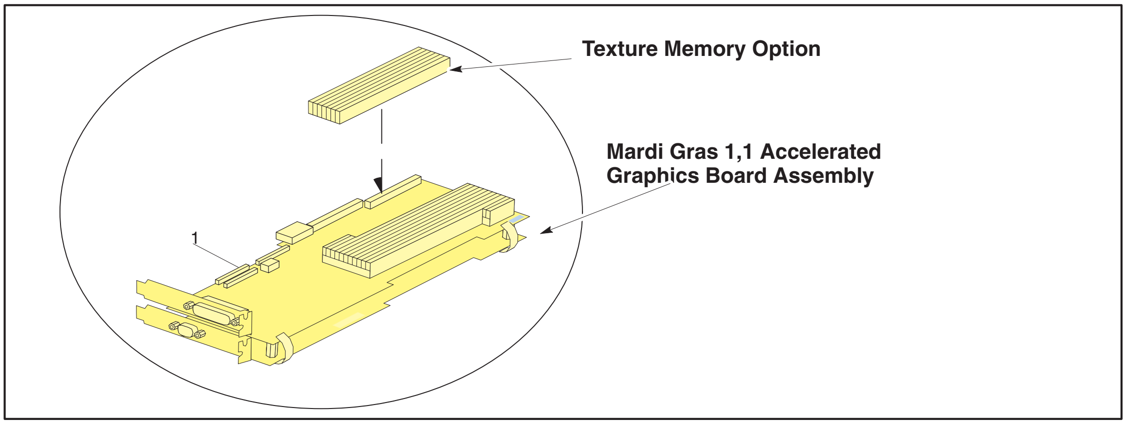


Figure 8-65 Mardi Gras Texture Memory Option (MG 1,1)

2.2.4.6 Mardi Gras 1,0 Replacement, 2115457-5

Mardi Gras 1,0 Assembly Removal

(1) Remove 2 screws holding connector bracket to chassis.

(2) Using the nylon finger loops pull the board assembly from the midplane.

Mardi Gras 1,0 Assembly Installation

(1) Guide the circuit board into the midplane slots 5 and 6 as counted form the motherboard up (GIO64 bus slots GIO2A and GIO2B).

(2) Install 2 screws to hold the connector brackets to the chassis.

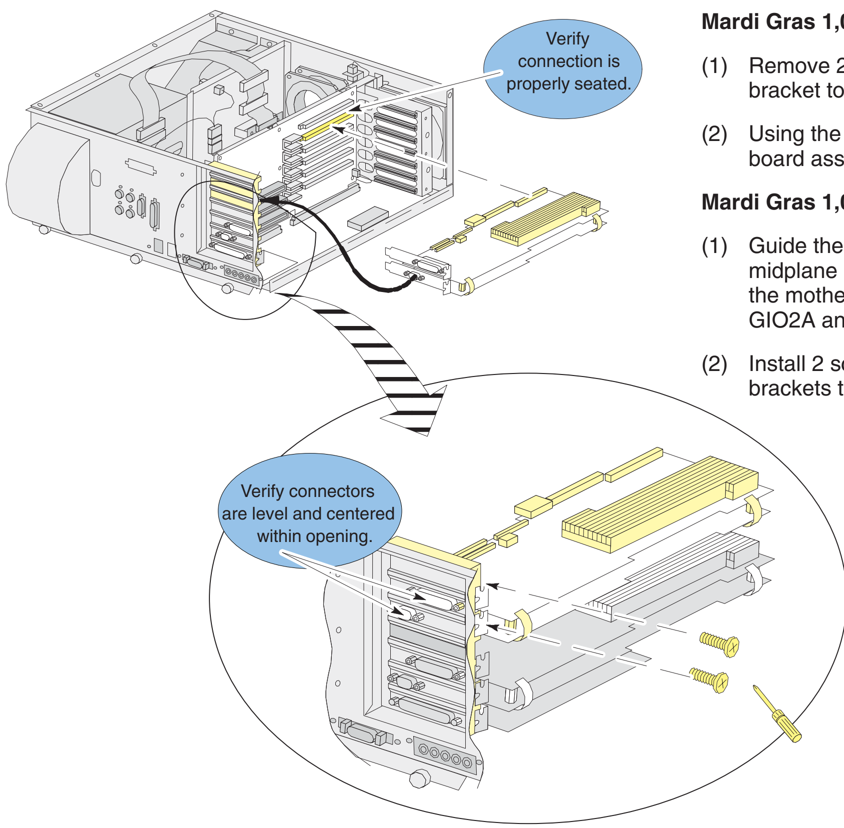
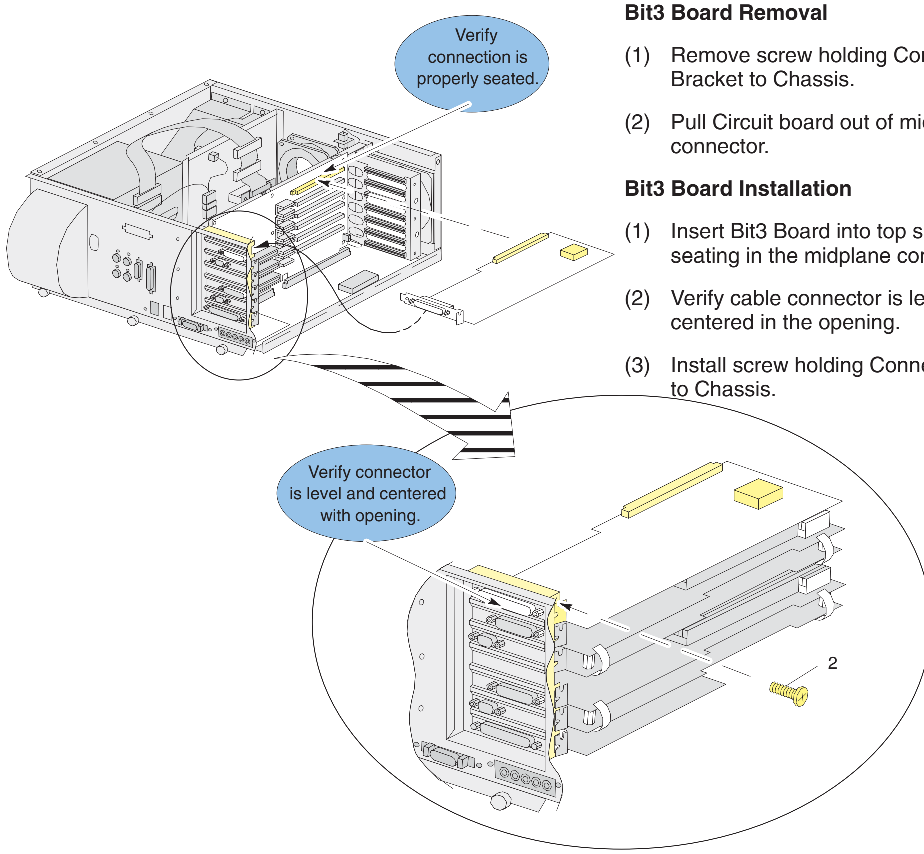


Figure 8-66 Mardi Gras 1,0 Removal and Installation

2.2.4.7 Bit3 (GIO64) Board Replacement, 2124215-2



Bit3 Board Removal

(1) Remove screw holding Connector Bracket to Chassis.

(2) Pull Circuit board out of midplane connector.

Bit3 Board Installation

(1) Insert Bit3 Board into top slot and verify seating in the midplane connector.

(2) Verify cable connector is level and centered in the opening.

(3) Install screw holding Connector Bracket to Chassis.

Figure 8-67 SGI Bit3 Board Removal/Installation

2.2.4.8 SGI Midplane Board Replacement, 2142755

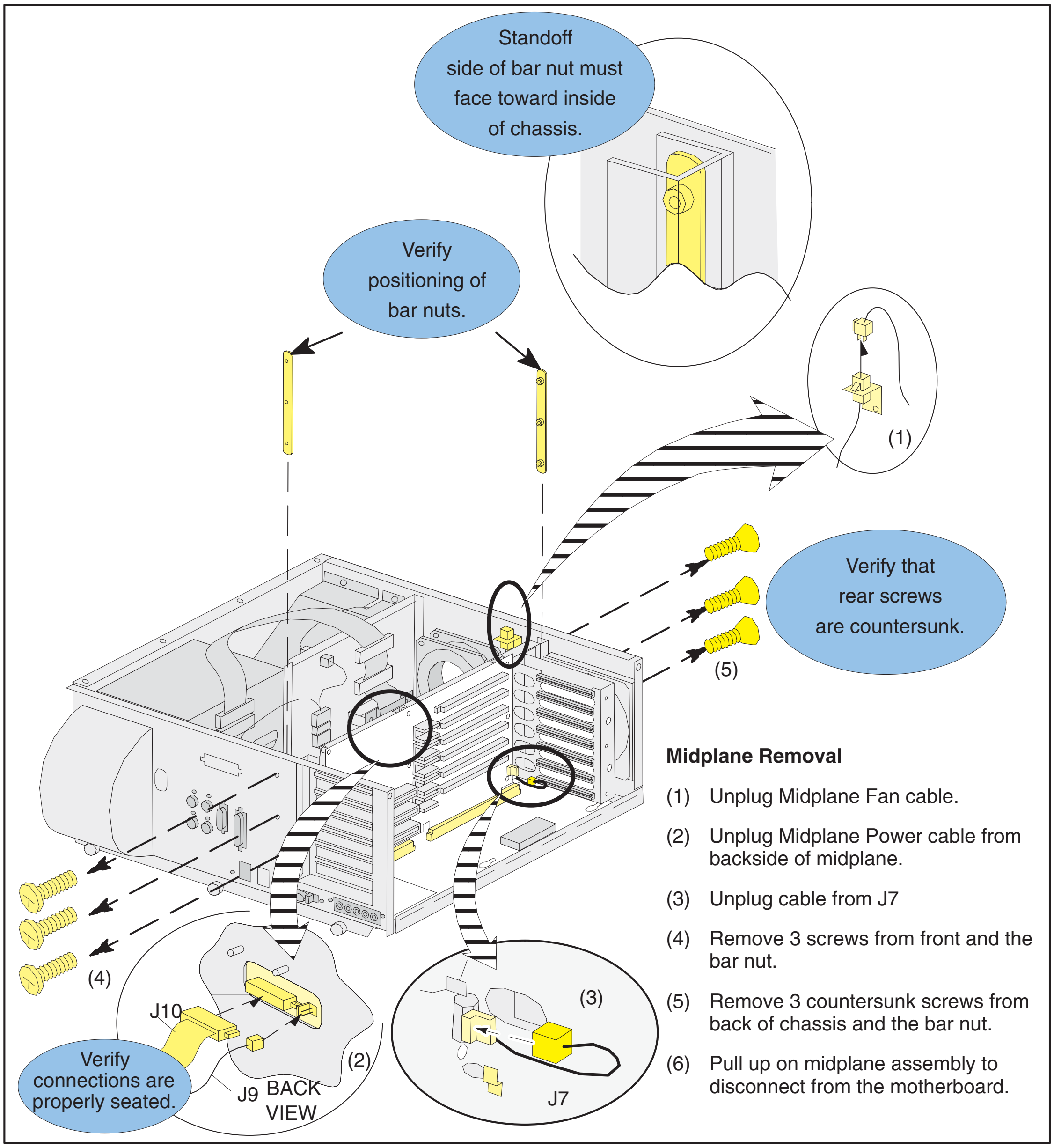


Figure 8-68 SGI Midplane Board

2.2.4.9 Midplane Circuit Board Removal

Midplane Circuit Board Removal

Follow instruction for Opening the SGI Chassis, Removal of expansion boards plugged into the midplane, and removal of the midplane assembly.

(7) Remove 9 screws holding Midplane Circuit Board to the sheet metal assembly.

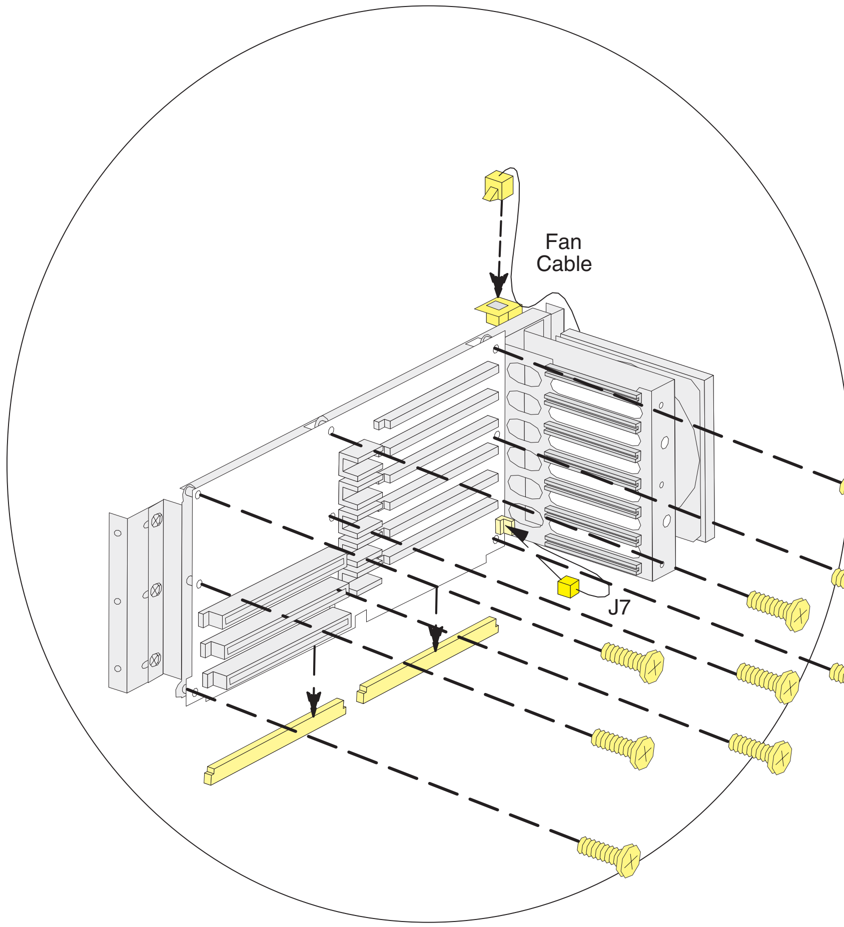


Figure 8-69 Midplane Circuit Board

2.2.4.10 Midplane circuit Board Installation

Midplane Circuit Board Installation

Do not tighten the hardware holding the Midplane Circuit Board to the sheet metal until after the assembly is installed/plugged into the motherboard.

(8) Install, but do not tighten, 9 screws holding Midplane Circuit Board to the sheet metal assembly.

(9) Place the assembly into the SGI chassis at an angle and position the Midplane edge connector over the mating connector on the motherboard.

(10) Carefully ALIGN and INSERT Midplane connector into mother board by pushing firmly on top edge of Midplane Assembly.

(11) Verify Connector Seating.

(12) Re–attach: Fan Cable, J7 and Power Connections to Midplane Assembly.
See page 461.

(13) Install 3 screws and front bar nut. Tighten hardware.

(14) Install 3 countersunk screws and rear bar nut. Tighten hardware.

(15) Temporarily install the MG 1,1 assembly and the Bit3 boards into the midplane. This will insure that the slot alignment relative to the midplane connectors is correct.

(16) Tighten the 9 screws that hold the Midplane Circuit Board to the Midplane Sheet Metal.

It may be necessary to remove the temporarily installed boards to tighten all of the hardware. Tighten the accessible hardware first.

(17) Replace the remaining circuit boards and reassemble chassis cover.

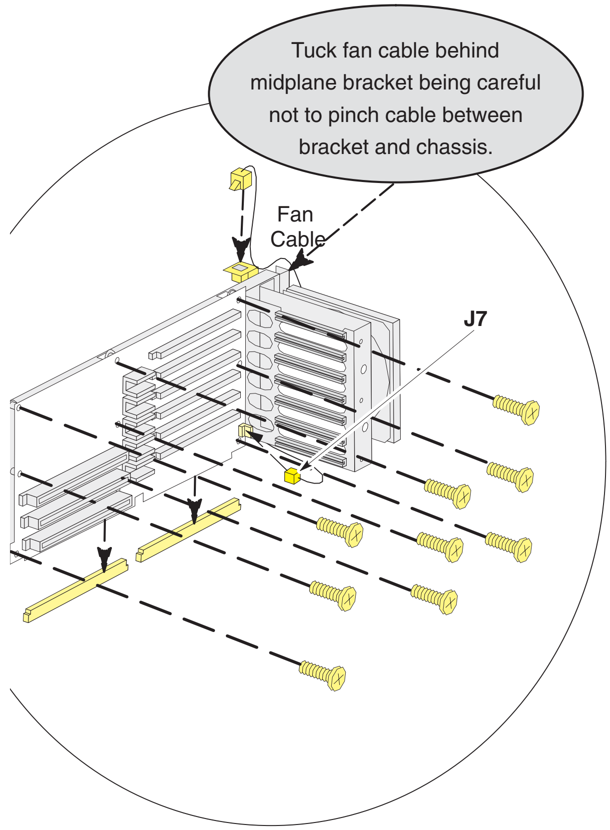


Figure 8-70 Mid Plane Circuit Board Installation

2.2.4.11 IP22 Motherboard, 2115457-14

1.) Follow instructions for removing: chassis cover page 454, Specialx board page 455, MG10 board set page 458, MG 11 board set page 455, and Bit3 board page 459.

2.) Remove the Midplane Assembly, follow the procedures in Figure 8-71 through Figure 8-82 that follow.

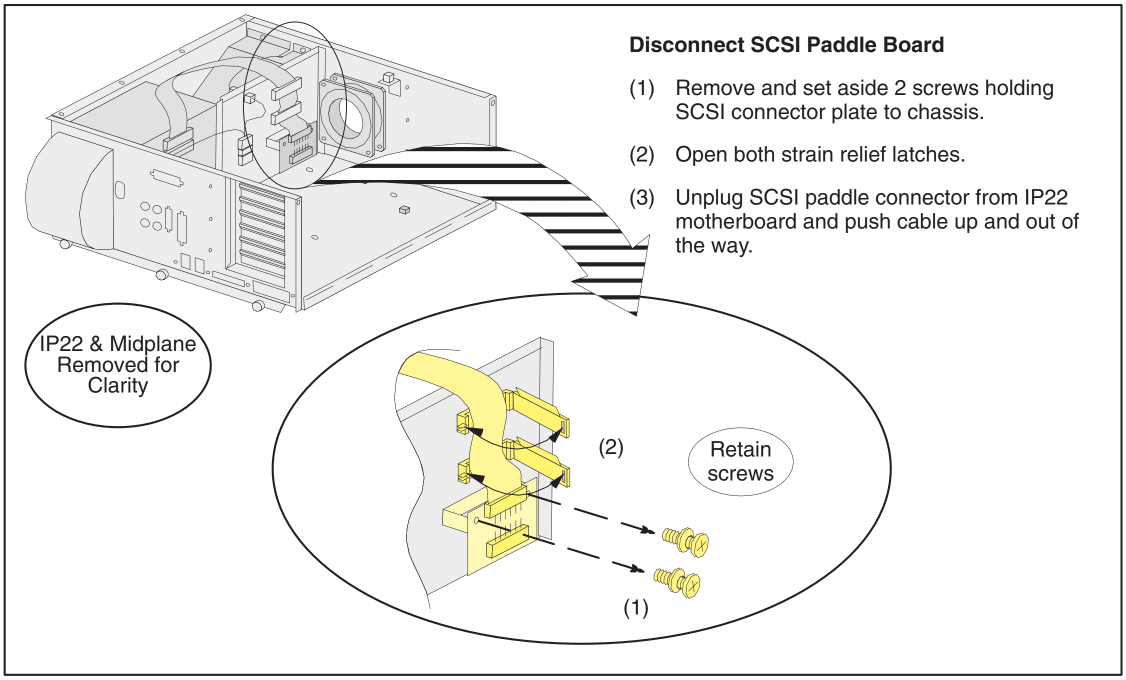


Figure 8-71 Disconnect SCSI Paddle Board

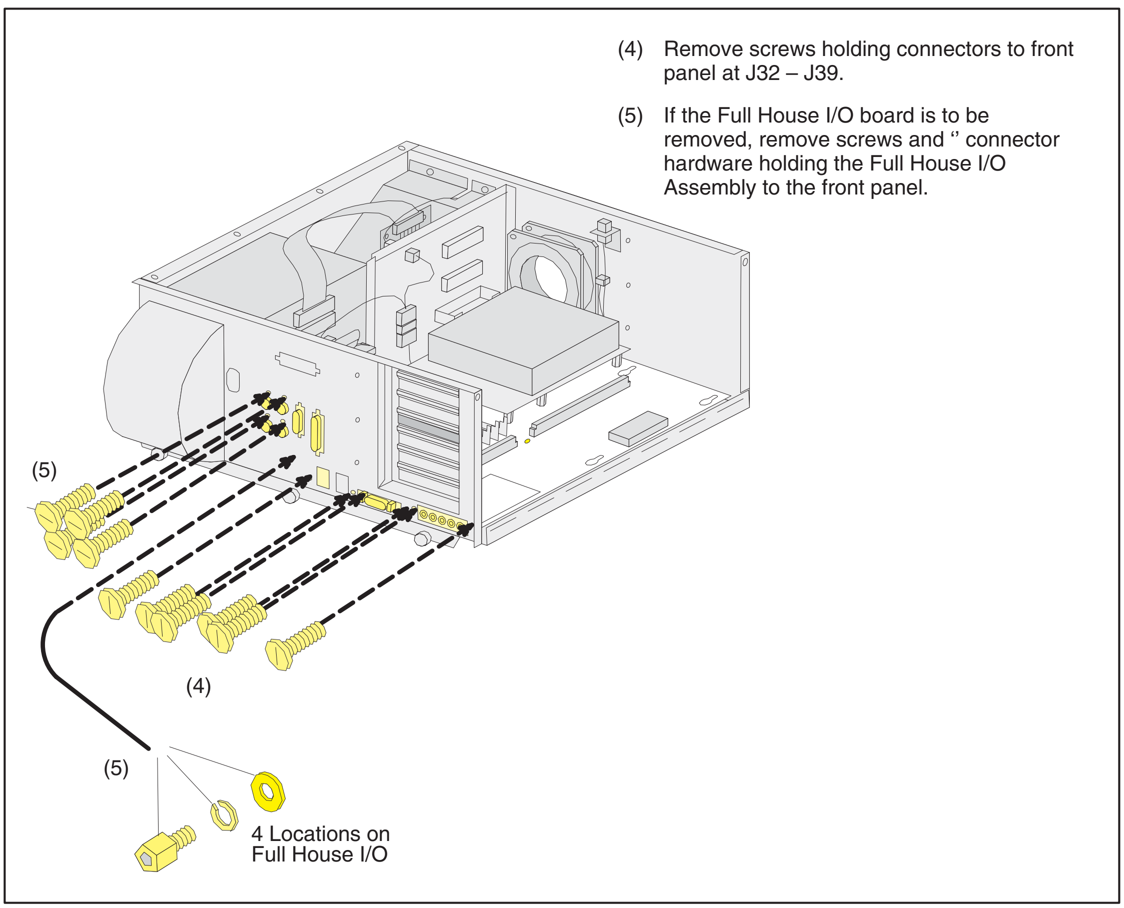


Figure 8-72 Remove IP22 Front Panel Hardware

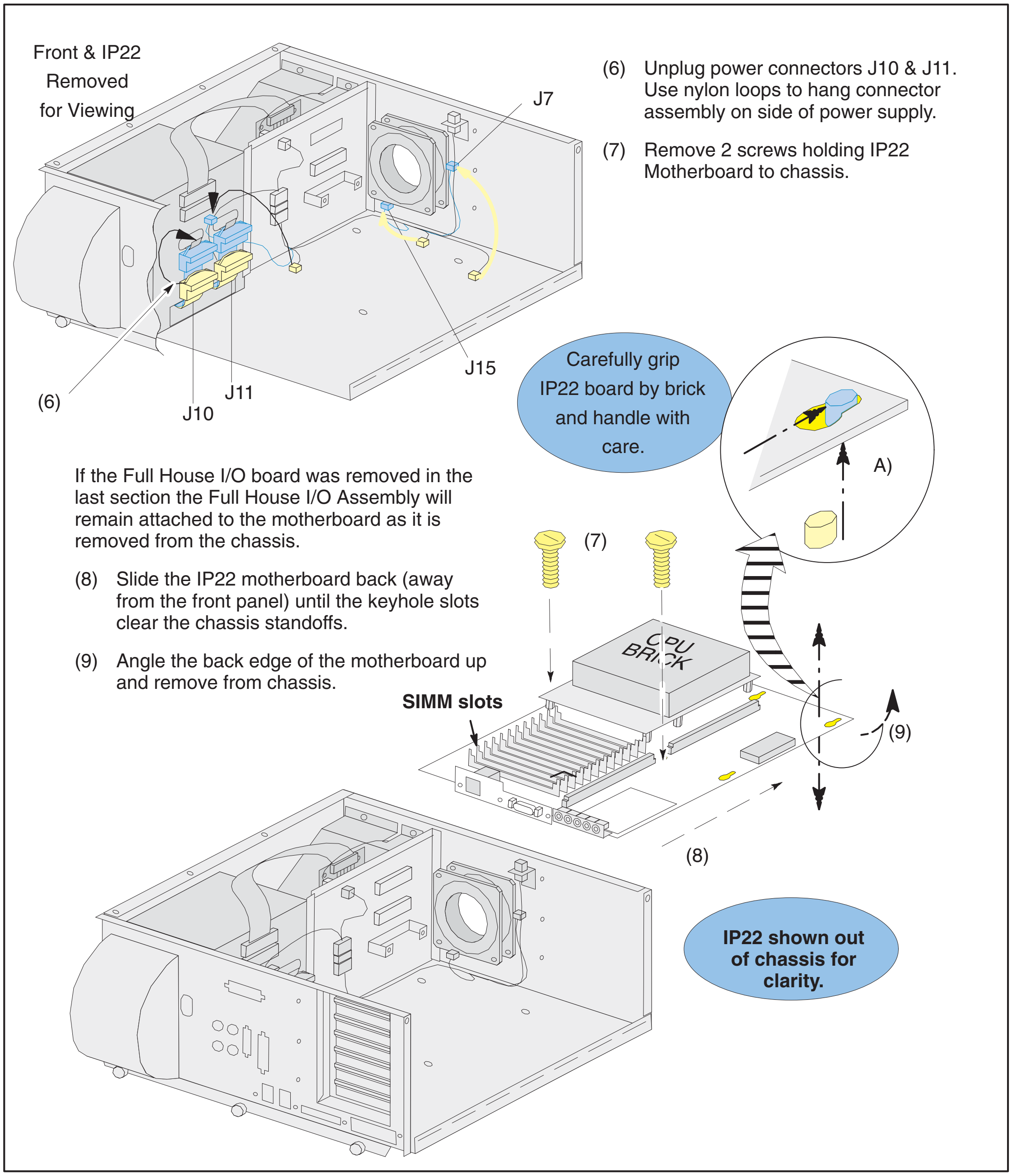


Figure 8-73 Remove IP22 From Chassis

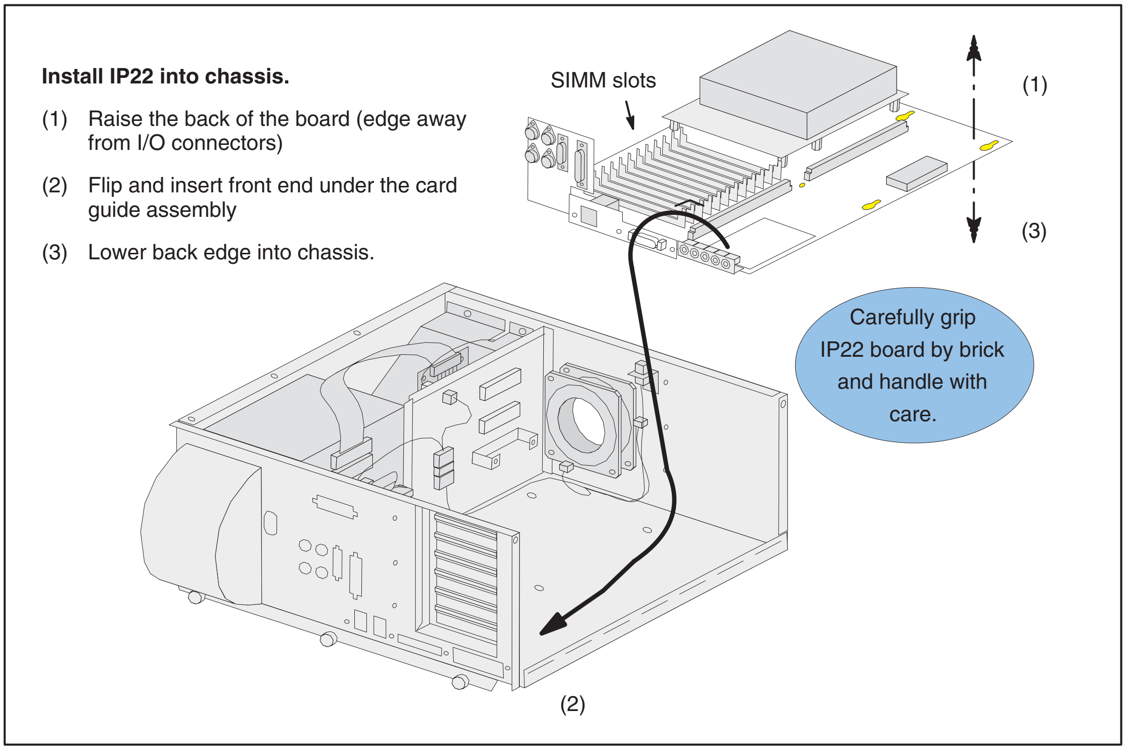


Figure 8-74 Install IP22 Motherboard into Chassis

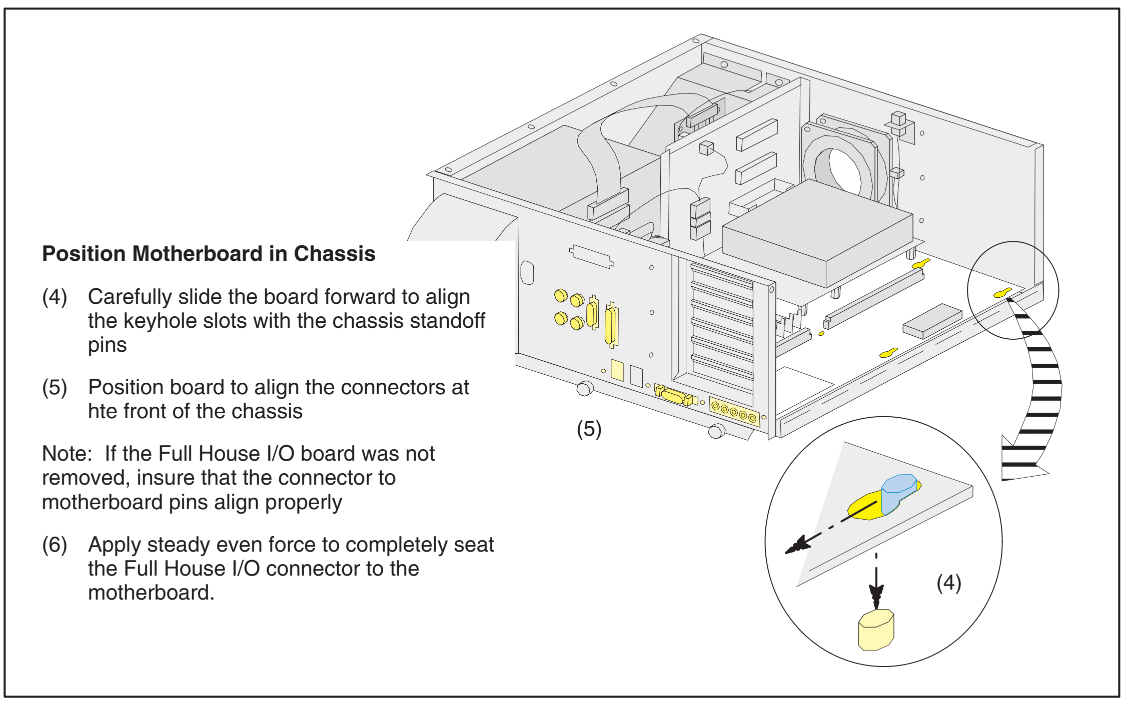


Figure 8-75 Position Motherboard in Chassis

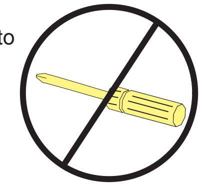
Start IP22 Connector Hardware

Loosely install (DO NOT Tighten) hardware to hold motherboard in chassis.

(7) IP22 into chassis,

(8) front panel J32–39 connectors,

(9) if the Full House I/O was removed, front panel to Full House I/O assembly



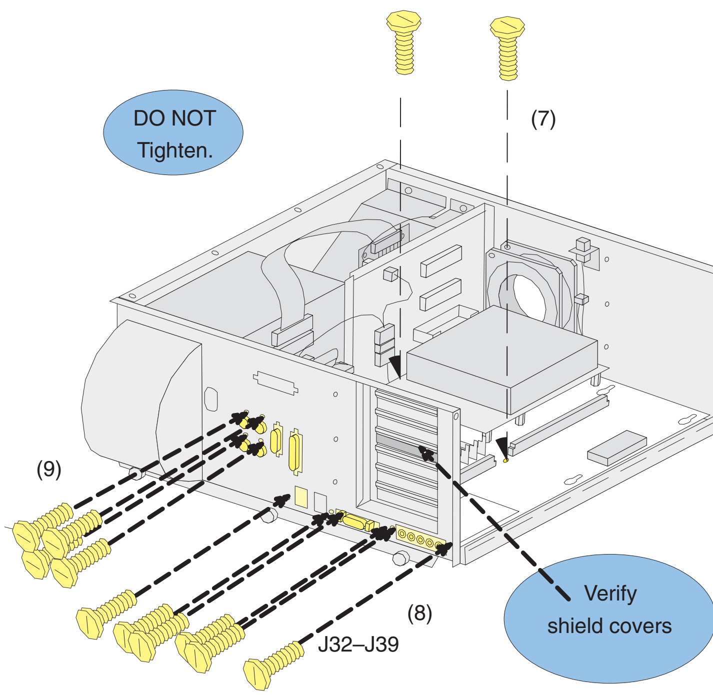


Figure 8-76 Start IP22 Connector Hardware

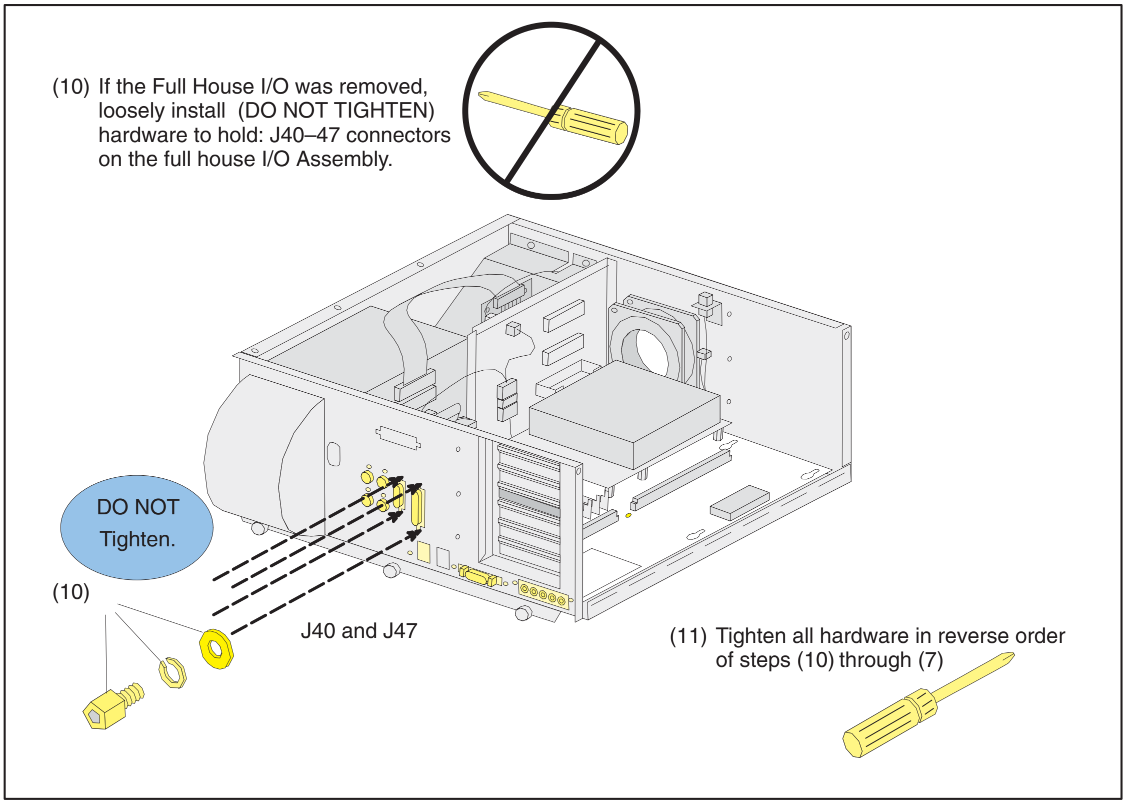


Figure 8-77 Install and Tighten Remaining IP22 Hardware

Connect IP22 Power Supply

(12) Connect J15 at rear of IP22 motherboard.

(13) Install Power Supply Paddle Board,

(14) and Power Supply Connectors to Paddle Board,

(15) and J105 cable to Power Supply Paddle Board.

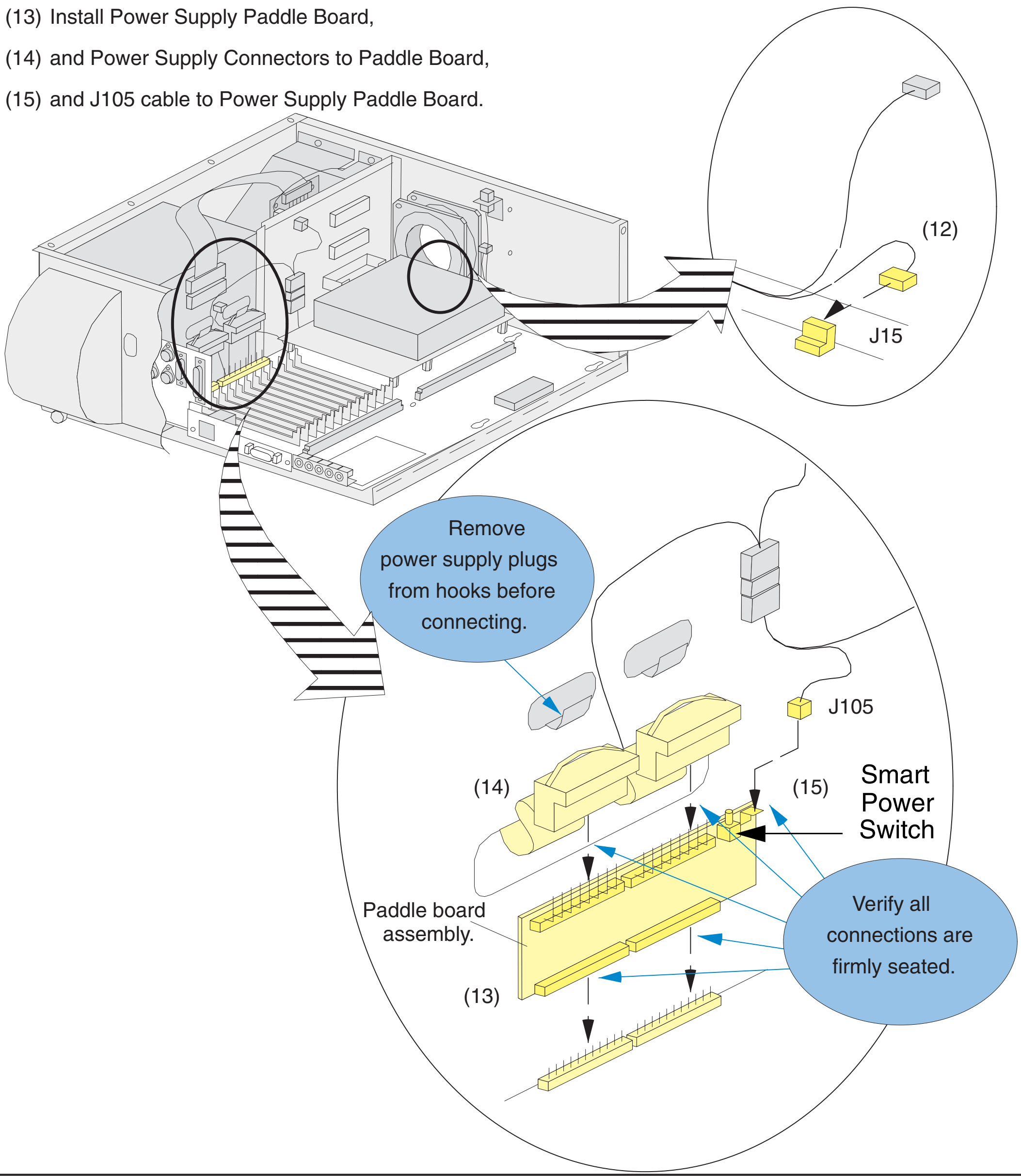


Figure 8-78 Install IP22 Power Connections

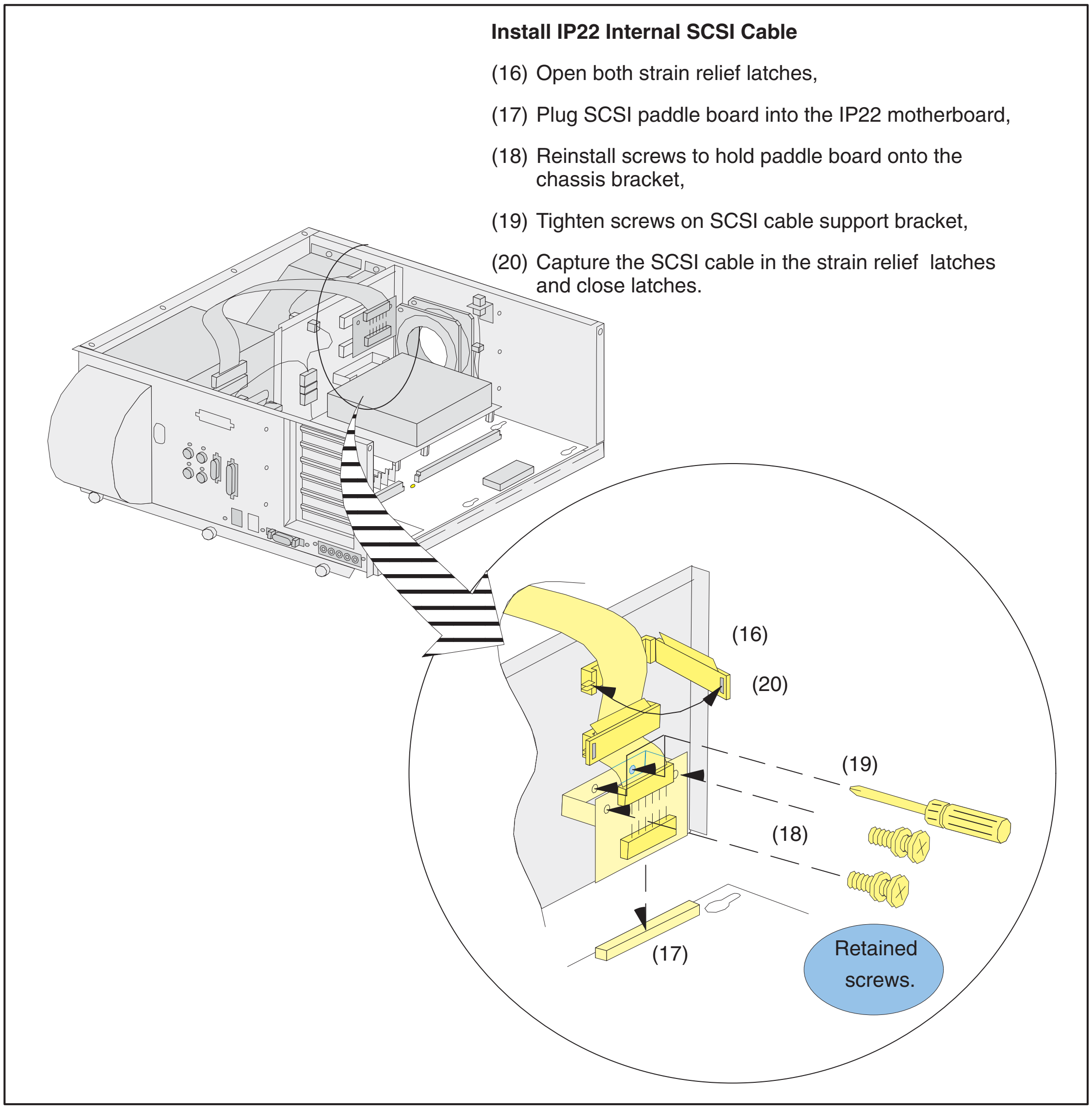


Figure 8-79 Install IP22 SCSI Cable

(21) Carefully align connectors at bottom of midplane, gently, but firmly push down to seat the connections.

(22) Reconnect fan cable to bulkhead and tuck wire behind fan.

(23) Tuck fan cable behind midplane bracket being careful not to pinch cable between bracket and chassis.

(24) Connect J7 plug.

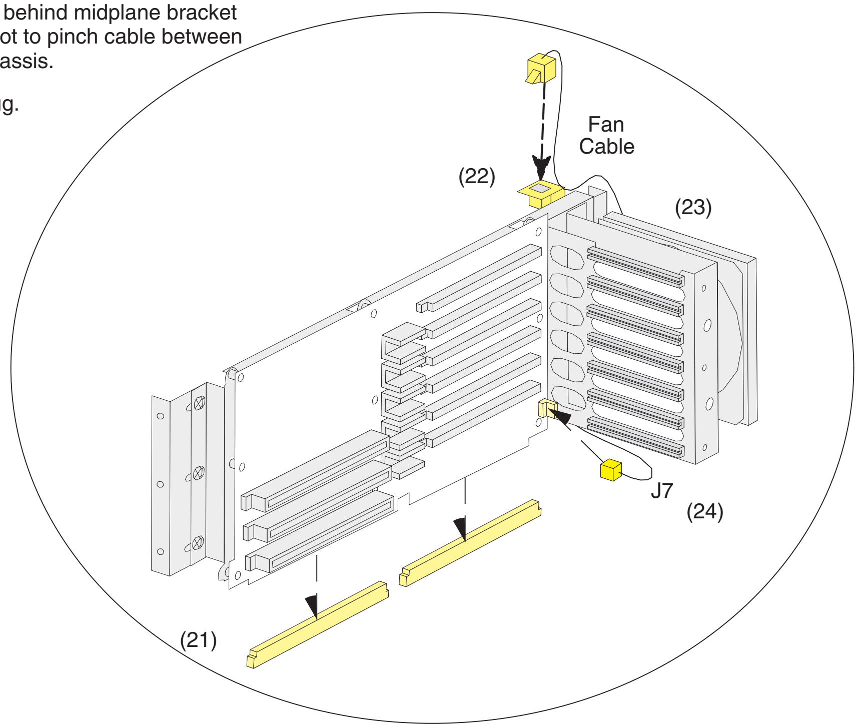


Figure 8-80 Re-Install Midplane Assembly

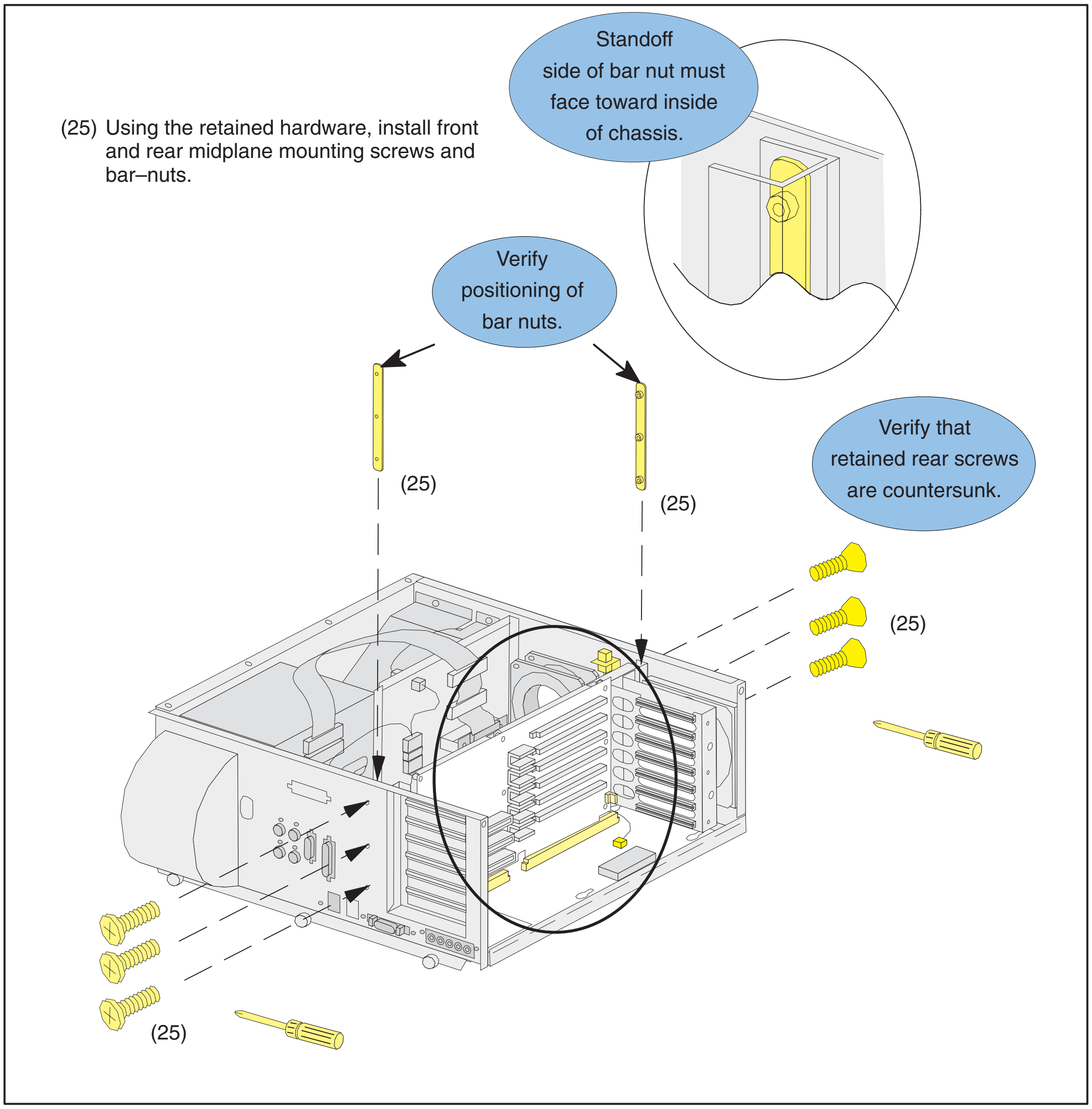


Figure 8-81 Re-Install Midplane Hardware

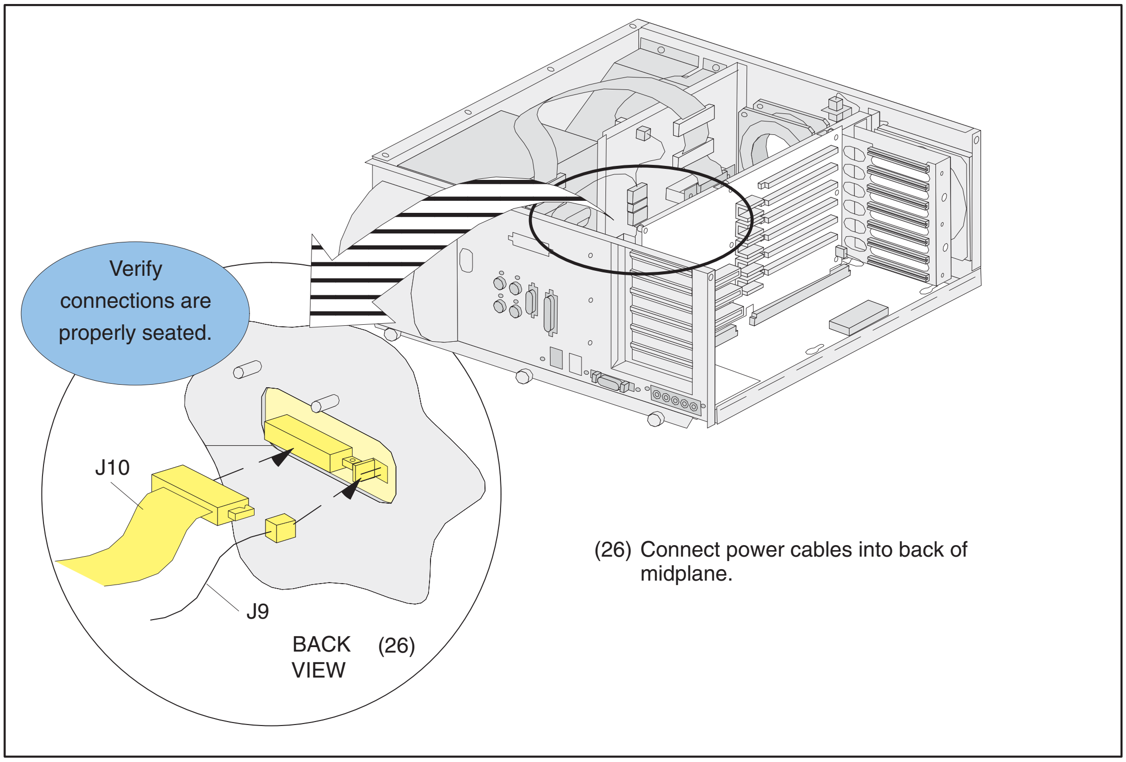


Figure 8-82 Re-Connect Midplane Power Connections

Note: Remember to Initialize Power

3.) After replacing the IP22 motherboard and re-connecting power; press the Smart Power Switch located on the power supply paddle board. See Figure 8-78. The paddle board is small circuit board that interfaces between the PS cable and motherboard. The green card cage LED should now be "on" and the fans operating. If you are experiencing difficulties, see Section 2.2.3.2 on page 428 for troubleshooting information.

4.) Verify the Indigo2 environment EPROMS are set properly. See Section 2.2.3.4 on page 433.

5.) After the system has booted to Unix, run the `setdate` command. This sets the OC date and synchronize the SBC.

2.3 DASM

2.3.1 DASM and Indigo2 (Only) Booting

When troubleshooting Indigo2 boot problems, always try turning off the DASM if the system complains that it can't find the boot disk/partition. There are Indigo2 PROM parameters that MUST be set properly in order to boot when a DASM is installed. Normally, these parameters will be set, but if an IP22 motherboard is swapped, you MUST set them yourself.

From the Indigo2 Command Monitor (firmware) prompt, use **`printenv`** to check for the following parameters:

```
> SystemPartition=scsi(1)disk(1)rdisk(0)partition(8)
> OSLoadPartition=scsi(1)disk(1)rdisk(0)partition(0)
> OSLoader=sash
> OSLoadFilename=/unix
```

Note: Only if any of these are NOT set, then set it using the **`setenv`** command from the Command Monitor (firmware) prompt (>) EXACTLY AS SHOWN:

```
> setenv SystemPartition scsi(1)disk(1)rdisk(0)partition(8)
> setenv OSLoadPartition scsi(1)disk(1)rdisk(0)partition(0)
> setenv OSLoader sash
> setenv OSLoadFilename /unix
```

The parameters will be stored in non-volatile memory "forever" or until the IP22 is swapped out.

It is also extremely important that the CT/i DASM (either VDB or LCAM) SCSI ID jumpers be set to SCSI ID 1 for proper booting and filming. Note that the digital LCAM used on CT/i is basically the same type as used on AW (except SCSI ID).

Note: The analog DASM/VDB used on CT/i (46-269566P2) has new firmware for CT/i performance and it is NOT the same DASM/VDB as used on AW (46-269566P1). Using an AW DASM/VDB on CT/i will result in mixed and missed film images.

WHERE FILMING ERRORS ARE LOCATED

To investigate a filming problem, look at the following logs:

```
/usr/g/service/log/gessy*.log
/var/adm/SYSLOG*
/usr/g/ctuser/logfiles/prslog
```

2.3.2 DASM Diagnostics

The DASM runs a power up **self-test** as well as an idle test loop (heartbeat) when on. See the manual that comes with all DASMs for more info on LED error status and heartbeat indications. When the DASM is failing, its two middle two LEDs flash an error code after all LEDs are momentarily flashed ON.

There is an application utility called "`showdasm`" that can be run from any shell to check basic communications with the DASM by retrieving its configuration. Note however that while there are active filming jobs, `showdasm` *will fail* with an "open failure" because the DASM device is opened exclusively by the filming print filter/manager.

A **SCSIbus0 reset popup ALERT** message is a clear indication of a physical DASM problem/ failure. This SCSIbus0 channel is dedicated to the DASM. The components in this chain include:

- DASM
- SCSI cable (SGI carrier to DASM)
- SCSI terminator module (on DASM)
- LED/switch/SCSI PWA inside SGI carrier
- SCSIbus0 ribbon cable inside SGI carrier
- ribbon-to-IP22 PWA inside SGI carrier
- IP22 motherboard (contains SCSI controller/termination

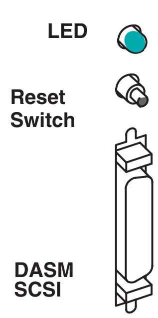


Figure 8-83 DASM LED, Reset Switch and SCSI Connector

Also, sometimes after a filming and/or **SCSIbus problem**/error, the DASM device can be confused and/or out of synchronization with the host SCSI driver and/or platform DASM manager. Usually a second or third attempt at running '**`showdasm`**' will re-synchronize SCSI communications.

While the **Analog DASM** is in its idle test/loop or when an image has been sent to the DASM, the Video Output should have either a continuously changing pattern or the last image sent. This may be checked for the Analog DASM by connecting a short piece of coaxial cable from the DASM Analog Video Output connector to the Green Video input on one of the display monitors, after disconnecting the MG Video Input cables.

2.3.3 DASM Specifications

Note: HOST CONTROL IS REQUIRED TO FILM WITH CT/i.

Camera installation FEs who are unable to successfully setup a camera on CT/i should contact their regional technical support group. CT/i has basically the same filming interface specifications as the GE Medical Systems Advantage Windows workstation, either analog (DASM/VDB) or digital (DASM/LCAM).

Note: The ANALOG (VDB) video output specs and serial interface were specified from, and are the same as, the Genesis interface output specs. This does NOT apply to the CT/i DIGITAL (LCAM) filming interface.

DASM/LCAM HOST CONTROL SERIAL LINK (DIGITAL DASM ONLY)

RS232 serial host control interface, 25-pin D-type connector

- pin 2 (TX)
- pin 3 (RX)
- pin 7 (GND)

Note: A null modem cable may be required (reverses pins 2–3) between some cameras

- baud rate = 1200
- start bits = 1
- stop bits =1
- parity = even
- end of message = CR
- protocol = ACK/NACK (3M M952)

2.3.4 DASM Status File

The LCAM Status file is used for ALARMS, ERRORS and other messages from the laser camera. Here are the error codes from `lc_msg_data.h`:

```
/*.......................str..................,num,sev*/
/* Status codes */
  ""                                             ,0,0,
  "1 Camera Is Ready"                            ,1,0,
  "2 Acquiring an image"                         ,2,0,
  "3 Opening the magazines"                      ,3,0,
  "4 Removing a film from supply magazine"       ,4,0,
  "5 Moving film to exposure area"               ,5,0,
  "6 Exposing film (no other operations can be performed)",6,0,
  "7 Closing the magazines"                      ,7,0,
  "8 Moving film to film processor"              ,8,0,
  "9 Unassigned status code"                     ,9,0,
/* Recoverable Alarm codes */
```

```
  "10 Supply Magazine Empty"          ,PRS_MEDIA_SUPPLY_EMPTY,ERR_FATAL,
  "11 Receive Magazine Full"          ,PRS_MEDIA_RECEIVE_FULL,ERR_FATAL,
  "12 Supply Magazine Full"          ,PRS_MEDIA_SUPPLY_MISSING,ERR_FATAL,
  "13 Receive Magazine Missing"
,PRS_MEDIA_RECEIVE_MISSING,ERR_FATAL,
  "14 Supply Drawer Open"              ,PRS_MEDIA_SUPPLY_OPEN,ERR_FATAL,
  "15 Receive Drawer Open"            ,PRS_MEDIA_RECEIVE_OPEN,ERR_FATAL,
  "16 Top Cover Open"                   ,PRS_TOP_COVER_OPEN,ERR_FATAL,
  "17 Film Processor Not Ready"
,PRS_FILM_PROCESSOR_NOT_READY,ERR_FATAL,
  "18 Docking Unit Not Ready"
,PRS_DOCKING_UNIT_NOT_READY,ERR_FATAL,
  "19 Unassigned alarm detected"
,PRS_UNDEFINED_ALARM_CODE_19,ERR_FATAL,
  "20 Film Transport Error..."          ,PRS_FIRST_FEED_ERROR,ERR_FATAL,
  "92 Camera Interface On Line ?"
,PRS_CAMERA_MMU_NO_RESPONSE,ERR_FATAL,
  "99 MMU timer started (952)"          ,99,0,
/* Status codes */
  "200 Camera Interface On Line ?"
,PRS_DASM_COMM_ERROR,ERR_FATAL,
  "201 Can't allocate camera after 15 mn
Trying...",PRS_PRINTER_BUSY_TIMEOUT,0,
  "202 Your Film Was Queued Trying To Allocate
Camera...",PRS_FILM_QUEUED,0,
  "203 Film Low..."                             ,PRS_MEDIA_SUPPLY_LOW,0,
  "204 Print Paused..."                         ,PRS_PRINT_PAUSED,0,
  "207 Camera Is busy at This Time Print
Paused...Restart",PRS_PRINTER_BUSY,0,
 "208 Can't Process Print Request at This Time Print Paused...Restart"
,PRS_BAD_PARAM,ERR_FATAL,
  "209 Time_out Print ...Restart"            ,PRS_PRINT_CYCLE_TIMEOUT,0,
  "210 Failed acquire "                     ,PRS_FAILED_ACQUIRE,ERR_FATAL,
  "212 Unsupported Format"                      ,212,0,
  "301 OK"                                      ,PRS_STATUS_OK,0,
  "Unknown Error returned"                      ,LAST_LC_MSG,0
```

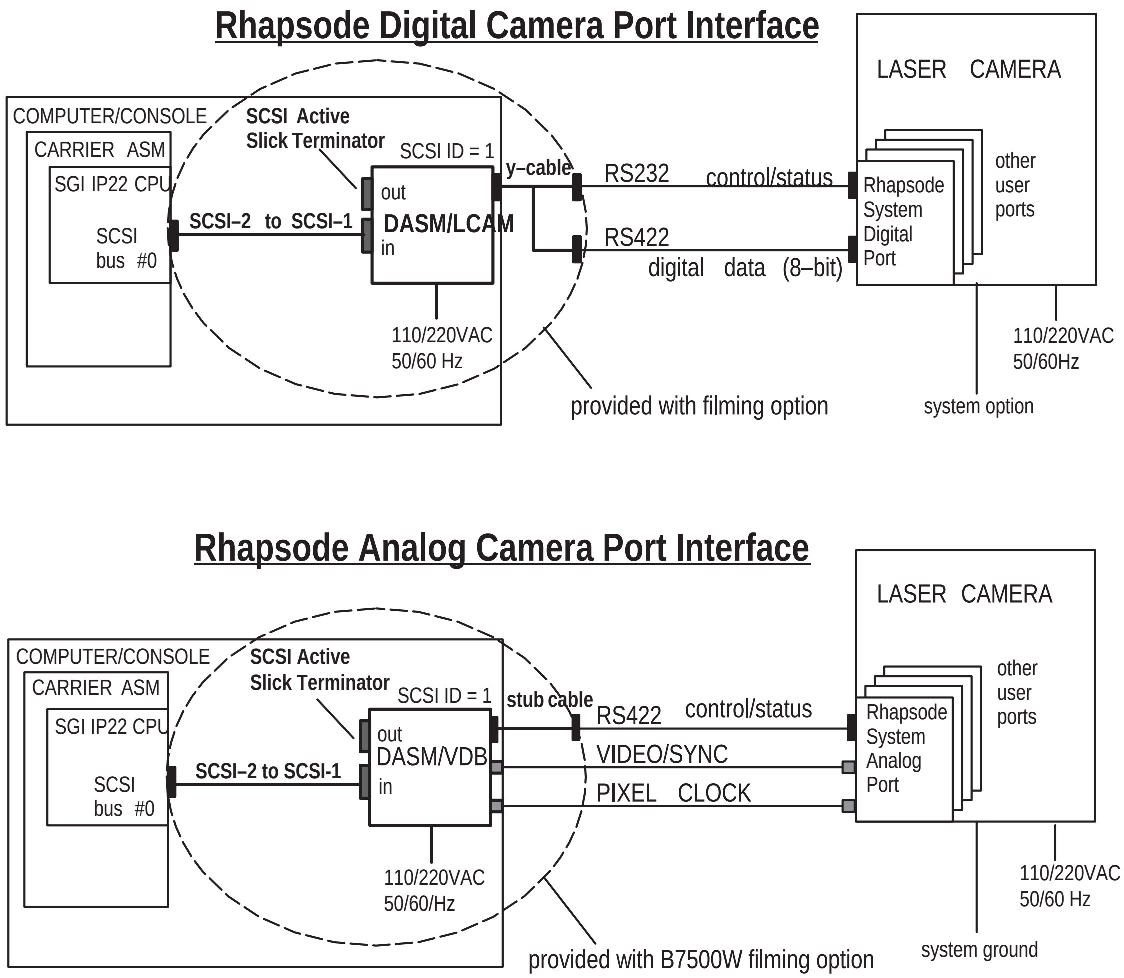
2.3.5 DASM/LCAM Image Data Interface

RS422/RS485 8-bit digital image data interface, 37-pin D-type connector (3M M952 defacto-industry standard digital interface)

- pixels: 512
- lines: 512
- bits/pixel: 8
- protocol: 3M M952

Note: The gray scale reference bar option at the left of the filmed images is *NOT* supported by the digital filming interface.

2.3.6 DASM Interfaces



***IMPORTANT NOTE: DO NOT USE THE 46–269566P1 DASM/VDB ON CT/i OR MIXED OR MISSED IMAGES MAY RESULT ON FILMS!**

Figure 8-84 Digital and Analog DASM Interfaces

2.3.7 DASM Jumpers

JUMPERS

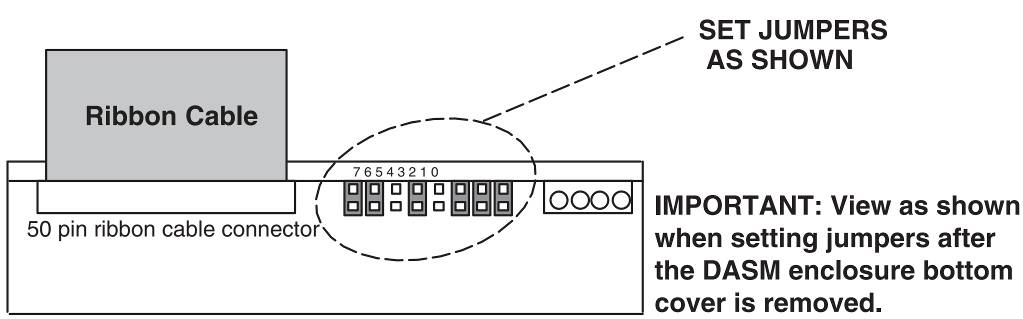


Figure 8-85 DASM Jumpers

2.3.8 DASM Video

DASM TIMING CHARACTERISTICS

ANALOG DASM VIDEO TIMING CHARACTERISTIC	60 HZ	50 HZ
pixel frequency:	24.192 Mhz	24.192 Mhz
pixel period:	41.336 nsec	41.336 nsec
horiz line freq:	33.6 Khz	33.6 Khz
horiz line width:	720 pixels	720 pixels
horiz active:	544 pixels	544 pixels
horiz blanking:	176 pixels	176 pixels
horiz front porch:	26 pixels	26 pixels
horiz sync:	76 pixels	76 pixels
horiz back porch:	74 pixels	74 pixels
vert frame freq:	60 Hz	50 Hz
vert frame time:	560 lines	672 lines
vert active:	524 lines	524 lines
vert blanking:	36 lines	148 lines
vert sync:	3 lines	3 lines
vert back porch:	30 lines	86 lines
vert front porch:	3 lines	59 lines
scanning format:	non-interlaced	non-interlaced

Table 8-11 Analog DASM Video Timing Specifications

DASM DISPLAY FORMATS

ANALOG DASM VIDEO DISPLAY FORMAT	
visible field:	544 pixels by 524 lines
image field:	512 pixels by 512 lines
grayscale field:	32 pixels by 16 level gray bar on left side of image
grayscale:	software selectable on/off
grayscale off value:	0 (black)
initial grayscale:	255 (white) at upper left corner
border field:	12 lines at bottom of visible field
border field value:	any 8-bit value, software programmable

Table 8-12 Analog DASM Video Display Formats

DASM SERIAL PORTS

ANALOG DASM HOST COMMUNICATIONS/CONTROL SERIAL PORT (ANALOG INTERFACE ONLY*)	
interface:	RS422
25D conn pinout:	pin 8 (RX+), pin 21 (RX-), pin 9 (TX+), pin 22 (TX-), pin 7 (GND)
baud rate:	1200 baud
word length:	8 bit, 1 start bit, 1 stop bit
parity:	even
type:	asynchronous

Table 8-13 Analog DASM Host Communications/Control Ports

Note: The CT/i digital DASM/LCAM serial control is standard RS232 on pins 2, 3, and 7. Some cameras may require a NULL MODEM cable and/or adapter.

2.3.9 Filming Interface Specifications (Video & Serial)

ANALOG DASM VIDEO OUTPUT

ANALOG DASM VIDEO OUTPUT (MEASURED INTO 75 OHMS AT BNC OUTPUT)	
amplitude:	1 volt peak-to-peak
video:	0.643V +/-10%
setup:	0.071V +/-10%
sync:	0.286V +/-10%
DAC resolution	8 bits
diff linearity:	+/- 1 LSB max
glitch area:	80 picovolt-seconds max, for any step size
rise/fall times:	> 10 nsec, 10%–90%
FS settling time:	7.5 nsec typical to 1 LSB
transfer func:	guaranteed monotonic
noise level:	> 5.0 millivolt peak-to-peak, combined sync/async noise
DC offset:	+/- 1VDC referenced to ground

Table 8-14 DASM Video Output Specifications

ANALOG DASM PIXEL CLOCK OUTPUT

ANALOG DASM PIXEL CLOCK OUTPUT	
logic family:	F series TTL
output low level:	0.8VDC max
output high level:	2.0VDC min
output period:	41.336 nsecs +/-10%
transition times:	10 nsec max, 10%-90%

Table 8-15 Analog DAS Pixel Clock Output

2.3.10 DASM Diagnostics

```
diagnostic(s): (OC) hinv, scsistat, showdasm, clrsp, rqs, rsp
error log(s): (OC)
```

2.3.10.1 DASM LEDs

DASM green LEDs viewed from front of DASM and air vents at bottom.

Note: The “RDY” and “XFR” LED's only exist on the analog VDB DASM.

```
-----------------------------------------

          o RDY
         o XFR
                    o    o    o    o
                   PWR  CPU  SCSI PIF
-----------------------------------------
DASM air inlet vents
-----------------------------------------
```

- PWR - on whenever DASM power applied (+5VDC)
- CPU - flashes idle heartbeat at 1 CPS or indicates CPU activity
- SCSI - flashes when OC and DASM communicate over the SCSIbus
- PIF - flashes when the DASM and camera communicate over the serial port
- RDY - analog VDB only, indicates an image is ready to be “grabbed” by the camera video/ analog input port
- XFR - analog VDB only, indicates an image is being “grabbed” by the camera video/analog input port

Make sure the DASM power is applied (green power LED) and that the DASM power up self-test completes successfully (flashing green CPU LED indicates idle heartbeat).
On analog VDB DASM only, the “RDY” and “XFR” LED's should toggle back and forth when filming is running. This toggling indicates that film sheet images are being output by the DASM (“RDY”) and then captured by the camera video/analog input port (“XFR”).

2.3.10.2 Checking DASM SCSIbus connection and DASM operation

A.) Use 'hinv' to check that the DASM was present at the last OC bootup.

Note: The DASM looks like a disk drive to the Irix OS.

DASM LINE FROM 'hinv' OUTPUT:

(other output)

Disk drive: unit 1 on SCSI controller 1

(other output)

B.) Use 'scsistat' to perform a “live” SCSIbus probe for the DASM device.

Note: Analog VDB and digital LCAM shown separately and the DASM firmware revisions should be as shown for CT/i & CT/i Pro.

ANALOG VDB LINE FROM '**scsistat**' OUTPUT:

(other output)

```
Device 1 1  Disk     CDA      DASM-VDB          FW Rev: 1.0e
```

(other output)

DIGITAL LCAM LINE FROM '**scsistat**' OUTPUT:

(other output)

```
Device 0 1  Disk      ANALOGIC DASM-LCAM-3M     FW Rev: 1.3
```

(other output)

C.) Use '**showdasm**' to perform an extended inquiry from the DASM device

Note: You must 'root' with 'ctuser' environment as shown below and the filming queue MUST be empty or fully paused or the '**showdasm**' will fail.

```
{ctuser@rhapby18}[1] showdasm
Could not initialize_scsi status = ffffffff
{ctuser@rhapby18}[2] su
Password:

{ctuser@rhapby18}[1] showdasm
Vendor: CDA  Device: DASM-VDB
Pif software rev: 1.0e  Krnl_rev: 2.1j
DRAM size: 1MB  SRAM size: 32KB I/O blocks: 2048  block size: 512
SCSI ID: 1   CMDBLK addr: 200000 Baud: 1200  RS232 ctl reg: hex 8e
Eprom checksum: hex 0038f90f       Internal checksum: hex 0038f770
RS232 Disabled DBUG Disabled           Power-on RAM tests Disabled
{ctuser@rhapby18}[2]
```

Any SCSIbus or device related errors will be logged to the shell window you're using, the OC console shell window, and will also be saved in the OC /var/adm/SYSLOG* Irix system log. The DASM device is /dev/dasm1 which is linked to /dev/scsi/sc1d1l0 (Octane).

If the above functions work, the DASM power, SCSIbus connections, and the host side DASM operation is all working properly. If not, you may have a problem with 'reconfig' (camera option, DASM type, etc.), SCSI cabling, or the DASM (it's usually NOT the DASM). Make sure you '**su**' from the 'ctuser' shell and that the filming queue is empty or fully paused or the 'scsistat' will show “EXCLUSIVELY OPEN” for the DASM line and the '**showdasm**' will fail to open the DASM device due to incorrect device permissions and environment variables.

2.3.10.3 Checking the DASM VDB serial port and video outputs:

Note: THE FILMING QUEUE MUST BE EMPTY OR FULLY PAUSED FOR THESE PROCEDURES TO WORK. YOU SHOULD OPEN A SHELL AS 'CTUSER' AND THEN 'SU' TO BECOME 'ROOT' SO YOU HAVE THE CORRECT COMMAND SEARCHPATHS AND DEVICE PERMISSIONS FOR THESE TESTS.

- DASM/VDB RS422 25-pin D-Type Socket Connector
- Jumper Pin 8 (RX+) To Pin 9 (TX+) For Loopback Test ONLY
- Jumper Pin 21 (RX-) To Pin 22 (TX-) For Loopback Test ONLY
- VDB RS422 LOOPBACK TEST

1.) CONNECT loopback jumpers on correct DASM pins per above.

2.) CLEAR the DASM camera response buffer status:
```
{ctuser@rhap1}[1] clrsp /dev/dasm1
```

3.) DISPLAY the DASM response buffer status and see that it has been cleared (line 110 data all 0's and ascii all dots)
```
{ctuser@rhap1}[2] rsp /dev/dasm1
Byte 0: 02 IDLE
Byte 128: 00
index: 0000 hex   buffer size: 00f0 hex
110: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
```

4.) CONFIRM that status line 110 has been CLEARED

5.) ISSUE a single RQS (Request for camera status) command out of the DASM serial port.

```
{ctuser@rhap1}[3] rqs /dev/dasm1
```

6.) DISPLAY the DASM response buffer status

Note: The “IDLE ERROR” message below is normal since the DASM does not expect an “RQS” in response to an “RQS” but this “error” is OK while running this test.

```
{ctuser@rhap1}[4] rsp /dev/dasm1
Byte 0: 06 IDLE  ERROR
Byte 128: b0
index: 000f hex   buffer size: 00f0 hex
110: 69 52 51 53 2d 52 51 53 0d 26 0a 0b b0 0c 06 00  iRQS-iRQS.&..._
```

7.) INSPECT for TWO RQS entries at LINE 110

a.) If only ONE RQS, the loopback FAILED: DASM serial port, external serial cable, or camera serial port may be bad or parameters set incorrectly

b.) If TWO RQS entries, the loopback PASSED: There is nothing wrong with your DASM

8.) REPEAT the entire test sequence to verify the results

If the test FAILS, make sure that the jumpers are installed on the right pins and making good connections!

9.) MOVE loopback JUMPERS to the appropriate interface CABLE connector pins or sockets (you’ll need to know which pins at the camera end of the cable carry the signals shown above to jumper them). Repeat steps 1-8.

10.) RESET the DASM to a known state by cycling DASM power when done testing because the loopback may confuse the DASM firmware.

ANALOG VDB VIDEO OUTPUT TEST

Using a length of 75 ohm video coaxial cable with male BNC's at each end, you can connect the analog DASM/VDB video output BNC to the GREEN input BNC of either RGB CRT display (you MUST remove the RED and BLUE BNC's from the CRT input during this test). You can then use the DASM power up grey scale test screens and/or the last filmed image (if still in the DASM video buffer) to view the DASM/VDB video output quality directly on the RGB CRT.

2.3.10.4 Checking the DASM LCAM serial port and video outputs:

Note: THE FILMING QUEUE MUST BE EMPTY OR FULLY PAUSED FOR THESE PROCEDURES TO WORK. YOU SHOULD OPEN A SHELL AS 'CTUSER' AND THEN 'SU' TO BECOME 'ROOT' SO YOU HAVE THE CORRECT COMMAND SEARCHPATHS AND DEVICE PERMISSIONS FOR THESE TESTS.

TESTS

- DASM/LCAM RS232 25-pin D-Type Pin Connector (located at the end of the LCAM Y-cable assembly)
- JUMPER PIN 2 (TX) to PIN 3 (RX)
- LCAM RS232 LOOPBACK TEST (same as DASM/VDB serial loopback test above EXCEPT for serial pins and jumpers on the 25-pin Y-cable instead)
- LCAM DIGITAL IMAGE DATA OUTPUT (unfortunately, there is no good way to test this output in the field at this time except to connect to digital camera input)

2.4 Autovoice/Intercom

`diagnostic(s): (OC) hinv, dinc`

`error log(s): ()C) /var/adm/SYSLOG*`

The CT/i Dynaplan (scan status baragraph) and Autovoice control (SBC-STC autovoice play status handshakes) is now done over the OC-SBC-STC LAN connections and is no longer run over a serial port (as of software release 5.3). If OC **`'ping SBC'`** and SBC **`'ping STC'`** are successful, then the hardware path is functional.

2.4.1 Intercom/Interconnect Boards

2.4.1.1 Functional Overview

Autovoice messages are sent simultaneously to the gantry and table speakers and the console speaker, except when the talk button is depressed. The gantry microphone is disabled during autovoice. The user at the operator's console is always be able to talk to the patient via the intercom. The patient on the table can hear the operator at the console when the operator depresses the talk button. The autovoice message is disconnected when the talk button is depressed.

The user at the operator's console is always be able to hear the patient on the table through the intercom, even at the lowest volume setting, except when the talk button is depressed and while autovoice is being played. A volume control knob is provided at the console to regulate the sound volume of Autovoice messages played back to the gantry or table. The auto voice volume at the console is controlled by a graphical user interface tool on the computer screen. In addition, two other volume control knobs for the intercom system shall be provided to adjust sound level for the speakers at the gantry/table and the console.

Computer based training audio playback at the console only. Volume is controlled on screen. Computer based training audio cannot be played if the talk button is depressed or when autovoice is playing.

2.4.1.2 Console Intercom Board (2204382)

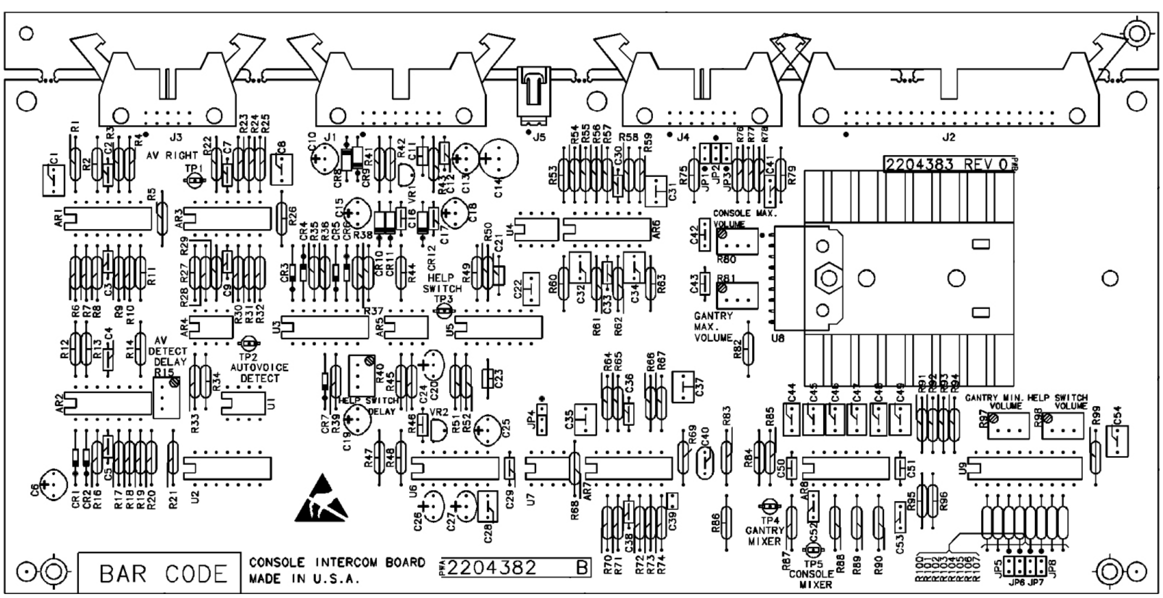


Figure 8-86 Console Intercom Board (2204382) Physical Layout

Reference Number	Description	Resistor Value Setting*
R80	Console Max Volume	1.5 k ohms (CW 7.5)
R81	Gantry Max Volume	1.5 k ohms (CW 7.5)
R97	Gantry Mic Min Volume	300 ohms (CCW 5.5)
R15	AutoVoice Detect Delay	150 k ohms (CCW 5)
R98	Help Switch Volume	500 k ohms (CCW 13.5)
R40	Help Switch Hold Delay	250 k ohms (CCW 8)
* The number of turns, as listed in parentheses, is a rough gauge only.		

Table 8-16 Intercom Potentiometer Values (2204382)

Description	Reference Number	Jumper Setting
Gantry Mic 1 Bias	JP1	OPEN
Gantry Mic 1 Ground	JP2	CLOSED
Gantry Mic 2 Bias	JP3	OPEN
AutoVoice Record	JP4	Pin 2 and 3 Connected
Gantry Gain (Mini)	JP5	OPEN
Gantry Gain (Rhapsode)	JP6	OPEN
Gantry Gain (Vectra)	JP7	CLOSED

Table 8-17 Intercom Jumper Settings (2204382)

2.4.1.3 Console Intercom Board (2167014)

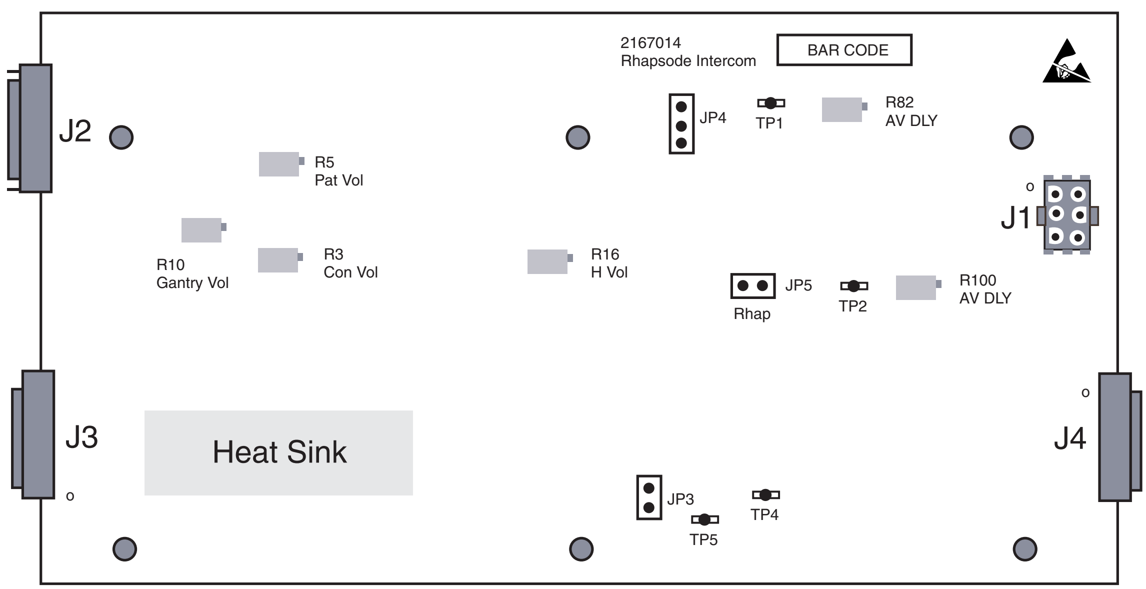


Figure 8-87 Console Intercom Board (2167014) Physical Layout

Theory of Operation

GANTRY MICROPHONE INPUT

Refer to schematic 2167014 sheet 3 to supplement the following discussion.

Patient voice signals from the Gantry Intercom circuit are supplied to the Console Intercom board using differential line driver amplifiers to help eliminate common mode noise, which may be induced in the interconnection cables. To complete the signal to noise improvement process, the differential voice signals must be received by a differential input amplifier which discriminates against any common mode signal.

Two sections of module U14 are used for impedance matching to the inputs at J3-11 and J3-30 and for establishing a local ground reference. Module U8 is the differential amplifier which provides for conversion from differential mode to single ended mode. Module U8 provides more than 70 dB of differential to common mode signal discrimination. A third section of U14 provides a gain of 2.2 and impedance matching to drive the High side of the console 5 k ohm volume control potentiometer through J2-11.

AUTOVOICE RIGHT

Refer to schematic 2167014 sheet 4 and 6 to supplement the following discussion.

Auto Voice signals at J4-3 are processed by three sections of U17 with unity gain to drive TP2 and the switching matrix, as found on schematic page 6 as signal AV_RIGHT.

AUTOVOICE LEFT

Refer to schematic 2167014 sheet 5 and 6 to supplement the following discussion.

Auto Voice signals at J4-2 are processed by three sections of U18, with unity gain to drive the High side of the 5k ohm Auto Voice Volume control through J2-5 as signal AVVOLPOT. A section of U11 provides a gain of 3.2 as signal AV_VOL to sheet 5 of the schematic.

The AV_VOL signal is fed into an active peak detector circuit formed by two sections of U11. The discharge time constant is adjusted by potentiometer R100. The resulting DC voltage is amplified by a third section of U11 to produce the “No Signal” = -5VDC, or the “600mv Signal” = +5VDC, control signal found at TP3. The DC signal is shifted by U7 to provide 5 volt drive for NOR gate U9 which provides a Low signal OC_CNTL to the switching logic on sheet 6.

CONTROL LOGIC:

Refer to schematic 2167014 sheet 6 to supplement the following discussion.

The normal state is:

- OC_CNTL High on U16 pin 5, which closes the signal path from patients speech into the console power amplifier.
- AV_CNTL High on U16 pin 6, which closes the signal path from the AV_RIGHT autovoice amplifier U17 pin 14 into the console power amplifier.
- CON_CNTL Low on U16 pin 16, which opens the signal path from the “Patient Volume Control” (PATVOLWIPER) into the patient power amplifier.

When AutoVoice appears:

- OC_CNTL goes Low on U16 pin 5 which opens the signal path from patients speech into the console power amplifier.
- AV_CNTL stays High on U16 pin 6 which closes the signal path from the AV_RIGHT autovoice amplifier U17 pin 14 into the con sole power amplifier.
- AV_CNTL High also drives U16 pin 15 High, which closes the signal path from the (AVVOLWIPER) autovoice volume control into the patient power amplifier.

- CON_CNTL Low on U16 pin 16, which opens the signal path from the "Patient Volume Control" (PATVOLWIPER) into the patient power amplifier.

Refer to schematic 2167014 sheet 5 to supplement the following discussion.

When the Talk button is pushed (Schematic Sheet 5). The N.O. Talk_Button signal between J2-3 and J2-4 Is supplied, limit ed and protected by resistors R14, R15,CR6 and CR7 on schematic sheet 5. The signal is then sent to Schmidt trigger U4. The output of U4 drives two sections of NOR gate U10. The out puts from these NOR gates provide drive

- OC_CNTL goes Low: This drives U16 pin 1 low which opens the signal path from the gantry speech amplifier (OCVOLWIPER) into the console power amplifier. This prevents "audio feedback" through the patient microphone.
- AV_CNTL goes Low: This drives U16 pin 6 low, which opens the signal path from the AV_RIGHT autovoice amplifier U17 pin 14 into the console power amplifier. It also drives U16 pin 15 low, which opens the signal path from the (AVVOLWIPER) autovoice volume control into the console power amplifier and the patient power amplifier.
- 3.4.3.3 CON_CNTL goes High: This drives U16 pin 16 High, which closes the signal path from the "Patient Volume Control" (PATVOLWIPER) into the patient power amplifier. This signal is supplied to the top of the PVC by amplifier U2 pin 14.

CONSOLE MICROPHONE PRE AMPLIFIER

Refer to schematic 2167014 sheet 8 to supplement the following discussion.

Voice signals from the Operator Console Microphone are brought to the input of module U2 through J2-14, J2-15 and J2-16. The signal amplitude at J2-15 is multiplied by ten times in amplifier U2 at Pin 8, for input to U13. Microphone Pre-amplifier U13 provides variable signal gain and compression to reduce variation in patient volume as the console operator moves around the console microphone.

Another section of U2 provides impedance matching at Pin 14, to drive the 5k ohm Patient Volume Control through J2-8. A third section of U2 provides impedance matching at Pin 7 for driving Balanced line driver U3 with output on pins on 1 and 8. VR1 is a +5 volt voltage regulator which determines the break point for signal compression in U13.

POWER AMPLIFIER

Refer to schematic 2167014 sheet 6 to supplement the following discussion.

Signals coming from the volume control wipers are switched by U16 and appear as inputs to the power amplifier section formed by U15 and U12. TP4 is connected to the output of U15 pin 1 and provides an opportunity to monitor the voice signals being sent from the patient. TP5 is connected to the output of U15 pin 7 and provides an opportunity to monitor the voice signals coming from the console. Both of these signals are imposed on the input terminals of power amplifier chip U12.

Signal OCSPK from U2 pin 4, drives the console speaker through J2-17. Signal PSPK from U12 pin 6 drives the patient speaker through J2-12.

CONNECTOR TO CONNECTOR FEED-THROUGH

Connectors J2, J3 and J4 provide for interconnection of a number of circuits which have little functional relationship to the intercom feature. These interconnection paths from connector to connector, are continuous conductors which can be checked by continuity measurement.

POWER SUPPLY

Refer to schematic 2167014 sheet 10 to supplement the following discussion.

Power for the board is obtained through connector J1. J1 pins 2 and 3 are connected to Analog ground. Pin 1 is connected to Logic ground. Pin 4 supplies +12 vdc. Pin 5 supplies + 5 vdc. Pin 6 supplies - 12 vdc. Module U1 is a voltage regulator which derives + 6 vdc, for Microphone bias, from the + 12 vdc supply.

Functional Test

PRE-SET POTENTIOMETERS:

- R3 Max (25 turns) CW, < 10 ohms between pot pins 1 and 3
- R5 Max (25 turns) CW, < 10 ohms between pot pins 1 and 3
- R10 Max (25 turns) CW, < 10 ohms between pot pins 1 and 3
- R100 Set to 150K ohms between pot pins 1 and 3

MEASURE SUPPLY CURRENT:

- +5 vdc supply Less than 1 ma
- +12 vdc supply 120 ma +/- 20 ma
- - 12 vdc supply 100 ma +/- 20 ma

ADJUST SUPPLY VOLTAGE TO:

- +5 vdc supply +/- 0.2 vdc (Across CR3)
- +12 vdc supply +/- 0.6 vdc (Across CR2)
- - 12 vdc supply +/- 0.6 vdc (Across CR1)
- +6 vdc regulator +/- 0.4 vdc (Across CR1)
- +5 vdc regulator +/- 0.4 vdc (Across CR34)

LOGIC TESTS:

Refer to schematic 2167014 sheet 5 to supplement the following discussion.

The following table shows the operation of the "Talk Button" logic with all ac signal sources removed.

Push-Button	J2-3	U9-4 (AV_CNTL)	U4-6 (CON_CNTL)	U9-1 (OC_CNTL)
Open	High	High	Low	High
Close	Low	Low	High	Low

Table 8-18 Talk Logic (2167014) on the intercom Board

AUTOVOICE SENSING:

This test confirms the action of a signal level sensing circuit. The test starts with no signal on J4-2. The DC voltage on TP 3 should be more negative than - 5 vdc. The voltage on U9-pin 1 should exceed +3.5 vdc. (Schematic Sheet 5)

Supply a 600 mv +/- 10% peak to peak, 1000 Hz sine wave to J4-2 (Auto Voice Left). The DC voltage on TP 3 should exceed +5vdc. The voltage on U9-pin 1 should be less than +0.25 vdc. Reduce signal level to 240mv+/- 10% peak to peak. J4-2 will change to negative in 2+/- 0.5 seconds.

GAIN TESTS:

The following gain tests are achieved by supplying 1000 Hz, 100 mv peak to peak Sine wave at the specified input with respect to analog ground. Output voltages are measured at the specified connector pin.

- J3-11 (Gantry Microphone Pre-Amplifier) to J2-11 gain = 2.1 +/- 10%
- J3-30 (Gantry Microphone Pre-Amplifier) to J2-11 gain = 2.1 +/- 10%
- J4-2 (Auto Voice Left to Vol Control) J2-5 Gain = 1 +/- 10%
- J4-3 (Auto Voice Right) Gain to TP 2 = 1 +/- 10%
- J4-3 (AutoVoice Right) to TP 4 gain =.9 +/- 10%, when AV_CNTL is High
- J4-3 (AutoVoice Right) to J2-17 gain = 25 +/- 10%, when AV_CNTL is High
- J2-6 (AVVOL.WIPER) to TP 5 gain = 6 +/- 10%, when AV_CNTL is High
- J2-9 (PATVOL.WIPER) to TP 5 gain = 10 +/- 10%, when CON_CNTL is High
- J2-12 (Patient vol control) to TP 4 gain =.33 +/- 10%, when OC_CNTL is High
- J2-12 (Patient vol control) to J2-17 gain = 10 +/- 10%, when OC_CNTL is High
- Adjust Potentiometer R3 to Max CCW position.
- J2-12 (Patient vol control) to J2-17 gain = 3.3 +/- 10%, when OC_CNTL is High

HIGH GAIN TESTS:

The following gain tests are achieved by supplying 1000 Hz, 10 mv peak to peak Sine wave at the specified input.

- J2-9 (PATVOL.WIPER) to J3-12 gain = 300 +/- 10%, when CON_CNTL is High
- J2-6 (AVVOL.WIPER) to J3-12 gain = 190 +/- 10%, when AV_CNTL is High
- Adjust Potentiometer R5 to Max CCW position.
- J2-6 (AVVOL.WIPER) to J3-12 gain = 60 +/- 10%, when AV_CNTL is High

ALC TEST:

- Supply a 10 mv peak to peak, 1000 Hz sine wave to J2-15. J2-15 to J2-8 gain = 7.5 +/- 20%.
- Supply a 100 mv peak to peak, 1000 Hz sine wave to J2-15. J2-15 to J2-8 gain = 1.5 +/- 20%.

Potentiometer settings

- R3 2.k ohms Console Max Volume
- R5 1.5k ohms Gantry Max Volume
- R10 500 ohms Gantry Min. Volume
- R100 150k ohms AutoVoice Detect delay

2.4.1.4 Console Intercom Board (2117167)

Please see Figure 8-89, Console Intercom Block Diagram on page 489 for an overview of the intercom interfaces and Console Intercom Board (2138289) Physical Layout on page 488 for the intercom board's layout (jumpers, switches, etc.).

OPERATING ENVIRONMENT

The console intercom PCB shall operate over the following range of environmental conditions:

- Temperature:0 degrees C to +40 degrees C
- Temperature Change:5 degrees C/Hr.
- Relative Humidity:10% to 80% (non-condensing)
- Altitude:-30 meters to +2133 meters
- Magnetic Field: 5 Gauss field

NON-OPERATING ENVIRONMENT

The console intercom PCB shall sustain no damage while in the following non-operating environment. (Power-off):

- Temperature: -34 to +50 degrees C
- Relative Humidity: 10%-90% (non condensing)

Console Electrical Interface

MICROPHONE SPECIFICATION

- Impedance 1k ohms
- Voltage range 2 - 10 volts, Current 650 mA
- All microphones are biased at the console.

SPEAKER SPECIFICATION

- Impedance 8 ohms
- Frequency Response 120hz - 12khz
- 5 watts rated power (4 watts on CT Mini)

VOLUME POT SPECIFICATION

- Impedance 5k ohms
- Slide bar noise 47mV

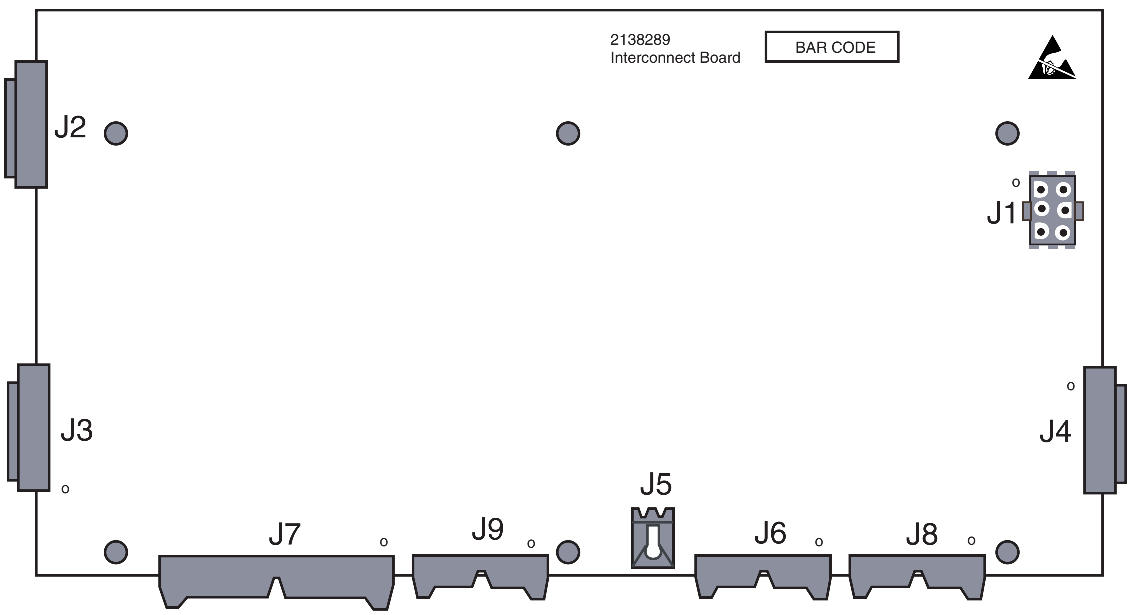


Figure 8-88 Console Intercom Board (2138289) Physical Layout

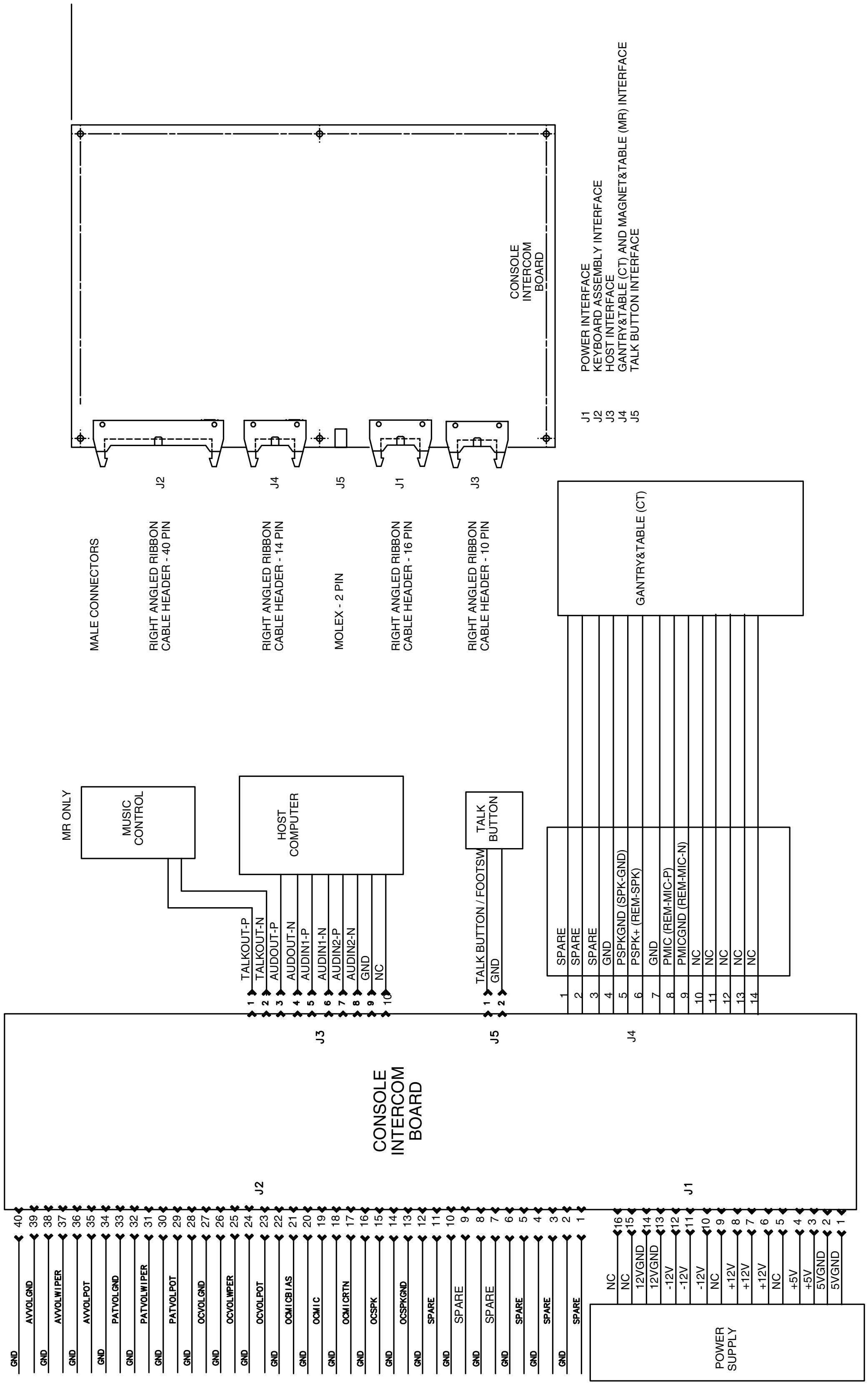


Figure 8-89 Console Intercom Block Diagram

SGI Computer Interface

Additional audio specifications are referenced in the Indigo Audio Technical Report, DM-INDAUD-TR, 1992. The SGI computer interface can be up to 80 feet cable distance.

ANALOG STEREO LINE LEVEL INPUT SPECIFICATION

- Impedance 5k ohms
- Voltage range: 1Vpp to 10Vpp

ANALOG STEREO LINE LEVEL OUTPUT SPECIFICATION

- Impedance 600 ohms nominal
- Voltage range: 6Vpp

Gantry Intercom Interface

The interface to the microphone is specified by the schematic 46-288766-S. The following pin-outs are called out on 46-288766-S. The input signals and pins are defined as follows:

Signal	Pin
REM-MIC-P	19
REM-MIC-N	20
SGND	3
SGND	6
SGND	9
SGND	12
-12VDC	23
+12VDC	18
SGND	24
SGND	25

Table 8-19 Microphone Pin-out

INTERCONNECT, JUMPERS, POTENTIOMETER SPECS

Because several signals require different signal modes, several are supported, Differential or single analog inputs or output is supported through jumpers. These may or may not be user selectable depending on your type of board. An example of how this is implemented can be seen in Figure 8-90.

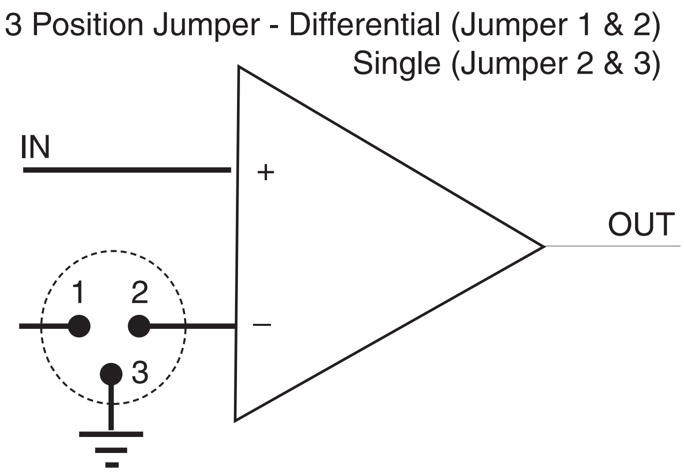


Figure 8-90 Differential and Single Ended I/O Selection

Connector/Signal Tables

J1 CONNECTOR

Pin Number	Signal Name	Source	Signal Type	Description
1, 2	+5GND	Power Supply	GND	Power for Console Intercom Board
3, 4	+5V	Power Supply	Power	Power for Console Intercom Board
5	NC			
6, 7, 8	+12V	Power Supply	Power	Power for Console Intercom Board and Gantry Intercom Board
9	NC			
10, 11, 12	- 12V	Power Supply	Power	Power for Console Intercom Board and Gantry Intercom Board
13, 14	12GND	Power Supply	GND	Power for Console Intercom Board and Gantry Intercom Board
15, 16	NC			

Table 8-20 J1 Connector

J2 CONNECTOR

Pin Number	Signal Name	Source	Signal Type	Description
1	Spare	Keyboard Cable	NC	Spare
2	GND	Keyboard Cable	GND	GND
3	Spare	Keyboard Cable	NC	Spare
4	GND	Keyboard Cable	GND	GND
5	Spare	Keyboard Cable	NC	Spare
6	GND	Keyboard Cable	GND	GND

Table 8-21 J2 Connector

Pin Number	Signal Name	Source	Signal Type	Description
7	Spare	Keyboard Cable	NC	Spare
8	GND	Keyboard Cable	GND	GND
9	Spare	Keyboard Cable	NC	Spare
10	GND	Keyboard Cable	GND	GND
11	Spare	Keyboard Cable	NC	Spare
12	GND	Keyboard Cable	GND	GND
13	OCSPKGND	Keyboard Cable	GND	Intercom OC Speaker Re turn
14	GND	Keyboard Cable	GND	GND
15	OCSPK	Keyboard Cable	Analog	Intercom OC Speaker
16	GND	Keyboard Cable	GND	GND
17	OCMICRTN	Keyboard Cable	Analog	Intercom OC Mic Return
18	GND	Keyboard Cable	GND	GND
19	OCMIC	Keyboard Cable	Analog	Intercom OC Mic
20	GND	Keyboard Cable	GND	GND
21	OCMICBIAS	Keyboard Cable	Analog	Intercom OC Mic Bias
22	GND	Keyboard Cable	GND	GND
23	OCVOLPOT	Keyboard Cable	Analog	Intercom OC Volume Pot
24	GND	Keyboard Cable	GND	GND
25	OCVOLWIPER	Keyboard Cable	Analog	Intercom OC Volume Wiper
26	GND	Keyboard Cable	GND	GND
27	OCVOLGND	Keyboard Cable	GND	Intercom OC Volume Return
28	GND	Keyboard Cable	GND	GND
29	PATVOLPOT	Keyboard Cable	Analog	Intercom Patient Volume Pot
30	GND	Keyboard Cable	GND	GND
31	PATVOLWIPER	Keyboard Cable	Analog	Intercom Patient Volume Wiper
32	GND	Keyboard Cable	GND	GND
33	PATVOLGND	Keyboard Cable	GND	Intercom Patient Volume Return
34	GND	Keyboard Cable	GND	GND
35	AVVOLPOT	Keyboard Cable	Analog	Intercom Autovoice Volume Pot
36	GND	Keyboard Cable	GND	GND
37	AVVOLWIPER	Keyboard Cable	Analog	Intercom Autovoice Volume Wiper
38	GND	Keyboard Cable	GND	GND
39	AVVOLGND	Keyboard Cable	GND	Intercom Autovoice Volume Return
40	GND	Keyboard Cable	GND	GND

Table 8-21 J2 Connector

J3 CONNECTOR

Pin Number	Signal Name	Source (S) Destination (D)	Signal Type	Description
1	TALKOUT-P	MR Music Control Card (D)	NC	Controls sound playback
2	TALKOUT-N	MR Music Control Card (D)	NC	Controls sound playback
3	AUDOUT-P	Computer (D)	Single Analog	Record analog - input to computer
4	AUDOUT-N	Computer (D)	GND	Record analog - input to computer
5	AUDIN1-P	Computer (S)	Single Analog	Left Channel, Gantry/Table/ Magnetic Speakers - out put from computer
6	AUDIN1-N	Computer (S)	GND	Left Channel, Gantry/Table/ Magnetic Speakers - out put from computer
7	AUDIN2-P	Computer (S)-CT	Single Analog	Right Channel, Console Speaker - output from computer - CT
8	AUDIN2-N	Computer (S)-CT	GND	Right Channel, Console Speaker - output from computer - CT
9	GND		GND	Ground
10	NC		NC	
Note: The TALKOUT differential signal will go active when the talk button is pressed or when an audio signal is detected on the AUDIN1 signals.				

Table 8-22 J3 Connector

J4 CONNECTOR

Pin Number	Signal Name	Source (S) Destination (D)	Signal Type	Description
1	Help SW GND		NC	Help SW Ground
2	Help SW		NC	Help SW
3	NC		NC	NC
4	PSPKGND		NC	Speaker
5	PSPKGND	Speaker (D)	gnd	Speaker
6	PSPK	Speaker (D)	signal	Speaker
7	GND	GND	gnd	GND
8	PMIC1	Mic (D)	Diff. Analog	Mic
9	PMICGND1	Mic (D)	Diff. Analog	Mic
10	PMICPWR1	Mic (D)	NC	Power for Mic
11	PMIC2	Mic (D)	NC	Mic
12	PMICGND2	Mic (D)	NC	Mic
13	PMICPWR2	Mic (D)	NC	Power for Mic
14	NC		NC	

Table 8-23 J4 Connector

J5 CONNECTOR

Pin Number	Signal Name	Source (S) Destination (D)	Signal Type	Description
1	TALK BUTTON	Talk Switch	Analog	Talk Switch
2	GND	Talk Switch	Gnd	Talk Switch

Table 8-24 J5 Connector

JUMPER SETTINGS

JUMPER #	DESCRIPTION	SETTING
JP1	GANTRY MIC 1 BIAS	OPEN
JP2	GANTRY MIC 2 BIAS	OPEN
JP3	GANTRY MIC 1 GROUND	OPEN
JP4	AUTOVOICE RECORD	POSITION 1 - PINS 1,2
JP5	GANTRY GAIN RHAPSODE	CLOSED
JP6	GANTRY GAIN VECTRA	OPEN
JP7	GANTRY GAIN CT MINI	OPEN
JP8	HELP SWITCH ENABLE	OPEN

Table 8-25 Jumper Settings

NOMINAL POTENTIOMETER SETTINGS

Reference	Description	Nominal Setting	Turns
R3	Console Max. Volume	1.5K ohm	0 CW
R5	Gantry Max. Volume	1.5K ohm	4 CW
R10	Gantry Min. volume	350 ohm	3 CCW
R16	Help Switch Volume	500K ohm	6 CW
R82	Help Switch Delay	250K ohm	9 CW
R100	A/V Detect Delay	150K ohm	10.25 CW

Table 8-26 NOMINAL POTENTIOMETER SETTINGS

2.4.2 Autovoice/Intercom Volume

Some of the processing for AutoVoice comes from the host's motherboard. If there is an autovoice problem, you may want to interrupt system boot up [Esc], Enter Command Monitor, then type:

```
ide audiofield
```

2.4.3 Intercom Volume Verification

To adjust the Gantry Speaker Volume, adjust the left most volume thumb wheel while speaking into the console microphone.

To adjust the Console Speaker volume, adjust the center volume thumb wheel while speaking into the Gantry microphone.

2.4.4 Autovoice Volume Verification

To adjust the Gantry Speaker volume, adjust the right volume thumb wheel while autovoice is playing, and check the volume for the gantry speaker.

To adjust the Console Speaker Volume, bring up the Autovoice volume control from the Tool chest.

- Adjust the RIGHT Channel volume only, this is the only volume control.
- The LEFT Channel must be kept locked at the maximum.

2.5 Video Monitors

2.5.1 CT/i Display Monitor Characteristics & Timing Parameters

The following tables define the video signal timing for the CT/i image and operator display video outputs. Both channels are 1280 x 1024 RGB color at 76Hz (FUTS Release) or 72HZ (2nd Release and forward), 1 Volt peak-to-peak video at 75 ohms.

CHARACTERISTICS

CT/i Display Monitor Video Characteristics	
Parameter	72 Hz
Active Pixel Format	1280 x 1024
Field/Frame	non-interlaced
Refresh Rate	72.239 Hz
Pixel Clock Freq, Period	129.25 MHz, 7.737 ns
Horizontal Freq, Period	12.998 usec, 1680 pixels
Horizontal Active	9.903 usec, 1280 pixels
Horizontal Front Porch	0.232 usec, 30 pixels
Horizontal Sync	1.083 usec, 140 pixels
Horizontal Back Porch	1.780 usec, 230 pixels
Horizontal Blanking	3.095 usec, 400 pixels
Vertical Freq, Period	13.843 msec, 1065 lines
Vertical Active	13.310 msec, 1024 lines
Vertical Front Porch	38.99 usec, 3 lines
Vertical Sync	38.99 usec, 3 lines
Vertical Back Porch	454.93 usec, 35 lines
Vertical Blanking	532.92 usec, 41 lines
Equalization Pulses	yes (@horiz rate)
Serration Pulses	no

Table 8-27 CT/i Display Monitor Video Characteristics & Timing

VIDEO LEVELS

Video Output	Video Level	Sync Level	Blanking Level
Red	0.714 Vp-p	none	0.054 volts
Blue	0.714 Vp-p	none	0.054 volts
Green	0.714 Vp-p	0.286 volts	0.054 volts

Table 8-28 DASM Red/Green/Blue Output Level Specifications

Note: If the display monitor RGB is tapped off using an RGB “splitter” for any reason, a commercial, high quality, splitter device and good quality (low loss) 75 ohm video cables are required to maintain display monitor and remote destination video quality. Reference remote gray scale monitor option (GEMS B7530RB).

2.5.2 Setting up the Color Monitor

OVERVIEW

The light output from all color monitors is lower than the output from black and white monitors, e.g. HiSpeed/HiLight. For this reason you need to be very careful when setting up the monitor brightness and contrast for CT/i. Initially, the systems are set to factory defaults, but these can be adjusted. Refer to the “HiSpeed CT/i Installation Manual” for details on how to adjust the Brightness and Contrast for these monitors.

The technologist may perceive that the image on the monitor is “softer” than the image on the film, (i.e. they like the film, but they would like the image on the monitor to look like their film in terms of contrast and brightness). By now, you’ve probably guessed that due to the light output of the color monitor, you need to make the adjustment for Brightness and Contrast so that the technologist can see anatomical structure (window width) at the right amount of brightness (window level).

You can type < `confidence` > in a Unix shell then select the monitor icon to have the host help you make some adjustments to the monitor.

2.5.2.1 Sony Trinitron Artifacts (Horizontal Lines)

Due to the Sony Trinitron picture tube design used in the Display Monitors on CT/i, an artifact on the display is seen as two equal distance horizontal lines.

- This artifact will NOT appear when using the Optional B&W image monitor.
- This artifact will NOT appear on films.

2.5.2.2 Phillips monitor built-in adjustments

1.) Press the button that looks like an incomplete diamond.
2.) Press + or - until the function you want to adjust is shown on the screen.
3.) Press the incomplete diamond <> to confirm this choice.
4.) Press + or - until the function's value that you want is shown on the screen. Select RGB to adjust Color Temperature.
5.) Select the up arrow or press the right arrow to store this value and exit.

2.5.2.3 SONY monitor built-in adjustments

For further information on monitor settings, see the Operator’s Manual for the CRT Monitor that was delivered with the monitor, or review the information contained in the System Installation Manual.

1.) Press the menu button (that looks like a monitor).
2.) Press the up or down arrow button until the appropriate menu item is highlighted on the screen.

3.) Press the menu button to confirm this choice.
4.) Press the up arrow or down arrow to select the item to adjust, and then the < or > until the item's value that you want is selected on the screen. Select the COLOR menu to adjust Color Temperature. (6500K is the desired setting.)
5.) Select the menu button twice to store this value and exit

2.6 Keyboard Replacement

To replace the keyboard, power down the console/computer, install the keyboard, then power the console back up.

Note: The keyboard will not work UNTIL power is recycled.

2.7 Serial Expander

```
diagnostic(s): (OC) hinv, scsistat, dinc
error log(s): (OC) /var/adm/SYSLOG*
```

2.7.1 General Serial Expander Information

As of CT/i Pro SW release 5.3, the Central Data Serial Expander is now only used for the trackball (port #4) and the SBC serial link (port #3) and ports #1 and #2 are RS232 spares. The devices are linked by the install/reconfig software as follows:

TRACKBALL:

```
lrwxr-xr-x  1 root   sys 12 Sep 11 11:06 /dev/input/trakb -> /dev/ttyd043
```

SBC SERIAL LINK:

```
lrwxr-xr-x  1 root   sys 12 Sep 11 11:06 /dev/ttya1 -> /dev/ttyd042
```

CENTRAL DATA SERIAL EXPANDER "SPARE" RS232 PORTS #1 AND #2:

`/dev/ttyd040` (spare RS232 serial port #1)

`/dev/ttyd041` (spare RS232 serial port #2)

To use the spares for troubleshooting trackball or SBC link problems, the respective RJ45 serial cable can be moved from the normal serial port to a spare serial port. Use the '**dinc**' utility as shown below to test the device on the new serial port to determine if the device or the port is

the problem (or something else).

When the Octane system boots up, the serial expander device driver will probe for the expander device. If it finds a functioning unit on the SCSIbus, it will post an "attach" message. This message appears on the Octane screen during bootup and in the /var/adm/SYSLOG* as shown below:

```
Mar 31 07:05:10 2A:rhap12 unix: STS: Config device ST-1400B
```

If you do not see the above message at bootup or in the /var/adm/SYSLOG*, the serial expander box is not powered, connected to SCSIbus1, or is faulty. The green LED on the backside of the serial expander box should be solid on. If it is flashing a "code", the power up self test has failed.

The DIP switches on the serial expander box must be set to SCSI ID 4 with termination DISABLED for CT/i Pro.

2.7.2 Using the SGI IRIX HINV Command

The Irix 'hinv' command shows all devices that were detected at bootup only. The serial expander device shows up in the 'hinv' list as follows:

(other output not shown)

```
Comm device: unit 4 on SCSI controller 1
Comm device: unit 4, lun 1 on SCSI controller 1
```

(other output not shown)

2.7.3 Using the GEMS SCSISTAT Command

The GEMS 'scsistat' command issues a *LIVE* SCSI inquiry to both SCSIbus's and all active SCSI devices. All active devices will respond. The Central Data SCSI-serial expander device shows in the 'scsistat' output as follows:

(other output not shown)

```
Device 1 4  Comm      CenData  ST-1400B         FW Rev: V6.4
```

(other output not shown)

If the device does not show up as above, make sure the green DC power LED on the rear of the Central Data module is on solid (a flashing code will indicate a power up self test failure). Also, make sure the SCSIbus cable chain between the Octane external SCSI connector, through the DASM (if present), and to the Central Data module is securely connected.

2.7.4 Using the DINC Serial Utility

The '**dinc**' serial terminal utility software is provided on the CT/i Pro system. The '**dinc**' utility does not require any of the setup files that the uucp '**cu**' function requires. The '**dinc**' utility can be used to confirm Central Data serial port functionality and can also be used for troubleshooting. There are 4 serial ports on the Central Data ST-1400B serial expander as follows:

The `/dev/input/trakb` is linked to `/dev/ttyd043` (Central Data port #4).

The '**cu sbc**' function uses `/dev/ttyd042` (Central Data port #3)

Ports #1 and #2 of the Central Data serial expander are not used.

Example: Using 'dinc' to SBC console

Example of using '**dinc**' to connect to SBC console serial port:

```
{ctuser@engbay27}[1] su
Password:
{ctuser@engbay27}[1] cd /usr/local/STS/dinc
{ctuser@engbay27}[2] dinc
usage: dinc [[-1278ENOhsi] [[baudrate]] [port]
  -1          one stop bit
  -2          two stop bits
  -7          7 bit characters
  -8          8 bit characters
  -E          even parity
  -N          no parity
  -O          odd parity
  -h          enable hardware flow control mode
  -s          disable software flow control
  -i          do not init port on start or reset port on exit

{ctuser@engbay27}[3] dinc -1 -7 -E 9600 /dev/ttyd042
```

```
------ DINC --- port=/dev/ttyd042 ------
  9600 BAUD 7 EVEN 1 SWFC=ON  HWFC=OFF
CAR=OFF DTR=ON  RTS=ON  CTS=OFF DSR=OFF
Type ~? for help.

RP27_sbc0 login: root
Password:
Sep  8 15:41:25 RP27_sbc0 login: ROOT LOGIN console
Last login: Fri Sep  4 16:16:04 from RP27_oc0
SunOS Release 4.1.1_U1 (GOS_IG) #4: Thu Jun 18 15:22:19 CDT 1998
root @ RP27_sbc0 1: exit
root @ RP27_sbc0 2: logout
RP27_sbc0: root logged out on Tue Sep  8 15:41:28 CDT 1998

RP27_sbc0 login: ~.
Closed connection.
```

2.7.5 The SGI IRIX SYSLOG

The Irix SCSI device driver will ALWAYS detect and log an error message in /var/adm/SYSLOG* if there are any SCSIbus or SCSI device hardware errors. If there are no SCSI errors logged here, the SCSIbus and devices are GOOD. Note that attempting to mount/attach/label the MOD without media inserted can result in NORMAL "error" messages in the SYSLOG. SCSI error messages may contain one of the following device ID forms:

`/dev/dsk/dks(X)d(Y)s(Z)` (example: /dev/dsk/dks0d1s0 = system disk root)

`/dev/rdsk/dks(X)d(Y)s(Z)` (example: /dev/rdsk/dks0d1s1 = system disk swap)

`/dev/scsi/sc(X)d(Y)l(Z)` (example: /dev/scsi/sc1d6l0 = CDROM drive)

where,

(X) is the host CPU SCSIbus controller/channel# (0=drive bays,1=external bus)

(Y) is the SCSI device ID# (1=System Disk, 2=Option Disk, 3=Maxoptix MOD,
4=Central Data Serial, 5=Pioneer MOD, 6=CDROM)

(Z) is the device partition# being accessed when the error occurred (0-8)

SCSI device error messages always contain the SCSIbus# and SCSI ID#. Messages will also be posted for "normal" events like trying to access a removable media device without media or with bad media (MOD or CDROM). Messages will also be posted if any SCSI device retries occur. If more than 1 SCSI device has errors, you may have a general SCSIbus problem (check cables, connectors, terminator, device jumpers, DC voltage levels, cooling fans, etc).

2.8 INSITE Modem

```
diagnostic(s): (OC) hinv, dinc
error log(s): (OC) /var/adm/SYSLOG*
```

2.9 Service Key

Use the 'check security' function on the examRxDisplay desktop to check for the presence and validity of the security key. As of software release 5.3 (NexGen second release), the security is now on serial port#2 of the Octane computer (`/dev/servKey -> /dev/ttyd2`).

Section 3.0
Scan Recon Computer

3.1 CT/i Scan Reconstruction Overview

The HiSpeed CT/i Scan Recon Computer (SRC) or Unit (SRU) is located inside the console and incorporates the following hardware and software features:

- Has Motorola MVME 167, VME, 9U, **Single Board Computer** (SBC). It manages the SRU operations to accumulate and preprocess scan data.
- Provides a new Scan Chassis with new **VME back-plane** and power supply very similar to the RP 2.x systems.
- Provides a backward compatible **FEP board** with jumper selectable coaxial or fiber optic DAS data inputs. HiSpeed CT/i uses only the fiber-optic input for DAS data. The FEP or Front End Processor collects the raw data from the DAS, offset corrects and view compresses it for each image.
- Has one **Image Generator (IG) circuit board**. The IG or Image Generator is used to perform convolution and back projection for reconstructing axial or helical images.
- Has **Bit3, High Speed Serial board** for transfer of images from the Scan Chassis to the SGI OC computer and provides the SRU communication interface between the SGI and SBC processors.
- Board Level Diagnostics (BLDs).
- Power-Up tests.
- “IMAGE GEN TEST” that uses the application reconstruction software to process a pre-determined set of reconstructions from “canned” scan data. Instead of an image, the test provides checksum verification of the reconstructions for comparison to expected values.

3.2 About The Scan Reconstruction Subsystem

3.2.1 Scan Chassis (Front View)

Figure 8-91 shows the locations of the Scan Recon Chassis main components. A poorly seated board can cause the system to hang or fail to boot.

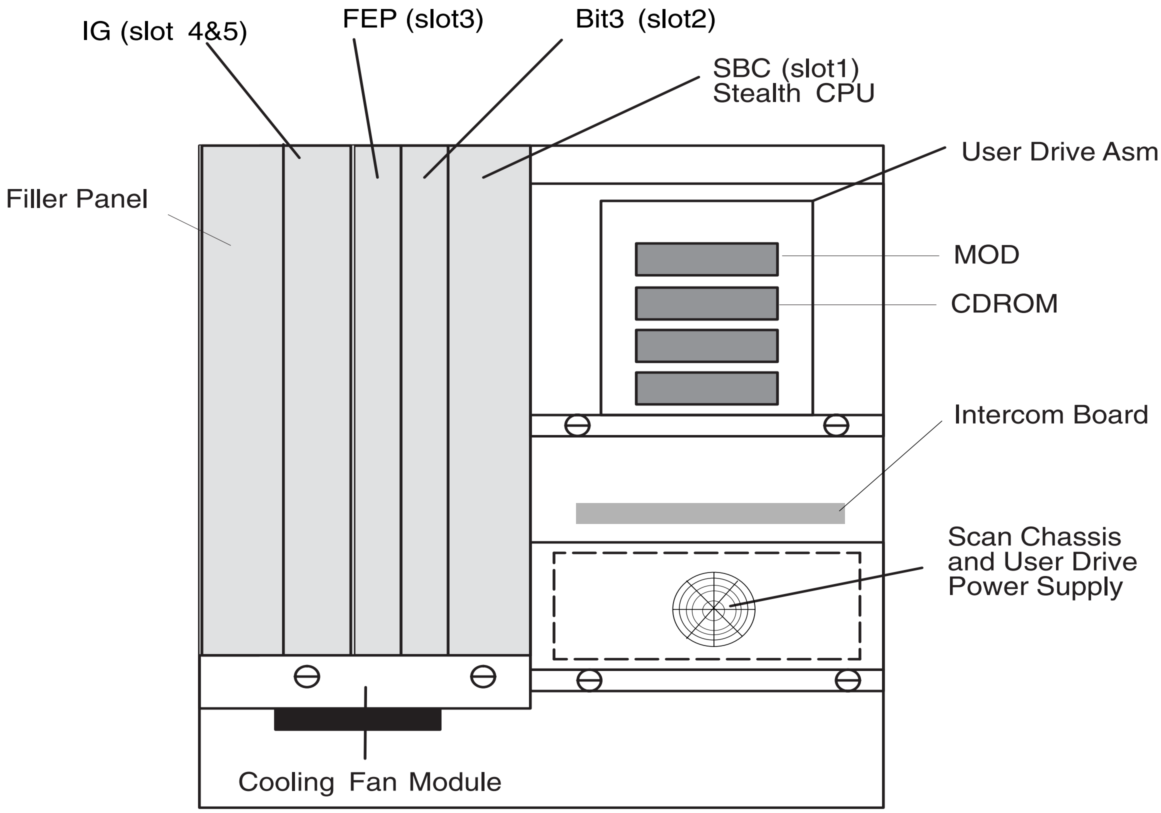


Figure 8-91 HiSpeed CT/i, Scan Recon Chassis Front View

SRC BOARD INSERTION PROCEDURE:

1.) Visually inspect the metal shield around the P1 and P2 VME connectors. If the shield is bent or separated from the plastic by more than 1/32 of an inch, replace the board.

NOTICE **If the shield is damaged or separated, the card cannot be seated in the chassis. Forcing the card into the chassis can damage the back-plane connectors.**

2.) If the connector is OK, insert the board into the card guides and snug up against the back-plane. Do not apply enough force to seat the board yet.

3.) With one thumb, align the top retaining screw with the hole in the chassis frame to keep the screw from interfering with the chassis frame.

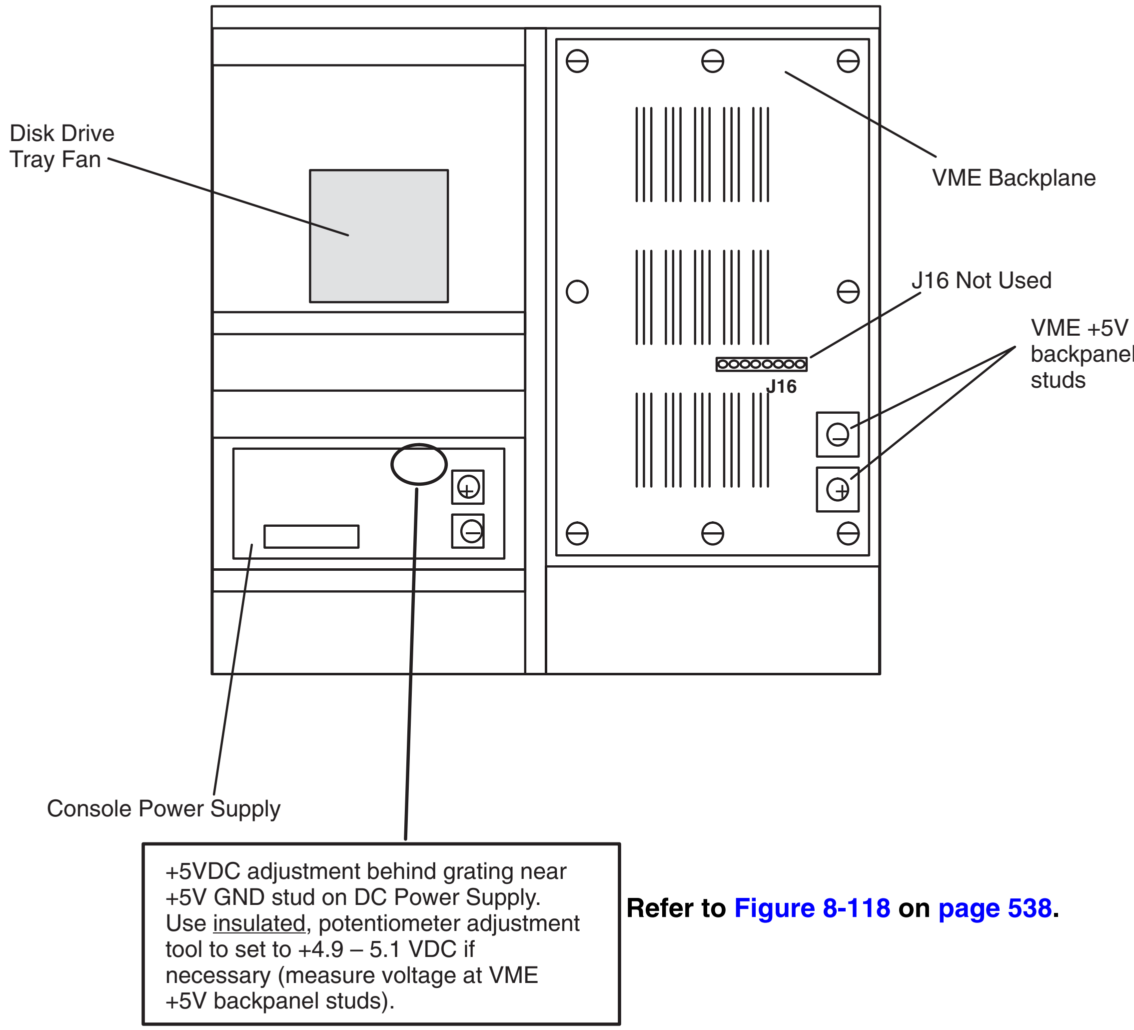
4.) With the other hand placed in the center of the board (near the screw for the center stiffener) shove the board into the connectors.

NOTICE **Do not press against the fiber optic connector when inserting the FEP. This may damage the connector.**

5.) Screw in the top and bottom retaining screws until tight.

6.) By pressing on the faceplate with one hand near the top ejector and the other hand near the bottom ejector, shove the board again to complete the insertion.

3.2.2 Scan Chassis (Rear View)



Refer to Figure 8-118 on page 538.

Figure 8-92 Scan Recon Chassis, Rear View

3.2.3 Scan Chassis VME Back-plane (inside)

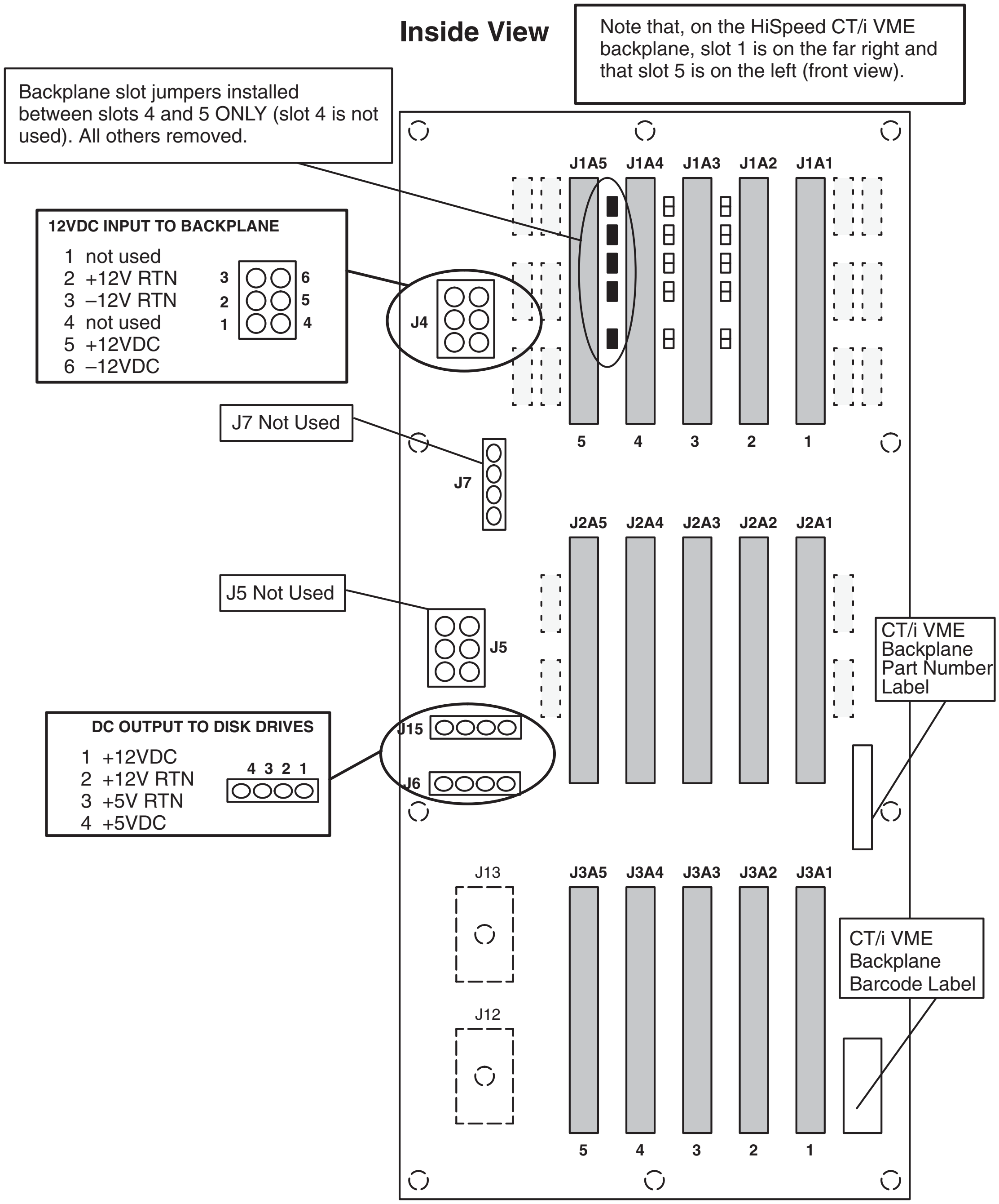


Figure 8-93 Scan Recon Chassis VME Back-plane, Inside View

3.2.4 Scan Chassis VME BACKPLANE (outside)

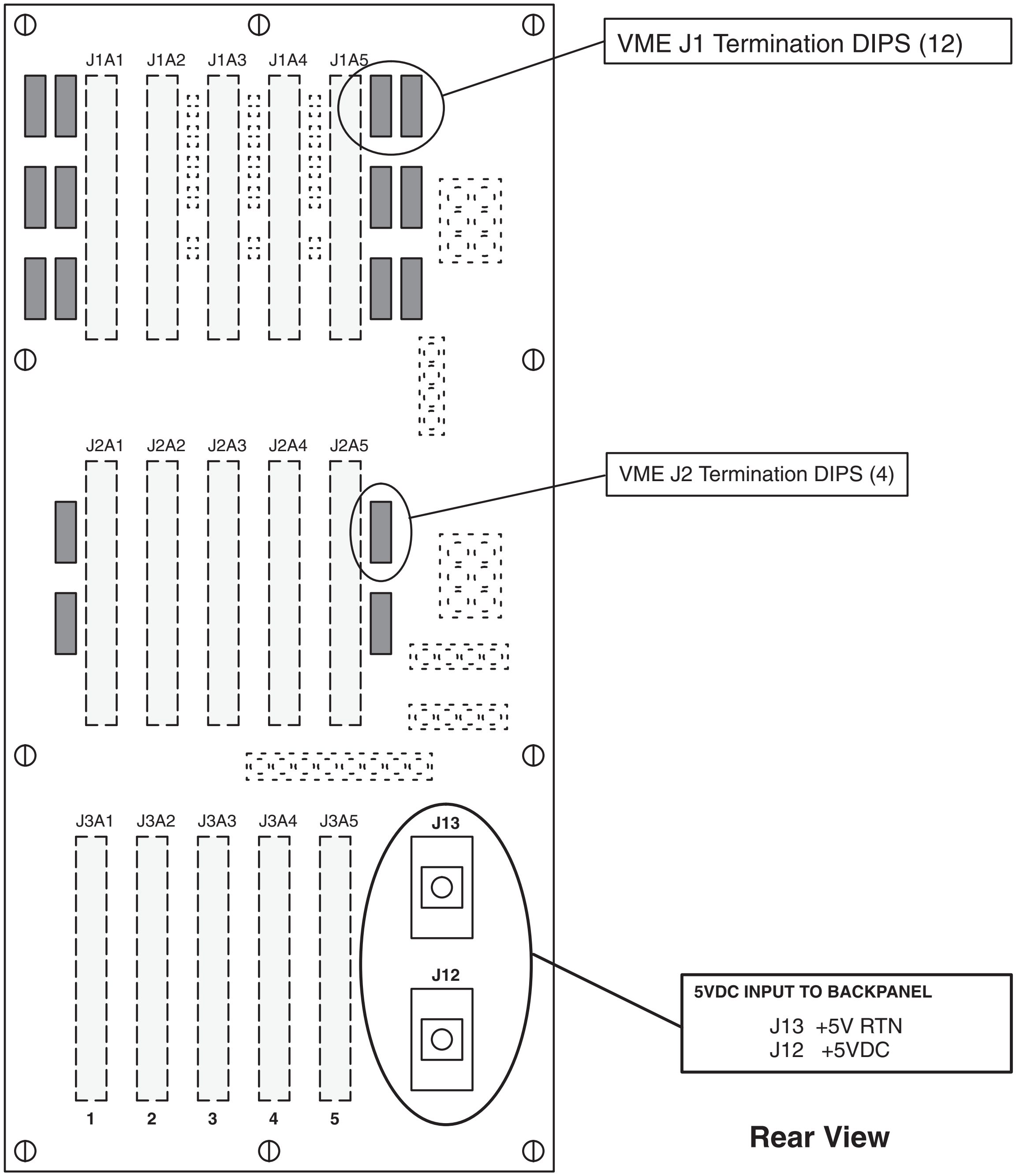


Figure 8-94 Scan Recon Chassis VME Back-plane, Outside View

3.3 Scan Recon Unit Board Replacement

3.3.1 Board Removal

1.) Shutdown Applications software
2.) Halt Unix on the SBC or verify Unix is not running.

NOTICE **Prevent permanent damage to the static-sensitive boards. Attach the anti-static wrist strap to your wrist, and to a bare metal grounding point on the scan chassis before you continue.**

3.) Remove front cover to the console.
4.) Turn OFF the chassis power switch.
5.) Unscrew the board retaining screws and lift up on the thumb release levers.
6.) Slide out the board and place it in an anti-static bag.

3.3.2 Board Installation

1.) Check the new board's switch and/or jumper settings.
2.) Visually inspect the metal shield around the P1 and P2 VME connectors. If the shield is bent or separated from the plastic by more than 1/32 of an inch, replace the board.

NOTICE **If the shield is damaged or separated, the card cannot be seated in the chassis. Forcing the card into the chassis can damage the back-plane connectors.**

3.) If the connector is OK, insert the board into the card guides and snug up against the back-plane. Do not apply enough force to seat the board yet.
4.) With one thumb, align the top retaining screw with the hole in the chassis frame to keep the screw from interfering with the chassis frame.
5.) With the other hand placed in the center of the board (near the screw for the center stiffener) shove the board into the connectors.

NOTICE Do not press against the fiber optic connector when inserting the FEP. This may damage the connector

6.) Screw in the top and bottom retaining screws until tight.
7.) By pressing on the faceplate with one hand near the top ejector and the other hand near the bottom ejector, shove the board again to complete the insertion.
8.) Turn ON the chassis power.
9.) Verify the board functions correctly by performing the same scenario for why it was replaced.

3.4 Drive Assemblies

There are two possible configurations of the user drive assembly, depending on whether you have an Octane or Indigo2 based computer system. In both cases the user drive assembly is used to house both the Optical and CD-ROM drives.

3.4.1 User Drive Assembly (Indigo2)

This user drive assembly houses OC image, SBC sw/scan data, optical and CD-ROM disk drives. Besides carrying the optical and CD-ROM drives, this "common" assembly holds hardware that is used by each main console subsystem (host and recon). Information about these disks drives can be found in the following sections:

- OC image disk, see 2.2.2.1 on page 420.
- SBC software and scan data disk, 3.5.2 on page 511 and 3.5.3 on page 513 respectively.

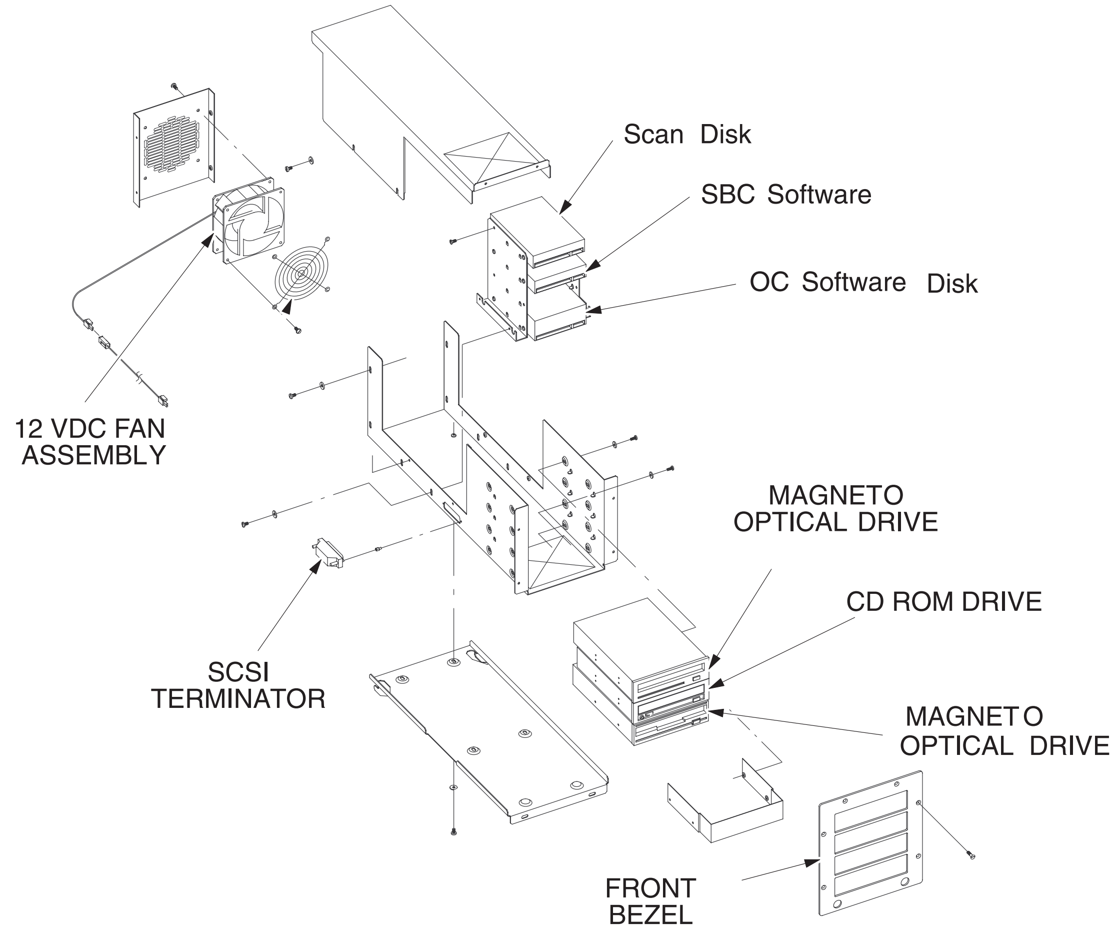


Figure 8-95 User Drive Assembly (Indigo2)

3.4.2 User Drive Assembly (Octane)

The Octane user drive assembly differs slightly from previous releases. Only the optical and CD-ROM drives are now housed within the user drive assembly. OC image disk(s) are no longer located here. They are now contained within one of the Octane computer drive bays.

3.4.3 Hard Drive Assembly (Octane)

The hard drive assembly is new for Octane but its contents are not. The hard drive assembly is an EMC compliant sub-assembly housed outside the EMC compliant scan reconstruction assembly. The SBC Scan data disks are now house here.

3.4.4 Using the CT Stealth OS SBC Disk Format Utility

3.4.4.1 Important Background Information

The CT VME 68040 SBC runs a custom version of SUNOS 4.1.3 that was ported specifically for GEMS. Many refer to this OS as "StealthOS". In particular, the `/usr/etc/format` utility in StealthOS does NOT operate like the original SUNOS `/usr/etc/format` utility.

Specifically, the way that “bad block lists” are managed by the StealthOS `/usr/etc/format` program is quite different than SUNOS. Also, the correct way to reformat a disk drive is different. So, DO NOT rely on SUN Administration manuals, SUN man pages, or SUN experiences for the use of `/usr/etc/format` on GEMS StealthOS systems.

The SUN `/usr/etc/format` program was written before intelligent imbedded SCSI disk drives existed so it attempted to create and manage "defect lists" which it then wrote to a special SUNOS area on the disk. But, all newer intelligent imbedded SCSI disks (ALL disks used on ALL StealthOS SBC's) store and manage their own defect lists (with no help from `/usr/etc/format`).

Therefore, ignore all StealthOS `/usr/etc/format` messages about "no defect list found" since this is completely normal and meaningless. As you'll read below, it's what you do just prior to reformatting a StealthOS SCSI disk that determines how defects are processed.

3.4.4.2 The STEALTHOS /USR/ETC/FORMAT Program

The primary uses of `/usr/etc/format` under StealthOS are:

A - Labeling and partitioning disk drives with a "SUN" label as required prior to a load-from-cold (done automatically by the CT install scripts).

B - Attempting to repair bad blocks reported by "fsck" at bootup or runtime 'medium errors' reported in /var/adm/messages (see info later on AWRE/ARRE).

C - Analyzing or testing an SBC disk drive (either with non-destructive or destructive type READ and/or WRITE/READ testing).

D - Viewing the manufacturers bad block list (original) or the "bad block growth" list (extract) which determines how the NEXT format command will cause the disk drive to process defects.

This document will focus only on bad block management and reformatting properly with StealthOS `/usr/etc/format`.

The StealthOS SCSI driver enables features in the SBC SCSI disks that AUTOMATICALLY attempt to reallocate (slip) any sectors experiencing WRITE or READ errors at runtime. All officially supported StealthOS SCSI drives have this capability (AWRE/ARRE) and it is enabled by all StealthOS versions since RP1.1 software (including all subsequent RP and CT/i SBC releases). A list of "officially supported" disk drive model numbers for your particular version of SBC software can be viewed on the SBC in `/etc/format.dat`.

There may be times when an error is still reported even when the automatic reallocation is enabled. This can occur when the bad sector is automatically slipped to a good spare but the drive was unable to successfully recover the user data from the bad sector and copy to the new sector. In this case, the bad sector is gone but the user data in that block is corrupted/missing.

Although not usually needed for any system since RP1.1 or later, to attempt the repair of reported 'medium error' (bad block), use the following procedure for any StealthOS SBC disks:

Note: You must know the bad block# before you can attempt to repair it (slip the bad block to a good spare and attempt to recover the user data). The new bad block is either reported by 'fsck' at SBC bootup or will be posted in the SBC `/var/adm/messages` **VMUNIX** log file.

3.4.4.3 STEALTH OS (SBC) Medium Error Bad Block Repair

Note: THIS'REPAIR' PROCEDURE IS ONLY PROVIDED TO ALLOW YOU TO ATTEMPT TO REASSIGN A NEW SBC DISK BAD BLOCK AND POSSIBLY PREVENT/DELAY A REFORMAT AND/OR LFC. IT MAY NOT ALWAYS WORK IF THE BAD BLOCK USER DATA CANNOT BE RECOVERED AFTER THE BLOCK HAS BEEN REMAPPED. SOMETIMES, EVEN THIS IS NOT A PROBLEM UNLESS THE BAD BLOCK (AND USER DATA) WAS INSIDE AN IMPORTANT SYSTEM OR APPLICATION FILE. EVERY SITUATION VARIES. BECAUSE STEALTHOS ENABLES AWRE/ARRE IN THE DRIVE, A MEDIUM ERROR WILL USUALLY MEAN THAT THE BAD BLOCK WAS SLIPPED BUT THE USER DATA RECOVER FAILED.

1.) Assume that a "new" bad block has been reported on a StealthOS SBC disk by 'fsck' at bootup (system disk) or posted in /var/adm/messages (scan data disk). Assume that the error indicates block 4167 is bad (the 'info' value in the "medium error" message).

```
sd0: device error: ASC_ASCQ_SK: 110003, 'medium error'
```

```
: asc=0x11, ascq=0x0, info=4167
: cdb=0x8 0x0 0x10 0x3e 0x7e 0x0
```

2.) If the new bad block is not in the `/` or `/usr` partition, you should be able boot the SBC to single user mode using `'b -s'` from the MC68040> prompt. If the new bad block is in `/` or `/usr`, you may have to boot the SBC MUNIX as if you were performing an SBC LFC in order to use the `/usr/etc/format` program to repair the system disk (don't forget to prepare the OC for booting SBC MUNIX by running the 'loadSBC' script as root on the OC with the App SW CD inserted in the OC CDROM drive). Once you've booted the SBC single user or booted SBC MUNIX, then use the following example to run `/usr/etc/format`:

EXAMPLE

```
ct01_sbc0 login: root
Nov  9 18:23:11 ct01_sbc0 login: ROOT LOGIN console
Last login: Tue Dec  5 23:51:12 on console
SunOS Release 4.1.1_U1 (GOS_IG) #1: Mon Nov 20 09:48:32 CST 1995
ct01_sbc0# /usr/etc/format
Searching for disks...done
```

AVAILABLE DISK SELECTIONS:

```
0. sd0 at 0 slave 0
sd0: <ST31051N cyl 2624 alt 2 hd 4 sec 64>
1. sd2 at 0 slave 8
sd2: <ST32171N cyl 3298 alt 2 hd 5 sec 237>
Specify disk (enter its number): 0
selecting sd0: <ST31051N>
[disk formatted, defect list found]
```

FORMAT MENU:

```
disk       - select a disk
type       - select (define) a disk type
partition  - select (define) a partition table
current    - describe the current disk
format     - format and analyze the disk
repair     - repair a defective sector
show       - translate a disk address
label      - write label to the disk
analyze    - surface analysis
defect     - defect list management
backup     - search for backup labels
quit
format> repair
Enter block number of defect: 4167
Ready to repair defect, continue? y
```

```
Repairing block 4167  (16/1/7)...done
format>
```

If there are no error messages above, the new bad block was successfully remapped and the user data was successfully copied to the new block. You are done. If the block remapped but the user data copy failed, the bad block is processed but you now have corrupted or no data at the new block. The system may or may not boot and startup applications. If it does, the bad block was most likely in a file that is not used. If the system does NOT boot or applications do not start normally, you may need to LFC the SBC to restore the missing data block.

3.4.4.4 Properly Formatting a STEATHOS SBC Disk Drive

You should very rarely need to do this but if you do, follow the procedure below. The key point here is that if there are any “new bad blocks” (shown by the `/usr/etc/format 'extract'` function), you MUST issue the 'extract' and commit it just prior to formatting the drive. This will ensure that the manufacturers original defects AND any new bad blocks are remapped during the format process. After the format completes, the `/usr/etc/format` program will automatically run 2 surface analysis passes across the drive media looking for any other bad blocks (and will automatically remap them if any are found).

Note: If there are no new bad blocks shown when the 'extract' is done, then there have been NO new bad blocks.

EXAMPLE

```
ct01_sbc0# /usr/etc/format
Searching for disks...done
```

AVAILABLE DISK SELECTIONS:

```
0. sd0 at 0 slave 0
 sd0: <ST31051N cyl 2624 alt 2 hd 4 sec 64>
1. sd2 at 0 slave 8
 sd2: <ST32171N cyl 3298 alt 2 hd 5 sec 237>
Specify disk (enter its number): 0
selecting sd0: <ST31051N>
[disk formatted, defect list found]
```

FORMAT MENU:

```
disk       - select a disk
type       - select (define) a disk type
partition  - select (define) a partition table
current    - describe the current disk
format     - format and analyze the disk
repair     - repair a defective sector
show       - translate a disk address
label      - write label to the disk
analyze    - surface analysis
defect     - defect list management
```

```
backup     - search for backup labels
quit
format> defect
```

DEFECT MENU:

```
restore  - set working list = current list
original - extract manufacturer's list from disk
extract  - extract working list from disk
add      - add defects to working list
delete   - delete a defect from working list
print    - display working list
dump     - dump working list to file
load     - load working list from file
commit   - set current list = working list
create   - recreates maufacturers defect list on disk
quit
defect> extract
Extracting defect list...Extraction complete.
Working list updated, total of 2 defects.
defect> q
Warning: working defect list modified; but not committed.
Do you wish to commit changes to current defect list? y
Current Defect List updated, total of 2 defects.
Disk must be reformatted for changes to take effect.
```

FORMAT MENU:

```
disk       - select a disk
type       - select (define) a disk type
partition  - select (define) a partition table
current    - describe the current disk
format     - format and analyze the disk
repair     - repair a defective sector
show       - translate a disk address
label      - write label to the disk
analyze    - surface analysis
defect     - defect list management
backup     - search for backup labels
quit
format> format
Ready to format. Formatting cannot be interrupted
and takes 11 minutes (estimated). Continue? y
Beginning format. The current time is Mon Nov  9 18:34:18 1998
Formatting...done
```

```
Verifying media...
 pass 0 - pattern = 0xc6dec6de
 pass 1 - pattern = 0x3f3f3f3f
Total of 0 defective blocks repaired.
format>
```

3.5 SBC Disk Drives

```
diagnostic(s): (SBC) 166bus>ioi, fwrv, 166diag>st, format
error log(s): (SBC) /var/adm/messages
```

3.5.1 SBC Disk Diagnostics

To probe the SBC SCSIbus at any time (either rlog'd into the SBC from the SBC or when connected to the SBC serial port over the **'cu sbc'** link from OC), use the **'fwrev'** command as root.

The following tests/diagnostics can be run on the SBC (with applications down and SBC vmunix down and using the **'cu sbc'** serial link to SBC):

To probe the SBC SCSIbus with **'vmunix'** down, use the '**ioi'** command at the 166bug> or 166diag> prompts.

To run built-in board levbel SBC diagnostics with **'vmunix'** down, use the **'st'** (serftest) command at the `166diag>` prompt.

3.5.2 SBC Software Disk

3.5.2.1 ST39216N, SBC Software

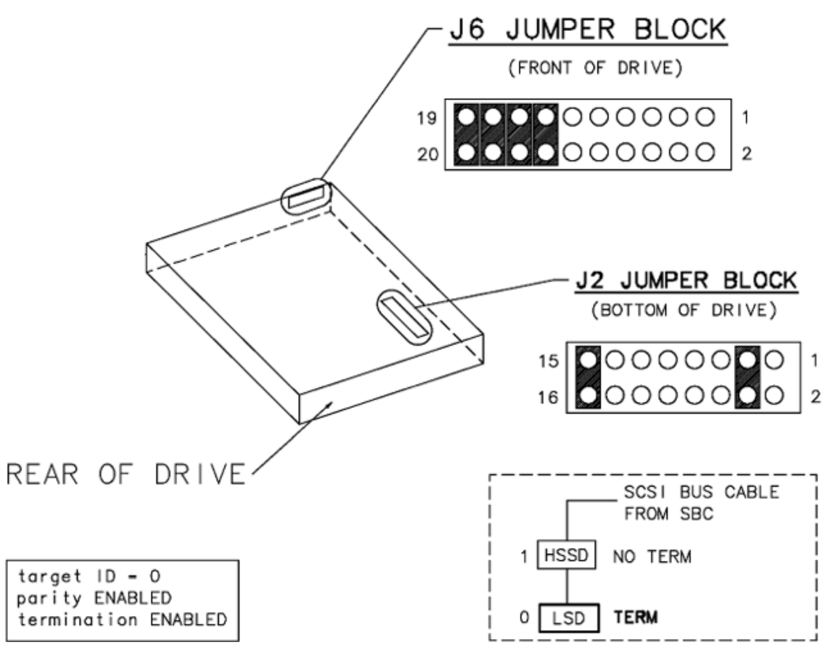


Figure 8-96 ST39216N (2226715-2) SBC (9.2 GB) Software Disk Jumpers

3.5.2.2 ST31051N, SBC Software

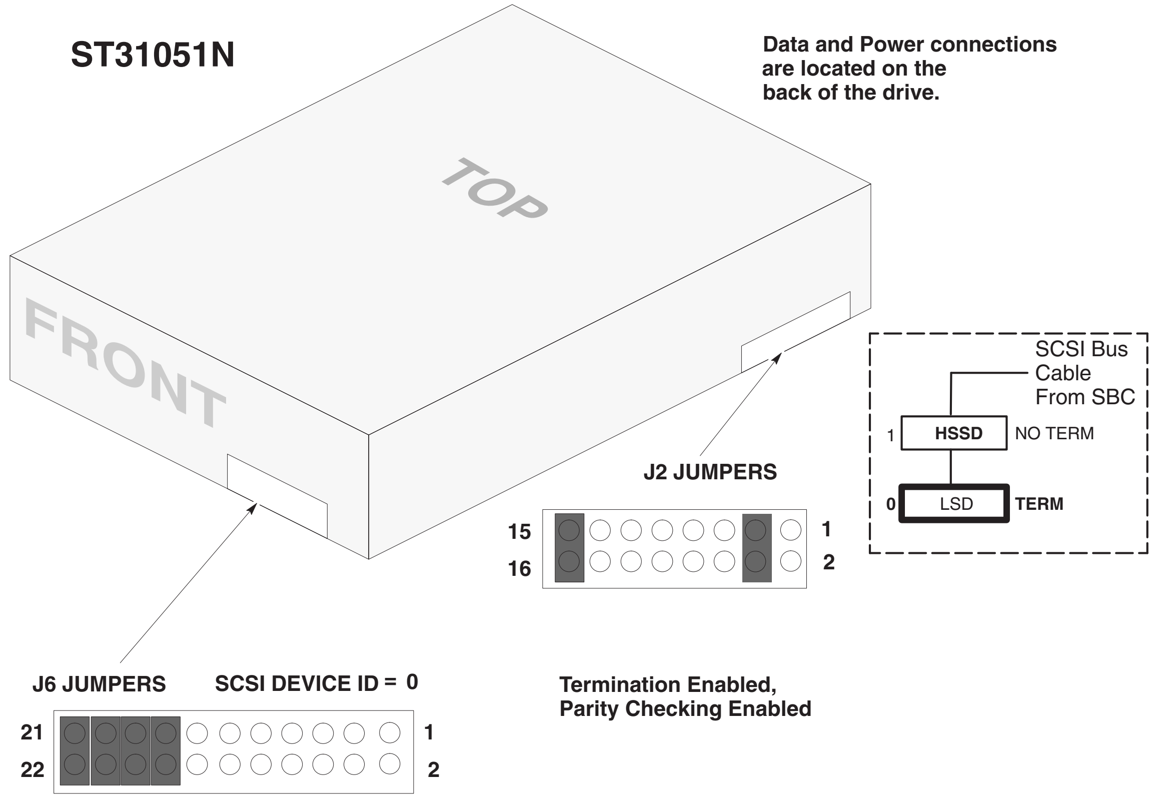


Figure 8-97 ST31051N (2157681) SBC (540 MB) Software Disk-Switches

3.5.2.3 ST32272N - SBC System Disk

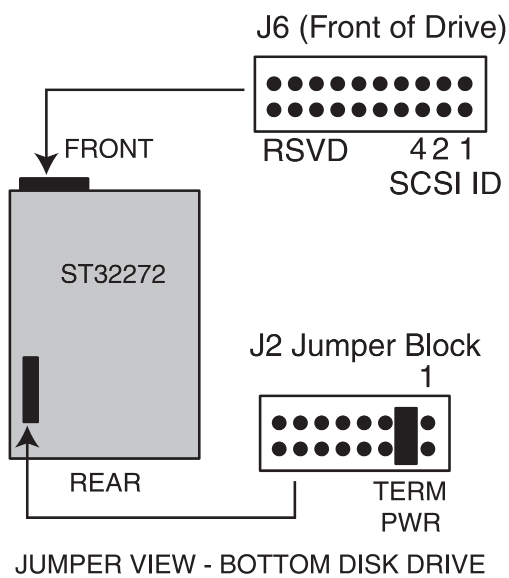


Figure 8-98 Jumpers (Bottom Disk Drive)

3.5.3 SBC Scan Data Disk

3.5.3.1 ST39216N - Scan Data

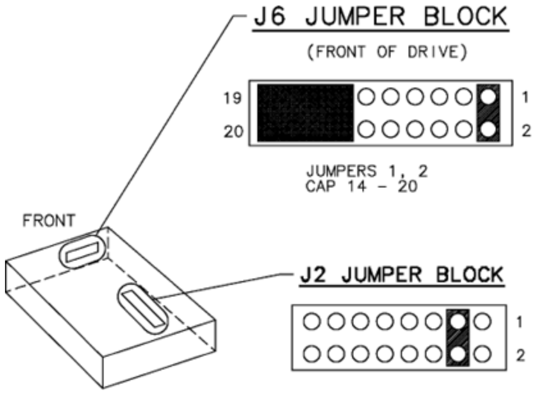


Figure 8-99 SBC 9.2GB Scan Data Disk, ST39216N (2226715-2)

3.5.3.2 ST32171N - Scan Data

Note: The ST32171N Disk is not compatible with CT/i Software Revisions prior to the 3.6 SW Release.

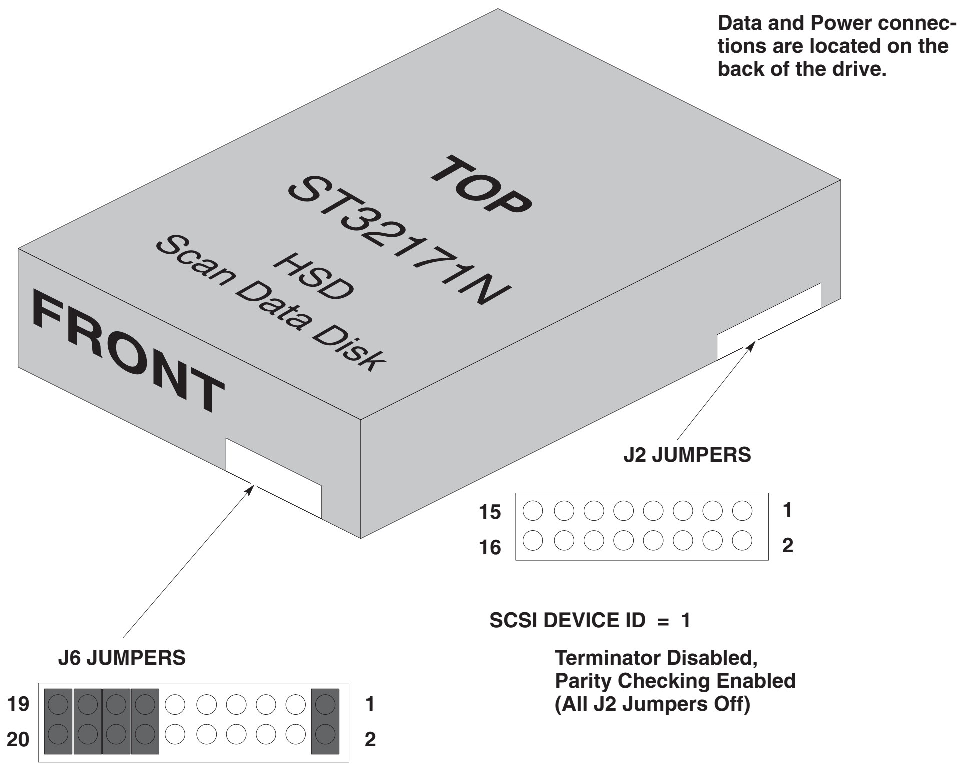


Figure 8-100 SBC 2GB Scan Data Disk, ST32171N (2180995)

3.5.3.3 ST31250N/ND and ST32550N/ND Configuration

Figure 8-101 illustrates ST31250N/ND and ST32550N/ND jumper connectors.

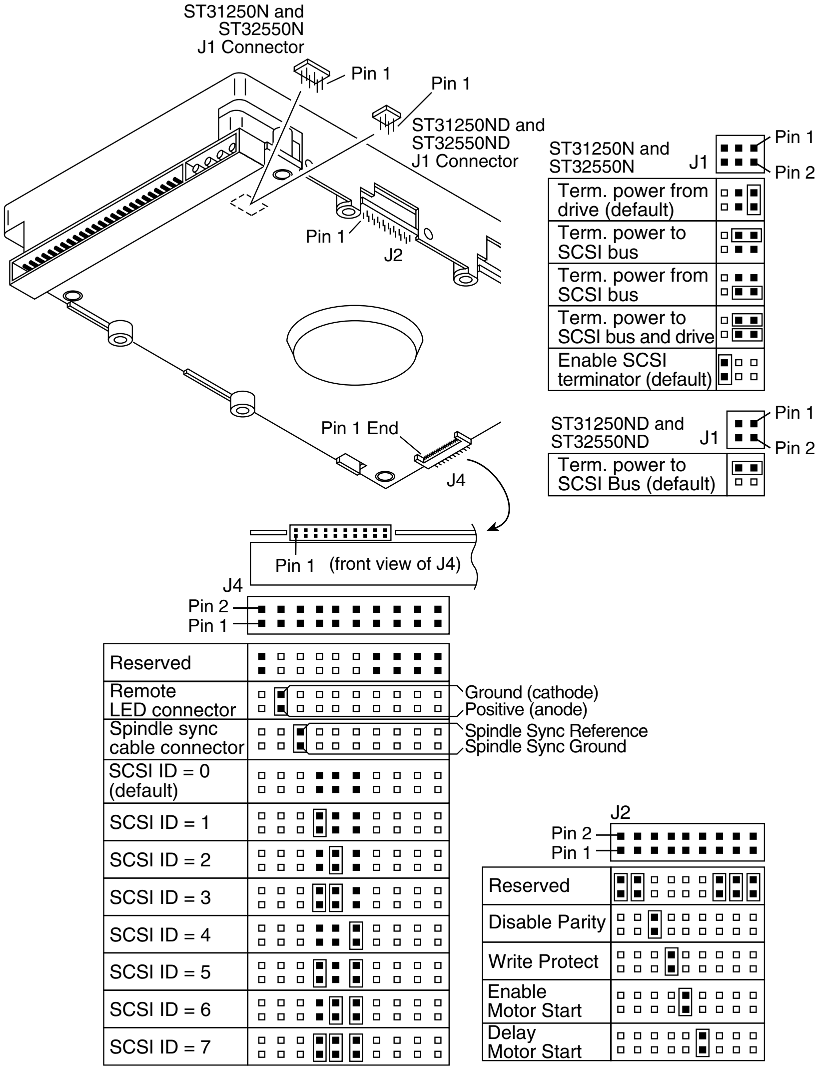


Figure 8-101 ST31250N/ND and ST32550N/ND jumper connectors

	Block	Pins	Function
ST31250N and ST32550N	J1	1 & 2	Terminator power from the drive (drive supplies terminator power to its own terminators).
		1 & 3	Terminator power to the SCSI bus (drive supplies power to an external terminator)
		2 & 4	Terminator power from the SCSI bus (drive receives terminator power from the SCSI bus–usually the host controller)
		1 & 2 & 3 & 4	Terminator power to the SCSI bus and drive (drive supplies termination power to its own terminators and to the SCSI bus)
		5 & 6	Enable SCSI termination. Jumper installed enables the internal termination on the drive. Jumper removed disables the drive's internal terminators.

3.5.3.4 ST32272N - Scan Data Disk

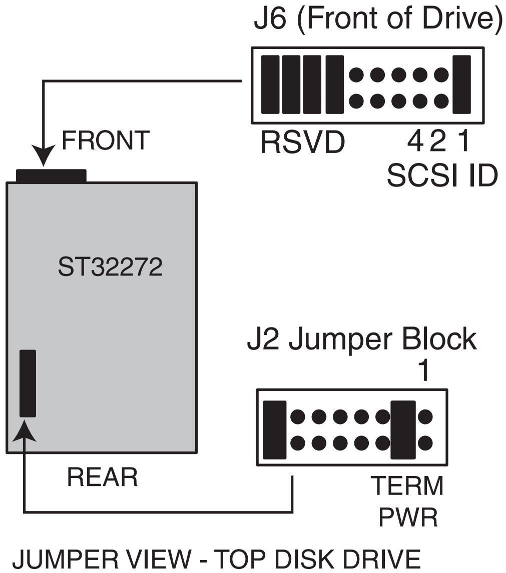


Figure 8-102 Jumpers (Top Disk Drive)

3.5.4 SBC Software/Scan Data Disks

3.5.4.1 SBC Software Disk ST32272N

Note: The ST32272N Disk is not compatible with CT/i software revisions prior to the 4.1 release (Indigo2 based console systems) or the 5.2 release (Octane based console systems).

The ST32272N Disk is used for both the SBC software (LSD) and the ScanData (HSSD).

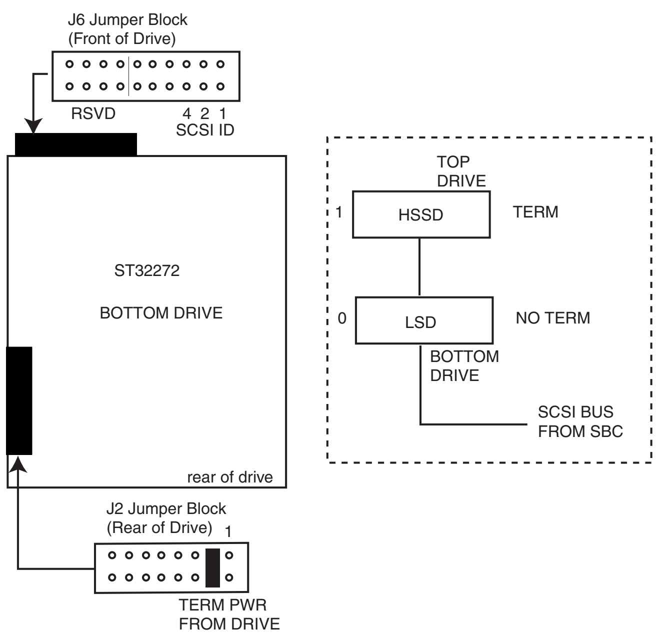


Figure 8-103 2GB SBC Software Disk Jumpers ST32272N/2205622

3.5.4.2 SBC ScanData Disk ST32272N

Note: The ST32272N Disk is not compatible with CT/i software revisions prior to the 4.1 release (Indigo2 based console systems) or the 5.2 release (Octane based console systems).

The ST32272N Disk is used for both the SBC software (LSD) and the ScanData (HSSD).

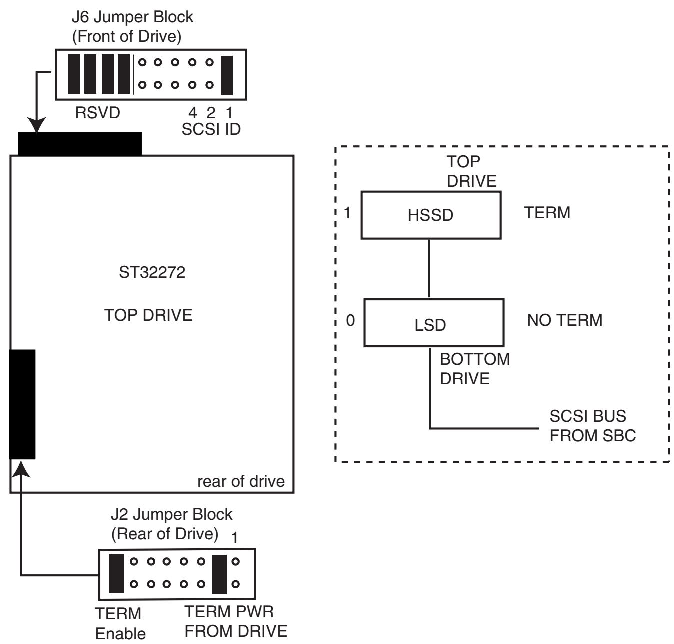


Figure 8-104 2GB SBC ScanData Disk Jumpers ST32272N/2205622

3.6 Single Board Computer (SBC)

```
diagnostic(s): (SBC) 166diag>st, 166bug>ioi
error log(s): (SBC) /var/adm/messages
```

3.6.1 "Motorola MVME166" SBC Firmware Version

3.6.1.1 Updating Firmware

The CT/i "Motorola MVME166" SBC requires version 3.x firmware. This version incorporates a BIT3 driver to allow remote boot during HiSpeed CT/i installation and SBC software loads. GE SBC FRUs are shipped with version 1.5 and must be checked and updated as needed depending on the application.

- CT RP2.X uses MVME166 with version 2.2 firmware (Genesis/SBC)
- MR Signa uses MVME166 with version 1.5 firmware (Genesis)
- CT/i SBC requires MVME166 with version 3.X firmware.

Using re-flashing code found in directory **`/stand`**, the firmware version is checked and updated as necessary. This software tool ensures that the proper version is being used (e.g. MR looks for ver. 1.5, CT RP looks for ver. 2.2, and CT/i for version 3.x). To use the tool, type:

`/stand/flash166`

The CT /stand/flash166 program re-flashes firmware to version 3.X on your CT/i.

3.6.1.2 Prior To Replacing a SBC or Disk Drive

Do not swap out both the SBC and its system disk at the same time. If you do, the system disk will NOT contain the code it needs to re-flash the new or spare SBC "MVME 166 version 1.5 firmware" FRU. Which it must have to preform a load form cold (LFC) of the SBC over the BIT3 connection. Swap these FRUs ONE-AT-A-TIME to identify the real source of the problem, *either,* the disk or the MVME166.

The following options are available only if you do not swap out both FRUs at the same time.

- If you swap the 166 first and the system disk operates, software can then re-flash the 1.5 firmware to 3.x
- If the disk turns out to be the bad FRU, the original SBC can be re-install without delay because it will still have the necessary 3.X firmware flashed into memory.

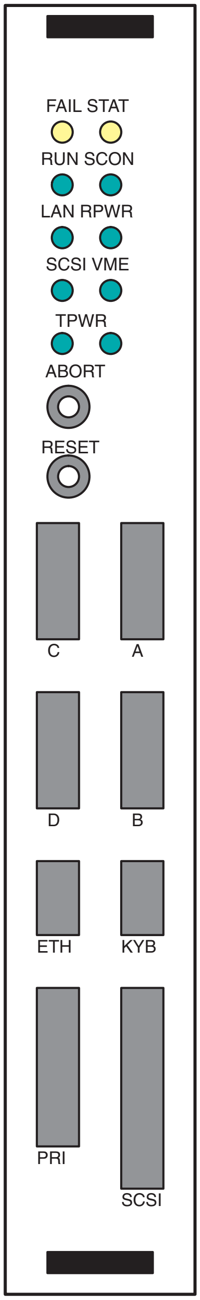
Note: It is still necessary to do preform a SBC Software "Load From Cold" if you replace a disk.

SBC DIAGNOSTICS

Test the Single Board Computer while system software is not running.

1.) Exit GUI diagnostics and bring up a Unix shell.
2.) Press UTILITIES then APPLICATIONS SHUTDOWN to stop GE software.
3.) Enter '**`cu sbc`**' to transfer from the host to the SBC and login as root.
4.) Enter '**`halt`**' and at MC68040> prompt on SBC, enter '**`x`**.'
5.) At `166-Bug>` prompt on SBC, enter '**`sd`**.'
6.) At `166-Diag>` prompt on SBC, enter '**`st`**' to run built-in self tests on SBC.
7.) To probe the SBC SCSIbus with **`'vmunix'`** down, use the **`'ioi'`** command at the 166bug> or 166diag> prompts.

3.6.2 SBC (MVME166/68040 “Stealth”) CPU LEDs and Connectors



SBC (MVME166–9U Stealth CPU) LED Descriptions:

FAIL	YELLOW LED indicating board RESET in progress (or board failure if it stays on).
STAT	YELLOW LED indicating processor is is in a HALT condition.
RUN	GREEN LED indicating processor is in a RUN condition (firmware or software).
SCON	GREEN LED indicating SBC is the VME SYSTEM CONTROLLER.
LAN	GREEN LED indicating the onboard ETHERNET is active (transferring data).
RPWR	GREEN LED indicating REMOTE POWER (+12VDC) present to the onboard Ethernet 15–pin connector (fused thru F1 on the SBC) for external transceivers.
SCSI	GREEN LED indicating the onboard SCSI is active (transferring data).
VME	GREEN LED indicating VME data transfers active to/from the SBC.
TPWR	GREEN LED indicating SCSI TERMINATOR power is present to the external SCSI ribbon cable (fused thru F2 on the SBC).

SBC (MVME166–9U Stealth CPU) Front Panel SWITCH Descriptions:

ABORT	Momentary pushbutton which HALTS the SBC and enters the onboard firmware monitor (166–Bug>).
RESET	Momentary pushbutton which RESETS the SBC and causes a power–up reset and boot cycle.

SBC (MVME166–9U Stealth CPU) CONNECTOR Descriptions:

Serial A	25–pin RS232C serial port A (to OC GAI panel, the “tip” line)
Serial B	25–pin RS232C serial port B (not used)
Serial C	25–pin RS232C serial port C (not used)
Serial D	25–pin RS232C serial port D (not used)
Ethernet	15–pin IEEE 802.3 Ethernet (to external transceiver to OC network)
Keyboard	SUN3 compatible keyboard port (not used)
Printer	Centronics parallel printer port (not used)
SCSI	50–pin SCSI bus connector (to subsystem LSD and HSD disk drives)

Figure 8-105 SBC MVME166/68040 “Stealth” CPU LEDS and Connectors

3.6.3 SBC board (MVME166 68040 CPU)

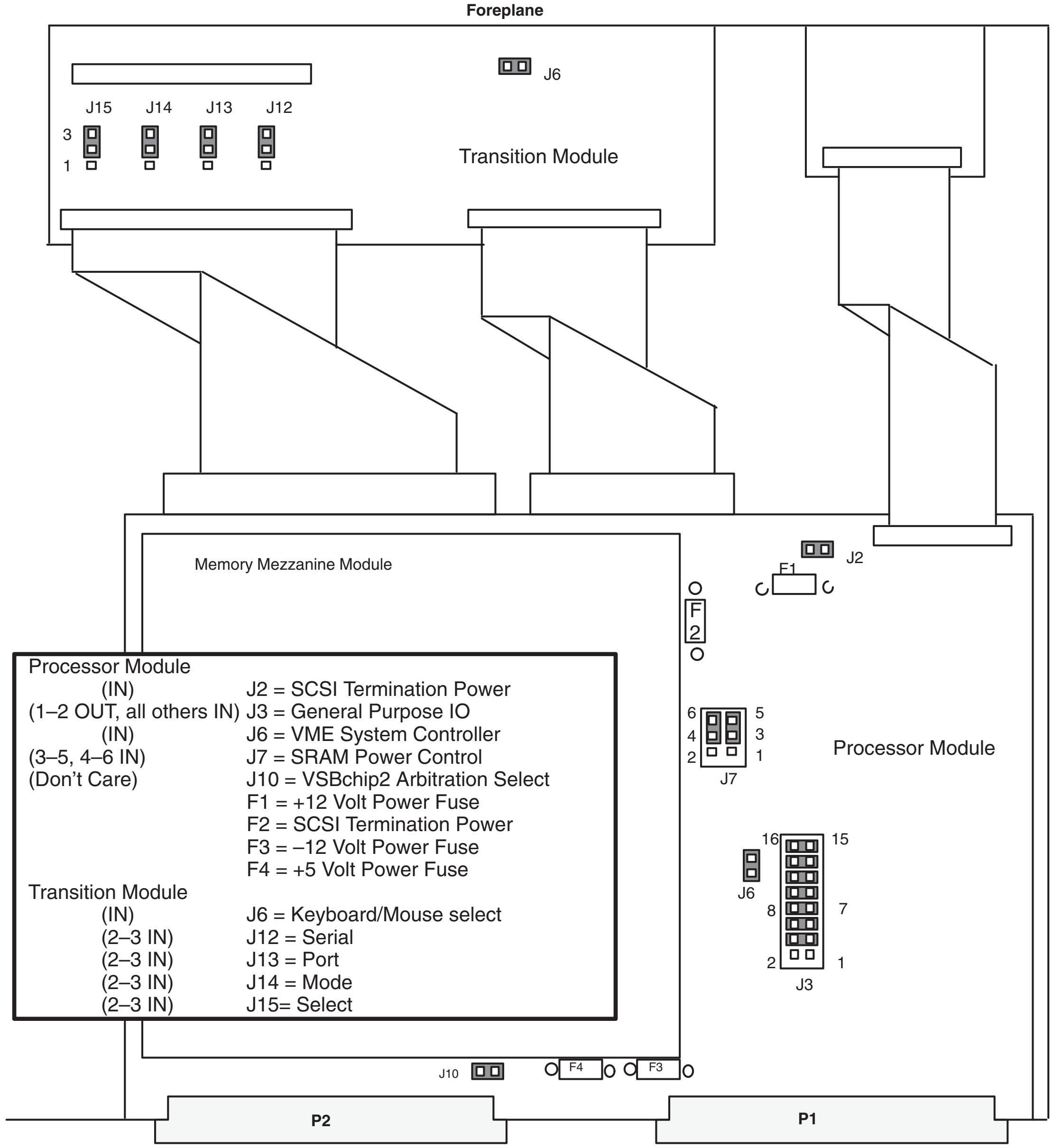
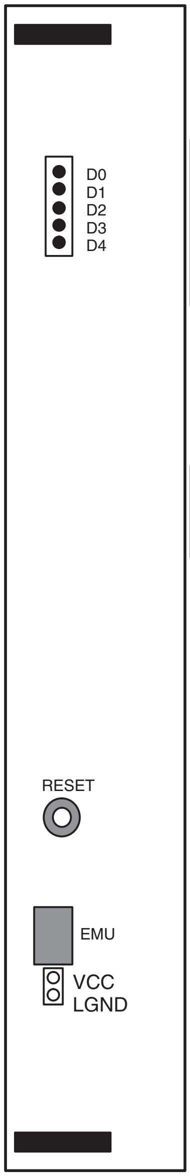


Figure 8-106 SBC board (MVME166 68040 CPU)

3.7 Image Generator (IG)

3.7.1 IG board Layout



IG (GEMS Image Generator) LED Descriptions:

D0 D0 thru D4 indicate the self–test
D1 being run on the IG board during
D2 power–on or reset and then generates
D3 an up/down chasing pattern when the
D4 IG is running normally.

IG (GEMS Image Generator) Front Panel SWITCH Descriptions:

RESET Momentary pushbutton that immediately resets the IG, causing the self–test to run.

IG (GEMS Image Generator) TEST POINT Descriptions:

VCC Onboard test point for +5VDC on the IG board.
LGND Onboard test point for LOGIC GROUND on the IG board.

IG (GEMS Image Generator) CONNECTOR Descriptions:

EMU Connector for DSP emulator (factory use only).

Figure 8-107 Image Generator (IG) Board

3.7.2 IG Jumpers, Switches and LEDs

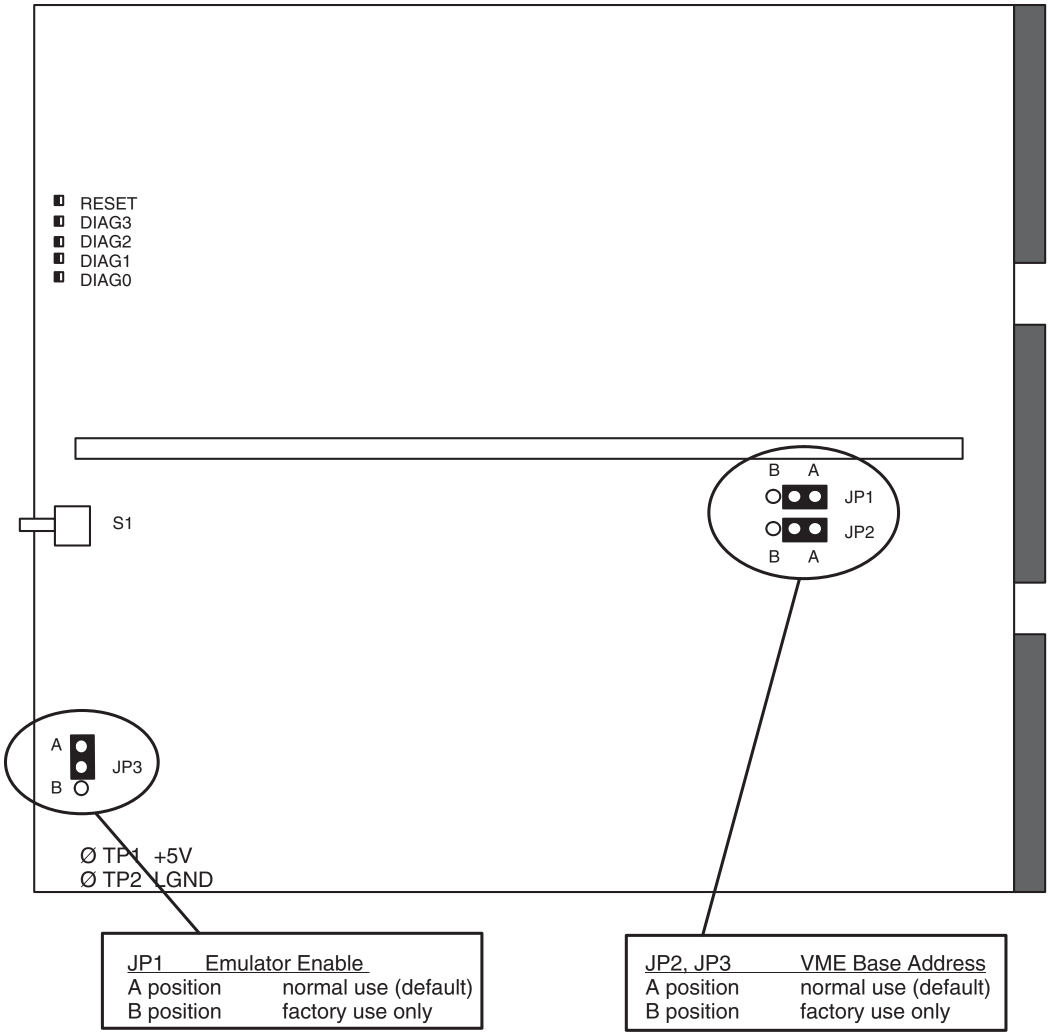


Figure 8-108 Image Generator (IG) Board

3.7.3 IG Diagnostics

Image Generator (IG) diagnostics consist of power-up self-tests (BST), board level diagnostics (BLD), one data path diagnostic and reconstruction tests. Test descriptions include: test coverage, coverage exceptions, test usage and test error messages.

3.7.3.1 IG Board Self Test (BST)

IG Board Self Test (BST) - Flash EPROM based code which runs at power-up or reset. This test does an initial validation of hardware.

General Test Strategy

Stress the system - The SBC and IG Board's DSPs DMA functions are used whenever possible.

Prevent the Unix operating system and diagnostic test code from crashing in the event of a bus error.

How to start

The Board Self Test is a FLASH EPROM based code which runs after a IG board reset. Test execution does not occur until after the DSPs transfer the BST code from FLASH to Local Memory. The BST has four main functions which are listed below:

Hardware Initialization

Hardware initialization for the IG board consists of downloading the configuration code to the Xilinx FPGA and enabling the ASICs for use. The Xilinx is part of the Communication Port Interface (CPI) which interfaces the DSP with the APU and PRAU ASICs.

Hardware Validation

The BST is responsible for performing a functional validation of all hardware accessible to the DSPs. Each DSP executes an identical test of the hardware except DSP 0. This DSP is required to test the PRAU ASIC and coordinate the creation of several partial images. Overall testing takes approximately 15 seconds to complete. This is an unacceptable amount of time for applications to wait. The solution was to create a FAST BST which is a subset of these tests. This BST takes only a second and touches a large portion of the board. Implementation consists of writing a code to Dual Port Memory before performing a board reset. See section for more information about the devices tested and the differences between the BSTs.

Note: The terms fast and normal resets are sometimes substituted for the terms Fast and Normal BSTs since applications considers the BST part of the board reset.

Notification of Test Results

The Board Self Test reports test results in the following ways:

LEDs

LEDs provide a visual means to read the status of the IG Board Self Test. During a board reset, all 5 LEDs located on the edge of the IG board are lit. Once the board reset is released, the top most LED (Reset) should extinguish and the bottom 4 LEDs should follow. During testing, DSP0, which is the only DSP in control of the LEDs, indicates the test it's presently performing by writing the binary test value to the lower LEDs. At the completion of all 8 self tests, DSP 0 summarizes the results as follows in the order given:

- One or more DSPs fail to complete its test – display the last binary test value on the LEDs
- One or more DSPs detect a failure – blink the binary test value of the first test to fail
- All DSPs pass their BST – “Walk” 1 lit LED up and down the group until applications is loaded

See Figure 8-109 for additional information.

P3 Connector

Manufacturing has the ability to read the BST results at the P3 connector. The status is the same 4 bit binary code displayed on the LEDs with the following exceptions:

- 0 indicates one of the following: board reset active, BST executing, or DSP 0 is hung.
- 0xF (15) indicates the test completed successfully. The “walking” bit pattern is not used.
- 2 through 13 indicates a test has failed or a DSP other than 0 has hung. See Figure 8-109 for

a definition of codes.

You can also access this BST status code through the VMEbus Status Register.

Dual Port and Local Memory

Both applications and diagnostics require access to the BST results. This is accomplished by reserving 24 32-bit consecutive words in Dual Port Memory. Each DSP uses 3 of these words to report test status. The first word is a bit map indicating the tests which have completed or are running. The second is also a bit map indicating the failed tests, and the third is reserved for a TESTING, FAILED, or PASSED status code. Additional error information is stored in Local Memory by each DSP. This information can be logged via the system error log. Access to this data is available via an IOPB.

IG Board

LEDs 3 2 1 0	Code	Description
○ ○ ○ ○	0	BST Initialized
○ ○ ○ ●	1	Dual Port Memory Test
○ ○ ● ○	2	Checksum of Flash Code
○ ○ ● ●	3	Checksum of Local Mem Code
○ ● ○ ○	4	Stack Memory Test
○ ● ○ ●	5	CPU Test
○ ● ● ○	6	Global Bus Registers Test
○ ● ● ●	7	FIFO Status Test
● ○ ○ ○	8	DSP–to–DSP Comms Test
● ○ ○ ●	9	Image Memory Test
● ○ ● ○	A	CPI Test
● ○ ● ●	B	APU ASIC Test
● ● ○ ○	C	PRAU ASIC Test
● ● ○ ●	D	Backprojection Test
● ● ● ○	E	Reserved
● ● ● ●	F	Board Reset

Key:
● – LED On
○ – LED Off

MODES

1. Reset – All 5 LEDs are lit.
2. BST in progress – Diag LEDs are sequencing through test codes.
3. DSP hung – Diag LEDs display the test code of last test to run.
4. BST Failed – Diag LEDs blink the test code of the first failed test.
5. BST Passed – Walking lit LED

IG Board Self Test LED Result Codes

Figure 8-109 IG Board Self Test LED Result Codes

IG Board Self Test (BST) Coverage

The following section describes the individual tests performed by the IG BST. Figure 8-110 and Figure 8-111 indicate the test coverage for the NORMAL and FAST BSTs. The shaded areas in these diagrams are not touched by the test.

- **Dual Port Memory** – validated by performing a write and read test using multiple test patterns.
- **FLASH EPROM Memory** – insure the Board Self Test code stored in FLASH has not been corrupted by calculating the check-sum and comparing it to the expected value.
- **Loaded Image** – calculate check-sum for various sections of the BST code stored in Local Memory and compare it to the expected value.
- **CPU** – execute various instructions and verify the results. Further testing includes validating

the DMAs and other DSP functions.

- **Local Memory** – validated by performing a write and read test using multiple test patterns. This is performed only on memory NOT utilized for the BST code and on memory reserved for the stack.
- **Registers** – each register is tested by performing a write and read test using multiple test patterns. The Global Status Register's Global Bus Error, Strobe 1 Access Out of Bounds, and Strobe 0 Access Out of Bounds are tested by writing to undefined address spaces and verifying the correct error bits are set. Further testing consists of verifying the fault was cleared after reading the VMEbus Status Register.
- **Communication ports** – ports 0, 2, 3, and 5 are validated during this test by transferring packets to and from the adjacent DSPs. The transfer of data requires the DMA and interrupt hardware to be fully functional.
- **Input and Output Data FIFOs** – testing is limited to status checking ONLY. This includes the Full and Empty Output Data FIFO; Full and Empty Input Data FIFO; and bus error status bits.
- **Image Memory** – validation consists of testing the crossbar selection bits on the Global Command Register and performing a write and read data test of both Image Memories. For FAST BST, each DSP will ONLY test 100 words of memory.
- **Communication port interface (CPI)** – download the configuration code to the Xilinx FPGA after verifying the check-sum and check that all data was transferred. Perform a PRAU Register read request and verify the correct amount of data was received. Testing requires the DMA and interrupt hardware to be fully functional.
- **APU ASIC** – Validate all registers and projection memory contained in the APU ASIC by first verifying the S&T and ATR Registers are set to zero after a reset, and second, by performing a write and read test using multiple test patterns. For a FAST BST, use only 2 test patterns to validate operation.
- **PRAU ASICs** – Validate the PRAU Register by first verifying the register is set to zero after a reset, and second, by performing a write and read test using multiple test patterns. For a NORMAL BST only, several Image Memory scans are made in a special diagnostic mode which tests the Pixel Requester Bus and the PRAU ASICs ability to scan Image Memory properly.
- **Back projection** – The back projection functionality is tested by creating 6 partial images (256 projections each) using all 8 DSPs simultaneously and validating the check-sum. The projection data is generated from a sine function.
- **Global Bus Arbitration** – This hardware is indirectly tested by allowing all DSPs access to the Global Bus while performing the tests listed above. The fact that the BST completed indicates the arbiter is functioning correctly.

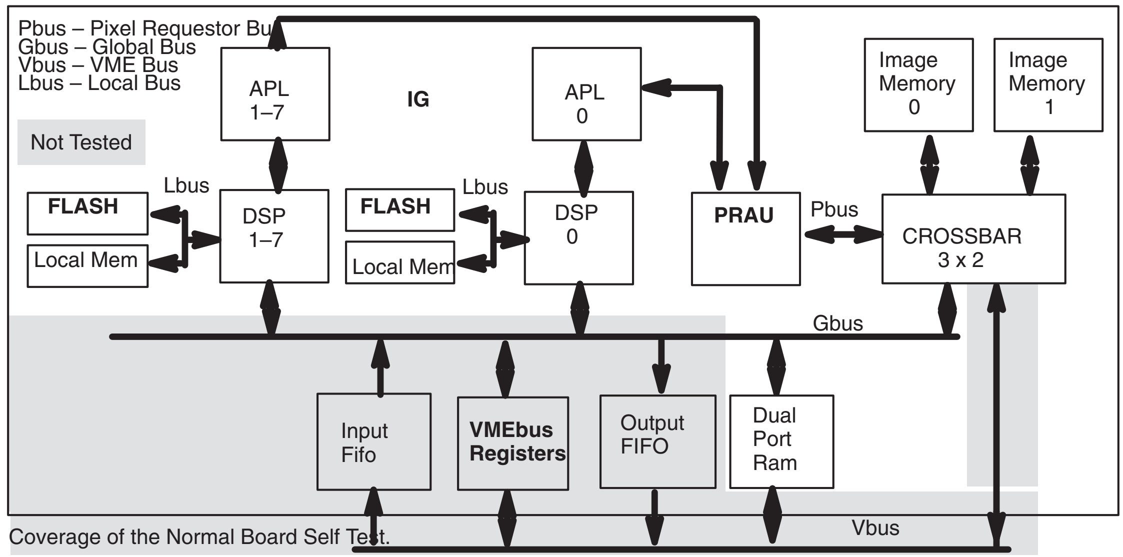


Figure 8-110 Test Hardware Functional Block Diagram Normal Board Self Test

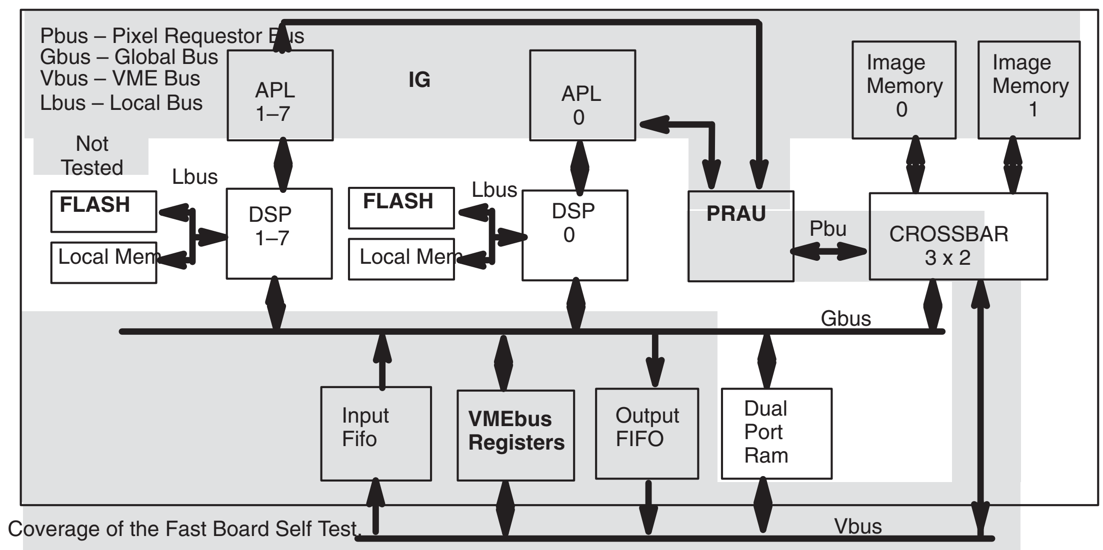


Figure 8-111 Test Hardware Functional Block Diagram FAST Board Self Test

Coverage Exceptions

The BST is limited to testing devices connected to the Global, Local, and Pixel Requester buses. **Devices not tested are:**

- Input and Output Data FIFO integrity and interrupt mechanisms
- VMEbus interfaces
- IOPB interrupts
- VMEbus Map register
- VMEbus Status Register
- VMEbus Command Register
- The Local Memory used to store the BST code is only partially validated by performing a check-sum

Test Usage

This test runs automatically at power-up and after a board reset. Test results are automatically logged when UNIX is booted and when the IG Board Level Diagnostic, SBC to IG Datapath, or RP Self-test is run.

Test Error Messages

The text for IG BST messages comes from the `igDiagUtil_ermes.di` file. The text for IG Memory errors can be found in the `igst_ermes.di` file. The IG BLD and Datapath Diagnostics text can be found in the `igDiag_ermes.di` and `igDiagUtil_ermes.di` files.

3.7.3.2 About IG Failure Codes

```
IG? DSP?: CPU error.
The DSP failed the CPU verification test.
Probable cause: IG board
CPU error code: 0x??
```

This error message is logged from the CPU test code responsible for performing an operational verification of the DSP. The CPU error code given with this message further defines the type of failure found. Replace the IG if it fails Image Generation when there is no higher level cause for the board's failure.

DSP Error Abbreviations

`ff` – Integer register error
`fe` – Floating point register error
`fd` – Auxiliary register error
`fc` – Address generation error
`fb` – ALU error
`fa` – Status register error
`f9` – Shifter or instruction set error
`f8` – Multiplier error
`c1` – Local memory error
`f7` – RAM 0 memory error
`f6` – RAM 1 memory error
`f5` – Stack error
`f3` – Parallel instruction set error
`f2` – Address mode error
`f1` – DMA error
`f0` – Timer error
`ef` – Interrupt error
`c3` – Dual Port RAM error

IOPB Processing

Application and diagnostic requests are sent from the SBC to the IG board via an IOPB. Processing of an IOPB is as follows:

- SBC writes the request to Dual Port Memory
- The SBC Interrupts the DSP indicating a request is present
- The selected DSP reads the request and performs the desired action
- The results of the request are stored in Dual Port Memory
- The selected DSP returns a complete interrupt via the VMEbus
- SBC reads the results from Dual Port Memory

The SBC has two ways to interrupt the DSPs. The first is a single address in Dual Port Memory which interrupts all 8 DSPs. The SBC chooses the desired DSP by writing its number (Called the

Global Go Word) to this VMEbus address. Each DSP in turn reads this location via the Global Bus to determine which DSP was selected. The DSP with the matching number responds by writing 0x00FF to this address and copying the request to its Local Memory for processing. An alternative way to interrupt the DSPs is to write the desired DSPs number to 1 of 8 Dual Port Memory addresses which interrupts only the desired DSP. Except for the Dual Port Memory addresses used, processing is identical to a Global IOPB interrupt. The following is a partial list of the types of requests sent via an IOPB.

- Setup the Image Memory's crossbar
- Load a block of memory via the Input Data FIFO
- Read a block of memory via the Output FIFO
- Perform a checksum of a block of memory
- Read or write the Global Command Register
- Execute the downloaded applications code stored at a given address
- Read error log

3.7.3.3 Image Generator (IG) Board Level Diagnostic

Image Generator (IG) Diagnostics fall into four categories:

- **IG Board Level Diagnostics (BLD)** – Validates the operation of the IG board and VME interfaces. Test is SBC and IG Flash EPROM based.
- **IG Datapath** – An extensive test of the VME interface between the SBC and IG board.
- **Image Generation Test** – This test reconstructs an image using known data for the purpose of validating datapaths and the back projector.
- Validate multiple devices simultaneously to duplicate applications use of the board.

Board Level Diagnostic (BLD)

The IG Board Level Diagnostic (BLD) is a UNIX and IG FLASH EPROM based test which validates the integrity of the IG board. Validation is performed in two steps:

- Runs the Board Self Test – This is IG FLASH EPROM based code which validates the internal operation of the IG hardware. See the previous section of this manual for more information.
- Validates VME interfaces – This is SBC based code for validating all devices connected to the VMEbus. Implementation requires only 1 DSP to coordinate the test between the SBC and IG board. To enhance hardware validation, the test is repeated using a different DSP until all DSPs have been tested.

Tests which require a coordinated effort between the SBC and IG BST use the IOPB mechanism described below for communications.

At the conclusion of the BLD, a fast reset is performed to place the board back into a known state.

BLD Coverage

The diagnostic coverage of this test is listed below and illustrated in Figure 8-112.

- Validate the operation of the **VMEbus Command Registers'** reset bits by toggling each bit and verifying the DSP reset.
- Executes a **Normal IG BST** and report all errors to the error log. See the IG BST section of this document for more information concerning the coverage of this test.
- Validate the **Global IOPB interrupt operation** by sending a crossbar request to each DSP via the Global IOPB interrupt. Verify the correct response is returned in the allotted time. This diagnostic leaves the crossbar setup for VMEbus access to Image Memory.

- Validate the functionality of the **local IOPB interrupts** by sending a NOP instruction to each DSP via the Local IOPB interrupt. Verify the correct response is returned in the allotted time.
- Validate the **Input and Output Data FIFO** integrity and VMEbus statuses. Testing consists of sending a test packet to the IG DSPs Local Memory via the Input Data FIFO (IDF) and reading it back via the Output Data FIFO (ODF). The SBC compares the received packet to the packet sent. If a mismatch is detected, the data stored in Local Memory is checked via a check-sum to determine which FIFO failed.
- Validate the operation of the **VMEbus Status Registers'** VME address Out of Bound bits by writing to undefined address spaces and verifying the correct error bits are set.
- Validate the operation of the VMEbus Status Register's FIFO status and error bits by reading and writing the FIFOs and verifying the correct bits are set.
- Validate **Image Memory** by performing a write and read data test. Testing validates the VMEbus access of the Image Memory and the operation of the crossbar. Additional test patterns are used if the SLOW BLD option is selected.
- Validate the operation of **Dual Port Memory** by performing a write and read data test via the VMEbus. Additional test patterns are used if the SLOW BLD option is selected.
- Validate Dual Port **Memory's VMEbus and Global Bus arbitration** by simultaneously having the SBC and a DSP perform a write and read test.

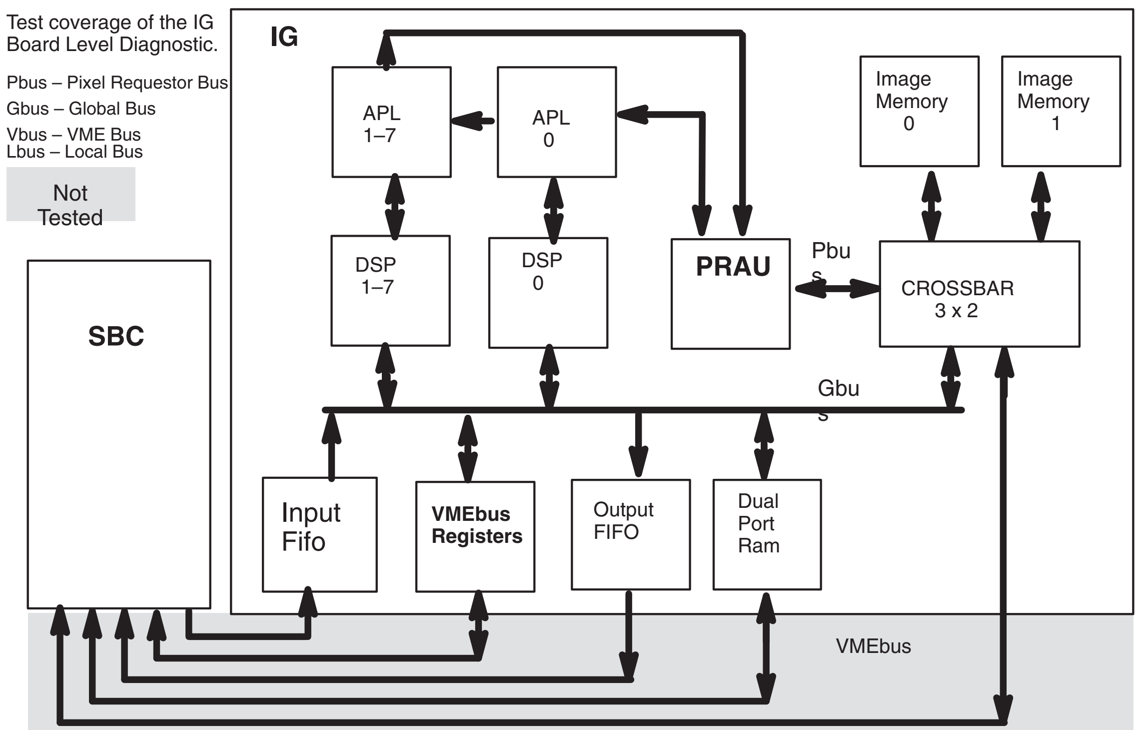


Figure 8-112 Test Hardware Functional Block Diagram

Coverage Exceptions

All board areas are tested.

Test Usage

These tests are prescribed via the Diagnostic Interface. The user has the option of selecting the SLOW BLD option and the number of passes. The test results table consists of the number of passes, failures, and the number of errors logged. The user also has the option of stopping the test at any given time. It is suggested that the test be run with SLOW BLD option enabled and 1 pass selected. This test can also be executed in a stand-alone environment.

3.7.3.4 IG Data Path Test

This test is identical to the BLD test described in the previous section except:

- The IG board is reset only twice during this diagnostic. Once at the beginning of the test and once at the end. A Fast Board Self Test is performed in both cases except when executed from the RP Self-test. In this case, a Normal BST is performed at the beginning of the test versus a Fast.
- VMEbus Status and Command Registers are not tested.

Test Coverage

Since this test performs a Fast BST when not run from RP Self-test, the coverage is slightly less then that described for a BLD. This test also does not validate the VMEbus Command and status bits.

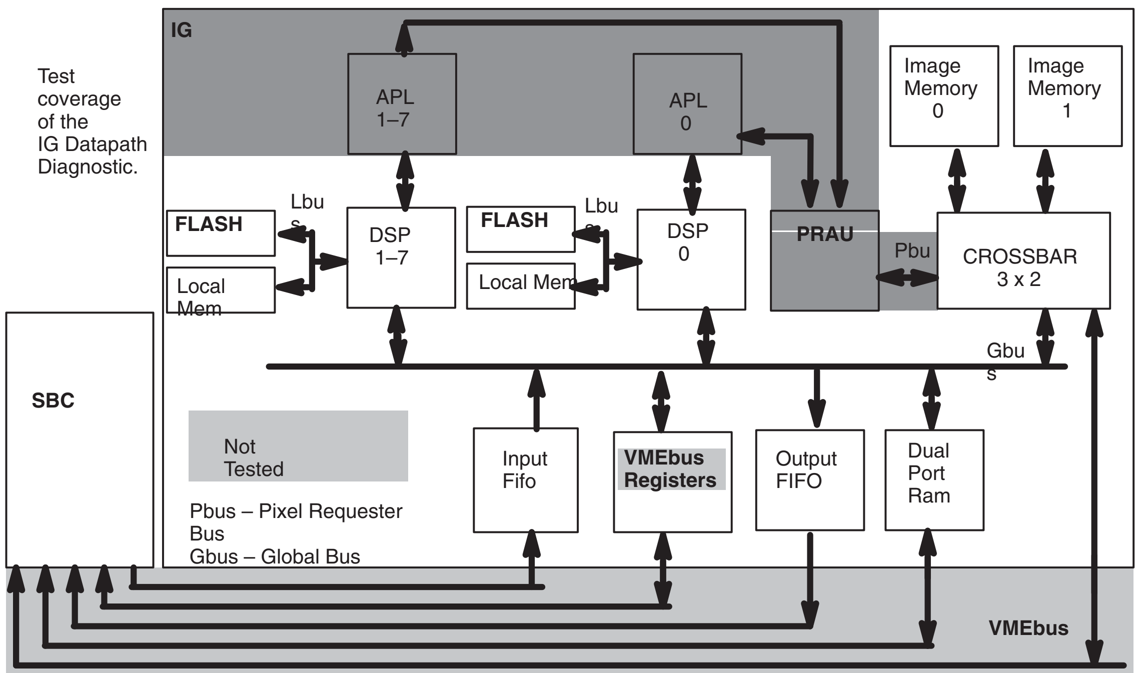


Figure 8-113 SBC to IG Data Path Test Hardware Functional Block Diagram

Coverage Exceptions

In addition to test coverage exceptions listed for the IG BLD, the following devices are not covered under this diagnostic:

- VMEbus Command and Status Registers
- Items not covered for a Fast BST (unless executed from RP Self-test)

Test Usage

You may run this test from 1 minute to 72 hours in 1 minute increments. A test duration of 1 minute should be a sufficient amount of time to find hard failures. Additional time may be required for intermittent faults.

3.7.3.5 Image Generation Test

Things that the IG does:

- Changes axial scan data to images
- If host is Octane, Back Projection portion of axial recon is done by host
- Creates scout images
- Performs PPSCANs
- Performs Views versus Channels diagnostic

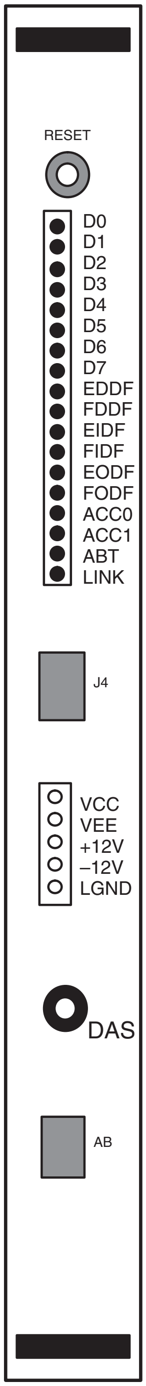
IMAGEGENTEST AND DOD KERNEL

`imageGenTest` will fail for systems that are using the DOD (Department Of Defense) reconstruction kernel. This is because the checksums are different for the images generated with and without the DOD recon kernel.

A new set of files for comparison will be included and correctly link on a future software release. Refer to Section 37.0 on page 358 of Chapter 7 for information on running Image Generation Test.

3.8 Front End Processor

3.8.1 FEP 46-327036 (Front End Processor)



FEP (GEMS Front End Processor) LED Descriptions:

D0	D0 thru D3 indicates the self–test being run
D1	on DSP0 during power–up or reset and then
D2	generates an up/down chasing pattern when
D3	DSP0 has successfully completed self–test.
D4	D4 thru D7 indicates the self–test being run
D5	on DSP1 during power–up or reset and then
D6	generates an up/down chasing pattern when
D7	DSP1 has successfully completed self–test.
EDDF	Indicates an EMPTY DAS DATA FIFO condition during operation.
FDDF	Indicates a FULL DAS DATA FIFO condition during operation.
EIDF	Indicates an EMPTY INPUT DATA FIFO condition during operation.
FIDF	Indicates a FULL INPUT DATA FIFO condition during operation.
EODF	Indicates an EMPTY OUTPUT DATA FIFO condition during operation.
FODF	Indicates a FULL OUTPUT DATA FIFO condition during operation.
ACC0	Indicates DSP0 local memory ACCESS.
ACC1	Indicates DSP1 local memory ACCESS
ABT	Indicates ABORT relay status is OPEN (scanning is NOT enabled).
LINK	Fiber–Optic receiver detects a valid light source (fiber connection).

FEP (GEMS Front End Processor) Front Panel SWITCH Descriptions:

RESET	Momentary pushbutton immediately resets the FEP, causing the self–test to run.

FEP (GEMS Front End Processor) TEST POINT Descriptions:

VCC	Onboard test point for +5V on the FEP from console power supply
VEE	Onboard test point for –5V (generated on the FEP).
+12V	Onboard test point for +12V DC on the FEP.
–12V	Onboard test point for –12V DC on the FEP.
LGND	Onboard test point for LOGIC GROUND.

FEP (GEMS Front End Processor) CONNECTOR Descriptions:

J4	Connector for DSP emulator (factory use only).
DAS	Fiber–Optic connector for DAS DATA fiber–optic cable from Gantry/SCOM.
AB	Connector for scan abort daisy chain cable from PDU to STC.

Figure 8-114 Front End Processor (FEP)

3.8.2 FEP Board Layout

The 46-327036G2 or G3 version of the FEP contains two jumpers.

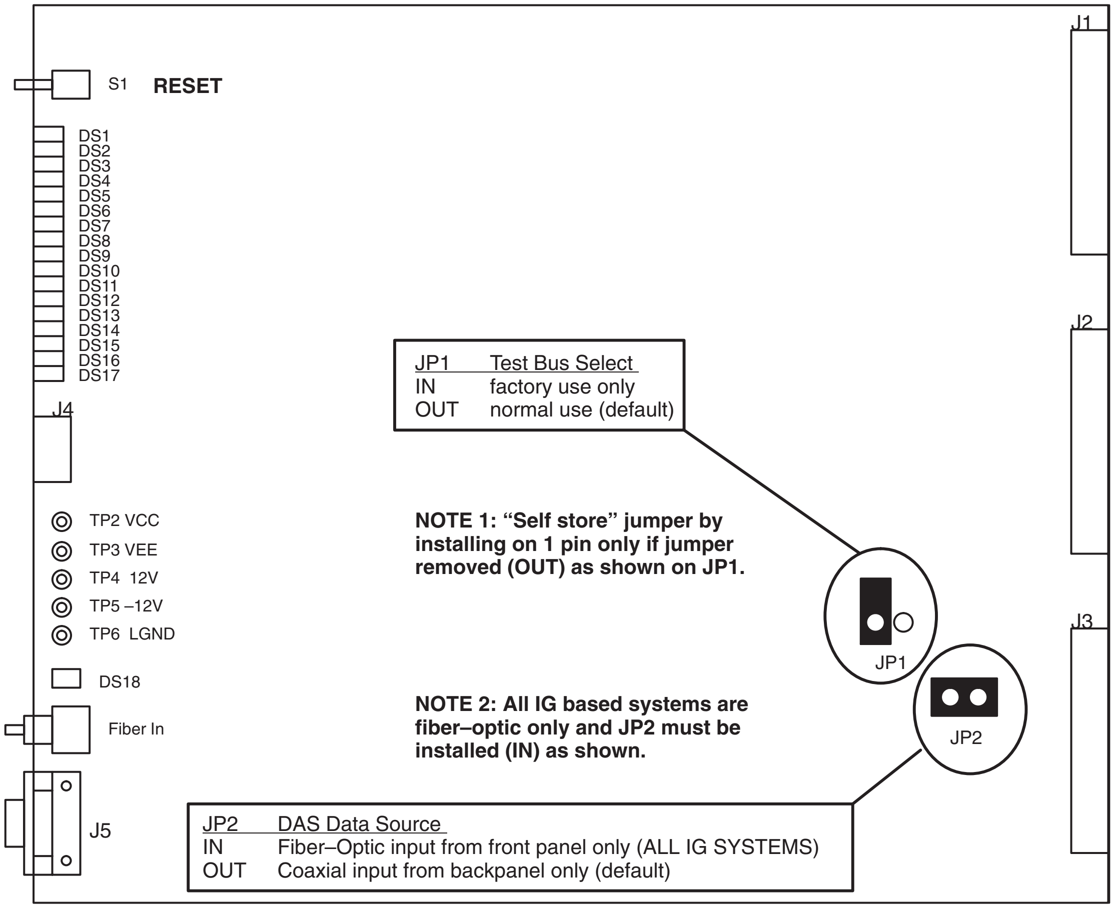


Figure 8-115 Front End Processor (FEP) Layout

Jumper	Name/condition	Meaning
JP1	TBS	Test Bus select. Selects the bus source of the JTAG test: either the 74ACT8990 test bus controller, or the XDS510 test bus interface.
JP1	Removed	74ACT8990 is test bus source **(DEFAULT)**
JP1	Installed	XDS510 is test bus source **(ENG TEST ONLY)**
JP2	DDS	AS Data Source select. Selects the source of the DAS data from the RPSCOM: either a BNC on the RP chassis back-plane, or a fiber optic receiver on the FEP front face.
JP2	Store position	BNC is DAS data source **(DEFAULT)**
JP2	Installed	Fiber optic is DAS data source

Table 8-29 46-327036G2 FEP BOARD JUMPERS

3.8.3 FEP LEDs

The front of the FEP board contains 18 LEDs. Hardware sets LEDs DS9 through DS18 to indicate current board status. The meaning of DS9 through DS18 never changes. The DSPs set DS1 through DS8 during board operation. DSP0 controls D0 through D3 and DSP1 controls D4 through

D7. The FEP has three operating modes: (1) Power-up, (2) Diagnostics, and (3) Applications. The meaning of D0 through D7 changes with the FEP mode.

Reference Designator	Color	Name	Meaning
ds1	Green	`d0`	Diagnostic/status LED #0, mapped to Internal Command Register Bit 24 and controlled by DSP0
ds2	Green	`d1`	Diagnostic/status LED #1, mapped to Internal Command Register Bit 25 and controlled by DSP0
ds3	Green	`d2`	Diagnostic/status LED #2, mapped to Internal Command Register Bit 26 and controlled by DSP0
ds4	Green	`d3`	Diagnostic/status LED #3, mapped to Internal Command Register Bit 27 and controlled by DSP0
ds5	Green	`d4`	Diagnostic/status LED #4, mapped to Internal Command Register Bit 28 and controlled by DSP1
ds6	Green	`d5`	Diagnostic/status LED #5, mapped to Internal Command Register Bit 29 and controlled by DSP1
ds7	Green	`d6`	Diagnostic/status LED #6, mapped to Internal Command Register Bit 30 and controlled by DSP1
ds8	Green	`d7`	Diagnostic/status LED #7, mapped to Internal Command Register Bit 31 and controlled by DSP1
ds9	Green	`eddf`	Empty DAS Data FIFO
ds10	Green	`fddf`	Full DAS FIFO
ds11	Green	`eidf`	Empty VME Input Data FIFO
ds12	Green	`fidf`	Full VME Input Data FIFO
ds13	Green	`eod`	Empty VME Output Data FIFO
ds14	Green	`fodf`	Full VME Output Data FIFO
ds15	Green	`acc0`	DSP#0 currently accessing local SRAM
ds16	Green	`acc1`	DSP#1 currently accessing local SRAM
ds17	Green	`abt`	Abort line open (scanning not enabled)
ds18	Green	Link Present	Fiber optic receiver detects a valid light source. **"This does not indicate the presence of a valid RPSCOM-FEP TAXI LINK or valid DAS DATA".**

Table 8-30 FEP BOARD LED Descriptions

3.8.4 FEP Power-up Mode

During Power Up Diagnostics (PUD, which takes about 2 seconds) LEDs D0 thru D7 flash a count down sequence from FF hex (all on) to 00h (all off) indicating the test in progress. If the PUD passes, two 4-LED "cylon" (or Knight Rider) patterns will be present, one for DSP0 and one for DSP1. If either DSP fails its PUD, one of the following 4-LED failure codes rapidly flash to indicate the failing test:

D3	D2	D1	D0	Description
ON	ON	ON	ON	Not Used
ON	ON	ON	off	DSP1 Diagnostic Initialization Failure
ON	ON	off	ON	DSP1 EPROM Check sum Failure
ON	ON	off	off	DSP1 Loaded Image Checksum Failure
ON	off	ON	ON	DSP1 to DSP1 Interprocessor Communication (semaphore) Failure
ON	off	ON	off	DSP1 Stack Space Failure
ON	off	off	ON	Not Used
ON	off	off	off	Not Used
off	ON	ON	ON	Not Used
off	ON	ON	off	DSP1 CPU Functionality Failure
off	ON	off	ON	DSP1 Register Access Failure
off	ON	off	off	DSP1 Dual Port RAM Access Failure
off	off	ON	ON	DSP1 Global Bus Error Detection Failure
off	off	ON	off	DSP1 Scan Control (SDV/ODV) Failure
off	off	off	ON	DSP1 View Loop Failure

Table 8-31 FLASHING LED PATTERN ON FEP

3.8.5 FEP Diagnostics Mode

During diagnostic mode scanning, LEDs D0 thru D7 represent the following:

INDICATOR	DESCRIPTION
D0	DSP0 Heartbeat. DSP0 alive as long as it flashes once per second.
D1	DSP0 Diagnostic Mode Enabled. ON means the software loaded diagnostics into DSP0 and awaits next command
D2	DSP0 Diag Exec Processing. ON while the diagnostic executive in DSP0 processes a diagnostic command
D3	DSP0 FEPCtl Processing. On while the FEP control processing in DSP0 processes a control command
D4	DSP1 Heartbeat. DSP1 alive as long as it flashes once per second.
D5	DSP1 Diagnostic Mode Enabled. ON means the software loaded diagnostics into DSP1 and awaits next command.
D6	DSP1 Diag Exec Processing. ON while the diagnostic executive, in DSP1, processes a diagnostic command
D7	DSP1 FEPCtl Processing. On while the FEP control processing in DSP1 processes a command

Table 8-32 FEP DIAGNOSTICS MODES

3.8.6 FEP Applications Mode

During applications mode, LEDs D0 thru D7 on the FEP indicate the following:

INDICATOR	DESCRIPTION	
DO	DSP0 Heartbeat. Indicates DSP0 alive when it flashes once per second.	
D1	DSP0 FEP Mode Active. ON means the DSP0 received a scan mode, and is actively processing a scan	
D3 & D2	DSP0 Mode Type. Indicates DSP0 processing mode (below)	
D3	D2	DSP0 Mode
off	off	Not Used
off	ON	Scout
ON	off	Axial
ON	ON	Others (RTS non proprietary Diags, NDC)
D4	DSP1 Heartbeat. Indicates DSP1 alive when it flashes once per second.	
D5	DSP1 FEP Mode Active. ON means the DSP1 received a scan mode, and is actively processing a scan	
D7 & D6	DSP1 Mode Type. Indicates DSP1 processing mode (below)	
D6	D7	DSP1 Mode
off	off	Not Used
off	ON	Scout
ON	off	Axial
ON	ON	Others (RTS non proprietary Diags, NDC)

Table 8-33 APPLICATIONS MODE ON FEP

3.8.7 FEP Test Points

The front of the FEP contains the following five test points:

TEST POINT	NAME	MEANING
TP2	VCC	+5V Logic Supply
TP3	-5V	-5V ECL Logic Supply
TP4	+12V	+12V Backplane Voltage
TP5	-12V	-12V Back-plane Voltage
TP6	LGND	Logic Ground

Table 8-34 FEP TEST POINTS

3.8.8 FEP Switches

The FEP contains one switch. Use switch S1 to power-up reset the FEP.

3.9 Bit3 VME Interface Board, 2124215 and 2235744-2

For diagnostic and troubleshooting information, refer to Diagnosing BIT3 Subsystem on CT/I 5.x (Octane) on page 391 and Diagnosing BIT3 Subsystem on CT/i 3.X/4.X (INDIGO2) on page 442.

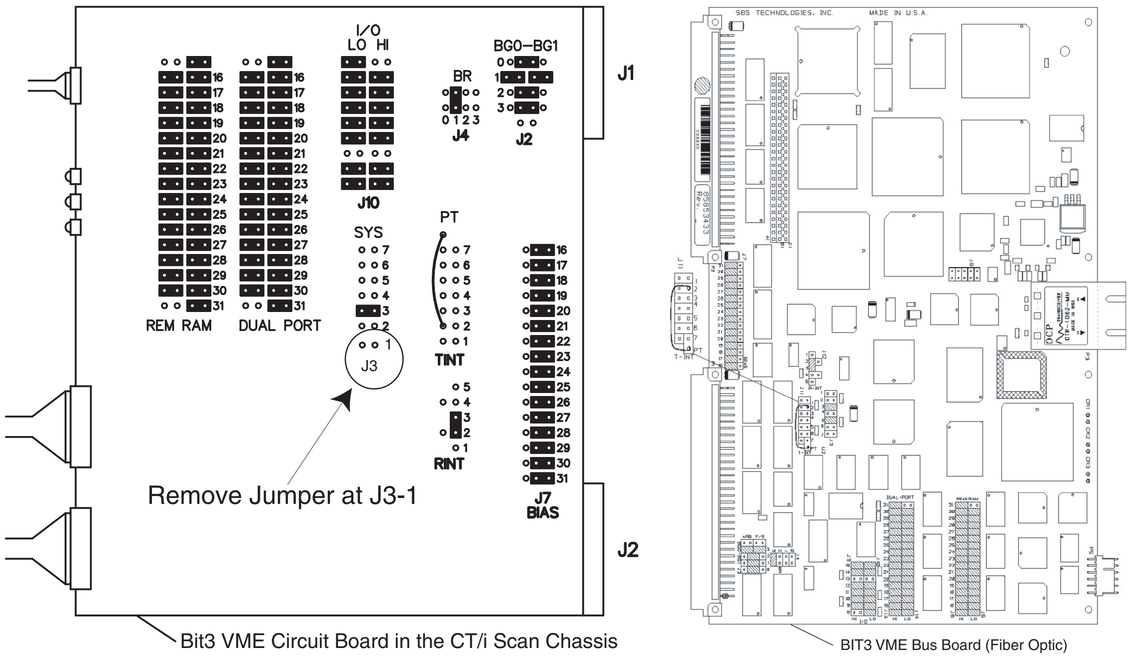


Figure 8-116 Bit3 CT/i VME Board Jumpers

Note: Always remove jumper located at position J3-1, see Figure 8-116, because it will cause the Octane computer to generate an operating (IRIX) “panic error” message during boot-up. This jumper is not used in the Indigo or Octane computer and is therefore unnecessary. It can be removed completely and discarded (thrown away).

3.10 Ethernet Transceivers

In addition to preforming as an electrical interface to the network, the ethernet transceiver also indicates the relative health of the network. If the network isn’t working, the LED on the ethernet transceiver will not be lit. Check the LED indicator on the MVME166 of the SBC board. If there is no green light ON under the RPWR label, check the 1 amp fuse F1 on the MVME166. The SBC requires 5V, 12VP and 12VN whose source is console power supply.

Note: When you receive a new transceiver, always check the heartbeat switch. The heartbeat test ON may cause the system ethernet to malfunction.

SQE SWITCH

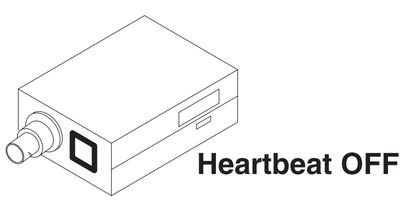


Figure 8-117 SQE Switch

The SQE switch enables and disables the heartbeat test. Disable the heartbeat test during normal operation. To configure the heartbeat (SQE) switch:

- Slide the SQE switch to ON, to enable the heartbeat (SQE) test.
- Slide the switch to the OFF, to disable the heartbeat (SQE) test.

USING PING TO CHECK NETWORK CONNECTIONS

PING is a basic command you use to check whether another device on the network is on or reachable. It tests the lowest level of the network. It tests whether the circuit is complete. It does not test how well the DICOM software communication is working.

To use Ping, you type the ping command and what you want to ping. You identify the what by its AE title or IP address. You get this information from the site's system administrator. If already setup, then ` **/usr/etc/netstat -i**' on the scanner will display the hostnames.

Example: **ping** <***hostname***> (or *IP address*)

or

/usr/etc/ping -q -f -s 2048 -c 100 <***hostname***>

When troubleshooting the Network, attach a laptop to the site's network rather than to the CT/i scanner. Use the laptop to:

- verify that you have the correct IP addresses and AE Titles for all the DICOM devices the unit uses
- determine if the laptop can successfully Ping a device that the scanner cannot
- check the network to scanner connection (transceiver) and communication

3.11 Power Supplies

3.11.1 Power Supply Requirements (FEP, IG, BIT3 & CPU):

The FEP, IG, BIT3, CPU boards (attached to VME back-plane) use the following voltages:

3.11.1.1 +5V Supply

- The P1, P2, and P3 VME connectors supply the voltage to the digital logic.
- Recommended DC operating voltage: 4.75V to 5.25V.
- The absolute maximum voltage which, if exceeded, damages the FEP: -0.5V to 7.0V, relative to LGND.
- The FEP draws no more than 8 Amps under normal operating conditions.
- The FEP draws a maximum current of 13 Amps.

3.11.1.2 +12V Supply

- The P1 VME connector supplies voltage to the FLASH RAMs during re-programming.
- Normal operating voltage: 11.4V to 12.6V.
- The absolute maximum voltage range: -2.0V to 13.5V, relative to LGND.
- The FEP draws no more than 16 milliampere under normal operating conditions.
- The FEP draws a maximum current of 36 milliampere.

3.11.1.3 -12V Supply

- The P1 VME connector supplies the voltage to the DC/DC converter for conversion to -5V.
- Normal operating voltage: -11.4V to -12.6V.
- The absolute maximum voltage range: -13.5V to 2.0V, relative to LGND.
- The FEP draws no more than 80 milliampere under normal operating conditions.
- The FEP draws a maximum current of 115 milliampere.

3.11.2 Power Supply Replacement

3.11.2.1 Remove Scan Chassis Power

1.) Shutdown Applications Software and Halt Unix on the SBC:

   a.) Start at the applications level, and select the UTILITIES soft key to display the **Utilities Main** menu.

   b.) Select SHUTDOWN.

2.) Unplug the console power cable.

NOTICE **Prevent permanent damage to the static-sensitive boards. Attach the anti-static wrist strap to your wrist, and to a bare metal grounding point on the scan chassis before you continue.**

3.11.2.2 Remove power supply

1.) From the back of the console, remove cable assemblies from supply.

2.) From the front of the console:

   a.) Loosen, and remove, the top 2 screws from the metal plate.

   b.) Remove the metal plate.

3.) Remove the incoming line power from the AC input terminals.

4.) Slide the tray assembly out of the scan chassis and lay it on a flat surface.

5.) Remove the 4 screws that fasten the power supply to the tray.

3.11.2.3 Install new power supply

1.) Use 4 screws to fasten the new power supply to the tray.

2.) Slide the tray assembly into the scan chassis.

3.) Reattach cables.

4.) Reconnect the incoming line power to the AC terminals.

5.) Restore AC power to the console.

6.) Verify power supply voltage.

7.) Replace the metal plate and reassemble the console.

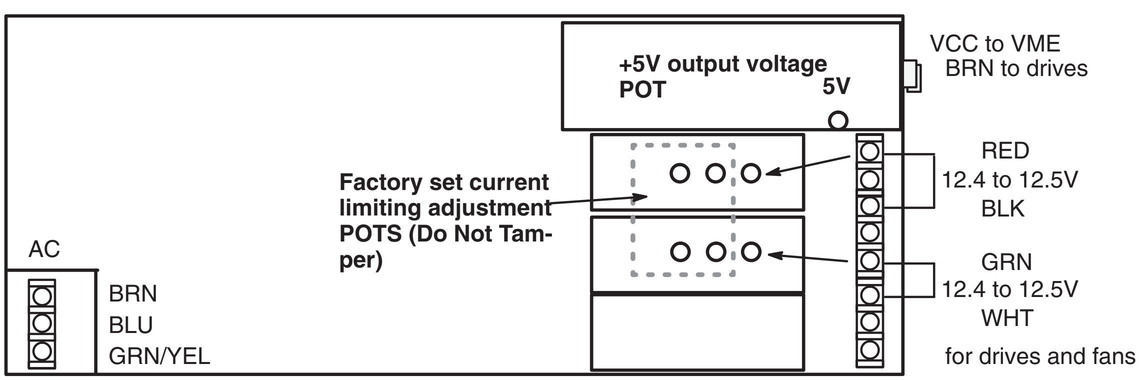


Figure 8-118 Console Power Supply

GE MEDICAL SYSTEMS

GE MEDICAL SYSTEMS-AMERICAS: FAX 414.544.3384
P.O. BOX 414; MILWAUKEE, WISCONSIN 53201-0414, U.S.A.

GE MEDICAL SYSTEMS-EUROPE: FAX 33.1.40.93.33.33
PARIS, FRANCE

GE MEDICAL SYSTEMS-ASIA: FAX 65.291.7006

GE Medical Systems

gehealthcare.com

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GE Medical Systems
CT/i System Service Manual - Advanced

Chapters 9 and 10

High Voltage & X-Ray, and DAS & Detector

Chapter 9
HV and X-Ray

Section 1.0
High Voltage Replacement Verification

NOTICE **Please perform the retests listed below when you replace or adjust a high voltage part. System Functional Test means scan the first six series using PROTOCOL LIST 20.8 called the System Scanning Test; how to scan with protocols begins on page 64.**

HV System Components	Task	Verification Test
Tube	Pull and complete tube data for old tube and install new tube. See page 565. Complete verification tests and reset Smart Trend Baseline if applicable (5.x SW). Refer to Chapter 6, page 264 for resetting Smart Trend baseline.	Refer to Chapter 3 to verify mA meter, new Auto mA Cal (Seed Shift only), (10) slices of micro phonics, Alignments, calibrate HV Tank Feed back, kV meter*, heat soak and season the tube, verify mA and kV, QCal, Cals, "N" number check, then do image series.
Rotor Controller Assembly aka: CTVRC Module	Replace faulty assembly. Refer to page 587.	System Scanning Test on page 69.
HV Cable	Replace faulty cable. See page 582 for the Anode Cable and page 584 for the Cathode cable.	Auto mA Cal on page 558.
HV Tank (Anode or Cathode)	Refer to page 579 for Anode Tank and page 580 for the Cathode Tank.	Calibrate HV Tank Feedback on page 553. Verify kV meter on page 547. Auto mA Cal on page 558. kV and mA verification on page 556. Rise and Fall Time Accuracy on page 558. System Scanning Test on page 69.
HV Inverter (Anode or Cathode)	Refer to page 582 for the Anode and Cathode Inverter.	Auto mA Cal on page 558. kV and mA verification on page 556. System Scanning Test on page 69.

Table 9-1 High Voltage System Retest Matrix

HV System Components	Task	Verification Test
Measurement Board	Replace faulty board. Refer to page 578	Calibrate HV Tank Feedback on page 553. Verify kV meter on page 547. Auto mA Cal on page 558. kV and mA verification on page 556. Rise and Fall Time Accuracy on page 558. System Scanning Test on page 69.
mA Board	Replace faulty board. Refer to page 592 or page 594 if HEMRC ma Control	Verify mA meter on page 549. Auto mA Cal on page 558. kV and mA verification on page 556. Exposure Time Accuracy, and System Functional Test on page 69
kV Board	Replace faulty board. Refer to page 585.	Calibrate HV Tank Feedback on page 553. Verify kV meter on page 547. Auto mA Cal on page 558. kV and mA verification on page 556. Rise and Fall Time Accuracy on page 558. System Scanning Test on page 69.
HEMRC Control Board	Replace faulty board. Refer to page 586. Board shown on page 596	Select KV & mA under Trouble Shoot; refer to information that begins on page 624 System Scanning Test on page 69.

Table 9-1 (Continued)High Voltage System Retest Matrix

Section 2.0
Access HV Maintenance through Service Desktop

1.) Display the Service Desktop Main Menu.
2.) Select REPLACEMENT PROCEDURES.
    Refer to page 545.
3.) Proceed to the next page.

Section 3.0
Advanced Replacement Procedures Menu

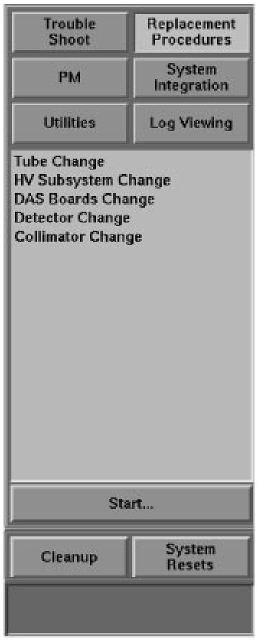


Figure 9-1 Advanced Replacement Menu Screen

Section 4.0
Advanced Replacement HV Subsystem Menu

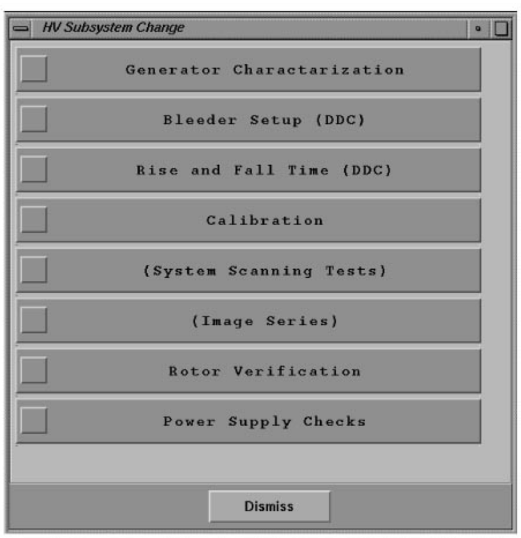


Figure 9-2 Advanced Replacement HV Subsystem Menu Screen

Section 5.0
Generator Characterization

Page 545 shows the Advanced HV Subsystem Change Selection that displays the tools/diags lists, shown on page 545. Use these tools to set up and calibrate the generator hardware, to accurately generate x-rays at the given kV/mA.

Use the Generator Characterization Program to update the “small spot” and “large spot” characterization files, to provide a starting point for the closed loop mode of the generator. This iterative process requires several scans at a different KV/MA/spot size. It calculates corrections, repeats the scan until the results fall within tolerance, then updates the characterization file.

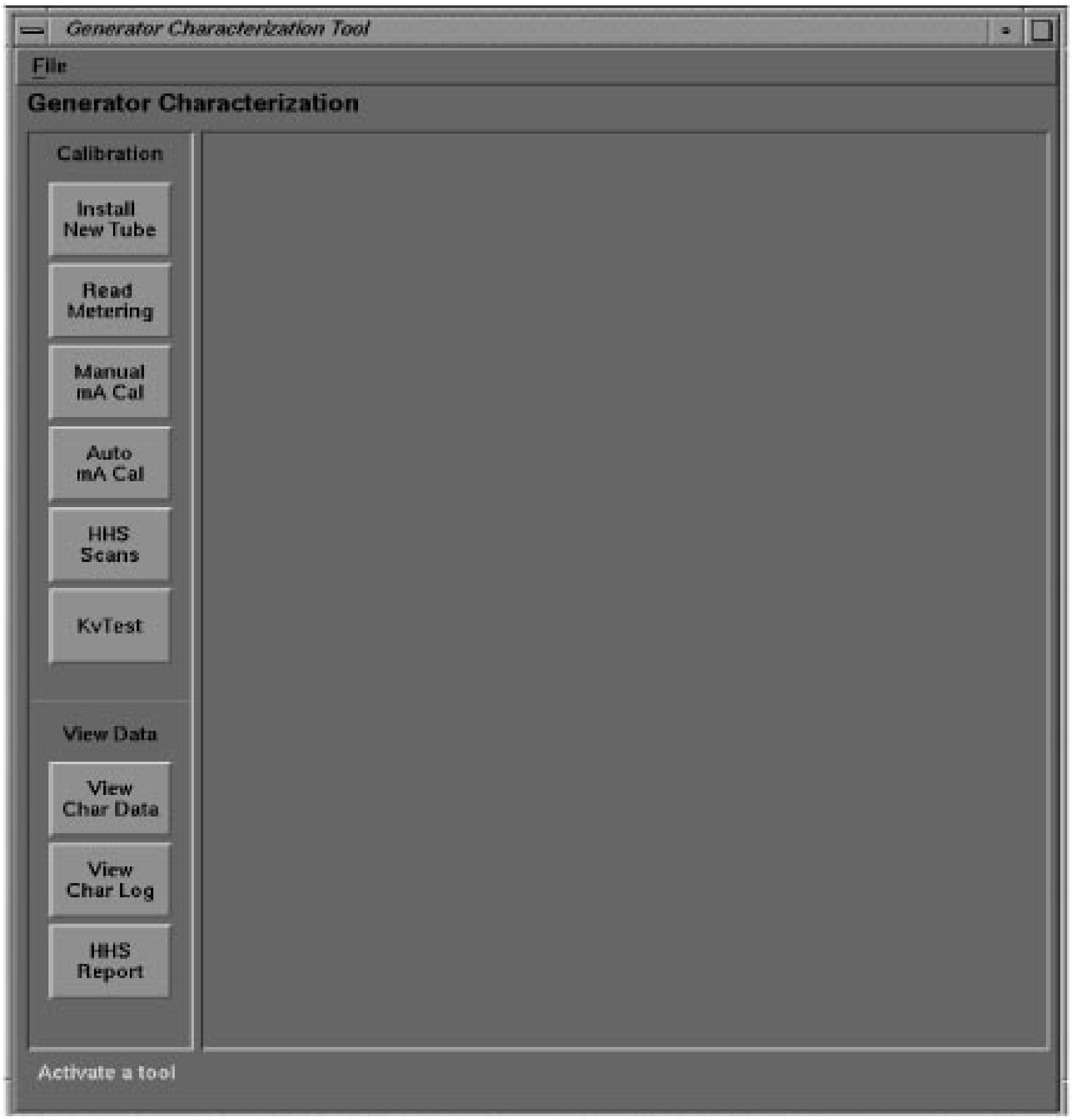


Figure 9-3 Generator Characterization Menu Screen

Section 6.0
Auto mA Calibration Status Screen

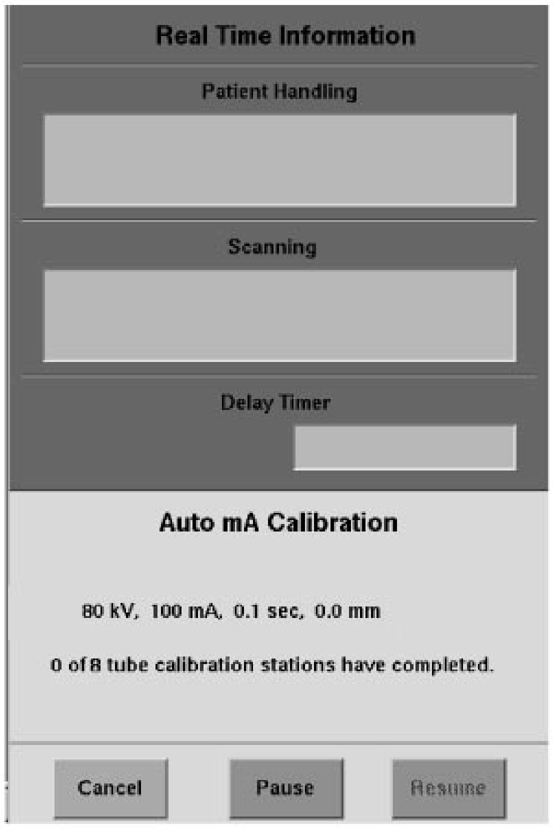


Figure 9-4 Auto mA Calibration Status Screen

Section 7.0
Verify kV Meter

This section describes the calibration check of system internal kV metering circuits.

1.) Select READ METERING (Page 546).
2.) Select RUN (Page 548) to start the test.
  - During the test, the firmware reads the metering circuits in the OFF state, then reads the metering circuits in the ON state, and finally reports the readings to the plasma display.
  - Page 548 shows the displayed Anode kV, Cathode kV and Total kV values for "Circuit OFF" and "Circuit ON".
3.) Compare the data in the "Delta" column on the Read Meter screen (Figure 9-5) to the data in the "Limit" column.

Note: "Delta" = DVM – A/D.

Circuit OFF	Circuit ON
Anode kV = 0 ±0.5	Anode kV = 50 ±7.5
Cathode kV = 0 ±0.5	Cathode kV = 50 ±7.5
Total kV = 0 ±0.5	Total kV = 100 ±15.0

Table 9-2 Generator Characterization test specifications

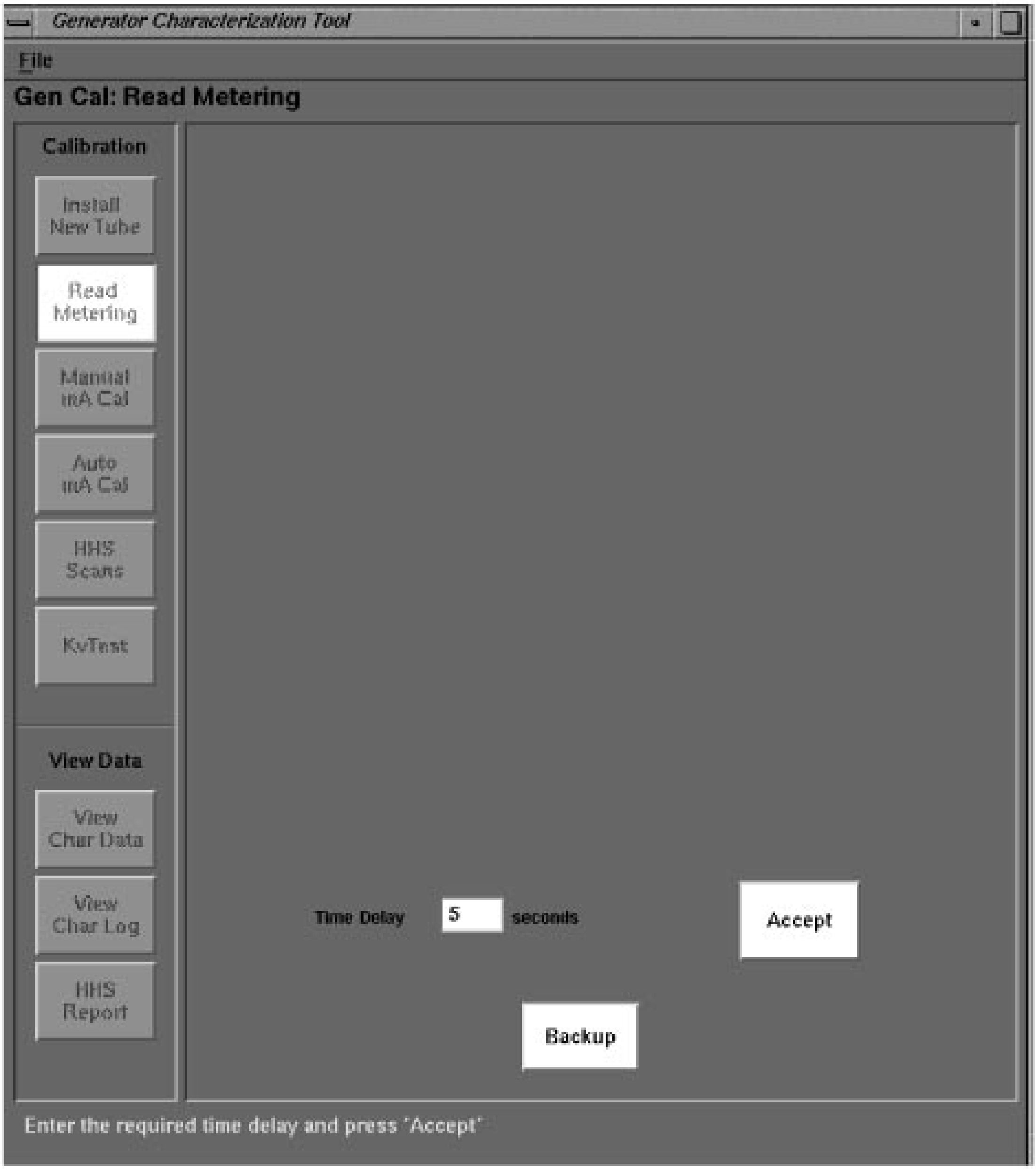


Figure 9-5 Read Meter Screen

Section 8.0
Verify mA Meter

This section describes the calibration check system internal mA metering circuits.

DANGER **NEVER PLACE ANY PART OF YOUR BODY INTO THE GANTRY WITHOUT DISABLING THE AXIAL DRIVE AND VERIFYING THAT IT IS DISABLED. BOTH OF THE AXIAL DRIVE STATUS LEDS MUST BE OUT. DO NOT SERVICE THE GANTRY IF EITHER LED IS ON. FAILURE TO HEED THIS WARNING MAY RESULT IN PERSONAL HARM, HARM TO OTHERS OR DEATH. BE SURE THAT YOU HAVE READ DIRECTION 46-018302, CT HISPEED ADVANTAGE SAFETY GUIDELINE MANUAL OR HAVE VIEWED 46-018308 SAFETY VIDEO TAPE PRIOR TO SERVICING THE GANTRY.**

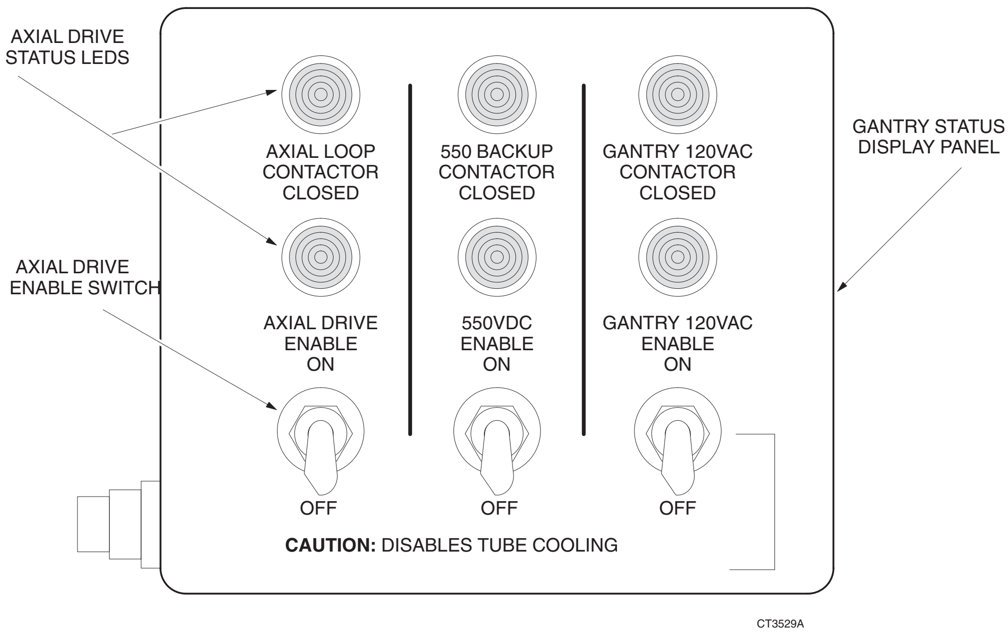


Figure 9-6 Axial Drive Status and Control Panel

1.) Inside the Gantry:
   a.) Switch OFF the 550 V enable on Gantry Control Panel.
   b.) Switch OFF the axial drive.
   c.) Rotate the Anode tank to the 2 o’clock position.
   d.) Pin the Gantry.
2.) Display Generator Characterization Menu.
3.) Select READ METERING (Figure 9-5)

Note: On the display, type/enter a time delay in seconds, to provide enough time for you to walk from the console to the DVM, and record the reading. The test will not begin until this time delay expires. Once it begins, the test enables the meter circuit for only 4 seconds.

4.) Use a DVM as an mA meter; connect it to the hardware on the **anode side**:
   a.) Connect the black lead to TP8 (ACAL1) on the mA board.
   b.) Connect the red lead to TP11 (ACAL2) on the mA board.

5.) Locate the Anode HV Tank Measurement board (Figure 9-7). If TP5 has a wire from the harness, you don't need the following test lead. If TP5 does NOT have a wire from the harness, install a test lead as follows:

a.) Use a jumper with a 68 ohm, 5 watt, resistor in series.

b.) Connect one end to TP11 (ACAL 2) on the mA board.

c.) Connect the other end to TP5 on the Anode Tank Measurement board (Figure 9-7).

6.) On the Display, select the ACCEPT softkey.

7.) Record the displayed, and measured, Anode mA values for "Circuit OFF" and "Circuit ON".

Note: If your system has the test wire to TP5 included in the harness, the Cathode side should read approximately 19 mA during "Circuit On".

8.) Disconnect the test equipment from the Anode side, if used.

Note: When you exit Generator Characterization, this test may generate `kV board tube spit counter = x` error messages.

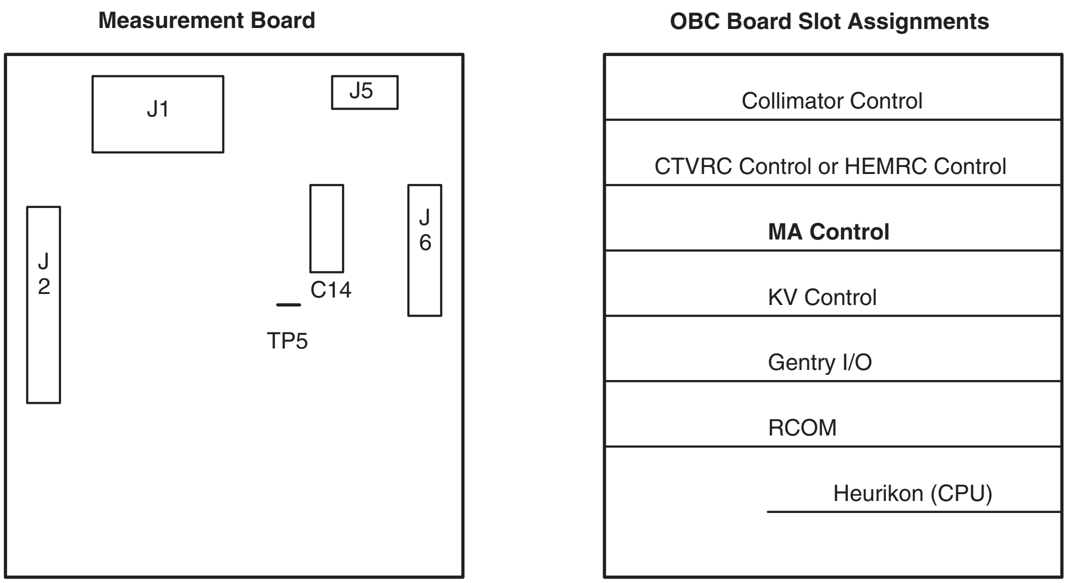


Figure 9-7 Tank Measurement board

9.) Use a DVM as an mA meter:

a.) Connect the black lead to TP9 (CCAL1) on the mA board (page 592).

b.) Connect the red lead to TP14 (CCAL2) on the mA board.

10.) Locate the Cathode HV Tank Measurement board (Figure 9-7). If TP5 has a wire from the harness, you don't need the following test lead. If TP5 does NOT have a wire from the harness, install a test lead as follows:

a.) Use a jumper with a 68 ohm, 5 watt, resistor in series.

b.) Connect one end to TP14 (CCAL 2) on the mA board (page 592).

c.) Connect the other end to TP5 on the Measurement board.

11.) On the Display, Select the ACCEPT softkey.

12.) Record the displayed, and measured, Cathode mA values for "Circuit OFF" and "Circuit ON".

Note: If your system has the test wire to TP5 included in the harness, the Anode side should read approximately 20mA during "Circuit On".

13.) Disconnect the test equipment from the Cathode side if used.

Note: When you exit Generator Characterization, this test may generate `kV board tube spit counter = x` error messages.

Section 9.0
Set Calseed Values

Use the following sequence to run a partial Install New Tube. Use this procedure to set the calseed values on a new system.

1.) Display the Generator Characterization Menu.
2.) Select INSTALL NEW TUBE.
3.) The system prompts you to verify the tube type.
4.) Verify the number corresponds to your tube type, answer **Y** (yes) or **N** (no):

SOFTWARE TOKEN	HOUSING #	INSERT #
12-MX_135CT	46-274800G1	46-274600G1
13-MX_165CT	46-309500G2	46-309300G1
14-MX_165CT_I	46-309500G2	46-309300G2
15-MX_200CT	2137130-2	2120785

Table 9-3 Software Tokens for Various Tube Housings & Inserts (CalSeed)

5.) Press START SCAN when it illuminates.
The system automatically runs the program and updates the display:
`- seed filament current shift scans -`

Note: ABORT the program after the seed filament current shift scans and before the ductility warm-up.

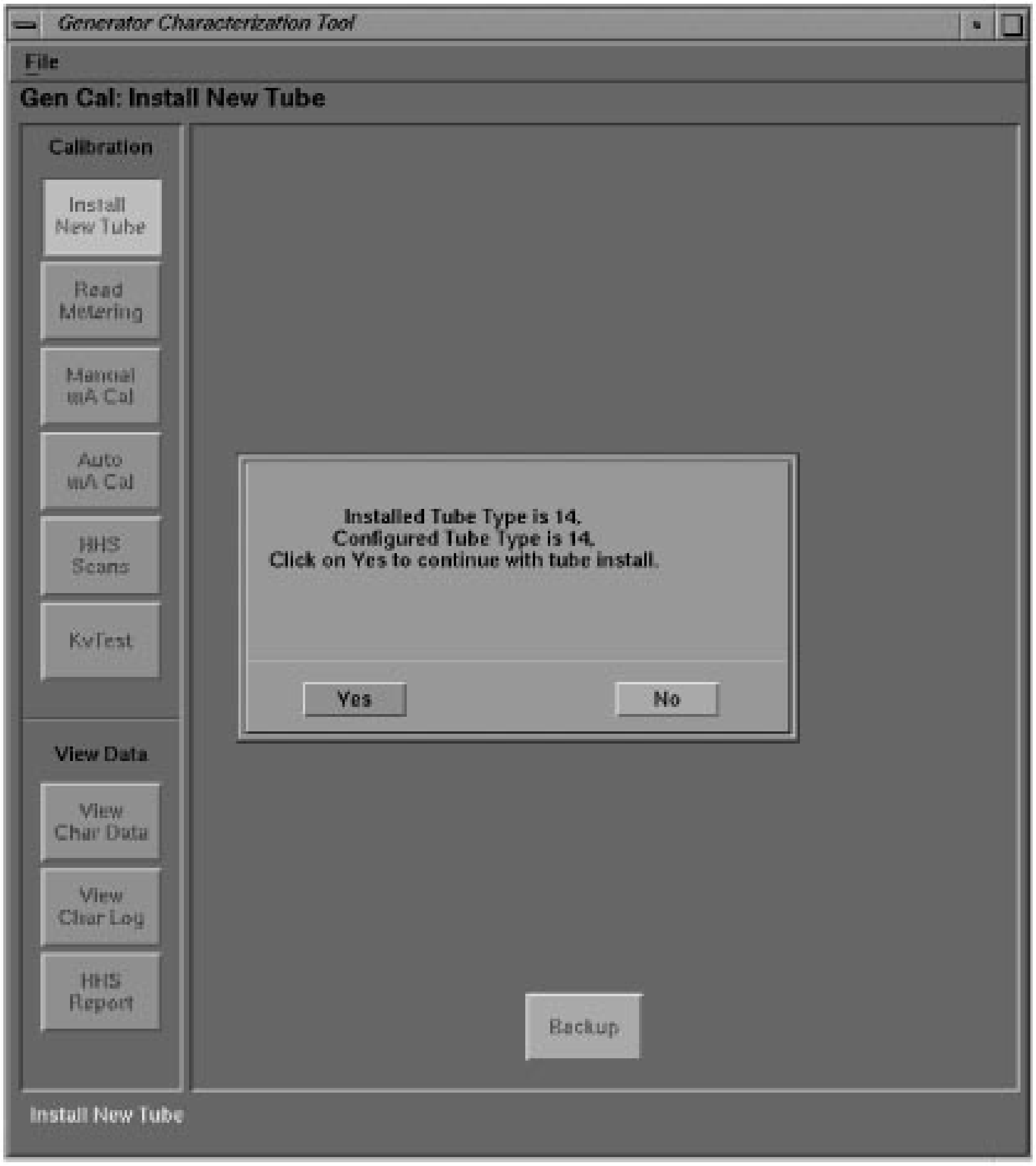


Figure 9-8 Install New Tube Screen

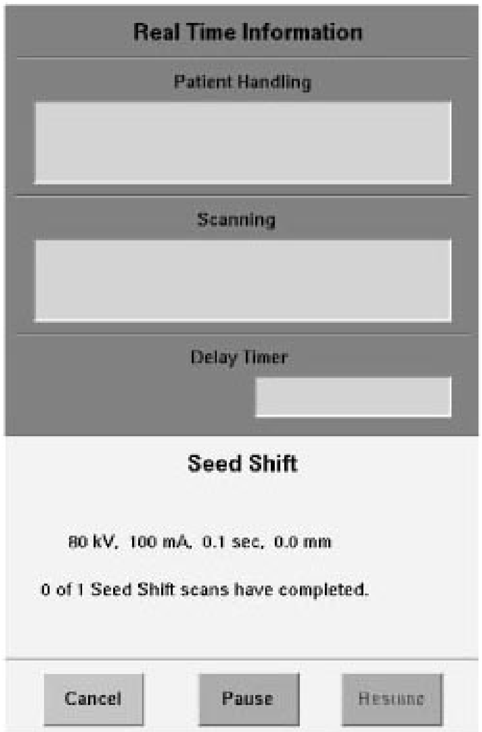


Figure 9-9 Seed Shift Real TIme Information Screen

Section 10.0
KV Gain Pots Adjustment

10.1 Install HV Divider

1.) Switch OFF "X-Ray and Drives" power on the gantry.
2.) Inside the Gantry:
  a.) Switch OFF the HVDC (550) enable.
  b.) Switch OFF the axial drive.
  c.) Rotate the Tube to the 3 o'clock position
  d.) Pin the Gantry.
  e.) Switch OFF the 120 Vac Gantry power.
  f.) Refer to Figure 9-10. Install the HV Divider between Tube and Tanks.

Note: Place the HV Divider on a table or tube hoist, so the cables reach the tube.

3.) Refer to Figure 9-10. Add a ground wire (minimum size of AWG 12) from Tube ground to Bleeder ground.

⚠ **CAUTION** **Performix tube unit MUST be grounded to the gantry during testing!**

4.) Switch ON the 120 Vac Gantry power.
5.) Switch ON the HVDC (550) enable.
6.) Switch ON "X-Ray and Drives" power on the gantry.
7.) Reset the hardware.

NOTICE **Incorrect installation of anode and cathode HV cables can destroy the Performix tube unit!**

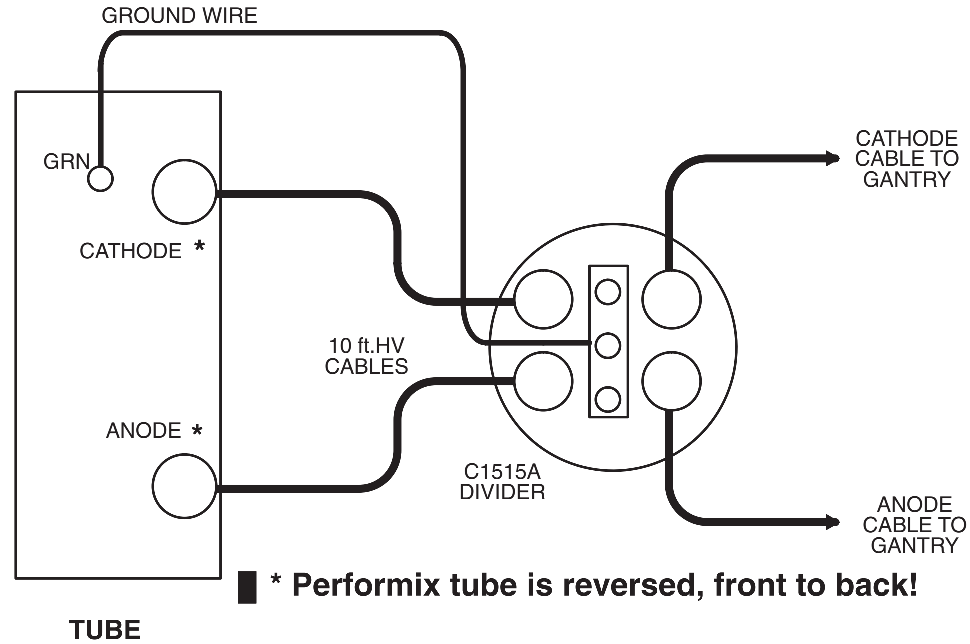


Figure 9-10 HV Divider Installation

10.2 Setup Instrumentation

Use an oscilloscope* with 10X probes:

1.) Connect channel one to the anode output of the divider.
Connect the scope ground to bleeder ground.

2.) Connect channel two to the cathode output of the divider.
Connect the scope ground to bleeder ground.

Note: In order to minimize bleeder-induced ripple on the kV waveform, connect a 30 foot Belden shielded twisted pair cable between the scope probes and the bleeder.

3.) Trigger channel one, positive, DC couple, trigger mode normal.

4.) Channel one and two, 10v/div, time base 200ms.

5.) Invert Channel two.

*If possible, try to plug the scope into the table 120 Vac outlet, to reduce the noise on the scope waveforms. Use an extension cord if necessary.

Note: Determine the type of kV board in the OBC.
This procedure lists the names of components on the 46-321064G1-D kV board without brackets.
This procedure lists the names of components on the 46-321198G1 or 2143147 kV board in [brackets].

10.3 Calibrate the Cathode

1.) Display the Generator Installation and Verification menu.

2.) Select BLEEDER SETUP (DDC).

3.) Verify/Set-up the following DDC parameters:

- STATIC X-RAY ON
- 1 SECOND
- 1 SCAN
- FOCAL SPOT LARGE
- 100 KV
- 50 MA
- MONITOR ENABLE

4.) Select ACCEPT RX

The Computer Displayed reading specification for the Cathode kV and Anode kV equals 50 ±0.5 kV.

Note: If you use scope cursors to window the trace, position the Left Vertical Cursor to the Right of the Rising Edge of the waveform. Position the Right Vertical Cursor to the Left of the Falling Edge of the Waveform.

5.) Adjust the Cathode pot on the kV board, until the scope reading for the Cathode kV, and the displayed reading for the Cathode kV in the message log, fall within ±0.5kV of each other.

6.) Use the pot, labeled CA [CAKV, R316], on the kV board, to adjust the scope reading.
- CCW decreases the scope kV.
- CW increases the scope kV.
- 1/2 turn equals approximately 0.5 kV.

7.) Record the results on **FORM 4879**.

10.4 Calibrate the Anode

1.) Display the Generator Installation and Verification menu.

2.) Select BLEEDER SETUP (DDC).

3.) Verify/Set-up the following DDC parameters:
- STATIC X-RAY ON
- 1 SECOND
- 1 SCAN
- FOCAL SPOT LARGE
- 100 KV
- 50 MA
- MONITOR ENABLE

4.) Select ACCEPT RX

The Computer Displayed reading specification for the Cathode kV and Anode kV is 50 ±0.5 kV.

Note: If you use scope cursors to window the trace, position the Left Vertical Cursor to the Right of the Rising Edge of the waveform. Position the Right Vertical Cursor to the Left of the Falling Edge of the Waveform.

5.) Adjust the Anode pot on the kV board, until the scope reading for the Anode kV, and the displayed reading for the Anode kV in the message log, fall within ±0.5kV of each other.

6.) Use the pot, labeled AN [ANKV, R318], on the kV board, to adjust the scope reading.
- CCW decreases the scope kV.
- CW increases the scope kV
- 1/2 turn is approximately 0.5 kV.

7.) Record the results on **FORM 4879**.

10.5 Measure Total kV

1.) Display the Generator Installation and Verification menu.
2.) Select BLEEDER SETUP (DDC).
3.) Verify/Set-up the following DDC parameters:
   - STATIC X-RAY ON
   - 1 SECOND
   - 1 SCAN
   - FOCAL SPOT LARGE
   - 100 KV
   - 50 MA
   - MONITOR ENABLE
4.) Change the oscilloscope to add ch.1 and ch.2, to read total kV from the HV divider.
5.) Channel one and two, 20v/div, time base 200ms, trigger ch. one, positive.
6.) Select ACCEPT RX
7.) Record the scope reading, and the Avg. kV displayed in the message log, in **FORM 4879**.
8.) Display the Generator Characterization menu.
9.) Toggle the softkey MONITOR ENABLE **OFF**, so the message log no longer displays kV and mA readings.

Section 11.0 Verify kV Meter

Use this procedure to verify the calibration of the internal kV metering circuits.

1.) Display the Generator Installation and Verification menu.
2.) Select MA METER CHECK
3.) Select ACCEPT
   - The test begins after the time delay expires.
   - Once the test begins, the software enables the meter circuit for 4 seconds.
4.) Record the displayed Anode kV, Cathode kV and Total kV values in the **FORM 4879** “Circuit OFF” and “Circuit ON” table.
5.) Select DISMISS

REMOVE THE EXTERNAL HV DIVIDER

1.) Switch OFF “X-Ray and Drives” power on the gantry.
   - Press the EMERGENCY OFF button on the Gantry Mounted Table controls.

   -or-

   - Press X-RAY AND DRIVES OFF on the REM box.
2.) Refer to Figure 9-6 on page 549. Inside the Gantry:
   a.) Switch OFF the HVDC (550) enable.
   b.) Switch OFF the 120 Vac Gantry power.
   c.) Remove the HV Divider between the Tube and Tanks (Figure 9-10).
   d.) Reconnect the HV cables for normal operation.

NOTICE **Incorrect installation of anode and cathode HV cables can destroy the Performix tube!**

e.) Re-apply paper toweling around tube locking ring to absorb excess oil.
f.) Un-pin the gantry.
g.) Switch on the 120 Vac gantry power.
h.) Switch ON the 550v enable.
i.) Switch ON the Axial Drive enable.

3.) Switch ON X-RAY AND DRIVES.
- Press the RESET button on the Gantry Mounted Table controls.

-or-

- Press X-RAY AND DRIVES ON on the REM box.

4.) Reset the hardware.

Section 12.0
Create KV Test Baseline

To save time on future tube change, create a kV Test Baseline on the freshly calibrated system.

1.) Display the Generator Installation and Verification menu
2.) Select KV TEST.
3.) Select TUBE AT TOP
4.) Turn OFF the axial drive switch.
5.) Remove the mylar window from the gantry.
6.) Select CREATE BASELINE
The system acquires four air scans.
7.) Place the kVp Evaluation Tool, PN 2108232, on the detector window.
Refer to Figure 9-11 for a guide to Evaluation Tool placement.
8.) Select CONTINUE
The system acquires four kV scans.
9.) Remove the KVp Evaluation tool from the gantry.

Upon completion, the system saves the kV Baseline with the system characterization values on the system state MOD.

Tool Placement

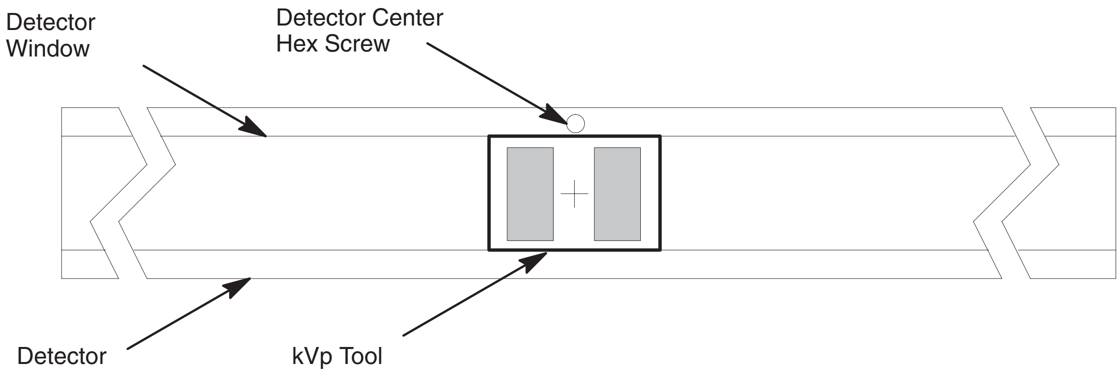


Figure 9-11 kVp Evaluation Tool Placement

Place the kV Test Tool on the detector window, with the print side up. Position the tool over the center of the detector, ±1/2 inch. Align the cross-hair on the tool with the center detector Hex Screw (5th from either visible side).

Section 13.0
Install New Tube Program

Use this program to complete Auto mA Cal on a new tube. **Run this program only on a new tube**.

Refer to Figure 9-6

1.) Display the Generator Characterization Menu.

2.) Select the INSTALL NEW TUBE softkey.

Note: The system automatically warms up the tube.

3.) The system prompts you with the tube type.
Verify the number corresponds to your tube type; answer **Y** or **N**.

Software Token	Housing #	Insert #
12-MX_135CT	46-274800G1	46-274600G1
13-MX_165CT	46-309500G2	46-309300G1
14-MX_165CT_I	46-309500G2	46-309300G2
15-MX_200CT	2137130-2	2120785

Table 9-4 Tube Type Table (SW Tokens for Various Housings & Inserts)

4.) Press START SCAN when it flashes, to automatically run the program and update the display:

`– seed filament current shift scans –`

Section 14.0
Auto mA Calibration

Run this program when you replace the X-Ray tube, or the system requires re-calibration.

1.) Select AUTO MA CAL.

Note: The software automatically warms up the tube.

2.) Press START SCAN when it flashes, to automatically run the program and update the display:

`– Ductility warm-up -`

`– Auto mA Cal -`

3.) The system displays the final filament currents on the screen.

Section 15.0
KV Rise and Fall Times

Note: Determine the type of kV board in the OBC.

This procedure lists the names of components on the 46-321064G1-D kV board without brackets.

This procedure lists the names of components on the 46-321198G1 or 2143147 kV board in [brackets].

1.) In the OBC, connect a scope to the KV board.
   - Channel 1: Exposure Command EXCM, TP22 [TP5].
     Scope ground to LGND, TP6 [TP3].
     2v/div
   - Channel 2: Total kV KVTB, TP30 [TP11].
     (At this test point KV = 20KV per volt.)
     Scope ground to AGND TP39 [SGND, TP12].
     1v/div
2.) Set the Scope Time base to 200 usec.
   Positive or Negative trigger as required.
3.) Select PROTOCOL NAME.
4.) Select LOAD.
5.) Select RISE FALL (DDC)

Technique		Rise		Fall	
kV	mA	**Record** Delay ms	Limit	**Record** Delay ms	Limit
80	400		0 +1.9 ms	Test not required!	N/A
140	40	Test not required!	N/A		-0 +0.5 ms

Table 9-5 kV Rise and Fall Time Record Table

Note: See pages 560 and 561 for measurement clarification.

Section 16.0 Measure Rise Time

1.) Refer to XREF Verify/Set-up the following DDC parameters:
   - STATIC X-RAY ON
   - 1 SECOND
   - FOCAL SPOT LARGE
   - 80 KV
   - 400 MA

Note: Measure rise time only on the 80kV/400mA scan.

2.) Select ACCEPT RX
3.) Select PAUSE after the start of scan, to prevent the scope from displaying the fall time.
4.) After you record the rise time, select the RESUME to initiate the fall time scan.
5.) Record the delay between the rise of the EXCM signal, and the 75% threshold crossing of the **selected kV** (on **FORM 4879**).
   - Do not include the waveform overshoot.
   - The 75% point for 80kV equals 60kV

Note: Refer to Figure 9-12 for measurement clarification.

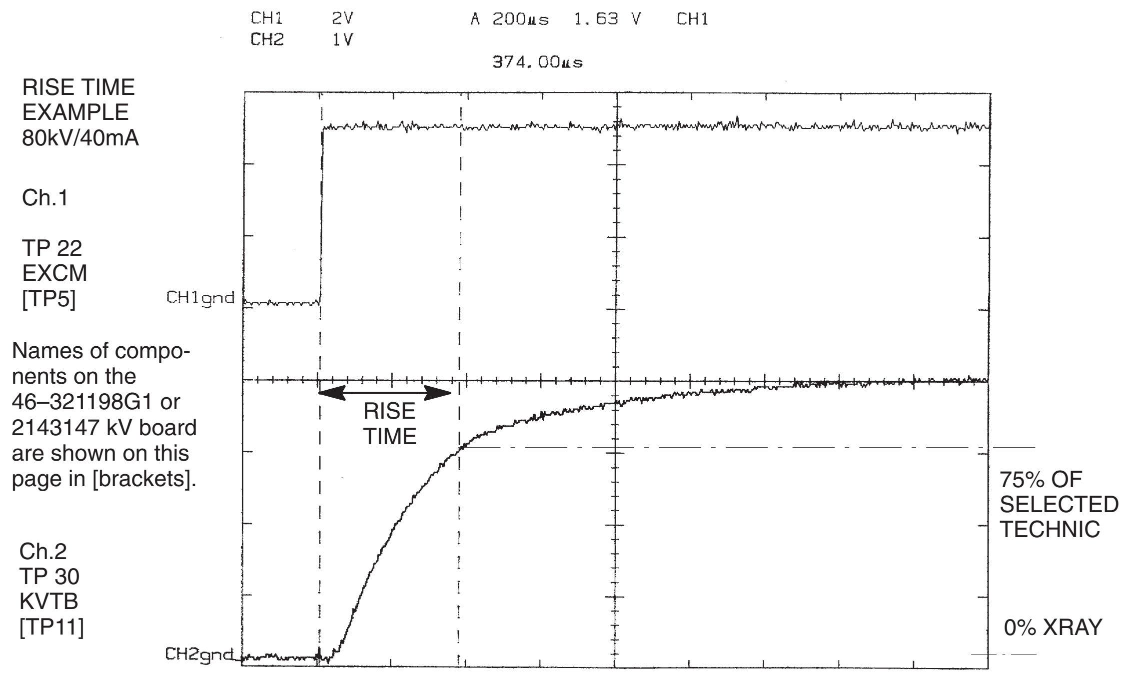


Figure 9-12 Rise Time Measurement

Note: The 75% point for:

- 80kV equals 60kV
- 100kV equals 75kV
- 120kV equals 90kV
- 140kV equals 105kV

Section 17.0 Measure Fall Time

1.) Refer to XREF. Verify/Set-up the following DDC parameters:
- STATIC X-RAY ON
- 1 SECOND
- FOCAL SPOT LARGE
- 140 KV
- 40 MA

Note: Measure fall time only on the 140kV/40mA scan.

2.) Record the delay between the fall of the EXCM signal, and the 75% threshold crossing of the **selected kV** (on **FORM 4879**).
- Do not include the waveform overshoot.
- The 75% point for 140kV equals 105kV

q Check box when complete.

Leave the scope connected for the next test.

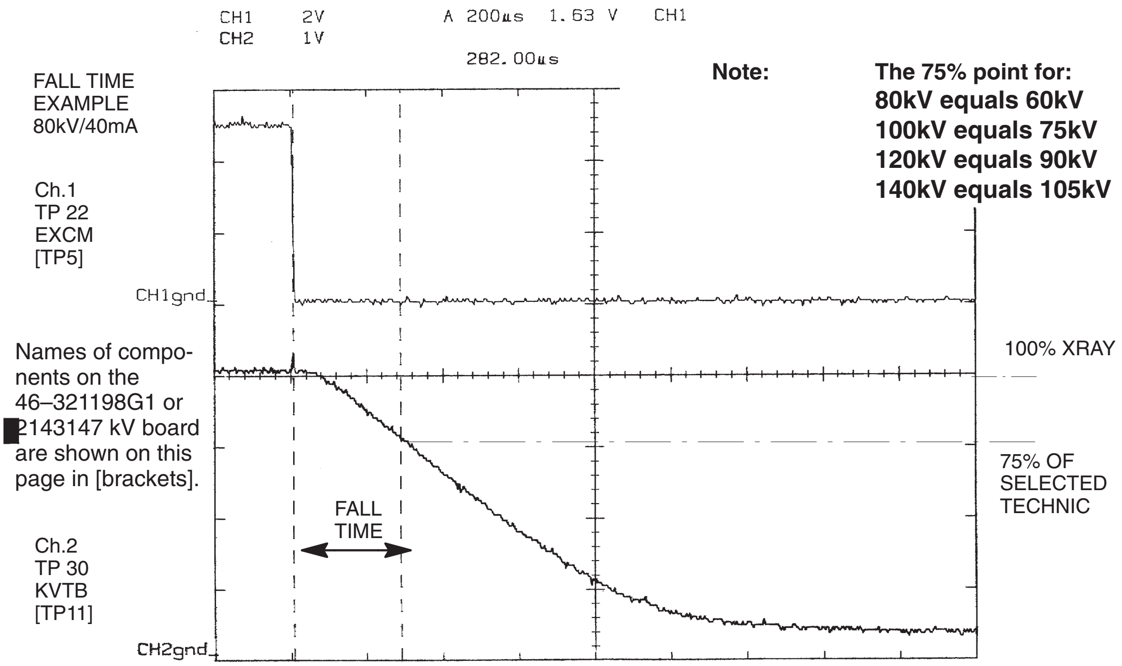


Figure 9-13 Fall Time Measurement

Section 18.0
Verify Internal Scan Timer

1.) Display the Generator Characterization menu (Figure 9-3).
2.) Toggle the softkey MONITOR ENABLE to **ON**, to display the scan time in the message log.
3.) In the OBC, connect a scope to the kV board, as follows:
   a.) Channel 1, Exposure Command (EXCM, TP22). Scope ground to (LGND, TP6) [TP3]. 2v/div
   b.) Channel 2, Total kV (KVTB, TP30) [TP11]. Scope ground to (AGND, TP39) [SIG, TP12]. 1v/div
   c.) Set the Scope Time base to 200msec, positive trigger.
4.) Use DDC to take a stationary, 1.0 sec, 0mm, 100kV, 40mA, Large Focal Spot scan.
5.) Record the measured scan time from the oscilloscope and the displayed scan time from the message log. Spec limits are as follows:

Note: Scope Exposure Duration = 0.96 to 1.04 s.
Displayed Exposure Duration = 0.99 to 1.02 s.

6.) Display the Generator Characterization menu.
7.) Toggle the softkey MONITOR ENABLE **OFF**, to stop the scan time display in the message log.
   Failure to turn the MONITOR ENABLE **OFF** results in the system message log filling with exposure information.
8.) Disconnect the scope from the kV board.
9.) Replace the OBC cover**.**

Section 19.0
Tube Usage Statistics

The display tool, *tube Usage*, displays a list of currently viewable tube usage files. The system stores these files when you save the mechanical characterization.

1.) Display the Service Desktop
2.) Select TROUBLE SHOOT
3.) Select X-RAY GENERATION
4.) Select TUBE USAGE.
5.) Select START.
6.) The shell window displays a tube list, with the corresponding installation dates. Go to next page.
7.) The screen prompts: `Tube to view::`
    - Type **1** **ENTER** to display the corresponding list of tube usage statistics.
    - Type/enter **Q** to quit.
8.) Select EXIT.

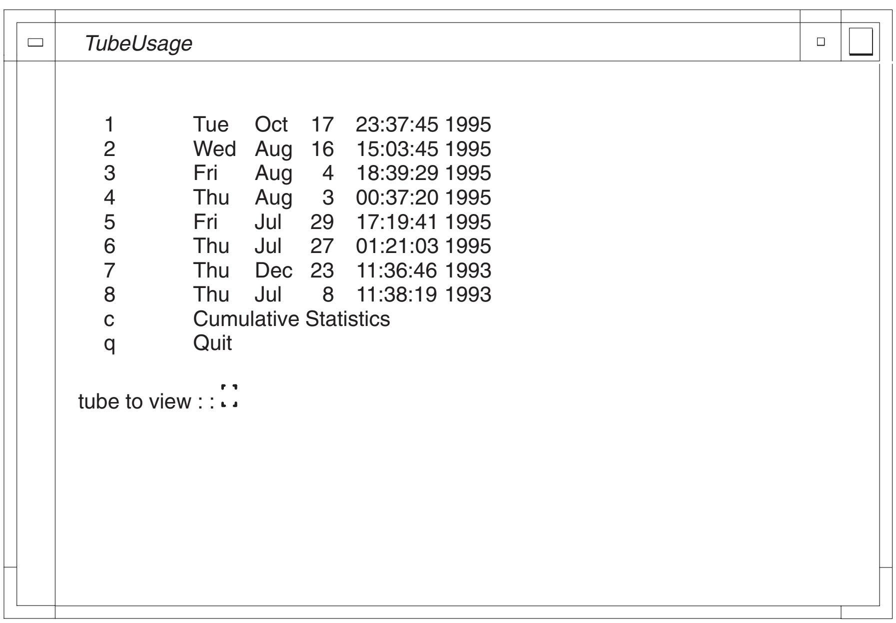


Figure 9-14 Tube Usage Initial Screen

```
Tube  Date Installed
      1     Wed   Oct   5     20:06:111996
      2     Fri   Jul   29    22:00:551996
      3     Tue   Jun   29    20:42:021996
      c     Cumulative Statistics
      q     Quit
Choose the tube # to display the following:
tube to view: 1
Hospital: Name Address
      Suite RPXX2
```

Installed: Wed Oct 5 20:06:11 CDT 1996

Removed:

Last Scan: Thu Jan 5 08:29:17 CST 1996

Housing Model #: 46-309500G2

Housing Serial #: 89851EC0

Insert Model #: 46-309300G2

Insert Serial #: 316778TU0

Failure Code:

Mode: Patient Scans

Number Slices: 52805 (total gantry rotations with X-Ray on)

Number KW Slices: 89761 (count 1 slice for <=24kW, 2 slices for >24kW scans)

KW Hours: 495.66 (total Kilowatt usage for an hour)

Scan Seconds: 69785.19 (total number of seconds of scanning) (1)

Mode: Non-Patient Scans

Number Slices: 8074 (total gantry rotations with X-Ray on)

Number KW Slices: 8501 (count 1 slice for <=24kW, 2 slices for >24kW scans)

KW Hours: 45.23 (total Kilowatt usage for an hour)

Scan Seconds: 11302.74 (total number of seconds of scanning)

After the first tube change with an RP1.4 software (or later release), the tube usage screen displays the following information:

Mode: Patient Scans

Number Slices: 52805 (total gantry rotations with X-Ray on)

Number KW Slices: 89761 (count 1 slice for <=24kW, 2 slices for >24kW scans)

KW Hours: 495.66 (total Kilowatt usage for an hour)

Scan Seconds: 69785.19 (total number of seconds of scanning) (1)

Mode: Non-Patient Scans

Number Slices: 8074 (total gantry rotations with X-Ray on)

Number KW Slices: 8501 (count 1 slice for <=24kW, 2 slices for >24kW scans)

KW Hours: 45.23 (total Kilowatt usage for an hour)

Scan Seconds: 11302.74 (total number of seconds of scanning)

Mode: Patient Scans Smart mA

Number Slices: 52805 (total gantry rotations with X-Ray on)

Number KW Slices: 89761 (count 1 slice for <=24kW, 2 slices for >24kW scans)

KW Hours: 400.54 (total Kilowatt usage for an hour)

Scan Seconds: 69785.19 (total number of seconds of scanning) (1)

Mode: Non-Patient Scans Smart mA

Number Slices: 8074 (total gantry rotations with X-Ray on)

Number KW Slices: 8501 (count 1 slice for <=24kW, 2 slices for >24kW scans)

KW Hours: 45.23 (total Kilowatt usage for an hour)

Scan Seconds: 11302.74 (total number of seconds of scanning)

The Smart mA information duplicates the non Smart mA information, except the kW Hours indicate any reduction in power, due to SmartScan use.

Section 20.0
Change Tube (New Tube) Program

New Tube prepares the system to store tube usage statistics, for trend analysis and tube warranty purposes.

1.) If applications are running, shutdown applications to Unix level:

    On the Service Desktop, select REPLACEMENT PROCEDURES, then select SHUTDOWN APPLICATIONS.

2.) Type **su** **ENTER**

3.) Type/enter the password:

    root password

4.) Type **newTu** **ENTER**

5.) Refer to the list on page 570, and type/enter the failure code for the defective tube in the "Tube Unit Failure Code" field on the screen.

    failcode

6.) Refer to the Tube Housing; type/enter the new Tube's Insert Serial Number in the appropriate field on the screen.

    Insert Serial Number

7.) Refer to the Tube Housing; type/enter the new Tube's Housing Serial Number in the appropriate field on the screen.

    Housing Serial Number

8.) Click OK to accept these changes.

9.) Shutdown and Reboot:

    Type **shutdown**.

Section 21.0
Install New Tube Program

Run this program when you change the tube:

1.) Select REPLACEMENT PROCEDURES.
2.) Select INSTALL NEW TUBE.

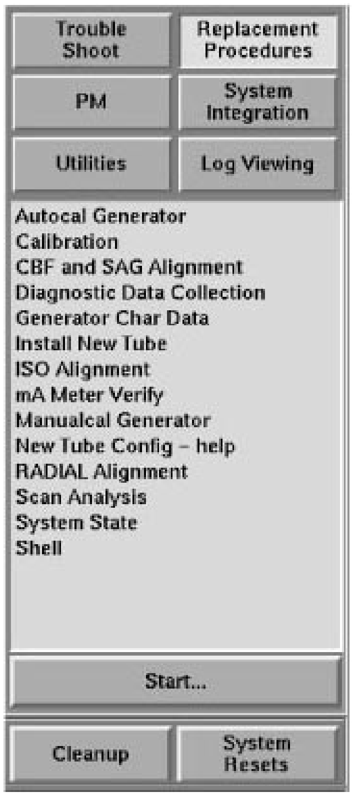


Figure 9-15 Service desktop, Replacement Procedures

Section 22.0
46-309500G1 X-Ray Tube Replacement

NOTICE **Always use the HV Bleeder when told to and verify that the calibration of the HV Bleeder you use is current. Miss-calibration can lead to premature loss of Xray Tubes or other damage.**

DANGER **THE GANTRY CONTAINS ELECTRICAL AND MECHANICAL HAZARDS. MAKE SURE YOU TURN OFF BOTH THE LOOP CONTACTOR AND GANTRY HVDC (550) ENABLE SWITCH BEFORE YOU ACCESS THE GANTRY. ALSO, MAKE SURE YOU READ DIRECTION 46-018302, CT HISPEED ADVANTAGE SAFETY GUIDELINES MANUAL OR VIEW THE 46-018308 CT HISPEED ADVANTAGE SAFETY VIDEO PRIOR TO SERVICING THE GANTRY SUBSYSTEMS.**

22.1 Tube Change Flow Diagram

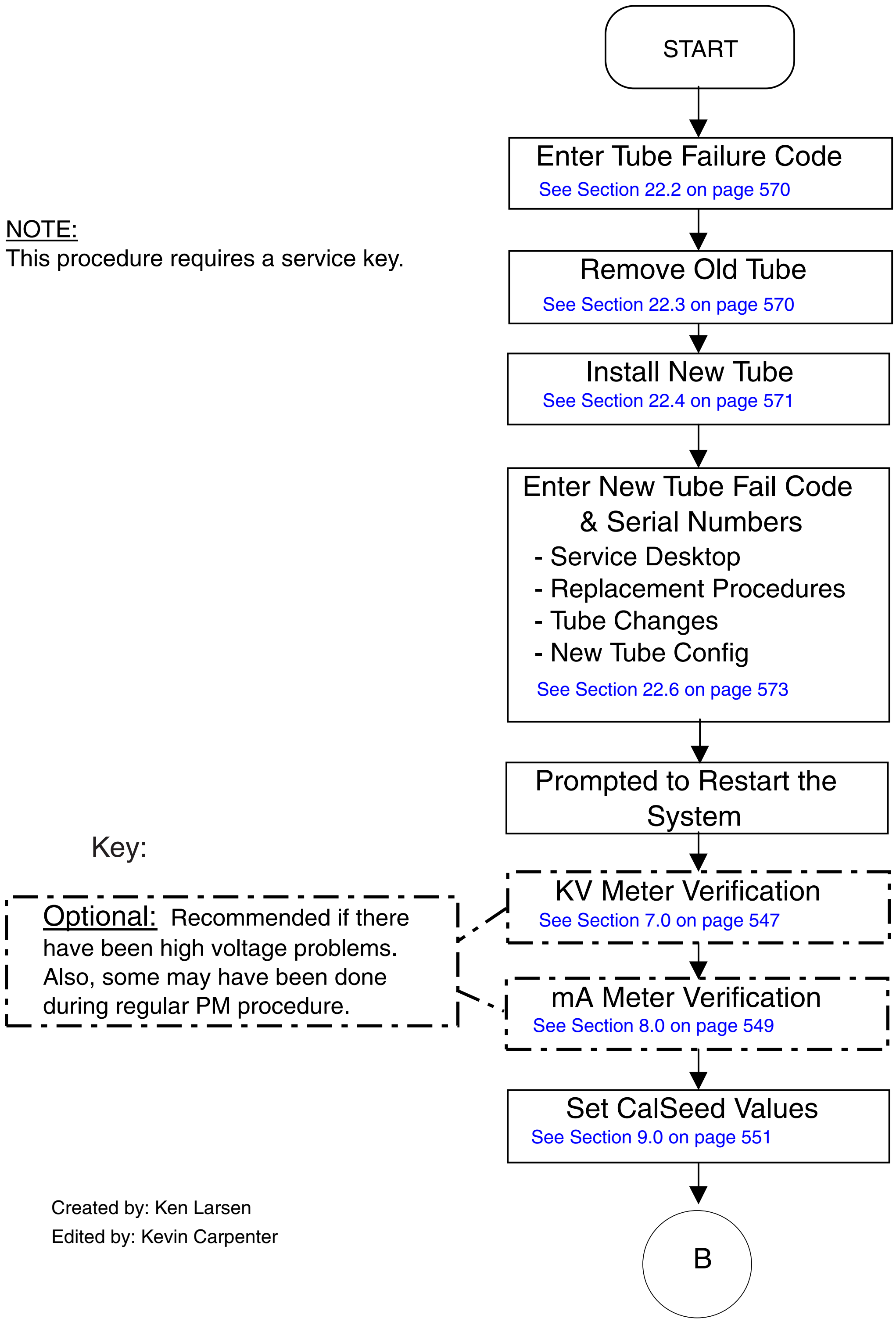


Figure 9-16 Tube Change Diagram (Part 1)

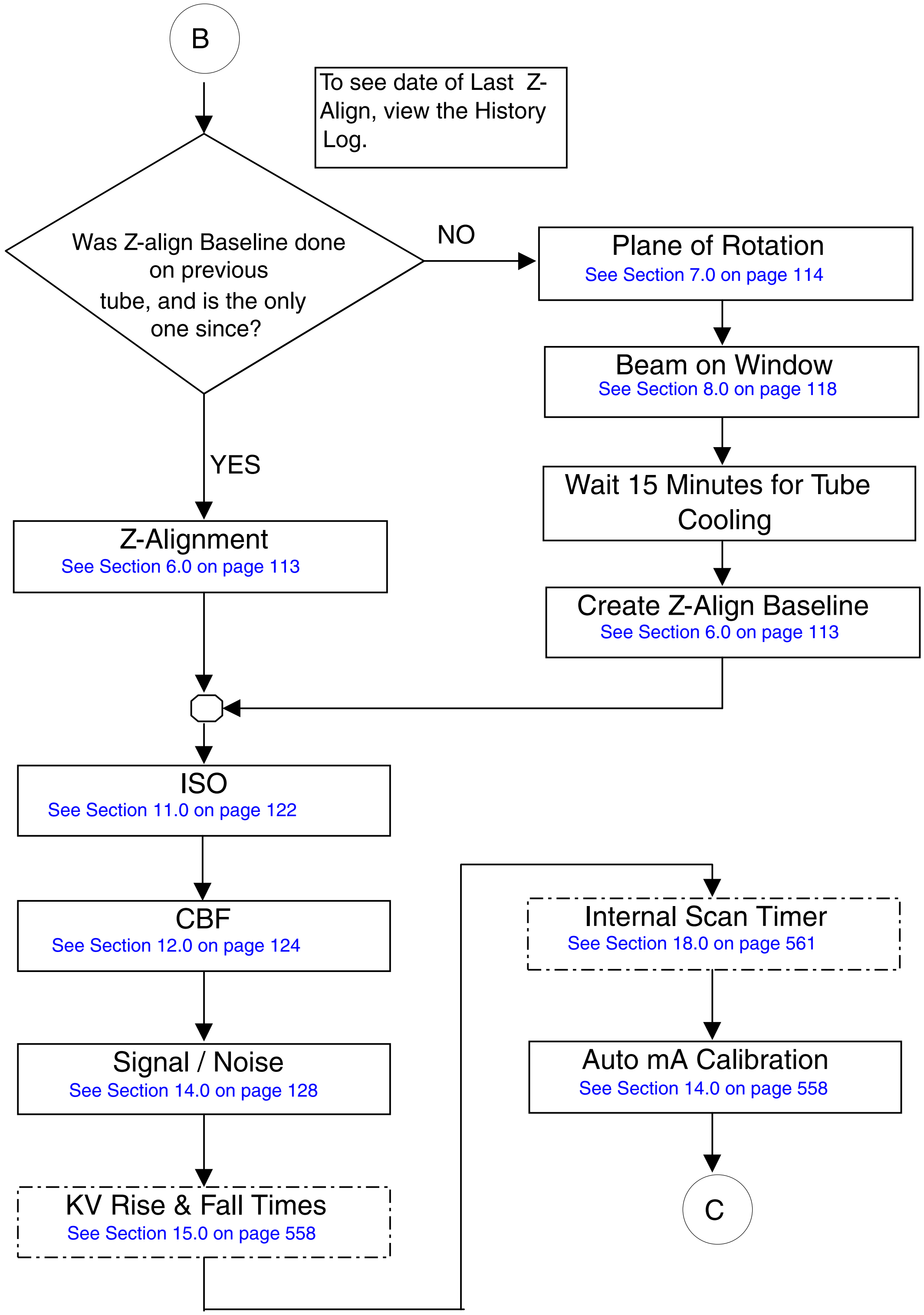


Figure 9-17 Tube Change Diagram (Part 2)

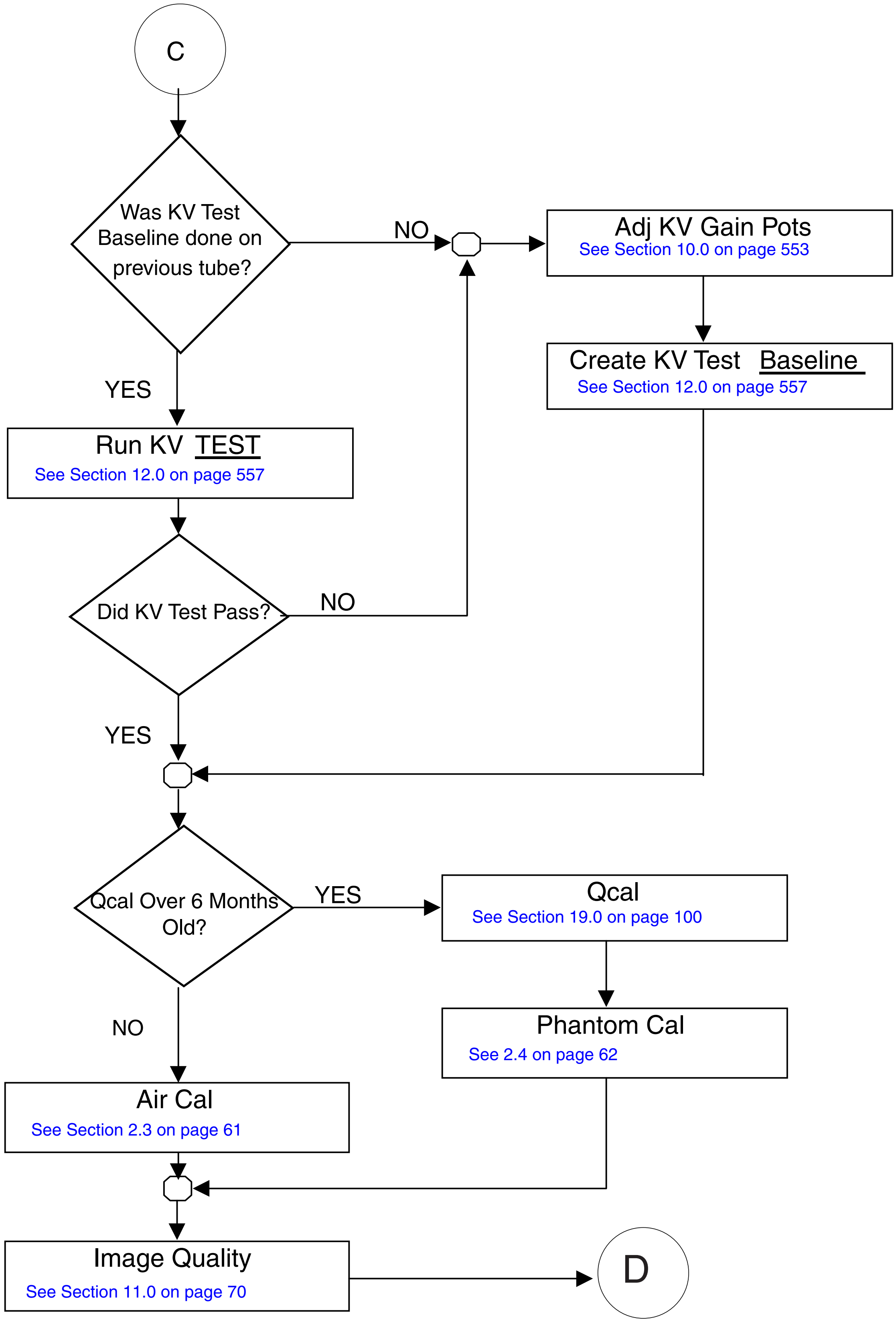


Figure 9-18 Tube Change Diagram (Part 3)

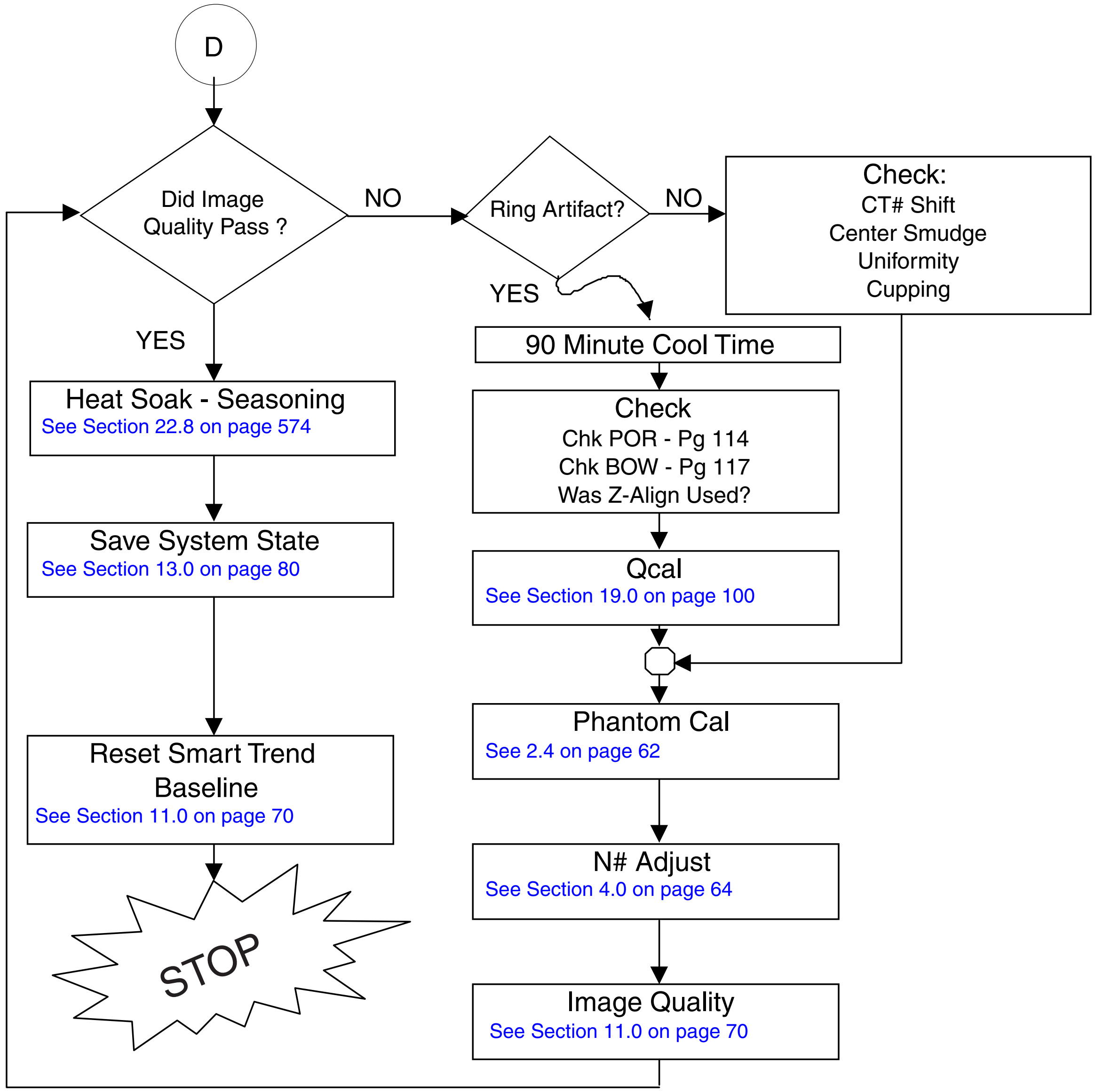


Figure 9-19 Tube Change Diagram (Part 4)

22.2 Tube Failure Codes

Use the following Tube Failure Codes when you report a tube change:

AI: Image Artifact
BG: Broken Glass
CA: Casing Arcing
CB: Casing Bubbles/Particles Seen
CL: Casing Oil Leak
GS: Grid Short
OC: Other-Cathode Related
OE: Tube Loss Due to Failure Elsewhere
OF: Open Filament
OG: Arcing
OH: Other-Housing Related
OL: Generator Overload
OR: Other-Rotor Related
PF: Overheat/Pump Failure
PT: Pulled Tube (No Failure)
RF: Frozen Rotor
RN: Noisy Rotor
SD: Shipping Damage/Error
SS: Stator Open/Stator Short
XL: Low X-Ray Output

22.3 Remove Old Tube

WARNING **MAKE SURE YOU ENGAGE THE LOCKING PIN BEFORE YOU REMOVE THE DETECTOR. FAILURE TO LOCK THE GANTRY COULD RESULT IN INJURY, SHOULD THE GANTRY SUDDENLY MOVE AND STRIKE YOU.**

1.) Turn off facility power to PDU.

DANGER **USE TAG AND LOCK OUT PROCEDURE.**

2.) Remove, and set aside, both gantry side covers.
3.) Turn off the AXIAL DRIVE ENABLE and 550 VDC ENABLE switches on the status control box on the right side of the Gantry.
4.) Rotate the Gantry until the failed tube unit reaches the 3 o'clock position.
5.) **Engage rotational lock**.
6.) Remove the retainer bolt, raise the gantry crane to its uppermost position, and insert the locking pin.
7.) Turn off the Gantry Power.
8.) Disconnect the system stator cable from the 4 pin mate-n-lok connector on the front of the tube unit.
9.) Disconnect the 12 pin tube I.D. system cable, from the top of the tube unit.
10.) Disconnect the 4 pin mate-n-lok pump and fan power system cable

11.) Disconnect the ground strap from the top of the tube unit.

12.) Remove the anode and the cathode cable:

- Loosen each cable's locking ring with the spanner wrench.
- Pull each cable terminal out of its receptacle.
- Ground the end of the cables to the Gantry frame.
- Wipe up any oil that drips from the cable terminal.
- Use paper towels to soak up any oil in the wells.

13.) Attach the hoist to the crane and tube.

CAUTION **Remove the mounting bars in the following (lower/upper) order to lessen the risk of injury to your hand.**

14.) Unfasten two 3/8-16 hex nuts, and remove the lower mounting bar.

15.) Unfasten two 3/8-16 hex nuts, and remove the upper mounting bar.

16.) Lower the defective tube unit to the floor, and rest the tube unit on its fans.

22.4 Install New Tube

1.) Allow the tube unit to warm to room temperature before you install it.

2.) Inspect the port. **Make sure the tube contains the fixed moly filter**:

- One end of the moly filter has 2 dots on it.
- If you don't see the dots, DO NOT USE THE TUBE.

Make sure the surface of the port is smooth and clean. If you see particles, clean the window with a lint free wipe. A blemish may cause image artifacts.

3.) Turn off facility power to the PDU.

DANGER **USE TAG AND LOCK OUT PROCEDURE.**

WARNING **SEVERE INJURY POSSIBLE.**

IF YOU TURN THE TORQUE WRENCH MORE THAN 90° (¼ TURN) WHILE APPLYING FINAL TORQUE, THE MOUNTING THREADS ARE STRIPPED.

DO NOT USE THIS TUBE.

NOTICE **Potential for IQ artifacts**

When attaching the mounting plate on the tube, be careful with the Copper Filter. It should be free of debris, scratches and dust. Particles create artifacts in the image. Use a lint-free wipe to clean, if necessary.

4.) Attach the mounting plate from the old tube to the new tube using the four NEW M10 bolts that come with the tube.

Note: Do NOT use Loctite.

a.) Finger tighten all four (4) bolts.

b.) Tighten all four bolts to the pre-load torque specified in Table 9-6.

This seats each bolt, enabling you to visually ensure that the mounting holes are not stripped while applying final torque.

15 lb-ft	20 N-m	180 in-lbs	210 kg-cm

Table 9-6 M10 Bolt “Pre-Load” Torque Specification

c.) Set final torque, specified in Table 9-7, on all four bolts.

30 lb-ft	41 N-m	360 in-lbs	420 kg-cm

Table 9-7 M10 Bolt Final Torque Specification

5.) Use the hoist to lift the new tube unit:

a.) Position the four 1" tube support bracket mounting holes over the four threaded studs on the gantry tube support plate.

b.) Position the 3 tube support bracket alignment pins into the alignment holes on the Gantry Tube Support Plate.

Note: To ease installation, fasten the top mounting bar to the rotating structure first. Then attach the bottom mounting bar.

c.) Fasten the lower and upper and mounting bars to the rotating structure with the 3/8-16 hex nuts, and torque to 30±1 ft.-lbs.

6.) Attach the system stator cable to the 4 pin, mate-N-Lok connector to the front of the tube unit, near the oil pump.

7.) Attach the tube I.D. cable to the 12 pin mate-N-Lok connector on top of the tube.

8.) Attach the tube pump and fan power cable to the 4 pin mate-N-Lok connector.

9.) Fasten the grounding strap to the 1/4-20 ground stud on top of the tube unit.

10.) Remove the plastic cap plug from each cable receptacle on the tube unit.

Take care not to lose the rubber quad rings.

11.) Lightly wet the new rubber quad ring with transformer oil (917).

12.) Return the quad ring to its slot at the top of the receptacle retaining ring.

13.) Pour transformer oil (917) into the receptacle to a depth of **10 mm (0.375 in)**.

14.) Align the cable terminal orienting key with the notch in the receptacle.

15.) Slowly insert the cable, to engage the connector pins, and seat the cable in the well.

- Tighten the cable locking ring.
- Rotate the cable strain relief for a clean cable dress.
- Use the spanner wrench to tighten the locking ring.
- Use a torque wrench to tighten the locking ring to **11±1 ft.-lb**.

⚠ **NOTICE** **Do not over tighten the locking ring. Over tightening can deform the cable plug sealing surfaces, break the oil seal between receptacle and housing, twist the receptacle, and disrupt internal wiring.**

- Back off on the cable locking ring without disturbing the cable plug.

- Re-tighten the locking ring, and torque to **7±1 ft lbs**.

Note: Use the spanner wrench with a torque wrench when you tighten the high-voltage cables on the tube unit.

⚠ **WARNING IF YOU GET OIL ON YOUR HANDS, WASH THEM NOW!**

16.) Carefully wipe up all excess oil.
17.) Disconnect hoist from tube and crane.
- Return the crane to the normal storage position.
- Make sure you install, and tighten, the crane storage retaining bolt.

18.) Check for oil leaks:
- Wrap rags or paper towels around the cable horns, and tape them into place.
- Manually rotate the tube to the 6 o'clock position.
- Return the tube to the 3 o'clock position
- Remove the toweling and wipe up all excess oil.
- Wipe off the cable horns, locking rings, and strain reliefs with a rag dampened with alcohol.
- Repeat with a dry rag.
- Wrap the cable strain reliefs and locking rings with a single layer of absorbent paper tissue.
- You can use two inch wide strips cut from a paper napkin.
- Wrap the bottom edge of the paper around the top end of the cable horn, and tape it into place.
- Extend the top edge of the paper over the top of the locking ring, and tape it to the plastic cable strain relief.
- Remove paper after leak check.

19.) Turn on gantry power, and wait at least 10 minutes to warm up the filament.
20.) Shutdown, and restart the software after every tube change, so the system can interrogate the configuration file that contains tube I.D. resistor.

22.5 Shutdown the system

Note: Type the boldface characters, and press the **ENTER** key. "Type/enter" also means type the boldface characters, and press the **ENTER** key

1.) Shutdown the system to the PROM monitor level.
2.) Start up the system from PROM monitor level.
3.) Boot Unix, but type/enter **N**, when prompted `You have 10 seconds to stop application software start up`.

22.6 Run New Tube

New Tube prepares the system to store tube usage statistics, for trend analysis and tube warranty purposes.

1.) If applications are running, shutdown applications to Unix level: On the Service Desktop, select REPLACEMENT PROCEDURES, then select SHUTDOWN APPLICATIONS.
2.) Type **su** **ENTER**
3.) Type/enter the password:

root password

4.) Type **newTu** ENTER

5.) Refer to the list on page 570, and type/enter the failure code for the defective tube in the "Tube Unit Failure Code" field on the screen.
failcode

6.) Refer to the Tube Housing; type/enter the new Tube's Insert Serial Number in the appropriate field on the screen.
Insert Serial Number

7.) Refer to the Tube Housing; type/enter the new Tube's Housing Serial Number in the appropriate field on the screen.
Housing Serial Number

8.) Click OK to accept these changes.

9.) Shutdown and Reboot:
Type **shutdown**

22.7 Align the xray beam

1.) Plane of Rotation (POR): Starts on page 114
2.) X-Ray Beam on Window (BOW): Starts on page 118
3.) Isocenter: Starts on page 122
4.) Center Body Filter (CBF) and SAG: Starts on page 124

22.8 Calibrate the generator

1.) kV Meter Verification: Starts on page 547
2.) mA Meter Verification: Starts on page 549
3.) Install NEW TUBE program: Starts on page 558
4.) Auto mA Calibration: Starts on page 558
5.) KV Rise and Fall Times: Starts on page 558
6.) Verify Internal Scan Timer: Starts on page 561
7.) Heat Soak and Seasoning: Procedure is shown next. Theory on page 351.

Note: When you run Tube Heat Soak and Stability, monitor for high voltage overcurrents, shoot threws and spits. If these errors occur during the seasoning steps, stop the series and return to the next lower kV seasoning cal series. Then proceed to the failing station, and scan until error free. If the errors continue, repeat the process up to four times before pulling the tube.

22.9 Season the Tube

⚠ **NOTICE** **Use this software only on MX_165_CT_I and MX_200CT Tubes**

Run this series of scans on every new **MX_165_CT_I or MX_200CT** tube per the following tables. You may also **run this series on a tube that causes scan aborts or image streaks**.

Display the GENERATOR CHARACT menu, and select AUTO MA CAL. Abort the scanning procedure after the system completes the Ductility warm-up.

If you recently completed the ductility warm-up and auto mA cal, do not perform the ductility warm-up again (unless you allowed the tube to cool).

1.) Select: REPLACEMENT PROCEDURES

2.) Select: DIAGNOSTIC DATA COLLECTION
3.) Select Protocol Name: HEATSOAKG for (MX 165) or HSSGEMINI for (MX 200)
4.) Select Scan Speed: 1.0
5.) Select Aperture: 0
6.) Select Filter: BOWTIE
7.) Select Spot Size: LARGE
8.) Select: AUTOSCAN

All scans are done on the Large Focal Spot and One Second Rotation.

Select the parameters from the following table for the MX_165_CT_I tube. Start with Scan Group Warm-up scans.

Scan Group	Pre-Group Delay	Minimum ISD	Exposure Duration	No. of Exp.	kV	mA
Warm-Up	2	2	2	15	80	100
Heat Input	2	2	3	24	80	300
Anode Soak 1	3	5	3	25	120	200
Anode Soak 2	2	1	1	9	80	300
Casing Soak 1	12	12	2	90	80	300
Casing Soak 2	7	7	1	10	120	200
Seasoning 1	5	5	0.1	5	90	50
Seasoning 2	5	5	0.1	5	100	50
Seasoning 3	5	5	0.1	5	110	50
Seasoning 4	5	5	0.1	5	120	50
Seasoning 5	5	5	0.1	5	130	50
Seasoning 6	5	5	0.1	5	135	50
Seasoning 7	5	5	0.1	10	140	50
Seasoning 8	5	5	0.1	10	145	50
Seasoning 9	5	5	0.1	5	150	50

Table 9-8 MX_165_CT_I SCAN TABLE

Note: The Pre-Group Delay is not a soft-key. Wait 30 seconds between groups.

Select the parameters from the following table for the MX_200_CT tube. Start with Scan Group Warm-up scans.

Scan Group	Pre-Group Delay	Minimum ISD	Exposure Duration	No. of Exp.	kV	mA
Warm-Up	0	2	2	15	80	100
Heat Input	2	2	3	24	80	300
Anode Soak 1	2	5	3	17	100	320
Anode Soak 2	2	3	1	10	100	320
Casing Soak	60	16	2	90	100	270
Seasoning 1	5	5	1	5	100	50
Seasoning 2	5	5	1	5	100	50

Table 9-9 MX_200_CT SCAN TABLE

Scan Group	Pre-Group Delay	Minimum ISD	Exposure Duration	No. of Exp.	kV	mA
Seasoning 3	5	5	1	5	110	50
Seasoning 4	5	5	1	5	120	50
Seasoning 5	5	5	1	5	130	50
Seasoning 6	5	5	1	5	135	50
Seasoning 7	5	5	1	10	140	50
Seasoning 8	5	5	1	10	145	50
Seasoning 9	5	5	1	15	150	50

Table 9-9 MX_200_CT SCAN TABLE (CONTINUED)

Note: The Pre-Group Delay is not a soft-key. Wait 30 seconds between groups.

9.) Select RUN

Note: When you run Tube Heat Soak and Stability, monitor for high voltage over currents, shoot threws, and spits. If these errors occur during the seasoning steps, stop the series and return to the next lower KV seasoning cal series. Then proceed to the failing station, and scan until error free. If the errors continue, repeat the process up to four times before pulling the tube.

When you successfully complete the high voltage stability test:

1.) Torque the locking ring at the tube end of each H.V. cable to 7 ± 1 ft. lbs. or 84 ± 12 in. lbs.
2.) Tape more rags or paper towels around the tube locking rings to absorb excess oil.
3.) Remove the rags or paper towels after several revolutions of the gantry.

22.10 Check Exposure Time Accuracy

Begin at the top level Service Screens, and execute the following sequence of soft-keys:

1.) Select: REPLACEMENT PROCEDURES
2.) Select: DIAGNOSTIC DATA COLLECTION
3.) Select: MONITOR ENABLE, to display scan times in the message log.
4.) Change to the ExamRx Desktop
5.) Select NEW PATIENT and prescribe the following scans.

22.11 Check Scout Scan Time

Use ExamRx to take scout scans with the following distances. (Total of six scans.)

1.) Distance in mm: 20, 25, 30, 40, 150, 300, 480
2.) Use 120kV and 40mA
3.) Record the scan time, displayed in the message log, on the HHS data sheet.

22.12 Check Axial and Helical Scan Time

Use normal applications (ExamRx) to acquire Axial Scans with the following parameters:

1.) Use 120kV and 40mA
2.) Use the following scan time and FOV:

Selected Time	FOV
0.6 sec	small FOV
0.6 sec	large FOV
1.0 sec	large FOV
2.0 sec	large FOV
3.0 sec	large FOV
4.0 sec	large FOV

Table 9-10 Axial Scan Time Parameters

Use normal applications (ExamRx) to acquire Helical Scans with the following parameters:

1.) Use 120kV and 40mA
2.) Use 10mm Scan Thickness
3.) Use the following scan time and location:

Selected Time	Scan Location
15.0 sec	S70-170
28.0 sec	S135-I135
30.0 sec	S145-I145

Table 9-11 Helical Scan Time Parameters

Note: When you complete the scan time tests, switch back to the Service Desktop to display the Diagnostic Data Collection screen, and toggle the MONITOR ENABLE OFF. Otherwise, the message log fills with kV, mA and scan statistics.

Signal to Noise Tube Output Check (Tube Output):

1.) Display the Service Desktop.
2.) Select REPLACEMENT PROCEDURES.
3.) Select DIAGNOSTIC DATA COLLECTION — START.
4.) Select PROTOCOL NAME.
5.) Load protocol **ta_to.**
6.) Check Run Description Tube Align – Tube Output.
7.) Select ACCEPT RX — Press START SCAN when it lights.

RECORD TUBE OUTPUT SCAN # — Exam ____ Series ____ Image ____

22.13 Calibrate the System

1.) Q-Cal: Calibrations refer to Chapter 2, page 59.
2.) Reset Smart Trend Baseline if applicable (5.x SW), refer to Chapter 6, page 264.
3.) Crosstalk Calibration: Calibrations refer to Chapter 2, page 59.
4.) Calibration Procedure, Standard Cals: Calibrations refer to Chapter 2, page 59.
5.) N# Check: Calibrations refer to Chapter 2, page 59.
6.) QA Noise Test: Refer to the QA section in the 2142707-100 Technical Reference manual
7.) Thermal Test: Starts below.

Note: Run the thermal test only if you suspect cold tube/hot tube image problems.

22.14 Perform Thermal Test

The thermal test requires a cold X-Ray tube. Allow the tube to cool a minimum of 60 minutes before you start this procedure. Replace the tube if the mean values vary more than 4 counts.

1.) Scan the water portion of the QA phantom with the following parameters:
   - small FOV: 1 slice at 1mm thickness
   - 120 kV: 20 slices at 10mm thickness
   - 200 mA: 1 slice at 1mm thickness, 4.0 sec (from a cold tube)
2.) Check images for thermal bands, and artifacts that vary with temperature.
3.) Use a 14cm x 14cm ROI box, centered at 0.0, 0.0cm.
4.) Compare the ROI mean values of the two 1mm slice images.
   The ROI mean values should fall within 4 counts of each other.
5.) Compare the ROI mean values of the twenty 10mm slice images.
   The ROI mean values should fall within 4 counts of each other.

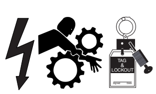
Section 23.0
46-297460P1 Tube Stud Replacement

1.) Remove the Tube. Follow all the Safety Instructions listed in the Tube Removal procedure.
2.) Use double 3/8-16 nuts and a 9/16 wrench, or an adjustable wrench, to remove the damaged stud(s)
3.) Apply Loctite 271 to one end of the replacement stud.
4.) Fasten the end with the Loctite into the rotating base casting.
5.) Use double 3/8-16 nuts and a 9/16 wrench, or an adjustable wrench, to adjust the stud to a height of 1.90 ±0.04 inches from the surface of the rotating base casting.
6.) Replace the Tube.
7.) Follow all the Safety Instructions listed in the Tube Installation procedures.

Section 24.0
45554264 or 2100553 Transformer Tank Measurement Board

1.) Position table to lowest elevation.
2.) Turn off facility power to PDU.

DANGER **USE TAG AND LOCK OUT PROCEDURE.**



3.) Remove, and set aside, both gantry side covers.
4.) Turn off all 3 switches on the status control box on right side of Gantry.
5.) Lift top cover, and engage prop rod.
6.) Remove Scan window.
7.) Open front cover.
8.) Rotate Measurement Board to 3:00 position.
9.) Unplug connectors J1, J2, J5, and J6
10.) Remove 6 screws and washers that fasten measurement board to High Voltage Supply.

11.) Carefully pry measurement board off High Voltage Tank.

12.) Replace Measurement Board.

Note: Carefully align connector pins from Interface Measurement Board to Round Interface Board on High Voltage Supply.

13.) Reassemble Gantry.

14.) Refer to Retest Verification Table at the beginning of this chapter.

Section 25.0
46-296701P1 38V Filament Supply

Note: Input leads: Black lead to transformer terminal one (1); White lead to transformer terminal two (2).

1.) Move table to lowest elevation.

2.) Turn off facility power to PDU. Use tag and lockout procedures.

3.) Remove gantry side covers.

4.) Turn off all three (3) switches on status control box on right side of gantry.

5.) Lift top cover and engage prop rod.

6.) Remove scan window.

7.) Open Front Cover.

8.) Rotate gantry until Filament Power assembly reaches the 3 o'clock position.

9.) Engage gantry rotational lock.

10.) Loosen four (4) captive screws on Filament Power assembly cover, and remove cover.

11.) Measure the voltage across the filter capacitor, to verify the bleeder resistor has dissipated energy to a safe level.

12.) Unsolder Black and White wires from terminals one (1) and two (2) on the transformer.

13.) Disconnect Red lead from fuse.

14.) Remove the screw from the negative terminal to disconnect the Black lead from the filter capacitor.

15.) Remove and save four (4) bolts fastening Filament Power Supply to gantry and, remove supply.

16.) Install new supply.

17.) Reassemble Gantry.

Section 26.0
45561210 or 2161307 Anode Transformer Tank

1.) Position table to lowest elevation.

2.) Turn off facility power to PDU.

DANGER **USE TAG AND LOCK OUT PROCEDURE.**

3.) Remove and set aside both gantry side covers.

4.) Turn off all 3 switches on the status control box on right side of Gantry.

5.) Open top cover and engage prop rod.

6.) Remove and set aside Gantry scan window.

7.) Open front cover.
8.) Rotate gantry until the Anode H.V. Supply reached the 9 o'clock position.
9.) Engage gantry rotational lock.
10.) Loosen two (2) wing handled screws holding STC Chassis in place.
11.) Rotate card cage 90 degrees to lock into position.
12.) Use spanner wrench to remove high voltage cable connector from H.V. Transformer Tank.
Use rags or paper toweling to wipe excess oil from H.V. cable connector and tank well.
13.) Ground the end of the HV cable to the Gantry frame to ensure no voltage.
14.) Disengage gantry rotational lock and rotate gantry until the Anode H.V. Supply reaches the 3 o'clock position.
15.) Remove cables J1 and J2 from measurement PWB on HV Transformer Tank.
16.) Remove four screws fastening cover to inverter assembly, and remove cover.
17.) Measure voltage on two large capacitors, to verify 0 volts.
18.) Disconnect J1 connector from side of inverter assembly.
19.) Disconnect J6 connector from gate driver PWB.
20.) Verify no voltage on the 550VDC cable.
21.) Carefully disconnect four fiber optic cables from gate driver board.
22.) Disconnect 550VDC cable from capacitor PWB.
23.) Cut Ty-raps from side plate of inverter, and remove all cables from the inverter assembly.
24.) Remove two (2) inverter output leads from H.V. Transformer Tank locations P1 and P2.
25.) Remove the four (4) 3/8 bolts from inverter baseplate which fasten inverter assembly to H.V. Transformer Tank.
26.) Remove inverter assembly from gantry.
27.) Attach hoist to boom arm in gantry.
28.) Attach hoist to lifting bracket on bottom of Anode H.V. Transformer Tank.
Remove slack from hoist chain.
29.) Using tool remove four (4) bolts which fasten transformer tank to rotating base.
30.) Use hoist to lower transformer tank to floor.
31.) Install new transformer tank.

Note: When installing four (4) 3/8” tank mounting bolts torque to 25 ft-lbs. (3.462 m-kg).

32.) Before you install the HV Cable Connector, add 20 cc of dielectric oil to the HV Connector well in the HV Transformer Tank.
33.) Use the spanner wrench to securely tighten high voltage cable connector.
- Wipe excess oil with rags or paper towels.
- After you complete the installation, rotate the gantry to verify cable clearance, and to drain excess oil from H.V. cable strain relief.
- Again wipe excess oil with rags or paper towels.
34.) Reassemble Gantry.
35.) Refer to Retest Verification Table at the beginning of this chapter.

Section 27.0
45561211 Cathode Transformer Tank

1.) Turn off facility power to PDU.

⚠ **DANGER** **USE TAG AND LOCK OUT PROCEDURE.**

2.) Remove, and set aside, both gantry side covers.
3.) Turn off all 3 switches on the status control box, on right side of Gantry.
4.) Open top cover, and engage prop rod.
5.) Remove, and set aside, Gantry scan window.
6.) Open front cover.
7.) Rotate gantry until the Cathode HV transformer tank reaches the 3 o'clock position.
8.) Engage gantry rotational lock.
9.) Verify 550 VDC or 120 VAC is not present.
10.) Use the spanner wrench to remove the high voltage cable connector from the high voltage transformer tank.
- Ground the ends of the H.V. cable to the Gantry frame, to ensure no voltage exists at the end of the cable.
- Use rags or paper towels to wipe excess oil from the High Voltage Cable Connector and tank well.

11.) Remove cables J1, J2 and J6 from the measurement PWB.
12.) Remove four screws fastening the cover to the inverter assembly
Remove cover.
13.) Measure voltage on the two large capacitors to verify 0 volts.
14.) Disconnect J1 connector from side of inverter assembly.
15.) Disconnect J6 connector from gate driver PWB.
16.) Carefully disconnect four fiber optic cables from gate driver board.
17.) Verify 550 VDC or 120 VAC is not present.
18.) Disconnect 550VDC cable from capacitor PWB.
19.) Cut Ty-raps from side plate of inverter.
20.) Remove all cables from the inverter assembly
21.) Remove two (2) inverter output leads from H.V. Transformer Tank locations P1 and P2.
22.) Remove the four (4) 3/8 bolts from the inverter baseplate, that fasten the inverter assembly to the H.V. Transformer Tank.
23.) Remove the inverter assembly from the gantry:
- Attach the hoist to the boom arm in the gantry.
- Attach the hoist lifting chain to the eyebolt on the transformer tank.
- Remove slack from the hoist chain.

24.) Remove the four (4) bolts which fasten transformer tank to the rotating base.
25.) Use the hoist to lower the transformer tank to the floor.
26.) Install the new transformer tank.

Note: Install four (4) 3/8 tank mounting bolts, and torque to 25 ft-lbs (3.462 m-kg).

27.) Before you install the HV Cable Connector, add 20 cc of dielectric oil to the HV Connector well in the HV Transformer Tank.
- Use the spanner wrench to securely tighten the cable connector.
- Wipe excess oil with rags or paper towels.
- After you complete the installation, rotate the gantry to verify cable clearance, and to drain excess oil from H.V. cable strain relief.
- Again wipe excess oil with rags or toweling.

28.) Reassemble Gantry.
29.) Refer to Retest Verification Table at the beginning of this chapter.

Section 28.0
Anode or Cathode Inverter
45435960 or 46-297703P1(anode)
45435962 or 46-297703P2 (cathode)

1.) Position table to lowest elevation.
2.) Turn off facility power to PDU. Use tag and lockout procedure.
3.) Remove, and set aside, both gantry side covers.
4.) Turn off all 3 switches on the status control box on right side of Gantry.
5.) Open top cover, and engage prop rod.
6.) Remove, and set aside, Gantry scan window.
7.) Open front cover.
8.) Rotate Gantry until the inverter reaches the 3:00 o'clock position.
9.) Engage gantry rotational lock.
10.) Remove four screws fastening cover to inverter assembly, and remove cover.
11.) Measure voltage on the two large capacitors to verify 0 volts.
12.) Disconnect J1 connector from side of inverter assembly.
13.) Disconnect J6 connector from gate driver PWB.
14.) Carefully disconnect four fiber optic cables from gate driver PWB.
15.) Disconnect 550 VDC cable from capacitor PWB.
16.) Cut Ty-raps from side plate of inverter.Remove all cables from inverter assembly.
17.) Remove two inverter output leads from Transformer Tank locations P1 and P2.
18.) Remove four (4) 3/8 bolts from inverter baseplate, which fastens inverter assembly to H.V. Transformer Tank.
19.) Remove inverter assembly from gantry.
20.) Install new inverter assembly.

Note: Use Loctite 242, and torque the four (4) 3/8 tank mounting bolts to 25 ft-lbs (3.462 m-kg).

21.) Reassemble Gantry.
22.) Refer to Retest Verification Table at the beginning of this chapter.

Section 29.0
46-195120G16 HV Anode Cable

1.) Position table to lowest elevation.
2.) Turn off facility power to PDU.

DANGER **USE TAG AND LOCK OUT PROCEDURE.**

3.) Remove, and set aside, both gantry side covers.
4.) Turn off all 3 switches on the status control box, on right side of Gantry.

5.) Open top cover, and engage prop rod.
6.) Remove, and set aside, Gantry scan window.
7.) Open front cover.
8.) Rotate Gantry until the tube reaches the 12 o'clock position.
9.) Lock Gantry into the 12 o'clock position.
10.) On left side of the Gantry:
   - Loosen the two wingnuts on the STC Chassis
   - Rotate the STC Chassis 90 degrees, then lock it into position, to gain access to Anode HV Transformer Tank
11.) Place a rag around the Anode HV Cable connector opening on the transformer tank.
12.) Use the spanner wrench to loosen the Anode HV cable connector.
   - Remove the cable from the HV transformer tank.
   - Ground the end of the cable to the Gantry Frame to verify no voltage.
13.) Wipe excess oil from HV cable connector, and out of the HV transformer tank connector well, before you rotate the Gantry.
14.) Rotate the Gantry until the tube reaches the 3:00 position.
15.) Engage rotational lock
16.) Use the spanner wrench to remove the Anode HV Cable connector from X-Ray tube HV connector well.
17.) Unscrew two cable clamps to remove Anode HV Cable from rotating base.
18.) Cut all remaining ty-raps, and remove the cable from the Gantry.
19.) Add 20 ml (0.7 oz.) of dielectric oil to the anode HV connector well of the X-Ray Tube.
20.) Place a rag around the tube connector well, to absorb excess oil when you insert the Anode cable.
21.) Insert the Anode HV Cable connector into the Anode HV well of the X-Ray tube.
22.) Tighten the nut with the spanner wrench.
23.) Wipe excess oil from the tube.
24.) Unlock, and rotate the Gantry, until the tube reaches the 12 o'clock position.
25.) Lock Gantry in place.
26.) Add 20 ml (0.7 oz) of dielectric oil to the well of the HV Anode transformer tank.
27.) Place a rag around transformer tank connector well, to absorb excess oil when you insert the Anode cable.
28.) Insert the Anode HV cable connector into Anode HV transformer well.
29.) Tighten the nut with the spanner wrench.
30.) Use the old screws and cable clamps to attach the new Anode cable to the rotating base.
31.) Wipe excess oil from the transformer.
32.) Use ty-raps to dress the cables, so a minimum loop exists on each end.
33.) Tighten ty-raps on rotating base.
34.) Wrap a rag or paper towel around both ends of the Anode HV cable and tape them into place.
35.) Unlock the Gantry.
36.) Rotate the STC into its original position, and tighten down the two wing nuts to secure it into place.
37.) Rotate Gantry by hand through two complete revolutions.

   Check to make sure the Anode cable does not catch on any stationary components during each revolution.
38.) Remove the rags, and wipe up any excess dielectric oil from the HV Anode Cable ends.

39.) Restore power to the Gantry.
40.) Rotate the Gantry at a speed of 1 Revolution per second, for several revolutions.
41.) Stop the gantry and shut down power.
42.) Turn off facility power to PDU.
Use tag and lockout procedures.
43.) Turn off all 3 switches on the status control box, on the right side of Gantry.
44.) Check the tube and transformer tank wells for oil leaks.
45.) Reassemble Gantry.
46.) Refer to Retest Verification Table at the beginning of this chapter.

Section 30.0
46-195120G16 HV Cathode Cable

1.) Position table to lowest elevation.
2.) Turn off facility power to PDU.

DANGER **USE TAG AND LOCK OUT PROCEDURE.**

3.) Remove, and set aside, both gantry side covers.
4.) Turn off all 3 switches on the status control box, on right side of Gantry.
5.) Open top cover, and engage prop rod.
6.) Remove, and set aside, Gantry scan window.
7.) Open front cover.
8.) Rotate Gantry until the tube reaches the 12 o'clock position.
9.) Lock Gantry into the 12 o'clock position.
10.) Place a rag around the Cathode HV Cable connector opening on the transformer tank.
11.) Use the spanner wrench to loosen the Cathode HV cable connector.
- Remove the cable from the HV transformer tank.
- Ground the end of the cable to the Gantry Frame to verify no voltage.

12.) Wipe excess oil from HV cable connector, and out of the HV transformer tank connector well, before you rotate the Gantry.
13.) Rotate the Gantry until the tube reaches the 3:00 position.
14.) Engage rotational lock
15.) Use the spanner wrench to remove the Cathode HV Cable connector from X-Ray tube HV connector well.
16.) Unscrew the two cable clamps to remove Cathode HV Cable from the rotating base.
17.) Cut all remaining ty-raps, and remove the cable from the Gantry.
18.) Install new ty-raps (Qty 2) using existing screws.

Note: Leave ty-raps loose around cable until final adjustment.

19.) Add 20 ml (0.7 oz.) of dielectric oil to the Cathode HV connector well of the X-Ray Tube.
20.) Place a rag around the tube connector well, to absorb excess oil when you insert the Cathode cable.
21.) Insert the Cathode HV Cable connector into the Cathode HV well of the X-Ray tube.
22.) Tighten the nut with the spanner wrench.
23.) Wipe excess oil from the tube.

24.) Unlock, and rotate the Gantry, until the tube reaches the 12 o'clock position.
25.) Lock Gantry in place.
26.) Add 20 ml (0.7 oz) of dielectric oil to the well of the HV Cathode transformer tank.
27.) Place a rag around transformer tank connector well, to absorb excess oil when you insert the Cathode cable.
28.) Insert the Cathode HV cable connector into Cathode HV transformer well.
29.) Tighten the nut with the spanner wrench.
30.) Use the old screws and cable clamps to attach the new Cathode cable to the rotating base.
31.) Wipe excess oil from the transformer.
32.) Use ty-raps to dress the cables, so a minimum loop exists on each end.
33.) Tighten ty-raps on rotating base.
34.) Wrap a rag or paper towel around both ends of the Cathode HV cable and tape them into place.
35.) Unlock the Gantry.
36.) Rotate Gantry by hand through two complete revolutions.
Check to make sure the Cathode cable does not catch on any stationary components during each revolution.
37.) Remove the rags, and wipe up any excess dielectric oil from the HV Cathode Cable ends.
38.) Restore power to the Gantry.
39.) Rotate the Gantry at a speed of 1 Revolution per second, for several revolutions.
40.) Stop the gantry and shut down power.
41.) Turn off facility power to PDU.
Use tag and lockout procedures.
42.) Turn off all 3 switches on the status control box, on the right side of Gantry.
43.) Check the tube and transformer tank wells for oil leaks.
44.) Reassemble Gantry.
45.) Refer to Retest Verification Table at the beginning of this chapter.

Section 31.0
46-321064G1 or 46-321198G1 or 2143147 kV Board

1.) Remove and set aside the right gantry side covers.
2.) Turn off all 3 switches on the status control box.
3.) Rotate the Gantry until the OBC reaches the 3 o'clock position.
4.) Engage rotational lock.
5.) Put on grounding wrist strap.
6.) Loosen the 8 captive screws, and remove the OBC Front Cover.
7.) Rotate the ejectors, and remove the defective board from the card cage.
8.) Place the board in an Anti-Static bag.
9.) Install the new board.
10.) Reassemble Gantry.
11.) Refer to Retest Verification Table at the beginning of this chapter.

Section 32.0
46-2144699 HEMRC Assembly

1.) Move table to lowest elevation.
2.) Turn off facility power to PDU.

DANGER **USE TAG AND LOCK OUT PROCEDURE.**

3.) Remove gantry side covers.
4.) Turn off all three (3) switches on status control box, on right side of gantry.
5.) Lift top cover, and engage prop rod.
6.) Remove scan window.
7.) Open Front Cover.
8.) Rotate gantry until Filament Power assembly reaches the 3 o'clock position.
9.) Engage gantry rotational lock.
10.) Loosen four (4) captive screws on Filament Power assembly cover.
11.) Remove cover.
12.) Measure voltage across filter capacitor to verify the bleeder resistor has dissipated energy to a safe level.
13.) Remove 550 Volt leads from HEMRC control board.
14.) Remove 2 Feedback connectors (J3, J9) from Gate Driver/Filter Board.
15.) Remove 120 Volt AC connector (J6) from Board.
16.) Disconnect Stator Cable connector from X-Ray Tube (J10).
17.) Remove hardware that fastens support bracket to back of HEMRC base.
18.) Remove, and keep 4 bolts, that fasten HEMRC to Filament Power assembly.
19.) Remove and replace HEMRC.
20.) Reassemble Gantry.
21.) Put on grounding wrist strap.
22.) Loosen the 8 captive screws, and remove OBC Front Cover.
23.) Rotate the ejectors, and remove the defective board from the card cage.
24.) Place the board in an Anti-Static bag.
25.) Install the new board.
26.) Reassemble Gantry.
27.) Refer to Retest Verification Table at the beginning of this chapter.

Section 33.0
46-2179860 HEMRC Control Board

1.) Remove, and set aside, the right gantry side covers.
2.) Turn off all 3 switches on the status control box.
3.) Rotate the Gantry until the OBC reaches the 3 o'clock position.
4.) Engage rotational lock.
5.) Put on grounding wrist strap.
6.) Loosen the 8 captive screws, and remove OBC Front Cover.

7.) Rotate the ejectors, and remove the defective board from the card cage.
8.) Place the board in an Anti-Static bag.
9.) Install the new board.
10.) Reassemble Gantry.

Section 34.0
46-288858G1 or 2138293 CTVRC Control Board

1.) Remove, and set aside, the right gantry side covers.
2.) Turn off all 3 switches on the status control box.
3.) Rotate the Gantry until the OBC reaches the 3 o'clock position.
4.) Engage rotational lock.
5.) Put on grounding wrist strap.
6.) Loosen the 8 captive screws, and remove OBC Front Cover.
7.) Rotate the ejectors, and remove the defective board from the card cage.
8.) Place the board in an Anti-Static bag.
9.) Jumper Position for the 2138293 board only:

Note: Install 2138293 CTVRC Board jumper JP1 in jumper position A for DCRGS (Compatible with 46-288858G1 board) and in jumper position B for URDCS.

10.) Install the new board.
11.) Reassemble Gantry.

Section 35.0
45435961 or 2122768 CTVRC Power Module

1.) Move table to lowest elevation.
2.) Turn off facility power to PDU.

DANGER **USE TAG AND LOCK OUT PROCEDURE.**

3.) Remove gantry side covers.
4.) Turn off all three (3) switches on status control box, on right side of gantry.
5.) Lift top cover, and engage prop rod.
6.) Remove scan window.
7.) Open Front Cover.
8.) Rotate gantry until Filament Power assembly reaches the 3 o'clock position.
9.) Engage gantry rotational lock.
10.) Loosen four (4) captive screws on Filament Power assembly cover.
11.) Remove cover.
12.) Measure voltage across filter capacitor to verify the bleeder resistor has dissipated energy to a safe level.
13.) Remove 550 Volt leads from CTVRC control board.
14.) Remove 2 Feedback connectors (J3, J9) from Gate Driver/Filter Board.
15.) Remove 120 Volt AC connector (J6) from Board.

16.) Disconnect Stator Cable connector from X-Ray Tube (J10).
17.) Remove hardware that fastens support bracket to back of CTVRC base.
18.) Remove, and keep 4 bolts, that fasten CTVRC to Filament Power assembly.
19.) Remove and replace CTVRC.
20.) Reassemble Gantry.

Section 36.0
46-288858G1 CTVRC Board

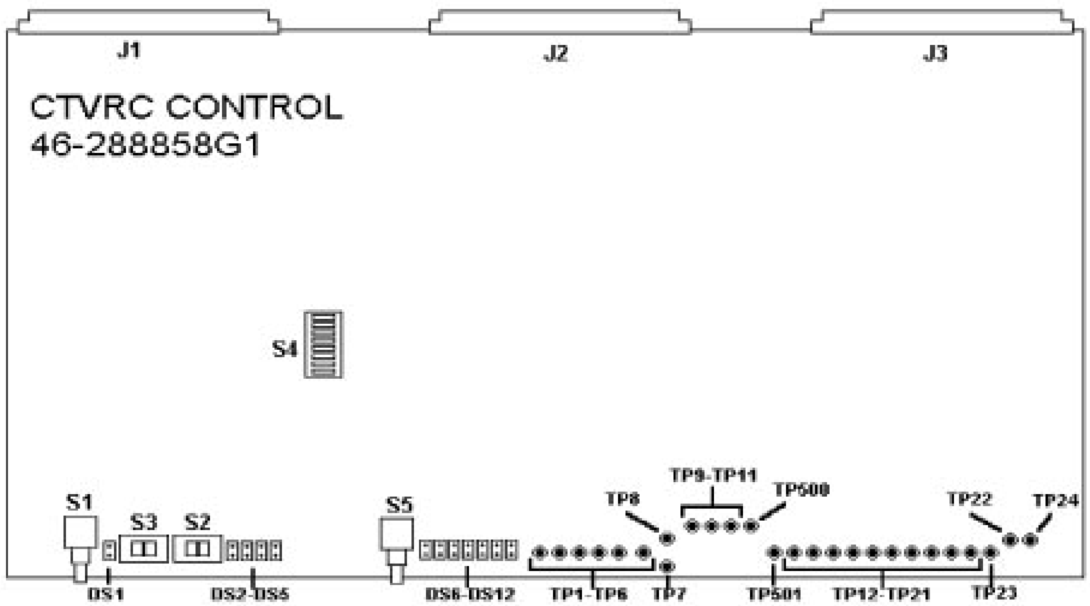


Figure 9-20 46-288858G1, CTVRC Control Board Test Points

36.1 CTVRC Board Test Points

- TP1: +5V — +5 V Supply voltage
- TP2: FA — Drive frequency of the left (main) inverter
- TP3: LPW — Pulse width generated for the left (main) inverter
- TP4: FB — Drive frequency of the right (auxiliary) inverter
- TP5: RPW — Pulse width generated for the right (auxiliary) inverter
- TP6: LGND — Logic ground
- TP7: LGARD — Guard ring tied to logic ground. Not used.
- TP8: SGARD — Guard ring tied to signal ground. Not used.
- TP9: CUR — DAC “A” (Average Current Command) Scale: 2 A / V
- TP10: PWR — DAC “B” (PWM Voltage Command) Scale: 10% / V
- TP11: +10V — +10 V Reference Supply
- TP12: DCV — DC rail monitor voltage. Scale: 100 V / V
- TP13: DCHI — Hi-side capacitor voltage. Scale: 50 V / V
- TP14: DCLO — Lo-side capacitor voltage. Scale: 50 V / V
- TP15: LAC — Left (main) inverter AC current. Scale: 5A/ V

- TP16: LCUR — Main (black) inverter average current. Scale: 2.5 A / V
- TP17: RAC — Right (auxiliary) inverter AC current. Scale: 5 A / V
- TP18: RCUR — Auxiliary (green) inverter average AC current. Scale: 2.5 A / V
- TP19: WCUR — White wire average current. Scale: 2.5 A / V
- TP20: STMP — Simulated stator temperature rise. Scale: 20 C / V
- TP21: CLR — Closed loop reference voltage. Scale: 10% / V
- TP22: +15V — +15 V Supply Voltage
- TP23: SGND — Signal Ground
- TP24: -15V — -15 V Supply Voltage
- TP500: MUX — Analog signal as selected by the muxes
- TP501: SGND — Signal Ground

36.2 46-288858G1 CTVRC Board Switch Settings

- S1: BOOST — (Mom.) Increases output power for accelerating or braking while in manual mode.
- S2: FOR / REV — Selects forward or reverse in manual mode.
- S3: AUTO / MAN — Selects automatic or manual mode.
- S4: INSITE — ASCII code for board version (Insite Switch)
- S5: RESET — (Mom.) Resets all command, fault and interrupt latches on this board.

36.3 46-288858G1 CTVRC Board LEDs

- DS1: (YEL) TEST — Indicates the CTVRC control is in manual (test) mode.
- DS2: (YEL) REV — Indicates reverse drive (braking) has been selected.
- DS3: (GRN) ON — Indicates the CTVRC is on.
- DS4: (YEL) LOI — Indicates there is less than 0.625 A in the three stator wires.
- DS5: (YEL) LOV — Indicates the DC Rail is less than 450 V.
- DS6: (RED) HIV — Indicates a DC Rail overvoltage (> 670 V) was detected.
- DS7: (RED) HCV — Indicates a capacitor overvoltage (> 375 V) was detected.
- DS8: (RED) LSTU — Indicates a shoot-thru was detected in the left (main) inverter.
- DS9: (RED) LSHT — Indicates a short was detected in the left (main) inverter.
- DS10: (RED) RSTU — Indicates a shoot-thru was detected in the right (auxiliary) inverter.
- DS11: (RED) RSHT — Indicates a short was detected in the right (auxiliary) inverter.
- DS12: (RED) OVRT — Indicates the simulated stator temperature exceeded 160 C rise.

Section 37.0
2138293 CTVRC Board

This latest version of the CTVRC Control board is backward compatible and is a replacement for the 46-288858G1 with the configuration jumper (JP1) set in the “A” position.

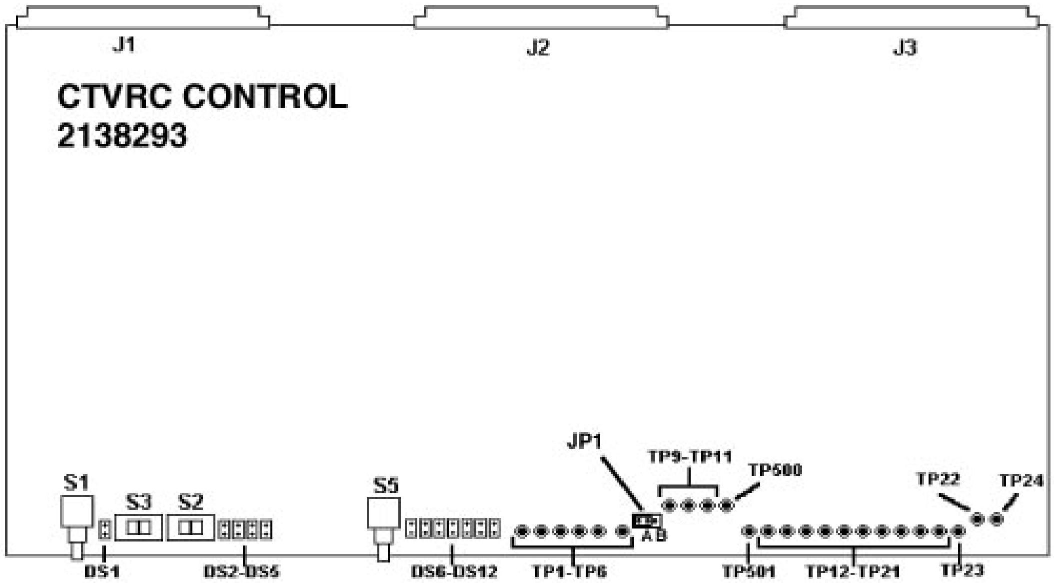


Figure 9-21 2138293, CTVRC Board Test Points

37.1 CTVRC Board Test Points

- TP1: (RED) +5V — +5 V Supply voltage
- TP2: (YEL) FA — Drive frequency of the left (main) inverter
- TP3: (YEL) LPW — Pulse width generated for the left (main) inverter
- TP4: (YEL) FB — Drive frequency of the right (auxiliary) inverter
- TP5: (YEL) RPW — Pulse width generated for the right (auxiliary) inverter
- TP6: (BLK) LGND — Logic ground
- TP7: LGARD — Guard ring tied to logic ground. Not used.
- TP8: SGARD — Guard ring tied to signal ground. Not used.
- TP9: (YEL) CUR — DAC “A” (Average Current Command) Scale: 2 A / V
- TP10: (YEL) PWR — DAC “B” (PWM Voltage Command) Scale: 10% / V
- TP11: (YEL) +10V — +10 V Reference Supply
- TP12: (YEL) DCV — DC rail monitor voltage. Scale: 100 V / V
- TP13: (YEL) DCHI — Hi-side capacitor voltage. Scale: 50 V / V
- TP14: (YEL) DCLO — Lo-side capacitor voltage. Scale: 50 V / V
- TP15: (YEL) LAC — Left (main) inverter AC current. Scale: 5A/ V
- TP16: (YEL) LCUR — Main (black) inverter average current.Scale: 2.5 A / V
- TP17: (YEL) RAC — Right (auxiliary) inverter AC current. Scale: 5 A / V
- TP18: (YEL) RCUR — Auxiliary (green) inverter average AC current. Scale: 2.5 A / V
- TP19: (YEL) WCUR — White wire average current. Scale: 2.5 A / V
- TP20: (YEL) STMP — Simulated stator temperature rise. Scale: 20 C / V
- TP21: (YEL) CLR — Closed loop reference voltage. Scale: 10% / V
- TP22: (RED) +15V — +15 V Supply Voltage
- TP23: (BLK) SGND — Signal Ground
- TP24: (WHT) -15V — -15 V Supply Voltage
- TP500: (YEL) MUX — Analog signal as selected by the muxes
- TP501: (BLK) SGND — Signal Ground

37.2 CTVRC Board LEDs

- DS1: (YEL) TEST — Indicates the CTVRC control is in manual (test) mode.
- DS2: (YEL) REV — Indicates reverse drive (braking) has been selected.
- DS3: (GRN) ON — Indicates the CTVRC is on.
- DS4: (YEL) LOI — Indicates there is less than 0.625 A in the three stator wires.
- DS5: (YEL) LOV — Indicates the DC Rail is less than 450 V.
- DS6: (RED) HIV — Indicates a DC Rail overvoltage (> 670 V) was detected.
- DS7: (RED) HCV — Indicates a capacitor overvoltage (> 375 V) was detected.
- DS8: (RED) LSTU — Indicates a shoot-thru was detected in the left (main) inverter.
- DS9: (RED) LSHT — Indicates a short was detected in the left (main) inverter.
- DS10: (RED) RSTU — Indicates a shoot-thru was detected in the right (auxiliary) inverter.
- DS11: (RED) RSHT — Indicates a short was detected in the right (auxiliary) inverter.
- DS12: (RED) OVRT — Indicates the simulated stator temperature exceeded 160 C rise.

37.3 CTVRC Board Jumper Setting (JP1)

The maximum output of the PDU can be determined by the OBC by reading the location of this jumper.

Jumper position:

A.) DCRGS — Selects voltage limits for systems with a DCRGS.
B.) URDCS — Selects voltage limits for systems with Unregulated HVDC Supply.

37.4 CTVRC Board Switch Settings (2138293)

- S1: BOOST — (Mom.) Increases output power for accelerating or braking while in manual mode.
- S2: FOR / REV — Selects forward or reverse in manual mode.
- S3: AUTO / MAN — Selects automatic or manual mode.
- S4: - - - — Not Used.
- S5: RESET — (Mom.) Resets all command, fault and interrupt latches on this board.

Section 38.0
46-288886G1 or 2154834 mA Circuit Board Replace

1.) Remove, and set aside, the right gantry side covers.
2.) Turn off all 3 switches on the status control box.
3.) Rotate the Gantry until the OBC reaches the 3 o'clock position.
4.) Engage rotational lock.
5.) Put on grounding wrist strap.
6.) Loosen the 8 captive screws, and remove the OBC Front Cover.
7.) Rotate the ejectors, and remove the defective board from the card cage.

8.) Place the board in an Anti-Static bag.
9.) Install the new board.
10.) Reassemble Gantry.
11.) Refer to Retest Verification Table at the beginning of this chapter.

Section 39.0
46-288886G1 mA Board

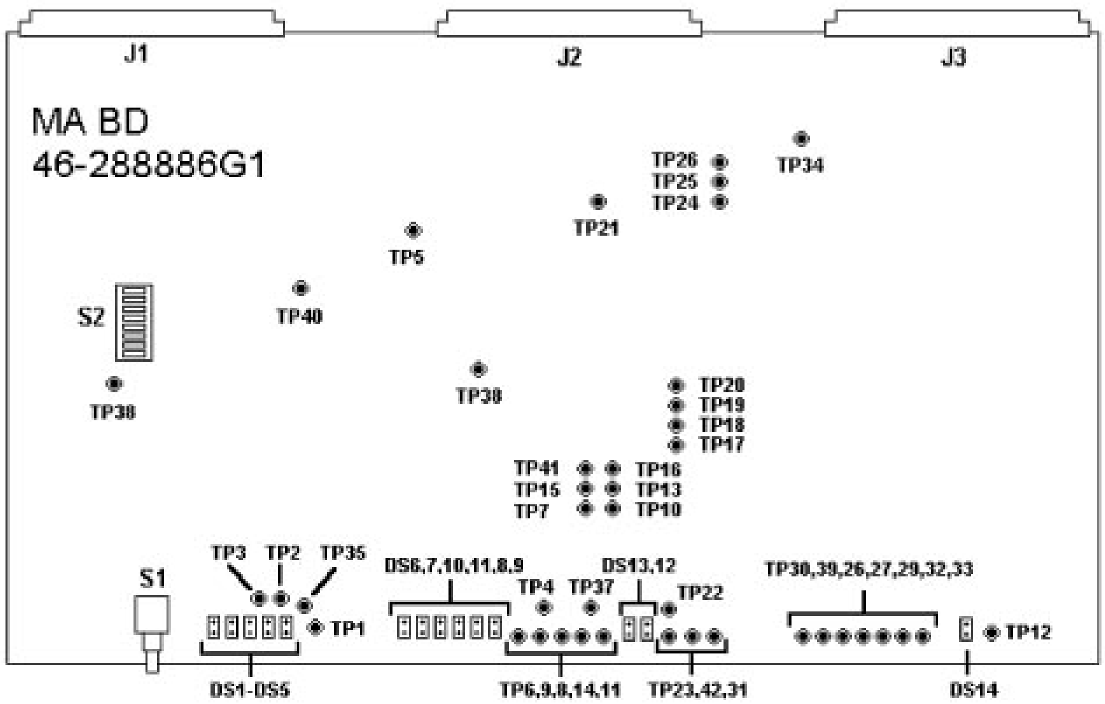


Figure 9-22 46-288886G1, mA Board Test Points

39.1 mA Board LEDs

- DS1: CLOOP
- DS2: INVEN
- DS3: ANC
- DS4: MA BAL
- DS5: CAO
- DS6: FIL FLT
- DS7: INV FLT
- DS8: FIL UC
- DS9: OFIL
- DS10: SH FIL
- DS11: FIL OC
- DS12: SMS
- DS13: 1FIL
- DS14: INV ON

39.2 mA Board Switch Settings

- S1: Reset
- S2: Board Rev in ASCII (Insite Switch)

39.3 46-288886G1 mA Board Test Points

- TP1: +5 REF
- TP2: SGND
- TP3: FERR
- TP4: CAMA
- TP5: FSIG
- TP6: LGND
- TP7: -10REF
- TP8: ACAL1
- TP9: CCAL1
- TP10: ANMA
- TP11: ACAL2
- TP12: FSHG
- TP13: MAFB
- TP14: CCAL2
- TP15: MADMD
- TP16: +24V
- TP17: OFIL
- TP18: IFL1
- TP19: EILOC
- TP20: FILSH
- TP21: +30V
- TP22: FCMD
- TP23: FCUR
- TP24: +15V
- TP25: FIL UC
- TP26: +30F
- TP27: FIL CT
- TP28: PD
- TP29: FIL2
- TP30: PS
- TP31: FGND
- TP32: FIL1
- TP33: FSH
- TP34: IPUL
- TP35: +5LED
- TP36: CLOOP
- TP37: -15V
- TP38: +15AV
- TP39: FD
- TP40: FDMD
- TP41: MAMUX
- TP42: FGND

Section 40.0
2154834 HEMRC mA Control Board

The 2154834 mA Board performs similarly to the previous mA board except the closed loop control changed to Cathode mA instead of Anode mA as in the previous HSA & CT/i design. This change is required for compatibility with the Performix Xray tube. The board is BACKWARD COMPATIBLE and is a replacement for the 46-288886G1 mA board.

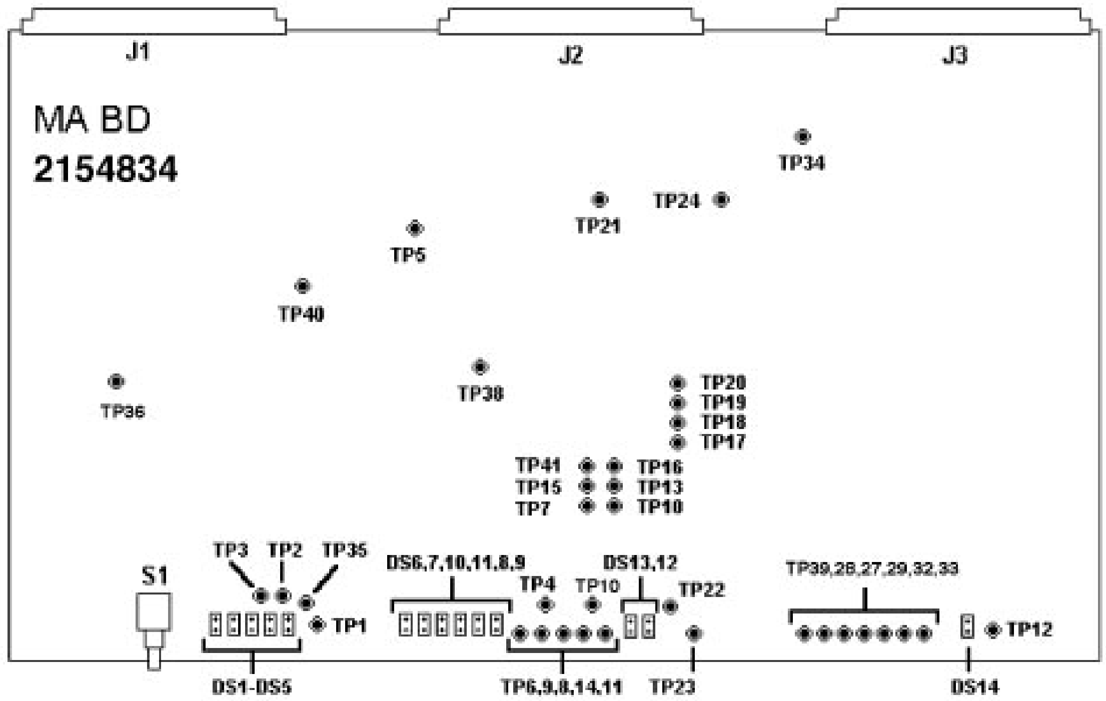


Figure 9-23 2154834, HEMRC mA Board

40.1 2154834mA Board LEDs

- DS1: (GRN) CLOOP mA Loop is Closed
- DS2: (GRN) INVEN Inverter is Enabled
- DS3: (RED) ANO Anode OverCurrent Fault Detected in mA Monitoring
- DS5: (RED) CAO Cathode OverCurrent Fault Detected in mA Monitoring
- DS6: (RED) FIL FLT Filament Inverter Fault Detected (includes Inverter Fault Filament Undercurrent, Filament Overcurrent Open Filament and Shorted Filament)
- DS7: (RED) INV FLT Inverter Fault Detected
- DS8: (RED) FIL UC Filament Undercurrent Fault Detected
- DS9: (RED) OFIL Open Filament Fault Detected
- DS10: (RED) SH FIL Shorted Filament Fault Detected
- DS11: (RED) FIL OC Filament Overcurrent Fault Detected
- DS12: (RED) IFLT Filament Inverter Fault Detected (same as DS5 except on Inverter section of board)
- DS13: (GRN) SMSP Small Focal Spot is Selected
- DS14: (GRN) INV ON Inverter is On

40.2 mA board Test Points

- TP1: +5 V — +5 V Reference Supply
- TP2: SGND — Signal ground
- TP3: FERR — Filament Error output from AR3, P7 (0.5 Volt / Amp)
- TP4: CAMA — Cathode HV supply mA feedback (1 Volt / 100mA)
- TP5: FSIG — Filament Demand output from AR7, P1 (1 Volt / Amp)
- TP6: LGND — Logic Ground
- TP7: -10REF — -10 V Reference Supply
- TP8: ACAL1 — Put an Ammeter between ACAL1 and CCAL2 as part of anode meter cals (200 mA scale)
- TP9: CCAL1 — Put an Ammeter between CCAL1 and CCAL2 as part of cathode meter cals (200 mA scale)
- TP10: ANMA — Anode HV supply mA feedback (1 Volt / 100mA)
- TP11: ACAL2 — Put an Ammeter between ACAL1 and ACAL2 as part of anode meter cals (200 mA scale)
- TP12: FSHG
- TP13: MAFB — mA feedback into multiplying DAC U44, P17
- TP14: CCAL2 — Put an Ammeter between CCAL1 and CCAL2 as part of cathode meter cals (200 mA scale)
- TP16: +24 V — +24 V Supply
- TP20: FILSH — Filament Short Signal
- TP21: +30 V — +30 V Reference Supply
- TP22: FCMD — Filament Command output from AR12, P1 (1 Volt / Amp)
- TP23: FCUR — Filament Feedback into PWM U67, P1 (1 Volt / Amp)
- TP24: +15 V — +15 V Reference Supply
- TP27: FIL CT — Filament Waveform into the Center Tap of the Filament Transformer
- TP28: PD — Switching Interval Waveform from the PWM
- TP29: FIL2 — Filament Inverter Q12 drain Voltage
- TP31: FGND — GND Tied to End of Guard Band
- TP32: FIL1 — Filament Inverter Q14 Drain Voltage
- TP33: FSH — Filament Current – DC Level
- TP35: +5LED — +5 V LED Chassis Supply
- TP36: +5 V — +5 V Chassis Supply
- TP37: -15V — -15 V Reference Supply
- TP38: +15AV — +15 V Reference Supply
- TP39: FD — Tie this Test Point high to disable fault generation
- TP40: FDMD — Filament Demand
- TP41: MAMUX — mA MUX Selection Output
- TP42: FGND — GND Tied to End of Guard Band

40.3 mA Board Switch Settings

S1: RESET - Manual reset for the board

Section 41.0
2179860 HEMRC Control Board

The HEMRC Control Board (High Efficiency Motor Rotor Control), performs three main functions. It provides an interface between the OBC and the HEMRC, HVDC Bus voltage monitoring, and a CAN interface between the OBC and future subsystems. This board replaces the CTVRC Control Board 2138293 or 46-288858G1 in systems using a HEMRC. The board is NOT BACKWARD COMPATIBLE on systems using the CTVRC assembly.

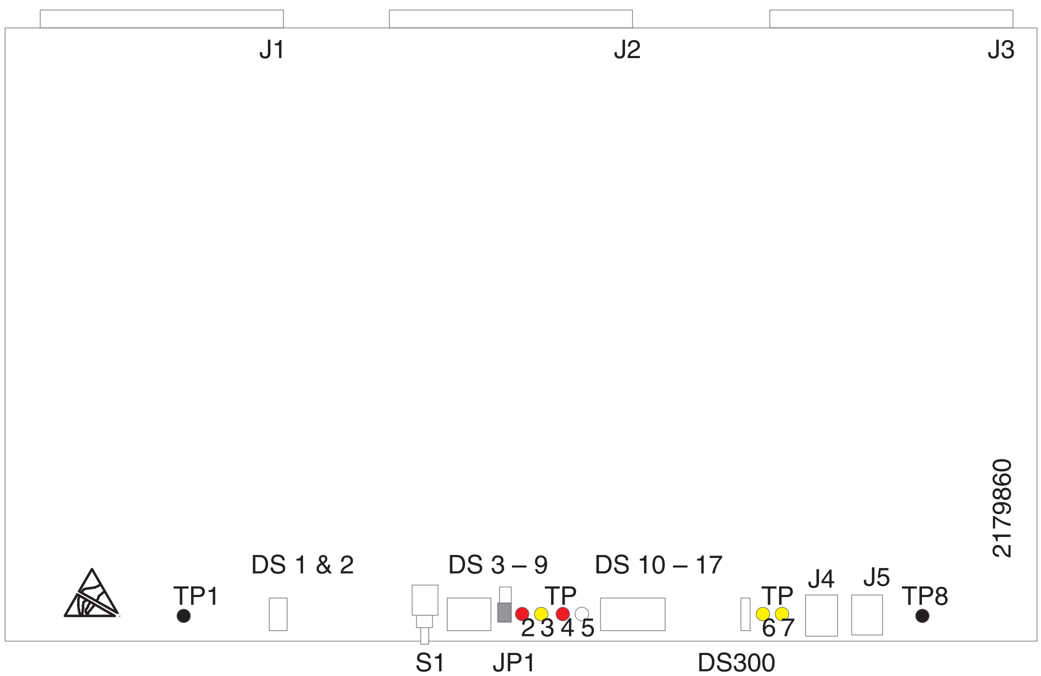


Figure 9-24 2179860 HEMRC Control Board

41.1 HEMRC Board Test Points

• TP1: (BLK) LGND	Logic ground
• TP2: (RED) +5V	+5 V supply voltage
• TP3: (YEL) MUX	Analog signal as selected by the muxes
• TP4: (RED) +15V	+15 V supply voltage
• TP5: (WHT) -15V	-15 V supply voltage
• TP6: (YEL) +10V	+10 V Reference
• TP7: (YEL) DCV	DC rail monitor voltage. Scale: 100 V / V
• TP8: (BLK) SGND	Signal ground

41.2 HEMRC Board LEDs

- DS1: (YEL) LORPM Indicates HEMRC output frequency is below programmed threshold
- DS2: (YEL) LOV Indicates the DC Rail is less than 470 V.
- DS3: (RED) HIV Indicates a DC Rail overvoltage (> 670 V) detected.
- DS4: (RED) GFLT Indicates a fault on a Gantry CAN based subsystem.
- DS5: (GRN) G1TX Indicates Gantry CAN 1 is transmitting.
- DS6: (GRN) G2TX Indicates Gantry CAN 2 is transmitting.
- DS7: (GRN) GRX Indicates GCAN is receiving.
- DS8: (GRN) HRX Indicates HEMRC CAN is receiving.
- DS9: (RED) HFLT General. Function defined by firmware.
- DS10: (GRN) General. Function defined by firmware.
- DS11: (GRN) General. Function defined by firmware.
- DS12: (GRN) General. Function defined by firmware.
- DS13: (GRN) General. Function defined by firmware.
- DS14: (GRN) General. Function defined by firmware.
- DS15: (GRN) General. Function defined by firmware.
- DS16: (GRN) General. Function defined by firmware.
- DS17: (GRN) General. Function defined by firmware.
- DS300: (GRN) G12V Indicates that GCAN_+12V_ISO is present.

41.3 HEMRC Board Jumper Setting (JP1)

The maximum output of the PDU can be determined by the OBC by reading the location of this jumper. This jumper location indicates whether the PDU has a DCRGS or not!

JUMPER POSITION

A = Selects voltage limits for systems with a DCRGS. (This is the default shipping position).

B = Selects voltage limits for systems with an Unregulated HVDC Supply.

41.4 HEMRC Board Jumper Plug

The jumper plug is a four position “shorting” plug that is installed in either the J4 or J5 CAN loopback connector. This jumper plug location selects whether the unit is in the normal or diagnostic CAN mode.

JUMPER PLUG

J5 = (Normal) Selects normal CAN operation where the HEMRC CAN and Gantry CAN are connected to their respective CAN networks. (This is the default shipping position).

J4 = (Loopback) Selects diagnostic CAN mode where the HEMRC CAN and Gantry CAN networks are connected together.

41.5 HEMRC Board Switch Function

S1: RESET Resets all command, fault and interrupt latches on this board, and also creates a GCAN_RESET signal which is sent to downstream controllers via the control interface bus connections.

Section 42.0
2145832 HEMRC Interface Board

The HEMRC (High Efficiency Motor Rotor Control) Interface Board provides a transition point for terminating existing gantry harness connections at J3 and J9. The board also provides the input means for the system to monitor the HVDC Bus and AC distribution. The board is located in the HEMRC assembly with a location designation of CT2 A2 A6 A2.

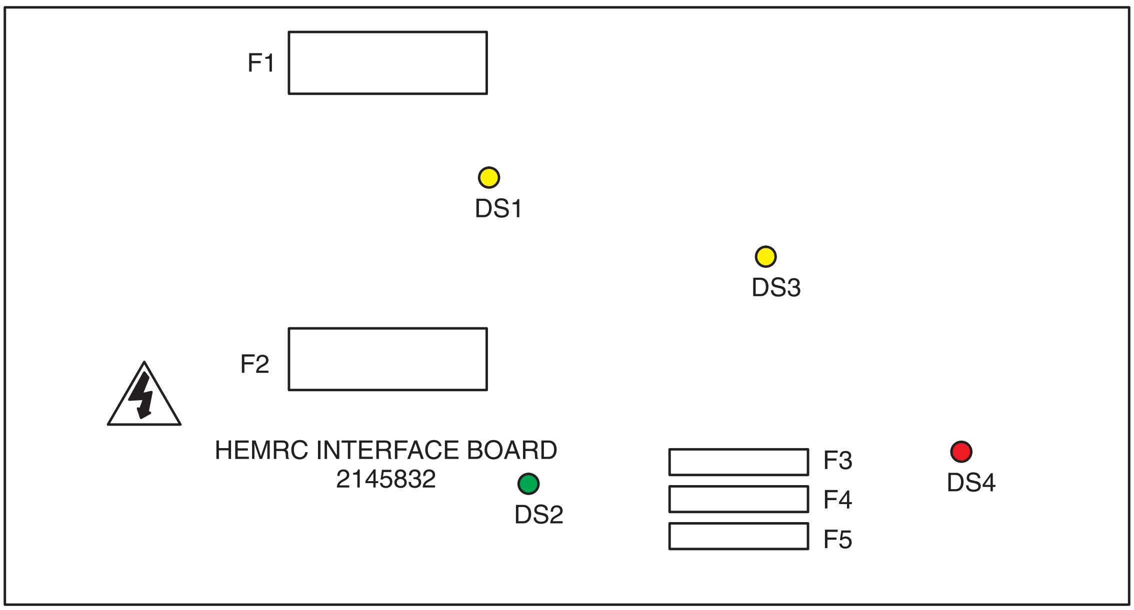


Figure 9-25 2145832 HEMRC Interface Board

42.1 HEMRC Interface Board Test Points

CAUTION **There are no test points on this board! All active circuitry is high impedance and tied to hazardous voltages. It must not be probed!**

CAUTION **The Chopper Control circuit is referenced to the DC- rail at all times. This is a potentially lethal voltage source. DO NOT connect to ground!**

42.2 HEMRC Interface Board LEDs

- DS1: (YEL) Indicates the HVDC- to HEMRC AC Drive.
- DS2: (GRN) Indicates 120 Vac is applied to the board.
- DS3: (YEL) Indicates the AC Drive DC+ and DC- are energized.
- DS4: (RED) Indicates a fault was detected in the Chopper Control.

Section 43.0 HEMRC Interface Board Fuses

- F1: (20A, 700 Vdc) HVDC- to HEMRC AC Drive.
- F2: (20A, 700 Vdc) HVDC+ to HEMRC AC Drive.
- F3: (3A, 250 Vac) 120 Vac to Collimator power supply.
- F4: (8A, 250 Vac slo-blo) 120 Vac to Filament power supply.
- F5: (8A, 250 Vac slo-blo) 120 Vac to HEMRC AC Drive Isolation Transformer.

Section 44.0 46-321064G1 kV Control Board

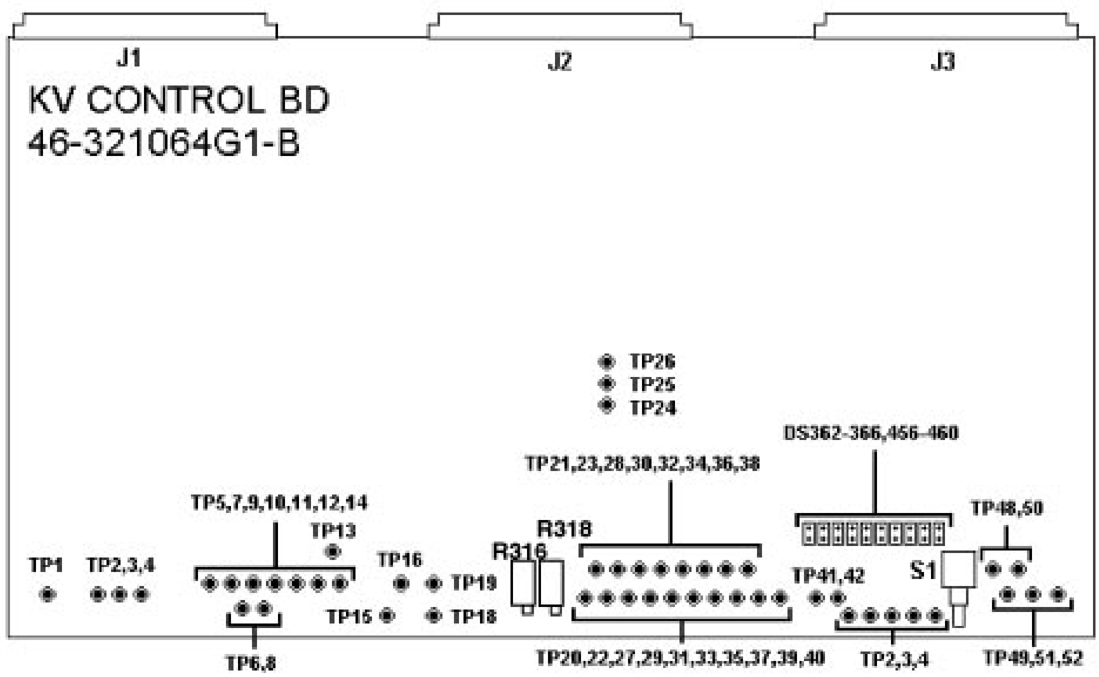


Figure 9-26 46-321064G1 kV Control Board

44.1 kV Control Board 46-321064G1 Test Points

- TP1 LGND
- TP2 SPIT: (TTL 400uS duration); This signal indicates a tube spit, measure for 400uS duration (LOW = Spit). i.e. rapid decay of the kV.
- TP3 AGND: (Analog 0 Volts); Analog ground.
- TP4 AGND: (Analog 0 Volts); Analog ground.
- TP5 PWMA: (TTL 39KHz to 66KHz during scan); This signal is the pulse width modulation for the anode, measure for 39KHz to 66KHz during a scan.
- TP6 LGND: (TTL 0 Volts); Logic ground.
- TP7 VPHA: (Analog) This signal is the voltage for an approximate pre-scan kV value for the phase control of the anode. The approximate pre-scan values for kV are:
  80kV = 2.81 VDC
  100kV = 1.97 VDC
  120kV = 1.20 VDC
  140kV = 0.521 VDC
- TP8 AGND: (Analog 0 Volts); Analog ground.
- TP9 VCAN: (Analog) This signal is the anode voltage control error signal.
- TP10 KVCM: (Analog 0 Volts to 10 Volts); This signal is the kV command voltage from the CPU. The approximate values for kV are:
  80kV = - 5.32 VDC
  100kV = - 6.65 VDC
  120kV = - 7.99 VDC
  140kV = - 9.32 VDC
- TP11 PWMC: (TTL 39KHz to 66KHz) This signal is the pulse width modulation for the cathode, measure for 39KHz to 66KHz during a scan.
- TP12 VDRV: (Analog) This signal is input voltage for the voltage controlled oscillator. The approximate pre-scan values for kV are:
  80kV = 3.50 VDC
  100kV = 2.76 VDC
  120kV = 2.00 VDC
  140kV = 1.25 VDC
- TP13 VREF: (Analog 10 Volts); This is the voltage reference for the board.
- TP14 VCO: (TTL) This signal is the voltage controlled oscillator.
- TP15 VCNT: (Analog) This voltage control signal is sent to the Voltage Controlled Oscillator with the approximate pre-scan value. The pre-scan values are:
  80kV = - 5.32 VDC
  100kV = - 4.16 VDC
  120kV = - 3.03 VDC
  140kV = - 1.97 VDC
- TP16 VPHC: (Analog); This signal is the cathode phase control voltage. The approximate pre-scan values for kV are:
  80kV = 3.80 VDC
  100kV = 2.65 VDC
  120kV = 1.51 VDC
  140kV = 0.380 VDC
- TP17: N/A
- TP18 SAW: (Analog 39KHz to 66KHz during scan)
  This signal is the sawtooth and is 39KHz to 66KHz during a scan.

- TP19 VER: (Analog) This signal is the cathode kV voltage error.
- TP20 INVEN: (TTL) This signal is the inverter enable signal from the CPU to enable the inverter.
- TP21 LU: (TTL) Use the Left Upper test point to check the fiber optic cables on the kV board by grounding TP-21.
- TP23, TP42, TP36, TP34, TP-28: One at a time, make sure the LEDs light in the proper sequence on the inverter gate driver board.
- TP22 EXCM: (TTL) Exposure Command from the Gentry I/O board.
- TP23 LL: (TTL) Use the Left Lower test point to check the fiber optic cables on the kV board by grounding TP-21,
- TP23, TP42, TP36, TP34 and TP-28: One at a time. Make sure the LEDs light in the proper sequence on the inverter gate driver board.
- TP24 ASPIT: (Anode Spit - Analog) When the fall time of the
- anode kV is fast enough, a large signal will pass thru the cap.
- TP25 CSPIT: (Cathode Spit - Analog) When the fall time of the cathode is fast enough, a large signal will pass thru the cap.
- TP26 kVMUX: (Analog) This is a MUXed signal for:

  Anode, Cathode kV
  Anode, Cathode current
  Total kV
  +10V Reference
  VDrive (TP 12)
- TP27 EXEN: (Exposure Enable - TTL) This is a signal from the rotor to indicate that the rotor is up to speed.
- TP28 ARL: (TTL). Use the Anode Right Lower test point to check the fiber optic cables on the kV board by grounding TP-21, TP23, TP42, TP36, TP34 and TP-28 one at a time. Make sure the LEDs light in the proper sequence on the inverter gate driver board.
- TP29 LEFT: (Left IBGTs - TTL) This is the control signal for the left IGBTs on the inverter.
- TP30 KVTB: (kV Total - Analog) This is the signal of the anode and cathode total kV.
- TP31 TRIG: (Trigger - TTL) Use this signal as a convenient way to trigger a scope to the start of a scan. Performing this action is the same as inverter on.
- TP32 HVON (High Voltage ON - TTL) This signal indicates when the anode or cathode kV exceeds 75% of requested kV.
- TP33 CART: (Cathode Right IGBT - TTL) This signal controls the right IGBT of the cathode inverter.
- TP34 ARU: (TTL) Use the Anode Right Upper test point to check the fiber optic cables on the kV board by grounding TP-21, TP23, TP42, TP36, TP34 and TP-28 one at a time. Make sure the LEDs light in the proper sequence on the inverter gate driver board.
- TP35 ANRT: (Anode Right IGBT - TTL) This signal controls the right IGBTs of the anode inverter.
- TP36 CRL: (TTL) Use the Cathode Right Lower test point to check the fiber optic cables on the kV board by grounding TP-21, TP23, TP42, TP36, TP34 and TP-28 one at a time. Make sure the LEDs light in the proper sequence on the inverter gate driver board.
- TP37 CAKV: (Analog1 Volt = 10kV Cathode kV) Use this test point to measure the specified cathode kV signal to the cathode inverter.
- TP38 LGND: Logic ground.
- TP39 AGND: (Analog 0 Volts); Analog ground.
- TP40 VERR: Analog Voltage Error.
- TP41 ANKV: (Analog 1 Volt = 10kV Anode kV) Use this test point to measure the specified

anode kV signal to the anode inverter.

- TP42 CRU: (TTL) Use the Cathode Right Upper test point to check the fiber optic cables on the kV board by grounding TP-21, TP23, TP42, TP36, TP34 and TP-28 one at a time. Make sure the LEDs light in the proper sequence on the inverter gate driver board.
- TP43 LGND: Logic ground.
- TP44 CAOC: (Analog 1 Volt = 10A Cathode Overcurrent) Use this test point to measure the specified current thru the cathode tank.
- TP45 ANOC: (Analog 1 Volt = 10A Anode Overcurrent) Use this test point to measure the specified current thru the anode tank.
- TP46 CAST: (Analog) Cathode Shoot-Thru.
- TP47 ANST: (Analog) Anode Shoot-Thru.
- TP48 FLT: (TTL Faults) The following errors are OR'ed at this node:
  Anode, Cathode Overcurrent
  Anode, Cathode Shoot-Thru
  Anode, Cathode Overvoltage
- TP49: +15-A
- TP50 FLTS: (TTL Faults) The following errors are OR'ed at the node:
  Anode, Cathode Overcurrent
  Anode, Cathode Shoot-Thru
  Anode, Cathode Overvoltage
  Inverter wiring interlock
  Fault reset
  Spit (TP2)
  Spit 32
- TP51: +5V
- TP52: -15-A

44.2 kV Control Board 46-321064G1 Switch Settings

S1: Reset

44.3 kV Control Board 46-321064G1 LEDs

- DS364: SPIT
- DS365: INVON
- DS366: HVON
- DS367: CAOV
- DS368: PNOV
- DS421: CAOC
- DS422: CAST
- DS423: ANOC
- DS424: ANST
- DS425: INTLK

44.4 kV Control Board 46-321064G1 Pots

R316 CAKV: Adjusts the gain of the cathode kV feedback. Factory adjusted for unity gain. Field adjusted during HV PS cal procedure. Range: approximately ±20%.

R318 ANKV: Adjusts the gain of the anode kV feedback. Factory adjusted for unity gain. Field adjusted during HV PS cal procedure. Range: approximately ±20%.

Section 45.0
46-321198G1 or 2143147 kV Control Board

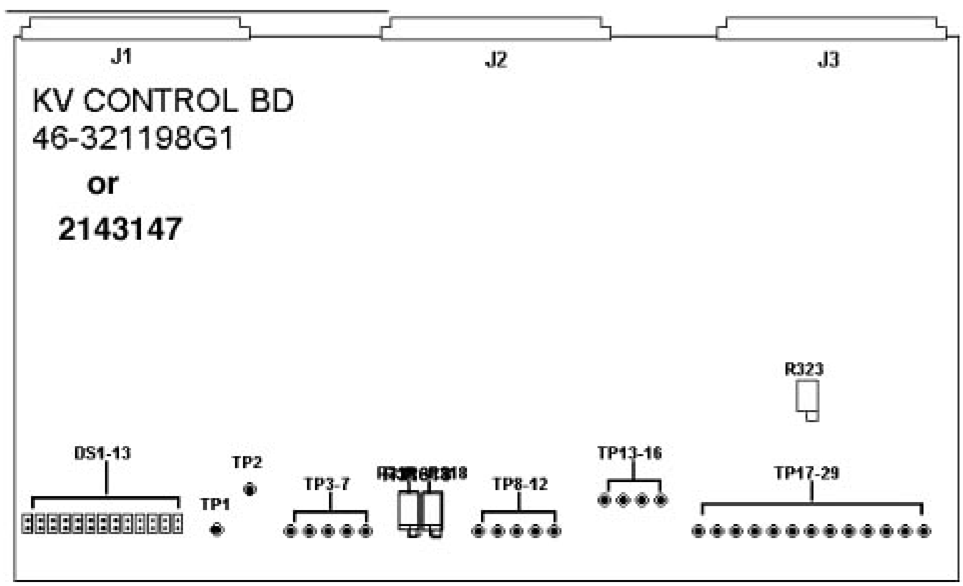


Figure 9-27 46-321198G1 or 2143147 kV Control Board

45.1 kV Control Board 46-321198G1 or 2143147 kV Test Points

- TP1 HVON: Indicates the kV feedback equals or exceeds 75% of command.
- TP2 +5V: +5V (VCC) logic power
- TP3 LGND: Logic ground
- TP4 TRIG: A “1” indicates the selected inverter(s) is (are) turned on.
- TP5 EXCM: Indicates an exposure command is being received from RCOMM bd
- TP6 EXEN: Indicates exposures are not disabled by the CTVRC, I/O or mA bds
- TP7 SPIT: Indicates a spit has been detected and recovery is in process.
- TP8 KVCM: kV command. Scale: 15kV/V
- TP9 ANKV: Anode kV feedback. Scale: 10 kV/V
- TP10 CAKV: Cathode kV feedback. Scale 10kV/V
- TP11 KVTB: Total kV feedback. Scale 20 kV/V
- TP12 SGND: Signal ground
- TP13 MUX: Analog MUX output as selected by firmware
- TP14 +10V: +10V reference
- TP15 -15V: -15V supply voltage

- TP16 +15V: +15V supply voltage
- TP17 KVERR: Integrated kV error signal
- TP18 PCNT: Average inverter duty cycle. Scale: 12%/V - 10%
- TP19 SGND: Signal ground
- TP20 ANOC: Anode inverter current. Scale: 25 A/V
- TP21 CAOC: Cathode inverter current. Scale: 25 A/V
- TP22 APH: Anode inverter duty cycle. Scale: 20% /V - 100%
- TP23 CPH: Cathode inverter duty cycle. Scale: 200% - 20%/V
- TP24 VCNT: Frequency control voltage. Scale: 19.5 kHz + 2.2kHz/V
- TP25 LGND: Logic ground
- TP26 SAW: Sawtooth (5 to 10V) at double the inverter frequency, nominally 39 to 61 kHz
- TP27 FREQ: 5V square wave at double the inverter frequency.
- TP28 APLSA: “1” indicates an “ON” pulse of the anode inverter
- TP2 CPLSA: “1” indicates an “ON” pulse of the cathode inverter

45.2 kV Control Board 46-321198G1 or 2143147 kV LEDs

- DS1 SPRT: Indicates the maximum spit rate has been exceeded.
- DS2 GFLT: Indicates a “GO” fault has occurred.
- DS3 ANST: Indicates an anode shoot-through has occurred.
- DS4 CAST: Indicates a cathode shoot-through has occurred.
- DS5 ANOC: Indicates an anode overcurrent has occurred.
- DS6 CAOC: Indicates a cathode overcurrent has occurred.
- DS7 ANOV: Indicates an anode overvoltage has occurred.
- DS8 CAOV: Indicates a cathode overvoltage has occurred.
- DS9 AINT: Indicates the anode inverter interlock is open.
- DS10 CINT: Indicates the cathode inverter interlock is open.
- DS11 OVRV: Indicates the kV feedback has exceeded the upper limit of the load regulator. May be ignored if on after power up or hardware reset.
- DS12 HVND: Indicates anode and/or cathode kV feedback signals exceed 10 kV.
- DS13 INON: Indicates the selected inverter(s) is (are) turned on.

45.3 kV Control Board 46-321198G1 or 2143147 kV Switch Settings

S1: InSite readable dip switch set for the ASCII equivalent of the board assembly version.

45.4 kV Control Board 46-321198G1 or 2143147 Adjustments

- R316 CAKV: Adjusts the gain of the cathode kV feedback. Factory adjusted for unity gain. Field adjusted during HV PS cal procedure. Range: approximately ±20%.
- R318 ANKV: Adjusts the gain of the anode kV feedback. Factory adjusted for unity gain. Field adjusted during HV PS cal procedure. Range: approximately ±20%.
- R323 (FREQ): Factory adjusted for minimum frequency of 39.0 kHz ±1.0 kHz at TP27 (FREQ) with TP24 (VCNT) set to 0V. Should not require field adjustment.

Section 46.0
46-288512G1 Gentry I/O Board

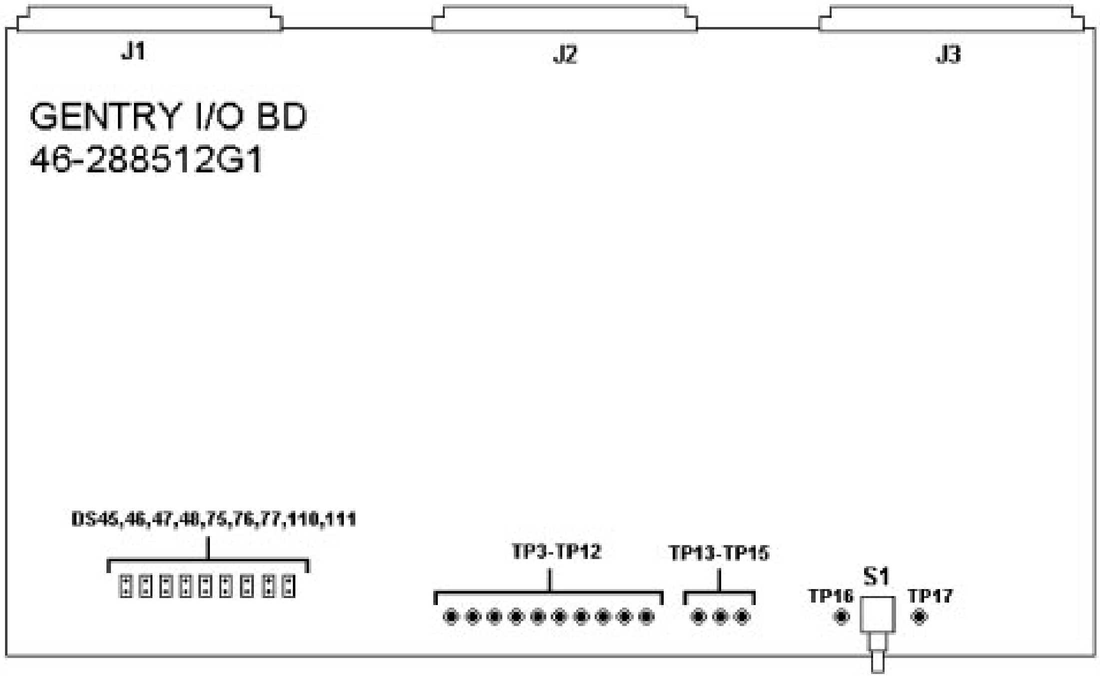


Figure 9-28 46-288512G1 Gentry I/O Board

46.1 Gentry I/O Board Test Points

- TP3: +5V
- TP4: LGND
- TP5: SGND
- TP6: +15V
- TP7: -15V
- TP8: +12V
- TP9: -12V
- TP10: +10V
- TP11: +24V
- TP12: A/DIN
- TP13: SOUT
- TP14: -SIN
- TP15: SRTN
- TP16: +SIN
- TP17: PGND

46.2 Gentry I/O Board LEDs

- DS45: ECMD INTR
- DS46: BCTR INTR
- DS47: DTHTR INTR
- DS48: BCTR CLS
- DS75: AXLTS ON
- DS76: ADC CMPLT
- DS77: BTMR EXP
- DS110: TPRLY ON
- DS111: INTR CLS

46.3 Gentry I/O Board Switch Settings

S1: Reset

Section 47.0
46-264888G1 Relay Control Board

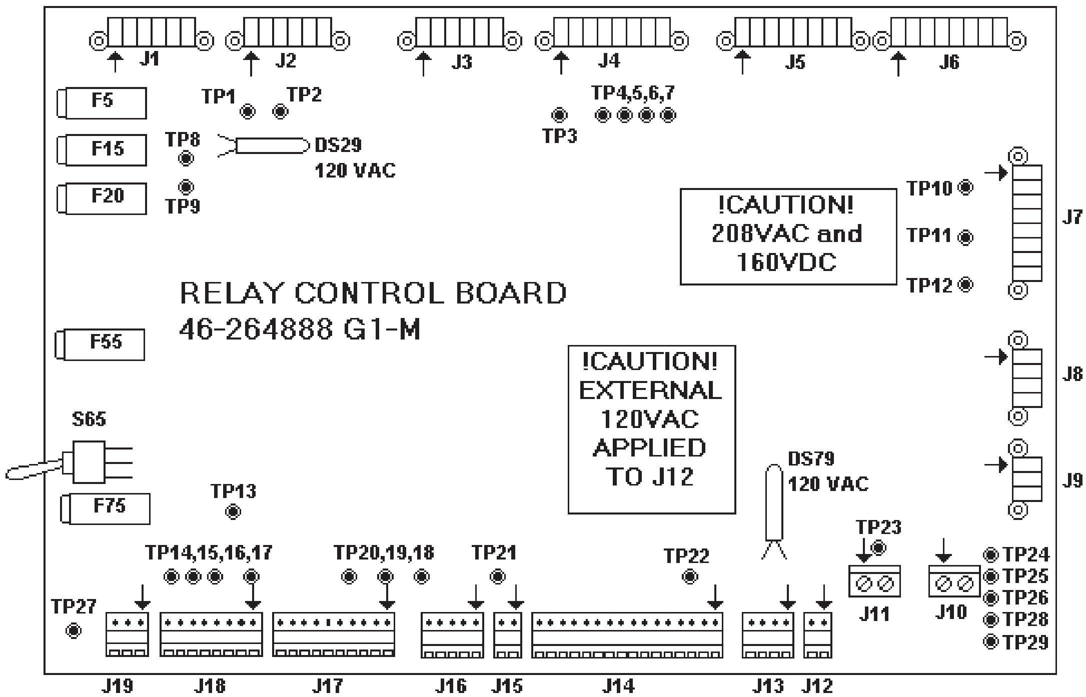


Figure 9-29 46-264888G1 Relay Control Board

47.1 Relay Control Board Test Points

- TP1 Analog 120/110 VAC: Will be 120/110 VAC when the axial is enabled. Will be at 0 VAC if the servo over current relay opens to disable the servo.
- TP2 Analog 120/110 VAC: Coil Power at 120/110 VAC for the servo enable contactor K2. At 120/110VAC for the servo to be able to drive gantry axial rotation.
- TP3 Analog 120/110 VAC: 120/110 VAC coil power for circuit breaker K1 (Table vertical drive and Gantry tilt) and K2 (Table horizontal drive). Present when drives are “ON” at the REM box.
- TP4 Analog 120/110 VAC: Power to turn on the SRU and the Operator console contactors. 120/110 VAC when “Data Processing power” is on at the REM box.
- TP5 Analog 120/110 VAC: Power for DCRGS contactor K1. Should be at 120/110 VAC if Gantry I/O, 550 enable switch in gantry and REM box drives are “ON”.
- TP6 Analog 120/110 VAC: Power for DCRGS contactor K1. Requires power at TP5, key switch closure and activation of circuit breaker assembly contactors K2, K1, and K4.
- TP7 Analog 120/110 VAC: Same function as TP6 if relay control board K33 is functioning properly.
- TP8 Analog 120/110 VAC: Input for 120/110 contactor coil power from 32kVA transformer.

- TP9 Analog 120/110 VAC: Coil Power at the RCB. 120 VAC for 60 Hz. and 110 VAC for 50Hz.
- TP10 Analog 208 VAC: 208 VAC line to line on these points when circuit breaker contactor K1 is on to produce 160 VDC power for table vertical drive and gantry tilt. “0” VAC when drives are OFF.
- TP11 Analog 208 VAC: 208 VAC line to line on these points when circuit breaker contactor K1 is on to produce 160 VDC power for table vertical drive and gantry tilt. “0” VAC when drives are OFF.
- TP12 Analog 208 VAC: 208 VAC line to line on these points when circuit breaker contactor K1 is on to produce 160 VDC power for table vertical drive and gantry tilt. “0” VAC when drives are OFF.
- TP13 Analog +24VDC 4: +24 VDC (18 - 28 DVC).
- TP14 Analog +24VDC: Loop On status, +24 when the servo is enabled.
- TP15 Analog +24VDC: OK to turn on Gantry Servo Output. Should be at +24 VDC for Gantry Axial Rotation to be enabled.
- TP16 Analog +24VDC: Hospital “X-Ray ON” control power. At +24 VDC when the external hospital light should be ON.
- TP17 Analog +24VD: Drives On, should be at +24 when the REM box “X-Ray/Drives On” is activated.
- TP18 Analog +24VDC: At +24 VDC when “X-Ray Drives On” is activated at the REM box. At 0 VDC when drives are Off. This is the pilot function for Drives On
- TP19 Analog +24VDC: At +24 VDC when Data Processing Power is activated at the REM box. At 0 VDC when Off. This is the pilot function for PDU On.
- TP20 Analog +24VDC: Read back for 550 enabled. At +24 VDC when DCRGS contactor K1 is on and 550 VDC is enabled.
- TP21 Analog +24VDC: Table and Console E-Stop operates properly at +24 VDC except when table or console E-Stop is engaged.
- TP22 Analog +24VDC: +24 VDC power supply output. At +24 VDC at all times.
- TP23 Analog +160 VDC: At 160 when drives are on.

47.2 Relay Control Board Switch

S65 **DCRGS 550 V** Backup Contactor Enable Test Switch

47.3 Relay Control Board Fuse

- F5 1A for **spare** 24 V output
- F15 2A for 120 Vac **24 hour power** for contactors
- F20 0.25 A for **primary power to T40** (24V step down transformer)
- F55 1.5A for **bridge output** (24 V supply)
- F75 1A for 24 V output to **Gentry I/O**

Section 48.0
Interface Measurement Board

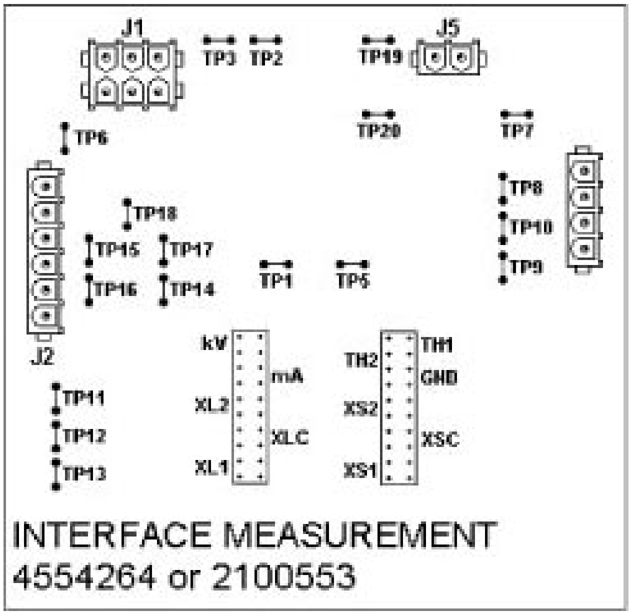


Figure 9-30 Interface Measurement Board

48.1 Interface Measurement Board Test Points

- TP1 KV: Sensed kV signal - Scale 10 kV / V
- TP2 KV: kV signal to OBC
- TP3 KV GND: kV return signal to OBC
- TP4
- TP5 MA: Sensed kV signal - Scale 100 kV / V
- TP6 MA GND: mA return signal to OBC
- TP7 PS1: Pressure Switch 1
- TP8 PS2: Pressure Switch 2
- TP9 TH1: Thermistor 1 (not used)
- TP10 TH2: Thermistor 2 (not used)
- TP11 XS2: Small Filament 2
- TP12 XSC: Small Filament Common
- TP13 XS1: Small Filament 1
- TP14 XL2: Large Filament 2
- TP15 XLC: Large Filament Common
- TP16 XL1: Large Filament 1
- TP17 MAout: mA signal to OBC
- TP18 GND: Tank ground
- TP19 kV: Sensed kV signal - Scale 10 kV / V
- TP20 GND: Tank ground

48.2 HEMRC Fuse replacement

46-170021P104 (10 amp.) Located on the HEMRC Resistor Panel Asm.

46-170021P43 (3 amp.) Located on the HEMRC Interface Board

46-170021P15 (8 amp.) Located on the HEMRC Interface Board

46-170021P101 (20 amp.) Located on the HEMRC Interface Board

1.) Position table to lowest elevation.
2.) Shut down system and turn off facility power to PDU

DANGER USE TAG AND LOCK OUT PROCEDURE.

3.) Remove both gantry side covers.
4.) Turn off all three (3) switches on status control box (right side of gantry).
5.) Open top cover and engage “Prop Rod”.
6.) Remove gantry scan window.
7.) Open front cover.
8.) Rotate gantry until the HEMRC is in a convenient working position.
9.) Engage gantry rotational lock.
10.) Remove five (5) bolts fastening cover to HEMRC assembly and remove cover.
11.) Short each end of fuse to ground using a shorting bar or conductor to ensure all capacitors in the circuit have been discharged.
12.) Remove fuse and confirm that it has opened.
13.) Replace fuse.
14.) Replace HEMRC cover and secure with five (5) mounting bolts.
15.) Disengage gantry rotational lock.
16.) Close front cover.
17.) Reinstall gantry scan window.
18.) Close Top Cover.
19.) Turn On all three (3) switches on status control box.
20.) Replace both gantry side covers.
21.) Remove “Tag Out” lock and turn on facility power.
22.) Proceed with system test.

48.3 46-2185277 Fuse Block on the HEMRC Resistor Panel Asm

1.) Position table to lowest elevation.
2.) Shut down system and turn off facility power to PDU

DANGER USE TAG AND LOCK OUT PROCEDURE.

3.) Remove both gantry side covers.
4.) Turn off all three (3) switches on status control box (right side of gantry).
5.) Open top cover and engage “Prop Rod”.
6.) Remove gantry scan window.
7.) Open front cover.
8.) Rotate the gantry until the HEMRC is in a convenient working position.

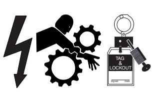
9.) Engage gantry rotational lock.
10.) Remove five (5) bolts fastening cover to HEMRC assembly and remove cover.
11.) Short each end of fuse to ground using a shorting bar or conductor to ensure all capacitors in the circuit have been discharged.
12.) Remove 10-amp fuse.
13.) Note position of three (3) wires attached to fuse holder. (Blk, Orn and Orn)
14.) Remove three (3) wires noted in step 13.
15.) Remove two (2) pan head screws attaching fuse holder.
16.) Remove and replace fuse holder (46-2185277).
17.) Mount fuse holder using two (2), 4 mm x 10 mm pan head screws. Torque to 1.25 ft/lbs.(1.7 Newton Meters).
18.) Replace leads removed in step 14 as noted in step 13.
19.) Replace 10 Amp fuse.
20.) Replace HEMRC cover and secure with five (5) mounting bolts.
21.) Disengage gantry rotational lock.
22.) Close front cover.
23.) Reinstall gantry scan window.
24.) Close Top Cover.
25.) Turn on all three (3) switches on the status control box.
26.) Replace both gantry side covers.
27.) Remove “Tag Out” lock and turn on facility power.
28.) Proceed with system test.

Section 49.0
46-2183892 HEMRC Dropping Resistors

Located on the HEMRC circuit board bracket.

1.) Position table to lowest elevation.
2.) Shut down system and turn off facility power to PDU

DANGER **USE TAG AND LOCK OUT PROCEDURE.**



3.) Remove both gantry side covers.
4.) Turn off all three (3) switches on status control box (right side of gantry).
5.) Open top cover and engage “Prop Rod”.
6.) Remove gantry scan window.
7.) Open front cover.
8.) Rotate gantry until the HEMRC is in a convenient working position.
9.) Engage gantry rotational lock.
10.) Remove five (5) bolts fastening cover to HEMRC assembly and remove cover.
11.) Remove electrical connection to interface board.
12.) Remove M4 x 0.7 hex nuts holding the dropping resistor to the HEMRC circuit board bracket.
13.) Replace dropping resistor and fasten with M4 x 0.7 hex nuts and torque to 1.25 ft/lbs. (1.7 Newton meters).
14.) Replace electrical connection to interface board.

15.) Replace HEMRC cover and secure with five (5) mounting bolts.
16.) Disengage gantry rotational lock.
17.) Close front cover.
18.) Reinstall gantry scan window.
19.) Close Top Cover.
20.) Turn On all three (3) switches on the status control box.
21.) Replace both gantry side covers.
22.) Remove “Tag Out” lock and turn on facility power.
23.) Proceed with system test.

Section 50.0
46-2184701-2 HEMRC Braking Resistors

Located on the HEMRC Resistor Panel Asm.

1.) Position table to lowest elevation.
2.) Shut down system and turn off facility power to PDU

DANGER **USE TAG AND LOCK OUT PROCEDURE.**

3.) Remove both gantry side covers.
4.) Turn off all three (3) switches on status control box (right side of gantry).
5.) Open top cover and engage “Prop Rod”.
6.) Remove gantry scan window.
7.) Open front cover.
8.) Rotate gantry until the HEMRC is in a convenient working position.
9.) Engage gantry rotational lock.
10.) Remove five (5) bolts fastening cover to HEMRC assembly and remove cover.
11.) Remove electrical connection from defective resistor.
12.) Remove mounting bolt extending through defective resistor.
13.) Replace resistor and secure with bolt, flat, conical, lock washers and nut supplied with resistor.
14.) Position resistor tabs at about 45 degrees above mounting pan.
15.) Use a drop of Loctite 242 to lock nut in place on mounting bolt.
16.) Tighten nut to 4.35 ft/lbs.(5.9 Newton meters).
17.) Replace electrical connections to resistor.
18.) Replace HEMRC cover and secure with five (5) mounting bolts.
19.) Disengage gantry rotational lock.
20.) Close front cover.
21.) Reinstall gantry scan window.
22.) Close Top Cover.
23.) Turn On all three (3) switches on the status control box.
24.) Replace both gantry side covers.
25.) Remove “Tag Out” lock and turn on facility power.
26.) Proceed with system test.

Section 51.0
46-2145832 HEMRC Interface Board

Located on the HEMRC circuit board bracket.

1.) Position table to lowest elevation.
2.) Shut down system and turn off facility power to PDU

DANGER **USE TAG AND LOCK OUT PROCEDURE.**

3.) Remove both gantry side covers.
4.) Turn off all three (3) switches on status control box (right side of gantry).
5.) Open top cover and engage “Prop Rod”.
6.) Remove gantry scan window.
7.) Open front cover.
8.) Rotate gantry until the HEMRC is in a convenient working position.
9.) Engage gantry rotational lock.
10.) Remove five (5) bolts fastening cover to HEMRC assembly and remove cover.
11.) Remove electrical cable connections from the interface board.
12.) Remove six (6) interface board mounting screws and washers. (Pan head 4 mm x 10 mm screw and 4.3 mm x 9 mm flat washer)
13.) Replace HEMRC interface board (46-2145832) and attach with six (6) mounting screws and washers. Torque to 1.25 ft/lbs. (1.7 Newton Meters).
14.) Reinstall cable interconnections to interface board.
15.) Replace HEMRC cover and secure with five (5) mounting bolts.
16.) Disengage gantry rotational lock.
17.) Close front cover.
18.) Reinstall gantry scan window.
19.) Close Top Cover.
20.) Turn On all three (3) switches on the status control box.
21.) Replace both gantry side covers.
22.) Remove “Tag Out” lock and turn on facility power.
23.) Proceed with system test.

Section 52.0
46-297104P1 HEMRC Detector Heater Power Supply

Located on the HEMRC power supply bracket. (23-volt unregulated power supply)

1.) Position table to lowest elevation.
2.) Shut down system and turn off facility power to PDU

DANGER **USE TAG AND LOCK OUT PROCEDURE.**

3.) Remove both gantry side covers.
4.) Turn off all three (3) switches on status control box (right side of gantry).
5.) Open top cover and engage “Prop Rod”.

6.) Remove gantry scan window.
7.) Open front cover.
8.) Rotate gantry until the HEMRC is in a convenient working position.
9.) Engage gantry rotational lock.
10.) Remove five (5) bolts fastening cover to HEMRC assembly and remove cover.
11.) Note and record position of four (4) wires attached to detector heater power supply. (Red and Black DC leads Vs Black and White AC leads)
12.) Remove leads identified in step 11.
13.) Remove four (4) M4 x 0.7 metric nuts which attach supply to bracket.
14.) Remove and replace 46-297104 P1 power supply module.
15.) Replace four (4) M4 x 0.7 metric nuts which attach supply to bracket. Torque to 1.25 ft/lbs. (1.7 Newton Meters).
16.) Reassemble electrical connections removed in step 12.
17.) Replace HEMRC cover and secure with five (5) mounting bolts.
18.) Disengage gantry rotational lock.
19.) Close front cover.
20.) Reinstall gantry scan window.
21.) Close Top Cover.
22.) Turn On all three (3) switches on the status control box.
23.) Replace both gantry side covers.
24.) Remove “Tag Out” lock and turn on facility power.
25.) Proceed with system test.

Section 53.0
46-215802 HEMRC Step-up Transformer

Located under the HEMRC power supply bracket.

1.) Position table to lowest elevation.
2.) Shut down system and turn off facility power to PDU

⚠ **DANGER** **USE TAG AND LOCK OUT PROCEDURE.**

3.) Remove both gantry side covers.
4.) Turn off all three (3) switches on status control box (right side of gantry).
5.) Open top cover and engage “Prop Rod”.
6.) Remove gantry scan window.
7.) Open front cover.
8.) Rotate gantry until the HEMRC is in a convenient working position.
9.) Engage gantry rotational lock.
10.) Remove five (5) bolts fastening cover to HEMRC assembly and remove cover.
11.) Remove two (2) M4 x 0.07 nuts securing HEMRC resistor pan assembly.
12.) Remove three (3) M8 x 1.25 x 16 mm metric hex head screws holding HEMRC/power supply bracket along with detector heater and collimator power supplies.
13.) Clip and remove cable ties as required to move power supply assembly.
14.) Fold power supply assembly out of the way without disturbing electrical connections.

15.) Remove four (4) 6 mm x 12 mm metric hex head screws holding HEMRC/transformer bracket and remove bracket.
16.) Note and record position of four (4) wires attached to HEMRC step-up transformer.
17.) Remove leads identified in step 16.
18.) Remove four (4) Hex M6 standoffs (14.3 hex x 97.75 Lg.–both ends threaded.) securing transformer to base plate.
19.) Remove and replace transformer (46-2150802).
20.) Replace four (4) hex M6 standoffs (14.2 hex x 97.75 Lg.–both ends threaded.) securing transformer to base plate. Use Loctite 242 and torque to 10.5 ft/lbs. (14.2 Newton Meters).
21.) Reassemble electrical connections removed in step 16
22.) Replace HEMRC/transformer bracket using four (4) 6 mm x 12 mm metric hex head screws. Use Loctite 242 and torque to 4.35 ft/lbs. (5.9 Newton Meters).
23.) Reposition power supply assembly without disturbing electrical connections.
24.) Replace three (3) M8 x 1.25 x 16 mm metric hex head screws holding HEMRC/power supply bracket along with detector heater and collimator power supplies. Use Loctite 242 and torque to 10.5 ft/lbs. (14.2 Newton Meters).
25.) Replace cable ties (46-208758 P2) as required.
26.) Reposition HEMRC resistor pan assembly and secure with two (2) M4 x 0.07 nuts. Torque to 1.25 ft/lbs. (1.7 Newton Meters).
27.) Replace HEMRC cover and secure with five (5) mounting bolts.
28.) Disengage gantry rotational lock.
29.) Close front cover.
30.) Reinstall gantry scan window.
31.) Close Top Cover.
32.) Turn On all three (3) switches on the status control box.
33.) Replace both gantry side covers.
34.) Remove “Tag Out” lock and turn on facility power.
35.) Proceed with system test.

Section 54.0
46-296701P1 Filament Power Supply

Located on the HEMRC PS bracket.

1.) Position table to lowest elevation.
2.) Shut down system and turn off facility power to PDU

⚠ **DANGER** **USE TAG AND LOCK OUT PROCEDURE.**

3.) Remove both gantry side covers.
4.) Turn off all three (3) switches on status control box on (right side of gantry).
5.) Open top cover and engage “Prop Rod”.
6.) Remove gantry scan window.
7.) Open front cover.
8.) Rotate gantry until the HEMRC is in a convenient working position.
9.) Engage gantry rotational lock.

10.) Remove five (5) bolts fastening cover to HEMRC assembly and remove cover.
11.) Remove two (2) M4 x 0.07 nuts securing HEMRC resistor pan assembly.
12.) Remove three (3) M8 x 1.25 x 16 mm metric hex head screws holding HEMRC/power supply bracket along with detector heater and collimator power supplies.
13.) Clip and remove cable ties as required to move power supply assembly.
14.) Fold power supply assembly out of the way without disturbing electrical connections.
15.) Note and record position of four (4) wires attached to filament power supply.
16.) Remove leads identified in step 14.
17.) Remove four (4) M6 x 1 metric hex nuts holding the filament power supply to the base plate.
18.) Remove and replace unregulated filament power supply (46-296701 P1).
19.) Replace four (4) M6 x 1 metric hex nuts which secure power supply to base plate. Torque to 4.35 ft/lbs. (5.9 Newton Meters).
20.) Reassemble electrical connections removed in step 15.
21.) Reposition power supply assembly without disturbing electrical connections.
22.) Replace three (3) M8 x 1.25 x 16 mm metric hex head screws holding HEMRC/power supply bracket along with detector heater and collimator power supplies. Use Loctite 242 and torque to 10.5 ft/lbs.(14.2 Newton Meters).
23.) Replace cable ties (46-208758 P2) as required.
24.) Reposition HEMRC resistor pan assembly and secure with two (2) M4 x 0.07 nuts. Torque to 1.25 ft/lbs. (1.7 Newton Meters).
25.) Replace HEMRC cover and secure with five (5) mounting bolts.
26.) Disengage gantry rotational lock.
27.) Close front cover.
28.) Reinstall gantry scan window.
29.) Close Top Cover.
30.) Turn On all three (3) switches on the status control box.
31.) Replace both gantry side covers.
32.) Remove “Tag Out” lock and turn on facility power.
33.) Proceed with system test.

Section 55.0
46-2185251 SCR Module

Located under the HEMRC Resistor Panel Asm.

1.) Position table to lowest elevation.
2.) Shut down system and turn off facility power to PDU

⚠ **DANGER** **USE TAG AND LOCK OUT PROCEDURE.**

3.) Remove both gantry side covers.
4.) Turn off all three (3) switches on status control box (right side of gantry).
5.) Open top cover and engage “Prop Rod”.
6.) Remove gantry scan window.
7.) Open front cover.
8.) Rotate gantry until the HEMRC is in a convenient working position.

9.) Engage gantry rotational lock.
10.) Remove five (5) bolts fastening cover to HEMRC assembly and remove cover.
11.) Disconnect cable from Connector J7 on the HEMRC Interface board.
12.) Remove five (5) M4 x 0.07 nuts securing HEMRC resistor panel assembly.
13.) Clip and remove cable ties as required to move resistor panel assembly.
14.) Invert resistor pan assembly without damaging electrical connections.
15.) Note and record position of three (3) wires (Red, Pur and Wht) attached to SCR.
16.) Note and record position of two (2) wires (Blk and Wht) attached to SCR.
17.) Note and record position of SCR terminals.
18.) Remove leads identified in steps 15 and 16.
19.) Remove two (2) 4 mm X 10 mm pan head screws which mount the SCR.
20.) Clean SCR mounting surface on resistor mounting panel, using a dry tissue to remove thermal compound.
21.) Prepare new SCR (46-2185251) by coating mounting surface with thermal compound (46-170212P1).
22.) Mount SCR in position recorded in step 17. Attach with two (2) 4 mm X 10 mm pan head screws. Torque each screw to 1.25 ft/lbs.(1.7 Newton Meters).
23.) Replace leads removed in steps 15 and 16.
24.) Reposition HEMRC resistor pan assembly and secure with five (5) M4 x 0.07 nuts. Torque to 1.25 ft/lbs.(1.7 Newton Meters).
25.) Connect cable to Connector J7 on the HEMRC Interface board.
26.) Replace cable ties (46-208758 P2) as required.
27.) Replace HEMRC cover and secure with five (5) mounting bolts.
28.) Disengage gantry rotational lock.
29.) Close front cover.
30.) Reinstall gantry scan window.
31.) Close Top Cover.
32.) Turn On all three (3) switches on the status control box.
33.) Replace both gantry side covers.
34.) Remove “Tag Out” lock and turn on facility power.
35.) Proceed with system test.

Section 56.0
46-2115199 HEMRC Module Replacement

Located on the HEMRC mounting plate

1.) Position table to lowest elevation.
2.) Shut down system and turn off facility power to PDU

⚠ **DANGER** **USE TAG AND LOCK OUT PROCEDURE.**

3.) (Use tag and lockout procedure).
4.) Remove both gantry side covers.
5.) Turn off all three (3) switches on the status control box (right side of gantry).
6.) Open top cover and engage “Prop Rod”.

7.) Remove gantry scan window.
8.) Open front cover.
9.) Rotate gantry until the HEMRC is in a convenient working position.
10.) Engage gantry rotational lock.
11.) Remove five (5) bolts fastening cover to HEMRC assembly and remove cover.
12.) Disconnect cable from Connector J7 on the HEMRC Interface board.
13.) Remove five (5) M4 x 0.07 nuts securing HEMRC resistor panel assembly.
14.) Clip and remove cable ties as required to move resistor panel assembly.
15.) Invert resistor pan assembly without damaging electrical connections.
16.) Note and record position of seven (7) wires on HEMRC, TB3 (Grn, Red, Blk, Pnk, Pnk, Brn and Yel)
17.) Note and record position of six (6) wires on HEMRC, TB2 (Pur, Blk, Blk, Blk,Yel and Orn)
18.) Note and record position of nine (9) wires on HEMRC, Tbx (Yel, Brn, Orn, Orn, Pur, Pur, Blk, Wht and Grn).
19.) Mark and remove leads identified in steps 16, 17 and 18.
20.) Remove four (4) M4 x 0.7 metric hex nuts which secure the HEMRC to the mounting plate.
21.) Remove and replace HEMRC module (46-2115199).
22.) Attach four (4) M4 x 0.7 metric hex nuts securing the HEMRC to the mounting plate. Torque each nut to 1.25 ft/lbs.(1.7 Newton Meters).
23.) Replace leads removed in step 19.
24.) Reposition HEMRC resistor pan assembly and secure with five (5) M4 x 0.07 nuts. Torque to 1.25 ft/lbs.(1.7 Newton Meters).
25.) Connect cable to Connector J7 on the HEMRC Interface board.
26.) Replace cable ties (46-208758 P2) as required
27.) Replace HEMRC cover and secure with five (5) mounting bolts.
28.) Disengage gantry rotational lock.
29.) Close front cover.
30.) Reinstall gantry scan window.
31.) Close top cover.
32.) Turn On all three (3) switches on the status control box.
33.) Replace both gantry side covers.
34.) Remove “Tag Out” lock and turn on facility power.
35.) Proceed with system test.

Section 57.0
46-2147062 Bridge Rectifier

Located on the HEMRC mounting plate

1.) Position table to lowest elevation.
2.) Shut down system and turn off facility power to PDU

DANGER **USE TAG AND LOCK OUT PROCEDURE.**

3.) Remove both gantry side covers.
4.) Turn off all three (3) switches on status control box (right side of gantry).

5.) Open top cover and engage “Prop Rod”.
6.) Remove gantry scan window.
7.) Open front cover.
8.) Rotate gantry until the HEMRC is in a convenient working position.
9.) Engage gantry rotational lock.
10.) Remove five (5) bolts fastening cover to HEMRC assembly and remove cover.
11.) Note and record position of bridge rectifier terminals and polarity mark.
12.) Note and record the position of four (4) leads connected to bridge rectifier. (Red, Blk, Pur and Orn)
13.) Remove the four leads from bridge rectifier.
14.) Remove M4 x 0.7 hex nut holding the bridge rectifier to the HEMRC mounting plate.
15.) Remove bridge rectifier.
16.) Clean bridge rectifier mounting surface on mounting plate, using a dry tissue to remove thermal compound.
17.) Prepare new bridge rectifier (46-2147062), by coating mounting surface with thermal compound (46-170212 P1).
18.) Mount bridge rectifier in position recorded in step 11. Attach with M4 x 0.7 hex nut. Torque nut to 1.25 ft/lbs.(1.7 Newton Meters).
19.) Replace leads removed in step 13 as noted in step 12.
20.) Replace HEMRC cover and secure with five (5) mounting bolts.
21.) Disengage gantry rotational lock.
22.) Close front cover.
23.) Reinstall gantry scan window.
24.) Close top cover.
25.) Turn On all three (3) switches on the status control box.
26.) Replace both gantry side covers.
27.) Remove “Tag Out” lock and turn on facility power.
28.) Proceed with system test.

Section 58.0
46-2168593 HEMRC Filter Board

Located on the HEMRC mounting plate.

1.) Position table to lowest elevation.
2.) Shut down system and turn off facility power to PDU.

⚠ **DANGER** **USE TAG AND LOCK OUT PROCEDURE.**

3.) Remove both gantry side covers.
4.) Turn off all three (3) switches on status control box (right side of gantry).
5.) Open top cover and engage “Prop Rod”.
6.) Remove gantry scan window.
7.) Open front cover.
8.) Rotate gantry until the HEMRC is in a convenient working position.
9.) Engage gantry rotational lock.

10.) Remove five (5) bolts fastening cover to HEMRC assembly and remove cover.
11.) Note and record position of cable shield lead fastened to mounting screw.
12.) Note and record the position of two (2) leads from filter board, attached to HEMRC AC Drive TB1. (Orn and Pur)
13.) Remove the two (2) leads from TB1.
14.) Remove four (4), 4 mm x 10 mm pan screws which mount the HEMRC Filter Board to mounting plate.
15.) Mount HEMRC filter board (46-2168593) using four (4), 4 mm x 10 mm pan head screws. Replace cable shield connection noted in step 11. Torque screws to 1.25 ft/lbs.(1.7 Newton Meters).
16.) Replace leads removed in steps 13 as noted in step 12.
17.) Replace HEMRC cover and secure with five (5) mounting bolts.
18.) Disengage gantry rotational lock.
19.) Close front cover.
20.) Reinstall gantry scan window.
21.) Close Top Cover.
22.) Turn On all three (3) switches on the status control box.
23.) Replace both gantry side covers.
24.) Remove “Tag Out” lock and turn on facility power.
25.) Proceed with system test.

Section 59.0
KV Related Problems

59.1 KV Troubleshooting Theory

59.1.1 Reported kV vs. Actual Tube kV

The KV (and mA) reported to the software, (which includes the reported KV to the console), does not have to be what is actually across the X-Ray Tube. The KV reported to the software does not have to be what is seen by the bleeder.

Reported KV comes from the Anode and Cathode KV Test points on the KV Control Bd. Because of KV closed loop regulation, this test point (on a normally operating scanner) should never be different from "SELECTED VALUE" (+/- 2.999%). This is the node that the loop uses to regulate. Because the gain of the electronic monitoring devices between the x-ray tube and this test point may not be 1, the KV reported here **IS NOT THE ACTUAL KV SEEN ACROSS THE X-RAY TUBE!** It is what the system **THINKS** is the actual tube voltage. The purpose of kv gain adjustment is to get a gain of one between x-ray tube and TP10.

Note: It is uncommon, but possible to get the kv gain pots out of adjustment as much as +/- 15kv.

The purpose of the KV Feedback Gain Pot is to ensure a gain of one in the feedback circuit. A gain of one will ensure that the voltage across the x-ray tube (and bleeder) is what gets reported to the KV Feedback Test Point. The closed loop regulates to these test points, if these test points are wrong the system will change inverter current to compensate for the wrong KV.

Improperly adjusted KV Gain Pots can result in the Kv being off as much as +/-30KV total from what the system (software) thinks is across the tube. This means that the system is cooling for 120KV and actual KV across the tube can be as high as 150KV. Tube life would be very low with this scenario.

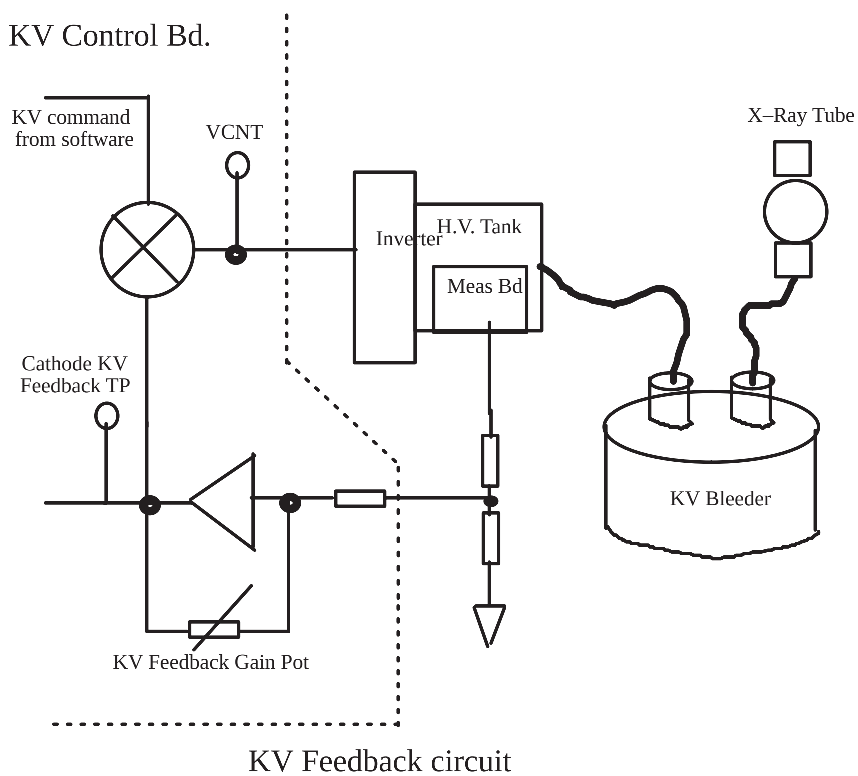


Figure 9-31 Why Reported KV may not be the Actual KV Across the Tube

EXAMPLE: KV Feedback Gain Pot is adjusted for a gain of 0.90. A KV command of 100 KV is received (50KV anode, 50KV cathode). With 50KV across KV Bleeder (as read with the scope), and a gain of 0.90 the KV Test Point will only see 45KV. The error mux. will command a higher inverter current until KV Test Point is 50KV. HOWEVER the KV across the bleeder (x-ray tube) is really 55.5KV.

Tweaking the KV Gain Pot for a gain closer to one will cause the error mux to reduce the inverter current, therefore compensating for the KV Test Point. The KV Gain Pots are adjusted correctly when the KV across the bleeder is the same as the KV Test Points.

59.1.2 KV Gain Pot Adjustment

59.1.2.1 Purpose of This Information

To reinforce that the kv feedback test points DO NOT reflect actual kv across the tube. The purpose of the kv gain pot is to ensure that there is a gain of 1 (one) between the tube kv measurement (meas bd) and tube kv reporting (kv test points.)

The reason that kv test points (on a normally operating system) will never be different than commanded, is that this is the node the closed loop uses to regulate the kv. If this test point is wrong the system will change inverter current to compensate. It takes milliseconds to do this, therefore it looks like these test points never change.

Another reason for this documentation is to emphasize how far off the kv gain pots can be adjusted and the system still think that the kv across the tube is what the cooling algorithm thinks it is.

59.1.2.2 Definitions

“Turns cw” The kv gain pot was turned fully ccw, then turned clockwise one turn at a time.

“bleeder” KV bleeder installed in system. This is actual kv across the tube.

“kvan” “kvca” anode and cathode test points on the kv control board.

NOTE that one turn cw (from fully ccw) will bring the gain closer to one, resulting in the bleeder voltage come up closer to the test point. This is true up until 15 turns when the gain is less than one. Now the actual kv across the tube is GREATER THAN the test points (measurement gain less than one).

59.1.2.3 Summary

A properly adjusted kv gain pot should be in the neighborhood of about 15 turns.

Example: KV FEEDBACK POT Values

ANODE FULLY CCW (starting pt)			CATHODE FULLY CCW (starting pt)		
TURNS CW	**BLEEDER**	**KVAN**	**TURNS CW**	**BLEEDER**	**KVCA**
2	44.854	5.9823	2	45.036	6.0004
3	45.751	5.9095	3	45.438	6.0192
4	47.008	5.9711	4	46.473	6.0306
5	48.266	5.9639	5	47.47	6.0232
6	49.543	5.9631	6	48.576	6.0274
7	50.614	5.9601	7	49.731	6.0244
8	51.613	5.9493	8	50.84	6.0235

Table 9-12 Typical KV Feedback Pot Values

ANODE FULLY CCW (starting pt)			CATHODE FULLY CCW (starting pt)		
TURNS CW	BLEEDER	KVAN	TURNS CW	BLEEDER	KVCA
9	52.615	5.9554	9	51.835	6.0238
10	53.705	5.9445	10	52.749	6.0327
11	54.883	5.9449	11	54.065	6.0235
12	56.103	5.9442	12	55.337	6.0241
13	57.32	5.9361	13	56.49	6.0225
14	58.315	5.9324	14	57.861	6.0207
15	59.532	5.931	15	58.917	6.0309
16	60.527	5.9238	16	60.06	6.0192
17	60.763	5.9081	17	61.354	6.0263
18	62.041	5.9359	18	62.695	6.0324
19	63.041	5.9328	19	63.68	6.0213
20	64.108	5.9361	20	65.114	6.0253
21	65.136	5.9479	21	66.429	6.0213
22	66.118	5.955	22	67.334	6.0327
23	67.122	5.9636	23	68.731	6.0293
24	68.134	5.9706	24	69.827	6.0303
25	69.164	5.986	25	70.917	6.0238
26	70.081	5.9981	26	71.974	6.0275
27	71.157	5.9385	27	73.147	6.0297
28	72.092	5.9854	28	74.256	6.0235
29	73.171	5.9808	29	74.961	6.0241
30	74.155	5.9801	30	75.041	6.0244

Table 9-12 Typical KV Feedback Pot Values (Continued)

59.1.3 SW & HW Tools Available for Troubleshooting

59.1.3.1 Diagnostics

KV & mA (X-Ray) Results Screen on the Trouble Shoot menu is the ****Primary tool** for KV related problems other than Overcurrents or Shoot-through. Overcurrents or Shoot-through will terminate scans, resulting in no data collection.

OBC BLDs

59.1.3.2 Schematics

DIRECTION 46-018303

- KV Control Bd. (Newer Style Bd.) Schematics 46-321198-S
- KV Inverter Gate Driver Bd. Schematics 46-264662-S
- KV Inverter Capacitor Bd. and Schematics 46-264664-S
- OBC Backplane list
- Gantry Rotating Member Block Diagram

- Gantry Rotating Interconnect
- X-Ray Tube 120vac

59.1.3.3 Equipment Needed

- Direction 46-018318 CT HiSpeed Advanced Functional Interconnect.
- Bleeder
- Bleeder /oscilloscope combination can cause aliasing with the bleeder KV signal, resulting in KV ripple as high as 20KV.
- Multi-meter
- oscilloscope

59.1.4 Explanation of kV/MA Results Screen

This is an example of a results screen from a 120kv, 200mA, 10 second scan.

(This example is from a "**newer style**" (46-321198 KV Control bd.) (with 550vdc DCRGS)

High voltage status				
No.	**Device**	**Average Value**	**Selected Value**	**Last Sample**
1.	Total KV:	119.4 KV	120.0 KV	119.4 KV
2.	Cathode KV:	59.7 KV	60.0 KV	59.7 KV
3.	Anode KV:	60.1 KV	60.0 KV	60.1 KV
4.	Cathode MA:	193.7 mA	200 mA	193.7 mA
5.	Anode MA:	193.7 mA	200 mA	193.7 mA
6.	Cathode inverter current:	30.7 A	–	30.7 A
7.	Anode inverter current:	30.7 A	–	30.7 A
8.	Approx. KV inverter frequency(VCNT):		(1.6V)	xx.x KHz
9.	Cathode inverter duty cycle:	100%	–	100%
10.	Anode inverter duty cycle:	83%	–	83%
11.	Rail voltage:	540 V	550 V	540 V
12.	Exposure duration:	–	10000 mS	10001 mA
13.	Exposure number:	–	1	1

Table 9-13 Kv/mA Results Screen - 46-321198 KV Control bd.

59.1.4.1 Header Explanation

No. Device	Average Value	Selected Value	Last Sample

AVERAGE VALUE: is the average taken over the duration of the scan (see EXPOSURE DURATION).

SELECTED VALUE: is the value prescribed by the user, or the value required to perform the scan.

LAST SAMPLE: is the last value read during an ACTIVE exposure.

Total KV Explanation:	**119.4KV**	**120.0KV**	**119.4KV**

On the 46-321198G1-F board this signal comes from TP11. It is an opamp sum of Anode KV (TP9) and Cathode KV (TP10). Because of KV closed loop regulation, this test point (on a normally operating scanner) should never be different from “SELECTED VALUE” (+/- 2.999%). **DO NOT TROUBLESHOOT “Total KV” low** (or high). Instead troubleshoot either the anode or the cathode being low (or high), they are the inputs to this value.

“Total KV” gets reported to the software through the Gentry I/O and OBC Backplane.

Cathode KV Explanation:	**59.7KV**	**60.0KV**	**59.7KV**

On the 46-321198G1-F board this signal comes from TP10. Because of KV closed loop regulation, this test point (on a normally operating scanner) should never be different from “SELECTED VALUE” (+/- 2.999%). This is the node that the loop uses to regulate. Because the gain of the electronic monitoring devices between the x-ray tube and this test point may not be 1, the KV reported here **IS NOT THE ACTUAL KV SEEN ACROSS THE X-RAY TUBE!** It is what the system **THINKS** is the actual tube voltage. The purpose of kv gain adjustment is to get a gain of one between x-ray tube and TP10.

Note: It is uncommon, but possible to get the kv gain pots out of adjustment as much as +/- 15kv.

Inverter current is commanded by (VCNT). Compare these three readings (cathode KV, Cathode inverter current and (VCNT)) and troubleshoot. Nominal values are attached.

Anode KV Explanation	**60.1KV**	**60.0KV**	**60.1KV**

On the 46-321198G1-F board this signal comes from TP9. Because of KV closed loop regulation, this test point (on a normally operating scanner) should never be different from “SELECTED VALUE” (+/- 2.999%). This is the node that the loop uses to regulate. Because the gain of the electronic monitoring devices between the x-ray tube and this test point may not be 1, the KV reported here **IS NOT THE ACTUAL KV SEEN ACROSS THE X-RAY TUBE!** It is what the system **THINKS** is the actual tube voltage. The purpose of kv gain adjustment is to get a gain of one between x-ray tube and TP9 (NOTE: It is uncommon, but possible to get the kv gain pots out of adjustment as much as +/- 15kv).

Inverter current is commanded by (VCNT). Compare these three readings (Anode KV, Anode inverter current and (VCNT)) and troubleshoot. Nominal values are attached.

Cathode MA Explanation:	**193.7mA**	**200.0mA**	**193.7mA**

This value comes from the mA Control Bd. 46-288886 TP4, thru the backplane and Gentry I/O bd. Since the cathode is in series with the anode, TP4 should be the same value as the anode mA. The scale is 1v/100mA.

In closed loop mode TP4 should be commanded mA. In open loop mode the value should be less (whatever is in GenCalSeed). TP4 is actually the cathode high voltage tank secondary amperage, the x-ray tube is the load for the secondary. MA Meter Verification verifies that the measurement electronics have a gain of one and that reported mA is actual ma.

Cathode high voltage tank secondary amperage (and x-ray tube mA) is the direct result of filament heating, for improper mA include filament function while troubleshooting. If mA is out of tolerance (3%), check KV values for large errors, verify mA metering, and run the Filament Functional test. BLDs can help also.

If cathode and anode mA are different, suspect mA measurement electronics (use mA Meter Test), or suspect a shattered x-ray tube insert shorting out the filament (cathode) or the anode.

Anode MA Explanation	**193.7mA**	**200.0mA**	**193.7mA**

This value comes from the mA Control Bd. 46-288886 TP10, thru the backplane and Gentry I/O bd. Since the anode is in series with the cathode, TP10 should be the same value as the cathode mA. The scale is 1v/100mA.

In closed loop mode TP10 should be commanded mA. In open loop mode the value should be less (whatever is in GenCalSeed). TP10 is actually the anode high voltage tank secondary amperage, the x-ray tube is the load for the secondary. MA Meter Verification verifies that the measurement electronics have a gain of one and that reported mA is actual ma.

Anode high voltage tank secondary amperage (and x-ray tube mA) is the direct result of filament heating, for improper mA include filament function while troubleshooting. If mA is out of tolerance (3%), check KV values for large errors, verify mA metering, and run the Filament Functional test. BLDs can help also.

Cathode Inverter Current Explanation: 30.7A – 30.7A

This value comes from the KV Control Bd. 46-321198G1-F TP21, thru the backplane and Gentry I/O bd. The scaling is 25A/volt. Values over 8 amps will result in an overcurrent error. Refer to direction 46-018318 Function Interconnect Drawings (little yellow/orange spiral bound notebook, rev 4) page 32, KV LOOP FUNCTIONAL INTERCONNECT. Locate the “OVERCURRENT” toroid/transformer. This toroid monitors the current leaving the inverter and going to the tank primary.

Anode Inverter Current Explanation: 30.7A – 30.7A

This value comes from the KV Control Bd. 46-321198G1-F TP20, thru the backplane and Gentry I/O bd. The scaling is 25A/volt. Values over 8 amps will result in an overcurrent error. Refer to direction 46-018318 Function Interconnect Drawings (little yellow/orange spiral bound notebook, rev 4) page 32, KV LOOP FUNCTIONAL INTERCONNECT. Locate the “OVERCURRENT” toroid/transformer. This toroid monitors the current leaving the inverter and going to the tank primary.

Approx. KV Inverter Frequency (VCNT) Explanation: (1.60V) xx.xKHz

This value comes from the KV Control Bd. 46-321198G1-F TP24, thru the backplane and Gentry I/O bd. This is the input voltage to the voltage controlled oscillator. the operating range is from 0-5v which will give a frequency range of 19.5khz to 31.5khz.

A (VCNT) of 0.2v is a command for a lower frequency, a lower frequency will allow more current through the primary resulting in more KV output. To summarize, a (VCNT) 0f 0.2v is max current command, should have max inverter current, should have max KV.

A (VCNT) of 5v (or more) is a command for a higher frequency, a higher frequency will allow less current through the primary resulting in less KV output. To summarize, a (VCNT) 0f 4.99v is min. current command, should have min. inverter current, should have min. KV.

(VCNT) is an composite signal generated from the difference between kv command and kv feedback. This error signal is also an input into (VCNT).

Compare these three readings (cathode KV, Cathode inverter current and (VCNT)) and troubleshoot. Nominal values are attached.

Cathode inverter duty cycle Explanation: 100% – 100%

This value comes from the KV Control Bd. 46-321198G1-F TP23, thru the backplane and Gentry I/O bd. The system uses duty cycle to regulate at the lower mAs more than it uses frequency. At the higher mAs the system uses frequency to regulate more than it uses duty cycle. Compare the cathode duty cycle to the anode duty cycle.

Anode Inverter Duty Cycle Explanation: 83% – 83%

This value comes from the KV Control Bd. 46-321198G1-F TP22, thru the backplane and Gentry I/O bd. The system uses duty cycle to regulate at the lower mAs more than it uses frequency. At the higher mAs the system uses frequency to regulate more than it uses duty cycle. Compare the cathode duty cycle to the anode duty cycle.

Note: The following statement is **only true** for the older KV board 46-321064G1 and inverters tuned to 19.1 Khz and 18.6 Khz. The anode duty cycle should never reach 100% and rarely gets past 95%.

At 95% and at a max (VCNT) command, the system is out of energy, therefore you should only see these percentages at 140kv, 340ma. When the system is out of energy, the kv will start caving in. Also at mAs higher than 100ma, the anode duty cycle should never exceed the cathode duty cycle.

IF THIS SCENARIO HAPPENS the system is running out of energy. Most likely due to an IGBT not firing.

FOR KV BOARDS OTHER THAN 46-321064G1: The duty cycle can achieve 100% on either the cathode or anode inverter. This should be considered normal operation for the new inverters.

Rail Voltage Explanation:	**540V**	**550V**	**540V**

This value comes from the CTVRC Control Bd. 46-288858 TP12, through the backplane and Gentry I/O bd.

Exposure Duration, Number, and Status Register Explanation

Exposure duration:	–	10000mS	10001mS
Exposure number:	–	1	1
Status register (Address = FFCFF9H):		–	8FH

This is an example of a results screen from a 120kv, 200mA, 10 sec. scan.

(This example is from a “**older style**” (46-321064) KV Control bd.) (with 550vdc DCRGS)

High voltage status				
No.	**Device**	**Average Value**	**Selected Value**	**Last Sample**
1.	Total KV:	119.4KV	120.0KV	119.4KV
2.	Cathode KV:	59.7KV	60.0KV	59.7KV
3.	Anode KV:	60.1KV	60.0KV	60.1KV
4.	Cathode MA:	193.7mA	200mA	193.7mA
5.	Anode MA:	193.7mA	200mA	193.7mA
6.	Cathode inverter current:	30.7A	–	30.7A
7.	Anode inverter current:	30.7A	–	30.7A
8.	KVVCO input	(Pre-exposure	value	x.xxv):1.60V
11.	Rail voltage:	540V	550V	540V
12.	Exposure duration:	–	10000mS	10001mA
13.	Exposure number:	–	1	1
14.	Status register (Address=FFCFF9H):	–	xFH	

Table 9-14 Kv/mA Results Screen - 46-321064 Kv Control Bd.

59.1.4.2 Header Explanation

See previous section

Total KV Explanation

On the 46-321064 board this signal comes from TP30 (KVTB). It is an op-amp sum of Anode KV (TP41) and Cathode KV (TP37). For further explanation refer to the previous section

Cathode KV Explanation

On the 46-321064 board this signal comes from TP37 (CAKV). For further explanation refer to the previous section.

Anode KV Explanation

On the 46-321064 board this signal comes from TP41 (ANKV). For further explanation refer to the previous section.

Cathode mA Explanation

Refer to the previous section.

Anode mA Explanation

Refer to previous section

Cathode Inverter Current Explanation

On the 46-321064 board this signal comes from TP44 (CAOC). For further explanation refer to the previous section.

Anode Inverter Current Explanation

On the 46-321064 board this signal comes from TP45 (ANOC). For further explanation refer to the previous section

KV VCO input Explanation

On the 46-321064 board this signal comes from TP15 (VCNT). For further explanation refer to the previous section.

THE OLDER STYLE KV BD (46-321064) DOES NOT REPORT DUTY CYCLE TO THE SOFTWARE

Filament Current, Rail Voltage, Exposure Duration and Exposure Number, refer to the previous section.

59.1.5 Tube Spit Explanation

HOW DOES THE SYSTEM DETERMINE WHEN A TUBE SPIT HAPPENS?

The answer: when a fast fall time is detected on the kv wave shape.

If a tubes spits most of the time, when does CTi stop scanning because of tube spits? The answer is, “when spits affect image quality or cause equipment damage.”

The KV Control Bd. monitors the KV via the KV feedback test points. A tube spit will cause KV to drop at a very fast rate. An integrator circuit used on the KV Control Bd. monitors the KV feedback, and whenever this integrator detects a fast fall time (an integrator has little impedance to fast frequencies), it considers the event as a tube spit. The KV Control Bd. will then turn off x-rays for approximately 100ms to allow the X-Ray Tube to recover.

WHY IS SCANNING STOPPED AFTER 32 SPITS HAVE BEEN DETECTED?

The answer is to preserve image quality.Overcurrents and Shoot-through protection is used to protect the system against damage. Spit detection is not used to protect the system against damage.

It has been determined that more than 32 spits per scan second may affect image quality, therefore the system will stop scanning when the system has counted more than 32 spits per scan second.

On an older style Kv Control Bd. (46-321064), scanning will abort when 32 spits occur in a rolling 1 second window, for a net rate of 27.1 to 40.7 spits/sec.

On the newer style KV Control Bd. (46-321198), scanning will abort when 9 spits occur in a rolling 0.26 second window for a net rate of 34.3 spits/sec.

59.1.6 Bleeder Ripple / Oscilloscope Aliasing

The current bleeder in use today was designed for line frequency type generators and has a band width of 720hz. The CT/i high voltage subsystem operates at approximately 19.khz to 31khz. The problem is, if you connect a scope probe directly to the bleeder, it has a tendency to amplify high frequency ripple. The bleeder requires the cable capacitance to roll off the high end properly. Also, there are actually a couple different bleeders out there under the same model #. They were never intended to deal with signals above 720 Hz (for line frequency machines). As a result the high frequency response is unreliable.

This is not to say avoid using the present bleeder for measuring high voltage. The Bleeder is what GEMS uses for measuring high volt. HOWEVER when using the bleeder it is very difficult deter mine what is real ripple and what is aliasing.

Refer to the wave shape in *Section 60.0* ***starting on*** *page 633* for examples of normal bleeder wave shapes.

59.2 Troubleshooting KV Related Problems

59.2.1 "Total KV" low (or high)

DO NOT TROUBLESHOOT "Total KV" low (or high). Instead troubleshoot either the anode or the cathode being low (or high), they are the inputs to this value.

59.2.2 Determining which style KV Control Bd. is in the system.

The easiest way to tell is from the diagnostic results screen. The older style KV Control Bd. DOES NOT have a duty cycle field. Where to start for low cathode or anode KV problems

(NEWER STYLE KV CONTROL BD. 46-321198)

Determine which style KV Control Bd. is in the system. The easiest way to tell is from the diagnostic results screen. The older style KV Control Bd. DOES NOT have a duty cycle field.

The primary tool to use for low kv is the results screen from diags/ X-Ray Generation/ KV & mA (X-Ray). There is a sample of this screen on page 624 for the newer style KV Control Bd. (46-321198). For the older style KV Control Bd. (46-321064), there is a sample of this screen on page 627.

Do not troubleshoot low kV problems at techniques lower than 120KV, 200mA. The system is capable of delivering 48KW. An inverter IGBT not firing is the major reason for low KV. With only one side of the inverters delivering power, the system can scan at techniques lower than 24KW (half power) without any indications or errors. At lower techniques the system can maintain KV and mA all day long without any errors or image quality problems. However, when techniques higher than 24KW are requested, the system cannot deliver the power, therefore the kv starts caving in.

Compare KV, Inverter current and VCNT then troubleshoot. Nominal values are in section 10 of this service note.

VCNT is a low voltage, KV is low, inverter current is low and VCNT is a low value (more inverter current command), suspect an IGBT not firing.

CHECK: Fiber Optic Sequencing (diags/ X-Ray Generation/ KV Loop/ H.V. Fiber optics).

If LEDs do not light on the Inverter Gate Driver BD., check the vdc test points on Gate Bd., check for command from KV Control Bd. via looking for red light at the end of the fiber optic cable.

CHECK: IGBT Resistance.

CHECK: Inverter fuse (most true if KV is 0).

CHECK: Frequency sweep the tanks (low probability as FRU at fault).

When dealing with low KV related problems, the MOST LIKELY FRUs at fault are:

- KV Control BD. Inverter

- Tank X-Ray Tube
- Fiber Optic Sequencing Inverter fuse

The Inverter fuse can be eliminated fast with an ohm or voltage check.

The KV Control Bd., Inverter and Tank can be isolated by comparing the result screen values.

Note: Use KV & mA (X-Ray) results screen to trouble shot.

Note: A frequency sweep of the tanks is recommended

Fiber Optic sequencing can be checked with a test.

X-Ray Tubes cause overcurrents/shoot-thrus more often than low KV problems.

59.2.3 Where to start for high cathode or anode KV problems

(NEWER STYLE KV CONTROL BD. 46-321198)

The primary tool to use for high kv is the results screen from diags/ X-Ray Generation/ KV & mA (X-Ray). There is a sample of this screen in previous sections for the newer style KV Control Bd. (46-321198). For the older style KV Control Bd. (46-321064), there is a sample of this screen in previous sections.

On the newer style KV Control Bd. 46-321198, the anode or cathode tries to make up for a low KV on the “other” side. For example if a cathode inverter IGBT was not firing, then at higher techniques the anode would command a higher inverter current, resulting in a higher anode KV. The total kv would the be whatever commanded KV was. Commanded Kv was 120Kv 300 mA and cathode was only 40kv, then the anode would try to command 80kv.

To summarize, on the newer style KV Control Bd. a high kv reading is usually the result of that side trying to make up for the “broken” side. Trouble shoot the other side having a low kv.

A system with high Total KV is usually the fault of the KV Control Bd.

59.2.4 Where to start for low cathode or anode KV problems

(OLDER STYLE KV CONTROL BD. 46-321064)

Determine which style KV Control Bd. is in the system. The easiest way to tell is from the diagnostic results screen. The older style KV Control Bd. DOES NOT have a duty cycle field

The primary tool to use for low kv is the results screen from diags/ X-Ray Generation/ KV & mA (X-Ray). There is a sample of this screen in previous section for the newer style KV Control Bd. (46-321198). For the older style KV Control Bd. (46-321064), there is a sample of this screen in previous sections

Do not troubleshoot low Kv problems at techniques lower than 120KV, 200mA. The system is capable of delivering 48KW. An inverter IGBT not firing is the major reason for low KV. With only one side of the inverters delivering power, the system can scan at techniques lower than 24KW (half power) without any indications or errors. At lower techniques the system can maintain KV and mA all day long without any errors or image quality problems. However, when techniques higher than 24KW are requested, the system cannot deliver the power, therefore the kv starts caving in.

Compare KV, Inverter current and VCNT then troubleshoot. Nominal values are in a later section.

VCNT is a low voltage, KV is low, inverter current is low and VCNT is a low value (more inverter current command), suspect an IGBT not firing.

If LEDs do not light on the Inverter Gate Driver BD., check the vdc test points on Gate Bd., check for command from KV Control Bd. via looking for red light at the end of the fiber optic cable.

CHECK: Inverter fuse (most true if KV is 0).

When dealing with **low KV related problems**, the **MOST LIKELY FRUs** at fault are:

- KV Control BD

- Inverter
- Inverter fuse
- Tank
- X-Ray Tube

The Inverter fuse can be eliminated fast with an ohm or voltage check.

The KV Control Bd., Inverter and Tank can be isolated by comparing the result screen values.

Note: Use KV & mA (X-Ray)) results screen to trouble shot.

X-Ray Tubes cause overcurrents/shoot-thrus more often than low KV problems.

59.2.5 Where to start for high cathode or anode KV problems

(OLDER STYLE KV CONTROL BD. 46-321064)

The primary tool to use for high kv is the results screen from diags/ X-Ray Generation/ KV & mA (X-Ray). There is a sample of this screen in previous sections

On the older style KV Control Bd. 46-321064, the anode follows the cathode. For example if a cathode inverter IGBT was not firing, then KV VCO would go low commanding a higher cathode inverter current, since the anode follows the cathode the anode inverter current would go to max also, resulting in a max KV on the anode side.

The readings from the results screen may be a low cathode KV with a high anode KV, in this case the problem is not a high anode kv, the problem would be low cathode KV.

If it were the anode inverter IGBT not firing, then the anode side would stay low and the cathode would not try to make up for difference.

A system with high Total KV or high cathode KV is usually the fault of the KV Control Bd.

59.2.6 Overcurrents sense the current leaving inverters & going in tanks

Where to start: The primary tool is KV & mA (X-Ray) results

This screen is the place to start, compare KV, Inverter Current and VCNT.

HOWEVER a scanner error (Overcurrent) can terminate scanning before any useful information can be collected, this will make KV & mA (X-Ray) results screen useless. What can be done is to reduce the KV and mA technique as low as possible (60KV, 10mA is the lowest) until the system will work long enough to give some useful information.

When dealing with overcurrents, the MOST LIKELY FRUs at fault are:

- X-Ray Tubes
- Tank
- HV Cables
- KV Control BD.
- Fiber Optic Sequencing

Note: Use very low values in KV & mA (X-Ray)

The KV Control Bd., Inverter and Tank can be isolated by comparing the result screen values.

The KV Control Bd. can cause overcurrents and shoot-thrus, but seldom do.

59.2.7 Shoot-Through Senses Current Entering & Leaving Inverters

Where to start: The primary tool is X-Ray Generation/ KV & mA (X-Ray) results.

This screen is the place to start, compare KV, Inverter Current and VCNT.

HOWEVER a scanner error (Shoot-Thru) can terminate scanning before any useful information can be collected, this will make KV & mA (X-Ray) results screen useless. What can be done is to reduce the KV and mA technique as low as possible (60KV, 10mA is the lowest) until the system will work long enough to give some useful information.

When dealing with Shoot-Thrus, the MOST LIKELY FRUs at fault are:

- X-Ray Tubes
- Inverter
- Fiber Optic Sequencing
- KV Control BD.

Note: Use very low values in KV & mA (X-Ray)

The KV Control Bd., Inverter and Tank can be isolated by comparing the result screen values.

The KV Control Bd. can cause overcurrents and shoot-thrus, but seldom do.

59.2.8 Tube Spit Troubleshooting

Tube spit rate can be seen by using viewStats. This is what you'll see in viewStats.

```
Fri Apr 28 1995
481 spits in 684 scan seconds :: 7.032164 per 10 scan seconds
```

`481 spits in 684 scan seconds::` can be used for trending.

`x.xxxx per 10 scan seconds:` is an indication of tube condition right now.

As spits per 10 seconds approaches 1, there is a good chance that the tube can be "saved" with a heat soak and "Install New Tube".

When the spit rate reaches about 9 to 12 spits per 10 scan seconds, it is about a 50-50 chance that the tube can be "saved".

Above 9 to 12 spits per 10 scan seconds, it is worth a try, but good luck.

Section 60.0
Kv Reference Material

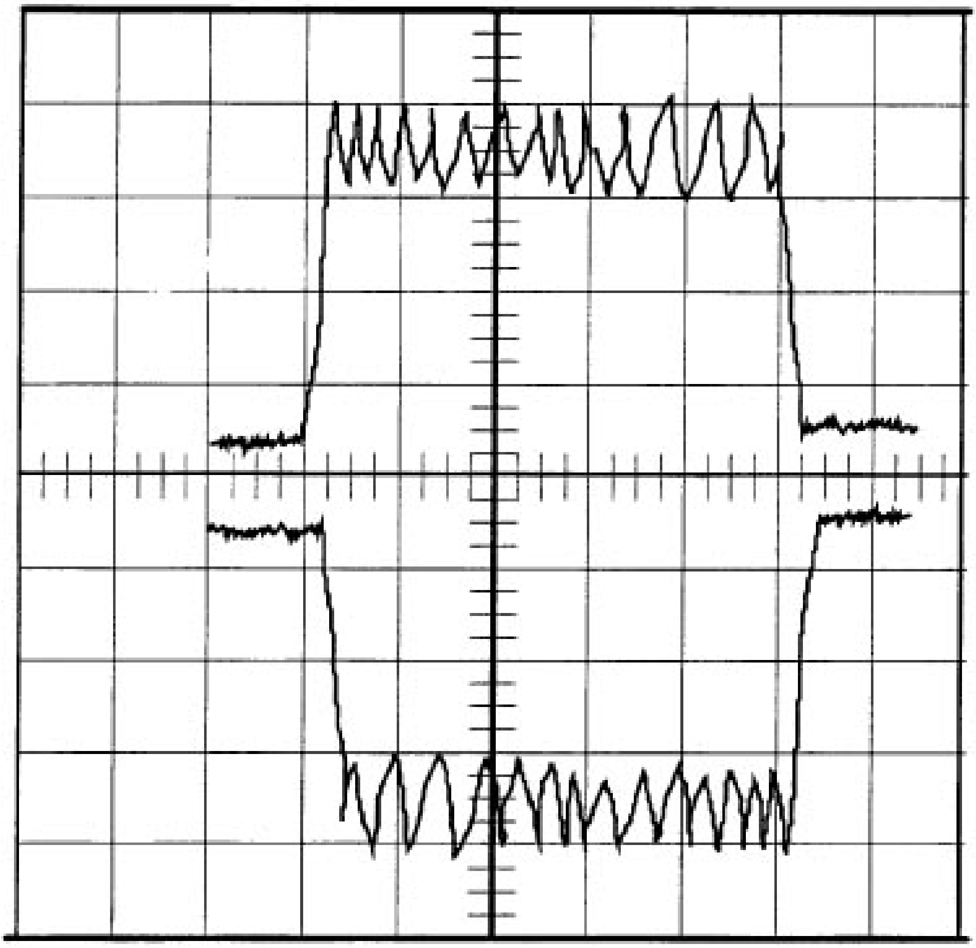


Figure 9-32 KV Ripple @ 0.2 second scope trace

```
A:20V=0.2S B:20V=0.2S
140KV 340ma Bleeder (slow sweep)
This is a normal picture.
```

This is a good example of the scope aliasing the inverter ripple. NOTE: that the ripple can be as high as 20KV per side. Although aliasing will indicate something at higher frequencies, it is not a true waveform.

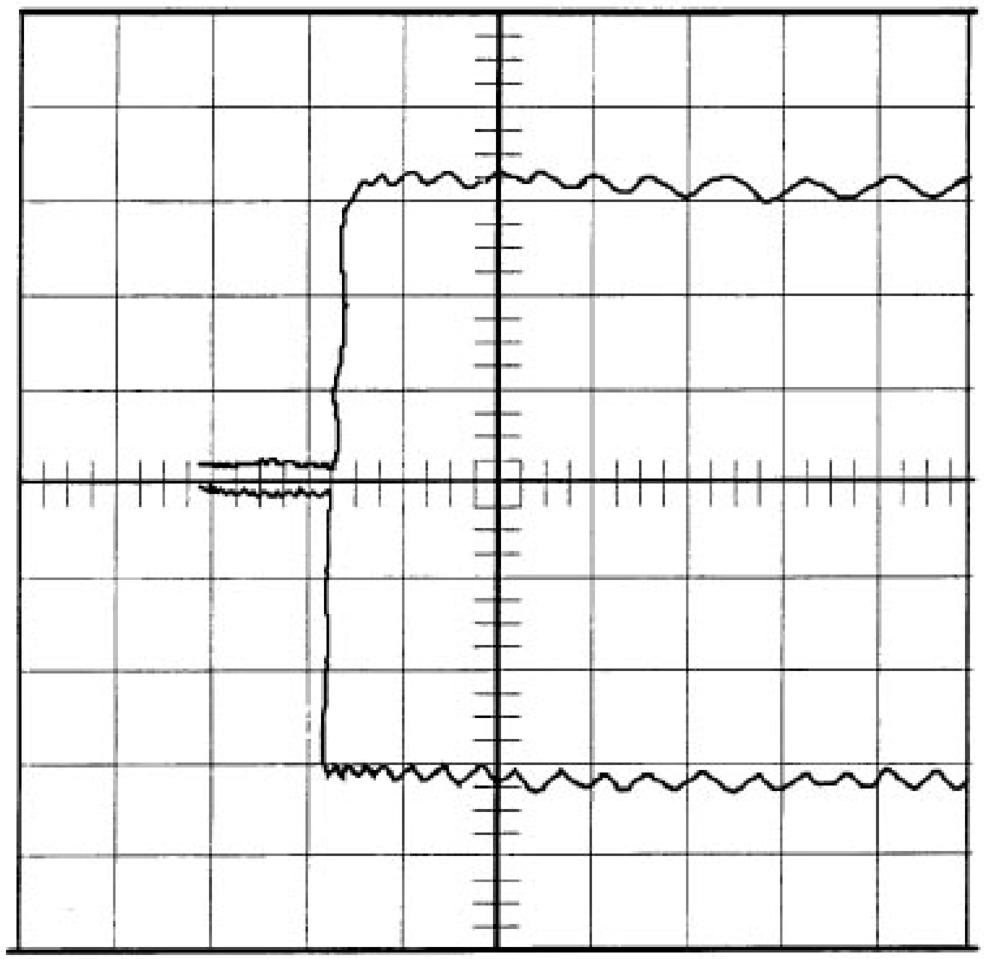


Figure 9-33 KV Ripple @ 0.1 second scope trace

```
A:20V=0.1S B:20V=0.1S
140KV 340ma Bleeder (slow sweep)
This is a normal picture.
```

This is a good example of the scope aliasing the inverter ripple. This picture is the same picture as is in the previous section, the only difference is the scope time base. NOTE: that the ripple can be as high as 20KV per side. Although aliasing will indicate some thing at higher frequencies, it is not a true waveform.

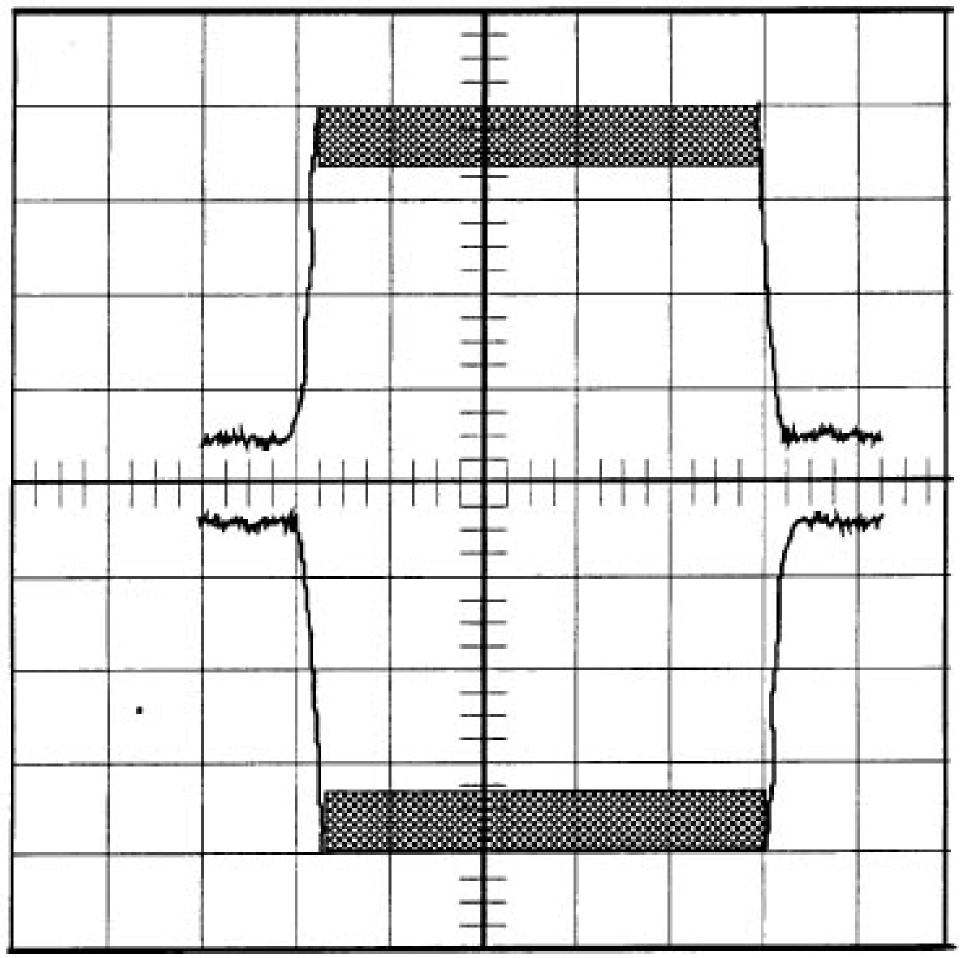


Figure 9-34 KV Ripple @ 0.2 sec. scope trace w/scope in “peak or “envelope” mode

```
A:20V=0.2S B:20V=0.2S
140KV 340ma Bleeder (slow sweep)
This is a normal picture, w/scope in “peak” or  “envelope” mode.
```

This is a good example of the scope aliasing the inverter ripple. NOTE: that the ripple can be as high as 20KV per side. Although aliasing will indicate something at higher frequencies, it is not a true waveform.

RESULTS SCREEN VALUES FOR:

kV Control Bd. (46-321064 or 46-321198)

- with a DCRGS (regulated 550VDC)
- with an HSA tube

80 kV mA	Total kV	Cathode kV	Anode kV	Anode mA	Cathode Inv I	Anode Inv. I	Vcnt V	Cathode. % Duty	Anode % Duty
10	798	39.9	39.9	9. 5	2.3	2.1	3.95	29	26
40	79.8	39.9	39.8	38.7	6.7	6.5	3.31	51	48
100	798	39.9	39.9	96.6	154	15.3	3.09	59	55
200	797	39.7	40	193.6	30.5	30.5	2.29	97	76
300	798	39.1	40.7	291.2	45.7	45.6	1.31	100	75
350	797	39.1	40 8	339.5	53.3	53.1	1.05	100	74
400	798	38.9	40.8	387.2	60.8	60.5	0.9	100	73

Table 9-15 80 kV, Result Screen

100 kV mA	Total kV	Cathode kV	Anode kV	Anode mA	Cathode Inv I	Anode Inv. I	Vcnt V	Cathode % Duty	Anode % Duty
10	99 7	49.8	49.8	9.5	2.4	2.3	3.83	32	30
40	997	49.6	49.8	38.7	6.9	6.7	3.05	60	57
100	997	49.8	49.8	96.7	15.5	15.4	2.83	68	63
200	99.7	49.5	50.1	193.6	30.7	30.5	1.97	100	79
300	99.7	49	50.5	291.3	45.9	45.7	1.17	100	77
350	997	49	50.6	339.8	53.4	53.3	0.9	100	76
400	99.6	49.1	50.6	387.3	60.9	606	0.79	100	74

Table 9-16 100kV, Result Screen

120 kV mA	Total kV	Cathode kV	Anode kV	Anode mA	Cathode Inv I	Anode Inv. I	Vcnl V	Cat. % Duty	Anode % Duty
10	119.6	59.8	59.6	9.5	2.5	2.3	3.7	38	34
40	119.6	59.8	59.8	38.7	7	6.9	2.13	71	68
100	119.6	59.8	59.8	Q6.7	15.5	15.5	2.55	78	73
200	119.6	59.2	60.3	193.7	30.7	30.6	1.6	100	83
300	119.6	59.2	60.4	291.4	46	458	0.93	100.--	80
350	119.6	59.2	60.3	340	53.6	53.4	0.76	100	79
400	119.7	59	60.8	387.5	61.1	60.8	0.6	100	78

Table 9-17 120 kV, Result Screen

140 kV mA	Total kV	Cathode kV	Anode kV	Anode mA	Cathode Inv I	Anode Inv. I	Vcnl V	Cat.ode % Duty	Anode % Duty
10	139.5	69.6	69.9	9.4	2.8	2.5	3.27	61	41
40	139.5	69.7	69.7	38.7	6.9	6.7	2.38	83	80
100	1395	69.7	69 7	96 7	15.6	15.5	2.21	90	85
200	1395	69.1	70.2	193.7	30.7	30.7	1.12	100	87
300	1394	69.2	70.2	291.6	46	45.9	0.62	100	85
340	139.6	69.2	70.3	330.4	52.1	51 9	0.49	100	84

Table 9-18 140 kV, Result Screen

RESULTS SCREEN VALUES FOR:

kV Control Bd. (2143147)

- with a Compact PDU (unregulated HVDC)
- with an HSA tube
- at Nominal Line Voltage

MA	TOTAL KV	CATHODE KV	ANODE KV	ANODE MA	CATHODE INV I	ANODE INV I	VCNT V	CATHODE % DUTY	ANODE % DUTY	HVDC
10	79.6	39.9	39.8	9.7	2.45	2.50	4.17	22	18	674
20	79.6	39.9	39.8	19.8	4.15	4.02	3.87	32	27	671
40	79.6	39.9	39.8	39.9	7.27	7.07	3.60	42	37	666
100	79.6	39.9	39.8	100.0	16.35	16.30	3.35	51	46	652
200	79.6	39.9	39.8	200.0	32.05	32.00	2.73	72	67	646
300	79.6	39.7	39.8	299.5	47.3	47.6	1.94	100	90	639
350	79.5	39.4	40.2	349.5	55.0	55.5	1.61	100	90	635
400	79.5	39.4	40.2	398.4	62.7	63.2	1.38	100	91	635

Table 9-19 KV Control Board (2143147) Values w/Nominal Line Voltage @ 80KV

MA	TOTAL KV	CATHODE KV	ANODE KV	ANODE MA	CATHODE INV I	ANODE INV I	VCNT V	CATHODE % DUTY	ANODE % DUTY	HVDC
10	99.4	49.8	49.7	9.7	2.5	2.5	4.09	25	20	676
20	99.4	49.8	49.7	19.8	4.3	4.2	3.70	39	33	670
40	99.4	49.8	49.7	39.8	7.4	7.2	3.45	47	42	663
100	99.4	49.8	49.7	100.0	16.35	16.30	3.14	58	53	651
200	99.5	49.8	49.7	200.0	32.0	32.0	2.58	78	72	646
300	99.4	49.5	49.9	299.5	47.5	47.6	1.78	100	89	637

Table 9-20 KV Control Board (2143147) Values w/Nominal Line Voltage @ 100kV

MA	TOTAL KV	CATHODE KV	ANODE KV	ANODE MA	CATHODE INV I	ANODE INV I	VCNT V	CATHODE % DUTY	ANODE % DUTY	HVDC
350	99.4	49.4	50.2	349.4	55.3	55.6	1.46	100	90	632
400	99.4	49.3	50.2	398.4	63.0	63.3	1.23	100	90	627

Table 9-20 KV Control Board (2143147) Values w/Nominal Line Voltage @ 100kV (Continued)

MA	TOTAL KV	CATHODE KV	ANODE KV	ANODE MA	CATHODE INV I	ANODE INV I	VCNT V	CATHODE % DUTY	ANODE % DUTY	HVDC
10	119.3	59.7	59.7	9.7	2.6	2.6	4.00	29	23	675
20	119.3	59.8	59.7	19.8	4.52	4.32	3.52	45	39	672
40	119.3	59.8	59.7	39.9	7.5	7.4	3.30	53	47	661
100	119.3	59.7	59.7	100.0	16.4	16.3	2.91	66	60	653
200	119.3	59.7	59.7	200.5	32.2	32.1	2.37	86	78	645
300	119.3	59.3	60.2	300.2	47.8	47.9	1.52	100	88	634
350	119.3	59.3	60.2	349.7	55.5	55.6	1.26	100	88	628
400	119.3	59.1	60.1	398.8	63.3	63.5	1.07	100	89	625

Table 9-21 KV Control Board (2143147) Values w/Nominal Line Voltage @ 120kV

MA	TOTAL KV	CATHODE KV	ANODE KV	ANODE MA	CATHODE INV I	ANODE INV I	VCNT V	CATHODE % DUTY	ANODE % DUTY	HVDC
10	139.2	69.7	69.6	9.7	2.8	2.7	3.89	33	26	677
20	139.2	69.7	69.6	19.8	4.6	4.4	3.32	52	46	672
40	139.2	69.7	69.6	39.9	7.7	7.5	3.09	60	53	662
100	139.2	69.7	69.7	100.1	16.4	16.4	2.68	74	68	650
200	139.2	69.6	69.7	200.3	32.3	32.1	2.06	100	85	643
300	139.1	69.2	70.1	300.6	48.0	48.0	1.24	100	88	629
340	139.2	69.2	70.2	340.1	54.2	54.2	1.06	100	87	626
xxx	xxxx	xxxx	xxxx	xxxx	xxxx	xxxx	xxxx	xxxx	xxxx	xxxx

Table 9-22 KV Control Board (2143147) Values w/Nominal Line Voltage @ 140kV

RESULTS SCREEN VALUES FOR:

kV Control Bd. (2143147)

- with a Compact PDU (unregulated HVDC)
- with an HSA tube
- at Low Line Voltage (approximately 441VAC line input)

MA	TOTAL KV	CATHODE KV	ANODE KV	ANODE MA	CATHODE INV I	ANODE INV I	VCNT V	CATHODE % DUTY	ANODE % DUTY	HVDC
10	79.6	39.9	39.8	9.7	2.5	2.5	4.08	25	21	589
20	79.6	39.9	39.8	19.8	4.1	4.1	3.75	37	31	585
40	79.6	39.9	39.8	40.0	7.2	7.1	3.45	47	42	584
100	79.6	39.9	39.8	100.1	16.4	16.3	3.20	56	51	576
200	79.6	39.8	39.8	200.1	32.1	32.0	2.42	83	78	562
300	79.6	39.5	40.2	300.6	47.8	47.8	1.55	100	90	552
350	79.7	39.4	40.3	350.4	55.6	55.7	1.26	100	91	543
400	79.6	39.3	40.3	399.0	63.7	63.4	1.05	100	90	536

Table 9-23 KV Control Board (2143147) Values with Low Line Voltage @ 80KV

MA	TOTAL KV	CATHODE KV	ANODE KV	ANODE MA	CATHODE INV I	ANODE INV I	VCNT V	CATHODE % DUTY	ANODE % DUTY	HVDC
10	99.5	49.8	49.7	9.7	2.6	2.6	3.98	29	24	590
20	99.5	49.8	49.8	19.8	4.3	4.2	3.55	44	38	587
40	99.5	49.8	49.7	39.9	7.5	7.3	3.2	55	50	582
100	99.5	49.8	49.7	100.0	16.4	16.3	2.96	64	59	574
200	99.5	49.7	49.8	200.0	32.1	32.1	2.19	93	84	560
300	99.5	49.4	50.3	300.2	47.8	47.8	1.30	100	90	544
350	99.5	49.3	50.2	350.4	55.7	55.8	1.05	100	89	534
400	99.3	49.2	50.0	398.4	63.2	63.5	0.89	100	89	529

Table 9-24 KV Control Board (2143147) Values with Low Line Voltage @ 100kV

MA	TOTAL KV	CATHODE KV	ANODE KV	ANODE MA	CATHODE INV I	ANODE INV I	VCNT V	CATHODE % DUTY	ANODE % DUTY	HVDC
10	119.4	59.8	59.7	9.7	2.8	2.7	3.87	33	27	589
20	119.3	59.8	59.7	19.8	4.5	4.3	3.32	52	46	584
40	119.4	59.8	59.7	40.0	7.7	7.5	2.95	65	59	585

Table 9-25 KV Control Board (2143147) Values with Low Line Voltage @ 120kV

MA	TOTAL KV	CATHODE KV	ANODE KV	ANODE MA	CATHODE INV I	ANODE INV I	VCNT V	CATHODE % DUTY	ANODE % DUTY	HVDC
100	119.4	59.8	59.7	100.2	16.4	16.4	2.68	74	69	573
200	119.4	59.4	60.1	200.2	32.3	32.2	1.78	100	86	555
300	119.4	59.2	60.2	300.2	48.0	47.9	1.03	100	83	537
350	119.3	59.2	60.3	350.3	55.9	55.8	0.81	100	88	528
400	119.3	59.2	60.2	399.4	63.8	63.5	0.63	100	88	515

Table 9-25 KV Control Board (2143147) Values with Low Line Voltage @ 120kV (Continued)

MA	TOTAL KV	CATHODE KV	ANODE KV	ANODE MA	CATHODE INV I	ANODE INV I	VCNT V	CATHODE % DUTY	ANODE % DUTY	HVDC
10	139.2	69.7	69.7	9.7	2.9	2.8	3.70	39	32	587
20	139.3	69.7	69.7	19.8	4.5	4.4	3.08	61	54	581
40	139.2	69.7	69.7	39.9	7.6	7.5	2.54	79	72	580
100	139.2	69.7	69.7	100.2	16.5	16.4	2.34	86	79	568
200	139.2	69.2	70.2	200.3	32.4	32.2	1.18	100	87	548
300	139.2	69.2	70.2	300.7	48.1	47.9	0.56	100	88	526
340	139.2	69.2	70.0	340.0	54.4	54.1	0.42	100	89	516
xxx	xxxx	xxxx	xxxx	xxxx	xxxx	xxxx	xxxx	xxxx	xxxx	xxxx

Table 9-26 KV Control Board (2143147) Values with Low Line Voltage @ 140kV

RESULTS SCREEN VALUES FOR:

kV Control Bd. (2143147)

- with a Compact PDU (unregulated HVDC)
- with an PERFORMIX tube
- at Nominal Line Voltage

TABLES ARE NOT AVALIBLE FOR PERFORMIX TUBES AT THIS TIME.

Section 61.0
MA Related Problems and Troubleshooting Theory

61.1 mA Meter Verification Theory

The purpose of mA Meter Verification is to ensure that the mA Measurement Electronics has a gain of one between actual tube current and reporting to the firmware. This firmware value would then be reported to the console.

The method outlined in various GEMS documents is to install an ISO compliant, calibrated ammeter across two test points. This meter effectively shorts out a 680ohm resistor. The next step is to install a 62ohm resistor from one side of the ammeter to the mA test point (TP5) on the measurement Bd.

TP5 is a voltage representation of tube current. This voltage is feed to the mA Control Bd. in the OBC and the resultant mA is fed to the firmware and can be read from a mA test point. This is where the gain of one is needed, between the Measurement Bd. TP5 and the mA Control Bd. mA Test Point.

When ACCEPT is touched on the console screen and after a time delay Q1 is turned on (+)15vdc (0r -15vdc) is applied thru the ammeter (not the shorted out 680ohm resistor), thru the 68ohm resistor to TP5. 15vdc over 62ohm comes out to 242mA, add in the other resistances in the measurement circuit an it comes out to be about 140mA to 200mA.

It really doesn't matter what the actual mA is, all that matters is that the console reading (firmware value from the mA Test Points) MATCHES what the ISO compliant, calibrated ammeter says.

61.2 SW and HW Tools Available for Troubleshooting

61.2.1 Schematics

- Direction 46-018303
- mA / Filament Control Bd. Schematics 46-288886-S
- OBC Backplane list
- Gantry Rotating Member Block Diagram
- Gantry Rotating Interconnect
- X-Ray Tube 120vac

61.2.2 Equipment

- Direction 46-018318 CT HiSpeed Advanced Functional Interconnect
- Multi-meter
- O'Scope

61.3 Explanation of Cathode, Anode mA Screen

Cathode MA:	193.7mA	200.0mA	193.7mA

This value comes from the mA Control Bd. 46-288886 TP4, thru the backplane and Gentry I/O bd. Since the cathode is in series with the anode, TP4 should be the same value as the TP10 (anode mA, they are in series). The scale is 1v/100mA.

In closed loop mode TP4 should be commanded mA. In open loop mode the value should be less (whatever is in GenCalSeed). TP4 is actually the cathode high voltage tank secondary amperage, the x-ray tube is the load for the secondary. MA Meter Verification verifies that the measurement electronics have a gain of one and that reported mA is actual ma.

Cathode high voltage tank secondary amperage (and x-ray tube mA) is the direct result of filament heating, for improper mA include filament function while troubleshooting. If mA is out of tolerance (3%), check KV values for large errors, verify mA metering, and run the Filament Functional test. BLDs can help also.

If cathode and anode mA are different, suspect mA measurement electronics (use mA Meter Test), or suspect a shattered x-ray tube insert shorting out the filament (cathode) or the anode.

Anode MA:	**193.7mA**	**200.0mA**	**193.7mA**

This value comes from the mA Control Bd. 46-288886 TP10, thru the backplane and Gentry I/O bd. Since the cathode is in series with the anode, TP10 should be the same value as the TP4 (cathode mA, they are in series). The scale is 1v/100mA.

In closed loop mode TP10 should be commanded mA. In open loop mode the value should be less (whatever is in GenCalSeed). TP10 is actually the anode high voltage tank secondary amperage, the x-ray tube is the load for the secondary. MA Meter Verification verifies that the measurement electronics have a gain of one and that reported mA is actual ma.

Anode high voltage tank secondary amperage (and x-ray tube mA) is the direct result of filament heating, for improper mA include filament function while troubleshooting. If mA is out of tolerance (3%), check KV values for large errors, verify mA metering, and run the Filament Functional test. BLDs can help also.

If cathode and anode mA are different, suspect mA measurement electronics (use mA Meter Test), or suspect a shattered x-ray tube insert shorting out the filament (cathode) or the anode.

61.4 Troubleshooting Cathode / Anode mA

Refer to section 4.0 Reference Materials for the following explanation. When using diagnostic mA Loop, Filament Drive for troubleshooting, monitor TP5 (sheet 5 Filament Signal (1v=1a)), monitor TP40 (Sheet 5 Filament Demand (1v=1a), monitor TP22 (sheet 11 Fill Command (1v=1a)). Compare TP5, TP40 and TP22 with TP23 (sheet11 Filament Feedback (1v=1a)).

The following test points are a result of the proper filament command, they may be useful in troubleshooting. TP10 (sheet 9 Anode mA (1v=100ma), TP13 (sheet 9 MAFB), TP15 (sheet 5 mA Demand (this is a dynamic signal)) and TP3 (sheet 5 Filament Error (1v = 0.5a). When troubleshooting these signals, do it at higher techniques and longer scan times (approx 2 seconds). At lower techniques and shorter times, the system has the ability to compensate for errors. Also do tests in both closed loop mA and open loop mA. Open loop mA should result in less filament drive and therefore less mA, approx 50 to 100ma less.

61.5 Open or Shorted Filament

Refer to following sections Reference Materials for the following explanation. When using diagnostic mA Loop, Filament Drive for troubleshooting, monitor TP5 (sheet 5 Filament Signal (1v=1a)), monitor TP40 (Sheet 5 Filament Demand (1v=1a), monitor TP22 (sheet 11 Fill Command (1v=1a)). Compare TP5, TP40 and TP22 with TP23 (sheet11 Fil Feedback (1v=1a)).

If the FIlament Select Relay is open, TP23 will have the wrong value.

The following test points are a result of the proper filament command, they may be useful in troubleshooting. TP10 (sheet 9 Anode mA (1v=100ma), TP13 (sheet 9 MAFB), TP15 (sheet 5 mA Demand (scale =? this is a dynamic signal)) and TP3 (sheet 5 Filament Error (1v = 0.5a). When troubleshooting these signals, do it at higher techniques and longer scan times (approx 2 seconds). At lower techniques and shorter times, the system has the ability to compensate for errors. Also do tests in both closed loop mA and open loop mA. Open loop mA should result in less filament drive and therefore less mA, approx 50 to 100ma less.

61.6 Reference Material

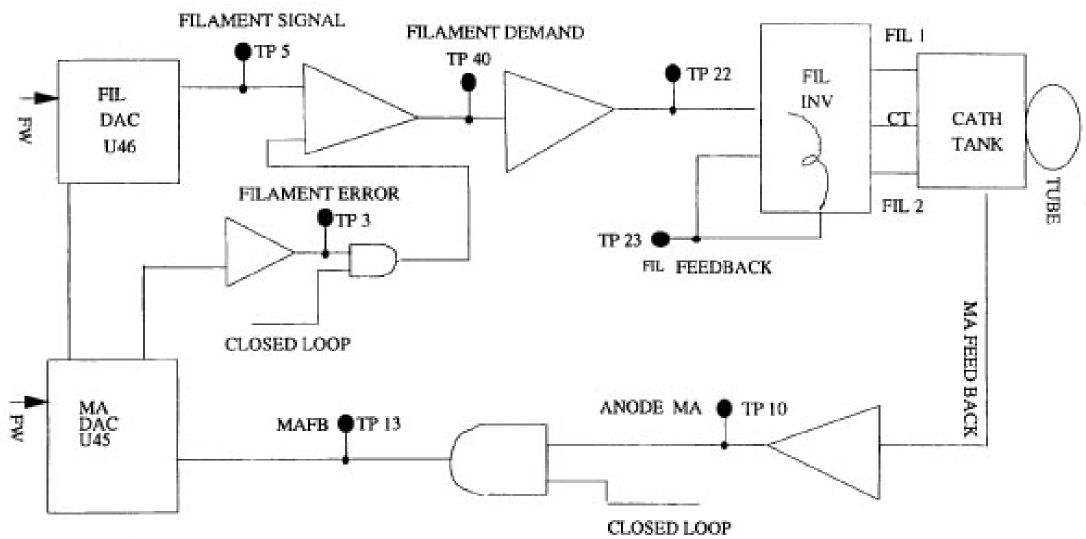


Figure 9-35 Simplified schematic

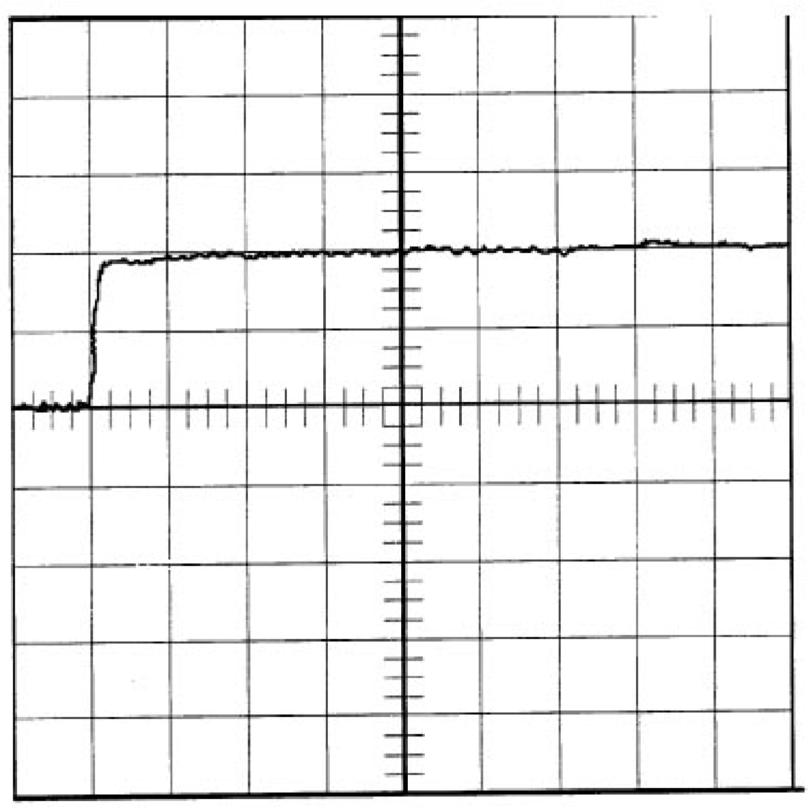


Figure 9-36 Normal mA wave shape for 80KV, 100mA.

```
channel A:0.5 v Board: ma Control 46-288886 scope: TEK 224
 Vert: 0.5v Horz:5mS TP 10 to TP 2
 80KV 100mA CLOSED LOOP mA TP10 (anma) to TP2 (sgnd)
```

This is a picture of the mA feedback read off of the mA Control Bd, with **CLOSED LOOP** mA selected. **There** should not be a great difference between CLOSED LOOP mA and OPEN LOOP mA. IF there is, it indicates that CLOSED LOOP is trying to make up for a problem, investigate for root cause.

Anode and cathode mAs are in series. Therefore anode and cathode mA wave shapes are duplicates.

Comment: *If not, either Ohm's law has been redefined for series circuits, or there is a problem. I would verify for a measurement problem first.*

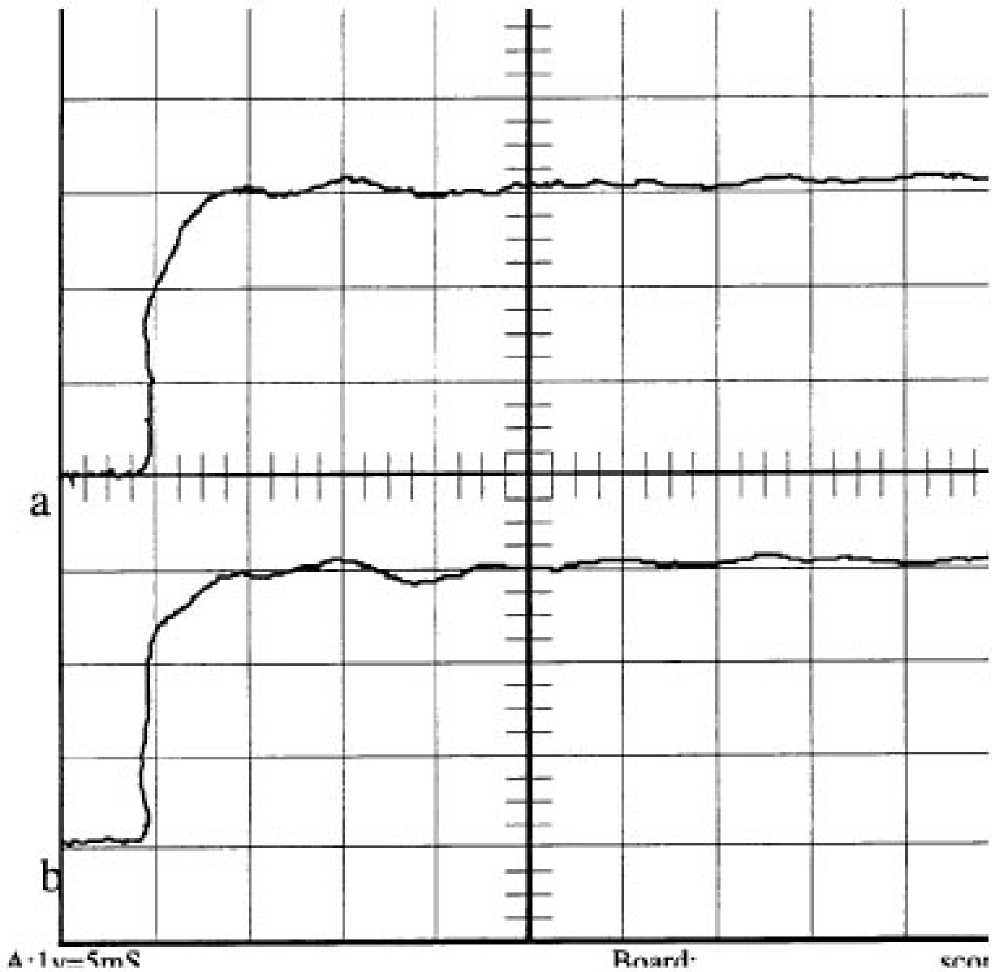


Figure 9-37 Normal mA wave shape for 80KV, 320mA.

```
channel A: 1v=5mS Board: mA control 46-288886 scope:TEK 244
B:1v=5mS Vert:1v Horz:5mS.
ch 1 (a)= 80KV 320mA CLOSED LOOP mA TP10 (ANMA) to TP2 (sgnd)
ch 2 (b)= 80Kv 320mA CLOSED LOOP mA TP4 (CAMA) to TP2 (sgnd)
```

This is a picture of the mA feedback read off of the mA control bd.

There should not be a great difference between CLOSED LOOP mA and Open loop mA. If there is, it indicates that CLOSED LOOP is trying to make up for a problem, investigate for root cause.

Anode and cathode mAs are in series. Therefore anode and cathode mA waveshapes are duplicates. If not, either Ohm's law has been redefined for series circuits, or there is a problem. (I would verify for a measurement problem first.)

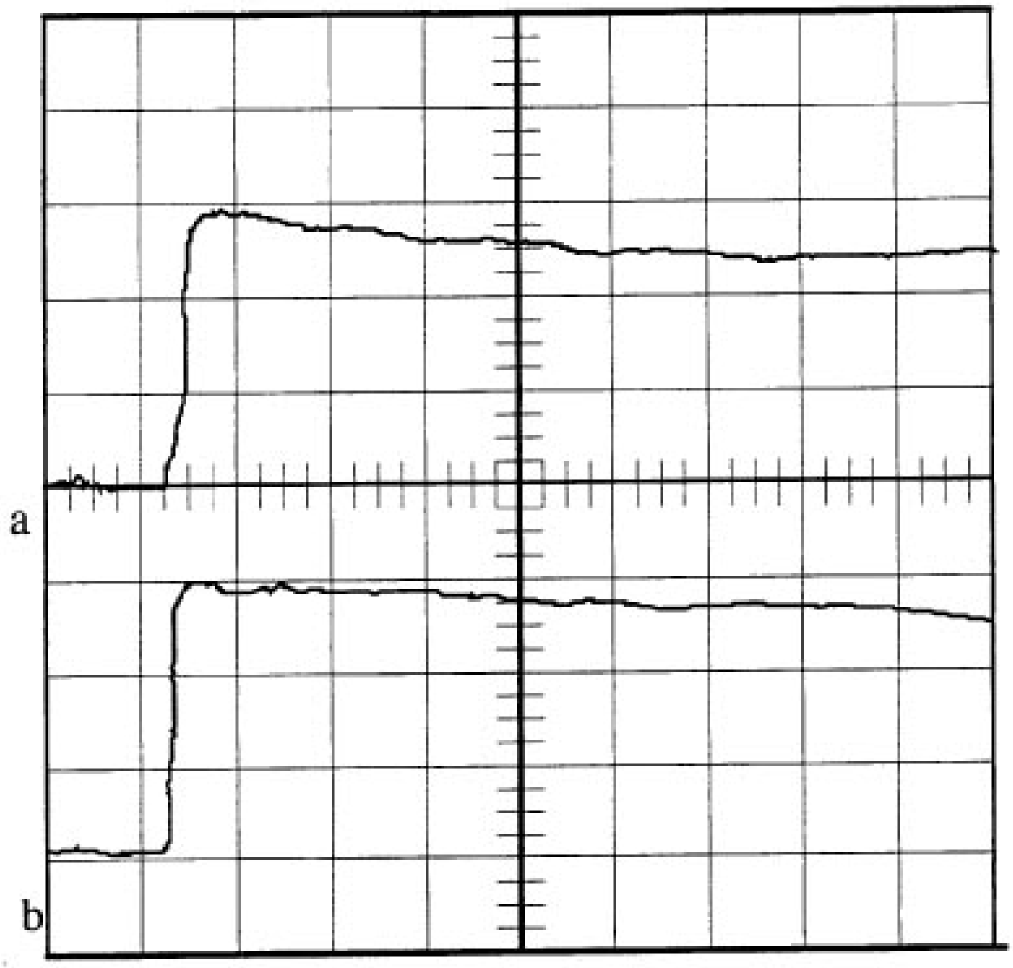


Figure 9-38 Normal mA waveshape for 80KV, 320mA OPEN LOOP MODE

```
channel a:1v=50mS Board scpoe:TEK 224
b:1v=50mS ma control Vert:1v
46-288886 Horz: 50mS
ch1 (a) =80KV 320mA OPEN LOOP mA TP10 (anma) to TP2 (sgnd)
ch2 (b) =80KV 320mA OPEN LOOP mA TP4 (cama) to Tp2 (sgnd)
```

IN OPEN LOOP MODE THERE IS A POSSIBILITY OF EXCEEDING 400mA. DO NOT RUN THE mA HIGHER THAN 350mA IN OPEN LOOP MODE. EXCEEDING 400mA WILL DAMAGE THE H.V. SUBSYSTEM.

This is a picture of the mA feedback read off of the mA Control Bd, with OPEN LOOP mode selected.

Compare this picture with CLOSED LOOP mA at the same technique. Note that in OPEN LOOP mode the mA is not regulated to a perfect 320 mA.

There should not be a great difference between CLOSED LOOP mA and OPEN LOOP mA. If there is, it indicates that CLOSED LOOP is trying to make up for a problem, investigate for root cause

Section 62.0
Rotor Related Problems and Troubleshooting Theory

62.1 Safety / Tag and Lockout

The 550 VDC that the DCRGS generates is actually +275 VDC and -275 VDC. It is not referenced directly to ground. However the building conduit that contains the 480 VDC feed will act as an AC capacitor allowing the AC component of the 550 VDC to seek an asymmetrical ground. This is a return path for current in the event you come into contact with the 550 VDC and ground. 550 VDC is not forgiving when physical contact is made.

A large majority of electrical accidents could of been prevented with the proper use of Tag and Lockout. Tag and Lockout Procedures are company policy. It is also common sense. Use it.

62.2 SW & HW Tools Available for Troubleshooting

SCHEMATICS (PLEASE SEE DIRECTION 46-018303)

- Gentry I/O (46-288512S) schematics
- CTVRC (46-288858S) schematics
- Stator Filter bd (46-288922) schematics
- CTVRC Power Amp Driver / Filter Bd. (46-264886S) schematics

62.3 Results Screens and User Selections

62.3.1 Rotor Speed

Description: This selection enables the user to choose one of the three rotor rotation speeds. This can be misleading since some X-Ray tubes are limited to one or two speeds. In this case, an invalid choice is automatically changed to the next valid selection. When choosing a speed, the user should be aware of the higher power demand for acceleration and braking as the speed is increased.

Selection: Low, Medium, or High

62.3.2 Rail Voltage

Description: The selection of rail voltage enables the user to operate the rotor normally (550VDC) or decrease the voltage (50VDC) while sever faults are present such as a capacitor overvoltage, shorts or shoot-thrus. This 50VDC operation helps prevents further damage to the equipment by reducing the amount of power available. The 0VDC mode operates the rotor as though power was present without enabling rail voltage. **Note:** The 50VDC is a differential voltage which can float 500 volts above ground. Always follow all safety precautions when working on the CTVRC inverter.

Selection: 0, 50, or 550VDC

62.3.3 Test Duration

Note: Test duration on high voltage diagnostics should never be less than 1 second. There is not time for the firmware to collect enough data for interpretation.

Description: This selection controls the duration of the run cycle. The total test duration is the sum of the selected duration and the time required to complete the accelerate and brake cycle. A 10 second minimum run duration enables the stator time to cool between the acceleration and brake cycle.

Range: 10 to 1800 Seconds

62.3.4 Rotor Loop

Description: This selection opens or closes the rotor control loop. Closing the loop enables the CTVRC to regulate the stator current to the expected value; however, problems can be concealed in this mode. Open loop enables the user to see subtle problems in performance. During 0 or 50VDC operation, the open loop mode should be selected to prevents the CTVRC from saturating the output control signals.

Note: A larger stator current warning level tolerance is used for open loop operation.

Selection: Closed or Open

62.3.5 CTVRC Operating Mode

Description: This field indicates the present operating mode. They are as follows:

- ACCEL – Rotor is in the acceleration cycle.
- RUN – Rotor is in the run cycle.
- BRAKE – The rotor is being de-accelerated.
- IDLE – CTVRC amplifier is disabled.

Error Conditions: The run cycle is bypassed during an abort condition.

Test duration on high voltage diagnostics should not be less than 1 second.

62.3.6 Green Stator Current

Description: This is the CTVRC inverter current and **NOT** the current flowing within the tube stator. The CTVRC inverter is connected to the tube through a 2:1 transformer which drops the current to half of the indicated value. Currents monitored for the various cycles are listed separately. They are as follows:

Present current: This is the present value of the green inverter current.

Average accel current: This is the average of all green inverter current samples taken during the acceleration cycle.

Average run current: This is a rolling average of the run cycle current level. On the 8th sample, the data is removed and the average value used as sample 1. This prevents the accumulation of large quantities of data for long run times.

Average brake current: This is the average of all green inverter current samples taken during the brake cycle.

Error Conditions: For more information, see the troubleshooting section.

Current values come from: CTVRC Power Amp Load Current Transformer.

Refer to Direction 46-018318 Rotor Control

Test duration on high voltage diagnostics should not be less than 1 second.

62.3.7 Black Stator Current

Description: This is the CTVRC inverter current and **NOT** the current flowing within the tube stator. The CTVRC inverter is connected to the tube through a 2:1 transformer which drops the current to half of the indicated value. Currents monitored for the various cycles are listed separately. They are as follows:

Present current: This is the present value of the black inverter current.

Average accel current: This is the average of all black inverter current samples taken during the acceleration cycle.

Average run current: This is a rolling average of the run cycle current level. On the 8th sample, the data is removed and the average value used as sample 1. This prevents the accumulation of large quantities of data for long run times.

Average brake current: This is the average of all black inverter current samples taken during the brake cycle.

Error Conditions: For more information, see the troubleshooting section.

Current values come from: CTVRC Power Amp Load Current Transformer.

Refer to Direction 46-018318 Rotor Control

Test duration on high voltage diagnostics should not be less than 1 second.

62.3.8 White Stator Current

Description: This is the current flowing in the white inverter lead which is common to both the black and green inverter leads. The current indicated is twice that of the current flowing in the white lead of the rotor stator. The approximate value is the square root of the sum of the black and green currents.

Error Conditions: In the event the white lead is open, the white current level will be 0 while the black and green current are normal. The rotor does not rotate in this state.

Test duration on high voltage diagnostics should not be less than 1 second.

Current values are a sum of black and green wire current.

62.3.9 Stator Temperature Rise

Description: This is the estimated temperature rise of the X-Ray tube stator. Values are calculated by the CTVRC board based on the green and black inverter currents. Values are given in volts and centigrade.

Error Conditions: Values which exceed the operating limits will disable the CTVRC inverters and therefore the test. Allowing the tube to cool for several minutes and running this test should not cause this error to occur unless an inverter current error exists or the CTVRC board temperature simulator is at fault.

Note: Running many test iterations in rapid succession can cause excessive heating of the CTVRC inverter and tube.

62.3.10 Rail Voltage

Description: This is the rail voltage as monitored by the CTVRC board. Value is calculated from the instantaneous sum of the high and low capacitor voltages.

Error Conditions: The rail voltage measured should remain close to that selected. For 550VDC operation, values less than 540VDC MAY indicate a DCRGS fault or a short on the rail. Scope this value at the CTVRC control board for ripple. For 50VDC operation, a dip in voltage may occur during testing. This is normal.

Voltage values come from: a voltage divider on the CTVRC Gate Driver Bd. to the OBC CTVRC Control Bd.

Refer to Direction 46-018318 550V_Monitor

Test duration on high voltage diagnostics should not be less than 1 second.

62.3.11 High Side Capacitor Voltage

Description: This is one of two CTVRC inverter capacitor voltages connected in series across the rail. This particular capacitor has one terminal connected directly to the high rail side. The normal value should be half the indicated rail voltage.

Error Conditions: Voltages which differ greatly from half the rail voltage indicate a CTVRC inverter fault or a control board monitoring error.

Voltage values come from: a voltage divider on the CTVRC Gate Driver Bd. to the OBC CTVRC Control Bd.

Refer to Direction 46-018318 550V_Monitor

62.3.12 Low Side Capacitor Voltage

Description: This is one of two CTVRC inverter capacitor voltages connected in series across the rail. This particular capacitor has one terminal connected directly to the low rail side. The normal value should be half the indicated rail voltage.

Error Conditions: Voltages which differ greatly from half the rail voltage indicate a CTVRC inverter fault or a control board monitoring error.

Voltage values come from: a voltage divider on the CTVRC Gate Driver Bd. to the OBC CTVRC Control Bd.

Refer to Direction 46-018318 550V_Monitor (Page 57 in Rev 6)

62.3.13 Current Command Voltage

Description: This is the output voltage of the current command DAC. Voltage corresponds to the expected inverter current level. Scaling is 1V = 2A of expected current.

Error Conditions: Values which differ more than several percent of the expected indicate a fault with the CTVRC board. Run the CTVRC BLD if a current DAC fault is suspected.

62.3.14 Pulse Width Command Voltage

Description: This is the output voltage of the pulse width DAC. During closed loop operation, the voltage changes to compensate for inverter current errors. Inverter current regulation is based on the larger of the two stator currents.

Error Conditions: During closed loop operation, low current voltages should force this voltage higher than the expected to compensate. Failure to see this change indicates possible errors with the CTVRC board. This is also true if the currents are high and the pulse width voltage fails to drop below the expected. Run the CTVRC BLD if a pulse width DAC fault is suspected.

62.3.15 CTVRC Reference Voltage

Description: This reference voltage is derived on the CTVRC board and is the operating standard for all rotor functions performed. There should be **NO** fluctuations in this value.

Error Conditions: An error message is logged if this voltage varies outside the operating limits. This voltage can also be viewed during the Power Supplies test and is checked during the CTVRC BLD. Errors indicate a fault with the CTVRC control board or a Gentry I/O monitoring error.

62.3.16 CTVRC Operating Frequency

Description: This is the expected operating frequency of rotor as determined by the Rotor characterization file.

Error Conditions: The operating frequency should agree with that measured.

62.3.17 Status Register

Description: This is the address and data contained in the CTVRC status register.

Error Conditions: See the CTVRC schematic for more information about each bit.

62.3.18 Fault Register

Description: This is the address and data contained in the CTVRC fault register.

Note: Test duration on high voltage diagnostics should never be less than 1 second. There is not time for the firmware to collect enough data for interpretation.

Faults such as stator shorts, a capacitor overvoltage, rail overvoltage, shoot-thrus, and over temperature values cause an immediate shutdown of hardware and the termination of the test.

The 50VDC operating mode is very useful in isolating faults while minimizing further damage. However, at this reduced power setting, errors such as shoot-thrus, shorts, a capacitor overvoltage, etc. may not be directly indicated. These errors can be found as follows:

1.) Shoot-thrus – This error can be caused by a shorted capacitor or IGBT. Conformation of this type of fault is done by verifying 1 of the 2 capacitor voltages is approximately equal to the rail voltage.
2.) Shorts – Compare the two stator currents. The stator which indicated a short should have a higher current value than the stator operating normally. If this is true, the tube can safely be disconnected and the test rerun to determine if fault is inverter or tube based.
3.) Capacitor overvoltage – Check the capacitor voltages to see if one or the other is approximately equal to the rail voltage. If true, see note above on shoot-thrus.
4.) Over temperature – At this reduced power level, only a small increase in temperature should be noticed.

Some failures occur only with higher voltages. It is possible to isolate between the X-Ray tube and the CTVRC inverter with 550VDC by disconnecting the tube from the circuit and running the test.

WARNING THIS PROCEDURE SHOULD ONLY BE USED AS A LAST RESORT SINCE EXCESSIVE INVERTER VOLTAGES CAN EXIST.

62.4 Troubleshooting

62.4.1 Where to Start

"Rotor Manual Functional Test" results screen and results screen are the primary tools for troubleshooting the CTVRC subsystem.

62.4.2 Shoot - Through

62.4.2.1 Shoot - Thru (Operator Induced)

Error #183142 ctvrc fault: left / right (main) inverter short. This error can be operator induced. When the rotor is in a braking mode (reverse acceleration) due to some action on the part of the operator and the operator then resumes scanning at the time the CTVRC is braking. The CTVRC will then try to forward accelerate the rotor which will cause a large current spike through the shoot-thru detector. This can be proven by verifying operator caused aborts in the errors logs around the time of the error. No parts are required, this is an OBC firmware bug.

62.4.2.2 Shoot - Thru (Component Failure)

Error #183142 CTVRC fault: left / right (main) inverter short. Compare the two stator currents using the results screen. The stator which indicated a short should have a higher current value than the stator operating normally. However errors may abort scanning resulting in no information available to the results screen. If this happens, try rotor test in 50vdc mode.

Most likely FRU is the CTVRC Power Amp. This can be proven by "ohming" our C3 and C4 on the power amp, a resistance check on the IGBT (looking for a short). Disconnect the tube from the circuit and running the test. "Open Wire Detected" may replace the Shoot - Thru.

WARNING THIS PROCEDURE SHOULD ONLY BE USED AS A LAST RESORT SINCE EXCESSIVE INVERTER VOLTAGES CAN EXIST WITH THE LOAD (X-RAY TUBE) DISCONNECTED).

62.4.2.3 Stator Wires Reports Low/No Current

Stator Current values come from: CTVRC Power Amp Load Current Transformer.

Get the stator values from the rotor functional screen. Compare the average current with the Current Command Voltage and Pulse Width Command Voltage to verify CTVRC Control Bd. These two signal should be trying to compensate for the lower current. IF these signals are not compensating, suspect bad CTVRC Control Bd., run OBC BLDs.

Check the command LEDs that are on the power amp Filter Driver Bd. Use manual mode from the OBC CTVRC Control Bd, or Rotor Functional test. The LEDs should light when a command is received. If the LEDs do not light, check the DC voltages on the power amp Filter Driver Bd, check the cables, the command from the OBC CTVRC Control Bd. Also check the 550vdc and 120vac coming into the power amp.

The rotor can be run up manually using the OBC CTVRC Control Bd. The procedure is to bring up the 550VDC manually. On the OBC CTVRC Control Bd. S3 in the “MAN” mode, S2 in “FOR” (forward) mode and use S1 “BOOST” to accel the tube. S2 in the “REV” mode and use S1 “BOOST” to brake the tube.

62.4.2.4 Rotor Overcurrent

Stator Current values come from: CTVRC Power Amp Load Current Transformer.

Get the stator values from the rotor functional screen. Compare the average current with the Current Command Voltage and Pulse Width Command Voltage to verify CTVRC Control Bd. These two signal should be trying to compensate for the lower current. IF these signals are not compensating, suspect bad CTVRC Control Bd., run OBC BLDs.

Run a frequency sweep on the rotor. Black Stator currents should peak at between 130hz and 170hz. The Green Stator current peaks 10hz higher than the Black Stator.

If the Frequency Sweep fails the problem is either the power amp or the tube. A different tube could be hung next to the gantry for comparison.

An example of a good frequency sweep and a failed frequency sweep is in the Reference section.

62.4.2.5 30 Amp Fuses Blow

These fuses can blow from too much current entering the CTVRC Power Amp as well as too much current LEAVING the Power Amp.

If accompanied by “SRU Mains Low”, suspect blown 480vac feed fuses to the DCRGS.

The large capacitance on the CTVRC Power Amp will try to hold up the 550vdc buss if voltage is lost for any reason. The capacitance on the two inverters will try to hold up the voltage also, but their capacitance is smaller and the fuses are larger. With this combination the CTVRC Power Amp fuses will blow more often while holding up the 550vdc buss.

If the fuses are blowing because of a CTVRC sub-system problem, the problem is most likely the Power Amp itself. The power amp would detect shorts in the stator harness or x-ray tube.

Loose 500vdc connections will cause the fuses to blow. Check connections.

Check connections on the slip ring, kv inverters, and DCRGS.

62.4.2.6 30A & KV Inverter and/or DCRGS Feed Fuses Blow

These fuses can blow from too much current entering the CTVRC Power Amp as well as too much current LEAVING the Power Amp. However. if the KV Inverter fuses or the fuses that feed the DCRGS are blowing, most likely the is a problem with the 550vdc. In this case refer to the DCRGS Service section.

62.4.2.7 High / Low Capacitor Voltages

Voltages which differ greatly from half the rail voltage indicate a CTVRC Inverter fault or a control bd monitoring error. It could also indicate a shorted rectifier in the DCRGS

What To Do:

1.) Measure with a voltmeter to verify imbalance
2.) Remove 550vdc from the CTVRC and see if imbalance goes away.
3.) Reference to the DCRGS Service Section.

```
Suite: MCT1     Host: OBC     Proc: Generator
Error: 183139     File: Method: No Method     Line: 449
Function: XRAY GENERATION : Tube Rotor Control
Scan: 3445/1/1     Type: Static
Exception Level: Pri/Most Severe Time: 12:02:57:395
Log Series: 24
CTVRC fault detected: Capacitor overvoltage.
DC Rail capacitor voltage > 375V.
Fault latch address = ffb82b hex, Bit = D6
```

62.4.2.8 Low or No DC Rail Voltages (550VDC)

The DCRGS subsystem only monitors the 550vdc rail. It has nothing to do with the generation of 550vdc. If the CTVRC subsystem 550vdc monitoring circuit has malfunctioned, it will be reported as a 550vdc problem.

What To Do:

1.) Check 550vdc fuses on the CTVRC.
2.) Check 120vac control power to CTVRC.
3.) Check monitoring test points on the OBC CTVRC Control bd.For 550vdc / DCRGS problems refer to the DCRGS Service section.

Determine if the plot just generated indicates unacceptable rotor wobble by comparing examples of “Rotor Wobble Plot”. (See Figure 9-39 and Figure 9-40).

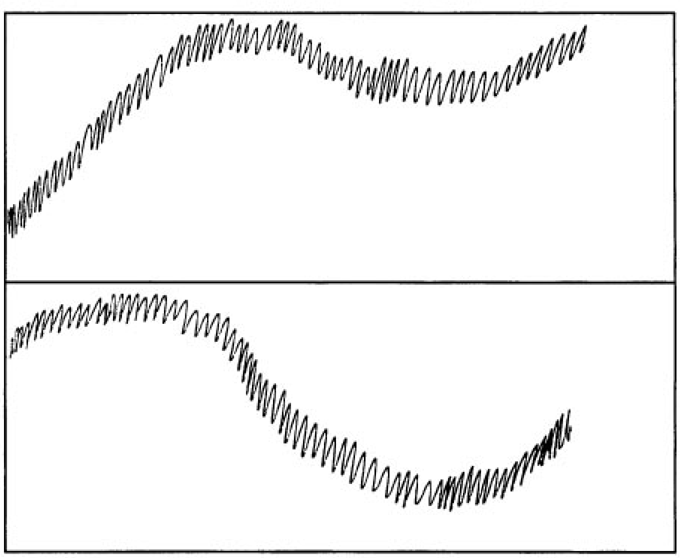


Figure 9-39 Rotor Wobble Plot 1

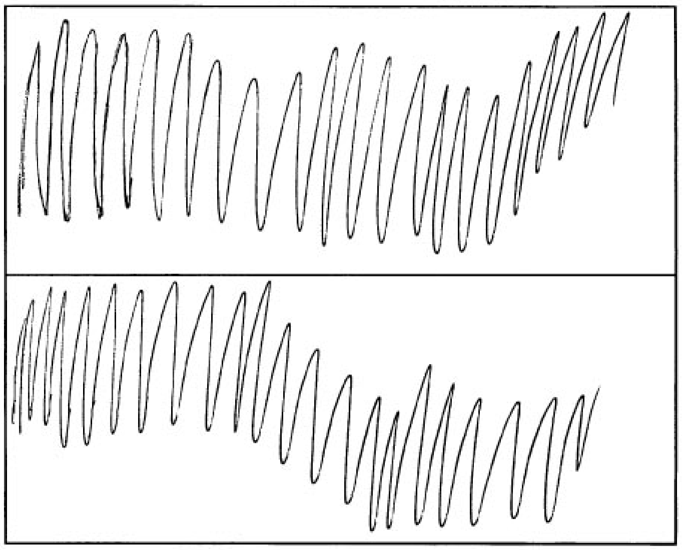


Figure 9-40 Rotor Wobble Plot 2

Section 63.0
DCRGS Related Problems

63.1 Safety / Tag and Lockout

The 550vdc that the DCRGS generates is actually +275vdc and (-)275vdc. It is not referenced directly to ground. However the building conduit that contains the 480vac feed will act as an AC capacitor allowing the AC component of the 550vdc to seek an asymmetrical ground. This is a return path for current in the event you come into contact with the 550vdc and ground. 550vdc is not forgiving when physical contact is made.

A large majority of electrical accidents could of been prevented with the proper use of Tag and Lockout. Tag and Lockout Procedures are company policy.

Comment: *It is also common sense. Use it.*

63.2 SW and HW Tools Available for Troubleshooting

DIRECTION 46-018303 REVISION

- Relay Control Bd. interconnects
- PDU Transformer Fuse Panel 480VAC fuses that feed the DCRGS
- 550VDV Backup Contactor Interlocks
- Rotor Control and 550V Monitor Functional Interconnect.
- Mains Under voltage and 550V Monitor
- High Voltage Test Mode 550VDC Control
- Gentry I/O schematics for the 24vdc control side of 550vdc command.
- CTVRC Bd. schematics Use for 550vdc monitoring

Relay Control Bd. schematics - Use for the 24vdc pilot relay and 120vac side of 550vdc command.

63.3 Problem Determination

63.3.1 Error 183182

SAMPLE OF ERROR

```
Suite: CCT2 Host: OBC Proc: Generator Error: 183182
File: Method: No Method Line: 615
Function: SYSTEM POWER CONTROL : 550VDC Control
Scan: 65001/1/0 Type: Axial
Exception Level: Pri/Most Severe Time: 06:54:22:395 Log Series: 212
I/O board fault detected.
I/O board pilot relays are closed. PDU pilot relay not energized.
BCSTAT = 0, 550 VDC not active.
```

EXPLANATION OF ERROR 183182

Refer to the Gentry I/O schematics (sheet 7). BCSTAT is an indicator of current flowing through the input of U287 (opto-isolator). Reference Direction 46-018318 550vdc Backup Contactor Interlocks (skinny yellow paper spiral bound notebook) to determine current flow path. Reference Direction 46-018318 550vdc Backup Contactor Interlocks (skinny yellow paper spiral bound notebook) and follow “550man” and “550vdc overdrive enable” to trouble-shoot if contactor will not come up in manual mode.

A LITTLE MORE EXPLANATION

U287 opto-isolator checks the current in the pilot relay loop ONLY. The current in the actual DCRGS contactor is not monitored. +24vdc comes from the Relay Control Board to the Gentry I/O, then back out to the Relay Control Board to pull in the Back-up Contactor Pilot relays.

OBJECTIVE WHILE TROUBLE-SHOOTING

Find out why U287 opto-isolator IS NOT detecting current in the pilot relay loop when the firmware has enabled K251 and K245 (Gentry I/O bd).

63.3.2 Error 183158

```
Suite: CCT2 Host: OBC Proc: Generator 183158
Function: SYSTEM MONITORING : 550V Monitor
Scan: 4055/2/1 Type: Static
Exception Level: Pri/Most Severe Time: 07:50:48:810 Log Series: 32
DC Rail not sensed in 2 seconds.
Pilot relays on I/O board closed, pilot relay in PDU energized.
Voltage sensed by CTVRC board = 1 V. 0VDC SOMETIMES
```

EXPLANATION OF ERROR 183158

The 550vdc generated by the DCRGS was not detected as being greater than SPEC within 2 seconds.

SPECIAL NOTE

If accompanied with error 184186 (no 550vdc) suspect 480vac feed fuses.

OBJECTIVE WHILE TROUBLE-SHOOTING

There are three scenarios why the 550VDC wasn't detected as being greater than SPEC within 2 seconds.

- **Scenario 1:**
  DCRGS 480vac Backup contactor not commanded to close, or being commanded to open when 550vdc is needed.
- **Scenario 2:**
  550vdc not up at all, or not enough.
- **Scenario 3:**
  550vdc monitoring circuit not working.

TROUBLE-SHOOTING

Error 183158 indicates that U287 opto-isolator (on the Gentry I/O Bd.) has detected current going to the DCRGS Back-up Contactor pilot relays (on the Relay Control Bd in the PDU), but no 550vdc has resulted within 2 seconds.

Error 183158 can be divided into three areas, control problem, DCRGS/Load problem, or a 550vdc monitor circuitry problem.

The Back Up Contactor command control circuitry can be broken down into two circuits. The primary side which involves the Gentry I/O and the Back-up Contactor pilot relays on the PDU Relay Control Bd. The primary side is a 24vdc control signal. Because the Gentry I/O has detected current in this part of the circuitry, the problem most likely is not with this part of the control circuitry

The so called secondary side would be the 120vac to the DCRGS Backup Contactor coil.

The monitor circuit for the 550vdc is on the CTVRC Power Amp, and OBC CTVRC Control Bd.

63.3.3 ERROR 183158

- **Scenario 1:**
  DCRGS 480vac Backup contactor not receiving the command to close, or being commanded to open when 550vdc is needed.
- **Contactor comes in and drops out with an applications plasma command**.

  Bring up the 550vdc through the plasma screen. (Diagnostics, or tube warm-up, etc.). If the 550vdc contactor come in and drops out right away, the problem is most likely with the DCRGS not delivering 550vdc or the monitoring circuit not detecting 550vdc. The 550vdc contactor will be commanded to drop out after 2 seconds if the firmware doesn't detect voltage. If the contactor does drop out after a few seconds, the first thing to check are the fuses on the CTVRC Power Amp (550vdc input to the monitoring circuitry). If these are blown, the monitoring circuit on the CTVRC will not detect voltage, therefore producing this error message.

  Try bringing up the 550vdc using diagnostics (Diags/X-Ray Generation/550vdc/Back-up contactor) and command a long enough time to measure the 550vdc. (Diags will allow the 550vdc to stay up if not detected by the ctvrc.)

  If the contactor doesn't come in, proceed to "*CONTACTOR DOESN'T COME IN AT ALL WITH A PLASMA COMMAND*".

  If 550vdc can be produced proceed to "*CONTACTOR COMES IN AND 550VDC IS PRESENT*".

  If 550vdc is not produced with the contactor in, proceed to "*CONTACTOR COMES IN AND NO 550VDC*".
- **Contactor doesn't come in at all with a plasma command**.

  If the DCRGS 550vdc contactor doesn't come in at all. Since U287 has detected current in the "primary" (24vdc control), the problem most likely is with the "secondary" (120vac) side of the control circuitry. This scenario can be verified by using the test switch on the PDU Relay

Control Bd. If the “secondary” is the problem, then using the test switch will not bring in the contactor. Items to check are:

- 120vac to the Relay Control Bd.
- All interlocks: Key switch
- X-ray/Drives are on
- 550vdc enable on the Gantry Safety Switch
- Gantry cover interlocks
- Relay Control Bd

• **Contactor doesn’t come in at all with a plasma command** and/but **will come in with the relay control bd test switch**.

This is an unlikely scenario because it indicates that the 24vdc primary command circuitry is not working. If this circuitry is not working, U287 would detect no current and flag error 183182 “PDU pilot relay not energized”. If this condition is verified to be true refer to direction 46-018318 550VDC BACKUP CONTACTOR INTERLOCKS for the schematics.

• **Contactor doesn’t come in at all with a plasma command** and/but **will not come in with the relay control bd test switch**.

Use the Manual 550vdc switch on the Relay Control Bd. to bring up 550vdc manually. If the DCRGS 480vac Backup contactor does not come in the problem is most likely the 120vac “secondary”. Items to check are:

- 120vac to the Relay Control Bd.
- All interlocks: Key switch
- X-ray/Drives are on
- 550vdc enable on the Gantry Safety Switch
- Gantry cover interlocks
- Relay Control Bd

• **Contactor comes in and no 550vdc**

Note: Must be done using diags or with the test switch on the Relay Control bd, otherwise firmware will command to turn contactor off if 550vdc is not detected.)

Use the 550vdc test switch on the Relay Control Bd, if the 550vdc contactor comes in and stays in but no 550vdc is produced at the output of the DCRGS, then the problem is most likely no power into the DCRGS or a DCRGS failure, or the load (gantry components) is loading down the DCRGS. Items to check are:

- 480vac power into the DCRGS.
- Fuses that feed the DCRGS
- Disconnect the load at TS1, TS2 and retry

• **Scenario 3:**
550vdc monitoring circuit not working.

• **Contactor comes in and 550vdc is present**

Note: Use test switch on Relay Control Bd.

This scenario indicates that the monitoring circuitry is not working. The first thing to check are the 550vdc fuses on the CTVRC power amp. The second thing is to check the test points on the OBC CTVRC Control board. These are the test points that firmware monitors for 550vdc within two seconds.

63.3.4 Error 184186 (SRU Indicates Mains Low)

```
764841201 1 1 Mon Mar 28 07:53:21 1994 184186
GECT STC Scan Controls
MainsMonitor.c 1.8 No Method 210
```

```
Function: SYSTEM MONITORING : Mains Undervoltage
Scan: 4055/2/0 Type: None/Unknown
Exception Level: Pri/Soft Time: 07:53:19:370 Log Series: 24
SRU indicates mains low.
HW ERROR: Power input problem.
EN 341
```

EXPLANATION OF ERROR 184186

Refer to Direction 46-018318 Mains Undervoltage for the monitor part. Refer to direction 46-018303 PDU_TRANSFORMER_FUSE_PANEL. The Mains Undervoltage monitor the voltage at PDU fuses A5F10, F11, F12. If this voltage is low, or if a phase is missing (i.e. blown fuse), the system will report this error.

SPECIAL NOTE

If accompanied with error 183158 (no 550vdc) suspect 480vac feed fuses.

OBJECTIVE WHILE TROUBLESHOOTING

Find out why the STC Axial board detected a “Mains Undervoltage” from the DCRGS.

TROUBLE-SHOOTING

Refer to schematics 46-018303), this is the schematic for the small 1/10 amp fuses that feed the monitoring circuit. Schematic 46-018303 this is the schematic of the 80amp fuses that feed the DCRGS. Mains Undervoltage in direction 46-018318 will give the schematic for the interconnects to the Axial bd.

63.3.5 Resistance Chart

63.3.5.1 DCRGS Side of TS1 and TS2 (Slip Ring Load Connected)

Forward bias (red meter lead on TS1)

- TS1 to TS2 227 ohm

Reverse bias (black meter lead on TS1)

- TS2 to TS1 180 ohm

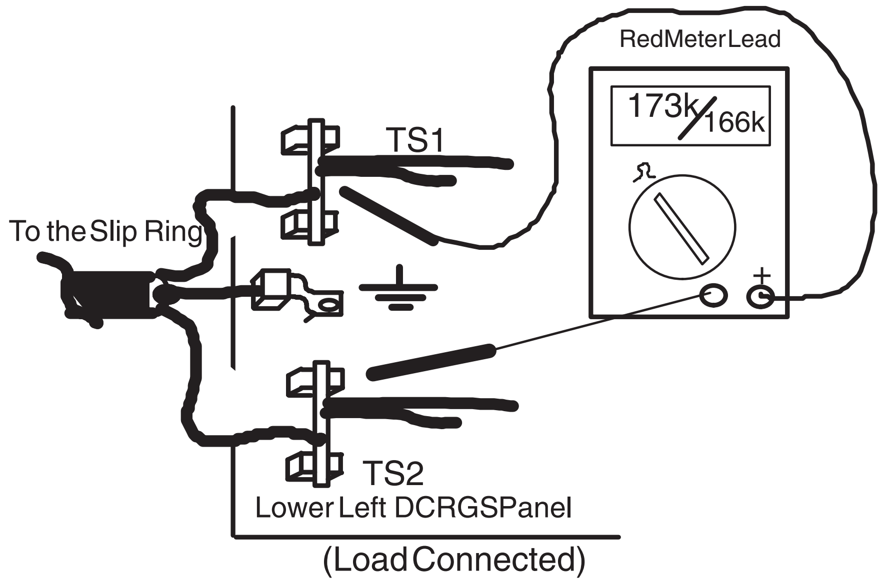


Figure 9-41 DCRGS Resistance Values between TS1 and TS2 (Load Connected)

Forward bias (red lead on TS2)

- TS1 to gnd173k
- TS2 to gnd166k

Reverse bias (black lead on TS2)

- TS1 to gnd 166k
- TS2 to gnd 173k

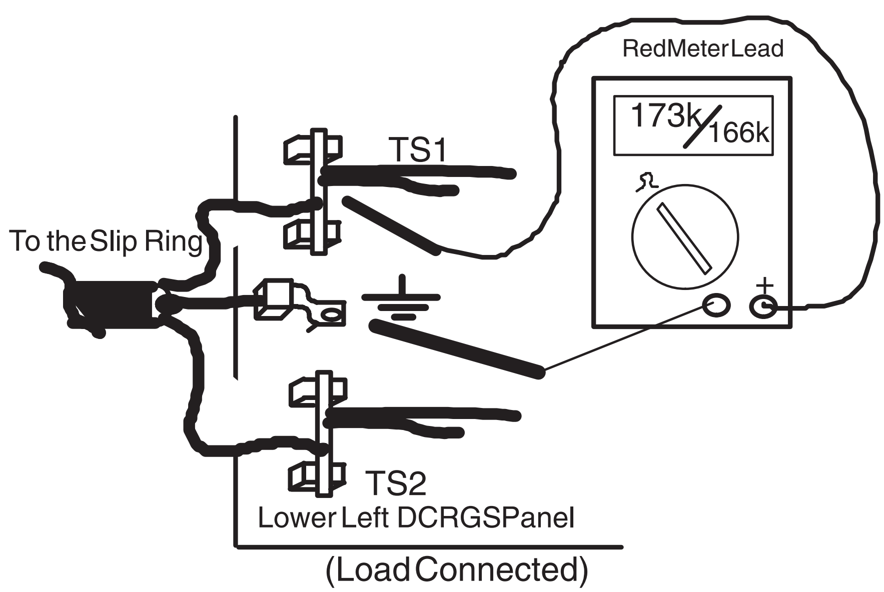


Figure 9-42 DCRGS Resistance Values between TS1 and GND (Load Connected).

63.3.5.2 DCRGS Side of TS1 and TS2 with slip ring load disconnected

1c) Forward bias (red meter lead on TS1)

- TS1 to TS2199 ohm

Reverse bias (black meter lead on TS1)

- TS1 to TS2 195 ohm

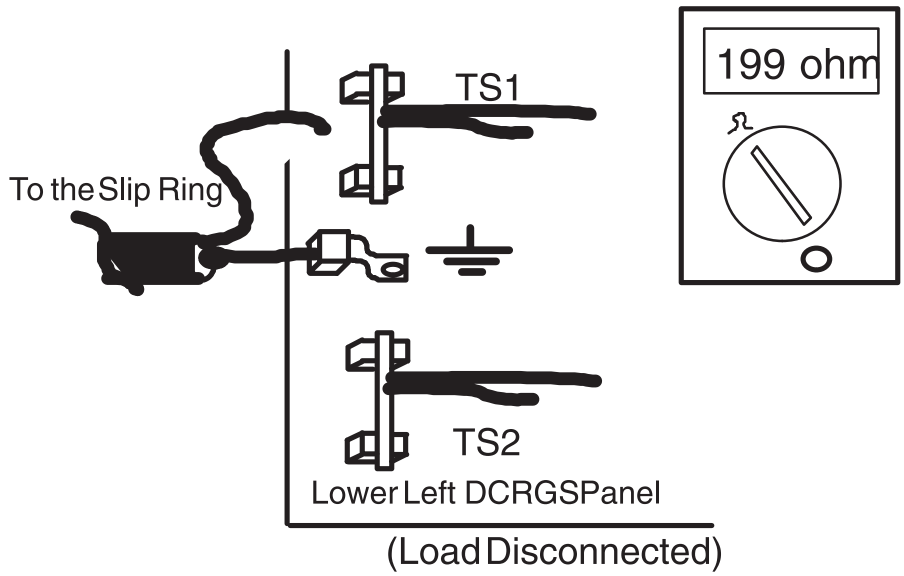


Figure 9-43 DCRGS Resistance Values between TS1 & TS2 (Load Disconnected).

Forward bias (red meter lead on TS1/TS2)

- TS1 to gnd500k
- TS2 to gnd500k

Reverse bias (black meter lead on TS1/TS2)

- TS1 to gnd 170.4k
- TS2 to gnd 170.4k

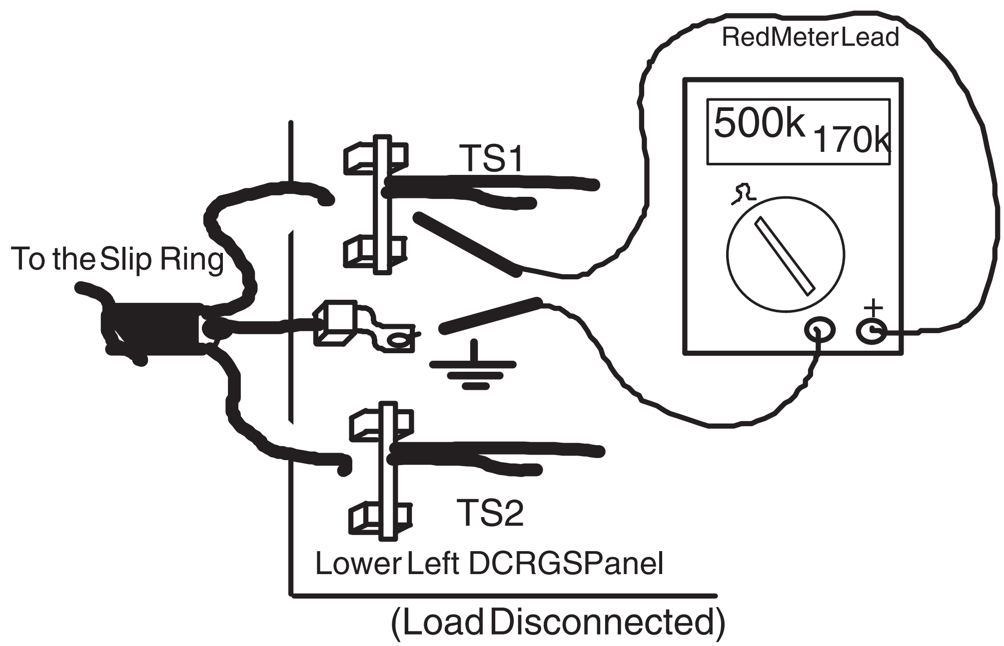


Figure 9-44 DCRGS Resistance Values between TS1 & GND (Load Disconnected).

63.3.5.3 Load Resistance from the PDU Load Disconnected

(i.e. with the big black and red wire disconnected and measuring the resistance of the big red and black wire.)

Forward Bias (with the smaller chassis ground connected)	Reverse Bias
• red 550vdc to blk 550vdc = 18k (approx 20 sec. to stabilize)	• 18k (4.7k after 4 minutes)

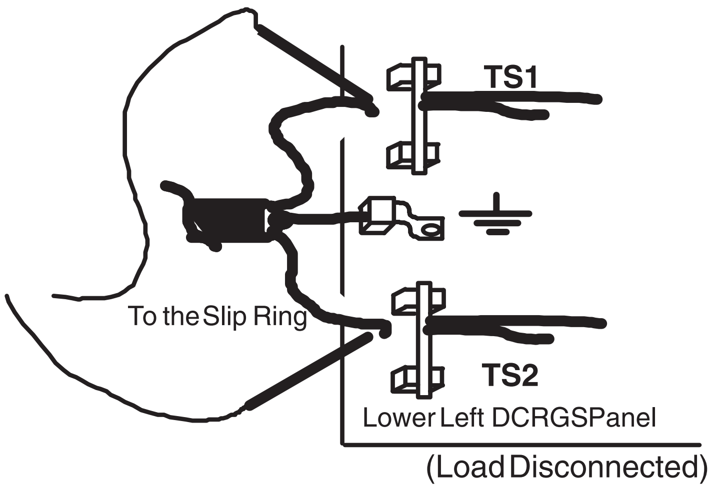


Figure 9-45 PDU Resistance Values between TS1 and TS2 (Load Disconnected).

Forward Bias (with the smaller chassis ground connected)	Reverse Bias
• red 550vdc to ground = 255k	• 255k
• blk 550vdc to ground = 255k	• 255k

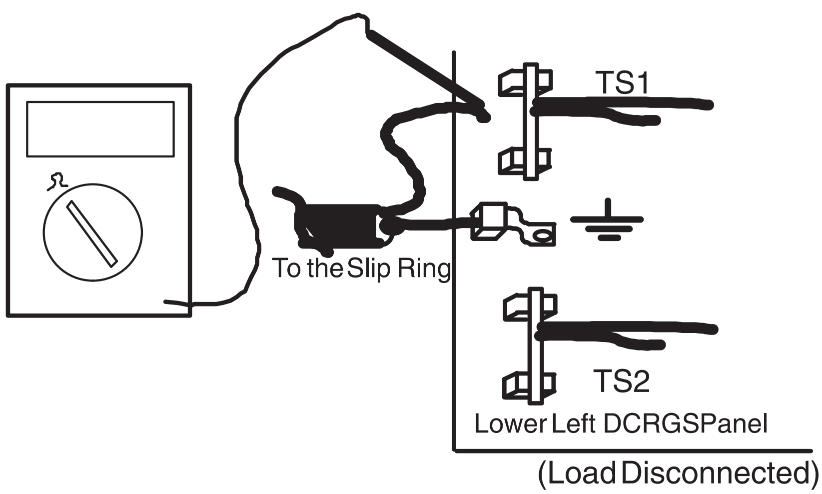


Figure 9-46 PDU Resistance Values between TS1 and GND (Load Disconnected).

Note: With the ground wire (shield) disconnected, GROUND TO GROUND = 0.5 OHM (meter lead resistance).

With the ground wire (shield) disconnected, 2a and 2b reading are the same.

63.3.5.4 Output of Contactor (DCRGS side)

Forward Bias (red lead on T1)

- (2) T1 to (4) T2 = 2.4meg
- (2) T1 to (6) T3 = 2.4meg

Reverse Bias (black lead on T1)

- 2.4meg
- 2.4meg

Forward Bias (red lead on T2)

- (4) T2 to (6) T3 = 2.4meg

Reverse Bias (black lead on T2)

- 2.4meg

Forward Bias (red lead on T1, 2, and 3)

- (2) T1 to chassis ground = 1.1meg
- (4) T2 to chassis ground = 1.1meg
- (6) T3 to chassis ground = 1.1meg

Reverse Bias (black lead on T1, 2 and 3)

- 1.1meg
- 1.1meg
- 1.1meg

63.3.5.5 Input to Contactor (Wall power side)

Forward Bias (red lead on L1)

- (1) L1 to (3)L2 = 0.5ohm (meter lead resistance)
- (1) L1 to (5)L3 = 0.5ohm (meter lead resistance)

Reverse Bias (black lead on L1)

- 0.5ohm (meter lead resistance)
- 0.5ohm (meter lead resistance)

Forward Bias (red lead on L2)

- (3)L2 to (5)L3 = 0.5ohm (meter lead resistance)

Reverse Bias (black lead on L2)

- 0.5ohm (meter lead resistance)

Forward Bias (red lead onLT1, 2, and 3)

- (1) L1 to chassis ground = 1.7meg
- (3) L2 to chassis ground = 1.7meg
- (5) L3 to chassis ground = 1.7meg

Reverse Bias (black lead on L1, 2 and 3)

- 1.7meg
- 1.7meg
- 1.7meg

63.3.6 Resistance on the Ring Itself

Load disconnected from the DCRGS at the DCRGS

Note: Measure on the inside of the slip ring.

Forward bias (red lead on ring 7)

- ring 7 to 9 = 27.8meg
- ring 7 to gantry ground = 71 ohm

Reverse bias (black lead on ring 7)

- ring 7 to 9 = 4.38meg
- ring 7 to gantry ground = 71 ohm

Forward bias (red lead on ring 7)

- ring 7 to 10 = 71 ohm
- ring 7 to 11 = 71 ohm

Reverse bias (black lead on ring 7)

- ring 7 to 10 = 71 ohm
- ring 7 to 11 = 71 ohm

Forward bias (red lead on ring 7)

- ring 7 to 13 = 254k
- ring 7 to 14 = 254k

Reverse bias (black lead on ring 7)

- ring 7 to 13 = 254k
- ring 7 to 14 = 254k

Forward bias (red lead on ring 8)

- ring 8 to gantry ground = open
- ring 8 is open to all other rings(11,12,13, and 14)

Reverse bias (black lead on ring 8)

- ring 9 to gantry ground = open

• ring 8 is open to all other rings(11,12,13, and 14)	• (reverse bias)
Forward bias (red lead on ring 9)	**Reverse bias (black lead on ring 9)**
• ring 9 to gantry ground = 4.38meg	• ring 9 to gantry ground = 27meg
Forward bias (red lead on ring 10)	**Reverse bias (black lead on ring 10)**
• ring 10 to gantry ground = 0.5 ohm (lead resistance)	
• ring 10 to gantry ground = 0.5 ohm (lead resistance)	• (reverse bias)
Forward bias (red lead on ring 11)	**Reverse bias (black lead on ring 11)**
• ring 11 to 10 = 0.5 ohm (lead resistance)	• ring 11 to 10 = 0.5 ohm (lead resistance)
• ring 11 to gantry ground = 0.5 ohm (lead resistance)	• ring 11 to gantry ground = 0.5 ohm (lead resistance) (reverse bias)
Forward bias (red lead on ring 9)	**Reverse bias (black lead on ring 9)**
• 9 to 11,12,13,14 = 4meg	• 9 to 11,12,13,14 = 28meg

63.3.7 Control Board LEDs

LED STATUS

DCRGS Status	Power On	Intl Open	Cap Unbalanced	O/C Under volt	Phase Loss
Power first applied to DCRGS after having been off.	**ON**	**-------------------- Can be any condition ------------------**			
480vac applied and 550vdc on	**ON**	off	off	off	off
550vdc turned off (120vac power still applied to control board)	**ON**	off	off	off	**ON**

Figure 9-47 Control Board LEDs

Section 64.0 Performix X-Ray Tube

64.1 Performix Tube Theory of Operation

The Performix Tube uses a three-phase stator, which requires a HEMRC (High Efficiency Motor Rotor Controller) assembly and HEMRC Control board instead of the CTVRC Control board and assembly.

64.2 HEMRC Control Board (HCB)

The HEMRC Control board replaces the CTVRC Control board (2138293 or 46-288858G1) in systems that use the Performix tube.

The HEMRC Control board, (2179860) resides in slot A2 of the OBC (CT2 A2 A5 A1). The HEMRC Control board performs three main functions. It provides:

- The interface between the OBC and the HEMRC

  Page 667 has a Functional Interconnect for the hard wire control signals.
- HVDC Bus voltage monitoring
- A CAN **(Controller Area Network)** interface between the OBC and future subsystems

The HEMRC Control board Theory of Operation begins on page 664.

64.3 HEMRC Assembly

The HEMRC Assembly contains an Interface Board, AC Drive, Chopper Resistor Assembly, harness and assorted power supplies. The HEMRC Assembly also contains the Detector Heater, Collimator and Filament Power Supplies, which operate in the same manner as their HSA (CTVRC) counterparts.

The HEMRC Assembly replaces the CTVRC Assembly in systems that use the Performix tube.

The HEMRC Assembly Theory of Operation begins on page 683.

Page 684 contains a block diagram of the HEMRC assembly.

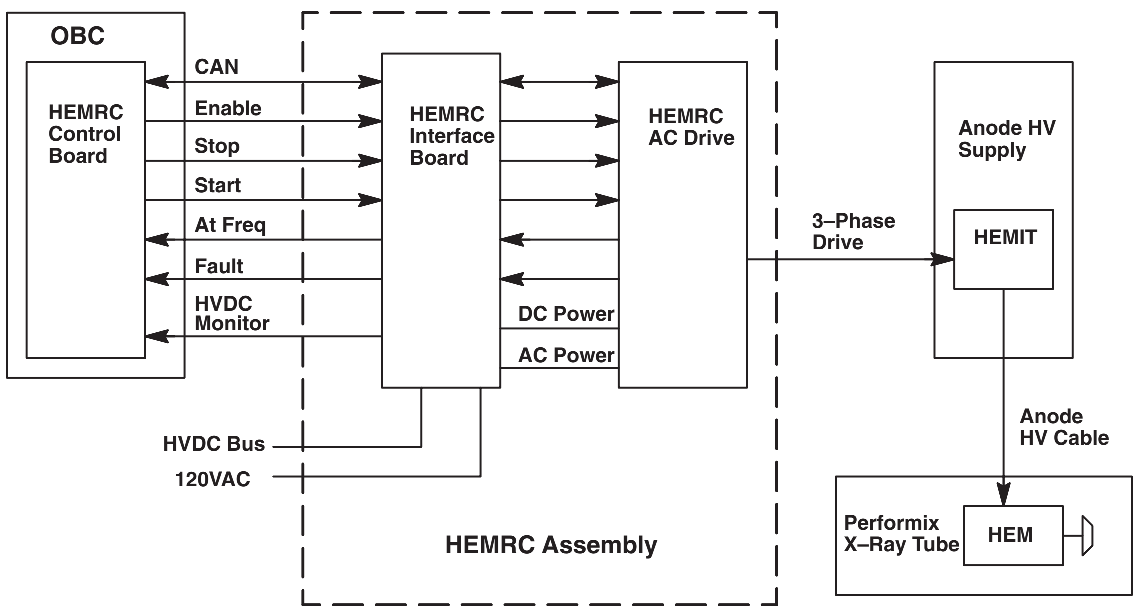


Figure 9-48 HEMRC Hardware Block Diagram

64.4 General HEMRC Function

The Rotor Control function provides three phase motor power to the rotating anode of the Performix x-ray tube. In addition to the HEMRC Control board, the Performix tube requires a HEMRC AC Drive, HEMRC Interface board, HEMIT (High Efficiency Motor Isolation Transformer), and controlling firmware. The rotor control firmware controls the acceleration, run, and deceleration cycles of the rotating anode. The rotor control firmware resides on the OBC CPU board and communicates directly with the HEMRC Control board (HCB), in slot A2. The HEMRC Control board communicates with the HEMRC AC Drive through the HEMRC Interface board (HIF), located on the HEMRC assembly. The drive converts input power to three phase anode HEM (High Efficiency Motor Drive). This motor drive power passes through the HEMIT, located in the Anode High Voltage Transformer, before it reaches the HEM in the Performix Tube.

Refer to Figure 9-49. Control signals travel from the HCB to the HEMRC through the gantry harness. The HEMRC Interface Board routes the signals to the HEMRC AC Drive. The HEMRC AC Drive sends Stator power through a low voltage, shielded stator cable to the HEMIT. The HEMIT provides 1:1 HV isolation and couples the stator power to the Performix tube through the Anode High Voltage cable.

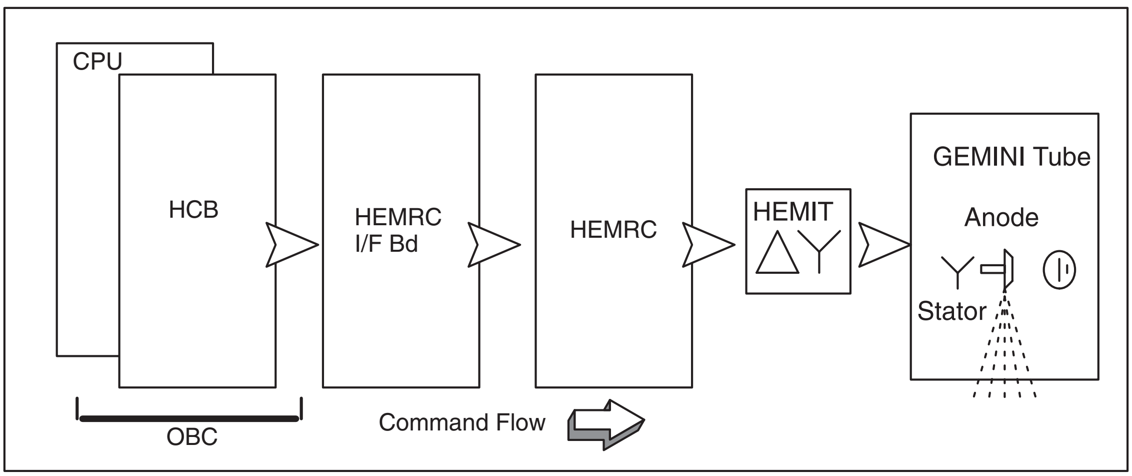


Figure 9-49 HEMRC Functional Command Flow Diagram

64.5 HEMRC Control Board – Theory of Operation

The HEMRC Control board (HCB), located in slot A2 of the OBC, uses the VME bus to communicate with the OBC CPU. The HCB uses a CAN (Controller Area Network) serial interface, called the HCAN (HEMRC CAN), and discrete signals to communicate with the HEMRC. Figure 9-50 on page 667 shows the discrete control signals. The CAN interface and discrete control signals enter the HIF (HEMRC Interface) for distribution to the HEMRC. The HIF also provides an analog feedback voltage to the HEMRC Control board in proportion to the voltage on the CT HVDC bus.

The HCB also contains a separate CAN serial interface, called the GCAN (Gantry CAN), and discrete control signals to communicate with other subsystems in the CT gantry subsystem. The HCB communicates with the HIF and gantry subsystems through shielded cables from the J3 connector of the HEMRC Control board.

Both the HEMRC CAN and the Gantry CAN originate in the HCB. The HCB supplies isolated 12V power (1.2A capability) and Fault circuitry to the GCAN, and accommodates up to six (6) CAN nodes (future) on the network.

Fault detectors and fault status feedback from the HEMRC and Gantry CAN based subsystems (if used) monitor hardware operation. The HCB notifies the CPU when it detects a fault, or a monitored value falls out of tolerance.

64.5.1 VME Interface

The VME interface contains the logic to perform address and data latching, address decoding, and VME handshaking, according to timing specified in PAL documentation, 2147462PDL. All signals pass through the standard VME connector, J1.

Of the seven interrupts defined for the VME bus, the HCB uses level 1, level 2, and level 4 interrupts. Other boards on the OBC also use the level 1 interrupt, which is wire OR-ed on the backplane.

- **IRQ1**: Interrupt level 1 indicates the presence of a hard failure. A fault signal from the HEMRC or the HIV (High Voltage DC Bus Over voltage signal) generates a level 1 interrupt. Firmware can mask a HEMRC fault with the HEMRC_FLT_EN signal, to prevent HCB tie ups.
- **IRQ2**: The HCB uses Interrupt level 2 during HCAN and GCAN communications.
- **IRQ4**: Interrupt level 4 indicates the occurrence of a transition a state change or the presence of a Gantry CAN fault. Firmware can mask the GCAN fault.

64.5.2 Command I/O

In normal operation, the OBC CPU sends state commands through the command registers located at address FFB821H and FFB823H. The OBC CPU also uses the command register located at FFB823H to provide Board Level Diagnostic (BLD) features.

The registers located at FFB829H, FFB82BH, and FFB82DH report Status.

Section 64.5.14 contains the Command Register assignments.

64.5.3 Reset Push-Button

The HCB contains a manual board reset push button. Pushing the on-board reset <u>does not</u> have the same effect as receiving a RACKRST or SYSRST from the VME. The RACKRST and SYSRST also reset the GCAN.

64.5.4 Clocks

U2, U3, and U4 on the HCB generate 244Hz and 15.26Hz clocks from the 16MHz clock. The HVDC Bus monitoring circuit uses the 244Hz clock and the HEMRC CAN Interface circuit uses the 15.26 Hz clock.

64.5.5 Voltage Reference

The Voltage Reference circuitry produces a test reference voltage used to test the HVDC Bus monitoring circuit during board level diagnostic (BLD) tests.

64.5.6 HEMRC CAN (HCAN)

The HCB uses the 82527 CAN protocol controller (U54) and CAN bus interface circuitry to communicate with the HEMRC Assembly. The HCB communicates with the OBC CPU through the address and data bus, R_W*, and HEMRC_CAN_CS* signals. The 82527 communicates with the HEMRC through the HCAN bus interface, connected to the TX0 and RX0 pins. The 82527 interrupts the OBC CPU with the HEMRC_CAN_IRQ* signal to indicate a status change of the 82527. The HEMRC_CAN_IRQ* generates a level 2 interrupt to the OBC CPU through the status register located at FFB82DH, bit 0.

The 82527 uses the 16MHz clock for timing. The board RESET* signal resets the 82527.

U66 and U67 optically isolate the HEMRC CAN bus from the HEMRC Control board circuitry. U77, the CAN transceiver chip, resides on the isolated side of the CAN interface. The optically isolated side of the HEMRC CAN receives power from an isolated 12 volt, 125 mA supply located on the HEMRC. VR3 regulates this 12 volt supply down to 5 volts. This isolated 5 volt supply provides power to the isolated side of the HEMRC CAN interface. The HEMRC CAN output signals are HCH and HCL. R94 provides the required CAN bus termination for the HEMRC Control board end of the HEMRC CAN bus.

DS8 illuminates whenever the 82C250 receives data over the HEMRC CAN bus.

64.5.7 OBC to HEMRC Interface Overview

The communications between the OBC and HEMRC consist of:

- Bidirectional CAN serial communications bus: a 125 Kbaud bidirectional serial link, used to convey commands and status information between the HEMRC and OBC
- Fault signal from the HEMRC to OBC
- At-speed signal from the HEMRC to the OBC
- Enable signal from the OBC to the HEMRC
- Three-wire start-stop signal

The opto-isolated Enable, Start, and Stop signals from the OBC to the HEMRC provide a contact closure as an input to the HEMRC. (Figure 9-50 on page 667.) The Enable contacts close electrically to enable the HEMRC, the Start contacts close electrically to start the HEMRC, and the Stop contacts close electrically to enable the HEMRC to run and open electrically to stop the HEMRC. The Enable, Start, and Stop opto-isolators carry 10mA with less than a 3V drop when closed, and withstand 5V when the contacts open.

The fault signal from the HEMRC to the OBC consists of the HEMRC_FLT_NC, HEMRC_FLT_NO, and HEMRC_FLT_SPD_RTN signal wires. Figure 9-50 contains a block diagram of the connections between the HCB and HEMRC. Refer to the schematic for actual component values. Use the components in Figure 9-50 for functional reference only. The circuit uses drives with a normally-open fault contact. If either the HEMRC_FLT_NC signal wires or HEMRC_FLT_NO signal wires open electrically, the HCB generates a fault condition. If the HEMRC_FLT_SPD_RTN signal wires open while the HEMRC_FLT_NC and HEMRC_FLT_NO signal wires are connected, the HCB does NOT sense a HEMRC fault condition. The HEMRC_FLT_SPD_RTN connects to chassis ground to provide a redundant signal return path for the fault signal. In addition, if the HEMRC_FLT_SPD_RTN wire opens, the HCB will not sense an at-speed condition, which indicates the x-ray tube anode failed to reach a safe speed to allow x-ray exposure.

The HEMRC fault feedback circuit uses three signals from the HEMRC (fed back through the HEMRC Interface board), HEMRC_FLT_NC, HEMRC_FLT_NO, and HEMRC_FLT_SPD_RTN. Under a no-fault operating condition, the HEMRC_FLT_NC and HEMRC_FLT_NO signals connect electrically, and the HEMRC_FLT_NO and HEMRC_FLT_SPD_RTN signals do not connect electrically, which creates a logic high signal to the input of U17 pin 1, that indicates a no-fault condition. During a fault condition, no electrical connection exists between the HEMRC_FLT_NC and HEMRC_FLT_NO signals, and the HEMRC_FLT_NO and HEMRC_FLT_SPD_RTN signals connect electrically to create a logic low to the input of U15 pin 1, which indicates a fault condition.

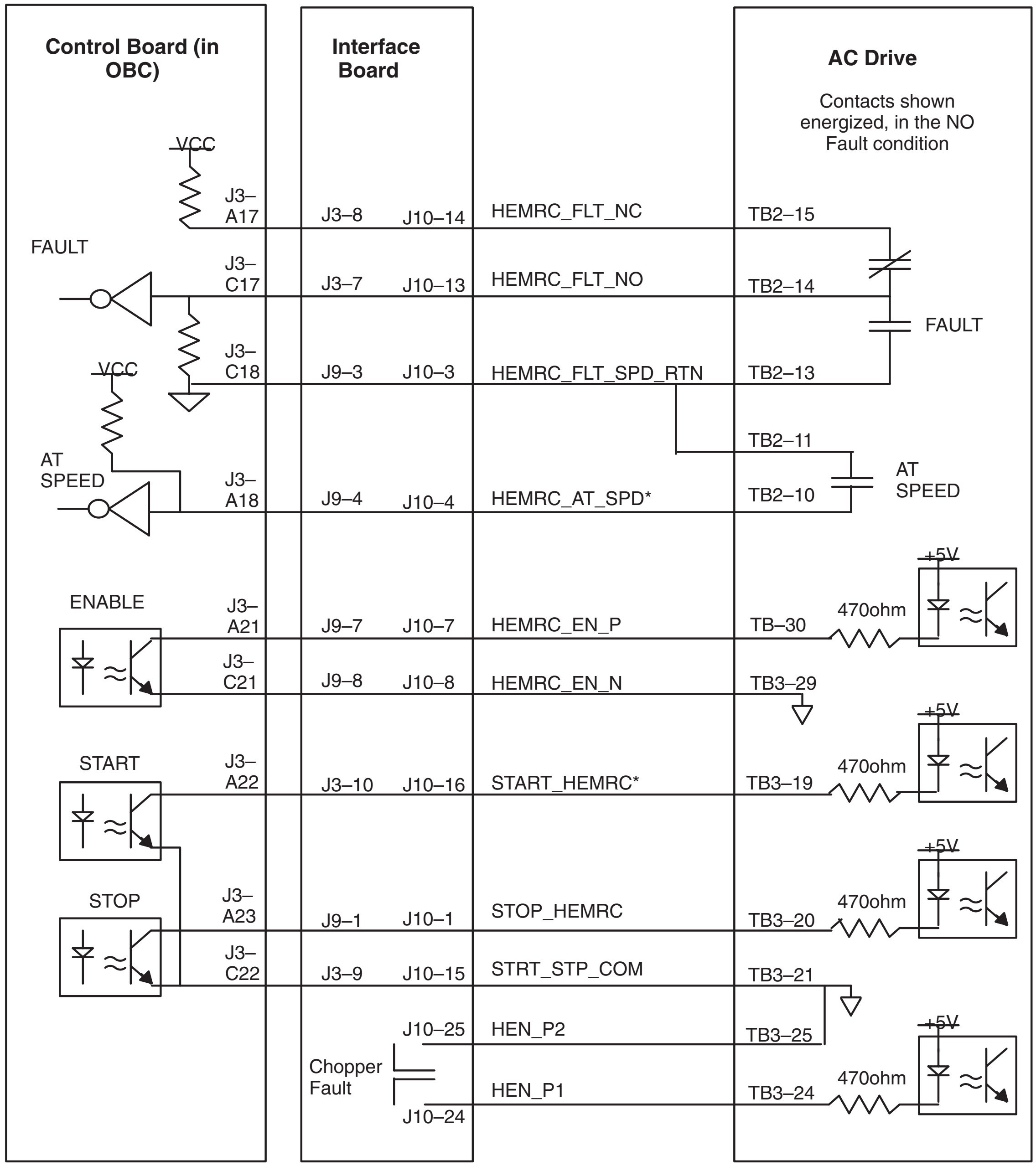


Figure 9-50 Hard wire Control Signals, Functional Interconnect

64.5.8 Fault Circuitry

This feedback uses three signals to allow a broken wire to be detected as a fault condition. If the HEMRC_FLT_NC wire breaks, the input to U15 pin 1 goes low to indicate a fault condition (regardless of the integrity of the two remaining signals). If the HEMRC_FLT_NO wire breaks, the the input to U15 pin 1 goes low to indicate a fault condition (regardless of the integrity of the two remaining signals). If the HEMRC_FLT_SPD_RTN signal wire breaks, the drive uses the remaining HEMRC_FLT_NC and HEMRC_FLT_NO signals to indicate a fault condition. With the three signal design, no possible combination of broken wires could prevent the detection of a HEMRC fault condition.

In a fault condition, U17 pin 2 goes to a logic high, which clears the D flip-flop U20 and causes U20 pin 12 to go high and indicate a fault with the signal HEMRC_FLT. During a fault condition, U41 pin 10 goes to a logic low, which combines with the ROT_EN signal to disable the HEMRC.

When the fault clears, the FLTRST signal resets the HEMRC_FLT.

Under a no-fault condition, the HEMRC enables when the ROT_EN signal and the output U41 pin 10 (no fault) go to a logic high, creating a low impedance between the HEMRC_EN_P and HEMRC_EN_N output signals, which turns the opto-isolator U73 “ON”.

The HEMRC_EN_P and HEMRC_EN_N output signals pass through the HEMRC Interface board on route to the HEMRC. The center of schematic sheet eight contains the HEMRC at-speed indication circuit. When the HEMRC reaches its programmed speed, it closes a contact between the HEMRC_AT _SPD* and HEMRC_FLT_SPD_RTN signals, to create a logic high to U17 pin 4. The next clock pulse clocks this signal into D flip-flop U6, which sends the AT_SPEED signal to a logic high to generate a momentary pulse on the STAT_CHG signal, which in turn sends a level 4 interrupt to the OBC CPU.

Because D flip-flop U6 receives a 15.26Hz clock pulse, the OBC CPU has enough time to respond to the level 4 interrupt and read the status of the HEMRC Control board before the status changes again. This clocking scheme also prevents the generation of simultaneous interrupts. The HEMRC_AT_SPD* signal combines with the INTLK* signal to prevent x-ray exposure from occurring before the HEMRC reaches its pre-programmed speed. The EXPEN signal must equal a logic high before x-ray exposure can occur.

64.5.9 HEMRC Stop and Start

The discrete start and stop signals to the HEMRC are opto-coupled logic signals. The HEMRC_STOP* signal must equal a logic high for the HEMRC to start acceleration, continue acceleration or run. The logic high HEMRC_STOP* signal creates a low impedance between the STOP_HEMRC and STRT_STP_COM output signals, which permits the HEMRC to accelerate or run.

When the HEMRC_START signal goes to a logic high (causing a low impedance between the START_HEMRC* and STRT_STP_COM outputs) and the HEMRC_STOP* signal equals a logic high, the HEMRC begins to accelerate (if it hasn’t already done so). Once acceleration begins, the HEMRC continue to advance along its acceleration profile, or continues to run, regardless of the logic condition of the HEMRC_START signal. The HEMRC begins to decelerate (if it is running) whenever the HEMRC_STOP* signal goes to a logic low.

64.5.10 Gantry CAN

The HEMRC circuitry supports the use of the 82527 CAN protocol controllers (U53 and U62) and CAN bus interface circuitry to communicate with the CAN based gantry subsystems. The Gantry CAN interface uses two CAN protocol controllers on the CAN bus. The 82527s communicate with the OBC CPU through the address and data bus, R_W*, GCAN1_CAN_CS*, and GCAN2_CAN_CS* signals. The 82527s communicate with the gantry subsystems through the CAN bus interface, connected to the TX0 and RX0 pins. A status change of the 82527 CAN causes the GCAN1_IRQ* and GCAN1_IRQ* signals to generate a level 2 interrupt to the OBC CPU, visible at status register location FFB82DH.

The 82527s use the 16MHz clock for timing. The board RESET* signal resets the 82527s.

U62, U63, U54, and U65 optically isolate the Gantry CAN (GCAN) bus from the HEMRC Control board (HBC) circuitry. The CAN transceiver chips, U74 and U75, on the isolated side of the CAN interface, receive power from an isolated five volt supply produced by DC-DC converter U82 on the HEMRC Control board. The GCAN output signals are GCH and GCL. R103 provides the required CAN bus termination for the HEMRC Control board end of the gantry CAN bus when you connect GCR to GCH with the jumper plug on J4 and J5.

When the HEMRC Control board transmits on the gantry CAN bus, DS5 (G1TX) or DS6 (G2TX) illuminates to indicate transmission activity. During reception, DS7 (GRX) illuminates.

The discrete signals for the Gantry CAN interface are GCAN_RST, GCAN_FLT and FAULT2. When GCAN_RESET goes to a logic high, it uses RS485 transceiver (DS3695) U79 to drive GCAN_RST_P high and GCAN_RST_N low.

The Gantry CAN fault feedback circuitry consists of two parts.

- The primary system uses the 82C250 CAN transceiver U72. When a fault condition exists on one of the gantry CAN subsystems, GCAN_FLT_P goes high relative to GCAN_FLT_N, and Rxd and Txd equal a logic zero. This condition causes a logic high on the GCAN_FLT signal which in turn generates a level 4 interrupt whenever the GCAN_FLT_EN signal equals a logic high. DS4 (GFLT) illuminates whenever a gantry CAN fault condition exists.
- The second fault uses a loop through signal of the GCAN_+12V_ISO signal through other modules. This signal enters the HCB at J2-A8 (FAULT_SENSE). If this signal goes low for any reason, the FAULT2 and GCAN_FLT signals go to a logic high. The firmware can also break the loop-through line by driving DRV_GCAN_FLT* to a logic zero, which creates a GCAN_FLT and FAULT2 signal that firmware can read back.

The GCAN_FLTRST signal resets both kinds of CAN faults. At 3.0 mS must elapse between the release of a Driven GCAN Fault (DRV_GCAN_FLT*) and the Gantry CAN Fault Reset (GCAN_FLTRST), to allow time for the solid state relay to switch.

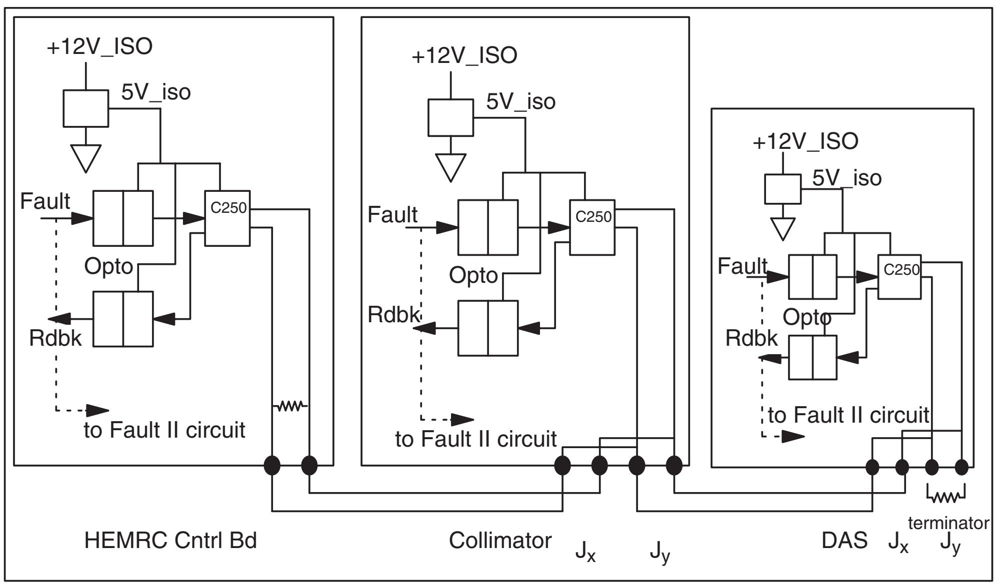


Figure 9-51 GCAN Bus Primary Fault Signal Path

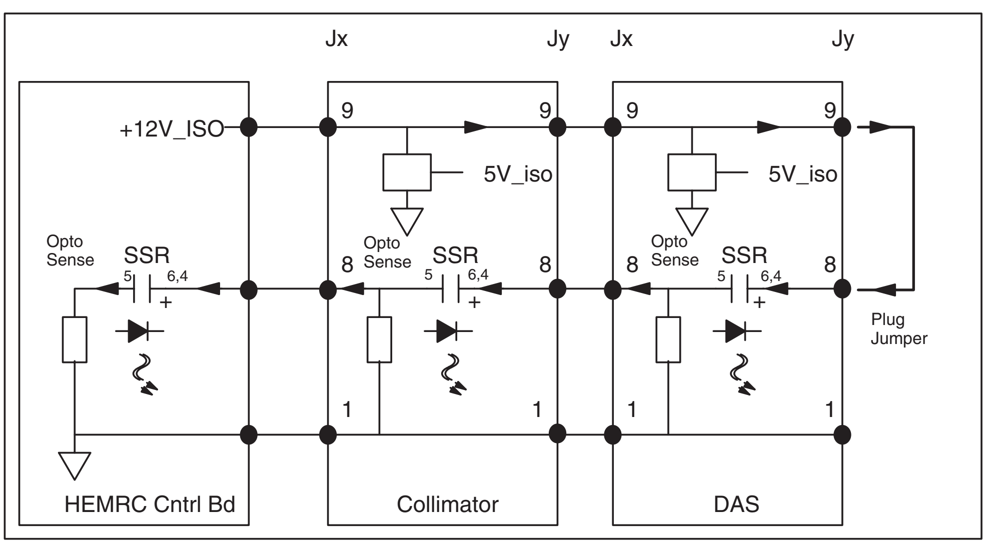


Figure 9-52 GCAN Bus Secondary Fault Signal Path

64.5.11 DC Bus Voltage Monitoring

The system uses the HBC to monitor the HVDC Bus that feeds the HEMRC. The HEMRC Interface board provides the HVDC Bus feedback to the HEMRC Control board (HCB).

The differential amplifier AR4 senses the feedback voltage. The HEMRC Interface board contains the one Meg-ohm of the input impedance. Resistor networks R147 & R148 and R143 & R144 limit the common mode voltage injected at the op-amp. You can monitor the voltage at “DCV” (TP 9); scaled 100V/V. The MUX periodically sends this value to the CPU for it to read.

The circuit compares this voltage to a fixed ~470V limit, and samples the result at a 244 Hz rate. Whenever the result changes state, it generates a level 4 interrupt to signal the CPU that the bus is responding to commands &/or faults. The CPU can then take appropriate action.

The voltage also passes through a filter with an approximately 20 millisecond time constant, for comparison to an upper limit. Jumper JP1, selects the upper limit. Position “A” selects an upper limit of ~670V for use by systems with a DCRGS PDU. Position “B” sets the upper limit to ~800V, for use by systems with an unregulated HVDC Supply in their PDU. The position of jumper JP1 produces a signal, DCR*, which the CPU reads at address FFB829H.

If a failure of the HVDC Supply occurs, the circuit exceeds the upper voltage limit, and generates an abort. This condition produces a level 1 interrupt, described in the following MUX_IRQ section.

64.5.12 MUX_IRQ

The MUX_IRQ function consists of an analog multiplexer, used to feed the HVDC Bus voltage feedback signal and test reference voltage back to the OBC through RC_MUX and the IRQ generation circuits.

You can monitor the multiplexer output at test point “MUX” (TP3). VR2 and AR3 generate the +10V and -10V reference voltages. The CPU reads the value of the +10V reference through the multiplexer. The CPU detects scaling errors in the system by comparing its value to an external reference.

Two fault conditions, High DC Bus Voltage and HEMRC Fault, generate a level 1 interrupt. Firmware can use the HEMRC_FLT_EN signal to mask the HEMRC_FLT signal. The firmware masks the interrupt during power-up reset conditions.

A High DC Bus Voltage fault generates a “KILLBC*” signal that immediately disables the back-up contactor supplying the inverter power. This fault may indicate a loss of control in the DC bus regulator, and the existence of a potential hazard.

When it receives a level 1 interrupt, the CPU interrogates the board status registers to determine what fault occurred, then it disables the HEMRC and resets the interrupt and fault latches while it posts error messages.

Three state transition conditions generate a level 4 interrupt:

- STAT_CHG
- VCHG
- GCAN Fault Firmware can use GCAN_FLT_EN to mask GCAN_FLT

When it receives a level 4 interrupt, the CPU interrogates the board status registers to determine the appropriate action.

A status change in one of the CAN protocol controller devices generates a level 2 interrupt. The HEMRC_CAN_IRQ*, GCAN1_IRQ*, and GCAN2_IRQ* signals indicate a status change in the corresponding CAN protocol control device.

When it receives a level 2 interrupt, the CPU interrogates the register at location FFB82DH to determine the appropriate action.

64.5.13 CAN Loopback

When you place the four position shorting plug in the J5 position, it connects the HEMRC CAN to the external HEMRC CAN bus and the Gantry CAN to the external GCAN bus. When you install the connector, the NORMAL signal equals a logic “1”. Register location 0FFB829 contains the status information.

Move the connector to the J4 position to place the HEMRC Control board in diagnostic CAN mode. This mode connects HEMRC CAN bus output to the GCAN bus on the HERMC Control board, and disconnects the external HERMC CAN bus from the circuit board while leaving the GCAN externally connected. This mode permits the read-back of the HERMC CAN bus output by the Gantry CAN bus and the read-back of the Gantry CAN bus output by the HEMRC CAN. The NOT_NORMAL signal equals a logic “1” when you place the connector in the J4 position. Register location 0FFB829 contains the status information.

The opto-coupler across the HEMRC_ISO_+12V provides read-back to the firmware to assist in troubleshooting an error in the CAN readback circuitry. The firmware only senses the presence or absence of the voltage; it tell whether the voltage falls inside or outside the tolerance.

64.5.14 Memory Maps

ADDRESS	READ/ WRITE	BIT(S)	FUNCTION/ SIGNAL NAME	DESCRIPTION
FFB801	R	All	Insite Info	Board #, character “?” and digit 1 Value = F2H.
FFB803	R	All	Insite Info	Board #, digits 2&3. Value = 17H.
FFB805	R	All	Insite Info	Board #, digits 4&5. Value = 98H.
FFB807	R	All	Insite Info	Board #, digits 6&7. Value = 60H
FFB809	R	All	Insite Info	Board Group #. Value = 01H
FFB80B	R	All	Insite Info	Board Version letter. Example: Value = 42H, ASCII code for “B”. Refer to assembly drawing for specific version information.
FFB80D through FFB81F				Not used.

Table 9-27 Memory Map of Insite Registers

ADDRESS	READ/ WRITE	BIT(S)	FUNCTION/ SIGNAL NAME	DESCRIPTION
FFB821	R/W	7	ROT_EN	A “1” turns the HEMRC control on.
		6	GCAN_RST	Setting this bit to a “1” initiates a reset of the peripherals on the Gantry CAN bus.
		5	INTLK*	Setting this bit to a “1” disables the broken wire interlock to the EXPEN signal. Intended as a diagnostic tool only.
		4	not used	
		3	MUXENA	Analog MUX Selection. MUXENA = “1” selects the analog MUX. MUXENA = “0” is not used but is provided for firm ware compatibility with previous circuit boards and for possible expansion of MUX channels.

Table 9-28 Memory Map of Command Registers

ADDRESS	READ/ WRITE	BIT(S)	FUNCTION/ SIGNAL NAME	DESCRIPTION
		2,1,0	2,1,0	MUXENA concatenated with bits 2,1,0 form the MUX ad dress space. Only the portion of the address space with MUXENA = “1” is used. Bits 2,1,0 select the channel with in the MUX. The four bit concatenated MUX address space and associated signals are listed below. Code Signal Selected x0H MUXENA = “0”, not used x1H MUXENA = “0”, not used x2H MUXENA = “0”, not used x3H MUXENA = “0”, not used x4H MUXENA = “0”, not used x5H MUXENA = “0”, not used x6H MUXENA = “0”, not used x7H MUXENA = “0”, not used x8H DC rail monitor voltage. Scale: 100 V/V x9H Signal Ground. 0 V xAH Signal Ground. 0 V xBH TESTREF analog voltage. DAC output used to test HVDC bus feedback circuitry. Scale: 1V/V. xCH Signal Ground. 0 V xDH Signal Ground. 0 V xEH Signal Ground. 0 V xFH +10 V Reference. Scale: 0.5 V/V

Table 9-28 Memory Map of Command Registers (Continued)

ADDRESS	READ/ WRITE	BIT(S)	FUNCTION/ SIGNAL NAME	DESCRIPTION
FFB823	R/W	7	not used	
		6	HVDC_MON_TST	A “1” injects 98.6% of the TESTREF DAC (B825) output into the HVDC bus monitor.
		5	not used	
		4	DRV_GCAN_FLT*	A low creates a GCAN Fault.
		3	HEMRC_FLT_EN	A “0” disables the HEMRC_FLT signal from generating a level 1 interrupt. A “1” allows the HEMRC_FLT to generate a level 1 interrupt. The status of HEMRC_FLT can be read at status register (B829) regardless of the state of HEMRC_FLT_EN.

Table 9-29 Memory Map of Diagnostic Command Register

ADDRESS	READ/ WRITE	BIT(S)	FUNCTION/ SIGNAL NAME	DESCRIPTION
		2	GCAN_FLT_EN	A “0” disables the GCAN_FLT gantry CAN fault signal from generating a level 4 interrupt. A “1” allows the GCAN_FLT to generate a level 4 interrupt. The status of GCAN_FLT can be read at status register (B829) regard less of the state of GCAN_FLT_EN.
		1	HEMRC_STOP*	A “0” commands the HEMRC to decelerate. A “1” is required to allow the drive to accelerate or run.
		0	HEMRC_START	A “1” commands the HEMRC to accelerate. HEMRC_STOP* must be a “1” for acceleration. This bit does not need to be maintained at “1” to keep the HEMRC running.

Table 9-29 Memory Map of Diagnostic Command Register (Continued)

Note: **Command Register FFB823 is intended for diagnostic use only. Application code must set all bits to “0” before turning the HEMRC on.**

ADDRESS	READ / WRITE	BIT(S)	FUNCTION/ SIGNAL NAME	DESCRIPTION
FFB825	W	All	DAC “A” Data	TESTREF voltage command. 00H - FFH corresponds to 0V – +10V.
FFB827	W	All	not used but available for DAC “B” Data expansion	

Table 9-30 Memory Map of DAC Command Registers

ADDRESS	READ/ WRITE	BIT(S)	FUNCTION/ SIGNAL NAME	DESCRIPTION
FFB829	R	7	FAULT 2	A “1” indicates that the loop back signal has been broken somewhere in the system. (See Figure **XREF**.)
		6	AT_SPEED	A “1” indicates the HEMRC is at or above programmed speed.
		5	DCR*	A “1” indicates jumper JP1 is in the Unregulated HVDC Supply selection state. This changes the limits for the HVDC Bus Overvoltage detector.
		4	GCAN_FLT	A “1” indicates a fault on one or more of the gantry CAN based devices.
		3	ROTINT4	A “1” indicates the HEMRC Control board has issued a level 4 interrupt.

Table 9-31 Memory Map of Status Register and GCAN Fault Reset

ADDRESS	READ/ WRITE	BIT(S)	FUNCTION/ SIGNAL NAME	DESCRIPTION
		2	LOV	A “1” indicates the DC Rail is less than 450V. This bit is continuously updated every 4 msec. A change of state generates a level 4 interrupt.
		1	NORMAL	A “1” indicates that the CAN mode connector is in the normal CAN mode location. This is required for CAN communication to the HEMRC and any other CAN based subsystems on the gantry.
		0	NOT_NORMAL	A “1” indicates that the CAN mode connector is in the diagnostic CAN mode location. In this position, CAN communications are looped back between the HEMRC CAN network and the Gantry CAN network.
FFB829	W	N/A	GCAN_FLTRST	A write to this address causes the gantry CAN fault signal to be reset provided a fault condition no longer exists. Any data value can be used for this write.

Table 9-31 Memory Map of Status Register and GCAN Fault Reset (Continued)

ADDRESS	R/W	BIT(S)	FUNCTION/ SIGNAL NAME	DESCRIPTION
FFB82B	R	7	HIV	A latched “1” indicates an overvoltage was sensed on the HVDC Bus. (if JP1=“A”, DCV > 670 V else if JP1=“B”, DCV > 800V.) This generates a level 1 interrupt and disables the HEMRC and back-up contactor.
		6	not used	
		5	not used	
		4	not used	
		3	not used	
		2	not used	
		1	HEMRC_12V_F LT	A “0” indicates that the HEMRC_ISO_+12V is non-zero. It does not guarantee that it is at +12V.
		0	HEMRC_FLT	A “1” indicates a fault condition exists on the HEMRC. This is latched on the HEMRC.
FFB82B	W	N/A		A write to this address will clear all the above latched status flags, provided the originating fault has cleared.
FFB82D	W	N/A		A write to this address will clear the level 4 interrupt re quest latch.

Table 9-32 Memory Map of Status Register and Fault Reset

ADDRESS	R/W	BIT(S)	FUNCTION/ SIGNAL NAME	DESCRIPTION
FFB82F	W	N/A		A write to this address will clear the level 1 interrupt re quest latch.

Table 9-32 Memory Map of Status Register and Fault Reset (Continued)

ADDRESS	R/W	BIT(S)	FUNCTION/ SIGNAL NAME	DESCRIPTION
FFB82D	R	7	not used	
		6	not used	
		5	not used	
		4	not used	
		3	not used	
		2	GCAN2_IRQ*	A “0” indicates the Gantry CAN protocol controller #2 (82527) is requesting interrupt service.
		1	GCAN1_IRQ*	A “0” indicates the Gantry CAN protocol controller #1 (82527) is requesting interrupt service.
		0	HEMRC_CAN_IRQ*	A “0” indicates the HEMRC CAN protocol controller (82527) is requesting interrupt service.
FFB82D	W	N/A		A write to this address will clear the level 4 interrupt re quest latch.

Table 9-33 Memory Map of CAN Interrupt Status Register and IRQ4 Reset

ADDRESS	R/W	BIT(S)	FUNCTION/ SIGNAL NAME	DESCRIPTION
FFB82F	W	N/A		A write to this address clears the level 1 interrupt request latch.

Table 9-34 Memory Map of IRQ1 Reset

PIN NUMBER	ROW A – SIGNAL NAME	ROW B – SIGNAL NAME	ROW C – SIGNAL NAME
1	D00	BBSY*	D08
2	D01	BCLR*	D09
3	D02	ACFAIL*	D10
4	D03	BG0IN*	D11
5	D04	BG0OUT*	D12
6	D05	BG1IN*	D13
7	D06	BG1OUT*	D14
8	D07	BG2IN*	D15

Table 9-35 Pin Assignments J1/P1 Connector – VME Bus Interface

PIN NUMBER	ROW A – SIGNAL NAME	ROW B – SIGNAL NAME	ROW C – SIGNAL NAME
9	LGND	BG2OUT*	LGND
10	SYSCLK	BG3IN*	SYSFAIL*
11	LGND	BG3OUT*	BERR*
12	DS1*	BR0*	SYSRESET*
13	DS0*	BR1*	LWORD*
14	WRITE*	BR2*	AM5
15	LGND	BR3*	A23
16	DTACK*	AM0	A22
17	LGND	AM1	A21
18	AS*	AM2	A20
19	LGND	AM3	A19
20	IACK*	LGND	A18
21	IACKIN*	SERCLK	A17
22	IACKOUT*	SERDAT*	A16
23	AM4	LGND	A15
24	A07	IRQ7*	A14
25	A06	IRQ6*	A13
26	A05	IRQ5*	A12
27	A04	IRQ4*	A11
28	A03	IRQ3*	A10
29	A02	IRQ2*	A09
30	A01	IRQ1*	A08
31	-12V	+5VSTDBY	+12V
32	+5V	+5V	+5V

Table 9-35 Pin Assignments J1/P1 Connector – VME Bus Interface (Continued)

PIN NUMBER	ROW A – SIGNAL NAME	ROW B – SIGNAL NAME	ROW C – SIGNAL NAME
1	+5LED	+5V	+5V
2	LGND	LGND	LGND
3		<RESERVED>	RACKRST*
4			
5			
6			
7			
8			
9			
10			

Table 9-36 J2/P2 Connector – Interboard Connections

PIN NUMBER	ROW A – SIGNAL NAME	ROW B – SIGNAL NAME	ROW C – SIGNAL NAME
11			
12	LGND	LGND	LGND
13	+5V	+5V	+5V
14			
15			
16			
17			
18			RC-MUX
19			KILLBC*
20			EXPEN
21	EXPCMD		
22	LGND	LGND	LGND
23			VREF
24			DCV
25			
26	SGND	SGND	SGND
27	+15V	+15V	+15V
28	SGND	SGND	SGND
29	-15V	-15V	-15V
30	SGND	SGND	SGND
31	LGND	LGND	LGND
32	+5V	+5V	+5V

Table 9-36 J2/P2 Connector – Interboard Connections (Continued)

PIN NUMBER	ROW A – SIGNAL NAME	ROW B – SIGNAL NAME	ROW C – SIGNAL NAME
1			
2			
3			
4	GCAN_+12V_ISO		IGND
5	GCAN_H	OPEN	GCAN_L
6	Connected to Pins B and C via backplane	Connected to Pin A via back plane	Connected to Pin A via back plane
7			
8	FAULT_SENSE	Connected to Pin A via back plane	Connected to Pin A via back plane
9	GCAN_FLT_P	OPEN	GCAN_FLT_N
10	OPEN	OPEN	OPEN
11	DAS_TRIG+		DAS_TRIG-

Table 9-37 J3/P3 Connector - External Connections

PIN NUMBER	ROW A – SIGNAL NAME	ROW B – SIGNAL NAME	ROW C – SIGNAL NAME
12			
13			
14	GCAN_RST_N		GCAN_RST_P
15	EXP_CMND_N		EXP_CMND_P
16	TRIG_N		TRIG_P
17	HEMRC_FLT_NC		HEMRC_FLT_NO
18	HEMRC_AT_SPD*		HEMRC_FLT_SPD_RTN
19			
20	HEMRC_CAN_H		HEMRC_CAN_L
21	HEMRC_EN_P		HEMRC_EN_N
22	START_HEMRC*		STRT_STP_COM
23	STOP_HEMRC		
24	DCRV-		DCRVM-
25	HEMRC_ISO_+12V		HEMRC_ISO_RTN
26			
27			
28			
29	PGND	PGND	PGND
30	+24V	+24V	+24V
31	+24V	+24V	+24V
32	PGND	PGND	PGND

Table 9-37 J3/P3 Connector - External Connections (Continued)

64.6 HEMRC Error Messages

The HEMRC Control function within the OBC may detect certain error conditions, generally related to communication or functional interfaces to the AC Drive. Many of these messages contain variable fields. (In the following listing of possible error messages a `%d` represents a numeric value, `%b` represents a variable text string, and `%xh` represents a data value.)

MESSAGE NUMBER	MESSAGE TEXT
185500	`HEMRC CAN chip interrupt can not be cleared. Disabling CAN chip.` `Interrupt code: %d  Disable code: %d`
185501	`HEMRC CAN Bus error.` `HEMRC serial communications are down.`
185502	`Software error detected while waiting for HEMRC semaphore.` `HEMRC message not sent.`

Table 9-38 HEMRC Error Messages

MESSAGE NUMBER	MESSAGE TEXT
185503	Software error detected while pending on event flags. CAN message request not processed. VRTX error: *%d*
185504	Software error: Timeout waiting for free HEMRC CAN Buffer. HEMRC message not sent.
185505	Could not send HEMRC message in the allotted time. HEMRC serial link is down. Possible causes: HEMRC CAN jumper, HEMRC fuse, interconnects, CAN devices, ... Retries used: *%d*
185506	Invalid response received when sending HEMRC message. HEMRC message not sent. Retries used: *%d*
185507	Software error: Invalid response received from CAN ISR. HEMRC message not sent. ISR response code: *%d*
185508	Software error: Invalid packet sent to HEMRC. HEMRC error response code: *%d* Service=*%d* Class=*%d* Instance=*%d* Attribute=*%d* Data=*%d*
185509	Software error: Invalid parameter. HEMRC message not sent.
185510	Software error: Parameter mismatch. The parameter value read from the HEMRC does not match the value written. Parameter number: *%d* Expected value: *%d* Actual value: *%d*
185511	HEMRC fault could not be reset.
185512	The HEMRC CAN communications initialization failed.
185513	The rotor configuration file may be incompatible with the present HEMRC firmware version. Rotor operation may be impaired. HEMRC Firmware version: *%d.%02d*
185514	The rotor configuration file may be incompatible with the present HEMRC drive type. Rotor operation may be impaired. HEMRC drive type: *%d*
185515	*%b* error: Reset state found active.
185516	HEMRC CAN chip
185517	Gantry CAN chip #1
185518	Gantry CAN chip #2
185519	Hardware error: *%b* control line as read from the HEMRC Control Board in unexpected state *%d*, expected state *%d*.
185520	Hardware error: *%b* control line as read from the HEMRC Drive was found in unexpected state *%d*, expected state *%d*.
185521	“Enable”
185522	“Start” ”
185523	“Run/Stop”

Table 9-38 HEMRC Error Messages (Continued)

MESSAGE NUMBER	MESSAGE TEXT
185524	The HEMRC is not running after being enabled. Unknown source of failure. HEMRC status: *0x%06x*
185525	Software error detected while waiting for HEMRC semaphore. CAN loopback test not run. VRTX error: *%d*
185526	HEMRC CAN Bus Error. An abnormal number of errors have occurred on the HEMRC CAN bus. Check diagnostic jumper on HEMRC board and wire connections.
185527	CAN Loopback Test Aborted. Message Transmit Error Message was not transmitted successfully from *%b* to *%b*. TxOk signal did not go high.
185528	CAN Loopback Test Aborted. Message Receive Error Message sent from *%b* was not successfully received by *%b*. New Data signal did not go high.
185529	CAN Loopback Test Aborted. *%b* received message successfully, but did not generate interrupt.
185530	CAN Loopback Test Identifier Error. *%d* of the transmissions from *%b* to *%b* resulted in conflicting message Identifiers.
185531	CAN Loopback Test Data Length Code Error. *%d* of the transmissions from *%b* to *%b* resulted in conflicting values for the data length code segment of the Message Configuration Register.
185532	CAN Loopback Test Message Lost Error. Lost *%d* of the messages transmitted from *%b* to *%b*.
185533	CAN Loopback Test Error. *%d* of the bytes transmitted from *%b* to *%b* contained an error.
185534	CAN Loopback Test Error. *%d* messages transmitted by *%b*, but *%b* only received *%d* messages.
185535	CAN Loopback Test Error. *%d* packets sent from *%b* to *%b* contained errors.
185536	Firmware Error: Unable to install interrupt handler.
185537	CAN Bus Error. An abnormal number of errors have occurred on the CAN bus. Error detected by *%b*.
185538	Firmware Error: Bus Fault Flag is set, but none of the CAN chips are bus off.
185539	Firmware Error: Unable to post HEMRC message semaphore.
185540	Firmware Error: Bad interrupt code found in interrupt register during loopback test. Interrupt Code: *%d*

Table 9-38 HEMRC Error Messages (Continued)

MESSAGE NUMBER	MESSAGE TEXT
185541	Hardware Error: Interrupts are occurring but there are no interrupt codes in any of the CAN chip interrupt registers.
185542	CAN Warning Status. An abnormal number of errors have occurred on the CAN bus. Error detected by *%b*. Status Register Value: *%xh*
185543	Hardware Error: *%b* chip interrupt cannot be cleared. Disabling chip. Interrupt Code: *%d*
185544	Error: The loopback test found the first error while sending a message from *%b* to *%b*. Expected Value: *%d* Actual Value: *%d*
185545	Hardware Error: Message Object 15 generated an interrupt, but Interrupt Pending was not set.
185546	The HEMRC is not running after being disabled. HEMRC status: *0x%06x*
185547	Hardware error: The At Frequency line from the HEMRC is stuck high. HEMRC Control Board latch address: 0xFFB829 Bit: D6 HEMRC status: OFF
185548	The HEMRC operating frequency is below the minimum value. Actual frequency: *%d.%03d* Minimum frequency limit: *%d.%03d* At Frequency signal: *%b* Expected: *%b* Rotor State = *%b*
185549	The At Frequency signal does not agree with the drive output frequency. Possible causes: Interconnection, HEMRC drive, HEMRC control board. Drive freq: *%d.%03d* Minimum freq. limit: *%d.%03d* At Frequency signal: *%b* Expected: *%b* Rotor State = *%b*
185550	Active
185551	Inactive
185552	The CAN test jumper on the HEMRC Control Board is in the wrong position for this test. Place the jumper in the diagnostic position and rerun test. HEMRC address: 0xFFB829 Bit: D0
185553	The HEMRC drive detected a Line Loss. This occurs when the input power to the drive falls below 85%% of the nominal Bus voltage. Possible causes: X-Ray tube stator, HEM-IT, HEMRC power supply, ... Alarm status: *%xH*
185554	The HEMRC drive is being re-initialized due to the detection of error *F%d*. Status: *%d* (1=OK)
185555	The CAN test jumper on the HEMRC Control Board is in the DIAGNOSTIC position. Place jumper in the NORMAL mode in order to scan.

Table 9-38 HEMRC Error Messages (Continued)

MESSAGE NUMBER	MESSAGE TEXT
185556	The CAN test jumper on the HEMRC Control Board is missing. Place jumper in the NORMAL mode in order to scan.

Table 9-38 HEMRC Error Messages (Continued)

64.6.1 Switches, Test Points, LEDs, Jumpers and Adjustments

SWITCHES

SW #1	LABEL	DESCRIPTION
S1	RESET	(Mom.) Resets all command, fault and interrupt latches on this board and also creates a GCAN_RESET signal which is sent to downstream controllers via the control interface bus connections.

Table 9-39 HEMRC Switches

TP#	Color	Label	Description
TP1	Black	LGND	Logic ground
TP2	Red	+5V	+5V supply voltage
TP3	Yellow	MUX	Analog signal as selected by the muxes
TP4	Red	+15V	+15V supply voltage
TP5	White	-15V	-15V supply voltage
TP6	Yellow	+10V	+10V Reference
TP7	Yellow	DCV	DC rail monitor voltage. Scale: 100V/V
TP8	Black	SGND	Signal ground

Table 9-40 HEMRC Test Points

LED	Color	Label	Description
DS1	Yellow	LORPM	HEMRC output frequency below programmed threshold
DS2	Yellow	LOV	DC Rail less than 470V
DS3	Red	HIV	DC Rail overvoltage (>670V) detected
DS4	Red	GFLT	Fault on GCAN based subsystem
DS5	Green	G1TX	GCAN1 transmitting
DS6	Green	G2TX	GCAN2 transmitting
DS7	Green	GRX	GCAN receiving
DS8	Green	HRX	HEMRC CAN receiving
DS9	Red	HFLT	Fault on the HEMRC
DS10	Green	—	General – Function defined by firmware
DS11	Green	—	General – Function defined by firmware
DS12	Green	—	General – Function defined by firmware
DS13	Green	—	General – Function defined by firmware

Table 9-41 HEMRC LEDS

LED	Color	Label	Description
DS14	Green	—	General – Function defined by firmware
DS15	Green	—	General – Function defined by firmware
DS16	Green	—	General – Function defined by firmware
DS17	Green	—	General – Function defined by firmware
DS300	Green	G12V	GCAN_+12V_ISO present

Table 9-41 HEMRC LEDS (Continued)

JUMPERS

JP#	Label	Description
JP1	Position A	Selects voltage limits for systems with a DCRGS PDU. (Default shipping position.)
JP1	Position B	Selects voltage limits for systems with an unregulated HVDC Supply

Table 9-42 HERMAC Jumpers

ADJUSTMENTS

none

64.6.2 Precautions

ESD can damage devices on the HEMRC Control board. This damage may not be immediately apparent, but may show up in the future as degraded operational performance. Never handle this board unless you are wearing a properly grounded ESD prevention wrist strap. Pay careful attention to ESD packaging and handling procedures to insure the long term reliability of this assembly.

64.6.3 Default jumper configuration

The default configuration for shipment: JP1= "A" and jumper plug in J5 (Normal CAN Mode). Secure the jumper plug to the header with a ty-rap. (See assembly drawing for details).

64.7 HEMRC Assembly – Theory of Operation

64.7.1 HEMRC AC Drive – (CT2 A2 A6 A1)

The HEMRC AC Drive is a customized version of a commercially available Allen-Bradley Model 1336 variable frequency AC motor drive. It contains its own microprocessor, power supplies and a three phase full bridge inverter. The AC Drive communicates with the OBC CPU through a CAN (Controller Area Network) serial bus.

Note: Through special arrangement with Allen-Bradley, the AC Drive uses a derivative of their PROPRIETARY protocol for maximum communication speed and efficiency.

The OBC/HCB firmware controls all sequence operations of the drive. The drive's internal CPU controls lower level detail functions and fault protection. In addition to the CAN communication, the HCB uses discrete signals to control the drive. These signals include Enable, Start, Stop, At Speed, and fault signals. The AC Drive sends an isolated 12V supply to the HCB, which it uses to power the opto-isolators.

During normal operation the AC Drive outputs a 3-phase voltage produced by variable pulse width switching of the drive's IGBT inverter. The peak voltage of this output equals either the HVDC (High Voltage DC) bus voltage or the rectified 380V from transformer T1 (described in Section 64.7.5, on page 687), whichever is greater. However, independent of bus voltage, the drive uses PWM switching to maintain the commanded RMS 3 phase output voltage and frequency. The HCB firmware modifies the commands to the drive as required to supply the current needed for acceleration, run and deceleration of the X-Ray tube.

64.7.2 HEMRC Interface Board (CT2 A2 A6 A2)

The following reference information comes from the HEMRC Interface Board Test Specification, 2145832TST. Refer to the 2145832TST document for any updated information, or for more complete discussion of the Interface Board functions.

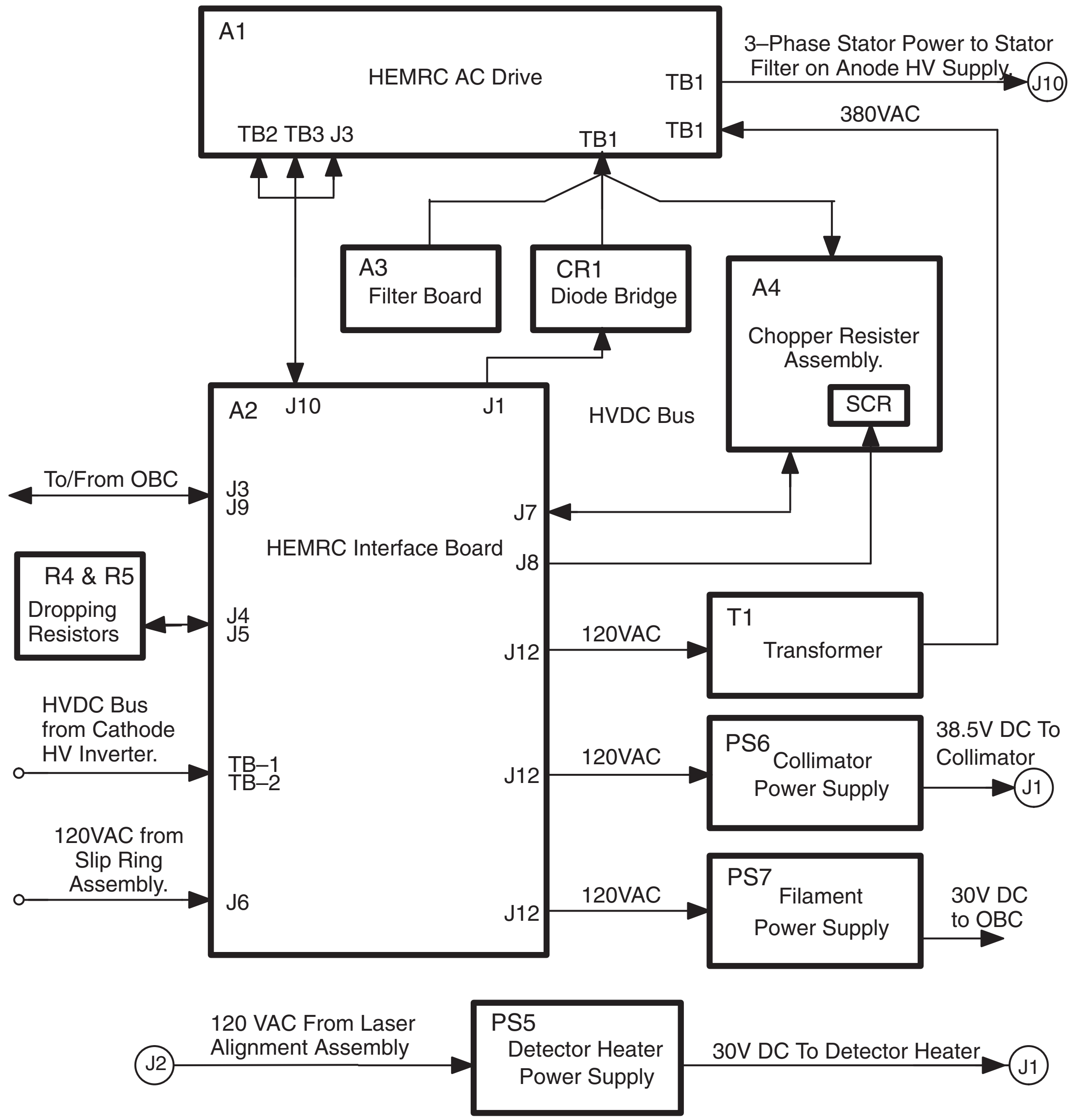


Figure 9-53 HEMRC Assembly Block Diagram

64.7.2.1 Wiring Harness Adapter

The Interface Board provides a transition point for terminating existing gantry harness connections at J3 & J9. These signal lines, originally used for CTVRC control have been reassigned to the HEMRC control. Table 9-43 contains a table of signal names used in the CT/i and HSA Tube system with a CTVRC, and the corresponding names used by the CT/i and Performix Tube system with a HEMRC.

HSA TUBE SYSTEM			GEMINI TUBE SYSTEM				
CTVRC CONTROL BOARD J3-	OLD SIGNAL NAME	CTVRC POWER AMP CONN.	HEMRC CONTROL BOARD J3-	NEW SIGNAL NAME	HEMRC I/ F BD INPUT CONN.	HEMRC I/ F BD OUTPUT CONN.	HEMRC AC DRIVE I/O CONN.
A17	LLEDL	J3-8	A17	HEMRC_FLT_NC	J3-8	J10-14	TB2-15
C17	ULEDL	J3-7	C17	HEMRC_FLT_NO	J3-7	J10-13	TB2-14
A18	LLEDR	J9-4	A18	HEMRC_AT_SPD*	J9-4	J10-4	TB2-10
C18	ULEDR	J9-3	C18	HEMRC_FLT_SPD_RTN	J9-3	J10-3	TB2-11 & 13
A20	STI1_L	J3-4	A20	HEMRC_CAN_H	J3-4	J10-12	J3-1
C20	STI2_L	J3-3	C20	HEMRC_CAN_L	J3-3	J10-11	J3-6
A22	LDI1_L	J3-10	A22	START_HEMRC*	J3-10	J10-16	TB3-19
C22	LDI2_L	J3-9	C22	STRT_STP_COM	J3-9	J10-15	TB3-21
A21	STI1_R	J9-7	A21	HEMRC_EN_P	J9-7	J10-7	TB3-30
C21	STI2_R	J9-8	C21	HEMRC_EN_N	J9-8	J10-8	TB3-29
A23	LDI1_R	J9-1	A23	STOP_HEMRC	J9-1	J10-1	TB3-20
C23	LDI2_R	J9-2	C23		J9-2	J10-2	
A25	DCRVM+	J9-5	A25	HEMRC_ISO_+12V	J9-5	J10-5	J3-4
C25	DCRV+	J9-6	C25	HEMRC_ISO_RTN	J9-6	J10-6	J3-3
A24	DCRV-	J3-6	A24	DCRV-	J3-6		
C24	DCRVM-	J3-5	C24	DCRVM-	J3-5		
				HEN_P1		J10-9	TB3-24
				HEN_P2		J10-10	TB3-25

Table 9-43 HSA to CT/i Signal Name Translation

64.7.2.2 HVDC Sensing

HVDC enters the board at TB1 & TB2, passes through fuses F1 & F2 and outputs to the AC Drive at J1. Fuses F1 & F2 provide isolation between the HVDC bus and the AC Drive in the event of a component failure. LED DS1 illuminates to indicate the presence of voltage.

The HEMRC Interface Board provides the HVDC Bus monitor input. The resistors R1 thru R5 form the input network of a differential amplifier circuit, located on the HEMRC Control Board. The output of this network drives a set of fault detectors read by the OBC CPU to monitor bus status.

R6 thru R10, along with CR1 & U1, form a threshold detector circuit. U1, an optically coupled, normally closed, solid-state relay enables the chopper regulator when the HVDC bus voltage falls below 500V. U1 switches (nominally) between 500 and 550 volts.

Capacitors C1, C2, & C3 provide common mode and differential mode EMI filtering.

64.7.2.3 Chopper Control

DANGER THE CHOPPER CONTROL CIRCUIT ON THE INTERFACE BOARD IS REFERENCED TO THE DC – RAIL AT ALL TIMES. THIS IS A POTENTIALLY LETHAL VOLTAGE SOURCE. DO NOT CONNECT TO GROUND.

The Interface board Chopper circuit helps to dissipate excess energy in the AC Drive's internal DC bus. During braking of the x-ray tube rotor, the system HVDC bus turns off, which in turn directs U1 to enable the chopper as discussed above. As the tube decelerates, its motor acts as a generator, which converts some of the kinetic energy to current. The AC Drive channels this current into its internal DC bus, which cause the voltage on the bus to rise. If the bus voltage exceeds 810V the drive disabled itself and aborts the braking process. When the braking process aborts, the rotor coasts to a stop. The chopper limits the bus voltage to approximately 750V to prevent the tube from coasting.

The AC Drive's DC voltage powers the circuit at J7. Two 7500 ohm 40W chassis mounted dropping resistors, connected at J4 & J5, limit the power supply current to <50 mA. CR4 regulates the nominal 15V to power the Chopper Control circuit. LED DS3 illuminates to indicate the presence of circuit power.

When U1 enables the chopper, the open collector of AR1-1 floats, which in turn enables the operation of comparator circuit of AR1-2. The voltage sensed at J7-1 is scaled and compared to a fixed 5V reference provided by VR1. When the bus voltage exceeds 750V, AR1-2 goes high, driving Q3 and turning on IGBT Q1. An external 100 ohm, 1000W shunt resistor, is connected through a fuse (DC+ and J7-5) to the collector of Q1. When Q1 turns on, this shunt resistor is applied to the DC bus and discharges the excess energy. When the voltage falls below 700V, AR1-2 goes low and driver transistor Q2 turns off Q1, which disconnects the shunt resistor.

The AR1-13 circuit detects the on state of Q1. Normally, Q1 stays on for a few milliseconds at a time. If it stays on too long, Q1 can damage the shunt resistor. Therefore, when the collector of Q1 goes low for more than ~1ms, AR1-13 floats high to release the RC timer of R26 & C10. If this condition lasts for more than ~130ms, AR1-14 goes high, which generates a fault condition.

Pins 4 & 5 of J8 are normally jumpered together so the normal low state of AR1-14 turns "on" the normally open solid-state relay U2, and closes its output "contact". The output of U2 passes through J10-9 & 10 to the AC Drive. When this circuit opens, the drive detects an error condition and aborts all operation. It also notifies the system of the fault.

The detected fault also turns on Q4, to generate a pulse from T1 at J8-1 & 3 which fires the gate of an external SCR. The SCR is connected between an 8 ohm tap on the shunt resistor and the DC-. When the SCR fires, the surge current blows the chopper's input fuse and isolates the fault from the HVDC bus supply.

64.7.2.4 AC Distribution

120Vac enters the board at J6 and illuminates LED DS2. Fuse F3 feeds the collimator power supply through J12-1 and fuse F4 feeds the filament power supply through J12-3. Fuse F5 feeds the isolation transformer, which supplies standby & braking power through J12-5 to the HEMRC AC Drive.

64.7.3 Filter Board – (CT2 A2 A6 A3)

The filter board adds differential mode and common mode capacitance to the AC Drive internal DC bus to reduce the electrical noise created by the switching IGBTs. This board is required for EMI/ EMC compatibility.

64.7.4 Chopper Resistor Assembly – (CT2 A2 A6 A4)

The chopper resistor assembly provides a high power dissipation load to the AC Drive bus, if required during x-ray rotor braking. The chopper resistor configuration resembles the shunt regulator. The Interface Board contains the actual chopper switching element (an IGBT).

When the x-ray tube induction motor brakes, it can momentarily generate a current. When this happens, the AC Drive converts some of the rotational energy to electrical energy and returns it to the internal DC bus causing a rise in the bus voltage. If the DC bus voltage exceeds ~750V, the chopper IGBT turns on and discharges the excess energy through resistors A4R1 & A4R2. The IGBT turns off when the voltage drops below ~700V. This process continues as long as necessary to keep the bus voltage below ~750V. Normally this action occurs for less than 5 seconds during the brake cycle. At all other times the IGBT remains off and essentially “disconnects” the resistors from the bus. The intermittent duty cycles permits the use of resistors with a much lower power rating than a continuous duty cycle would require.

Because the circuit uses the intermittent duty rated resistors A4R1 & A4R2, it contains fuse A4F1 to isolate the resistors from the bus, in the event of a control failure. If a fault occurs, A4SCR1 fires and crowbars the bus. The anode of A4SCR1 connects to a tap on resistor A4R1, nominally set to 8 ohms from the fused end. When the SCR fires, the high current load it creates causes fuse A4F1 to open and disconnect the resistor assembly from the bus, to isolate the fault.

64.7.5 Step-up Transformer – (CT2 A2 A6 T1)

500VA isolation transformer, T1, is configured as a nominal 115:380 V step-up transformer. T1 provides the 24 hour power to the AC Drive, needed to maintain communication with the HCB/OBC. Diodes inside the AC Drive rectify the ~380Vac create a nominal 500 Vdc bus (no load, with 120 Vac input). DC to DC converters inside the drive develop power for its internal logic from this bus. During extended periods of running the rotor, the system main HVDC bus turns off, and T1 becomes the course of continuing power for the Drive. T1 always provides the power during rotor braking.

64.7.6 Bridge Rectifier – (CT2 A2 A6 CR1)

Bridge Rectifier CR1 connects in series between the system main HVDC and the AC Drive internal bus to provide an alternate power source for the drive. The drive internal bus voltage always equals the greater of either the main HVDC or the T1 voltage.

Because the drive bus remains energized at all times, but the main HVDC bus only energizes during rotor acceleration, exposures and 1 minute hold-up times, CR1 isolates the main HVDC bus from the drive’s internal bus. CR1 prevents the drive bus from feeding back to the main HVDC bus and accidently energizing the gantry sliprings and PDU.

64.7.7 Dropping Resistors – (CT2 A2 A6 R4 & R5)

Chassis mounted dropping resistors R4 & R5 provide the power supply from the AC Drive internal bus to the Chopper Control on the Interface Board. The Chopper Control supply is referenced to the HVDC bus return, NOT to ground. NEVER reference this voltage to ground.

DANGER **THE CHOPPER CONTROL CIRCUIT ON THE INTERFACE BOARD IS REFERENCED TO THE DC – RAIL AT ALL TIMES. THIS IS A POTENTIALLY LETHAL VOLTAGE SOURCE. DO NOT CONNECT TO GROUND.**

64.7.8 Connectors

The HEMRC Assembly has many connections to the CT/i system. Unless otherwise indicated, for ease of installation and field upgrade of existing systems, these connections use Mate-N-Lok connectors. Connector designations follow the labelling conventions used in the previous system configurations. Many of the external connections are made directly to sub-components on the assembly and use that sub-component's location identifier in its label.

The following sections group the connections into External Connections and Internal Connections.

64.7.8.1 External Connections

CATHODE INVERTER

Pin#	Signal	Description
1	HVDC-	HVDC- from Cathode Inverter EMC Filter

Table 9-44 A2 TB1 – From Cathode Inverter (10-32 stud)

Pin#	Signal	Description
1	HVDC+	HVDC+ from Cathode Inverter EMC Filter

Table 9-45 A2 TB2 – From Cathode Inverter (10-32 stud)

HERMAC

Pin#	Signal	Description
1	—	No Connection
2	—	No Connection
3	HEMRC_CAN_L	Bidirectional CAN data line (low)
4	HEMRC_CAN_H	Bidirectional CAN data line (high)
5	DCRVM-	HVDC+ Rail Voltage Monitor to control board
6	DCRV-	HVDC- Rail Voltage Monitor to control board
7	HEMRC_FLT_N O	Normally open fault signal to control board
8	HEMRC_FLT_NC	Normally closed fault signal to control board
9	STRT_STP_CO M	Common return for START_HEMRC* and STOP_HEMRC signals from the control board
10	START_HEMRC*	Start command from the control board

Table 9-46 A2 J3 – To/From HEMRC Control Board in the OBC

SLIPRING

Pin#	Signal	Description
1	120VAC	120Vac from the gantry slipring
2	0VAC	AC Neutral from the gantry slipring
3	—	No Connection

Table 9-47 A2 J6 – From 120 VAC Sliprings

HERMAC CONTROL BOARD

Pin#	Signal	Description
1	STOP_HEMRC	Stop command from the control board
2	spare	Unused
3	HEMRC_FLT_SPD_RTN	Common return for _FLT_ and HEMRC_AT_SPD* signals to the control board
4	HEMRC_AT_SPD*	At speed signal to the control board
5	HEMRC_ISO_+12V	AC Drive 12V power supply to the control board
6	HEMRC_ISO_RTN	Return for the 12V power supply to the control board
7	HEMRC_EN_P	High side of enable signal from the control board
8	HEMRC_EN_N	Low side of enable signal from the control board
9	—	No Connection
10	—	No Connection

Table 9-48 A2 J9 – To/From HEMRC Control Board in the OBC

DETECTOR HEATER

Pin#	Signal	Description
1	DET.(+)	23 to 32 VDC positive DC
2	DET.(-)	23 to 32 VDC return

Table 9-49 J1 – Output from Detector Heater Power Supply, PS5

Pin#	Signal	Description
1	Htr_120	120V_SW) Switched 120VAC from Laser Alignment Assembly
2	Htr_0	0VAC return to Laser Alignment Assembly

Table 9-50 J2 – Input to Detector Heater Power Supply, PS5

COLLIMATOR

Pin#	Signal	Description
1	OUT+	38.5 VDC positive DC to Collimator
2	-OUT	38.5 VDC return to Collimator
3	OUT+	38.5 VDC positive DC to Collimator
4	-OUT	38.5 VDC return to Collimator

Table 9-51 (Collimator) J1 – Output from Collimator Power Supply, PS6

FILAMENT

Pin#	Signal	Description
Push-On	30VDC	30 VDC positive to mA Control Board in the OBC
Ring Term	30VRTN	30 VDC return to mA Control Board in the OBC

Table 9-52 (No Connector) – Output from Filament Power Supply, PS7

STATOR

Pin#	Signal	Description
1	BLK	Phase 2 to Stator Filter on Anode HV Supply
2	WHT	Phase 1 to Stator Filter on Anode HV Supply
3	GRN	Phase 3 to Stator Filter on Anode HV Supply
4	SHLD	Cable Shield to Stator Filter on Anode HV Supply

Table 9-53 CT2 A2 A3 J10 – Stator Cable Output to Stator Filter (& HEMIT)

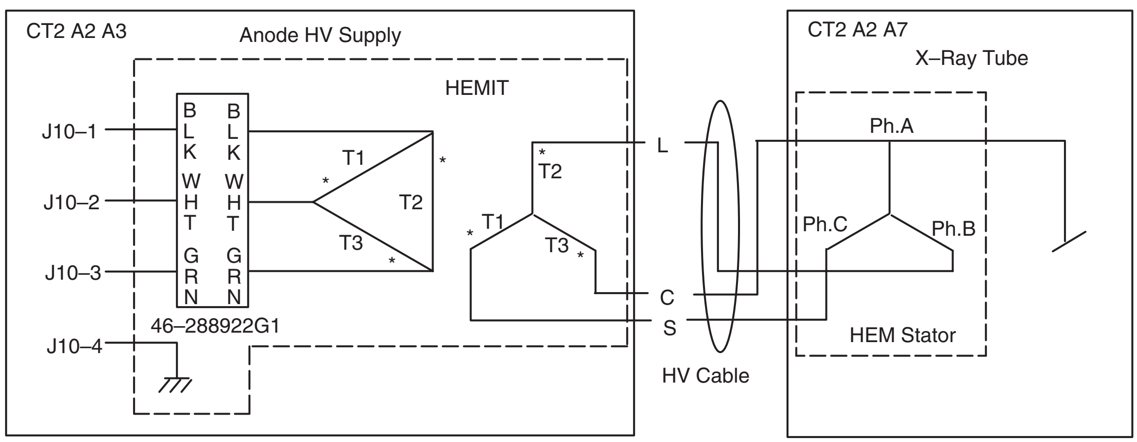


Figure 9-54 HEMIT Wiring Diagram

64.7.8.2 Internal Connections

HERMAC

Terminal	Signal	Description
S	H4	380 Vac input from transformer, T1
T	H1	380 Vac return from transformer, T1
DC+	DC+	Bidirectional DC bus connection from CR1+, Filter Board, and Chopper Resistor Assembly
DC-	DC-	Bidirectional DC bus connection from CR1-, Filter Board, and Chopper Resistor Assembly
U	BLK	Phase 2 to Stator Filter on Anode HV Supply
V	WHT	Phase 1 to Stator Filter on Anode HV Supply

Table 9-54 A1 TB1 – HEMRC AC Drive Power Connections, A1

Terminal	Signal	Description
W	GRN	Phase 3 to Stator Filter on Anode HV Supply

Table 9-54 A1 TB1 – HEMRC AC Drive Power Connections, A1 (Continued)

“TO” HERMAC INTERFACE BOARD

Terminal	Signal	Description
10	HEMRC_AT_SPD*	At speed signal to the OBC via I/ F Board
11	HEMRC_FLT_SPD_RTN	Common return for _FLT_ and HEMRC_AT_SPD*
signals to the OBC via I/F Board	13	HEMRC_FLT_SPD_RTN
Jumper to terminal 11	14	HEMRC_FLT_NO
Normally open fault signal to the OBC via I/F Board	15	HEMRC_FLT_NC
Normally closed fault signal to the OBC via I/F Board		

Table 9-55 A1 TB2 – To HEMRC Interface Board, A2

“FROM” HERMAC INTERFACE BOARD

Terminal	Signal	Description
19	START_HEMRC*	Start command from the OBC via I/F Board
20	STOP_HEMRC	Stop command from the OBC via I/F Board
21	STRT_STP_COM	Common return for START_HEMRC* and STOP_HEMRC signals from the OBC via I/F Board
24	HEN_P1	High side of auxiliary enable from the I/F Board
25	HEN_P2	Low side of auxiliary enable from the I/F Board
29	HEMRC_EN_N	Low side of enable signal from the OBC via I/F Board
30	HEMRC_EN_P	High side of enable signal from the OBC via I/F Board

Table 9-56 A1 TB3 – From HEMRC Interface Board, A2

"TO/FROM" HERMAC INTERFACE BOARD

Pin#	Signal	Description
1	HEMRC_CAN_H	Bidirectional CAN data line (high) to the OBC via I/F Board
3	HEMRC_ISO_RTN	Return for the 12V power supply to the OBC via I/F Board
4	HEMRC_ISO_+12V	AC Drive 12V power supply to the OBC via I/F Board
6	HEMRC_CAN_L	Bidirectional CAN data line (low) to the OBC via I/F Board

Table 9-57 A1 J3 -– To/From HEMRC Interface Board, A2

DIODE BRIDGE

Pin#	Signal	Description
1	HVDC	Fused HVDC+ to bridge diode
2	HVDC	Fused HVDC+ to bridge diode
3	—	No connection
4	—	No connection
5	HVDC_RTN	Fused HVDC- to bridge diode
6	HVDC_RTN	Fused HVDC- to bridge diode

Table 9-58 A2 J1 - To chassis mounted Diode Bridge, CR1

RESISTOR R4

Pin#	Signal	Description
1	none	One end of dropping resistor R1
2 to 9	—	No connection
10	none	Other end of dropping resistor R1

Table 9-59 A2 J4 – To Resistor, R4

RESISTOR R5

Pin#	Signal	Description
1	none	One end of dropping resistor R2
2 to 9	—	No connection
10	none	Other end of dropping resistor R2

Table 9-60 A2 J5 – To Resistor, R5

"TO/FROM" CHOPPER RESISTOR ASSEMBLY

Pin#	Signal	Description
1	DCOUT+	AC Drive internal DC+
2	—	No connection
3	DCFUSED	Fused AC Drive internal DC+
4	—	No connection
5	CHOP_R	Chopper power resistor assembly
6	—	No connection
7	—	No connection
8	DCOUT-	AC Drive internal DC-

Table 9-61 A2 J7 - To / From Chopper Resistor Assembly, A4

"TO SCR" CHOPPER RESISTOR ASSEMBLY

Pin#	Signal	Description
1	GATE	To Gate of SCR
2	—	No connection
3	GATE_RTN	To auxiliary Cathode of SCR
4	none	Interlock loopback to J8-5
5	none	Interlock loopback to J8-4

Table 9-62 A2 J8 - To SCR on Chopper Resistor Assembly, A4SCR1

HERMAC AC DRIVE

Pin#	Signal	Description
1	STOP_HEMRC	Stop command to the AC drive
2	spare	Unused
3	HEMRC_FLT_SPD_RTN	Common return for _FLT_ and HEMRC_AT_SPD* signals from the AC drive
4	HEMRC_AT_SPD*	At speed signal from the AC drive
5	HEMRC_ISO_+12V	AC Drive 12V power supply from the AC drive
6	HEMRC_ISO_RTN	Return for the 12V power supply from the AC drive
7	HEMRC_EN_P	High side of enable signal to the AC drive
8	HEMRC_EN_N	Low side of enable signal to the AC drive
9	HEN_P1	High side of auxiliary enable to AC drive
10	HEN_P2	Low side of auxiliary enable to AC drive
11	HEMRC_CAN_L	Bidirectional CAN data line (low)
12	HEMRC_CAN_H	Bidirectional CAN data line (high)
13	HEMRC_FLT_NO	Normally open fault signal from the AC drive

Table 9-63 A2 J10 – To/From HEMRC AC Drive, A1

Pin#	Signal	Description
14	HEMRC_FLT_NC	Normally closed fault signal from the AC drive
15	STRT_STP_COM	Common return for START_HEMRC* and STOP_HEMRC signals to the AC drive
16	START_HEMRC*	Start command to the AC drive
17	—	No connection
18	—	No connection
19	—	No connection
20	—	No connection

Table 9-63 A2 J10 – To/From HEMRC AC Drive, A1 (Continued)

POWER SUPPLIES

PIN#	SIGNAL	DESCRIPTION
1	Col_120	Fused 120Vac to collimator power supply, PS6
2	Col_0	0Vac to collimator power supply
3	Fil_120	Fused 120Vac to filament power supply, PS7
4	Fil_0	0Vac to filament power supply
5	Xform_120	(X3) Fused 120Vac to Isolation Transformer, T1
6	0VAC	(X1) 0Vac to Isolation Transformer

Table 9-64 A2 J12 - To Power Supplies, PS6, PS7, & T1

64.7.9 Test Points, LEDs, Fuses & Tap Adjustments

TEST POINTS

DANGER **VARIOUS COMPONENTS, INCLUDING THE CHOPPER RESISTOR ASSEMBLY AND HEMRC INTERFACE BOARD, ARE REFERENCED TO THE HEMRC AC DRIVE DC BUS AT ALL TIMES. THIS IS A POTENTIALLY LETHAL VOLTAGE SOURCE. DO NOT CONNECT TO GROUND.**

THE HEMRC INTERFACE BOARD CONTAINS NO TEST POINTS. ALL ACTIVE CIRCUITRY IS HIGH IMPEDANCE AND TIED TO HAZARDOUS VOLTAGES. DO NOT PROBE.

HERMAC LEDS

LED	COLOR	DESCRIPTION
A1 A1 DS1	Red	Fault condition detected by AC Drive
A1 A2 DS1	Yellow	Power applied to AC Drive
A2 DS1	Yellow	HVDC bus energized
A2 DS2	Green	120Vac applied to the Interface board

Table 9-65 LEDs

LED	COLOR	DESCRIPTION
A2 DS3	Yellow	AC Drive DC+ and DC- energized
A2 DS4	Red	Fault detected in the Chopper Control

Table 9-65 LEDs (Continued)

HERMAC FUSES

FUSE#	VALUE	DESCRIPTION
A2 F1	20A, 700Vdc	HVDC- to HEMRC AC Drive
A2 F2	20A, 700Vdc	HVDC+ to HEMRC AC Drive
A2 F3	3A, 250Vdc	120 Vac to Collimator power supply
A2 F4	8A, 350Vac Slo-Blo	120 Vac to Filament power supply
A2 F5	8A, 350Vac Slo-Blo	120 Vac to HEMRC AC Drive Isolation Transformer
A4 F1	10A, 700Vdc	DCIN+ to Chopper Resistor Assembly
PS5 F1	10A, 32V	Fused DC to Detector Heater
PS7 F1	15A, 250V	Fused DC to mA Board in the OBC

Table 9-66 Fuses

A4R1 & A4R2 TAP ADJUSTMENTS

Verify/Align the connection tabs and hardware of the chopper resistors A4R1 and A4R2 so they clear any sheet metal by at least 0.5in.

Adjust the tap band on chopper resistor A4R1 to 8 ohms, +/- 0.5 ohms, with respect to the end connected to fuse A4F1.

The tap band on chopper resistor A4R2 is not used, but you still must secure the band in place to prevent dielectric failure to the adjacent sheet metal. To minimize confusion, adjust the tap band to 8 ohms, +/- 0.5 ohms, with respect to the end connected to A2J7-5.

64.7.10 Error Messages

The HEMRC AC Drive contains an independent microprocessor controller. When the drive detects a fault, it sends a Fault Code to the HCB/OBC. The OBC, in turn, posts an error message in the log. Unfortunately, the OBC logs all AC Drive error messages as # 219800. However, the body of the message contains the actual Fault Code in the first line, as shown below.

HERMAC ERROR MESSAGES

MESSAGE NUMBER	MESSAGE TEXT
219800	The HEMRC AC Drive reported fault code: Fxx (Where xx equals the number in the following table.)

Table 9-67 HEMRC 219800 Error Message

Subsequent lines in the message contain unique description/interpretation information based on the actual Fault Code.

FXX	FAULT	DESCRIPTION
1	Undefined Fault	This fault code is undefined and may indicate a defective Drive. Retry operation. If problems persist contact the factory for instructions.
2	Auxiliary Fault	The interlock between the Chopper Control circuit on the HEMRC I/F Board and the Drive is open. Possible Chopper Control fault or connections to the I/F board. Also check the fuse on the chopper resistor pan.
3	Power Loss Fault	The Drive internal DC bus remained low for >500mS. Possible low voltage condition on 120 VAC in gantry or power interruption. Also may indicate excessive run or braking power required due to sluggish tube.
4	Undervoltage Fault	The Drive internal DC bus voltage dropped below 325V. Possible low voltage condition on 120 VAC in gantry or power interruption. Also, may indicate excessive run or braking power required due to sluggish tube.
5	Overvoltage Fault	The Drive internal DC bus voltage has exceeded 810V. Possible failure of HEMRC I/F Board Chopper Control or excessive motor regeneration from X-ray tube during braking.
6	Motor Stall Fault	The Drive output current has exceeded 12.6A for > 4 seconds. Possible X-ray tube frozen bearing or shorted stator or Anode HV cable. Also, possible defective HEMIT and/or stator cable.
7	Overload Fault	The Drive output current has exceeded 9.7A for an extended time. Possible X-ray tube sticky bearing or shorted stator or Anode HV cable. Also, possible defective HEMIT and/or stator cable.
8	Overtemp Fault	The Drive heatsink temperature has exceeded 90C (195F). Check for blocked or dirty heat sink fins. Also check if the gantry ambient temperature has exceeded 40C (104F).
9	Open Pot Fault	Potentiometer speed control is not used in this system. This fault code indicates a possible corrupted configuration parameter, defective Drive or Control Board. Retry operation.
10	Serial Fault	This fault code indicates a possible corrupted configuration parameter, defective Drive or Control Board. Retry Operation.
11	Op Error Fault	This fault code indicates a possible corrupted configuration parameter, defective Drive or Control Board. Retry Operation.
12	Overcurrent Fault	Check for a short circuit at the drive output or excessive load conditions at the motor.
13	Ground Fault	Check the motor and external wiring to the drive output terminals for a grounded condition.
14 to 18	Undefined Fault	This fault code is undefined and may indicate a defective Drive. Retry operation. If problems persist contact the factory for instructions.
19	Precharge Fault	This fault code indicates a possible hardware failure in the Drive. Retry operation. If problems persist, replace the Drive.

Table 9-68 HEMRC Fault Codes for the 219800 Error Message

FXX	FAULT	DESCRIPTION
20 21	Undefined Fault	This fault code is undefined and may indicate a defective Drive. Retry operation. If problems persist contact the factory for instructions.
22	Drive Fault Reset	Power up has occurred with an open Stop_HEMRC or closed Start_HEMRC* signal. Check Control Board and wiring between Drive and OBC.
23	Loop Overrun Fault	This fault code indicates a possible hardware failure in the Drive. Retry operation. If problems persist, replace the Drive.
24	Motor Mode Fault	This fault code indicates a possible hardware failure in the Drive. Retry operation. If problems persist, replace the Drive.
25	Undefined Fault	This fault code is undefined and may indicate a defective Drive. Retry operation. If problems persist contact the factory for instructions.
26	Power Mode Fault	This fault code indicates a possible hardware failure in the Drive. Retry operation. If problems persist, replace the Drive.
27	Undefined Fault	This fault code is undefined and may indicate a defective Drive. Retry operation. If problems persist contact the factory for instructions.
28	Timeout Fault	This fault code indicates a possible hardware failure in the Drive. Retry operation. If problems persist, replace the Drive.
29	Hertz Error Fault	This fault code indicates an operating frequency parameter was out of range. Possible corrupted configuration parameter, defective Drive or Control Board. Retry Operation.
30	Hertz Select Fault	This fault code indicates an operating frequency parameter was out of range. Possible corrupted configuration parameter, defective Drive or Control Board. Retry Operation.
31	Timeout Fault	This fault code indicates a possible hardware failure in the Drive. Retry operation. If problems persist, replace the Drive.
32	EEprom Fault	This fault code indicates a possible hardware failure in the Drive. Retry operation. If problems persist, replace the Drive.
33	Max Retries Fault	The Drive unsuccessfully tried to reset a fault. See message(s) above for original problem.
34	Run Boost Fault	Verify that the [Run Boost] parameter is less than or equal to the [Start Boost] parameter.
35	Negative Slope Fault	This fault code indicates a Volts / Hertz programming error. Possible corrupted configuration parameter, defective Drive or Control Board. Retry Operation.
36	Diag C Lim Fault	Check programming of [Cur Lim Trip En] parameter. Check for excess load, improper DC boost setting, DC brake volts set too high or other causes of excess current.
37	P Jump Error Fault	
38	Phase U Fault	A phase to ground short has been detected in the U phase. Check the wiring between the drive and HEMIT. Check HEMIT for grounded primary winding.

Table 9-68 HEMRC Fault Codes for the 219800 Error Message (Continued)

FXX	FAULT	DESCRIPTION
39	Phase V Fault	A phase to ground short has been detected in the V phase. Check the wiring between the drive and HEMIT. Check HEMIT for grounded primary winding.
40	Phase W Fault	A phase to ground short has been detected in the W phase. Check the wiring between the drive and HEMIT. Check HEMIT for grounded primary winding.
41	UV Short Fault	A phase to phase short has been detected between the U & V phases. Check the wiring between the drive and HEMIT. Check HEMIT for shorted primary.
42	UW Short Fault	A phase to phase short has been detected between the U & W phases. Check the wiring between the drive and HEMIT. Check HEMIT for shorted primary.
43	VW Short Fault	A phase to phase short has been detected between the V & W phases. Check the wiring between the drive and HEMIT. Check HEMIT for shorted primary.
44 45	Undefined Fault	This fault code is undefined and may indicate a defective Drive. Retry operation. If problems persist contact the factory for instructions.
46	Power Test Fault	This fault code indicates a possible hardware failure in the Drive. Check all connections to the Power/Driver Board. Retry operation. If problems persist, re place the Drive.
47	Transistor Saturation Fault	This fault code indicates a possible hardware failure in the Drive. Retry operation. If problems persist, replace the Drive.
48	Reprogram Fault	Reset the OBC or cycle power to the drive.
49	Undefined Fault	This fault code is undefined and may indicate a defective Drive. Retry operation. If problems persist contact the factory for instructions.
50	Poles Calc Fault	
51	Background 10ms Over	This fault code indicates a possible hardware failure in the Drive. Retry operation. If problems persist, replace the Drive.
52	Foreground 10ms Over	This fault code indicates a possible hardware failure in the Drive. Retry operation. If problems persist, replace the Drive.
53	EE Init Read	This fault code indicates a possible hardware failure in the Drive. Retry operation. If problems persist, replace the Drive.
54	EE Init Value	This fault code indicates a possible hardware failure in the Drive. Retry operation. If problems persist, replace the Drive.
55	Temp Sense Open	This fault code indicates a possible hardware failure in the Drive. Retry operation. If problems persist, replace the Drive.
56	Precharge Open	This fault code indicates a possible hardware failure in the Drive. Retry operation. If problems persist, replace the Drive.
57	Ground Warning	Check the HEMIT and external wiring to the drive output terminals for a grounded condition.
58	Blown Fuse Fault	This fault code indicates a possible hardware failure in the Drive. Retry operation. If problems persist, replace the Drive.

Table 9-68 HEMRC Fault Codes for the 219800 Error Message (Continued)

FXX	FAULT	DESCRIPTION
59 to 64	Undefined Fault	This fault code is undefined and may indicate a defective Drive. Retry operation. If problems persist contact the factory for instructions.
65	Adapter Frequency Error	This fault code indicates an operating frequency parameter was out of range. Possible corrupted configuration parameter, defective Drive or Control Board. Retry Operation.
66	EEprom Checksum Fault	This fault code indicates a possible hardware failure in the Drive. Retry operation. If problems persist, replace the Drive.
67	Undefined Fault	This fault code is undefined and may indicate a defective Drive. Retry operation. If problems persist contact the factory for instructions.
68	ROM or RAM Loss Fault	Internal power-up tests did not execute properly. Check Language Module. Retry operation. If problems persist, replace the Drive.
69 70	Undefined Fault	This fault code is undefined and may indicate a defective Drive. Retry operation. If problems persist contact the factory for instructions.

Table 9-68 HEMRC Fault Codes for the 219800 Error Message (Continued)

Chapter 10
Das and Detector

Section 1.0
Replacement Verification and Re-test

1.1 Preparation

Because DAS and detector tests use X-Ray to look for opens, lows and shorts, please verify the following condition before you begin collecting diagnostic data:

- Flex cables ON
- All covers ON
- Nothing in detector FOV

1.2 Tests

Note: Please perform the retests listed below when you replace or adjust a DAS or Detector FRU. System Functional Test means scan the first six series using PROTOCOL LIST 20.8 called the System Scanning Test; how to scan with protocols begins on page 63.

1.2.1 DAS

DAS Component	Task	Verification Test
Complete DAS, Chassis or Backplane	Replace faulty assembly and reset Smart Trend Baseline, page 264 if applicable (5.x SW)	1.) Check power supplies, 2.) Integrate DAS, Chapter 7 of Install Manual 3.) QCal/Cals, page 100 4.) Streak test 5.) System Functional Test (page 69)
HP-DAS Converter Board	Replace faulty board and reset Smart Trend Baseline, page 264if applicable (5.x SW)	DAS Tools: Offsets, Offset Drift, DC Cal, Micro phonics If Moved or replaced fewer than 12 filters: Streak test and image series If Moved or replaced more than 12 filters: QCal/Cals (page 100), Streak test (page 91), and run an image series.
E-DAS Timing Processor or Keyboard	Replace	Use DDC X-Ray Verification then Microphonic sIntegrate DAS and run an image series.

Table 10-1 DAS Retest Matrix

DAS Component	Task	Verification Test
E-DAS Right or Left Analog Box	Replace	Integrate DAS, Streak test (page 91), QCal (page 100), and run an image series.
E_DAS Filter Board	Replace faulty board	Do Air Cals and run an image series
E_DAS Cal/Aux Board	Replace faulty board	Do Air Cals and run an image series
E_DAS ADC Board	Replace faulty board	Do Air Cals and run an image series
DC Power Supplies **(+/- 15 volt and 5 volt)**	Replace 15, see page 712 See page 713 for the 5 volt power supplies	Check power supplies, Offsets, Offset Drift, DC Cal, Microphonics, Streak test, System Functional Test
Detector Heater **Power Supply**	Replace, see page 713	Check power supply Allow system to warm for two hours then acquire 10 scans using the System Test Protocol

Table 10-1 (Continued)DAS Retest Matrix (Continued)

1.2.2 Detector

Detector Component	Task	Verification Test
Detector	Replace faulty detector (page 714) and reset Smart Trend Baseline, page 264if applicable (5.x SW).	Integrate DAS, Chapter 7 of Install Manual Alignments: Chapter 3 of this manual (POR, BOW, Z-Align Base, Radial Alignment, ISO, CBF, SAG, Detector output, QCAL, Xtalk Cal, Alpha Vector Cal, Cals Streak test (page 91), N number check (page 64), and Image Series.
Flex Cables	Replace faulty cable(s), procedure on page 716 and reset Smart Trend Baseline, page 264 if applicable (5.x SW).	DAS Tools: Offsets, Microphonics, Streak test, Detector Output, If Moved or replaced fewer than 12 filters: Streak test and image series If Moved or replaced more than 12 filters: QCal/Cals (page 100), Streak test (page 91), and run an image series.

Table 10-2 Detector Retest Matrix

Section 2.0
Diagnostic Data Collection (DDC)

2.1 Overview

The DDC service tool is used to prescribe scans in order to test the performance of the system. This tool is very flexible since it provides the user with many options to prescribe different parameters and techniques. Some of the issues encountered for the new sub-second scanning feature are: the ability to prescribe scan speeds at 0.8 seconds, ability to determine when to allow sub-second scanning, ability to prescribe half-scans, ability to limit the inter scan delay, and ability to prescribe techniques for 48KW or 53KW.

Another feature of the DDC tool is the existence of set Protocols. When a protocol is selected, it overwrites the existing entries, and the software will not check to see if the protocol is valid. This means that the new changes created for sub-second scanning will not alter the results for existing protocols. This however, will cause problems if a new protocol is created for a configuration not supported by the system.

Another interesting fact of the current design is that there is no distinction made between an Axial and a Cine scan. To prescribe an Axial scan, the user must enter in the Scan Time field, the same or lesser value as in Scan Speed. If the Scan Time is grater than the Scan Speed, the scan will be a Cine Scan. It is not clear at this time if the interface needs to be more explicit, so this design will only restrict the Scan Speed when prescribing half-scans (these are only used for Axial scans).

2.2 DDC Scanning & Data Processing Requirements Matrix

Filter Type > Calibration	**Air**	**Bowtie**	**Large Bowtie**
None	1.) no cal vectors in scan file. 2.) CANNOT retro recon.	1.) no cal vectors in scan file. 2.) CANNOT retro recon.	1.) no cal vectors in scan file. 2.) CANNOT retro recon.
Air (To use this calibration re quires a valid QCal and Crosstalk vectors)	1.) only crosstalk and QCal vectors in scan file 2.) CANNOT retro recon.	1.) only crosstalk and QCal vectors in scan file 2.) CANNOT retro recon.	1.) only crosstalk and QCal vectors in scan file 2.) CANNOT retro recon.
Small (To use this calibration re quires a full set of valid Cals for small SFOV)	1.) all vectors in scan file 2.) CANNOT retro recon.	1.) all small CAL vectors in scan file 2.) CAN retro recon. Retro image marked “SMALL”	NOT VALID Should not be able to scan
Large (To use this calibration re quires the respective bow tie shapes to have a full set of valid Cals for large SFOV)	1.) all CAL vectors in scan file 2.) CANNOT retro recon.	1.) all bowtie CAL vectors in scan file 2.) CAN retro recon. Retro image marked “LARGE”	1.) all large bowtie CAL vectors in scan file 2.) CAN retro recon. Retro image marked “LARGE%”

Table 10-3 DDC SCANNING & DATA PROCESSING REQUIREMENTS MATRIX

2.3 Scanning with DDC

Select the SERVICE Desktop, SYSTEM INTEGRATION Softkey then DIAGNOSTIC DATA COLLECTION from the displayed menu: (see Figure 10-1)

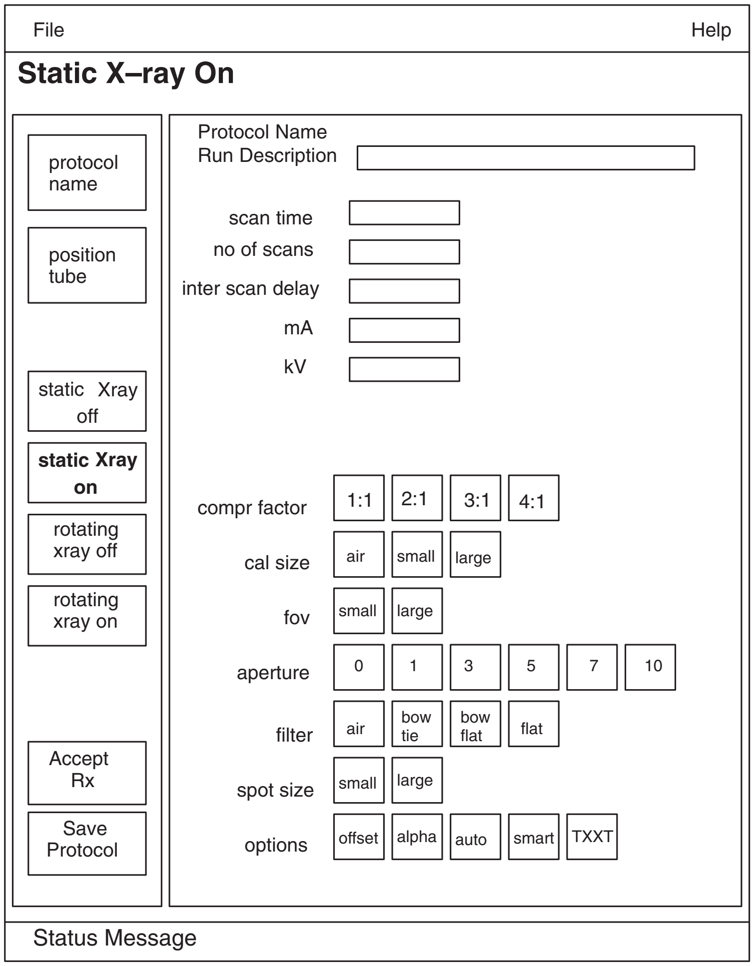


Figure 10-1 DDC Menu

SERVICE SCAN WITH DDC

1.) Select the DDC softkey (Diagnostic Data Collection) on the monitor display.
2.) Select the REAL TIME STATS softkey.
3.) Select the STATIC X-RAY ON softkey.
4.) Use the mouse and “click left” to select the Scan Time field. Type/enter 4 second scan time.
5.) Use default technique values of 40 mA and 80 kV.
6.) Select the AIR Filter softkey.
7.) Select the 5 (mm) Aperture softkey.
8.) Select the SMALL Spot Size softkey.
9.) Type/enter a Run Description name (arbitrary).
10.) Select the RUN softkey, and wait for the **START SCAN** Button on the keyboard to illuminate. Press the START SCAN key to initiate the scan.

11.) After the scan completes, the monitor displays the DD filename.
Record the filename.

12.) Select the DISMISS softkey.

Section 3.0 Data Plot

Display the Service Desktop, Utilities Menu and Select DD File Analysis:

1.) Select the DD File Analysis from the Service Desktop, Utilities Menu.

2.) Refer to page 705.
Select the DD File of interest from the List/Select window.

3.) Select SHOW VECTOR.

Note: Verify that the first channel is 6 and the total channels equal 747. Channels 1 – 5 and 748 – 750 are reference channels and are not normally displayed. Because the detector has 750 cells and six are for reference, it uses 744 cells for patient data.

	EDAS	HP DAS
Channels 6–64	66,000–81,000	25,000–112,500
Channels 689–747	66,000–81,000	25,000–112,500
Channels 65–688	35,000–44,000	10,000–65,000

Table 10-4 EDAS and HP DAS Channel Counts

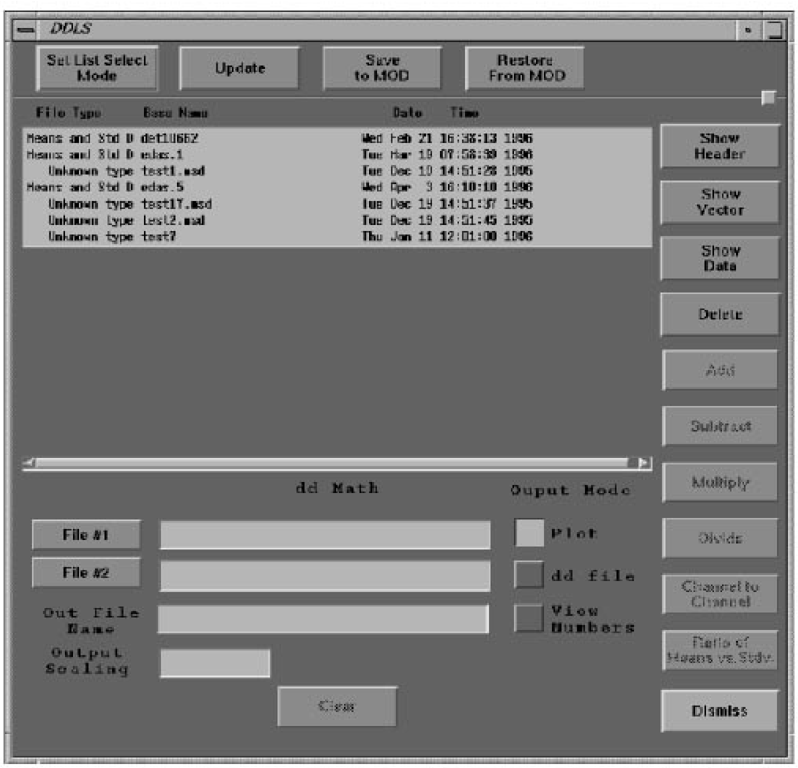


Figure 10-2 DD FILE ANALYSIS SCREEN

Section 4.0 Detector Slope Test

4.1 Overview

This is a description for the automated Detector Slope Test (DST) for HSA CT/i starting with the 3.6 software release. It also covers the manual (service) execution of this feature. The process followed after the Detector Slope Test reports (and logs) a questionable detector is covered in the Detector Slope Test Failure Procedure.

Over time, detectors used on the HSA CT/i system may experience a phenomena which causes non-uniform gain over the channel in the "Z" (front to back) direction. Given the right object, scan technique and this problem, an artifact could be introduced into the image.

The Detector Slope Test runs periodically to determine if the detector has degraded in such a way that it is at risk of creating an artifact related to this failure mode. The test includes taking a number of scans, processing those scans into vectors that simulate slices covering the detector, then processing those points to come up with two final vectors. These vectors are then compared to a set of limits. If the detector is within the limits, a new target date is set for running the test. If it is not within the limits, a message is posted on the screen instructing the operator to contact GE Service, a message is logged and the next target date is set for running the test.

The Detector Slope Test is also available to the service person to run manually, regardless of when it was run previously. It will execute the same test as if it was run automatically, including setting a new target date for the next execution of the test.

The Detector Slope Test (DST) process is software that runs on the OC Computer. FastCal is changed to check to see if it is time to run the Detector Slope Test. If it is, FastCal will initiate the DST process. If it is not time to run the DST again, it will just return and FastCal will be done. When selected on the Service Desktop, the test will run regardless of the date it was last run.

The DST process will only be started when the Detector Slope Test needs to run. It can be started by a button selection on the Service Desktop menu. It can also be started by FastCal. FastCal will decide if DST needs to run. If so, it will be started; if not, it will be skipped. By doing it this way, DST does not need to make any decisions about whether it is time for it to run.

Operator intervention will be required to start the scans, if the process is initiated from the service desktop menu. If initiated from FastCal, the scanning starts automatically without any operator intervention (similar to GenCal).

DST uses the service scanning modes for its scans. The Means and standard deviation vectors resulting from the scans are written in HSA CT/i DD File format in directory `/data`. The means from these scans are used for the DST algorithmic processing. The DD files are also saved until the next DST run.

When started from the Service Desktop, DST will prompt the operator to make sure there is no obstruction in the beam prior to starting the scans. If DST is started from the Service Desktop, the user interface screens will pop up on the Display Monitor (normally the right monitor) of the CT/i System. If DST is started by FastCal, the screens will appear on the ExamRx Monitor (normally the left monitor).

Error checks are made on the data gathered during the test. The first is a check for overranging. If overranging is found, the testing will be discontinued. There are also a variety of beam obstruction tests. See Section 4.6.5 for further details on these checks.

4.2 DST General Flow of Events

The following diagram is a general view of the flow of events in the Detector Slope Test. It is not exhaustive in error cases, but is only intended to give an overview of the steps in the process. It starts with the “Started by FastCal” event on the top left or the “Selection from Tools Screen” on the top right. It ends at “On Fail: Post Message” on the right.

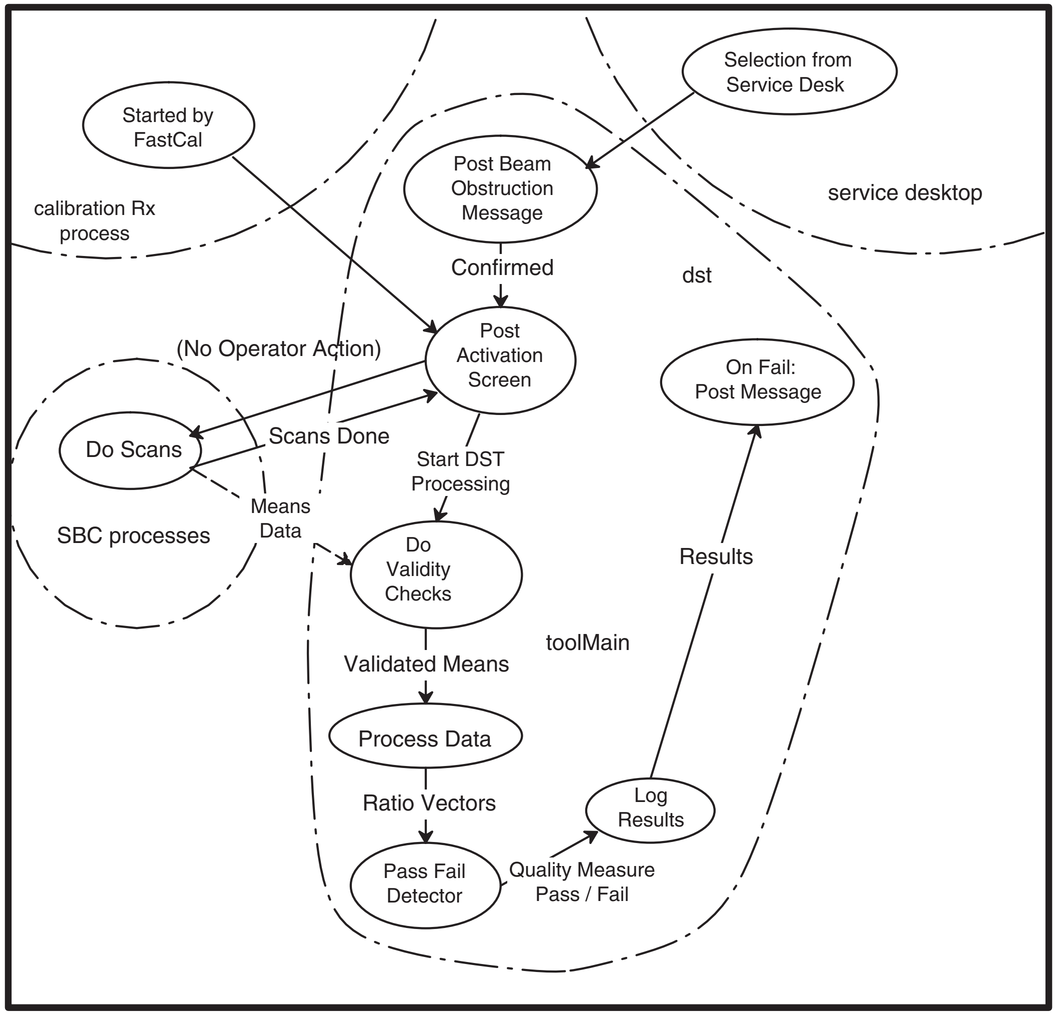


Figure 10-3 DST PROCESSING FLOW DIAGRAM

4.3 Detector Slope Test – Scanning

Each collimator mandrel type has a set of scans specific to it. The scanning process will check to see if a file exists indicating that the mandrel is the G1 version. If the file exists, the G1 mandrel scan set will be done. If the file does not exist, the G2 mandrel scan set will be done. (Note that either a G1 or G2 mandrel can be in a G2 or later collimator, so the collimator revision is not definitive in determining which mandrel is in the system).

These scans have varying apertures and aperture offsets. Only the Offset Corrected Means are required to be saved from the scans.

4.4 Detector Slope Test – Processing

A set of simulated detector slices is made by subtracting various combinations of the original scans. Again, only the Offset Corrected Means vectors are kept.

Two phantom scans are simulated over a set of simulated detector slices. One phantom scan is a simple ramp, referred to as the slope ratio, the other has a shape like “^”, referred to as the wedge ratio. Each of these two vectors is a ratio with the unweighted average of the same simulated slices.

There are four key error checks at the beginning of the processing phase. Reference Section 4.6.5. The first is to determine if the data has over-ranged. The Detector Slope Test will be “permanently” disabled if an over-range is found on a system. A check will then be made to determine if the data has been clipped (as occurs when using a G1 mandrel with a G2 scan set). If the clipping occurs, the file will be created that scanning keys on and processing will terminate without updating the latest test run date. The data will then be checked to make sure it meets the minimum acceptable range. The DST program must have all data within a certain relative range. If data is outside of this range, it will create a failure condition. This condition will cause the operator to be notified to call GE Service. Lastly, a check will be made to determine if there is obstruction in the beam. If the beam is obstructed, the test will terminate.

4.5 Detector Slope Test – Detection

This processing has been developed to take the ratio of the two vectors in and return a pass/fail as an output.

4.6 Detector Slope Test – Message Posting

If the result from the Detection Algorithm is “pass”, the quality measure will be logged to the system log. The quality measure will also be posted, along with the date and time, to the detector quality trend log. The two vectors that the Detection Algorithm based its evaluation on, will be saved on the disk (slopeRatio and wedgeRatio DdFiles). The first time DST is run, it will save these ratios as base files called “SlopeBase” and “WedgeBase”. These files will give a basis of comparison for any changes of the detector over time.

If the result from the Detection Algorithm is “fail”, the quality measure will be logged to the system log, indicating that the detector has failed this test. The quality measure will also be posted, along with the date and time, to the detector quality trend log. The two vectors that the Detection Algorithm based its evaluation on will be saved on the disk. A message will be posted to the operator instructing him or her to call GE Service (see Section 4.6.1 for the text of the message).

4.6.1 Detector Slope Failure Message

When the Detector Slope Test fails, a message will be posted to the operator requesting that they contact GE Service to follow-up on the automated test. The message reads:

```
The Image Slope Test has detected a condition that may result in an image
with unacceptable image quality. Please call GE Service and request a more
thorough evaluation of your systems condition.
```

The screen will look like: (Screen will be updated with the new message!)

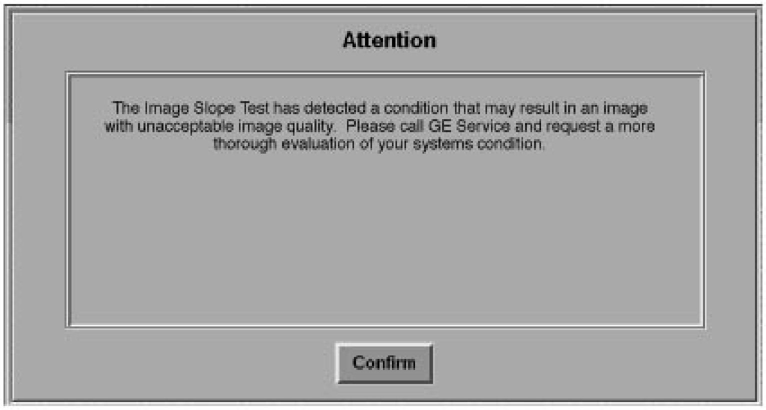


Figure 10-4 DETECTOR SLOPE TEST FAILURE WINDOW

4.6.2 Beam Obstruction Window

The FastCal Beam Obstruction screen looks similar to the screen below:

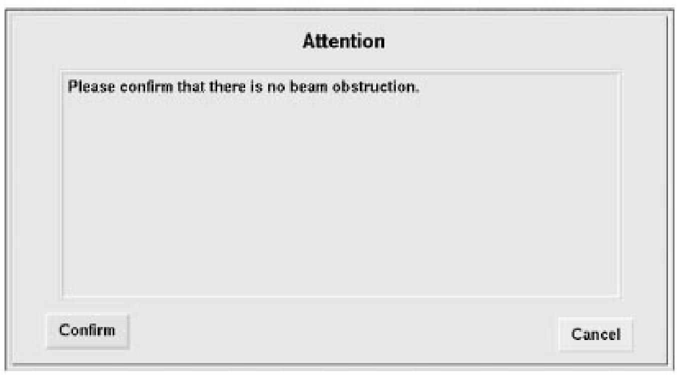


Figure 10-5 DETECTOR SLOPE TEST BEAM OBSTRUCTION WINDOW

4.6.3 Detector Slope Test Success Screen

The screen below will notify the service user when the DST successfully passes the test. This screen will appear only if the test is initiated from the service desktop menu. When initiated from FastCal, the success message will appear as an operator message on the bottom of the screen message area.

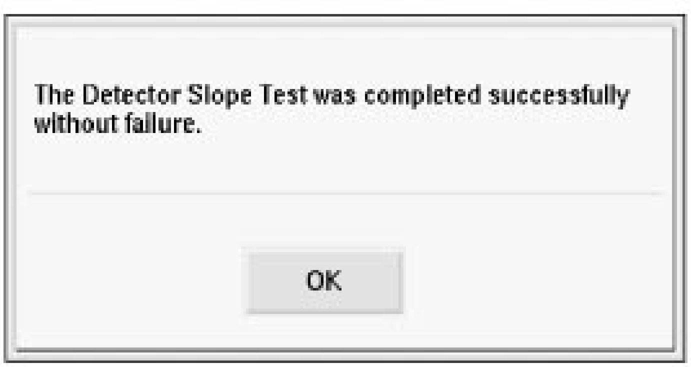


Figure 10-6 DETECTOR SLOPE TEST PASS WINDOW

4.6.4 Internal Error Handling and Recovery

Whenever a Detector Slope Test fails, the log file will have an entry added that has a failure code instead of the quality measure and a failed test code instead of the pass/fail indicator.

If processing is unable to access the DdFiles that scanning should have created, processing will terminate. A message indicating a file access error will be posted to the message log. The time to run the next pass of the Detector Slope Test will NOT be updated.

All processing errors are handled as a "normal" case. These types of errors will be logged and Detector Slope Test outcomes will be based on the error that occurred (either quit with no update on when the next test should run, or consider it a failed test with appropriate operator message and logging).

4.6.5 Validity Checking

There are four validity checks performed. Processing only continues if all four checks are passed. These checks are (and order is important):

1.) Overranged Data Check
2.) Mandrel Check
3.) Data Threshold Check
4.) Beam Obstruction Check.

CHECKS:

1.) Overranged Data Check

The Overranged Data Check is to verify that the detector has not been saturated with X-Ray. A message will be put in the message log and Detector Quality Trend Log indicating overranged data and processing will terminate. DST will be told that the test PASSed. The Detector Slope Test will be discontinued on these detectors by setting the next scheduled date for six months after the date the test is run. Only setting it six months out allows for the detector to be changed and have the Detector Slope Testing resume.

2.) Mandrel Check

The Mandrel Check is done to try to increase the likelihood that the test can be performed successfully. It is known that a G1 style mandrel will clip the beam when done on a G2 scan set. However, there are also other conditions that can clip the beam (Plane of Rotation misadjustment, etc.) that can also clip the beam. If beam clipping is found, scanning will be instructed to switch to the narrow (G1) scan set. This check is only done on G2 scan sets.

3.) Data Threshold Check

The detection algorithm expects that each vector is within 95% of the mean of the center (least offset) vectors. This check verifies that this is true. It is only run on G1 scan sets (the equivalent for G2 scan sets is the Mandrel Check above). If any vector is below 95% of the center, the detection algorithm results would not be meaningful, therefore the test terminates. Failure of this test will cause the operator to be prompted to contact GE Service to do the phantom test. A message will be logged to the message log, an entry will be made in the Detector Quality Trend Log and DST will be told the test FAILed. The date for the next scheduled test will be updated.

4.) Beam Obstruction Check

The Beam Obstruction check will always be performed. It should not fail on Detector Slope Tests started by FastCal because FastCal requires the operator to remove any beam obstructions before it will run. For Detector Slope Tests started from Tools, a prompt will be put up on the screen that requests the FE to make sure the beam is not obstructed. After all of the data is collected, this check will be preformed. If it is found to be blocked, the Detector Slope Test will log a message to the message log, add an entry in the Detector Quality Trend Log and DST will be told that the test PASSed.

This check is only done when the Detector Slope Test was initiated by Tools (because is uses data from the FastCal initiation of the test). If FastCal has never initiated the Detector Slope Test, the normalization vector will not exist, so it will return success. When FastCal initiates the Detector Slope Test, the 7mm scan offset closest to the table will be retained. This will be used to normalize the Tools data (since it is assumed not to be blocked). If any of the channels is below 95% after normalization, the detector will be said to be blocked.

4.6.6 Status Logging

After completion of the Detection Algorithm, the status will be logged to the message log and to a test history log. This will include the “quality measure” and the pass fail status.

The next scheduled date is also updated. Thirty days will be added to the current date and this will be the new value written to `/usr/g/service/state/.dst.date`. If there was any failure in the processing a message will be logged and the next scheduled date will be updated by 180 days.

4.6.7 What to do if Detector Slope Test Fails

If the Detector Slope Test fails, GE Service should be contacted to further evaluate the condition of the detector.

Simply because the Detector Slope Test fails does not imply that there is a problem with the detector. Several conditions can cause a failure report as the result of the test that requires a manual evaluation to determine the systems condition.

In order to correctly assess the condition of the system and the detector after a failure report from the Detector Slope Test, GE personnel should request the Detector Inspection Procedure (FMI 25226) from their respective FMI coordinator.

Schedule and complete the FMI and follow all of the procedures in order to complete the manual evaluation of the system and the detector.

Section 5.0
DAS and Detector Replacement Procedures

WARNING **WHEN WORKING IN THE GANTRY, MAKE SURE THAT AXIAL DRIVE ENABLE IS IN THE OFF POSITION AND THAT THE GANTRY IS LOCKED. FAILURE TO DO SO CAN RESULT IN SERIOUS INJURY TO YOURSELF OR OTHERS FROM BEING STRUCK BY GANTRY PARTS WHICH ARE IN MOTION.**

5.1 46-136343P12 DAS Power Switch

1.) Turn off facility power to PDU.

DANGER **USE TAG AND LOCKOUT PROCEDURES.**

2.) Remove gantry side covers.
3.) Turn off all three (3) switches on status control box, on right side of gantry.
4.) Rotate gantry until the OBC reaches the 3 o’clock position.
5.) Engage gantry rotational lock.
6.) Loosen four (4) captive screws on the DAS Power Supply cover.
7.) Remove the cover.
8.) Remove two (2) leads from DAS power supply switch.

9.) Remove mounting hardware from switch.
10.) Remove switch from base of DAS power supply assembly.
11.) Install new switch.
12.) Reassemble Gantry.

5.2 DAS 15VDC Power Supplies

WARNING **WHEN WORKING IN THE GANTRY, MAKE SURE THAT AXIAL DRIVE ENABLE IS IN THE OFF POSITION AND THAT THE GANTRY IS LOCKED. FAILURE TO DO SO CAN RESULT IN SERIOUS INJURY TO YOURSELF OR OTHERS FROM BEING STRUCK BY GANTRY PARTS WHICH ARE IN MOTION.**

1.) Move table to lowest elevation.
2.) Turn off facility power to PDU.

DANGER **USE TAG AND LOCKOUT PROCEDURES.**

3.) Remove gantry side covers.
4.) Turn off all three (3) switches on status control box, on right side of gantry.
5.) Lift top cover, and engage prop rod.
6.) Remove scan window.
7.) Open front cover.
8.) Rotate Gantry until the OBC reaches the 3:00 position.
9.) Loosen four (4) captive screws on the DAS Power Supply cover.
10.) Remove the cover.

Note: Note the location of leads connected to the Power Supply.

11.) Unsolder Leads from Power Supply.
12.) Remove 4 standoffs that fasten the Power Supply to the assembly.
13.) Install new Power Supply.

Due to the voltage drop between the power supplies and the actual components of the DAS, it is necessary to verify the voltages at the DAS. These voltages may be measured on the analog controller board and at the filter box.

14.) Energize Gantry 24hr AC power.
15.) With the DAS load applied, remove the cover of the left half DAS to gain access to the Analog Controller board or AUX/CAL if EDAS
16.) Connect the multimeter to the +15V and AGND test points on the Analog Controller, and adjust the +15 VDC potentiometer (P1) on the triple supply for +15.0 VDC. -OR- Connect the multimeter to the -15V and AGND test points on the Analog Controller, and adjust the -15 VDC potentiometer (P2) on the triple power supply for -15.0 VDC.

Turn the voltage adjustment pot for a voltage output between 15.0 and 15.25 VDC.

	at power supply	**at HP DAS**
+15V analog	+15. 0 to +15.2V	+15.0 V
-15V analog	-15.0 to -15.2V	-15.0 V

17.) Remove power from Gantry (tag and lockout procedures).
18.) Reassemble Gantry.

5.3 DAS 5VDC Power Supplies

1.) Move table to lowest elevation.
2.) Turn off facility power to PDU.

DANGER USE TAG AND LOCKOUT PROCEDURES.

3.) Remove gantry side covers.
4.) Turn off all three (3) switches on status control box, on right side of gantry.
5.) Lift top cover, and engage prop rod.
6.) Remove scan window.
7.) Open front cover.
8.) Rotate Gantry until the OBC reaches the 3:00 position.
9.) Loosen four (4) captive screws on the DAS Power Supply cover.
10.) Remove the cover.

Note: Pay attention to the location of Leads connected to Power Supply.

11.) Unsolder Leads from Power Supply.
12.) Remove 4 standoffs that fasten the Power Supply to the assembly.
13.) Install new Power Supply.

Due to the voltage drop between the power supplies and the actual components of the DAS, it is necessary to verify the voltages at the DAS. The HPDAS voltages may be measured on the analog controller board and at the filter box. The half gain EDAS voltages are measured on the AUX/CAL board.

14.) Turn ON the Gantry AC power at the A1 panel, and service switch box.
15.) With the DAS load applied, remove the cover of the left half DAS to gain access to the Power Filter Asm and Analog Controller board if an HP DAS.
16.) **For EDAS:** Turn the voltage adjustment pot on the power supply regulator PWB for a voltage output between 5.00 and 5.25 VDC.

For analog +5: Connect the DVM to the +6V and AGND test points at the power filter box and adjust the 6 VDC potentiometer (R10) on the+6V supply for +5.8 VDC.

For analog -5: Connect the digital multimeter to the -6V and AGND test points at the power filter box, and adjust the 6 VDC potentiometer (R10) on the -6V supply for -5.8 VDC.

For 5 V digital: Using the 5V digital and DGND test points on the Analog Controller board, adjust the 5 VDC pot on the triple power supply for +5.0 V.

If the measurements at the power supply exceed the limits shown, check for loose connections, solder joint integrity, wire damage, etc.

	at power supply	**at HP DAS**
+5V digital	+5.0 to +5.3V	+5.0 V
+6V digital	+5.8 to +6.1V	+5.8 V
-6V digital	-5.8 to -6.1V	-5.8V

17.) Remove power from Gantry (tag and lockout procedures).
18.) Reassemble Gantry.

5.4 46-297104P1 Detector Heater Power Supply

1.) Move table to lowest elevation.
2.) Turn off facility power to PDU.

DANGER USE TAG AND LOCKOUT PROCEDURES.

3.) Remove gantry side covers.
4.) Turn off all three (3) switches on status control box, on right side of gantry.
5.) Lift top cover, and engage prop rod.
6.) Remove scan window.
7.) Open Front Cover.
8.) Rotate gantry until Filament Power assembly reaches the 3 o'clock position.
9.) Engage gantry rotational lock.
10.) Loosen four (4) captive screws on Filament Power Assembly cover.
11.) Remove cover.
12.) Measure voltage across filter capacitor to verify the bleeder resistor has dissipated energy to a safe level.
13.) Unsolder Black and White leads from transformer terminals 1 and 3.
14.) Remove two screws that fasten the Red and Black leads to the filter capacitor.
15.) Remove and save the four (4) screws that fasten the Detector Heater Power Supply to the gantry.
16.) Install new power supply.

Note: Input leads: Black to transformer terminal one (1); White to transformer terminal three (3.) Output leads: Red to Filter Capacitor Terminal marked (+); Black to remaining capacitor terminal.

17.) Detector Heater P.S. output check:
- Connect positive voltmeter lead to the red lead on output filter capacitor.
- Connect negative voltmeter lead to the black lead on output filter capacitor.
- Energize Gantry 24hr AC power.
- The voltage reading should be 23–32 VDC.

18.) Remove power from Gantry (tag and lockout procedures).
19.) Reassemble Gantry. Let the detector warm up for two hours.

5.5 Detector

1.) Turn off facility power to PDU.

DANGER USE TAG AND LOCKOUT PROCEDURES.

2.) Open all gantry covers.

DANGER ALWAYS TURN OFF THE GANTRY SCAN DRIVE BEFORE YOU PERFORM A PROCEDURE ON THE GANTRY WITH THE GANTRY COVERS OPEN. FAILURE TO HEED THIS WARNING CAN RESULT IN SERIOUS INJURY OR LOSS OF LIFE.

3.) Turn off the Scan Drive power switch.
4.) Rotate the gantry until the detector reaches 45 degrees.
5.) Lock the gantry in the 45 degree position.

WARNING **MAKE SURE YOU ENGAGE THE LOCKING PIN BEFORE YOU REMOVE THE DETECTOR. FAILURE TO LOCK THE GANTRY COULD RESULT IN INJURY, SHOULD THE GANTRY SUDDENLY MOVE AND STRIKE YOU.**

6.) Remove the DAS covers.

7.) Disconnect the 3 Heater Controller Cables between the detector and the detector heater.

8.) Install the hoist.

- a.) Connect the hook on the hoist to the hole drilled into the main detector plate, in the center of the detector.
- b.) Take up any slack on the hoist chain.

9.) Remove the Detector:

Note: Label Cables — Remember to label the cables before you disconnect them, so you can restore them to their original configuration.

- a.) Carefully label and disconnect all 47 flex cables from the Detector and DAS.
- b.) Remove 2 detector mounting nuts from each detector mount, and set aside for reuse.
- c.) Carefully swing the detector off its mounts, and out into the gantry.

  Take care not damage or lose any of the mounting hardware.
- d.) Carefully lower the detector onto a skid.
- e.) Take care not to set the detector on any of its cables.
- f.) If necessary, change any damaged mounting hardware.

10.) Install the new detector:

- a.) Install the hoist.
- b.) Connect the hook on the hoist to the hole drilled into the main detector plate, in the center of the detector.
- c.) Take up any slack on the hoist chain.
- d.) Make sure all the mounting hardware is present and in place.
- e.) Swing the detector into position.
- f.) Replace the washer and nuts, but do not tighten the mounting bolts.
- g.) Mechanically align the detector to the base plate.

  Check and adjust 3.5 inches across the full arc of the detector.

11.) Check detector isolation:

- a.) Connect one lead of a DVM to the Frame of the detector, and one lead to the gantry rotating base.
- b.) Maker sure you measure infinite resistance.
- c.) If you measure any resistance, check to make sure each mounting bolt has a fiber washer.

12.) Tighten the mounting nuts:

- a.) Torque each of the inside detector mounting nuts to 25 ft-lbs.
- b.) Hold the lower mounting nut, to prevent it turning, while you tighten the outside nut to 25 ft-lbs.

NOTICE Avoid Damage **Do not over tighten the mounting nuts. Because the detector no longer has shocks, excess force can twist and damage it.**

13.) Connect all 47 flex cables between the Detector and DAS.

14.) Reassemble Gantry.

Perform the following verification/checks and resets after you have complete detector installation:

- Alignments: Start on page 109.

- Perform a complete image calibration, QCAL, Cross Talk, Afterglow, Air CALs, Phantom CALs, and N number correction: Start on page 59.
- Reset Smart Trend Baseline if applicable (5.x SW), on page 264 in Chapter 6.
- Image Series: Refer to Chapter 2.

5.6 46-241601G2 Flex Circuit Assembly

1.) Position table to lowest elevation.
2.) Turn off facility power to PDU.
Use tag and lockout procedure.
3.) Remove gantry side covers.
4.) Turn off all 3 switches on status control box, on right side of Gantry.
5.) Open top cover, and engage prop rod.
6.) Remove scan window.
7.) Open front cover.
8.) Rotate Gantry until tube reaches the 9:00 position, and lock into position.

NOTICE **Prevent permanent damage to the static-sensitive boards Attach the anti-static wrist strap to your wrist, and to a bare metal grounding point on the chassis before you continue.**

9.) Remove DAS and detector covers from Gantry (4 pcs).
10.) Locate the defective flex circuits.

Note: Pay attention to the connector locations for each flex circuit.

11.) Take care when you remove the flex circuit from the detector:
   a.) Pull the flex circuit connector from the pin header on back of the detector.
   b.) Remove the other end of the flex circuit connector from the DAS circuit card, and slide it out of the flex guide.
12.) Replace defective flex circuit.
13.) Reassemble Gantry.

Section 6.0
HP-DAS “Sometimes” Puts Bad Data in RCOM after Power Cycle

After the HP-DAS is power cycled, bad data will sometimes be put into the RCOM buffers. This problem will cause the system to report an HP-DAS, 253823, parity error when attempting to scan.

- It does not always happen.
- You can prevent the error by doing a Hardware Reset after an HPDAS Power Cycle.

GE MEDICAL SYSTEMS

GE MEDICAL SYSTEMS-AMERICAS: FAX 414.544.3384
P.O. BOX 414; MILWAUKEE, WISCONSIN 53201-0414, U.S.A.

GE MEDICAL SYSTEMS-EUROPE: FAX 33.1.40.93.33.33
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GE Medical Systems
CT/i System Service Manual - Advanced

Chapters 11, 12, 13, Index and Glossary

Gantry, Table and Power Distribution Unit (PDU)

Chapter 11
Gantry

Purpose: This chapter provides test and part replacement information necessary to maintain operational performance of the CT/i HiSpeed Advantage. Please perform the re-tests listed in each section, when you replace or adjust any gantry assembly and/or part. System Functional Test means scan the first six series using PROTOCOL LIST 20.8 called the System Scanning Test; how to scan with protocols begins on page 63.

Section 1.0
OBC

1.1 Replacement Verification and Re-Test

RCOM FRU/COMPONENTS RETEST MATRIX

Communication FRU	Task	Verification Test
RCOM	To replace faulty FRU: RCOM on page 723 RPSCOM on page 728 Buffer or TAXI Boards, begin your search on page 776 Manual check for TAXI errors: Open a Unix shell and, type: `cu sbc` type: `nbsClient stc,` type: `s sr0` Verify fewer than 2 TAXI Link failures occur in 80 slices, **CTRL + C** [to exit], type: `nbsClient obcr` type: `s scr0` Verify fewer than 2 TAXI Link failures occur in 80 slices, **CTRL + C** [to exit], ~. [to exit].	1.) Hardware Reset 2.) Acquire 10 scouts: (120kv/ 40ma., 1000mm table movement), 3.) Acquire 100 axials: (120kv/ 80ma., 1 sec. scan, 1 sec. ISD), 4.) Acquire 1 helical: (120 kv/ 40ma., 30 sec. scan), 5.) Acquire 10 axial scans: (120kv/400ma.) 6.) Verify NO TAXI Link errors To check Rcom/Scom Statistics enter: **FSST; 12; 3**

Table 11-1 Retest Matrix for Gantry/RCOM Component

GANTRY FRU/COMPONENTS RETEST MATRIX

OBC COMPONENTS	TASK	VERIFICATION TEST
Collimator Board	Replace faulty board	System Functional Test.
CTVRC (CT Voltage Rotor Controller) Board	Replace faulty board and set jumper correctly!	System Functional Test on page 69 and X-Ray Functional Test on page 356
Gentry I/O Board	Replace faulty board, page 723	1.) Verify mA meter on page 549 2.) Verify kV meter on page 556 3.) kV and mA verification on page 547 4.) Exposure time accuracy on page 67 5.) System Scanning Test on page 69 6.) X-Ray Functional Test on page 356
OBC Heurikon CPU Board	Replace faulty board, page 723	1.) Verify OBC Node DIP switches on page 726 2.) Perform System Functional Test on page 69
OBC Backplane	Check empty chassis power supplies – all slots	1.) Verify mA meter on page 549 2.) Verify kV meter on page 556 3.) Auto mA Cal on page 558 4.) System Scanning Test on page 69

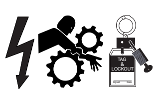
Table 11-2 Retest Matrix for Gantry/OBC Components

1.2 Replacement Procedures

1.2.1 46-220234P3 OBC Fan

1.) Move table to lowest elevation.
2.) Turn off facility power to PDU.

DANGER **USE TAG AND LOCK OUT PROCEDURE.**



3.) Remove gantry side covers.
4.) Turn off all three (3) switches on status control box, on right side of gantry.
5.) Lift top cover, and engage prop rod.
6.) Remove scan window.
7.) Open front cover.
8.) Rotate the Gantry until the OBC reaches the 3 o'clock position.
9.) Engage rotational lock.
10.) Disconnect Power Cord from Fan.
11.) Remove and keep, 4 screws that fasten Fan and Grill in place.
12.) Transfer grill to new fan, if necessary, and install new Fan.
13.) Reassemble Gantry.

1.2.2 46-264700G1 RCOM Bd

1.) Turn off facility power to PDU.

DANGER **USE TAG AND LOCK OUT PROCEDURE.**

2.) Remove, and set aside, both gantry side covers.
3.) Turn off all 3 switches on the status Control box, on right side of Gantry.
4.) Rotate Gantry until the OBC reaches the 3 o'clock position
5.) Engage the rotational lock.
6.) Wear grounded wrist strap.
7.) Loosen the 8 captive screws, and remove OBC Front Cover.
8.) Rotate the ejectors, and remove the defective board from the card cage.
9.) Place the board in an Anti-Static bag.
10.) Install the new board.
11.) Reassemble Gantry.

1.2.3 46-296377P1 OBC Heurikon Board

NOTICE **Wear a grounded wrist strap when you handle a circuit board.**

1.) Remove and set aside gantry side covers.
2.) Turn off all 3 switches on status control box.
3.) Rotate the Gantry until the OBC reaches 3 o'clock position.
4.) Engage rotational lock.
5.) Put on grounding wrist strap.
6.) Loosen 2 captive screws, and remove the OBC Front Cover.
7.) Remove the ribbon cable from the serial port on front panel.
8.) Remove board from slot A7 by loosening 2 retaining screws and rotating ejectors.
9.) Place Board in Anti-Static bag.
10.) Install new board.
11.) Reassemble Gantry.

1.2.4 46-288512G1 Gentry I/O Board

1.) Remove and set aside the right gantry side covers.
2.) Turn off all 3 switches on the status control box.
3.) Rotate the Gantry until the OBC reaches the 3 o'clock position.
4.) Engage rotational lock.
5.) Put on grounding wrist strap.
6.) Loosen the 8 captive screws, and remove the OBC Front Cover.
7.) Rotate the ejectors, and remove the defective board from the card cage.
8.) Place the board in an Anti-Static bag.
9.) Install the new board.
10.) Reassemble Gantry.

1.2.5 46-264660G1 OBC Backplane

1.) Move table to lowest elevation.
2.) Turn off facility power to PDU.

DANGER **USE TAG AND LOCK OUT PROCEDURE.**

3.) Remove gantry side covers.
4.) Turn off all three (3) switches on status control box, on right side of gantry.
5.) Lift top cover, and engage prop rod.
6.) Remove scan window.
7.) Open front cover.
8.) Rotate Gantry until the OBC reaches the 3:00 position.
9.) Put on grounding wrist strap.
10.) Loosen 8 captive screws, and remove the OBC Front Cover.
11.) Rotate the ejectors and remove the Circuit Boards from slots A1 through A6.
12.) Place Boards in Anti-Static bag.
13.) Remove, and keep, ribbon cable from the front of the CPU Board.
14.) Loosen the 2 screws on the front plate of the CPU Board assembly.
Remove the CPU Board from slot A8, and place it in a static bag.
15.) Remove ribbon cable, J9, from the OBC backplane, inside the Card Cage.
16.) Rotate Gantry until the OBC reaches the 6:00 position.
Disconnect 5 connectors, J1 through J5, attached to Control assembly.
17.) Remove the J9 connector, that supplies AC Power to the OBC.
18.) Remove the ty-raps that fasten the cables to control assembly brackets.
19.) Remove 4 screws that fasten the control assembly bracket to the OBC and Casting.
20.) Remove the control assembly and bracket from the Gantry.
21.) Remove, and keep, 2 screws that fasten the Collimator Connector Bulk Head to the OBC assembly.
22.) Remove 8 Fiber Optic Link Connectors from OBC Backplane.
23.) Remove DC Power Connector, J6, from OBC Backplane.
24.) Remove 2 coaxial cables, J4 and J5, from OBC Backplane.
25.) Remove Ground Strap between OBC Backplane and Chassis.
26.) Remove 6 (six) 96 Position Connectors from J3 on slots A1 through A6 on OBC Backplane.
27.) Remove, and keep, 2 screws that fasten the Harness Ground Bracket to the OBC Chassis.
28.) Disconnect DAS DC Power Connector J
29.) Remove Filament Power Connector J17 from OBC Backplane.
30.) Use a short screw driver to remove 8 screws that fasten the OBC Backplane to the Card Cage.
31.) Remove Backplane.
32.) Replace Backplane.

Note: Align Backplane — Install Backplane and replace all 8 screws but do not tighten. Install Circuit Boards in A1 and A6 to align Backplane to Card Cage, then securely tighten 8 mounting screws.

33.) Reassemble Gantry.

NOTICE **Carefully check connector locations before you install the connectors.**

1.2.6 46-297603G1 OBC Thermistor

NOTICE
Avoid Pinched Leads

Take care to route the thermistor leads so the fan housing does not pinch leads.

1.) Move table to lowest elevation.
2.) Turn off facility power to PDU.

DANGER

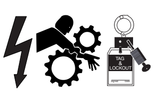
USE TAG AND LOCK OUT PROCEDURE.

3.) Remove gantry side covers.
4.) Turn off all three (3) switches on status control box, on right side of gantry.
5.) Lift top cover, and engage prop rod.
6.) Remove scan window.
7.) Open front cover.
8.) Rotate Gantry until the OBC reaches the 3:00 position.
9.) Loosen 4 captive screws on DAS PS Assembly, and remove cover.
10.) Remove, and set aside, the Power Cord from the Fan.
11.) Remove, and keep, 2 screws that fasten the cables to the fan housing.
12.) Remove, and keep, 6 screws that fasten the fan housing to the OBC Chassis. Three Screws are located in the base of the DAS PS Assembly. The remaining 3 screws are located on the opposite side of the Card Cage.
13.) Remove, and set aside, the Fan Housing.
14.) Detach J8 Connector from OBC Chassis.
15.) Unscrew Thermistor from OBC Chassis.
16.) Replace Thermistor.
17.) Reassemble Gantry.

1.2.7 46-297445P1 OBC Ribbon Cable

1.) Turn off facility power to PDU.

DANGER



USE TAG AND LOCK OUT PROCEDURE.

2.) Remove, and set aside, gantry side covers.
3.) Turn off all three switches on status control box.
4.) Rotate the Gantry until the OBC reaches the 3 o'clock position.
5.) Engage rotational lock.

NOTICE

Wear a grounded wrist strap when you handle a circuit board.

6.) Loosen 8 screws, and remove the OBC Front Cover.
7.) Remove Circuit Board from slot 6 on the OBC.
8.) Place Board in Anti-Static bag.
9.) Use the ejectors to remove the Ribbon Cable connectors from the front of CPU card and OBC Backplane.
10.) Open the Ribbon Cable Clamp, located next to the CPU Card.
11.) Replace the Ribbon Cable.
12.) Reassemble Gantry.

1.3 OBC Heurikon CPU Board

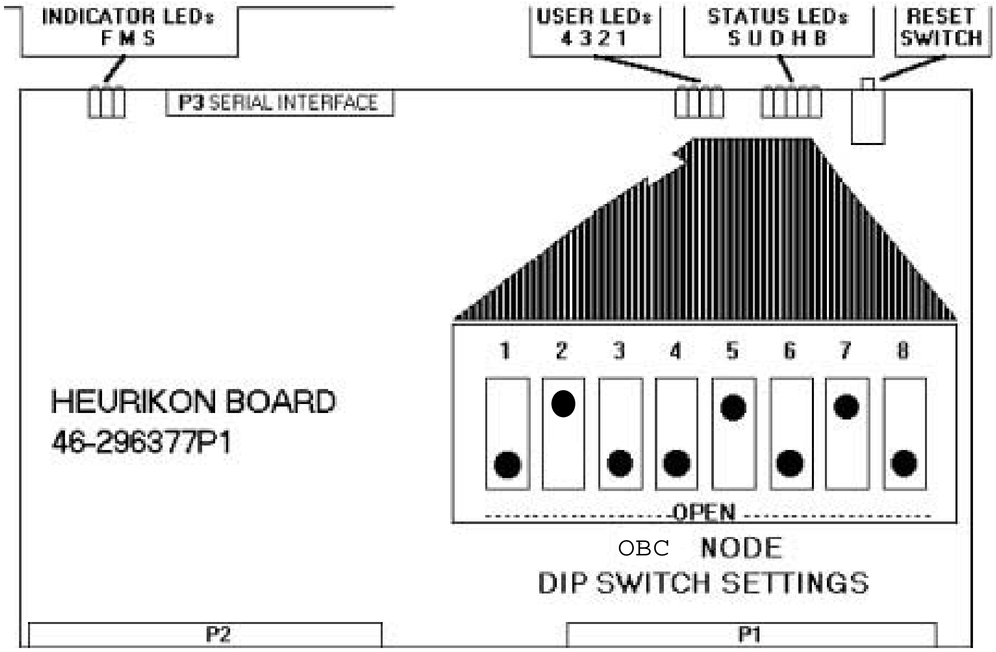


Figure 11-1 46-296377 Heurikon CPU OBC

1.3.1 Test Points

No serviceable test points.

1.3.2 Heurikon CPU Board LEDs

Indicator LEDs

- S = Slave
- M = Master
- F = Fail

User LEDs

- 1 = User LED1 - MSB
- 2 = User LED2
- 3 = User LED3
- 4 = User LED4 - LSB

The Heurikon CPU self test section contains additional node specific troubleshooting information for the User LEDs (see ETC, STC & OBC “Heurikon” CPU - Power-up Self Tests on page 223 of Chapter 6).

1.3.3 Status LEDs

- B = Bus: Another VMEBus master has control of the bus.
- H = Halt: The MPU has halted.
- D = DMAC: The DMAC has control of the local bus.
- U = User: The MPU is in the user state.
- S = Super: The MPU is in the supervisor state.

1.3.4 Heurikon CPU Board Switch Settings

Reset Switch – Resets the CPU board and initiates the self test.

1.3.5 Configuration DIP Switch

Used to configure the CPU board to a specific node. The correct setting of this DIP switch can be found by selecting the Board Layout button below.

Additional information is available in the Heurikon CPU self test section (see ETC, STC & OBC “Heurikon” CPU - Power-up Self Tests on page 223 of Chapter 6).

Section 2.0 STC

2.1 Replacement Verification and Re-Test

2.1.1 RPSCOM

Communication FRUs	Task	Verification Test
RPSCOM	To replace faulty FRU: RCOM on page 723 RPSCOM on page 728 Buffer or TAXI Boards, begin your search on page 776 Manual check for TAXI errors: Open a Unix shell and, type: **`cu sbc`** type: **`nbsClient stc,`** type: **`s sr0`** Verify fewer than 2 TAXI Link failures occur in 80 slices, **CTRL + C** [to exit], type: **`nbsClient obcr`** type: **`s scr0`** Verify fewer than 2 TAXI Link failures occur in 80 slices, **CTRL + C** [to exit], ~. [to exit].	1.) Hardware Reset 2.) Acquire 10 scouts: (120kv/ 40ma., 1000mm table movement), 3.) Acquire 100 axials: (120kv/ 80ma., 1 sec. scan, 1 sec. ISD), 4.) Acquire 1 helical: (120 kv/ 40ma., 30 sec. scan), 5.) Acquire 10 axial scans: (120kv/ 400ma.) 6.) Verify NO TAXI Link errors To check Rcom/Scom Statistics enter: **FSST; 12; 3**

Table 11-3 Retest Matrix for RPSCOM Communication FRU

2.1.2 Gantry (Stationary) Parts Retest Matrix

STC COMPONENTS	TASK	VERIFICATION TEST
STC Heurikon CPU Board	Replace faulty board, page 729	1.) Verify STC Node DIP Switches on page 731. 2.) Perform System Functional Test on page 69.
Axial Board	Replace faulty board	Exposure time accuracy on page 67, and System Functional Test on page 69.
LAN Board	Replace faulty board	Perform board level diagnostics, and System Functional Test.
STC Backplane	Check empty chassis power supplies – all slots.	System Functional Test on page 69.

Table 11-4 Retest Matrix for STC Components

2.2 Replacement Procedures

2.2.1 46-220234P3 STC Fan

1.) Turn off facility power to PDU.

DANGER USE TAG AND LOCK OUT PROCEDURE.

2.) Remove, and set aside, Left Side Cover.
3.) Loosen 2 wing nuts that fasten the card cage in place.
4.) Swing the card cage 90 degrees, to the fully open position.
5.) Unplug power cable from fan.
6.) Remove 4 screws that fasten fan to card cage.
7.) Remove fan and grill from card cage.
8.) Install the new fan with old grill.

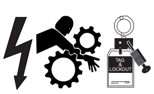
Note: Use push-to-release tab to rotate card cage back into place.

9.) Reassemble Gantry.

2.2.2 46-321246G1 RPSCOM Board

1.) Move table to lowest elevation.
2.) Turn off facility power to PDU.

DANGER USE TAG AND LOCK OUT PROCEDURE.



3.) Remove gantry side covers.
4.) Turn off all three (3) switches on status control box, on right side of gantry.
5.) Loosen 2 screws that fasten the front cover to the STC.
6.) Remove cover.
7.) Loosen the two wingnuts on the STC Chassis, rotate the STC Chassis 90 degrees, then lock it into position.

NOTICE **Wear a grounded wrist strap when you handle a circuit board.**

8.) Use the extractors to remove the defective PWB.

9.) Install new SCOM board assembly.

Note: Use push-to-release tab to rotate card cage back into position.

10.) Reassemble Gantry.

2.2.3 46-297475G1 SCOM Board

1.) Turn off facility power to PDU.

DANGER **USE TAG AND LOCK OUT PROCEDURE.**

2.) Remove left Side Cover.

3.) Loosen 2 screws that fasten the front cover to the STC.

4.) Remove cover.

5.) Loosen the two wingnuts on the STC Chassis, rotate the STC Chassis 90 degrees, then lock it into position.

NOTICE **Wear a grounded wrist strap when you handle a circuit board.**

6.) Use the extractors to remove the defective PWB.

7.) Place the board in an Anti-Static bag.

8.) Install the new SCOM board assembly.

Note: Use push-to-release tab to rotate card cage back into position.

9.) Reassemble Gantry.

2.2.4 46-296377P1 STC Heurikon Board

1.) Move table to lowest elevation.

2.) Remove gantry side covers.

3.) Turn off all three (3) switches on status control box, on right side of the gantry.

4.) Open Gantry Left Side Cover.

5.) Loosen 2 screws that fasten the front cover to the STC.

6.) Remove cover.

7.) Loosen the two wingnuts on the STC Chassis, and rotate the STC Chassis 90 degrees, then lock it into position.

8.) Remove, and set aside, Scom Circuit Board.

9.) Unscrew the top and bottom fasteners, then remove the board.

10.) Reassemble Gantry.

Note: Use push-to-release tab to rotate card cage back into position.

2.2.5 46-136343P12 STC Assembly Switch

1.) Turn off facility power to PDU.

DANGER **USE TAG AND LOCK OUT PROCEDURE.**

2.) Remove left Side Cover.

3.) Loosen 2 screws that fasten the front cover to the STC.

4.) Remove cover.

5.) Loosen the two wingnuts on the STC Chassis, and rotate the STC Chassis 90 degrees, then lock it into position.

6.) Remove two leads from the back of the toggle switch, on the right side of the STC Chassis.

7.) Remove the retaining nut from the front of the toggle switch.

8.) Remove the defective switch from the chassis.

9.) Install the new switch.

10.) Reassemble Gantry.

Note: Use push-to-release tab to rotate card cage back into position.

2.2.6 46-264806G1/G2 Axial Control Board

1.) Move table to lowest elevation.

2.) Remove gantry side covers.

3.) Turn off all three (3) switches on status control box on the right side of the gantry.

4.) Remove left Side Cover.

5.) Loosen 2 screws that fasten the front cover to the STC.

6.) Remove cover.

7.) Loosen the two wingnuts on the STC Chassis, and rotate the STC Chassis 90 degrees, then lock it into position.

NOTICE **Wear a grounded wrist strap when you handle a circuit board.**

8.) Remove the BNC Connector from front of Axial Control PWB.

9.) Use the extractors to remove the defective PWB.

10.) Install new **Axial Control board**.

Note: Use push-to-release tab to rotate the card cage back into position.

11.) Reassemble Gantry.

2.2.7 46-264802G1 STC Backplane

1.) Move table to lowest elevation.

2.) Remove gantry side covers.

3.) Turn off all three (3) switches on status control box, on the right side of the gantry.

4.) Turn off facility power to PDU.

DANGER **USE TAG AND LOCK OUT PROCEDURE.**

5.) Remove left Side Cover.

6.) Loosen 2 screws that fasten the front cover to the STC.

7.) Remove cover.

8.) Loosen the two wingnuts on the STC Chassis, and rotate the STC Chassis 90 degrees, then lock it into position.

NOTICE **Wear a grounded wrist strap when you handle a circuit board.**

9.) Remove the BNC Connector from front of Axial Control PWB.

10.) Use the extractors to remove the PWB.

11.) Use the extractors to remove the SCOM PWB.

12.) Remove the CPU PWB:
   a.) Loosen the two (2) captive screws in the front bezel.
   b.) Use the extractors to remove the CPU PWB from the card cage.
   c.) Store all circuit boards in static bags.
13.) Remove J4 connector from the Backplane inside the card cage.
14.) Press release tab and rotate card cage into its retracted position.
   a.) Loosen 2 screws that fasten the rear cover to the STC.
   b.) Remove cover.
15.) Remove J5 and J6 BNC Connector from Backplane.
16.) Remove three 96 pin connectors from Backplane.
17.) Remove the Ground Strap between the backplane and chassis.
18.) Remove the two screws from the bottom of the card cage, that fasten the harness bracket to the chassis.
19.) Use a short screwdriver to remove the eight (8) 6-32 screws that fasten the Backplane to the card cage.
20.) Remove the Backplane PWB from the STC assembly.
21.) Remove the two (2) terminator assemblies from the old Backplane.
22.) Transfer the two (2) terminator assemblies to the new Backplane.
23.) Install the new backplane.
24.) Reassemble Gantry.

2.3 STC Heurikon CPU Board

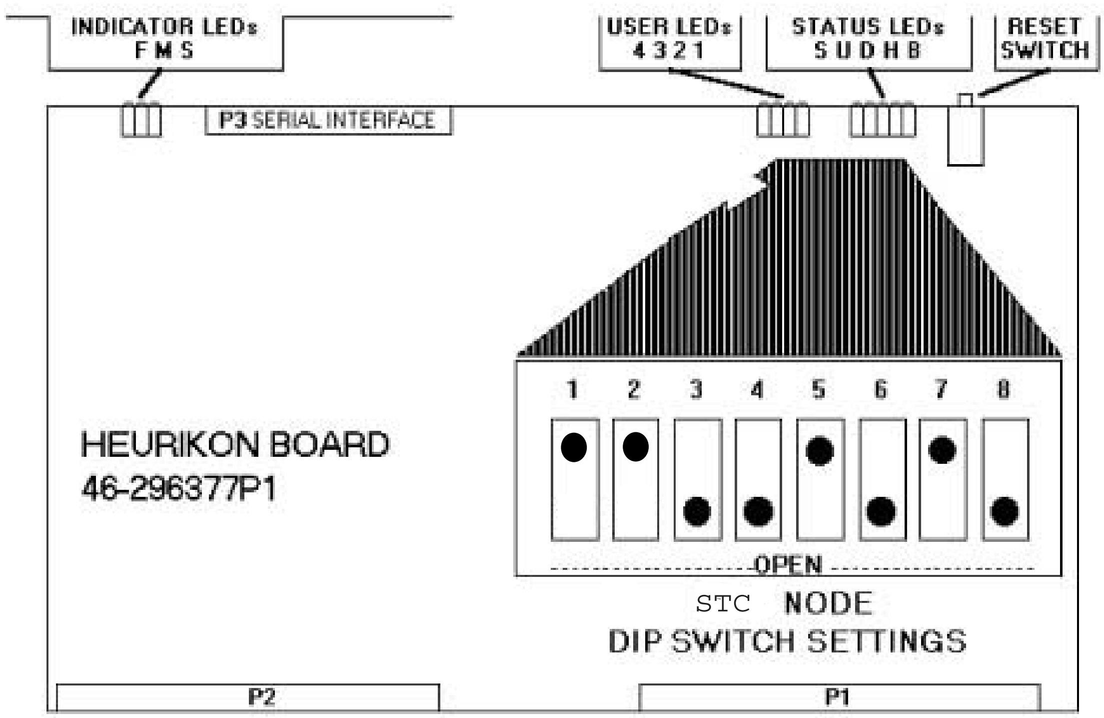


Figure 11-2 46-296377 Heurikon CPU STC

2.3.1 Test Points

No serviceable test points.

2.3.2 Heurikon CPU Board LEDs

Indicator LEDs

- S = Slave
- M = Master
- F = Fail

User LEDs

- 1 = User LED1 - MSB
- 2 = User LED2
- 3 = User LED3
- 4 = User LED4 - LSB

The Heurikon CPU self test section contains additional node specific troubleshooting information for the User LEDs (see ETC, STC & OBC “Heurikon” CPU - Power-up Self Tests on page 223 of Chapter 6).

2.3.3 Status LEDs

- B = Bus: Another VMEBus master has control of the bus.
- H = Halt: The MPU has halted.
- D = DMAC: The DMAC has control of the local bus.
- U = User: The MPU is in the user state.
- S = Super: The MPU is in the supervisor state.

2.3.4 Heurikon CPU Board Switch Settings

Reset Switch – Resets the CPU board and initiates the self test.

2.3.5 Configuration DIP Switch

Used to configure the CPU board to a specific node. The correct setting of this DIP switch can be found by selecting the Board Layout button below.

Additional information is available in the Heurikon CPU self test section (see ETC, STC & OBC “Heurikon” CPU - Power-up Self Tests on page 223 of Chapter 6).

2.4 46-288170G1 Elevation/Tilt Amplifier

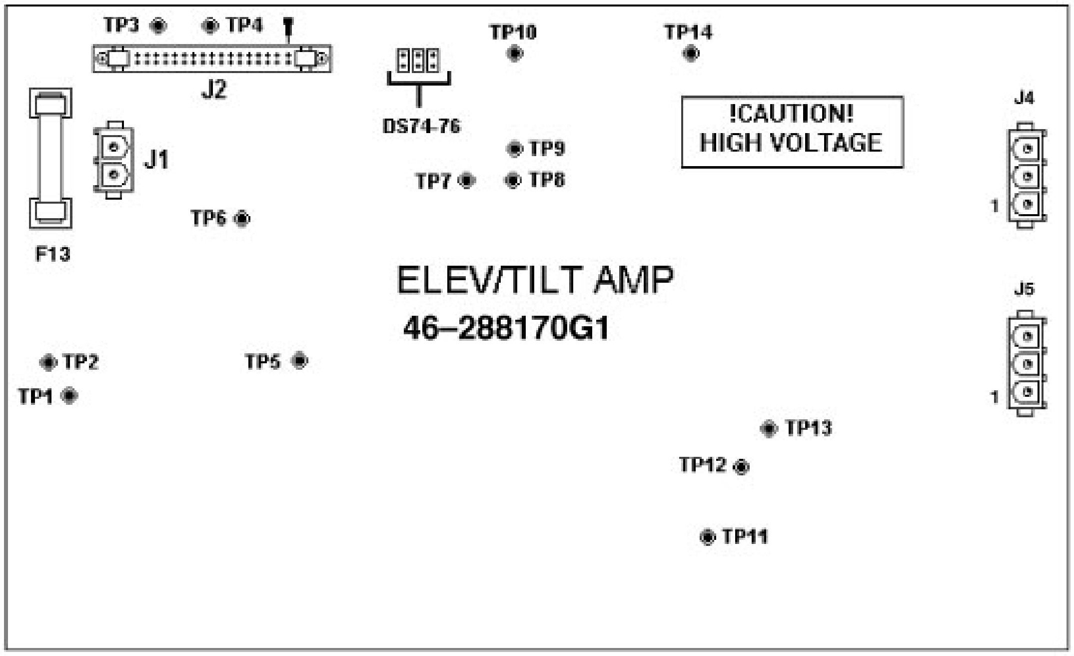


Figure 11-3 46-288170G1 Elevation/Tilt Amplifier

2.4.1 Elevation/Tilt Amplifier Board LEDs

LEDs exist to give a “feel” for the state of the circuitry on the amplifier board. All of the following LEDs pulse at various duty cycles (with the exception of DS PS) so they vary in intensity from a fully on LED.

- DS74: On when the lower FET (Q20) is commanded off. This LED indicates the presence of logic drive to the isolated FET drive circuitry, not the state of the Q20's gate direction.
- DS75: On when the lower FET (Q150) is commanded off. This LED indicates the presence of logic drive to the isolated FET drive circuitry, not the state of the Q150's gate direction.
- DS76: On when +24V is present on the amplifier board. If +24V is not present, the DC-DC converter will not work, the relay cannot operate and there isn't any drive voltage present of the bridge.
- DS125: ON when DCCLK pulses are present at the amplifier. These pulses are needed for the DC-DC converter to operate.

2.4.2 Elevation/Tilt Amplifier Board Switch Settings

None

2.4.3 Elevation/Tilt Amplifier Board Test Points

- TP1 ISO3: Positive side of the isolation supply for the lower FETs of the “H-Bridge” Q20 and Q150.
- TP2 GND: Low side of the amplifier bridge. It also acts as ISO3 RTN, This is the high voltage supply return.
- TP3 AGND: Analog ground.
- TP4 +24V: 24 volts for the relays and the DC-DC converter.

- TP5 ISO1: Positive side of the isolation supply for Q55 FET drive, measured with respect to TP9.
- TP6 P-IL: Elev/Tilt Pulse-IL*1- Goes low when the amplifier bridge is at or above 6A (nominal).
- TP7 SCKT: Short Circuit signal. This goes low when the bridge current goes above 12A (nominal).
- TP8: This test point is the output of the push-pull driver which drives the DC-DC converters' isolation transformer primaries.
- TP9 RTN1: Low side of the isolation supply for Q55 FET drive.
- TP10 PGND: This is the +24V return.
- TP11 HV: This is the high side of the amplifier bridge. This is at about +170V.
- TP12 RTN2: Low side of the isolation supply for Q130 FET drive.
- TP13 ISO2: Positive side of the isolation supply for Q130 FET drive, measured with respect to TP12.
- TP14 VMTR: The scaled, filtered and buffered E/T motor voltage.

Section 3.0 Intercom

3.1 Replacement Verification and Re-Test

Gantry FRU	Task	Verification Test
Microphone	Replacement procedure on page 735	Play canned AUTOVOICE messages. Adjusting volume is explained on page 482.
Intercom Board	Replace board, see page 723.	To check and adjust the pots for the intercom, see page 136 Play canned AUTOVOICE messages.

Table 11-5 Retest Matrix for Gantry Intercom Components

3.2 Replacement Procedures

3.2.1 46-297488P1 Front Cover Microphone

1.) Move table to lowest elevation.
2.) Turn off facility power to PDU.

DANGER **USE TAG AND LOCK OUT PROCEDURE.**

3.) Remove gantry side covers.
4.) Turn off all three (3) switches on status control box, on right side of gantry.
5.) Lift top cover, and engage prop rod.
6.) Remove scan window.
7.) Open Front Cover.
8.) Remove defective microphone:

a.) Locate the Front microphone interface PWB, inside the front cover, just below the laser window:

b.) Loosen the retaining screws, and remove the red and black leads from the terminal block.

c.) Remove (2) nuts from the two microphone mounting studs

d.) Remove the microphone by pulling the device straight out from the front side of the front cover.

9.) Install new microphone:

a.) Thread the red and black leads through the bottom mounting hole in the front cover.

b.) Position the red and black leads in the slotted cutout in the bottom mounting hole.

c.) Insert the two studs on the microphone into the two mounting holes in the front cover.

d.) Thread the red and black leads through the center hole on the microphone interface PWB.

e.) Position the PWB over the two mounting studs on the inside of the cover, and install the two mounting nuts.

f.) Connect the red lead to position 1 and the black lead to position 2 on the terminal block.

10.) Reassemble Gantry.

3.2.2 46-288766G1 Intercom Circuit Board

1.) Move table to lowest elevation.

2.) Turn off facility power to PDU.

DANGER USE TAG AND LOCK OUT PROCEDURE.

3.) Remove gantry side covers.

4.) Turn off all three (3) switches on status control box, on right side of gantry.

5.) Lift top cover, and engage prop rod.

6.) Remove scan window.

7.) Open rear gantry cover.

8.) Remove 25 pin sub D connector from Intercom PWB.

Note: Pay attention to the configuration before you remove the connectors and leads from the assembly. Reassemble with the same configuration.

9.) Remove two AC leads from line filter.

10.) Remove DC connector (J3) from Assembly.

11.) Remove four (4) screws that fasten the assembly to the gantry.

12.) Remove the intercom assembly from gantry

NOTICE Wear a grounded wrist strap when you handle a circuit board.

13.) Remove the four (4) nuts that fasten the Intercom PWB cover to the bracket.

14.) Remove the cover with the PWB from the assembly.

15.) Remove four (4) screws that fasten the PWB to its cover, and remove PWB.

16.) Install the new Intercom PWB.

17.) Reassemble Gantry.

3.2.3 46-297488P1 Rear Microphone

1.) Move table to lowest elevation.

2.) Turn off facility power to PDU.

DANGER USE TAG AND LOCK OUT PROCEDURE.

3.) Remove gantry side covers.
4.) Turn off all three (3) switches on status control box, on right side of gantry.
5.) Lift top cover, and engage prop rod.
6.) Remove scan window.
7.) Open rear cover.
8.) Locate Rear Microphone Interface PWB on the inside of the rear cover near the top of the cone:
    a.) Loosen the retaining screws, and remove the red and black leads from the terminal block.
    b.) Remove two (2) nuts from studs.
    c.) Remove the microphone by pulling the device straight out from the outside of the rear cover.
9.) Install the new microphone:
    a.) Thread the red and black leads through the bottom mounting hole in the rear cover, and position them in the slotted cutout.
    b.) Insert the two studs on the microphone into the two mounting holes in the rear cover.
    c.) Thread the red and black leads through center hole on the microphone interface PWB.
    d.) Position the PWB over the two mounting studs on the inside of the cover, and fasten it into place with the two mounting nuts.
    e.) Connect the red lead to position 1 and the black lead to position 2 on the terminal block.
10.) Reassemble Gantry.

3.2.4 Remote Intercom X Board

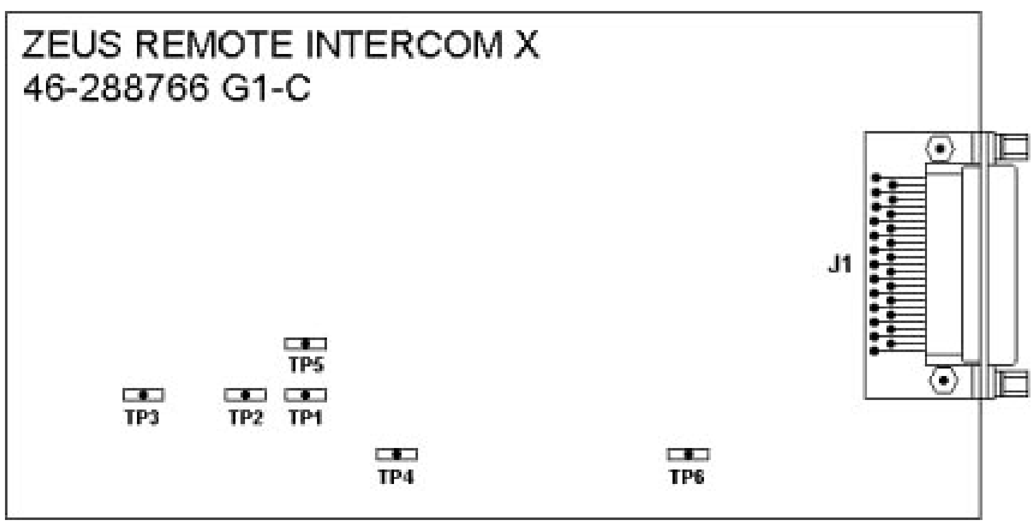


Figure 11-4 REMOTE INTERCOM X BOARD

3.2.5 Remote Intercom X Board Test Points

- TP1: Noise Cancel circuit output.
- TP2: Level Control amp output.
- TP3: Line Driver circuit input.
- TP4: Line Driver circuit output.
- TP5: Automatic Gain Control (AGC) for Level Control amp.
- TP6: Regulator circuit output.

3.2.6 Remote Intercom X Board LEDs

None

3.2.7 Remote Intercom X Board Switch Settings

None

Section 4.0
Axial

4.1 Replacement Verification and Re-Test

Gantry FRU	Task	Verification Test
Axial Motor or Drive Belt	To replace the motor, refer to page 740. To replace the belt, refer to page 738. To set C-Pulse, home flag physical position, refer to page 135	Run Microphonics, a test under DAS Tools, found under Trouble Shoot. System Scanning Test on page 69
Axial Motor Encoder	To set C-Pulse, home flag physical position, refer to page 135 To replace the encoder, refer to page 741	System Scanning Test on page 69
Idler Assembly	To set C-Pulse, home flag physical position, refer to page 135 To replace the idler, refer to page 739	System Functional Test on page 69
Azimuth Board	To set C-Pulse, home flag physical position, refer to page 135 To replace the azimuth board, refer to page 742	Perform Axial Encoder check (refer to page 135) and exercise alignment lights.
Axial Brake	To replace the brake, see page 740. Axial Brake check is on page 136	Exercise alignment lights, and acquire several 1 second scans.

Table 11-6 Retest Matrix for Gantry Axial Drive Components

4.2 Replacement Procedures

4.2.1 46-198495P1 Axial Drive Belt Replacement Procedure

1.) Position Table to lowest elevation.
2.) Turn off facility power to PDU.

DANGER **USE TAG AND LOCK OUT PROCEDURE.**

3.) Remove, and set aside, both gantry side covers.
4.) Turn off all 3 switches on the status control box, on right side of Gantry.
5.) Lift top cover, and engage prop rod.
6.) Remove scan window.
7.) Open Rear and Front Covers.
8.) Rotate Gantry until the tube reaches the 12 o'clock position.
9.) Lock Gantry into the 12 o'clock position.
10.) Remove, and set aside, the Anode Inverter Cover. (Right side of Gantry)
11.) Remove, and set aside, the Axial Brake assembly.
12.) Loosen three 3/8-16 bolts on Idler assembly, to relieve drive belt tension.
13.) Remove old belt by cutting or reversing step 14.
14.) Replace belt: Place the belt around the tube, followed by the CTVRC, the cathode tank and inverter, both DAS. Work the belt over the Anode inverter and tank, and finally, over the OBC.
15.) Replace Anode Inverter Cover.
16.) Position the Drive Belt on the gear ring. Then, fold the belt, and pull it through the opening in frame for the axial motor shaft. Loop the gear belt over the brake shaft and around drive pulley.
17.) Position the drive belt on and around the gear ring, the idler pulleys, and the drive pulley.
18.) Adjust the idler assembly until the belt feels snug.
19.) Rotate Gantry by hand through 3 complete revolutions.
20.) Make the final tension adjustment on the drive belt: Turn the Idler Pulley Adjustment screw until the spring has compressed an additional 0.285 inches.
21.) Replace the Axial Brake assembly.
22.) Turn power back on.
23.) Adjust the C-pulse:
   a.) Rotate the gantry until the tube reaches the 12 o'clock position.
   b.) Pin gantry
   c.) Slightly loosen the 3 synclamps that hold the axial motor encoder in place.

Note: Do loosen Synclamps — Do NOT loosen synclamps too much, or you will have trouble maintaining the encoder position when you tighten the synclamps.

   d.) Place a torpedo level on the machined surface beneath the collimator and hand rotate the gantry until the tube is exactly level and at the top.
   e.) Rotate the Axial Motor Encoder until the C-Pulse Light on the Status Control box remains constantly lit.
   f.) Tighten the three synclamps to fasten the encoder in the current position. Ensure that the C pulse remains lit as you tighten the encoder.

24.) Unlock Gantry.
25.) Reassemble Gantry.

4.2.2 46-296058G1 Drive Belt Idler Pulley

1.) Position table to lowest elevation.
2.) Turn off facility power to PDU.

DANGER **USE TAG AND LOCK OUT PROCEDURE.**

3.) Remove, and set aside, both gantry side covers.
4.) Turn off all 3 switches on the status control box, on right side of Gantry.
5.) Lift top cover, and engage prop rod.
6.) Remove scan window.
7.) Open front cover.
8.) Relieve tension on drive belt by loosening three 3/8 -16 bolts.
9.) Loosen three 3/8-16 bolts on Idler assembly, to relieve drive belt tension.

STATIONARY PULLEY REMOVAL SEQUENCE:

1.) Remove Shoulder Belt.
2.) Replace pulley and shoulder belt.
3.) Apply Loctite 242 to bolt threads and torque to 25 ft-lbs. (0.024 m-kg)

Note: G3 model gantries have an access hole in the frame. In the event the pulley mounting bolt set screw breaks, use this hole to gain access to the set screw. Although restrictive, you may find it easier to extract the broken bolt this way, than by using an EZ out tool.

TENSIONING PULLEY REMOVAL SEQUENCE:

1.) Loosen five 3/8-16 bolts, and remove Tensioning assembly from Gantry.
2.) Loosen the internal hex screw from the back side of the Tensioning assembly, and remove the Pulley from the Tensioning assembly.
3.) Attach new Pulley to Tensioning assembly.
4.) Reattach Tensioning assembly to Gantry.
5.) Check the position of the Drive Belt on and around the Gear ring, the idler pulleys, and the drive pulley on the axial drive motor.
6.) Adjust the idler assembly until the belt feels snug.
7.) Rotate the Gantry by hand through 3 complete revolutions.
8.) To make the final tension adjustment on the drive belt: Turn the Idler Pulley Adjustment screw until the spring has compressed an additional 0.285 inches.
9.) Adjust the C-pulse:
    a.) Rotate the Gantry until the tube reaches the 12 o'clock position.
    b.) Slightly loosen the 3 synclamps that hold the axial motor encoder in place.

Note: Do NOT loosen synclamps too much, or you will have trouble maintaining the encoder position when you tighten the synclamps.

    c.) Place a torpedo level on the machined surface beneath the collimator and hand rotate the gantry until the tube is exactly level and at the top.
    d.) Rotate the Axial Motor Encoder until the C-Pulse Light on the Status Control box remains constantly lit.
    e.) Tighten the three synclamps to fasten the encoder in the current position. Ensure that the C pulse remains lit as you tighten the encoder.

10.) Reassemble Gantry.

4.2.3 46-297875P1 Axial Brake

1.) Remove, and set aside, both gantry side covers.
2.) Turn off all three switches on the status control box on right side of Gantry.
3.) Open top cover, and engage prop rod.
4.) Remove, and set aside, Gantry scan window.
5.) Open rear cover.
6.) Engage Manual Indexer Pin Lock to prevent gantry rotation.
7.) Disconnect electrical connection to Axial Brake.
8.) Remove, and keep, three set screws on aluminum hex nut, on end of brake shaft.
9.) Remove hex nut and key.
10.) Remove four bolts holding brake to torque plate.
11.) Remove brake.
12.) Mount replacement brake onto torque plate.
13.) Restore power to energize the brake before you fasten the hex nut to the brake shaft and aperture plate:
    a.) Turn on ONLY the gantry 120VAC on the status control box, on the right side of gantry.
    b.) Once the brake energize, align the aperture plate with the hex nut on the brake shaft.
14.) Insert key.
15.) Adjust the hex nut until the dimension from the outer surface of the brake to the outer surface of the hex nut equals 0.370 ± 0.010 inches.
16.) Sparingly apply 242 Loctite to the three set screws, and torque them to 25-30 in-lbs. (0.288 m-kg).
17.) Reassemble Gantry.

4.2.4 46-296158P1 Axial Motor

1.) Position table to lowest elevation.
2.) Turn off facility power to PDU.

⚠ **DANGER** **USE TAG AND LOCK OUT PROCEDURE.**

3.) Remove, and set aside, both gantry side covers.
4.) Turn off all 3 switches on the status control box on right side of Gantry.
5.) Open top cover, and engage prop rod.
6.) Remove, and set aside, Gantry scan window.
7.) Open front and rear covers.
8.) Remove, and set aside, Axial Brake.
9.) Disconnect the Axial Motor Encoder electrical connections.
10.) Loosen three screws of Idler Tensioning assembly
11.) Loosen the Tensioning Bolt until the spring is unloaded.
12.) Remove, and set aside, electrical box cover on motor.
    a.) Remove, and keep, all 3 cable conductors.
    b.) Cut Ty-raps to free cable from motor.
13.) Rotate the gantry to position the sagittal laser bracket near the axial motor.
14.) Remove, and keep, the bottom two motor bolts.
15.) Attach the hoist to the gantry:

a.) Fasten the hoist strap around the motor, centered on the housing.

b.) Remove all slack from the hoist strap.

16.) Remove remaining two bolts securing motor.

17.) Swing motor around pivot sector, and out of Gantry, then lower it to the floor.

18.) Use the hoist to move the new motor into position.

a.) Sparingly apply 242 Loctite to the four mounting bolts

b.) Torque the mounting bolts to 25 ft-lbs. (.024 m-kg)

19.) Reinstall power cables

Fasten the cable to the motor with large Ty-raps.

20.) Install Brake Assembly.

21.) Position the drive belt on, and around, the gantry gear ring, the two idler pulleys, and the drive pulley on the axial drive motor.

a.) Perform preliminary idler assembly adjustment to snug up the belt.

b.) Rotate the Gantry by hand through three (3) complete revolutions.

c.) **Retension drive belt:** Turn the idler pulley adjustment screw until the spring has compressed an additional 0.285 inches.

22.) Torque tensioning adjustment plate bolts to 25 ft-lbs. (.024 m-kg)

23.) Adjust the C-pulse:

a.) Rotate the Gantry until the tube reaches the 12 o'clock position.

b.) Slightly loosen the 3 synclamps that hold the axial motor encoder in place.

Note: Do Not Loosen Synclamps

Do NOT loosen synclamps too much, or you will have trouble maintaining the encoder position when you tighten the synclamps.

c.) Place a torpedo level on the machined surface beneath the collimator, and level the gantry.

d.) Rotate the Axial Motor Encoder until the C-Pulse Light on the Status Control box remains constantly lit.

e.) Tighten the 3 synclamps to fasten the encoder in the current position.

24.) Reassemble Gantry.

4.2.5 46-296854P1 Axial Motor Encoder

1.) Position table to lowest elevation.

2.) Turn off facility power to PDU.

DANGER **USE TAG AND LOCK OUT PROCEDURE.**

3.) Remove, and set aside, both gantry side covers.

4.) Turn off all three (3) switches on status control box, on right side of gantry.

5.) Open top cover, and engage prop rod.

6.) Remove, and set aside, Gantry Scan Window.

7.) Open Front Cover.

8.) Remove plastic cover from access hole in aluminum encoder fixturing plate.

9.) Rotate gantry to align the coupling screw with the access hole in the aluminum encoder fixturing plate

10.) Loosen the coupling screw.

11.) Loosen and remove the three (3) synclamps that fasten the encoder to the aluminum fixturing plate.

12.) Remove the Encoder from fixturing plate.

13.) Disconnect the Encoder electrical connection.

14.) Replace and assemble the Encoder.

15.) Adjust the C-pulse:

a.) Rotate the Gantry until the tube reaches the 12 o'clock position.

b.) Slightly loosen the 3 synclamps that hold the axial motor encoder in place.

Note: Do NOT loosen synclamps too much, or you will have trouble maintaining the encoder position when you tighten the synclamps.

c.) Place a torpedo level on the machined surface beneath the collimator, and level the gantry.

d.) Rotate the Axial Motor Encoder until the C-Pulse Light on the Status Control box remains constantly lit.

e.) Tighten the 3 synclamps to fasten the encoder in the current position.

16.) Reassemble Gantry.

4.2.6 46-186462G1 Azimuth Circuit Board

1.) Position table to lowest elevation.

2.) Turn off facility power to PDU.

DANGER **USE TAG AND LOCK OUT PROCEDURE.**

3.) Remove, and set aside, both gantry side covers.

4.) Turn off all three switches on the status control box, on right side of Gantry.

5.) Open top cover, and engage prop rod.

6.) Remove, and set aside, Gantry scan window.

7.) Open front cover.

8.) Open rear Cover.

9.) Rotate gantry until the tube reaches the 12 o'clock position.

10.) Remove the two screws that fasten the PWB cover to the right side of the upper front main bearing frame.

11.) Remove the PWB cover.

12.) Remove the left side slip ring cover (as viewed from rear facing forward).

NOTICE **Wear a grounded wrist strap when you handle a circuit board.**

13.) Disconnect cable from J1 on circuit board.

14.) From the rear of the gantry:

a.) Remove the two (2) screws that fasten the Board mounting bracket to the main bearing frame.

b.) Remove the assembly from the gantry.

15.) Remove the three (3) screws that fasten the PWB to the bracket, and remove the PWB.

16.) Install the new PWB.

17.) Adjust PWB- center flag in the optical detector slot on the PWB.

a.) Adjust flag depth to 0.1 inches ± 0.02 in. from the bottom of the slot in the optical detector.

b.) Radially center the flag between the slot openings.

18.) Reassemble Gantry.

Section 5.0
Power

5.1 Replacement Verification and Re-Test

Gantry FRU	Task	Verification Test
DC Power Supply (+12 V)	Refer to 3.6 on page 134. Measure unloaded and loaded voltages	System Functional Test on page 69
Collimator Power Supply	Refer to 3.10 on page 135. Measure unloaded and loaded voltages	System Functional Test (page 69)
Detector Heater Power Supply	Refer to 3.7 on page 135. Measure unloaded and loaded voltages	System Functional Test, DAS Tools – Aux channel test to confirm 34 degrees C. -or- in shell, enter: **FSST; 12; 5; 3**
Filament Power Supply	Refer to 3.8 on page 135. Measure unloaded and loaded voltages	Auto mA Cal, verify kV and mA, and System Functional Test

Table 11-7 Retest Matrix for Gantry Power Supplies

GANTRY PARTS RETEST MATRIX

STC COMPONENTS	TASK	VERIFICATION TEST
DC Power Supply (+5 volt, +/- 15 volt, +24 volt)	Measure unloaded and loaded voltages, see page 133	System Functional Test on page 69.

Table 11-8 Retest Matrix for STC Power Supply Components

OBC COMPONENTS	TASK	VERIFICATION TEST
DC Power Supply (+5 volt,+/- 15 volt, +24 volt)	To replace, refer to page 747. Measure unloaded and loaded voltages, see page 133	System Functional Test on page 69.

Table 11-9 Retest Matrix for OBC Power Supply Components

5.2 Replacement Procedures

5.2.1 46-297335P2 OBC Power Line Filter

1.) Turn off facility power to PDU.

DANGER **USE TAG AND LOCK OUT PROCEDURE.**

2.) Remove, and set aside, gantry side covers.
3.) Remove, and set aside, scan window.
4.) Lift Top Cover, and engage prop rod.
5.) Turn off Axial Drive Switch.
6.) Remove Front Cover.
7.) Rotate Gantry until the OBC reaches the 3:00 position.
8.) Locate the Line Filter, attached to the OBC assembly, between the X-Ray Tube and OBC Assemblies.

Note: Line Filter Connections — Pay attention to the wire locations on the Line Filter.

9.) Verify the wires attached to the line filter have no voltage.
10.) Disconnect 4 leads from Line Filter.
11.) Remove, and keep, 2 screws that fasten the Line Filter to OBC assembly.

CAUTION **Do not connect any leads to Ground Terminal on Line Filter.**

12.) Replace Line Filter.
13.) Reassemble Gantry.

5.2.2 46-297335P2 STC Power Line Filter

1.) Turn off facility power to PDU.

DANGER **USE TAG AND LOCK OUT PROCEDURE.**

2.) Remove left Side Cover.
3.) Loosen 2 screws that fasten the front cover to the STC.
4.) Remove cover.
5.) Loosen the two wingnuts on the STC Chassis, and rotate the STC Chassis 90 degrees, then lock it into position.

Note: Line Filter Connections — Pay attention to the wire locations on the Line Filter on the right side of the STC Chassis.

6.) Remove leads from line filter.
7.) Remove two (2) 8-32 screws that fasten the line filter to STC Chassis.
8.) Remove the defective filter.
9.) Install new filter.
10.) Reassemble Gantry.

Note: Use push-to-release tab to rotate card cage back into position.

5.2.3 46-170021P52 3A, 250V DAS Fuse

1.) Move table to lowest elevation.
2.) Turn off facility power to PDU.

DANGER USE TAG AND LOCK OUT PROCEDURE.

3.) Remove gantry side covers.
4.) Turn off all three (3) switches on status control box, on right side of gantry.
5.) Lift top cover, and engage prop rod.
6.) Remove scan window.
7.) Open front cover.
8.) Rotate Gantry until the OBC reaches the 4:00 position.
9.) Locate defective Fuse, and remove cap from fuseholder.
10.) Replace Fuse.
11.) Reassemble Gantry.

5.2.4 46-170021P30 2A, 350V DAS Fuse

1.) Move table to lowest elevation.
2.) Turn off facility power to PDU.

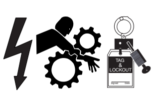
DANGER USE TAG AND LOCK OUT PROCEDURE.

3.) Remove gantry side covers.
4.) Turn off all three (3) switches on status control box, on right side of gantry.
5.) Lift top cover, and engage prop rod.
6.) Remove scan window.
7.) Open front cover.
8.) Rotate Gantry until the OBC reaches the 4:00 position.
9.) Locate defective Fuse, and remove cap from fuseholder.
10.) Replace Fuse.
11.) Reassemble Gantry.

5.2.5 46-170021P15, P52 & P74 Filament Power Asm Fuse

1.) Move table to lowest elevation.
2.) Turn off facility power to PDU.

DANGER USE TAG AND LOCK OUT PROCEDURE.



3.) Remove gantry side covers.
4.) Turn off all three (3) switches on status control box, on right side of gantry.
5.) Lift top cover, and engage prop rod.
6.) Remove scan window.
7.) Open Front Cover.
8.) Rotate gantry until Filament Power assembly reaches the 3 o’clock position.
9.) Engage gantry rotational lock.
10.) Loosen four (4) captive screws on the Filament Power Assembly cover.
11.) Remove cover.

12.) Locate fuses on inside surface of filament power assembly.
13.) Remove cap from fuse holder, and remove defective fuse.
14.) Replace Fuse.
15.) Reassemble Gantry.

5.2.6 46-170021P74 OBC Tube Cooling Fuse

1.) Position table to lowest elevation.
2.) Turn off facility power to PDU.

DANGER USE TAG AND LOCK OUT PROCEDURE.

3.) Remove, and set, aside both gantry side covers.
4.) Turn off all 3 switches on the status control box, on right side of Gantry.
5.) Open top cover, and engage prop rod.
6.) Remove, and set aside, Gantry scan window.
7.) Open front cover.
8.) Rotate Gantry until the OBC reaches the 3 o'clock position.
9.) Engage gantry rotational lock.
10.) Remove the defective fuse from the fuse holder, located at the top of the control assembly.
11.) Replace fuse.
12.) Reassemble Gantry.

5.2.7 54358P25 120VAC Inverter Fuse (Anode or Cathode)

1.) Turn off facility power to PDU.

DANGER USE TAG AND LOCK OUT PROCEDURE.

2.) Open Gantry
3.) Rotate the Gantry until the Inverter reaches the 3 o'clock position
4.) Unscrew the fuseholder cover.
5.) Replace the Fuse and fuseholder cover.
6.) Reassemble Gantry.

5.2.8 54367P50 550VDC Inverter Fuse (Anode or Cathode)

1.) Turn off facility power to PDU.

DANGER USE TAG AND LOCK OUT PROCEDURE.

2.) Open Gantry
3.) Rotate the Gantry until the Inverter reaches the 3 o'clock position
4.) Loosen 4 screws, and remove the Inverter Cover
5.) Verify Capacitor Voltage on C5 and C6 equals 0 VDC.
6.) Remove 2 screws, 2 lockwashers and 2 flat washers that fasten the fuse to the brass standoffs.
7.) Replace the Fuse.
8.) Reassemble Gantry.

5.2.9 54358P18 600V Inverter Fuse (Anode or Cathode)

1.) Turn off facility power to PDU.

DANGER USE TAG AND LOCK OUT PROCEDURE.

2.) Open Gantry

3.) Rotate the Gantry until the Inverter reaches the 3 o'clock position

4.) Unscrew the fuseholder cover.

5.) Replace the Fuse and fuseholder cover.

6.) Reassemble Gantry.

5.2.10 46-296317P1 OBC Power Supply

1.) Move table to lowest elevation.

2.) Turn off facility power to PDU.

DANGER USE TAG AND LOCK OUT PROCEDURE.

3.) Remove gantry side covers.

4.) Turn off all three (3) switches on status control box, on right side of gantry.

5.) Lift top cover, and engage prop rod.

6.) Remove scan window.

7.) Open front cover.

8.) Rotate gantry until the OBC reaches the 2 o'clock position.

9.) Engage rotational lock.

10.) Remove two (2) nuts that fasten the Z Bracket, near the center of the power supply, and remove the bracket.

11.) Verify no voltage exists on the leads connected to the power supplies.

12.) Remove all leads from each end of the power supply.

13.) Remove two (2) screws that fasten the power supply to the OBC.

14.) Remove the defective supply.

15.) Install new power supply.

- Make sure voltage selection jumper, located near AC input terminals, is in the 115 VAC Position.
- Take care to correctly reconnect the leads to the power supply.

16.) Reassemble Gantry.

5.2.11 46-296317P1 STC Power Supply

1.) Turn off facility power to PDU.

DANGER USE TAG AND LOCK OUT PROCEDURE.

2.) Remove left Side Cover.
3.) Loosen 2 screws that fasten the front cover to the STC.
4.) Remove cover.
5.) Loosen the two wingnuts on the STC Chassis, and rotate the STC Chassis 90 degrees, then lock it into position.
6.) Remove seven leads from DC output (bottom) of Power Supply.
7.) Remove the small connector with the single black lead (J1) from the DC output area of Power Supply (bottom).
8.) Remove two (2) AC input leads from the Power Supply.
9.) Remove two (2) nuts that fasten the power supply Z bracket.
10.) Remove the bracket.
11.) Remove two (2) screws that fasten the Power Supply to the front flange on the card cage.
12.) Remove the defective Power Supply.
13.) Replace the Power Supply.
14.) Reassemble Gantry.

Note: Take care to correctly replace the power supply lead connections. Use push-to-release tab to rotate card cage back into position.

5.2.12 46-251198P37 Collimator Power Supply

1.) Move table to lowest elevation.
2.) Turn off facility power to PDU.

DANGER USE TAG AND LOCK OUT PROCEDURE.

3.) Remove gantry side covers.
4.) Turn off all three (3) switches on status control box, on right side of gantry.
5.) Lift top cover, and engage prop rod.
6.) Remove scan window.
7.) Open Front Cover.
8.) Rotate gantry until Filament Power assembly reaches the 3 o'clock position.
9.) Engage gantry rotational lock.
10.) Loosen four (4) capture screws on Filament Power assembly cover.
11.) Remove cover.
12.) Unsolder Black and White leads from terminals one (1), and (2), on the Power Supply Transformer.
13.) Unsolder Red and Black leads from (+) and (-) power supply output terminals.
14.) Remove and save four (4) screws mounting collimator power supply to gantry.
15.) Install new supply.

Note: Input leads: Black to transformer terminal one (1) White to transformer terminal two (2) Output leads: Red to PWB terminal (+) OUT Black to PWB terminal (-) OUT

16.) Collimator P.S. output checks:
- Connect positive voltmeter lead to +OUT terminal on power supply.

- Connect negative voltmeter lead to -OUT terminal on power supply.
- Energize 24hr AC Pwr on Gantry.
- Turn voltage adjustment pot to adjust the output voltage to 39.5 ±.5 VDC.
- Remove power from Gantry (tag and lockout procedures).

17.) Reassemble Gantry.

5.2.13 46-170021P52 Collimator Fuse

No specific procedure at this time. Please use good service practices.

5.2.14 46-296885P1 Communication Power Supply

1.) Move the table to its lowest elevation.

2.) Turn off facility power to PDU.

DANGER USE TAG AND LOCK OUT PROCEDURE.

3.) Remove gantry side covers.

4.) Turn off all three (3) switches on the status control box.

5.) Lift top cover, and engage prop rod.

6.) Remove scan window.

7.) Open rear gantry cover.

8.) Remove 25 pin sub D connector from Intercom PWB.

9.) Remove two AC leads from line filter.
Pay attention to the location of each lead when you remove it.

10.) Remove DC connector (J3) from assembly.

11.) Remove four (4) screws mounting assembly to gantry and remove assembly from gantry.

12.) Remove three (3) connectors from power supply.
Pay attention to the location of each connector when you remove it.

13.) Remove four (4) screws that fasten the power supply to the bracket

14.) Remove the defective supply, and replace it with the new supply.

15.) Communication P.S. output check:

a.) Connect positive voltmeter lead to pin 1 of J3.

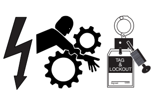
b.) Connect negative voltmeter lead to pin 2 of J3.

c.) Turn ON Gantry 24hr AC power.

d.) Turn the voltage adjustment pot, to adjust the output voltage to equal 12.0 ±0.25 VDC.

e.) Remove power from Gantry

DANGER USE TAG AND LOCK OUT PROCEDURE.



16.) Reassemble the Gantry.

5.2.15 46-188067P1 OBC Detector Heater Relay

1.) Position table to lowest elevation.
2.) Turn off facility power to PDU.

DANGER USE TAG AND LOCK OUT PROCEDURE.

3.) Remove, and set aside, both gantry side covers.
4.) Turn off all 3 switches on the status control box, on right side of Gantry.
5.) Open top cover, and engage prop rod.
6.) Remove, and set aside, Gantry scan window.
7.) Open front cover.
8.) Rotate Gantry until the OBC reaches the 3 o'clock position.
9.) Engage gantry rotational lock.
10.) Remove connectors J1 through J5 from control assembly.
11.) Remove 4 screws that fasten the control assembly to the back of the OBC.
12.) From the bottom of the control assembly, locate the detector heater relay, (normally open) and remove the four (4) leads attached to positions 1 through 4.
13.) Remove two (2) nuts that fasten the relay to the chassis.
14.) Remove defective relay, and install new relay.
15.) Reassemble Gantry.

5.2.16 46-251198P37 Collimator Power Supply

Located on the HEMRC power supply bracket. (38-volt unregulated power supply)

1.) Position table to lowest elevation.
2.) Shut down system and turn off facility power to PDU (Use tag and lockout procedure).
3.) Remove both gantry side covers.
4.) Turn off all three (3) switches on status control box (right side of gantry).
5.) Open top cover and engage "Prop Rod".
6.) Remove gantry scan window.
7.) Open front cover.
8.) Rotate gantry until the HEMRC is in a convenient working position.
9.) Engage gantry rotational lock.
10.) Remove five (5) bolts fastening cover to HEMRC assembly and remove cover.
11.) Note and record position of four (4) wires attached to collimator power supply. (Red and Black DC leads Vs. Black and White AC leads)
12.) Remove leads identified in step 11.
13.) Remove four (4) M4 x 0.7 metric nuts which attach supply to bracket.
14.) Remove and replace power supply module (46-251198 P37).
15.) Replace four (4) M4 x 0.7 metric nuts which attach supply to bracket. Torque to 1.25 ft/lbs. (1.7 Newton Meters).
16.) Reassemble electrical connections removed in step 12.
17.) Replace HEMRC cover and secure with five (5) mounting bolts.
18.) Disengage gantry rotational lock.
19.) Close front cover.
20.) Reinstall gantry scan window.

21.) Close top cover.
22.) Turn On all three (3) switches on the status control box.
23.) Replace both gantry side covers.
24.) Remove "Tag Out" lock and turn on facility power.
25.) Proceed with system test.

Section 6.0 Tilt

6.1 Replacement Verification and Re-Test

Gantry FRU	Task	Verification Test
Tilt Gear Reducer Box	To replace, see page 754.	Characterize Tilt (see page 142), Tilt Gantry, and Characterize Limits (page 146).
Tilt Motor	Replace faulty motor. Refer to page 755	Verify full range of tilt operation
Tilt Gas Springs	Refer to page 751 to replace gas springs	Tilt Gantry full range and no leakage.

Table 11-10 Retest Matrix for Gantry Tilt Components

6.2 Replacement Procedures

6.2.1 46-296209P1 Tilt Gas Spring

Note: Due to alignment and dimensional variations, you may have to remove the Gantry Front cover to replace a Gantry Gas Spring. Because the front cover weighs almost 90 kg (200 lbs), the task requires three people for a short period of time.

1.) Position table to lowest elevation.
2.) Remove, and set aside, both gantry side covers.
3.) Turn off the 550VDC (HVDC) and Axial Drive Enable switches, on the status control box, on the right side of the gantry.
4.) Remove scan window.
5.) Engage Indexer Pin, to lock gantry rotation.
6.) On the right side of the gantry:
   a.) Remove the two (2) flat head screws that fasten the spacer block to the Base's hard stop.
   b.) Leave the spacer sitting in position for the time being.
7.) Use the tilt buttons to tilt the gantry to +30 degrees, or until the front stop on the pivot sector touches the loose spacer block left in position on the Base, as described in step 6.
8.) Work through the opening in the side of the Pivot Sector Support Base.

   Manually turn the end of the tilt-gearbox drive shaft anti-clockwise, until the pivot sector touches the loose spacer block left in position on the Pivot Sector Support Base, as described in step 6.
9.) Stop tilting, and remove the spacer block.

CAUTION **System specifications require the gantry to tilt a minimum of ±30.25 degrees. The gantry meets this requirement with the spacer block in the correct position. When you remove the spacer block, the gantry can tilt forward more than 30.25 degrees.**

HOWEVER, when you remove the spacer block, the gas springs should prevent the gantry hardstops from meeting. DO NOT force the gantry hard stops to meet, or you may damage, and rupture, the gas springs.

10.) As you continue to manually tilt, **remember** to monitor Gas Spring Length, and watch out for Front Cover Interference:

a.) Gas Spring Length – Wiggle both gas springs as the two hard stops approach each other. Stop when one of the gas springs feels unloaded reaches full extension.

CAUTION **IMPORTANT: Monitor both gas springs to ensure they don't extend past their fully extended length. DO NOT force the gantry hard stops to meet, or you may damage, and rupture, the gas springs.**

b.) Front Cover interference – Due to dimensional variation in the Front Cover assembly, the Front Cover and/or Front Cover hinge doors may come in contact with the Gantry Base before you can tilt the Gantry far enough to exchange a gas spring. Monitor the relationship between the front fiberglass covers and the Gantry Base (on both sides of the Gantry) while you tilt the Gantry. If interference occurs, STOP TILTING.

11.) If you do NOT encounter interference, proceed to step 14.

12.) **IF** you encounter interference:

a.) Return gantry to an upright position (zero degrees of tilt)

b.) Remove Front Cover; proceed to step 13.

13.) Remove Front Cover:

a.) Return gantry to zero degrees of tilt.

b.) Lift top cover, and engage prop rod.

c.) Open front cover.

d.) Disconnect the electrical connections near the top hinge, on the left side of the gantry.

e.) Remove the Front Cover: (requires three people)

- Two people lift and hold the front cover.
- The third person removes the two hinge pins on the left side of the gantry.

f.) Place the front cover horizontally on a clean, protected surface.

- Spread a blanket, or other protective covering on the floor.
- Take care not to scratch the surface of the cover.

g.) Disengage the top cover prop rod, and gently lower the top cover down onto the gantry.

CAUTION **DO NOT ROTATE THE GANTRY WITH THE TOP COVER PROPPED OPEN. If you leave the top cover propped open, it may fall rearward when you tilt the Gantry.**

h.) Repeat steps 7 to 10.

14.) Remove the two (2) retaining clips from the ends of the gas spring that has reached full extension – even if it you do not plan to replace it.

Note: Remove the retaining clips before you remove the gas spring.

15.) Grasp the gas spring near the top end fitting, and firmly pull away from the gantry. Once you free the top end fitting, continue pulling to free the bottom end fitting.

16.) If you need to remove the second gas spring, continue to tilt the gantry until the second gas spring reaches full extension.
   a.) Remove the two (2) retaining clips from the ends of the gas spring.
   b.) Grasp the gas spring near the top end fitting, and firmly pull away from the gantry. Once you free the top end fitting, continue pulling to free the bottom end fitting.
17.) Clean and inspect the ball studs.
   a.) Remove and replace worn or damaged ball studs.
   b.) Use Loctite 242 on the replacement ball stud(s), and torque to 40 ft-lbs. (0.0384 m-kg)
18.) Inspect the new gas spring(s), and tighten the end fittings, if necessary.

Note: If you removed both gas springs, replace the first gas spring on the side with the greatest distance between ball studs. Then, manually tilt the gantry toward zero degrees, until the second gas spring fits onto the other two ball studs.

19.) Manually tilt the gantry until the gas spring fits onto the two (2) ball studs.
   a.) Install the gas spring in the rod down/tube up position.
   b.) Firmly press the top and bottom end fitting around its ball stud.
   c.) Replace the retaining clip on each end fitting, taking care to position the clip through the opening in each end fitting.
20.) Verify system meets the ±30.25° tilt requirement with the new gas spring(s):
   - Try to insert the spacer block between the base and pivot sector hardstops.
   - **Acceptable:** If you CANNOT insert the spacer block between the two hardstops, then the gas spring installation length is ACCEPTABLE.
   - **Unacceptable:** If you CAN insert the spacer block, add a spacer.
      A.) Increase the gas spring length with a flat washer.
      B.) Washer: P/N 1000904P489 – 0.328 ID/ 0.500 OD/ 0.062 Thk
      C.) Remove one end fitting and install one flat washer, as a spacer.
      D.) Apply Loctite 242 and tighten the end fitting.
      E.) Adjust the tilt and install the gas spring.
      F.) Again, use the spacer block to test tilt capability.
21.) After you install the new gas spring(s):
   a.) Return the gantry to the zero degree tilt position, as indicated by the tilt indicator bracket, on the right side of the Gantry.
   b.) Use the two (2) flat head screws to refasten the spacer block to the Base hard stop.
22.) If you removed the Front Cover:
   a.) Lift the Top Cover, and engage the prop rod
   b.) Replace Front Cover, and reconnect the electrical connection
   c.) Adjust/align Front Cover
   d.) Lower Top Cover
23.) Check Gantry tilt with new gas spring(s):
   a.) **Remember** to CLOSE THE TOP COVER, if you propped it open to replace the Front Cover.
   b.) Tilt the gantry to +30 degrees and -30 degrees, as indicated by the display monitor located on top of the Front Cover.
24.) Reassemble Gantry.
25.) Disengage the Indexer Pin to unlock gantry rotation.
26.) Turn ON the 500VDC (HVDC) and Axial Drive Enable switches, on the status control box, on the right side of the gantry.

6.2.2 46-278120P1 Gear Reducer

1.) Position table to lowest elevation.
2.) Turn off facility power to PDU.

⚠ **DANGER** **USE TAG AND LOCK OUT PROCEDURE.**

3.) Remove, and set aside, both gantry side covers.
4.) Turn off all three switches on the status control box, on right side of Gantry.
5.) Open top cover and engage prop rod.
6.) Remove, and set aside, Gantry scan window.
7.) Open front cover.
8.) Install shipping brackets to prevent the gantry from tilting.
  - Gantry must be at 0 degrees tilt to install brackets.
  - Install three (3) bolts in each bracket.
9.) Remove, and set aside, tilt motor.
10.) Remove one (1) tilt chain adjusting bolt and washer.
11.) Remove two (2) tilt chain locking bolts from the adjusting block.
12.) Place the chain on the ground, and remove the chain from the drive pulley.
13.) Remove, and keep, four (4) bolts that fasten the gear reducer to the gantry base.
14.) Remove the Pot assembly set screw, on the output end of the gear box.
15.) Slide the Pot assembly off the output shaft of the Gear Reducer Box.
16.) Remove the Gearbox from the Gantry.
17.) Remove the two set screws that fasten the drive pulley to the output shaft of the gear reducer, and remove the drive pulley.
18.) Fasten the old drive pulley to the output shaft of the new Gear Reducer.
  a.) Position the gears toward the pulley.
  b.) Apply 242 Loctite to the two set screws, and torque into place.
19.) Position the new gear reducer on the gantry base:
  a.) Place the chain around the drive pulley and the two (2) idler pulleys.
  b.) Apply Loctite 242 to the two (2) bolts.
  c.) Loosely attach the tilt chain adjustment block to the pivot sector with the two bolts.
20.) Replace the tilt chain adjusting bolt and washer.
  Tighten chain tension to 5 ±1 lbs.
21.) Tighten tilt chain locking bolts to 25 ft-lbs.
22.) Replace Tilt Motor.
23.) Loosely install set screw on output end of new gear reducer.
24.) Slide Pot Assembly on output shaft of Gear box.
25.) Position pot shaft into the end of the gear reducer output shaft.
26.) Apply Loctite 242 to the set screw, and torque into place.
27.) Remove, and set aside, Gantry shipping brackets.
28.) Position Gantry to 0 degrees, to adjust potentiometer.
  Align Gantry with the tilt indicator on the right side of Gantry Base.
29.) Reassemble Gantry.

6.2.3 46-296363P1 Tilt Motor

1.) Position table to lowest elevation.
2.) Turn off facility power to PDU.

DANGER USE TAG AND LOCK OUT PROCEDURE.

3.) Remove, and set aside, both gantry side covers.
4.) Turn off all 3 switches on the status control box, on right side of Gantry.
5.) Open top cover, and engage prop rod.
6.) Remove, and set aside, Gantry scan window.
7.) Open front cover.
8.) Remove, and set aside, the center and right inside Base Covers, and the table footswitch cover.
9.) Disconnect the Tilt Motor Power harness from the table connection point.
10.) Cut the Ty-raps, and remove the harness.
11.) Remove the three bolts that fasten the motor to the gear reducer.

Note: One of the four mounting holes is not used.

12.) Slide the defective Motor off the gearbox.
13.) Replace the Motor:
   a.) Reconnect the harness to table
   b.) Ty-rap the harness to the base.
14.) Install the key into the motor shaft.
15.) Align key with keyway in Gearbox.

NOTICE Completely insert the Key into the motor shaft. No overhang allowed.

16.) Apply a light coat of Loctite 242 to the three bolts, and use them to attach the Motor to the Gear Box.
17.) Reassemble Gantry.

6.2.4 46–297036G1 Tilt Potentiometer

1.) Position table to lowest elevation.
2.) Remove, and set aside, the right gantry side cover.
3.) Tilt the gantry to the upright, 0 degree, position.
4.) Turn off all 3 switches on the status control box, on right side of Gantry.
5.) Remove power to the Tilt Potentiometer:
   a.) Remove the left table base cover.
   b.) Turn off all 3 service switches in the rear end of the table.
6.) Remove the defective tilt pot:
   a.) Unfasten the four screws, and remove the guard bracket from the pot.
   b.) Disconnect the power leads from the tilt pot.
   c.) Loosen the set screw that ties the tilt pot to the Gear Reducer Output.
   d.) Loosen and/or remove the synclamps, as needed, to remove the pot.
7.) Replace the pot:
   a.) Reconnect the power leads.
   b.) Place the pot on the end of the Gear Reducer Output shaft.
   c.) Apply Loctite–242 to the set screw.
   d.) Replace the set screw, and tighten into place.

8.) Turn on the rear table “24 hour power” switch, to apply 10 volts across the pot.
   a.) Connect a DVM across the pot.
   b.) Rotate the tilt pot until the meter displays 5.0 DCV.
9.) Tighten the three synclamps to fasten the tilt pot into place.
10.) Replace the Tilt Pot guard.
11.) Turn on all 3 service switches in the rear end of the table.
12.) Reassemble the Table and Gantry.

Section 7.0 High Voltage

7.1 Replacement Verification and Re-Test

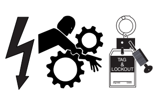
High Voltage FRU	Task	Verification Test
Tube Cooling Relay	For replacement, see 7.2 on page 756	• Plane of Rotation (POR): Chapter 3, Section 7.0. • X-Ray Beam on Window (BOW): Chapter 3, Section 8.0 • Isocenter: Chapter 3, Section 11.0 • Center Body Filter (CBF) and SAG: Chapter 3, Section 12.0

Table 11-11 Retest Matrix for High Voltage Components

7.2 Replacement Procedure(s) - 46-297396P1 Tube Cooling Relay

1.) Position table to lowest elevation.
2.) Turn off facility power to PDU.

DANGER **USE TAG AND LOCK OUT PROCEDURE.**



3.) Remove, and set aside, both gantry side covers.
4.) Turn off all 3 switches on the status control box, on right side of Gantry.
5.) Open top cover, and engage prop rod.
6.) Remove, and set aside, Gantry scan window.
7.) Open front cover.
8.) Rotate Gantry until the OBC reaches the 3 o’clock position.
9.) Engage gantry rotational lock.
10.) Remove connectors J1 through J5 from the control assembly.
11.) Remove 4 screws that fasten the control assembly to the back of the OBC.
12.) From the bottom of the control assembly:
   a.) Locate the (normally closed) tube cooling relay
   b.) Remove the four (4) leads attached to positions 1 through 4.
   c.) Remove two (2) nuts that fasten the relay to the chassis.
   d.) Remove the defective relay.
13.) Install the new relay.

14.) Reassemble Gantry.

Section 8.0 Balancing - Replacement Procedure

8.1 46-196464P1 Steel Weight (small), 46-327263P1 (large)

Use the Gantry Static Balance Procedure when you add or replace steel weights. Refer to 8.2 on page 757.

8.2 Gantry Static Balance Procedure

1.) Leave gantry 120 AC enabled.
2.) Disable HVDC and Axial Drive.
3.) Axial brake must be energized (Released).
4.) Take 8 force readings at the locations. See Figure 11-5.
5.) If the FD - Fu is less than 6 lbs for all four positions then the gantry is balanced.
6.) If FD - Fu is greater than 6 lbs, balance weights must be added or removed at one or more of the 4 positions. See Figure 11-6. Balance weight part number 46-297604P1. Small balance weights are approximately 2 lbs each.

Note: Various plate sizes are available

- For 1 washer plate use a 0.75LG cap screw 46-208561P39.
- For 2 washer plate use a 1.00LG cap screw 46-208561P41.
- For 3 washer plate use a 1.25LG cap screw 46-208561P43.
- For 4 washer plate use a 1.50LG cap screw 46-208561P42.

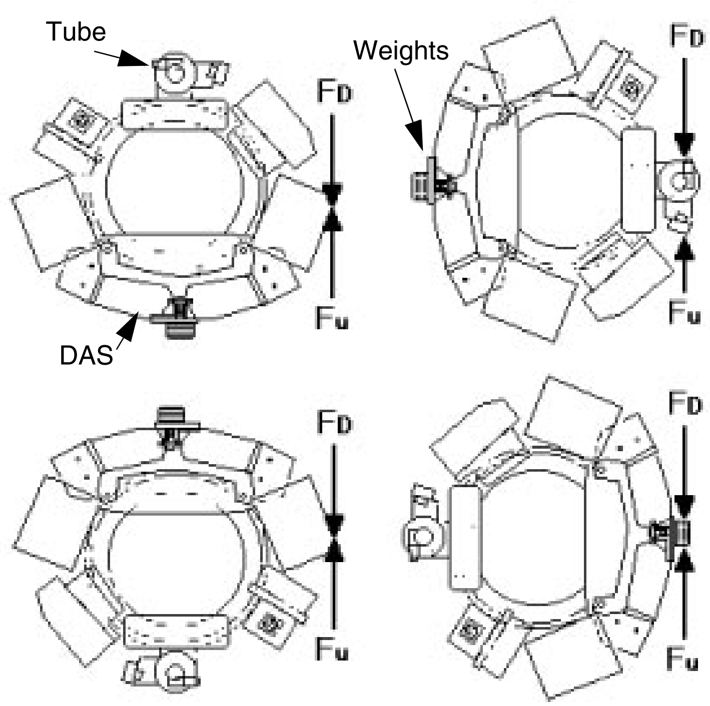


Figure 11-5 Gantry Static Balance Force Measurements

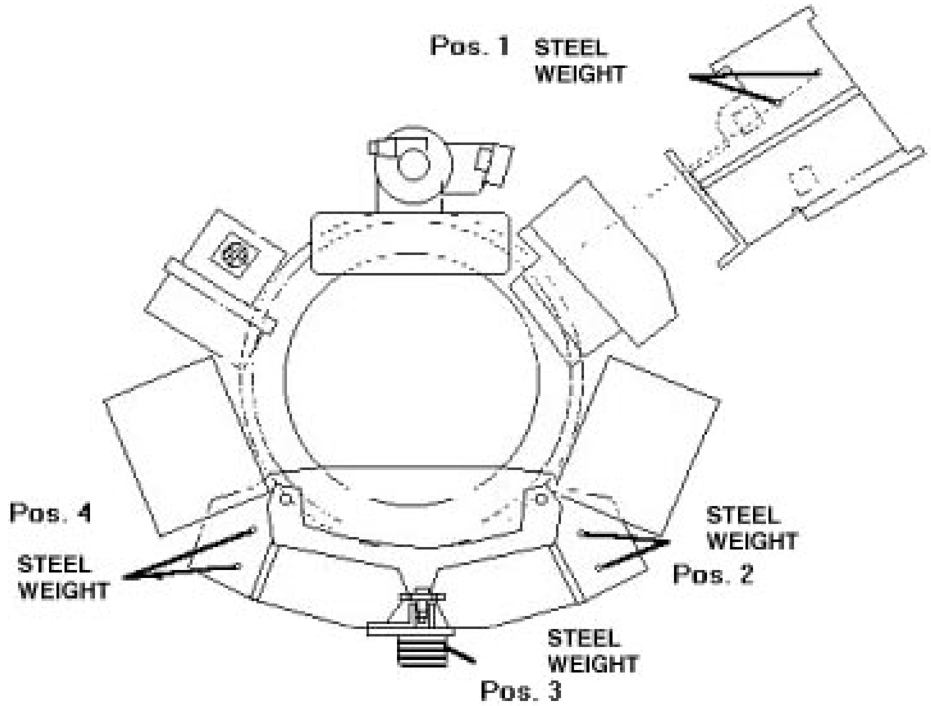


Figure 11-6 Gantry Static Balance Weight Locations

Section 9.0 Collimator

9.1 Replacement Verification and Re-Test

Gantry FRU	Task	Verification Test
Collimator	Refer to page 759 and reset Smart Trend Baseline if applicable (5.x SW), on page 264.	Characterize Collimator (page 145), X-ray Alignments, Chapter 3 Q-Cal, Cals, (page 100) System Functional Test (page 69)

Table 11-12 Retest Matrix for Gantry Components

9.2 Replacement Procedures

9.2.1 46-296263P1 Aperture Microstepper Drive

1.) Position table to lowest elevation.
2.) Turn off facility power to PDU.

9.2.2 46-296664P1 Filter Stepper Motor Drive

1.) Position table to lowest elevation.
2.) Turn off facility power to PDU.

9.2.3 46-296300G5 Collimator Assembly

The cost reduced collimator, 46-296300G5, replaces the G4 version in forward production. (See Table 11-13)

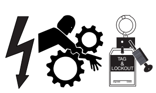
- The system requires CT/i 3.5 or CT/i 4.0 software to support the G5 collimator operation.
- Systems with G1, G2 or G3 collimators: You **must** install RP1.4 or RP2.2 release software and calibrate the system when you replace one of the older collimators with the G4 version. Failure to update the software could result in Image Quality and dose issues.
- The Calibration procedures remain the same, but the presence of the SmartBeam filter increases the required number of scans per kV. The system automatically adjusts the screens to show the correct number of scans.

MODEL NUMBER	COMMENTS
46-296300G1	Original HiSpeed Advantage Collimator
46-296300G2	Aperture modified to reduce x-ray scatter; backwards compatible with G1
46-296300G3	EMC compatible and excess parts removed from collimator assembly; backwards compatible with G1 and G2
46-296300G4	More durable Aluminum Carbon (AlC) SmartBeam filter replaces Teflon filter in collimator; backwards compatible with G1, G2 and G3 collimators
46-296300G5	New filter drive mechanism, higher torque aperture stepping motor. **NOT BACKWARD COMPATIBLE!**

Table 11-13 HSA Collimator History

1.) Position table to lowest elevation.
2.) Turn off facility power to PDU.

DANGER **USE TAG AND LOCK OUT PROCEDURE.**



3.) Remove, and set aside, gantry side covers.
4.) Turn off all three switches on status control box, on right side of Gantry.
5.) Lift Top Cover, and engage prop rod.
6.) Remove scan window
7.) Open Front Cover.
8.) Rotate Gantry until the Collimator reaches the 3 o'clock position.
9.) Disconnect the electrical connections to the Collimator.
    a.) Disconnect Connectors J2, J3, and J4 at OBC.
    b.) Disconnect the J1 harness from left side of Collimator.
10.) Remove, and keep, the two nuts from the adjustment bracket, located on the right side of the collimator. Remove hardware.
11.) Remove, and keep, the four (4) 1/4-20 bolts, washers, and lock washers that fasten the collimator in place.
12.) The Collimator weighs 42lbs (19kgs). Take care when you remove it.
13.) Replace Collimator.
14.) Collimator Characterization:
    a.) Display the Applications menu.
    b.) Select UTILITIES.
    c.) Select CT TOOLS.
    d.) Select MECHANICAL CHARACTERIZATION.

e.) Select COLLIMATOR DATA.

f.) Compare the collimator characterization values on the display screen to the values listed on the collimator characterization label.

g.) IF the values don't match:

- Type/enter the values listed on the collimator characterization label.
- Select UPDATE TABLE, and wait for the system to display a `successful completion` message.

ALIGN COLLIMATOR:

1.) Verify the table has been withdrawn from the tube and the detector field of view. Nothing should obstruct the attenuation of the X-Ray beam to detector window.

2.) Available functions for CBF and SAG alignment:

- Default CBF Scan (Also used for SAG calculation).
- List/Select DD Files.
- Store DD Result File.

3.) Results display for CBF and SAG:

- Success of failure of scan performed.
- Collimator adjustment recommendations (if required).
- SAG measurement value.

4.) Display the tube alignment screen, and select the following softkeys to perform a CBF and SAG scan/calculation:

a.) Select CBF and SAG

b.) Select CBF SCAN (Push scan enable when lit)

Note: Last Air.DD Filename — The last ISO Air.DD filename on the CBF and SAG screen becomes the default isocal during CBF and SAG calculation. The CBF and SAG screen also displays the average of the small and large spot isocenter value. Select the Large Air ISO Air CAL.

5.) Upon completion of the scan, the system calculates and displays the SAG value, and the average centroid value for the CBF, along with any collimator movement recommendations

6.) If the Calculated CBF values fail to meet the 373.75 ±0.2 specification, adjust the collimator by the amount displayed on the CRT:

a.) Wait 5 minutes between exposures.

b.) Mount the gauge and its nonmagnetic holding fixture to the special mounting bracket on the collimator.

c.) Zero out the gauge.

d.) Loosen the four collimator mounting bolts, located in the four slotted holes on the collimator.

e.) Adjust the 3/4 inch bolt (located at the end of the collimator) to move the collimator right or left by the required amount. The CRT displays CBF adjustments in an up and down direction format:

- UP = Move Collimator to the right, and
- DOWN = Move collimator to the left.

f.) Bolt down the collimator and torque the four lock bolts to 5 ±0.5 ft. lbs. (60 ±6 in. lbs).

g.) Repeat the CBF scan and collimator adjustment until the calculated CBF values meet the 373.75 ±0.2 specification

h.) Remove the gauge and holding fixture from the Gantry.

i.) Record the CBF and SAG values on the data sheet.

7.) Reassemble Gantry.

9.2.4 46-321276G1 321276G1/G2 46-321276G1 Collimator II Bd

1.) Remove, and set aside, the right gantry side covers.
2.) Turn off all 3 switches on the status control box.
3.) Rotate the Gantry until the OBC reaches the 3 o'clock position.
4.) Engage rotational lock.

NOTICE **Wear a grounded wrist strap when you handle a circuit board.**

5.) Put on grounding wrist strap.
6.) Loosen the 8 captive screws, and remove OBC Front Cover.
7.) Rotate the ejectors, and remove the defective board from the card cage.
8.) Place the board in an Anti-Static bag.
9.) Install the new board.
10.) Reassemble Gantry.

Section 10.0 Laser

10.1 Replacement Verification and Re-Test

Gantry FRU	Task	Verification Test
Laser Control Assembly (46-297157G1) (46-288308G1)	Refer to page 761 for replacement procedure Perform Hardware Reset to turn fans and pumps on.	Verify Laser, see page 137 Listen for tube fan, Detector heater control, and Tube pump operation
Laser Lights	Refer to page 762 for replacement procedure	Refer to page 137 for Alignment procedure, characterize alignment, and tomographic plane indication.

Table 11-14 Retest Matrix for Gantry Components

10.2 Replacement Procedures

10.2.1 46-288308G1 Laser Control Board

NOTICE **Wear a grounded wrist strap when you handle a circuit board.**

1.) Move table to lowest elevation.
2.) Turn off facility power to PDU.

DANGER **USE TAG AND LOCK OUT PROCEDURE.**

3.) Remove gantry side covers.
4.) Turn off all three switches on status control box, on right side of Gantry.
5.) Lift top cover, and engage prop rod.
6.) Remove scan window.

7.) Open front cover.
8.) Rotate the OBC to the Gantry 3 o'clock position and engage the gantry rotational lock.
9.) Remove connectors J1 through J5 from the control assembly.
10.) Remove 4 screws that fasten the control assembly to the back side of the OBC.
11.) From the bottom of the control assembly:
   a.) Locate the laser control PWB.
   b.) Remove the 4 screws that fasten the PWB to the Control assembly.
12.) Remove the J1 connector from the laser control PWB.
13.) Remove defective laser control PWB.
14.) Install new laser control PWB.
15.) Reassemble Gantry.

10.2.2 Alignment Lights

46-297301G3 OR 46-327058G3 (BOTTOM)
46-297301G2 OR 46-327058G2 (LEFT)
46-297301G1 OR 46-327058G1 (RIGHT)
12V OR 5V DIODE LASER

WARNING **NEVER STARE INTO THE LASER ALIGNMENT LIGHT BEAM. FAILURE TO FOLLOW THIS INSTRUCTION COULD RESULT IN INJURY TO THE EYES.**

1.) Position the table to lowest elevation.
2.) Turn off facility power to PDU.

DANGER **USE TAG AND LOCK OUT PROCEDURE.**

3.) Remove, and set aside, gantry side covers.
4.) Turn off all three (3) switches on status control box, on right side of gantry.
5.) Lift top cover, and engage prop rod.
6.) Remove scan window.
7.) Open Front Cover.
8.) Disconnect the electrical connection to the defective laser assembly.
9.) Remove the three (3) bolts holding the laser assembly in place.
10.) Remove defective assembly, and install the new laser assembly.
11.) Reconnect electrical connections.
12.) Align Laser light.
13.) Reassemble Gantry.

Section 11.0
Display

11.1 Replacement Verification and Re-Test

Gantry FRU	Task	Verification Test
Gantry Display Assembly	Replace faulty assembly on page 763	Verify self test and counts change for table up/down, in/out and gantry tilt.

Table 11-15 Retest Matrix for Gantry Components

11.2 Replacement Procedure(s) -46-296341G1 Gantry Display Assembly

11.2.1 Remove Gantry Display

1.) Remove the four 1/4x20 machine screws holding gantry module
2.) Disconnect the display cable, and remove the gantry display from the support arms
3.) Remove the two nylon 8/32 x 0.500 machine screws that fasten the cover plate in place.
4.) Remove the cover plate.
5.) Remove the four 1/4x20 x 3.25 machine screws that fasten the support arms and standoffs.
6.) Remove the support arms and standoffs
7.) Remove the two cable clamps from the left support arm.
8.) Remove the two bearings on the pin from the support arm.
Repeat this task on both arms.

11.2.2 Install Gantry Display

1.) Install the two bearings on the pin, and install the pin on the support arm.
Repeat this assembly procedure on both support arms.
2.) Attach two cable clamps to the left support arm.
3.) Fasten the support arms and the standoffs to the Gantry with four 1/4x20 x 3.25 machine screws.
4.) Fasten the cover plate to the support arms with two nylon 8/32 x 0.500 machine screws
5.) Route and connect the display cable.
6.) Attach the gantry display to the support arms.
Make sure you firmly seat the display.
7.) Fasten the Gantry module to the display assembly with four 1/4x20 x 0.75 machine screws.

Section 12.0
Slip Ring

12.1 Troubleshooting

12.1.1 Theory of Troubleshooting

12.1.1.1 Theory of LAN

Please reference the FUNCTIONAL Interconnect (46-018318), the section on “CONTROL NETWORK”. There are two types of “CONTROL NETWORK”. The older style has a LAN Card in the SBC (connected to the Max-HIB). The newer system configurations have a HAWK CLA board and an Ethernet thick wire to Thinwire Conversion Kit.

Regardless of style of ethernet, LAN Watchdog Timeouts are caused by the following components:

- LAN card in the ETC.
- LAN card in the STC.
- LAN card in the SBC (if supplied with one) or the Hawk CLA ethernet connections in an RP.
- Ethernet cable between the SRU (or SBC), STC, and ETC.
- Ethernet cable T-connectors, terminators, or wrong impedance terminator.
- Improperly grounded terminators can cause LAN Watchdog timeouts.

Note: There is only one grounded terminator.

12.1.1.2 Theory of TAXI Link

TAXI Link failures are generated by the Receive TAXI chip (Rx) on the TAXI boards. The Rx converts the data coming across the ring into a byte containing 11 bits each. 60% of the possible bytes that can be generated are valid, the other possible combinations are not valid. A TAXI Link failure means that one of the invalid combinations has been generated.

TAXI Link failures are a function of the ring and directly associated components, i.e. Terminator and Buffer boards, TAXI boards, the ring itself, brushes, dust, power supplies or grounding.

RCOM or SCOM boards may be involved when dealing with TAXI Link failures. To get TAXI Link stats, **`nbsClient stc`** (or **`nbsClient obcr`**) from the SBC. Once there, **`s sr0`** will get you the data. **CTRL + C** will get you out of nbsClient. All nbsClients must be done from the SBC.

WARNING USE NBSCLIENT CORRECTLY nbsClient is a single user function. If you are **`nbsClient`** to one of the controllers when the SBC needs it (i.e. during scanning) the system will crash applications. Ensure the customer is not scanning when using nbsClient. The single user function nbsClient will crash the system if accessed while scanning.

12.1.1.3 Troubleshooting Flow

1.) Read General Precautions section.
2.) Determine error code that is causing problem (see examples). Get error codes from:
   - System message log.
   - Run time stats.
   - `nbsClient`.
3.) For LAN Watchdog Timeout go to LAN Watchdog Timeout.
4.) For other communication errors proceed to TAXI Link Failures section.

LAN Watchdog Timeout

1.) Do Visual Checks for LAN Watchdog Timeout.
   - Check LAN card seating.
   - Check coax connections.
   - Check terminator impedance and connection.
   - Check terminator grounding (grounded one side only).
2.) Determine which controller is at fault.
   - OBC, STC, ETC or all 3 controllers timing out.
3.) Swap the LAN card of the problem controller and see if problem follows.
4.) If timeouts are on all three controllers the problem could be the first controller in line, i.e. the SBC. Time-outs on all three controllers indicate a problem to something that is common to all three controllers.
   - Could be a problem on the first controller (SBC), or
   - A problem with the LAN coax.
     - Suspect coax connections.
     - Suspect terminator impedance and connection.
     - Terminator grounding (grounded one side only).
     - Bypass existing coax by laying new coax on the hospital floor.
5.) Check power supply per Power and Grounding Checks.
6.) Replace LAN cards and/or coax.
7.) Check controller temperature. (This happens most often on the STC.)

TAXI Link Failures

1.) Determine configuration (refer to 12.1.2 on page 766).
2.) Establish a baseline using either METHOD A (refer to 12.2.1.1 on page 769) or METHOD B (refer to 12.2.1.2 on page 770) (METHOD A is recommended).
3.) Do visual checks.
4.) Go to minimum system and verify with ping, METHOD A (refer to 12.2.1.1 on page 769).
5.) Check power and grounding.
6.) Vacuum slip ring (Refer to PM procedure CTPM-1321) and Repeat Baseline (refer to 12.2.1 on page 769) to see if error goes away.
7.) If stationary, jumper out ring using coax.
8.) Swap buffer boards.

9.) Swap brushes (left to right).

10.) Order parts.

Only as a Last Resort

Alcohol clean the slip ring. Refer to Slip Ring Alcohol Cleaning in Slip Ring Service Actions.

12.1.2 Gantry Model Numbers and Allowable Configurations

Type	HSA only Non-EMC	RP, HSA, CT/i Non-EMC	EMC CT/i	Lightspeed S/A Style Slipring Assemblies
(RP)SCOM	46-321246G1	46-321300G1	46-321300G1	ALL
RPSCOM JMPR	N/A	JP600 out	JP600 in	Ref bd. type
RCOM	46-297474G1	46-297474G1	2126034	Ref bd. type
RCOM JMPR	N/A	N/A	JP1 in	Ref bd. type
BFR/TERM	Old BFR Set	Old BFR Set	New BFR Set	S/A BFR Set

Table 11-16 Allowable Configurations of Boards and Jumpers

"OLD BFR SET" MEANS:

Stationary Buffer	46-321056G1
Rotation Buffer	46-321058G1
Stationary Terminator	46-321052G1
Rotation Terminator	46-321054G1

"NEW BFR SET" MEANS

Buffer (Rot & Sta)	2118208
Terminator (Sta & Rot)	2118209

"S/A BFR SET" MEANS

Buffer (Rot & Sta)	2253794
Terminator (Sta & Rot)	2238323

12.1.3 Effect of Having RPSCOM JP600 or RCOM JP1 in Wrong Position

TAXI chips are similar to a modem with a certain baud rate. Just as two modems connected to each other must be configured to the same baud rate in order to function, so must the TAXI chips. The RPSCOM jumper JP600 and RCOM jumper JP1 are used to change the baud rate of the TAXI chips. When the jumpers are out, the baud rate of all TAXI chips are set to 55.00 Mbaud. When the jumpers are in, the RCOM Tx and RPSCOM Rx TAXI chips (outbound link) are set to 55.068 Mbaud, while the others (inbound link) stay at 55.00 Mbaud. Experience has shown the TAXI chips will work if a Tx is set at 55.000 Mbaud and a Rx at 55.068 Mbaud or vice versa. These settings are out of the tolerance specified by the manufacturer of the TAXI chip and all TAXI Link violations may result.

12.1.4 Changes to the RCOM, RPSCOM, and Slip Ring for EMC Compliance

EMC (Electromagnetic Compliant) CT systems have notable changes in packaging. This describes changes required for the slip ring communications subsystem and the RCOM and SCOM interface boards.

12.1.4.1 RCOM/SCOM Bd. Changes (Configurations & EMC Systems)

RCOM

RCOM board assembly 2126034 (which contains the RCOM board 46-288854G1-D) is required for EMC CT systems. The RCOM assembly that includes the version D RCOM board (assembly includes 2-TAXI daughter boards and RARQ board) has changed from 46-297474G1 to 2126034. The "D" version is reflected in the IN-SITE device on the board. The version D RCOM board is required for all EMC CT systems, but is also compatible with all HSAs to date. The two differences between the old and new versions are the additions of a configurable oscillator to stagger frequencies on the slip ring by 68.8 Khz to spread RF emission power, and the routing of +/- 12V power to the P2 connector for the rotating buffer board.

The RCOM board will have a single jumper. There are temporary rework instructions to modify older RCOMs to version D. The jumper JP1 (not labeled on boards reworked to version D) must be removed for operation in standard HSA systems, and in this state all RCOMs are compatible with all SCOMs and RPSCOMs. EMC CT systems require the jumper to be installed. If the jumper on the RCOM is installed, then the jumper must be installed on the RPSCOM as well (see RPSCOM below). In this state, older SCOMs or RPSCOMs are not compatible, only RPSCOMs 46-321300 can be used. However, if the jumpers are placed in the wrong positions, the scanner will probably still work. Every test we've run so far has shown this. But the AMD TAXI chip components on the boards would be grossly out-of-spec!

RPSCOM

The RPSCOM (46-321246G1) is not compatible with RP-EMC systems. This board is compatible with all older HSA systems. The new board is 46-321300G1. This board can output DAS data via star AP parallel format, FEP TAXI coax format, and FEP TAXI fiber-optic interface. Because of this, the board is compatible with all older HSA systems. EMC CT systems will require use of the fiber interface which will require the new RPSCOM and 46-327036G2 FEP boards. All HSAs today are shipping with the new 46-327036G2 FEP assembly that has the fiber interface option. To select the fiber channel for data input on the FEP, jumper J2 needs to be installed on the FEP.

As with the version D RCOM, there is a staggered frequency jumper on the new 46-321300 RPSCOM. With the jumper JP600 removed, the board is compatible with all older systems. The jumper must be installed for EMC CT, and then must run with a 2126034 RCOM assembly (version D RCOM board) or higher with the RCOM jumper installed as well. INSITE will read 46-321300 rev B on the new RPSCOM.

12.1.4.2 Slip Ring Changes (For EMC and S/A Systems)

Slip Ring Buffer Boards

For EMC the existing buffer boards (46-321056 and 46-321058) are replaced by buffer "modules" (2118208) on EMC CT gantries. Instead of two different components, there is a single component type used in two places. One module is inverted from the other such that the "goes outa" from one goes to the "goes into" at the other end. The modules are shielded and the ground screws are an important electrical component to circuit function. These modules will work only with EMC CT systems, and are NOT backward compatible.

The S/A buffer boards are designed for use with the S/A ring assembly only. They are not backward compatible with any ETC design. Both rotating and stationary boards are identical. Note, the rotating buffer board has the ground strap connected to the rotating frame. The Stationary board ground strap is electrically connected via the mounting screws. **THE STATIONARY GROUND STRAP SHOULD NOT BE CONNECTED AND SHOULD BE TY-WRAPPED OUT OF THE WAY.**

Slip Ring Terminator Boards

The two terminator boards on current HSA systems (46-321052 and 46-321054) are replaced with resistor blocks (2118209). These blocks are shipped to GE from our slip ring supplier with the ring. The part is a 20 ohm, 1/8 watt, 1% resistor on FR4 circuit board material with eyelets to screw onto a ring.

The stationary resistor attaches to the stationary brush block on the power brush block side of the gantry to ring studs #4 and #5. The rotating resistor attaches to the ring inside diameter opposite the buffer board, to ring stud #2 and #3. These mounting instructions are silk-screened onto the resistor blocks. These resistors will work only with EMC CT systems.

S/A style slipring uses 2 terminator boards (2238323). This is a small circuit board with a single precision 24 ohm, 1% surface mount resistor. They are mounted the same as the ETC.

Slip Ring

The EMC slip ring incorporates an “image plane” in the slip ring approximately 1/8 inch below the six communications tracks. Running a differential buffer board set, on rings without this plane will yield unpredictable results (TAXI violations). The rings are easy to tell apart since the EMC ring has a metal shield on the inside face of the ring.

The S/A ring is was designed EMC compliant. There is no need for an "Image Plane" in this design. Also the S/A ring is shielded with copper. The S/A rings themselves are machined, solid brass.

Lightspeed Style S/A Slip Ring

The S/A slipring is the exact same ring used on the Lightspeed QXI systems. This ring is easily identified by it’s blue color. Additionally the HSDCD portion of the ring is NOT used in this application. Also there is a modified power jumper harness installed on the stationary side. This is necessary to convert the white molex to subminiature "D" connection. There are no known compatibility issues related to RCOM, SCOM, RPSCOM version boards. All noted configuration settings above are applicable. All Service procedures, tools, diagnostics, and error reporting are the same as the ETC style rings. This is simply an alternate assembly.

12.1.5 General Precautions

- DO NOT use vacuum brush on anything else except the slip ring. Avoid contamination.
- DO NOT touch brush tips or slip ring. The contamination from body oil, etc. can damage them.
- DO NOT file or clean the brush tips, under any circumstance (per vendor warranty). The brush tips will wear at slightly different rates and may appear out of alignment. This is expected.
- DO NOT bend or deform brushes or spring material (per ETC vendor warranty).
- DO NOT apply side forces to the S/A brush tips. They will break.
- DO NOT work on the ring or brushes unless the system A1 power is tagged and locked.
- DO NOT use anything else other than the GE-approved alcohol to clean the ring.
- DO NOT get cleaning solution on the brushes.

- DO NOT use any other cleaning solution than what is specified in the cleaning procedure.
- Remove brushes for vacuuming and any other type of cleaning.
- Remove brushes before cleaning the ring with alcohol.
- Vacuuming is a PM action. Refer to PM procedure CTPM-1321.
- Deviation from these procedures will result in increased material costs.
- S/A rings have a NORMAL discoloration. Brush material build-up provides its own lubrication.

12.2 Service Procedures

12.2.1 Baseline

12.2.1.1 Method A — TAXI Link Error Troubleshooting

The following procedure is used to troubleshoot TAXI Link and other data-related problems between the stationary buffer and rotating buffer board on the CT/i.

There are two purposes for this procedure.

- Establish a baseline prior to troubleshooting.
- Verify if the problem has been eliminated after a correction was applied.

Procedure

1.) Access to two UNIX shells is required.
  - **L1B** could be one shell and opening a shell under Utilities could be the other.
  - CT/i open up two unix shells.
  - Telnet in from an AW.

2.) Open Utilities and download the diagnostic firmware.

  `CT/i`

3.) After the firmware has been successfully downloaded enter **`ping -s obcr`** into one of the UNIX shells. This will send recursive data packets across the slip ring to the OBC and back to the host for error detection.

  Verify that the return message from the **`ping -s`** command is, on average, less than 20ms.

4.) Use the diagnostics to rotate the gantry for at least 10 minutes at speed of one second. To accomplish this use the axial control loop diagnostic.

  Enter 600 seconds for rotation duration and one second for gantry speed.

5.) On the unused UNIX shell check for TAXI Link errors by using nbsClient on both the OBCR and the STC (refer to 12.2.4.2 on page 772). This should be zero because downloading the diagnostic firmware resets the error registers.

  a.) Check for TAXI Link errors after five minutes and again after ten minutes of rotation.

  b.) If there are no errors, repeat steps 4 and 5 one more time. If there are still no errors, it probably is not worth proceeding with component replacement or service actions.

  c.) If there are errors, record this number as it will be used to determine if repairs are successful.

  d.) Remember that the TAXI Link register is reset if the firmware is downloaded or a system hardware reset is commanded.

6.) Now that the baseline is established, troubleshooting can begin. Refrain from performing parallel steps. Replace one board or service action at a time, then repeat steps 4 and 5.

Note: If slip rings are cleaned or if data brushes were replaced, there is a burn-in period of approximately ten minutes. Therefore, use nbsClient to monitor the rate of TAXI Link error growth that should slow down to zero during the second ten minutes of testing.

Other Use for ping -s

`ping -s` used on a non-rotating gantry is useful to eliminate the slip rings as a source for problems and also for troubleshooting other components in the serial communication area. Use steps 3 and 5 to interrogate the TAXI Link error register. If errors are present, rotate the gantry 90 degrees and repeat test. If the errors are still present, there is a 99% chance that the slip rings are not causing the problem.

12.2.1.2 Method B

Develop a baseline.

1.) **`nbsClient`** to controller (refer to 12.2.4.2 on page 772).
2.) **`s sr0 c`** will clear out stats.
3.) Do 30 scans.
4.) Check s sr0 stats to see how many if any TAXI Link errors were created.
    - Do this while stationary,
    - Rotating fast or slow,
    - High x-ray technique and low.

    Determine which method creates TAXI Link failures and work with that particular method.
5.) Repeat every time a service action is done. This way one can tell if a service action had any effect.

Note: If TAXI Link errors occur while the gantry is stationary, jumper out the slip ring with 50ohm coax cables to help troubleshoot.

or

Black holes are transmit errors between stc cpu and obcr cpu and will accompany TAXI Link errors.

There is no absolute limit. What is important is if TAXI Link stops scanning.

12.2.2 Visual Checks

- Excess dust build-up (Vacuum first only).
- Check terminator boards (sometimes missing).
- Terminator boards mounted correctly with correct screws.
- Coax connector pins OK and connected.
- Damaged boards/ring/brushes.

12.2.3 Power and Grounding Checks

- Ground screws on the buffer board.
- Ring 12 grounded.
- DC power supplies on both buffers.
- OBC, STC, non-EMC slipring power supplies.
- Strap to brush block.
- Stationary Buffer Power Harness Adaptor connections.(S/A only)

12.2.4 Software and Hardware Tools Available for Troubleshooting

12.2.4.1 Ping

To isolate rough spots in the rotation, execute an extended version of the ping command while the FE manually rotates the gantry. The output will demonstrate where the rough spots exist. Any cleaning efforts should be concentrated in those areas. This test requires two people, the FE and a person at the console end to indicate where the rough spots occur.

Perform the following command:

```
insite @ HSA1_SBC0 130: ping -s obcr 16 100
24 bytes from 192.9.224.2: icmp_seq=27. time=20. ms
24 bytes from 192.9.224.2: icmp_seq=28. time=20. ms
24 bytes from 192.9.224.2: icmp_seq=29. time=20. ms
24 bytes from 192.9.224.2: icmp_seq=30. time=20. ms
24 bytes from 192.9.224.2: icmp_seq=31. time=20. ms
24 bytes from 192.9.224.2: icmp_seq=32. time=20. ms
24 bytes from 192.9.224.2: icmp_seq=33. time=20. ms
24 bytes from 192.9.224.2: icmp_seq=34. time=1600. ms
24 bytes from 192.9.224.2: icmp_seq=35. time=18120. ms
24 bytes from 192.9.224.2: icmp_seq=36. time=27180. ms
24 bytes from 192.9.224.2: icmp_seq=37. time=31260. ms
24 bytes from 192.9.224.2: icmp_seq=38. time=35260. ms
24 bytes from 192.9.224.2: icmp_seq=39. time=34260. ms
24 bytes from 192.9.224.2: icmp_seq=40. time=33300. ms
24 bytes from 192.9.224.2: icmp_seq=41. time=22320. ms
24 bytes from 192.9.224.2: icmp_seq=42. time=11320. ms
24 bytes from 192.9.224.2: icmp_seq=43. time=1360. ms
24 bytes from 192.9.224.2: icmp_seq=44. time=20. ms
24 bytes from 192.9.224.2: icmp_seq=45. time=20. ms
24 bytes from 192.9.224.2: icmp_seq=46. time=20. ms
24 bytes from 192.9.224.2: icmp_seq=47. time=20. ms
24 bytes from 192.9.224.2: icmp_seq=48. time=20. ms
```

EXPLANATION

The packet internet groper is used to determine if another user is still active. The ping command works by sending out an internet control message protocol (ICMP) echo request. If the destination user is reachable, then an ICMP echo will be returned immediately. The sending user will continue to send this ping/ICMP request until terminated or it times out. By using the `-s` option you get a verbose statistics printout of each ping event. Pay particular attention to the response time of the packet to determine network response. In this case, problems with the rings were discovered and marked by the FE during rotation. The 16 used in the command line sets up a 16 data byte packet (if needed, it could be larger). The 100 used in the command line sends out the ping request 100 times.

SOLUTION

Cleaned/cleared slip rings of debris in vicinity where network response rate degraded.

12.2.4.2 Using nbsClient

nbsClient to the OBC

1.) On the SBC, type **`nbsClient obcr`**.
2.) While in nbsClient type **`s sr0`** (0 = zero).
3.) At the bottom of the response:

```
 |sr0: timeout Error Statistics:
961003|0925|06588|  INIT timeouts          =        0   DATA
   |      |      |  timeouts           = 0
   |      |      |sr0: Interrupt Statistics:
   |      |      |  Level 6 interrupts    =    84590   DAS/AP FIFO
   |      |      |  full        = 1
   |      |      |  TAXI link failures     =      157
   |      |      |sr0: Miscellaneous Statistics:
   |      |      | Black hole occurrences = 1 Total words thrown
   |      |      |  away = 1
   |      |      |  Most words thrown away =   1   Errored DATA
   |      |      |  frames     = 0
```

type **CTRL + C** to get out of nbsClient.

nbsClient to the STC

1.) On the SBC, type **`nbsClient stc`**.
2.) While in nbsClient type **`s sr0`** (0 = zero).
3.) At the bottom of the response:

```
 |sr0: timeout Error Statistics:
961003|0925|06588|  INIT timeouts          =        0   DATA
   |      |      |  timeouts           = 0
   |      |      |sr0: Interrupt Statistics:
   |      |      |  Level 6 interrupts    =    84590   DAS/AP FIFO
   |      |      |  full        = 1
   |      |      |  TAXI link failures     =      157
   |      |      |sr0: Miscellaneous Statistics:
   |      |      | Black hole occurrences = 1 Total words thrown
   |      |      |  away = 1
   |      |      |  Most words thrown away =  1   Errored DATA
   |      |      |  frames     = 0
```

type **CTRL + C** to get out of nbsClient.

12.2.4.3 Minimum Operational System Set

There is no minimum system for the STC.

Whereas the minimum system set for the OBC is the RCOM and CPU.

12.2.4.4 Jumper Out Ring Using Coax

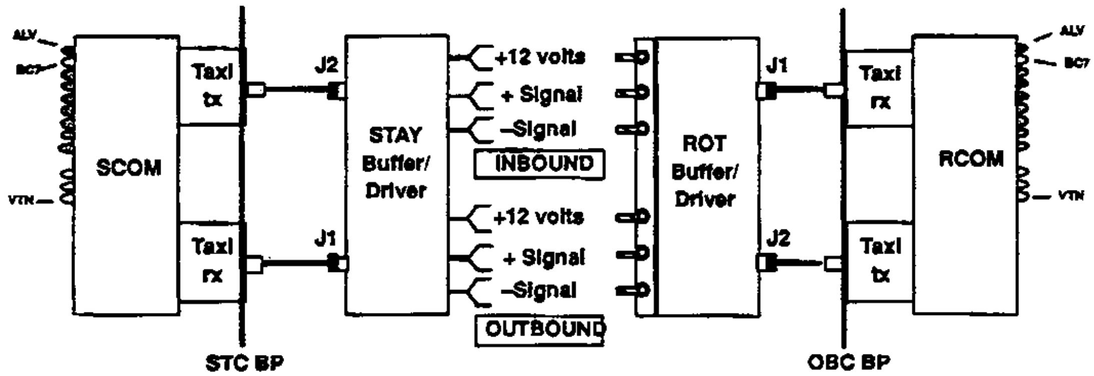


Figure 11-7 Slip Ring Communications Bypassing Routines

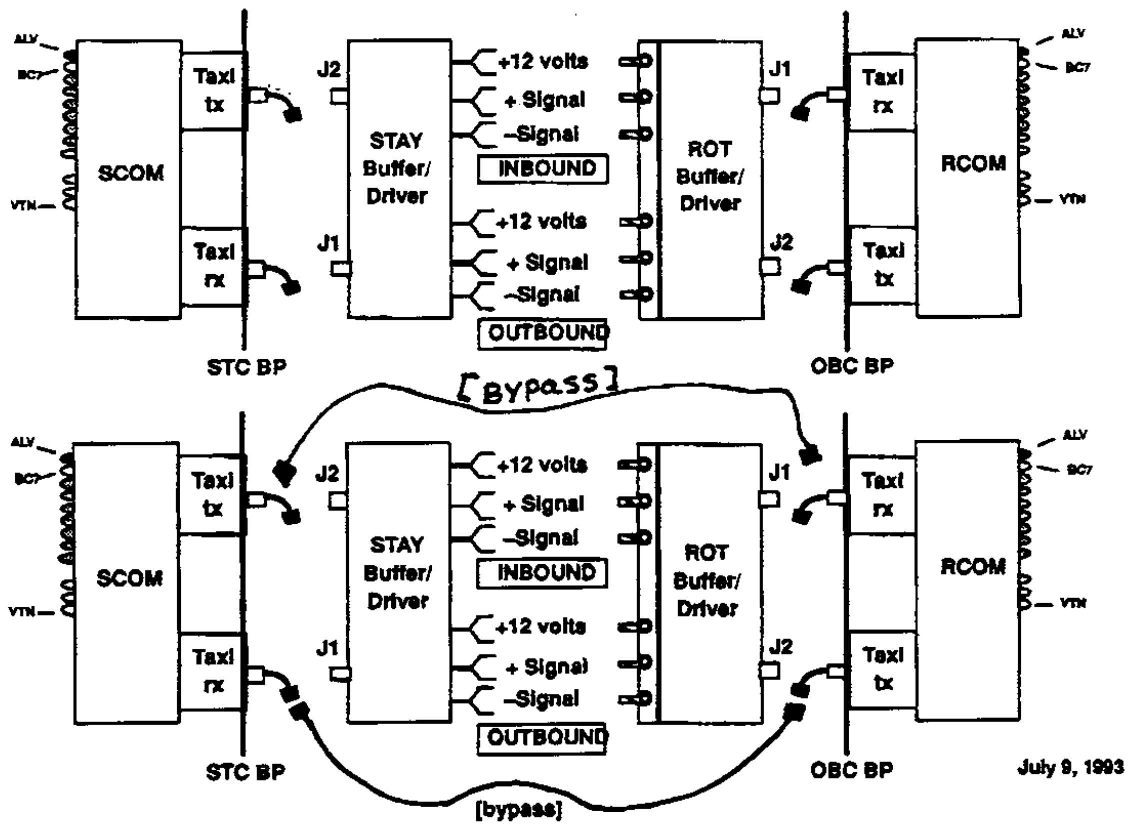


Figure 11-8 Normal Indications

12.2.4.5 Slip Ring Service Actions

Why Clean with Alcohol or Sand with Cratex Crayons

1.) Communication failures (TAXI Link Failures) after vacuuming failed.
2.) Cratex sanding is for mechanical damage to the ring **ONLY** (e.g. pits and arc marks).

Alcohol Clean

Alcohol cleaning should be done as a corrective (repair) action only. Alcohol clean is not a PM action and should only be done if necessary and only AFTER vacuuming the Slip Ring and Gantry has not corrected the problem.

1.) Vacuum ring per PM procedure.
2.) Check baseline.
3.) Remove all of the power and signal brush blocks (the alcohol will contaminate the brushes). Refer to the General Service Manual 2152918-100 for the proper removal procedure.
4.) Use specified alcohol (46-183039p1) and allow to AIR DRY for 15 min. DO NOT use the alcohol prep pads found in hospitals. They are often not PURE alcohol, and can contaminate the slip ring and brushes.
5.) Reinstall the removed components. Reference the appropriate section in the General Service Manual 2152918-100. The proper replacement procedure is critical to the life of the slip ring components.

Note: Use the proper alcohol.

Only Use GE Approved Alcohol 46-183039P1

Cratex

The cratex should be used to only fix pit and arc marks.

1.) Using Cratex fine abrasive stick (46-297961P2) attempt to smooth out the pitted area or areas with deposits on the slip ring. ONLY use Cratex on the ring that is in need of repair. If the area(s) are still not smooth, use Cratex medium abrasive stick (46-297961P1). After using the medium, repeat procedure with the fine.

Note: Smoothing out the track surfaces is a time consuming task and if not done properly and completely will result in either permanent damage or will cause another arc. Do it right the first time.

Note: Removing the “clogged” end of the cratex stick with a coarse file will help speed up the process of smoothing out pitted areas.

2.) When done with the Cratex sticks it is very important to remove ALL traces of abrasive with a thorough alcohol cleaning.
3.) Replace all of the removed components Reference the appropriate section in the General Service Manual 2152918-100. The proper replacement procedure is critical to the life of the slip ring components.

Removal/Installation/Replacement

Reference the appropriate section in the General Service Manual 2152918-100. The proper replacement procedure is critical to the life of the slip ring components.

12.2.5 The Handling and Removal of Slip Ring Brush Debris

The ETC slip ring brushes, made of Polymet AG material and used on this scanner, contain silver and a trace of cadmium. The S/A slip ring brushes are made of Carbon and Carbon Silver. Refer to the corresponding MSDS for a description of these materials.

CAUTION **Take care to avoid contact, inhalation and ingestion of slip ring debris whenever you work with slip ring components. Take the following steps when you handle slip ring material:**

1.) Wear Neoprene or nitrile gloves, to limit irritation and ingestion of metallic dust.
   - Do NOT remove gloves near an exposed slip ring. The powder inside the gloves can contaminate the ring.
   - Gloves: Large (Qty 100) 46-194427P347.
   - Gloves: XL (Qty 100) 46-194427P348.

2.) Use a HEPA (High Efficiency Particulate Air) vacuum cleaner to remove residual brush debris.
   - HEPA vacuum Cleaner: 46-297933P1.
   - HEPA filter: 46-297948P1.

3.) Use the HEPA vacuum cleaner to remove all existing brush debris from the brush blocks, brackets and slip ring covers *before* you service the slip ring brush assemblies.

4.) Use the HEPA vacuum cleaner to remove all existing brush debris from the gantry base and floor *after* you reassemble the slip ring covers.

5.) Wash your hands *thoroughly* after you service any slip ring components.

Note: All exposure surveys, conducted during service procedures and system operation, reported levels of silver and cadmium well below the occupational exposure limits for these materials.

12.3 Replacement Verification and Re-Test

Communication FRUs	Task	Verification Test
Slip Ring Assembly	To replace faulty Slip Ring Assembly, refer to page 784	Acquire rotating scans at: 100kV/ 100mA. and 120kV/400mA.
Brush Blocks	Replace faulty power (page 778) or signal (page 782) block and align correctly.	System Functional Test on page 69
Buffer, TAXI Boards, and Slip Ring Communication rings	To replace faulty FRU: RCOM on page 723 RPSCOM on page 728 Buffer or TAXI Boards, begin your search on page 776 Manual check for TAXI errors: Open a Unix shell and, type: `cu sbc` type: `nbsClient stc,` type: `s sr0` Verify fewer than 2 TAXI Link failures occur in 80 slices, **CTRL + C** [to exit], type: `nbsClient obcr` type: `s scr0` Verify fewer than 2 TAXI Link failures occur in 80 slices, **CTRL + C** [to exit], ~. [to exit].	1.) Hardware Reset 2.) Acquire 10 scouts: (120kv/ 40ma., 1000mm table movement), 3.) Acquire 100 axials: (120kv/ 80ma., 1 sec. scan, 1 sec. ISD), 4.) Acquire 1 helical: (120 kv/ 40ma., 30 sec. scan), 5.) Acquire 10 axial scans: (120kv/400ma.) 6.) Verify NO TAXI Link errors To check Rcom/Scom Statistics enter: **FSST; 12; 3**

Table 11-17 Retest Matrix for Communication FRUs

12.4 Replacement Procedures

12.4.1 46-321058G1 or 46-264698G1 or 2253794 Rotating Buffer Board

Note: Three (3) types of buffer and terminator PWBs exist. Check the part numbers of the PWBs on your system before ordering replacement parts.

CAUTION **Read The Handling and Removal of Slip Ring Brush Debris on page 775 before you start this procedure.**

1.) Position table to lowest elevation.
2.) Turn off facility power to PDU.

DANGER **USE TAG AND LOCK OUT PROCEDURE.**

3.) Remove, and set aside, both gantry side covers.
4.) Turn off all three (3) switches on the status control box, on right side of Gantry.
5.) Open top cover, and engage prop rod.

6.) Remove, and set aside, Gantry scan window.
7.) Open rear cover.

NOTICE **Wear a grounded wrist strap when you handle a circuit board.**

8.) Locate the rotating buffer board on the inside surface of the slip ring.
The Buffer board has two coax cables connected to it.
9.) Disconnect the two (2) coax cables connected to J1 and J2 on the Buffer PWB.
10.) Remove the seven (7) 4-40 screws, and one (1) 1/4-20 ground screw, that fasten the Buffer PWB to the slip ring.
11.) Remove the Buffer PWB from Gantry.
12.) Install the new Rotating Buffer PWB.
13.) Reassemble Gantry.

12.4.2 46-321054G1 or 46-264702G1or 2238323 Rotating Terminator Board

Note: Three (3) types of buffer and terminator PWBs exist. Check the part numbers of the PWBs on your system before YOU order replacement parts.

CAUTION **Read The Handling and Removal of Slip Ring Brush Debris on page 775 before you start this procedure.**

1.) Position table to lowest elevation.
2.) Turn off facility power to PDU.

DANGER **USE TAG AND LOCK OUT PROCEDURE.**

3.) Remove, and set aside, both gantry side covers.
4.) Turn off all three (3) switches on the status control box, on right side of Gantry.
5.) Open top cover, and engage prop rod.
6.) Remove, and set aside, Gantry scan window.
7.) Open rear cover.

NOTICE **Wear a grounded wrist strap when you handle a circuit board.**

8.) Locate the rotating terminator PWB on the inside surface of the slip ring.
The Terminator PWB has no cables attached to it.
9.) Remove the seven (7) 4-40 screws, and one (1) 1/4-20 ground screw, that fasten the terminator PWB to the slip ring.
10.) Remove the terminator PWB from Gantry.
11.) Install the new rotating terminator board.
12.) Reassemble Gantry.

12.4.3 46-321056G1 or 46-264696G1 or 2253794 Stationary Buffer Board

Note: Three (3) types of buffer and terminator PWBs exist. Check the part numbers of the PWBs on your system before ordering replacement parts.

CAUTION **Read The Handling and Removal of Slip Ring Brush Debris on page 775 before you start this procedure.**

1.) Position table to lowest elevation.
2.) Turn off facility power to PDU.

DANGER **USE TAG AND LOCK OUT PROCEDURE.**

3.) Remove, and set aside, both gantry side covers.
4.) Turn off all 3 switches on the status control box, on right side of Gantry.
5.) Open top cover, and engage prop rod.
6.) Remove, and set aside, Gantry scan window.
7.) Open rear cover.
8.) Remove slip ring covers.

NOTICE **Wear a grounded wrist strap when you handle a circuit board.**

9.) Disconnect three cables at connectors J1, J2 and J3.
10.) Remove eight (8) 4-40 screws that fasten the Stationary Buffer PWB to the brush block assembly.
11.) Remove the Stationary Buffer PWB from the Gantry.
12.) Install the new Stationary Buffer PWB.
13.) Reassemble Gantry.

12.4.4 46-321052G1 or 46-264700G1 or 2238323 Stationary Terminator Board

Note: Three (3) types of buffer and terminator PWBs exist. Check the part numbers of the PWBs on your system before ordering replacement parts.

CAUTION **Read** **The Handling and Removal of Slip Ring Brush Debris on page 775** **before you start this procedure.**

1.) Position table to lowest elevation.
2.) Turn off facility power to PDU.

DANGER **USE TAG AND LOCK OUT PROCEDURE.**

3.) Remove, and set aside, both gantry side covers.
4.) Turn off all three (3) switches on the status control box, on right side of Gantry.
5.) Open top cover, and engage prop rod.
6.) Remove, and set aside, Gantry scan window.
7.) Open rear cover.
8.) Remove slip ring covers.

NOTICE **Wear a grounded wrist strap when you handle a circuit board.**

9.) Remove seven (7) 4-40 screws that fasten the terminator PWB to the brush block assembly.
10.) Remove the defective terminator PWB from Gantry.
11.) Replace terminator PWB.
12.) Reassemble Gantry.

12.4.5 46-297837G1 Power Brush Block Assembly (ETC Style)

1.) Position table to lowest elevation.
2.) Turn off facility power to PDU.

DANGER **USE TAG AND LOCK OUT PROCEDURE.**

3.) Remove, and set aside, both gantry side covers.
4.) Turn off all three (3) switches on the status control box, on right side of Gantry.
5.) Open top cover, and engage prop rod.
6.) Remove, and set aside, Gantry scan window.
7.) Open rear cover.
8.) Remove the slip ring covers.

CAUTION **Read The Handling and Removal of Slip Ring Brush Debris on page 775 before you start this procedure.**

9.) The Gantry contains three (3) brush blocks: two for signal and one for power.
   a.) Remove the stationary buffer board to access and remove either of the signal brush blocks. Refer to the corresponding procedures for a detailed explanation.
   b.) Remove the grounding strap from at the power brush block bracket before you remove the power brush block.
   c.) If you plan to replace the power brush block, remove the grounding strap from the power brush block and the power brush block bracket.
10.) Remove Brush Block:
   a.) Remove two (2) of the four (4) bolts, one on the top of the block and one on the bottom.
   b.) With one hand, hold the brush block against the brackets, and remove the remaining screws with the other hand.
   c.) Remove brush block from the brackets.
11.) Install Brush Block:
   a.) Hold the Brush Block in position, and align the brush tips with the slip ring tracks.
   b.) Compress the brush tip springs against the slip ring tracks, and hold the brush block against the brackets.
   c.) Install all four (4) screws, but do not tighten.
   d.) Adjust the brush blocks to center the top and bottom of the brush tips.
   e.) Tighten all four (4) screws to 22 ±3in-lbs. (0.254 ±0.035 m-kg)
12.) Manually rotate Gantry while you check that the brush tips are centered between the track barriers.
13.) Reassemble Gantry.

12.4.6 S/A Pwr Brush Asm 2238140-2, High 2257517, Low 2257520 Replacement

1.) Position table to lowest elevation.
2.) Turn off facility power to PDU.

DANGER **USE AND FOLLOW TAG AND LOCK OUT PROCEDURES.**

3.) Remove, and set aside, both gantry side covers.
4.) Turn off all three (3) switches on the status control box, on right side of Gantry.
5.) Open top cover, and engage prop rod.
6.) Remove, and set aside, Gantry scan window.
7.) Open rear cover.
8.) Remove the slip ring covers.

CAUTION Avoid inhalation

Read **The Handling and Removal of Slip Ring Brush Debris on page 775** **before you start this procedure.**

9.) The Gantry contains three (3) brush blocks: two for signal and one for power.
   - a.) Remove the stationary buffer board or stationary terminator to access and remove either of the signal brush blocks. Refer to the corresponding procedures for a detailed explanation.
   - b.) Remove the grounding strap at the power brush block bracket before you remove the power brush block.

10.) Remove Brush Block:
   - a.) Remove two (2) of the four (4) bolts, one on the top of the block and one on the bottom.
   - b.) With one hand, hold the brush block against the brackets, and remove the remaining screws with the other hand.

11.) Remove brush block from the brackets. (Skip step 12 if replacing entire assembly)

12.) Remove and install replacement modules:
   - a.) Place brush block on a hard surface with the tips in the up position. There are five high power modules and three low power modules.
   - b.) To remove a module, push the tab on the module out of the locking position. While pushing the tab out of the locking slot, pull the module up. Only pull the module out 2 mm, or just enough to keep the tab from snapping back into the slot. Repeat this procedure on the other side of the module.

Figure 11-9 Power Module Push Tab

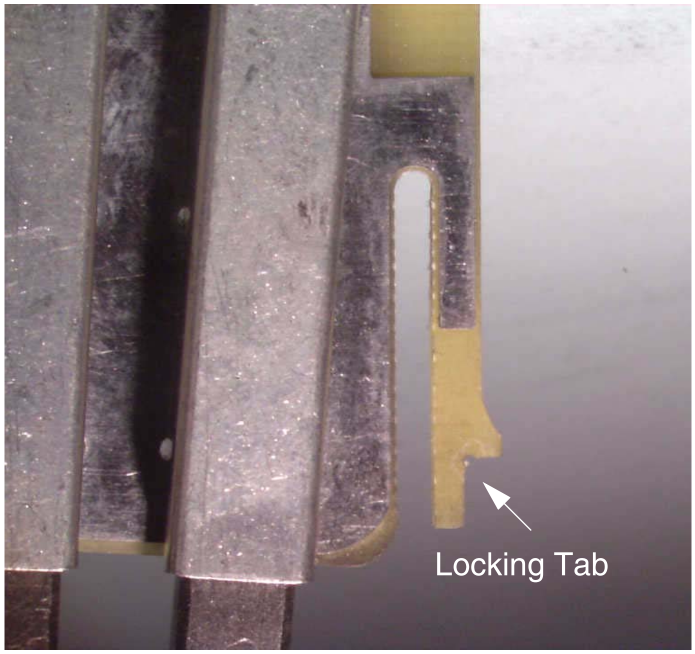


Figure 11-10 Power Module Tab

c.) With the tabs on each side of the module out of the locking slot, pull the module straight out of the brush block housing.

Figure 11-11 Module Removal

d.) Align the replacement module in slots and push into brush block housing until the module tabs snap into the locking holes.

13.) Install Brush Block:

a.) Hold the Brush Block in position, and align the brush tips with the slip ring tracks.

b.) Compress the brush tip springs against the slip ring tracks, and hold the brush block against the brackets.

c.) Install all four (4) screws, but do not tighten.

- d.) Adjust the brush blocks up against the set screws on the bracket to center the top and bottom of the brush tips on the ring.
- e.) Tighten all four (4) screws to 22 ±3in-lbs. (0.254 ±0.035 m-kg)

14.) Manually rotate Gantry while you check that the brush tips are centered between the track barriers.

15.) Reassemble Gantry.

12.4.7 46-297839G1 Signal Brush Block (ETC Style)

CAUTION **Read** **The Handling and Removal of Slip Ring Brush Debris on page 775** **before you start this procedure.**

1.) Position table to lowest elevation.

2.) Turn off facility power to PDU.

DANGER **USE TAG AND LOCK OUT PROCEDURE.**

3.) Remove, and set aside, both gantry side covers.

4.) Turn off all three (3) switches on the status control box, on right side of Gantry.

5.) Open top cover, and engage prop rod.

6.) Remove, and set aside, Gantry scan window.

7.) Open rear cover.

8.) Remove the slip ring covers.

9.) The Gantry contains three (3) brush blocks: two for signal and one for power.

- Remove the stationary buffer board to access and remove either of the signal brush blocks. Refer to the corresponding procedures for a detailed explanation.
- Remove the grounding strap from the power brush block before you remove the power brush block.

10.) Remove Brush Block:

- a.) Remove two (2) of the four (4) bolts, one on the top of the block and one on the bottom.
- b.) With one hand, hold the brush block against the brackets, and remove the remaining screws with the other hand.
- c.) Remove brush block from the brackets.

11.) Install Brush Block:

- a.) Hold the Brush Block in position, and align the brush tips with the slip ring tracks.
- b.) Compress the brush tip springs against the slip ring tracks, and hold the brush block against the brackets.
- c.) Install all four (4) screws, but do not tighten.
- d.) Adjust the brush blocks to center the top and bottom of the brush tips.
- e.) Tighten all four (4) screws to 22 ±3in-lbs. (0.254 ±0.035 m-kg)

12.) Manually rotate Gantry while you check that the brush tips are centered between the track barriers.

13.) Reassemble Gantry.

12.4.8 S/A Signal Brush Block 2238141-2 or Signal Brush Module 2254362

CAUTION Avoid inhalation

Read **The Handling and Removal of Slip Ring Brush Debris on page 775** **before you start this procedure.**

1.) Position table to lowest elevation.
2.) Turn off facility power to PDU.

DANGER **USE AND FOLLOW TAG AND LOCK OUT PROCEDURES.**

3.) Remove, and set aside, both gantry side covers.
4.) Turn off all three (3) switches on the status control box, on right side of Gantry.
5.) Open top cover, and engage prop rod.
6.) Remove, and set aside, Gantry scan window.
7.) Open rear cover.
8.) Remove the slip ring covers.
9.) The Gantry contains three (3) brush blocks: two for signal and one for power.
   - Remove the stationary buffer board to access and remove either of the signal brush blocks. Refer to the corresponding procedures for a detailed explanation.
   - Remove the grounding strap from at the power brush block bracket before you remove the power brush block.
10.) Remove Brush Block:
   a.) Remove two (2) of the four (4) bolts, one on the top of the block and one on the bottom.
   b.) With one hand, hold the brush block against the brackets, and remove the remaining screws with the other hand.
   c.) Remove brush block from the brackets.
11.) Remove and install replacement modules:
   a.) Place the signal brush block on a hard surface with the tips down and remove the two screws that attach the module to the brush block body.

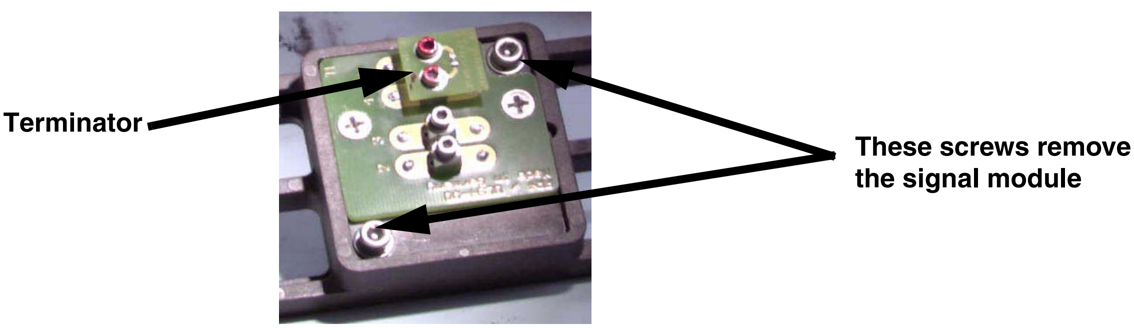


Figure 11-12 Signal Brush Block Module

   b.) Insert the replacement module and tighten the screws.
12.) Install Brush Block:
   a.) Hold the Brush Block in position, and align the brush tips with the slip ring tracks.
   b.) Compress the brush tip springs against the slip ring tracks, and hold the brush block against the brackets.
   c.) Install all four (4) screws, but do not tighten.
   d.) Adjust the brush blocks up against the set screws on the bracket to center the top and

bottom of the brush tips on the ring.

- e.) Tighten all four (4) screws to 22 ±3in-lbs. (0.254 ±0.035 m-kg)

13.) Manually rotate Gantry while you check that the brush tips are centered between the track barriers.

14.) Reassemble Gantry.

12.4.9 46-297840G1 or 46-296001G1or 2239006 Slip Ring Asm (ETC and S/A)

Note: The slip ring is encased in plastic and the plastic should be removed last. The plastic is to protect the ring from contamination.

Slip ring replacement requires two people, plus a third person for approximately five hours.

CAUTION **Read** **The Handling and Removal of Slip Ring Brush Debris on page 775** **before you start this procedure.**

1.) Position table to lowest elevation.

2.) Turn off facility power to PDU.

DANGER **USE TAG AND LOCK OUT PROCEDURE.**

3.) Remove, and set aside, both gantry side covers.

4.) Turn off all three (3) switches on the status control box, on right side of Gantry.

5.) Open top cover, and engage prop rod.

6.) Remove, and set aside, Gantry scan window.

7.) Open Rear cover.

- a.) Locate left hinge arm inside the rear cover, and disconnect the two (2) electrical connections leading to the top and rear covers.
- b.) Disconnect the two (2) wires leading to the rear cover lamp.
- c.) Disconnect the harness leading to the rear cover microphone.

8.) Remove the top cover: (Requires 2 people)

- a.) Engage the prop rod.
- b.) Remove the retaining clip from the end of the top cover gas spring.
- c.) Grasp the gas spring, and firmly pull outward.
- d.) Let the gas spring swing down and hang from the rear cover support.
- e.) Attach the retaining clip to the gas spring for safe keeping.
- f.) Remove the rear cover portion of the middle hinge that holds the top cover in place.
- g.) Move the top cover to the right, remove it from the rear cover, and set aside.

9.) Remove the Rear Cover: (Requires 3 people)

- a.) Grab the bottom of the cover on each side near the swing doors.
- b.) Lift on the Rear Cover Assembly, while the third person guides the pivot pins out of the pivot arm on each side of the cover.
- c.) Place the Rear Cover on top of a blanket to prevent scratching.
- d.) Lean the Cover against the wall, or if space permits, lay the cover horizontally on a blanket on the floor.

10.) Remove the two (2) Slip Ring Covers.

CAUTION **Read** **The Handling and Removal of Slip Ring Brush Debris on page 775** **before you start this procedure.**

NOTICE **Wear a grounded wrist strap when you handle a circuit board.**

11.) Remove the two BNC Connectors, J1 and J2, from Rotating Buffer Board.
12.) Remove the Rotating Buffer Board and the Rotating Terminator Board:
    a.) Remove seven (7) 4-40 socket head cap screws and one (1) 1/4 -20 screw from each board.
    b.) Remove the boards from the slip ring, and place them in anti-static bags.
13.) Remove two (2) Lexan terminal covers that cover the electrical junctions between the gantry and the slip ring.
    a.) Remove the two (2) 1/4 -20 screws that fasten each cover in place.
    b.) Remove both covers, and set aside, for reuse.
14.) Disconnect all cable connections on the inside of the Slip Ring, including the two (2) Ground Strap connections.
15.) Remove the Stationary Terminator Board:
    a.) Remove the seven (7) 4-40 socket head cap screws.
    b.) Remove the board and place it in an anti-static bag.
16.) Remove the Stationary Buffer Board:
    a.) Disconnect (3) BNC connectors, J1,J2 and J3.
    b.) Remove the eight (8) 4-40 socket head cap screws.
    c.) Remove the board and place it in an anti-static bag.
17.) Remove, and set aside, the one power and two signal Brush Blocks.
    a.) Disconnect the Power Brush electrical connections
    b.) Disconnect the grounding strap.
    c.) Four bolts fasten each Brush Block in place
    Detailed Brush Block instructions begin on page 778.
18.) Remove six (6) Brush Block Brackets:
    a.) Remove the shoulder screw and socket head cap screw from each bracket.
    b.) Remove each bracket, and set aside for reuse.
19.) Open the front cover and remove two (2) access panels.

Note: A cutout on the Slip Ring, and a pilot on the bearing flange, hold the Slip Ring in place, as long as you apply pressure against the back face of the Slip Ring.

20.) Remove eight bolts from the Slip Ring: (Requires 2 people)
    a.) Rotate the gantry to position one bolt in each of the two (2) access holes.
    b.) Remove the bolts from the gantry access holes, and set aside for reuse.
    c.) Second person: Push against the back face of the Slip Ring to hold it in place while your partner removes the bolts
    d.) Rotate the Gantry, by hand, until another set of bolts appear in the access holes.

e.) Remove the bolts, set aside, and repeat the procedure to remove the a total of 8 bolts. See Figure 11-13

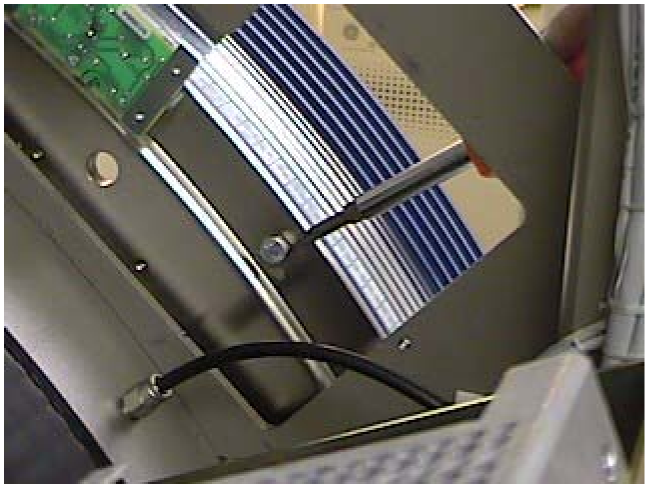


Figure 11-13 Slipring Anchor bolt access.

21.) Remove the Slip Ring assembly:

a.) Stand on either side of the Slip Ring assembly.

b.) Grasp the Slip Ring with both hands.

c.) Pull the Slip Ring off the bearing flange.

⚠ **NOTICE** **The Slip Ring is fragile, so handle it with extreme care. DO NOT STAND THE SLIP RING ON END; LAY IT FLAT.**

22.) Replace the Slip Ring:

a.) Hold the Slip Ring next to Bearing flange.

b.) Rotate the bearing, and/or Slip Ring, to align the black line on the inside of Slip Ring with the small hole on the bearing flange pilot.

23.) Fasten Slip Ring to Bearing Flange: (Requires 2 people)

a.) Second person: Press the Slip Ring flush against the Bearing Flange while your partner replaces the bolts. Start four (4) dry bolts by hand, finger tight, every other hole.

b.) Apply a small amount of 242-Loctite to the other four (4) Slip Ring bolts.

c.) Rotate the gantry, by hand, and replace the next set of bolts.

d.) Tighten each wet bolt with 242-Loctite to ((5 ft.-lbs (0.0048 m-kg)) at 1 to 2 ft-lb. increments. Apply torque evenly. This will help seat the ring properly.

e.) Rotate the gantry by hand, and one at a time, remove each of the first four (4) dry bolts, then apply 242-Loctite as above.

f.) Repeat the replacement and torquing procedure above for the remaining four (4) bolts.

24.) Recheck the torque of all eight (8) bolts: 5 ft-lbs. (0.0048 m-kg)

Note: Use three people to replace the Rear Cover.

25.) Reassemble Gantry.

Section 13.0 Safety

13.1 Replacement Verification and Re-Test

Gantry FRU	Task	Verification Test
Axial Enable Interlock	Replace faulty Scan Switch Assembly, see page 788	Test Axial Enable by exercising alignment lights. They won't come on if axial functions are inoperable.

Table 11-18 Retest Matrix for Gantry Components

13.2 Replacement Procedures

13.2.1 46-136334P23 F/C Interlock Switch

1.) Position table to lowest elevation.

2.) Turn off facility power to PDU.

⚠ **DANGER** **USE TAG AND LOCK OUT PROCEDURE.**

3.) Remove, and set aside, gantry side covers.

4.) Turn off all three (3) switches on status control box, on right side of gantry.

5.) Lift top cover, and engage prop rod.

6.) Remove scan window.

7.) Open Front Cover.

8.) Disconnect the two electrical connections from the front cover interlock limit switch.

9.) Loosen nut, and remove limit switch.

10.) Install the new limit switch, and adjust the distance between the push roller and metal support bracket to 0.94 inches.

11.) Reconnect two (2) electrical connections:

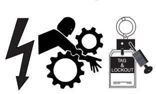
- Black wire to N.O.
- Red wire to Comm.

12.) Reassemble Gantry.

13.2.2 46-229342P1 Rear Cover Lamp

1.) Turn off facility power to PDU.

⚠ **DANGER** **USE TAG AND LOCK OUT PROCEDURE.**



2.) Remove, and set aside, both gantry side covers.

3.) Turn off all 3 switches on the status Control box, on right side of Gantry.

4.) Lift Top Cover and engage prop rod.

5.) Remove scan window.

6.) Open Rear Cover.

7.) Locate the lamp, in the upper center portion of the rear cover.

8.) Disconnect 2 wires leading to the Lamp.
9.) Remove, and keep, the Retaining Ring from the inside of the Cover.
10.) Replace the Lamp and the retaining ring.
11.) Reassemble Gantry.

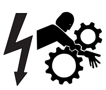
13.2.3 46-296993G1 Scan Switch Assembly

1.) Turn off facility power to PDU.

DANGER **USE TAG AND LOCK OUT PROCEDURE.**

2.) Remove right side cover.
3.) Disconnect main harness connectors, J1 and J2, from the back side of the scan switch assembly.
4.) Locate the lower screw of the scan switch assembly mounting flange, behind the bracket containing the revolution counter and service switch connector.
   a.) Carefully set the revolution counter assembly aside.
   b.) Leave the revolution counter leads connected for later reassembly.
   c.) Remove the upper screw from the scan switch assembly
   d.) Remove the defective assembly from the gantry.
5.) Install the new scan switch assembly.

DANGER **BEFORE YOU TURN ON POWER, SWITCH ALL THE SCAN SWITCH ASSEMBLY SWITCHES TO THE OFF POSITION.**



6.) Reassemble Gantry.

Section 14.0 Gantry Misc.

14.1 Replacement Verification and Re-Test

Gantry FRU	Task	Verification Test
Mylar Window	To replace, see page 789	Perform (10) slices at 10mm aperture.

Table 11-19 Retest Matrix for Gantry Components

14.2 Replacement Procedures

14.2.1 2101967 Scan Window

14.2.1.1 Remove Scan Window

1.) Grab the window at the top and pull firmly downward.
2.) Continue to pull until the top of the scan window makes contact with the bottom portion of the scan window.
3.) Hold the top and bottom portions of the scan window together, grasp both sides of the scan window, move them together and lightly pull upward, until you can free the window from between the front and rear covers.

Figure 11-14 Scan Window Removal

14.2.1.2 Install Scan Window

1.) Close the front and rear covers.
2.) Deform the scan window, as shown in Figure 11-15, and nest the scan window at the bottom of the opening between the front and rear covers, (Figure 11-16) with the rivets in the 6 o'clock position.
3.) After you complete the initial seating of scan window, let the window slowly unfold, and work both sides of the window into position, starting at the bottom and finishing at the top.
4.) Make sure you position the window with the rivets at the 12 o'clock position, and the mylar window slit at either the 3 or 9 o'clock position.

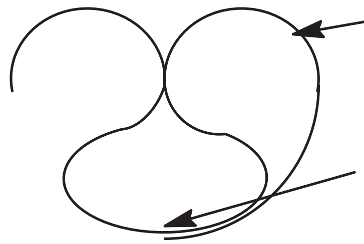


Figure 11-15 Install Scan Window

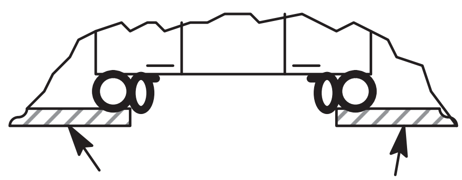


Figure 11-16 Scan Window Nested Between Front and Rear Cover

14.2.2 46-220247P2 Top Cover Fan

1.) Turn off facility power to PDU.

DANGER USE TAG AND LOCK OUT PROCEDURE.

2.) Remove, and set aside, both gantry side covers.
3.) Turn off all three (3) switches on status control box, on right side of gantry.
4.) Open top cover, and engage prop rod.
5.) Remove, and set aside, gantry scan window.
6.) Open Rear Cover.
7.) Locate the defective fan(s).
8.) Disconnect the power from the defective fan.
9.) Loosen, and/or remove, the three (3) screws and washers that fasten the fan into place.
10.) Remove the fan from the top cover.
11.) Remove the fan guard from defective fan.
12.) Attach the fan guard to the new fan.
13.) Install new fan in the top cover.
14.) Reconnect the electrical connection.
15.) Reassemble Gantry.

14.2.3 46-297738P1 Top Cover Gas Spring

1.) Turn off facility power to PDU.

DANGER USE TAG AND LOCK OUT PROCEDURE.

2.) Remove, and set aside, both gantry side covers.
3.) Turn off all three (3) switches on the status control box on right side of Gantry.
4.) Open top cover, and engage prop rod.
5.) Remove and keep the retaining clip from each end of the top cover gas spring.

6.) Grasp the defective gas spring and firmly pull outward.
7.) Install the replacement gas spring and clip into place.
8.) Reassemble Gantry.

14.2.4 46-297797G1 Top Cover Thermostat Assembly

1.) Turn off facility power to PDU.

⚠ **DANGER** **USE TAG AND LOCK OUT PROCEDURE.**

2.) Remove, and set aside, both gantry side covers.
3.) Turn off all three (3) switches on status control box, on right side of gantry.
4.) Open top cover, and engage prop rod.
5.) Remove, and set aside, gantry scan window.
6.) Open Rear Cover.
7.) Locate thermostat assembly, on the left side of the gantry top cover.
8.) Disconnect two (2) electrical connections, one from each side of the thermostat assembly.
9.) Unfasten four (4) screws, and remove the defective thermostat assembly.
10.) Replace the thermostat assembly.
11.) Reconnect the two electrical connections.
12.) Reassemble Gantry.

Chapter 12
Table

Note: Please perform the retests listed below when you replace or adjust a table part.

Section 1.0
Replacement Verification and Re-Test

Table Components	Task	Verification Test
DC Power Supply (+/- 15 volt, +24 volt)	Measure loaded voltages, see page 133	Perform System Scanning Test, (page 69)
DC Drive Power Supply (+24 volt)	Measure loaded voltages, see page 133	1.) Confirm full cradle travel and Gantry tilt capability. 2.) Perform System Scanning Test, (page 69)
Longitudinal (Cradle) Pot Assembly (page 817) Encoder cable	Replace, Install or Adjust (page 817)	1.) Remove the Pot (page 817) 2.) Install the Pot (page 817) 3.) Check Pot Voltage (page 812 and 817) 4.) Perform System Scanning Test, (page 69)
Longitudinal (Cradle) Encoder	Replace, Install or Adjust (page 812)	1.) Remove the Encoder. (page 812) 2.) Install the Encoder. (page 812) 3.) Adjust C-Pulse. (page 812) 4.) Check Pot Voltage (page 812 and 817) 5.) Perform System Scanning Test, (page 69)
Cradle Drive Amplifier	Replace faulty board, (see page 803)	Perform System Scanning Test, (page 69)
Cradle Assembly	Replace cradle (page 803)	Perform System Scanning Test, (page 69) Repeat with maximum weight
Cradle Drive Assembly	Replace Cradle Drive Assembly (page 803)	1.) Characterize cradle (see page 145) 2.) Perform System Scanning Test, (page 69)
Elevation Encoder or Timing Belt	1.) Replace Elevation Encoder (page 805) 2.) Set C-Pulse (page 806)	1.) Characterize Elevation (page 142) 2.) Perform System Scanning Test, (page 69)

Table 12-1 Table Component Replacement Verification

Table Components	Task	Verification Test
Actuator (Elevation) Limit Switch	Replace, Install or Adjust (page 796)	1.) Characterize Elevation (page 142) 2.) Perform System Scanning Test, (page 69)
Table Elevation Actuator	Replace, Install (page 797)	1.) Characterize Elevation (page 142) 2.) Perform System Scanning Test, (page 69)
ETC board	Replace, install faulty board (page 806)	1.) Check Characterization Limits (page 146) (for tilt and table elevation interference) 2.) Perform System Scanning Test, (page 69)
DC Power Supply (+170 Volt)	Measure unloaded and loaded voltages. Measure on the Servo Amp Assembly.	Exercise tilt and elevation function.
Interference Matrix Switch (Elevation Limit Switches)	Replace, install faulty switch (page 810)	1.) Check Characterization Limits (page 146) (for tilt and table elevation interference) 2.) Perform System Scanning Test, (page 69)
Gas Springs	Replace, install both Springs (page 808)	1.) Clean excess oil from springs, and exercise table elevation full range with maximum load. 2.) Perform System Scanning Test, (page 69)
Left Control Panel	Replace, Install control panel (page 818)	1.) Test function of all push buttons 2.) Perform System Scanning Test, (page 69)
Right Control Panel	Replace, Install control panel (page 819)	1.) Test function of all push buttons 2.) Perform System Scanning Test, (page 69)
Home Latch Assembly	Replace, Install assembly (page 810)	1.) Verify Cradle latches and unlatches. 2.) Perform System Scanning Test, (page 69)
E-Stop Switches	Replace, Install Assembly	1.) Emergency Stop Check (page 138) 2.) Perform System Scanning Test, (page 69)
Table Side Cover Tape Switch	Replace, Install Assembly (page 821)	1.) Emergency Stop Check (page 138) 2.) Perform System Scanning Test, (page 69)
Leg Tape Switches	Replace, Install faulty switch (page 811)	1.) Emergency Stop Check (page 138) 2.) Perform System Scanning Test, (page 69)

Table 12-1 Table Component Replacement Verification (Continued)

Table Components	Task	Verification Test
Left or Right Top Cover	Replacement or removal during installation (page 820)	1.) Emergency Stop Check (page 138) 2.) Perform System Scanning Test, (page 69)
Left or Right Side Panel	Replacement or removal during installation (page 820)	1.) Emergency Stop Check (page 138) 2.) Perform System Scanning Test, (page 69)
ETC SCA-LAN Board	Replace, Install faulty board (page 819)	Perform System Scanning Test, (page 69)
ETC Heurikon Board	Replace, Install faulty board (page 809)	1.) Verify ETC Node DIP switch. (page 823) 2.) Perform System Scanning Test, (page 69)

Table 12-1 Table Component Replacement Verification (Continued)

Section 2.0 Table Replacement Procedures

2.1 46-297093P1 AC/DC Power Switch Replacement

1.) Raise the table to maximum height.
2.) Turn off the Table breaker in the PDU, to remove power from the entire table.
3.) Remove the left Base Covers.
4.) Locate the Power Assembly:
   a.) Remove the screws from the outlet cover.
   b.) Move the cover aside, to gain access to the power switches.
   c.) Loosen the nut that fastens the defective power switch to the plate.
   d.) Remove the switch from the plate

Note: Pay attention to the location of the wires on the defective switch, before you remove them. Restore the wires to their original configuration on the replacement switch.

5.) Loosen the screw terminals, and remove the wires from the defective switch.
6.) Transfer the wires to the same location on the replacement switch.
7.) Reassemble the Table, and replace the covers.
8.) Turn on the Table breaker in the PDU to restore power.

2.2 Actuator Cover

1.) Raise the table to maximum height.
2.) Remove the Base Covers, the right Table Side Covers and right Side Panels.
3.) Locate the Actuator cover:
   a.) Remove both of the clips that fasten the spring pin to the actuator cover.
   b.) Slide out the pin, to release the spring.
   c.) Remove the two screws that fasten the cover hinge to the U-bracket.

   d.) Remove the Actuator cover from the table.
4.) When you install the actuator cover:
   a.) Center the cover in the rear leg opening.
   b.) Tighten the hinge screws
   c.) Seat the spring in the notch on the spring pin.
5.) Reassemble the Table, and replace the covers.

2.3 46-296561P10 Actuator Limit Switch

1.) Raise the table to maximum height.
2.) Turn off the Table breaker in the PDU, to remove power from the entire table.
3.) Remove the Base Covers.
4.) Remove the Actuator Cover, if you plan to replace or adjust the Upper Limit Switch. (Refer to section 2.2, on page 795)
5.) Remove the wires from the switch.
6.) To remove the Upper Limit Switch:
   a.) Loosen the screw that fastens the switch in place.
   b.) Slide the switch up, along the actuator until free.
   c.) Remove the wires from the terminals
7.) To remove the Lower Limit Switch:
   a.) Remove the screw and two washers from the switch.
   b.) The nut plate is trapped, but may slide down the actuator.
   c.) Remove the wires from the terminals
8.) To install either limit switch:
   a.) Slide the switch into place on the actuator.
   b.) Connect the wires to the W and C terminals.
   c.) Do not tighten the screw until you adjust the switch position.
9.) To adjust the Upper Limit Switch:
   a.) Locate the calibration plate, on the right side of the base frame.
   b.) Loosen the two Cal plate screws.
   c.) Move the plate to its horizontal position, and tighten the screws.
   d.) Restore power, and elevate the table until the distance between the bottom of the Cal plate, at the UPPER LIMIT SWITCH position, and the center mark on the upper, rear leg pivot pin equals 36.63 ±±0.03.
   e.) Remove power, and attach a continuity device to the W and C terminals.
   f.) Loosen the clamping screw on the upper limit switch.
   g.) Slide the switch upward until at least 1 of the magnet tubes appears below the switch.
   h.) Slowly slide the switch downward until the switch opens, then tighten the screw.
10.) To adjust the Lower Limit Switch:
   a.) Locate the calibration bar, on the right, rear corner of the base frame.
   b.) Remove the Cal bar from its storage position.
   c.) Reposition the bar in the storage hole, to make it parallel to the rear surface of the base frame, then tighten the screw.
   d.) Restore power, and elevate the table until the distance between the bottom outer edge of the Cal bar and the center mark on the upper, rear leg pivot pin equals 14.51 ±0.03.

e.) Remove power, and attach a continuity device to the W and C terminals.
f.) Loosen the clamping screw on the switch, and slide the switch downward as far as possible.
g.) Slowly slide the switch upward, until the switch opens, then tighten the screw.

11.) Return the calibration plate and bar to their storage positions.
12.) Reassemble the Table, and replace the covers.
13.) Turn on the Table breaker in the PDU to restore power.

2.4 46-296561P11 or 2100671 Actuator Magnet Rod

1.) Raise the table to maximum height.
2.) Turn off the Table breaker in the PDU, to remove power from the entire table.
3.) Remove the Base Covers, the right Side Panels and the Actuator Cover.
Refer to Actuator Cover section 2.2, on page 795.
4.) Loosen and remove the two nuts that fasten the Magnet Rod to its plate at the upper end of the actuator.
5.) Slide the Magnet Rod out of the actuator.
6.) Replace the Magnet Rod, and adjust the Upper and Lower Actuator switches.

Note: Because magnet strengths vary, follow the procedure that starts on page 796, to adjust both Actuator Limit Switches.

7.) Reassemble the Table, and replace the covers.
8.) Turn on the Table breaker in the PDU to restore power.

2.5 46-296561P1 or 2103043 Table Elevation Actuator

To prevent the table from moving during the change of the Elevation Actuator, a service tool has been developed that fits over the elevation gas springs and blocks the table in the full up position. A modified Table Elevation Actuator replacement procedure has been developed and is included in this Service Note.

Elevation Gas Spring Service Support part number: 2144261

1.) Fully elevate the table using the table control panel; THERE SHOULD BE NO WEIGHT ON THE CRADLE.
2.) Turn OFF the table breaker in the PDU to remove power from the whole table.
3.) Remove the Base Covers and the right Side Panels.
4.) Remove the Actuator Cover U–bracket by removing four screws.
5.) Remove spring clip from end of cross bar that holds the lower end of the Actuator Cover Spring. Slide cross bar out of the Actuator cover, releasing the spring tension. Remove pin and cover from table.
6.) Disconnect the motor wires, taking note of wire positions and the ground strap. Cut the ty-raps and remove the clamp holding the limit switch wires to the Actuator. Disconnect wires from limit switch, taking note of wire positions.

⚠ **CAUTION** **When manually adjusting the table height in the next step to unload the elevation actuator, it is easy to overshoot the desired position and place the elevation actuator in compression.**

7.) When the table is fully elevated, the Actuator is in tension due to the force produced by the gas springs. The table must be elevated further with the hex drive on the end of the motor, see Figure 12-4, on page 800 (Observe arrow marking on motor cover end plate for "Up" direction). Raise the table until the Actuator is unloaded, which occurs when the Elevation Actuator can be moved "side to side" on the clevis pin in the lower mounting block. If you continue to turn the hex drive past this point you will no longer be able to move the Elevation Actuator from side to side.

8.) Place the "Elevation Gas Spring Service Support" onto the lower gas spring mount by sliding the service support down the top surface of the gas springs. See Figure 12-1, on page 799. Insure that Service Support, top surface, at arrows, is touching the Elevation Gas Cylinders for proper installation.

⚠ **DANGER** **PERSONAL INJURY OR EQUIPMENT DAMAGE POSSIBLE IF PROP IS NOT POSITIONED CORRECTLY. INSURE THAT PROP, TOP SURFACE, AT ARROWS, IS TOUCHING ELEVATION GAS CYLINDER FOR PROPER INSTALLATION.**

⚠ **CAUTION** **Insure that the Elevation Actuator lower clevis pin can easily be rotated before moving to the next step. If the Lower Clevis Pin can be rotated by hand, it indicates that the Elevation Actuator has no load applied to it.**

9.) Remove the 'E' clip from the elevation actuator lower clevis pin. Using only your fingers, remove the clevis pin from the lower mounting block.

10.) Loosen and remove the locknut from the upper Actuator mounting pin. Note the positions of the two bumper washers and spacer, and remove them. See Figure 12-1, on page 799, and Figure 12-2, on page 799.

11.) Lift the lower end of the elevation actuator out of the lower clevis mounting block. Once the lower end clears the clevis block, move the elevation actuator towards the 'foot' end of the table while supporting the top of the elevation actuator to prevent it from falling. Lower the top end which will allow the elevation actuator to be moved out the 'foot' end of the table.

12.) Adjust the length of the new Elevation Actuator to be the same as the one just removed. The easiest way to adjust the length is to turn the rod end of the Elevation Actuator counter clockwise. See Figure 12-4, on page 800.

13.) Insert the new Elevation Actuator into the table by reversing the removal process in step 11. Insert the top mount clevis pin. When installing the Upper Mounting Pin, make sure there is one bumper washer on either side of the rod end. On the right side, the bumper washer fits over the shoulder on the hex–side of the pin; on the left side, the spacer fits inside the bumper spacer, and spaces the rod end away from the mounting clevis. Torque the locknut to 100 ft-lbs.

14.) Fine tune the length of the actuator to get exactly the same length by adjusting the hex drive on the actuator motor. Make sure that the lower pin is coated with Molykote grease (Stock Code, 81596). Support the lower end of the Elevation Actuator and insert the clevis pin by hand into the lower clevis mounting block. Do not force the pin into the mount and actuator. If the length is adjusted correctly, the pin will slide in easily. Insure that the E–ring is correctly installed.

⚠ **CAUTION** **Failure to remove the Elevation Gas Spring Service Support may result in damage to the table, if the elevation drive is activated with the Service Support in place.**

15.) Remove the "Elevation Gas Spring Service Support" from the gas spring mount by lifting the top edge and sliding the service support up the top surface of the gas springs and out of the table.

16.) Reconnect the motor wires, noting the polarity. Ty-rap the limit switch harness in place, making sure to connect the wires to the W and C terminals of each switch.

17.) Adjust both Actuator Limit Switches per their procedure.

Note: Rough adjustment of the actuator limit switches may be made by measuring the limit switches on the old actuator and setting the new switches accordingly.

18.) Refit the Actuator Cover, make sure to center the cover in the rear leg opening before tightening the actuator cover "U" bracket, and replace the cover cross–bar and retainer clip while positioning the spring in the cross–bar's notch. Refit the Side Panels (make sure that the safety ground strap is installed properly), Side Cover and Base Covers.

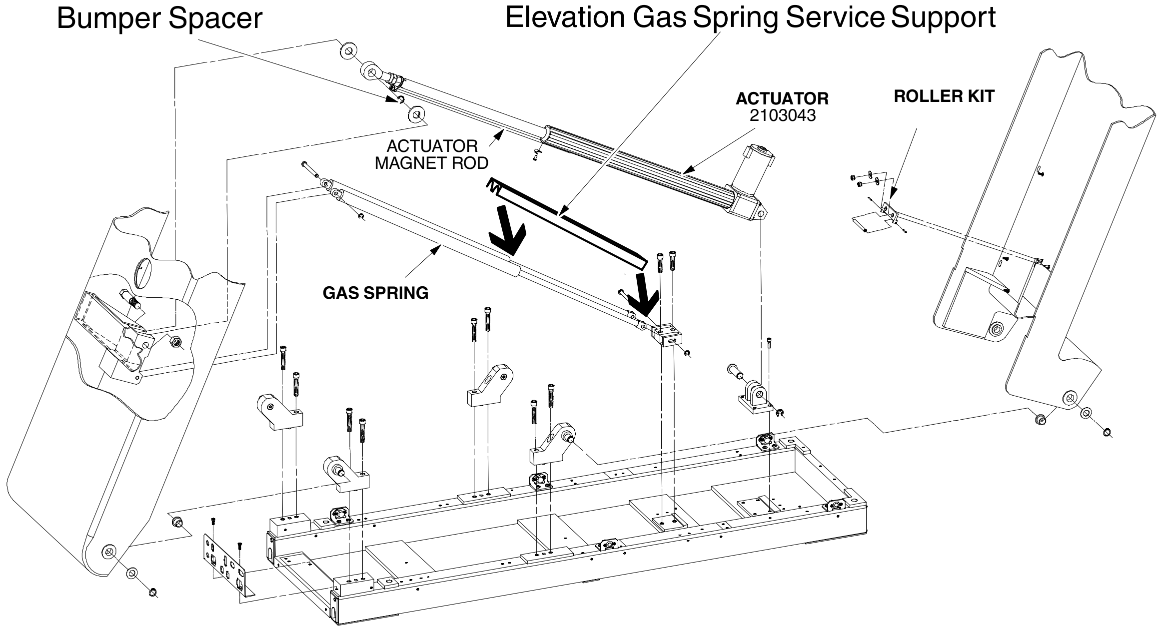


Figure 12-1 Table Base Layout

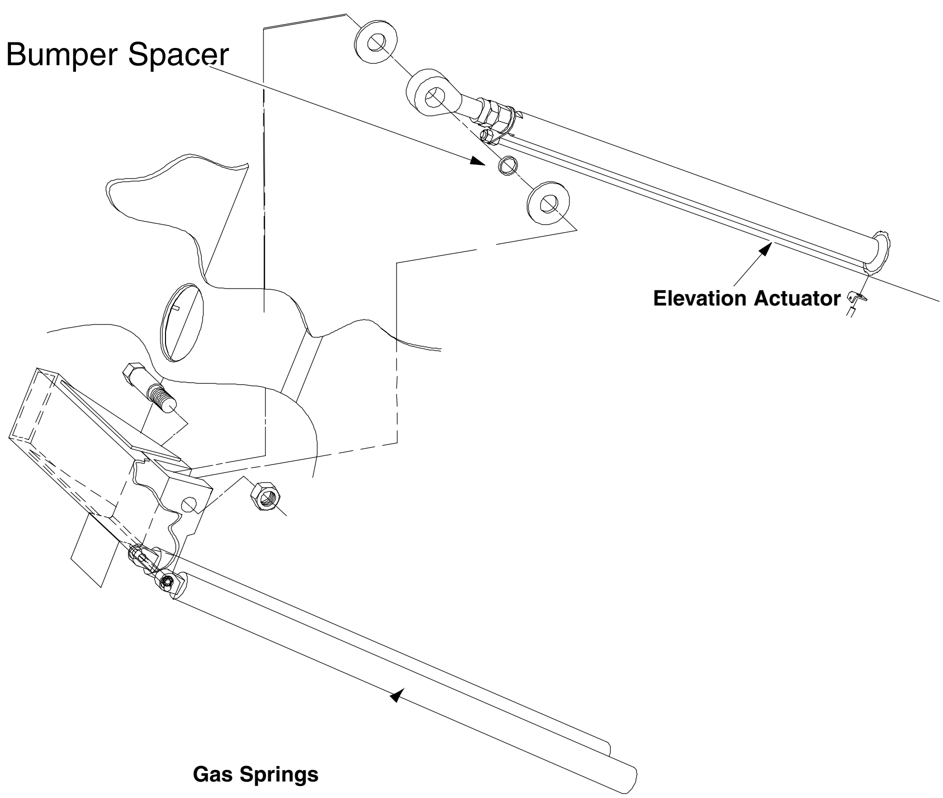


Figure 12-2 Elevation Actuator Upper Mounting Detail

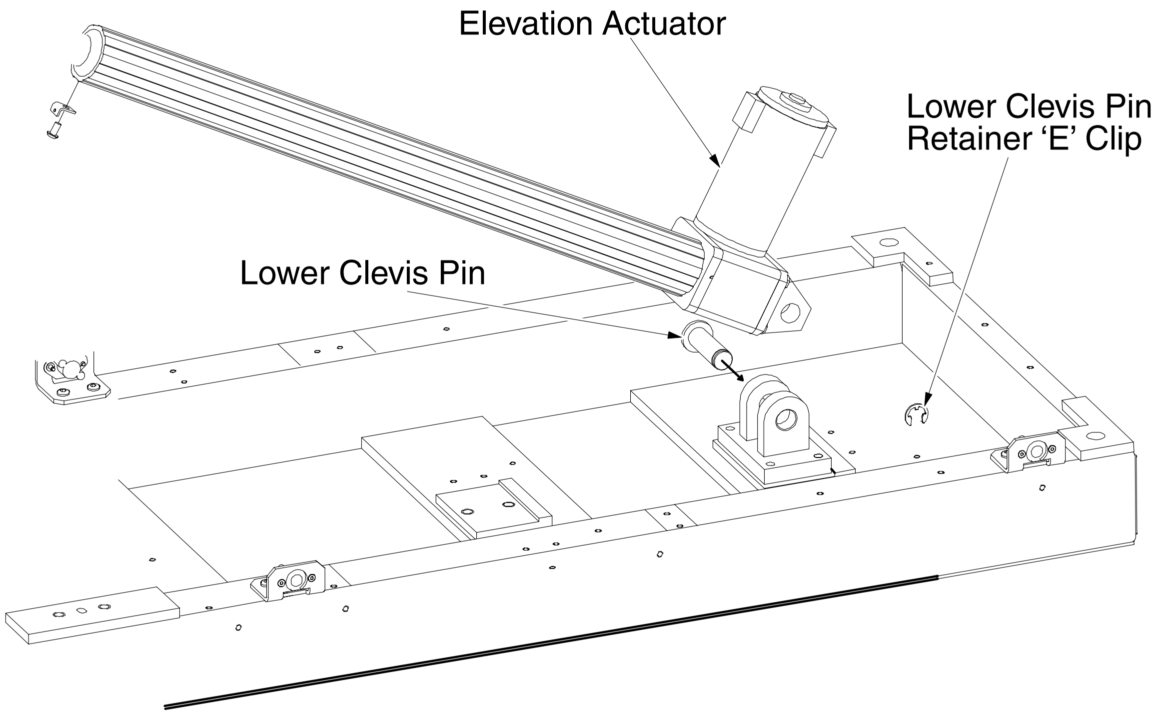


Figure 12-3 Elevation Actuator Lower Mounting Detail

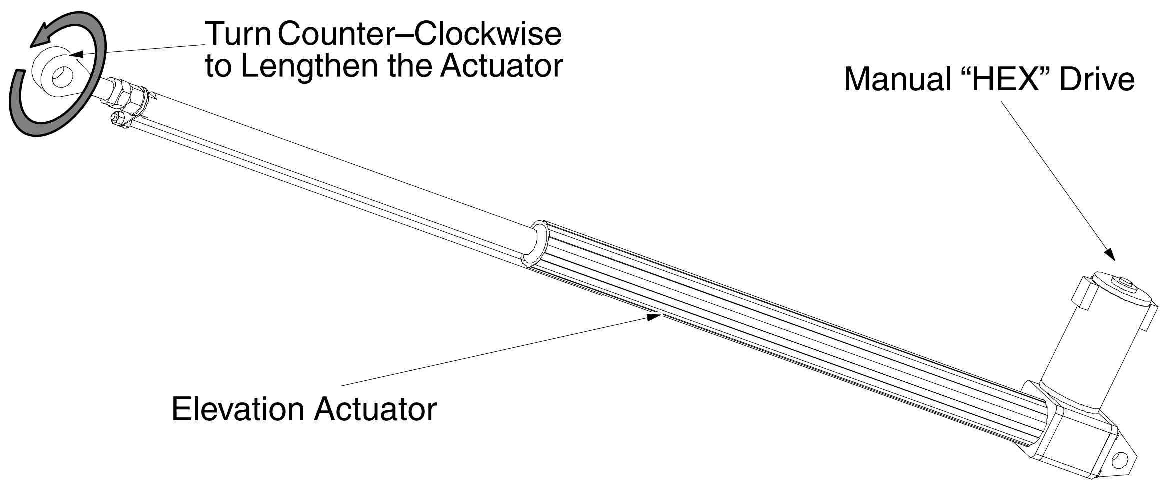


Figure 12-4 Adjusting Elevation Actuator Length

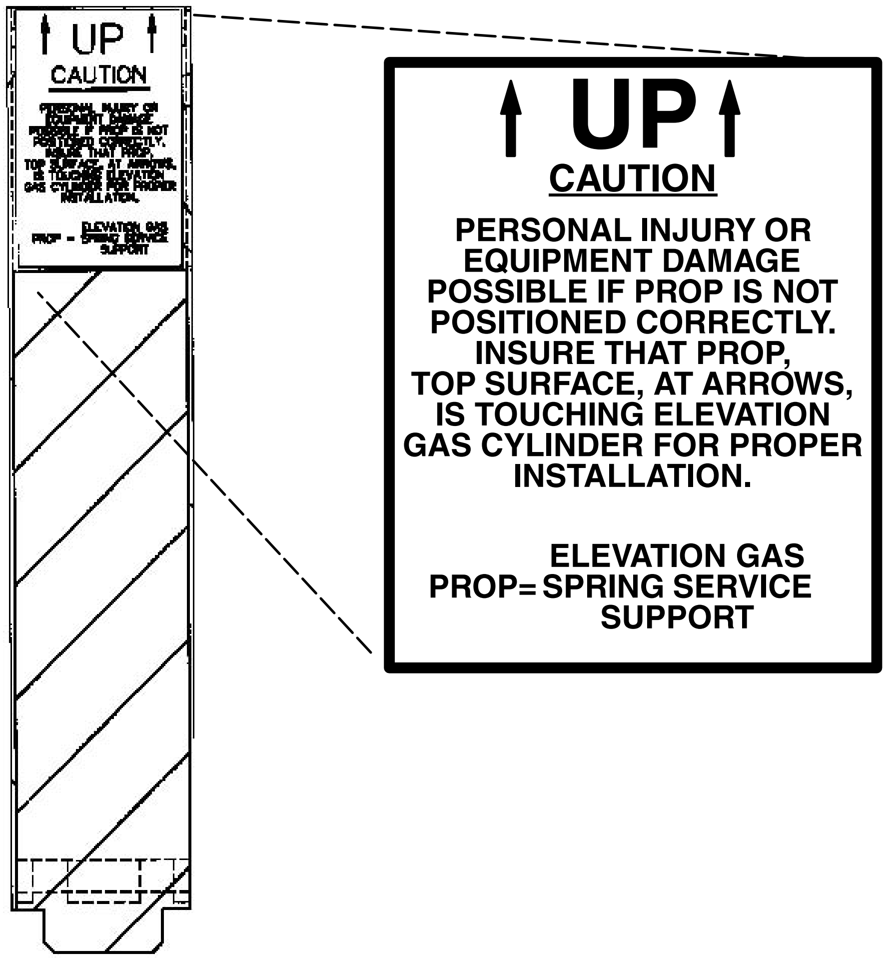


Figure 12-5 Elevation Gas Spring Service Support Caution Label

2.6 46-221532P18 Cradle Drive Belt

1.) Raise the table to maximum height.
2.) Turn off the Table breaker in the PDU, to remove power from the entire table.
3.) Remove the Cradle Drive Assembly, and set it on a work surface.
    Cradle Drive Assembly procedure begins on page 803.
4.) Remove the belt from the pulleys:
    a.) Loosen the four screws that fasten the motor in place.
    b.) Slide the motor toward the drive roller, to remove tension from the belt.
    c.) Remove the belt.
5.) After you install the new belt:
    a.) Slide the motor away from the drive roller, until the belt deflects 0.050/0.060 inches with 4 - 6 oz applied at its mid-span.
    b.) Tighten the four screws, to fasten the motor with tension on the belt.
6.) Replace the Cradle Drive Assembly.
7.) Reassemble the Table, and replace the covers.
8.) Turn on the Table breaker in the PDU to restore power.

2.7 46-170047P7 Elevation Encoder Belt

1.) Raise the table to maximum height.
2.) Turn off the Table breaker in the PDU, to remove power from the entire table.
3.) Remove the Base Covers and left Side Panels.
4.) Remove the elevation encoder belt from the sprockets:
   a.) Loosen the two screws that fasten the Elevation Encoder Assembly to the base.
   b.) Slide the encoder assembly toward the large sprocket, to remove tension from the encoder belt.
   c.) Remove the belt.
5.) After you install the new belt,
   a.) Align the splice on the belt with the mark on the large sprocket.
   b.) Slide the Elevation Encoder Assembly away from the large sprocket, until the belt deflects 0.250 inches with 32 – 35 oz. applied at its mid-span.
   c.) Tighten both screws, to fasten the encoder assembly in place, with tension on the belt.
6.) Loosen the coupler screw, and adjust the Encoder Table Elevation C-pulse.
   - Encoder Table Elevation procedure begins on page 806.
   - The C-Pulse adjustment procedure begins on page 806.
7.) Loosen, and adjust, the switch cam.
   Table Interference Matrix Switch procedure begins on page 810.
8.) Reassemble the Table, and replace the covers.
9.) If necessary, turn on the Table breaker in the PDU to restore power.

2.8 46-297332P1 50Ohm BNC T-Connector

1.) Locate the T-connector on the SCA-LAN PWB, on the table ETC PWB.
2.) Rotate the knurled ring to remove, and install, the BNC T-Connector.
3.) Use the plastic support to position the right-angle BNC connectors.

2.9 46-297350P1 Upper Pin Actuator Bumper Disk

The front leg attachment point for the Actuator contains two Bumper Disks.

Refer to the Table Elevation Actuator procedure, on page 797.

2.10 46-297253P1 ETC Cradle Support

The ETC PWA contains the Cable Support, which positions the BNC-T connector.

Refer to the SCA-LAN Bd ETC procedure, on page 819.

2.11 46-297576P1 Cal Pin

Use the Cal Pin to lock the cradle/carriage into position, at specific locations.

- Remove the right Table Side Covers, and Cradle Drive Cover, to access the Cal pin.
- Store the Cal pin in the bottom of the right z-channel, beneath the Cradle Drive Cover.

The following procedures use the Cal pin:

- Home Switch: start on page 809
- Home Latch Assembly: starts on page 810
- Longitudinal Encoder Assembly: starts on page 812
- Longitudinal Limit Switch: starts on page 816
- Longitudinal Encoder Pot Assembly: starts on page 817

2.12 46-297420G1 Cradle Assembly

1.) Raise the table to maximum height.
2.) Drive the Cradle/carriage to the latched, home position, before you try to remove the assembly.
    An unlatched Cradle/carriage assembly could quickly move toward the gantry, and damage the longitudinal encoder assembly.
3.) Remove, and keep, the six plug buttons that cover the cradle bolt holes.
4.) Loosen, and remove, the six screws, located beneath the plug buttons.
5.) Lift the Cradle upward, to remove the assembly from the table.
6.) To install the Cradle:
    a.) Position the rear end of the Cradle over the carriage.
    b.) Align the holes in the cradle to the threads in the carriage.
    c.) Start, but do not tighten, all six screws.
    d.) Look through the gantry bore at the end of table.
    e.) Laterally position the Cradle on the drive rollers, until the same gap exists between the sloped sides of the Cradle and the guide rollers.
    f.) Tighten the six screws to 13 ft-lbs.
    g.) Replace the plug buttons.
7.) Make sure the cradle/carriage does not rest against the latch.

2.13 46-264370G1 Cradle Drive Amplifier

1.) Raise the table to maximum height.
2.) Turn off the Table breaker in the PDU, to remove power from the entire table.
3.) Remove the left Base Covers.
4.) Loosen the captive screws, or remove the four screws, that fasten the servo amp cover in place.
5.) Remove the servo amp cover, and set aside.
6.) Locate the Cradle Drive Amplifier, toward the outside of the table:
    a.) Detach all of the wire connectors.
    b.) Remove the seven screws that fasten the Amp to its mounting bracket.
7.) Remove, and replace, the defective Cradle Drive Amplifier.
8.) Reassemble the Table, and replace the covers.
9.) Turn on the Table breaker in the PDU to restore power.

2.14 46-296235G1 Cradle Drive Assembly

1.) Raise the table to maximum height.
2.) Turn off the Table breaker in the PDU, to remove power from the entire table.
3.) Remove the right Table Side Covers.

4.) Latch the carriage, and remove the Cradle Assembly.
Cradle Assembly procedure begins on page 803.
5.) Loosen the two captive screws, and remove the Cradle Drive Cover.
  a.) Disconnect the ground strap from the Z-Channel
  b.) Unplug the J21 wire connector from the left side.
  c.) Unplug the J18 wire connectors from the right side.

Note: An unlatched Cradle/carriage assembly could quickly move toward the gantry, and damage the longitudinal encoder assembly.

6.) Move the carriage toward the gantry:
  a.) Hold the carriage in place with one hand.
  b.) Manually unlatch the carriage with the other hand.
  c.) Slowly move the carriage toward the gantry, until it meets the bumper stop.
7.) Remove the Cradle Drive Assembly:
  a.) Tilt the front roller of the Cradle Drive downward.
  b.) Lift the entire assembly up, then backward, and then down.
  c.) Gently remove the assembly from the bottom of the table.

Note: Take care not to disturb the Longitudinal Encoder cable.

8.) After you reassemble the Cradle Drive:
  a.) Center the Cradle between the guide rollers
  b.) Torque the six screws to 13 ft-lbs.
  Cradle Assembly procedure begins on page 803.
9.) Reassemble the Table, and replace the covers.
10.) Turn on the Table breaker in the PDU to restore power.

2.15 46-296594P1 Cradle Drive Cover

1.) Raise the table to maximum height.
2.) Loosen the two captive screws at the rear of the cover.
3.) Slide the cover forward, until it clears the two mounting pins.

2.16 46-278446P1 Electrical Outlet

1.) Raise the table to maximum height.
2.) Turn off the Table breaker in the PDU, to remove power from the entire table.
3.) Remove the left Base Covers.
4.) Locate the Power Assembly:
  a.) Remove the screws from the outlet cover.
  b.) Pull out the cover to access the wire connections.

Note: Pay attention to the location of the wires on the defective outlet, before you remove them. Restore the wires to their original configuration on the replacement outlet.

5.) Loosen the screw terminals, and remove the wires from the hot, neutral and ground terminals.
  - White wire to silver screw
  - Black wire to brass screw
  - Green & yellow wire to green (ground) screw
6.) Remove, and replace, the defective outlet.

7.) Transfer the wires to the same locations on the replacement outlet.
8.) Reassemble the Table, and replace the covers.
9.) Turn on the Table breaker in the PDU to restore power.

2.17 46-296633G1 Elevation Encoder Assembly

1.) Raise the table to maximum height.
2.) Turn off the Table breaker in the PDU, to remove power from the entire table.
3.) Remove the Base Covers, and the left Side Panels.
4.) Locate the table harness at the rear of the ETC mounting panel:
   a.) Disconnect the J15 Encoder cable from the table harness.
   b.) Cut the ty-wraps, to free the cable.
5.) Loosen, and remove, the two screws that fasten the Elevation Encoder Assembly to the table base.
6.) Slide the Assembly toward the large sprocket, to relieve belt tension.
7.) Remove the belt from the encoder assembly.
8.) Remove the defective Encoder Assembly from the table.
9.) Install the replacement Elevation Encoder Assembly:
   a.) Replace the belt
   b.) Slide the Elevation Encoder Assembly away from the large sprocket, until the belt deflects 0.250" with 32 – 35 oz. applied at its mid-span.
   c.) Tighten both screws, to fasten the encoder assembly in place, with tension on the belt.
10.) Adjust the encoder C-Pulse position.
   - Encoder Table Elevation (C-pulse) procedure begins on page 806.
   - The C-Pulse adjustment procedure begins on page 806.
11.) Adjust the Table Interference Matrix Switch.
   Interference Matrix Switch procedure begins on page 810.
12.) Characterize the elevation axis.
   The Mechanical Characterization procedure begins on page 140.
13.) Reassemble the Table, and replace the covers.
14.) Turn on the Table breaker in the PDU to restore power.

2.18 46-288170G1 Elevation/Tilt Amplifier

1.) Raise the table to maximum height.
2.) Turn off the Table breaker in the PDU, to remove power from the entire table.
3.) Remove the left Base Covers.
4.) Loosen the captive screws, or remove the four screws, that fasten the servo amp cover in place.
5.) Remove the servo amp cover, and set aside.
6.) Locate the Elevation/Tilt Amplifier, toward the inside of the table.
   a.) Detach all of the wire connectors.
   b.) Remove the seven screws that fasten the Amp to its mounting bracket.
7.) Remove, and replace, the defective Elevation/Tilt Amplifier.
8.) Reassemble the Table, and replace the covers.
9.) Turn on the Table breaker in the PDU to restore power.

2.19 46-296854P1 Table Elevation Encoder

1.) Raise the table to maximum height.
2.) Turn off the Table breaker in the PDU, to remove power from the entire table.
3.) Remove the Base Covers, and the left Side Panels.
4.) Locate the table harness at the rear of the ETC mounting panel:
    a.) Disconnect the J15 Encoder cable from the table harness.
    b.) Cut the ty-wraps, to free the cable.
5.) Locate the Elevation Encoder Assembly:
    a.) Loosen the screw on the encoder-side of the flexible coupling. (*if* the encoder assembly has a flexible coupling)
    b.) Loosen the two set screws that fasten the thumb wheel to the encoder shaft. (*if* the encoder assembly has a thumb wheel)
6.) Remove the encoder assembly:
    a.) Locate the three servo clamps that fasten the encoder to the mounting block.
    b.) Turn each servo clamp 1/2 turn ccw.
    c.) Pull the encoder away from the block.
    d.) Slide the thumb wheel and the spacer off the shaft.
    e.) Remove the encoder from the table.
7.) Install the replacement encoder assembly:
    a.) Place the spacer on the Encoder shaft.
    b.) Insert the shaft through the block and thumb wheel.
    c.) Firmly seat the encoder in the block.
    d.) Let the Encoder cable hang down (±45 degrees) while you tighten the three servo clamps.
    e.) Press the thumb-wheel against the spacer, and the Encoder, while you tighten the two set screws in the thumb-wheel.
8.) C-Pulse Adjustment:
    a.) Locate the Calibration plate, on the right side of the base frame.
    b.) Loosen the two screws on the Cal plate, move the plate to its horizontal position, then tighten the screws.
    c.) Turn on the Table breaker in the PDU to restore power.
    d.) Restore table power, and elevate the table until the distance between the bottom of the Cal plate, at the C-pulse position, and the center mark on the upper, rear leg pivot pin equals 30.44 ±.01 inches.
    e.) Turn the thumb-wheel, to rotate the encoder shaft, and light the C-Pulse LED on the ETC PWB.
    f.) Tighten the flexible coupler screw to clamps the Encoder shaft in the C-Pulse position.
    g.) Verify the C-Pulse LED remains lit.
    h.) Return the Cal plate to its storage position.
9.) Reassemble the Table, and replace the covers.

2.20 46-264368G1 ETC Board

1.) Raise the table to maximum height.
2.) Remove the Base Covers.
3.) Turn off the Table breaker in the PDU, to remove power from the entire table, or turn off the

breaker switch at the rear of the table.

NOTICE **Prevent permanent damage to the static-sensitive boards. Attach the anti-static wrist strap to your wrist and to a bare metal grounding point on the table before you continue.**

4.) Locate the Table ETC board in the bottom of the table.
   a.) Remove, and keep, 12 screws that fasten the ETC board to the standoffs.
   b.) Detach all connectors from the ETC board.
   c.) Detach **ETC** Board from the CPU Board.
5.) Replace ETC Board:
   a.) Connect ETC Board to CPU Board.
   b.) Fasten to the standoffs with 12 screws.
   c.) Attach all connectors.
6.) Reassemble the Table
7.) Restore power to the table, and replace the covers.

2.21 46-229455P1 ETC Fan

1.) Raise the table to maximum height.
2.) Turn off the Table breaker in the PDU, to remove power from the entire table.
3.) Remove the Base Covers.
4.) Unplug the Fan connector.
5.) Pay attention to the position of the Fan connector (lower left corner) and the direction of air flow (toward the CPU PWA).
6.) Remove three screws that fasten the Fan to the power assembly bracket.
7.) Remove the two screws that fasten the guard to the defective Fan.
8.) When you install the replacement Fan, position it with the connector in the lower left corner and the air flow indicator pointing toward the CPU PWA.
9.) Turn on the Table breaker in the PDU to restore power.
10.) Reassemble the Table, and replace the covers.

2.22 46-297664P1 Filler Cover Spring

No specific procedure at this time. Please use good service practices.

2.23 46-170021P15 Servo Amp Fuse

1.) Raise the table to maximum height.
2.) Turn off the Table breaker in the PDU, to remove power from the entire table.
3.) Remove the left Base Covers.
4.) Loosen the captive screws, or remove the four screws, that fasten the servo amp cover in place.
5.) Remove the servo amp cover, and set aside.
6.) Remove the plastic Fuse cover, if present.
7.) Remove the defective fuse from its holder.
8.) Install the new fuse.
9.) Reassemble the Table, and replace the covers.

10.) Turn on the Table breaker in the PDU to restore power.

2.24 46-297155G2 Long Grounding Strap

Fasten this Ground Strap between the table base frame and the Power Assembly ground bar.

2.25 46-297155G1 Short Grounding Strap

Use this Ground Strap in ten locations, between:

- The Cradle Drive and the left z-channel
- Both z-channels and the front leg
- The front leg and the base frame
- The rear leg and the base frame
- Each Side Panel and the corresponding z-channel
- The Actuator and the base frame

2.26 46-296891P1 Gas Spring Replacement

1.) Raise the table to maximum height.
2.) Turn off the Table breaker in the PDU, to remove power from the entire table.
3.) **Important:** Remove all objects from the cradle.
4.) Remove the Base Covers, the right Side Panels.
5.) The gas springs keep the Actuator under tension, even when you raise the table to maximum height. To remove tension from the Actuator:
   - Raise the table past its upper height limit with a power supply, or by manually turning the hex drive on the end of the motor.
   - Raise the table to fully extend the gas springs, while the lower gas spring remains at the back of the slot in the lower mounting block.
6.) Loosen and remove the lock nut from the upper Actuator mounting pin.
   If you cannot easily remove the upper Actuator mounting pin, return to the previous step, and raise the table to remove the gas spring tension.

Note: **Important:** Loosen *both* actuator mounting pins before you remove either one.

7.) Remove the E-Rings from the top and bottom Gas Spring mounting pins.
8.) Remove the pins from their mounting blocks.
9.) Remove both Gas Springs.

⚠ **WARNING** **DO NOT THROW DEFECTIVE GAS SPRINGS IN THE TRASH! COMPACTING A PRESSURIZED GAS SPRING MAY CAUSE IT TO EXPLODE. RETURN BOTH GAS SPRINGS TO:**

GEMS Recycling Center
Ace Warehouse
Attn.: Paul Neumiller
2200 E. College Ave.
BLDG. 11
Cudahy, WI, 53110

10.) When you install the replacement Gas Springs:

a.) Position the Gas Spring pressure chambers UP, or closest to the front table leg.

b.) Take care to install the E-ring correctly.

11.) Reassemble the Table, and replace the covers.

12.) Turn on the Table breaker in the PDU to restore power.

2.27 46-296377P1 ETC Heurikon Board

1.) Raise the table to maximum height.

2.) Turn off the Table breaker in the PDU, to remove power from the entire table, or turn off the breaker switch at the rear of the table.

3.) Remove the Base Covers.

NOTICE **Prevent permanent damage to the static-sensitive boards. Attach the anti-static wrist strap to your wrist and to a bare metal grounding point on the table before you continue.**

4.) Locate the ETC chassis assembly:

a.) Loosen the captive screw that fastens the ETC chassis to the center of the table base.

b.) Pivot the chassis panel until the Heurikon PWA reaches the horizontal position.

c.) Loosen, and remove, the screw that fastens the Heurikon PWA to the ETC chassis.

d.) Disconnect the Heurikon PWA from the ETC PWA.

e.) Slide the defective Heurikon PWA out of the card guide.

5.) Install the replacement Heurikon PWA.

6.) Reassemble the Table, and replace the covers.

7.) Turn on the Table breaker in the PDU to restore power.

2.28 46-136334P23 Home Position Switch

1.) Raise the table to maximum height.

2.) Turn off the Table breaker in the PDU, to remove power from the entire table.

3.) Remove the right Table Side Covers, and Cradle Drive Cover, to access the Cal pin.

4.) Install, and tighten, the Cal pin about one-half inch from the home position, to hold the carriage in place.

Note: Pay attention to the location of the wires on the defective switch, before you remove them. Restore the wires to their original configuration on the replacement switch.

5.) Disconnect the wires from the Home Position switch terminals

6.) Remove the nut that fastens the switch in place.

7.) Remove the defective switch from its bracket.

8.) Install the replacement switch

Make sure you connect the wires to the COM and N.O. terminals.

9.) Adjust the position of the switch in the bracket:

a.) Loosen the two nuts that fasten the bracket in place

b.) Move the bracket until the switch actuates when the carriage reaches 0.50 ±0.03 inches from the Home Position

10.) Return the Cal pin to its storage position.

11.) Reassemble the Table, and replace the covers.

12.) Turn on the Table breaker in the PDU to restore power.

2.29 46-296233G1 Home Latch Assembly

1.) Raise the table to maximum height.
2.) Turn off the Table breaker in the PDU, to remove power from the entire table.
3.) Remove the right Table Side Covers, and Cradle Drive Cover, to access the Cal pin.
4.) Install, and tighten, the Cal pin in the home position, to hold the carriage in place.
5.) Remove the Cradle Assembly and the right side Rail Cover.
    Cradle Assembly procedure begins on page 803.
6.) Disconnect the harness at the terminal strip.
7.) Remove the four screws that fasten the Latch Assembly to the z-channel.
8.) Remove the defective Latch assembly.
9.) Install the replacement Latch Assembly, but do not tighten the four screws.
10.) Adjust the position of the entire Latch Assembly in its slots:
    a.) Adjust the Latch assembly until the distance between the carriage latch block, and the forward edge of the latch bar opening, equals 0.050 ±0.005 inches.
    b.) Maintain this distance while you tighten the four screws.
    c.) Adjust the set screw in the latch clevis block, until the outer edge of the latch bar overlaps the outer edge of the carriage latch block by 0.050 ±0.00 inches.
    d.) Maintain this distance while you tighten the jam nut.
11.) Make sure the solenoid plunger bottoms out:
    a.) Adjust the position of the solenoid bracket, until the clearance between the outer edge of the latch bar and the outer edge of the carriage latch block equals 0.050 ±0.005 inches.
    b.) Maintain this distance while you tighten the two screws.
12.) Adjust the position of the spring bracket, until the spring has 0.125 inches pre-load, when the latch bar rests against the set screw.
    Maintain this distance while you tighten the two screws.
13.) Install the Cradle.
    Cradle Assembly procedure begins on page 803.
14.) Return the Cal pin to its storage position.
15.) Reassemble the Table, and replace the covers.
16.) Turn on the Table breaker in the PDU to restore power.

2.30 46-136334P57 Interference Matrix Switch

1.) Raise the table to maximum height.
2.) Remove the left Base Covers.
3.) Turn off the Table breaker in the PDU, to remove power from the entire table.
4.) Remove the Servo Amp Assembly.
    The Servo Amp replacement procedure begins on page 819.
5.) Remove the two screws that fasten the interference switches to the bracket.

Note: Pay attention to the location of the wires on the defective switch, before you remove them. Restore the wires to their original configuration on the replacement switch.

6.) Detach the wires from the terminals of the defective switch.
7.) Remove the defective switch and install the replacement switch.
8.) Fasten the switches to the bracket, but not tighten the screws.

9.) Adjust the switch-to-cam clearance:
   a.) Position the switches until the roller lever comes in contact with the switch housing.
   b.) Adjust the switch position until a 0.035/0.045 inch gap exists between the roller and the outside cam surface.
   c.) Maintain this distance while you tighten the two screws.
10.) Turn on the Table breaker in the PDU to restore power.
11.) Adjust the actuation point:
   a.) Locate the Cal plate, on the right side of the base frame
   b.) Loosen the two Cal plate screws, move the plate to the horizontal position, then tighten the Cal plate screws.
   c.) Elevate the table until the distance between the bottom of the Cal plate, at the INT. MTX SWITCH S2 position, and the center mark on the upper, rear leg pivot pin equals 29.14 ±0.03 inches.
12.) Adjust the cam:
   a.) Loosen the cam clamping screw.
   b.) Rotate the cam until switch S2 (closest to center of the table) rides on the outside surface of the cam.
   c.) Turn the cam CW (seen from the left side of the table) until S2 actuates.
   d.) Tighten the cam clamping screw, and verify the setting.
13.) Reassemble the Table, and replace the covers.

2.31 46-297687P1 Intercom Speaker

1.) Raise the table to maximum height.
2.) Turn off the Table breaker in the PDU, to remove power from the entire table.
3.) Remove the right Side Panels.
4.) Remove the two nuts that fasten the speaker cover to the speaker.
5.) Remove the two nuts that fasten the Speaker and grill in place.
6.) Remove the defective speaker, and install the replacement speaker.
7.) Reassemble the Table, and replace the covers.
8.) Turn on the Table breaker in the PDU to restore power.

2.32 46-297805G1 Tape Switch Jumper Plug

Use the Jumper Plug to simulate the presence of a Side Cover Tape Switch, when you remove the corresponding cover from the table.

- Refer to the Table Side Covers procedures.
- The Table Side Cover descriptions begin on page 820

2.33 46-297698P1 Leg Tape Switch

1.) Raise the table to maximum height.
2.) Turn off the Table breaker in the PDU, to remove power from the entire table.
3.) Remove the right Side Panels.
4.) Disconnect the Tape Switch from the harness.

5.) Remove the Tape Switch, and the adhesive, from the rear leg.
6.) Thoroughly clean the mounting surface with alcohol.
7.) Remove the protective strip from the adhesive and press the switch firmly in place.
8.) Replace the Side Panels.
9.) Turn on the Table breaker in the PDU to restore power.
10.) Test the Tape Switch for proper operation.

2.34 46-296234G1 Longitudinal Encoder Assembly

Purpose: The purpose of the longitudinal encoder assembly adjustment is to set the encoder C-Pulse and the cradle pot to 0.8v at the cradle home position. This allows for the potentiometer to read safely across its available mechanical range during winding/unwinding of the encoder cable. If allowed to exceed its mechanical limits, the potentiometer will operate nonlinearly on an intermittent basis. This will cause cradle potentiometer/encoder mismatch errors.

The factory replacement assemblies will arrive with:

- an initial three turns of pre-load spring tension applied,
- all of the cable wire wound on the take up spool, and
- the cable eyelet anchored to the hex spacer.

2.34.1 Replacement Encoder Assembly Pre-work

1.) Refer to Figure 12-6. Loosen the replacement encoder potentiometer's clamp and the encoder's clamp.

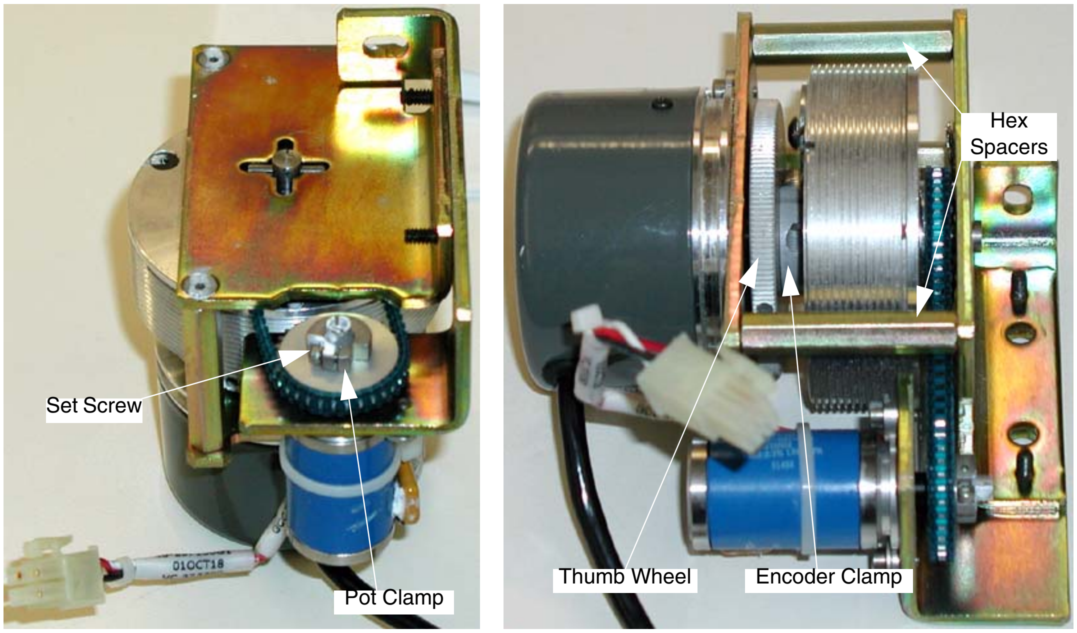


Figure 12-6 Replacement Encoder Assembly Prep.

2.) Remove the potentiometer clamp, and, using a small flat blade screwdriver, carefully spread the pulley's clamping splines. Do not spread them too far (see Figure 12-7) as the fragile aluminum will fracture. Work to break the hold of the white glyptol and to provide a little extra clearance around the pot shaft.

Remove clamp and spread splines *slightly,* as shown. Do not spread them too far, as the fragile aluminum will fracture.

Figure 12-7 Spread Pot Pulley Splines.

3.) Do not reinstall the clamp.
4.) Using a DMM, measure resistance between pins 2 and 1 (GND). This is where the capacitor is soldered.
5.) Using a small flat blade screwdriver, adjust the pot to approximately 5 K Ohms.
6.) Set aside the new assembly.

2.34.2 Defective Encoder Assembly Removal

1.) Raise the table to maximum height.
2.) Turn off the table breaker in the **PDU**, to remove power from the entire table.
3.) Remove the right table side covers and cradle drive cover to access the cal pin.
4.) Remove the cradle assembly and the right side rail cover.
Cradle assembly procedure begins on page 803.
5.) Remove the two lock nuts that fasten the longitudinal encoder assembly cover in place.
6.) Remove the two screws that fasten the front end cover in place.
7.) Grasp the carriage with one hand, while manually unlatching the carriage with the other hand.
8.) Slowly move the carriage assembly toward the gantry until it rests against the bumper stop.

⚠ NOTICE **Releasing the cradle carriage assembly before it rests against the bumper stop will damage the longitudinal encoder assembly.**

9.) Detach and store the stranded steel cable from the carriage by:
 a.) firmly holding the eyelet on the encoder cable,
 b.) removing the shoulder screw and spacer from the carriage, and
 c.) easing the cable forward until in a position to ty-wrap it to the hex spacer.

⚠ NOTICE **Maintain at least 2 pounds of tension on the cable. If tension is released and spool is allowed to unwind, the encoder assembly will be damaged.**

d.) Fasten the eyelet to the hex spacer with a ty-rap, to maintain the initial three turns of pre-load on the spool.

10.) Locate the right z-channel:

a.) Unplug the encoder J16 connector from the table harness.

b.) Unplug the pot connection at J17.

11.) Remove the two screws that fasten the encoder assembly to the table.

12.) Remove the defective longitudinal encoder assembly.

2.34.3 Replacement Encoder Assembly Installation

1.) When installing the replacement encoder assembly:

a.) Ensure that the cable maintains the initial three turns of pre-load tension on the spool.

Factory replacement assemblies arrive with the initial three turns of pre-load applied and the eyelet anchored to the hex spacer.

b.) Do not reinstall the pot sprocket clamp at this time.

c.) Connect J16 and J17.

d.) Attach a DMM to terminals #2 and #1 (GND) of the pot (see Figure 12-8) and measure the resistance between these two points. The value should be approximately 5 K Ohms. If it is not, adjust the pot shaft to produce this reading.

NOTICE **At no time should the pot resistance drop below 0 Ohms during the course of this installation.**

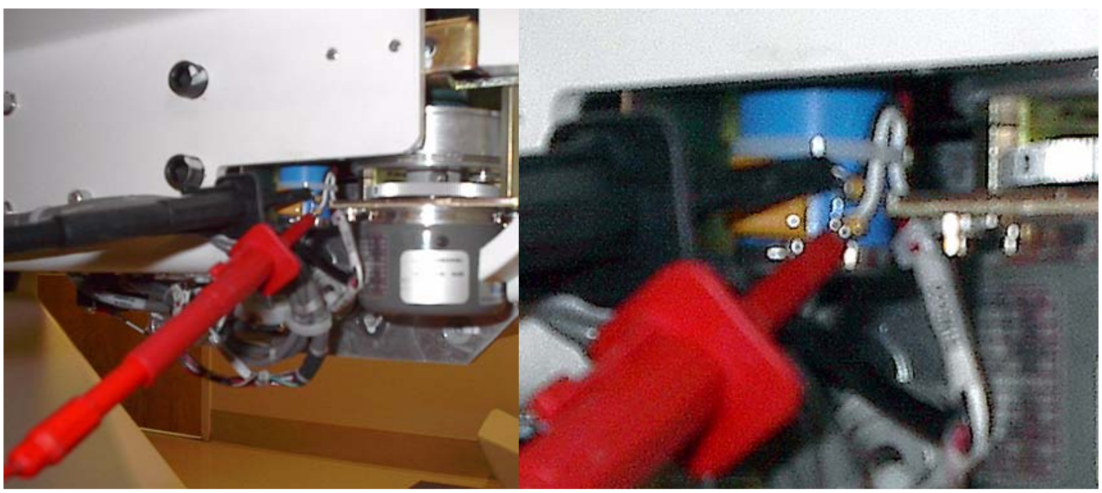


Figure 12-8 Attach DMM to pins 2 and 1 (GND)

e.) Position the DMM in such a way as to make the displayed resistance value visible throughout the remainder of this procedure.

2.) Fasten the cable to the carriage with the shoulder screw and spacer.

3.) While observing the read out on the DMM, slowly move the carriage to the home position, then install and tighten the Cal pin to fasten the carriage in place.

NOTICE **At no time should the pot resistance drop below 0 Ohms during the course of this installation.**

4.) Switch your DMM to voltage measurement.

5.) Turn on the table breaker in the **PDU** to restore power.

6.) Using a small pair of needle nose pliers, carefully squeeze the splines on the pot pulley to allow the clamp to easily slip over the splines.

7.) Reinstall the clamp such that proper access exists to the set screw. Do not tighten.

8.) Adjust the pot:
   a.) If not already in place, attach a DMM to terminals 2 and 1 (GND) of the pot.
   b.) Turn the pot shaft with a small screwdriver until the DMM displays 0.80 ±0.01 VDC.
   c.) Maintain the voltage displayed while tightening the pot clamp.
   d.) Do not remove the DMM at this time.

9.) Adjust the C-Pulse:
   a.) If not already done, loosen the clamp that fastens the cable spool to the encoder shaft.
   b.) Turn the encoder thumb-wheel to light the C-pulse **LED** on the **ETC** PWA.
   c.) Tighten the clamp and verify the C-pulse **LED** remains lit.

10.) Check for increase in pot voltage:
   a.) Hold the carriage assembly in position with one hand, while you remove the cal pin with the other hand.
   b.) Continue to hold onto the carriage assembly while manually releasing the home position latch.
   c.) Move the carriage towards the gantry and observe that the pot voltage increases.

⚠ **NOTICE** **The pot will be damaged if moved beyond the 0 VDC point.**

11.) Return the carriage to the home position and lock it in place.

12.) Reinstall the encoder cover, front end cover and the cradle.

13.) Characterize the longitudinal axis.

14.) Store the cal pin, reassemble the table, and replace the covers.

2.34.4 Encoder Cover

Two encoder covers currently exist in the field. That shown in Figure 12-9 is what is currently on most installed base tables. This cover has two main inadequacies that often lead to premature failure of the encoder assembly it is intended to protect:

1.) The encoder cable is unprotected from the point of exit from the cover to the beginning of the table's Z-channel. This makes it susceptible to the introduction of customer obstructions like blankets and patient straps.

2.) The cover does not allow for sufficient clearance for the potentiometer. Excessive friction with the cover will cause the pot value to vary resulting in intermittent, or even hard, cradle potentiometer/encoder mismatch errors.

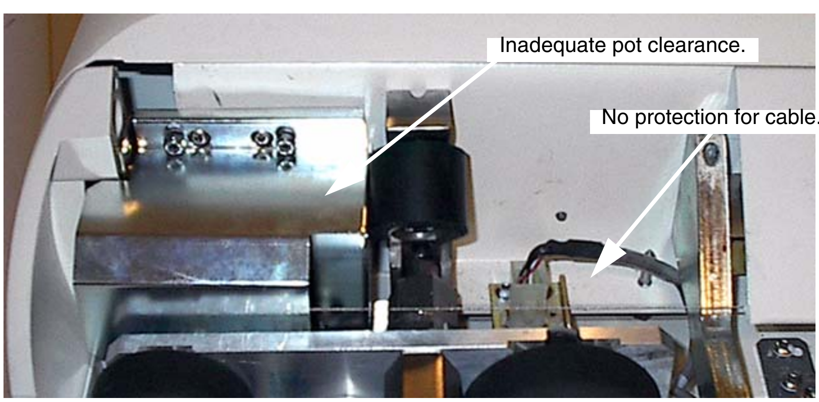


Figure 12-9 Old Encoder Cover

These issues, identified and addressed with a Six Sigma project, were resolved with a new cover as shown in Figure 12-10.

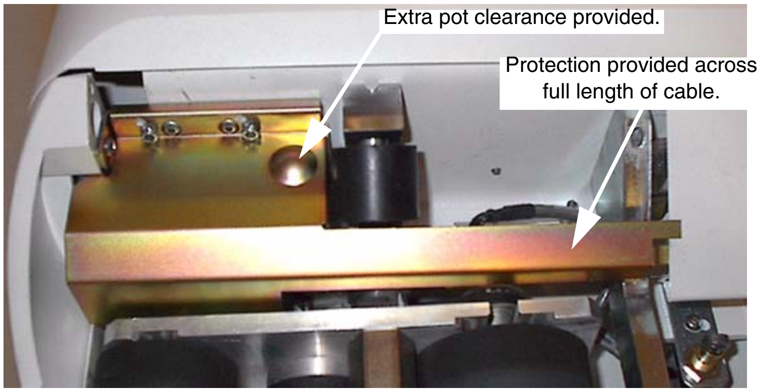


Figure 12-10 New Encoder Cover: 2261675

It is highly recommended that the field order and replace any of the old style covers present on installed base systems. These new covers were cut into forward production with the advent of the LCC table. All LightSpeed MDAS Ultra/Plus and LightSpeed 16 systems should have this cover installed.

The new covers are fully compatible with all tables from the HighSpeed Z-series systems through the LightSpeed16 systems.

2.35 46-136334P23 Longitudinal Limit Switch

1.) Raise the table to maximum height.
2.) Turn off the Table breaker in the PDU, to remove power from the entire table.
3.) Remove the Cradle Drive Cover.

Note: Pay attention to the location of the wires on the defective switch, before you remove them. Restore the wires to their original configuration on the replacement switch.

4.) Disconnect the wires from the limit switch terminals
5.) Remove the nut that fastens the switch in place.
6.) Remove the defective switch from its bracket.
7.) Install the replacement switch
    Make sure you connect the wires to the COM and N.O. terminals.
8.) Adjust the position of the switch in the bracket with the two nuts, so that it actuates when the cradle/carriage is at its maximum travel position, as determined by the Cal pin.
9.) Return the Cal pin to its storage position. Refit the Side Cover and Cradle Drive Cover.
10.) Reassemble the Table, and replace the covers.
11.) Turn on the Table breaker in the PDU to restore power.
12.) Adjust the position of the switch in the bracket:
    a.) Loosen the two nuts that fasten the bracket in place
    b.) Move the Cradle/carriage assembly to the maximum travel position, and fasten into position with the Cal pin.
    c.) Move the bracket until the switch actuates with the carriage in the maximum travel position.
13.) Return the Cal pin to its storage position.
14.) Reassemble the Table, and replace the covers.

15.) Turn on the Table breaker in the PDU to restore power.

2.36 46-278575P1 Cradle Drive Motor

1.) Raise the table to maximum height.
2.) Turn off the Table breaker in the PDU, to remove power from the entire table.
3.) Follow the procedure to remove the Cradle Drive Assembly, and set it on a work surface.
Cradle Drive Assembly procedure begins on page 803.
4.) Disconnect the motor wires from the terminal strips.
5.) Remove the four sets of screws and washers that fasten the motor to the cradle drive frame.
6.) Loosen the two set screws that fasten the pulley to the motor shaft, and slide the pulley off the shaft.
7.) Remove the defective motor.
8.) After you install the replacement motor:
    a.) Align the motor pulley with the drive roller pulley
    b.) Position one of the pulley set screws over the shaft flat.
    c.) Adjust the position of the motor until the belt deflects 0.050/0.060 inches when you apply 4 – 6 oz of pressure to the mid-span of the belt,
    d.) Maintain belt tension, while you tighten four screws, to fasten the motor in place.
9.) Reassemble the Table, and replace the covers.
10.) Turn on the Table breaker in the PDU to restore power.

2.37 46-297036G2 Longitudinal Encoder Pot Assembly

1.) Raise the table to maximum height.
2.) Remove the Cradle Drive Cover.
3.) Install the Cal pin at the home position, then remove the Cradle Assembly.
Cradle Assembly procedure begins on page 803.
4.) Turn off the Table breaker in the PDU, to remove power from the entire table.
5.) Remove the two lock nuts that fasten the Longitudinal Encoder Assembly Cover in place.
6.) Remove the two screws that fasten the Front End Cover in place.
7.) Remove the pot assembly:
    a.) Loosen the clamp that fastens the pot sprocket to the pot shaft.
    b.) Disconnect the Pot Assembly from the table harness, at J17.
    c.) Loosen the two servo clamps.
    d.) Slide the defective Pot Assembly downward, and off the sprocket.
    e.) Take care not to lose the plastic spacer on the pot shaft.
8.) After you install the replacement Pot Assembly:
    a.) Make sure the sprocket comes in contact with the plastic spacer.
    b.) Attach a DVM to terminals #2 and #1 (GND) of the pot.
    c.) Turn the pot shaft with a small screwdriver, until the DVM displays 0.80 ±0.01 VDC.
    d.) Maintain the voltage display, while you tighten the pot clamp.
9.) Return the Cal pin to its storage position.
10.) Reassemble the Table, and replace the covers.
11.) Turn on the Table breaker in the PDU to restore power.

2.38 46-296317P1 Quad Output Power Supply

1.) Raise the table to maximum height.
2.) Turn off the Table breaker in the PDU, to remove power from the entire table.

Note: Pay attention to the location of the wires on the defective supply, before you remove them. Restore the wires to their original configuration on the replacement supply.

3.) Remove the Table Drive Power Supply:
   a.) Locate the Quad Output PS. (lower power supply)
   b.) Disconnect the power input and out wires from the terminals
   c.) Locate, and remove the 2 screws that fasten the Quad Output PS to the right side of the Power Assembly main bracket.
   d.) Slide the defective power supply out of the assembly.
4.) Install the replacement Quad Output PS.
5.) Reassemble the Table, and replace the covers.
6.) Turn on the Table breaker in the PDU to restore power.

2.39 46-170053P11 Elevation and Cradle Amplifier Relay

1.) Raise the table to maximum height.
2.) Turn off the Table breaker in the PDU, to remove power from the entire table.
3.) Remove the left Base Covers.
4.) Loosen the captive screws, or remove the four screws, that fasten the servo amp cover in place.
5.) Remove the servo amp cover, and set aside.
6.) Slide the wire retainer to the side, and pull the defective relay from its socket.
7.) Install the replacement relay.
8.) Reassemble the Table, and replace the covers.
9.) Turn on the Table breaker in the PDU to restore power.

2.40 46-327096G1 Right Base Cover

1.) Raise the table to maximum height.
2.) Turn each of four captive Dzus fasteners one-quarter turn CCW, to release the cover.
3.) Turn each of four captive Dzus fasteners one-quarter turn CW, to fasten the cover.
   The screwdriver slot in the fastener appears horizontal when you successfully engage the fastener.

2.41 46-296909G1 Left Control Panel

1.) Raise the table to maximum height.
2.) Turn off the Table breaker in the PDU, to remove power from the entire table.
3.) Remove the Table Side Covers.
4.) Disconnect the ribbon cable(s) and J1 connector, from the control panel
5.) Remove the four sets of screws, washers, and spacers, and the two fish-paper spacers, that fasten the control panel to the table.
6.) After you install the replacement Control Panel and tighten the hardware, make sure the panel floats freely.

7.) Reassemble the Table, and replace the covers.
8.) Turn on the Table breaker in the PDU to restore power.

2.42 46-296909G2 Right Control Panel

1.) Raise the table to maximum height.
2.) Turn off the Table breaker in the PDU, to remove power from the entire table.
3.) Remove the Table Side Covers.
4.) Disconnect the ribbon cable(s) and J1 connector, from the control panel
5.) Remove the four sets of screws, washers, and spacers, and the two fish-paper spacers, that fasten the control panel to the table.
6.) After you install the replacement Control Panel and tighten the hardware, make sure the panel floats freely.
7.) Reassemble the Table, and replace the covers.
8.) Turn on the Table breaker in the PDU to restore power.

2.43 46-264832P1 ETC SCA-LAN Board

1.) Raise the table to maximum height.
2.) Remove the Base Covers.

NOTICE **Prevent permanent damage to the static-sensitive boards. Attach the anti-static wrist strap to your wrist and to a bare metal grounding point on the table before you continue.**

3.) Turn off the Table breaker in the PDU, to remove power from the entire table, or turn off the breaker switch at the rear of the table.
4.) Locate the ETC chassis assembly:
   a.) Loosen the captive screw that fastens the ETC chassis to the center of the table base.
   b.) Pivot the chassis panel until the Heurikon PWA reaches the horizontal position.
   c.) Disconnect the BNC-T connector from the SCA-LAN Board.
   d.) Loosen, and remove, the 3 screws that fasten the SCA-LAN Board to the ETC Board.
   e.) Lift the defective SCA-LAN board straight away from the ETC Board, to unplug its connector.
5.) Install the replacement SCA-LAN Board.
6.) Reassemble the Table, and replace the covers.
7.) Turn on the Table breaker in the PDU to restore power.

2.44 46-297059G1 Servo Amplifier Assembly

1.) Raise the table to maximum height.
2.) Turn off the Table breaker in the PDU, to remove power from the entire table.
3.) Remove the left Base Covers.
4.) Loosen the captive screws, or remove the four screws, that fasten the servo amp cover in place.
5.) Remove the servo amp cover, and set aside.
6.) Detach all the connectors from both servo amps.
7.) Loosen the four screws that fasten the servo amp bracket to the base frame.

8.) Slide the entire assembly out, and up, to remove it from the table.
9.) When you install the replacement Servo Amp Assembly, take care to reconnect all the connectors in their original configuration.
10.) Reassemble the Table, and replace the covers.
11.) Turn on the Table breaker in the PDU to restore power.

2.45 Table Side Cover

1.) Raise the table to maximum height.
2.) Turn each of two Dzus fasteners one-quarter turn CCW, and remove, to release the cover.
3.) Tip the bottom of the cover outward slightly, then lift the cover upward to clear the four pins in the z-channel.
4.) If you plan to use any of the table drives:
    a.) Remove the Tape Switch Jumper Plug from its storage position,
    b.) Plug the Tape Switch Jumper into the harness connector.
5.) When you install the Side Cover:
    a.) Return the jumper plug to its storage position.
    b.) The screwdriver slot in the fastener aligns with the Side Cover when you successfully engage the fastener.
6.) Test the Side Cover Tape Switch operation.
7.) Reassemble the Table, and replace the covers.

2.46 Table Side Panel

1.) Raise the table to maximum height.
2.) Remove the Table Base Covers and Table Side Covers.

Note: Pay attention to the orientation of the ground strap terminal, before you remove it. Orient the terminal in the same direction when you replace it.

3.) Remove the ground strap connection from the z-channel
4.) Each panel has two flat-head screws that fasten the Pivot Tube to its bracket.
    Remove, and keep, the flat head screws
5.) When you install the Side Panel, make sure the pivot points move without interference. If the pivot points cannot move freely:
    a.) Loosen the two screws that fasten the upper mounting bracket in place.
    b.) Slide the bracket in its slots, until the side panel pivot points move freely.
6.) Reassemble the Table and replace the covers.

2.47 46-296316P1 Table Drive Power Supply

1.) Raise the table to maximum height.
2.) Turn off the Table breaker in the PDU, to remove power from the entire table.
3.) Remove the Base Covers.
4.) Remove the power supply hold down bracket.
5.) Remove the screws attaching the outlet cover and move the cover aside to gain access to the power supplies.

6.) The Table Drive PS is the upper power supply. Disconnect the wires from both ends of the power supply.
7.) Remove the upper two attaching screws from the right side of the Power Assembly main bracket attaching the Table Drive PS, and slide the power supply out.
8.) To install the Table Drive PS, reverse above steps, making sure to connect the wires correctly.
9.) Refit the Base Covers.
10.) Reassemble the Table, and replace the covers.
11.) Turn on the Table breaker in the PDU to restore power.

2.48 46-297698P2 Table Side Cover Tape Switch

1.) Raise the table to maximum height.
2.) Turn off the Table breaker in the PDU, to remove power from the entire table.
3.) Remove the Table Side Covers.
4.) Remove the two shoulder screws that fasten the connector to the bracket in the Side Cover.
5.) Remove the terminals from the connector.
6.) Cut the ty-raps that fasten the Tape Switch wire to the four brackets.
7.) Remove the Tape Switch:
    a.) Peel the adhesive away from the cover.
    b.) Pull the wires through the holes in the cover.
    c.) Remove all the adhesive from the cover.
    d.) Thoroughly clean the mounting surface with alcohol.
8.) When you install the replacement Tape Switch:
    a.) Orient the shorter wires toward the connector bracket, when you route the wires through the holes in the cover.
    b.) Remove the protective strip from the adhesive.
    c.) Press the switch firmly into place.
    d.) Replace the side cover.
9.) Turn on the Table breaker in the PDU to restore power.
10.) Test the Tape Switch operation.

Section 3.0
Table Component Details

3.1 46-264370G1 Cradle Amplifier

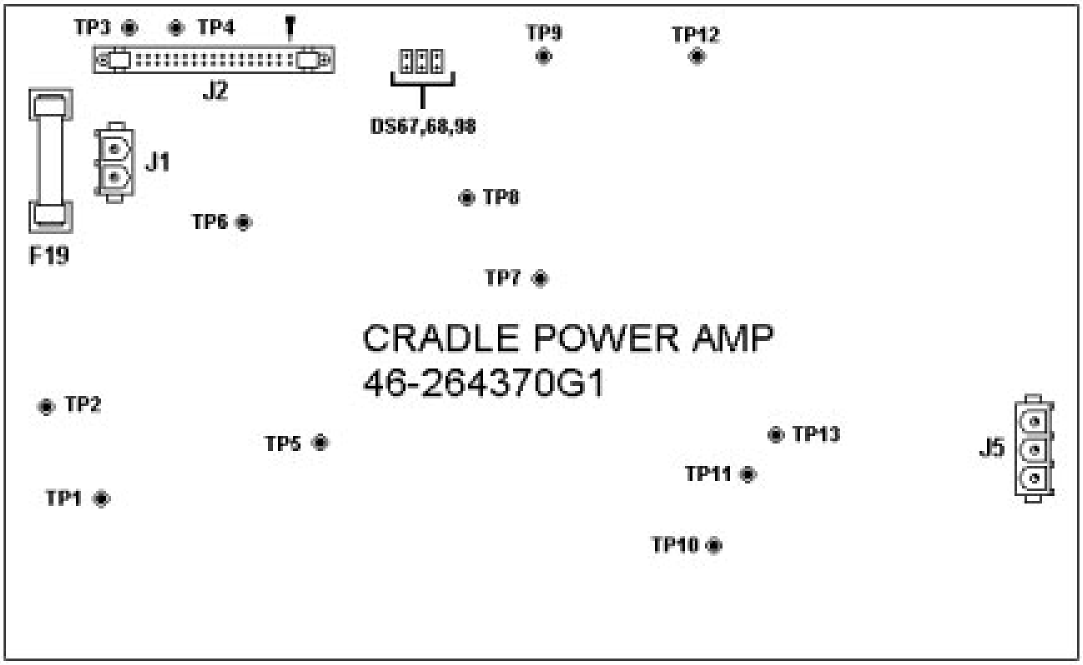


Figure 12-11 46-264370G1 Cradle Amplifier

3.1.1 Cradle Amplifier Board Test Points

- TP1 ISO3: High side of isolation supply for the lower FETs. (Above 14V)
- TP2 GND: Low side of the amplifier bridge. High power return.
- TP3 AGND: Analog ground.
- TP4 +24V: Supplied from the ETC board.
- TP5 ISO1: High side of the isolation supply for the flowing gate drive of FET Q55. Measured with respect to TP8.
- TP6 CRADLE PULSE-IL*1: Goes low when the amplifier bridge current is at or above 6A nominal
- TP7: Output of the push-pull driver of the isolation transformer for the DC-DC converters.
- TP8 RTN1: Low side of the isolation supply for Q55 FET drive.
- TP9 PGND: 24V return.
- TP10 HV: Positive side of the amplifier bridge. This is nominally +24V.
- TP11 RTN2: Low side of the isolation supply for Q130 FET drive.
- TP12 VMTR: The scaled, filtered and buffered cradle motor voltage.
- TP13 ISO2: Positive side of the isolation supply for Q130 FET drive, measured with respect to TP8.

3.1.2 Cradle Amplifier Board LEDs

LEDs exist to give a "feel" of the state the circuitry on the amplifier board. The following LEDs receive pulses at various duty cycles (with the exception of DS PS) so they vary in intensity from a fully on LED.

- DS67: On when the lower FET is commanded off. This LED indicates the presence of logic drive to the isolated FET drive circuitry, not the state of the Q14's gate direction.
- DS68: On when +24V is present on the amplifier board. If +24V is not present, the DC-DC converter will not work, the relay cannot operate and there isn't any drive voltage present at the bridge.
- DS98: On when the lower FET is commanded off. This LED indicates the presence of logic drive to the isolated FET drive circuitry, not the state of the Q155's gate directly.
- DS124: On when DCCLK pulses are present at the amplifier. These pulses are needed for the DC-DC converter to operate.

3.1.3 Cradle Amplifier Board Switch Settings

None

3.2 ETC Heurikon CPU Board

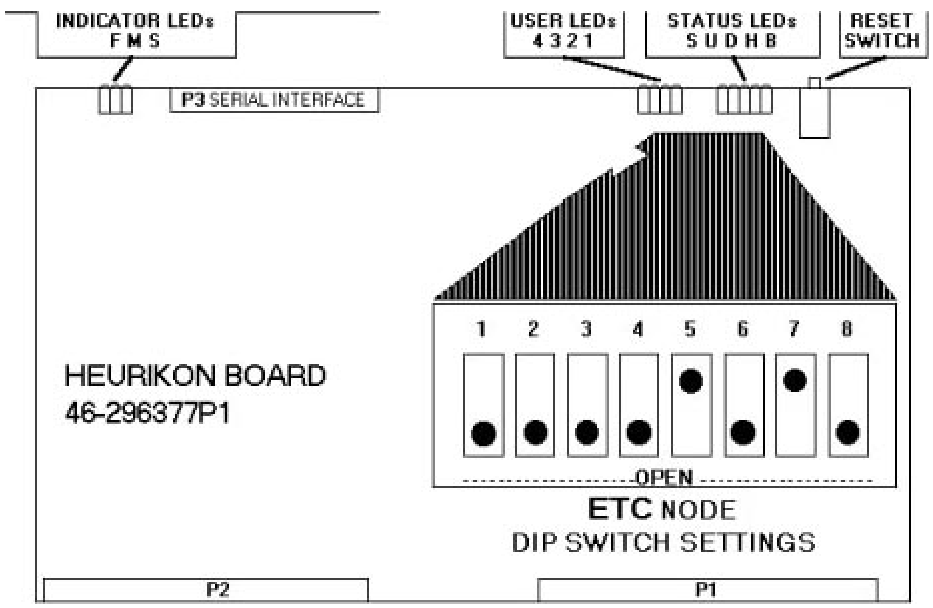


Figure 12-12 46-296377 Heurikon CPU ETC

3.2.1 Test Points

No serviceable test points.

3.2.2 Heurikon CPU Board LEDs

Indicator LEDs

- S = Slave
- M = Master
- F = Fail

User LEDs

- 1 = User LED1 - MSB
- 2 = User LED2
- 3 = User LED3
- 4 = User LED4 - LSB

The Heurikon CPU self test section contains additional node specific troubleshooting information for the User LEDs (see ETC, STC & OBC “Heurikon” CPU - Power-up Self Tests on page 223 of Chapter 6).

3.2.3 Status LEDs

- B = Bus: Another VMEBus master has control of the bus.
- H = Halt: The MPU has halted.
- D = DMAC: The DMAC has control of the local bus.
- U = User: The MPU is in the user state.
- S = Super: The MPU is in the supervisor state.

3.2.4 Heurikon CPU Board Switch Settings

Reset Switch – Resets the CPU board and initiates the self test.

3.2.5 Configuration DIP Switch

Used to configure the CPU board to a specific node. The correct setting of this DIP switch can be found by selecting the Board Layout button below.

Additional information is available in the Heurikon CPU self test section (see ETC, STC & OBC “Heurikon” CPU - Power-up Self Tests on page 223 of Chapter 6).

3.3 46-288170G1 Elevation/Tilt Amplifier

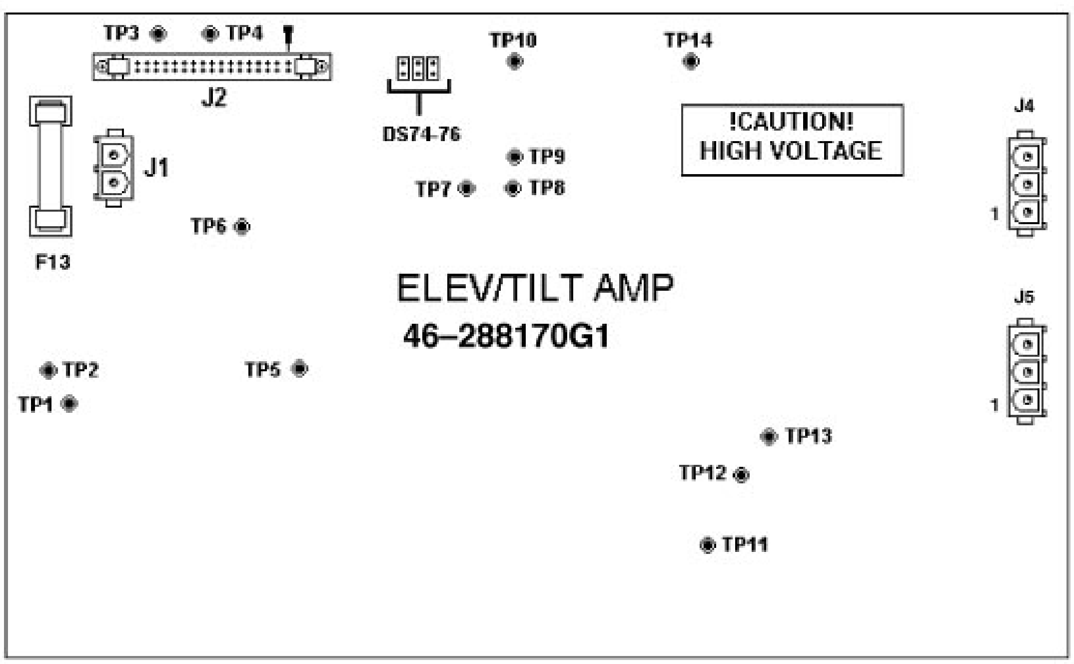


Figure 12-13 46-288170G1 Elevation/Tilt Amplifier

3.3.1 Elevation/Tilt Amplifier Board LEDs

LEDs exist to give a “feel” for the state of the circuitry on the amplifier board. All of the following LEDs pulse at various duty cycles (with the exception of DS PS) so they vary in intensity from a fully on LED.

- DS74: On when the lower FET (Q20) is commanded off. This LED indicates the presence of logic drive to the isolated FET drive circuitry, not the state of the Q20's gate direction.
- DS75: On when the lower FET (Q150) is commanded off. This LED indicates the presence of logic drive to the isolated FET drive circuitry, not the state of the Q150's gate direction.
- DS76: On when +24V is present on the amplifier board. If +24V is not present, the DC-DC converter will not work, the relay cannot operate and there isn't any drive voltage present of the bridge.
- DS125: ON when DCCLK pulses are present at the amplifier. These pulses are needed for the DC-DC converter to operate.

3.3.2 Elevation/Tilt Amplifier Board Switch Settings

None

3.3.3 Elevation/Tilt Amplifier Board Test Points

- TP1 ISO3: Positive side of the isolation supply for the lower FETs of the “H-Bridge” Q20 and Q150.
- TP2 GND: Low side of the amplifier bridge. It also acts as ISO3 RTN, This is the high voltage supply return.
- TP3 AGND: Analog ground.
- TP4 +24V: 24 volts for the relays and the DC-DC converter.
- TP5 ISO1: Positive side of the isolation supply for Q55 FET drive, measured with respect to TP9.
- TP6 P-IL: Elev/Tilt Pulse-IL*1- Goes low when the amplifier bridge is at or above 6A (nominal).
- TP7 SCKT: Short Circuit signal. This goes low when the bridge current goes above 12A (nominal).
- TP8: This test point is the output of the push-pull driver which drives the DC-DC converters' isolation transformer primaries.
- TP9 RTN1: Low side of the isolation supply for Q55 FET drive.
- TP10 PGND: This is the +24V return.
- TP11 HV: This is the high side of the amplifier bridge. This is at about +170V.
- TP12 RTN2: Low side of the isolation supply for Q130 FET drive.
- TP13 ISO2: Positive side of the isolation supply for Q130 FET drive, measured with respect to TP12.
- TP14 VMTR: The scaled, filtered and buffered E/T motor voltage.

Section 4.0 Troubleshooting - Table Velocity Errors

4.1 Problem

Occasionally, HiSpeed Advantage installed base owners have reported cradle velocity errors.This occurs while driving into the gantry, and with the cradle loaded down by a patient. There has also been reports of a potentiometer to encoder correlation error, but this error is more likely caused by a problem with the longitudinal encoder assembly, specifically the pot. or pot. drive belt and sprockets.

The most likely cause for the velocity error is an out-of-adjustment clutch on the cradle drive assembly. This clutch is adjusted to slip when 36–39 pounds is exerted horizontally on the cradle while driving into the gantry. When the clutch slips, the velocity of the cradle will be far enough out of normal range to trigger an error, which stops the drive. Ideally, this would not occur within the normal operating range of less than 36 pounds. However, when the clutch is out of adjustment it will slip at lower drive forces that are within the normal range of operation. A one–direction roller-clutch, inside the clutch assembly, prevents any slipping when driving out of the gantry.

Although traction problems between the drive roller and cradle could exist, they are unlikely due to the rough bottom surface of the cradle, and due to the weight of the patient maintaining the contact between the cradle and roller. Another unlikely cause would be roller smoothness; the harder cradle surface is intentionally molded with a rough surface, which slightly distorts the roller's softer rubber surface, creating the high coefficient of friction.Generally, traction problems only occur when there is no patient weight to keep the cradle in contact with the roller. In this case, the shimming between the cradle and the carriage should be reviewed.

4.2 Solution

During the manufacturing of the clutch friction discs, a burr on the inside diameter of the disc was created, that relaxes after a period of time, causing the clutch to go out of adjustment after leaving the factory. One of three courses of action car be followed, depending on the amount of time available for repair, availability of new parts, and availability of a force gauge:

- The existing clutch can be adjusted. This is the quickest procedure, since it does not require the cradle drive to be removed from the table. However, this is a two-person procedure, and requires a force gauge. Also, since the burrs have not been removed, the adjustment may not be maintained for a long period of time.
- The entire clutch can be removed and replaced with a new pre-adjusted unit, FRU 46-296368G1. This is the next quickest procedure, but requires a new, factory–adjusted clutch. The force gauge is optional, but the cradle drive must be removed from the table.
- The clutch can be disassembled, the burrs removed, and then re–assembled and adjusted. This is the most time consuming procedure, but does not require a new clutch. However, this is a two-person procedure, and requires a force gauge, and cradle drive removal.

4.3 Tools Required - For Clutch Adjustment

4.3.1 Clutch Adjustment

- Push force gauge, 0-50# or 0-100# (P/N 46-308109P2)
- 5/64” and 1/8” hex key wrenches
- 1 1/4” open-end wrench or channel-lock pliers with this capacity
- Loctite 242

4.3.2 Clutch Replacement

- 1/8” hex key wrench
- Loctite 242
- Torque wrench, 0-100 in-lbs
- Push force gauge, 0-50# or 0-100# (OPTIONAL)

4.3.3 Clutch Repair

- Push force gauge, 0-50# or 0-100#
- 5/64” and 1/8” hex key wrenches
- 1-1/4” open-end wrench or channel-lock pliers with this capacity.
- Loctite 242
- Sandpaper, 220 grit

4.3.4 Procedures

Figure 12-14 is provided on the last page as a reference drawing for the clutch assembly. Please review Figure 12-14 to familiarize yourself with the various parts of the clutch assembly before beginning any procedure.

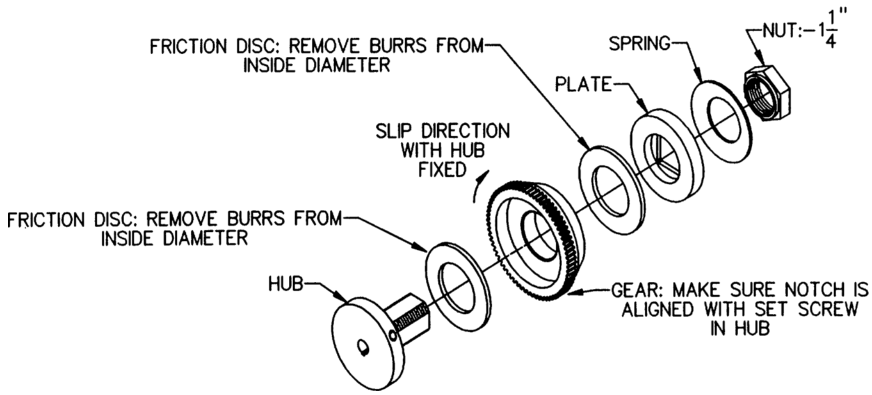


Figure 12-14 Table Clutch Assembly

4.3.5 Clutch Adjustment

1.) Place at least 100# on the cradle, toward the gantry end.

2.) Remove the cradle drive cover from the bottom of the table.

3.) Locate the clutch on the left end of the drive roller. Loosen the two set screws securing the 11/4" hex nut with the 5/64" hex wrench. If necessary, release and move the cradle to rotate the drive roller and clutch to gain access to the set screws.

4.) Position the cradle about 3 feet from home. Tighten the hex nut a small amount (1/4 flat), and then measure the driving force into the gantry with the force gauge. Drive the cradle with the table-side controls at the fast speed, while the FE reaches through the gantry with the force gauge pushing on the end of the cradle. Push hard enough for the clutch to slip, and note the reading on the gauge. Insure that the drive roller is stationary (i.e., not slipping on the cradle bottom), and that the end of the clutch (i.e., hex nut) is rotating when the measurement is taken. If the roller is slipping on the cradle, then add more weight to the cradle.

5.) For proper adjustment, the gauge reading should be as close to 40# as possible, but must **not** exceed 40#. An ideal range is 36-39#. Repeat 4.) until the correct force is measured. Loctite and tighten the set screws and verify the reading again.

6.) A check must now be made to see if the cradle releasing solenoid and gear rack are properly adjusted. Removing the cradle will make this check easier to perform, and more accurate; follow the procedure in section 2.12 for removing the cradle. When the solenoid is energized, the gear rack is engaged in the clutch gear, and allows the cradle to be driven. The engagement of the rack in the gear must not have any backlash, nor can the solenoid plunger be excessively extended out of the solenoid body. When correctly set, the solenoid plunger will be within 0.010" of bottoming-out in the body, when there is no backlash at the rack/gear interface. Adjust the solenoid bracket so that the plunger is bottomed when the solenoid is energized, and then move the bracket forward (toward the gantry) until there is no backlash between the rack and gear, as checked at four, 90 degree apart, positions on the gear. The solenoid should be energized so that all the looseness is removed from the linkage; if energizing is not possible, be sure to push on the plunger itself (not the pin or link) when checking the adjustment.

7.) Refit the cradle drive cover.

4.3.6 Clutch Replacement (FRU 46-296368G1)

1.) Follow the procedure in section 2.13 and section 2.14 for removing the cradle drive from the table.

2.) On the left side of the cradle drive assembly, mark the position of the solenoid bracket. Remove the two screws holding the solenoid bracket, and withdraw the solenoid plunger and remove the washer and spring. Allow the plunger and arm to hang from the gear rack. To avoid unplugging the solenoid or damaging the wires, re-attach the solenoid bracket to the cradle drive.

3.) In order to remove the clutch from the drive roller shaft, the set screw in the clutch hub must be loosened. The set screw can only be reached when a notch in the face of the gear is aligned with the set screw. Loosen the two set screws securing the 1¼” hex nut with the 5/64” hex wrench. If necessary, release and move the cradle to rotate the drive roller and clutch to gain access to the set screws. Now loosen the hex nut until the gear can be easily rotated (in one direction only) on the clutch. Align the notch in the gear with the set screw in the hub.

4.) Loosen the set screw with the 1/8” hex wrench, and slide the entire clutch off the shaft. Be sure to keep the key.

5.) The new clutch should have the set screw already aligned with the notch. (DO NOT LOOSEN THE TWO SET SCREWS OR MOVE THE HEX NUT; the slip force setting will be lost, and the “CLUTCH ADJUSTMENT PROCEDURE” above will have to be performed.) Make sure the key is fully seated in the shaft, and then slide the clutch onto the shaft. The clutch is correctly positioned when the end of the shaft is flush with the clutch hub. Loctite and tighten the set screw to 65 in-lbs.

6.) Reassemble the solenoid bracket, plunger, washer and spring. Position the bracket at the location marked during disassembly. The solenoid bracket will be further adjusted in Step 9.).

7.) Refit the cradle drive assembly according to the procedure in section 2.13 and section 2.14.

8.) If a force gauge is available, check, and adjust if needed, the slip force using the technique and values in step 4.) and 5.) in the “CLUTCH ADJUSTMENT PROCEDURE” above.

9.) A check must now be made to see if the cradle releasing solenoid and gear rack are properly adjusted. Removing the cradle will make this check easier to perform, and more accurate; follow the procedure in section 2.12 for removing the cradle. When the solenoid is energized, the gear rack is engaged in the clutch gear, and allows the cradle to be driven. The engagement of the rack in the gear must not have any backlash, nor can the solenoid plunger be excessively extended out of the solenoid body. When correctly set, the solenoid plunger will be within 0.010" of bottoming-out in the body, when there is no backlash at the rack/gear interface. Adjust the solenoid bracket so that the plunger is bottomed when the solenoid is energized, and then move the bracket forward (toward the gantry) until there is no backlash between the rack and gear, as checked at four, 90 degree apart, positions on the gear. The solenoid should be energized so that all the looseness is removed from the linkage; if energizing is not possible, be sure to push on the plunger itself (not the pin or link) when checking the adjustment.

10.) Refit the cradle drive cover.

4.3.7 Clutch Repair

1.) Follow the procedure in section 2.13 and section 2.14 for removing the cradle drive from the table.

2.) Locate the clutch on the left end of the drive roller. Loosen the two set screws securing the 11/4” hex nut with the 5/64” hex wrench. Remove the hex nut.

3.) from the clutch, along with the spring washer, hub plate, friction washer, gear with one-way bearing, and the second friction washer. Do not remove the clutch hub itself.

4.) Inspect the inside diameter of both friction washers for burrs. Remove any burrs with the sandpaper. Clean the dust and particles from the washers and then re-assemble in reverse order; hand tighten the hex nut. Note that the friction washers are centered by the roller clutch that is pressed into the gear.

5.) Refit the cradle drive assembly according to the procedure in section 2.13 and section 2.14.

6.) Position the cradle about 3 feet from home. Tighten the hex nut a small amount (1/4 flat), and then measure the driving force into the gantry with the force gauge. Drive the cradle with the table-side controls at the fast speed, while the FE reaches through the gantry with the force gauge pushing on the end of the cradle. Push hard enough for the clutch to slip, and note the reading on the gauge. Insure that the drive roller is stationary (i.e., not slipping on the cradle bottom), and that the end of the clutch (i.e., hex nut) is rotating when the measurement is taken. If the roller is slipping on the cradle, then add more weight to the cradle.

7.) For proper adjustment, the gauge reading should be as close to 40# as possible, but must **not** exceed 40#. An ideal range is 36-39#. Repeat Step 4) until the correct force is measured. Loctite and tighten the set screws and verify the reading again.

8.) A check must now be made to see if the cradle releasing solenoid and gear rack are properly adjusted. Removing the cradle will make this check easier to perform, and more accurate; follow the procedure in section 2.12 for removing the cradle. When the solenoid is energized, the gear rack is engaged in the clutch gear, and allows the cradle to be driven. The engagement of the rack in the gear must not have any backlash, nor can the solenoid plunger be excessively extended out of the solenoid body. When correctly set, the solenoid plunger will be within 0.010" of bottoming-out in the body, when there is no backlash at the rack/gear interface. Adjust the solenoid bracket so that the plunger is bottomed when the solenoid is energized, and then move the bracket forward (toward the gantry) until there is no backlash between the rack and gear, as checked at four, 90 degree apart, positions on the gear. The solenoid should be energized so that all the looseness is removed from the linkage; if energizing is not possible, be sure to push on the plunger itself (not the pin or link) when checking the adjustment.

9.) Refit the cradle drive cover.

Chapter 13
Power Distribution Unit (PDU)

Section 1.0
PDU Safety Warning

DANGER **EXERCISE EXTREME CARE WHEN SERVICING THE PDU. 480 VAC AND OTHER LETHAL VOLTAGES ARE PRESENT AT VARIOUS POINTS WITHIN THE PDU AT ALL TIMES. THEREFORE, CONSIDER ALL POINTS WITHIN THE PDU AS ELECTRICALLY HAZARDOUS. DO NOT PERFORM ANY WORK IN THE PDU UNLESS IT IS DE-ENERGIZED. FAILURE TO HEED THIS WARNING MAY RESULT IN PERSONAL INJURY OR DEATH.**

IN ADDITION, DO NOT WORK ON THE PDU UNTIL YOU HAVE EITHER READ THE DIRECTION 212152915-100, HSA CT/I SAFETY GUIDELINES MANUAL OR HAVE VIEWED THE CT HISPEED ADVANTAGE SAFETY VIDEO. FAILURE TO HEED THIS WARNING MAY RESULT IN PERSONAL INJURY OR DEATH.

Section 2.0
PDU Replacement Verification and Re-Test

PDU COMPONENTS	TASK	VERIFICATION TEST
DCRGS (present on older systems)		Perform System Scanning Test, (page 69)
Complete PDU		1.) System continuity and ground checks per the HiSpeed CT/i Installation Manual, Direction 2136597-100 2.) Perform System Scanning Test, (page 69). (Verify room warning light function while doing the System Scanning Test)
Gantry Servo Amplifier	Refer to page 855	Perform System Scanning Test, (page 69)
Axial Interface Board	Replace board page 840	Perform System Scanning Test, (page 69)

Table 13-1 PDU Re-Test Matrix

PDU COMPONENTS	TASK	VERIFICATION TEST
Relay Control Board	Replace faulty board, page 863	1.) Test Service Switch Box Functions by switching state of switches in the box and verifying. 2.) Perform System Scanning Test, (page 69). (Verify room warning light function while doing the System Scanning Test)
Unregulated HVDC Supply Bridge Rectifier, Capacitors, Inductor	Replace Faulty Component	Perform System Scanning Test, (page 69)

Table 13-1 PDU Re-Test Matrix (Continued)

Section 3.0 Component Locations

3.1 GPDU (Model 2113764) Major Component Locations

Figure 13-1 shows the GPDU (Global Power Distribution Unit) component area designators. This manual uses the component designators as abbreviations for the various components. For example, we refer to the Circuit Breaker Panel as the “A3” panel.

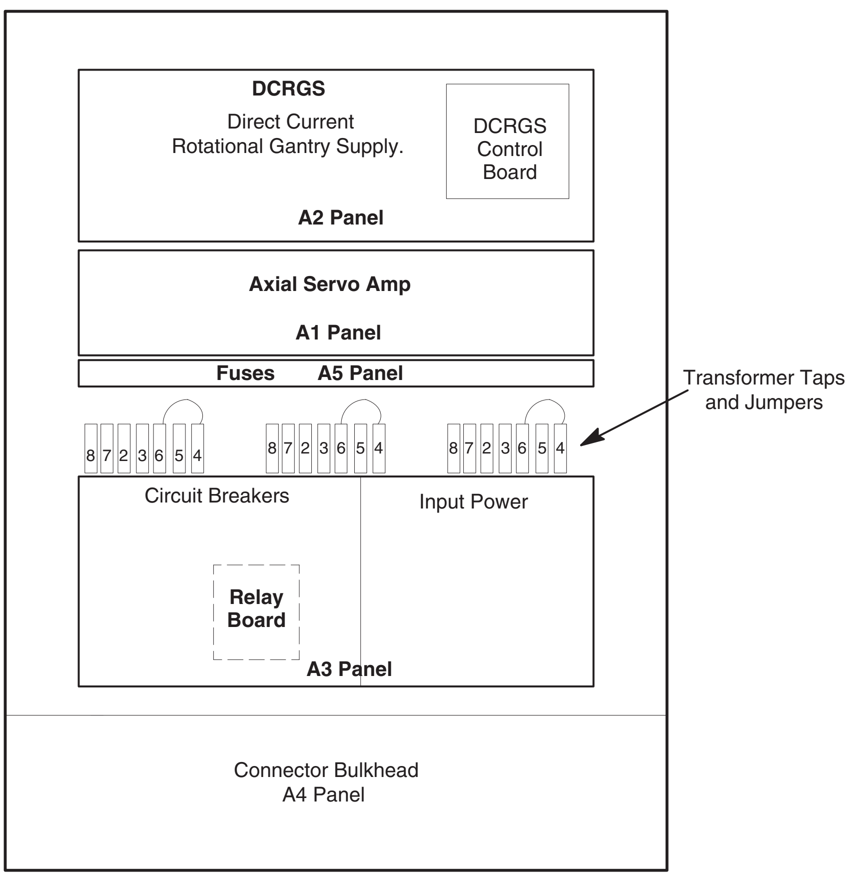


Figure 13-1 GPDU Area Designators

3.2 CPDU (Model 2133533) Major Component Locations

Figure 13-2 shows the CPDU (Compact Power Distribution Unit) component area designators. This manual uses the component designators as abbreviations for the various components. For example, we refer to the Circuit Breaker Panel as the “A3” panel.

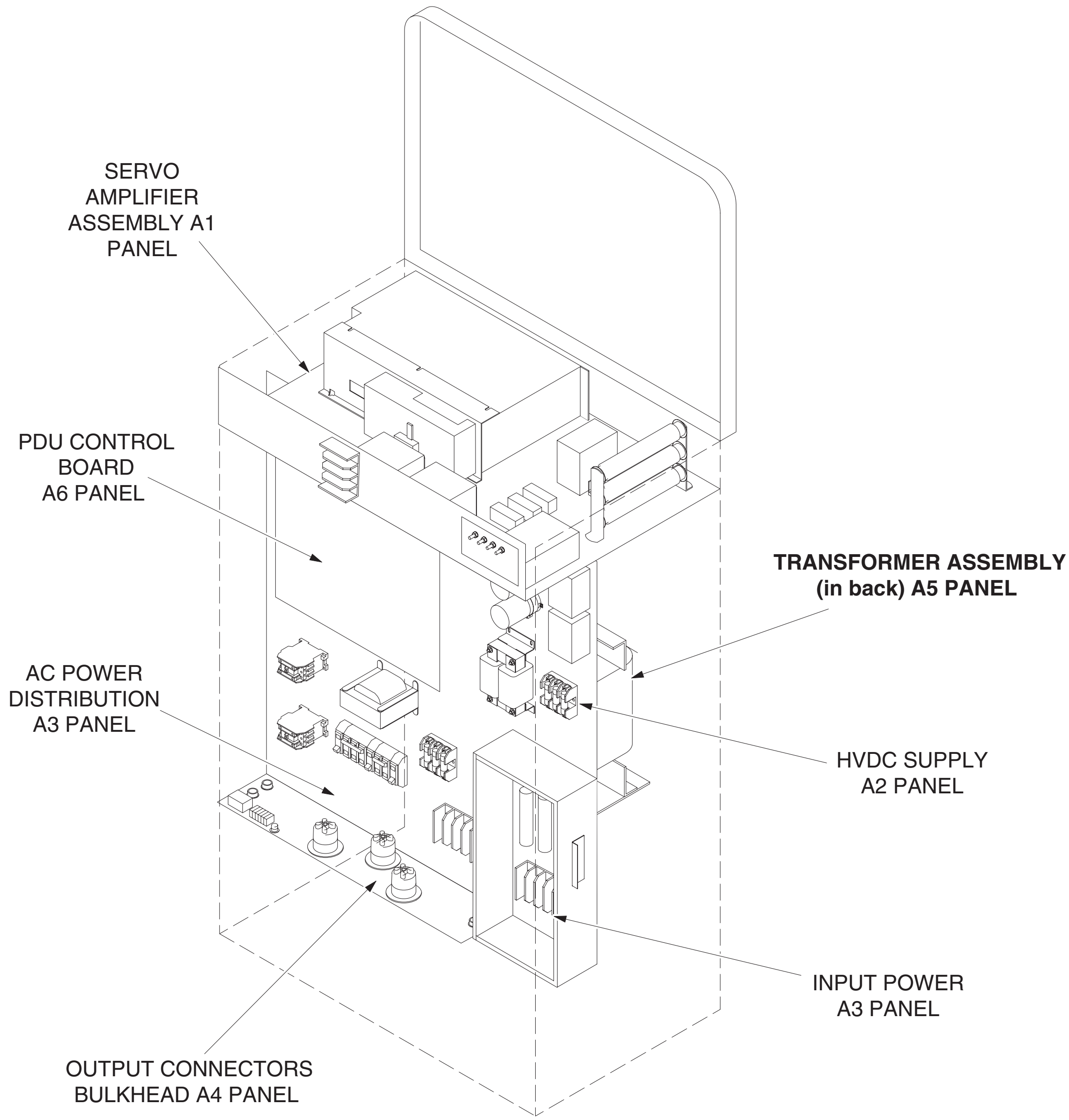


Figure 13-2 CT CPDU Area Designators

Section 4.0 Ground Bus Continuity Check (GPDU Model 2113764)

If problems arise which point to a possible system grounding problem, perform a ground continuity check as follows:

1.) Remove front cover from PDU A3 (circuit breaker) panel (Figure 13-1).

2.) Remove ribbon cable connector from DCRGS.
3.) Verify **less than 1 ohm of resistance** exists between the following ground connections:
   - Wall ground connection and PDU cabinet
   - PDU A2 chassis to the PDU cabinet
   - PDU A3-TS5 busbar to the PDU cabinet
   - PDU A4-TS3 busbar to the PDU cabinet
   - PDU A5-TS1 busbar to the PDU cabinet
4.) Reconnect DCRGS ribbon cable connector
5.) Leave the PDU A3 cover (circuit breaker) panel off for now.

Section 5.0
PDU Component Details

5.1 Line Transformer Settings

Use this section to verify or change transformer settings.

WARNING **TURN OFF, TAG AND LOCK MAIN WALL POWER BEFORE CHANGING TAPS. FAILURE TO DISCONNECT POWER AT MAIN INPUT MAY RESULT IN ELECTROCUTION**

CAUTION **Do not touch meter or leads when wall power is on.**

1.) Use a 0 – 750 AC voltmeter of 3/4% accuracy to measure the line-to-line voltages at A5PT4, PT5 and PT6.
   - Verify the highest line-to-line voltage does not exceed 1.02 times the lowest voltage.
   - **Example:** IF the lowest voltage equals 474, the highest voltage should not exceed 474 x 1.02 = 483.5 volts.
2.) **CPDU 213353:** Refer to Table 13-2, and Figure 13-3.
   Determine the nearest nominal line, and verify the tap connections match.
3.) **GPDU 2113764:** Refer to Table 13-3, and Figure 13-1.
   Determine the nearest nominal line, and verify the tap connections match.
4.) Verify that the No Load Line to Line Voltage never falls outside the corresponding minimum and maximum values listed in Table 13-2 and Table 13-3.

No Load Line to Line Voltages		Tap Connections
Nominal	Maximum Range (± 8%)	Phase A, B, C Connections
480V	442 to 518	3 – 4
460V	423 to 497	3 – 5
440V	405 to 475	3 – 6
420V	386 to 454	2 – 4
400V	368 to 432	2 – 5
380V	350 to 410	2 – 6

Table 13-2 PDU 2133533 LINE TAP CONNECTION TABLE

No Load Line to Line Voltages		Tap Connections (All 3 phases must be same configuration)		
Nominal	Maximum Range (±8%)	Phase A Connection	Phase B Connection	Phase C Connection
500V	460 to 540	4 – 5	4 – 5	4 – 5
480V	442 to 518	4 – 6	4 – 6	4 – 6
460V	423 to 497	3 – 5	3 – 5	3 – 5
440V	405 to 475	3 – 6	3 – 6	3 – 6
420V	386 to 454	2 – 6	2 – 6	2 – 6
400V	368 to 432	3 – 7	3 – 7	3 – 7
380V	350 to 410	2 – 7	2 – 7	2 – 7
360V	331 to 389	2 – 8	2 – 8	2 – 8

Table 13-3 PDU 2113764 LINE TAP CONNECTION TABLE

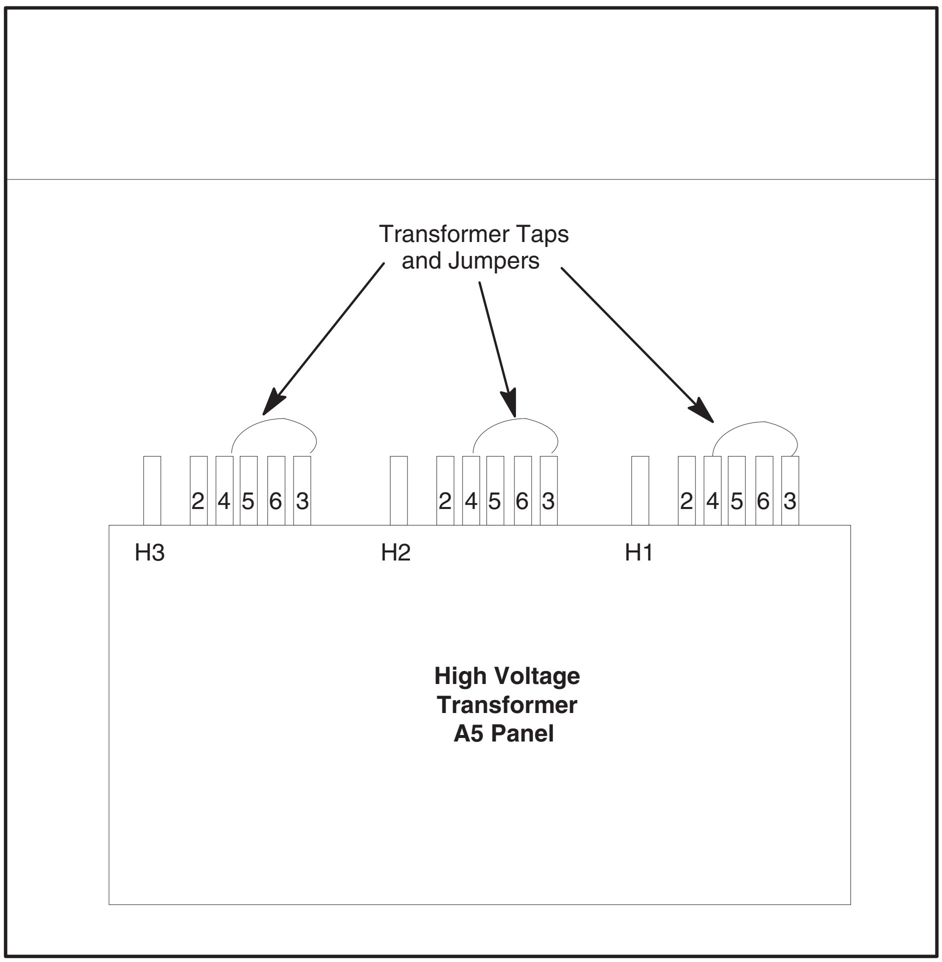


Figure 13-3 PDU 2133533 tap positions (Rear)

5.2 Westamp Servo Amp Hardware

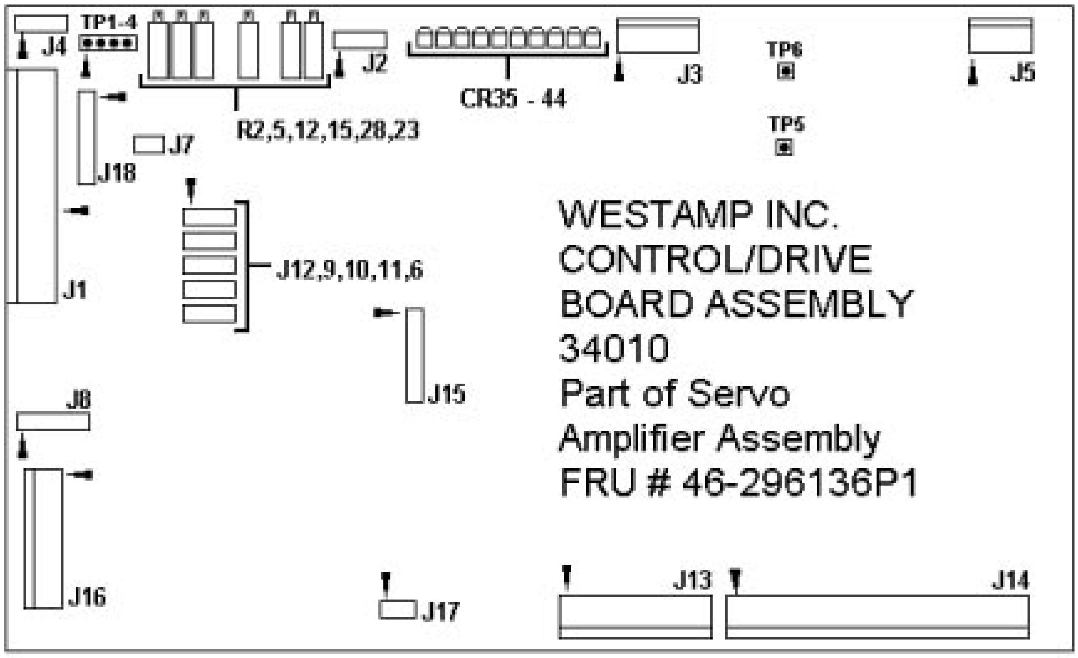


Figure 13-4 Westamp Servo Amp

5.2.1 Westamp Servo Amp Internal Test Points (Non-Accessible)

- TP1: Tachometer input
- TP2: Signal input
- TP3: Output of velocity control amplifier
- TP4: Output of current error amplifier
- TP5: Base drive oscillator
- TP6: Base drive oscillator

5.2.2 Westamp Servo Amp LEDs

- CR35: NO FAULT
- CR36 REM LIM:
  - J1-8, remote enable/disable (non-latching)
  - J1-6, A limit switch (non-latching)
  - J1-7, B limit switch (non-latching).
- CR37 TEMP: Indicates excessive ambient temperature.
- CR38 OV VOLT: Bus over-voltage.
- CR39 RMS: Indicates excessive current draw.
- CR40 LOGIC: Indicates ±12 VDC logic supply.
- CR41 GND FAULT: Indicates output short to ground.
- CR42 SURGE: Indicates excessive transistor current or shorted outputs.
- CR43: PWR ON
- CR44 BUS: Indicates DC bus power, even with logic power off.

5.3 Axial Servo Amp Control Board – Allen Bradley Servo Amp

5.3.1 LEDs – Allen Bradley Servo Amp

5.3.1.1 Current Foldback

The red current foldback LED illuminates during current foldback circuitry operation, to indicate the circuit exceeded the time vs. current overload rating of the power transistors. The intensity of the LED varies in proportion to the amount of overload.

Current Foldback LED illuminates when:

1.) The logic supply (±12 VDC, +5V) circuits have malfunctioned (fuse blown, etc.) or the AC input at A3TB1-9,10,11 is incorrectly wired.
2.) The output current exceeds its time-current rating.
   a.) The acceleration/deceleration command to the servo requires peak current for an excessive amount of time.
   b.) The gain pot is set too high, causing excessive peak currents.
   c.) The machine friction, inertial load and/or viscous loading is excessive.
   d.) A short circuit exists across the servo output terminals.
   e.) Excessive current ripple, caused by lower than minimum load inductance.

5.3.1.2 Enable (EN)

The application of an Enable signal by the system illuminates the Green enable LED.

ENABLE LED EXTINGUISHES WHEN:

1.) The system has not enabled the servo.
2.) The enable wiring to the servo is open.
3.) The axial interface board enable relay has malfunctioned.
4.) The system has detected a system malfunction that will not allow the servo to be enabled.
5.) Power has not been applied to the servo.
6.) The logic supply (±12 VDC, +5 VDC) circuits have malfunctioned (fuse blown, etc.) or the AC input at A3TB 1-9,10,11 is incorrectly wired.

ENABLE LED ILLUMINATES, BUT SERVO DOES NOT ENABLE WHEN:

1.) A servo malfunction has occurred but is not annunciated by the LED indicators. Check the status of the fault contact output.
2.) A component malfunction exists in the enable circuit.
3.) The servo circuit breaker (CB1) is tripped.
4.) Servo fan/contactor fuse has blown. Verify that the fan operates when you apply power.
5.) The a line/DB contactor malfunction occurred inside the servo.
6.) The servo logic supplies are not operational
   a.) The logic supply fuses are blown.
   b.) Logic supply AC voltage is missing
7.) UV LED illuminates

   Enable signal exists prior to power application to the power transformer (wait at least 1 second after the applying the 120 VAC, three-phase power to enable the servo). Also wait at least 1 second after a servo reset cycle to enable the servo.

5.3.1.3 Motor Overload (MOD)

The red motor overload LED illuminates when the circuit exceeds the motor overload detection circuit trip point. IMPORTANT: The overload indication may take several minutes to reset. If cycling the power does not reset the overload indication, remove power from the controller for about 3 minutes, then retry.

Motor Overload LED illuminate when:

1.) The logic supply (±12 VDC, +5VDC) circuits malfunction (fuse blown, etc.) or the AC input at A3TB1-9,10,11 is incorrectly wired.
2.) The circuit exceeds the continuous rating of the servomotor.
    a.) The programmed duty cycle requires an RMS torque that exceeds the motor rating.
    b.) The machine frictional force or alignment incorrectly set.
    c.) A mechanical limit, or stop, on the machine causes the motor to develop excessive torque.
    d.) You set the servo peak current pot higher than the motor rating.
    e.) The motor is partially demagnetized, causing excessive current draw.

5.3.1.4 Over Temperature (OT)

The red LED illuminates when the heatsink overtemperature device, inside the servo, trips.

Over temperature LED illuminates when:

1.) The logic supply (±12 VDC, +5VDC) circuits malfunction (fuse blown, etc.) or the AC input at A3TB1-9,10,11 is incorrectly wired.
2.) The heatsink thermal overload trips. If the surface temperature of the exposed heat sink adjacent to the ground stud measures more than 90 degrees Celsius, check for the following conditions:
    a.) Cabinet ambient temperature too high
    b.) Machine duty cycle RMS current requirements exceed the continuous rating of the servo.
    c.) Integral fan failure
    d.) Limited, or blocked, airflow access to the servo.
    e.) The input line voltage exceeds the maximum controller input voltage.

5.3.1.5 Overvoltage (OV)

This red LED illuminates when the DC power bus exceeds 265VDC.

Overvoltage LED illuminate when:

1.) The logic supply (±12 VDC, +5VDC) circuits malfunction (fuse blown, etc.) or the AC input at A3TB1-9,10,11 is incorrectly wired.
2.) Check shunt regulator resistors fuse on the servo panel.
3.) Motor armature shorts to ground.
4.) The power bus voltage exceeds 265VDC
    a.) The logic or shunt regulator board malfunctions, and incorrectly senses the bus voltage.
    b.) The input line voltage exceeds the maximum controller input voltage.
    c.) Shunt regulator board or transistor malfunction.

5.3.1.6 Transistor Overcurrent (TOC)

This red LED illuminates when the circuit exceeds the peak current rating of the power transistor(s).

Transistor overcurrent LED illuminates when:

1.) The logic control board malfunctions.

2.) The transistor base driver board fails.
3.) The logic control board common is not properly grounded.
4.) The logic supply (±12VDC, +5VDC) circuits malfunction (fuse blown, etc.) or the AC input at A3TB1-9,10,11 is incorrectly wired.
5.) The circuit exceeds the peak rating of the power transistors.
   a.) Grounded, or short circuited, servo output or servomotor
   b.) Power transistor module malfunction.

5.3.1.7 Undervoltage (UV)

This red LED illuminates when the DC power bus drops below 75 VDC (±20%) and the servo is enabled, or the logic supplies drop 10% below their nominal value.

Undervoltage LED illuminates when:

1.) The power bus voltage drops below 75 VDC.
   a.) The line/DB contactor (inside the servo) fails to energize, or drops out.
   b.) The line/DB contactor fuse opens.
   c.) The input line voltage drops below the minimum.
   d.) The power bus capacitor malfunctions.
   e.) The servo circuit breaker (CB1) trips.
   f.) The three-phase input line opens.
   g.) The transformer malfunctions, or provides incorrect line voltage.
2.) The logic supplies drop 10% below their nominal value.
   a.) The input line voltage drops below the minimum.
   b.) The transformer auxiliary logic supply winding opens.
   c.) The logic supply (±12 VDC, +5VDC) circuits malfunction (fuse blown, etc.) or the AC input at A3TB1-9,10,11 is incorrectly wired.
3.) The enable LED also illuminates, when:
   a.) The enable signal exists prior to power application to the transformer (wait at least 1 second after applying the 120 VAC, three-phase power to enable the controller). Also wait at least 1 second after a servo reset cycle, to enable the servo.
   b.) The controller circuit breaker opens

5.3.2 Test Points

None

5.3.3 Switches Axial Servo Amp Control Board – AB

None

5.4 Axial Interface Board – AB Hardware

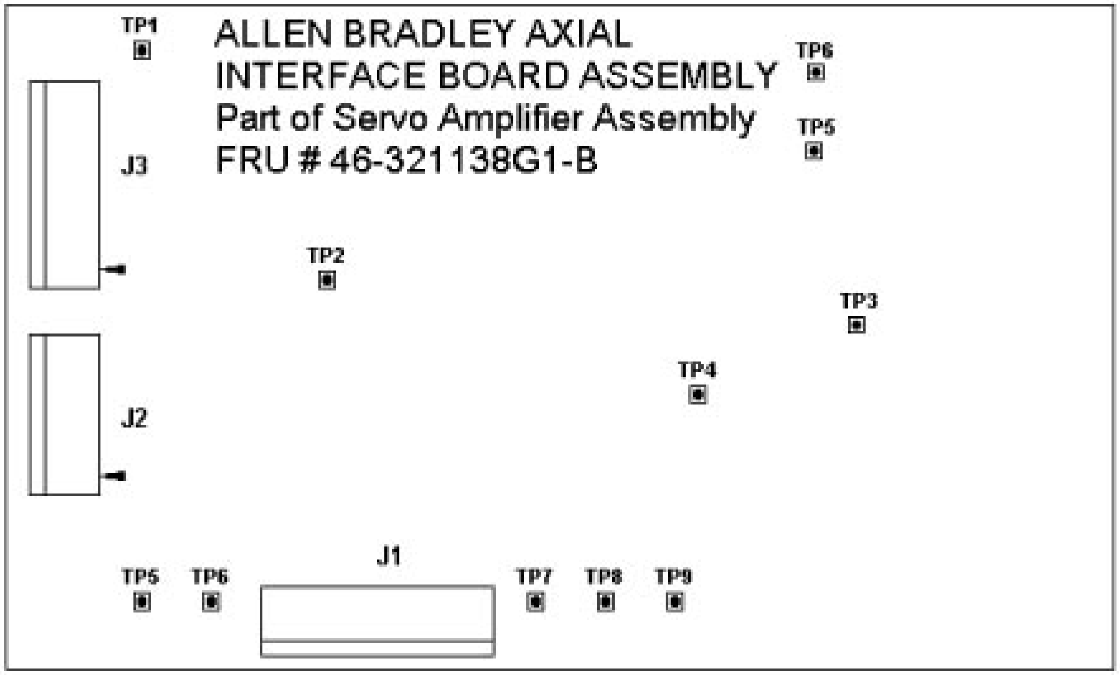


Figure 13-5 Axial Interface Board – AB

5.4.1 Test Points

- TP1 AGND: Analog Ground and Neutral of transformer input
- TP2: +12VOutput of +12 volt regulator
- TP3: -12VOutput of -12 volt regulator
- TP4 ENABLE: Collector of Enable Opto-Isolator. When the opto-isolator is off (not enabled), the voltage at TP4 should fall within 50 millivolts of the supply test point. When the opto-isolator is on (enabled), the voltage at this test point should read about 7 volts less than the supply voltage.
- TP5 I_RTN: Amplifier current output readback return
- TP6 I_OUT: Amplifier current output readback
- TP7 +15V: Output of +15 volt regulator
- TP8 -15V: Output of -15 volt regulator
- TP9 RESET: Collector of Reset Opto-Isolator. When the opto-isolator is off, the voltage at TP9 should fall within 50 millivolts of the supply voltage. When the opto-isolator is on (reset), the voltage at this test point should read about 7 volts less than the supply voltage.

5.4.2 Axial Interface Board – AB LEDs

None

5.4.3 Axial Interface Board – AB Switch Settings

None

5.5 SCR Firing Board

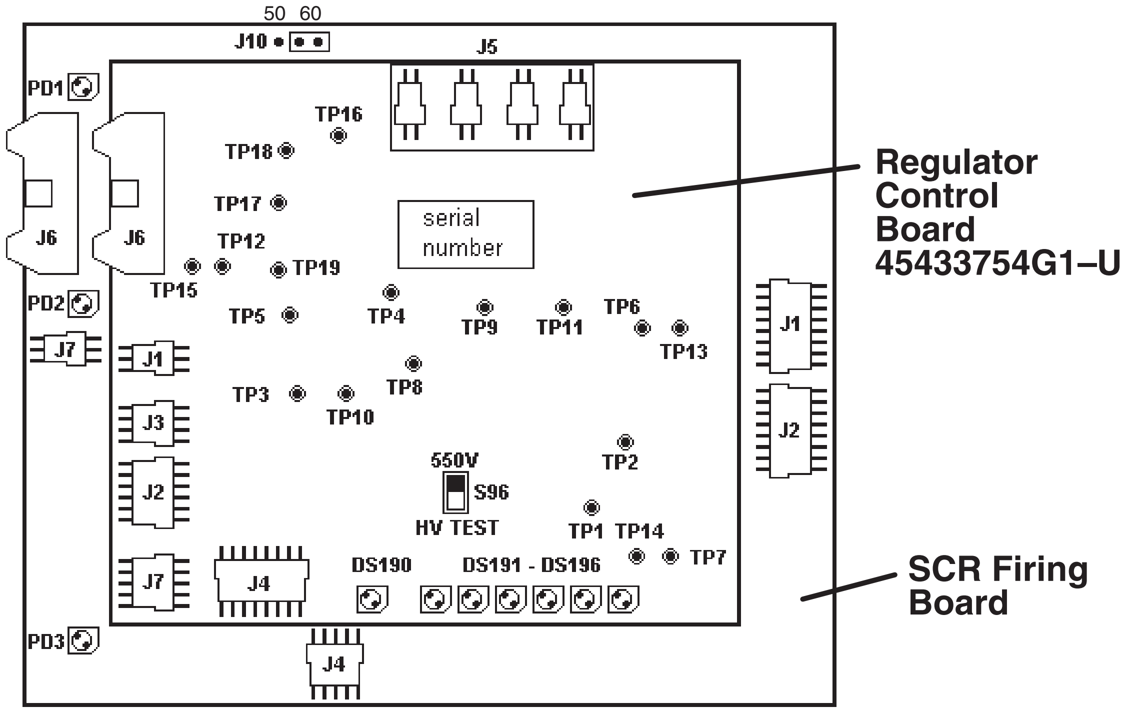


Figure 13-6 SCR Firing Board

5.5.1 SCR Firing Board Test Points

None

5.5.2 SCR Firing Board LEDs

Condition/state of LEDs with backup contactor OFF when 120 VAC power is initially applied to SCR Regulator board.

- PD1 PHASE LOSS:yellow on
- PD2 INHIBIT: yellow on
- PD3 12V POWER OK:green on

5.5.3 SCR Firing Board Switch Settings J10: 50/60Hz Jumper

CONDITIONS:

1.) Turn on 550 VDC supply back up contactor. Yellow, PHASE LOSS LED on SCR Firing Board to be “OFF” (If ON then input fuse may be open or phase sensing leads at J 7B on the SCR Firing Board may be faulty). A large number of 2M ohm resistors in phase sensing leads to J7B on the SCR Firing Board have failed open.

- All yellow LEDs must now be off.
- Once back up contactor de-energized, all yellow LEDs on the regulator board must be OFF except for the PHASE LOSS LED.

Note: The CAP UNBALANCE (DS 193), OVER CURRENT (DS 194), and UNDER VOLT (DS 195) are latched faults that are reset by back up contactor energizing.

Note: Refer to Figure 13-6 TP2 and TP3 DCRGS Control Board for the time to reach 435 volts after Phase Loss lamp turned off.

2.) During the no load condition, TP3 should have no missing pulses, and its peak to peak voltage should equal 1000 ±200mV. Refer to No load current at TP3. A full cycle contains sets of six pulses, and each set may have two pulses that are higher or lower than the other two. This trace detects open SCRs.

5.6 46-264884G1 DCRGS Control Board

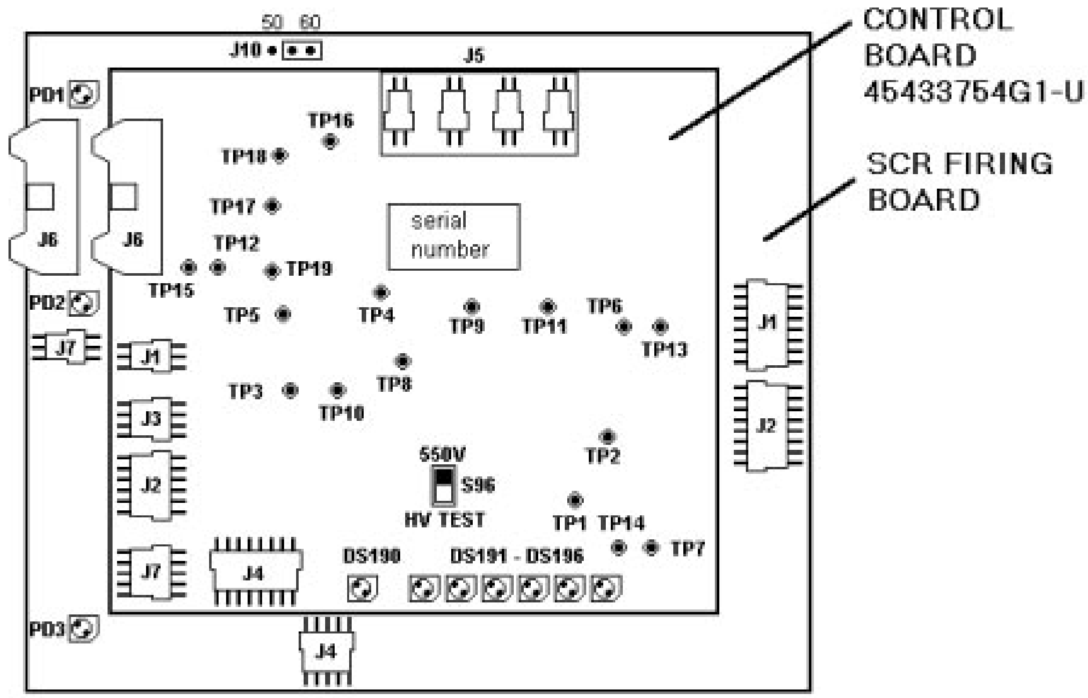


Figure 13-7 46-264884G1 DCRGS Control Board

5.6.1 DCRGS Control Board LEDs

Condition/state of LEDs with backup contactor OFF when 120 VAC power is initially applied to SCR Regulator board.

- DS190 LINE VOLTAGE LOW: yellow off
- DS191 22V POWER ON: green on
- DS192 INTLK OPEN: yellow off
- DS193 CAP UNBALANCE: yellow off (on most units)
- DS194 OVER CURRENT: yellow on (on most units)
- DS195 UNDER VOLT: yellow on (on most units)
- DS196 PHASE LOSS: yellow on

5.6.2 DCRGS Control Board Switch Settings

S96 550V/HVTEST: Normal switch position 550V

5.6.3 DCRGS Control Board Test Points

- TP1 CBV: CAPACITOR BALANCE VOLTAGE 50V/V
- TP2 DUTV: OUTPUT VOLTAGE 200V/V
- TP3 OUTCUR: OUTPUT CURRENT 10 AMPS/VOLT
- TP4: GND
- TP5 UVRST: UNDER VOLTAGE RESET
- TP6: PH-LOSS
- TP7 I2*: A low signal on this line prevents gating of the SCRs. When this signal is taken from a low to a high state the gate firing board limits the rate of delay angle change. The actual firing angle changes from 120 degrees to the final value over a specified period of time. VIL = 1.0V, Max VIH 10.0V min.
- TP8 DLYCMD: DELAY ANODE COMMAND, 33 DEG/V 120 DEG = 5.0V
- TP9: VR2.5
- TP10: VR7.5
- TP11 INPV: PEAK RECT. LINE VOLTAGE 200V/V
- TP12: 12R
- TP13: AGND
- TP14: AGND
- TP15: AGND
- TP16 +30V: +30V ±2V
- TP17 +12V: +12V ±0.5V
- TP18 +5V: +5V ±0.05V
- TP19 +22V: +22.2V ±1V with respect to TP13

5.7 CPDU Control Board (2139289)

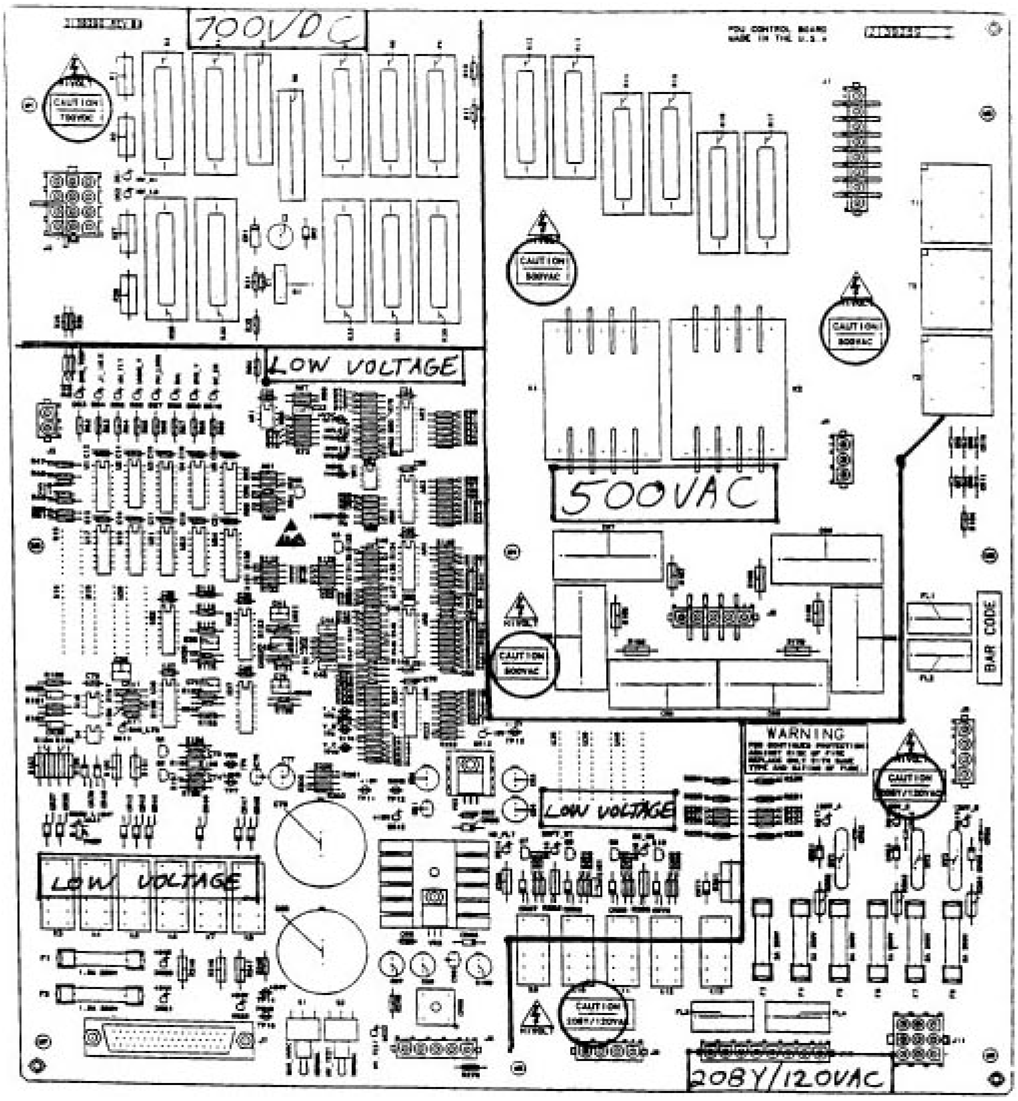


Figure 13-8 CPDU CONTROL BOARD

5.7.1 X-ray & Drives Logic

The X-Ray & Drives Logic provides Emergency Stop functions for the system. The free-running oscillator circuit of U35 drives the gantry reset pushbutton lamps through Q4. Two flashing rates are provided. The slower rate of ~20% duty cycle (~450 ms “on” / 1800 ms “off”) indicates all drives and X-Rays (HVDC Bus) have been disabled. The faster rate of ~55% duty cycle (~450 ms “on” / 350 ms “off”) indicates only the table drive has been disabled in response to a tape switch actuation. Pressing one of the reset buttons on the gantry resets either condition and enables all drives and X-Ray. When enabled, diode CR40 bypasses Q4 and turns the gantry reset lights on steady.

A new feature has been added to this version control board. JP1 has been added to allow field programming the hospital room lights to toggle with either X-ray or the HVDC Bus (Prep). The “X-Ray” position is the normal state in the U.S. The “PREP” position is provided for use in the U.K. or wherever the customer prefers.

5.7.2 HVDC Bus Contactor Control

The HVDC Bus contactors are controlled by the pilot relays K9 thru K12. K9 is normally energized and provides a master disable function whenever a board level fault is detected.

K11 provides selection between the normal HV (550-770V) bus voltage or the “HV_TEST” voltage of ~70 Vdc. (This historically has been referred to as the “50V” mode). With switch S2 in the “Normal” position, bus voltage is selected externally. S2 allows for local override to force the lower voltage “HV_TEST” mode. The circuit of U2 latches the selection once the bus is energized. While in the idle state DS23 will track the input selection. However, once the bus is energized, DS23 indicates the status of K11.

If no faults are present, a back-up contactor command from the generator causes soft-start pilot relay K10 to pull-in. This in turn causes either K1 or K2 to close and charge the main output capacitors to the appropriate level. After a delay of approximately 1.2 seconds, K12 is energized. If K11 has selected the higher voltage mode, this will energize the main back-up contactor. The coil resistance of K12 in parallel with R261 causes sufficient current in the command line (BUCSW) satisfy the current detector in the generator. A fault condition will inhibit K12 and thereby cause the generator to sense a loss of “BCSTAT”. However, depending on the nature of the fault, some HVDC Bus voltage may be present from the soft-start process and may be reported by the generator.

5.7.3 HVAC Contactors & Monitoring

Contactors K1 & K2 are used to energize the HVDC Bus. When K1 is energized, the 494Y/285 Vac source at J5 is passed through soft-start resistors R12 thru R17 to output connector J1. When K2 is energized, the 52Y/30 Vac source at J4 is passed through R13, R15, & R17 to J1. J1 is connected to the AC input of the bus bridge rectifier.

The sensing transformers T1, T2 & T3 and their output circuit provide a miniaturized representation of the bus voltage to AR4-4. If the sensed voltage does not exceed an appropriate threshold value when the control delay of 200 ms times out, a “MAINS_FAULT” is detected. This may happen if the input fuses are blown or a short circuit exists on the bus. Q3 changes the scaling at AR4-4 to accommodate the two different operating voltages. The two thresholds are representative of ~234 Vdc and ~39 Vdc on the bus.

5.7.4 HVDC Monitoring

The HVDC Bus voltage is fed back to the board at J2. Filter capacitance discharge resistors R3, R4, R29 & R30 assure a minimum discharge rate for safety purposes. However, for normal system operation, the circuit of Q1 provides 15 times faster discharge. This is necessary for adequate system response to an operator “RESUME”.

The three stages of buffer amplifier AR2 provide scaled reference signals for monitoring the bus. Scaling is 100V/V. Total voltage as well as Lo-side and Hi-side voltage is measurable at TP1, TP2, & TP3, respectively. The total voltage is compared to a fixed limit of ~830V at AR4-6. Each half is compared to a fixed limit of ~430V. Any overvoltage causes an immediate abort. In addition, the difference between Hi-side and Lo-side capacitor voltages is continuously monitored by AR3 & AR5. An imbalance greater than ~140V will cause an imbalance fault to be detected.

5.7.5 HVDC Enable Timing

At power up, R118 & C38 initiate a reset pulse to clear all fault latches. In normal operation the arrival of a “BUCSW” command also generates a reset pulse. In this way fault indicating LEDs remain lighted for serviceability until the next bus command.

Once all faults have been cleared, the slow-start timing of the HVDC Bus is controlled by the circuit of U37. The initial “Control_Delay” of 200 milliseconds disables the “MAINS_FAULT” detection until the bus can charge normally. The second stage of U37 provides an additional 1 second delay and the comparator circuit of AR1 assures the DC bus is at least 510V before the main back-up contactor is closed.

U4 and U17 provide redundant paths to disable the bus in the event of a fault.

The power transformer contains normally closed thermal cutouts in its secondary winding. These are connected in series and monitored at J3. An overtemperature will cause an immediate abort.

5.7.6 LVAC Distribution & Monitoring

208Y/120 Vac power for the system is monitored through J11. MOVs RV1, 2, & 3 provide surge and transient protection for the connected loads. Each MOV is protected by an individual fuse and DS17, 18, & 19 provide visual confirmation of an OK status.

Fuses F4, F6 and connector J6 are provided to power cabinet fans, if required. F3 and J9 provide 120Vac to the external control transformer used to power this board. A tap in its primary at 115Vac is fed back to power various contactors.

Differential buffer amplifiers, AR6-1, 7, & 8 provide scaled line voltage signals at TP8, 9, & 10. Scaling is 20V/V. All three phases are used to detect an undervoltage condition via AR5-2. If the incoming line drops uniformly below ~85% of nominal, an undervoltage signal is sent to the system via Q2.

Frequently the system will be provided with UPS back-up of the computer sub-system. In this case phases A & C will be powered by the UPS and will not be subject to interruption. However, phase B will follow the incoming wall power. AR5-1 will generate a “PH_LOSS” fault if the incoming power is interrupted. This will cause an immediate abort of the HVDC Bus and provide the generator subsystem with a loss of “BCSTAT” rather than a “KV Out of Tolerance” error.

5.7.7 Power Supplies

This board utilizes on-board power supplies for all logic and control functions. The power supplies are fed by a remote isolation transformer at J8 which provides a center-tapped 40VAC, at ~2.5 amps. The on-board supplies are split into three categories as follows:

Relay circuits:

- +24 Vdc (unregulated)
- Power Ground (PGND)

Analog Circuits:

- + / - 15 Vdc
- Signal Ground (SGND)

Digital circuits:

- +15 Vdc
- Logic Ground (LGND)

A mixture of star and daisy chain connections is used to tie all power supply grounds together. Power ground (PGND) is referenced to chassis via zero ohm resistor, R270. Signal ground (SGND) is then referenced to PGND via zero ohm resistor, R249 and logic ground (LGND) is referenced to SGND via zero ohm resister, R200.

5.7.8 Connector & Pin Assignments

J1 / P1 – 50 / 494 VAC OUTPUT

PIN#	SIGNAL	DESCRIPTION
1	ILK1	Connector interlock to J2. (Jumper required between P1-1 and P1-2.)
2	no name	1 k pull-up to VDD
3	–	No connection
4	RF	50 / 494 VAC Phase C Output
5	–	No connection
6	SF	50 / 494 VAC Phase B Output
7	–	No connection
8	TF	50 / 494 VAC Phase A Output

Table 13-4 J1 / P1 – 50 / 494 VAC Output

J2 / P2 – HVDC BUS MONITORING INPUT

PIN#	SIGNAL	DESCRIPTION
1	no name	Connector Interlock feedback. (Jumper required between P2-1 and P2-4.)
2	–	No connection
3	NCAPA	HVDC Bus negative rail, ~ -350Vdc w.r.t. gnd.
4	ILK1	Connector interlock to J1.
5	–	No connection
6	–	No connection
7	–	No connection
8	–	No connection
9	–	No connection
10	PCAPA	HVDC Bus positive rail, ~ +350Vdc w.r.t. gnd.
11	–	No connection
12	0VCAPA	HVDC Bus capacitor mid-point, (not solidly referenced to 0V)

Table 13-5 J2 / P2 – HVDC Bus Monitoring Input

J3 / P3 – TEMPERATURE SWITCH INPUT

PIN#	SIGNAL	DESCRIPTION
1	no name	1 k pull-up to VDD
2	no name	Overtemperature feedback from temperature switch(es)

Table 13-6 J3 / P3 – Temperature Switch Input

J4 / P4 – 50 VAC INPUT

PIN#	SIGNAL	DESCRIPTION
1	52VA	52 VAC Phase A Input
2	52VB	52 VAC Phase B Input
3	52VC	52 VAC Phase C Input

Table 13-7 J4 / P4 – 50 VAC Input

J5 / P5 – 494 VAC INPUT

PIN#	SIGNAL	DESCRIPTION
1	494VA	494 VAC Phase A Input
2	–	No connection
3	494VB	494 VAC Phase B Input
4	–	No connection
5	494VC	494 VAC Phase C Input

Table 13-8 J5 / P5 – 494 VAC Input

J6 / P6 – 120 VAC OUTPUT POWER TO FANS

PIN#	SIGNAL	DESCRIPTION
1	no name	120V_A for fan #1, fused at 2 amps (F4).
2	0VAC	0VAC for fan #1.
3	0VAC	0VAC for fan #2.
4	no name	120V_C for fan #2, fused at 2 amps (F6).

Table 13-9 J6 / P6 – 120 VAC Output Power to Fans

J7 / P7 – SYSTEM INTERCONNECT

PIN#	SIGNAL	DESCRIPTION
1	XRAYLITE	Switched 24V (+24A) from STC when exposure is active.
2	KNEWVAC	Switched 24V to Gantry power contactor, A3K2.
3	KNEWCONT	Switched 24V (+24A) from Gantry Service Switch.
4	REGCOM	External line regulator disable, contact common.
5	REGNO	External line regulator disable, normally open contact.
6 & 7	DRIVEON	Switched 24V (+24B) to Drives contactor, A3K1.
8	TABLEOFF	Switched 24V (+24B) from table tape switches.
9	BUCONT	Switched 24V (+24A) from OBC I/O.
10	CLSELOOP	Switched 24V (+24B) from STC.
11,12,13	FOUR	Switched 24V (+24B) from E-Stop switches.
14	LITESHI	Switched 24V (+24B) to Gantry ready lites.

Table 13-10 J7 / P7 – System Interconnect

PIN#	SIGNAL	DESCRIPTION
15,16,17	DRIVON	Switched 24V (+24B) from Gantry reset switches.
18,..,25	+24B	+24Vdc to gantry for drives circuitry. Fused at 1.5A (F1).
26,..,33	PGND	Power ground. Referenced to chassis ground via zero ohm resistor, R270.
34,..,37	+24A	+24Vdc to gantry for Exposure Control circuitry. Fused at 1.5A (F2).
38	–	No connection
39	HV_MODE_RTN	“50V” Mode select from STC.
40	HV_MODE	Pull-up from VDD to STC.
41	MAINS_UV+	Pull-up from VDD to STC.
42	MAINS_UV-	Switched pull-down to STC undervoltage receiver.
43	550 MAN	Switched 24V (+24A) to Gantry Service Switch to manually turn-on the HVDC Bus.
44	HSPRLY	Switched 24V (+24B) to room lite relay. If JP1 is in position “A”, this follows the status of “XRAYLITE”, above. If JP1 is in the “B” position, this follows the status of the HVDC Bus.
45	REGNC	External line regulator disable, normally closed contact.
46	DRRDBK	Normally closed contact to ETC drives readback sensor.
47	–	No connection
48	–	No connection
49	–	No connection
50	DRRDBKRN	Common contact to ETC drives readback sensor.

Table 13-10 J7 / P7 – System Interconnect (Continued)

J8/ P8 – 40 VAC INPUT FROM CONTROL TRANSFORMER

PIN#	SIGNAL	DESCRIPTION
1	20V_A	20VAC, ph.A from Control Transformer
2	–	No connection
3	PGND	Transformer center tap. 0V reference for all power supplies. Tied to chassis ground via 0 ohm resistor, R270.
4	–	No connection
5	20V_B	20VAC, ph.B (=-A) from Control Transformer

Table 13-11 J8/ P8 – 40 VAC Input from Control Transformer

J9 / P9 – 120 VAC POWER TO CONTROL TRANSFORMER

PIN#	SIGNAL	DESCRIPTION
1	120V_AF	120V_A to Control Transformer, fused at 2 amps (F3).
2	0VAC	0VAC to Control Transformer.
3	GND	Not used.
4	115VAC	115VAC from tap in Control Transformer Primary.

Table 13-12 J9 / P9 – 120 VAC Power to Control Transformer

J10 / P10 – CONTACTOR INTERFACE OUTPUT

PIN#	SIGNAL	DESCRIPTION
1	115VAC	115VAC to Key Switch
2	SW1	Switched 115VAC from Key Switch
3	BUA2K1	Controlled 115VAC to Back-up Contactor, A2K1.
4	0VAC	0VAC to Back-up Contactor
5	0VAC	0VAC to Axial Servo Loop Contactor, A1K2.
6	LOOPHI	Controlled 115VAC to Axial Servo Loop Contactor
7	OVRLD	From Axial Servo Overload Relay
8	115VAC	115VAC to Axial Servo Overload Relay (OVRLDHI)

Table 13-13 J10 / P10 – Contactor Interface Output

J11 / P11 – 208Y / 120 VAC INPUT

PIN#	SIGNAL	DESCRIPTION
1	0VAC	0VAC from Neutral Bus
2	–	No connection
3	120V_B	120VAC, ph B from main 30A fuse.
4	GND	Green/Yel chassis ground bond from ground bus.
5	–	No connection
6	–	No connection
7	120V_A	120VAC, ph A from main 30A fuse via UPS interface connection.
8	–	No connection
9	120V_C	120VAC, ph C from main 30A fuse via UPS interface connection.

Table 13-14 J11 / P11 – 208Y / 120 VAC Input

5.7.9 Test Points, Switches, Jumpers, Leds & Fuses

TEST POINTS

TP#	Color	Label	Description
TP1	Yel	HVDC	HVDC Bus Voltage. Scale: 100V / V
TP2	Yel	HVLO	Lo-side capacitor voltage. Scale: 100 V / V
TP3	Yel	HVHI	Hi-side capacitor voltage. Scale: 100 V / V
TP4	Blk	SGND	Signal ground.
TP5	Yel	10VREF	+10V Reference.
TP6	Yel	V_A	120 Vac, ph. A. Scale: 20 V / V
TP7	Yel	V_B	120 Vac, ph. B. Scale: 20 V / V
TP8	Yel	V_C	120 Vac, ph. C. Scale: 20 V / V
TP9	Red	VDD	+15 Vdc Logic power.

Table 13-15 CPDU CONTROL BOARD (2139289) Test Points

TP#	Color	Label	Description
TP10	Blk	LGND	Digital Logic ground.
TP11	Red	+15V	+15 Vdc Analog power.
TP12	Blk	SGND	Analog Signal ground.
TP13	Wht	-15V	-15 Vdc Analog power.
TP14	Wht	+24V	+24 Vdc Relay power.
TP15	Blk	PGND	Relay Power ground.

Table 13-15 CPDU CONTROL BOARD (2139289) Test Points (Continued)

SWITCHES

SW#	Label	Description
S1	MAN_HVDC	Manually turns the HVDC Bus on if the gantry service switch is enabled.
	NORMAL	Normal mode. Allows system to control HVDC Bus.
S2	HVDC_TEST	Forces selection of HV_TEST mode when HVDC Bus is energized.
	NORMAL	Normal mode. Allows system to control HVDC Bus mode.

Table 13-16 CPDU CONTROL BOARD (2139289) Switches

JUMPERS

PIN#	SIGNAL	DESCRIPTION
JP1	“A” (XRAY)	Programs the hospital room light to follow the system “X-RAY ON” status. This is the default shipping position.
JP1	“B” (PREP)	Programs the hospital room light to follow the HVDC Bus / Generator Ready, i.e., PREP condition.

Table 13-17 CPDU Control Board (2139289) Jumpers

LEDS

LED#	Color	Label	Description
DS1	Yel	HV_HI	Indicates the HVDC Bus Hi-side capacitance is charged.
DS2	Yel	HV_LO	Indicates the HVDC Bus Lo-side capacitance is charged.
DS3	Red	OVR_TEMP	Indicates the Over Temperature Interlock at J3 is open.
DS4	Red	INLK	Indicates the Connector Interlock at J1 or J2 is open.
DS5	Red	MN_FLT	Indicates the AC voltage at J1 was under limit after 0.5 sec.
DS6	Yel	UNDR_V	Indicates the 208Y/120 Vac is below 85% of nominal.
DS7	Red	PH_LOSS	Indicates the 120 Vac, ph. B is below 70% of nominal.
DS8	Red	BAL	Indicates the HVDC Bus Hi-side and Lo-side capacitors were not uniformly charged.
DS9	Red	OVR_V	Indicates an overvoltage condition existed on the HVDC Bus.
DS10	Yel	DC_EN	Indicates the HVDC Bus is enabled.
DS11	Grn	GAN_LTS	Indicates the Gantry ready lights are on.

Table 13-18 CPDU Control Board (2139289) LEDs

LED#	Color	Label	Description
DS12	Grn	-15V	Indicates the -15 Vdc supply is on.
DS13	Grn	+15V	Indicates the +15 Vdc supply is on.
DS14	Grn	NO_FLT	Indicates no faults have been detected.
DS15	Yel	SOFT_ST	Indicates the Soft Start / 50 V pilot relay is energized.
DS16	Yel	DC_ON	Indicates the Back-up Contactor pilot relay is energized.
DS17	Grn	120V_A	Indicates the 120 Vac, ph. A is on & fuse F5 is OK.
DS18	Grn	120V_C	Indicates the 120 Vac, ph. C is on & fuse F7 is OK.
DS19	Grn	120V_B	Indicates the 120 Vac, ph. B is on & fuse F8 is OK.
DS20	Grn	+24B	Indicates the +24 Vdc power supply is on & fuse F1 is OK.
DS21	Grn	+24A	Indicates the +24 Vdc power supply is on & fuse F2 is OK.
DS22	Grn	+24V	Indicates the +24 Vdc power supply is on.
DS23	Grn	HV_TEST	Indicates the HV_TEST mode is selected. (“50V” mode)

Table 13-18 CPDU Control Board (2139289) LEDs (Continued)

FUSE

F#	Rating	Description
F1	1.5A, 250V	+24B circuit. 24Vdc off-board to E-Stop circuitry.
F2	1.5A, 250V	+24A circuit. 24Vdc off-board to Exposure Enable/Back-up Contactor circuitry in gantry.
F3	2A, 250V	120Vac for this board's control transformer & power supplies.
F4	2A, 250V	120Vac to fan #1.
F5	8A, 250V	120Vac to “A” phase MOV, (RV1).
F6	2A, 250V	120Vac to fan #2.
F7	8A, 250V	120Vac to “C” phase MOV, (RV2).
F8	8A, 250V	120Vac to “B” phase MOV, (RV2).

Table 13-19 CPDU Control Board (2139289) Fuses

Section 6.0
Replacement Procedures

6.1 46-170026P14 32kVA Transformer, CR1-CR3

1.) Turn off facility power to PDU.

WARNING **USE TAG AND LOCKOUT PROCEDURES.**

2.) Disconnect wires from C4, C5 and C6.
3.) Remove, and keep, 3 screws from the bar.
4.) Remove screw next to the defective Varistor.
5.) Replace Varistor.
6.) Reassemble PDU and restore facility power.

6.2 46-170026P17 32 kVA Transformer, CR4-CR6

1.) Turn off facility power to PDU.

WARNING **USE TAG AND LOCKOUT PROCEDURES.**

2.) Remove screws, nuts and washers, and disconnect wires.
3.) Leave the wires attached to the bar, and remove the bar.
4.) Remove screws that fasten the defective varistor to the fuse panel.
5.) Replace Varistor.
6.) Reassemble PDU, and restore facility power.

6.3 46-296221P15 32 kVA Transformer, F10 – F12

1.) Turn off facility power to PDU.

WARNING **USE TAG AND LOCKOUT PROCEDURES.**

2.) Locate the defective fuse, and remove its bolt and washer.

NOTICE **Do not try to pry out the fuse.**

3.) Replace Fuse.
4.) Reassemble PDU, and restore facility power.

6.4 46-170021P85 32 kVA Transformer, F13

1.) Turn off facility power to PDU.

WARNING **USE TAG AND LOCKOUT PROCEDURES.**

2.) Locate the defective fuse, and remove its bolt and washer.

NOTICE **Do not try to pry out the fuse.**

3.) Replace Fuse.

4.) Reassemble PDU, and restore facility power

6.5 46-313346P1 32 kVA Transformer, F14 – F16

1.) Turn off facility power to PDU.

WARNING **USE TAG AND LOCKOUT PROCEDURES.**

2.) Locate and pry out the defective fuse.

3.) Replace Fuse.

4.) Reassemble PDU, and restore facility power.

6.6 46-170021P86 32 kVA Transformer, F4 – F6

Same procedure as for Fuses 14 to 16.

6.7 46-170021P84 32 kVA Transformer, F7 – F9

Same procedure as for Fuses 14 to 16.

6.8 46-170021P44 32 kVA Transformer, F17 – F19 (Top Board)

1.) Turn off facility power to PDU.

WARNING **USE TAG AND LOCKOUT PROCEDURES.**

2.) Use the **fuse puller** to remove the defective fuse from its fuseholder.

3.) Replace Fuse.

4.) Reassemble PDU and restore facility power.

6.9 46-170021P96 32 kVA Transformer, F20 – F21 (Top Board)

Same procedure as for Fuses 17 to 19.
45433453 AC Inductor

1.) Turn off facility power to PDU.

WARNING **USE TAG AND LOCKOUT PROCEDURES.**

2.) **Safety Check:** Verify no voltage exists at the 3 Phase Input Terminals to the PDU Isolation Transformer Fuse Panel.

- Locate the terminal strip on the DCRGS assembly.

- Check for voltage between:
  - T1 and T2
  - T1 and ground
  - T2 and ground
  - If you measure ANY voltage, do not proceed until you remove all power to the PDU.

3.) Remove 3 leads from Backup Contactor Terminals:
- T1
- T2
- T3

4.) Remove 3 wires from SCR Bridge Terminals:
- TS1-L1
- TS1-L2
- TS1-L3

5.) Unplug connector J2 from DCRGS Regulator Control Board.
6.) Unplug Connector J7B from SCR Bridge Gate Driving Board.
7.) Loosen 4 screws and washers that fasten the defective AC Inductor to the DCRGS main panel.
8.) Replace the AC Inductor.
9.) Reassemble PDU, and restore facility power.
46-296136P1 Westamp Amplifier

1.) Turn off facility power to PDU.

WARNING USE TAG AND LOCKOUT PROCEDURES.

2.) Disconnect, and set aside, the J1 Cable.
3.) Disconnect the wires from TB-401, and move to one side.
4.) Remove 3 screws from each mounting bracket at the top of the amplifier.
5.) Loosen the 2 screws on amplifier's left side mounting bracket.
6.) Remove the 2 screws from the right side mounting bracket.
7.) Remove, and replace, the defective Amplifier.
8.) Reassemble PDU, and restore facility power

6.10 46-297910P1 Allen Bradley Servo Amplifier

1.) Turn off facility power to PDU.

WARNING USE TAG AND LOCKOUT PROCEDURES.

2.) Remove the following connectors from the axial interface PWB:
- J1
- J2
- J3

3.) Remove the 4 nuts that fasten the axial interface PWB cover to the Servo assembly.
4.) Remove the Servo assembly cover.
5.) Remove 4 position option connectors from the Logic Control PWB.
6.) Remove 3 screws that fasten the Logic Control PWB cover to the Servo panel.
7.) Loosen 3 screws that fasten the Logic Control PWB cover to the Servo amplifier.
8.) Lift and remove the Logic Control PWB cover.

9.) Remove the following connectors from the left side of the Logic Control PWB:
- TB1
- TB2

10.) Remove, and keep, the 7 leads from the terminal strip on the right side of the Servo amplifier.
11.) Locate the ground stud in the lower right corner of the Servo amplifier.
- Remove the nut attaching the ground lead to the ground stud
- Remove the ground lead from the stud.

12.) Remove, and keep, the 3 nuts that fasten the Servo amplifier to the Servo panel.
13.) Remove the defective Servo amplifier.
14.) Replace the Servo amplifier.
15.) Reassemble PDU, and restore facility power.

6.11 46-288748G1 Axial Drive Filter Board Replacement

1.) Turn off facility power to PDU.

WARNING USE TAG AND LOCKOUT PROCEDURES.

2.) Remove, and set aside, the J1 cable.
3.) Remove, and keep, 4 screws from the standoff.

Note: Spacers will fall out when you remove the screws.

4.) Replace the defective Axial Drive Filter Board.
5.) Reassemble PDU, and restore facility power

6.12 46-297803P1 PDU Fan Assembly

1.) Turn off facility power to PDU.

WARNING USE TAG AND LOCKOUT PROCEDURES.

2.) Detach power connector from Fan Assembly.
3.) Hold the Fan assembly with one hand.
Remove the 4 screws that fasten the Fan assembly to the top of the PDU
4.) Use a 7/64 hex driver, and a screwdriver to remove the four nuts and screws that fasten the defective fan to the grill.
5.) Replace fan.
6.) Reassemble PDU, and restore facility power

6.13 54261P92 Backup Contactor

1.) Turn off facility power to PDU.

WARNING USE TAG AND LOCKOUT PROCEDURES.

2.) **Safety Check:** Verify no voltage exists at the 3 Phase Input Terminals to the PDU Isolation Transformer Fuse Panel.
- Locate the terminal strip on the DCRGS assembly
- Check for voltage between:

- T1 and T2
- T1 and ground
- T2 and ground

- If you measure ANY voltage, do not proceed until you remove all power to the PDU.

3.) Remove 3 wires from Backup Contactor Input Terminals:
- L1
- L2
- L3

4.) Remove 3 wires from Backup Contactor Output Terminals:
- T1
- T2
- T3

5.) Remove, and keep, 3 screws and washers that fasten the Backup Contactor to the DCRGS Main Panel.

6.) Remove and replace the defective Backup Contactor.

7.) Reassemble PDU, and restore facility power.

6.14 53214P33 PDU Capacitor (DCRGS)

1.) Turn off facility power to PDU.

⚠ **WARNING** **USE TAG AND LOCKOUT PROCEDURES.**

2.) **Safety Check:** Verify no voltage exists at the 3 Phase Input Terminals to the PDU Isolation Transformer Fuse Panel.
- Locate the terminal strip on the DCRGS assembly.
- Check for voltage between:
  - T1 and T2
  - T1 and ground
  - T2 and ground
- If you measure ANY voltage, do not proceed until you remove all power to the PDU.

3.) Remove the 4 screws and washers on the right side of the DCRGS Panel, that fasten the swing-out panel in place.

Swing out the Main Panel.

4.) **Safety Step:** Verify no voltage exists on Capacitors C1and C2.

5.) Remove the leads from the terminals of the defective Capacitor.

6.) Loosen the Capacitor Bracket.

7.) Replace Capacitor.

Note: Ensure correct polarity of capacitor.

8.) Reassemble PDU, and restore facility power.

6.15 46-296409P1 6100 microfarad, 350V Electrolytic Capacitor

WARNING REMOVE POWER AND WAIT 45 SECONDS TO ALLOW THE CAPACITOR TO PARTIALLY DISCHARGE. WEAR SAFETY GLASSES.

1.) Turn off facility power to PDU.

WARNING USE TAG AND LOCKOUT PROCEDURES.

2.) Remove, and keep, 2 screws that fasten the cover in place.
3.) Remove, and set aside, the cover.
4.) Wear your Safety Glasses while you short the capacitor terminals with a screwdriver, to completely discharge the capacitor.
5.) Remove 2 wires from the capacitor terminals.
6.) Loosen the screw in the mounting clamp.
7.) Remove, and replace, the defective capacitor.
8.) Reassemble PDU and restore facility power.

6.16 PDU Fuse Replacement Procedures

6.17 54367P40 DCRGS Fuse

1.) Turn off facility power to PDU.

WARNING USE TAG AND LOCKOUT PROCEDURES.

2.) **Safety Check:** Verify no voltage exists at the 3 Phase Input Terminals to the PDU Isolation Transformer Fuse Panel.
   - Locate the terminal strip on the DCRGS assembly
   - Check for voltage between:
     - T1 and T2
     - T1 and ground
     - T2 and ground
   - If you measure ANY voltage, do not proceed until you remove all power to the PDU.
3.) Locate and replace defective fuse.
4.) Reassemble PDU, and restore facility power.

6.17.1 46-170021P29 F1-F3 Circuit Breaker Assembly

1.) Turn off facility power to PDU.

WARNING USE TAG AND LOCKOUT PROCEDURES.

2.) Remove, and keep, the 4 screws that fasten the cover in place.
3.) Lift off the cover, and set it aside.
4.) Gently pry defective fuse out of holder.
5.) Replace fuse.
6.) Reassemble PDU and restore facility power.

6.17.2 46-297069P1 F4-F6 Circuit Breaker Assembly

Same procedure as for F1 to F3

6.17.3 46-170021P74 10A Servo Assembly Fuse

1.) Turn off facility power to PDU.

WARNING USE TAG AND LOCKOUT PROCEDURES.

2.) Twist the fuseholder cap one-quarter turn, and pull.
3.) Remove the cap and fuse from the fuseholder.
4.) Pull the defective fuse out of the cap.
5.) Insert new fuse into cap, and replace in fuseholder.
6.) Reassemble PDU, and restore facility power.

6.17.4 46-170021P94 3A Servo Assembly Fuse

1.) Turn off facility power to PDU.

WARNING USE TAG AND LOCKOUT PROCEDURES.

2.) Twist the fuseholder cap one-quarter turn, and pull.
3.) Remove the cap and fuse from the fuseholder.
4.) Pull the defective fuse out of the cap.
5.) Insert new fuse into cap, and replace in fuseholder.
6.) Reassemble PDU, and restore facility power.

6.17.5 46-221905P2 1 Pole 15A Circuit Breaker (CB16, 18, 20, 21, 22)

1.) Turn off facility power to PDU.

WARNING USE TAG AND LOCKOUT PROCEDURES.

2.) Remove and keep the 4 screws that fasten the cover in place.
3.) Lift off the cover and set it aside.
4.) Remove the wires from the defective Circuit Breaker.
5.) Lift up the wire terminal end of the breaker, and remove the defective breaker.
6.) Replace the Circuit Breaker.
7.) Reassemble PDU and restore facility power.

6.17.6 46-221905P21 2 Pole 15A Circuit Breaker (CB14)

Same procedure as for CB16 to 22.

6.17.7 46-221905P3 1 Pole 20A Circuit Breaker (CB19)

Same procedure as for CB16 to 22.

6.17.8 46-221905P32 3 Pole 25A Circuit Breaker (CB1, 6, 11)

Same procedure as for CB16 to 22.

6.17.9 46-221905P32 3 Pole 25A CB w/ AB modification (CB1, 6, 11, 23)

Same procedure as for CB16 to 22.

6.17.10 46-221905P35 3 Pole 40A Circuit Breaker (CB23)

Same procedure as for CB16 to 22.

6.18 46-296138P1 Contactor

1.) Turn off facility power to PDU.

WARNING USE TAG AND LOCKOUT PROCEDURES.

2.) Remove 4 sets of hardware, and set aside the cover.

3.) Loosen the retaining screws, and remove the wires from the following terminals:
- 1
- 3
- 5
- 13
- 14
- A1
- A2

4.) Loosen the following terminal screws:
- 2
- 4
- 6

5.) Remove, and set aside, Contactor mounting hardware.

6.) Replace defective Contactor.

7.) Reassemble PDU, and restore facility power.

6.19 45433455 DC Inductor

1.) Turn off facility power to PDU.

WARNING USE TAG AND LOCKOUT PROCEDURES.

2.) **Safety Check:** Verify no voltage exists at the 3 Phase Input Terminals to the PDU Isolation Transformer Fuse Panel.
- Locate the terminal strip on the DCRGS assembly
- Check for voltage between:
  - T1 and T2
  - T1 and ground
  - T2 and ground
- If you measure ANY voltage, do not proceed until you remove all power to the PDU.

3.) Remove the following wires from the DCRGS assembly:
- TS1-3
- TS2-3

4.) Remove the following wires from the from SCR Bridge:

- TS2-plus
- TS2-minus

5.) Unplug connector J3 from the DCRGS Regulator Control Board.
6.) Remove 4 screws and washers that fasten the defective DC Inductor to the DCRGS Main Panel.
7.) Replace the DC Inductor.
8.) Reassemble PDU, and restore facility power.

6.20 46-296127P1 Elapsed Time Indicator

1.) Turn off facility power to PDU.

WARNING USE TAG AND LOCKOUT PROCEDURES.

2.) Remove the screw the top and bottom of the Indicator panel.
3.) Detach the Harness from the back of Panel.
4.) Remove the Indicator assembly panel from the PDU.
5.) Disconnect the wires from the back of Indicator.
6.) Remove, and keep, 3 sets of screws and nuts, that fasten the Indicator assembly to the panel.
7.) Remove the defective Indicator from the panel.
8.) Replace the Indicator.
9.) Reassemble PDU, and restore facility power.

6.21 46-327154P1 Servo Output Inductor

1.) Turn off facility power to PDU.

WARNING USE TAG AND LOCKOUT PROCEDURES.

2.) Remove, and keep, 4 screws.
3.) Remove hardware from the terminals, and push aside wire end.
4.) Replace Inductor.
5.) Reassemble PDU, and restore facility power.

6.22 46-229342P2 Green 125V Lamp Assembly (DS1-DS6)

1.) Turn off facility power to PDU.

WARNING USE TAG AND LOCKOUT PROCEDURES.

2.) Locate, and remove the Spade Terminals from the Lamp assembly
3.) Pry the retaining ring off the back of the Lamp.
4.) Reassemble PDU, and restore facility power.

6.23 46-229342P2 DS1-DS4 Lamp Panel Assembly

1.) Turn off facility power to PDU.

WARNING USE TAG AND LOCKOUT PROCEDURES.

2.) Locate, and disconnect the wires the from back of the defective Lamp.
3.) Discard the ring and defective lamp.
4.) Replace the lamp.
5.) Reassemble PDU, and restore facility power.

6.24 46-222200P1 OverLoad Relay Element

1.) Turn off facility power to PDU.

WARNING USE TAG AND LOCKOUT PROCEDURES.

2.) Remove (2) screws which fasten the heater element to the K1 relay.
3.) Remove the defective heater element.
4.) Fasten the new heater element to K1 with the screws you removed from the defective heater element.
5.) Press the K1 reset button to enable the relay.
6.) Reassemble PDU, and restore facility power.

6.25 45433456 Output Current Transformer

1.) Turn off facility power to PDU.

WARNING USE TAG AND LOCKOUT PROCEDURES.

2.) **Safety Check:** Verify no voltage exists at the 3 Phase Input Terminals to the PDU Isolation Transformer Fuse Panel.
   - Locate the terminal strip on the DCRGS assembly
   - Check for voltage between:
     - T1 and T2
     - T1 and ground
     - T2 and ground
   - If you measure ANY voltage, do not proceed until you remove all power to the PDU.
3.) Unplug connector J7 from the DCRGS Control Board.
4.) Remove the wire from DCRGS TS1-2.
5.) Remove the wire from the defective Output Current Transformer (T1).
6.) Loosen 2 sets of screws and washers that fasten the defective Transformer (T1) to the DCRGS Main Panel.
7.) Replace Transformer.
8.) Reassemble PDU, and restore facility power.

6.26 46-170021P14 F-F2 Filter PWB

1.) Turn off facility power to PDU.

WARNING USE TAG AND LOCKOUT PROCEDURES.

2.) Remove 4 sets of hardware that fasten Circuit Breaker Cover in place.
3.) Remove cover, and set aside.
4.) Gently pry defective fuse from holder.
5.) Replace fuse.
6.) Reassemble PDU and restore facility power.

6.27 46-264888G1 Relay Control Board

Note: PRECONDITION: Check the Label and follow safety instructions.

1.) Turn off facility power to PDU.

WARNING USE TAG AND LOCKOUT PROCEDURES.

2.) Wait at least 1 minute for the capacitors to discharge.
3.) Remove 4 sets of hardware that fasten the Front Cover in place.
4.) Remove, and set aside, Front Cover and hardware.
5.) Detach all connectors.
6.) Remove, and set aside, Loopback Connector.
7.) Unplug all cables.
8.) Remove 4 spacers and 7 screws that fasten the defective Relay Control Board to the Relay Control panel.
9.) Replace Relay Control board
10.) Reassemble PDU, and restore facility power.

6.28 46-186852P2 Axial Drive Relay Contactor

1.) Turn off facility power to PDU.

WARNING USE TAG AND LOCKOUT PROCEDURES.

2.) Locate the Axial Drive Relay contactor:
   - Remove leads from the 6 contact terminals.
   - Remove 2 leads from the relay coil.
   - Remove 2 leads from the auxiliary contact.
3.) Remove, and keep, the screws that fasten the base of the contactor in place.
4.) Remove, and keep, 2 screws that fasten the auxiliary contact to the main contactor body.
5.) Replace the defective Relay.
6.) Reassemble PDU, and restore facility power.

6.29 53396P01 Resistor

1.) Turn off facility power to PDU.

WARNING USE TAG AND LOCKOUT PROCEDURES.

2.) **Safety Check:** Verify no voltage exists at the 3 Phase Input Terminals to the PDU Isolation Transformer Fuse Panel.
   - Locate the terminal strip on the DCRGS assembly
   - Check for voltage between:
     - T1 and T2
     - T1 and ground
     - T2 and ground
   - If you measure ANY voltage, do not proceed until you remove all power to the PDU.

3.) Remove the 4 screws and washers on the right side of the DCRGS Panel, that fasten the swing-out panel in place.
   Swing out the Main Panel.

4.) **Safety Step:** Verify no voltage exists on Capacitors C1and C2.

5.) Remove the leads from the ends of the defective resistors.

6.) Remove, and replace, the defective resistor.

7.) Reassemble PDU, and restore facility power.

6.30 46-221454P77 25Ω 50W, 1% Wirewound Resistor

1.) Turn off facility power to PDU.

WARNING USE TAG AND LOCKOUT PROCEDURES.

2.) Locate the defective resistor:
   - Unsolder the 2 leads from the defective resistor.
   - Remove, and keep, the 2 screws that fasten the resistor in place.

3.) Replace Resistor.

4.) Reassemble PDU, and restore facility power.

6.31 46-296415P2 3Ω 300W, 10% Ribwound Resistor

1.) Turn off facility power to PDU.

WARNING USE TAG AND LOCKOUT PROCEDURES.

2.) Remove the screws, nuts and washers that fasten the wires to the defective resistor's terminals.

3.) Remove the nut from the long resistor mounting bolt, that fastens the mounting bracket to the right edge of the resistor.

4.) Remove the bolt from the resistor bracket.

Note: Pay attention to the location of the centering and insulating washers. Restore them to the same configuration when you replace the resistor.

5.) Remove the defective resistor:
   - Slide the resistor, centering washers and insulating washers to one side to remove them.
   - Note the placement of the washers.

6.) Replace Resistor.

7.) Reassemble PDU, and restore facility power.

6.32 45433454 SCR Bridge

1.) Turn off facility power to PDU.

WARNING **USE TAG AND LOCKOUT PROCEDURES.**

2.) **Safety Check:** Verify no voltage exists at the 3 Phase Input Terminals to the PDU Isolation Transformer Fuse Panel.
- Locate the terminal strip on the DCRGS assembly
- Check for voltage between:
  - T1 and T2
  - T1 and ground
  - T2 and ground
- If you measure ANY voltage, do not proceed until you remove all power to the PDU.

3.) Remove wires from the following terminals:
- TS1-L1
- TS1-L2
- TS1-L3
- TS2-plus
- TS2-minus

4.) Locate the SCR Bridge Gate Firing Board, and disconnect J4.
5.) Remove the12 sets of screws and washers that fasten the SCR Bridge to the DCRGS Main Panel.
6.) Remove, and replace, the defective SCR Bridge.
7.) Reassemble PDU, and restore facility power.

6.33 45433770 DCRGS Replacement

CAUTION **Prevent permanent damage to the static-sensitive boards. Attach the anti-static wrist strap to your wrist, and to a bare metal grounding point on the PDU before you continue.**

1.) Turn off facility power to PDU.

WARNING **USE TAG AND LOCKOUT PROCEDURES.**

2.) **Safety Check:** Verify no voltage exists at the 3 Phase Input Terminals to the PDU Isolation Transformer Fuse Panel.
- Locate the terminal strip on the DCRGS assembly
- Check for voltage between:
  - T1 and T2
  - T1 and ground
  - T2 and ground
- If you measure ANY voltage, do not proceed until you remove all power to the PDU.

3.) Disconnect Wires:
- Three wires from Back up Contactor on DCRGS ASM
- Three wires from F1, F2 and F3
- Two wires to Back up Contactor coil (A2 & A1)
- Two wires to auxiliary contact of Back up Contactor (13 & 14)

- Disconnect three input wires from 480 Volt (VAC) fuses on DCRGS ASM
- Disconnect 550 Volt Output Wire from TS1-1.
- Disconnect 550 Volt Return Wire from TS2-1.
- Disconnect 550 Volt Cable Shield from DCRGS ASM Ground Stud
- Disconnect 120 Volt (AC) Terminal from SCR Bridge (Gate Firing Bd.)
- DCRGS STC Interface Cable from DCRGS Regulator Control Bd. (J4).

4.) Loosen 8 screws that fasten the DCRGS assembly to the PDU Cabinet.

5.) Replace the DCRGS.

6.) Set the 50/60 Hz jumper at J10 of the Enerpro SCR Bridge (Gate Firing Board) to the proper position.

6.34 45433754 DCRGS Control Board

1.) Turn off facility power to PDU.

WARNING **USE TAG AND LOCKOUT PROCEDURES.**

2.) **Safety Check:** Verify no voltage exists at the 3 Phase Input Terminals to the PDU Isolation Transformer Fuse Panel.

- Locate the terminal strip on the DCRGS assembly
- Check for voltage between:
  - T1 and T2
  - T1 and ground
  - T2 and ground
- If you measure ANY voltage, do not proceed until you remove all power to the PDU.

CAUTION **Prevent permanent damage to the static-sensitive boards. Attach the anti-static wrist strap to your wrist and to a bare metal grounding point on the PDU before you continue.**

3.) Remove the 480 Volt Safety Cover from DCRGS Regulator Control Board.

4.) Unplug the following connectors from the DCRGS Regulator Control Board:

- J1
- J2
- J3
- J4
- J5
- J6
- J7

5.) Remove the 4 screws and washers that fasten the DCRGS Control Board to the SCR Bridge.

6.) Remove and replace the defective DCRGS Control Board.

7.) Reassemble PDU, and restore facility power.

Section 7.0 X-Ray Warning Light (Configurations)

The CT/i system provides for an alternate configuration of the hospital x-ray warning light. The reason for an alternate configuration exists because of Regulations for X–Ray Warning Lights in the United Kingdom. One requirements is that the System shall give a warning light (indication) when the X–Ray Tube is in a state of readiness to produce X–Rays.

CT–I systems have a jumper (JP1) on the PDU Control Board that will permit Service Personnel to select Standard Hospital Warning Light functionality (default) or the UK required Warning Light functionality. With the jumper JP1 in position A the Hospital Warning Light will be active when X–Rays are ON. And when the jumper is in position B the Warning Light will be active when the High Voltage D.C. Buss is ON as required by the UK regulations.

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Appendix A Torque

Section 1.0 Recommended Torque Wrench Practices

1.) Never use a torque wrench to loosen a tightened fastener.
Permanent damage of the internal mechanism can occur due to excessive strain.

2.) Always approach the specified torque slowly. This is not a speed wrench.
   a.) Hand location is important. Position one hand at the axis of rotation and one hand on the tool handle. This give the user stability and accurate torque repeatability.
   b.) Always approach the desired torque evenly and slowly. If the desired torque is 66 N-m on 4 bolts, then tighten each bolt 50 to 70% of desired value. Then set the wrench to the required torque and tighten slowly until the wrench “Just Clicks”.

3.) Always release the tension on the torque wrench to prevent “spring set” on adjustable or “clicker” type torque wrenches. This will ensure correct torque settings throughout the range of the tool.

4.) Always allow the tool to reach room temperature.
   - Spring tension is the basis of “Clicker” type torque wrenches.
   - A spring’s tension changes with temperature.

5.) Calibrate the tool on a regular schedule.
Follow established local calibration processes.

6.) Do not drop or shock the tool.
Internal damage can occur. Calibration should be performed to ensure accuracy.

7.) Do not attempt to straighten a bent “Beam” or non adjustable wrench. Replace it.

8.) Never use a “Universal Joint” with a torque wrench.
The angle of the universal joint can change the torque value by more than 50%.

9.) Always use the torque wrench with a 90 degree angle whenever possible.
   a.) Figure A-3 illustrates the effects not being perpendicular.
   b.) The 25 degree tilt is the physical limit of a Bondhus Ball End Hex key.
   c.) Use the specified torque value for the HV tank mounting fasteners. Do not attempt to calculate the sin angle correction.
   There is less than 2% error for up to 10 degrees of tilt from the desired angle.
   d.) Minimize the angle as much as possible.

10.) Always clean fastener threads to reduce friction.
Fasteners should thread easily using finger pressure.
   a.) Replace fasteners or clean threads using a tap or die, compressed air, brass brush.
   b.) Never use a tap to clean thread inserts. It will damage them requiring replacement.

11.) Never lubricate a fastener unless specifically instructed.
Loctite is considered in the design. It must be used when specified.

12.) Replace Nylon nuts if they are finger loose.

13.) **ALL FASTENERS HAVE A TORQUE REQUIREMENT.** DEFAULT TABLES SHOULD BE USED **ONLY** IF THE SERVICE DOCUMENTATION DOES NOT SPECIFY A TORQUE VALUE.

Section 2.0
General Torque Cross Reference

Table A-1 and Table A-2 are provided as default references only. Use the appropriate replacement procedure to verify the correct torque requirement for each specific fastener.

Note: The *Illustrated Parts List* contains Engineering drawings that should also be used as a reference. These drawings call out specific instructions as notations, where needed.

NOTICE **Use Table A-1 and Table A-2 only as a last resort. If the Service documentation does not contain specific torque values, the default values can then be assumed to apply.**

All Fasteners use either "flat and lock washers" or loctite. These items must be used as specified.

FASTENER SIZE	TOOL SIZE HEX KEY	TOOL SIZE SOCKET	TORQUE IN N-M	TORQUE IN LBF-FT	TORQUE IN LBF-IN
M3	2.5 mm	5.5 mm	1	-	8.9
M4	3 mm	7 mm	2.3	1.7	20.4
M6	5 mm	10 mm	7.9	5.8	70
M8	6 mm	13 mm	19	14	168
M10	8 mm	16 mm	38.4	28.3	-
M12	10 mm	18 mm	66.4	48.9	-
M16	14 mm	24 mm	160	117.8	-

Table A-1 Default Torque Values as Specified by GEMS CT for Lightspeed Plus and Forward

FASTENER SIZE	TORQUE IN STEEL	TORQUE IN ALUMINUM	TOOL SIZE HEX KEY	TOOL SIZE SOCKET
3/8 - 16	25 +/- 2 Lbf-Ft 33.9 +/- 2.7 N-m	20 +/- 2 Lbf-Ft 27.1 +/- 2.7 N-m	5/16	9/16
1/4 - 20	8 +/- 1 Lbf- Ft 10.85 +/- 1.36 N-m	5 +/- 0.5 Lbf-Ft 6.8 +/- 0.7 N-m	3/16	7/16
8 - 32	20 +/- 2 Lbf-In 2.26 +/- 0.23 N-m	15 +/- 2 Lbf-In 1.7 +/- 0.23 N-m	9/64	5/16

Table A-2 Default Torque Values as Specified by GEMS CT for Lightspeed QXI and Previous

Many service operations on this CT scanner require a torque wrench. The use of a torque wrench may appear complicated because there are several standards and metrics. Using conversion factors and the conversion chart below can simplify that task.

First, only use a calibrated torque wrench. Use a torque wrench that is on a Calibration schedule and is approved by GEMS-AM Service. The kit that can be used that is on a regular Calibration schedule is kit number 46-268445G1. This torque wrench kit has wrenches that measure inch pounds and foot pounds.

Second, make any necessary conversions for the torque wrench you are using.The units of measure are typically marked on most torque wrenches. To make conversions to Kgcm and Newton meters, use the following conversion table, or calculate using conversion factors.

Kilogram Centimeter (Kgcm)	Inch lbs (in-lbs)	Foot lbs (Ft-lbs)	Newton Meters (Nm)
1	0.868	-	-
2	1.74	-	-
3	2.6	-	-
4	3.5	-	-
5	4.3	-	-
6	5.2	-	-
7	6.1	-	-
8	6.9	-	-
9	7.8	-	-
10	8.7	-	0.98
20	17.4	1.4	1.96
30	26.0	2.2	2.94
40	34.7	2.9	3.92
50	43.4	3.6	4.90
60	52.0	4.3	5.88
70	60.8	5	6.86
80	69.4	5.8	7.85
90	78.1	6.5	8.83
100	86.8	7.2	9.81
200	173.6	14.5	19.61
300	260.4	21.7	29.42
400	347.2	28.9	39.23

Table A-3 Torque Conversion Cross Reference

TORQUE CONVERSION FACTORS

- To convert Kgcm to foot-lbs, multiply Kgcm by 0.07233
- To convert Kgcm to inch-lbs, multiply Kgcm by 0.8679
- To convert Kgcm to N-m, multiply Kgcm by 0.0981
- To convert N-m to inch-lbs, multiply N-m by 8.8508
- To convert N-m to foot-lbs, multiply N-m by 0.73756
- To convert foot-lbs to N-m, multiply lbf-ft by 1.3558
- To convert inch-lbs to N-m, multiply lbf-in by 0.11298

Section 3.0
Torque Formula

T = R x F x sin (angle)

Where: T = Torque in N-m

R = Distance from axis of rotation

F = Force Applied

Sin(90) = 1

From this formula we can see that it is necessary to apply the force at a 90 degree angle to the axis of rotation to achieve accurate fastener torque. This same principle can be applied when using accessories with the torque wrench. See Figure A-1 and Figure A-2.

Note: The length of a standard square drive extension has no effect on torque since it is along the axis of rotation. See Figure A-3.

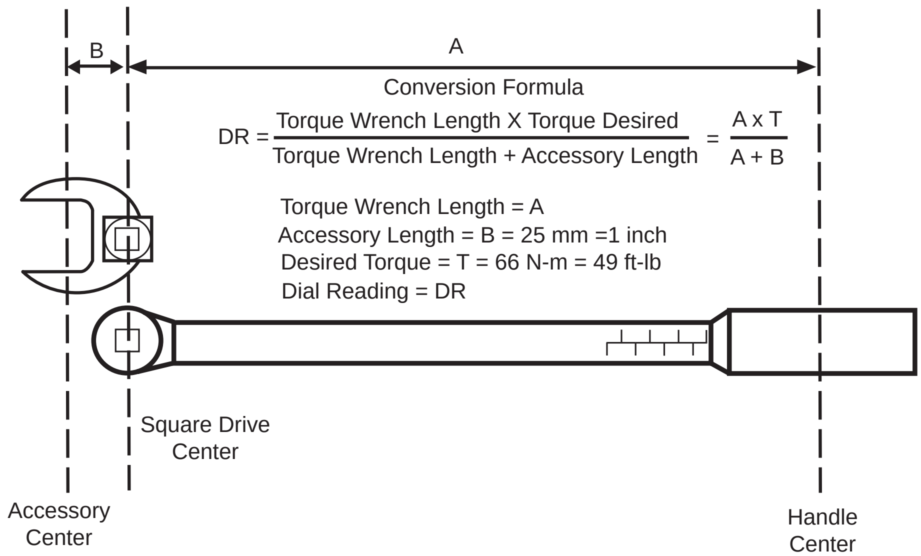

Figure A-1 Formula to adjust for Straight Line Accessory

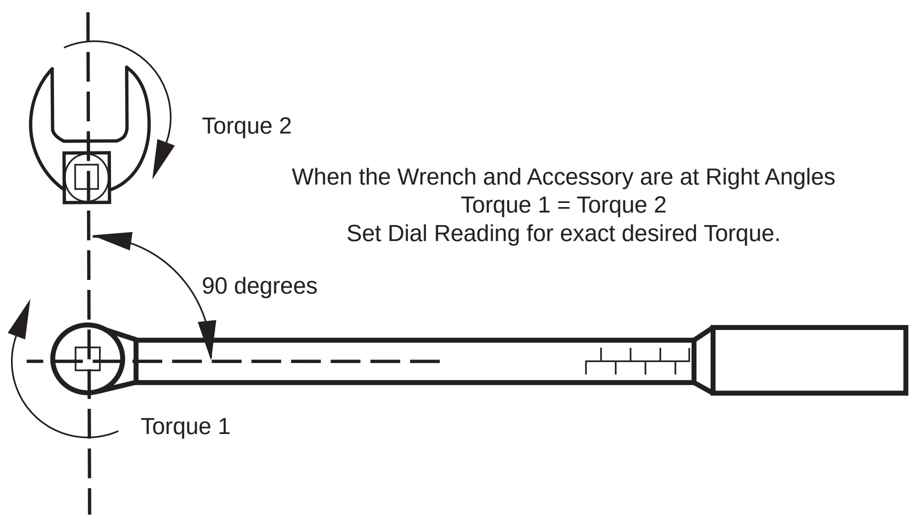

Figure A-2 Formula for 90 Degree Accessory Usage

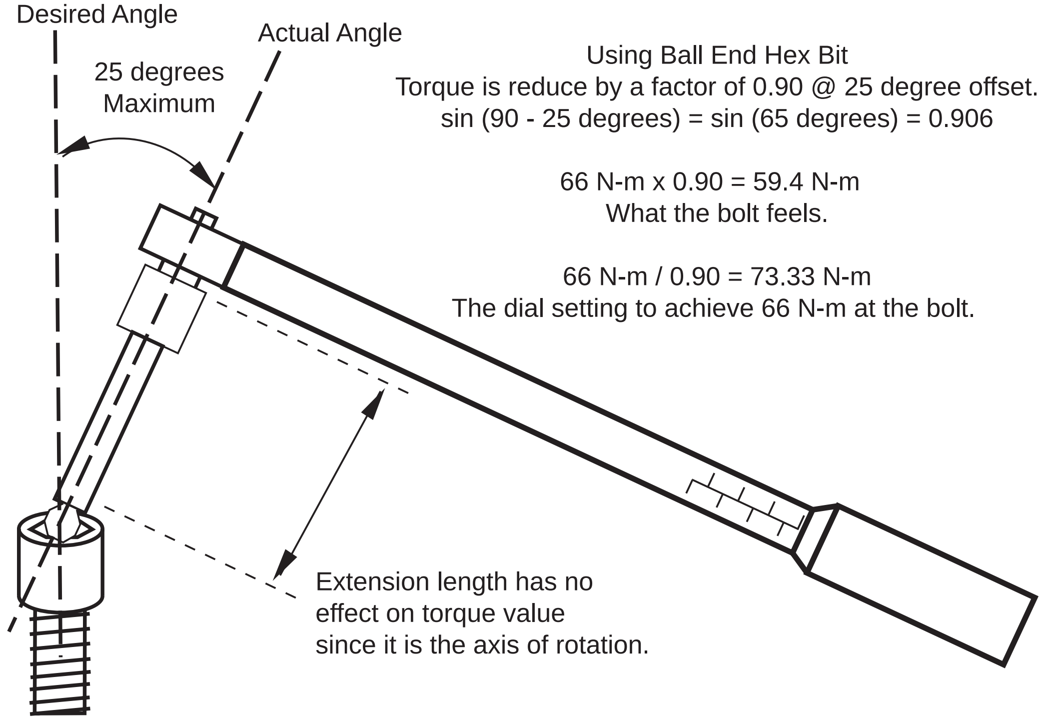

Figure A-3 Formula when not at 90 Degree to Axis of Rotation

Section 4.0
Torque Wrench Accuracy

It needs to be clearly understood that “torque” is an indirect measure of tension or “preload force”. The components of a bolted joint can be defined as,

- Preload force (Fp), bolt stretch.
- Tension force (Ft), resistance of bolted materials.
- Clamping force (Fc), difference of preload and tension forces.
- Shear force (Fs), sideways or sliding force of bolted materials.

Therefore, Fc = Fp - Ft

With shear force, a properly designed and tightened joint, the friction between the bolted materials absorbs the stress and the bolt itself feels little to no load.

There are other factors that need to be considered as well. Fastener material has a large effect on torque versus preload force. Lubricants can also significantly change the effects of torque versus preload force. Anti-seize compounds can reduce the needed torque up to 20%.

In short, torque measurement is an economical method of achieving a properly tensioned joint. Other methods are available, but training needs and tool expense increase.

CT Engineering has taken into account the variability of using torque wrenches. The design standard applied is a safety factor of 8 on all fasteners, after the “G Force” load is calculated for each component. This is to ensure clamping force is maintained without exceeding the strength of the fastener.

Various studies have been performed on the effectiveness of torque wrench accuracy. The following conclusions have been made.

PRELOAD MEASURING METHOD	ACTUAL PRELOAD FORCE ERROR
“Feeling”	> 35%
Torque Wrench	+/- 25%
Angle Torquing	+/- 15%
Indicating Washer	+/- 10%
Fastener Elongation	+/- 5%
Strain Gauge	+/- 1%

Table A-4 Torque Method Accuracy

As demonstrated in Table A-4, not using a torque wrench is the worst case event.

The “Feeling” method also changes with the tool. A ¼” drive “feels” different than a ½” drive.

Glossary

TERM	MEANING
10-BASE2 or 5 or T 100-BASE T	A 10BASE connection can transfer data between networked computers at up to 10 Mbps. 10Base2 is thin coaxial and segments must be no longer than 185 m (607 ft.). 10Base5 is thick coaxial and segments must be no longer than 500 m (1640 ft.). 10BaseT is twisted pair wiring; use Category 5 or better. The Octane can support 100BASE T which transfers at 100 Mbps.
140 SPECint92	The computer industry has developed a standard measure of integer, floating point, and other system performance to better compare actual system performance under real conditions, unlike the older MIPS or MEGAFLOPS ratings. The SPECint92 is a standard measurement of integer performance across various computing systems. The CT/i SGI host computer (200 Mhz IP22) is rated at 140 SPECint92. For comparison, the HSA Z-series computer is around 4 SPECint92 and HSA-RP 1.x/2x computer is around 12 SPECint92.
A/D, ADC	Analog to Digital Converters are used to convert real world values such as temperature, power, sound into something a computer can use. Devices that detect these entities create analog electrical signals. Transducers convert sound to electricity. Thermistors convert temperature. Analog to Digital Converters convert the analog electrical signals to digital quantities.
AE Title	Application Entity is the DICOM name for a machine with a DICOM purpose on a network. The site's network administrator assigns a specific title to each application entity. You must carefully enter this information with the same capitalization as it is given to you.
AfterGlow or Alpha-Cal	Afterglow is a correction to the output of each detector cell. The light does not completely stop between views, so a percentage of the previous reading for each cell is subtracted from the subsequent view.
AiM	Application Integration Mechanism is a simple mechanism to enable new modules to share data and messages with older modules. The idea is to limit the interactions between the to-be-integrated application and the “integrated” ones. It is proposed as an efficient way to add new applications to the SdC platform, such as an existing *Advantage Windows* system. The design paradigm of AiM is referred to as **weakly coupled** design.
Air Cals	Air Calibration This calibration is a series of scans that are taken of only air. The images are reconstructed and the CT numbers adjusted to give a number of -1000 for each pixel.
ALARA	ALARA is an acronym for As Low As Reasonably Achievable which is a safety reminder to use the least power necessary to get a diagnostically useful image.
API	Application Programming Interface is the network software libraries or subroutines from which an application writer can call upon for various services.
ARP	Address Resolution Protocol is a network protocol that maps ethernet addresses to IP ones.
ASIC	Application Specific Integrated Circuit.
AUI	Attachment Unit Interface. An IEEE 802.3 connecting the Media Access Unit (MAU) to the networked device. It also refers to the connector that attaches the host port to an AUI cable.
AW	Advantage Windows workstation is a stand alone image work station used with 5.X Signa, HLA, HSA and CT/i systems. The system is Sun Computer based and the software was developed in France.
back projection	Mathematically summing and averaging all the data for a given pixel from every CT view during acquisition. The recon processor creates three dimensional voxels in two dimensions which eventually become the display image pixels. 1: Given a filtered projection, this term refers to the process of smearing the projection back across an image matrix.

TERM	MEANING
Bit 3	The Bit 3 board is the manufacturer's name. The board is used to send data between the Silicon Graphics (SGI) Computer and the Scan Recon Computer (sbc).
BOW	Beam On Window is the alignment of the X-Ray beam to the window on the detector.
bps	bits per second
bulkhead	The bulkhead refers to a panel where peripherals, laptops, modems, network can be connected.
bus	A parallel communications pathway composed of a group of wires, or of traces on a board or within a chip. The same bus can be used for different signals when tristate IC's are used because those not needed can be turned off; their output is changed to high impedance.
byte	A byte is eight bits numbered 0 through 7, bit 0 is the least significant bit (LSB). A byte is the smallest unit stored by a computer. Its location has one unique address. The VME standard divides all locations into four groups that share the same last two digits, 00, 01, 10, 11, in their address.
cat	UNIX command used to create or print files on the screen or to a file or device
CBF	Center Body Filter This is the alignment of the focal spot of the X-Ray tube to the center of the body filter in the collimator. This is the left/right alignment when looking at the gantry from the table.
CCITT	Consultative Committee for International Telephone and Telegraph is an organization that sets worldwide voice and data communications standards.
CDROM	Compact Disk Read Only Memory is an off-the-shelf 4X CDROM drive. It is used to load software and play the Sherlock Operator's Manual.
CGI	Common Gateway Interface is an API developed for the Internet. A CGI could convert a WORD 6.0 document into a web page (HTML) or return user input on a web form to a WWW server or enable a computer to access the Internet through a firewall.
client	The computer or application that uses computer services provided by another computer or application. Each can then be optimized for their task.
CMOS	Complementary Metal Oxide Semiconductors are densely populated Integrated Circuits (IC's). They tend to need less power than TTL IC's. Nominal operating levels are 0 - 0.8 V for Low and 3.4 - 5 V for High. TTL compatible CMOS recognizes 2.4 V as High. CMOS chips are readily damaged by ESD.
control bus	A control bus carries signals used to initiate memory and data I/O operations.
CPDU	Compact Power Distribution Unit Originally this was called the CRPDU. It is a cabinet used to supply power to the entire system.
CPU	Central Processing Unit. The CPU or host is contained on the SBC board.
CQA	Customer Quality Assurance is a report by a customer to complain about the quality of a GE Medical Systems product. Strict procedures are followed to resolve the complaint to the customer's and government's satisfaction.
cron	A cron is a UNIX process that runs at regular intervals when the system is not busy with higher priority tasks. Looking for scheduled patient data on the network is a cron task.
CT/i	Computer Tomography / interactive is used to classify the current premium CT Scanner.
CTS	Serial control signal from the DCE that stands for Clear To Send.
CUP	Common Unix Platform is a foundational software library that CT and MR share. CUP monitor is used to control the most fundamental processes like the startup and shutdown of the scanner.
DA, DAC	Digital to Analog Converter
daemon	A daemon is a UNIX background software process. The routing daemon maintains a routing table or database used to select the appropriate network interface when transmitting packets. This routing table contains a single entry for each route to a specific network or host.
DAS	Data Acquisition System is used to collect the data from the detector, convert it to digital, and send it to the Front End Processor.

TERM	MEANING
DASM	The DASM is the interface to a camera for filming the images. The DASM takes a single image and transmits it digitally or in analog form depending on the type of DASM. Data Acquisition System Manager: Analogic Corporation device converts SCSI bus signals into different signal formats for interfacing to a diagnostic film system. Several forms: DASM-VDB has SCSI input, composite video/digital control signal output that connects to 3M KEIB and VEIB interfaces. DASM-LCAM has SCSI input, digital output for images and control compatible with 3M digital interface cameras.
DAT	Signal abbreviation for Data. The VME Data bus transfers are bidirectional because the Master Controller may command either a Read or Write. Other data buses are one directional and carry a circuit board's output to its destination(s). The Vector Parameter bus, aka Scan Control bus, and the I and Q Data buses are the other major DAT buses.
datagram	The smallest unit of network data
DCD	Serial control signal from the DCE that stands for Data Carrier Detect.
DCE	Data Communication Equipment is an EIA term that refers to a digital device designed to emulate or provide a transmission connection, such as a modem. RS-232 signals move in one prescribed direction relative to the DCE or DTE.
DCM	The DICOM Command Manager (DCM) is software that provides the Application Programming Interfaces (APIs) that implement DICOM tasks. A DICOM task initializes the DCM kernel on the AK server which will communicate with the remote DICOM station using DCM APIs.
DICOM	Digital Imaging and COmmunication in Medicine (DICOM) is a computer file and protocol standard used by the medical imaging industry to enable transfer of computerized data between various medical scanners and devices that use that data to analyze it, print it, store it, schedule patients, share information and remotely view diagnostic images. The practical emphasis has been on medical device manufacturers to conform so that their patient data from one particular modality are readable by computers, workstations, printers, medical scanning devices from many vendors. A DICOM task will initialize the DCM kernel on the AK server which will communicate with the remote DICOM client station using DCM APIs. The software structure that enables a communication link between a server and client is called a socket. Digital Imaging and COmmunication in Medicine is a global standard for enabling the sharing of medical images and files within a modality no matter who made the equipment.
DMA	Direct Memory Access provides fast transfers between circuit board memory and its destination. The DMA controller relieves the CPU of managing I/O operations between RAM and disk or A/D devices. It is used to transfer completed axial images from the SRC to the OC.
DNS	Domain Name Service is a software protocol that translates Internet location names which are easier to remember to their IP addresses.
domain	The domain name identifies the machine/computer on a network.
DOS MODE	MODs labeled (formatted) for storing images have a DOS like structure. MODs formatted for software have a UNIX structure. There are some DOS MODE commands in `/usr/g/bin` to help you view and copy files between the Image Archive media and the system. The size of DICOMDIR indicates how much space images are taking on the MOD. You must use Image Works to DETACH it then do another dmls in a shell to see an updated size.
DRAM	Dynamic Random Access Memory
DSP	Digital Signal Processor is an integrated circuit (IC) that performs special function digital calculations.
DSR	Serial control signal from the DCE that stands for Data Set Ready.
DTE	Data Terminal Equipment is an EIA term that refers to a digital device designed or configured to provide data, such as a computer or peripheral. RS-232 signals move in one prescribed direction relative to the DCE or DTE.
DTR	Serial control signal from the DTE that stands for Data Terminal Ready.

TERM	MEANING
ECL	Emitter Coupled Logic; a family of IC's used for high-speed signal transfer applications. It is faster than TTL. It requires voltages of -5 and -2 V which are labeled 5VN and 2VN. ECL differential signals are parallel terminated.
ecomm	ecomm is the communication layer/library used by the CTi software processes. Event router is the CTi software process that uses ecomm communications to receive then forward (route) events to registered receivers.
EFS	Extent File System (EFS) was used on R3.5 and earlier for SGI IRIX OC disks. Starting with R3.6, the OC disk uses the XFS system. The SBC disks still use EFS.
EIA	Electronic Industries Association is a US government department that provides the latest electronic related standards for engineers and manufacturers.
EMC	Electro-Magnetic Compatibility describes an electronic device that resists other and curbs its own electromagnetic influence.
EPROM	Erasable Programmable Read Only Memory uses ultraviolet light through a window on the chip to erase it.
ESD	ElectroStatic Discharge. Always use a known working (tested) wrist strap grounded to the unit before you touch any part with electronic components. There are several special grounding plugs on the frame for this. It is highlighted with a yellow icon label. Place the removed part in an anti-static bag or on a grounded pad. Protect it from further damage.
ETC	Enhanced Table Controller manages table/cradle movement and gantry tilt.
ethernet	Ethernet describes a hardware protocol for transferring data on a local area network (LAN). Ethernet cable can be coaxial, twisted pair or fiber optic.
Ethernet Address	Every system on an Ethernet network must have a unique Ethernet address. The physical Ethernet address of your system is the unique number assigned to the Ethernet board in the host. This unique number is assigned to the manufacturer of your Ethernet hardware by the IEEE. This is not to be confused with the IP address, which can be set arbitrarily.
FEP	FEP: In CT, a digital circuit board introduced with the CT/i configuration. Front End Processor collects the raw data from the DAS, and offset corrects and view compresses it for each image. It is located in the console in the chassis with the Single Board Computer. Originally the board design provided a co-axial, serial data interface for incoming DAS data. Later revisions of the board provide both a co-axial and a fiber-optic interface for incoming DAS data.
FIFO	First In, First Out; a memory device in which the first piece of data stored in the buffer is the first removed; can be used as a buffer to align outputs.
firewall	A firewall is a computer that prevents unauthorized access to the network upon which it resides. A correctly configured internal computer can reach outside the firewall. See 'proxy.'
FPGA	Field Programmable Gate Array is a standardized ASIC. It's a digital component that is designed and programmed to perform a specialized board function.
FPR	Field Problem Report is a means to formally report a potential safety or regulatory problem to headquarters.
FRU	Field Replaceable Unit A GE Acronym for items which can be replaced by field personnel.
ftp	File Transfer Protocol is a TCP/IP standard that is used to move files between computers on a network. It is particularly needed between dissimilar computers. It also describes Internet sites that use this protocol. Popular Web browsers and PC applications eliminate the need for you to know the FTP commands by simplifying the interface. They can usually be listed with the `help' command.
gateway	A gateway is a program or computer that handles moving data from one network to another. It often refers to communications between different kinds of networks. It handles client input and output for the server. The Gateway Host Name is also the AE Title.

TERM	MEANING
Gentry I/O	Generator / Gantry I/O is located in the On Board Computer Chassis. It performs miscellaneous gantry and generator functions.
GND	Ground is used both as a signal reference and a power return path.
GSB	Gantry Service Box Located on the right side of the gantry. It can be used to turn off Gantry 24 hour power, the Axial Drive and the HVDC voltage. LED's indicate status of each function.
HAS*	High Address Strobe indicates that the eight most significant bits (23:16) of an address will be transferred. Address Strobe, AS*, transfers the first 16 bits (15:0). Used to transfer VME data.
HHCS	Hex Head Cap Screw
HIS	Hospital Information System describes a computer system that retrieves and stores patient personal data and their diagnostic images on a network. Some of these HIS systems are compatible with our scanner. When the CTi host application called Work list Server, or WLServer, conforms with the HIS, then that patient data can be shared across the network.
HSD	High Speed Disk holds scan data
HSHC	Hex Socket Head Cap screw
HSSD	CTi Scan Data Disk is used for saving raw data as it comes from the DAS. Located in the console.
HTML	Hyper Text Markup Language is an Internet standard that decrees how a web page should be tagged in order to display information as intended or to go to another place on the Internet or to start a particular function. HTML is evolving. It is readable by both computers and people.
HTTP	Hyper Text Transfer Protocol is an information serving protocol that helps make the Internet possible because it is generic, stateless and object oriented means to transfer files.
hypertext	Hypertext describes the kind of information that the Internet supplies; beside text, there are sounds, voice recordings, maps, pictures, animations, videos, 3D simulations, live interactive games and conversations, links to other information sources.
ICD	Inspection Certification Document arrives with new equipment. It is used to prove the unit was tested. A SHIPMENT and INSTALLATION card accompany it. They are submitted to headquarters upon those events to track the location of the unit.
ICMP	ICMP is the error and control message protocol used by the Internet protocol family. It is used by the kernel to handle and report errors in protocol processing. It may also be accessed through a `raw socket' for network monitoring and diagnostic functions. ICMP is used internally by the protocol code for various purposes including routing, fault isolation, and congestion control. Receipt of an ICMP redirect message will add a new entry in the routing table or modify an existing one. ICMP messages are routinely sent by the protocol code.
IF or I/F	InterFace. An interface is a circuit needed to connect either two different devices or families of circuits. It solves a problem. An interface may prepare and protect circuits; it may decode, deliver, translate signals.
IG	Image Generator is used to perform convolution and back projection for reconstructing axial or helical images.
InterNIC	The Internic provides the primary directory and IP address registration services for the American part of the Internet.
IP	Internet Protocol (IP) describes globally used computer communications applications like ping, telnet, and ftp. These are not specific to Ultrasound or GE Medical Systems. IP is the internetwork datagram delivery protocol that is central to the Internet protocol family. Programs may use IP through higher-level protocols such as the Transmission Control Protocol (TCP) or the User Datagram Protocol (UDP), or may interface directly using a ``raw socket.'' `pings' have an IP and ICMP header.

TERM	MEANING
IP Address	Every computer on the Internet has a unique IP Address consisting of four 8 bit integers (bytes) separated by dots. Each part can be number from 0 to 255. One portion identifies the host and another the network. That portion can be from one to three contiguous parts. IP Address allocation is managed by a central authority.
IPC	IPC (Inter Process Communication) is the exchange of data between two software processes, either within the same computer or over a network. It implies a protocol that guarantees a response to a request. Examples are CT/i or Unix sockets, RISC OS' messages and Microsoft Windows' DDE
IRIX	IRIX is a UNIX-based operating system from Silicon Graphics (SGI) that is used in its computer systems from desktop to supercomputer. It is an enhanced version of UNIX System V Release 4. IRIX integrates the X Window system with OpenGL, creating the first real-time 3-D X environment.
ISDN	Integrated Services Digital Network is a telecommunication media that US phone companies are beginning to offer. It transfers data through existing phone lines five time faster than V.32bis modems. It is already in use in Europe.
ISO	ISO Alignment is the alignment of the focal spot of the tube to the center channel of the detector. This alignment is left/right when viewing the gantry from the table.
ISR	Interrupt Service Routines are needed in a real-time (VME) system to notify, respond, or process new conditions then get out of the way of the next interrupt. It resets a device, starts a task, reads or writes data, tells the CPU of a user request, a software error, a hardware fault.
kernel	Describes the portion of a computerized machine that controls it. Sometimes it means the hardware, the Central Processing Unit (CPU), that controls all the Input/Output (I/O) and coordinates the operation of all hardware; sometimes it means the software that does this. Since it involves both, one cannot do its job without the other, kernel really means the controlling hardware and software.
LAN	Local Area Network A network for transferring data or images that is confined to a small area. Usually within the same building.
LSB	Least Significant Bit. Bus names include the number of signals that comprise that bus. The number that appears after the colon is the LSB of that bus. The following example has eight signal lines. EXAMPLE: BUSNAME(7:0)
LSD	Local SCSI Disk 450MB hard disk used to hold the UNIX and scan recon software for the Single Board Computer. Located in the console.
LUT	Look-Up Table is memory under VME control that quickly adjusts parameters for a specific system control or performs a mathematical function via mapping.
malloc errors	This is a fatal situation for software; if it could not correctly allocate memory space for an operation, the system cannot continue.
MBD	Modem Back Door offers another way for InSite to access the scanner when the PPP connection does not work.
mean	The arithmetic average of all values in a set.
memory map	Each component on a board has its own unique address in the VME memory map. Each BE board has a range of VME addresses assigned to it. The boards reside in the VME memory map in two different areas: the short I/O space and the extended memory space.
MFM	Message Format Manager (MFM) is the AKSERVER (software) component that translates data to DICOM format so that it can be sent to another DICOM device on the network.
MG1,0	Mardi Gras 1,0 board is the display board used to setup and perform scanning. The 1 means 1 compute engine (processor), and the 0 means 0MB of texture RAM used for pixel interpolation.
MG1,1	Mardi Gras 1,1 board is the display board used to display images. The first 1 means 1 compute engine (processor), and the second 1 means 1MB of texture RAM used for pixel interpolation.

TERM	MEANING
MNP	Microcom Networking Protocol compresses uncompressed files as they are transferred through a modem.
MOD	Magneto Optical Disk is a storage device that can be recycled. It's used to store system software, files, and images.
modem	Device used to transmit digital information across phone lines. It is an abbreviation for Modulator-Demodulator.
MSB	Most Significant Bit. Bus names include the number of signals that comprise that bus. The number that appears before the colon is the MSB of that bus. The following example has eight signal lines. EXAMPLE:
MTM	Message Transfer Manager, a DICOM term
MTU	Internet datagrams can be fragmented and reassembled during their transmission. If the datagram is larger than the maximum transmission unit (MTU) of the network, it is fragmented on output.
MUX	Multiplexer selects one of multiple inputs to be routed to one output.
mv	UNIX command to move a file to another location or to rename it.
Name	For the network configuration, you must enter the DICOM Archive or Print application's name exactly as the site's network administrator has named the DICOM device so that all software on the network can recognize it properly. One device can have more than one DICOM application so there can be more than one Name and AE Title associated with any particular DICOM computer.
NDIS	Network Device Interface Specification describes 3Com and Microsoft drivers needed to make TCP/IP networking happen.
Net Mask	A Net Mask is an IP Address filter that eliminates communication/noise from network devices of no interest to your machine
Network Protocol	CT/i makes use of a Point to Point Protocol (PPP) to communicate with the OnLine Centers. The PPP allows standard TCP/IP connectivity tools to be used as if the modem connection where part of a TCP/IP based network. Multiple levels of access security are used to insure that unauthorized users cannot access the system. For PPP to work correctly, a unique IP address must be assigned to either the CT/i modem or to the SGI computer gateway.
Network Type Support	**100BASE T** describes the speed and hardware that can be used to connect computers. The Indigo2 supports either AUI or 10BASE T. It does not support 100BASE T. The Octane however supports 10BASE T and 100BASE T depending on what it senses when it boots. SGI configuration settings for networking are in file `/etc/inetd.conf`. To reset the network when applications are down: enter: **`killall -v -HUP inetd`**
NFS	Network File System describes a computer system that can use or supply other computer systems even if they are dissimilar. NFS consists of client (user) and server (supplier) systems. An NFS server can export local directories for remote clients to use. A NFS client can then use those remote files. Filesystem describes filesystems that are exported from one host and mounted on other hosts across a network. NFS enables you to access files and directories located on remote systems on the network as if they were located on your local system.
NIS	Network Information Services is an NFS service that supports distributed databases for maintaining administrative files for the network, like passwords, host addresses. Network Information Services (NIS) provides a centralized database of information about systems on the network. This service can be used to look up the hostname or IP address of a particular system on the network.
NVRAM	Non Volatile Random Access Memory is used to hold important system info.

TERM	MEANING
OBC	On Board Computer, the CPU that is on the rotating frame. It is used to monitor and control the components on the rotating frame.
OBCR	On Board Computer (Remote) Same as the OBC. Used when pinging the OBC.
OC	Operator's Console Computer is the Silicon Graphics Computer.
OE	Output Enable signal
packet	A packet is a group of binary digits representing data and control which is sent in a well defined format over a network.
Partition	A *disk partition* can be used as a file system, a logical volume, or raw disk space.
P-Cal	Phantom Cal: The phantoms are made of water (CT# 0) or teflon (CT# ~100). Large medium and small phantoms are scanned and the images generated. Then an adjustment is made to give each pixel the correct CT#. This is applied to all images scanned.
ping	A command you use to check whether another device on the network is on or reachable. Example: `ping hostname (or IP adr). You identify the network host or gateway by name or IP address. You get this information from the site's system administrator.
PLD	Programmable Logic Device is also an ASIC
POR	Plane Of Rotation This is the physical alignment of the focal spot of the tube with the aperture of the collimator. The alignment is towards or away from the table.
Port	For network configuration, enter the number that the administrator has assigned for the DICOM application.
POSIX	Portable Operating System Interface for UNIX (POSIX) is an IEEE standard that defines the language interface between application programs and the UNIX operating system. Adherence to the standard ensures compatibility when programs are moved from one UNIX computer to another. POSIX is primarily composed of features from UNIX System V and BSD UNIX.
PPP	Point to Point protocol enables a computer to access a network with a telephone, a fast modem and a service provider.
PROM	Programmable Read Only Memory is programmed by burning fusible links inside the chip. Once burned, they cannot be changed.
protocol	A recipe of software, parameters and settings that will enable two computers to communicate.
proxy	A network proxy enables a computer user to communicate across a firewall of an intranet whose access from the outside world is guarded by that firewall. Business employees need to configure their web browser software proxies for various protocols used to access Internet information is various ways, http being the most common. Home users who have an independent service provider do not need or use proxies.
Q-Cal	This is a calibration that compensates for the tungsten plates in the detector not being perfectly parallel.
Radial Alignment	Radial Alignment This is the alignment of the detector so that both ends are equidistant from the focal spot of the tube.
RAM	Random Access Memory
reconfig	A shell started program with a GUI that changes system parameters. To start Reconfig, Shutdown Applications (on Utilities Service Menu), become su at root, enter: reconfig. Make required time zone, operation, site preferences, network, hardware configuration changes with the GUI. To restart Applications, select YES to reboot prompt or enter: st&
register	A digital, electronic device for temporary storage of a value.

TERM	MEANING
repeater	A network repeater is a device to connect two or more devices to a subnet; the last port on a repeater can be used to connect multiple hubs. A repeater conditions the signal and with the hub port can extend the physical distance between devices. This is important because there are limits to how far a cable length can be effective.
REQ	Request signal
RI	Ring Indicator is a serial control signal from the DCE.
RIS	Radiology Information System describes a computer system that retrieves and stores patient personal data and their diagnostic images on a network. Some of these RIS systems are compatible with the CTi. When the CTi host application called Worklist Server, or WLServer, conforms with the RIS, then that patient data can be shared across the network.
ROI	Region of Interest
router	A router is a device that determines what path network traffic will take to reach its destination. It extends a local area network (LAN) to create a larger internetwork. It uses the routing information inside the data and the criteria programmed into it to make decisions on how to most efficiently route the data.
routine	A specialized software program or module. This system uses Activity Manager and Delivery routines.
Routing Table	A file that identifies network interfaces; it details the names and IP addresses of all the routers and gateways in the network.
RS-232	Electronic Industries Association (EIA) standard for serial data transmission that prescribes signals by voltage level and pin location.
RS-422	EIA standard for the serial exchange of digital data between two pieces of electronic equipment that uses a balanced, or differential, interface. It uses relative differences between a positive and negative signal without reference to a common ground. This enables greater speed and immunity to noise or EMI.
RTS	Serial control signal from the DTE that stands for Request To Send.
RxD	Serial data from the DCE to the DTE that stands for Received Data. It is input to the host from a peripheral or modem.
S/A	Slipring vendor short name.
SARQ	Stationary Automatic Retry Query Small board used in transmitting data across the slip rings to the rotating part of the gantry. It generates an ECC error code used to verify data integrity. It is located in the STC chassis.
sash	Standalone shell can be started from the SGI command monitor prompt, reached by interrupting the CTi boot and selecting 5. You can use sash to find and load files and devices, files outside the reach of the command monitor, the SGI PROM, meaning files in IRIX or SBC Unix.
SBC	Single Board Computer Motorola 166 CPU board. It communicates with the Silicon Graphics Computer in the console and tells the ETC, STC, and OBC what to do. It also oversees image reconstruction, then sends the completed images to the Silicon Graphics Computer.
SCP	Service Class Provider describes a DICOM task/device that allows other devices on the network to query the SCP for images or data. A SCP task listens on the specified port for the Application Entities (AE) that it has been configured to hear. SCP is like a server.
SCSI	Small Computer System Interface is a peripheral interface standard commonly used for hard disk drives and some printers to speed up data transfer.
SCU	Scan Control Unit is a term for the chassis that contains the SBC, FEP, IG and Bit 3 boards, the boards that reconstruct scan data into image files.

TERM	MEANING
SCU	Service Class User describes a DICOM task/device that uses another unit on the network to store or print images or get patient information so that the technologist does not have to key it in. SCU is like a client
SdC	Station de Consultation French for Advantage Windows workstation. Some of the software for the CT/i system was imported from these work stations so the acronym SdC is still used.
semaphore	A software object that handles device reservations for tasks.
server	A server is a computer system or application that provides the programs and disk space that a client computer or application possibly somewhere else on the network uses. The communication link between a server and client is called a socket.
SGI	Silicon Graphics Incorporated Company makes the Silicon Graphics Computer which is why it is referred to as the SGI computer.
socket	The software structure that enables a communication link between any two network computer processes, like a server and client, is called a socket. You need an IP address and a port to establish a socket. The verb 'bind' is often used in connection to socket.
Software Level - Application	Applications Level is the software level where the scanner specific software has been initialized and the system could be used to: scan, archive, display, film, etc.
Software Level - Boot	Boot level is where no software is running other than what can be run out of CPU firmware. This was often referred to as `Prom Monitor' or `Boot Prompt' or Single User Mode.
Software Level - Operating (Irix)	Operating is the software level in between `Boot Level' and `Applications Level'. This is often referred to as the Operating System level. The HiSpeed CT/i system will normally start and login as user `ctuser' leaving the User Interface ready for selection of Irix and Unix Commands or start-up of the Scanner Applications Software.
SOP	Service Object Pair, a DICOM software term. Server Object Pair is Service Class User plus Service Class Provider. Client/Server
SRC or SRU	Scan Recon Chassis, Scan Recon Computer, Scan Recon Unit describe the chassis which contains the SBC, FEP, IG and Bit 3 boards.
STC	STationary Computer used to monitor ETC and OBC status. Controls communications between the SBC and the ETC & OBC. Also monitors the axial rotation of the gantry.
subnet	A subnet is a group of connected computers or hosts. The network portion of their IP addresses would match, but the host portion would be unique.
System State	Program available on Service PM menu that enables you to save and restore protocols, calibration, configuration, Auto Voice, Display Preferences, and characterization of the Table, Gantry, and InSite features. This should be done with a Max Optics MOD. Mark this MOD so that no one will use it for Image Archive. The LABEL instruction under that feature will reformat your System State MOD into a DOS MODE format, destroying it.
task	The smallest complete unit of software. A task can use and wait for system resources without explicit concern for other tasks.
TCP	Transmission Control Protocol (software) assumes the datagram service it is layered above is unreliable. A checksum over all data helps TCP implement reliability. Using a window-based flow control mechanism that makes use of positive acknowledgements, sequence numbers, and a retransmission strategy, TCP can usually recover when datagrams are damaged, delayed, duplicated or delivered out of order by the underlying communication medium. If the local TCP receives no acknowledgements from its peer for a period of time, as would be the case if the remote machine crashed, the connection is closed and an error is returned to the user. If the remote machine reboots or otherwise loses state information about a TCP connection, the connection is aborted and an error is returned to the user.
TCP/IP	Transmission Control Protocol/Internet Protocol is a common standard for transferring data across the Internet.

TERM	MEANING
telnet	Telnet is another TCP/IP standard; telnet is a protocol that enables your computer to logon to a remote computer and query that computer for its information or use its programs.
Termination	Termination is required at both ends of a SCSI bus.
TRAM	Texture Random Access Memory. Memory on the MG1,0 and MG1,1 boards used to perform pixel interpolations and hold same image data.
tristate	Describes electronic device whose output may be HIGH, LOW, or high impedance meaning not driven This makes it possible to use the same bus for different purposes. It also is used as a verb to mean to disconnect the unused circuitry by making it's connection high impedance.
TTL	Transistor to Transistor Logic is low with voltage levels from 0 to 0.8 V, and high at levels of 2.4 to 5 V. This is also called Vcc, digital logic, and 5V.
TxD	Transmitted Data, serial data from the DTE to DCE. It is serial data from the host to a peripheral or modem.
udp	user datagram protocol, a network term
UID	Unique IDentifier
URL	Uniform Resource Locator is a way to define a resource location on a network. It describes the type of service (http, ftp, or telnet, and its exact location by network, if different, its directory and its file name. Format: protocol://computer[:port]/path/filename Example: http://www.microsoft.com
V.32	A CCITT standard for 4800 and 9600 baud modem communications. V.32 modems transfer data at 9600 bps unless phone line quality is bad. Until it improves, the modem transfers at 4800 bps.
V.32bis	A CCITT standard for modem communications that extends the V.32 connection rate range in the following steps: 4800, 7200, 9600, 12 k, and 14.4 k bps. These modems fall back one speed at a time as phone line quality worsens, or up one as it improves.
V.34	A CCITT standard for modem communications that extends the V.32 connection rate to 28.8 k bps. With data compression, this rate can theoretically go to 115.2 kbps but the condition of most phone company links prevents that from happening. This standard was previously known as V.Fast and V32terbo.
V.42	A CCITT standard for modem communication that improves throughput by correcting errors and compressing data
V.Everything	A CCITT standard for modem communications that improves throughput by adapting to the modem to which it connects and using optimal protocols.
VLSI	Very Large Scale Integration of electronic circuits on one chip.
VME ASIC	The SBC CPU has a master ASIC that implements the VMEbus interface standard. It contains a DMA controller, local and global interrupt handlers, and the VMEbus R/W logic. The other boards have a slave VME Interface ASIC to communicate with that master ASIC.
VME_ADR	Thirty-one lines of three state driven one directional signals that identify the devices that will receive or place data on the bus. All devices are memory mapped.
VME_AM	Address Modifier; a VMEbus signal that broadcasts information about the address during the address load cycle such as whether it is short (16 bits), standard (24), or extended (32 bits long). It can be used to identify a sequential transfer which is not to be interrupted until the entire data block is transferred. Six lines are reserved for this purpose.
VME_AS*	Address Strobe is a three state driven signal whose falling edge indicates the master has placed a stable, valid address and modifier onto the bus. Besides ADR, an address consists of AM, LWORD*, and IACK*.
VME_BERR*	VMEbus Error is generated by any slave board if the data size is wrong or an error occurred in a transfer; it is generated by the CPU bus timer if a data transfer fails to occur.

TERM	MEANING
VME_DAT	Thirty-two lines of three state driven bidirectional data used to transfer information between the CPU and the other boards on the VMEbus.
VME_DS0* or 1	Data Strobe is a high current, three state VMEbus signal driven by the VME host and interrupt handlers. The falling edge of a Data Strobe informs when data should be read or written. When combined with LWORD and ADR01, they also indicate the size and type of data transfer.
VME_DTACK*	Data Transfer Acknowledge signal is driven low by a slave or interrupter. During a write cycle, DTACK* is asserted after the slave has received data on the bus. During a read or interrupt acknowledge cycle, it is asserted to tell the master it has placed the requested data on the bus.
VME_IACK*	Interrupt acknowledgement is accomplished by a VME daisy chain. IACK jumpers should be open or removed if there is a board in its associated Back End slot. One must be installed to continue the interrupt path if there is no board in a slot.
VME_IRQ0n*	Interrupt Requests; see ISR also. These seven lines are monitored by the MVME166 for signals from the other boards that indicate that an I/O process is waiting, that no device responded to a command, that a voltage or output is wrong. The highest numbered request line has the highest priority. Software assigns the priorities and what appropriate routine should be implemented.
VME_LWORD*	Long WORD select is a three state VMEbus address signal driven low by the VME host and used with ADR01, DS0*, DS1* to indicate a 32-bit data transfer.
VME_SYSRESET*	A control signal that resets every board. This happens when the unit is powered ON, or the RESET switch on the CPU is pressed.
VMEbus	VersaModule Eurocard bus; an IEEE backplane standard that prescribes how data transfers will be managed. The VMEbus can handle 8, 16, and 32 bit transfers. It has multiprocessing and interrupt capability. The maximum data transfer rate is 40 MB/sec.
X Window	X Window is a windowing system developed at MIT, which runs under all major operating systems. X lets users run applications on other computers in the network and view the output on their own screen.
xfs	Starting with R3.6, the host uses the XFS filesystem rather than EFS. XFS uses database journaling technology to provide high reliability and rapid recovery. Recovery after a system crash is completed within a few seconds, without the use of a filesystem checker such as the fsck command. Recovery time is independent of filesystem size. XFS is designed to be a very high performance filesystem. Under certain conditions, throughput exceeds 100 MB per second.
Y/C	An abbreviation for a composite video signal that carries color, sync and brightness information. The Y portion carries the sync and brightness and can be used for black and white as well as color video. It is called luminance. It is formed by combining the Red, Green, Blue signals from their source in this proportion: Y=0.59G+0.3R+0.11B. The C signal carries color information that is called chrominance or chroma. It synchronizes with the horizontal frequency.
Z-Alignment	After changing a tube, both the BOW (beam on window) and POR (plane of rotation) need to be done. Since the collimator & detector have not changed position, the X-Ray tube only needs to be adjusted toward or away from the table. (Assumes the collimator & detector are in the correct position.) The Z-Align can do this with one adjustment instead of two.

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GE MEDICAL SYSTEMS

GE MEDICAL SYSTEMS-AMERICAS: FAX 414.544.3384
P.O. BOX 414; MILWAUKEE, WISCONSIN 53201-0414, U.S.A.

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